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By

AKHILESH MANI & RUCHIRA AHLAWADI

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PUBLISHERS' NOTE

The present work is a humble attempt to satisfy the needs of students appearing for UGC NET/SET Examinations. Syllabus and Guidelines for Paper II have been given in the book. After theory and MCQs, model papers are provided for practice. We have also incorporated questions asked in previous NET examinations. The book in the present form shall serve as a reliable evocative book for those preparing for NET/SET examinations.

Any suggestions for the improvement shall be thankfully acknowledged.

— With Best Wishes

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UGC-NET NEW SYLLABUS

The Test will consist of two papers as given below:

Paper	Marks	Number of Questions	MCQs	Duration
I	100	50	The questions will be generic in nature, intending to assess the teaching/research aptitude of the candidate. It will primarily be designed to test reasoning ability, comprehension, divergent thinking and general awareness of the candidate.	03 hours without any break. All the questions will be compulsory.
II	200	100	This will be based on the subject selected by the candidate.	

The Test will be conducted in Computer Based Test (CBT) mode only.

The candidates who desire to appear in the test may see the detailed **Information Bulletin** for UGC-NET which will be available on the website www.nta.ac.in and www.ntanet.nic.in.

SYLLABUS

1. EMERGENCE OF PSYCHOLOGY

Psychological thought in some major Eastern Systems: Bhagavad Gita, Buddhism, Sufism and Integral Yoga. Academic psychology in India: Pre-independence era; post-independence era; 1970s: The move to addressing social issues; 1980s: Indigenization; 1990s: Paradigmatic concerns, disciplinary identity crisis; 2000s: Emergence of Indian psychology in academia. Issues: The colonial encounter; Post colonialism and psychology; Lack of distinct disciplinary identity.

Western: Greek heritage, medieval period and modern period. Structuralism, Functionalism, Psychoanalytical, Gestalt, Behaviorism, Humanistic-Existential, Transpersonal, Cognitive revolution, Multiculturalism. Four founding paths of academic psychology - Wundt, Freud, James, Dilthey. Issues: Crisis in psychology due to strict adherence to experimental-analytical paradigm (logical empiricism). Indic influences on modern psychology.

Essential aspects of knowledge paradigms: Ontology, epistemology, and methodology. Paradigms of Western Psychology: Positivism, Post-Positivism, Critical perspective, Social Constructionism, Existential Phenomenology, and Co-operative Enquiry. Paradigmatic Controversies. Significant Indian paradigms on psychological knowledge: Yoga, Bhagavad Gita, Buddhism, Sufism, and Integral Yoga. Science and spirituality (*avidya* and *vidya*). The primacy of self-knowledge in Indian psychology.

2. RESEARCH METHODOLOGY AND STATISTICS

Research: Meaning, Purpose, and Dimensions.

Research problems, Variables and Operational Definitions, Hypothesis, Sampling.

Ethics in conducting and reporting research

Paradigms of research: Quantitative, Qualitative, Mixed methods approach

Methods of research: Observation, Survey [Interview, Questionnaires], Experimental, Quasi-experimental, Field studies, Cross-Cultural Studies, Phenomenology, Grounded theory, Focus groups, Narratives, Case studies, Ethnography

Statistics in Psychology: Measures of Central Tendency and Dispersion. Normal Probability Curve. Parametric [t-test] and Non-parametric tests [Sign Test, Wilcoxon Signed rank test, Mann-Whitney test, Kruskal-Wallis test, Friedman]. Power analysis. Effect size.

Correlational Analysis: Correlation [Product Moment, Rank Order], Partial correlation, multiple correlation.

Special Correlation Methods: Biserial, Point biserial, tetrachoric, phi coefficient.

Regression: Simple linear regression, Multiple regression.

Factor analysis: Assumptions, Methods, Rotation and Interpretation.

Experimental Designs: ANOVA [One-way, Factorial], Randomized Block Designs, Repeated Measures Design, Latin Square, Cohort studies, Time series, MANOVA, ANCOVA. Single-subject designs.

3. PSYCHOLOGICAL TESTING

Types of tests

Test construction: Item writing, item analysis

Test standardization: Reliability, validity and Norms

Areas of testing: Intelligence, creativity, neuropsychological tests, aptitude, Personality assessment, interest inventories

Attitude scales – Semantic differential, Staples, Likert scale.

Computer-based psychological testing

Applications of psychological testing in various settings: Clinical, Organizational and business, Education, Counseling, Military. Career guidance.

4. BIOLOGICAL BASIS OF BEHAVIOR

Sensory systems: General and specific sensations, receptors and processes

Neurons: Structure, functions, types, neural impulse, synaptic transmission. Neurotransmitters.

The Central and Peripheral Nervous Systems – Structure and functions. Neuroplasticity.

Methods of Physiological Psychology: Invasive methods – Anatomical methods, degeneration techniques, lesion techniques, chemical methods, microelectrode studies. Non-invasive methods – EEG, Scanning methods.

Muscular and Glandular system: Types and functions

Biological basis of Motivation: Hunger, Thirst, Sleep and Sex.

Biological basis of emotion: The Limbic system, Hormonal regulation of behavior.

Genetics and behavior: Chromosomal anomalies; Nature-Nurture controversy [Twin studies and adoption studies]

5. ATTENTION, PERCEPTION, LEARNING, MEMORY AND FORGETTING

Attention: Forms of attention, Models of attention

Perception:

Approaches to the Study of Perception: Gestalt and physiological approaches

Perceptual Organization: Gestalt, Figure and Ground, Law of Organization

Perceptual Constancy: Size, Shape, and Color; Illusions

Perception of Form, Depth and Movement

Role of motivation and learning in perception

Signal detection theory: Assumptions and applications

Subliminal perception and related factors, information processing approach to perception, culture and perception, perceptual styles, Pattern recognition, Ecological perspective on perception.

Learning Process:

Fundamental theories: Thorndike, Guthrie, Hull

Classical Conditioning: Procedure, phenomena and related issues

Instrumental learning: Phenomena, Paradigms and theoretical issues; Reinforcement: Basic variables and schedules; Behaviour modification and its applications

Cognitive approaches in learning: Latent learning, observational learning.

Verbal learning and Discrimination learning

Recent trends in learning: Neurophysiology of learning

Memory and Forgetting

Memory processes: Encoding, Storage, Retrieval

Stages of memory: Sensory memory, Short-term memory (Working memory), Long-term Memory (Declarative – Episodic and Semantic; Procedural)

Theories of Forgetting: Interference, Retrieval Failure, Decay, Motivated forgetting

6. THINKING, INTELLIGENCE AND CREATIVITY

Theoretical perspectives on thought processes: Associationism, Gestalt, Information processing, Feature integration model

Concept formation: Rules, Types, and Strategies; Role of concepts in thinking Types of Reasoning

Language and thought

Problem solving: Type, Strategies, and Obstacles

Decision-making: Types and models

Metacognition: Metacognitive knowledge and Metacognitive regulation

Intelligence: Spearman; Thurstone; Jensen; Cattell; Gardner; Stenberg; Goleman; Das, Kar & Parrila

Creativity: Torrance, Getzels & Jackson, Guilford, Wallach & Kogan

Relationship between Intelligence and Creativity

7. PERSONALITY, MOTIVATION, EMOTION, STRESS AND COPING

Determinants of personality: Biological and socio-cultural

Approaches to the study of personality: Psychoanalytical, Neo-Freudian, Social learning, Trait and Type, Cognitive, Humanistic, Existential, Transpersonal psychology.

Other theories: Rotter's Locus of Control, Seligman's Explanatory styles, Kohlberg's theory of Moral development.

Basic motivational concepts: Instincts, Needs, Drives, Arousal, Incentives, Motivational Cycle.

Approaches to the study of motivation: Psychoanalytical, Ethological, S-R Cognitive, Humanistic

Exploratory behavior and curiosity

Zuckerman's Sensation seeking

Achievement, Affiliation and Power

Motivational Competence

Self-regulation

Flow

Emotions: Physiological correlates

Theories of emotions: James-Lange, Canon-Bard, Schachter and Singer, Lazarus, Lindsley.

Emotion regulation

Conflicts: Sources and types

Stress and Coping: Concept, Models, Type A, B, C, D behaviors, Stress management strategies [Biofeedback, Music therapy, Breathing exercises, Progressive Muscular Relaxation, Guided Imagery, Mindfulness, Meditation, Yogasana, Stress Inoculation Training].

8. SOCIAL PSYCHOLOGY

Nature, scope and history of social psychology

Traditional theoretical perspectives: Field theory, Cognitive Dissonance, Sociobiology, Psychodynamic Approaches, Social Cognition.

Social perception [Communication, Attributions]; attitude and its change within cultural context; prosocial behavior

Group and Social influence [Social Facilitation; Social loafing]; Social influence [Conformity, Peer Pressure, Persuasion, Compliance, Obedience, Social Power, Reactance], Aggression. Group dynamics, leadership style and effectiveness. Theories of intergroup relations [Minimal Group Experiment and Social Identity Theory, Relative Deprivation Theory, Realistic Conflict Theory, Balance Theories, Equity Theory, Social Exchange Theory]

Applied social psychology: Health, Environment and Law; Personal space, crowding, and territoriality.

9. HUMAN DEVELOPMENT AND INTERVENTIONS

Developmental processes: Nature, Principles, Factors in development, Stages of Development. Successful aging.

Theories of development: Psychoanalytical, Behavioristic, and Cognitive

Various aspects of development: Sensory-motor, cognitive, language, emotional, social and moral.

Psychopathology: Concept, Mental Status Examination, Classification, Causes

Psychotherapies: Psychoanalysis, Person-centered, Gestalt, Existential, Acceptance Commitment Therapy, Behavior therapy, REBT, CBT, MBCT, Play therapy, Positive psychotherapy, Transactional Analysis, Dialectic behavior therapy, Art therapy, Performing Art Therapy, Family therapy.

Applications of theories of motivation and learning in School

Factors in educational achievement

Teacher effectiveness

Guidance in schools: Needs, organizational set up and techniques

Counselling: Process, skills, and techniques

10. EMERGING AREAS

Issues of Gender, Poverty, Disability, and Migration: Cultural bias and discrimination. Stigma, Marginalization, and Social Suffering; Child Abuse and Domestic violence.

Peace psychology: Violence, non-violence, conflict resolution at macro level, role of media in conflict resolution.

Wellbeing and self-growth: Types of wellbeing [Hedonic and Eudemonic], Character strengths, Resilience and Post-Traumatic Growth.

Health: Health promoting and health compromising behaviors, Life style and Chronic diseases [Diabetes, Hypertension, Coronary Heart Disease], Psychoneuroimmunology [Cancer, HIV/AIDS]

Psychology and technology interface: Digital learning; Digital etiquette: Cyber bullying; Cyber pornography: Consumption, implications; Parental mediation of Digital Usage.





EMERGENCE OF PSYCHOLOGY

PSYCHOLOGICAL THOUGHT IN SOME MAJOR EASTERN SYSTEMS

What do we mean by Indian Psychology?

By Indian psychology we mean an approach to psychology that is based on ideas and practices that developed over thousands of years within the Indian sub-continent. In other words, we use the word 'Indian' to indicate and honour the origin of this approach to psychology—the origin of the underlying philosophy, the conceptual framework, the methods of enquiry, and the technology of consciousness that it uses to bring about psychological change and transformation. It may be useful to make explicit that we do not use the word 'Indian' to localize or limit the scope of this approach to psychology; we do not mean, for example, 'the psychology of the Indian people', or 'psychology as taught at Indian universities'. We hold that Indian psychology as a meta-theory and as an extensive body of related theories and practices has something essential and unique to contribute to the global civilization as a whole.

It may also be useful to make explicit that this volume is not about the past, but about the present and the future. You will look in vain for chapters about the history of Indian philosophy or religion as they developed over the ages. Many such texts are already available, but this is not one of them. This volume has contributions that demonstrate how ideas and practices from the Indian tradition can be used to tackle issues in contemporary psychology and constructively inform its disciplinary practice by helping theory building and application.

Psychology as taught at present, all over the world, is still amazingly uncultural. This is rather remarkable if we consider the intensity and ease of international communications, and the fact that it is almost half a century since the political decolonization of Asia and Africa was completed. Though the large component of European and American thought

in psychology is understandable historically, it is not any longer excusable. For it is not that the rest of the world has not thought about human nature, and it is definitely not that contemporary psychology has found the one and only correct way of doing so. In this context, one could argue that Indian psychology will be relevant particularly to Asian, African, or Latin-American countries which share alternative non-Western world views about mind, psyche and various psychological phenomena such as healing, health, self, or personality; but we strongly believe that in spite of all cultural differences, there is a large common core to human nature, and that, to the extent that Indian psychology deals with that common core, it should be of interest to all members of the human family.

In short, we do not look at Indian psychology as something that belongs only to India or the past, but as a rich source of psychological insight and know-how that can be utilised to create a better future for the whole of humanity.

What the Indian Civilization can Contribute to Psychology

The unique contribution which the Indian civilization can make to modern psychology can be looked at as consisting of three distinct elements – a sophisticated and well-worked out, psychology-based meta-theoretical frame-work, a wide repertoire of psychological practices, and a rich treasury of psychological theories. These three are, obviously, closely interconnected, and it may be clear that none of them can be fully understood without a fairly complete understanding of the other two. Yet, as language is inevitably linear, we will give here a separate short introduction to each of them.

A Psychology-friendly Meta-theoretical Framework

The first major contribution the Indian civilization can make to psychology is a psychology-friendly meta-theoretical framework. To delineate the underlying theory, the basic 'paradigm' of the Indian tradition is, of course, a pretentious undertaking fraught with possibilities of error. The Indian civilization is immensely complex, and, given the abundance of different – often contrary – voices it harbours within itself, it is hard to state anything about it that cannot be contradicted with a striking counter-example. And yet, it is useful to give it a try, for the simple reason that without this background it is impossible to fully understand its psychological practices and its theories.

When one looks at the Indian civilization as it developed over the ages, it becomes quickly clear that within it there exists such a huge variety of distinct cultural traditions, that one may doubt whether it actually makes sense to speak of a single Indian tradition and whether it would not be more accurate to speak of Indian traditions in the plural. The doubt is understandable, but we would contend that in case of the Indian tradition, singularity and multiformity are not necessarily mutually exclusive. A rich variety of expressions does not preclude the possibility of a common thread, a single foundation supporting the variety, and we are inclined to think that especially in India such a common core indeed does exist. In fact, the idea of a single truth supporting a variety of manifestations is itself one of the core-characteristics of the deep view of reality that underlies the whole wide gamut of Indian traditions. One of the most-often-quoted aphorisms expressing this acknowledgment of divergent views in spite of a single underlying reality is probably: *ekam sad vipra bahudha vadanti*, which means, 'the truth is One, but the wise call it by different names'. An interesting aspect of this saying is that the differences are not described as errors: it is the wise that give different names to the

one truth. Moreover, one would miss the point if one were to take this saying as no more than a polite exhortation for religious tolerance. It rests on a deep, psychological understanding of the human condition, which says that reality as it really is, will always remain beyond our limited mental capacity to grasp, and that each individual can perceive of that reality only as much as their individual capacity and inclination will allow.

There is another ancient saying which goes a step further. It deals with the different perceptions that arise from affirmative and agnostic approaches to reality. It says – and one can immediately see how close some ancient Indian thinkers came to postmodern constructivism – that not only the name we give to an experience, but even the experience itself is determined by our 'set'. *The Taittiriya Upanisad* (2.6.1), for instance, says, *asann eva sa bhavati, asad brahmeti veda cet, asti brahmeti ced veda, santam enam tato viduh*, meaning, 'whoever envisages it as existence becomes (or realizes) it as existence, and whoever envisages it as non-being becomes (or realizes) that non-existence'. It may be noted that in the Indian tradition such differences are not attributed only to the different cultural priming; they are attributed primarily to the different type, level and quality of the internal state of the observer. And this brings us to what might well be described as the most important difference between the Indian and the Western paradigm.

The differences. Western psychology is largely confined to two dimensions which are both fully accessible to the ordinary waking consciousness – the physical and the social. Genetics, neurophysiology and the cognitive sciences are typical for sub-disciplines with a focus on the physical dimension, and the various offshoots of psychoanalysis, social constructivism and cross-cultural psychology could be considered typical for those who focus on social factors. Between the two, there is still, in spite of many attempts at 'softening' psychology, a widespread tendency to take the physical dimension more seriously than the social. Even in the field of consciousness studies, the existence of physical reality tends to be taken for granted, while the ontological 'reality' of consciousness and subjective experience is open for discussion. Their apparent existence needs some kind of justification, and both are commonly considered epiphenomenal products of material processes. Related to this, in terms of epistemology, the ordinary waking consciousness is considered the only acceptable state for the researcher to be in, and a clear rational mind is taken as the ultimate arbiter of truth. In fact, non-ordinary states of awareness are primarily associated with drugs and somewhat frivolous new-age activities. Finally, in terms of practical methodology, objectivity is taken as the ultimate ideal, and first-person, subjective observations are taken seriously only if they are embedded in statistics and third-person objective measures to counteract their inherent weaknesses. Obviously all this is a simplification and there are exceptions to this pattern – one could, for example, think of phenomenology – but still, a strong physicalist bias, an absolute faith in the ordinary waking consciousness and a total reliance on objective methods are so much part of mainstream psychology that amongst psychologists, they are commonly considered indispensable elements of the scientific method.

The intellectual tradition of India starts from radically different assumptions. Ontologically, the most fundamental reality is not matter, but spirit; or more precisely, the indivisible unity of *saccidananda*, of absolute existence, consciousness and delight. In other words, the Indian tradition includes psychological phenomena like consciousness and joy as core-elements of reality, and in fact it takes not physics, but 'knowledge of the self' (*adhyatma-vidya*) as the fundamental science. Accordingly, the possibility and cosmic importance of an absolutely silent, transcendent consciousness are hardly ever

doubted, while there are major schools of thought that do doubt the importance and even the reality of the material pole of existence. While Western science has come to terms with the fact that there are many different types of physical energies and substances, of which some are not directly perceptible by the human senses, the Indian tradition takes it for granted that there are also various types and levels of non-physical existence – entire inner 'worlds' which are not directly perceptible to the ordinary waking consciousness, but that are ontologically as real, or even more real than the ordinary physical world. These non-physical realities are considered to be intermediate planes of conscious existence between the absolute, silent consciousness of the transcendent and the apparent unconsciousness of matter. As a result, physical and social factors are accepted as part of causal networks, but not as the full story – events are thought to be influenced by a wide variety of forces that include factors belonging to non-physical realities. Similarly, epistemologically, a rational mind is appreciated and cultivated, but it is understood that there are higher sources of knowledge and the possibility of a direct, intuitive apprehension of truth. Finally, objective, sense-based knowledge is considered a minor form of knowledge (or even ignorance, *avidya*) and an immense collective effort has gone into the development of processes that can make us more open to the subtle worlds, and especially to the pre-existing inner knowledge, *vidya*.

It may be clear that these two basic views of reality lead to a very different sense of what psychology is about, how it is to be conducted, and what can be expected from it. For those under the influence of the physicalist worldview, psychology deals either with outer behaviour or with mental processes that happen within the neuro-physiological apparatus of individual human beings; even those who stress social influences, tacitly assume that such influences are transferred by physical means. It is taken for granted that consciousness, whether individually or socially determined, depends on working neural systems. Non-physical realities are illusionary and parapsychological phenomena are 'anomalous'. For an eternal soul there is no place (except as a belief of others, not as an 'objective' reality that exists in itself). Methodologically, one has to rely on statistics and sophisticated third-person methods of research. In terms of application, one aims at (behaviourally verifiable) changes in others.

For those under the influence of the Indian system, consciousness is primary. It is taken to be all-pervasive, and as existing within space and time, as well as beyond both. The borders of the individual are porous, and the individual consciousness is found to extend through space and time, to others, to all kinds of inner worlds, and even to what is beyond all manifestation. As a result, non-physical realities and parapsychological phenomena fit perfectly within this explanatory framework, and there is no difficulty accepting an eternal soul as our real self. For research in Indian psychology, sophisticated first person methods are the natural first choice. In terms of application, Indian psychology aims primarily at the mastery and transformation of oneself.

When one lists these differences in this manner, the two systems seem to belong to different worlds, and not only serious misunderstandings, but even a certain mutual distrust appears almost inevitable. Historically this has indeed been the case. In the Indian tradition, right from the *Upanisads* and the stories of the *Puranas*, the basic ontological and epistemological assumptions of modern psychology are looked at as beginners' errors, remnants of an ordinary, naive way of looking at the world that stand in the way of a deeper understanding of how the human mind, consciousness in general, and even the physical reality actually work. Seen from the other side, from the perspective of main-

stream psychology, giving up its positivist, constructivist, or agnostic assumptions looks like a return to a superstitious past, a giving up of the most valuable accomplishments of the European Enlightenment, a recipe for disaster.

Roads to reconciliation. There are several factors that may, however, help to overcome these difficulties. The first is that the inability of modern science to deal effectively with non-physical realities and 'the divine', may not be intrinsic to science as such. Future generations, who are likely to have a more globally informed cultural background, may ascribe this inability largely to the vagaries of European history. It might well be found that in the early years of modern science, Europe left these inner realms aside, not because it is intrinsically too difficult to research them in an intelligent and open-minded manner, but simply because they were too encrusted in the religious environment of the time. It is true that neither alchemy, nor the later efforts of parapsychology have led to sufficiently concrete results to convince the sceptics; but that might well be because their studies were hampered on the one side by the lack of a sufficiently supportive philosophical framework, and on the other by their failure to develop effective powers within the inner realms they purported to study.

As we will try to present in this volume, the Indian tradition might be able to provide both. Though the Indian civilization has had its own difficulties – 800 years of foreign interference not the least of them – such a dramatic split between the physical and the inner domains is not part of the Indian story. In fact, the social structures and mental attitudes supporting spiritual pursuits in India are much closer to those of European science than to those of European religion. Even Sankara – who arguably comes closest to what in the Christian tradition would have been called a church-father, given his role in founding centres of religious authority and power – in the end puts personal experience (*anubhava*) above tradition. In his *Bhagavad Gita Bhasya* he says, for example (18, 66), 'Even a hundred scriptural passages will not become authoritative when they, for instance, announce that fire is cool or dark'. The methods of yoga and meditation are nowadays primarily looked at soteriologically, that is, as a means for salvation, as a means to arrive at *samadhi* or *nirvana*—at least if they are not seen as a means to arrive at physical health and the survival of a corporate lifestyle. In the culture of origin, however, they are part of a coherent knowledge system and they are clearly looked at as a way to arrive at reliable knowledge. This is most clear in the case of *jñānāyoga* (the yoga of knowledge); but one can easily discern elements of the pursuit of truth even in *karma*- and *bhaktiyoga* (the paths of works and devotion), which also, in their own way, have methods to reduce the distortions of perception and affect that are part of the ordinary human consciousness.

The good news then is that modern scientific and ancient Indian approaches to psychology may not be so much contradictory as complementary. It is true that they are based on different ontological and epistemological assumptions, that they use different methods, and to some extent, that they look at different sides of the human enterprise, but in the end, they are based on the same human urge for true knowledge, pure love, effective power and happiness. It may not be easy to come to mutual respect and understanding, but the effort will be worth it, for our preoccupation with knowledge and power in the physical domain has not solved humanity's problems. On a global scale, suffering due to poverty, violence and disease is still rampant, and we have added a considerable risk of sudden environmental self-destruction. One could well argue that the one thing we need most at present is a more comprehensive understanding of our own nature. As

editors of this volume, we would like to argue that Indian psychology can make a valuable contribution to that endeavour.

Psychological Practices

According to a survey commissioned by the *Yoga Journal*, there were in February 2008, some 15.8 million practitioners of (*hatha*) yoga in the USA alone, and amongst the rest of the adult population, another 8 per cent, or eighteen million people, were 'very or extremely interested in yoga'. Over the years, thousands of researches on yoga and meditation have been conducted, but according to the latter, this research is as yet rather imbalanced. Most research is conducted with beginning practitioners, and the vast majority of researches have been carried out with not more than three -basic techniques – *hathayoga*, *vipassana* and *Transcendental Meditation* (TM). Almost all research is, moreover, in a mode that cultural anthropologists would call *etic*, rather than *emic*. In other words, the research is done from an outsider's, rather than from an insider's perspective; the techniques are decontextualized, and their effectiveness is measured in terms that belong to the theoretical framework of mainstream psychology. This is in itself not surprising, for measurement involves the use of standards, and in science these standards have to come from previously conducted research. But the result is that the effects of yoga and meditation have been measured almost exclusively on variables like blood pressure, anxiety, depression and extroversion, which have little to do with what would have been considered relevant in the culture of origin, such as equanimity, compassion, wisdom and detachment.

While reflecting on the scope of existing research on yoga and meditation, there is another issue that warrants careful consideration. It is true that India has developed an astounding variety of structured methods to 'do' yoga and meditation. There can also be no doubt that it is worth studying these techniques, and that one should not do this only by *etic*, but also, or even especially, by *emic* approaches. The methods of yoga should be understood on their own terms, and ideally not only in their gross 'effectiveness' but in terms of the underlying spiritual and psychological processes. But even a sympathetic, insider's look at these techniques will not give us the whole story. Amongst the Indian psychological practices that could benefit humanity, there are not only such formalised methods and techniques, but there is also an implicit, informal know-how that is orally transmitted from teacher to student within the *guru-sisya parampara* (the -master-disciple relationship), or passed down from generation to generation in the form of social institutions, customs, and culturally prescribed – but individually adopted and adapted – attitudes and inner gestures. When we look at yoga not only as a way to find the Divine but also as a way to bring our entire life more in harmony with the highest we can conceive and experientially 'realize', then it becomes clear why these informal, implicit aspects of yoga play such a big role in the Indian civilization, and why they are so interesting for modern psychology. An anecdote from E. Richard Sorenson (2008) may illustrate the point. Sorensen relates an experience he had in a Tibetan monastery where most of the monks were young, and where he had noticed earlier that the novices were always 'eagerly rushing to share whatever special tidbit [*sic*] might have come their way (whether material or ideational)'. As he relates:

One day, while having lunch with a group of novices, a burst of mirth snared my attention. An adolescent novice had just selected, as if solely for himself, the largest apple off a plate. Bursts of laughter from the others, no verbal comment, just hilarity,

as several then did much the same, usually with some special fillip or perspective of their own. There was no obligation to be either different or the same ... they were just nuzzling at a trait all had seen outside.

The interesting part is that amongst these youngsters, there were no pejorative remarks or outbursts of self-righteous indignation. Egoism was for them not something natural and tempting, yet socially unacceptable, but an utterly hilarious trait they had so far noticed only in the behaviour of people outside their own community. Presuming there is no major genetic difference in such matters, it is clearly worthwhile to study what it is exactly that made sharing the natural baseline for these children. It seems extremely unlikely that such a fundamental difference can be brought about by formal exercises or explicit instructions.

Regarding the spiritual core of the Indian psychological tradition, there is amongst professional psychologists a similar tendency to focus on formal practices and specialised techniques. Yet, in the *Yoga Sutras* of Patañjali, the undisputed authority on *rajayoga*, only one of its many slokas deals with *asanas* (yogic postures), and the *Bhagavad Gita* hardly mentions strongly structured practices at all. Even in our times, some of the greatest sages of modern India, like Ramakrishna Paramahansa, Sai Baba of Shirdi, Ramana Maharshi and Sri Aurobindo, did not advocate the use of highly structured and formalized techniques at all. They worked instead through a focussed, specialized application of – in itself quite simple – psychological processes and powers. There is an enormous variety of those, and even though all the great gurus had their own favourites – for example, Ramakrishna's absolute devotion to the Divine Mother, or Ramana's sustained and unremitting focus on the question, 'Who am I?' – they typically adjusted their method of teaching to the needs of each disciple at any given moment.

The literature contains many different lists of desirable inner attitudes and gestures. Typical examples might be: a silent, non-judgemental self-observation; a growing surrender to the highest one can conceive; a sustained aspiration towards the Divine (whether in terms of knowledge, work, love, or oneness); a systematic development of traits like equanimity, calm, patience, vigilance, kindness, compassion, love, joy, harmony, oneness, wideness; small inner gestures of self-giving, consecration, openness, silence, surrender; the relocation of the centre of one's consciousness inwards and upwards. As yet, it is hard to say with certainty, whether such non-sectarian, informal 'paths' will dominate the future of Indian psychology, or the more formalized 'techniques' that have played such a big role in the preservation of the tradition into the present. What seems clear to us is that there is an urgent need for research in both.

Psychological Theories

Indian psychology has dealt with most areas in which mainstream psychology is interested, and in many of them it has something unique to add. As we will see, there is a special, common quality to the contributions it can make to all these different fields. If we start with the structure of the personality then we find that the Indian tradition has developed the concept of *ahamkara*, which stands somewhere in between the Western concepts of ego and self-concept. But besides this egoic centre, which belongs to the ordinary waking consciousness, the Indian tradition has also developed a detailed nomenclature for many other, more subtle and non-egoic centres of consciousness; and it has even worked out, especially in certain Buddhist schools, how a consciousness can exist without any centre whatsoever. Similarly, the Indian tradition has found below the

surface of our waking consciousness not only the dark 'unconscious' that depth psychology has explored, but a whole range of subtle *kosas* or layers of consciousness, that each have their own characteristic nature. It has even worked out many different ways of 'realizing' in one's experience (or perhaps one should rather say, in one's being) a Transcendent beyond all nature. It has found that all these inner layers, types, and centres of conscious existence have their specific influences on the surface personality, and that a direct access to them can, with sufficient training, enable levels of freedom, peace, joy, compassion, and understanding much beyond what is possible in the ordinary waking state.

In the field of cognition, we see a similar pattern. On the one hand, there is a detailed theoretical understanding of ordinary, sense-based cognition, mostly described as a system of *pramana*, or knowledge-producing events. Different schools developed somewhat different theories about these matters – and they made much of their differences – but there is actually quite a large common base. It is noteworthy that the philosophical school of the *Nyaya-Vaisesika*, which specialised in issues of epistemology and methodology, came to conclusions that are similar to modern thought in many respects. However, just as we saw in the field of Self and Personality, the ordinary sense-based cognition was not enough for the Indian tradition, and it developed besides a detailed understanding of intuition, inspiration, revelation, and various other types of 'intuitive knowledge' for which there are not always equivalent terms in English. The sheer complexity of the terminology, the subtle but significant differences between the various terms, and the stress on concrete methods to develop and refine these various forms of intuitive knowledge may give an idea not only of the enthusiasm and energy with which these possibilities have been explored, but also of the rigour, precision and attention for detail with which this work was undertaken. The study, cultivation and perfection of these subtle, not sense-based forms of cognition, might well deserve to become one of the major thrust areas of Indian psychology, as their development may lead to the creation of appropriate research methodologies for a whole new field of psychology.

Another major area of interest in psychology is that of emotion and motivation. To fully understand the various Indian theories of emotion, one has to go back to what should perhaps count as one of the greatest discoveries of the Indian tradition – the idea that the nature of ultimate reality can be described as an indivisible unity of *Sat*, *Cit* and *Ananda*, or Existence, Consciousness, and Delight. While in mainstream psychology, it is generally presumed that happiness is dependent on the satisfaction of individual needs and desires, this theory asserts that delight is inherent in existence, even though it can be clouded in humans by their 'ignorance' (*avidya*). In other words, suffering is attributed to ego-bound deformations and limitations of the over-individualized human consciousness. Seen from this angle, the satisfaction of desires may give temporary relief, but the road to lasting and unconditional happiness and wisdom runs through detachment from the ego, and a rediscovery of one's knowledge of, love for, and oneness with the ultimate reality. With the ideal of perfect detachment and complete equanimity, a series of intriguing questions arises, which have occupied some of the best minds in Indian history, regarding the possibility of action for someone who has completely overcome all desire, preference and attachment. If such 'unmotivated' action is possible – and most schools of Indian thought agree that it is – then what kind of action can that be?

In the field of emotion, a special place deserves to be given to the ideas of Bharata (400–200 BCE) on aesthetic enjoyment. Bharata starts with the fascinating question,

why people enjoy watching tragic plays in spite of the fact that they already know beforehand that watching such plays will make them cry. Considering the emotions that spectators and actors suffer and savour, he arrives at the subtle theory of *rasa*, the basic 'taste', which triggers the original delight hidden in all things.

It is commonly held, especially amongst American authors (for example, Walsh & Shapiro, 2006) that there are several areas of psychology where the Indian tradition has little to contribute. Though this is in itself perfectly plausible, it does not hold for the most commonly quoted examples. In all four areas where Walsh and Shapiro, for example, think that 'the meditative traditions' have little to contribute – child-development, psychodynamics, psychopathology and psycho-pharmacology – there is in fact detailed theoretical knowledge available within Ayurvedic and Siddha literature. In all four fields, the Indian tradition has paid attention to very similar social and physical factors as are taken into account in Western psychology; but there is, besides, an additional interest in influences on more subtle planes. In child-development, for example, influences from previous lives and the unique 'soul-quality' of the child, the *svabhava*, are acknowledged as major contributors to the child's character and development. A similar multilevel understanding is part of the Indian way of looking at psychopathology, and the developmental stages later in life – the four *asramas*. The misconception that there is no Indian contribution to any of these fields, is in all likelihood due to the same peculiar way in which Western psychology has studied the Indian tradition that we mentioned earlier. Mainstream academics have either looked at the decontextualized techniques of yoga and meditation, or at the other extreme, at equally decontextualized philosophical systems. The surrounding culture, as actually practiced, and the mediating theories – which are both very well developed in India – have so far not received the attention they deserve.

Indian Psychology Applied

Psychology is very much an applied science; and fields like pedagogy, education, social work, human resource development, organisational behaviour and therapy can all be looked at as specialized fields of applied psychology. At present, practitioners in these fields often experience a certain tension between the official theory, which prescribes well-defined, explicit methods and procedures, and experience, which tells them that a more personalized, eclectic and intuitive approach works better. The informal experience seems to come closer to reality than the formal theory. Interestingly, there is substantial statistical evidence to support this view. Bruce Wampold (2001), for example, has collected massive meta-analytical data to show that hardly any of the efficacy in psychotherapy can be ascribed to specific methods and theories. According to his findings, the efficacy in therapy is almost entirely determined by factors that either belong to the environment in which the therapy takes place or that are internal to the therapist and the client. While this is hard to understand from within the medical model that is used in most mainstream therapy research, it is in full harmony with Indian psychology, where the focus is on the work people have to do on themselves. The guide—whether he plays the role of pedagogue, teacher, human resource professional or therapist—guides by example, and perhaps even by direct influence. He shows that growing up is both worthwhile and feasible; that difficult material can be learned; that it is possible to master complex social situations in a constructive manner; and that life's problems can be solved. In a very deep sense, the guide guides mainly by sharing who he is in the essence of his being and what matters is how he expresses that essence in his life. In many forms of

therapy-training this is recognized, and undergoing therapy oneself is then an essential part of the training-process. But in Indian psychology, which is built on self-knowledge rather than on knowledge of statistically generalized others, the demand for self-work, for understanding and mastering one's own self, goes further—both in breadth and in depth. In breadth, because it extends to all fields of applied psychology, and in depth because a good guide is supposed to show that it is possible, in the words of Sri Aurobindo, to 'transcend and integrate': A good guide should be able to show that it is possible to go beyond one's limitations; to live from a higher consciousness; to act from less selfish motives; to work more in harmony with the whole.

The practical application of Indian psychology differs from present-day mainstream psychology in a manner that parallels the differences we found in the various areas of theory formation. There is on the one hand the same constant attention to the multidimensional nature of the personality—a multidimensionality that is not limited to the physical and the social, but that extends to, or rather starts with, the spiritual. And there is on the other hand, a constant awareness that each individual is ultimately unique. Both attention points come together in the important concepts of *svabhava* and *svadharma*—the recognition that individuals have not only their own true nature, their own unique set of qualities, but also their own truth of action, their own rules of conduct. Both concepts are based on the underlying sense that the individual is not just a cluster of self-concepts and tendencies to behave according to pre-established patterns, but a spiritual being, a soul who has taken birth for a definite purpose, a purpose which it has to find and fulfil.

ACADEMIC PSYCHOLOGY IN INDIA

Pre-independence Era

The term Indian Psychology refers to the Psychologically relevant materials in ancient Indian thought. Usually this term does not cover modern developments in Psychology in India.

Modern Psychology at the beginning of the century emphasized sensation, perception and psychologists in India took out Indian theories of sensation and perception from the classics and created an Indian Psychology. For example Indian theories emphasise the notion that in perception the mind goes out through the senses and assumes the shape of the objects. In 1934, Jadunath Sinha wrote a book on Indian theories of perception. As soon as Western Psychologists started studying cognition, Indian Psychologists started looking for Indian theories of cognition. In 1958, Jadunath Sinha wrote a book on Cognition. Later on modern Psychology started emphasising emotions, and in 1981, Jadunath Sinha wrote a book on Emotions and the Will.

The major part of ancient Indian scriptures (Hindu, Buddhist and Jain) emphasise self-realization, samadhi or nirvana. After 1960 Humanistic Psychology emerged and Psychologists became interested in paranormal dimensions of growth. Maslow's theory of self-actualization and transcendental self-actualization established the link to the major part of ancient Indian theories and methods and almost the whole of ancient Indian writings became psychologically relevant. Psychology of Consciousness, Parapsychology, Psychology of Mysticism, Psychology of Religion and Transpersonal Psychology borrow extensively from Indian writings. The terms Oriental Psychology, Buddhist Psychology, Yoga

Psychology, Jain Psychology, etc. are frequently found in modern psychological literature now. Many book lists in Psychology now include books on Yoga, Buddhism and Zen. There seems to be a paradigm shift in Western Psychology, a shift from the notion of mental disease and healing to personal growth, the reference point shifting from the statistical average or "normal" to the ideal or upper limits of man's potentiality.

The rudiments of the theory of consciousness can be traced back to the Indus valley civilization (6000 to 1500 B.C.). Artifacts of a man sitting in Padmasana have been obtained in excavations. The Swastika symbol was used in Indus valley script. Buddhist thought and methods (6th century B.C.) are in line with the objective spirit of modern science and the law of parsimony of science and Buddhism can be easily incorporated into a scientific framework. The Psychological relevance of the four noble truths and eight-fold path and Sunya vada of Buddhism and Buddhist techniques of meditation are of considerable relevance in modern Psychology. Similarly Jain scriptures also are found to be relevant to Psychology in more than one way. The Vedas date from about 1500 B.C. However, Upanishads (appendices to the Vedas, which date from 600 B.C.) which describe the Vedanta philosophy and provide the theoretical foundation of Jnana Yoga are of more direct relevance to Psychology. The Bhagavat Gita gives a quintessence of Indian way of life and philosophy and it describes the four yogas, Karma, Bhakthi, Raja and Jnana. Several books have come on the psychological relevance of Gita. Maslow's theory of Meta-motivation is very similar to the concept of Nishkama karma outlined in the Gita.

Patanjali's Ashtanga Yoga is a very systematic presentation of Raja yoga. Both Bhagavat Gita and Ashtanga Yoga are supposed to have been written around the turn of B.C. to A.D. Sankara's writings (8th century A.D.) on the different yogas as well as his Advaita philosophy are considered as classics in the area and are of great value to the Psychology of consciousness as well as personal growth. Modern interest in relaxation can be traced to studies on Savasana. Rising popularity of meditation practice links Psychology to Oriental religious practices and philosophy.

Indian literature on aspects of consciousness is vast, considering the classics and their commentaries. Mental states have been analyzed, classified and differentiated in detail. Similarly paranormal powers (siddhis) have been classified in detail. The process of personal growth and obstacles to growth have been examined thoroughly. There is a great deal of maturity resulting from long experience in these areas reflected in the writings. Indian theories of linguistics, social behavior, crime, etc. are all based on the holistic approach and the broad-based intuitive understanding of behavior in contradistinction to Western theories which are piece-meal, analytic and situation specific. The increasing importance given to the holistic approach and need for synthesis makes it possible to integrate modern Western Psychology with ancient Indian thoughts as well as methods.

The psychosomatic relationship was well known and salient in ancient times. The very first invocatory stanza of Ashtangahridaya (the main text in Ayurveda, written in 4th century A.D.) describes how emotions like desires lead to both physical and mental diseases.

Many attempts are being made to integrate ancient Indian Psychology with modern Western Psychology. More than 40 books have appeared in the field of Indian Psychology. There is a journal of Indian Psychology published from Andhra University which has an Institute of Yoga and Consciousness. At least five persons have developed personality

inventories based on the Triguna theory (Satwa, Rajas and Tamas) of Kapila (Sankhya philosophy, 6th century B.C.)

Post-independence Era

Soon after India gained independence from the British rule, psychology witnessed an explosive growth with departments of psychology opening up in old universities as well as in a continually widening circle of new universities and institutes of technology and management. There was a similar explosion in the number of research publications in Indian as well as international journals. Despite all the exceptional growth of the field, there has been acute restlessness about the significance of the accomplishments.

Dalal (2002) complains about the

growing disillusionment with applicability of western theories and their mindless testing in India. Their failure to resolve inner conflicts of cherishing Indian cultural values at the personal level and maintaining western orientation at professional level was reflected in their methodologically sophisticated but socially irrelevant research. Western psychological theories and research were not effective in understanding the Indian social reality. As a result, Indian psychologists became increasingly marginalised in society.

Regardless of such restlessness, psychological research has continued at an ever-increasing pace. As far as theories are concerned, there are notable contributions that have followed both traditional Indian as well as a few Western models. A brief overview of the more important contributions is in order.

Theoretical Contributions Following the Traditional Indian Lines

A natural reaction to finding the cultural misfit and redundancy of imported models is to turn to the rich intellectual heritage of one's own culture. A clarion call in this direction was given by Durganand Sinha (1965) asking for the integration of modern psychology with Indian thought. In a national conference in 2002 well over 150 psychologists proposed the "Pondicherry Manifesto of Indian Psychology" (The full text of the Pondicherry Manifesto of Indian Psychology is available on the following link on the Internet:

www.infinityfoundation.com/mandala/i_es/i_es_corne_manifesto_frameset.htm) which repeated

Sinha's call in following words:

By Indian psychology we mean a distinct psychological tradition that is rooted in Indian ethos and thought, including the variety of psychological practices that exist in the country... Indian models of psychology would have enormous implications for health psychology, education, organizational management and human and social development. Emphasis on Indian psychology would provide a comprehensive foundation and a refreshing new and indigenous orientation to all other branches of psychology.

Two rather distinct but related lines of development in "Indian Psychology" can be identified which signify elements of resistance and protest. The first one mainly involves interpretation of traditional approaches in light of modern perspectives. In such works, attempt is usually made to explain the relevance of traditional concepts and methods with explanations given in currently popular idiom. Foundational issues underlying theory

building in terms of ontological and epistemological issues are discussed in Rao et al. (2008) and Cornelissen et al.

There is a wide range of studies that explain how insights of traditional Indian origin contribute to the understanding of specific psychological issues. Notable in this context are publications in the fields of consciousness, self, emotion, and perception and cognition.

Aside from such theory building efforts based on traditional foundations, there are efforts toward the empirical validation of theories with the use of Western-style tests and measurements. Several measures have been developed to assess personality typology based on the Samkhya concepts of the three strands of Prakrti, the principle of materiality. A remarkable effort was made by Pande and Naidu (1992) to empirically examine a set of propositions from the theory of acting without attachment to results of one's action described in the Bhagavad-Gita. Then, they developed and validated a measure for an attitude of non-attachment, and correlated the strength of such attitude with various indices of mental health.

The dominant approach to psychological knowledge in the Indian tradition is, however, grounded in a different world view in which focus on self and self-development is valued, and the success of a theory is judged in terms of the usefulness of applications in existential benefits and spiritual progress. With the divorce of religion and science in the history of Europe typified by Galileo's inquisition, spirituality was driven into the religious camp, and it became an anathema for the "science" of psychology in the West. Most Indian psychologists simply followed this trend. But things have changed more recently; research on meditation has become common in contemporary psychology, and yoga has become a household word. In this context, theories and methods of traditional Indian origin are being recovered and critically examined and articulated in the contemporary context.

First of all, there is a burgeoning body of literature on various techniques of meditation and the measurement of their success. A review of this literature with specific reference to Indian approaches may be found in Rao (2011) and Salagame (2011). Paranjpe (2008) has adopted a case-study approach; he has examined the life history of a modern sage, Sri Ramana Maharshi, to see how the traditional Advaitic method of meditation is modified and practiced in modern times, and the kind of transformation it can lead to. Similarly, he has examined the life history of B.G. Tilak, a modern exponent of the Path of Action (karma yoga) and tried to assess how and how far he brought into action the principles he preached. Such use of case studies for validation of theories fits the distinctive character of the Indian tradition where a personal application of psychological models is crucial. The worldwide popularity of Yoga and meditation indicates the relevance of the Indian approaches where similar goals are valued.

Patañjali's theory of Yoga provides the backbone of an ambitious and continuing program for the assessment of psychosomatic benefits of Yogic practices. Literature reporting the results of numerous studies is available from the website (See http://www.svyasa.org/research/research_publication.asp) of the Swami Vivekananda Yoga University, which has become the hub of research on Yoga. Another similar source of information about ongoing research focused on psychological theories of Indian origin is the website of the Indian Psychological Institute, which is closely associated with Sri Aurobindo Ashram in Pondicherry (See <http://ipi.org.in/>).

Theoretical Contributions Following Modern Western Lines

Since the growth of knowledge is socially conditioned, the developments of psychology in India including its theories and concepts need to be appreciated in the local and global historical and sociocultural matrix in which the country has been positioned. Being a developing country with millennia-old culture, a richly diverse society, and a two-century-long colonial past, India is currently aspiring to emerge as a self-reliant and economically strong nation. Faced with the challenge of socioeconomic transformation, the country has been engaged in efforts toward industrialization, modernization, and globalization. Navigating through this difficult terrain has been a complex challenge and has shaped the developments in the academic disciplines also.

India has been engaged with rapid growth of higher education and rapid expansion of professional institutions to meet the increasing demand for trained personnel in various service sectors like health, administration, banking, police, military, and management. The cultural complexity of India due to diversity in ecology, language, religion, family structure, and uneven introduction of technology has put a challenge before the planners to ensure social welfare through democratic processes. This context has shaped the course of higher education in general and psychology in particular.

The challenge to relate India's past and modern psychology was a main concern in the early period and continued since then. In fact, the search for a distinct identity for psychology in the Indian context has remained a key issue. Theoretical innovations came from the real world as well as the difficulties faced in applying psychological knowledge to the diverse problems faced in the Indian conditions. In the course of the disciplinary journey, the theoretical-conceptual ambience of the works of Indian psychologists has undergone several shifts in themes, alignments, and emphases.

The teaching and research in modern psychology began largely as an extension of the EuroAmerican tradition in the British period. Its initial emphasis was not so much on questioning and doubting the Western concepts and methods, but on preserving the essential configuration of the discipline and keeping it as similar to the one in the Western world, as possible. We also note that there existed a tradition of British psychoanalysts who tried to offer interpretations of the Indian psyche to justify the British rule. (Citing Christine Hartnack's work *Vahali* (2011) has discussed at length how early British psychoanalysts tried to create universal psychoanalytic conceptualizations that explain away Indian experiences of selfhood, or view them as essentially inferior, less worthy or simply pathological, or otherwise deficient. It acted as a tool to justify social oppression and colonial rule).

In the mainstream critical paradigmatic questions of ontology and epistemology were sidelined to give space for the newly emerging positivist scientific enterprise. We find that the Western model of research and teaching provided the initial necessary direction to Indian psychologists. Dependence on Western thought was deemed legitimate owing to unexamined theoretical suppositions about the universality of psychological knowledge. It was largely in the 1970s that many psychologists raised the issue of insufficient and inadequate attention to socialpsychological problems. The Indian psychologists realized that they had been indifferent to the vast and rich collection of knowledge inherent in the Indian texts. However, the commitment to scientific inquiry was venerated and debates related to the philosophy of science and related arguments could not receive due attention until the 1980s. Also, there has been a constant pressure for problem solving and application in the social world (Sinha 1986).

A scrutiny of the published literature suggests that the theoretical endeavors of Indian psychologists have taken many forms. They have been more interested in the use of theory as a heuristic device for problem solving in relation to the changing features of the social reality (e.g., technology, economy, media, environment, migration, and education) rather than formal theorization. The mismatch between western theories/concepts and Indian reality has led not only to the introduction of new concepts but also to the modification of constructs in vogue to accommodate newer aspects of reality as applicable in the Indian milieu characterized by a mix of factors demanding a balancing act between the age old traditions of oral culture, cosmological worldview, and hierarchical social order on the one hand and modern influences which emphasize more on technology, social and geographical mobility, secularism, and materialism, on the other. There are growing attempts to test the assumptions and predictions of psychological theories in the Indian context. While culturally informed studies are on rise, full scale reconceptualization or indigenous theorization has been limited.

Early Efforts

Keeping in mind the colonial background of modern learning in India, the mandate of initiating scientific psychology in the prevalent Wundtian tradition and subsequently in the behaviorist tradition was a natural choice. The eagerness to attain an independent identity for the discipline constituted the package of academic delivery consisting of empirical work, positivist metatheory, a universalistic stance, and the presumption of cultural immunity of psychological concepts and theories.

In terms of the institutional structure, psychology was earlier a part of philosophy departments. Separate psychology departments were started largely between 1940 and 1960. In order to maintain a separate identity, the teaching and research practices opted to fashion themselves as differently as possible from the parental discipline of philosophy and similar to the physical and natural science disciplines. To this end, they over-emphasized experimental psychology and psychometry – the distinctive features of the new science – and made them the core of psychology curricula which continues till today. Indeed, empiricism and quantification made possible the flourishing of an empiricist-positivist brand of psychology aimed at generating and testing nomothetic laws as objectively as possible.

However, it will be a mistake to ignore another feature of the academic prowess of the first generation of Indian psychologists. They were also cognizant of their cultural roots and tried to address the academic as well as nonacademic audiences. In a pioneering work entitled *The Science of the Emotions*, Bhagwan Das (1908) presented a rich account of the Indian science of affect. G. S. Bose wrote about the Upanishads and mythology in Bangla language and had an academic fascination for psychoanalysis. Early researchers did attend to the theoretical issues and noted the importance of traditional knowledge but did not reject modern theories.

It is interesting to note that even during the early part of the development in India, many indigenous lines of inquiry were also prevalent. Examples that highlight this trend include Asthana's (1950) work on Samkhya theory of personality and Indra Sen's (1986) elaboration of the integral psychology of Sri Aurobindo. E.G. Parameshwaran started research on the Triguna (sattva, rajas, and tamas) theory which has been followed by several studies. We also find works on the Indian typology of personality and tantra. Some

notable works were undertaken from the Western tradition for further study. For instance, Asthana (1960) proposed that perceptual distortion is the function of the valence which an object acquires from the field structure in which it exists. In this way he tried to resolve the differences between gestalt and learning theories and incorporated Lewin's field theory. In the area of learning theories, the S-R theory was challenged by Kothurkar (1968).

Thus, we see a dual focus of the researchers. One set of works was focused on the study of phenomena pertaining to sensation, perception, psychophysics, and reaction time in the natural science tradition with the spirit of creating a universal theory which would be modern in its texture. Another set took a theoretically relevant initiative to interpret various Indian phenomena in the light of Western theories and constructs and vice versa, while also trying to develop theories based on traditional Indian concepts. The scholars were cognizant of the possibility of indigenous knowledge resources but considered scientific enlightenment as more important. The coexistence of the two traditions which had some overlap but maintained separate identities was gradually replaced by a move that led to a greater gap between the Indian ethos and the academic pursuit of psychology. The assumptions of universally shared computational notion of the mind/brain and strong empiricism were very powerful in creating the boundaries of the discipline and furnishing the criteria of inclusion and exclusion.

Under the prevailing academic culture, Indian psychologists' theoretical engagement remained confined to attempts at enlarging the scope of a set of explanatory (independent) variables that may enhance predictive power in accounting for a set of chosen (dependent) variables. Thus mapping quantitative variations in psychological variables was the main research strategy. Other theoretical and methodological approaches (e.g., Indian, spiritual, qualitative, and discursive) were marginalized on account of their doubtful scientific status and consequently underrated as knowledge claims.

Against this backdrop, it was natural that testing (western/universal) psychological ideas on Indian samples or creating the Indian version of Western tests/tools/concepts became a major preoccupation. This led to proliferation of adaptation and adoption of tests in different areas like intelligence and personality. The practice of psychology remained Western in content and orientation, and used Western academic developments as standards for comparison. This was done as an authentic and legitimate academic practice and got reflected in teaching programs and choice of research topics. The traditional Western models from behaviorism, schema theory, Gestalt school, Pavlovian theory, and information processing theory were popular. Similarly, theories by Cattell, Eysenck, Erikson, McClelland, Piaget, and Herzberg served as some of the dominant frameworks of Indian psychologists for conceptualizing psychological issues and explanations. The researchers were dominated by the mindset that psychological characteristics are stable and reliable dispositions subject to quantification. On the other hand, cognitive processes were handled in terms of concrete and manipulable entities. The positivist methodology was dominant and critical determinant of the choice of problems, variables, processes, and practices. Theory was treated as a copy or map of reality. A strong correspondence between the two was emphasized. There was explicit and implicit endorsement of biology as the ultimate, and reductionism became a strong belief. Affective and social phenomena were mere derivatives of biological and cognitive processes which were foundational. All this was going on in terms of an image of science and scientific practices that did not bother about the dynamic nature of social reality and social conditioning of knowledge.

The “basic problem”, as Nandy and Kakar (1980) have observed, involved “dependence on conceptual frameworks which are not intrinsic to the experience of society” (p. 159).

A Socially Relevant Psychology

In the 1980s, several lines of investigation across many domains of social psychological processes showed that many of the phenomena reported in Western research literature required different explanations rooted in the Indian cultural milieu. Examples of this kind are found in the areas of social cognition. Thus predictions from attribution theory with socially and culturally specific causal categories were tested for understanding achievement, health, and other aspects of human behavior (Dalal 1988). The change took place when psychologists found the applications of Western theories/methods to be either ineffective or irrelevant in real life situations in India. A selective overview of some of these developments in key areas is presented below.

Human cognition. Researches on attention, emotion, and consciousness, by using multiple methods and approaches, have shown cross-cultural aspects of emotion as well as of meditation, in terms of underlying neurophysiology. There are also researches on philosophical aspects of cognition that view cognition and other mental phenomena as central to the functioning of all living beings. To put it another way, the fundamental principles governing cognition run from a single cell to human societies.

Planning is a key intellectual function. Extending the earlier work on PASS (Planning, Attention, Successive, and Simultaneous Processes) theory, Das et al. (2000) have brought out its significance in various cognitive functions. Srivastava and Misra (2007) have developed an indigenous conceptualization of intelligence and termed it integral intelligence. It has four dimensions: cognitive competence, social competence, competence in action, and emotional competence. The analysis of creativity has been undertaken from a culturally informed position (Misra et al. 2006).

Sinha was perhaps the first one in India to emphasize the role of sociocultural context in understanding cognitive development. Following this tradition, R.C. Mishra (1997) has been investigating the ways in which basic cognitive processes like perception and memory get shaped and manifested under diverse eco-cultural settings. Pirta (2011) has investigated native cognition in Himalayas and has endeavored to develop a bio-ecological framework integrating ecology, biology, and behavior.

Attitude and social cognition. With the political independence of the country in 1947, the highest challenge was that of the problem of communal tension arising out of partition. This led to studies of conflicts, prejudice, stereotypes, discrimination, and violence. Since then it has remained a productive area of research. These studies were directed toward measuring attitudes, stereotypes, and cognitions and relating them to a number of contextual and dispositional variables. Also, there was the challenge of social and national development of reality and psychology played the role of identifying the facilitators of, and resistances to, the process of development.

In interesting rumor studies, Prasad (1935) examined the responses to an earthquake in Bihar, and published a comparative analysis of earthquake rumors which provided basis for cognitive dissonance theory.

Ashis Nandy's *The Intimate Enemy* (Nandy 1983) and *Illegitimacy of Nationalism* and Sudhir Kakar's *The Colors of Violence* (Kakar 1995) have touched on the cultural/historical aspects of selfhood and intergroup relations, indicating the need to attend to macro aspects to capture and understand the psychosocial systems. Further, such works encour-

aged psychologists to employ other methods to understand human behavior. A major programmatic and cross-cultural work based on studies in the Netherlands and India is by De Ridder and Tripathi (1992) recognized the prominence of group norms in intergroup behavior. They pointed out that norm violation by one group leads to a chain of negative reactions by both groups and, if this sequence continues, it is likely to escalate violent behavior.

Singh (2011) has systematically examined judgment and decision-making within the framework of information integration theory. His work spanning over a period of more than 3 decades has found that Indians use averaging, subtracting, multiplying, and dividing rules and their combinations. Such cognitive algebra, however, represents causal beliefs and not mathematical calculations.

Research in the area of distributive justice has focused on principles of distributive justice. L. Krishnan (2005) has analyzed the Indian notion of *dana* (charity) and has drawn attention to its nuances with respect to deservingness in the Indian tradition.

The above works demonstrate a significant shift in research thinking and execution, where the psychologist seeks a realistic appraisal of the problems in the Indian context. This socially relevant focus helps in understanding everyday social issues, caste, and religious identities, intergroup behavior, justice, and nation building. The need for Indian psychologists to be rooted within the sociocultural and historical contexts was and is repeatedly emphasized.

Psychology of poverty and deprivation: The study of poverty and deprivation has been an important area of research where researchers in different parts of the country (e.g., Rath at Bhubaneswar, A.K. Singh at Ranchi, D. Sinha at Allahabad, L.B. Tripathi and G. Misra at Gorakhpur) moved in many directions and have mapped the diverse effects of poverty, social disadvantage, and deprivation. Most of these studies have situated deprivation in the experiential-environmental context and have traced its detrimental influences back to aspects of development. The detrimental effects of poverty are accentuated by the unfavorable proximal environment of the child. Thus, intervention should address not only the cognitive-attentional drawbacks of the children but also the conditions prevailing in the family and school settings. They should be planned to create in the people a sense of empowerment to effect change in their life conditions. Unfortunately, the planning rooted in the Western model of development often ignores the traditional attitudes, beliefs, and values, and considers them antithetical to development ideology. There is growing evidence that social-psychological problems of Indian society are now being increasingly addressed by psychologists.

The challenge of achievement. The economic and social development was an important concern for a developing country like India. The theoretical analysis by McClelland underscored the significance of achievement motivation (*n-Ach*) as a driving engine for development. The lack of emphasis on individualistic and competitive spirit and independence were identified as the main causes of underachievement. This became the basis for a major intervention program at Kakinada in Tamil Nadu, as reported in *Motivating Economic Achievement*. It provided impetus for promoting entrepreneurship. The relevance of achievement motivation theory was, however, Indian challenged. The perceived value of various achievement goals is determined by the expectations of significant others. The concepts of “extension motivation”, “dependency proneness”, “achievement value”, and “dissatisfaction-based achievement motivation” are important contributions.

Organizational behavior. Rapid industrialization in the 1960s led to recognition of the need to study the labor-management relationship and organizational effectiveness. Chakraborty (1995) has brought into focus the critical role of values in managerial transformation, as well as ethics in management. R. Gupta (2002) emphasized the need to go beyond the American and Japanese models of organizational behavior, and develop models specific to the Indian conditions. The concept of “nurturant task leadership” proposed by J.B.P. Sinha (1980) was an innovation showing the need for developing a theory relevant to culture-specific aspects of organizational behavior. It emphasized nurturance, dependency, personalized relationship, and status consciousness from the Indian cultural context and combined them with the contingency approach and the principle of reinforcement.

Individualism-collectivism, self, and identity. Indian scholarship has shown that the elements of Indian selfhood are complex and it is difficult to categorize it as either individualist or collectivist. Sinha and Tripathi (1994) see that there is the presence of individualist as well as collectivist aspects of self indicating a kind of “coexistence of opposites.” Mascolo et al. (2004) have demonstrated multifacetedness of the representation and experience of Indian self. They propose four ways of conceptualizing selfhood: independent, interdependent, relational, and encompassing. In addition, there are text-based, theoretical, in-depth, and extensive analyses – like the ones on the Indian views of self and identity, concept of self in the Sufi tradition (Beg 1970) – which offer insights to selfhood and identity embedded in the Indian traditions in which higher or “spiritual self” occupy important place.

Sinha and Pandey (2007) have proposed that Indian people function with diverse mindsets in different contexts. Thus, they manifest a materialist mindset in multinational organizations, and dependence prone or collectivist mindset in family owned, bureaucratic, and/or traditional organizations. They noted that Indians are holistic in terms of combining excellence in work, personalized relationships, abstract thinking, emotionality, rationality, and spirituality in those organizations that valued both performance and people. Materialistic mindset was associated with manipulative behavior, and a holistic mindset with a proactive stance that manifested in innovative and extraordinary performance under inspiring superiors; both materialistic and holistic mindsets were instrumental to success at work, in different organizational contexts. The use of mindset varied depending on the cultural context.

Cultural psychology of emotions. In this area, certain indigenous concepts with culturally specific implications such as *lajja* (shame) (Menon and Shweder 1994) and *bhakti* (devotion to God) (Paranjpe 1998) have been recovered and elaborated. At another level, depersonalized, transcendental and transformative aspects of the *rasa* experience have been delineated (Paranjpe and Bhatt 1997).

Health, human development and well-being: The Indian ideas and concepts like *ahamkara*, *anasakti* have been explored, as are implications for health of various issues particularly relevant for the Indian context, for example, experience of crowding, notions of health and well-being, and Hindu parents’ ethno theories. Neki (1973) has tried to build therapeutic interventions for the promotion of mental health and well-being using yoga and has come up with a model called Guru-Chela therapy which involves the teacher-disciple relationship developed in the Indian tradition. Kakar’s *Shamans, Mystics and Doctors* has become a classic which talks about indigenous healing practices. It clearly indicates the role of traditional healers in maintaining mental health in traditional soci-

eties. In *The Inner World*, Kakar has tried to present the interplay of the universal processes of development and the specific aspects of Indian social reality. He comprehensively tries to weave the story of development, health, passion and relationship by drawing from various sources including religious ideals, traditions, and institutions that constitute a society.

These dimensions of psychology are leading psychologists to develop theories and concepts which do not take from any Western thought, but derive entirely from Indian traditions of thought.

Move Toward Indigenization

The indigenous thought systems remained neglected because there was a strong aversion toward them owing to doubts regarding their scientific status, contemporaneous relevance, and ontological suppositions. Psychological theories and constructs were taken as intrinsically biological, materialistic/objective in content, and quantitative in methodological approach. Therefore psychology, like other natural and physical sciences, was thought to be culture and psychological processes as distributed/shared uniformly across diverse cultures and sub-cultures.

This spurred the need for a radical change in cross-cultural psychology's universalist stance, and its almost exclusive focus on the discovery of panhuman patterns of behavior. "There was an implicit assumption that the definition of... concepts and their measurement as proposed by the Western research workers will also hold good in our cultural context". The signs of efforts to outgrow the alien frame were noticed in the 1970s. The search for a new identity became a major question. Culture-specific concepts, and a search for culturally appropriate methods and tools were emphasized and the relevance of culture was realized. In this context, the interaction with cross-cultural and cultural psychologists has provided important impetus. Gradually, blending scientific ways with indigenous concepts emerged as an important academic agenda. Ramanujan (1990) has emphasized context sensitivity as the key feature of Indian way of thinking. Critical reflections have drawn attention to the limitations of natural science-based approaches and to new possibilities. It was realized that an understanding of Indian social reality would benefit from indigenous psychological knowledge and the discipline should contribute to the programs of socioeconomic development.

Rao has discussed human cognitive processes from the perspective of *Samkhya Yoga* system. According to this system, there are two principles that govern our existence – *purusa* (consciousness) and *prakrti* (matter). *Purusa* is pure consciousness and has no quality or characteristics of its own; it is inert and formless. *Prakrti*, on the other hand, is the material basis of our being. In Yoga, *citta* denotes the functional mind, which comprises of not only the cognitive processes, the ego and the senses, but also contains instinctual tendencies (*vasnas*) inherited from previous lives and the effect of past actions in the present life (*samskaras*). They influence our cognitions and predispose us to behave in certain ways.

In Yoga, *citta* controls our actions. Information processing in *citta* may take place at three levels that is, *buddhi*, *ahamkara* (egoic self), and *manas*. *Manas*, the central processing unit, selects information provided by the sensory system and processes them. *Ahamkara* (the emotional self) appropriates the processed information from *manas* and considers it as required by the perceiving person. And, *buddhi* decides and plans the actions and reactions in an appropriate manner.

The above discussion indicates the gradual inclination of Indian psychologists to move from Western theories, models, and modes of research toward an indigenous approach to theories and methods. We note the continued use of Western theories; yet they are consistently being tested in the Indian context. A beginning toward cultural sensitivity in the form of culture-based concepts, theories, and methods has been made. Today an Indian psychology is emerging which promises a broad theoretical foundation for the exploration of human consciousness (Yoga), and radical psychological transformation. Its applications are found in modern areas like organizational behavior. Accounts of the states and contents of mental functions regulating responsible human conduct available in vast Indian texts and practices is being rediscovered in a more contemporary context. It emphasizes a kind of perspective which is practical path or life ways that allow pursuit of balanced living and enjoying well-being and equanimity through self transformation and personal growth. The universal consciousness and transcendence requires methods that combine sensory, mental and spiritual tools and innovative theoretical paradigms. The recent publications on Indian psychology clearly indicate a serious move in the direction of creating and using *samvada* (dialogue) for better understanding. The conceptual network is being extended. The current theoretical developments such as feminism, subaltern studies, critical theory, and post modernism are providing new ways of engaging with reality. The life world is being appreciated in newer ways and options are generated. There have been enabling moments that have helped Indian psychologists to go beyond the constraints. Promising elements of critique as well as reconstruction are seen.

WESTERN: GREEK HERITAGE, MEDIEVAL PERIOD AND MODERN PERIOD

Who are we? How can we be happy? Does the universe have a purpose? Greek philosophers approached the big questions of life sometimes in a genuine scientific way, sometimes in mystic ways, but always in an imaginative fashion. Pythagoras considered a charlatan for claiming the doctrine of reincarnation, a half-naked Socrates haranguing people in the street with provocative and unanswerable questions, Aristotle tutoring great generals: these are examples of how Greek thinkers dared to question traditional conventions and to challenge the prejudices of their age, sometimes putting their own lives at stake. Greek Philosophy as an independent cultural genre began around 600 BCE, and its insights still persist to our times.

The Pre-Socratics

About 600 BCE, the Greek cities of Ionia were the intellectual and cultural leaders of Greece and the number one sea-traders of the Mediterranean. Miletus, the southernmost Ionian city, was the wealthiest of Greek cities and the main focus of the “Ionian awakening”, a name for the initial phase of classical Greek civilization, coincidental with the birth of Greek philosophy.

The first group of Greek philosophers is a triad of Milesian thinkers: Thales, Anaximander, and Anaximenes. Their main concern was to come up with a cosmological theory purely based on natural phenomena. Their approach required the rejection of all traditional explanations based on religious authority, dogma, myth and superstition. They

all agreed on the notion that all things come from a single “primal substance”: Thales believed it was water; Anaximander said it was a substance different from all other known substances, “infinite, eternal and ageless”; and Anaximenes claimed it was air.

Observation was important among the Milesian school. Thales predicted an eclipse which took place in 585 BCE and it seems he had been able to calculate the distance of a ship at sea from observations taken at two points. Anaximander, based on the fact that human infants are helpless at birth, argued that if the first human had somehow appeared on earth as an infant, it would not have survived: therefore, humans have evolved from other animals whose offspring are fitter. The science among Milesians was stronger than their philosophy and somewhat crude, but it encouraged observation in many subsequent thinkers and was also a good stimulus to approach in a rational fashion many of the traditional questions that had previously been answered through religion and superstition. The Ionian rational view caused nothing but perplexity among some of their powerful neighbours such as the Babylonians and Egyptians, which were nations based on theocratic governments where religion played an important political and social role.

Pythagoras is considered one of the Ionian thinkers but outside the Milesian school: he was originally from Samos, an offshore Ionian settlement. His approach combines science with religious beliefs, something that would have caused horror among the Milesian school. His philosophy has a dose of mysticism, probably an influence of the Orphic tradition. Mathematics, in the sense of demonstrative deductive arguments, begins with Pythagoras: he is credited as the author of the first known mathematical formulation, the theorem which states that the square of the longest side of a right triangle equals the sum of the squares of the other two sides. Deductive reasoning from general premises seems to have been a Pythagorean innovation.

Democritus

Atomism began with Leucippus and Democritus. Among the ancient schools, this approach is the closest to modern science: they believed that everything is composed of atoms, which are indestructible and physically indivisible. They were strict determinists, who believed that everything happens in accordance with natural laws and the universe, they said, has no purpose and is nothing more than a mixture of infinite atoms being shuffled and re-shuffled according to the indifferent rules of nature. What is interesting about this school is that it attempted to understand the universe as objectively as possible and minimize intellectual deviations in favour of cultural and mystic prejudices.

The Rise of Athens: the Sophists & Socrates

About 500 BCE, the Greek city-states or poleis were still largely divided. They had a common language and culture, but they were very often rivals. Some years earlier, Athens implemented a socio-political innovation by which all free male citizens had equal rights regardless of their origin and fortune. They named it democracy. Before the time of democracy, government decision-making was in the hands of a few, often aristocratic and noble families. Democracy allowed all free citizens to be part of the important decisions of the polis. They could engage in the discussions held during deliberative assembly and tribunals, their voices could be heard everywhere and had the same value as any other voice. In this context, speech was king: being able to discuss different topics effectively

and to persuade others, granted a competitive advantage. This was true not only of citizens actively involved in politics, but for any other citizen. During court hearings, for example, prosecutor and accused had to appear in court in person, never through lawyers, and the failure or success of the process relied largely on rhetorical skills and any citizen could be subject to a court hearing. This period, therefore, saw the beginning of the Sophist school.

The Sophists were intellectuals who taught courses in various topics, including rhetoric, a useful skill in Athens. Because they taught in return for a fee, the Sophists' schools were only attended by those who could afford it, usually members of the aristocracy and wealthy families. This was a time of profound political and social change in Athens: democracy had replaced the old way of doing politics and many aristocrats whose interests were affected were trying to destroy the democracy; the rapid increase of wealth and culture, mainly due to foreign commerce, undermined traditional beliefs and morals. In a way, the Sophists represented the new political era in Athenian life, especially because they were linked with the new educational needs.

Caught in the clash between cultural conservatism and innovation, we find a peculiar character: Socrates, the pivotal figure in Greek philosophy and the wisest among Greeks at his time according to the oracle of Delphi. Like the Sophists, Socrates enjoyed teaching, but unlike the Sophists he never requested a fee in return and lived a life of austerity. He either underestimated or ignored most of the topics that were popular among his predecessors. Before the time of Socrates, philosophers' main concern had been the physical world and how to explain it naturally. However, Socrates set in motion a new approach by focusing entirely on moral and psychological questions. His methodology sought to define key questions such as: what is virtue? what is patriotism? what do you mean by morality? As a result of this, most of his debates ended up with even more questions, the central issue unanswered, and the disputers' ignorance on many topics revealed, since he invariably proved that the words being used by his contenders were actually abstract terms with an empty meaning.

By combining a humble spirit (he never claimed to be any wiser than anyone else) and a strict agnosticism (he said he knew nothing) with a method that challenged conventional assumptions and an intolerance for unclear thinking, Socrates gradually earned enemies from various sectors of Athenian society. He was, consequently, put on trial and condemned to death. However, Athenians did not like to condemn a citizen to death, therefore, this was merely a formal sentence and he was offered the possibility to escape. He refused to do so and obeyed the jury's decision: a mixture containing poison hemlock took away his life, but his example granted him immortality.

Plato & Aristotle

Plato and Aristotle are the two most important Greek philosophers. Their work has been the main focus of interest for students of philosophy and specialists. This is partly because, unlike most of their predecessors, what they wrote survived in an accessible form and partly because Christian thought, which was the dominant thought in the Western world during the Middle Ages and early modern age, contained a high dose of Platonic and Aristotelian influence.

Plato was a student of Socrates who left Athens disgusted by the death of his teacher. After travelling for many years, he returned to Athens and opened his famous Academy.

He is the best known Greek philosopher; the triumph of his work has been so complete and influential in western philosophy, that the famous quote from Alfred North Whitehead, although an exaggeration, is not far from the truth: "The safest general characterization of the European philosophical tradition is that it consists of a series of footnotes to Plato."

Plato had many philosophical interests including ethics and politics but he is best known for his metaphysical and epistemological ideas. One of his most influential insights is the Theory of Ideas: to Plato, notions like virtue, justice, beauty, goodness, etc., would not be possible unless we had some direct knowledge of these things in an earlier existence. We are born into this world with an imperfect memory of these Forms. In that ideal world of Ideas, one can experience the real Forms which are perfect and universal. Our world is an imperfect parody of the Platonic flawless and superior world of Ideas. A knowledge of these Forms is possible only through long and arduous study by philosophers but their eventual enlightenment will qualify them, and they alone, to rule society.

Aristotle, a student of Plato for almost 20 years, was the tutor of Alexander the Great. Aristotle's interests covered a wide scope: ethics, metaphysics, physics, biology, mathematics, meteorology, astronomy, psychology, politics and rhetoric, among other topics. Aristotle was the first thinker who systematically developed the study of logic. Some of the components of Aristotelian logic existed long before Aristotle such as Socrates' ideas on exact definition, argumentative techniques found in Zeno of Elea, Parmenides and Plato, and many other elements traceable to legal reasoning and mathematical proof. Aristotle's logic system consists of five treatises known as the *Organon*, and although it does not exhaust all logic, it was a pioneering one, revered for centuries and regarded as the ultimate solution to logic and reference for science. Aristotle's contribution in logic and science became an authority and remained unchallenged as late as the modern age: we can recall Galileo who, after careful observation during the Renaissance, came to the conclusion that most of the Aristotelian physics and astronomy was not in line with the empirical evidence and yet, Galileo's ideas were widely rejected by his contemporary Aristotelian scholars. Even during the most obscure times during the Middle Ages, a copy of the *Organon*, or maybe fragments of it, could be found in all prestigious libraries.

Hellenistic Philosophy

During the Hellenistic age, four philosophical schools flourished: the Cynics, Sceptics, Epicureans and Stoics. During this time, political power was in the hands of the Macedonians. Therefore, Greek philosophers abandoned their political concerns and focused on problems of the individual. Instead of trying to come up with plans to improve society, their interest was how to be happy or virtuous.

The Cynics rejected all types of conventions: marriage, manners, religion, housing, and even decency. The Sceptic philosophical school systematized old doubts: the senses caused troubles to most philosophers except some rare exceptions like Plato who simply denied the cognitive value of perception in favour of his world of ideas. On top of the scepticism of the senses, the Sceptics added moral and logical scepticism. Epicureanism claimed that life was about pursuing this world's pleasures. They only believed in the material world, a belief which attracted the opposition of the Stoics. Stoics said that everything that happens is due to divine providence, therefore, whatever misfortune occurs, a stoic will accept it without complaint. Stoics rejected Aristotle's views on the

relevance of bodily and material goods to human happiness. Achieving happiness, stoics said, is not important, what is actually important is to pursue happiness since the outcome of our attempt is not fully under our own control.

Legacy

While Rome was expanding, Greece started to decline. The western Mediterranean was left untouched by Alexander the Great. After the first and second Punic Wars (264-241 and 218-201 BCE), Rome neutralized Carthage and controlled Syracuse (the two leading city-states of the western Mediterranean), and continued its expansion by conquering the Macedonian monarchies during the second century BCE followed by Spain, France, and Britain. Paradoxically, despite its expansion and military superiority, Rome's influence in the cultural life of Greece was not significant. On the contrary, the influence of Greece on Roman culture was deep and long-lasting. Roman gods were identified with the Olympian deities, Hellenic art, literature, architecture, philosophy and even the language captivated most educated Romans. Rome was superior to Greece in building roads, implementing social cohesion, creating effective systematic legal codes and military tactics. However, Roman science, art and philosophy were heavily influenced by the Greek tradition.

Given this Roman admiration of all things Greek it is, therefore, no wonder that one of the most important Roman philosophers, Plotinus (204-270 CE), is the founder of Neo-Platonism. Plotinus lived during a time of political disaster in Rome. Roman rulers were placed and removed at will by the army in return for favours. German tribes from the north and Persians from the east profited from this scenario: the Roman army was more concerned with domestic political struggle rather than defending the borders and their ineptitude in defence was complete. Pestilence reduced the population, unsuccessful military campaigns increased expenditure and taxes while resources diminished and the entire Roman fiscal system crashed. The world showed few signs of hope during the time of Plotinus, which could explain why the ideal and eternal world of Platonic ideas was an appealing refuge. This shift of attention from the Real World to the Other World was also adopted by pagans and Christians alike whose philosophies revolved around the idea of an eternal and heavenly afterlife. The resemblances between Platonic and Christian thought are so strong that Christian theologians used many ideas of Plotinus to build their philosophy.

Platonism played a central role in shaping Christian theology. Christian religion developed during the time of Rome and combined Platonism, some philosophical beliefs from the Stoics and Orphism, esoteric aspects traceable to cults of the Near East, and morals and history acquired from Judaism. Even Saint Augustine refers to Plato's ideas as "the most pure and bright in all philosophy". Christianity has undergone many changes during its long history and it is important to note that during the Middle Ages its philosophy revolved largely around ideas derived directly from the Greeks.

Across the millennia, the voices of the Greek philosophers have been shaping our minds, our institutions, our leaders and our civilization as a whole. These Greek thinkers have unquestionably proved that the same problem can be approached in different ways, that common sense is not as common as we like to believe, that considering unfamiliar possibilities can enlarge our thought and that imagination and ideas can be immortal.

Structuralism

Structuralism in psychology is considered as a theory of consciousness which was suggested by Wilhelm Wundt and developed by his student Edward Titchener. The theory came to be in the 20th century: where its reliability was debated and challenged by the growing scientific community at that time. Structuralism is also considered as a school of psychology which seeks to analyze the components of an adult mind. It seeks to analyze the simplest thoughts of a mind that bring about the more complex experience that we go through in our day to day life.

There are various sources such as books and articles that speak about structuralism. One such source is an article “How structuralism and functionalism influenced early psychology” written by Kendra Cherry. The article informs us that in the early 20 the century Psychology was separated from biology. At that time there was a raging debate in the scientific community on how the human mind and behavior worked. This questions led to the establishment of two major schools of Psychology. They included Structuralism and Functionalism. Structuralism was the first school of thought. Many if not all structuralism components were idealized by (Wilhelm Wundt) who was also the founder of the first psychological lab.

Later on, one of his student (Edward B. Titchener) went on to formally establish structuralism as a theory. However, Edward’s ideas had misrepresented the teachings of Wundt. Almost immediately after the establishment of structuralism other ideas emerged such as functionalism from thinkers like Charles Darwin (Cherry, 2016). Furthermore, we learn that structuralism was the first school of psychology and it focused on breaking down the mental process into basic elements. Researchers tried to learn the basic elements of the mind through a method known as introspection.

A second source of the formation, background and development of structuralism is an article “Structuralism” written by Richard Hall. Richard informs us that in the past many advances in science were occurring due to the concept of “elementism”. “Elementism” referred to the conception of complex phenomena in terms of underlying elements. It was at this moment that what psychologist refer to as the first school of psychology was established. A psychologist called Wilhelm Wundt started the first psychological laboratory in Leipzig, Germany. Hall further informs us that the school of psychology that Wundt championed was called Structuralism. It led many people to refer to Wundt as the father of Structuralism.

Structuralism is fundamentally defined as the study of the human conscious (Hall, 2001). The rationale behind it is that the human conscious could be broken down into basic conscious elements. Most of the experiments conducted in Wundt’s laboratory involved cataloging primary conscious elements. To research on the basic elements, structuralism relied on a method called introspection. An example is how someone can describe the basic elements of an orange (cold, juicy). Introspection involved describing each basic elements separately from the complex entity. Through the use of this method, Wundt was able to catalog different human experience in mind.

Although structuralism was established as a psychological theory, it faced a lot of criticism through the times. Many psychologist failed to accept the theoretical background of Structuralism. The experimental methods that was used to study the structures of the mind were too subjective. Moreover, we also learn that through using introspection, this led to unreliability of the data gotten. Others critics also argued that structuralism was

concerned with internal human behaviors. Internal human behaviors are considered non-observable which cannot be accurately measured.

Moreover, we also learn that structuralism faced more limitations such as not having its principal theory supported by most psychologist in the scientific world. In the present times, Structuralism is considered to be dead in psychology. Hall (2001) informs us that one reason why Structuralism faced criticism was because of a methodological flaw in Wundt's structuralism. The theory relied on introspection which lacked subject agreement and reliability. In psychology, it is important that many observers agree independently on phenomena. When it comes to Wundt's Structuralism experiment, his observes were student's trained by him. Wundt was also the one who resolved any disagreement of concepts during the experiments. The use of trained observers is opposed to the current practice of psychology.

However, the existence of criticism was not enough to undermine the strength of structuralism. Structuralism was important because it was the first school of thought. Structuralism led to the development of experimental psychology. Structuralism has been dead for many years since the passing of Wundt. Other sources differ on how Structuralism developed. The last alternative narrative of how structuralism was formed is that Structuralism was a theory that was introduced by psychologist Wilhelm Wundt and was later on popularized by Edward Titchener. An article written and submitted to the journal of psychologist informs us that, an example of Structuralism is a fleece blanket, it can be considered as warm, fuzzy, soft and green. The breakdown of a complex component such as the fleece blanket to its basic elements (soft, fuzzy) is what's considered as structuralism. Another example is how an apple can be described as red, crisp and sweet (Editor, 2016). Structuralism was only interested in showing the basic elements of something and not the complex ideas. The person describing the apple or fleece blanket can only describe it to its most basic elements.

In conclusion, structuralism dictates that the total sum of parts that have been broken down are what make up the whole "something." Wundt mainly formed structuralism to focus on understanding the fundamental component of the human mind. Through the use of different processes such as introspection, he was able to conduct experiments to the conscious mind. This way Wundt subjectively identified what makes them experience those thoughts. However, the structural school lost considerable influence when Titchener died. In the end structuralism led to the development of other theories such as behaviorism, functionalism and Gestalt psychology.

Functionalism

Functionalism formed as a reaction to structuralism and was heavily influenced by the work of William James and the evolutionary theory of Charles Darwin. Functionalists sought to explain the mental processes in a more systematic and accurate manner.

Rather than focusing on the elements of consciousness, functionalists focused on the purpose of consciousness and behavior. Functionalism also emphasized individual differences, which had a profound impact on education.

Some of the important functionalist thinkers included William James, John Dewey, Harvey Carr and John Angell.

Strengths and Criticisms of Functionalism

Functionalism was criticized perhaps most famously by Wundt. "It is literature. It is beautiful, but it is not psychology," he said of functionalist William James' *The Principles of Psychology*.

Functionalism was an important influence on psychology. It influenced the development of behaviorism and applied psychology. Functionalism also influenced the educational system, especially with regards to John Dewey's belief that children should learn at the level for which they are developmentally prepared.

Psychoanalytical

Psychoanalysis was founded by Sigmund Freud (1856-1939). Freud believed that people could be cured by making conscious their unconscious thoughts and motivations, thus gaining insight.

The aim of psychoanalysis therapy is to release repressed emotions and experiences, i.e., make the unconscious conscious. It is only having a cathartic (i.e., healing) experience can the person be helped and "cured."

Psychoanalysis Assumptions

1. Psychoanalytic psychologists see psychological problems as rooted in the unconscious mind.
2. Manifest symptoms are caused by latent (hidden) disturbances.
3. Typical causes include unresolved issues during development or repressed trauma.
4. Treatment focuses on bringing the repressed conflict to consciousness, where the client can deal with it.

Gestalt

Gestalt psychology is a school of thought that looks at the human mind and behavior as a whole. When trying to make sense of the world around us, Gestalt psychology suggests that we do not simply focus on every small component.

Instead, our minds tend to perceive objects as part of a greater whole and as elements of more complex systems. This school of psychology played a major role in the modern development of the study of human sensation and perception.

A Brief History of Gestalt Psychology

Originating in the work of Max Wertheimer, Gestalt psychology formed partially as a response to the structuralism of Wilhelm Wundt.

While Wundt was interested in breaking down psychological matters into their smallest possible part, the Gestalt psychologists were instead interested in looking at the totality of the mind and behavior. The guiding principle behind the Gestalt movement was that the whole was greater than the sum of its parts.

The development of this area of psychology was influenced by a number of thinkers, including Immanuel Kant, Ernst Mach, and Johann Wolfgang von Goethe.

The development of Gestalt psychology was influenced in part by Wertheimer's observations one day at a train station. He purchased a toy stroboscope which displayed pic-

tures in a rapid sequence to mimic the appearing movement. He later proposed the concept of the Phi phenomenon in which flashing lights in sequence can lead to what is known as apparent motion. In other words, we perceive movement where there is none. Movies are one example of apparent motion. Through a sequence of still frames, the illusion of movement is created.

"The fundamental 'formula' of Gestalt theory might be expressed in this way," Max Wertheimer wrote. "There are wholes, the behavior of which is not determined by that of their individual elements, but where the part-processes are themselves determined by the intrinsic nature of the whole. It is the hope of Gestalt theory to determine the nature of such wholes."

Major Gestalt Psychologists

There were a number of thinkers who had an influence on Gestalt psychology. Some of the best-known Gestalt psychologists included:

1. **Max Wertheimer.** Regarded as one of the three founders of Gestalt psychology, Wertheimer is also known for his concept of the phi phenomenon. The phi phenomenon involves perceiving a series of still images in rapid succession in order to create the illusion of movement.
2. **Kurt Koffka.** Known as one of the three founders of Gestalt psychology, Kurt Koffka had diverse interests and studied many topics in psychology including learning, perception, and hearing impairments.
3. **Wolfgang Kohler.** Also a key founding figure in the history of the Gestalt movement, Kohler also famously summarized Gestalt theory by saying, "The whole is different than the sum of its parts." He was also known for his research on problem-solving, his criticisms of the introspection used by the structuralists to study the human mind, and his opposition to behaviorism.

Gestalt Laws of Perceptual Organization

Have you ever noticed how a series of flashing lights often appears to be moving, such as neon signs or strands of Christmas lights? According to Gestalt psychology, this apparent movement happens because our minds fill in missing information. This belief that the whole is greater than the sum of the individual parts led to the discovery of several different phenomena that occur during perception.

Gestalt Psychology

In order to better understand how human perception works, Gestalt psychologists proposed a number of laws of perceptual organization, including the laws of similarity, *Pragnanz*, proximity, continuity, and closure.

The law of similarity suggests that similar items tend to be grouped together. If a number of objects in a scene are similar to one another, you will naturally group them together and perceive them as a whole. For example, a series of circles or squares stacked together will be viewed as a series of columns rather than just individual shapes.

The law of proximity suggests that objects near each other tend to be viewed as a group. If you see a number of people standing close together, for example, you might immediately assume that they are all part of the same social group.

At a restaurant, for example, the host or hostess might assume that people seated next to each other in the waiting area are together and ask if they are ready to be seated. In

reality, they may only be sitting near each other because there is little room in the waiting area or because those were the only open seats.

Gestalt psychology also helped introduce the idea that human perception is not just about seeing what is actually present in the world around us. Much of what we perceive is heavily influenced by our motivations and expectations.

Behaviorism

Behaviorism, also known as behavioral psychology, is a theory of learning based on the idea that all behaviors are acquired through conditioning. Conditioning occurs through interaction with the environment. Behaviorists believe that our responses to environmental stimuli shape our actions.

According to this school of thought, behavior can be studied in a systematic and observable manner regardless of internal mental states. Basically, only observable behavior should be considered—cognitions, emotions, and moods are far too subjective.

Strict behaviorists believed that any person can potentially be trained to perform any task, regardless of genetic background, personality traits, and internal thoughts (within the limits of their physical capabilities). It only requires the right conditioning.

A Brief History of Behaviorism

Behaviorism was formally established with the 1913 publication of John B. Watson's classic paper, "Psychology as the Behaviorist Views It." It is best summed up by the following quote from Watson, who is often considered the "father" of behaviorism:

"Give me a dozen healthy infants, well-formed, and my own specified world to bring them up in and I'll guarantee to take any one at random and train him to become any type of specialist I might select—doctor, lawyer, artist, merchant-chief and, yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations, and race of his ancestors."

Simply put, strict behaviorists believe that all behaviors are the result of experience. Any person, regardless of his or her background, can be trained to act in a particular manner given the right conditioning.

From about 1920 through the mid-1950s, behaviorism grew to become the dominant school of thought in psychology. Some suggest that the popularity of behavioral psychology grew out of the desire to establish psychology as an objective and measurable science. Researchers were interested in creating theories that could be clearly described and empirically measured but also used to make contributions that might have an influence on the fabric of everyday human lives.

Types of Conditioning

There are two major types of conditioning:

1. **Classical conditioning** is a technique frequently used in behavioral training in which a neutral stimulus is paired with a naturally occurring stimulus. Eventually, the neutral stimulus comes to evoke the same response as the naturally occurring stimulus, even without the naturally occurring stimulus presenting itself. The associated stimulus is now known as the conditioned stimulus and the learned behavior is known as the conditioned response.

2. **Operant conditioning** (sometimes referred to as instrumental conditioning) is a method of learning that occurs through reinforcements and punishments. Through operant conditioning, an association is made between a behavior and a consequence for that behavior. When a desirable result follows an action, the behavior becomes more likely to occur again in the future. Responses followed by adverse outcomes, on the other hand, become less likely to happen again in the future.

Top Things to Know About Behavioral Psychology

1. **Learning can occur through associations.** The classical conditioning process works by developing an association between an environmental stimulus and a naturally occurring stimulus. In physiologist Ivan Pavlov's classic experiments, dogs associated the presentation of food (something that naturally and automatically triggers a salivation response) with the sound of a bell, at first, and then the sight of a lab assistant's white coat. Eventually, the lab coat alone elicited a salivation response from the dogs.

2. **Different factors can influence the classical conditioning process.** During the first part of the classical conditioning process, known as acquisition, a response is established and strengthened. Factors such as the prominence of the stimuli and the timing of presentation can play an important role in how quickly an association is formed.

When an association disappears, this is known as extinction, causing the behavior to weaken gradually or vanish. Factors such as the strength of the original response can play a role in how quickly extinction occurs. The longer a response has been conditioned, for example, the longer it may take for it to become extinct.

4. **Learning can also occur through rewards and punishments.** Behaviorist B.F. Skinner described operant conditioning as the process in which learning can occur through reinforcement and punishment. More specifically, by forming an association between a certain behavior and the consequences of that behavior, you learn. For example, if a parent rewards their child with praise every time they pick up their toys, the desired behavior is consistently reinforced. As a result, the child will become more likely to clean up messes.

5. **Reinforcement schedules are important in operant conditioning.** This process seems fairly straight forward – simply observe a behavior and then offer a reward or punishment. However, Skinner discovered that the timing of these rewards and punishments has an important influence on how quickly a new behavior is acquired and the strength of the corresponding response.

Continuous reinforcement involves rewarding every single instance of a behavior. It is often utilized at the beginning of the operant conditioning process. But as the behavior is learned, the schedule might switch to one of partial reinforcement. This involves offering a reward after a number of responses or after a period of time has elapsed. Sometimes, partial reinforcement occurs on a consistent or fixed schedule. In other instances, a variable and unpredictable number of responses or time must occur before the reinforcement is delivered.

6. **Several thinkers influenced behavioral psychology.** In addition to those already mentioned, there are a number of prominent theorists and psychologists who left

an indelible mark on behavioral psychology. Among these are Edward Thorndike, a pioneering psychologist who described the law of effect, and Clark Hull, who proposed the drive theory of learning.

7. **There are a number of therapeutic techniques rooted in behavioral psychology.** Though behavioral psychology assumed more of a background position after 1950, its principles still remain important. Even today, behavior analysis is often used as a therapeutic technique to help children with autism and developmental delays acquire new skills. It frequently involves processes such as shaping (rewarding closer approximations to the desired behavior) and chaining (breaking a task down into smaller parts and then teaching and chaining the subsequent steps together). Other behavioral therapy techniques include aversion therapy, systematic desensitization, token economies, modeling, and contingency management.
8. **Behavioral psychology has some strengths.** Behaviorism is based on observable behaviors, so it is sometimes easier to quantify and collect data when conducting research. Effective therapeutic techniques such as intensive behavioral intervention, behavior analysis, token economies, and discrete trial training are all rooted in behaviorism. These approaches are often very useful in changing maladaptive or harmful behaviors in both children and adults.
9. **It also has some weaknesses.** Many critics argue that behaviorism is a one-dimensional approach to understanding human behavior. They suggest that behavioral theories do not account for free will and internal influences such as moods, thoughts, and feelings. Also, it does not account for other types of learning that occur without the use of reinforcement and punishment. Moreover, people and animals can adapt their behavior when new information is introduced even if that behavior was established through reinforcement.
10. **Behavioral psychology differs from other perspectives.** One of the major benefits of behaviorism is that it allowed researchers to investigate observable behavior in a scientific and systematic manner. However, many thinkers believed it fell short by neglecting some important influences on behavior. Freud, for example, felt that behaviorism failed by not accounting for the unconscious mind's thoughts, feelings, and desires that influence people's actions. Other thinkers, such as Carl Rogers and the other humanistic psychologists, believed that behaviorism was too rigid and limited, failing to take into consideration personal agency. More recently, biological psychology has emphasized the power the brain and genetics play in determining and influencing human actions. The cognitive approach to psychology focuses on mental processes such as thinking, decision-making, language, and problem-solving. In both cases, behaviorism neglects these processes and influences in favor of studying just observable behaviors.

Humanistic-Existential

Humanistic psychology can be traced to Abraham Maslow as the founding father, but through time has become closely associated with Carl Roger's Person-Centered Therapy (or Client-Centered Therapy). However, humanistic psychology today is much broader and more complex than Maslow and Roger's foundational approach.

Today, it is common for people to view existential psychotherapy as one of the humanistic psychotherapies. There are many reasons for this assimilated view of these therapies. For one, since their emergence these two approaches have been in close dialogue. Secondly, there have been many attempts to blend existential and humanistic therapy (i.e., Bugental's existential-humanistic psychotherapy). Finally, both approaches share many of the same values.

Similarities to Existential Psychology

Both approaches are phenomenological. While the term phenomenology is a complicated term which many psychologists and philosophers disagree about, the essence of what it means for these approaches is that they value personal experience and subjectivity. Psychology, in its attempt to become a science, has developed a preference for the objective. While phenomenological approaches don't discount the importance of objective approaches, they would focus on the limitations of objectivity. This, in part, means objective knowledge is only one part of the big picture.

The "here-and-now" or the therapeutic moment is a shared value of these approaches. While the past is important, it is also important not to forget the present. Included in the here-and-now is a commitment to understanding, processing, and valuing the therapeutic relationship. This relationship is seen as being a real relationship under unique constraints, boundaries, and contexts. In other words, while many psychoanalytic approaches see the therapy relationship as primarily a product of transference, humanistic and existential approaches focus on the real in the relationship in addition to the transference/countertransference patterns.

Both approaches value self-awareness. In the more general sense, this is shared with all the depth psychotherapies. However, there is another unique aspect to self-awareness within humanistic and existential thought. Self-awareness in the more general sense refers to an understanding of the self which is primarily seen as accumulated life experience and unconscious knowledge. In humanistic and existential thought, self-awareness is also deeply concerned with the human condition and how this impacts the individual self.

Humanistic and existential approaches both value the basic goodness in people and the human potential. Part of the therapy process is understood as freeing the individual up to embrace their basic goodness and potential. In doing this, it is believed they will be happier and satisfied with life.

Differences from Existential Psychology

While both approaches believe in the human potential and goodness, existentialism has focused more on the potential for evil and human limitation. This is more of a distinction of process than basic values. In other words, humanistic psychology typically espouses a similar position to existentialism, but humanistic therapists have not spent as much time dwelling in the shadow or daimonic. This distinction should not be minimized despite the shared foundation of their beliefs. Through time, humanistic psychology has been unfairly characterized as being overly "warm and fuzzy." Many people have shied away from this theoretical approach because of the perception that it does not deal with the reality of the human condition. Conversely, existentialists often get accused of spending too much time in the dark places and being rather morbid. Neither characterization is

accurate, yet they have influenced who has been drawn to the different theoretical positions and how they have developed over time.

An important discussion between Carl Rogers and Rollo May highlights and extends these differences. The discussion began with an article published by Carl Rogers in the Association for Humanistic Psychology's Perspectives. It was followed by a later article published by May (1982) in the Journal of Humanistic Psychology along with a reply by Rogers (1982; both articles were republished in Miller, 1992). For Rogers, human evil is distinct from human nature. It is located in the culture. For May, people innately have both the potential for good and for evil. For Rogers, and many humanistic psychologists, evil is an external reality which impacts individuals through culture and socialization. May voices concern for this partially because he does not believe this adequately deals with our own potential for evil.

In this same dialogue, May (1982) points out another important distinction which sometimes arises between humanistic and existential therapists. May voices some concern that in the extreme focus on the client and the empathetic response in client-centered therapy, there is some cost to a deeper genuine engagement that requires the therapist to focus on their own subjective experience. Stated differently, humanistic psychologists may focus on the client, at times, to the expense of acknowledging their own experience. In doing so, the client is deprived of the opportunity for a deeper engagement with the therapist as a subjective self.

These two theories have different philosophical roots. Humanistic thought is not as closely associated with humanistic philosophy as existential psychology is to existential philosophy. In fact, the confusion between humanistic psychology and humanism in the historical sense is rather significant. Generally speaking, phenomenological, continental, and existential philosophies have influenced both humanistic and existential psychology more than humanistic philosophy and humanism. In stating this, it still must be acknowledged that there are many broad approaches to humanistic psychology.

Humanistic psychology has tended to focus more on the art of therapy, the subjective, and intentionality, to the neglect of the science of therapy, the objective, and human limitation. While many humanistic thinkers would rightly challenge this statement, when compared to existential thought, there is a strong case for this statement. Existentialism tends to be more balanced. Its values are consistent with humanistic psychology's focus, but it creates more space for science, objectivity, and human limitation.

Lastly, it could be maintained that existential thought maintains a more flexible framework to integrate in other approaches. This is even true in regards to the solution-focused therapies. While personally I have some strong concerns about cognitive-existential approaches, at the same time I consider it a strength of existential thought that it is adaptable enough to allow for this. Existential psychology is more commonly used as a frame which integrates other depth approaches. While this can still be done with humanistic psychology, it is not as naturally as adaptable.

In the end, it is the adaptability and balance which are the strengths of existential psychology. Many times in my career I've re-evaluated where my therapy foundation lies. I have considered whether Jungian, relational psychoanalysis, or humanistic approaches would be a better fit for my values. In the end, it is also this adaptability and balance which keeps me convinced that the existential approach is the more appropriate foundation when compared to other depth psychotherapy approaches.

Transpersonal

Transpersonal psychology is one of the lesser known fields in psychology. It began in the late 1960s, as an attempt to establish a “fourth force” in psychology, following the psychodynamic, behaviourist and humanistic approaches. To a large extent, it was an outgrowth of humanistic psychology—in fact, one of the best known humanistic psychologists, Abraham Maslow, was a pioneer of the transpersonal approach. Transpersonal psychology was strongly influenced by the “human potential” and counterculture movements of the 1960s, and the wave of psycho-experimentation it involved, through psychedelic substances, meditation and other consciousness-changing practices. You could see transpersonal psychology as an attempt to understand the different states of consciousness—and the different views of reality—which were revealed through this experimentation. At the same time, it was an attempt to integrate the ideas and insights of western psychology with the insights of eastern spiritual traditions, such as Buddhism and Hindu Vedanta and Yoga, particularly their examination of “higher” states of consciousness, and “higher” stages of human development. In Abraham Maslow’s words, the role of transpersonal psychology was to explore the “farther reaches of human nature.”

This is one of the reasons why Transpersonal Psychology appeals to me so strongly - because one of its central principles is that what we think of as a “normal” state of being is in some ways limited. It recognises that there are more expansive and more intense states of awareness which we can experience in certain circumstances. (I call these “awakening experiences.”) It suggests that what other psychologists might view as “optimum” human psychological functioning—e.g. freedom from anxiety and irrational negative thought-patterns, an optimistic outlook, a strong sense of identity—is by no means the end point of our development. There are potentially higher functioning states in which our perception becomes intensified, we experience an increased sense of connection to nature and to other human beings, become more compassionate and altruistic, have a wider sense of perspective and live more authentically, and so on.

I have had many experiences of these states myself, and have examined many cases of them in other people. My book *Waking From Sleep* is a study of temporary awakening experiences, while my book *Out of the Darkness* is a study of people who have undergone a permanent shift into a higher-functioning “wakeful” state following periods of intense turmoil. I think it’s important to view these states from a psychological perspective, rather than through the prism of Eastern spiritual traditions. I have found that by far the majority of these states occur outside the context of those traditions. They don’t often occur to people while they are meditating or doing yoga, or to people who would describe themselves as Buddhists or spiritual seekers. They most frequently occur in the midst of everyday life, while people are walking in the countryside, running or swimming, watching an arts performance, or while they are in the midst of stress and psychological turmoil. Most of the people I interviewed for my book *Out of the Darkness* had no background—or even any interest—in spiritual traditions or practices. So rather than investigating spiritual traditions, or arguing about whether there is such a thing as a “perennial philosophy”—that is, a common mystical core underlying all different spiritual traditions—I believe it’s much more fruitful for transpersonal psychologists to look outside these traditions, at the awakening experiences which many people are having in a completely secular context.

Other psychological approaches touch on these states, but tend to deny their validity. For example, in Freudian psychology “awakening experiences” are treated as a form of

regression, back to the state oneness and well-being which we experienced in our mother's womb, or in early childhood. Psychiatrists and neuro-psychologists tend to see them as a kind of aberration, caused by unusual neurological functioning. Transpersonal psychology is different in that it views these states not as aberrational but revelatory, not as sub-normal but super-normal, as a glimpse of a more intense reality rather than an illusory one. It believes that they bring significant insights into our true nature and into reality itself, and offer a glimpse of our potential as human beings.

While transpersonal psychology has been on the periphery for a long time, its significance may be increasing. There are many important contemporary trends in psychology—and science in general—which relate very strongly to transpersonal theory. Mindfulness is certainly very closely connected to transpersonal psychology, as is contemporary interest in consciousness, and the recent renewal of research into the therapeutic properties of psychoactive substances. There are many areas of positive psychology which intersect with the concerns of transpersonal psychology too, such as the study of altruism, well-being and states of flow. Some time ago, it was easy to dismiss TP as too theoretical and "unscientific," but in recent years there has been an increased focus on research, both qualitative and quantitative. As a result, transpersonal psychology appears to be moving closer to the mainstream.

Although transpersonal psychology was formulated as an approach during the 1960s, its roots go back even further—as far back as the great pioneering psychologist William James, who laid out the principles of so many psychological approaches. Over hundred years ago, James expressed the essential insight and inspiration of transpersonal psychology in the following passage:

Our normal waking consciousness, rational consciousness as we call it, is but one special type of consciousness, whilst all about it, parted from it by the filmiest of screens, there lie potential forms of consciousness entirely different. We may go through life without suspecting their existence; but apply the requisite stimulus, and at a touch they are there in all their completeness, definite types of mentality which probably somewhere have their field of application and adaptation. No account of the universe in its totality can be final which leaves these other forms of consciousness quite disregarded.

Certainly no account of human psychology can be complete if it disregards these states, or if it explains them away as aberrations. Transpersonal psychology's importance lies in its recognition that we are not all that we could be, that the world as we perceive it does not necessarily represent the world as it is, and that our glimpses of a more expansive and higher-functioning state don't have to be temporary—they can become our permanent state.

Cognitive Revolution

Behaviorism's emphasis on objectivity and focus on external behavior had pulled psychologists' attention away from the mind for a prolonged period of time. The early work of the humanistic psychologists redirected attention to the individual human as a whole, and as a conscious and self-aware being. By the 1950s, new disciplinary perspectives in linguistics, neuroscience, and computer science were emerging, and these areas revived interest in the mind as a focus of scientific inquiry. This particular perspective has come to be known as the cognitive revolution (Miller, 2003). By 1967, Ulric Neisser pub-

lished the first textbook entitled *Cognitive Psychology*, which served as a core text in cognitive psychology courses around the country (Thorne & Henley, 2005). Although no one person is entirely responsible for starting the cognitive revolution, Noam Chomsky was very influential in the early days of this movement. Chomsky (1928–), an American linguist, was dissatisfied with the influence that behaviorism had on psychology. He believed that psychology's focus on behavior was short-sighted and that the field had to re-incorporate mental functioning into its purview if it were to offer any meaningful contributions to understanding behavior.

A photograph shows a mural on the side of a building. The mural includes Chomsky's face, along with some newspapers, televisions, and cleaning products. At the top of the mural, it reads "Noam Chomsky." At the bottom of the mural, it reads "the most important intellectual alive."

Noam Chomsky was very influential in beginning the cognitive revolution. In 2010, this mural honoring him was put up in Philadelphia, Pennsylvania.

European psychology had never really been as influenced by behaviorism as had American psychology; and thus, the cognitive revolution helped reestablish lines of communication between European psychologists and their American counterparts. Furthermore, psychologists began to cooperate with scientists in other fields, like anthropology, linguistics, computer science, and neuroscience, among others. This interdisciplinary approach often was referred to as the cognitive sciences, and the influence and prominence of this particular perspective resonates in modern-day psychology.

Cognitive Psychology

Cognitive psychology is radically different from previous psychological approaches in that it is characterized by both of the following:

It accepts the use of the scientific method and generally rejects introspection as a valid method of investigation, unlike phenomenological methods such as Freudian psychoanalysis.

It explicitly acknowledges the existence of internal mental states (such as belief, desire, and motivation), unlike behaviorist psychology.

Cognitive theory contends that solutions to problems take the form of algorithms, heuristics, or insights. Major areas of research in cognitive psychology include perception, memory, categorization, knowledge representation, numerical cognition, language, and thinking.

Multiculturalism

Culture has important impacts on individuals and social psychology, yet the effects of culture on psychology are under-studied. There is a risk that psychological theories and data derived from white, American settings could be assumed to apply to individuals and social groups from other cultures and this is unlikely to be true. One weakness in the field of cross-cultural psychology is that in looking for differences in psychological attributes across cultures, there remains a need to go beyond simple descriptive statistics. In this sense, it has remained a descriptive science, rather than one seeking to determine cause and effect. For example, a study of characteristics of individuals seeking treatment for a binge eating disorder in Hispanic American, African American, and Caucasian American individuals found significant differences between groups. The study

concluded that results from studying any one of the groups could not be extended to the other groups, and yet potential causes of the differences were not measured.

This history of multicultural psychology in the United States is a long one. The role of African American psychologists in researching the cultural differences between African American individual and social psychology is but one example. In 1920, Cecil Sumner was the first African American to receive a Ph.D. in psychology in the United States. Sumner established a psychology degree program at Howard University, leading to the education of a new generation of African American psychologists. Much of the work of early African American psychologists (and a general focus of much work in first half of the 20th century in psychology in the United States) was dedicated to testing and intelligence testing in particular. That emphasis has continued, particularly because of the importance of testing in determining opportunities for children, but other areas of exploration in African-American psychology research include learning style, sense of community and belonging, and spiritualism.

The American Psychological Association has several ethnically based organizations for professional psychologists that facilitate interactions among members. Since psychologists belonging to specific ethnic groups or cultures have the most interest in studying the psychology of their communities, these organizations provide an opportunity for the growth of research on the impact of culture on individual and social psychology.

Before the time of Wundt and James, questions about the mind were considered by philosophers. However, both Wundt and James helped create psychology as a distinct scientific discipline. Wundt was a structuralist, which meant he believed that our cognitive experience was best understood by breaking that experience into its component parts. He thought this was best accomplished by introspection.

William James was the first American psychologist, and he was a proponent of functionalism. This particular perspective focused on how mental activities served as adaptive responses to an organism's environment. Like Wundt, James also relied on introspection; however, his research approach also incorporated more objective measures as well.

Sigmund Freud believed that understanding the unconscious mind was absolutely critical to understand conscious behavior. This was especially true for individuals that he saw who suffered from various hysterias and neuroses. Freud relied on dream analysis, slips of the tongue, and free association as means to access the unconscious. Psychoanalytic theory remained a dominant force in clinical psychology for several decades.

Gestalt psychology was very influential in Europe. Gestalt psychology takes a holistic view of an individual and his experiences. As the Nazis came to power in Germany, Wertheimer, Koffka, and Köhler immigrated to the United States. Although they left their laboratories and their research behind, they did introduce America to Gestalt ideas. Some of the principles of Gestalt psychology are still very influential in the study of sensation and perception.

One of the most influential schools of thought within psychology's history was behaviorism. Behaviorism focused on making psychology an objective science by studying overt behavior and de-emphasizing the importance of unobservable mental processes. John Watson is often considered the father of behaviorism, and B.F. Skinner's contributions to our understanding of principles of operant conditioning cannot be underestimated.

As behaviorism and psychoanalytic theory took hold of so many aspects of psychology, some began to become dissatisfied with psychology's picture of human nature. Thus, a humanistic movement within psychology began to take hold. Humanism focuses on the potential of all people for good. Both Maslow and Rogers were influential in shaping humanistic psychology.

During the 1950s, the landscape of psychology began to change. A science of behavior began to shift back to its roots of focus on mental processes. The emergence of neuroscience and computer science aided this transition. Ultimately, the cognitive revolution took hold, and people came to realize that cognition was crucial to a true appreciation and understanding of behavior.

EARLY SCHOOLS OF PSYCHOLOGY: STILL ACTIVE AND ADVANCED BEYOND EARLY IDEAS

School of Psychology	Description	Earliest Period	Historically Important People
Psychodynamic Psychology	Focuses on the role of the unconscious and childhood experiences in affecting conscious behavior.	Very late 19th to Early 20th Century	Sigmund Freud, Erik Erikson
Behaviorism	Focuses on observing and controlling behavior through what is observable. Puts an emphasis on learning and conditioning.	Early 20th Century	Ivan Pavlov, John B. Watson, B.F. Skinner
Cognitive Psychology	Focuses not just on behavior, but on mental processes and internal mental states.	1920s	Ulric Neisser, Noam Chomsky, Jean Piaget, Lev Vygotsky
Humanistic Psychology	Emphasizes the potential for good that is innate to all humans and rejects that psychology should focus on problems and disorders.	1950s	Abraham Maslow, Carl Rogers

Four Founding Paths of Academic Psychology

Psychology is a relatively young science with its experimental roots in the 19th century, compared, for example, to human physiology, which dates much earlier. As mentioned, anyone interested in exploring issues related to the mind generally did so in a philosophical context prior to the 19th century. Two men, working in the 19th century, are generally credited as being the founders of psychology as a science and academic discipline that was distinct from philosophy. Their names were Wilhelm Wundt and William James. This section will provide an overview of the shifts in paradigms that have influenced psychology from Wundt and James through today.

Wundt

Wilhelm Wundt (1832–1920) was a German scientist who was the first person to be referred to as a psychologist. His famous book entitled *Principles of Physiological Psychology* was published in 1873. Wundt viewed psychology as a scientific study of conscious experience, and he believed that the goal of psychology was to identify components of consciousness and how those components combined to result in our conscious experience.

Wundt used introspection (he called it “internal perception”), a process by which someone examines their own conscious experience as objectively as possible, making the human mind like any other aspect of nature that a scientist observed. Wundt’s version of introspection used only very specific experimental conditions in which an external stimulus was designed to produce a scientifically observable (repeatable) experience of the mind (Danziger, 1980). The first stringent requirement was the use of “trained” or practiced observers, who could immediately observe and report a reaction. The second requirement was the use of repeatable stimuli that always produced the same experience in the subject and allowed the subject to expect and thus be fully attentive to the inner reaction. These experimental requirements were put in place to eliminate “interpretation” in the reporting of internal experiences and to counter the argument that there is no way to know that an individual is observing their mind or consciousness accurately, since it cannot be seen by any other person. This attempt to understand the structure or characteristics of the mind was known as structuralism. Wundt established his psychology laboratory at the University at Leipzig in 1879. In this laboratory, Wundt and his students conducted experiments on, for example, reaction times. A subject, sometimes in a room isolated from the scientist, would receive a stimulus such as a light, image, or sound. The subject’s reaction to the stimulus would be to push a button, and an apparatus would record the time to reaction. Wundt could measure reaction time to one-thousandth of a second.

However, despite his efforts to train individuals in the process of introspection, this process remained highly subjective, and there was very little agreement between individuals. As a result, structuralism fell out of favor with the passing of Wundt’s student, Edward Titchener, in 1927 .

James

William James (1842-1910) was the first American psychologist who espoused a different perspective on how psychology should operate. James was introduced to Darwin’s theory of evolution by natural selection and accepted it as an explanation of an organism’s characteristics. Key to that theory is the idea that natural selection leads to organisms that are adapted to their environment, including their behavior. Adaptation means that a trait of an organism has a function for the survival and reproduction of the individual, because it has been naturally selected. As James saw it, psychology’s purpose was to study the function of behavior in the world, and as such, his perspective was known as functionalism. Functionalism focused on how mental activities helped an organism fit into its environment. Functionalism has a second, more subtle meaning in that functionalists were more interested in the operation of the whole mind rather than of its individual parts, which were the focus of structuralism. Like Wundt, James believed that introspection could serve as one means by which someone might study mental activities, but James also relied on more objective measures, including the use of various recording devices, and examinations of concrete products of mental activities and of anatomy and physiology.

Freud

Perhaps one of the most influential and well-known figures in psychology’s history was Sigmund Freud. Freud (1856–1939) was an Austrian neurologist who was fascinated by patients suffering from “hysteria” and neurosis. Hysteria was an ancient diagnosis for

disorders, primarily of women with a wide variety of symptoms, including physical symptoms and emotional disturbances, none of which had an apparent physical cause. Freud theorized that many of his patients' problems arose from the unconscious mind. In Freud's view, the unconscious mind was a repository of feelings and urges of which we have no awareness. Gaining access to the unconscious, then, was crucial to the successful resolution of the patient's problems. According to Freud, the unconscious mind could be accessed through dream analysis, by examinations of the first words that came to people's minds, and through seemingly innocent slips of the tongue. Psychoanalytic theory focuses on the role of a person's unconscious, as well as early childhood experiences, and this particular perspective dominated clinical psychology for several decades.

Freud's ideas were influential, and you will learn more about them when you study life-span development, personality, and therapy. For instance, many therapists believe strongly in the unconscious and the impact of early childhood experiences on the rest of a person's life. The method of psychoanalysis, which involves the patient talking about their experiences and selves, while not invented by Freud, was certainly popularized by him and is still used today. Many of Freud's other ideas, however, are controversial. Drew Westen (1998) argues that many of the criticisms of Freud's ideas are misplaced, in that they attack his older ideas without taking into account later writings. Westen also argues that critics fail to consider the success of the broad ideas that Freud introduced or developed, such as the importance of childhood experiences in adult motivations, the role of unconscious versus conscious motivations in driving our behavior, the fact that motivations can cause conflicts that affect behavior, the effects of mental representations of ourselves and others in guiding our interactions, and the development of personality over time. Westen identifies subsequent research support for all of these ideas.

More modern iterations of Freud's clinical approach have been empirically demonstrated to be effective. Some current practices in psychotherapy involve examining unconscious aspects of the self and relationships, often through the relationship between the therapist and the client. Freud's historical significance and contributions to clinical practice merit his inclusion in a discussion of the historical movements within psychology.

Dilthey

Wilhelm Dilthey was a German philosopher who lived from 1833–1911. Dilthey is best known for the way he distinguished between the natural and human sciences. Whereas the main task of the natural sciences is to arrive at law-based causal explanations, the core task of the human sciences is the understanding of the organizational structures of human and historical life. It will be shown that this distinction is not so sharp as to rule out causal explanations in human sciences such as psychology and history; it merely delimits the scope of such explanations. Dilthey's aim was to expand Kant's primarily nature-oriented Critique of Pure Reason into a Critique of Historical Reason that can also do justice to the social and cultural dimensions of human experience. Understanding the meaning of human historical events requires being able to organize them in their proper contexts and to articulate the structural uniformities that can be found in this way. Dilthey's reflections on the human sciences, historical contextualization and hermeneutics influenced many subsequent thinkers such as Husserl, Heidegger, Cassirer, Gadamer and Ricoeur. Recently, there has been some attention given to the ways in

which Dilthey's empirical approach to experience influenced Carnap in his early attempts to overcome metaphysics.

Issues: Crisis in Psychology due to Strict Adherence to Experimental-analytical Paradigm (Logical Empiricism)

Permanent Crisis

One of the key features of the methodological self-consciousness of psychology that has accompanied it since its first steps is a permanent sense of crisis invariably registered since the times of W. James (James, 1890; etc.). The following are the commonly noted symptoms of the crisis:

- the disjointed character of the psychological science, a lack of conceptual unity;
- lack of a single universally shared theory;
- the mosaic-like and unsystematic character of psychological knowledge;
- lack of universal criteria of verification and authenticity of knowledge;
- non-cumulative nature of knowledge, with each new psychological trend declaring all the preceding psychology to be a collection of misconceptions and artifacts;
- separation of the whole personality into memory, thought, perception, attention and other mental functions, each leading a strange independent existence;
- various "parallelisms"-psycho-physical, psycho-physiological, psycho-biological and psycho-social – which psychology feels unable to resolve for itself, "puzzles," to borrow T. Kuhn's words (Kuhn, 1962); and others.

True, at least three circumstances mitigate the permanent sense of crisis. First, in terms of their attitude to methodology psychologists may be divided into four main categories: (a) those who are largely indifferent to general methodological issues; (b) methodological rigorists who adhere to traditional, largely Positivist, research standards; (c) methodological anarchists generally sharing Paul Feuerabend's "anything goes" credo; and (d) methodological liberals who combine post-modernists and more traditional research approaches. By far the largest number of psychologists belongs to the first group, which makes the psychological community not very sensitive to the "eternal" methodological problems of psychology. Secondly, psychologists have got used to its permanent crisis which they often perceive as the normal state of their "abnormal" science. This sentiment, among other things, strengthens the long-standing conviction of the "special path" of psychology whereby its status cannot be assessed by comparing it to the natural sciences regarding their dissimilarity as a manifestation of the crisis of psychology.

Third, psychology's methodological discourse expresses a kind of "poetics" of crisis, perceived in a positive light as something that stimulates the development of the psychological science, makes it more perceptive of methodological problems, etc., which prevents a sense of permanent crisis degenerating into a sense of permanent frustration. Studiedly optimistic statements such as "it is unclear whether there was a crisis or not, but it's good it's over" (McGuire, 1976) have gained some currency.

Even so, in spite of the circumstances that mitigate the sense of crisis much of the psychological community is highly sensitive to it, and not so much to the symptoms mentioned above but to the lack of progress in overcoming them. Assessments of the general methodological status of psychology given by W. James, K. Buhler and others

more than a hundred years ago do not differ much from the modern assessments. Thus, any present-day psychologist would probably go along with W. James' contention that psychology is reminiscent of physics before Galileo: there is not a single universally recognized fact nor a single universally shared generalization (James, 1890). Lack of progress in overcoming the crisis puts into question the progress of psychological science in general, questions that cannot be overcome even by such glib statements as the following: "a century of psychological research has advanced tremendously our understanding of human behavior". The crisis is compounded by the fact that in addition to the above mentioned cognitive symptoms which manifest the state of psychological knowledge there are social symptoms that reflect the status of the psychological community and its social environment. The "vertical" disunity of psychology - with various schools such as cognitivism, behaviorism, and psychoanalysis, "horizontal" division into natural-scientific and humanitarian psychology and "diagonal" division into research and practical psychology is compounded by the watering down of the foundations of scientific rationality, which in turn affects the cognitive status of psychology.

COGNITIVE SYSTEMS IN PSYCHOLOGY

Types of cognitive systems	Constituent characteristics	Examples
Meta-theories	Understanding and explaining the psyche, approach to its study, etc.	Cognitivism, behaviourism, psychoanalysis, etc.
Paradigms	Vision of the subject matter of psychology, methods of solving its "puzzles", etc.	Natural-scientific and humanitarian psychology
Sociodigms	Interdisciplinary communities	Research and practical psychology
Metadigms	Types of rationality	Western science, Eastern science, para-science, religion, etc.

"Horizontal" Rupture

The division of psychology into various schools that look like psychological empires is one of the commonly known symptoms of the permanent crisis of psychology. The number of "psychological empires keeps growing as new "states within the psychological state" such as social constructionism, dialectical psychology, etc. are added to the traditional ones such as cognitivism, behaviorism, and psychoanalysis (Smith, 2001). The proliferation of schools of thought that cannot find a common language (Ohlsson, 2007) is a great irritant to psychologists who seek to unite these schools in larger "types of psychology" such as "German-Austrian" and "North-American psychology". It multiplies the number of horizontal demarcation lines within the psychological science. Such demarcations of the psychological science are reproduced and consolidated in psychological practice because its main areas of research give rise to corresponding trends in practical psychology. The feeble unification trends, far from diminishing, tend to increase entropy because their product as a rule is new, independent research currents that bear little resemblance to their "progenitors".

The main challenges in the way of integration of psychology spring not so much from the multiplicity of schools and trends arising on the basis of psychological meta-theories, as from their incommensurability, to use T. Kuhn's term. Each "psychological empire"

lives according to its own laws including the rules of generating psychological knowledge, criteria of its authenticity, verification methods, etc. Meanwhile, the psychological meta-theories that underlie research trends are "more than theories". C.H. Franklin rightly notes that psychological theories are rules which prescribe what the researcher should do and how, i.e. psychological theories effectively perform paradigmatic functions.

Naturally, most psychological theories generate a huge body of empirical studies carried out on their basis and such publications as Psychological Abstracts bring together both under the heading of "theory". And, as is well known, the results of empirical studies inspired by this or that psychological theory are usually interpreted on the basis of that theory, thus repeatedly reproducing the "methodological circle" described by K. Danziger. As a result the so-called "crucial experiments" designed to establish which of the conflicting theories is the right one turn out to be largely ineffective because empirical data are interpreted in terms of one of the theories being "verified," which inevitably they confirm. Contrary to Thomas Huxley's dismal statement that "the great tragedy of science is that an ugly fact can kill a beautiful theory" facts per se are unable not only to "kill" a psychological theory, but even to seriously damage it.

To add to the picture, the majority of psychological theories have an "upper" and "lower" tie-ins: the "upper" being the underlying philosophical ideas of the nature of man and society and the "lower" having to do with the personal and psychological traits of their authors. As L. Hjelte and D. Ziegler stress, all the theories of personality are based on certain philosophical premises regarding the nature of man. For example, psychoanalysis adheres to a "gloomy concept of the human nature" (Ibid.) which flows from Freud's study of mentally disturbed people, and humanitarian psychology is wedded to the idea that man is inherently good and capable of self-improvement (Ibid.). L. Hjelte and D. Ziegler emphasize the "lower-end" tie-in of psychological theories regarding the need to explain it as a general methodological principle and claiming that in order to understand the principles preached by this or that student of personality one needs information about his/her religious and social-economic status, the size of his/her family and the sequence of their birth, relations with parents, education and professional experience (Ibid.). Earlier the idea was expressed by B. Eiduson who believed that the theories of human nature are less an intellectual means of expressing the objective reality than the psychological traits of its authors. And J. Richards stressed that the theories any science develops bear the footprints of the personality of their authors, but that there is not a single scientific discipline in which the link stands out in such bold relief as in psychology (Richards, 1987). Not surprisingly, to borrow an apt metaphor from Agnew and Pyke, scientific research is similar to a love affair, and rejection of a once recognized theory is akin to jilting a sweetheart and requires more than just negative information about the sweetheart (Agnew, Pyke). As a result, psychological theories are not just "more", but "much more" than theories, at least much more than the term means in natural sciences, and the theoretical disunity of psychology is something more than the coexistence of various concepts within that science.

In addition to the "upper" and "lower" tie-ins, one can identify many intermediate ties of psychological theories, for example, their social and cultural roots. As noted by K. Huffman, M. Vernoy, J. Vernoy, "Until relatively recently, most psychological research was conducted in Europe and North America. It is from this research that theories of human behavior have been formulated". European theories have pronounced distinctions from North American ones (greater emphasis on social determination of human

behavior, etc.), and, for example, the theories born behind the Iron Curtain, such as the theories of L. Vygotsky and others, are even more peculiar. But even in the absence of an Iron Curtain, the social and cultural features of psychological science are very pronounced. For example, this science in Japan reveals some striking differences from its West European or North American counterparts (Takasuna, 2007). In turn, West European, for example German-Austrian, psychology differs from that in North America (Toomela, 2007), and the popular thesis that science has no national borders, while useful in ideological terms, significantly distorts the true state of affairs when applied to social and humanitarian disciplines.

The incompatibility (to use T. Kuhn's term) of the theoretical systems of psychological knowledge is connected with its pronounced non-cumulative character. To be sure, even the natural-scientific disciplines, which are relatively well-off in cognitive terms, are cumulative only to a limited degree while the concept of the cumulative nature of science seems to have been discarded after its very foundations were undermined by the works of T. Kuhn, I. Lakatos and others. However, the non-cumulative character of the psychological science is also "much more than non-cumulative character": new systems of psychological knowledge are typically built "from scratch" while all the preceding psychological knowledge is either ignored or at best is treated as a collection of instructive mistakes. Although social science, as stressed by D.T. Campbell, needs mutual criticism, repeat analysis of the phenomena revealed and cross-validation more than physical science because the representatives of social sciences and humanities have more limited possibilities of isolating the phenomena studied experimentally; the data they obtain are generated by people who have a great interest in obtaining a certain result; they work in areas in which interest in a certain result is so great that objective description of facts becomes an insignificant motive, etc.

The boundaries between schools in psychology are not of course impenetrable and the mutual cross-pollination and eclecticism of explanatory schemes, inevitable in any science, contribute to overcoming them. As noted by K. Huffman, M. Vernoy and J. Vernoy, "most psychologists take an eclectic approach, using principles and techniques from different perspectives". That mitigates, but does not eliminate the "horizontal" cleavage of psychological science which still lends a mosaic and unsystematic shape to psychological knowledge.

Indic Influences on Modern Psychology

The outline begins with some distinctions between psychology, psychotherapy and psychiatry. Then there is a quick and extremely brief overview of the basic historical trends in psychology. Finally, there is a list with brief descriptions of some of the major individuals who have brought Indic influences to bear on psychology.

Psychology, Psychotherapy and Psychiatry

"Psychology", as taught in universities throughout the world, refers to an academic discipline, and a science which was established in the late 19th century. Several times on the Yoga Psychology list, there has been reference to "science" as if the only real science is that which deals with physics and the brain. However, the research methodology of psychology has been dealing with human experience for over 120 years, even if often or mostly in a reductionist fashion. Psychological science includes a wide variety of disciplines, including developmental psychology, personality psychology, social psychology,

comparative or evolutionary psychology, and among the applied psychological disciplines, educational, sports and neuropsychology. Typically, at least in the United States, a psychologist has a Ph.D. (Doctorate)

"Psychiatry" is a subspecialty of the medical profession. A practicing psychiatrist usually has at least an M.D., though sometimes a Ph.D. as well. Psychiatrists are primarily focused on psychopathology, treating mental illness. Historically (until approximately 50 years ago) this meant serious mental illness – the psychoses, manic-depression, major depression, obsessive-compulsive disorder, etc. With the increasing monetarization of the profession, there has been an widening array of so-called disorders which have increased to the extent that virtually nobody may be said to be free of mental illness.

"Psychotherapy" refers to a healing practice whose beginnings are associated for the most part with the psychoanalysis of Sigmund Freud, who was by training a neurologist, not a psychologist. For the first 50 years of the 20th century, most practitioners of psychotherapy were psychiatrists, which means their training was in the field of medicine. By the 1970s, psychoanalysis had begun to fall out of favor, and the biological approach (drugs, essentially, though electro-convulsive therapy has had a resurgence as well) has gradually come to predominate, not in a small way because of the influence of the pharmaceutical companies. Psychiatrists receive very little training in psychology. Psychologists make use of psychoanalytic ideas and practices, though there are uniquely psychological contributions to therapy such as behavioral therapy, cognitive-behavioral therapy and humanistic therapy, to name the most widely used approaches.

Historical Overview

The psychologist Cyril Burt once summarized the modern history of psychological science in this way: "Psychology first lost its soul, then it lost its mind, until it was finally in danger of losing consciousness altogether". The official beginning of the science of psychology is 1879, when Wilhelm Wundt established the first psychological laboratory in Leipzig, Germany. The first formal school of psychology was the introspectionist school. Previously, psychology had been the domain of the philosophers. In an attempt to separate philosophy and science, psychologists no longer referred to the "soul". The "introspection" of this early school of psychology was an extremely superficial and reductionist cataloguing of sensory experience. There was little progress made during the 30 or 40 years this school was in existence. (Though William James wrote his famous "Principles of Psychology" during this period, he failed to establish a separate school or approach to psychology).

The second psychological school represented a radical break from the Introspectionists, and was initiated in 1913 by John Watson, the first behaviorist. He called for the elimination of all references to internal experience (this is what Burt was referring to as psychology "losing its mind"). B. F. Skinner, several decades later, took this a step further, referring to the brain as a "black box", and refusing acknowledgment of any sort of consciousness at all.

In the late 1950s, the cognitive "revolution" was initiated, often considered to be officially begun with George Miller's article, "The Magic Number 7, plus or minus 2" (this refers to the amount of "bits of information" which the average human being can carry in their memory - the reference to the ancient magic number "7" was made consciously, though the idea that mystical associations were to be taken seriously was strenuously

avoided. You might say that psychological science was finally beginning to regain consciousness at this time!

Since the 1950s, there have no longer been separate competing "schools" of psychology, a problem which has plagued the field of psychotherapy, which has over 400 separate and non-integrated theories of how therapy works. The above-mentioned divisions of psychology - developmental, personality, social, etc. - are considered specialty areas of one discipline, not competing schools of thought, much like nuclear physics and astrophysics, as opposed to the conflicting theories of cognitive therapy and psychoanalysis.

Generally speaking, university departments of psychology hold the purely scientific disciplines in much higher esteem than the applied disciplines of psychotherapy, educational psychology, etc. The names "Freud" and "Jung" are rarely even mentioned past the introductory courses of psychology. Many people who are familiar with popular versions of psychology have heard of the "four forces of psychology" (psychoanalysis, behavior therapy, humanistic therapy, transpersonal therapy). Technically, these are 4 divisions of psychotherapy, and have little connection with the world of scientific psychology. Even within the world of mainstream psychotherapy, the field of transpersonal psychotherapy is considered a fringe discipline. This should be kept in mind for the rest of this outline, as it is in the area of transpersonal theory that the Indic traditions have had the most direct influence.

Indic Influences on Modern Psychology

Gustav Fechner

(I'm writing this partly from memory, and will try to find more in the future. I invite anyone who has more knowledge of Fechner's work to send corrections and additional information).

Fechner was a physiologist with interests in mystical traditions. I believe his interests included Indian philosophy, but I'm not positive. Here is an interesting quotation which shows a surprisingly modern - or even post-modern - approach to science:

"That gravitation extends throughout the whole world is a matter of faith; that laws which are traceable in our limited realm extend limitlessly in space and time is a matter of faith; that there are atoms and lightwaves is a matter of faith; the beginning and the goal of history are matters of faith; even in geometry there are things we take upon faith, such as the number of the dimensions of space and the definition of parallel lines. Indeed, strictly speaking everything is a matter of faith which is not directly experienced... Ultimately the best faith is that which is least contradictory in itself and to all knowledge and to our practical interest.

Franz Brentano

Brentano was a philosopher among whose students was Sigmund Freud. Brentano's student Carl Stumpf, was a psycholgoist who taught Brentano's ideas to the phenomenological philosopher Edmund Husserl. (Again, I'm stating this from memory, so please if you are interested or know more about this, send me information.) I believe that Brentano was familiar with Indian philosophy, though I'm not positive. Certainly, he must have been familiar with the Romantic ideas of the "Unconscious", which according to J.J. Clarke were definitely inspired by readings of various Indian texts. It is extremely important, I believe, in revising our understanding of the history of psychotherapy, to recognize

the extent to which the idea of the Unconscious, popularized by Freud, can be traced by to its Indic roots. Clarke notes that Carus and von Hartmann, two pre-Freudian popularizers of the notion of the Unconscious, were influenced by Vedanta.

Brentano was a phenomenologist, and believed that observation rather than experimentation should be the method of scientific psychology. "The proper subject of psychology is mental activity, for example, the mental act of seeing rather than the mental content (that which is seen)".

William James

James was familiar with Vedanta and other branches of Indian philosophy. According to Clarke, he had something of a 'love/hate' relationship with Vedantic ideas, though he accorded them a partially sympathetic hearing in brief sections of his famous "Varieties of Religious Experience". Members of the forum with more knowledge of Indic influences on James are encouraged to send in this information.

Roberto Assagioli

Assagioli, a psychiatrist and the founder of the psychotherapeutic technique known as "Psychosynthesis", made an assiduous study of the Upanisads, the Yoga Sutras, and various Buddhist texts. He openly acknowledges his debt to these writings. Although a contemporary of Freud and Jung, he did not attain widespread recognition until the 1960s. This recognition came primarily from the fringes of the psychotherapy world, particularly among those already interested in Indian religion and philosophy.

In the 1920s, J.H. Schultz, a psychiatrist, developed his "Autogenic training", a fairly widely used system of visualization and relaxation. Although it is largely derived from Yogic practices, most contemporary practitioners are unaware of its Indic roots. This would be an especially fruitful area to bring out, as Autogenic training is an accepted practice among mainstream therapists.

1914. Caroline Rhys Davids: published *Buddhist Psychology*, declaring that "Buddhist thought is very largely an inquiry into mind and its activities" and has much to teach the West.

1918. Friedrich Heiler, an: orientalist said that Zen meditation should be seen as a mental health technique.

Carl Jung

There are many studies of Eastern influences on Jung, among the most notable are Coward's "Jung and Eastern Thought". Jung engaged in extensive studies of Taoism, Mahayana and Vajrayana Buddhism, as well as Vedanta and the Yoga-Sutras. His ambivalence toward these teachings is well-known. He emphatically denied the possibility of experiencing a "universal consciousness", or anything resembling what is known in Vedanta as "the Self". This was not from a sophisticated Buddhist position, but rather from the perspective of European philosophy.

D.T. Suzuki is an interesting example of influences from the other direction. According to Clarke, he was influenced by William James' pragmatism, emphasis on experience and phenomenological analysis of mysticism.

1930s-1940s. Gordon Allport, a personality psychologist at Harvard, met regularly with Swami Akhilananda of the Vedanta Society of Boston to discuss the Upanishads, Yoga Sutras and other Indian texts.

1950s. Michael Murphy visits the Sri Aurobindo Ashram, is deeply inspired, and returns to California to start Esalen, where many of the ideas for the humanistic and transpersonal psychology movements were developed. It was also during the 1950s that D.T. Suzuki came from Japan to California, which led to a dramatic increase in American interest in Zen Buddhism.

1951. Christman Humphreys, the leader of the Buddhist Society of England (I'm not sure that name is correct) said that "the West has more to learn from Buddhism on psychology than it yet knows".

1955. Hubert Benoit, a French psychiatrist, wrote on psychoanalysis and Zen.

The 1960s. Due in no small part to the influence of "mind-expanding" drugs, many individuals became interested in Asian spirituality. A significant number of them traveled East and later brought the fruits of their searches back to the United States. At the same time, a large number of Asian teachers were becoming popular in the West, including Swami Satchitananda, Kirpal Singh, Nahanaponika Thera, Swami Rama, Thich Nhat Hanh, Chogyam Trungpa, Maharishi Mahesh Yogi, Swami Muktananda, Pir Vilayat Kahn, the Karmapa, and many others. Of course, it was Richard Alpert, now known as Ram Das, who along with Timothy Leary was responsible for the widespread use of LSD. Alpert, a psychologist teaching at Harvard, studied with Neem Karoli Baba, a North Indian Guru, along with Jeffrey Miller, who is now known as Surya Das, a widely-respected teacher of the Dzogchen tradition of Tibetan Buddhism. Some of the other American individuals who are now recognized as qualified roshis, swamis and tulkus include Sivananda Radha, Jiyu Kennett Roshi, Jack Kornfield, Robert Frager, Richard Baker Roshi, and others.

1960. Erich Fromm writes on psychoanalysis and Zen.

1965. Medard Boss, a psychiatrist interested in phenomenological and existential thought, visits India to study Yoga, is deeply impressed, and on his return to Europe writes on Yoga and psychotherapy

Karen Horney: a psychiatrist who was a leading psychoanalyst had with personal interest in Zen, and visited Japan near the end of her life to study with a teacher, but didn't incorporate it in her psychotherapy practice

Leaders of humanistic psychology in 1960s:

Carl Rogers, Rollo May and Abraham Maslow - All were indirectly influenced by Eastern thought and practices: According to Rollo May: "Eastern thought never suffered the radical split between subject and object that has characterized Western thought". Maslow referred often to Vedanta in his writings on "Being" vs doing. Rogers was influenced by Taoism in the development of his notion of a natural, "organism" process of self-actualization.

1969. The beginning of the transpersonal psychology movement.

Although there had been Indic influences on psychology, psychiatry and psychotherapy since the late 1800s, the late 1960s and early 1970s saw a virtual explosion of interest in meditation and Eastern spirituality in general. This was true in the popular culture (sorry for being Americo-centric here, but it is the area I know best - if you have information regarding other countries, please send it in) but spread into the sciences as well. Many young people who went to India, Burma, Thailand and other Asian countries in the 1960s returned in the 1970s to receive first-rate scientific training in psychology. One of the major sources of scientific research was the Maharishi Mahesh International University (MIU) founded by individuals who had studied Transcendental Meditation (TM) with

the Maharishi. Over 508 research studies have been conducted on Tm over 2 decades. Some of the more well-known researchers include David Orme Johnson, Richard Wallace, and Charles Alexander. They have done excellent work, and have received over \$2,500,000 in research grants from the National Institutes of Health.

I should mention that there is a strong Buddhist influence in transpersonal psychology. Several years ago, I took a quick look at the major authors who have appeared in the *Journal of Transpersonal Psychology*, and I would estimate that close to 65% are students of Buddhist meditation (some names include Jack Kornfield, Ken Wilber, Daniel Goleman, John Welwood, Claudio Naranjo, Jack Engler, Roger Walsh, Deane Shapiro, Mark Epstein, all students of Zen, Vipassana or Tibetan Buddhism).

Some of the other major figures in transpersonal psychology include:

Charles Tart, an academic psychologist, who published "Altered States of Consciousness" in 1969, which contained an attempt to bring together systems thinking and ideas from teachings on Hindu and Buddhist meditation.

Claudio Naranjo, a psychiatrist, began writing in the late 1960s on: Gestalt therapy and Buddhist meditation.

Robert Ornstein, a psychologist, in 1971 published "The Psychology of Consciousness", which incorporates ideas from Sufism, Buddhism, Taoism and Hinduism.

Allan Weinstock, a psychologist, (named "Swami Ajaya by the Indian Yoga teacher Swami Rama) wrote in 1976 "The Evolution of Consciousness and Psychotherapy". This book is particularly notable as it was published before Ken Wilber's first book, and includes the idea, derived independently from Wilber, of a spectrum of consciousness. However, the ideas in the book are much more closely drawn from Yoga and Vedanta, with the latter being fully acknowledged. Indian "psychology" is depicted as far more subtle and sophisticated than the work of Freud or Jung. Also, unusual in transpersonal writings, Indian psychology is taken to deal with virtually all of the areas which many modern writings feel are better handled by Freud and Jung.

Daniel Goleman, a psychologist and former chief editor of the periodical "Psychology Today", wrote "The Varieties of Meditative Experience" in 1977. Goleman is worth dwelling on for a bit. A friend of Richard Alpert, he went to India to study Hindu and Buddhist meditation. He returned to study psychology at Harvard. He wrote his dissertation studying under Herbert Benson. I'm not absolutely sure of this, but if my memory is correct, it was Goleman who, in what I think was a spark of inspiration with decidedly mixed results, suggested that meditation could be presented in a palatable form to modern secularized individuals as a form of "stress management". He took ideas developed in the 1920s by the physiologist Walter Cannon and later refined by Hans Selye in the 1950s as the "stress cycle" (dealing with the continual arousal of the autonomic nervous system leading to nervous exhaustion). He then suggested that the technique of meditation primarily served to deactivate this stress cycle. At the time, I thought this was a brilliant way of bringing meditation into the mainstream. However, over the years, I've come to see that the result is that meditation has been reduced to something negligible compared to its original purpose. Goleman, for example, defines meditation in "Varieties" as "an attitude of attentional manipulation". Anyone with any meditative experience knows that bringing this sort of attitude to the practice of meditation is likely to lead to headaches and irritability rather than a transformation of consciousness! In any case, the whole development of stress management and the wider disciplines of health psychology,

behavioral medicine and complementary medicine, needs a whole study. Each of these fields has been deeply impacted by Indian practices and ideas, and many of the new centers for alternative and complementary medicine are making big efforts to hide this connection.

One notable exception to this trend is Jon Kabat Zinn. He has been using Buddhist meditation to treat pain patients for over 20 years at the Massachusetts Medical Center in Worcester Massachusetts. He has seen over 10,000 individuals, most of who were referred because conventional medical treatments had been ineffective. He has published several excellent research studies in top-rated journals.

Ken Wilber published his first book, *The Spectrum of Consciousness*, in 1977. An overview of his work can be found on the ECIT website (Infinityfoundation.com/ECIT).

In 1977, biofeedback pioneers Elmer and Alyce Green published "Beyond Biofeedback", in which they described their ground-breaking studies of Swami Rama's ability to control parts of his nervous system which physiologists had previously believed to be beyond the control of the mind. Though they considered themselves Christians, the Greens quite openly acknowledged their life-long interest in and debt-of-gratitude to Indian philosophy and meditative practices.

Kenneth Pelletier, who in 1983 published an excellent book entitled "Toward a Science of Consciousness". I don't know if the Tucson conferences took their name from this book, but it remains a good collection of ideas that have yet to be worked out regarding the integration of Indian ideas and modern science, including not just psychology, but neuroscience, quantum physics, and other disciplines.

Jeremy Hayward, now the president of the late Chogyam Trungpa's "Naropa Institute", published in 1983 an excellent book called "Perceiving Ordinary Magic". A brilliant scientist who trained as a nuclear physicist and worked in molecular biology research with arch-materialist Francis Crick, he brings together ideas from Tibetan Buddhism, phenomenology and Whitehead's process philosophy to propose truly original and inspired solutions to dilemmas in the fields of physics, evolutionary biology, neuroscience and psychology, with some particularly interesting ideas for the psychology of perception.

In 1986, Guy Claxton, a British psychologist, publishes a book on the impact of spiritual traditions of the East on psychology and psychotherapy. Claxton is one of the minority of writers on East/West psychology, along with Charles Tart, who incorporates ideas from cognitive science with meditation.

In 1990, Crook and Fontana published an excellent book on Buddhist influences on modern psychology.

In 1991, Francisco Varela, Evan Thompson and Eleanor Rosch publish "The Embodied Mind". Varela is a cognitive scientist, as well versed in neuroscience as in Tibetan Buddhist philosophy and practice. Evan Thompson, the son of William Irwin Thompson, is a philosopher, and Eleanor Rosch is a cognitive scientist at the University of California at Berkeley. "The Embodied Mind", a highly original book, brings together phenomenological philosophy, cognitive science and Tibetan Buddhism.

1991. Robert Thurman, along with the Dalai Lama and scientists from several disciplines, including neuroscience, psychology, biology, among others, convene the first of several "Mind Science" conferences at Harvard.

I'd like to close with a few quotations that point out the limitations, and to some extent the dangers, of the current largely secularized approach to the integration of meditative practices and psychology. First, this lovely passage from the philosopher Jacob Needleman:

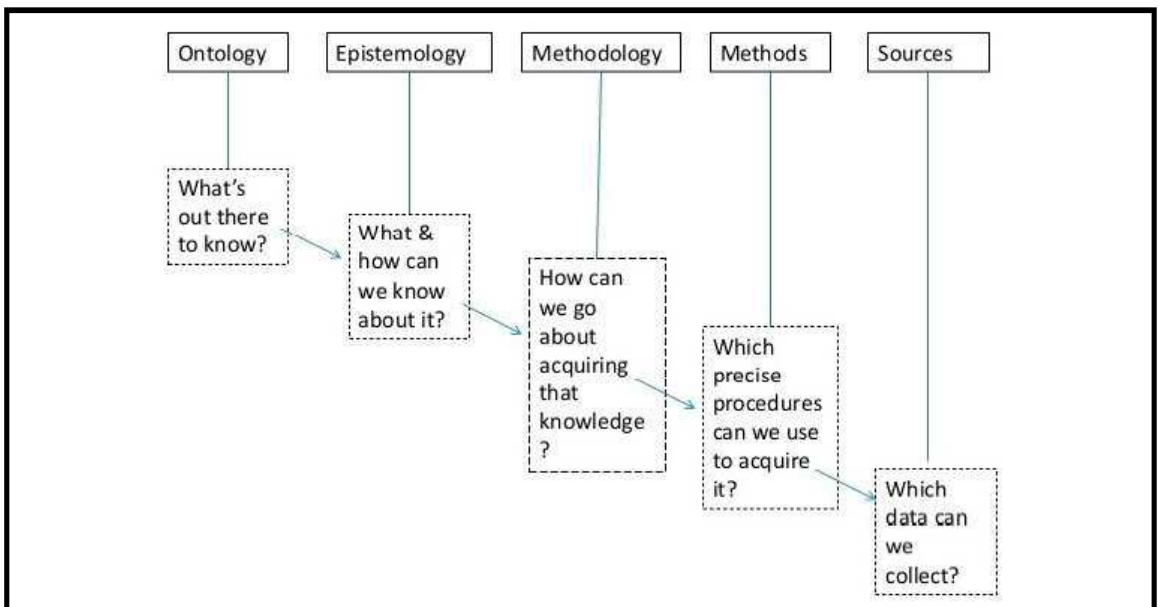
"What part of the mind then is being activated when one is thinking from these particular concepts? [new paradigm concepts such as are found in quantum physics]... What would be the role of an idea, like a holographic idea, in actually activating a different way of life? When Buddha came with the idea, or Christ came with the Christian formulation of the idea that the personality or the ego is not ultimately real, it didn't seem that more than a few people were able to bring that idea into their own tissues, their own blood, their feelings, and their life. [underlining added; D.S.] The problem is not having the right idea...but how to incarnate that idea in my actual life so that I am transformed in the light of that idea...We have never lacked great theory. What we suffer from, as the traditions tell us, is that the good, that I would, that I do not; and that which I do, that I would not... Simply holding a new paradigm is far from being that which can help me. It sometimes is the most dangerous of all things because this is where the danger of the mind is approached. It's not that the mind is wrong; it's just that all the power of attention gets sucked into the mind. Then one is under the illusion that because something is understood with the mind, one can be what is understood."

ESSENTIAL ASPECTS OF KNOWLEDGE PARADIGMS

Ontology

Ontology is the branch of metaphysics which deals with the nature of being, existence as well as category of being and their relations. Ontology as a branch of philosophy is the science of what is, of the kinds and structures of objects, properties, events, processes and relations in every area of reality.

E.g. For a religious person, god exists but for an atheist the God may not exist. Therefore, to determine the ontological perspective of an individual, the culture, environment, education, interactions, society, religion and various other aspects play a vital role.



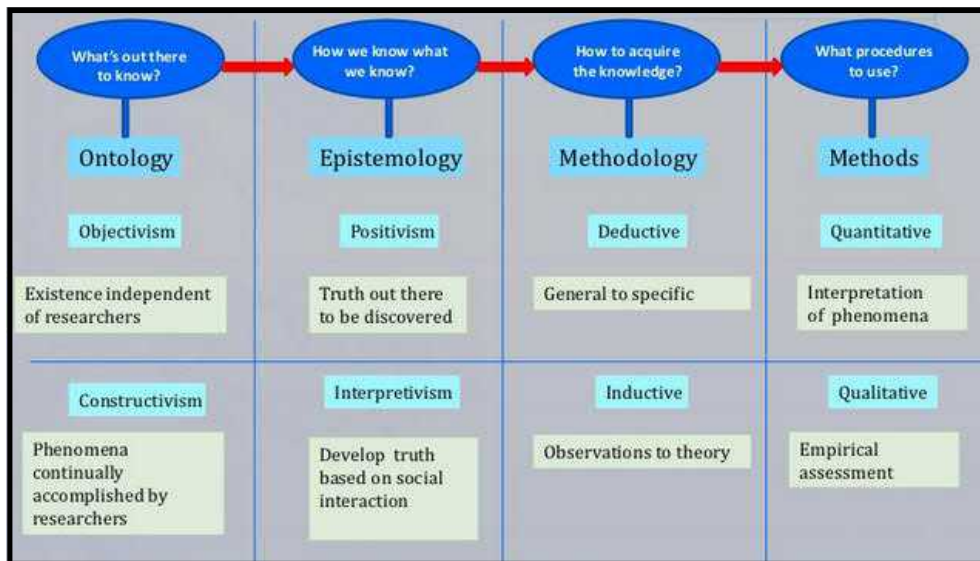
Epistemology: (Nature of knowledge, way of getting knowledge)

Epistemology is the branch of philosophy which deals with the way of obtaining valid knowledge. It is concerned with the nature and scope of knowledge.

E.g. For hard-core natural scientists, knowledge derived from experiment is valid. (i.e. positivism)

For those who believe in theology, knowledge in religious books is valid. (i.e. Theology)

For marxist and feminists knowledge is situated.



Methodology

Methodology is a body of rules, methods and postulates employed by a field of study in order to gain a valid knowledge about a subject matter. Methodology is the study of methods applied in research. E.g. Qualitative, quantitative, Mixed etc.

PARADIGMS OF WESTERN PSYCHOLOGY

Positivism

Positivism emerged as a philosophical paradigm in the 19th century with Auguste Comte's rejection of meta-physics and his assertion that only scientific knowledge can reveal the truth about reality. It was later formally established as the dominant scientific method in the early part of the 20th century by members of the Vienna Circle, including Gustav Bergmann, Rudolf Carnap, Herbert Feigl, Philipp Frank, Karl Menger, Otto Neurath, and Moritz Schlick.

The Vienna Circle sought to construct a unified scientific world-conception that rejects the use of philosophy as a means of learning about the true nature of reality. Unfortunately, it failed as a coherent philosophy of science because of a critical inconsistency between its theory of "reality" and its theory of "knowledge."

Positivism adopted David Hume's theory of the nature of reality (i.e., philosophical ontology). Hume believed that reality consists of atomistic (micro-level) and independent events. He believed in the use of the senses to generate knowledge about reality (i.e., scientific method). He thought that philosophical and logical reasoning could lead us to "see" nonexistent links between events occurring simultaneously. However, positivism also adopted Rene Descartes's epistemology (i.e., theory of knowledge). Descartes believed that reason is the best way to generate knowledge about reality. His deductive method implies that events are ordered and interconnected, and therefore reality is ordered and deducible. This internal inconsistency eventually undermined the validity of positivism.

The positivist paradigm asserts that real events can be observed empirically and explained with logical analysis. The criterion for evaluating the validity of a scientific theory is whether our knowledge claims (i.e., theory-based predictions) are consistent with the information we are able to obtain using our senses. Positivist research methodology (methodological individualism) emphasizes micro-level experimentation in a lab-like environment that eliminates the complexity of the external world (e.g., social, psychological, and economic linkages between unemployment, and crime or suicide). Policies are then prescribed based on conclusions derived via the "scientific method" (e.g., job training for the unemployed, antidepressants for the suicidal, and jail time for the criminal). Psychologists now realize that this yields results that have internal validity (i.e., the relations observed in the experiment are valid within that context). While the results obtained using experimental methods provide valuable insights into the nature of reality, those results may lack external validity. That is, the relations observed in the laboratory may not be the same in the more complicated external world where a much greater number of factors interact.

A positivist dealing with complex social problems such as unemployment and crime would be concerned with their visible manifestations (i.e., the unemployed individual or criminal who can be sensed or perceived) rather than with the underlying causal mechanisms that are invisible to us. Hence, positivist prescriptions tend to treat the symptoms rather than the root cause of the problem.

Positivism exerted an important influence on scientific practice in the social sciences for decades in the early 20th century. This was especially true in the natural sciences where laboratory experiments can closely approximate the real world environment, thus allowing for accurate predictions. In the social sciences, however, human volition and uncertainty make the laboratory experiment less reliable. Ultimately, its internal inconsistency resulted in the abandonment of positivism in favor of scientific approaches such as critical multiplism, which is based on the belief that no one approach is ever sufficient for developing a valid understanding of a phenomenon. The application of critical judgment in investigating multiple research questions using multiple measures, samples, designs, and analyses are necessary to permit a convergence on a valid understanding of a phenomenon.

Post-Positivism

In philosophy and models of scientific inquiry, postpositivism is a metatheoretical stance that critiques and amends positivism. While positivists emphasize independence between the researcher and the researched person (or object), postpositivists argue that theories, background, knowledge and values of the researcher can influence what is

observed. Postpositivists pursue objectivity by recognizing the possible effects of biases. While positivists emphasize quantitative methods, postpositivists consider both quantitative and qualitative methods to be valid approaches.

Critical Perspective

At this point you have learned about four different theoretical paradigms we use to understand communication. One problem with these approaches is they often lack an explicit critique of the status quo of communication. Put another way, they serve as a general approach to understand communication norms rather than challenge them. We all realize that there are communication realities in the world that are hurtful and oppressive to particular people, and that there are people in the world that use communication to serve their own needs and interests. How do we bring these communicative practices to light and work to change communication practices that are hurtful?

Byron Hurt is a modern theorist who uses film to critique how sexism impacts both men and women in our society. His cutting-edge film “Hip-Hop: Beyond Beats and Rhymes” looks at the Hip-Hop industry from a critical perspective, focusing on how it enables sexism against women while keeping men in narrowly defined gendered roles.

The Need for Critical Theories

The Critical Theories Paradigm helps us understand how communication is used to oppress, and provides ways to foster positive social change (Foss & Foss; Fay). Critical Theories challenge the status quo of communication contexts, looking for alternatives to those forms of oppressive communication. These theories differ from other theoretical approaches because they seek praxis as the overarching goal. Praxis is the combination of theory and action. Rather than simply seeking to understand power structures, critical theories actively seek to change them in positive ways. Easily identifiable examples of critical approaches are Marxism, postmodernism, and feminism. These critical theories expose and challenge the communication of dominant social, economic, and political structures. Areas of inquiry include language, social relationships, organizational structures, politics, economics, media, cultural ideologies, interpersonal relationships, labor, and other social movements.

Cultural Studies focus on understanding the real-life experiences of people, examining communication contexts for hidden power structures, and accomplishing positive social change as a result (Dines and Humez; Kellner). According to Kellner, cultural studies involves three interconnected elements necessary for understanding, evaluating and challenging the power dynamics embedded in communication—political economy, textual analysis, and audience reception.

Political economy focuses on the macro level of communication. Specifically, this part of cultural studies looks at the way media as text are situated in a given cultural context, and the political and economic realities of the cultural context. In the U.S., we would note that the political economy is one marked with gender, racial, and class inequities.

Textual analysis involves the process of deconstructing and analyzing elements of a media text. If you wanted to look at a magazine with a critical eye, you would pay attention to the visual elements (the pictures in the ads; the celebrity photos, and any other drawings, cartoons or illustrations), the verbal messages (the text of the ads, the copy, captions that accompany the photographs), and the relationship between the advertise-

ments and the copy. For example, is there an ad for Clinique eye shadow next to an article on the “hot new beauty tips for fall?” You would also want to pay attention to the representation of gender, race, and class identities as well. Are there any differences or similarities between the portrayal of white women and women of color? What sort of class identity is being offered as the one to emulate?

Audience reception asks us to consider the role of the text for the audience that consumes it. You might want to learn why people read particular magazines—what purpose does it fill, what is the social function of this text?

Origins of Critical Theories in Communication

Marxism is one of the earliest origins of critical theory. In addition, postmodernism, feminism, and postcolonialism have greatly influenced how critical theories have grown and expanded to challenge a greater number of social power structures. While each of these approaches examines a different area of oppression, all are critical approaches to enact great social changes, not only in western societies, but in cultures worldwide.

In the late 19th and early 20th centuries, Karl Marx’s ideas challenged the status quo of newly emerging industrial societies. As societies moved from agrarian-based economies to ones based in industrial manufacturing, there became an increasing division between the rich and the poor – much like the income inequality talked about so much today. Marx, in two of his most well-known works, *The Communist Manifesto* and *Capital*, argued that working class laborers were being oppressed by those in power, specifically the owners of manufacturing plants.

In any discussion of Postmodernism, another critical theoretical perspective, the difficulty of defining the term is invariably part of the discussion. Modern refers to just now (from *modo* in Latin) and post means after. Thus, this term translates into “after just now”—an idea that can be difficult to wrap our heads around. How do you, for example, point to or mark the period after just now?. In discussing the postmodern condition, Lyotard explained the relationship between those who have and don’t have social power: “The [decision makers] allocate our lives for the growth of power. In matters of social justice and scientific truth alike, the legitimation of that power is based on optimizing the system’s performance—efficiency”.

A third major influence on the development of the Critical Theories Paradigm comes from feminist theories. Feminist theories explore power structures that create and recreate gendered differentiations in societies. Critical feminist theories contend that gender relations are often oppressive to both men and women, and that they support an institution based on patriarchal values. Thus, critical feminist theories challenge dominant assumptions and practices of gender in ways that foster more equal and egalitarian forms of communication and social structures in society.

When discussing feminism and feminist theories we refer to a set of multiple and diverse theories. Feminist theories include a wide range of philosophical arguments, economic structures, and political viewpoints. Some of these include Marxist feminism, which focuses on the division of labor as a source of gender inequality, and liberal feminism, which asserts that men and women should have equal status in the culture—such as voting rights, educational and professional opportunities, and equal pay. Eco-feminism recognizes that all parts of the universe are interconnected and that oppression of women and other minorities is analogous to the oppression of the natural environment such as

in the cutting down of natural forests to meet consumer demands for paper goods, or the killing of animals for the eating of meat.

Critical Theories in Action

Whether we listen to music on our phones, watch TV, go to the movies, or read a magazine, most of us consume media. Have you ever stopped to think about who puts together those messages? Have you wondered what their goals might be and why they want to send the messages they do? One way we can use critical theories is to examine who owns what media to determine what they are trying to accomplish. For example, why does General Electric want to own companies like RCA and NBC? Why does a company like Seagram's want to buy MCA (Universal Studios) and Polygram records? What world-views are these companies creating in the media they produce? These are all questions for which we might consider using theories from the Critical Theories Paradigm. Did you know that in 1983 50 corporations controlled most of the U.S. media (papers, television, movies, magazines, etc.) and that by 2004 that number has dropped down to five corporations? Using Critical Theories Paradigm, we can begin to examine the messages that so few companies are constructing and their impacts on how we understand the world around us as shaped through these messages.

Other examples from the critical paradigm include works that examine gender, consumerism, advertising, and television. In her work, *Who(se) Am I? The Identity and Image of Women in Hip-Hop*, Perry examines the potential danger and damage to African-American women through their objectification in Hip-Hop videos. Carole A. Stable examines the labor and marketing practices of Nike in her article, *Nike, Social Responsibility, and the Hidden Abode of Production*. Clint C. Wilson II and Felix Gutierrez discuss the portrayal of people of color in advertising in their article, *Advertising and People of Color*, while Jackson Katz explores mask of masculinity with his film, *Tough Guise 2: The Ongoing Crisis of Violent Masculinity*. We use critical theories to reveal a vast range of possible ideological structures that create and foster dominant world-views, and to challenge and change those ideologies that oppress others.

Strengths

A significant strength of the Critical Theories Paradigm is that it combines theory and practice, seeking to create actual change from theoretical development. Rather than seeking prediction and control, or explanation and understanding, critical theories seek positive social change. The intent behind these theoretical perspectives is to help empower those whose world-views and ideological perspectives have not found equality in social contexts. At their best, critical theories have the potential to enact large-scale social change for both large and small groups of people.

Weaknesses

A potential weakness of critical theories is their dependence on social values. While empirical laws theories seek an objective reality, critical theories highlight subjective values that guide communication behaviors. When values conflict the question of, "whose values are better?" emerges. Because values are subjective, answering this question is often filled with much conflict and debate. The example of gay marriage highlights a current debate taking place over ideological values. How do we define marriage? And, whose definition is best?

Social Constructionism

This approach to psychology is inspired by the philosophical tradition developed by thinkers such as Buber, Kierkegaard, Nietzsche, Heidegger, Gadamer, de Beauvoir, Sartre, Marcel, Merleau-Ponty, and Levinas. Existential-phenomenology seeks to develop an in-depth, embodied understanding of human existence. It challenges approaches in psychology and psychiatry that view human beings in a reductionistic manner. The program distinguishes itself by using this philosophical tradition as a foundation for psychotherapy and growth, and as a source of insight into the richness and diversity of human psychological experience.

Existential-Phenomenological Psychology is humanistic in that it challenges the modern tendency to interpret the human condition through a set of narrow technological lenses. It engages with and appreciates the wisdom accumulated by the rich traditions of reflection on the human condition in the social sciences and humanities. It deepens our understanding of the experiences and perspectives of others through its focus upon the meanings that we make in our lives and the choices that are reflected in our understandings and actions. The phenomenological dimension nourishes an openness to understanding the lives and needs of others by helping us to identify and set aside our theoretical and ideological prejudgments as we approach our work as therapists. Finally, the existential-phenomenological approach is ethical in recognizing that the fundamental characteristic of being human is to be responsible to others.

Co-operative Enquiry

The core idea of cooperative inquiry, also known as collaborative inquiry, is to 'research with' rather than to 'research on' people. This means that all active participants are fully involved in research decisions as co-researchers.

Since its inception by John Heron in 1971 different methods developed, each dealing in slightly different ways with issues like self-fulfilling prophecy, group think or group bias, and how representative the resulting information is for people outside the research group.

The John Heron and Peter Reason's Approach

Cooperative inquiry creates a research cycle among four different types of knowledge: propositional knowing (as in contemporary science), practical knowing (the knowledge that comes with actually doing what you propose), experiential knowing (the feedback we get in real time about our interaction with the larger world) and presentational knowing (the artistic rehearsal process through which we craft new practices). The research process iterates these four stages at each cycle with deepening experience and knowledge of the initial proposition, or of new propositions, at every cycle.

Stage 1. The first reflection phase that determines topics and methods of inquiry. This phase involves primarily propositional knowing.

Stage 2. The first action phase, usually within the group, that tests the agreed actions, records outcomes from the testing, and observes if the actions conform to the original ideas from Stage 1. This stage involves primarily practical knowing by doing.

Stage 3. A second action phase, usually by individuals in their everyday life outside the group, where the experiences and the consequences of one's new inquiries in action can generate profound new feelings and awarenesses. In this stage, the experiences

may lead to new fields, actions and insights that depart from the original ideas. This stage involves primarily experiential knowing.

Stage 4. The second reflection phase when, in the group, co-researchers reflect on their experiences and the data collected in Stages 2 and 3. Now they may re-frame the original ideas and amend inquiry procedures. In this stage, co-researchers also decide whether to proceed to further cycles in the inquiry processes. This stage involves primarily presentational knowing, developing new images and ways of acting. This leads back to propositional knowing, if the inquiry group decides to start a next cycle.

Paradigmatic Controversies

Although modern psychology attempts to be a scientific endeavour, the field has a history of controversy. Some criticisms of psychology have been made on ethical and philosophical grounds. Some have argued that by subjecting the human mind to experimentation and statistical study, psychologists objectify persons. Because it treats human beings as things, as objects that can be examined by experiment, psychology is sometimes portrayed as dehumanizing, ignoring or down playing what is most essential about being human. This criticism has come from within the field as well, particularly by existential and humanistic psychologists.

A common criticism of psychology concerns its fuzziness as a science. Philosopher Thomas Kuhn suggested in 1962 that psychology is in a pre-paradigmatic state, lacking the agreement on facts found in mature sciences such as chemistry and physics. Because some areas of psychology rely on "soft" research methods such as surveys and questionnaires, critics have claimed that psychology is not as scientific as psychologists assume. Methods such as introspection and psychoanalysis, used by some psychologists, are inherently subjective. Objectivity, validity, and rigour are key attributes in science, and some approaches to psychology have fallen short on these criteria. On the other hand, greater use of statistical controls and increasingly sophisticated research design, analysis, and statistical methods, as well as a decline (at least within academic psychology departments) in the use of less scientific methods, have lessened the impact of this criticism to some degree.

Debates continue, however, such as the questioned effectiveness of probability testing as a valid research tool. The concern is that this statistical method may promote trivial findings as meaningful, especially when large samples are used. Psychologists have responded with an increased use of effect size statistics, rather than sole reliance on the traditional $p < .05$ decision rule.

There is also concern from researchers about a perceived scientific gap between empirically based practices. Exponents of evidence-based approaches to psychological practice say that "over the past several decades, the fields of clinical psychology, psychiatry, and social work have borne witness to a widening and deeply troubling gap between science and practice" and "less and less of what researchers do finds its way into the consulting room, and less and less of what practitioners do derives from scientific evidence." Moreover there are many "unvalidated and sometimes harmful psychotherapeutic methods" that have been widely adopted by the profession. However, "the fields of clinical psychology, psychiatry, and social work have recently placed increased emphasis on evidence-based mental health practices." Lilienfeld, Scott O. "SRMHP: Our Reason d'Être". Retrieved 13 February 2010.

SIGNIFICANT INDIAN PARADIGMS ON PSYCHOLOGICAL KNOWLEDGE

Yoga

Yoga psychology is the theoretical basis for yoga psychotherapy. Yoga psychotherapy plays a vital role in the successful treatment and management of physical, psychological and spiritual conditions. When combined with Western psychotherapeutic approaches it creates a powerful holistic approach to long-term healing.

One of the greatest things about yoga psychology and psychotherapy is that they not only provide a holistic path to self-healing, they also enable spiritual awakening. This is because in these systems illness can be viewed as a sacred path to higher awareness if one has the appropriate knowledge, techniques and support.

Modern medicine is realizing that a physical approach alone, e.g. prescription drugs, or a psychological approach alone, e.g., Cognitive Behavioral Therapy (CBT), delivers limited and often short-term benefits.

A holistic approach, which is sometimes called a bio-psycho-social-spiritual approach is preferred over the purely biomedical model and the purely psychological model. The biomedical model does not fully recognize the effects of the mind on the body. The purely psychological model does not fully grasp the impact of the body on the mind.

Yoga psychology is a modern term that is in evolution. I define it as: *The science of the embodied relationship between consciousness and mind.*

Yoga psychology sees body and mind as indivisible and gives maximum emphasis to the role of cultivating self-awareness as the foundation on which improved health and wellbeing is built.

Yoga psychology can also be called the science of the “subtle body,” the part of us that lies between and links the physical body and the spiritual, or causal body, the subtlest part of us.

The Four Dimensions of Yoga Psychology

In order to gain a larger vision of the theoretical basis of yoga psychology, we need a 4-dimensional approach that draws from various Indian philosophical systems, including Yoga, Samkhya, Tantra, and Vedanta.

These Indian philosophical traditions store a vast amount of information about the inner workings of the body, mind, and spirit. They give us luminous insights into the organs of the mind, how the organs of the body and mind interact with each other, and the various energies that animate them.

Yoga psychology describes how the idealized healthy body and mind function and how imbalances can occur leading to pathology and suffering. In its highest form, yoga psychology seeks to actualize a conscious relationship with the vast cosmic forces that lie beyond socially conditioned awareness.

Yoga psychotherapy shows us how to transform adversity and to use suffering and adversity to actualize this conscious relationship with those vast cosmic forces. From this perspective, we gain information about our individual karma. We also begin to see our unique life purpose and destiny from a higher perspective, one that links us to a greater and more universal destiny.

Bhagavad Gita

It is said that counseling and psychotherapy was developed in the western countries and imported in India from the western literatures. But before five thousand year ago, Lord Krishna had narrated Bhagavad Gita to Arjuna, which is the perfect example of its basic form. It is relevant in modern world as well.

Arjuna is a great warrior and was the 3rd of the pandava brothers. As the Mahabharata battle comes close, he is obsessed with self-doubt about the righteousness of the battle against his own relatives, his friends and family such as his teacher, Drona and grandfather Bhishma. It was then that lord Krishna took charge and did his counseling and explained him the inevitability and necessity of the Mahabharata battle to Arjuna. This conversation is a key part of the Mahabharata which is known as Bhagavad gita, and is considered as a holy scripture of Hinduism

Arjuna plays the role of the client in the Bhagavad Gita and his charioteer lord Krishna played the role of counselor. The gita is a narrative framework of a philosophical dialogue between arjuna and his charioteer Lord Krishna. And is the part of great epic Mahabharata.

General Appearance and Behavior

Arjuna is a warrior. At the time of Mahabharata (war), he was anxious, nervous, and sad and in state on inaction after seeing his uncle, gurus (teacher), nephews, friends, relatives etc. in the opposite (enemy) camp. On appearance, there were signs of fear, sorrow and guilt. He appeared depressed.

Speech

Arjuna articulated himself clearly. He asked questions spontaneously, although at slow rate and speed. He spoke softly throughout the conversation.

Affect and Mood

Arjuna's affect was depressed. He also appeared anxious and nervous.

Thought

Arjuna thought stream was decreased. He thought that- to win the war means killing all these people whom he loved and respected. He loses his will to fight, saying even if they wish to kill him; he does not wish to kill them. He is in the state of guilt and sadness. He drops his Gandiva (weapons) and turns to his charioteer, Lord Krishna, for help and guidance. He was obsessed with knowing the aftermath of war and was constantly asking the Krishna for war result. These thoughts were causing arjun to lack of motivation and confused. Death wish was detected.

Perception

Arjuna exhibits confusion and paralyzed to make decisions. Symptoms, such as misinterpretations, illusion, passivity phenomena, were elicited.

Cognition

Arjuna was alert and orientated to time and place. He was able to ask questions and recall his past without difficulties.

Insight and Judgment

When lord Krishna questioned about his condition, Arjuna accepted the fact that he confused, sad, anxious and discouraged. Lord Krishna helped a grief-stricken Arjun through dialogue of gita. He has cooperated with lord Krishna.

Summary and Diagnosis

Arjun is a great warrior who had been diagnosed with a depression with predominant guilt. The diagnosis explains his symptoms of persistent guilt, sorrow, anxiety, nervousness, confusion, negative thoughts and death wish. Hence, it is likely that he is in the state of acute depression with symptoms of anxiety. Taking the situation into consideration his condition is most likely to be a condition of maladjustments or faulty coping pattern.

Arjuna has become depressed and did not want to participate in mahabharata.,he is very anxious about the possibility of the destruction that comes to family and society in the aftermath of war. He becomes discourage, anxious, confused and lays down his weapons saying he will not fight.

What is the importance and relevance of this case study in our present life?

In the modern society battle field is the symbol of challenges of our life. We all have mountains of desires, dreams, duties and ambitions, obligations to family, society and self. Each of us has our own society, family and culture in which we embedded and through which our self, behavior, personality and behavior are conditioned. Our life is the perfect example of a battle between good and evil. Many types of question are come to our mind such as; who we are?, what is the purpose and meaning of our life?, what happen to us when we die? It is the existential question that is not directly come on the surface but run like a river beneath our busy life.

Many time we come across a situation where we are afraid to do the right thing because of social pressure and conformity. We face conflict with a choice regarding what we want to do over what we should do. Bhagavad Gita is a story that tells us about the importance of doing the right thing, of conquering our own darkness, regardless of the cost or struggle.

Many times in our life we confront with mountains of problems. We get confused and unable to make decisions. In that time we want someone to help us what to do because in that situation the problems we face seems beyond to our capacity to solve. According to mahatma Gandhi- "When doubts haunt me, when disappointments stare me in the face, and I see not one ray of hope on the horizon, I turn to Bhagavad-Gita and find a verse to comfort me; and I immediately begin to smile in the midst of overwhelming sorrow. Those who meditate on the Gita will derive fresh joy and new meanings from it every day."

The present world is filled with Arjunas and psychologist is now developing Bhagavad Gita as a great clinical tool to cope up with life challenges psychological problems.

Buddhism

Buddhism includes an analysis of human psychology, emotion, cognition, behavior and motivation along with therapeutic practices. A unique feature of Buddhist psychology is that it is embedded within the greater Buddhist ethical and philosophical system, and its

psychological terminology is colored by ethical overtones. Buddhist psychology has two therapeutic goals: the healthy and virtuous life of a householder (*samacariya*, "harmonious living") and the ultimate goal of nirvana, the total cessation of dissatisfaction and suffering (*dukkha*).

Buddhism and the modern discipline of Psychology have multiple parallels and points of overlap. This includes a descriptive phenomenology of mental states, emotions and behaviors as well as theories of perception and unconscious mental factors. Psychotherapists such as Erich Fromm have found in Buddhist enlightenment experiences (e.g. *kensho*) the potential for transformation, healing and finding existential meaning. Some contemporary mental-health practitioners such as Jon Kabat-Zinn increasingly find ancient Buddhist practices (such as the development of mindfulness) of empirically proven therapeutic value, while Buddhist teachers such as Jack Kornfield see Western psychology as providing complementary practices for Buddhists.

Sufism

Sufism is considered the inner dimension of Islam and is based on the teachings of the Prophet Mohammad (PBUH) and the holy Qur'an. Sufi psychology is the application of Sufism. Expressions of the basic concepts vary, sometimes radically, between Sunni and Shi'a Muslims and between the various Sufi orders.

The goal of Sufism and Sufi psychology is unified oneness, in which every aspect of one's being is unified in harmony. It is based on the concept *La ilaha illa Allah*, which means there is only one God. The Prophet taught through his actions that whatever exists in one individual has to come to one point and present itself. When this occurs, the absolute knowledge of oneness with Existence is attained. Called *Irfan*, derived from the word *ma'rifa* meaning cognition, Sufism is considered the science of the soul.

Integral Yoga

Integral yoga, also called supramental yoga, is the yoga-based philosophy and practice of Sri Aurobindo and The Mother (Mirra Alfassa). Integral yoga finds all life conscious or subconscious a yoga, defines the term yoga as a methodised effort towards self-perfection by the expression of the secret potentialities latent in the being and highest condition of victory in that effort – a union of the human individual with the universal and transcendent existence which is seen partially as expressions in humans and in the cosmos. As a yoga idea, that Spirit manifests itself in a process of involution. The reverse process of evolution is driven toward a complete manifestation of spirit.

According to Sri Aurobindo, the current status of human evolution is an intermediate stage in the evolution of being, which is on its way to the unfolding of the spirit, and the self-revelation of divinity in all things. Yoga is a rapid and concentrated evolution of being, which can take effect in one life-time, while unassisted natural evolution would take many centuries or many births. Sri Aurobindo suggests a grand program called *sapta chatusthaya* (seven quadrates) to aid this evolution.

Science and Spirituality (Avidya and Vidya)

The Upanishads identify two types of knowledge: the lower knowledge of the rituals, sacrifices, obligatory duties, occupational knowledge and the like, and the higher knowledge

of the Self (Atman) and the Supreme Self. The former is often equated with ignorance (avidya) and the latter with true knowledge (vidya).

Inferior knowledge or ignorance arises from our minds and their faculties. Their interaction with the phenomenal world is responsible for our experience of duality and delusion. From this perspective, even the knowledge we gain from the study of the scriptures, unless it is augmented by spiritual experience, is considered inferior.

Many spiritual teachers affirm that mere book knowledge does not help much in our liberation. Some even say that it becomes an obstacle and comparatively an empty mind is better suited for spiritual sadhana than an intellectual mind full of theories and concepts.

This knowledge is considered inferior, because it does not liberate us from the cycle of births and deaths. Instead, it leads to more attachment, karma, egoism, worldliness and more involvement with things and Nature.

All learned knowledge is accumulated knowledge and as long as the mind is clogged with worldly knowledge, peace and equanimity are not possible. In contrast, spiritual knowledge arises from the contact with the Self in a deeper state of self-absorption, at the end of a long spiritual journey and self-purification.

It arises in a state in which the mind and the senses are withdrawn and fully resting. Spiritual knowledge is transcendental knowledge which arises in a person who is self-aware without the involvement of his mind or its faculties.

The Upanishads state clearly that both types of knowledge are important. From the lower knowledge comes the discipline and the ability to practice the higher and realize the highest. They also affirm that knowledge of Brahman is the highest knowledge, because it is permanent, unchanging and indivisible, by knowing which there is no further knowing.

It is important to know that Brahman is not an object of knowledge, something that we need to know or gain or possess or remember but the goal of our spiritual effort to achieve liberation. In worldly life, we seek knowledge for some worldly gain.

We think knowledge in material terms, because we cannot conceptualize knowledge that arises spontaneously without the intervention of the mind, the senses and the ego.

To aspire for divine knowledge through our minds and senses would be equal to creating another distraction or activity in our lives, another mirage or delusion in which our egos indulge and lead us astray.

In spiritual world, we have to seek knowledge transcend it and enter into a state of awareness in which the distinction between the knower and the known are completely absent. It is a state in which knowing alone exists without the duality of these two.

Brahman is realized only in a transcendental state, when we manage to rise above the habitual movements of our minds and senses and cease all goal oriented effort to know him or experience him.

The paradox of pure knowledge

The problem and paradox of spiritual experience is that, from a mental perspective, in a state of duality, the knowledge of Brahman is indescribable, and in a state of unity it is unknowable.

When we are in a state of samadhi, we have no means of knowing what is happening, and when we come out of it we have no means of remembering what happened, because

our minds and senses were not involved in that experience. It is like the deep sleep state, beyond dreams and all mental formations, about which we have no idea.

Brahman is the absolute, infinite and complete being. In his indivisible and absolute state he is not subject to duality, because he is not united with Prakriti (nature) or its qualities (gunas) and elements (tattvas).

So the absolute Brahman does not know, other than by himself, that he knows or that he exists. It is a state of Being, in contrast to the state of Doing, in which he simply is without a second and without any instruments such as a body or senses. It is a purely subjective state in which there is no process, no otherness, no knowing, no doing and no remembering.

Such is the paradox of the state of Brahman. To know him is not to know him and not to be him in the present. To describe him from the memory of our experience is to negate his transcendental unity and continuity! I

In his absolute state Brahman has no sense of objectivity, no simultaneous experience of self and the not-self, but only a subjective sense of "I am" ness, because to be aware of the self means duality which is not the state of Brahman.

Why Brahman Manifests Creation

The only way the knowledge of Brahman becomes known is in a state of duality because if there is no creation, there is no object to be known, and there is no possibility of knowing. Brahman is the subject. In the beings, the same subject manifests as individual Selves. The purpose of objectivity is experience and enjoyment.

Therefore to make possible duality and create objectivity, Brahman subjects himself to the illusion of otherness and enters into a state of beingness, reflecting Himself in the triple qualities of sattva, rajas and tamas. His association with Prakriti (Nature) results in diversity of creation.

The Vedas affirm that creation exists for the enjoyment of the Self and the Supreme Self. The senses, the mind, the speech, the organs of action, intelligence and breath perform their duties to facilitate this enjoyment of the Self only. Unknowingly, they participate in the sacrifice of life.

Thus, to consciously experience the joy of seeing himself in innumerable forms, Brahman projects himself outwardly, as a reflection of Himself, to manifest worlds, beings and objects. He subjects them to varying degrees of knowledge, intelligence, ignorance and delusion, through which He perceives the field of His creation as a playground of innumerable realities.

This is the secret and purpose of God's creation. Brahman descends into our worlds, as limited beings, in order to experience in his limited state the state of knowing and becoming, which are inconceivable in an absolute state of oneness. He associates himself with his materiality (the primordial nature) and manifests all this as a part of his universal play (maha-lila).

Knowledge in its Absolute and Relative States

In contrast to the knowledge of Brahman, which is self illumined and exists by itself and unaware of any other, the knowledge that arises from duality is illumined by the mind and intelligence and exists only in relation to things and otherness. Thus our mental and

intellectual knowledge is objective knowledge, which is subject to duality, division and pairs of opposites.

Brahman is pure knowledge and awareness. In His manifested state, He becomes associated with diversity. In beings also the same conditions manifest. The Self, which is pure consciousness, remains enveloped by a field of modifications in the form of a body.

In this state, we experience knowledge as a condition or state or even a possession, different from us and other than us. In the state of duality, our knowledge does not truly represent us and cannot be considered perfect or complete. It is a mixture of light and darkness, subject to desires and attachment and guided by self-interest.

In our knowing there is a process, a movement, a conscious experience of knowing or not knowing, the division of time, a distinction between what we know and what we don't, what we can hold and what we cannot, a desire to know or not to know and an effort to know or resistance to know.

There is also a discontinuity in our knowing. In between two thoughts and two pieces of knowledge there is always a gap, a silence in which hides the unity of our existence. Our senses and minds are imperfect and unreliable as instruments of knowledge.

In Brahman there are such divisions and pairs of opposites. There is no effort to know, no desire to know, no process of knowing, no means of knowing, no experience of knowing, no contradictions of what is known and unknown and no division of time and movement.

It is a state of pure consciousness, the state of Being, the state of Truth which is veiled in us and which is denied to us by the power of maya. The purpose of our existence is to realize that consciousness and to become one with it.

The Journey Back to the Source

If we want to go back to our source, we have to overcome our limitations of Prakriti and duality and regain our true consciousness in the state of Brahman.

We need to experience our knowledge of oneness, withdrawing ourselves from the objects of the worlds, the senses and the mind and transcending our memory and phenomenal awareness.

As free individuals, we have the choice to choose our paths to arrive at the destination. We can continue with our limited existence as long as we want, for we have that freedom too. There is nothing wrong with human existence or human knowledge. It is neither inferior nor superior. It is definitely not an evil or sinful existence. It is an aspect of Brahman, a plane of existence that has its own significance in the order of the universe.

We are here because of the will of Brahman. We are divine in essential nature because our source is divine. If that is not so we would not be worshipping so many divinities who represent the plurality of our cosmos in our microcosm.

In its ultimate aspect, existence is divine. So are all beings and objects. Through them Brahman witnesses His own actions and manifestations. At the end of each cycle of creation, He withdraws everything into Himself and goes into sleep.

Knowing the distinction between true knowledge and false knowledge will help us to realize who we are, why we are here and what is the purpose of our creation and existence. It will help us to know the mysteries of Atman and Brahman. It reminds us what we have been and where we need to go.



RESEARCH METHODOLOGY AND STATISTICS

RESEARCH: MEANING, PURPOSE, AND DIMENSIONS

Research is defined as a careful consideration of study regarding a particular concern or a problem using scientific methods. According to the American sociologist Earl Robert Babbie, “Research is a systematic inquiry to describe, explain, predict and control the observed phenomenon. Research involves inductive and deductive methods.”

Inductive research methods are used to analyze the observed phenomenon whereas, deductive methods are used to verify the observed phenomenon. Inductive approaches are associated with qualitative research and deductive methods are more commonly associated with quantitative research.

One of the most important aspects of research is the statistics associated with it, conclusion or result. It is about the “thought” that goes behind the research. Research is conducted with a purpose to understand:

1. What do organizations or businesses really want to find out?
2. What are the processes that need to be followed to chase the idea?
3. What are the arguments that need to be built around a concept?
4. What is the evidence that will be required that people believe in the idea or concept?

Characteristics of Research

1. A systematic approach is followed in research. Rules and procedures are an integral part of research that set the objective of a research process. Researchers need to practice ethics and code of conduct while making observations or drawing conclusions.

2. Research is based on logical reasoning and involves both inductive and deductive methods.
3. The data or knowledge that is derived is in real time, actual observations in the natural settings.
4. There is an in-depth analysis of all the data collected from research so that there are no anomalies associated with it.
5. Research creates a path for generating new questions. More research opportunity can be generated from existing research.
6. Research is analytical in nature. It makes use of all the available data so that there is no ambiguity in inference.
7. Accuracy is one of the important character of research, the information that is obtained while conducting the research should be accurate and true to its nature. For example, research conducted in a controlled environment like a laboratory. Here accuracy is measured of instruments used, calibrations, and the final result of the experiment.

Following are the types of research:

1. **Basic Research.** Basic research is mostly conducted to enhance knowledge. It covers fundamental aspects of research. The main motivation of this research is knowledge expansion. It is a non-commercial research and doesn't facilitate in creating or inventing anything. For example, an experiment is a good example of basic research.
2. **Applied Research.** Applied research focuses on analyzing and solving real-life problems. This type of research refers to the study that helps solve practical problems using scientific methods. This research plays an important role in solving issues that impact the overall well-being of humans. For example, finding a specific cure for a disease.
3. **Problem Oriented Research.** As the name suggests, problem-oriented research is conducted to understand the exact nature of the problem to find out relevant solutions. The term "problem" refers to having issues or two thoughts while making any decisions.
For e.g. Revenue of a car company has decreased by 12% in the last year. The following could be the probable causes: There is no optimum production, poor quality of a product, no advertising, economic conditions etc.
4. **Problem Solving Research.** This type of research is conducted by companies to understand and resolve their own problems. The problem-solving research uses applied research to find solutions to the existing problems.
5. **Qualitative Research.** Qualitative research is a process that is about inquiry, that helps in-depth understanding of the problems or issues in their natural settings. This is a non-statistical research method.

Qualitative research is heavily dependent on the experience of the researchers and the questions used to probe the sample. The sample size is usually restricted to 6-10 people in a sample. Open-ended questions are asked in a manner that one question leads to another. The purpose of asking open-ended questions is to gather as much information as possible from the sample.

Following are the methods used for qualitative research:

1. One-to-one interview
2. Focus groups

3. Ethnographic Research
4. Content/ Text Analysis
5. Case study research

Quantitative Research

Qualitative research is a structured way of collecting data and analyzing it to draw conclusions. Unlike qualitative research, this research method uses a computational, statistical and similar method to collect and analyze data. Quantitative data is all about numbers.

Quantitative research involves a larger population as more number of people means more data. In this manner, more data can be analyzed to obtain accurate results. This type of research method uses close-ended questions because, in quantitative research, the researchers are typically looking at measuring the extent and gathering foolproof statistical data.

Online surveys, questionnaires, and polls are preferable data collection tools used in quantitative research. There are various methods of deploying surveys or questionnaires. In recent times online surveys and questionnaires have gained popularity. Survey respondents can receive these surveys on mobile phones, emails or can simply use the internet to access surveys or questionnaires.

Purpose of Research

There are three purposes of research:

1. **Exploratory Research.** As the name suggests, exploratory research is conducted to explore the research questions and may or may not offer a final conclusion to the research conducted. It is conducted to handle new problem areas which haven't been explored before. Exploratory research lays the foundation for more conclusive research and data collection. For example, a research conducted to know the level of customer satisfaction among the patrons of a restaurant.
2. **Descriptive Research.** Descriptive research focuses on throwing more light on current issues through a process of data collection. Descriptive studies are used to describe the behavior of a sample population. In descriptive research, only one variable (anything that has quantity or quality that varies) is required to conduct a study. The three main purposes of descriptive research are describing, explaining and validating the findings. For example, a research conducted to know if top-level management leaders in the 21st century possess the moral right to receive a huge sum of money from the company profit?
3. **Explanatory Research.** Explanatory research or causal research, is conducted to understand the impact of certain changes in existing standard procedures. Conducting experiments is the most popular form of casual research. For example, research conducted to understand the effect of rebranding on customer loyalty.

Research method is defined as the tools or an instrument that is used to accomplish the goals and attributes of conducting a research process. Think of the methodology as a systematic process in which the tools or instruments will be employed. There is no use of a tool if it is not being used efficiently.

To understand research methods we must first understand what research is. Research is a careful study of a specific problem or concern using scientific methods, tools, and

instruments. Remember as kids, we were given projects in elementary schools and we were asked to know about it in details. Research is the adult form of that elementary project, only a more precise and thorough version!

Research can be about anything, scientific research, basic research, applied research, market research- qualitative market research and quantitative market research, problem-oriented research, problem-solving research. To begin researching something, you need to have a problem or a concern that needs a solution.

Research really begins with asking the right research questions, what follows next is choosing appropriate research method to be able to investigate the problem or issue in the right direction and then finally analyze your findings or observations to draw appropriate conclusions.

When it comes to customers and related research, there should be no stone unturned. Specify and plan the customer's research, help make smart decisions about how to market and position your business efficiently. Choosing the right research method will help you get important answers about your clients. But before we go any further, first let us identify the research methods

Types of Research Methods and Research Example

Research methods are broadly classified as Qualitative Research and Quantitative Research.

Both research methods have distinctive properties and data collection methods. In this segment of the blog, we will learn more about both these research methods.

Qualitative Research Methods

Qualitative research is a research method that collects data using conversational methods, where participants involved in the research are asked open-ended questions. The responses collected are essentially non-numerical. This method not only helps a researcher understand “what” participant think but also, “why” they think in a particular way.

Here are the widely used qualitative research methods:

1. **One-to-one Interview.** This interview technique is systematically planned and as the name suggests is conducted with one participant at a given point in time. One-to-one interviews need a researcher to prepare questions in advance and to make sure the researcher asks only the most important questions to the participant. This type of interview lasts anywhere between 20 minutes to half an hour. During this time the researcher collects as many meaningful data as possible from the participants to draw inferences.
2. **Focus Groups.** Focus groups are small groups comprising of around 6-10 participants who are usually experts in the subject matter. A moderator is assigned to a focus group who facilitates the discussion amongst the group members. A moderator's experience in conducting focus group plays an important role. An experienced moderator can probe the participants by asking the correct research questions that will help him/her collect a sizable amount of information related to the research.
3. **Ethnographic Research.** Ethnographic research is an in-depth form of research where people are observed in their natural environment with any alterations. This method can prove to be a bit demanding in terms of a researcher getting adapted to

the natural environment of the target audience which could even be the Amazon rainforest! Geographic locations can be a constraint in this type of research method. Instead of conducting interviews a researcher needs to experience the settings in person to collect information.

4. **Text Analysis.** Text analysis is a little different from other qualitative research methods as it used to analyze the social life by decoding words, texts etc. through any available form of documentation. The researcher studies and understands the context in which the documents are furnished with the information and then tries to draw meaningful inferences from it. In modern times, researchers follow activities on a social media platform and try and understand the pattern of thoughts.
5. **Case Study Research.** Case study research, as the name suggests is used to study an organization or an entity. This research method has evolved over the years and is one of the most valuable qualitative research methods known to researchers. This type of research is used in fields like education sector, philosophical and psychological studies. This method involves a deep diving into the ongoing research and collects data.

Quantitative Research Methods

Quantitative research methods are the methods that deal with numbers and anything that can be dealt with a measurable form, in a systematic way of investigating the phenomenon. It is used to answer questions in terms of justifying relationships with measurable variables to either explain, predict or control a phenomenon.

There are three methods that are often used by researchers to conduct this type of research, they are:

1. **Survey Research.** The ultimate goal of survey research is to learn about a large population by deploying the survey. Gone are the days where a survey was carried out using a pen and a paper. Today, online surveys are a popular mode of research as they are convenient and can be sent in an email or made available on the internet. In this method, a researcher designs a survey with most relevant survey questions and deploys the survey. Once the researcher receives the responses he/she summarizes them to tabulate meaningful findings and data.
2. **Descriptive Research.** Descriptive research is a quantitative research method, which corresponds to identifying the characteristics of an observed phenomenon and collecting more information. This research method is designed to depict the participants in a very systematic and accurate manner. In simple words, descriptive research is all about describing the phenomenon, observing and drawing conclusions from it.
3. **Correlational Research.** Correlational research examines the relationship between two or more variables. Let us take an example to understand correlational research. Let us take an example to understand correlational research, Consider hypothetically, a researcher is studying a correlation between cancer and married women. Let us say married women have a negative correlation with cancer. In this example, there are two variables: cancer, and married women. When we say negative correlation it means women who are married are less likely to develop cancer. However, it doesn't mean that marriage directly avoids cancer.

Identifying Research Methodology

To choose the appropriate research methods, it is necessary to clearly identify the research objectives. Here is an example of some of the research objectives you can take into consideration for your business:

- To start with, find out the needs of your clients

- Know their preferences and understand what is important to them

- Find an appropriate way to make your customers aware of your products and services.

- Find ways to improve your products or services to suit the needs of your customers.

After identifying what you need to know, you should then ask what research methods will offer you that information.

Organize your questions within the framework of the 7 Ps of marketing that influences your company – product, price, promotion, place, people, processes and physical tests.

A well-organized customer research process produces valid, accurate, reliable, timely and complete results. Research results that rigorously reflect the opinions and needs of your clients will help you grow your sales and improve your operations. To obtain the results you need to establish and follow the processes that you have detailed out for your organization:

Set your goals. Consider the client's research objectives and define those that identify with yours and plan a strategy once you obtain the information. Make sure that your goals objectives smart do not presume their result, and define them intelligently and make sure you set achievable targets, smart goals, and objectives.

Plan your research. Good planning allows the use of creative and logical approaches to select the research methods that gather the information. Your plan will be influenced by the type and complexity of the information you need, the skills of your market research team, and how soon you need the information and your budget. Make an adequate strategic planning for your market research

Identify your list of questions and decide on the research methods that will best achieve your goals. Detail your research approach and some initial idea of how you will classify and analyze the data.

Collect and collate your results. Make a list of how you are going to carry out the research process, the data you need to collect and collection methods. This will help you keep track of your research processes and make sense of your findings. It will also allow you to verify that your research accurately reflects the opinions of your clients and your market. Create a record table with:

- The consumer research activity

- The necessary data

- The research method (s) for data collection

- The steps to follow for data analysis.

Remember, research is only valuable and useful when it is valid, accurate and reliable. Relying on imperfect research is dangerous; Incorrect results can lead to customer churn and a decrease in sales.

It is important to obtain information about how the collection of customer information was carried out, and to ensure that your data is:

- Valid – founded, logical, rigorous and impartial.

Accurate – free of errors and including required details.

Reliable – that can be reproduced by other people who investigate in the same way.

Timely – current and collected within an appropriate time frame.

Complete – includes all the data you need to support your business decisions.

Analyze and understand your research. The analysis of the data can vary from simple and direct steps to technical and complex processes. Adopt an approach, and choose the method of data analysis based on the research methods you have carried out.

Keep the findings ready. Choose a spreadsheet that allows you to easily enter your data. If you do not have a large amount of data, you should be able to manage them with the use of basic tools available in the software. If you have collected more complete and complex data, you may have to consider using specific programs or tools that will help you manage your data.

Review and interpret the information to draw conclusions. Once you have gathered all the data, you can scan your information and interpret it to draw conclusions and make informed decisions. You should review the data and then:

Identify the main trends and issues, opportunities and problems you observe, and write a sentence about each one.

Keep track of the frequency with which each of the main findings appears.

Make a list of your findings from the most common to the least common.

Evaluate and perform separately a list of the strengths, weaknesses, opportunities, and threats that have been identified in a SWOT analysis.

Preparation of conclusions and recommendations about your research.

Review your goals before making any conclusions about your research. Keep in mind if the process you have completed and the data you have gathered help answer your questions. Ask yourself if what your research revealed facilitates the identification of your conclusions and recommendations. Review your conclusions and, based on what you know now:

Choose some strategies that will help you improve your business

Act on your strategies

Look for gaps in the information, and consider doing additional research if necessary

Plan to review the results of the research, and consider efficient strategies.

RESEARCH PROBLEMS, VARIABLES AND OPERATIONAL DEFINITIONS, HYPOTHESIS, SAMPLING

Research Problem

Defining a research problem is the fuel that drives the scientific process, and is the foundation of any research method and experimental design, from true experiment to case study.

It is one of the first statements made in any research paper and, as well as defining the research area, should include a quick synopsis of how the hypothesis was arrived at. Operationalization is then used to give some indication of the exact definitions of the variables, and the type of scientific measurements used.

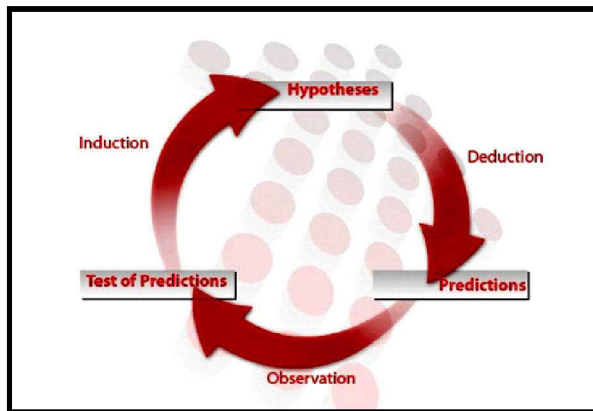
This will lead to the proposal of a viable hypothesis. As an aside, when scientists are putting forward proposals for research funds, the quality of their research problem often makes the difference between success and failure.

Structuring the Research Problem

Look at any scientific paper, and you will see the research problem, written almost like a statement of intent.

Defining a research problem is crucial in defining the quality of the answers, and determines the exact research method used. A quantitative experimental design uses deductive reasoning to arrive at a testable hypothesis.

Qualitative research designs use inductive reasoning to propose a research statement.



Defining a Research Problem

Formulating the research problem begins during the first steps of the scientific process.

As an example, a literature review and a study of previous experiments, and research, might throw up some vague areas of interest.

Many scientific researchers look at an area where a previous researcher generated some interesting results, but never followed up. It could be an interesting area of research, which nobody else has fully explored.

A scientist may even review a successful experiment, disagree with the results, the tests used, or the methodology, and decide to refine the research process, retesting the hypothesis.

This is called the conceptual definition, and is an overall view of the problem. A science report will generally begin with an overview of the previous research and real-world observations. The researcher will then state how this led to defining a research problem.

The Operational Definitions

The operational definition is the determining the scalar properties of the variables.

For example, temperature, weight and time are usually well known and defined, with only the exact scale used needing definition. If a researcher is measuring abstract concepts, such as intelligence, emotions, and subjective responses, then a system of measuring numerically needs to be established, allowing statistical analysis and replication.

For example, intelligence may be measured with IQ and human responses could be measured with a questionnaire from '1- strongly disagree', to '5 - strongly agree'.

Behavioural biologists and social scientists might design an ordinal scale for measuring and rating behavior. These measurements are always subjective, but allow statistics and replication of the whole research method. This is all an essential part of defining a research problem.

Examples of Defining a Research Problem

An anthropologist might find references to a relatively unknown tribe in Papua New Guinea. Through inductive reasoning, she arrives at the research problem and asks,

'How do these people live and how does their culture relate to nearby tribes?'

She has found a gap in knowledge, and she seeks to fill it, using a qualitative case study, without a hypothesis.

The Bandura Bobo Doll Experiment is a good example of using deductive reasoning to arrive at a research problem and hypothesis.

Anecdotal evidence showed that violent behavior amongst children was increasing. Bandura believed that higher levels of violent adult role models on television, was a contributor to this rise. This was expanded into a hypothesis, and operationalization of the variables, and scientific measurement scale, led to a robust experimental design.

Variable

Variables are anything that might impact the outcome of your study. An operational definition describes exactly what the variables are and how they are measured within the context of your study. For example, if you were doing a study on the impact of sleep deprivation on driving performance, you would need to operationally define what you mean by *sleep deprivation* and *driving performance*.

In this example you might define sleep deprivation as getting less than seven hours of sleep at night and define driving performance as how well a participant does on a driving test.

What is the purpose of operationally defining variables? The main purpose is control. By understanding what you are measuring, you can control for it by holding the variable constant between all of the groups or manipulating it as an independent variable.

A variable is something that can be changed, such as a characteristic or value. Variables are generally used in psychology experiments to determine if changes to one thing result in changes to another.

The Dependent and Independent Variables

In a psychology experiment:

The **independent variable** is the variable that is controlled and manipulated by the experimenter. For example, in an experiment on the impact of sleep deprivation on test performance, sleep deprivation would be the independent variable.

The **dependent variable** is the variable that is measured by the experimenter. In our previous example, the scores on the test performance measure would be the dependent variable.

Extraneous and Confounding Variables

The independent and dependent variables are not the only variables present in many experiments. In some cases, extraneous variables may also play a role. This type of variable is one that may have an impact on the relationship between the independent and dependent variables.

For example, in our previous description of an experiment on the effects of sleep deprivation on test performance, other factors such as age, gender and academic background may have an impact on the results. In such cases, the experimenter will note the values of these extraneous variables so this impact on the results can be controlled for.

There are two basic types of extraneous variables:

Participant Variables. These extraneous variables are related to individual characteristics of each participant that may impact how he or she responds. These factors can include background differences, mood, anxiety, intelligence, awareness and other characteristics that are unique to each person.

Situational Variables. These extraneous variables are related to things in the environment that may impact how each participant responds. For example, if a participant is taking a test in a chilly room, the temperature would be considered an extraneous variable. Some participants may not be affected by the cold, but others might be distracted or annoyed by the temperature of the room.

In many cases, extraneous variables are controlled for by the experimenter. In the case of participant variables, the experiment might select participants that are the same in background and temperament to ensure that these factors do not interfere with the results. If, however, a variable cannot be controlled for, it becomes what is known as a **confounding variable**. This type of variable *can* have an impact on the dependent variable, which can make it difficult to determine if the results are due to the influence of the independent variable, the confounding variable or an interaction of the two.

Operationally Defining a Variable

Before conducting a psychology experiment, it is essential to create firm operational definitions for both the independent variable and dependent variable. An operational definition describes how the variables are measured and defined within the study.

For example, in our imaginary experiment on the effects of sleep deprivation on test performance, we would need to create very specific operational definitions for our two variables. If our hypothesis is “Students who are sleep deprived will score significantly lower on a test,” then we would have a few different concepts to define. First, what do we mean by *students*? In our example, let’s define students as participants enrolled in a introductory university-level psychology course.

Next, we need to operationally define the *sleep deprivation* variable. In our example, let’s say that sleep deprivation refers to those participants who have had less than five hours of sleep the night before the test. Finally, we need to create an operational definition for the test variable. For this example, the test variable will be defined as a student’s score on a chapter exam in the introductory psychology course.

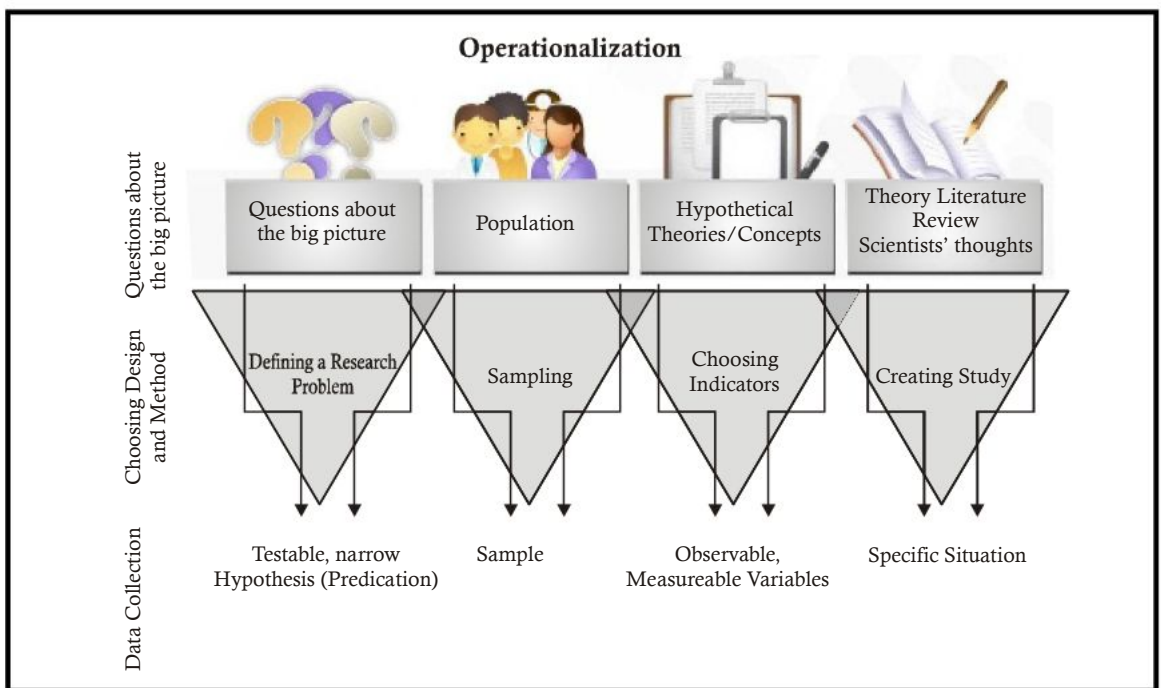
Students often report problems with identifying the independent and dependent variables in an experiment. While the task can become more difficult as the complexity of an experiment increases, there are a few questions you can ask when trying to identify a variable.

What is the experimenter manipulating? The things that change, either naturally or through direct manipulation from the experimenter, are generally the independent variables. What is being measured? The dependent variable is the one that the experimenter is measuring.

Operationalization

For experimental research, where interval or ratio measurements are used, the scales are usually well defined and strict.

Operationalization also sets down exact definitions of each variable, increasing the quality of the results, and improving the robustness of the design.



For many fields, such as social science, which often use ordinal measurements, operationalization is essential. It determines how the researchers are going to measure an emotion or concept, such as the level of distress or aggression.

Such measurements are arbitrary, but allow others to replicate the research, as well as perform statistical analysis of the results.

Fuzzy Concepts

Fuzzy concepts are vague ideas, concepts that lack clarity or are only partially true. These are often referred to as “conceptual variables”.

It is important to define the variables to facilitate accurate replication of the research process. For example, a scientist might propose the hypothesis:

“Children grow more quickly if they eat vegetables.”

What does the statement mean by ‘children’? Are they from America or Africa. What age are they? Are the children boys or girls? There are billions of children in the world, so how do you define the sample groups?

How is 'growth' defined? Is it weight, height, mental growth or strength? The statement does not strictly define the measurable, dependent variable.

What does the term 'more quickly' mean? What units, and what timescale, will be used to measure this? A short-term experiment, lasting one month, may give wildly different results than a longer-term study.

The frequency of sampling is important for operationalization, too.

If you were conducting the experiment over one year, it would not be practical to test the weight every 5 minutes, or even every month. The first is impractical, and the latter will not generate enough analyzable data points.

What are 'vegetables'? There are hundreds of different types of vegetable, each containing different levels of vitamins and minerals. Are the children fed raw vegetables, or are they cooked? How does the researcher standardize diets, and ensure that the children eat their greens?

The Importance of Operationalization

Of course, strictly speaking, concepts such as seconds, kilograms and centigrade are artificial constructs, a way in which we define variables.

Pounds and Fahrenheit are no less accurate, but were jettisoned in favor of the metric system. A researcher must justify their scale of scientific measurement.

Operationalization defines the exact measuring method used, and allows other scientists to follow exactly the same methodology. One example of the dangers of non-operationalization is the failure of the Mars Climate Orbiter.

This expensive satellite was lost, somewhere above Mars, and the mission completely failed. Subsequent investigation found that the engineers at the sub-contractor, Lockheed, had used imperial units instead of metric units of force.

A failure in operationalization meant that the units used during the construction and simulations were not standardized. The US engineers used pound force, the other engineers and software designers, correctly, used metric Newtons.

This led to a huge error in the thrust calculations, and the spacecraft ended up in a lower orbit around Mars, burning up from atmospheric friction. This failure in operationalization cost hundreds of millions of dollars, and years of planning and construction were wasted.

Hypothesis

A hypothesis (plural: hypotheses), in a scientific context, is a testable statement about the relationship between two or more variables or a proposed explanation for some observed phenomenon. In a scientific experiment or study, the hypothesis is a brief summation of the researcher's prediction of the study's findings, which may be supported or not by the outcome. Hypothesis testing is the core of the scientific method.

The researcher's prediction is usually referred to as the alternative hypothesis, and any other outcome as the null hypothesis – basically, the opposite outcome to what is predicted. (However, the terms are reversed if the researchers are predicting no difference or change, hypothesizing, for example, that the incidence of one variable will not increase or decrease in tandem with the other.) The null hypothesis satisfies the requirement for falsifiability: the capacity for a proposition to be proven false, which some

schools of thought consider essential to the scientific method. According to others, however, testability is adequate, on the grounds that if there is sufficient support for a hypothesis it is not necessary to be able to conceive of a contrary outcome.

A simple hypothesis might predict a causal relationship between two variables, meaning that one has an effect on the other. Here's an example: More hours spent studying for an exam result in higher grades. Hours spent studying, in this statement, is the independent variable and grades is the dependent variable. The independent variable is manipulated and the dependent variable is measured to see how it is affected as the independent variable changes.

A complex hypothesis is similar to a simple one but includes two or more independent variables or two or more dependent variables. In the first case, for example, the hypothesis might be that more hours studying and more classes attended lead to higher grades; in the second case, the hypothesis might be that more hours studying lead to higher grades and a shorter amount of time required to write the exam.

Hypotheses don't necessarily predict causality. In statistics, for example, a hypothesis might predict simple correlation – for example, that increased incidence of the independent variable is associated with a decrease in the dependent variable, although there is no supposition that one causes the other.

Sampling

The items selected constitute what is technically called a sample. The researcher must decide the way of selecting a sample or what is popularly known as the sample design. In other words, a sample design is a definite plan determined before any data are actually collected for obtaining a sample from a given population. Thus, the plan to select 12 to a city's 200 drug stores in a certain way constitutes a sample design. Samples can be either probability samples or non-probability samples. With probability samples each element has a known probability of being included in the sample but the non-probability samples do not allow the researcher to determine this probability. Probability samples are those based on simple random sampling, systematic sampling, stratified sampling, cluster/area sampling whereas non-probability samples are those based on consciences sampling, judgement sampling and quota sampling techniques. A brief mention of the important sample designs is described below.

Deliberate Sampling

This sampling method involves purposive or deliberate selection of particular units of the universe for constituting a sample which represents the universe. Therefore this is also known as purposive or non probability sampling.

When population elements are selected for inclusion in the sample based on the ease of access, it can be called convenience sampling. If a researcher wishes to secure data from, say, gasoline buyers, he may select a fixed number of petrol stations and may conduct interviews at these stations. This would be an example of convenience sample of gasoline buyers. At times such a procedure may give very biased results particularly when the population is not homogeneous. On the other hand, in judgement sampling the researcher's judgement is used for selecting items which he considers as representative of the population. For example, a judgement sample of college students might be taken to secure reactions to a new method of teaching. Judgement sampling is used quite fre-

quently in qualitative research where the desire happens to be to develop hypothesis rather than to generalise to larger populations.

Simple Random Sampling

In this sampling each and every item in the population has an equal chance of inclusion in the sample and each one of the possible samples, in case of finite universe, has the same probability of being selected. For example, if we have to select a sample of 300 items from a universe of 15,000 items, then we can put the names or numbers of all the 15,000 item on slips of paper and conduct a lottery. Using the random number tables is another method or random sampling. To select the sample, each item is assigned a number from 1 to 15,000. Then, 300 five digit random numbers are selected from the table. To do this we select some random starting point and then a systematic pattern is used in proceeding through the table. We might start in the 4th row, second column and proceed down the column to the bottom of the table and then move to the top of the next column to the right. When a number exceeds the limit of the numbers in the frame, in our case over 15,000, it is simply passed over and the next number selected that does fall within the relevant range. Since the numbers were placed in the table in a completely random fashion, the resulting sample is random. This procedure gives each item an equal probability of being selected. In case of infinite population, the selection of each item in a random sample is controlled by the same probability and that successive selections are independent of one another.

Systematic Sampling

In some instances the most practical way of sampling is to select every 15th name on a list, every 10th house on one side of a street and so on. Sampling of this type is known as systematic sampling. An element of randomness is usually introduced into this kind of sampling by using random numbers to pick up the unit with which to start. This procedure is useful when sampling frame is available in the form of a list. In such a design the selection process starts by picking some random point in the list and then every n th element is selected until the desired number is secured.

Stratified Sampling

If the population from which a sample is to be drawn does not constitute a homogeneous group, then stratified sampling technique is applied so as to obtain a representative sample. In this technique, the population is stratified into a number of non-overlapping subpopulations or strata and sample items are selected from each stratum. If the items selected from each stratum is based on simple random sampling the entire procedure, first stratification and then simple random sampling, is known as stratified random sampling.

Quota Sampling

In stratified sampling the cost of taking random samples from individual strata is often so expensive that interviewers are simply given quota to be filled from different strata, the actual selection of items for sample being left to the interviewer's judgement. This is called quota sampling. The size of the quota for each stratum is generally proportionate to the size of that stratum in the population. Quota sampling is thus an important form of non-probability sampling. Quota samples generally happen to be judgement samples rather than random samples.

Cluster Sampling and Area Sampling

Cluster sampling involves grouping the population and then selecting the groups or the groups or the clusters rather than individual elements for inclusion in the sample. Suppose some departmental store wishes to sample its credit card holders. It has issued its cards to 15,000 customers. The sample size is to be kept say 450. For cluster sampling this list of 15,000 card holders could be formed into 100 clusters of 150 card holders each. Three clusters might then be selected for the sample randomly. The sample size must often be larger than the simple random sample to ensure the same level of accuracy because in cluster sampling procedural potential for order bias and other sources of error is usually accentuated. The clustering approach can, however, make the sampling procedure relatively easier and increase the efficiency of field work, specially in the case of personal interviews.

Area sampling is quite close to cluster sampling and is often talked about when the total geographical area of interest happens to be big one. Under area sampling we first divide the total area into a number of smaller non-overlapping areas, generally called geographical clusters, then a number of these smaller areas are randomly selected, and all units in these small areas are included in the sample. Area sampling is specially helpful where we do not have the list of the population concerned. It also makes the field interviewing more efficient since interviewer can do many interviewer can do many interviews at each location.

Multi-Stage Sampling

This is a further development of the idea of cluster sampling. This technique is meant for big inquiries extending to a considerably large geographical area like an entire country. Under multi-stage sampling the first stage may be to select large primary sampling units such as states, then districts, then towns and finally certain families within towns. If the technique of random sampling is applied at all stages, the sampling procedure is described as multi-stage random sampling.

Sequential Sampling

This is somewhat a complex sample design where the ultimate size of the sample is not fixed in advance but is determined according to mathematical decisions on the basis of information yielded as survey progresses. This design is usually adopted under acceptance sampling plan in the context of statistical quality control.

ETHICS IN CONDUCTING AND REPORTING RESEARCH

When most people think of ethics (or morals), they think of rules for distinguishing between right and wrong, such as the Golden Rule ("Do unto others as you would have them do unto you"), a code of professional conduct like the Hippocratic Oath ("First of all, do no harm"), a religious creed like the Ten Commandments ("Thou Shalt not kill..."), or a wise aphorisms like the sayings of Confucius. This is the most common way of defining "ethics": norms for conduct that distinguish between acceptable and unacceptable behavior.

Most people learn ethical norms at home, at school, in church, or in other social settings. Although most people acquire their sense of right and wrong during childhood,

moral development occurs throughout life and human beings pass through different stages of growth as they mature. Ethical norms are so ubiquitous that one might be tempted to regard them as simple commonsense. On the other hand, if morality were nothing more than commonsense, then why are there so many ethical disputes and issues in our society?

One plausible explanation of these disagreements is that all people recognize some common ethical norms but interpret, apply, and balance them in different ways in light of their own values and life experiences. For example, two people could agree that murder is wrong but disagree about the morality of abortion because they have different understandings of what it means to be a human being.

Most societies also have legal rules that govern behavior, but ethical norms tend to be broader and more informal than laws. Although most societies use laws to enforce widely accepted moral standards and ethical and legal rules use similar concepts, ethics and law are not the same. An action may be legal but unethical or illegal but ethical. We can also use ethical concepts and principles to criticize, evaluate, propose, or interpret laws. Indeed, in the last century, many social reformers have urged citizens to disobey laws they regarded as immoral or unjust laws. Peaceful civil disobedience is an ethical way of protesting laws or expressing political viewpoints.

Another way of defining 'ethics' focuses on the disciplines that study standards of conduct, such as philosophy, theology, law, psychology, or sociology. For example, a "medical ethicist" is someone who studies ethical standards in medicine. One may also define ethics as a method, procedure, or perspective for deciding how to act and for analyzing complex problems and issues. For instance, in considering a complex issue like global warming, one may take an economic, ecological, political, or ethical perspective on the problem. While an economist might examine the cost and benefits of various policies related to global warming, an environmental ethicist could examine the ethical values and principles at stake.

Many different disciplines, institutions, and professions have standards for behavior that suit their particular aims and goals. These standards also help members of the discipline to coordinate their actions or activities and to establish the public's trust of the discipline. For instance, ethical standards govern conduct in medicine, law, engineering, and business. Ethical norms also serve the aims or goals of research and apply to people who conduct scientific research or other scholarly or creative activities. There is even a specialized discipline, research ethics, which studies these norms. See Glossary of Commonly Used Terms in Research Ethics.

There are several reasons why it is important to adhere to ethical norms in research. First, norms promote the aims of research, such as knowledge, truth, and avoidance of error. For example, prohibitions against fabricating, falsifying, or misrepresenting research data promote the truth and minimize error.

Second, since research often involves a great deal of cooperation and coordination among many different people in different disciplines and institutions, ethical standards promote the values that are essential to collaborative work, such as trust, accountability, mutual respect, and fairness. For example, many ethical norms in research, such as guidelines for authorship, copyright and patenting policies, data sharing policies, and confidentiality rules in peer review, are designed to protect intellectual property interests while encouraging collaboration. Most researchers want to receive credit for their contributions and do not want to have their ideas stolen or disclosed prematurely.

Third, many of the ethical norms help to ensure that researchers can be held accountable to the public. For instance, federal policies on research misconduct, conflicts of interest, the human subjects protections, and animal care and use are necessary in order to make sure that researchers who are funded by public money can be held accountable to the public.

Fourth, ethical norms in research also help to build public support for research. People are more likely to fund a research project if they can trust the quality and integrity of research.

Finally, many of the norms of research promote a variety of other important moral and social values, such as social responsibility, human rights, animal welfare, compliance with the law, and public health and safety. Ethical lapses in research can significantly harm human and animal subjects, students, and the public. For example, a researcher who fabricates data in a clinical trial may harm or even kill patients, and a researcher who fails to abide by regulations and guidelines relating to radiation or biological safety may jeopardize his health and safety or the health and safety of staff and students.

Codes and Policies for Research Ethics

Given the importance of ethics for the conduct of research, it should come as no surprise that many different professional associations, government agencies, and universities have adopted specific codes, rules, and policies relating to research ethics. Many government agencies, such as the National Institutes of Health (NIH), the National Science Foundation (NSF), the Food and Drug Administration (FDA), the Environmental Protection Agency (EPA), and the US Department of Agriculture (USDA) have ethics rules for funded researchers. Other influential research ethics policies include Singapore Statement on Research Integrity, the American Chemical Society, The Chemist Professional's Code of Conduct, Code of Ethics (American Society for Clinical Laboratory Science) American Psychological Association, Ethical Principles of Psychologists and Code of Conduct, Statements on Ethics and Professional Responsibility (American Anthropological Association), Statement on Professional Ethics (American Association of University Professors), the Nuremberg Code and the World Medical Association's Declaration of Helsinki.

The following is a rough and general summary of some ethical principals that various codes address:

Honesty. Strive for honesty in all scientific communications. Honestly report data, results, methods and procedures, and publication status. Do not fabricate, falsify, or misrepresent data. Do not deceive colleagues, research sponsors, or the public.

Objectivity. Strive to avoid bias in experimental design, data analysis, data interpretation, peer review, personnel decisions, grant writing, expert testimony, and other aspects of research where objectivity is expected or required. Avoid or minimize bias or self-deception. Disclose personal or financial interests that may affect research.

Integrity. Keep your promises and agreements; act with sincerity; strive for consistency of thought and action.

Carefulness. Avoid careless errors and negligence; carefully and critically examine your own work and the work of your peers. Keep good records of research activities, such as data collection, research design, and correspondence with agencies or journals.

Openness. Share data, results, ideas, tools, resources. Be open to criticism and new ideas.

Respect for Intellectual Property. Honor patents, copyrights, and other forms of intellectual property. Do not use unpublished data, methods, or results without permission. Give proper acknowledgement or credit for all contributions to research. Never plagiarize.

Confidentiality. Protect confidential communications, such as papers or grants submitted for publication, personnel records, trade or military secrets, and patient records.

Responsible Publication. Publish in order to advance research and scholarship, not to advance just your own career. Avoid wasteful and duplicative publication.

Responsible Mentoring. Help to educate, mentor, and advise students. Promote their welfare and allow them to make their own decisions.

Respect for colleagues. Respect your colleagues and treat them fairly.

Social Responsibility. Strive to promote social good and prevent or mitigate social harms through research, public education, and advocacy.

Non-Discrimination. Avoid discrimination against colleagues or students on the basis of sex, race, ethnicity, or other factors not related to scientific competence and integrity.

Competence. Maintain and improve your own professional competence and expertise through lifelong education and learning; take steps to promote competence in science as a whole.

Legality. Know and obey relevant laws and institutional and governmental policies.

Animal Care. Show proper respect and care for animals when using them in research. Do not conduct unnecessary or poorly designed animal experiments.

Human Subjects Protection. When conducting research on human subjects, minimize harms and risks and maximize benefits; respect human dignity, privacy, and autonomy; take special precautions with vulnerable populations; and strive to distribute the benefits and burdens of research fairly.

PARADIGMS OF RESEARCH

Three research approaches are advanced: (a) qualitative, (b) quantitative, and (c) mixed methods. Unquestionably, the three approaches are not as discrete as they first appear. Qualitative and quantitative approaches should not be viewed as rigid, distinct categories, polar opposites, or dichotomies. Instead, they represent different ends on a continuum (Newman & Benz, 1998). A study tends to be more qualitative than quantitative or vice versa. Mixed methods research resides in the middle of this continuum because it incorporates elements of both qualitative and quantitative approaches.

Often the distinction between qualitative research and quantitative research is framed in terms of using words (qualitative) rather than numbers (quantitative), or using closed-ended questions (quantitative hypotheses) rather than open-ended questions (qualitative interview questions). A more complete way to view the gradations of differences between them is in the basic philosophical assumptions researchers bring to the study, the types of research strategies used in the research (e.g., quantitative experiments or qualitative case studies), and the specific methods employed in conducting these strategies (e.g., collecting data quantitatively on instruments versus collecting qualitative data through observing a setting). Moreover, there is a historical evolution to both approaches—with the quantitative approaches dominating the forms of research in the social sciences from the late 19th century up until the mid-20th century. During the latter half of

the 20th century, interest in qualitative research increased and along with it, the development of mixed methods research. With this background, it should prove helpful to view definitions of these three key terms:

Quantitative

Qualitative research is an approach for exploring and understanding the meaning individuals or groups ascribe to a social or human problem. The process of research involves emerging questions and procedures, data typically collected in the participant's setting, data analysis inductively building from particulars to general themes, and the researcher making interpretations of the meaning of the data. The final written report has a flexible structure. Those who engage in this form of inquiry support a way of looking at research that honors an inductive style, a focus on individual meaning, and the importance of rendering the complexity of a situation.

Qualitative

Quantitative research is an approach for testing objective theories by examining the relationship among variables. These variables, in turn, can be measured, typically on instruments, so that numbered data can be analyzed using statistical procedures. The final written report has a set structure consisting of introduction, literature and theory, methods, results, and discussion. Like qualitative researchers, those who engage in this form of inquiry have assumptions about testing theories deductively, building in protections against bias, controlling for alternative explanations, and being able to generalize and replicate the findings.

Mixed Methods Approach

Mixed methods research is an approach to inquiry involving collecting both quantitative and qualitative data, integrating the two forms of data, and using distinct designs that may involve philosophical assumptions and theoretical frameworks. The core assumption of this form of inquiry is that the combination of qualitative and quantitative approaches provides a more complete understanding of a research problem than either approach alone.

METHODS OF RESEARCH

Research psychology encompasses the study of behavior for use in academic settings, and contains numerous areas. It contains the areas of abnormal psychology, biological psychology, cognitive psychology, comparative psychology, developmental psychology, personality psychology, social psychology and others. All branches of psychology can have a research component to them. Research psychology is contrasted with applied psychology.

Research in psychology is conducted in broad accord with the standards of the scientific method, encompassing both qualitative ethological and quantitative statistical modalities to generate and evaluate explanatory hypotheses with regard to psychological phenomena. Where research ethics and the state of development in a given research domain permits, investigation may be pursued by experimental protocols. Psychology tends to be eclectic, drawing on scientific knowledge from other fields to help explain and understand psychological phenomena. Qualitative psychological research utilizes a broad spectrum of observational methods, including action research, ethnography, exploratory

statistics, structured interviews, and participant observation, to enable the gathering of rich information unattainable by classical experimentation. Research in humanistic psychology is more typically pursued by ethnographic, historical, and historiographic methods.

The testing of different aspects of psychological function is a significant area of contemporary psychology. Psychometric and statistical methods predominate, including various well-known standardized tests as well as those created ad hoc as the situation or experiment requires.

Academic psychologists may focus purely on research and psychological theory, aiming to further psychological understanding in a particular area, while other psychologists may work in applied psychology to deploy such knowledge for immediate and practical benefit. However, these approaches are not mutually exclusive and most psychologists will be involved in both researching and applying psychology at some point during their career. Clinical psychology, among many of the various disciplines of psychology, aims at developing in practicing psychologists knowledge of and experience with research and experimental methods which they will continue to build up as well as employ as they treat individuals with psychological issues or use psychology to help others.

When an area of interest requires specific training and specialist knowledge, especially in applied areas, psychological associations normally establish a governing body to manage training requirements. Similarly, requirements may be laid down for university degrees in psychology, so that students acquire an adequate knowledge in a number of areas. Additionally, areas of practical psychology, where psychologists offer treatment to others, may require that psychologists be licensed by government regulatory bodies as well.

Quantitative psychology involves the application of statistical analysis to psychological research, and the development of novel statistical approaches for measuring and explaining human behavior. It is a young field (only recently have Ph.D. programs in quantitative psychology been formed), and it is loosely comprised of the subfields psychometrics and mathematical psychology.

Psychometrics is the field of psychology concerned with the theory and technique of psychological measurement, which includes the measurement of knowledge, abilities, attitudes, interests, achievement in particular degree or course, and personality traits (Carl Dellomos, 2009). Measurement of these unobservable phenomena is difficult, and much of the research and accumulated knowledge in this discipline has been developed in an attempt to properly define and quantify such phenomena. Psychometric research typically involves two major research tasks, namely: (i) the construction of instruments and procedures for measurement; and (ii) the development and refinement of theoretical approaches to measurement.

Psychology is a science, to be approached as such. Experiments should be designed using the scientific method.

Experimental Research

One of the most important things for you to learn in this course is the difference between correlational research and experimental research. Like correlational research, **experimental research** concerns relationships between variables. Unlike correlational research,

however, experimental research provides strong evidence for causal interpretations. Here we will focus on the two most important features of experimental research.

(i) Manipulation of an Independent Variable

One important feature that distinguishes experimental research from correlational research is that instead of simply measuring two variables, the researcher **manipulates** one of them. This means that the experimenter actually changes the value of that variable in a systematic way. This variable, which is called the **independent variable**, is the one that the researcher believes is the cause. The other variable, which the researcher believes is the effect, is called the **dependent variable**.

For example, you could do a correlational study on the relationship between noise level and concentration by going to a variety of places, measuring the noise levels there, and giving people a task that requires concentration. Or you could do an experiment by setting up a situation in which you could manipulate the noise level—perhaps by making it really loud for one group of people and really soft for another. And of course you could give them a task that requires concentration, and their performance on this task would be the dependent variable.

But what does this have to do with causation? The idea is that if changes in one variable occur reliably right after you manipulate another variable, then you can be fairly sure that your manipulation is causing the changes. But if you are not manipulating anything, it is harder to know what the cause is. Back to the noise example. If you turn up the noise level and people lose their concentration, then it seems clear that your turning up the noise level caused them to lose their concentration. But what if you do a correlational study and find that people tend to be less able to concentrate in noisier places? Well, it could be that the noise is causing the loss of concentration or it could be that people with good concentration tend to avoid noisy places (the directionality problem).

(ii) Control of Extraneous Variables

The second feature that distinguishes experimental research from correlational research is the **control of extraneous variables**. Extraneous variables are basically all variables other than those you are interested in for purposes of your research. In an experiment on the effects of noise on concentration, there is an infinite number of extraneous variables: age and sex of the research participants, whether or not they have eaten recently, the temperature of the room, the time of day,

To control extraneous variables means to keep their values or levels as similar as possible across the different values or levels of your independent variable. The different values or levels of your independent variable are also called **conditions**. For example, your noise experiment might have two conditions—noisy and quiet—and what you want is for things to be as similar as possible under these two conditions. You do not want old people in one condition and young people in the other, or men in one condition and women in the other, or hungry people in one condition and full people in the other, Why not? Because if there is a relationship between your independent and dependent variables, you cannot tell if it was because of your manipulation or if it was because of one of these extraneous variables. (It is the **third-variable problem** again.) If people in the noisy condition were hungry and people in the quiet condition were not, then perhaps that is why those in the noisy condition had more trouble concentrating. But if things are highly

similar under the different conditions, then you cannot “blame” the effect on anything other than the independent variable.

Confounding Variables

An extraneous variable that differs systematically across conditions is called a **confounding variable**. It is important to see the difference between extraneous variables and confounding variables. For example, in an experiment on the effectiveness of cognitive psychotherapy for treating depression, the independent variable is whether or not patients get the psychotherapy, and the dependent variable is how much they improve. Certainly, there will be lots of extraneous variables that are not really of primary interest: the sex of the patients, the age of the patients, their occupations, what they typically eat for breakfast, These extraneous variables are not a problem if they are similar in the two conditions. If both conditions contain men and women, young patients and old, doctors and plumbers, pancake eaters and coffee drinkers ... then there is no problem. In fact, in many situations, this kind of within-condition variability is considered good because then the results of the experiment can be said to apply to a wide variety of people. What is bad, however, is if any of these extraneous variables are confounding variables, in the sense that their average level varies *across* conditions. For example, if one condition contains a lot more men than women, then sex of the patients is a confounding variable. Now, if there is a difference between the conditions, it could be due to the psychotherapy or it could be a sex difference (*e.g.*, women tend to improve over time but men do not).

The Limitations of Experiments

The obvious advantage of experimental research is that it provides stronger evidence for causal claims. It does, however, have at least two limitations.

The first is that sometimes you cannot do an experiment because you cannot manipulate the independent variable, either for practical or ethical reasons. For example, if you are interested in the effects of a person’s culture on their tendency to help strangers, you cannot do an experiment. Why not? You cannot manipulate a person’s culture. Or if you are interested in how damage to a certain part of the brain affects behavior, you cannot do an experiment. Why not? You cannot go around damaging people’s brains to see what happens. In such cases, correlational research is the only alternative.

The second limitation of experimental research is that sometimes controlling extraneous variables means creating situations that are somewhat artificial. A good example is provided research on the effect of smiling on first impressions. To control extraneous variables, people are typically brought into a laboratory and asked standard questions about a small number of posed stimulus photographs. It is legitimate to ask, however, whether the effect of smiling is likely to be the same out in the “real world” where people are actually interacting with each other. For a good discussion of why this is not always a problem, though, see Stanovich’s (2007) discussion of the “artificiality criticism” of psychological research.

Quasi-Experiments

The prefix “quasi” means, in essence, “sort of.” So a **quasi-experiment** is a “sort of” experiment. Specifically, a quasi-experiment is a study that includes a manipulated independent variable but lacks important controls (*e.g.*, random assignment), or a study that

lacks a manipulated independent variable but includes important controls. So a quasi-experiment has some features of a well conducted experiment but not others.

Types of Quasi-Experiments

There are many types of quasi-experiments. Here we discuss just a few of the more common ones.

(i) Non-Equivalent Groups Design

A **non-equivalent groups** design includes an existing group of participants who receive a treatment and another existing group of participants to serve as a control or comparison group. Participants are not randomly assigned to conditions, but rather are assigned to the treatment or control conditions along with all the others in their existing group.

For example, imagine that we wanted to do a study to compare student performance in a cooperative learning section of Psych 144 with student performance in a standard lecture section. Imagine further that we scheduled two sections of the course, let students sign up for which one they wanted, and then taught one using cooperative learning and the other using standard lecture. Note that this study includes a manipulated independent variable, but it lacks random assignment of participants to conditions. The problem with this approach, of course, is that there might be differences between the two groups of students other than the style of teaching to which they were exposed. Perhaps the students who signed up for the earlier section are more “gung ho.” Or perhaps the students who signed up for the evening section are more likely to be working adults. Or perhaps the students in the 1:00 p.m. section tend to be drowsy after lunch. It is possible that differences in the dependent variable could have been caused by these differences rather than differences in teaching style.

You might wonder what the difference is between this and a plain old correlational study. The answer is “not much.” In fact, if you referred to this as a correlational study, we think few people would fault you. In general, however, non-equivalent groups are usually chosen to be as similar as possible to each other, which helps to control extraneous variables. For example, we probably would not use a daytime class as our cooperative learning group and an evening class as our standard lecture group. Instead, we would use another daytime class as our control group.

(ii) Pretest-Posttest Design

In a **pretest-posttest design**, a single group of participants is measured on the dependent variable both before and after the manipulation of the independent variable. Imagine that a group of 100 sixth graders is given a test of their attitudes toward drugs. This is the pretest. Then, a week later, a police officer comes to school and presents an anti-drug program (complete with “cool” decorated car and performing police dog). This is the treatment. Then, in another week, the students are given another test of their attitudes toward drugs. This is the posttest. Obviously, the substantive question here is whether the students’ attitudes toward drugs change after being presented with the anti-drug program.

The problem with pretest-posttest designs is that you cannot be completely sure that a change in the dependent variable was caused by the manipulation of the independent variable. For example, it is possible that something other than the anti-drug program might have occurred between the pretest and the posttest that influenced the students’

attitudes. It is even possible that having taken the pretest influenced their scores on the posttest.

The basic pretest-posttest design can be augmented by adding a control group. One way to do it would be to add a non-equivalent control group, for example, 100 sixth graders at another school who are not presented with the anti-drug program. Note that if this group also showed the same change in attitudes, it would not make sense to conclude that the change was due the program. Another way to do it would be to start with a larger sample and then randomly assign participants to either a treatment group or a true control group. A true control group is better than a non-equivalent control group because one can assume greater similarity between the control group and the treatment group. Note also that a pretest-posttest design with a true control group is an experiment and not a quasi-experiment.

(iii) Interrupted Time-Series Designs

A **time series** is simply a set of measurements of a variable taken at various points in time. For example, we could measure the moods of the students in our class each day throughout the semester, and we could see how people's moods changed (or did not change) over time. In an **interrupted time-series design**, a time series like this (the dependent variable) is interrupted (usually near the middle) by the manipulation of the independent variable. For example, if we were interested in the effect of the color of the classroom on students' moods, we could start measuring your moods each day from the beginning of September through Halloween. Then we could paint the room yellow (hoping to raise your moods) and continue measuring your moods each day from the beginning of November to the end of the semester. In analyzing the data, we would want to see whether there was an increase in your moods shortly after the painting of the room that continued to the end of the semester. The figure below shows some results that would confirm our hypothesis.

Note that this design is like a pretest-posttest design but with multiple pretests and multiple post-tests. The advantage of this approach is that it provides greater confidence that the change in the dependent variable was caused by the manipulation and is not just a random fluctuation. For example, if students' moods bounced around from week to week – high some weeks and low some weeks – then the change between Weeks 7 and 8 might just be one of these random fluctuations. However, the fact that students' moods were consistently around 6.00 and then jumped right after the manipulation to be consistently around 7.00 makes it clear that this the jump is not just a random fluctuation. The interrupted time series design also allows us to see how long the effect of the manipulation lasts. Like the pretest-posttest design, the interrupted time series design can be augmented with either a non-equivalent control group or a true control group.

Sometimes, the independent variable in an interrupted time series design is not manipulated; it just changes naturally. For example, research has shown that the number of suicides in the general population increases right after a particularly prominent suicide (e.g., the suicide of a famous person). This has been shown by looking at the number of suicides for several weeks or months before the prominent suicide and then seeing how it changes immediately after the prominent suicide. Note that this design includes no manipulation and also does not include any attempt to control extraneous variables. Yet if the number of suicides suddenly increases after a prominent suicide, this seems like

strong evidence of a causal connection – especially if this same result can be shown after many different prominent suicides.

(iv) Internal Validity

Many researchers (including us) see the difference between correlational studies, quasi-experiments, and experiments as one of degree rather than as one of kind. At one end of this continuum are ideal experiments in which only the independent variable differs across condition, so that it is perfectly clear that changes in the dependent variable were caused by the independent variable. But as we move from ideal experiments to less ideal ones to quasi-experiments to correlational studies, there are more and more variables that differ across conditions, which makes it more and more difficult to see whether it was the independent variable that was responsible for changes in the dependent variable.

Researchers actually have a name for this continuum: **internal validity**. To the extent that a study allows one to conclude that the independent variable affected the dependent variable, we say that it has good internal validity. So an ideal experiment has perfect internal validity, experiments usually have good internal validity, quasi-experiments have somewhat less internal validity, and correlational studies often have poor internal validity.

Field Study Methods

To observe the learning and teaching of generic skills in vocational and academic settings, we conducted intensive field studies of both kinds of classrooms. Our efforts to understand these classrooms focused on answering several questions:

What teaching practices do vocational and academic teachers use to teach generic skills and work-related attitudes?

How do the practices employed by vocational and academic teachers differ? Why?

What is the broader context within which these practices arise? How are the practices supported or hindered by elements of this context?

What positive aspects of the context are amenable to intervention – by staff development, policy, and so on?

We addressed these questions by examining teaching of generic skills in vocational and academic classrooms with a combination of research approaches that included case study, ethnographic, and survey methods. A case study approach permitted the intensive observation and analysis of a limited set of specifically targeted sites. We used ethnographic methods to deepen our understanding of what goes on in classrooms, particularly the interactions among teachers and students. By triangulating data collection methods—using participant observation, formal and informal interviews, student focus groups and survey, and collection of artifacts—we attempted to strengthen the reliability of our findings within classrooms.

As described in Section 1, our previous study of vocational classrooms produced a framework for analyzing the learning and teaching of generic skills. The present study further develops the framework and applies it to the analysis of a variety of vocational and academic classrooms that purport to engage students in learning generic skills and dispositions, but that differ in significant ways—the students, teacher, subject, and school. In both studies, we use theory and previous research to develop the framework and refine it

based on our classroom research experience. Our intent is not to test a particular theory but to use theory as an initial guide for data collection and for interpreting the results.

Our approach leaves open the possibility that factors not explicitly identified in our framework may be important in some classrooms and contexts in explaining the activities observed. It also permits the discovery of issues important to understanding classrooms that are not accounted for *a priori* by an analytic perspective. In our previous study (Stasz et al., 1990) this approach proved valuable, enabling us to identify an unexpected emphasis by teachers on shaping and supporting development of work-related attitudes in students.

This approach to studying classrooms also accepts the view that reality in a social setting—the classroom, in this case—is constructed by individuals through interactions that reflect their understanding about the world. When individuals share a classroom over time, they develop mutual understandings and expectations about the classroom. These constitute the classroom culture. The classroom is also embedded in a wide school and community culture that further affects teaching and learning.

Site Selection

In keeping with our goal to test the framework produced from our previous study (Stasz et al., 1990) in both vocational and academic settings, we used site selection criteria suggested by that study's findings. Initially we sought to identify classrooms of both sorts where teachers claimed to (1) teach problem solving skills, (2) value students' cooperation and responsibility for their own learning, (3) provide opportunities for project and group work, and (4) hold high expectations for students' performance.

We employed a "snowballing" or "chain-sampling" approach to locate potential sites (cf. Patton, 1980). We began by contacting high schools, skills centers, and community colleges throughout Los Angeles County based on leads from ongoing research and other professional contacts. We developed a slate of twelve candidate classrooms and secured permission for a prestudy visit to each. Two members of the research team conducted the visit, which included classroom observation, informal discussions with students, and an interview with the teacher. Based on a written report of the prestudy visit and subsequent discussion by the research team, we identified eight classrooms and negotiated entree with the school principals. (Again, one of these classrooms—an interior design class—was reported on in our preceding study.)

We rejected two candidate math classrooms because the curriculum relied on traditional classroom practices to help students acquire skills that were prerequisites for the next level. We also rejected a candidate science classroom when district budgetary problems led to overcrowding and delayed delivery of supplies, spoiling the teacher's planned use of cooperative projects. In the final candidate site, the teacher resigned before we could conduct a visit.

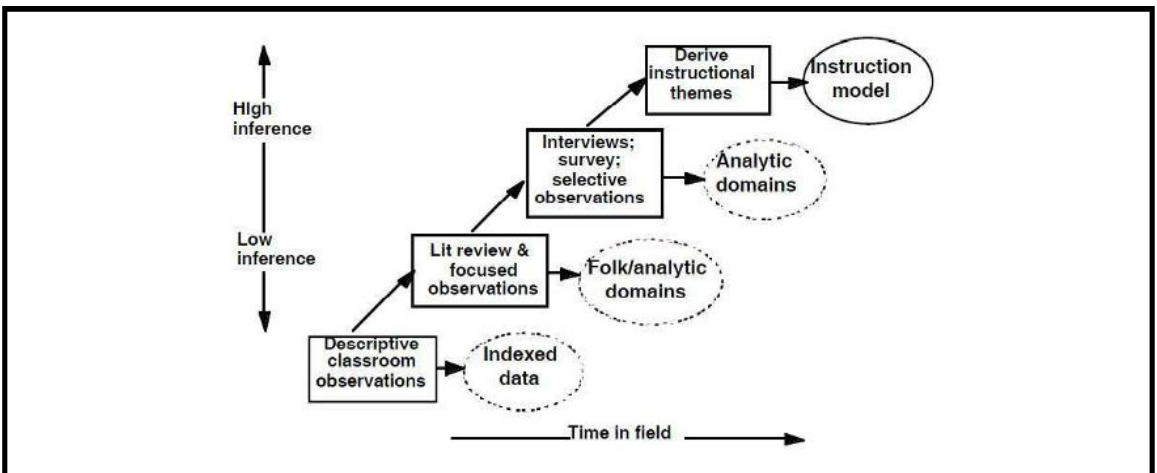
In spite of the selection process, observations culled from a single visit were insufficient to fully assess our prerequisite criteria. For instance, all the teachers in our sample *stated* that they held high expectations for students in all the classes they taught and nothing revealed during a single visit suggested otherwise. However, in the three classrooms that we judged not to work, we *observed* learning expectations for students to be low. This conclusion came as a result of many and repeated classroom observations and subsequent analysis. This, of course, reinforces the appropriateness of the extensive and in-depth ethnographic methods we employ to understand classrooms.

Analytic Procedures

The goal of the field study was to develop a model of generic skills instruction based on an understanding of classrooms that “worked”—*i.e.*, that appeared successful in the attempt to impart generic skills and attitudes. During the course of the present study we considered the three teachers sequentially and carried out intensive, six-week case studies of five classrooms (landscape/horticulture, English, electronics, and two chemistry) and shorter case studies of two vocational classrooms (architectural drawing; mechanical drawing/manufacturing). We targeted all three academic classrooms (English and the two chemistry classes) for intensive study since our previous research had been based solely on vocational classrooms. The intensive case study of the interior design classroom, as indicated above, was carried out as part of the preceding project.

Following ethnographic methods as described by Spradley (1980), the analytic process for understanding these classrooms was organized into five activities:

1. Descriptive observation and questioning, which identified key teacher practices and student behaviours.
2. Focused observations and structural questions, which expanded the list of practices and behaviors and their differences.
3. Selected observations and contrast questions, which clarified the differences that separate practices and behaviours.
4. Indexing data, which placed practices and behaviors in meaningful domains.
5. Analysis of relationships between domains and generation of themes, which composed the model of instruction.



RESEARCH APPROACH FOR DEVELOPING MODEL OF GENERIC SKILLS INSTRUCTION

Figure suggests how the analytic activities progressed over time (details of the figure will be explained in the following pages). We began our analysis during early stages of the fieldwork with “low inference” procedures so as not to bias or close off our observations: that is, we attempted to observe with little theoretical guidance, not inferring initially that we understood and could describe in theoretical terms what was occurring. Over time, increasingly inferential procedures became appropriate and were applied. As we became ethnographically familiar with classrooms, we returned to our conceptual frame-

work and explored cognitive science and motivation literature to derive hypotheses to explain what we were observing. We reviewed literature on work design and socio-technical systems to help us frame questions about the design of the classroom and the teacher role as we were beginning to understand these elements as explanatory of the classroom culture. Throughout the process we employed an iterative approach to develop deeper understanding of the classrooms (Spradley, 1980). The iteration involved blending and staging the data collection and analytic procedures as represented in the figure.

We conducted training sessions to prepare field-workers for each task. This training included practice opportunities prior to going into sites, modeling by an experienced member of the team, and feedback on field notes.

All five members of the fieldwork team collected data for each intensive case. Teachers were scheduled sequentially over a year, spring 1991 to summer 1992. Two field-workers visited each classroom twice weekly for six weeks and submitted field notes of their observations. The fieldwork team met weekly to review their notes, discuss role management (*i.e.*, how to be an effective classroom observer), review potentially useful literature, discuss observations and reconcile discrepancies between observers, and plan the week's observation agenda. After completing observations, we conducted formal interviews with teachers and administrators and focus group discussions with students. Before the focus group discussions, students completed the survey instrument.

Descriptive Observation

Experience with micro-ethnography has shown that nine-tenths of what goes on in a social scene is captured in interactions (Spradley, 1980). To collect this data, ethnographic fieldwork relies on an iterative strategy of observation and questioning. Repeated observation of interactions begins to expose the "meaning-making" behind behavior (Spradley, 1980; Geertz, 1973). The observer's understanding of this meaning is sharpened through questioning of the participants.

We observed students in various classroom activities, paying particular attention to their interactions and their context. At first we focused on making participants familiar and comfortable with field-workers while we became familiar with the classroom environment and identified key informants, individuals who would serve as our early guides, provide informal introductions to other students, and supply an "insider's" view into the inner workings of the classroom. We then structured our descriptive observations by "shadowing" these key informants, joining their work groups as an observer and questioning them during the course of their activities. In all cases the teacher was a particularly important informant, and one researcher in each classroom shadowed the teacher to observe specific teacher/student interactions. We collected classroom artifacts such as assignment sheets, lab experiments, and drafts of student work. Finally, through informal discussions with teachers and students, we attempted to gain understanding of the meanings that they attached to the activities and practices that we observed.

Indexing of Observational Data

At the close of descriptive observations in each case, we began indexing raw field data. Indexing involves identifying and tagging segments of field-note text.

Suspending observations for a week, as a group we developed a list of index terms as a first attempt at describing the classroom: Examples include classroom policies and practices, feedback, and infractions. This involved a process of reading and noticing the in-

interesting things in our data and variation within a classroom. In addition, the list included concepts or issues identified in our earlier study: Examples are problem solving and motivation/disposition. Over the course of the fieldwork and after several discussions, we reached consensus on a list of low-inference terms and definitions that each researcher-field-worker used to tag their field notes – *e.g.*, affect toward learning/classroom/teacher; extraordinary engagement, trust/respect, stress, acting out, and horse-play.

To achieve reliability in reading, noticing, and tagging, we calibrated the indexing task. Each field-worker indexed several field notes, written by different field-workers, and we compared our results. As a group we clarified definitions and identified previously neglected areas. We used *The Ethnograph*, a computer-based analytic tool, to help manage the indexing task. The computer program made it possible to number lines of text and then sort text lines by index terms. *The Ethnograph* permitted organization of our data set, some 1,000 plus pages of text, into relevant categories – using our index terms – while establishing an audit trail. It facilitated data reduction by permitting assignment of index terms to sections of text. The text then could be sorted according to these index terms to reveal patterns across the separate observations. As domains were identified, index codes could be recorded and merged. Thus, we were able to index complex passages of text while maintaining the whole and merge data as domains were developed.

Focused Observation

By the third and fourth week, we were able to account for a broad and varied range of classroom activities, practices, and behaviors, using our developing index terms. At this point we sought to understand how each of these worked or was employed. We moved from asking descriptive questions (*e.g.*, “what are you doing now?”) to posing structural questions that were directed at reasons for behaviors and the conditions in which they were considered appropriate (*e.g.*, to the teacher, “when do you use one-to-one instruction? when do you use cooperative groups?”). We planned for observation opportunities to obtain answers to structured questions by engaging in participant observation with a specific project group or to shadow the teacher during teaching rounds.

Selective Observation

During the final weeks of fieldwork we moved to selective observations to clarify the differences and relationships among core skills taught in the classroom, student behaviors, and practices teachers employed. During our weekly meetings we developed contrast questions that would look at two or three teaching practices and explore how they differed or resembled each other. Interviews with teachers and student focus group discussions provided a final opportunity to account for differences and relationships among similar key observed characteristics of classrooms.

On the final day of observations, we surveyed students to gather personal background information and data on classroom motivation and beliefs about their own self-efficacy for learning. The questionnaire was designed as a series of statements against which students could rate their own level of agreement on a scale of 1 (strongly disagree) to 4 (strongly agree). For example, the statement “Working in groups is a good way to learn” refers to students’ assessment of the educational benefit of group work.

Following the survey we conducted focus group discussions to understand student grades, expectations, and experiences. Two field-workers, one leading the discussion and the

other taking notes, held a discussion with a group of 10–12 students. Multiple groups were held simultaneously over 30–40 minutes. To encourage students to speak freely about the class, the teacher was absent from the classroom. During the weeks following classroom observation, we interviewed teachers and school administrators to better understand classroom and school contexts, respectively. Our formal interview instruments were designed to carefully assess respondents' perceptions about and experiences with the learning of generic skills and how they viewed school policies and teaching practices that potentially influence classroom instruction and curriculum. Because we had already spent many weeks in the classrooms, we were able to fold our observations into the interview questions, asking for explanation and insights. During interviews we asked respondents first about their perceptions, then sought specific examples or probed for issues known to affect vocational and academic settings.

Domain Analysis

Domain analysis involves understanding patterns of behavior or practices. A domain is a category of meaning that includes other smaller categories; for instance, the domain of *student goals* may include *easy grade*, *graduation credit*, *preparation for work*, etc. Similarly, the domain of *teacher tactics* includes subcategories such as *scaffolding*, *fading*, and *articulation*.

EXAMPLES OF CONTRAST QUESTIONS USED TO DEFINE TEACHING TECHNIQUES OBSERVED IN THE ENGLISH CLASSROOM

Contrast Question	Observation
When do scaffolding, fading, and articulation resemble each other?	Never.
How are the three different?	When scaffolding, the teacher verbalizes or performs a process. When fading, the teacher “hands off” verbalization or performance to the student. During articulation, the teacher prompts the student to verbalize understanding of a concept or process.
When are any two related?	Scaffolding and fading are used sequentially. Articulation might result from scaffolding, but also in response to other prompts, <i>e.g.</i> , a question. Teachers fade, students articulate.

As we progressed from focused observations to selective observations, our domains shifted from “folk domains” (terms used and recognized by participants – *e.g.*, *seat-of-the-pants instruction*, “*negative capability*,” *trial and error*) to more “analytic domains” (terms adopted or created by the research team – *e.g.*, *master/apprentice roles*, *work-related attitudes*, *generating alternative solutions*). We determined that nineteen domains sufficiently accounted for understanding the eight classrooms we studied (see Appendix C for full list). In Section 3 we compare the eight classrooms along these nineteen domains.

Generation of Themes

We began our final analytic procedure by comparing classrooms by domain. Establishing relations between domains in certain classrooms led to generation of themes (*e.g.*, the

importance of having instructional goals that explicitly target generic skills and work-related attitudes). We report these themes in the section that follows and use them to organize our descriptions and analyses of the successful classrooms in later sections. Taken together, these themes compose a model of generic skills instruction. In classrooms that are organized around these themes, the teaching and learning of generic skills and attitudes appears to be successful.

THREE EXAMPLES OF SHIFTING FROM FOLK DOMAINS TO ANALYTIC DOMAINS

Folk Domain	Analytic Domain	Research Literature
"Seat-of-the-pants" instruction	Master-apprentice role	Cognitive apprenticeship (Collins et al., 1989)
The use of "negative capability"	Work-related attitude	Generic skills (Stasz et al., 1990)
The skill of "trial and error"	Generate alternative solutions	Complex reasoning skills (Stasz et al., 1990)

We designed and conducted the study to ensure a high degree of reliability and validity of findings. In particular, we triangulated our data collection methods, required interobserver reliability among the five-member research team during within-site and cross-site analysis, staged analysis by suspending high-inference procedures until we had spent sufficient time in the field to understand fully how participants make meaning of their classroom activities and learning, and delineated characteristics of teachers and classroom practices to establish clear differences between generic skills classrooms and traditional classrooms. These methodological procedures were set to support translatability and comparability across sites. Whereas translatability and comparability are factors that contribute to generalizability in experimental studies, they are the foundation for generalizability and causal claims in ethnographic research.

Although the sample of eight classrooms is too small to permit statistical inference needed for a claim of generalizability, our findings hold across many observations of teacher and student behaviors in the eight classrooms that purport to focus on generic skills. Our level of analysis informs teaching practices in these classrooms. Moreover, because several classrooms we studied "did not work," we were able to use these cases to contrast with our findings of "what does work" and to generate an instructional model for generic skills.

Case Study

In the social sciences and life sciences, a *case study* (or *case report*) is a descriptive, exploratory or explanatory analysis of a person, group or event. An explanatory case study is used to explore causation in order to find underlying principles. Case studies may be prospective (in which criteria are established and cases fitting the criteria are included as they become available) or retrospective (in which criteria are established for selecting cases from historical records for inclusion in the study).

Thomas offers the following definition of case study: "Case studies are analyses of persons, events, decisions, periods, projects, policies, institutions, or other systems that are studied holistically by one or more methods. The case that is the *subject* of the inquiry will be an instance of a class of phenomena that provides an analytical frame – an *object* – within which the study is conducted and which the case illuminates and explicates."

Another suggestion is that *case study* should be defined as a *research strategy*, an empirical inquiry that investigates a phenomenon within its real-life context. Case study research can mean single and multiple case studies, can include quantitative evidence, relies on multiple sources of evidence, and benefits from the prior development of theoretical propositions. Case studies should not be confused with qualitative research and they can be based on any mix of quantitative and qualitative evidence. Single-subject research provides the statistical framework for making inferences from quantitative case-study data. This is also supported and well-formulated in (Lamnek, 2005): “The case study is a research approach, situated between concrete data taking techniques and methodologic paradigms.”

The case study is sometimes mistaken for the case method, but the two are not the same.

It is generally believed that the case-study method was first introduced into social science by Frederic Le Play in 1829 as a handmaiden to statistics in his studies of family budgets. *Les Ouvriers Europeens*. The use of case studies for the creation of new theory in social sciences has been further developed by the sociologists Barney Glaser and Anselm Strauss who presented their research method, Grounded theory, in 1967.

The popularity of case studies in testing hypotheses has developed only in recent decades. One of the areas in which case studies have been gaining popularity is education and in particular educational evaluation.

Case studies have also been used as a teaching method and as part of professional development, especially in business and legal education. The problem-based learning (PBL) movement is such an example. When used in (non-business) education and professional development, case studies are often referred to as *critical incidents*.

When the Harvard Business School was started, the faculty quickly realized that there were no textbooks suitable to a graduate program in business. Their first solution to this problem was to interview leading practitioners of business and to write detailed accounts of what these managers were doing. Cases are generally written by business school faculty with particular learning objectives in mind and are refined in the classroom before publication. Additional relevant documentation (such as financial statements, time-lines, and short biographies, often referred to in the case as “exhibits”), multimedia supplements (such as video-recordings of interviews with the case protagonist), and a carefully crafted teaching note often accompany cases.

Case study research excels at bringing us to an understanding of a complex issue or object and can extend experience or add strength to what is already known through previous research. Case studies emphasize detailed contextual analysis of a limited number of events or conditions and their relationships. Researchers have used the case study research method for many years across a variety of disciplines. Social scientists, in particular, have made wide use of this qualitative research method to examine contemporary real-life situations and provide the basis for the application of ideas and extension of methods. Researcher Robert K. Yin defines the case study research method as an empirical inquiry that investigates a contemporary phenomenon within its real-life context; when the boundaries between phenomenon and context are not clearly evident; and in which multiple sources of evidence are used.

Critics of the case study method believe that the study of a small number of cases can offer no grounds for establishing reliability or generality of findings. Others feel that the

intense exposure to study of the case biases the findings. Some dismiss case study research as useful only as an exploratory tool. Yet researchers continue to use the case study research method with success in carefully planned and crafted studies of real-life situations, issues, and problems. Reports on case studies from many disciplines are widely available in the literature.

This paper explains how to use the case study method and then applies the method to an example case study project designed to examine how one set of users, non-profit organizations, make use of an electronic community network. The study examines the issue of whether or not the electronic community network is beneficial in some way to non-profit organizations and what those benefits might be.

Many well-known case study researchers such as Robert E. Stake, Helen Simons, and Robert K. Yin have written about case study research and suggested techniques for organizing and conducting the research successfully. This introduction to case study research draws upon their work and proposes six steps that should be used:

- Determine and define the research questions
- Select the cases and determine data gathering and analysis techniques
- Prepare to collect the data
- Collect data in the field
- Evaluate and analyze the data
- Prepare the report

Step 1. Determine and Define the Research Questions

The first step in case study research is to establish a firm research focus to which the researcher can refer over the course of study of a complex phenomenon or object. The researcher establishes the focus of the study by forming questions about the situation or problem to be studied and determining a purpose for the study. The research object in a case study is often a program, an entity, a person, or a group of people. Each object is likely to be intricately connected to political, social, historical, and personal issues, providing wide ranging possibilities for questions and adding complexity to the case study. The researcher investigates the object of the case study in depth using a variety of data gathering methods to produce evidence that leads to understanding of the case and answers the research questions.

Case study research generally answers one or more questions which begin with “how” or “why.” The questions are targeted to a limited number of events or conditions and their inter-relationships. To assist in targeting and formulating the questions, researchers conduct a literature review. This review establishes what research has been previously conducted and leads to refined, insightful questions about the problem. Careful definition of the questions at the start pinpoints where to look for evidence and helps determine the methods of analysis to be used in the study. The literature review, definition of the purpose of the case study, and early determination of the potential audience for the final report guide how the study will be designed, conducted, and publicly reported.

Step 2. Select the Cases and Determine Data Gathering and Analysis Techniques

During the design phase of case study research, the researcher determines what approaches to use in selecting single or multiple real-life cases to examine in depth and

which instruments and data gathering approaches to use. When using multiple cases, each case is treated as a single case. Each case's conclusions can then be used as information contributing to the whole study, but each case remains a single case. Exemplary case studies carefully select cases and carefully examine the choices available from among many research tools available in order to increase the validity of the study. Careful discrimination at the point of selection also helps erect boundaries around the case.

The researcher must determine whether to study cases which are unique in some way or cases which are considered typical and may also select cases to represent a variety of geographic regions, a variety of size parameters, or other parameters. A useful step in the selection process is to repeatedly refer back to the purpose of the study in order to focus attention on where to look for cases and evidence that will satisfy the purpose of the study and answer the research questions posed. Selecting multiple or single cases is a key element, but a case study can include more than one unit of embedded analysis. For example, a case study may involve study of a single industry and a firm participating in that industry. This type of case study involves two levels of analysis and increases the complexity and amount of data to be gathered and analyzed.

A key strength of the case study method involves using multiple sources and techniques in the data gathering process. The researcher determines in advance what evidence to gather and what analysis techniques to use with the data to answer the research questions. Data gathered is normally largely qualitative, but it may also be quantitative. Tools to collect data can include surveys, interviews, documentation review, observation, and even the collection of physical artifacts.

The researcher must use the designated data gathering tools systematically and properly in collecting the evidence. Throughout the design phase, researchers must ensure that the study is well constructed to ensure construct validity, internal validity, external validity, and reliability. Construct validity requires the researcher to use the correct measures for the concepts being studied. Internal validity (especially important with explanatory or causal studies) demonstrates that certain conditions lead to other conditions and requires the use of multiple pieces of evidence from multiple sources to uncover convergent lines of inquiry. The researcher strives to establish a chain of evidence forward and backward. External validity reflects whether or not findings are generalizable beyond the immediate case or cases; the more variations in places, people, and procedures a case study can withstand and still yield the same findings, the more external validity. Techniques such as cross-case examination and within-case examination along with literature review helps ensure external validity. Reliability refers to the stability, accuracy, and precision of measurement. Exemplary case study design ensures that the procedures used are well documented and can be repeated with the same results over and over again.

Step 3. Prepare to Collect the Data

Because case study research generates a large amount of data from multiple sources, systematic organization of the data is important to prevent the researcher from becoming overwhelmed by the amount of data and to prevent the researcher from losing sight of the original research purpose and questions. Advance preparation assists in handling large amounts of data in a documented and systematic fashion. Researchers prepare databases to assist with categorizing, sorting, storing, and retrieving data for analysis.

Exemplary case studies prepare good training programs for investigators, establish clear protocols and procedures in advance of investigator field work, and conduct a pilot study in advance of moving into the field in order to remove obvious barriers and problems. The investigator training program covers the basic concepts of the study, terminology, processes, and methods, and teaches investigators how to properly apply the techniques being used in the study. The program also trains investigators to understand how the gathering of data using multiple techniques strengthens the study by providing opportunities for triangulation during the analysis phase of the study. The program covers protocols for case study research, including time deadlines, formats for narrative reporting and field notes, guidelines for collection of documents, and guidelines for field procedures to be used. Investigators need to be good listeners who can hear exactly the words being used by those interviewed. Qualifications for investigators also include being able to ask good questions and interpret answers. Good investigators review documents looking for facts, but also read between the lines and pursue collaborative evidence elsewhere when that seems appropriate. Investigators need to be flexible in real-life situations and not feel threatened by unexpected change, missed appointments, or lack of office space. Investigators need to understand the purpose of the study and grasp the issues and must be open to contrary findings. Investigators must also be aware that they are going into the world of real human beings who may be threatened or unsure of what the case study will bring.

After investigators are trained, the final advance preparation step is to select a pilot site and conduct a pilot test using each data gathering method so that problematic areas can be uncovered and corrected. Researchers need to anticipate key problems and events, identify key people, prepare letters of introduction, establish rules for confidentiality, and actively seek opportunities to revisit and revise the research design in order to address and add to the original set of research questions.

Step 4. Collect Data in the Field

The researcher must collect and store multiple sources of evidence comprehensively and systematically, in formats that can be referenced and sorted so that converging lines of inquiry and patterns can be uncovered. Researchers carefully observe the object of the case study and identify causal factors associated with the observed phenomenon. Renegotiation of arrangements with the objects of the study or addition of questions to interviews may be necessary as the study progresses. Case study research is flexible, but when changes are made, they are documented systematically.

Exemplary case studies use field notes and databases to categorize and reference data so that it is readily available for subsequent reinterpretation. Field notes record feelings and intuitive hunches, pose questions, and document the work in progress. They record testimonies, stories, and illustrations which can be used in later reports. They may warn of impending bias because of the detailed exposure of the client to special attention, or give an early signal that a pattern is emerging. They assist in determining whether or not the inquiry needs to be reformulated or redefined based on what is being observed. Field notes should be kept separate from the data being collected and stored for analysis.

Maintaining the relationship between the issue and the evidence is mandatory. The researcher may enter some data into a database and physically store other data, but the researcher documents, classifies, and cross-references all evidence so that it can be efficiently recalled for sorting and examination over the course of the study.

Step 5. Evaluate and Analyze the Data

The researcher examines raw data using many interpretations in order to find linkages between the research object and the outcomes with reference to the original research questions. Throughout the evaluation and analysis process, the researcher remains open to new opportunities and insights. The case study method, with its use of multiple data collection methods and analysis techniques, provides researchers with opportunities to triangulate data in order to strengthen the research findings and conclusions.

The tactics used in analysis force researchers to move beyond initial impressions to improve the likelihood of accurate and reliable findings. Exemplary case studies will deliberately sort the data in many different ways to expose or create new insights and will deliberately look for conflicting data to disconfirm the analysis. Researchers categorize, tabulate, and recombine data to address the initial propositions or purpose of the study, and conduct cross-checks of facts and discrepancies in accounts. Focused, short, repeat interviews may be necessary to gather additional data to verify key observations or check a fact.

Specific techniques include placing information into arrays, creating matrices of categories, creating flow charts or other displays, and tabulating frequency of events. Researchers use the quantitative data that has been collected to corroborate and support the qualitative data which is most useful for understanding the rationale or theory underlying relationships. Another technique is to use multiple investigators to gain the advantage provided when a variety of perspectives and insights examine the data and the patterns. When the multiple observations converge, confidence in the findings increases. Conflicting perceptions, on the other hand, cause the researchers to pry more deeply.

Another technique, the cross-case search for patterns, keeps investigators from reaching premature conclusions by requiring that investigators look at the data in many different ways. Cross-case analysis divides the data by type across all cases investigated. One researcher then examines the data of that type thoroughly. When a pattern from one data type is corroborated by the evidence from another, the finding is stronger. When evidence conflicts, deeper probing of the differences is necessary to identify the cause or source of conflict. In all cases, the researcher treats the evidence fairly to produce analytic conclusions answering the original “how” and “why” research questions.

Step 6. Prepare the Report

Exemplary case studies report the data in a way that transforms a complex issue into one that can be understood, allowing the reader to question and examine the study and reach an understanding independent of the researcher. The goal of the written report is to portray a complex problem in a way that conveys a vicarious experience to the reader. Case studies present data in very publicly accessible ways and may lead the reader to apply the experience in his or her own real-life situation. Researchers pay particular attention to displaying sufficient evidence to gain the reader’s confidence that all avenues have been explored, clearly communicating the boundaries of the case, and giving special attention to conflicting propositions.

Techniques for composing the report can include handling each case as a separate chapter or treating the case as a chronological recounting. Some researchers report the case study as a story. During the report preparation process, researchers critically examine the document looking for ways the report is incomplete. The researcher uses repre-

sentative audience groups to review and comment on the draft document. Based on the comments, the researcher rewrites and makes revisions. Some case study researchers suggest that the document review audience include a journalist and some suggest that the documents should be reviewed by the participants in the study.

Cross-cultural Research

Cross-cultural research is a scientific method of comparative research which focuses on systematic comparisons that compares culture to culture and explicitly aims to answer questions about the incidence, distributions, and causes of cultural variation and complex problems across a wide domain, usually worldwide. Such questions include:

What are the patterns of coherence and sources of coherence in the practices, beliefs, social roles, norms, expressions, and forms of organization and conflict in: (a) human communities? (b) Other forms of groups? (c) Other extra-community trajectories?

How much of that coherence is due to: (a) Common history, language, identity? (b) Common or recurrent modes of adaptation to recurrent human problems? (c) recurrent consistencies in how language, discourse and expression, social and (d) Roles, norms and organizations constructed into shared cultures?

What are the patterns of decoherence and disjuncture, misunderstanding and conflict that arise given the (a) multiplicity and overlapping of cultures? and (b) cleavages and disjuncture of cultures?

Cross-cultural research strives to arrive at reliably supported explanations of why things are the way they are:

To compare culture is not to deny their individual uniqueness. Cross-culturalists do not deny the uniqueness of culture; uniqueness and similarity are always present, simultaneously. Cross-cultural research deals with what is general, what is true for some or even all human cultures.

Results of Cross-cultural research are generalizable to all types of society from a small hamlet with population in the hundreds or a few thousands to urban societies with population in the hundreds of millions. It helps researchers to make general (cross-cultural) statements about the similarities and differences of cultures and to identify what may be universal and variable about human cultures, as well as to discover reasons why the variation exists.

Cross-cultural psychologists generally collect primary data, they do not make use of secondary data derived from national surveys on issues such as crime rate, gross national product, etc. however, their comparison tend to be limited often only to two-cultures (the investigator's own and one other).

Perspectives in Cross-Cultural Psychology

Basically, there are two types of perspectives in cross-cultural psychology: the Absolutist Approach and the Relativist Approach. The absolutists view in cross-cultural psychology is that psychological phenomena are basically the same in all cultures while the relativists view in cross-cultural psychology is that psychological phenomena should be studied only from "within" a culture where these phenomena occur.

Types of Cultures: Subcultures or Populations

Culture is a matrix of behaviours, beliefs, practices and values that typifies a particular group of people. It deals with a variety of things that influence all aspects of behaviour; emphasizes group influence rather than individual variability and essential part of the meaning-formation process.

Human communities have a variety of practices, beliefs, social roles, norms, expressions, forms of organization and conflicts that exhibit various sorts of internal coherence as well as cleavages within communities. These coherences and cleavages bear many close connections to the different historical experiences, physical and social environments in which people live. They include configurations of elements and characteristic ways of interrelating that are shared with neighbouring and interacting groups, and shared among dispersed groups that have common historical experiences and similarities, including common origin, common membership in historical civilizations, and languages that are mutually understood or that derive common families. Lines of cleavage, conflict, and marginality, of course, are part of cultural phenomena. Elements and relationships that individuals or communities have in common are shared in a variety of ways. Some, such as the more intensive patterns of interaction that derive from common residence, joint experience, and discourse in a common language or system of signs, are relatively well bounded. Other patterns of sharing or similarity derive from processes of dispersal: migration, Diasporas, the trajectory of lives lived through spatial movements, social mobility, careers, and distinctive histories. Interactions are by no means limited to localities, but to the trajectories of inhabitants who move through and between localities.

Cultures consist of shared constructions that emerge out of social interactions of sets of individuals who inhabit overlapping social and physical spaces. Coherence may be viewed as an emergent property, but may be present or absent to varying degrees and along varying dimensions or trajectories.

Three Central Questions Revolving Around the Study of Culture

Murdock and White, in creating the Standard Sample for Cross-Cultural Research, raised three central questions that revolve around the study of culture. First, what are the foci of study? Second, what are the issues of coherence or de-coherence within the foci studied? Third, how do these issues apply between foci?

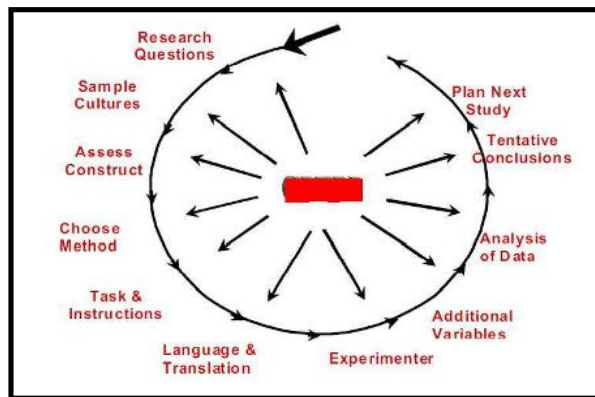
First, there is the problem of what it is that one wants to study in terms of foci: in their case, Murdock and White take up the issue of sampling the diversity of human communities in order to learn something about coherence or decoherence within and between communities. Comparing communities (through ethnographies) is not the only approach to cultural comparisons, but it is the one that focuses on clusters of people who inhabit, at least in part, specific and stable community sites which are often the loci of ethnography. Other standard samples could be those devoted to the study of: migration; institutions and organizations; particular types of cultures; subcultures or populations. Their insistence on this point is not on communities as representatives of larger cultures, but on communities at pinpointed times and places in which ethnographic study has been sufficiently well carried out. That is, it is possible to list, from the ethnography, what specific bundles of practices, beliefs, social roles, norms, expressions, forms of organization and conflicts (economic, political, legal, religious, expressive and artistic) are present in each of the specific times and places of the distinct ethnographic foci that constitute the sample.

Second, there are the problems and issues of coherence or de-coherence within the foci studied, as in the comparative study of human communities. The functionalist approach to ethnography was vastly over generalized in its heyday, which lasted from the 1920s through the 1970s. Murdock and White emphasize the weakness of the functionalist arguments that were commonly used by ethnographers to argue for the cultural coherence of the communities they studied. Functional linkage of traits asserted by an ethnographer in one society, for example, are often contradicted that one trait but not the other is present in a similar neighboring society, or by the fact that the traits do not correlate across societies.

Third, there is the problem and the issues of coherence or de-coherence between foci in the study of human communities due to common historical origin or experience. Murdock and White (1969:330) note that the conjectures of those anthropologists who attempt to reconstruct culture history on the basis of trait similarities are not notably superior to those of ethnographers who investigate functional relationships.

The Cross-Cultural Research Process

The research process in cross-cultural study is circular in nature (*as indicated in Figure*), with built-in evaluation mechanisms at each stage of the process, which may cause the researcher(s) to re-evaluate decisions made at previous stages. McGrath stated that cross-cultural research should be viewed as a series of “logically ordered – though chronologically chaotic – choices” about potential cultural *ki*’s.



CROSS-CULTURAL RESEARCH PROCESS

Types of Cross-cultural Research

There are several types of cross-cultural research. The first of which is Regional comparative cross-cultural research which is well represented by the works of Kroeber and Driver. This approach is an attempt to define classifications of cultures and to then make inferences about processes of diffusion within a cultural region. Regional comparativists are likely to know a lot about each society in a within-region comparison but may not know all the cases in the region that depends mostly on the size of the region.

Next is small-scale regional comparison advocated by Fred Eggan who called it “controlled comparisons” on the assumption that it may make it easier to rule out possible effects of similarity in history, geography, and language. His assumption was that researchers could readily discern what accounts for some aspect of cultural variation within

the region if history, geography and language were held constant. This is the approach toward smaller scale comparative studies. Eggan suggests the combination of the anthropological concepts of ethnology with structure and function, allowing the researcher to pose more specific questions on a broader range of subjects. An example of this type of study is Spoehr's study which examined the changes in kinship systems among the Creek, Chickasaw, and Choctaw, and other regional tribes of the Southeast after their removal to the Oklahoma reservations. Spoehr detailed these changes with an analysis of the historical factors responsible for them and the resulting processes.

Another type is large-scale within-region research (using data on all or most of the cultural traits in the region) which generally tries to arrive at classifications of cultures in order to make inferences about processes of diffusion and historical ancestry within-region comparativists are interested in trying to see how the cultures in the region are related to each other.

Holocultural analysis or worldwide cross-cultural analysis which developed out of the ancestry from Tylor to Sumner and Keller, and then to Murdock is another type of cross-cultural research. According to Levinson holocultural study "is usually designed to test or develop a proposition through the statistical analysis of data on a sample of ten or more non-literate societies from three or more geographical regions of the world". In this approach, cultural traits are taken out of the context of the whole culture and are compared with cultural traits in widely diverse cultures to determine patterns of regularities and differences within the broad base of the study. These approaches differ on the determination of what constitutes a unit of analysis.

Another type is coding which implies that data can be collected in two ways. First, data can be coded directly from ethnographic sources, or from ethnographic reports in the HRAF files. Second, is the usage of previously coded data from coded ethnographic sources or previous holocultural studies. Levinson and Malone suggest coding the dependent variables from HRAF files or ethnographic sources, and collecting independent variables from compendia of coded data.

Inferential studies are another type of cross-cultural research. These are used to answer important or challenging questions such as "*so what?*" or "What does it mean?"

The last is emic vs. etic models. Emic models view behaviour as culture-specific phenomenon which must be understood in the context of a particular culture while etic models view behaviour as universal phenomenon which must be understood in comparison to behaviour in other cultures. Thus, three approaches (or combinations thereof) are possible: Subjective single-culture, Objective single-culture and Objective cross-cultural.

Methodological Concepts in Cross Cultural Research

There are primarily two main methodological concepts in cross-cultural research. The first is that comparison is essential to anthropological research. Therefore to understand culture, societies must be compared. The comparative method is a search for comparable cultural patterns in multiple societies, particularly the comparison of cultural traits taken out of cultural context.

Secondly, all theories, despite fads or current trends require testing. Without comparison there is no way to evaluate if presumed cause and effect are related. This relates to the logical "if" – "then" inductive process. If cause is not present then the effect should not be present. Cross-cultural analyses are used to describe the range and distribution of

cultural variation existent in the ethnographies recorded as well as to test the hypotheses and theories that are proposed to explain the variation recorded.

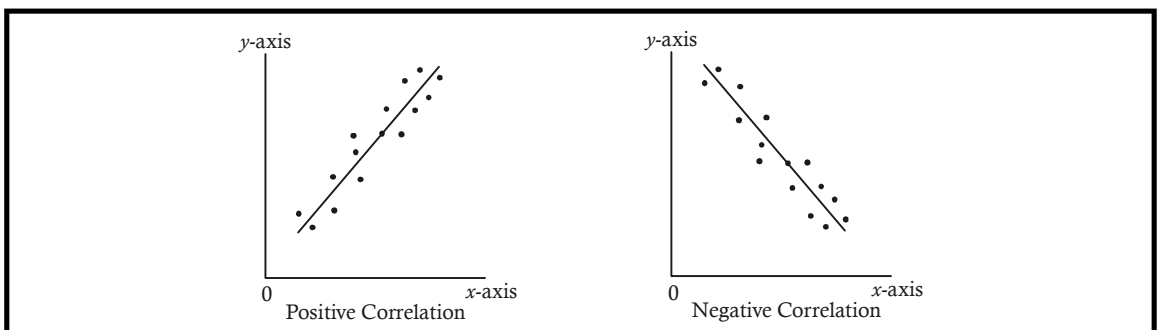
Methods of Observation in Cross Cultural Research and, Their Significance

Cross-cultural researchers usually employ standard principles of scientific method and research design-random sampling techniques which are supposed to be unbiased, reliable or repeatable measurements so that others can try to replicate their results and the researchers may determine how much confidence they can place in such results. These methods are case-study, survey, testing method, naturalistic-observation, laboratory-observation, correlational method, and experimental method.

The case-study method is a carefully drawn biography that may be obtained through interviews, questionnaires, and psychological tests. The survey method is a method of scientific investigation in which a large sample of people is questioned about their attitudes or behaviour. There are three types of survey research design. These are direct surveys, indirect surveys and focus group method. With direct surveys Interviewer maintains or can maintain a direct communication with the respondent and is able to provide feedback, repeat a question, or ask for additional information. Through indirect surveys, researcher's personal impact is very small because there is no direct communication between the respondent and the interviewer. The questions are typically written and handed in, mailed, or sent electronically to the respondents' homes, classrooms, or work places. The focus group methodology is a survey method used in academic & marketing research. The most common usage is one where a group responds to specific social, political, or marketing messages. The typical focus group contains 7-10 members who are experts, potential buyers, viewers or other customers.

The testing method entails the usage of psychological tests like intelligence, aptitude, and personality, to measure various traits and characteristics among a population by psychologists. Such testing instruments must be standardized, show face and concurrent validity, and reliability. They must have established norms as well.

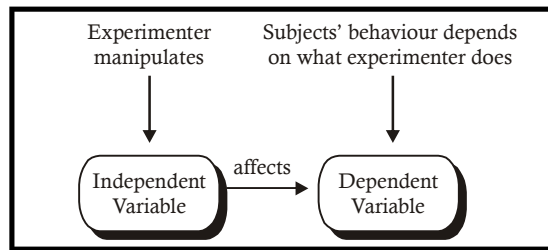
The naturalistic-observation is a scientific method in which organisms are observed in their natural environments while the laboratory-observation is a method where a place is found in which theories, techniques, and methods are tested and demonstrated.



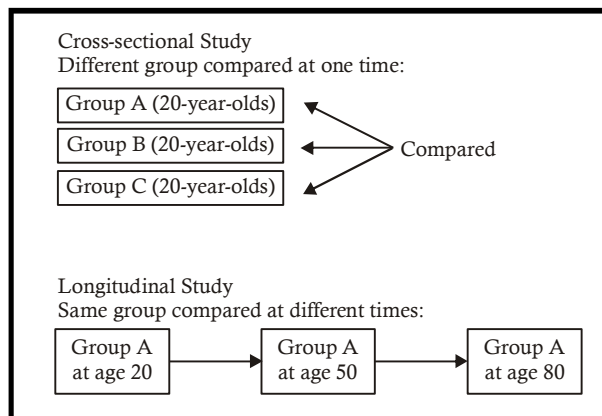
The correlational method is a scientific method that studies the relationships between cross-cultural variables. The Correlation coefficient is a number between +1.00 to -1.00 that expresses the strength and direction of the relationship between two vari-

ables. There are two major types of correlation: Positive correlation (which implies that increases in one variable are associated with increases in the other; decreases are likewise associated) and Negative correlation (which implies that increases in one variable are associated with decreases in the other).

The experimental method is used to hunt for causes of cultural variations in societies. The experimental variables consist of the independent variable which is a variable that an experimenter manipulates and the dependent variable which is a variable that an experimenter predicts will be affected by manipulations of the independent variable. The experimental design has three elements: Treatment, Experimental subject and Control subjects. The treatment refers to a condition (which may be placebo, blind or double-blind) received by participants so that its effects may be observed. Experimental subjects receive the treatment and the Control subjects do not receive the treatment but for whom all other conditions are comparable to those of experimental subjects.



All cross-cultural research methods may be employed in either Cross-Sectional Study or Longitudinal Study. Cross-sectional study is used to compare subjects of different ages at a given time but in longitudinal Study subjects are followed and periodically reassessed over a long period of time.

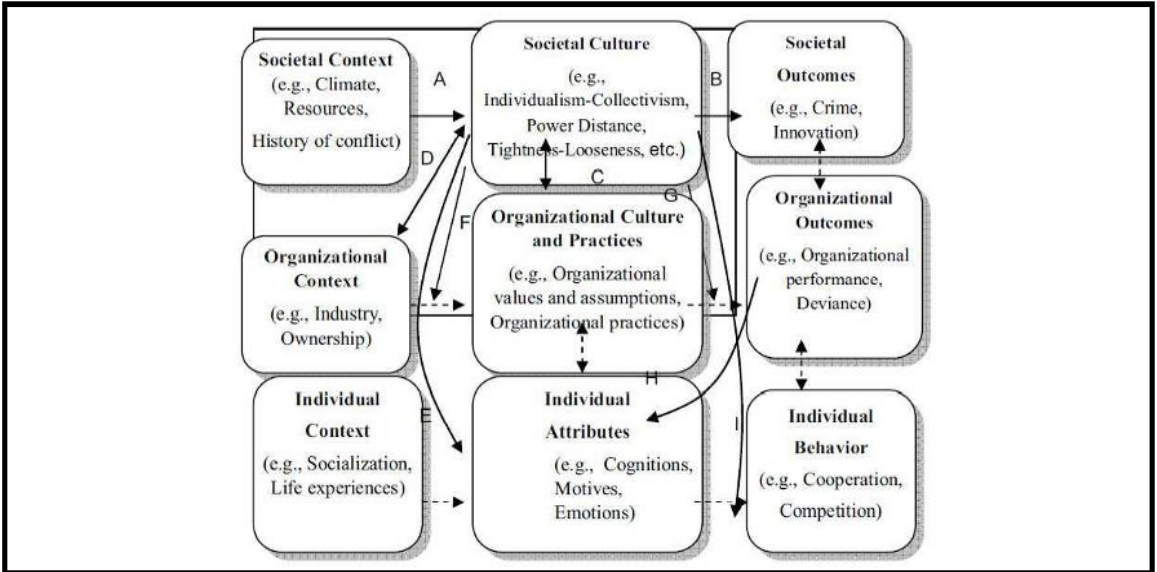


Features of Cross-cultural Research

Essential features of cross-cultural research are question, sampling, theories and hypotheses, operational definition, and principle of falsifiability. The research question has to be a clear, one sentence practical question which can be answered in a reasonable time frame (Figure below presents a typology of cross-cultural research questions)

Cross-cultural (worldwide comparative) researchers usually ask four kinds of questions. These are descriptive question, causal question, consequence question and non-directional relational question. Descriptive question usually deals with the prevalence or

frequency of a trait. Examples are: how common is the belief that sex is dangerous to one's health?



A TYPOLOGY OF CROSS-CULTURAL RESEARCH QUESTIONS

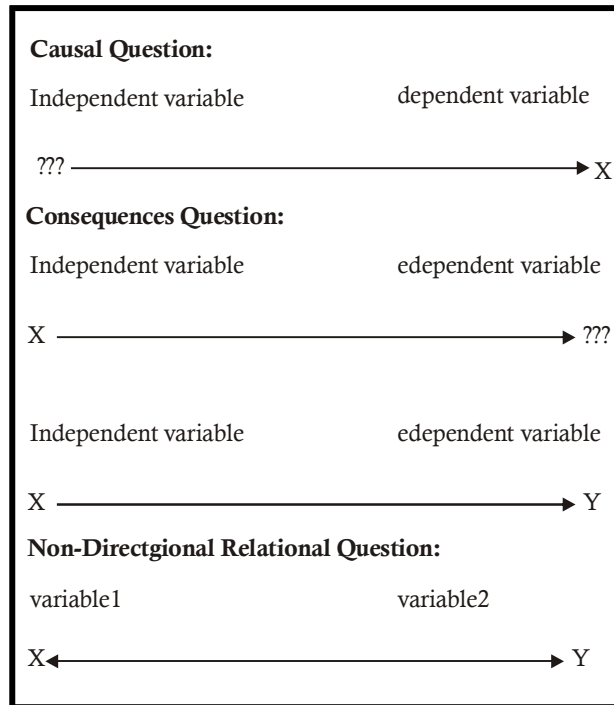
How common is it for a society to allow polygyny? Are peaceful societies common? Is the custom of marriage universal? What percentage of the world's societies practice polygyny? Which is the most important subsistence activity among food collectors – gathering, hunting, or fishing? How common is female infanticide?

Causal question seeks answers about the causes of a trait or custom. Examples of causal questions are: Why is polygyny permitted in most societies known to anthropology? Why do women (as opposed to men) do most of the agricultural work in some societies? Why is the extended family the customary form of household in many societies?

Consequence question asks about effects of the consequences or effects of a particular trait or custom. Examples of consequence questions are: What are the effects on infant care of high involvement of women in subsistence activities? Does punitive child training affect the frequency of warfare?

Non-directional relational question asks whether two traits are related or not, without implying cause or consequence. Rather than postulating causes or consequences, a researcher may simply ask how a particular aspect of culture may be associated with some other aspect(s). For example: Is there an association between most important subsistence activity and level of political complexity?

Causal question and Non-directional question are relational questions that ask how traits are related to other traits. Of the four types of questions, the causal or 'why' question is the most challenging because it rarely specifies what the researcher needs to do to answer the question. The descriptive question tells the researcher what to count. The consequence question can also be unspecific or open ended; the possible outcomes may not be specified in advance.



TYPES OF RELATIONAL QUESTIONS

The “consequence” and “relational” questions usually specify two sets of concrete phenomena that may be related to look at. If you want to know whether type of marriage has an effect on or is related to fertility, then there is the need to measure both variables (types of marriage and fertility). But the causal question does not tell the researcher where to look for causes. It only specifies what scientists call the *dependent* variable (the thing to be explained).

With regards to sampling, cross-cultural researchers must decide what societies to examine because no one can examine all cultures. The most important operating principles in a scientific test of hypothesis are: (1) to choose a sample that is representative of some universe of societies the researcher wants to generalize the results to; and (2) to use a large enough sample such that the results are likely to be true for the larger universe of cases. Theories and Hypotheses are ideas (implicit or explicit about how and why things are the way they are) underlying descriptive, consequences, causal or non-directional relational questions. Spelling out the assumptions and reasons for a researcher’s expectations can clarify his theorizing and also help derive hypotheses that could be tested. Theories are assumption of laws and statistical associations which do result in predictions called hypotheses. For instance:

Assumption 1. Males will be more aggressive than females.

Assumption 2. Individuals with higher levels of androgen will be more aggressive.

Cross-cultural analysts test hypotheses and draw statistical correlations based on the assumption of the existence of universal patterns.

Operational definition is a precise definition of a term in a hypothesis, which specifies the operations for observing and measuring the process or phenomenon being measured.

The principle of falsifiability assumes that a scientific theory must make predictions that are specific enough to expose the theory to the possibility of disconfirmation; that is, the theory must predict not only what will happen, but also what will not happen.

Other features include participants, replication (The ability to repeat, reproduce or copy a study), selection factor (The bias source that may occur when subjects are allowed to determine for themselves whether or not they will receive a treatment condition in a scientific study), and measures for each variable in the hypothesis. Devising a measure involves at least four steps; (1) theoretically defining the variable of interest (in words or mathematically); (2) operationally defining the variable, which means spelling out the “scale” that the researcher has devised for measuring it; (3) telling the coder where to find the required information and (4) pre-testing the measure to see if it can be applied generally to most cases. Designing a measure requires some trial-and-error. If the scale is too confusing or too hard to apply (because the required information is lacking), the measure needs to be rethought. For instance, the degree to which a society has extended family households needs to be defined. The researcher needs to state what an extended family means, what a household means, and how he will decide the “degree” to which a sample society has extended family households. Having defined the concepts, the researcher must specify how to measure the degree to which a society has extended family households.

Next is analysis of results which is used to determine if the predicted relationship actually exists in the data. It is precisely here that cross-cultural researchers usually resort to statistical tests of significance. There are several Statistical tests that show how “perfect” a result has to be for us to believe that there is probably an association between the variables of interest, that one variable generally predicts the other. Essentially, every statistical result is evaluated in the same objective way. The question is asked: *What is the chance that this result is purely accidental, that there is really no association at all between the two variables.* The answer always involves a probability value (or *p-value*), the likelihood that the observed result or a stronger one could have occurred by chance. So, if a result has a *p-value of less than .01*, this indicates that there is less than one chance in one hundred that the relationship observed is purely accidental. A *p-value of less than .01* is a fairly low probability; most social scientists conventionally agree to call any result with a *p-value of .05 or less* (five or fewer chances in one hundred) a *statistically significant* or probably true result.

STATISTICS IN PSYCHOLOGY

Measures of Central Tendency

We have studied how to collect raw data, its classification and tabulation in a useful form, which contributes in solving many problems of statistical concern. Yet, this is not sufficient, because practical purposes, there is need for further condensation, particularly when we want to compare two or more different distributions. We may reduce the entire distribution to one number which represents the distribution.

A single value which can be considered as typical or representative of a set of observations and around which the observations can be considered as Centered is called an ‘Average’ (or average value) or a Center of location. Since such typical values tends to lie centrally within a set of observations when arranged according to magnitudes, averages are called measures of central tendency.

In fact the distribution have a typical value (average) about which, the observations are more or less symmetrically distributed. This is of great importance, both theoretically and practically. Dr. A.L. Bowley correctly stated, "Statistics may rightly be called the science of averages."

The word average is commonly used in day-to-day conversations. For example, we may say that Vinod is an average boy of my class; we may talk of an average Indian, average income, etc. When it is said, "Vinod is an average student," it means is that he is neither very good nor very bad, but a mediocre student. However, in statistics the term average has a different meaning.

The fundamental measures of tendencies are:

- (1) Arithmetic mean
- (2) Median
- (3) Mode
- (4) Geometric mean
- (5) Harmonic mean
- (6) Weighted averages

However the most common measures of central tendencies or Locations are Arithmetic mean, median and mode.

ARITHMETIC MEAN

This is the most commonly used average which you have also studied and used in lower grades. Here are two definitions given by two great masters of statistics.

Horace Sacrist. "Arithmetic mean is the amount secured by dividing the sum of values of the items in a series by their number."

W.I. King. "The arithmetic average may be defined as the sum of aggregate of a series of items divided by their number."

Thus, the students should add all observations (values of all items) together and divide this sum by the number of observations (or items).

Ungrouped Data

Suppose, we have 'n' observations (or measures) $x_1, x_2, x_3, \dots, x_n$ then the Arithmetic mean is obviously $\frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$

We shall use the symbol $\bar{x}$ (pronounced as x bar) to denote the Arithmetic mean. Since we have to write the sum of observations very frequently, we use the usual symbol 'Σ' (pronounced as sigma) to denote the sum. The symbol x_i will be used to denote, in general the 'i'th observation. Then the sum, $x_1 + x_2 + x_3 + \dots + x_n$ will be represented by $\sum_{i=1}^n x_i$ or

x_i simply

Therefore the Arithmetic mean of the set $x_1 + x_2 + x_3 + \dots + x_n$ is given by,

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

This method is known as the "Direct Method".

Indirect Method (Assumed Mean Method)

$$\bar{u} = \frac{\sum u_i}{n} \text{ where } u = x_i - A$$

$$A = \text{Assumed Mean} = A + \bar{u}$$

Calculations:

$$\text{Let } A = 175 \text{ then}$$

$$\begin{aligned} u_i &= -65, -58, -46, +20, -80, -75, -75, +0, +75, +575 \\ &= 670 - 399 \\ &= 271/10 = 27.1 \end{aligned}$$

$$\begin{aligned} \bar{x} &= A + \bar{u} \\ &= 175 + 27.1 \\ &= 202.1 \end{aligned}$$

Short-cut Method

Sometimes the values of x are very big and in that case, to simplify the calculation the short-cut method is used. For this, first you assume a mean (called as the assumed mean). Let it be A . Now find the deviations of all the values of x from A . We now get a new variable $u_i = x_i - A$

Now find

$$\bar{u} = \frac{\sum u_i}{n} \text{ then } \bar{x} = A + \bar{u} \text{ or } \bar{x} = A + \bar{u}$$

Discrete Series. There is a difference in the methods for finding the arithmetic means of the individual series and a discrete series. In the discrete series, every term (*i.e.* value of x) is multiplied by its corresponding frequency ($f_i x_i$) and then their total (sum) is found ($\sum f_i x_i$). The arithmetic mean is then obtained by dividing the total frequency ($\sum f_i$) by the above sum so obtained.

Therefore, if the observations $x_1 + x_2 + x_3 + \dots + x_n$ are repeated $f_1 + f_2 + f_3 + \dots + f_n$ times, then we have :

$$\text{Arithmetic mean } (\bar{x}) = \frac{\sum f_i x_i}{\sum f_i} \text{ i.e. } \frac{f_1 x_1 + f_2 x_2 + f_3 x_3 + \dots + f_n x_n}{f_1 + f_2 + f_3 + \dots + f_n}$$

The formulae for Arithmetic mean by direct method and by the short-cut methods are as follows:

Direct Method Short-cut Method

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} \text{ or } \bar{x} = A + \bar{u} \text{ where } \bar{u} = \frac{\sum f_i u_i}{\sum f_i}$$

$$\text{and } u = x_i - A$$

$$\text{Therefore, } \bar{x} = A + \frac{\sum f_i u_i}{\sum f_i}$$

Mean for Grouped Data

Continuous Series. The procedure of finding the arithmetic mean in this series, is the same as we have used in the discrete series. The only difference is that in this series, we are given class-intervals, whose mid-values (class-marks) are to be calculated first.

$$\text{Formula, Arithmetic mean } (\bar{x}) = \frac{\sum f_i x_i}{\sum f_i}$$

where x = mid-value

Step-Deviation Method. Here all class intervals are of the same width say 'c'. This method is employed in place of the Short-cut method. We measure all the class-marks (mid values) from some convenient value, say 'A', which generally should be taken as the class-mark of a class of maximum frequency or of a class which is the middle one. All the class marks happen to be multiples of c, since all class intervals are equal. We consider class frequencies as if they are centered at the corresponding class-marks.

Theorem. If $x_1, x_2, x_3, \dots, x_n$ are n values of the class marks with frequencies $f_1, f_2, f_3, \dots, f_n$ respectively and if each x_i is expressed in terms of the new variable u_i by the relation $x_i = A + cu_i$ then, with the usual notation, we have $\bar{x} = A + c\bar{u}$

$$\text{where } \bar{u} = \frac{\sum f_i u_i}{\sum f_i} \text{ and } u_i = \frac{x_i - A}{c}$$

This method is also known as the "Coding method."

Properties of Arithmetic Mean

1. The sum of the deviations, of all the values of x , from their arithmetic mean, is zero.

$$\text{Justification : } \sum f_i (x_i - \bar{x}) = \sum f_i x_i - \bar{x} \sum f_i = 0$$

$$\text{Since } \bar{x} \text{ is a constant, } \sum \frac{f_i x_i}{f_i} = \sum f_i \bar{x} = \bar{x} \sum f_i$$

2. The product of the arithmetic mean and the number of items gives the total of all items.

$$\text{Justification : } \bar{x} = \frac{\sum f_i x_i}{\sum f_i} \quad \sum f_i x_i = \bar{x} \sum f_i$$

$$\text{or } \bar{x} = \frac{\sum x_i}{N} \quad \bar{x} N = \sum x_i$$

3. If $\bar{x}_1$ and $\bar{x}_2$ are the arithmetic mean of two samples of sizes n_1 and n_2 respectively then, the arithmetic mean $\bar{x}$ of the distribution combining the two can be calculated as

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

This formula can be extended for still more groups or samples.

$$\bar{x}_1 = \frac{\sum x_{1i}}{n_1} \quad \sum x_{1i} = n_1 \bar{x}_1$$

Justification: $\bar{x}_1 \mid \frac{x_{1i}}{n_1}$ $x_{1i} \mid n_1 \bar{x}_1$ = total of the observations of the first

Similarly $x_{2i} \mid n_2 \bar{x}_2$ = total of the observations of the first sample

The combined mean of the two samples

$$= \frac{\text{combined total}}{n_1 \sigma n_2}$$

$$\bar{x} = \frac{n_1 \bar{x}_1 \sigma n_2 \bar{x}_2}{n_1 \sigma n_2}$$

Merits

1. It is rigidly defined. Its value is always definite.
2. It is easy to calculate and easy to understand. Hence it is very popular.
3. It is based on all the observations; so that it becomes a good representative.
4. It can be easily used for comparison.
5. It is capable of further algebraic treatment such as finding the sum of the values of the observations, if the mean and the total number of the observations are given; finding the combined arithmetic mean when different groups are given etc.
6. It is not affected much by sampling fluctuations.

Demerits

1. It is affected by outliers or extreme values. For example, the average (A.) mean of 10, 15, 25 and 500 is $\bar{x} \mid \frac{10 \sigma 15 \sigma 25 \sigma 500}{4} \mid 137.5$

Now observe first three values whose A. mean is $\frac{10 \sigma 15 \sigma 25}{3} \mid 16.67$ (approx.)

Due to the outlier 500 the A. mean of the four numbers is raised to 137.5. In such a case A. mean is not a good representative of the given data.

2. It is a value which may not be present in the given data.
3. Many a times it gives absurd results like 4.4 children per family.
4. It is not possible to take out the averages of ratios and percentages.
5. We cannot calculate it when open-end class intervals are present in the data.

MEDIAN

It is the value of the size of the central item of the arranged data (data arranged in the ascending or the descending order). Thus, it is the value of the middle item and divides the series in to equal parts.

In Connor's words - "The median is that value of the variable which divides the group into two equal parts, one part comprising all values greater and the other all values lesser than the median." For example, the daily wages of 7 workers are 5, 7, 9, 11, 12, 14 and 15 dollars. This series contains 7 terms. The fourth term i.e. 11 is the median.

Median In Individual Series (ungrouped Data)

1. Set the individual series either in the ascending (increasing) or in the descending (decreasing) order, of the size of its items or observations.

2. If the total number of observations be 'n' then

A. If 'n' is odd,

The median = size of $\frac{n+1}{2}$ observation

B. If 'n' is even, the median

= $\frac{1}{2}$ size of $\frac{n}{2}$ Observations
 + size of $\frac{n+1}{2}$ Observations

Median In Discrete Series

Steps :

1. Arrange the data in ascending or descending order of magnitude.
2. Find the cumulative frequencies
3. Apply the formula :

A. If 'n' = f_i (odd) then,

Median = size of $\frac{n+1}{2}$ item

B. If 'n' = f_i (even) then,

Median = $\frac{1}{2}$ size of $\frac{n}{2}$ item
 + size of $\frac{n+1}{2}$ item

Median In Continuous Series (grouped Data)

Steps :

1. Determine the particular class in which the value of the median lies. Use $n/2$ as the rank of the median and not $\frac{n+1}{2}$
2. After ascertaining the class in which median lies, the following formula is used for determining the exact value of the median.

$$\text{Median} = l_1 + \frac{N/2 - c.f.}{f} (l_2 - l_1)$$

where, l_1 = lower limit of the median class, the class in which the middle item of the distribution lies.

l_2 = upper limit of the median class

c.f. = cumulative frequency of the class preceding the median class

f = sample frequency of the median class

It should be noted that while interpolating the median value of frequency distribution it is assumed that the variable is continuous and that there is an orderly and even distribution of items within each class.

Alternative Formula

Median

$$\begin{aligned}
 &= l_2 \rightarrow \frac{n/2 - c.f.}{f} (l_2 - l_1) \\
 &= 30 \rightarrow \frac{50 - 22}{44} (30 - 20) \\
 &= 30 \rightarrow \frac{28}{44} \cdot 10 \\
 &= 30 - 7 \\
 &= 23 \text{ marks}
 \end{aligned}$$

Note that, while calculating the median of a series, it must be put in the 'exclusive class-interval' form. If the original series is in inclusive type, first convert it into the exclusive type and then find its median.

Merits of Median

1. It is rigidly defined.
2. It is easy to calculate and understand.
3. It is not affected by extreme values like the arithmetic mean. For example, 5 persons have their incomes Rs 2000, Rs 2500, Rs 2600, Rs 3000, Rs 5000. The median would be Rs 2600 while the arithmetic mean would be Rs 3020.
4. It can be found by mere inspection.
5. It is fully representative and can be computed easily.
6. It can be used for qualitative studies.
7. Even if the extreme values are unknown, median can be calculated if one knows the number of items.
8. It can be obtained graphically.

MODE

It is the size of that item which possesses the maximum frequency. According to Professor Kenney and Keeping, the value of the variable which occurs most frequently in a distribution is called the mode.

It is the most common value. It is the point of maximum density.

Ungrouped Data

Individual series : The mode of this series can be obtained by mere inspection. The number which occurs most often is the mode.

Grouped Data**Steps :**

1. Determine the modal class which has the maximum frequency.
2. By interpolation the value of the mode can be calculated as -

$$\text{Mode} = l_1 \rightarrow \frac{f_1 - f_0}{2f_1 - f_0 - f_2} (l_2 - l_1)$$

where

l_1 = lower limit of the modal class

- l_2 = upper limit of the modal class
 f_1 = frequency of the modal class
 f_0 = frequency of the class preceding to the modal class
 f_2 = frequency of the class succeeding the modal class

Merits of Mode

1. It is simple to calculate.
2. In individual or discrete distribution it can be located by mere inspection.
3. It is easy to understand. Everyone is used to the idea of average size of a garment, an average American etc.
4. It is not isolated like the median as it is the most common item.
5. Like the Average mean, it is not a value which cannot be found in the series.
6. It is not necessary to know all the items. What we need the point of maximum density frequency.
7. It is not affected by sampling fluctuations.

Demerits

1. It is ill defined.
2. It is not based on all observations.
3. It is not capable of further algebraic treatment.
4. It is not a good representative of the data.
5. Sometimes there are more than one values of mode.

Empirical Relation between Mean, Median and Mode

A distribution in which the values of mean, median and mode coincide (*i.e.* mean = median = mode) is known as a symmetrical distribution. Conversely, when values of mean, median and mode are not equal the distribution is known as asymmetrical or skewed distribution. In moderately skewed or asymmetrical distribution a very important relationship exists among these three measures of central tendency. In such distributions the distance between the mean and median is about one-third of the distance between the mean and mode, as will be clear from the diagrams 1 and 2. Karl Pearson expressed this relationship as:

$$\text{Mode} = \text{mean} - 3 [\text{mean} - \text{median}]$$

$$\text{Mode} = 3 \text{ median} - 2 \text{ mean}$$

$$\text{and Median} = \text{mode} + \frac{2}{3} [\text{mean} - \text{mode}]$$

Knowing any two values, the third can be computed.

HARMONIC MEAN

Harmonic Mean is also one of the different kinds of average and it is applied for the situations when you have a desired rates of average.

The reciprocal of the arithmetic mean of the reciprocal is called as Harmonic Mean. It is calculated by dividing the number of observations by the sum of reciprocal of the observation.

The formula to find the harmonic mean is given by:

$$\text{Harmonic Mean (H)} = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \dots + \frac{1}{x_n}}$$

Where, n = Total number of numbers or terms.

$x_1, x_2, x_3, \dots, x_n$ = Individual terms or individual values.

GEOMETRIC MEAN

Geometric Mean is a measure of central tendency which is generally used to compare two set of values. It is evaluated by finding product of all the numbers and then taking their n th root. Geometric mean is different from arithmetic mean because it uses product of the observations rather than sum.

Formula for evaluating geometric mean is as follows, if we have n number of observations, then

$$\text{Geometric Mean} = \sqrt[n]{x_1 x_2 x_3 \dots x_n}$$

WEIGHTED MEAN

Weighted Mean is a measurement of central tendency. It represents the average of a given data. Weighted mean is similar to arithmetic mean or sample mean. Weighted mean is calculated when data is given in a different way than in arithmetic mean or sample mean.

Sometimes, there are frequencies assigned to each term in the given data, then we can no longer use simple arithmetic mean formula. Few values may occur more frequently than others, therefore corresponding frequencies are allotted to each value. These frequencies are often referred as weights. In this case, weighted mean formula is used.

Weighted mean formula is as follows:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

Where, $\bar{x}$ = weighted mean.

$x_i = x_1, x_2, x_3, \dots$ = Items given.

and $f_i = f_1, f_2, f_3, \dots$ = Frequencies corresponding to the given items

DISPERSION

Range

In any statistical series, the difference between the largest and the smallest values is called as the range.

$$\text{Thus Range (R)} = L - S$$

L = Largest value of the series
S = Smallest value of the series

Coefficient of Range

The relative measure of the range. It is used in the comparative study of the dispersion

$$\text{co-efficient of Range} = \frac{L - S}{L + S}$$

Mean Deviation

Average deviations (mean deviation) is the average amount of variations (scatter) of the items in a distribution from either the mean or the median or the mode, ignoring the signs of these deviations by Clark and Senkade.

Individual Series

Steps :

- (1) Find the mean or median or mode of the given series.
- (2) Using any one of the three, find the deviations (differences) of the items of the series from them.
i.e. $x_i - \bar{x}$, $x_i - \text{Me}$ and $x_i - \text{Mo}$.
Me = Median and Mo = Mode.
- (3) Find the absolute values of these deviations i.e. ignore their positive (+) and negative (-) signs.
i.e. $|x_i - \bar{x}|$, $|x_i - \text{Me}|$ and $|x_i - \text{Mo}|$.
- (4) Find the sum of these absolute deviations.
i.e. $|x_i - \bar{x}| + |x_i - \text{Me}|$ and $|x_i - \text{Mo}|$.
- (5) Find the mean deviation using the following formula.
M.D. (x) = $\frac{\sum |x_i - \bar{x}|}{n}$, (Me) = $\frac{\sum |x_i - \text{Me}|}{n}$
or (Mo) = $\frac{\sum |x_i - \text{Mo}|}{n}$

Note that :

- (i) generally M. D. obtained from the median is the best for the practical purpose.
- (ii) co-efficient of M. D. = $\frac{\bar{x}}{x} \left\{ \frac{(\text{Me})}{\text{Me}} \right\} \left\{ \frac{(\text{Mo})}{\text{Mo}} \right\}$

Variance

The term variance was used to describe the square of the standard deviation R.A. Fisher in 1913. The concept of variance is of great importance in advanced work where it is possible to split the total into several parts, each attributable to one of the factors causing variations in their original series. Variance is defined as follows:

$$\text{Variance} = \frac{\sum (x_i - \bar{x})^2}{n}$$

Standard Deviation (S.D.)

It is the square root of the arithmetic mean of the square deviations of various values from their arithmetic mean. It is denoted by s.d. or σ .

$$\begin{aligned} \text{Thus, s.d. (x)} &= \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} \text{ for the ungrouped data} \\ &= \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{n}} \text{ for the grouped data} \end{aligned}$$

$$\text{where } n = \sum f_i$$

Merits

- (1) It is rigidly defined and based on all observations.
- (2) It is amenable to further algebraic treatment.
- (3) It is not affected by sampling fluctuations.
- (4) It is less erratic.

Demerits

- (1) It is difficult to understand and calculate.
- (2) It gives greater weight to extreme values.

Note that variance $V(x) = \frac{(x_i - \bar{x})^2}{n}$ and

and s. d. (x) = and $\sqrt{\frac{(x_i - \bar{x})^2}{n}}$ and $\sqrt{\frac{f_i(x_i - \bar{x})^2}{n}}$

Then $V(x) = \frac{s^2}{x}$

Co-efficient of Variation (C.V.)

To compare the variations (dispersion) of two different series, relative measures of standard deviation must be calculated. This is known as co-efficient of variation or the co-efficient of s.d. Its formula is

$$C.V. = \frac{s}{\bar{x}} \times 100$$

Thus it is defined as the ratio s. d. to its mean.

Remark: It is given as a percentage and is used to compare the consistency or variability of two more series. The higher the C.V., the higher the variability and lower the C.V., the higher is the consistency of the data.

Combined Standard Deviation. If two sets containing n_1 and n_2 items having means $\bar{x}_1$ and $\bar{x}_2$ and standard deviations s_1 and s_2 respectively are taken together then,

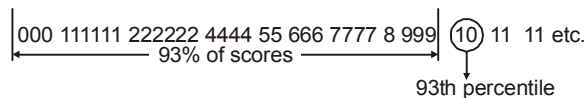
(1) Mean of the combined data is $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

(2) s.d. of the combined set is $s = \sqrt{\frac{n_1(s_1^2 + d_1^2) + n_2(s_2^2 + d_2^2)}{n_1 + n_2}}$

Where $d_1 = \bar{x}_1 - \bar{x}$ and $d_2 = \bar{x}_2 - \bar{x}$

Percentile

The n th percentile is that value (or size) such that $n\%$ of values of the whole data lies below it. For example, a score of 7% from the topmost score would be 93 the percentile as it is above 93% of the other scores.

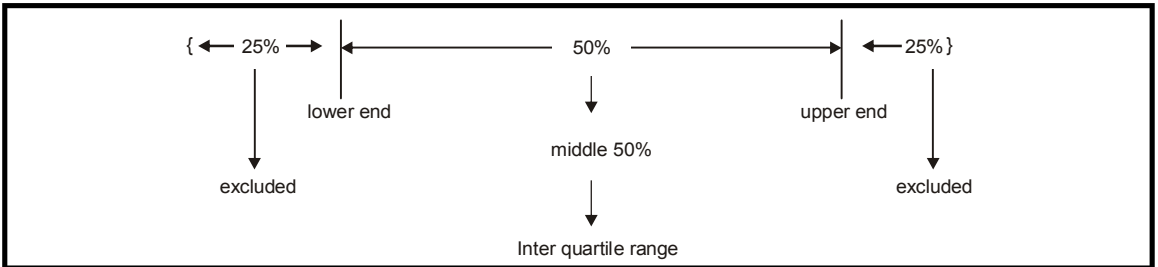
**Percentile Range**

It is used as one of the measures of dispersion. It is a set of data and is defined as $P_{90} - P_{10}$ where P_{90} and P_{10} are the 90th and 10th percentile respectively. The semi-percentile range,

i.e. $\frac{P_{90} - P_{10}}{2}$ can also be used but it is not common in use.

Quartiles and Interquartile Range

If we concentrate on two extreme values (as in the case of range), we don't get any idea about the scatter of the data within the range (*i.e.* the two extreme values). If we discard these two values the limited range thus available might be more informative. For this reason the concept of interquartile range is developed. It is the range which includes middle 50% of the distribution. Here $1/4$ (one quarter of the lower end and $1/4$ (one quarter) of the upper end of the observations are excluded.



Now the lower quartile (Q_1) is the 25th percentile and the upper quartile (Q_3) is the 75th percentile. It is interesting to note that the 50th percentile is the middle quartile (Q_2) which is in fact what you have studied under the title ' Median '. Thus symbolically

$$\text{Inter quartile range} = Q_3 - Q_1$$

If we divide ($Q_3 - Q_1$) by 2 we get what is known as Semi-Inter quartile range.

i.e. $\frac{Q_3 - Q_1}{2}$. It is known as Quartile deviation (Q. D or SI QR).

$$\text{Therefore Q. D. (SI QR)} = \frac{Q_3 - Q_1}{2}$$

Normal Probability Curve

The normal distribution developed by Gauss is a continuous distribution of maximum utility.

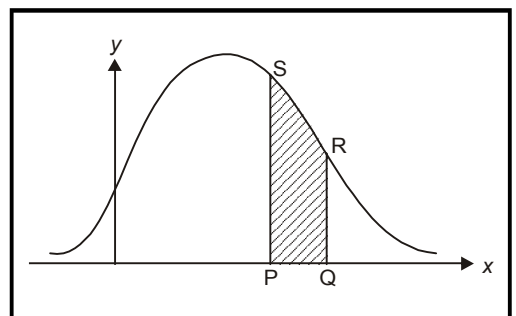
If we know a curve such that the area under the curve from $x = a$ to $x = b$ is equal to the probability that x will take a value between a and b and that the total area under the curve is unity, then the curve is called the probability curve.

If the curve is described by a relation $y = f(x)$ then $y = f(x)$ is called a probability density or simply probability function.

Among all the probability curves, the normal curve is the most important one. The corresponding function is called the normal probability function and the probability distribution is called the normal distribution. The normal distribution can be considered as the limiting form of the Binomial Distribution, however n , the number of trials, is very large and neither P nor q is very small.

The normal distribution is given by

$$y = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$



where y = ordinate, x = abscissa of a point on the curve, μ = the mean of x , σ = S. D. of x ,
 $x = a$ a constant = 3.1416 and e = a constant = 2.7183.

The Normal Curve

The shape of a normal curve is like a bell. It is symmetrical about the maximum ordinate. If P and Q are two points on the x -axis (see figure), the shaded area PQRS, bounded by the portion of the curve RS, the ordinates at P and Q and the x -axis is equal to the probability that the variate x lies between $x = a$ and $x = b$ at P and Q respectively. We have already seen that the total area under a normal curve is unity. Any probability distribution, defined this way is known as the normal distribution. The distribution can be completely known if we know the values of μ and σ . Therefore μ and σ are known as the parameters of the distribution. The normal distribution with mean μ and standard deviation σ is denoted by $N(\mu, \sigma)$.

Properties of the Normal Distribution (Normal Curve)

1. The normal curve is bell-shaped and symmetrical about the maximum ordinate at $x = \mu$, the mean. This ordinate divides the curve into two equal parts. The part on one side is the mirror image of the other side. It has the maximum height at $x = \mu$. Thus the mode of the distribution is also μ . The ordinate $x = \mu$ divides the whole area under the curve into two equal parts. Hence the median is also $x = \mu$. Thus for the normal distribution, the mean, mode and median coincide i.e. mean = median = mode = μ .
2. We know that the area under the normal curve is equivalent to the probability of randomly drawing a value in the given range. The area is the greatest in the middle, where the "hump" (where mean, mode and median coincide) and then thin out towards out on the either sides of the curve, i.e. tails, but never becomes zero. In other words, the curve never intersects x -axis at any finite point. i.e. x -axis is its Asymptote.
3. Since the curve is symmetrical about mean. The first quartile Q_1 and the third quartile Q_3 lie at the same distance on the two sides of the mean μ . The distance of any quartile from μ is 0.6745 units. Thus,

$$Q_1 = \mu - 0.6745\sigma \quad \text{or} \quad -\frac{2}{3}\sigma \quad \text{and}$$

$$Q_3 = \mu + 0.67566\sigma \quad \text{or} \quad +\frac{2}{3}\sigma$$

Hence, middle 50% observations lie between

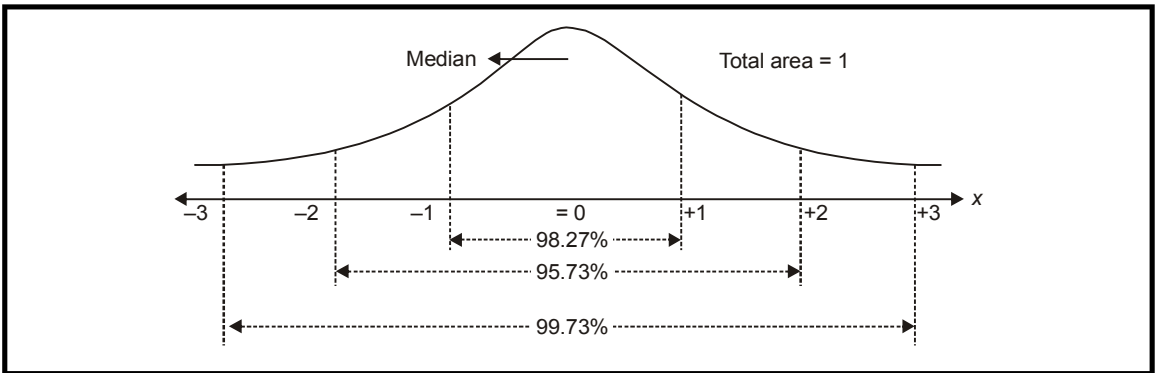
$$-\frac{2}{3}\sigma \quad \text{and} \quad +\frac{2}{3}\sigma$$

4. Since the normal curve is symmetrical its skewness is zero and kurtosis is 3. The curve is meso kurtic.
5. The mean deviation is approximately $\sigma/\sqrt{\pi}$.
6. As discussed earlier, the probability for the variable to lie in any interval (a, b) in the range of variable is given by the area under the normal curve, the two ordinates $x = a$ and $x = b$, and the x -axis

The area under the normal curve is distributed as follows :

1. The area between $x = -$ and $x = +$ is 18.27%
2. The area between $x = -2$ and $x = +2$ is 95.45%
3. The area between $x = -3$ and $x = +3$ is 99.73%

These areas are shown in the following figure.



i.e. $P(-1 < x < +1) = 68.27\%$

$P(-2 < x < +2) = 95.73\%$

$P(-3 < x < +3) = 99.73\%$

7. ± 1 are points of inflexions

Parametric [*t*-test]

Paired Sample *t*-Test

The paired sample *t*-test, sometimes called the dependent sample *t*-test, is a statistical procedure used to determine whether the mean difference between two sets of observations is zero. In a paired sample *t*-test, each subject or entity is measured twice, resulting in pairs of observations. Common applications of the paired sample *t*-test include case-control studies or repeated-measures designs. Suppose you are interested in evaluating the effectiveness of a company training program. One approach you might consider would be to measure the performance of a sample of employees before and after completing the program, and analyze the differences using a paired sample *t*-test.

Hypotheses

Like many statistical procedures, the paired sample *t*-test has two competing hypotheses, the null hypothesis and the alternative hypothesis. The null hypothesis assumes that the true mean difference between the paired samples is zero. Under this model, all observable differences are explained by random variation. Conversely, the alternative hypothesis assumes that the true mean difference between the paired samples is not equal to zero. The alternative hypothesis can take one of several forms depending on the expected outcome. If the direction of the difference does not matter, a two-tailed hypothesis is used. Otherwise, an upper-tailed or lower-tailed hypothesis can be used to increase the power of the test. The null hypothesis remains the same for each type of alternative hypothesis. The paired sample *t*-test hypotheses are formally defined below:

The null hypothesis (H_0) assumes that the true mean difference (μ_d) is equal to zero.

The two-tailed alternative hypothesis (H_1) assumes that μ_d is not equal to zero.

The upper-tailed alternative hypothesis (H_1) assumes that μ_d is greater than zero.

The lower-tailed alternative hypothesis (H_1) assumes that μ_d is less than zero.

The mathematical representations of the null and alternative hypotheses are defined below:

$$H_0: \mu_d = 0$$

$$H_1: \mu_d \neq 0 \quad (\text{two-tailed})$$

$$H_1: \mu_d > 0 \quad (\text{upper-tailed})$$

$$H_1: \mu_d < 0 \quad (\text{lower-tailed})$$

Note. It is important to remember that hypotheses are never about data, they are about the processes which produce the data. In the formulas above, the value of μ_d is unknown. The goal of hypothesis testing is to determine the hypothesis (null or alternative) with which the data are more consistent.

Assumptions

As a parametric procedure (a procedure which estimates unknown parameters), the paired sample t -test makes several assumptions. Although t -tests are quite robust, it is good practice to evaluate the degree of deviation from these assumptions in order to assess the quality of the results. In a paired sample t -test, the observations are defined as the differences between two sets of values, and each assumption refers to these differences, not the original data values. The paired sample t -test has four main assumptions:

The dependent variable must be continuous (interval/ratio).

The observations are independent of one another.

The dependent variable should be approximately normally distributed.

The dependent variable should not contain any outliers.

Level of Measurement

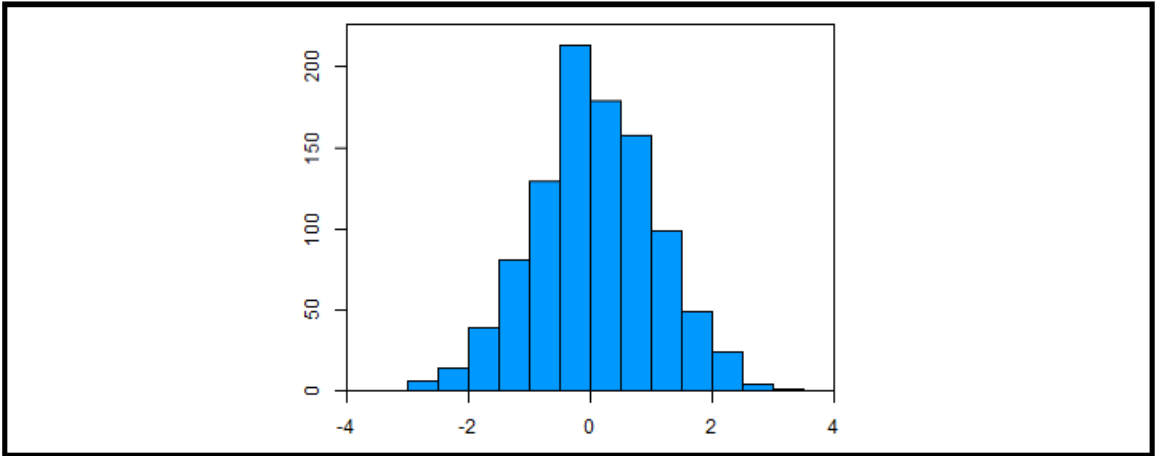
The paired sample t -test requires the sample data to be numeric and continuous, as it is based on the normal distribution. Continuous data can take on any value within a range (income, height, weight, etc.). The opposite of continuous data is discrete data, which can only take on a few values (Low, Medium, High, etc.). Occasionally, discrete data can be used to approximate a continuous scale, such as with Likert-type scales.

Independence

Independence of observations is usually not testable, but can be reasonably assumed if the data collection process was random without replacement. In our example, it is reasonable to assume that the participating employees are independent of one another.

Normality

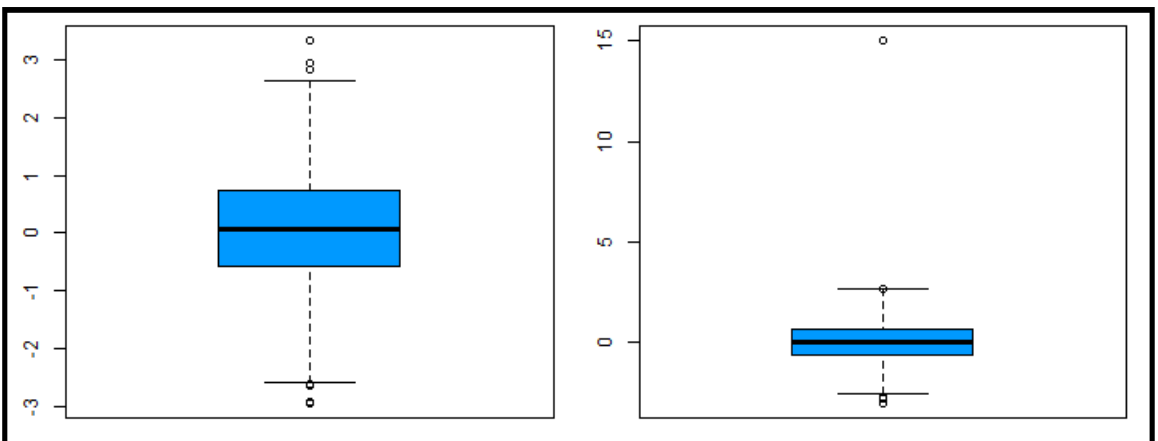
To test the assumption of normality, a variety of methods are available, but the simplest is to inspect the data visually using a tool like a histogram (Figure). Real-world data are almost never perfectly normal, so this assumption can be considered reasonably met if the shape looks approximately symmetric and bell-shaped. The data in the example figure below is approximately normally distributed.



HISTOGRAM OF AN APPROXIMATELY NORMALLY DISTRIBUTED VARIABLE

Outliers

Outliers are rare values that appear far away from the majority of the data. Outliers can bias the results and potentially lead to incorrect conclusions if not handled properly. One method for dealing with outliers is to simply remove them. However, removing data points can introduce other types of bias into the results, and potentially result in losing critical information. If outliers seem to have a lot of influence on the results, a nonparametric test such as the Wilcoxon Signed Rank Test may be appropriate to use instead. Outliers can be identified visually using a boxplot.



BOXPLOTS OF A VARIABLE WITHOUT OUTLIERS (LEFT) AND WITH AN OUTLIER (RIGHT).

Procedure

The procedure for a paired sample t -test can be summed up in four steps. The symbols to be used are defined below:

D = Differences between two paired samples

d_i = The i th observation in D

n = The sample size

$\bar{d}$ = The sample mean of the differences

s_d = The sample standard deviation of the differences

T = The critical value of a t -distribution with $(n - 1)$ degrees of freedom

t = The t -statistic (t -test statistic) for a paired sample t -test

p = The p -value (probability value) for the t -statistic.

The four steps are listed below:

1. Calculate the sample mean.

Undefined control sequence \cfrac

2. Calculate the sample standard deviation.

Undefined control sequence \cfrac

3. Calculate the test statistic.

Undefined control sequence \cfrac

4. Calculate the probability of observing the test statistic under the null hypothesis. This value is obtained by comparing t to a t -distribution with $(n - 1)$ degrees of freedom. This can be done by looking up the value in a table, such as those found in many statistical textbooks, or with statistical software for more accurate results.

$p = 2 \cdot \Pr(T > |t|)$ (two-tailed)

$p = \Pr(T > t)$ (upper-tailed)

$p = \Pr(T < t)$ (lower-tailed)

determine whether the results provide sufficient evidence to reject the null hypothesis in favor of the alternative hypothesis.

Interpretation

There are two types of significance to consider when interpreting the results of a paired sample t -test, statistical significance and practical significance.

Statistical Significance

Statistical significance is determined by looking at the p -value. The p -value gives the probability of observing the test results under the null hypothesis. The lower the p -value, the lower the probability of obtaining a result like the one that was observed if the null hypothesis was true. Thus, a low p -value indicates decreased support for the null hypothesis. However, the possibility that the null hypothesis is true and that we simply obtained a very rare result can never be ruled out completely. The cutoff value for determining statistical significance is ultimately decided on by the researcher, but usually a value of .05 or less is chosen. This corresponds to a 5% (or less) chance of obtaining a result like the one that was observed if the null hypothesis was true.

Practical Significance

Practical significance depends on the subject matter. It is not uncommon, especially with large sample sizes, to observe a result that is statistically significant but not practically significant. In most cases, both types of significance are required in order to draw meaningful conclusions.

Nonparametric Tests

Nonparametric, or *distribution free* tests are so-called because the assumptions underlying their use are “fewer and weaker than those associated with parametric tests”. To put it another way, nonparametric tests require few if any assumptions about the shapes of the underlying population distributions. For this reason, they are often used in place of parametric tests if/when one feels that the assumptions of the parametric test have been too grossly violated (e.g., if the distributions are too severely skewed). Discussion of some of the more common nonparametric tests follows.

The Sign test (for 2 repeated/correlated measures)

The sign test is one of the simplest nonparametric tests. It is for use with 2 repeated (or correlated) measures (see the example below), and measurement is assumed to be at least ordinal. For each subject, subtract the 2nd score from the 1st, and write down the *sign* of the difference. (That is write “-” if the difference score is negative, and “+” if it is positive.) The usual null hypothesis for this test is that there is no difference between the two treatments. If this is so, then the number of + signs (or - signs, for that matter) should have a **binomial distributional** with $p = .5$, and $N =$ the number of subjects. In other words, the sign test is just a binomial test with + and - in place of Head and Tail (or Success and Failure).

To carry out the sign test, we could let our statistic be the number of + signs, which is 3 in this case. The researcher did not predict a particular outcome in this case, but wanted to know if the two conditions differed. Therefore, the alternative hypothesis is **nondirectional**. That is, the alternative hypothesis would be supported by an extreme number of + signs, be it small or large. A middling number of + signs would be consistent with the null.

NUMBER OF BAR-PRESSES IN BRAIN STIMULATION EXPERIMENT

Subject	Bar 1	Bar 2	Difference	“Sign” of Difference
1	20	40	-20	-
2	18	25	-7	-
3	24	38	-14	-
4	14	27	-13	-
5	5	31	-26	-
6	26	21	+5	+
7	15	32	-17	-
8	29	38	-9	-
9	15	25	-10	-
10	9	18	-9	-
11	25	32	-7	-
12	31	28	+3	+
13	35	33	+2	+
14	12	29	-17	-

The sampling distribution of the statistic is the binomial distribution with $N = 14$ and $p = .5$. With this distribution, we would find that the probability of **3 or fewer** + signs is .0287. But because the alternative is nondirectional, or **two-tailed**, we must also take

into account the probability **11 or more** + signs, which is also .0287. Adding these together, we find that the probability of (3 or fewer) or (11 or more) is .0574. Therefore, if our pre-determined alpha was set at .05, we would not have sufficient evidence to allow rejection of the null hypothesis.

Tied Scores

If a subject has the same score in each condition, there will be no sign, because the difference score is zero. In the case of tied scores, some textbook authors recommend dropping those subjects, and reducing N by the appropriate number. This is not the best way to deal with ties, however, because reduction of N can result in the loss of too much power.

A better approach would be as follows: If there is only one subject with tied scores, drop that subject, and reduce N by one. If there are 2 subjects with tied scores, make one a + and one a -. In general, if there is an even number of subjects with tied scores, make half of them + signs, and half - signs. For an odd number of subjects (greater than 1), drop one randomly selected subject, and then proceed as for an even number.

Wilcoxon Signed-Ranks Test (for 2 repeated/correlated measures)

One obvious problem with the sign test is that it **discards** a lot of information about the data. It takes into account the direction of the difference, but not the *magnitude* of the difference between each pair of scores. The Wilcoxon signed-ranks test is another non-parametric test that can be used for 2 repeated (or correlated) measures when measurement is at least ordinal. But unlike the sign test, it *does* take into account (to some degree, at least) the magnitude of the difference. Let us return to the data used to illustrate the sign test. The 14 difference scores were:

-20, -7, -14, -13, -26, +5, -17, -9, -10, -9, -7, +3, +2, -17

If we sort these on the basis of their absolute values, we get the results shown in below Table. The statistic T is found by calculating the sum of the positive ranks, and the sum of the negative ranks. T is the smaller of these two sums. In this case, therefore, $T = 6$.

DIFFERENCE SCORES RANKED BY ABSOLUTE VALUE

Score	Rank	
+2	1	
+3	2	
+5	3	
-7	4.5	Sum of positive ranks = 6
-7	4.5	
-9	6.5	Sum of negative ranks = 99
-9	6.5	
-10	8	T = 6
-13	9	
-14	10	
-17	11.5	
-17	11.5	
-20	13	
-26	14	

If the null hypothesis is true, the sum of the positive ranks and the sum of the negative ranks are expected to be roughly equal. But if H_0 is false, we expect one of the sums to be quite small – and therefore T is expected to be quite small. The most extreme outcome favourable to rejection of H_0 is $T = 0$.

If we wished to, we could generate the sampling distribution of T (i.e., the distribution of T assuming that the null hypothesis is true), and see if the observed value of T is in the rejection region. This is not necessary, however, because the sampling distribution of T can be found in tables in most introductory level statistics textbooks. When I consulted such a table, I found that for $N = 14$, and $\alpha = .05$ (2-tailed), the *critical value* of $T = 21$. The rule is that if T is equal to or less than T critical, we can reject the null hypothesis. Therefore, in this example, we would reject the null hypothesis. That is, we would conclude that monkeys prefer stimulation in brain area B to stimulation in area A.

This decision differs from our failure to reject H_0 when we analysed the same data using the sign test. The reason for this difference is that the Wilcoxon signed-ranks test is more *powerful* than the sign test. Why is it more powerful? Because it makes use of more information than does the sign test. But note that it too does discard some information by using *ranks* rather than the scores themselves.

Tied Scores

When ranking scores, it is customary to deal with tied ranks in the following manner:

Give the tied scores the *mean of the ranks they would have if they were not tied*. For example, if you have 2 tied scores that would occupy positions 3 and 4 if they were not tied, give each one a rank of 3.5. If you have 3 scores that would occupy positions 7, 8, and 9, give each one a rank of 8. This procedure is preferred to any other for dealing with tied scores, because the sum of the ranks for a fixed number of scores will be the same regardless of whether or not there are any tied scores.

If there are tied ranks in data you are analysing with the Wilcoxon signed-ranks test, the statistic needs to be adjusted to compensate for the decreased variability of the sampling distribution of T . Siegel and Castellan (1988, p. 94) describe this adjustment, for those who are interested. Note that if you are able to reject H_0 without making the correction, then do not bother, because the correction will increase your chances of rejecting H_0 . Note as well that the problem becomes more severe as the number of tied ranks increases.

Mann-Whitney U Test (for 2 independent samples)

The most basic independent groups design has two groups. These are often called Experimental and Control. Subjects are randomly selected from the population and randomly assigned to two groups. There is *no basis for pairing scores*. Nor is it necessary to have the same number of scores in the two groups.

The *Mann-Whitney U test* is a nonparametric test that can be used to analyse data from a two-group independent groups design when measurement is at least ordinal. It analyses the *degree of separation* (or the amount of overlap) between the Experimental and Control groups.

The *null hypothesis* assumes that the two sets of scores (E and C) are samples from the same population; and therefore, because sampling was random, the two sets of scores *do not differ systematically* from each other.

The *alternative hypothesis*, on the other hand, states that the two sets of scores do differ systematically. If the alternative is directional, or one-tailed, it further specifies the direction of the difference (i.e., Group E scores are systematically higher or lower than Group C scores).

The statistic that is calculated is either U or U' .

U_1 = the number of Es less than Cs

U_2 = the number of Cs less than Es

U = the smaller of the two values calculated above

U' = the larger of the two values calculated above.

Calculating U Directly

When the total number of scores is small, U can be calculated directly by counting the number of Es less than Cs (or Cs less than Es). Consider the following example:

DATA FOR TWO INDEPENDENT GROUPS

Group E:	12	17	9	21	
Group C:	8	18	26	15	23

It will be easier to count the number of Es less than Cs (and vice versa) if we rank the data from lowest to highest, and rewrite it as shown in below table.

ILLUSTRATION OF DIRECT CALCULATION OF THE U STATISTIC.

Score	Group	Rank	E < C	C < E	
8	C	1	0		
9	E	2		1	
12	E	3		1	
15	C	4	2		$U = 7$
17	E	5		2	$U' = 13$
18	C	6	3		
21	E	7		3	
23	C	8	4		
26	C	9	4		
			13	7	

CHECK

Note that $U + U' = n_1 n_2$. This will always be true, and can be used to check your calculations. In this case, $U + U' = 7 + 13 = 20$; and $n_1 n_2 = 4(5) = 20$.

Calculating U with Formulae

When the total number of scores is a bit larger, or if there are tied scores, it may be more convenient to calculate U with the following formulae:

$$U_1 \leq n_1 n_2 \sigma \frac{n_1(n_1 + 1)}{2} \rightarrow R_1 \quad \dots(1)$$

$$U_2 \leq n_1 n_2 \sigma \frac{n_2(n_2 + 1)}{2} \rightarrow R_2 \quad \dots(2)$$

where n_1 = # of scores in group 1
 n_2 = # of scores in group 2
 R_1 = sum of ranks for group 1
 R_2 = sum of ranks for group 2

As before, U = smaller of U_1 and U_2 , and U' = larger of U_1 and U_2 .

For the data shown above, $R_1 = 2 + 3 + 5 + 7 = 17$; and $R_2 = 1 + 4 + 6 + 8 + 9 = 28$. Substituting into the formulae, we get:

$$U_1 \leq 4(5) \sigma \frac{4(4 \sigma 1)}{2} \rightarrow 17 \leq 13 \quad \dots(3)$$

$$U_2 \leq 4(5) \sigma \frac{5(5 \sigma 1)}{2} \rightarrow 28 \leq 7 \quad \dots(4)$$

Therefore, $U = 7$ and $U' = 13$.

Making a Decision

The next step is deciding whether to reject H_0 or not. In principle, we could generate a probability distribution for U that is conditional on the null hypothesis being true – much like we did when working with the binomial distribution earlier. Fortunately, we do not have to do this, because there are tables in the back of many statistics textbooks that give you the *critical values* of U (or U') for different values of n_1 and n_2 , and for various significance levels.

For the case we've been considering, $n_1 = 4$ and $n_2 = 5$. For a two-tailed test with $\alpha = .05$, the critical value of $U = 1$. In order to reject H_0 , the observed value of U would have to be *equal to or less than* the critical value of U . (Note that **maximum separation** of E and C scores is indicated by **$U = 0$** . As the E and C scores become more mixed, U becomes larger. Therefore, *small* values of U lead to rejection of H_0). Therefore, we would decide that we cannot reject H_0 in this case.

Tied Scores

According to Siegel and Castellan (1988), any ties that involve observations in the same group do not affect the values of U and U' . (Note that Siegel and Castellan refer to this test as the *Wilcoxon-Mann-Whitney Test*, and that they call the statistic W rather than U .) But if two or more tied ranks involve observations from both groups, then the values of U and U' are affected, and a correction should be applied. See Siegel & Castellan should you ever need more information on this, and note that the problem is particularly severe if you are dealing with the large-sample version of the test, which we have not yet discussed.

Kruskal-Wallis H-test (for k Independent Samples)

The Kruskal-Wallis H-test goes by various names, including *Kruskal-Wallis one-way analysis of variance by ranks*. It is for use with k independent groups, where k is equal to or greater than 3, and measurement is at least ordinal. (When $k = 2$, you would use the Mann-Whitney U-test instead.) Note that because the samples are independent, they can be of different sizes.

The null hypothesis is that the k samples come from the same population, or from populations with identical medians. The alternative hypothesis states that not all population medians are equal. It is assumed that the underlying distributions are continuous; but only ordinal measurement is required.

The statistic H (sometimes also called KW) can be calculated in one of two ways:

$$H \leq \frac{12}{N(N+1)} \sum_{i=1}^k n_i (\bar{R}_i - \bar{R})^2 \quad \dots(5)$$

or, the more common computational formula,

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1) \quad \dots(6)$$

where k = the number of independent samples

n_i = the number of cases in the i th sample

N = the total number of cases

R_i = the sum of the ranks in the i th sample

$\bar{R}_i$ = the mean of the ranks for the i th sample

$\bar{R} = \frac{N+1}{2}$ = the mean of all ranks

Example

A student was interested in comparing the effects of four kinds of reinforcement on children's performance on a test of reading comprehension. The four reinforcements used were: (a) praise for correct responses; (b) a jelly bean for each correct response; (c) reproof for incorrect responses; and (d) silence. Four independent groups of children were tested, and each group received only one kind of reinforcement. The measure of performance given below is the number of errors made during the course of testing.

DATA FROM 4 INDEPENDENT GROUPS

a	b	c	d
68	78	94	54
63	69	82	51
58	58	73	32
51	57	67	74
41	53	66	65
		61	80

The first step in carrying out the Kruskal-Wallis H-test is to rank order all of the scores from lowest to highest. This can be quite laborious work if you try to do it by hand, but is fairly easy if you use a spreadsheet program. Enter all scores in a single column, and enter a group code for each score one column over.

For example:

Group	Score
a	68
a	63
a	58
a	51
a	41
b	78
etc.	

When all the data are entered thus, sort the scores (and their codes) from lowest to highest. Now you can enter ranks from 1 to N (taking care to deal with tied scores appro-

priately). After the scores are ranked, you can sort the data the data by group code, and then calculate the sum and mean of the ranks for each group. I did this for the data shown above, and came up with the following ranks:

RANKS OF DATA				
a	b	c	d	
15	19	22	6	
11	16	21	3.5	
8.5	8.5	17	1	
3.5	7	14	18	
2	5	13	12	
		10	20	
40.0	55.5	97.0	60.5	Sum of Ranks
8.0	11.1	16.2	10.1	Mean of Ranks

CHECK

The sum of ranks from 1 to N will always be equal to $[N(N + 1)]/2$. We can use this to check our work to this point. We have 22 scores in total, so the sum of all ranks should be $[22(23)]/2 = 253$. Similarly, when we add the sum of ranks for each group, we get $40 + 55.5 + 97 + 60.5 = 253$. Therefore, the mean of all ranks = $253/22 = 11.5$.

Now plugging into Equation 3 shown above, we get $H = 4.856$. If the null hypothesis is true, and the k samples are drawn from the same population (or populations with identical medians), and if $k > 3$, and all samples have 5 or more scores, then the distribution of H closely approximates the chi-squared distribution with $df = k - 1$. The critical value of chi-squared with $df = 3$ and $\alpha = .05$ is 7.82. In order to reject H_0 , the obtained value of H would have to be equal to or greater than 7.82. Because it is less, we cannot reject the null hypothesis.

Sampling Distribution of H

As described above, when H_0 is true, if $k > 3$, and all samples have 5 or more scores, then the sampling distribution of H is closely approximated by the chi-squared distribution with $df = k - 1$. If $k = 3$ and the number of scores in each sample is 5 or fewer, then the chi-squared distribution should not be used. In this case, one should use a table of critical values of H .

Tied Observations

Tied scores are dealt with in the manner described previously (*i.e.*, they are given the mean of the ranks they would receive if they were not tied). The presence of tied scores does affect the variance of the sampling distribution of H . Siegel and Castellan show a correction that can be applied in the case of tied scores, but go on to observe that its effect is to increase the value of H . Therefore, if you are able to reject H_0 without correcting for ties, there is no need to do the correction. It should only be contemplated when you have failed to reject H_0 .

Multiple Comparisons

Rejection of H_0 tells you that at least one of the k samples is drawn from a population with a median different from the others. But it does not tell you which one, or how many are different. There are procedures for conducting multiple comparisons between treatments, or comparisons of a control condition to all other conditions in order to answer these kinds of questions. Should you ever need to use one of them, consult Siegel and Castellan.

The Jonckheere Test for Ordered Alternatives

The Jonckheere test for ordered alternatives is similar to the Kruskal-Wallis test, but has a more specific alternative hypothesis. The alternative hypothesis for the Kruskal-Wallis test states that all population medians are *not equal*. The more precise alternative hypothesis for the Jonckheere test can be summarised as follows:

$$H_1: \mu_1 < \mu_2 < \dots < \mu_k$$

where the μ 's are the population medians. This alternative is tested against a *null hypothesis of no systematic trend across treatments*.

The test can be applied when you have data for k independent samples, when measurement is at least ordinal, and when it is possible to specify *a priori* the ordering of the groups. Because the alternative hypothesis specifies the order of the medians, the test is one-tailed.

Siegel and Castellan use J to symbolise the statistic that is calculated. It is sometimes also called the "Mann-Whitney count". As this name implies, J is based on the same kind of counting and summing that we saw when calculating the U statistic via the direct method. The mechanics of it become somewhat complicated for the Jonckheere test, so we will not go into it here.

Friedman ANOVA

This test is sometimes called the *Friedman two-way analysis of variance by ranks*. It is for use with k repeated (or correlated) measures where measurement is at least ordinal. The null hypothesis states that all k samples are drawn from the same population, or from populations with equal medians.

Example

The table on the left (below) shows reaction time data from 5 subjects, each of whom was tested in 3 conditions (A, B, and C). The Friedman ANOVA uses ranks, and so the first thing we must do is rank order the k scores for each subject. The results of this ranking are shown in the table on the right, and the sum of the ranks (R_j) for each treatment is shown at the bottom.

It may be useful at this point to consider what kinds of outcomes are expected if H_0 is true. H_0 states that all of the samples (columns) are drawn from the same population, or from populations with the same median. If so, then the sums (or means) of the ranks for each of the columns should all be roughly equal, because the ranks 1, 2, and 3 would be expected by chance to appear equally often in each column. In this example, the expected R for each treatment would be 10 if H_0 is true. (In general, the expected sum of ranks for each treatment is $N(k+1)/2$.) The Friedman ANOVA assesses the degree to which the observed R 's depart from the expected R 's. If the departure is too extreme (or not likely due to chance), one concludes by rejecting H_0 .

RT DATA AND RANKS FOR 3 LEVELS OF A WITHIN-SUBJECTS VARIABLE

Subj	A	B	C	Subj	A	B	C
1	386	411	545	1	1	2	3
2	545	563	556	2	1	3	2
3	662	667	665	3	1	3	2
4	453	502	574	4	1	2	3
5	548	546	575	5	2	1	3
				R_i	6	11	13

The F_r statistic is calculated as follows:

$$F_r \sim \frac{12}{Nk(k+1)} \sum_{i=1}^k R_i^2 - 3N(k+1) \quad \dots(7)$$

where N = the number of subjects

k = the number of treatments

R_i = the sum of the ranks for the i th treatment

Critical values of F_r for various sample sizes and numbers of treatments can be found in tables. Note that when the number of treatments or subjects is large, the sampling distribution of F_r is closely approximated by the chi-squared distribution with $df = k - 1$. (Generally, use a table of critical values for F_r if it provides a value for your particular combination of k and N . If either k or N are too large for the table of critical values, then use the chi-squared distribution with $df = k - 1$.)

For the example we've been looking at, the critical values of F_r are 6.40 for $\alpha = .05$, and 8.40 for $\alpha = .01$. In order to reject H_0 , the obtained value of F_r must be equal to or greater than the critical value. Therefore, we would fail to reject H_0 in this case.

Tied Scores

If there are ties among the ranks, the F_r statistic must be corrected, because the sampling distribution changes. The formula that corrects for tied ranks is actually a general formula that also works when there are no ties. However, it is rather complicated, which is why the simplified version shown above is used when possible. The general formula is not shown here, but can be found in Siegel and Castellan (1988, p. 179), should you need it.

Multiple Comparisons

Siegel and Castellan (1988) also give formulae to be used for conducting multiple comparisons and/or comparisons of a control condition to each of the other conditions.

Large Sample Versions of Non-parametric Tests

You may have noticed that the tables of critical values for many nonparametric statistics only go up to sample sizes of about 25-50. If so, perhaps you have wondered what to do when you have sample sizes larger than that, and want to carry out a nonparametric test. Fortunately, it turns out that the sampling distributions of many nonparametric statistics converge on the normal distribution as sample size increases. Because of that, it is possible to carry out a so-called "large-sample" version of the test (which is really a z-test)

if you know the mean and variance of the sampling distribution for that particular statistic.

Common Structure of all *z*- and *t*-tests

As I have mentioned before, all *z*- and *t*-tests have a common structure. In general terms:

$$z \text{ (or } t) \mid \frac{\text{statistic} - (\text{parameter} \mid H_0 \text{ is true})}{\text{standard error of the statistic}} \quad \dots(8)$$

When the sampling distribution of the statistic in the numerator is normal, then if the true (population) standard error (SE) of the statistic is known, the computed ratio can be evaluated against the standard normal (*z*) distribution. If the true standard error of the statistic is not known, then it must be estimated from the sample data, and the proper sampling distribution is a *t*-distribution with some number of degrees of freedom.

Example: Large-sample Mann-Whitney *U* test

The following facts are known about the sampling distribution of the *U* statistic used in the Mann-Whitney *U* test:

$$U \mid \frac{n_1 n_2}{2} \quad \dots(9)$$

$$U \mid \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}} \quad \dots(10)$$

Furthermore, when both sample sizes are greater than about 20, the sampling distribution of *U* is (for practical purposes) normal. Therefore, under these conditions, one can perform a *z*-test as follows:

$$z_U \mid \frac{U - \mu_U}{\sigma_U} \quad \dots(11)$$

The obtained value of z_U can be evaluated against a table of the standard normal distribution. Alternatively, one can use software to calculate the *p*-value for a given *z*-score.

Example: Large-sample Wilcoxon signed ranks test

The following are known to be true about the sampling distribution of *T*, the statistic used in the Wilcoxon signed ranks test:

$$T \mid \frac{N(N+1)}{4} \quad \dots(12)$$

$$T \mid \sqrt{\frac{N(N+1)(2N+1)}{24}} \quad \dots(13)$$

If $N > 50$, then the sampling distribution of *T* is for practical purposes normal. And so, a *z*-ratio can be computed as follows:

$$z_T \mid \frac{T - \mu_T}{\sigma_T} \quad \dots(14)$$

The obtained value of z_T can be evaluated against a table of the standard normal distribution, or using software as described above.

Example: Large-sample Jonckheere test for ordered alternatives

The mean and standard deviation of the sampling distribution of J are given by the following:

$$J \rightarrow \frac{N^2 \sum_{i=1}^k n_i^2}{4} \quad \dots(15)$$

$$J \rightarrow \sqrt{\frac{1}{72} N^2 \sum_{i=1}^k n_i^2 (2n_i - 3)} \quad \dots(16)$$

where N = total number of observations

n_i = the number of observations in the i th group

k = the number of independent groups

As sample sizes increase, the sampling distribution of J converges on the normal, and so one can perform a z -test as follows:

$$z_J \rightarrow \frac{J - \mu_J}{\sigma_J} \quad \dots(17)$$

Example: Large Sample Sign Test

The sampling distribution used in carrying out the sign test is a binomial distribution with $p = q = .5$. The mean of a binomial distribution is equal to Np , and the variance is equal to Npq . As N increases, the binomial distribution converges on the normal distribution (especially when $p = q = .5$). When N is large enough (i.e., greater than 30 or 50, depending on how conservative one is), it is possible to carry out a z -test version of the sign test as follows:

$$z^2 \rightarrow \frac{(X - Np)^2}{Npq} \quad \dots(18)$$

You may recall that z^2 is equal to χ^2 with $df = 1$.

Therefore,

$$\chi^2_1 \rightarrow \frac{(X - Np)^2}{Npq} \quad \dots(19)$$

This formula can be expanded with what Howell calls "some not-so-obvious algebra" to yield:

$$\chi^2_1 \rightarrow \frac{(X - Np)^2}{Np} + \frac{(N - X - Nq)^2}{Nq} \quad \dots(20)$$

Note that X equals the **observed** number of p -events, and Np equals the **expected** number of p -events under the null hypothesis. Similarly, $N - X$ equals the **observed** number of q -events, and Nq = the **expected** number of q -events under the null hypothesis. Therefore, we can rewrite equation (20) in a more familiar looking format as follows:

$$\chi^2_1 \rightarrow \frac{(O_1 - E_1)^2}{E_1} + \frac{(O_2 - E_2)^2}{E_2} \quad \dots(21)$$

Advantages of Nonparametric Tests

Siegel and Castellan (1988, p. 35) list the following advantages of nonparametric tests:

1. If the sample size is very small, there may be no alternative to using a nonparametric statistical test unless the nature of the population distribution is *known exactly*.
2. Nonparametric tests typically make fewer assumptions about the data and may be more relevant to a particular situation. In addition, the hypothesis tested by the nonparametric test may be more appropriate for the research investigation.
3. Nonparametric tests are available to analyze data which are inherently in ranks as well as data whose seemingly numerical scores have the strength of ranks. That is, the researcher may only be able to say of his or her subjects that one has more or less of the characteristic than another, without being able to say *how much* more or less. For example, in studying such a variable as anxiety, we may be able to state that subject A is more anxious than subject B without knowing at all exactly how much more anxious A is. If data are inherently in ranks, or even if they can be categorized only as plus or minus (more or less, better or worse), they can be treated by nonparametric methods, whereas they cannot be treated by parametric methods unless precarious and, perhaps, unrealistic assumptions are made about the underlying distributions.
4. Nonparametric methods are available to treat data which are simply classificatory or categorical, *i.e.*, are measured in a nominal scale. No parametric technique applies to such data.
5. There are suitable nonparametric statistical tests for treating samples made up of observations from several different populations. Parametric tests often cannot handle such data without requiring us to make seemingly unrealistic assumptions or requiring cumbersome computations.
6. Nonparametric statistical tests are typically much easier to learn and to apply than are parametric tests. In addition, their interpretation often is more direct than the interpretation of parametric tests.

Note that the objection concerning “cumbersome computations” in point number 5 has become less of an issue as computers and statistical software packages become more sophisticated, and more available.

Disadvantages of Nonparametric Tests

In closing, I must point out that nonparametric tests do have at least two major disadvantages in comparison to parametric tests. First, **nonparametric tests are less powerful**. Why? Because parametric tests use more of the information available in a set of numbers. Parametric tests make use of information consistent with interval scale measurement, whereas parametric tests typically make use of ordinal information only. As Siegel and Castellan put it, “nonparametric statistical tests are wasteful.”

Second, parametric tests are much more flexible, and allow you to test a greater range of hypotheses. For example, factorial ANOVA designs allow you to test for interactions between variables in a way that is not possible with nonparametric alternatives. There are nonparametric techniques to test for *certain kinds of interactions under certain circumstances*, but these are much more limited than the corresponding parametric techniques.

Therefore, when the assumptions for a parametric test are met, it is generally (but not necessarily always) preferable to use the parametric test rather than a nonparametric test.

CORRELATIONAL ANALYSIS

It is a statistical method to determine the relationship between two or more variables. Its values can lie between +1 and -1. The direction of the relation is determined by sign. If increase in one variable leads to increase in dependent variable, then relation is positive. On the other hand, if increase in one variable leads to decrease in dependent variable, then value will be negative. In case no relation exists between two variables, correlation will be zero.

Uses of Correlation

1. Decision maker will come to know the nature of relationship between variables and the degree of the relation.
2. Getting a quantitative figure for correlation makes the decision making process objective.
3. It helps in forecasting and planning because changes in variables and its impact can be estimated beforehand.
4. It helps the researcher in identifying such factors which can stabilize the economy.

Precautions in Applying Correlation

1. It is difficult to distinguish between dependent variables and independent variables.
2. Both the correlated variables are affected by a third variable, which has not been taken into consideration by the researcher.
3. The correlation may be due to chance.
4. Very high degree of correlation between two variables does not necessarily indicate a cause and effect relationship between them.

Types of Correlation

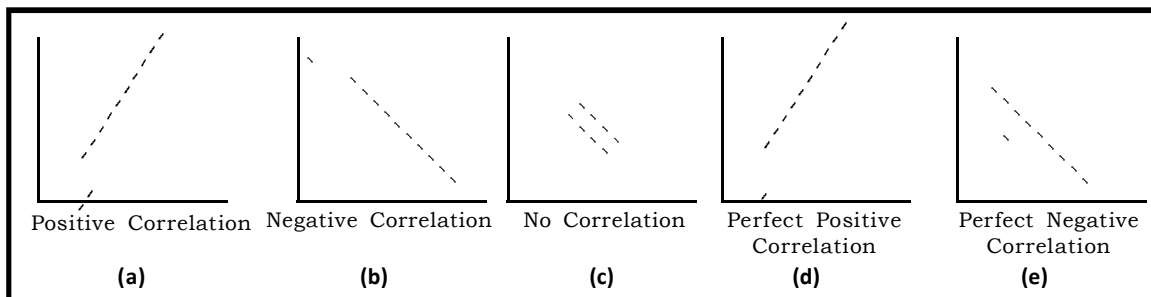
1. **Single and Multiple Correlation.** Only two variables are considered in single correlation, *i.e.*, one independent and another dependent variable. In case of multiple correlation, the relation between more than two variables is judged.
2. **Partial and Total Correlation.** In the case of partial correlation, relation of two or more variables are considered assuming other variables to be constant. Total correlation is based on all the variables without assuming any variable to be constant.
3. **Linear and Non-linear Correlation.** When variation in the values of two variables have a constant ratio, there will be linear correlation between them. In non-linear correlation, the amount of change in one variable does not bear a constant ratio to the amount of change in the other related variable.

Methods of Determining Correlation

(a) Scatter diagram

Different values are plotted in graphs. Dependent variables are represented on y-axis and independent variables are represented on x-axis. The diagram of dots, so obtained is called a scatter diagram (see figure). If the patterns of dots on a scatter diagram are upward sloping, then correlation is positive. If the patterns of dots on a scatter diagram

are downward sloping, then correlation is negative. If the dots do not show any trend, the two variables have no correlation.



SCATTER DIAGRAM

- (i) **Merits of Scatter diagram.** Scatter diagram is easy to understand. Even a layman who is not trained in mathematics or statistics can understand scattered diagram. If the relation is fairly apparent, then it can be shown through scattered diagram rather than by calculating through mathematical formula.
- (ii) **Demerits of scattered diagram.** This method does not give any quantitative figure of the degree of correlation. Hence comparison of different data and their correlation are not possible. Scattered diagram is not an objective method of measuring correlation.

(b) Karl Pearson's Coefficient of correlation

(i) Measure from mean

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

where x and y are the values of the variates.

If σ_x and σ_y denotes the standard deviation of the variates then r will be

$$r = \frac{\frac{1}{n} \sum (x - \bar{x})(y - \bar{y})}{\sqrt{\frac{1}{n} \sum (x - \bar{x})^2} \sqrt{\frac{1}{n} \sum (y - \bar{y})^2}}$$

$$= \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

where

$$\text{Cov}(x, y) = \frac{1}{n} \sum (y - \bar{y})(x - \bar{x}) = \frac{1}{n} \sum xy - \bar{x}\bar{y}$$

(ii) Change of Origin and Scale

$$\text{Let } u = \frac{x - a}{h}, \quad v = \frac{y - b}{k}$$

So that $x = a + hu$, $y = b + kv$

$$\bar{x} = a + h\bar{u}, \quad \bar{y} = b + k\bar{v}$$

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} = \frac{\sum (u - \bar{u})(v - \bar{v})}{\sqrt{\sum (u - \bar{u})^2 \sum (v - \bar{v})^2}}$$

Correlation coefficient is independent of the change of origin and scale.

Example:

A: 105 104 102 101 100 99 98 96 93 92

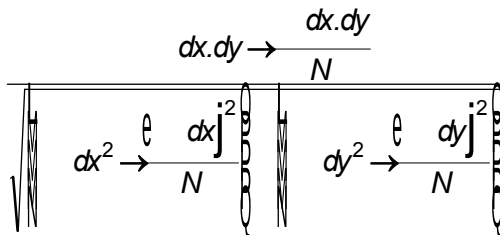
B: 101 103 100 98 95 96 104 92 97 94

Let x and y denote the values in the series A and B respectively and let u = x – 100, V = y – 96.

Sl. No.	x	y	u	v	uv	u ²	v ²
1	105	101	5	5	25	25	25
2	104	103	4	7	28	16	49
3	102	100	2	4	8	4	16
4	101	98	1	2	2	1	4
5	100	95	0	-1	0	0	1
6	99	96	-1	0	0	1	0
7	98	104	-2	8	-16	4	64
8	96	92	-4	-4	16	16	16
9	93	97	-7	+1	-7	49	1
10	92	94	-8	-2	16	64	4
N = 10			-10	20	72	180	180

$$r = \frac{72 - \frac{1}{10}(-10)(20)}{\sqrt{180 - \frac{1}{10}(-10)^2} \sqrt{180 - \frac{1}{10}(20)^2}} = 0.596$$

(iii) **Short cut method**



In the formula,

dx and dy are sum of deviations from the assumed mean of x and y respectively.

dx² and dy² are sum of square of the deviations from the assumed mean in x and y series respectively.

(iv) Correlation with frequency distribution

$$r = \frac{N \sum f dx \cdot dy \rightarrow \sum f dx \cdot \sum f dy}{\sqrt{N \sum f dx^2} \sqrt{N \sum f dy^2}}$$

(v) Properties of the coefficient of correlation

1. The 'value of r ' will lie between +1 and -1.
2. The 'value of r ' is independent of change of scale and origin of the variable x and y .
3. The 'value of r ' is the geometric mean of two regression co-efficients.

$$r = \sqrt{b_{xy} \cdot b_{yx}}$$

4. Probable error can be calculated

$$P.E. = 0.6745 \frac{1-r^2}{\sqrt{N}}$$

where 0.6745 is a constant number

r is Pearson coefficient of correlation

N is number of items.

5. Standard error can be calculated

$$SE = \frac{1-r^2}{\sqrt{N}}$$

(vi) Merits of Coefficient of correlation

1. In one figure of r , both the degree of correlation and its direction can be calculated.
2. It can be derived from the use of standard deviation.
3. It is closely related to regression analysis.
4. Probable error and standard error can be calculated from coefficient of correlation.

(vii) Limitations of coefficient of correlation

1. It assumes a linear relationship between independent and dependent variables, which may not always be true.
2. It is applicable only if the distribution is normal.
3. The value of the coefficient is unduly affected by extreme items.
4. It is difficult to calculate.
5. It is difficult to understand.

(c) Spearman's Rank Difference Method

Under this method, original values are not taken into account. Instead, the values are ranked.

- (i) **Formula.** Formula of Spearman's rank-correlation coefficient is

$$r = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

Where ' d ' is the difference between the ranks of paired items of x and y variables.

- (ii) **Correction for tied ranks.** In Spearman's rank-correlation method, the biggest item gets the first rank, the next to it gets second rank and so on. But difficulty may be encountered where two or more items are of equal value. In such cases, correction has to be applied. $\frac{m(m^2-1)}{12}$ has to be added to d^2 . 'm' is the number of times a particular value of x and y has been repeated. Separate correction has to be applied for each separate case of repetition of ranks.
- (iii) **Values of Spearman's rank-correlation.** If the value of d is zero for each individual, then the correlation is positive and will be equal to one.

If $d^2 \mid n(n^2-1)$ or, $d^2 \mid \frac{n(n^2-1)}{6}$ then the correlation is zero, which means that there is no correlation at all between the two variables.

Example:

Rank						
Sl. No.	x	y	x	y	d	d^2
1	105	101	1	3	-2	4
2	104	103	2	2	0	0
3	102	100	3	4	-1	1
4	101	98	4	5	-1	1
5	100	95	5	8	-3	9
6	99	96	6	7	-1	1
7	98	104	7	1	+6	36
8	96	92	8	10	-2	4
9	93	97	9	6	+3	9
10	92	94	10	9	+1	1
					0	$d^2 = 66$

$$r = 1 - \frac{6 \sum d^2}{n(n^2-1)} = 1 - \frac{6 \times 66}{10(10^2-1)} = 1 - \frac{396}{990}$$

$$r = 1 - 0.4 = 0.6$$

SPECIAL CORRELATION METHODS

Biserial Correlation Coefficient

Another measure of association, the biserial correlation coefficient, termed r_b , is similar to the point biserial, but pits quantitative data against ordinal data, but ordinal data with

an underlying continuity but measured discretely as two values (dichotomous). An example might be test performance vs anxiety, where anxiety is designated as either high or low. Presumably, anxiety can take on any value in between, perhaps beyond, but it may be difficult to measure. We further assume that anxiety is normally distributed. The formula is very similar to the point-biserial but yet different:

$$r_b = (Y_1 - Y_0) \cdot (pq/Y) / \sigma_y,$$

where Y_0 and Y_1 are the Y score means for data pairs with an x score of 0 and 1, respectively, $q = 1 - p$ and p are the proportions of data pairs with x scores of 0 and 1, respectively, and σ_y is the population standard deviation for the y data, and Y is the height of the standardized normal distribution at the point z , where $P(z' < z) = q$ and $P(z' > z) = p$. Since the factor involving p , q , and the height is always greater than 1, the biserial is always greater than the point-biserial.

Tetrachoric Correlation Coefficient

The tetrachoric correlation coefficient, r_{tet} , is used when both variables are dichotomous, like the phi, but we need also to be able to assume both variables really are continuous and normally distributed. Thus it is applied to ordinal vs. ordinal data which has this characteristic. Ranks are discrete so in this manner it differs from the Spearman. The formula involves a trigonometric function called cosine. The cosine function, in its simplest form, is the ratio of two side lengths in a right triangle, specifically, the side adjacent to the reference angle divided by the length of the hypotenuse. The formula is: $r_{tet} = \cos(180/(1 + \sqrt{BC/AD}))$.

Rank-Biserial Correlation Coefficient

The rank-biserial correlation coefficient, r_{rb} , is used for dichotomous nominal data vs rankings (ordinal). The formula is usually expressed as $r_{rb} = 2 \cdot (Y_1 - Y_0)/n$, where n is the number of data pairs, and Y_0 and Y_1 , again, are the Y score means for data pairs with an x score of 0 and 1, respectively. These Y scores are ranks. This formula assumes no tied ranks are present. This may be the same as a Somer's D statistic for which an online calculator is available.

Coefficient of Nonlinear Relationship (ETA)

It is often useful to measure a relationship irrespective of if it is linear or not. The eta correlation ratio or eta coefficient gives us that ability. This statistic is interpreted similar to the Pearson, but can never be negative. It utilizes equal width intervals and always exceeds $|r|$. However, even though r is the same whether we regress y on x or x on y , two possible values for eta can be obtained.

Again, the calculation goes beyond what can be presented here at the moment.

REGRESSION

Regression analysis is based on the relationship between two or more variables. The known variable is called as 'independent variable' and the variable we are trying to predict is called as 'dependent variable'. Regression analysis shows us how to determine both the nature and the strength of a relationship between independent variable and dependent variable.

According to M. M. Blair, “Regression analysis is a mathematical measure of the average relationship between two or more variables in terms of the original units of the data.”

The term ‘Regression’ was first used by Sir Francis Galton in 1877 in the study of heredity. Galton made a study that showed that the height of sons born to tall parents will tend to move back, or ‘Regress’ towards the mean height of the population. In statistics the term ‘Regression’ is used to denote backward tendency which means going back to average. In Economics regression analysis is used as the basic technique for measuring or estimating the relationship among economic variables that constitute the essence of economic theory and economic life.

Before developing the methods of regression it is necessary to distinguish between two types of relationships, linear and non-linear :

1. If the relationships between two (or more) variables is best described by means of a straight line (or a plane), it is called linear. Linear relationship implies a constant change in the dependent variable with respect to changes in the independent variables. It is extensively used for predictive purposes.
2. If the underlying relationship between two (or more) variables is best described by a curve the relationship is called non-linear. A non-linear relationship implies a varying absolute change in the dependent variable with respect to changes in the independent variable. Its predictive value is limited.

Types of Regression

1. **Simple Regression.** The regression analysis for studying only two variables at a time, is called the Simple Regression.
2. **Multiple Regression.** The regression analysis for studying more than two variables at a time is known as Multiple Regression.
3. **Linear Regression.** The task of bringing out linear relationships consists of developing methods of fitting a straight line or the regression line, as is often called, to the data on two variables.
4. **Non-linear Regression.** If the curve of regression is not a straight line, then it is called a Non-linear Regression.

Utility of Regression Analysis

Today, the regression analysis technique plays a vital role in managerial decisions. The importance of this technique can be stated through the following points:

1. This technique helps in determining the coefficient of correlation.
2. This technique establishes the rate of change in one variable, in terms of the changes in another variable.
3. Regression analysis helps prediction and thus it can estimate the value of unknown quantities.
4. This technique helps in economic and business investigations.
5. The study of regression analysis indicates the cause and effect relationship.

Difference between Correlation and Regression Analysis

1. In correlation, we cannot say that one variable is the cause and other the effect.

While in regression analysis one variable is taken as dependent and the other is independent—thus making it possible to study the cause and effect relationship.

2. Correlation coefficient is independent of change of scale and origin but regression coefficients are independent of change of origin but not of scale.
3. Correlation indicates the quantity of relation between variables but regression analysis indicates the nature of two variables.

Precautions in the Use of Regression Analysis

As the regression analysis has a predictive value, its limitations must be recognised and it must be used with caution. Regression Analysis suffers from the following limitations:

1. A value of Y cannot be predicted if the value of X is beyond the values on which the regression equation is based. For example, if the various values of X range from 200 to 1400, then a value of Y corresponding to $X = 1500$ should not be estimated.
2. In case of a time series, the value of Y which involves the prediction of a result for 'X', a future event, may not be relevant. For example, if the data relate to years 1971 to 1979 and a prediction for 1980 or any later date is desired, regression analysis may not be used satisfactorily.
3. The analysis is confined to data where the distribution of Y is normal.

FACTOR ANALYSIS

Factor analysis is a technique that is used to reduce a large number of variables into fewer numbers of factors. This technique extracts maximum common variance from all variables and puts them into a common score. As an index of all variables, we can use this score for further analysis. Factor analysis is part of general linear model (GLM) and this method also assumes several assumptions: there is linear relationship, there is no multicollinearity, it includes relevant variables into analysis, and there is true correlation between variables and factors. Several methods are available, but principal component analysis is used most commonly.

Types of Factoring

There are different types of methods used to extract the factor from the data set:

1. **Principal component analysis.** This is the most common method used by researchers. PCA starts extracting the maximum variance and puts them into the first factor. After that, it removes that variance explained by the first factors and then starts extracting maximum variance for the second factor. This process goes to the last factor.
2. **Common factor analysis.** The second most preferred method by researchers, it extracts the common variance and puts them into factors. This method does not include the unique variance of all variables. This method is used in SEM.
3. **Image factoring.** This method is based on correlation matrix. OLS Regression method is used to predict the factor in image factoring.
4. **Maximum likelihood method.** This method also works on correlation metric but it uses maximum likelihood method to factor.
5. **Other methods of factor analysis.** Alfa factoring outweighs least squares. Weight square is another regression based method which is used for factoring.

Factor Loading

Factor loading is basically the correlation coefficient for the variable and factor. Factor loading shows the variance explained by the variable on that particular factor. In the SEM approach, as a rule of thumb, 0.7 or higher factor loading represents that the factor extracts sufficient variance from that variable.

Eigenvalues. Eigenvalues is also called characteristic roots. Eigenvalues shows variance explained by that particular factor out of the total variance. From the commonality column, we can know how much variance is explained by the first factor out of the total variance. For example, if our first factor explains 68% variance out of the total, this means that 32% variance will be explained by the other factor.

Factor score. The factor score is also called the component score. This score is of all row and columns, which can be used as an index of all variables and can be used for further analysis. We can standardize this score by multiplying a common term. With this factor score, whatever analysis we will do, we will assume that all variables will behave as factor scores and will move.

Criteria for determining the number of factors. According to the Kaiser Criterion, Eigenvalues is a good criteria for determining a factor. If Eigenvalues is greater than one, we should consider that a factor and if Eigenvalues is less than one, then we should not consider that a factor. According to the variance extraction rule, it should be more than 0.7. If variance is less than 0.7, then we should not consider that a factor.

Rotation method. Rotation method makes it more reliable to understand the output. Eigenvalues do not affect the rotation method, but the rotation method affects the Eigenvalues or percentage of variance extracted. There are a number of rotation methods available: (1) No rotation method, (2) Varimax rotation method, (3) Quartimax rotation method, (4) Direct oblimin rotation method, and (5) Promax rotation method. Each of these can be easily selected in SPSS, and we can compare our variance explained by those particular methods.

Assumptions

No outlier. Assume that there are no outliers in data.

Adequate sample size. The case must be greater than the factor.

No perfect multicollinearity. Factor analysis is an interdependency technique. There should not be perfect multicollinearity between the variables.

Homoscedasticity. Since factor analysis is a linear function of measured variables, it does not require homoscedasticity between the variables.

Linearity. Factor analysis is also based on linearity assumption. Non-linear variables can also be used. After transfer, however, it changes into linear variable.

Interval Data. Interval data are assumed.

Key Concepts and Terms

Exploratory factor analysis. Assumes that any indicator or variable may be associated with any factor. This is the most common factor analysis used by researchers and it is not based on any prior theory.

Confirmatory factor analysis (CFA). Used to determine the factor and factor loading of measured variables, and to confirm what is expected on the basic or pre-

established theory. CFA assumes that each factor is associated with a specified subset of measured variables. It commonly uses two approaches:

- a. **The traditional method.** Traditional factor method is based on principle factor analysis method rather than common factor analysis. Traditional method allows the researcher to know more about insight factor loading.
- b. **The SEM approach.** CFA is an alternative approach of factor analysis which can be done in SEM. In SEM, we will remove all straight arrows from the latent variable, and add only that arrow which has to observe the variable representing the covariance between every pair of latents. We will also leave the straight arrows error free and disturbance terms to their respective variables. If standardized error term in SEM is less than the absolute value two, then it is assumed good for that factor, and if it is more than two, it means that there is still some unexplained variance which can be explained by factor. Chi-square and a number of other goodness-of-fit indexes are used to test how well the model fits.

EXPERIMENTAL DESIGNS

Analysis of variance (ANOVA) is the most efficient parametric method available for the analysis of data from experiments. It was devised originally to test the differences between several different groups of treatments thus circumventing the problem of making multiple comparisons between the group means using *t*-tests (7). ANOVA is a method of great complexity and subtlety with many different variations, each of which applies in a particular experimental context. Hence, it is possible to apply the wrong type of ANOVA in a particular experimental situation and, as a consequence, draw the wrong conclusions from the data.

The logic which underlies the method and the assumptions necessary to apply it to data. In addition, the simplest form of the analysis, the one-way ANOVA in a randomised design, was described and illustrated using a clinical experiment drawn from optometric research.

Statistical Models

We described a commonly used notation to describe the basic model of an ANOVA. The subscript '*i*' is used to denote the group or class (i.e. the treatment group), '*i*' taking the values '1 to *a*', whereas the subscript '*j*' designates the members of the class, '*j*' taking the values '1 to *n*' (hence, '*a*' groups and '*n*' replicates per group). Within class '*i*', the observations x_{ij} are assumed to be normally distributed about a mean μ with variance s^2 . This linear model can be written:

$$x_{ij} = \mu + \alpha_i + e_{ij}$$

Hence, an observed value x_{ij} is the sum of three parts: (1) the overall mean of the observations (μ) (2) a treatment or class deviation ' α_i ', and (3) a random element ' e ' drawn from a normally distributed population. The random element reflects the combined effects of natural variation between replications and errors of measurement. All the more complex types of ANOVA can be derived from this simple model by the addition of one or more further terms to equation 1.

STATISTICAL MODELS OF THE TYPES OF ANOVA DESCRIBED.
FOR DEFINITION OF TERMS SEE RELEVANT TEXT

Type of ANOVA	Y	Constant elements	Additional components	Interaction terms
One-way, fixed	x_{ij}	$\mu + e_{ij}$	a_i	
One-way, random	x_{ij}	$\mu + e_{ij}$	A_i	
Two-way, randomised blocks	x_{ij}	$\mu + e_{ij}$	$a_i + b_j$	
Three-way	x_{ijk}	$\mu + e_{ijk}$	$a_i + b_j + c_k + d_{ij}$	
Two-factor, factorial	x_{ijk}	$\mu + e_{ijk}$	$a_i + b_j$	$(ab)_{ij}$
Two-factor, factorial split-plot	x_{ijk}	$\mu + e_{ijk}$	$M_i + B_j + T_k + d_{ijk}$	$(MT)_{ik}$

In each case, the dependent variable (Y) can be considered to be the sum of the constant elements, the additional components and, where appropriate, the interaction terms in each row.

One-way ANOVA

We described a one-way ANOVA in a randomised design which compared the reading rates of three groups of subjects, viz., young normal subjects, elderly normal subjects and subjects with age-related macular degeneration. This ANOVA is described as a 'fixed effects' model in which the objective is to estimate the differences between the subject groups and these are regarded as 'fixed' or discrete effects. There is, however, an alternative model called the 'random effects' model in which the objective is to estimate the degree of variation of a particular measurement and in many circumstances to compare different sources of variation in space and time.

Example

Five measurements of intraocular pressure (IOP) were made on a subject, 1 min apart on 3 days chosen randomly. The objective was to determine, for individual subjects, the degree of variation in IOP from minute-to-minute compared with the variation between measurements made on different days. In this case, estimates of variability are the objective rather than the determination of fixed effects. Based on these estimates of variability, a suitable protocol for measuring IOP in a clinical context might be devised. For example, if IOP varied considerably from minute to minute but, on average, little between days, several measurements of the IOP on a single occasion might be an appropriate strategy. By contrast, if minute to minute variation was negligible but there was significant day to day variation, it might be better to measure IOP only once, but on a sample of days. This type of experiment may also be described as a 'hierarchical' or 'nested type' design, especially if each sample is composed of subsamples and these in turn are subsampled. In addition, variation may be spatial rather than temporal, e.g. visual function may be quantified at different locations on the retina. In this case, there may be variation between eyes, between different locations within the same retina, and between sequential measurements at the same location.

Model

A 'fixed effect' model in which the a_i are fixed quantities to be estimated. The corresponding 'random effects' model is similar, but the symbol A_i replaces a_i and represents the difference between the IOP of the i th patient and the mean. Hence, A_i is a random variable and the term e_{ij} refers to errors of measurement and to the fact that IOP is measured on a sample of occasions.

THE ONE-WAY ANOVA 'RANDOM EFFECTS MODEL' (HIERARCHICAL OR NESTED DESIGN) ON FIVE MEASUREMENTS OF IOP MADE ON A SUBJECT 1 MIN APART ON 3 SAMPLE DAYS (A)

Design					
Sample day	A1		A2		A3
Repeat	18		17		19
Measurements	19		18		18
	20		16		20
	19		17		20
	21		17		19
ANOVA table:					
Variation	SS	d.f.	MS	F	ExpMS
Between days	17.73	2	8.866	10.64	$\sigma_m^2 + 5\sigma_D^2$
Between repeat measurements within days	10.0	12	0.833		σ_m^2

ExpMS = expected mean square. Components of variance: between days (σ_D^2) = 1.607, between repeat measurements (σ_m^2) = 0.833.

ANOVA

The ANOVA appropriate to this design provides an F-test of whether there is significant variation between days. When the null hypothesis is false, however, the mean square between classes (in this case between days) estimates the quantity, $\sigma_m^2 + n_D \sigma_D^2$ and it is possible to calculate the 'components of variance' σ_m^2 and σ_D^2 from the ANOVA. These are estimates of the variance of the measurements made between days and between determinations on a single day. In the example quoted, the F-test suggests significant variation in IOP between days. In this context, however, the components of variance are more useful and indicate that the component between days is approximately twice as great as that on a single day.

It is important to identify whether a 'fixed' or 'random' effect model is appropriate in each case. This is particularly important in more complex factorial-type designs in which there may be a mixture of 'fixed' and 'random' effects. One way of deciding whether a factor is 'fixed' or 'random' is to imagine the effect of changing one of the levels of the factor. If this makes it a different experiment, for example, by substituting a different group of subjects, then it is a 'fixed' effect factor. By contrast, if we considered it the same experiment, for example, by substituting a different sample day, as in the example described above, it would be a 'random' effect factor.

Factorial ANOVA, split-plot design

In the 22 factorial described above, the experimental subjects were assigned at random to all possible combinations of the two factors. However, in some designs, the two factors are not equivalent to each other. A common case, called a split-plot design, arises when one factor can be considered to be a major factor and the other a minor factor.

Example

In optometry research, this situation could arise if measurements were made on both the right and left eyes of subjects employing two different subject groups. As an example,

an investigator wished to study whether IOP in right and left eyes was elevated in patients with high blood pressure. Alternatively, a measurement may be made on subjects within two treatment groups employing different methods or equipment. The problem that arises in these types of design is the dependence or correlation between the measurements made on a subject. In these experiments, the subject group would be the major factor while 'which eye' or 'which method' would be regarded as a minor factor. The difference between this and an ordinary factorial design is that previously, all treatment combinations were assigned at random to replicates whereas in a split-plot design, the subplot treatments are randomised over the whole large block but only within each main plot. In some applications, experimenters may subdivide the subplots further to give a split-split-plot design.

FACTORIAL ANOVA, SPLIT-PLOT DESIGN WITH TWO SUBJECT GROUPS (NORMAL AND ELEVATED BLOOD PRESSURE) WITH THREE SUBJECTS (P1, P2, P3) IN EACH GROUP AND THE LEFT (L) AND RIGHT (R) EYE STUDIED FROM EACH PATIENT. DATA ARE INTRAOCULAR PRESSURES

Design	P1	P2	P3	
Control				
R eye	17.3	16.9	14.7	
L eye	17.1	16.5	14.3	
ARMD				
R eye	21.4	24.3	21.4	
L eye	20.7	22.1	24.2	
ANOVA table:				
Variation	SS	d.f.	MS	F
Subject group	115.94	1	115.94	34.4**
Main plot error	13.48	4	3.37	
Right/Left eye	0.10	1	0.10	0.06 ns
Interaction	0.067	1	0.067	0.04 ns
Sub plot error	6.59	4	1.65	

** $p < 0.01$; ns = not significant.

Model

In this case, M represents main-plot treatments, B blocks and, T subplot treatments. The symbols ' i ' and ' j ' indicate the main-plots while ' k ' identifies the subplot within the main-plot. The two components of error e_{ij} and d_{ijk} represent the fact that the error variation between main-plots is likely to be different from that between subplots. For example, one might expect there to be less natural variation when comparing eyes within a subject (subplot) than the eyes of different subjects (main-plot). An alternative model suggested by would be to consider the x_{ij} as distributed about the same mean and variance but for any two members of the same class to be correlated, the degree of correlation being measured by the intra-class correlation coefficient.

ANOVA

The resulting ANOVA is more complex than that of a simple factorial design because of the different error terms. Hence, in a two-factor, split-plot ANOVA, two errors are calculated, the main-plot error is used to test the main effect of subject group while the subplot error

is used to test the main effect of eyes and the possible interaction between the factors. In the quoted example, there is a significant increase in IOP in patients with elevated blood pressure but no difference between eyes; the non-significant interaction suggesting that the elevation in IOP was consistent in both eyes. The subplot error is usually smaller than the main-plot error and also has more d.f. Hence, such an experimental design will usually estimate the main effect of the subplot factor and its interaction more accurately than the main effect of the major factor. Some experimenters will deliberately design an experiment as a 'split-plot' to take advantage of this property.

A disadvantage of such a design, however, is that occasionally, the main effect of the major factor may be large but not significant while the main effect of the minor factor and its interaction may be significant but too small to be important. In addition, a common mistake is for researchers to analyse a split-plot design as if it were a fully randomised two-factor experiment. In this case, the single pooled error variance will either be too small or too large for testing the individual treatment effects and the wrong conclusions could be drawn from the experiment. To decide whether a particular experiment is a split-plot it is useful to consider the following: (1) Are the factors equivalent or does one appear to be subordinate to the other? (2) Is there any restriction in how replicates were assigned to the treatment combinations? and (3) Is the error variation likely to be the same for each factor?

Caution should also be employed in the use of 'posthoc' tests in the case of a split-plot design. 'Posthoc' tests assume that the observations taken on a given subject are uncorrelated so that the subplot factor group means are not related. This is not likely to be the case in split-plot experiment because some correlation between measurements made on the right and left eye is inevitable. Standard errors appropriate to the split-plot design can be calculated and can be used, with caution, to make specific comparisons between the treatment means. However, a better method is to partition the sums of squares associated with main effects and interaction into specific contrasts and to test each against the appropriate error.

With a randomized block design, the experimenter divides subjects into subgroups called blocks, such that the variability within blocks is less than the variability between blocks. Then, subjects within each block are randomly assigned to treatment conditions. Compared to a completely randomized design, this design reduces variability within treatment conditions and potential confounding, producing a better estimate of treatment effects.

Randomized Block Designs

The table below shows a randomized block design for a hypothetical medical experiment.

Gender	Treatment	
	Placebo	Vaccine
Male	250	250
Female	250	250

Subjects are assigned to blocks, based on gender. Then, within each block, subjects are randomly assigned to treatments (either a placebo or a cold vaccine). For this design, 250 men get the placebo, 250 men get the vaccine, 250 women get the placebo, and 250 women get the vaccine.

It is known that men and women are physiologically different and react differently to medication. This design ensures that each treatment condition has an equal proportion of men and women. As a result, differences between treatment conditions cannot be attributed to gender. This randomized block design removes gender as a potential source of variability and as a potential confounding variable.

Repeated Measures Design

As you'd expect, repeated measures designs involve multiple measurements of each subject. That's no surprise, but there is more to it than just that. In repeated measures designs, the subjects are typically exposed to all of the treatment conditions. Surprising, right?

In this type of design, each subject functions as an experimental block. A block is a categorical variable that explains variation in the response variable that is not caused by the factors that you really want to know about. You use blocks in designed experiments to minimize bias and variance of the error because of these nuisance factors.

In repeated measures designs, the subjects are their own controls because the model assesses how a subject responds to all of the treatments. By including the subject block in the analysis, you can control for factors that cause variability between subjects. The result is that only the variability within subjects is included in the error term, which usually results in a smaller error term and a more powerful analysis.

The Benefits of Repeated Measures Designs

1. **More statistical power.** Repeated measures designs can be very powerful because they control for factors that cause variability between subjects.
2. **Fewer subjects.** Thanks to the greater statistical power, a repeated measures design can use fewer subjects to detect a desired effect size. Further sample size reductions are possible because each subject is involved with multiple treatments. For example, if an independent groups design requires 20 subjects per experimental group, a repeated measures design may only require 20 total.
3. **Quicker and cheaper.** Fewer subjects need to be recruited, trained, and compensated to complete an entire experiment.
4. **Assess an effect over time.** Repeated measures designs can track an effect overtime, such as the learning curve for a task. In this situation, it's often better to measure the same subject at multiple times rather than different subjects at one point in time for each.

Latin Square

In combinatorics and in experimental design, a Latin square is an $n \times n$ array filled with n different symbols, each occurring exactly once in each row and exactly once in each column. An example of a 3×3 Latin square is:

A	B	C
C	A	B
B	C	A

The name "Latin square" was inspired by mathematical papers by Leonhard Euler (1707–1783), who used Latin characters as symbols, but any set of symbols can be used:

in the above example, the alphabetic sequence A, B, C can be replaced by the integer sequence 1, 2, 3. Korean mathematician Choi Seok-jeong was the first to publish an example of Latin squares of order nine to construct a magic square in 1700, predating Leonard Euler by at least 67 years.

Cohort Studies

A study design where one or more samples (called cohorts) are followed prospectively and subsequent status evaluations with respect to a disease or outcome are conducted to determine which initial participants exposure characteristics (risk factors) are associated with it. As the study is conducted, outcome from participants in each cohort is measured and relationships with specific characteristics determined.

Advantages

- Subjects in cohorts can be matched, which limits the influence of confounding variables.

- Standardization of criteria/outcome is possible.

- Easier and cheaper than a randomized controlled trial (RCT).

Disadvantages

- Cohorts can be difficult to identify due to confounding variables.

- No randomization, which means that imbalances in patient characteristics could exist.

- Blinding/masking is difficult.

- Outcome of interest could take time to occur.

Fictitious Example

A cohort study was designed to assess the impact of sun exposure on skin damage in beach volleyball players. During a weekend tournament, players from one team wore waterproof, SPF 35 sunscreen, while players from the other team did not wear any sunscreen. At the end of the volleyball tournament players' skin from both teams was analyzed for texture, sun damage, and burns. Comparisons of skin damage were then made based on the use of sunscreen. The analysis showed a significant difference between the cohorts in terms of the skin damage.

Time Series

For many experimental psychologists, the go-to methodological designs are cross-sectional. Cross-sectional studies involve measuring the relationship between some variable(s) of interest at one point in time; some common examples include single-session lab studies and online surveys. These designs can be useful for isolating relationships between variables, establishing conditions of convergent and discriminant validity, and utilizing samples that are statistically representative of larger populations. Nevertheless, quantitative researchers have noted that attempts to measure and analyze interindividual variation are incomplete without an accompanying account of the underlying temporal dynamics that define these processes. This claim follows from the idea that cross-sectional designs, while potentially well-suited for large samples, are often underpowered, overgeneralized, and ill-approximated to the statistical assumptions implied by general

linear methods. For these reasons, psychological scientists should consider supplementing their methodological toolkits with time-series techniques to explicitly investigate the time-dependent variation that can be observed within individual subjects.

The purpose of this article is to briefly discuss the importance of time-series methods in experimental research and to acquaint the reader with some statistical techniques that are easily accessible and can be employed when testing hypotheses with time-series data.

Measuring Behavior as a Time Series

According to Daniel T. Kaplan and Leon Glass (1995), there are two critical features of a time series that differentiate it from cross-sectional data-collection procedures:

Repeated measurements of a given behavior are taken across time at equally spaced intervals. Taking multiple measurements is essential for understanding how any given behavior unfolds over time, and doing so at equal intervals affords a clear investigation of how the dynamics of that behavior manifest at distinct time scales.

The temporal ordering of measurements is preserved. Doing so is the only way to fully examine the dynamics governing a particular process. If we expect that a given stimulus will influence the development of a behavior in a particular way, utilizing summary statistics will completely ignore the temporal ordering of the data and likely occlude one's view of important behavioral dynamics.

Linear computations such as mean and variance merely describe global properties of a data set and thus may fail to capture meaningful patterns that only can be identified by looking at the sequential dependency between time points. Consequently, time-series techniques provide a valuable approach in studying psychological processes, which are, by their very nature, fundamentally embedded in time.

Analyzing Time-Series Data

Once you've collected a series of behavioral measurements on your variable(s) of interest, there are a variety of ways to explore and quantify the observed dynamics. Here are a few techniques that can be used to investigate patterns within time-series data:

Auto-correlation/Cross-correlation. An Auto-correlation reflects the magnitude of time dependency between observations within a time series. An Auto-correlation plot depicts correlations between measurements X_t and X_{t+n} , such that each value represents the extent to which any given behavior is related to previous behaviors within the series. A cross-correlation involves relating two time series that are shifted in time at lag n (i.e., X_t and Y_{t+n}), and can reveal, for example, whether one process tends to "lead" the other's behavior or whether they oscillate together.

Recurrence quantification analysis (RQA). RQA begins by simply plotting a time series against itself (i.e., X_t against X_t) and then quantifies whether certain states of the behavior remain stable or recur in time, as well as what percentage of the series is constituted by deterministic patterns. Cross-RQA also can be used to analyze the degree of recurrence and deterministic patterning between two processes, and it has been applied to the study of interpersonal coordination and postural control.

Phase space reconstruction (PSR). When obtaining a behavioral time-series, one of your goals could be to determine what variables are involved in producing particular patterns

of behavior and what the possible structure of the underlying dynamics may be. One way to accomplish this is to reconstruct the phase space, which is a multidimensional plot that represents all possible states within the process and can be used to approximate the number of variables involved in producing the observed behavioral changes. For example, we may interpret high trait self-esteem as representing a strong tendency for an individual to adopt and maintain positive self-evaluations. Collecting repeated measurements of state self-esteem and then performing a PSR could help describe the strength of that individual's tendency to retain a positive image of herself as well as reveal the compensatory dynamics that follow from a negative self-evaluative state.

Spectral analysis. Mathematically, any time series can be transformed into a linear composition of sine and cosine waves with varying frequencies. One goal in analyzing time-series data is often to find out what deterministic cycles (i.e., which of the component waves) account for the most variance within the series. Performing a spectral decomposition transforms a time series into a set of constituent sine and cosine waves that then are used to calculate the series' power spectral density function (PSD). Plotting the series' PSD reveals the squared correlations between each component frequency and the series as a whole, yielding a similarly intuitive interpretation to R^2 in multiple regression. In this vein, Gottschalk, Bauer, and Whybrow (1995) applied spectral analysis toward studying the changes in self-reported mood among bipolar patients and control subjects, finding that bipolar individuals tended to exhibit cyclical patterns of mood change that were significantly more chaotic and deterministic than the comparatively random fluctuations observed in control subjects.

Differential equation modeling. Essentially, differential equations allow one to study how different variables change with one another as well as how the state of one variable can be influenced by how it is changing. Derivative estimates of a single time series can be calculated by a number of different techniques from which differential equations then are constructed and tested based on the researcher's predictions about how those variables are related. An intuitive example of this might be in considering a committed romantic relationship, in which changes in one person's level of emotional satisfaction conceivably lead to changes in their partner's level of satisfaction and vice versa. Each partner's feelings might be coupled with the other's in a complex manner, such that differential equations could be used to model their emotional relationship and show how changes in one person's mood are inextricably linked with changes in the other's mood.

MANOVA

Multivariate ANOVA (MANOVA) extends the capabilities of analysis of variance (ANOVA) by assessing multiple dependent variables simultaneously. ANOVA statistically tests the differences between three or more group means. For example, if you have three different teaching methods and you want to evaluate the average scores for these groups, you can use ANOVA. However, ANOVA does have a drawback. It can assess only one dependent variable at a time. This limitation can be an enormous problem in certain circumstances because it can prevent you from detecting effects that actually exist.

MANOVA provides a solution for some studies. This statistical procedure tests multiple dependent variables at the same time. By doing so, MANOVA can offer several advantages over ANOVA.

In this post, I explain how MANOVA works, its benefits compared to ANOVA, and when to use it. I'll also work through a MANOVA example to show you how to analyze the data and interpret the results.

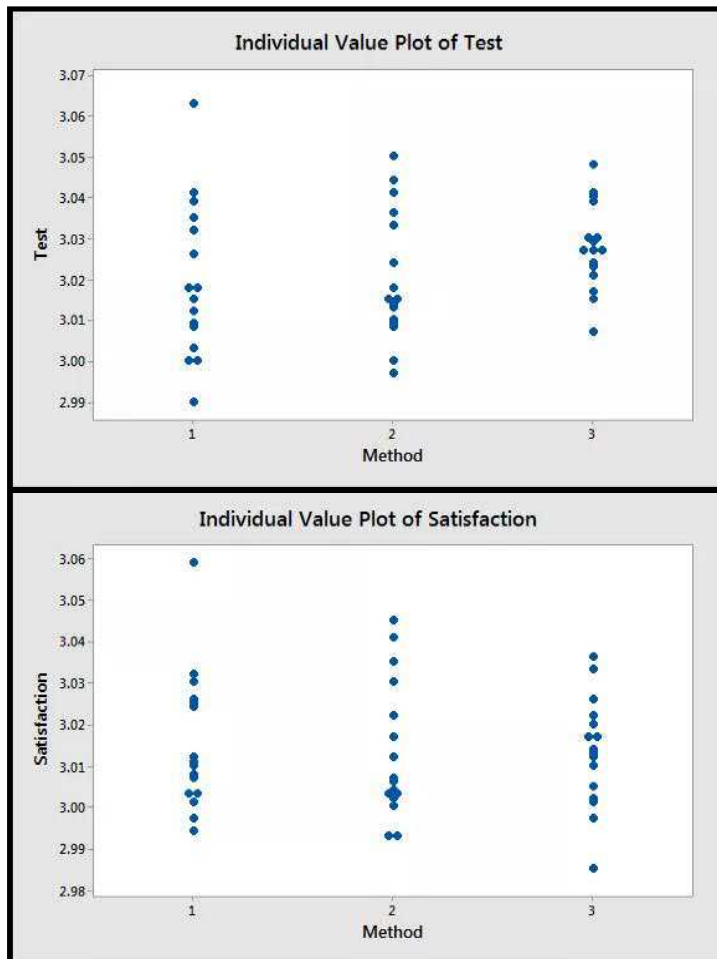
ANOVA Restrictions

Regular ANOVA tests can assess only one dependent variable at a time in your model. Even when you fit a general linear model with multiple independent variables, the model only considers one dependent variable. The problem is that these models can't identify patterns in multiple dependent variables.

This restriction can be very problematic in certain cases where a typical ANOVA won't be able to produce statistically significant results. Let's compare ANOVA to MANOVA.

Comparison of MANOVA to ANOVA Using an Example

MANOVA can detect patterns between multiple dependent variables. But, what does that mean exactly? It sounds complex, but graphs make it easy to understand. Let's work through an example that compares ANOVA to MANOVA.



Suppose we are studying three different teaching methods for a course. This variable is our independent variable. We also have student satisfaction scores and test scores. These variables are our dependent variables. We want to determine whether the mean scores for satisfaction and tests differ between the three teaching methods. Here is the CSV file for the MANOVA_example.

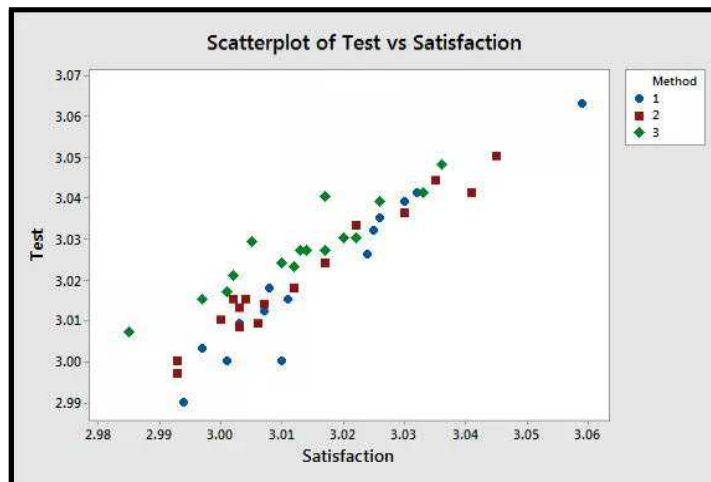
The graphs given previous display the scores by teaching method. One chart shows the test scores and the other shows the satisfaction scores. These plots represent how one-way ANOVA tests the data—one dependent variable at a time.

Both of these graphs appear to show that there is no association between teaching method and either test scores or satisfaction scores. The groups seem to be approximately equal. Consequently, it's no surprise that the one-way ANOVA P-values for both test and satisfaction scores are insignificant (0.923 and 0.254).

Case closed! The teaching method isn't related to either satisfaction or test scores. Hold on. There's more to this story!

How MANOVA Assesses the Data

Let's see what patterns we can find between the dependent variables and how they are related to teaching method. I'll graph the test and satisfaction scores on the scatterplot and use teaching method as the grouping variable. This multivariate approach represents how MANOVA tests the data. These are the same data, but sometimes how you look at them makes all the difference.



The graph displays a positive correlation between Test scores and Satisfaction. As student satisfaction increases, test scores tend to increase as well. Moreover, for any given satisfaction score, teaching method 3 tends to have higher test scores than methods 1 and 2. In other words, students who are equally satisfied with the course tend to have higher scores with method 3. MANOVA can test this pattern statistically to help ensure that it's not present by chance.

In your preferred statistical software, fit the MANOVA model so that Method is the independent variable and Satisfaction and Test are the dependent variables.

The MANOVA results are below.

General Linear Model: Test, Satisfaction versus Method						
MANOVA for Method						
s = 2 m = -0.5 n = 21.0						
Criterion	Test Statistic	F	Num DF	Denom DF	P	
Wilks'	0.51094	8.778	4	88	0.000	
Lawley-Hotelling	0.95577	10.275	4	86	0.000	
Pillai's	0.48977	7.297	4	90	0.000	
Roy's	0.95431					

Even though the one-way ANOVA results and graphs seem to indicate that there is nothing of interest, MANOVA produces statistically significant results—as signified by the minuscule P-values. We can conclude that there is an association between teaching method and the relationship between the dependent variables.

When MANOVA Provides Benefits

Use multivariate ANOVA when your dependent variables are correlated. The correlation structure between the dependent variables provides additional information to the model which gives MANOVA the following enhanced capabilities:

Greater statistical power. When the dependent variables are correlated, MANOVA can identify effects that are smaller than those that regular ANOVA can find.

Assess patterns between multiple dependent variables. The factors in the model can affect the relationship between dependent variables instead of influencing a single dependent variable. As the example in this post shows, ANOVA tests with a single dependent variable can fail completely to detect these patterns.

Limits the joint error rate. When you perform a series of ANOVA tests because you have multiple dependent variables, the joint probability of rejecting a true null hypothesis increases with each additional test. Instead, if you perform one MANOVA test, the error rate equals the significance level.

ANCOVA

Analysis of covariance (ANCOVA) is used in examining the differences in the mean values of the dependent variables that are related to the effect of the controlled independent variables while taking into account the influence of the uncontrolled independent variables.

The Analysis of covariance (ANCOVA) is used in the field of business. This document will detail the usability of Analysis of covariance (ANCOVA) in market research.

Analysis of covariance (ANCOVA) can be used to determine the variation in the intention of the consumer to buy a particular brand with respect to different levels of price and the consumer's attitude towards that brand.

Analysis of covariance (ANCOVA) can be used to determine how a change in the price level of a particular commodity will affect the consumption of that commodity by the consumers.

Analysis of covariance (ANCOVA) consists of at least one categorical independent variable and at least one interval natured independent variable. In Analysis of covariance

(ANCOVA), the categorical independent variable is termed as a factor, whereas the interval natured independent variable is termed as a covariate. The task of the covariate in Analysis of covariance (ANCOVA) is to remove the extraneous variation from the dependent variable. This is done because the effects of the factors are of major concern in Analysis of covariance (ANCOVA).

Analysis of covariance (ANCOVA) is most useful in those cases where the covariate is linearly related to the dependent variables and is not related to the factors.

Similar to Analysis of variance (ANOVA), Analysis of covariance (ANCOVA) also assumes similar assumptions. The following are the assumptions of Analysis of Covariance (ANCOVA):

The variance in Analysis of covariance (ANCOVA) that is being analyzed must be independent.

In the case of more than one independent variable, the variance in Analysis of covariance (ANCOVA) must be homogeneous in nature within each cell that is formed by the categorical independent variables.

The data should be drawn from the population by means of random sampling in Analysis of covariance (ANCOVA). Analysis of covariance (ANCOVA) assumes that the adjusted treatment means those that are being computed or estimated are based on the fact that the variables obtained due to the interaction of covariate are negligible.

The Analysis of covariance (ANCOVA) is done by using linear regression. This means that Analysis of covariance (ANCOVA) assumes that the relationship between the independent variable and the dependent variable must be linear in nature.

In Analysis of covariance (ANCOVA), the different types of the independent variables are assumed to be drawn from the normal population having a mean of zero.

The Analysis of covariance (ANCOVA) assumes that the regression coefficients in every group of the independent variable must be homogeneous in nature.

Analysis of covariance (ANCOVA) is applied when an independent variable has a powerful correlation with the dependent variable. But, it is important to remember that the independent variables in Analysis of covariance (ANCOVA) do not interact with other independent variables while predicting the value of the dependent variable. Analysis of covariance (ANCOVA) is generally applied to balance the effect of comparatively more powerful non interacting variables. It is necessary to balance the effect of interaction in Analysis of covariance (ANCOVA) in order to avoid uncertainty among the independent variables.

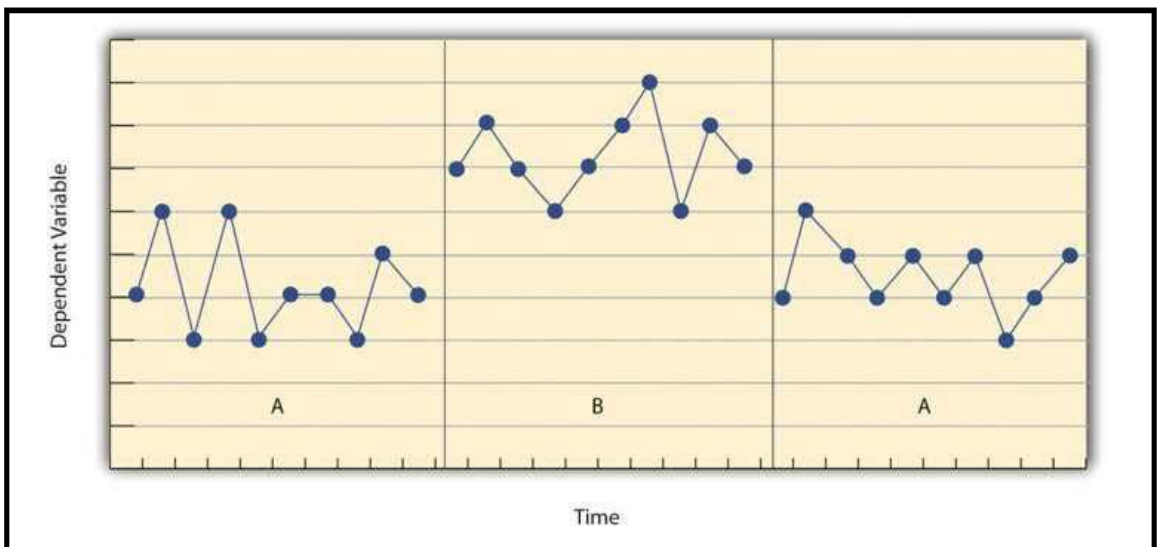
Analysis of covariance (ANCOVA) is applied only in those cases where the balanced independent variable is measured on a continuous scale.

Let us assume a researcher wants to determine the effect of an in-store promotion on sales revenue. In this case, Analysis of covariance (ANCOVA) is an appropriate technique because the change in the attitude of the consumer towards the store will automatically affect the sales revenue of the store in Analysis of covariance (ANCOVA). Therefore, in Analysis of covariance (ANCOVA), the dependent variable will be the sales revenue of the store. And the independent variable will be the attitude of the consumer in Analysis of covariance (ANCOVA).

Single-subject Designs

General Features of Single-Subject Designs

Before looking at any specific single-subject research designs, it will be helpful to consider some features that are common to most of them. Many of these features are illustrated in Figure, which shows the results of a generic single-subject study. First, the dependent variable (represented on the y -axis of the graph) is measured repeatedly over time (represented by the x -axis) at regular intervals. Second, the study is divided into distinct phases, and the participant is tested under one condition per phase. The conditions are often designated by capital letters: A, B, C, and so on. Thus Figure represents a design in which the participant was tested first in one condition (A), then tested in another condition (B), and finally retested in the original condition (A). (This is called a reversal design and will be discussed in more detail shortly.)



RESULTS OF A GENERIC SINGLE-SUBJECT STUDY ILLUSTRATING SEVERAL PRINCIPLES OF SINGLE-SUBJECT RESEARCH

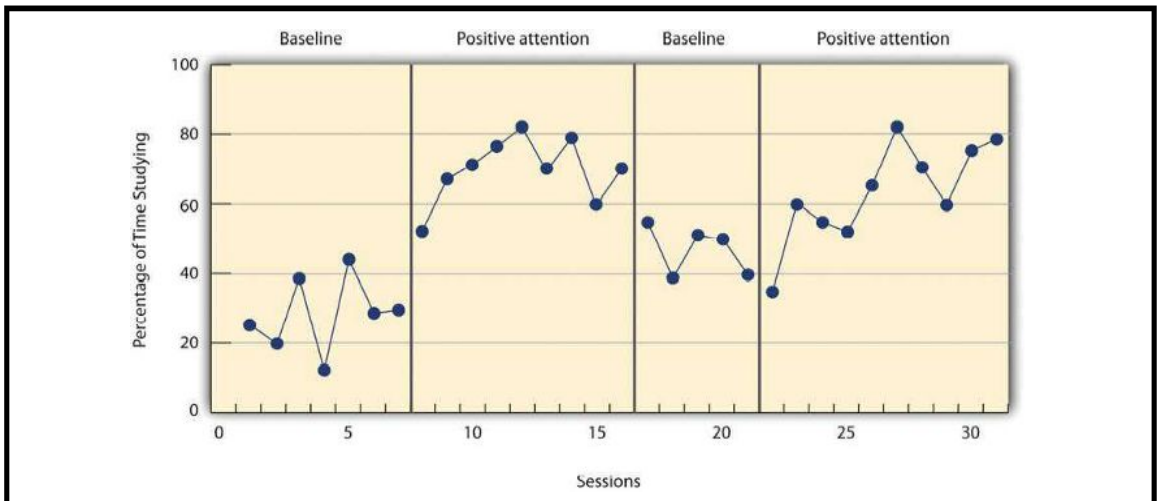
Another important aspect of single-subject research is that the change from one condition to the next does not usually occur after a fixed amount of time or number of observations. Instead, it depends on the participant's behaviour. Specifically, the researcher waits until the participant's behaviour in one condition becomes fairly consistent from observation to observation before changing conditions. This is sometimes referred to as the steady state strategy. The idea is that when the dependent variable has reached a steady state, then any change across conditions will be relatively easy to detect. Recall that we encountered this same principle when discussing experimental research more generally. The effect of an independent variable is easier to detect when the "noise" in the data is minimized.

Reversal Designs

The most basic single-subject research design is the reversal design, also called the ABA design. During the first phase, A, a baseline is established for the dependent variable.

This is the level of responding before any treatment is introduced, and therefore the baseline phase is a kind of control condition. When steady state responding is reached, phase B begins as the researcher introduces the treatment. There may be a period of adjustment to the treatment during which the behaviour of interest becomes more variable and begins to increase or decrease. Again, the researcher waits until that dependent variable reaches a steady state so that it is clear whether and how much it has changed. Finally, the researcher removes the treatment and again waits until the dependent variable reaches a steady state. This basic reversal design can also be extended with the reintroduction of the treatment (ABAB), another return to baseline (ABABA), and so on.

The study by Hall and his colleagues was an ABAB reversal design. Figure approximates the data for Robbie. The percentage of time he spent studying (the dependent variable) was low during the first baseline phase, increased during the first treatment phase until it leveled off, decreased during the second baseline phase, and again increased during the second treatment phase.



AN APPROXIMATION OF THE RESULTS FOR HALL AND COLLEAGUES' PARTICIPANT ROBBIE IN THEIR ABAB REVERSAL DESIGN

Why is the reversal – the removal of the treatment – considered to be necessary in this type of design? Why use an ABA design, for example, rather than a simpler AB design? Notice that an AB design is essentially an interrupted time-series design applied to an individual participant. Recall that one problem with that design is that if the dependent variable changes after the treatment is introduced, it is not always clear that the treatment was responsible for the change. It is possible that something else changed at around the same time and that this extraneous variable is responsible for the change in the dependent variable. But if the dependent variable changes with the introduction of the treatment and then changes back with the removal of the treatment (assuming that the treatment does not create a permanent effect), it is much clearer that the treatment (and removal of the treatment) is the cause. In other words, the reversal greatly increases the internal validity of the study.

There are close relatives of the basic reversal design that allow for the evaluation of more than one treatment. In a multiple-treatment reversal design, a baseline phase is followed by separate phases in which different treatments are introduced. For example, a

researcher might establish a baseline of studying behaviour for a disruptive student (A), then introduce a treatment involving positive attention from the teacher (B), and then switch to a treatment involving mild punishment for not studying (C). The participant could then be returned to a baseline phase before reintroducing each treatment—perhaps in the reverse order as a way of controlling for carry-over effects. This particular multiple-treatment reversal design could also be referred to as an ABCACB design.

In an alternating treatments design, two or more treatments are alternated relatively quickly on a regular schedule. For example, positive attention for studying could be used one day and mild punishment for not studying the next, and so on. Or one treatment could be implemented in the morning and another in the afternoon. The alternating treatments design can be a quick and effective way of comparing treatments, but only when the treatments are fast acting.



PSYCHOLOGICAL TESTING

TEST CONSTRUCTION

Definition

“An objective and standardized measure of a sample of behaviour.”

Standards of Test Construction and Test Use

A good test should be reliable and valid. Concerns relating to standards include user qualification, security of test content, confidentiality of test results, and the prevention of the misuse of tests and results.

Test Characteristics and Response sets.

Characteristics include

- i. a test of maximum performance (*e.g.*, achievement test) which tells us what a person can do.
- ii. a test of typical performance (*e.g.*, personality test) which tells us what a person usually does.
- iii. a speed test, in which response rate is assessed.
- iv. a mastery test assesses whether or not the person can attain a pre-specified mastery level of performance.

Response sets include: social desirability (giving responses that are perceived to be socially acceptable), acquiescence (agreeing or disagreeing with everything, and deviation (giving unusual or uncommon responses).

All of the above can threaten the validity of a given set of results.

Reliability

Classical Test Theory States

A test is reliable a) to the degree that it is free from error and provides information about examinees' "true" test scores and b) to the degree that it provides repeatable, consistent results.

Methods of Estimating Reliability

[we did not go over this as it involves calculating a reliability coefficient—a correlation coefficient—which we have not discussed yet]

Standard Error of Measurement

[we skipped over, as we are not ready to discuss yet]

Factors Affecting the Reliability Coefficient

Any factor which reduces score variability or increases measurement error will also reduce the reliability coefficient. For *e.g.*, all other things being equal, short tests are less reliable than long ones, very easy and very difficult tests are less reliable than moderately difficult tests, and tests where examinees' scores are affected by guessing (*e.g.* true-false) have lowered reliability coefficients.

Validity

Three major categories:

1. Content Validity

A test has content validity if it measures knowledge of the content domain of which it was designed to measure knowledge. Another way of saying this is that content validity concerns, primarily, the adequacy with which the test items adequately and representatively sample the content area to be measured. For *e.g.*, a comprehensive math achievement test would lack content validity if good scores depended primarily on knowledge of English, or if it only had questions about one aspect of math (*e.g.*, algebra). Content validity is primarily an issue for educational tests, certain industrial tests, and other tests of content knowledge like the Psychology Licensing Exam.

Expert judgement (not statistics) is the primary method used to determine whether a test has content validity. Nevertheless, the test should have a high correlation w/other tests that purport to sample the same content domain.

This is different from **face validity**: face validity is when a test appears valid to examinees who take it, personnel who administer it and other untrained observers. Face validity is not a technical sense of test validity; *i.e.*, just b/c a test has face validity does not mean it will be valid in the technical sense of the word. "*just cause it looks valid doesn't mean it is.*"

2. Criterion-related Validity

Criterion-related validity is a concern for tests that are designed to predict someone's status on an external criterion measure. A test has criterion-related validity if it is useful for predicting a person's behavior in a specified situation.

- a. **Concurrent vs. Predictive Validation.** First the term “validation” refers to the procedures used to determine how valid a predictor is. There are two types. In *concurrent* validation, the predictor and criterion data are collected at or about the same time. This kind of validation is appropriate for tests designed to assess a person’s current criterion status. It is good diagnostic screening tests when you want to diagnose. In *Predictive validation*, the predictor scores are collected first and criterion data are collected at some later/future point. This is appropriate for tests designed to assess a person’s future status on a criterion.
- b. **Standard Error of Estimate.** The standard error of estimate (s est) is used to estimate the range in which a person’s true score on a criterion is likely to fall, given his/her score as estimated by a predictor.
- c. **Decision-making.** In many cases when using predictor tests, the goal is to predict whether or not a person will meet or exceed a minimum standard of criterion performance — the criterion cutoff point. When a predictor is to be used in this manner, the goal of the validation study is to set an optimal predictor cutoff score; an examinee who scores at or above the predictor cutoff is predicted to score at or above the criterion cutoff.
- We then get
- true positives* (or valid acceptance): accurately identified by the predictor as meeting the criterion standard
- False positives* (or false acceptance): incorrectly identified by the predictor as meeting the criterion standard.
- True negative* (valid rejection): accurately identified by the predictor as not meeting the criterion standard.
- False negative* (invalid rejection): meets the criterion standard, even though the predictor indicated s/he wouldn’t.
- d. **Factors affecting the criterion-related validity coefficient.** This is about factors that potentially affect the magnitude of the criterion-related validity coefficient. We will not go into this as it relates to correlation.

3. Construct validity

A test has construct validity if it accurately measures a theoretical, non-observable construct or trait. The construct validity of a test is worked out over a period of time on the basis of an accumulation of evidence. There are a number of ways to establish construct validity.

Two methods of establishing a test’s construct validity are convergent/divergent validation and factor analysis.

a. Convergent/divergent validation

A test has *convergent validity* if it has a high correlation with another test that measures the same construct. By contrast, a test’s *divergent validity* is demonstrated through a low correlation with a test that measures a different construct. Note *this is the only case when a low correlation coefficient (b/w two test that measure different traits) provides evidence of high validity.*

The multitrait-multimethod matrix is one way to assess a test’s convergent and divergent validity. We will not get into this right now.

b. Factor analysis

Factor analysis is a complex statistical procedure which is conducted for a variety of purposes, one of which is to assess the construct validity of a test or a number of tests. We will get there.

c. Other methods of assessing construct validity

We can assess the *test's internal consistency*. That is, if a test has construct validity, scores on the individual test items should correlate highly with the total test score. This is evidence that the test is measuring a single construct also *developmental changes* tests measuring certain constructs can be shown to have construct validity if the scores on the tests show predictable developmental changes over time.

And *experimental intervention*, that is if a test has construct validity, scores should change following an experimental manipulation, in the direction predicted by the theory underlying the construct.

4. Relationship between reliability and validity

If a test is unreliable, it cannot be valid.

For a test to be valid, it must be reliable.

However, just because a test is reliable does not mean it will be valid.

Reliability is a necessary but not sufficient condition for validity!

Item Analysis

There are a variety of techniques for performing an item analysis, which is often used, for example, to determine which items will be kept for the final version of a test. Item analysis is used to help “build” reliability and validity “into” the test from the start. Item analysis can be both qualitative and quantitative. The former focuses on issues related to the content of the test, *e.g.* content validity. The latter primarily includes measurement of item difficulty and item discrimination.

Item difficulty an item's difficulty level is usually measured in terms of the percentage of examinees who answer the item correctly. This percentage is referred to as the item difficulty index, or “p”.

Item discrimination refers to the degree to which items differentiate among examinees in terms of the characteristic being measured (*e.g.*, between high and low scorers). This can be measured in many ways. One method is to correlate item responses with the total test score; items with the highest test correlation with the total score are retained for the final version of the test. This would be appropriate when a test measures only one attribute and internal consistency is important.

Interpretation of Test Scores**1. Norm-referenced Interpretation**

Involves comparing an examinee's score to that of others in a normative sample. It provides an indication of where the examinee stands in relation to others who have taken the test. Norm-referenced scores include developmental scores (*e.g.*, mental age scores and grade equivalents), which indicate how far along the normal developmental path an individual has progressed, and within-group norms, which provide a comparison to the scores of other individuals whom the examinee most resembles.

2. Criterion-referenced Interpretation

Involves interpreting an examinee's test score in terms of an external pre-established standard of performance.

Writing test items is a matter of precision, perhaps more akin to computer programming than to writing prose. A test item must focus the attention of the examinee on the principle or construct upon which the item is based. Ideally, students who answer a test item incorrectly will do so because their mastery of the principle or construct in focus was inadequate or incomplete. Any characteristics of a test item which distract the examinee from the major point or focus an item, reduce the effectiveness of that item. Any item answered correctly or incorrectly because of extraneous factors in the item, results in misleading feedback to both examinee and examiner.

A poet or writer, especially of fiction, relies on rich mental imagery on the part of the reader to produce an impact. For item writers, however, the task is to focus the attention of a group of students, often with widely varying background experiences, on a single idea. Such communication requires extreme care in choice of words and it may be necessary to try the items out before problems can be identified.

Items Writing

Essential Characteristics of Item Writers

Given a task of precision communication, there are several attributes or mind sets that are characteristics of a proficient item writer.

Knowledge and Understanding of the Material Being Tested

At the University level, the depth and complexity of the material on which students are tested necessitates that only faculty members fully trained in a particular discipline can write concise, unambiguous test items in that discipline. Further, the number of persons who can meaningfully critique test items, in terms of the principles or constructs involved, is limited. An agreement by colleagues to review each others' tests will likely improve the quality of items considerably prior to the first try-out with students.

Continuous Awareness of Objectives

A test must reflect the purposes of the instruction it is intended to assess. This quality of a test, referred to as content validity, is assured by specifying the nature and/or number of items prior to selecting and writing the items. Instructors sometimes develop a chart or test blueprint to help guide the selection of items. Such a chart may consider the modules or blocks of content as well as the nature of the skills a test is expected to assess.

In the case of criterion-referenced instruction, content validity is obtained by selecting a sample of criteria to be assessed. For content-oriented instruction, a balance may be achieved by selecting items in proportion to the amount of instructional time allotted to various blocks of material. An example of a test blueprint for a fifty-item test is shown below.

	Types of Tests	Reliability	Validity	Correlation	Total
Knowledge of terms	3	1	1	1	5
Comprehension of principles	3	4	4	4	15
Application of principles	2	4	6	5	17
Analysis of situations	1	2	2	2	7
Evaluation of solutions	-	2	2	2	6
Total	8	13	15	14	50

The blueprint specifies the number of items to be constructed for each cell of the two-way chart. For example, in the above test blueprint, four items are to involve the application of the principles of reliability.

Continuous Awareness of Instructional Model

Different instructional models require items of quite different characteristics for adequate assessment. For example, appropriate item difficulty in a mastery-model situation might be a maximum value of 20 (twenty-percent of the students answering incorrectly). On the other hand, items written for a normative model might have an appropriate average difficulty of the order of 30 to 40.

Ideally, item discrimination (the degree to which an item differentiates between students with high test scores and students with low test scores) should be minimal in a mastery-model situation. We would like to have all students obtain high scores. In the normative-model, item discrimination should be as high as possible in order that the total test differentiate among students to the maximum degree.

Understanding of the Students for Whom the Items are Intended

Item difficulty and discrimination are determined as much by the level of ability and range of ability of the examinees as they are by the characteristics of the items. Normative-model items must be written so that they provide the maximum intellectual challenge without posing a psychological barrier to student learning through excessive difficulty. In either the normative or mastery models, item difficulty must not be so low as to provide no challenge whatever to any examinee in a class.

It is generally easier to adjust the difficulty than to adjust the discrimination of an item. Item discrimination depends to a degree on the range of examinee ability as well as on the difficulty of the item. It can be difficult to write mastery-model items which do not discriminate when the range of abilities among examinees is wide. Likewise, homogeneous abilities make it more difficult to write normative-model items with acceptably high discriminations.

No matter what the instructional model or the range of abilities in a class, the only way to identify appropriate items is to select them on the basis of subjective judgment, administer them, and analyze the results. Then only items of appropriate difficulty and discrimination may be retained for future use.

Skill in Written Communication

An item writer's goal is to be clear and concise. The level of reading difficulty of the items must be appropriate for the examinees. Wording must not be more complicated than that used in instruction.

Skill in Techniques of Item Writing

There are many helpful hints and lists of pitfalls to avoid which may be helpful to the item writer. This is an area where measurement specialists may be particularly helpful. The remainder of this hand-out will be devoted to item-writing tips.

Writing Specific Types of Items

The remainder of this handbook will deal with skills helpful in writing specific types of items. There is an almost infinite variety to the forms test items may take. Test items are often grouped into two main categories: objective items and constructed-response items. Objective items are those in which the examinee recognizes a best answer from options presented in the item. Objective items include multiple-choice items, alternative-response items and matching items. Constructed-response items include restricted-response items, short-answer items, completion items and essay items.

Alternative-Response Items

An alternative-response item is a special case of the multiple-choice item format. There are many situations which call for either-or decisions, such as deciding whether a specific solution is right or wrong, whether to continue or to stop, whether to use a singular or plural construction, and so on. For such situations, the alternative response item is an ideal measuring device. Since only two options are possible, alternative-response items are generally shorter, and, therefore, require less reading time. Students may respond to more alternative-response items than other types of items in a given length of time.

A major disadvantage of alternative-response items is the fact that students have fifty-fifty probability of answering the item correctly by chance alone. When carefully written and pretested, alternative-response items may be written which exceed multiple-choice items in ability to discriminate. Generally, however, it is considered necessary to have a larger number of alternative-choice items than of other types of items in order to achieve a given level of test reliability.

There are two main types of alternative-response items. One type is essentially a two-option multiple-choice item. An example follows.

To determine the degree of relationship between two continuous variables, one must compute the: 1. product-moment coefficient of correlation. 2. rank-order coefficient of correlation.

Matching Items

A matching item consists of two columns: one column of stems or problems to be answered, and another column of responses from which the answers are to be chosen. Traditionally, the column of stems is placed on the left and the column of responses is placed on the right.

The Proof of the Item Writing is in the Item Analysis

Careful item writing can result in greatly improved tests. However, there can be no substitute for feedback to the item writer in the form of data from trial runs or actual usage. For even relatively small numbers of students, say twenty or thirty, useful data can be collected by means of optically-scanned sheets. Scan sheets may be picked up at the Scoring Office, 114 Computer Center.

After the test has been administered and the students have marked their responses to the items on the scan sheets, the sheets are returned to the Scoring Office, along with a correct-answer key. The instructor must indicate to the Scoring Office staff that he or she wishes to receive an item analysis. An item analysis is generally available by noon of the next day after the sheets are delivered to the Scoring Office.

TEST STANDARDIZATION

Reliability

If test scores are to be used for scientific purpose, they must be reliable and valid. Test scores are reliable when they are dependable, reproducible, and consistent. Confusing or tricky tests may mean different things to a tested at different times. Tests may be too short to be reliable, or scoring may be too subjective. If a test is inconsistent in its results when measurements are repeated or when it is scored by two people, it is unreliable. A simple analogy is a rubber yardstick. If we did not know how much it stretched each time we took a measurement, the results would be unreliable, no matter how carefully we had marked the measurement. We need reliable tests if we are to use the results with confidence.

Evaluation of Reliability

In order to evaluate reliability, we must secure two independent scores for the same individual on the same test—by treating halves of the test separately, by repeating the test, or by giving it in two different but equivalent forms. If we have such a set of paired scores from a group of individuals, we can determine the test's reliability. If the same relative score levels are preserved on the two measurements, the test is reliable. Some differences are to be expected, owing to errors of measurement, so that an index of degree of relationship between the two sets of scores is needed. This relationship is provided by the coefficient of correlation, already familiar to us as a measure of degree of correspondence between two sets of scores. The coefficient of correlation between the two sets of test scores is a reliability coefficient. Well-constructed psychological tests of ability usually have reliability coefficients of $r = .90$ or above.

Validity

Tests are valid when they measure what they are intended to measure. A college examination in economics full of trick questions might be a test of student intelligence rather than of the economics that was to have been learned in the course. Such an examination might be reliable, but it would not be a valid test of achievement for the course. A test of sense of humour, for example, might be made up of jokes that were hard to catch unless one were both very bright and very well read. Hence it might turn out to be a reliable test of something but still not be valid as a test of sense of humour.

Measurement of Validity

To measure validity, we must also have two scores for each person: the test score and some measure of what the test is supposed to be measuring. This measure is called a criterion. Suppose that a test is designed to predict success in learning to receive telegraphic code. To determine whether the test is valid, it is given to a group of individuals before they start their study of telegraphy. After they have been trained to receive coded

messages, the students are tested on the number of words per minute they can receive. This later measure furnishes an additional set of scores, which serves as a criterion. Now we can obtain a coefficient of correlation between the early test scores and the scores on the criterion. This correlation coefficient is known as a validity coefficient, and it tells something about how valuable a given test is for a given purpose. The higher the validity coefficient, the better the prediction that can be made from an aptitude test.

High validity coefficients are desirable if test scores are to be used to help an individual with an important decision such as vocational choice. But even relatively low validity coefficients may prove useful when large numbers of people are tested. For example, a battery of tests used for the selection of aircrew specialists in the Second World War proved effective in prediction job success, even though some of the validity coefficients for single tests were of very moderate size. Illustrative validity coefficients from this battery are shown in below table. Although no single test showed a validity above .49, the "composite" score derived from the battery of tests correlated .64 with the criterion.

VALIDITY COEFFICIENTS FOR PREDICTION OF SUCCESS IN PILOT TRAINING.

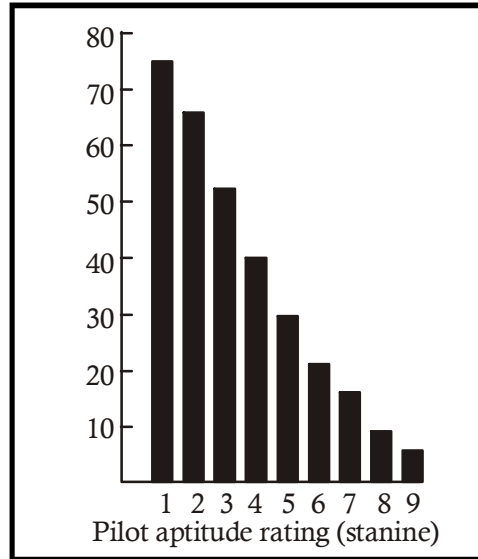
<i>Tests Among Those in Classification Battery</i>	<i>Validity Coefficient</i>
Printed tests with highest validity coefficients	
General information	.49
Instrument comprehension	.46
Mechanical principles	.42
Dial and table reading	.40
Spatial orientation II	.38
Apparatus tests with highest validity coefficients	
Complex coordination	.42
Discrimination reaction time	.41
Rudder control	.36
Two-hand pursuit	.35
Rotary pursuit	.31
Pilot stanine	
(a composite score)	.64

Test Scores (as a Basis for Prediction)

With high reliability and validity coefficients we know the test is satisfactory, but the problem of using the test in prediction still remains. The method of prediction most easily understood is the one based on critical scores. By this method, a critical point on the scale of scores is selected. Only those candidates with scores above the critical point are accepted—for pilot training, for admission to medical school, or for whatever purpose the testing may serve.

The pilot-selection program of the Air Force illustrates this use of critical scores. The composite scores (called stanines) give each candidate a pilot-prediction rating from 1 to 9. From next page figure shows that those with low stanines failed pilot training much more frequently than those with high stanines. After experience with the tests, the examiners eliminated those with stanines below 5 prior to training. Thus a stanine of 5 is a critical score. Had this critical score been adopted before training the candidates repre-

sented in below figure, only 17 per cent of those accepted would have failed to complete training. Those dropped would have been the group of low scorers, 54 per cent of whom failed elementary pilot training.



THE BASIS FOR A CRITICAL SCORE.

Test Norms

Test norms consist of data that make it possible to determine the relative standing of an individual who has taken a test. By itself, a subject's raw score (*e.g.*, the number of answers that agree with the scoring key) has little meaning. Almost always, a test score must be interpreted as indicating the subject's position relative to others in some group. Norms provide a basis for comparing the individual with a group.

Numerical values called centiles (or percentiles) serve as the basis for one widely applicable system of norms. From a distribution of a group's raw scores the percentage of subjects falling below any given raw score can be found. Any raw score can then be interpreted relative to the performance of the reference (or normative) group—eighth-graders, five-year-olds, institutional inmates, job applicants. The centile rank corresponding to each raw score, therefore, shows the percentage of subjects who scored below that point. Thus, 25 percent of the normative group earn scores lower than the 25th centile; and an average called the median corresponds to the 50th centile.

Another class of norm system (standard scores) is based on how far each raw score falls above or below an average score, the arithmetic mean. One resulting type of standard score, symbolized as z , is positive (*e.g.*, +1.69 or +2.43) for a raw score above the mean and negative for a raw score below the mean. Negative and fractional values can, however, be avoided in practice by using other types of standard scores obtained by multiplying z scores by an arbitrarily selected constant (say, 10) and by adding another constant (say, 50, which changes the z score mean of zero to a new mean of 50). Such changes of constants do not alter the essential characteristics of the underlying set of z scores.

The French psychologist Alfred Binet, in pioneering the development of tests of intelligence, listed test items along a normative scale on the basis of the chronological age

(actual age in years and months) of groups of children that passed them. A mental-age score (*e.g.*, seven) was assigned to each subject, indicating the chronological age (*e.g.*, seven years old) in the reference sample for which his raw score was the mean. But mental age is not a direct index of brightness; a mental age of seven in a 10-year-old is different from the same mental age in a four-year-old.

To correct for this, a later development was a form of IQ (intelligence quotient), computed as the ratio of the subject's mental age to his chronological age, multiplied by 100. (Thus, the IQ made it easy to tell if a child was bright or dull for his age.)

Ratio IQs for younger age groups exhibit means close to 100 and spreads of roughly 45 points above and below 100. The classical ratio IQ has been largely supplanted by the deviation IQ, mainly because the spread around the average has not been uniform due to different ranges of item difficulty at different age levels. The deviation IQ, a type of standard score, has a mean of 100 and a standard deviation of 16 for each age level. Practice with the Stanford-Binet test reflects the finding that average performance on the test does not increase beyond age 18. Therefore, the chronological age of any individual older than 18 is taken as 18 for the purpose of determining IQ.

The Stanford-Binet has been largely supplanted by several tests developed by the American psychologist David Wechsler between the late 1930s and the early 1960s. These tests have subtests for several capacities, some verbal and some operational, each subtest having its own norms. After constructing tests for adults, Wechsler developed tests for older and for younger children.

AREAS OF TESTING

Tests designed to measure capacities, that is, to predict what one can accomplish with training, are called aptitude tests; they include tests of general intelligence as well as tests of special abilities. Tests that tell what one can do now are achievement tests. An intelligence test that predicts how well you will do in college is an aptitude test; examinations given at the end of a course to see how much you have learned are achievement tests. Both are ability tests.

Tests of General Intelligence

The first tests designed to measure intelligence were developed by sir Francis Galton. But for present use Galton's tests are useless.

The intelligence test as we know it today was formulated by the French psychologist Alfred Binet (1857-1911). The French government asked Binet to devise a test that would detect those children too slow intellectually to profit from regular schooling. He assumed that intelligence should be measured by tasks requiring reasoning and problem solving, rather than perceptual-motor skills. In collaboration with Theodore Simon (1873-1961), another French psychologist, Binet published a scale in 1905, which he revised in 1908 and again in 1911. These Binet scales are the direct predecessors of contemporary intelligence tests.

Binet's Method--- (A Mental Age Scale)

Binet devised a scale of units of mental age. Average mental-age (MA) scores correspond to chronological age (CA), that is, to the age determined from the date of birth. A bright child's MA is above his CA; a dull child has an MA below his CA. The mental-age scale is

easily interpreted by teachers and others who deal with children differing in mental ability.

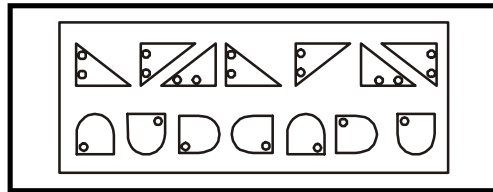
Item Selection

There are two chief ways to find items for which success is uninfluenced by special training. One way is to choose novel items with which an untaught child has as good a chance to succeed as one who has been taught at home or in school. Below figure illustrates novel items. In this particular case the child is asked to choose figures that are alike with the assumption that the designs are unfamiliar to all children. The second way is to choose familiar items, with the assumption that all those for whom the test is designed have had the requisite prior experience to deal with the items. The following problem provides an example of a supposedly familiar item:

Mark F if the sentence is foolish; mark S if it is sensible.

S F Mrs. Smith has had no children, and I understand that the same was true of her mother.

This item is “fair” only for children who know the English language, who can read, and who understand all the words in the sentence. For such children, detection of the fallacy in the statement becomes a valid test of intellectual ability.



NOVEL ITEMS USED IN INTELLIGENCE TESTS.

Many of the items on an intelligence test of the Binet type assume general familiarity. A vocabulary test, for example, appears in almost all the scales. Familiarity with the standard language of the test is necessarily assumed.

The intelligence test is in some respects a crude instrument, for its assumptions can never be strictly met. The language environment of one home is never exactly that of another, the reading matter available to the subjects differs, and the stress upon cognitive goals varies. Even the novel items depend upon perceptual discriminations that may be acquired in one culture or subculture and not in another. Despite the difficulties, items can be chosen that work reasonably well. The items included in contemporary intelligence tests are those that have survived in practice after many others have been tried and found defective. It should be remembered, however, that intelligence tests have been validated by success in predicting school performance within a particular culture.

Item Testing

It is not enough to look at an item and to decide that it requires intelligence to answer it successfully. Some “tricky” or “clever” items turn out to be poor because of the successes or failures that occur through guessing. More pedestrian items, such as matters of common information, sometimes turn out to be most useful. These are “fair” items if all have had a chance to learn the answers.

One method of testing an item is to study the changes in proportions of children answering it correctly at different ages. Unless older children are more successful than

younger ones in answering the item, the item is unsatisfactory in a test based on the concept of mental growth.

A second method of testing an item is to find out whether the results for it correspond to the results on the test as a whole. This can be done by correlating success and failure on the item with the score made on the remaining items. If all items measure something in common, then every single item ought to contribute a score that correlates with the total score.

These two requirements for an acceptable item (increase in percentage passing with age and correlation with total score) reflect both validity and reliability. The first requirement is an indirect way of guaranteeing validity, being based on the inference that what we mean by intelligence should distinguish an older child from a younger one; the second requirement is a guarantee of reliability through internal consistency of the measures.

Contemporary Binet Tests

The tests originally developed by Binet underwent several revisions in this country, the first by Goddard in 1911. For many years the best-known and most widely used revision was that made by Terman at Stanford University in 1916, commonly referred to as the Stanford-Binet. The test was revised in 1937, 1960, and 1972.

In the Binet tests, an item is age-graded at the level at which a substantial majority of the children pass it. The present Stanford-Binet has six items of varied content assigned to each year, each item when passed earning a score of two months of mental age.

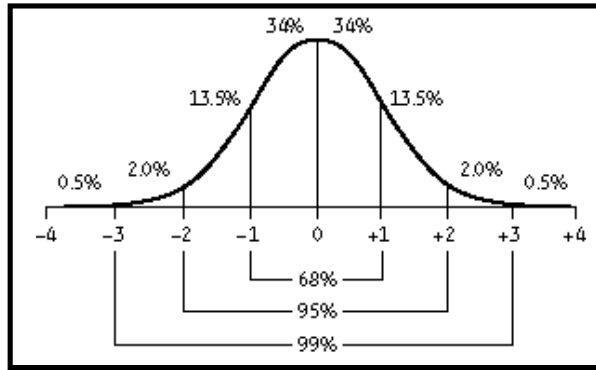
The procedure for testing is first to establish the child's basal mental age, the mental-age level at which he passes all items. Two months of mental age are then added for each item passed at higher age levels. Consider, for example, the child who passes all items at the mental-age level of six years. If the child then passes two items at the seven-year level, four months are added; passing an additional item at the eight-year level adds two more months. This particular child will have an earned mental age of six years and six months, regardless of chronological age. The test allows for some unevenness in development, so that two children can earn the same mental age by passing different items on the test.

David Wechsler published the first test for assessing intelligence in adults in 1939. The Wechsler Adult Intelligence Scale contains many items that assess nonverbal reasoning ability and therefore depends less on verbal ability than does the Stanford-Binet. It also provides separate scores of verbal intelligence and nonverbal or performance intelligence, as well as a score that indicates overall intelligence.

The term intelligence quotient, or IQ, is also used to describe the score on the Wechsler test. However, the Wechsler test presented scores based on a normal distribution of data rather than the intelligence quotient. The normal distribution is a symmetrical bell-shaped curve that represents how characteristics like IQ are distributed in a large population. In this scoring system, the mean IQ score is set at 100, and the standard deviation is set at 15. The test is constructed so that about two-thirds of people tested (68 percent) will score within one standard deviation of the mean, or between 85 and 115.

On the Wechsler test, the IQ score reflects where a person falls in the normal distribution of IQ scores. Therefore, this score, like the original Stanford-Binet IQ score, is a relative score, indicating how the test taker's score compares to the scores of other people. Most current intelligence tests, including the revised versions of the Stanford-Binet, now have scoring systems based on the normal distribution. About 95 percent of the popula-

tion will score between 70 and 130 (within two standard deviations from the mean), and about 99.7 percent of the population will score between 55 and 145 (within three standard deviations from the mean).



GROUP INTELLIGENCE TESTS

Individual intelligence tests can be given only by specially trained psychologists. Such tests are expensive and time-consuming to administer, and so educational institutions often use tests that can be given to a group of people at the same time. Commonly used group intelligence tests include the Otis-Lennon School Ability Test and the Lorge-Thorndike Intelligence Test.

Biological Tests of Intelligence

Some researchers have suggested that biological indices such as reaction time and perceptual speed relate to intelligence as measured by IQ tests:

Reaction time the amount of time a subject takes to respond to a stimulus, such as by pushing a button when a light is presented.

Perceptual speed the amount of time a person takes to accurately perceive and discriminate between stimuli. For example, a test of perceptual speed might require a person to determine which of two lines is shorter when pairs of lines flash very briefly on a screen.

The Influence of Culture

Many psychologists believe that cultural bias can affect intelligence tests, for the following reasons:

Tests that are constructed primarily by white, middle-class researchers may not be equally relevant to people of all ethnic groups and economic classes.

Cultural values and experiences can affect factors such as attitude toward exams, degree of comfort in the test setting, motivation, competitiveness, rapport with the test administrator, and comfort with problem solving independently rather than as part of a team effort.

Cultural stereotypes can affect the motivation to perform well on tests.

Characteristics of IQ Tests

Some characteristics of IQ tests are standardization, norms, percentile scores, standardization samples, reliability, and validity.

Standardization

Intelligence tests are standardized, which means that uniform procedures are used when administering and scoring the tests. Standardization helps to ensure that people taking a particular test all do so under the same conditions. Standardization also allows test takers to be compared, since it increases the likelihood that any difference in scores between test-takers is due to ability rather than the testing environment. The SAT and ACT are two examples of standardized tests.

Norms and Percentile Scores

Researchers use norms when scoring the tests. Norms provide information about how a person's test score compares with the scores of other test takers. Norms allow raw test scores to be converted into percentile scores. A percentile score indicates the percentage of people who achieved the same as or less than a particular score. For example, if someone answered twenty items correctly on a thirty-item vocabulary test, he receives a raw score of 20. He consults the test norms and finds that a raw score of 20 corresponds with a percentile score of 90. This means that he scored the same as or higher than 90 percent of people who took the same test.

Standardization Samples

Psychologists come up with norms by giving a test to a standardization sample. A standardization sample is a large group of people that is representative of the entire population of potential test takers.

Reliability

Most intelligence tests have good reliability. Reliability is a test's ability to yield the same results when the test is administered at different times to the same group of people.

Validity

Validity is a test's ability to measure what it is supposed to measure. Although intelligence tests cannot be considered good measures of general intelligence or general mental ability, they are reasonably valid indicators of the type of intelligence that enables good academic performance.

Critical Views on Intelligence Testing

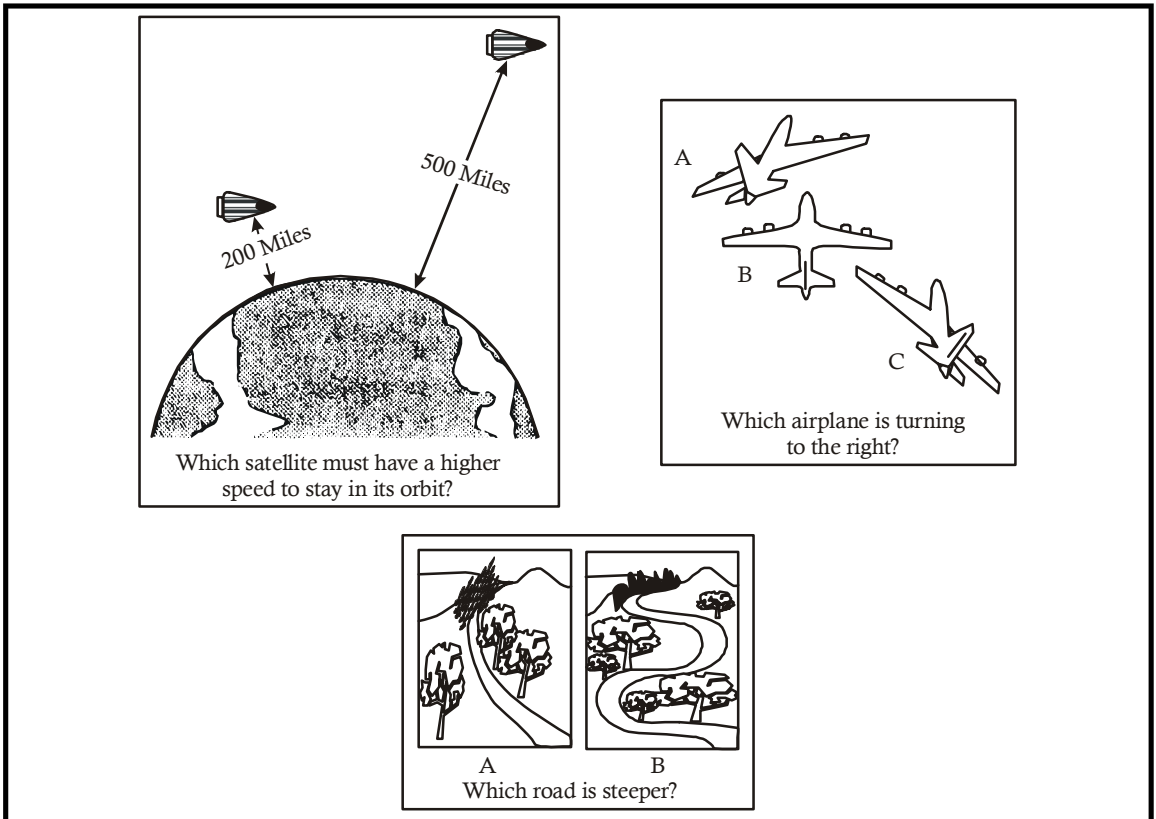
Critics of widespread intelligence testing point out that politicians and the public in general misuse and misunderstand intelligence tests. They argue that these tests provide no information about how people go about solving problems. Also, say the critics, these tests do not explain why people with low intelligence scores can function intelligently in real-life situations. Advocates of intelligence testing point out that such tests can identify children who need special help, as well as gifted children who can benefit from opportunities for success.

Aptitude Tests

Aptitude tests designed to predict performance over a broad range of abilities are called intelligence tests. Other aptitude tests measure more specific abilities; mechanical aptitude tests measure various types of eye-hand coordination; musical aptitude tests measure discrimination of pitch, rhythm, and other aspects of musical sensitivity that are

predictive of musical performance with training; and clerical aptitude tests measure efficiency at number-checking and other skills that have been found to be predictive of an individual's later achievement as an office clerk. Many aptitude tests have been constructed to predict success in specific jobs or vocations. Since the Second World War the armed forces have devised tests to select pilots, radio technicians, submarine crews, and many other specialists.

Aptitude is usually measured by a combination of tests. Pilot aptitude tests include not only measures of mechanical knowledge but also tests of spatial orientation, eye-hand coordination, and other skills. A combination of tests used for prediction is known as a test battery. Scores from individual tests are weighted to get the best possible prediction. Scores on the tests that predict well count more than scores on tests that predict less well. If an eye-hand coordination test predicts pilot success better than a spatial orientation test, scores in eye-hand coordination will be weighted more heavily than scores in spatial orientation.



ITEMS FROM A TEST TO MEASURE MECHANICAL COMPREHENSION.

Intelligence Quotient (IQ)

Terman adopted a convenient index of brightness that was suggested by the German psychologist William Stern (1871-1938). This index is the intelligence quotient, commonly known by its initials IQ. It expresses intelligence as a ratio of mental age to chronological age:

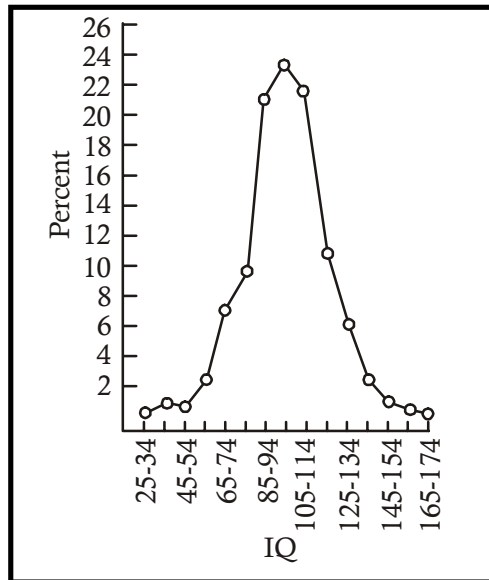
$$IQ = \frac{\text{Mental age (MA)}}{\text{Chronological age (CA)}} \times 100$$

The 100 is used as a multiplier to remove the decimal point and to make the IQ have a value of 100 when MA equals CA. It is evident that if the MA lags behind the CA, the resulting IQ will be less than 100; if the MA is above the CA, the IQ will be above 100.

INTERPRETATION OF INTELLIGENCE QUOTIENTS ON THE STANFORD-BINET.

<i>IQ</i>	<i>Verbal Description</i>	<i>Per cent in each Group</i>
Above 139	Very superior	1
120-139	Superior	11
110-119	High average	18
90-109	Average	46
80-89	Low average	15
70-79	Borderline	6
Below 70	Mentally retarded	3
		100%

The distribution of IQs follows the form of curve found for many differences among individuals, such as differences in height; this is the bell-shaped “normal” distribution curve shown in Figure. In this curve most cases cluster around a mid value, tapering off to a few at both extremes. The adjectives commonly used to describe the various IQ levels are given in above table.



A NORMAL DISTRIBUTION CURVE OF IQS

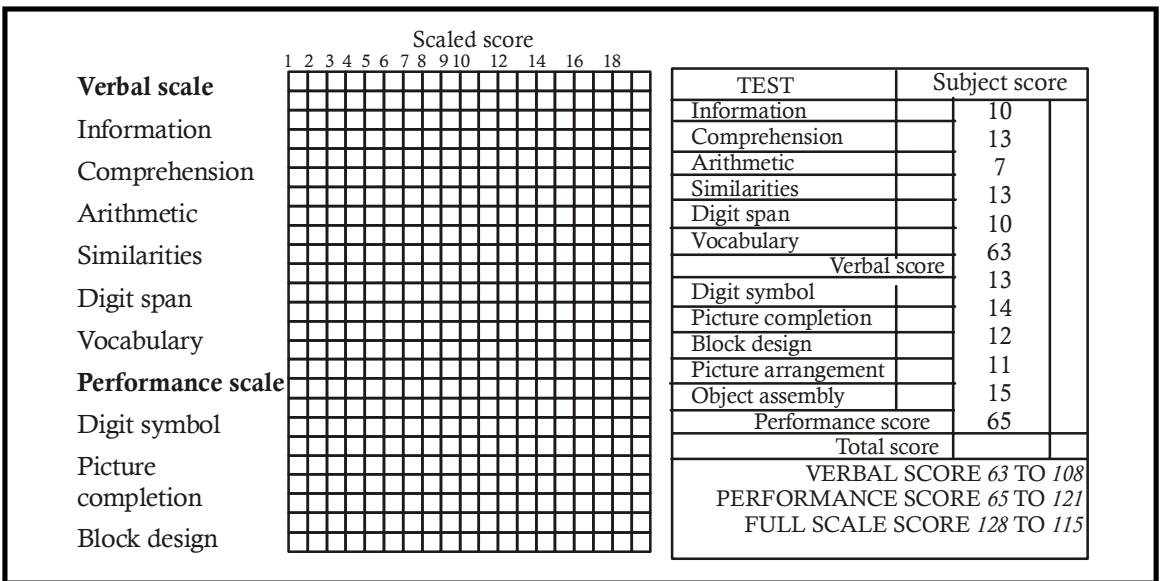
In the 1960 and subsequent revisions of the Stanford-Binet, the authors introduced a method of computing the IQ from tables. The meaning of an IQ remains essentially the same as before, but the tables permit corrections to allow the IQ at any age to be interpreted somewhat more exactly. It is now arranged so that for each age the IQ averages 100 and has a standard deviation of 16.

A modern IQ is merely a test score adjusted for the age of the person being tested. It is, therefore, no longer a “quotient” at all, but the expression IQ persists because of its familiarity and convenience.

Tests More Than One Scale

One way to obtain information on specific kinds of abilities, rather than a single mental-age score, is to separate the items into more than one group and to score the groups separately. The Wechsler Adult Intelligence Scale and the Wechsler Intelligence Scale for Children use items similar to those in the Binet tests, but they divide the total test into two parts – a verbal scale and a performance scale – according to the content of the items. A performance item is one that requires manipulation or arrangement of blocks, beads, pictures, or other materials in which both stimuli and responses are nonverbal. The separate scaling of the items within one test is convenient for diagnostic purposes. Below figure shows a test profile and how the test scores are summed to yield IQs.

In general, the full scale (verbal and performance) and the verbal scale of the Wechsler Scales yield scores corresponding closely to the scores of the Stanford-Binet. The verbal scale of the Wechsler correlates .77 with its performance scale.



PROFILE FOR WECHSLER ADULT INTELLIGENCE SCALE (WAIS)

TESTS COMPRISING THE WECHSLER ADULT INTELLIGENCE SCALE

Test	Description
Verbal Scale	
Information	Questions tap general range of information; <i>e.g.</i> , “How many weeks in a year ?”
Comprehension	Tests practical information and ability to evaluate past experience; <i>e.g.</i> , “How would you find your way out if lost in a forest ?”
Arithmetic	Verbal problems testing arithmetic reasoning.

<i>Test</i>	<i>Description</i>
Similarities	Asks in what way certain objects or concepts (<i>e.g.</i> , egg and seed) are similar; measures abstract thinking.
Digit Span	Series of digits presented auditorily (<i>e.g.</i> , 7-5-6-3-8) are repeated in a forward or backward direction. Tests attention and rote memory.
Vocabulary	Tests word knowledge.
Performance Scale	
Digit Symbol	A timed coding task in which numbers must be associated with marks of various shapes; tests speed; tests speed of learning and writing.
Picture Completion	The missing part of an incompletely drawn picture must be discovered and named; tests visual alertness and visual memory.
Block Design	Pictured designs must be copied with blocks, tests ability to perceive and analyse patterns.
Picture Arrangement	A series of comic strip type pictures must be arranged in the right sequence to tell a story; tests understanding of social situations.
Object Assembly	Puzzle pieces must be assembled to form a complete object such as a human profile or elephant; tests ability to deal with part-whole relationships.

Personality Tests

We have linked different ways of assessing personality with different theoretical conceptions of personality. This relationship, however, is a little forced. The various ways of looking at personality – trait, psychoanalytic, social learning, and humanistic theories—have led, because of the nature of their constructs, to different approaches to assessment. But methods developed by one group have been taken over and modified by other groups. Humanistic theorists, for example, may use projective techniques such as the Rorschach and TAT (designed originally to uncover hidden motives) to measure creative potential. The MMPI, a personality inventory that measures traits, had widespread use in research guided by all four theoretical viewpoints. Of these personality tests, projective tests have already been discussed. We shall take account of only MMPI under this heading.

MMPI

The MMPI was developed to aid clinicians in the psychiatric diagnosis of abnormal personality types. The test is composed of some 550 statements (about attitudes, emotional reactions, physical and psychological symptoms, and past experiences) to which the subject answers “true,” “false,” or “cannot say.” Some of the items are:

I have never done anything dangerous for the thrill of it.

I daydream very little.

My mother or father often made me obey even when I thought it was unreasonable.

At times my thoughts have raced ahead faster than I could speak them.

The responses are scored according to the correspondence between the answers given by the subject and those given by patients with different kinds of psychological disturbances.

Since the MMPI is derived from differences between criterion groups, it does not really matter whether what the person says about himself is true or not. The fact that he says it is important. If all schizophrenics answer true and all normal subjects false to the statement "My mother never loved me," their answers distinguish the two groups, regardless of how mother actually behaved. This is one of the advantages of a test based on the method of empirical construction as opposed to one where the test constructor assumes that certain answers indicate certain psychological disorders. Answering true to the statement "I think that most people would lie to get ahead" might be assumed to be a sign of paranoia. When this item was included in the MMPI it was found that patients diagnosed as paranoid were significantly less apt to respond true to this statement than normal individuals.

1. Like mechanics magazines.
2. I have a good appetite.
3. I woke up fresh and rested most mornings.
4. I think I would like the work of a librarian.
5. I am easily awakened by noise.
6. I like to read newspaper articles on crime
7. My hands and feet are usually warm enough.
8. My daily life is full of things that keep me interested.
9. I am about as able to work as I ever was.
10. There seems to be a lump in my throat much of the time.
11. A person should try to understand his dreams and be guided by or take warning from them.
12. I enjoy detective or mystery stories.
13. I work under a great deal of tension.
14. I have diarrhea once a month or more.
15. Once in a while I think of things too bad to talk about.

MMPI (Sample Page)

Although the original MMPI scales were derived from comparison groups suffering from personality disorders often severe enough to require hospitalization, they have been widely used in the study of other populations.

Attitude Scales

Various devices have been incorporated into personality inventories to compensate for the individual's "test-taking attitude," but none have proved completely satisfactory.

One way to avoid this problem is to have someone else evaluate the individual, either on the basis of what they know or by observing him in certain situations. To put this kind of judgment into a standardized form, rating scales are used. A rating scale is a device for recording judgments about a trait.

In order for the rating to be meaningful, the rater must (1) understand the scale; (2) be sufficiently acquainted with the person being rated so that meaningful judgments can be made; and (3) avoid the “halo effect” – the tendency to rate a person in a favourable direction on all traits because of a good impression made on one or two (or, conversely, to rate unfavourably because of a poor impression made on one or two traits).

Unless the rater knows the person being rated fairly well, or the behaviour being rated is very specific, ratings may be influenced by social stereotypes. That is, the rater may base his ratings as much on how he believes a “suburban housewife,” a “long-haired intellectual,” or a “high school athlete” acts and thinks as on the actual behaviours of the subject being rated. Despite such problems, descriptions of a person provided by different raters in different situations often yield good agreement. In one study, for example, the aggressiveness of a group of school boys between the ages of eight and twelve was rated by their peers and by observers who watched them playing games in the school yard. There was close agreement between the two sets of ratings.

SOME EXAMPLES OF RATING SCALES

How would you describe the individual's self-confidence?	Considers himself incapable of much success	Underestimates his own abilities	Knows just what he is capable of	Exaggerates his abilities	Judges himself capable of anything
Place a check at the point that describes the individual's poise.	Nervous and ill at ease	Somewhat tense easily upset	Average poise and security	Sure of himself adapts well to crises	Composed;
How would you rate the subject's emotional control	Very low	Below average	Average	Above average	Very high
Does the individual antagonize others?	Never	Rarely	Sometimes	Often	Always
How would you rate the parent's behaviour toward the child	No control of emotion in response to child's behaviour	Controlled more by emotion than reason in dealing with child	Emotion freely expressed but behaviour controlled	Remains calm and objective toward child despite provocation	Never shows any signs of emotion always controlled
Does the individual need constant prodding or proceed with his work without being told?	Needs much prodding in doing ordinary assignments	Needs occasional prodding	Does ordinary assignments of own accord	Completes suggested supplementary work	Seeks and sets for himself additional tasks

Four types of scales are generally used :

1. Nominal Scale

This is a very simple scale. It consists of assignment of facts/choices to various alternative categories which are usually exhaustive as well mutually exclusive. These scales are just numerical and are the least restrictive of all the scales. Instances of Nominal Scale are - credit card numbers, bank account numbers, employee id numbers etc. It is simple and widely used when relationship between two variables is to be studied. In a

Nominal Scale numbers are no more than labels and are used specifically to identify different categories of responses. Following example illustrates -

What is your gender?

- ☐ Male
- ☐ Female-

Another example is - a survey of retail stores done on two dimensions - way of maintaining stocks and daily turnover.

How do you stock items at present?

- ☐ By product category
- ☐ At a centralized store
- ☐ Department wise
- ☐ Single warehouse

Daily turnover of consumer is?

- ☐ Between 100 - 200
- ☐ Between 200 - 300
- ☐ Above 300

A two way classification can be made as follows

Daily/Stock Turnover Method	Product Category	Department wise	Centralized Store	Single Ware house
100 - 200				
200 - 300				
Above 300				

Mode is frequently used for response category.

2. Ordinal Scale

Ordinal scales are the simplest attitude measuring scale used in Marketing Research. It is more powerful than a nominal scale in that the numbers possess the property of rank order. The ranking of certain product attributes/benefits as deemed important by the respondents is obtained through the scale.

Example 1. Rank the following attributes (1 - 5), on their importance in a microwave oven.

1. Company Name
2. Functions
3. Price
4. Comfort
5. Design

The most important attribute is ranked 1 by the respondents and the least important is ranked 5. Instead of numbers, letters or symbols too can be used to rate in a ordinal scale. Such scale makes no attempt to measure the degree of favourability of different rankings.

Example 2. If there are 4 different types of fertilizers and if they are ordered on the basis of quality as Grade A, Grade B, Grade C, Grade D is again an Ordinal Scale.

Example 3. If there are 5 different brands of Talcom Powder and if a respondent ranks them based on say, "Freshness" into Rank 1 having maximum Freshness Rank 2 the second maximum Freshness, and so on, an Ordinal Scale results.

Median and mode are meaningful for ordinal scale.

3. Interval Scale

Herein the distance between the various categories unlike in Nominal, or numbers unlike in Ordinal, are equal in case of Interval Scales. The Interval Scales are also termed as Rating Scales. An Interval Scale has an arbitrary Zero point with further numbers placed at equal intervals. A very good example of Interval Scale is a Thermometer.

Illustration. How do you rate your present refrigerator for the following qualities.

Company Name	Less Known	1	2	3	4	5	Well Known
Functions	Few	1	2	3	4	5	Many
Price	Low	1	2	3	4	5	High
Design	Poor	1	2	3	4	5	Good
Overall Satisfaction	Very Dis-Satisfied	1	2	3	4	5	Very Satisfied

Such a scale permits the researcher to say that position 5 on the scale is above position 4 and also the distance from 5 to 4 is same as distance from 4 to 3. Such a scale however does not permit conclusion that position 4 is twice as strong as position 2 because no zero position has been established. The data obtained from the Interval Scale can be used to calculate the Mean scores of each attributes over all respondents. The Standard Deviation (a measure of dispersion) can also be calculated.

4. Ratio Scale

Ratio Scales are not widely used in Marketing Research unless a base item is made available for comparison. In the above example of Interval scale, a score of 4 in one quality does not necessarily mean that the respondent is twice more satisfied than the respondent who marks 2 on the scale. A Ratio scale has a natural zero point and further numbers are placed at equally appearing intervals. For example scales for measuring physical quantities like - length, weight, etc.

The ratio scales are very common in physical scenarios. Quantified responses forming a ratio scale analytically are the most versatile. Ratio scale possess all the characteristics of an interval scale, and the ratios of the numbers on these scales have meaningful interpretations. Data on certain demographic or descriptive attributes, if they are obtained through open-ended questions, will have ratio-scale properties.

Consider the following questions :

Q 1) What is your annual income before taxes? _____ \$

Q 2) How far is the Theater from your home ? _____ miles

Answers to these questions have a natural, unambiguous starting point, namely zero. Since starting point is not chosen arbitrarily, computing and interpreting ratio makes sense. For example we can say that a respondent with an annual income of \$ 40,000 earns twice as much as one with an annual income of \$ 20,000.

Self Rating Scales

1. Graphic Rating Scale

The respondents rate the objects by placing a mark at the appropriate position on a line that runs from one extreme of the criterion variable to another.

Example

0	1	5	7
(poor quality)	(bad quality)	(neither good nor bad)	(good quality)

BRAND 1

This is also known as continuous rating scale. The customer can occupy any position. Here one attribute is taken ex-quality of any brand of ice-cream.

poor	good
------	------

BRAND 2

This line can be vertical or horizontal and scale points may be provided. No other indication is there on the continuous scale. A range is provided. To quantify the responses to question that "indicate your overall opinion about ice-cream Brand 2 by placing a tick mark at appropriate position on the line", we measure the physical distance between the left extreme position and the response position on the line.; the greater the distance, the more favourable is the response or attitude towards the brand.

Its limitation is that coding and analysis will require substantial amount of time, since we first have to measure the physical distances on the scale for each respondent.

2. Itemized Rating Scales

These scales are different from continuous rating scales. They have a number of brief descriptions associated with each category. They are widely used in Marketing Research. They essentially take the form of the multiple category questions. The most common are - Likert, Semantic, Staple and Multiple Dimension. Others are - Thurston and Guttman.

a. Likert Scale

It was developed Rensis Likert. Here the respondents are asked to indicate a degree of agreement and disagreement with each of a series of statement. Each scale item has 5 response categories ranging from strongly agree and strongly disagree.

5	4	3	2	1
Strongly agree	Agree	Indifferent	Disagree	Strongly disagree

Each statement is assigned a numerical score ranging from 1 to 5. It can also be scaled as -2 to +2.

-2	-1	0	1	2
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For example quality of Mother Dairy ice-cream is poor then Not Good is a negative statement and Strongly Agree with this means the quality is not good.

Each degree of agreement is given a numerical score and the respondents total score is computed by summing these scores. This total score of respondent reveals the particular opinion of a person.

Likert Scale are of ordinal type, they enable one to rank attitudes, but not to measure the difference between attitudes. They take about the same amount of efforts to create

as Thurston scale and are considered more discriminating and reliable because of the larger range of responses typically given in Likert scale.

A typical Likert scale has 20-30 statements. While designing a good Likert Scale, first a large pool of statements relevant to the measurement of attitude has to be generated and then from the pool statements, the statements which are vague and non-discriminating have to be eliminated.

Thus, likert scale is a five point scale ranging from 'strongly agreement' to 'strongly disagreement'. No judging gap is involved in this method.

b. Semantic Differential Scale

This is a seven point scale and the end points of the scale are associated with bipolar labels.

1							7
Unpleasant	2	3	4	5	6		Pleasant
Submissive							Dominant

Suppose we want to know personality of a particular person. We have options:

1. Unpleasant/Submissive
2. Pleasant/Dominant

Bi-polar means two opposite streams. Individual can score between 1 to 7 or -3 to 3. On the basis of these responses profiles are made. We can analyse for two or three products and by joining these profiles we get profile analysis. It could take any shape depending on the number of variables.

Profile Analysis

-----/-----
 -----/-----
 -----/-----

Mean and median are used for comparison. This scale helps to determine overall similarities and differences among objects.

When Semantic Differential Scale is used to develop an image profile, it provides a good basis for comparing images of two or more items. The big advantage of this scale is its simplicity, while producing results compared with those of the more complex scaling methods. The method is easy and fast to administer, but it is also sensitive to small differences in attitude, highly versatile, reliable and generally valid.

c. Stapel's Scale

It was developed by Jan Stapel. This scale has some distinctive features:

- i. Each item has only one word/phrase indicating the dimension it represents.
- ii. Each item has ten response categories.
- iii. Each item has an even number of categories.
- iv. The response categories have numerical labels but no verbal labels.

For example, in the following items, suppose for quality of ice cream, we ask respondents to rank from +5 to -5. Select a plus number for words which best describe the ice cream accurately. Select a minus number for words you think do not describe the ice cream quality accurately. Thus, we can select any number from +5, for words we think are very accurate, to -5, for words we think are very inaccurate. This scale is usually presented vertically.

+5

+4

+3

+2

+1

High Quality

-1

-2

-3

-4

-5

This is a unipolar rating scale.

Multi Dimensional Scaling

It consists of a group of analytical techniques which are used to study consumer attitudes related to perceptions and preferences. It is used to study-

The major attributes of a given class of products perceived by the consumers in considering the product and by which they compare the different ranks.

- I. To study which brand competes most directly with each other.
- II. To find out whether the consumers would like a new brand with a combination of characteristics not found in the market.
- III. What would be the consumers ideal combination of product attributes.
- IV. What sales and advertising messages are compatible with consumers brand perceptions.

It is a computer based technique. The respondents are asked to place the various brands into different groups like similar, very similar, not similar, and so on. A goodness of fit is traded off on a large number of attributes. Then a lack of fit index is calculated by computer program. The purpose is to find a reasonably small number of dimensions which will eliminate most of the stress. After the configuration for the consumer's preference has been developed, the next step is to determine the preference with regards to the product under study. These techniques attempt to identify the product attributes that are important to consumers and to measure their relative importance.

This scaling involves a unrealistic assumption that a consumer who compares different brands would perceive the differences on the basis of only one attribute. For example, what are the attributes for joining M.Com course. The responses may be -to do PG, to go into teaching line, to get knowledge, appearing in the NET. There are a number of attributes, you can not base decision on one attribute only. Therefore, when the consumers are choosing between brands, they base their decision on various attributes. In practice, the perceptions of the consumers involve different attributes and any one consumer perceives each brand as a composite of a number of different attributes. This is a short-coming of this scale.

Whenever we choose from a number of alternatives, go for multi- dimensional scaling. There are many possible uses of such scaling like in market segmentation, product life cycle, vendor evaluations and advertising media selection.

The limitation of this scale is that it is difficult to clearly define the concept of similarities and preferences. Further the distances between the items are seen as different

Thurston Scales

These are also known as equal appearing interval scales. They are used to measure the attitude towards a given concept or construct. For this purpose a large number of statements are collected that relate to the concept or construct being measured. The judges rate these statements along an 11 category scale in which each category expresses a different degree of favourableness towards the concept. The items are then ranked according to the mean or median ratings assigned by the judges and are used to construct questionnaire of twenty to thirty items that are chosen more or less evenly across the range of ratings. The statements are worded in such a way so that a person can agree or disagree with them. The scale is then administered to assemble of respondents whose scores are determined by computing the mean or median value of the items agreed with. A person who disagrees with all the items has a score of zero. So, the advantage of this scale is that it is an interval measurement scale. But it is the time consuming method and labour intensive. They are commonly used in psychology and education research.

Guttman Scales/Scalogram Analysis

It is based on the idea that items can be arranged along a continuum in such a way that a person who agrees with an item or finds an item acceptable will also agree with or find acceptable all other items expressing a less extreme position. For example - Children should not be allowed to watch indecent programmes or government should ban these programmes or they are not allowed to air on the television. They all are related to one aspect.

In this scale each score represents a unique set of responses and therefore the total score of every individual is obtained. This scale takes a lot of time and effort in development.

They are very commonly used in political science, anthropology, public opinion, research and psychology.

The Q Sort technique

It is used to discriminate among large number of objects quickly. It uses a rank order procedure and the objects are sorted into piles based on similarity with respect to some criteria. The number of objects to be sorted should be between 60-140 approximately. For example, here we are taking nine brands. On the basis of taste we classify the brands into tasty, moderate and non tasty.

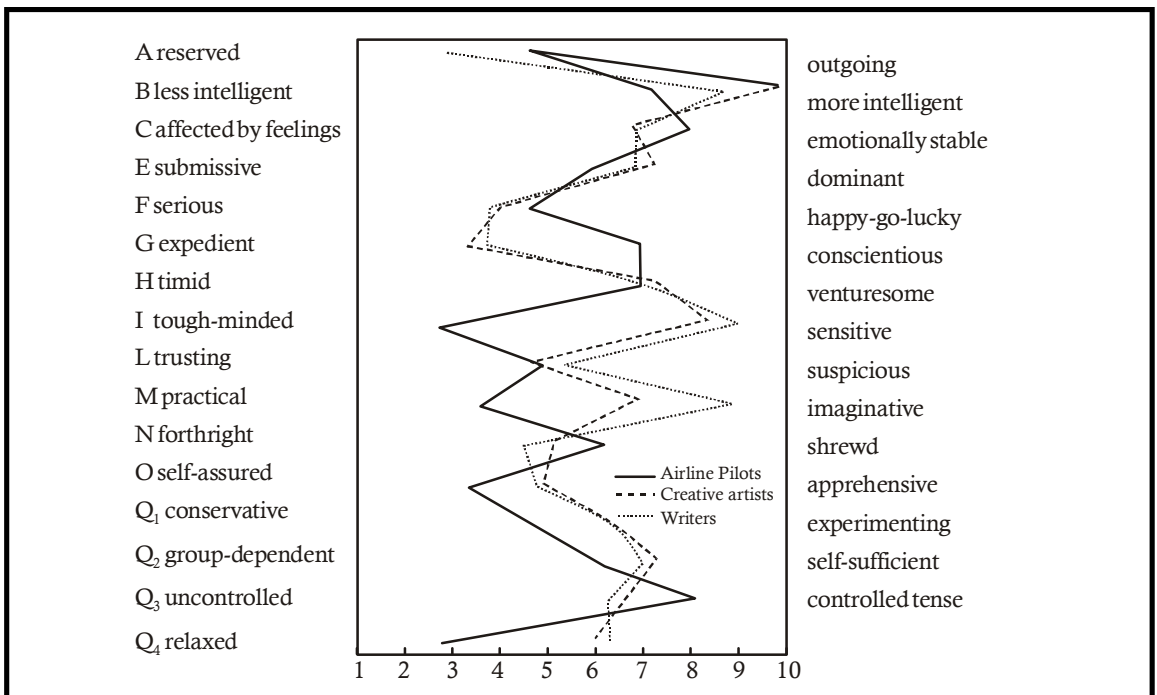
We can classify on the basis of price also-Low, medium, high. Then we can attain the perception of people that whether they prefer low priced brand, high or moderate. We can classify sixty brands or pile it into three piles. So the number of objects is to be placed in three piles-low, medium or high.

Thus, the Q-sort technique is an attempt to classify subjects in terms of their similarity to attribute under study.

Interest Inventories

A personality inventory is essentially a questionnaire in which the person reports reactions or feelings in certain situations. It resembles a very set or standardized kind of interview. A personality inventory asks the same questions of each person, and the answers are usually given in a form that can be easily scored—often by machine. A personality inventory may be designed to measure a single dimension of personality (such as introversion-extraversion) or it may measure several personality traits simultaneously, resulting in a profile of scores.

Different strategies have been used in constructing personality inventories. The Sixteen Factor Personality Questionnaire, whose results are shown in next page figure is based on factor analysis. By factor analysing the trait descriptions of many individuals, the author of this questionnaire isolated 16 factors as the basic personality traits. He then devised questions to measure each of them. For example, answering “No” to the question “Do you tend to keep in the background on social occasions?” would earn you a point toward the dominant side of the E factor. Note that the factors themselves are nameless statistical quantities; they could just as well be called A, B, C, and so forth, which is the way they were initially labelled. The trait names that were given to the factors are simply the experimenter’s best guess as to what a particular factor represents, based on the data that contribute to it and the real-life behaviour with which it correlates.



PERSONALITY PROFILES.

Each factor in above figure is given two names: one for a high score and another for a low score. By plotting an individual’s test score for each of the factors, we arrive at a personality profile – a kind of shorthand description of the individual’s personality. The

black profile in above figure shows the average test scores for a group of airline pilots, while the red and gray profiles give the average scores for a group of artists and a group of writers. We can see that artists and writers, as a group, differ significantly from pilots on a number of personality traits.

The Minnesota Multiphasic Personality Inventory (MMPI) is another personality inventory, which is based on the method of empirical construction rather than factor analysis. Instead of assuming specific personality traits and designing questions to measure them, many items were given to well-defined groups of individuals who were known to differ from the norm on some external criterion. Only those questions that discriminated significantly between groups were retained to form the inventory.

For example, to develop a scale of items that will distinguish between schizophrenic and normal individuals, the same series of questions are given to two groups; the criterion group consists of patients hospitalized with diagnosis of schizophrenia; the control group is an equal number of individuals who have never been diagnosed as having psychiatric problems. Only those test items that discriminate significantly between the two groups are retained for use in a scale for measuring schizophrenia. Questions that at face value might seem to distinguish normals from schizophrenics (*e.g.*, “I feel as if I am not a part of the world”) may or may not do so when put to the test. The method of empirical construction contrasts the responses of two distinct groups to ensure that the test item bears an actual (empirical) relationship to the personality characteristic being measured.

Interest Inventories to the Rescue

In 1927 EK Strong, a psychologist, developed the first interest inventory, a tool used to measure individuals' interests and compare them to the interests of those working in various occupations. It was called the Strong Vocational Interest Blank. This tool has seen many revisions, and name changes, over the years. Now called the Strong Interest Inventory (SII), it remains one of the most popular self assessment tools in use today. There are others as well, including the Kuder Occupational Interest Survey, Self-Directed Search and the Campbell Interest and Skill Survey.

How to Take an Interest Inventory

A career counselor or other career development professional can administer an interest inventory as part of a complete self assessment which should also look at things such as your personality type, skills and work values. You will be asked to complete a questionnaire containing a series of items about your likes and dislikes. These items may measure, for example, your interests regarding leisure activities, work-related activities, people with whom you prefer to work and school subjects. In order to get the most accurate results, it is important to respond to each item with as much honesty as possible. There are no right or wrong answers and no one will judge you.

When responding to items related to work-related activities, do not worry about whether or not you have training or a particular skill. For the purposes of completing an interest inventory, that is not important. All you are being asked at this point is whether that activity is of interest to you. There will be plenty of time later on, as you begin to explore your options, to decide whether or not you want to become skilled in a particular area.

Getting and Understanding Your Results

After completing an interest inventory, you will receive a report containing your results. The professional who administered the inventory should go over it with you and help you make sense of it. Your report should include a list of occupations that may be suitable for someone with your interests. Some of the occupations on the list may appeal to you while others will not. It is important to remember that just because an occupation shows up in the results of an interest inventory or other self assessment tool, that doesn't necessarily mean it's the best choice for you. It is imperative that you carefully explore any occupation you are considering regardless of what list it turns up on. Your interests may indicate that a particular occupation may be suitable for someone who shares your interests, but it may not be suitable for you for a variety of other reasons.

How to Discover Your Interests on the Cheap

If you want to try using an interest inventory on your own, you can take the Self-Directed Search (SDS). There is an online version published by PAR (Psychological Assessment Resources, Inc.) available for a small fee. After completing the assessment, you will receive a printable report containing a list of occupations that most closely match your interests. You can also try the O*Net Interest Profiler, a free online tool that can help you discover your interests.

Educational Measurement

The statements about amount or quantities are necessary for any measurement study. Variables have to be assessed in some clear manner, so that investigations can be repeated and confirmed by others. Occasionally variables can be grouped into classes or categories as when separating boys and girls for the study of sex differences.

Sometimes the variables are subject to ordinary physical measurement: for example, hours of sleep deprivation, dosage level of a drug, or time required to press a brake pedal when a light flashes. Sometimes variables have to be scaled in a manner that places them in some sort of order. For example, a psychotherapist in rating a patient's feelings of insecurity might use a five-point scale ranging from never to rarely, sometimes, often, and always. Usually, for purposes of precise communication, numbers are assigned to variables; in that case we are dealing with quantitative variables. The term measurement can be used whenever numerical values are assigned to independent and dependent variables, or indeed to any variable?

Experimental Design

The expression experimental design is used to describe any of the more formal patterns according to which experiments are planned. The same design might be used for an experiment in vision, one in learning, and perhaps one in psychotherapy. The total plan includes more than the design for it involves the substance of the particular experiment.

Graphical Representation

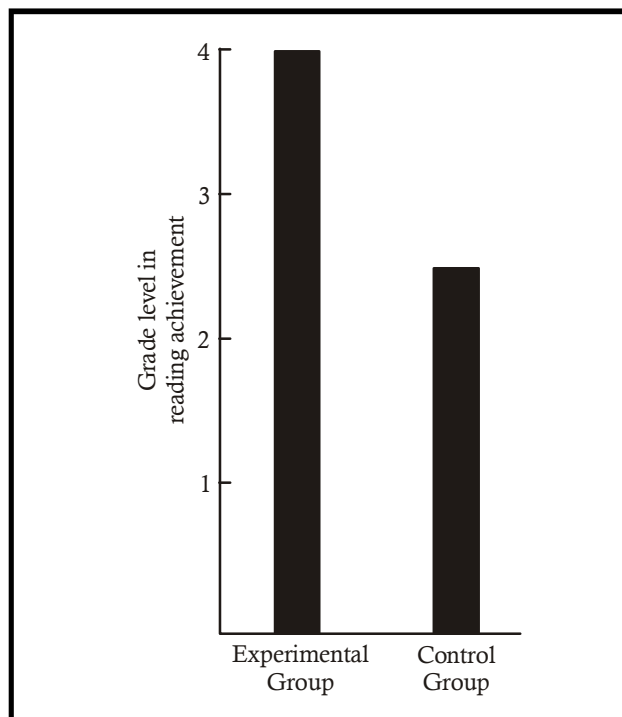
The simplest designs are those in which the investigator manipulates one variable (the independent variable) and studies its effects on another variable (the dependent vari-

able). The ideal is to hold everything constant except the variable in question, in order to obtain an assertion of the form, "With everything else constant, when X increases, Y also increases." Or, in other cases, "When X increases, Y decreases." Note that almost any content can fit into this kind of statement—the subjective experience of loudness related to the physical energy of the sound source, the rate of learning related to the age of the learner, the fear of snakes related to prior experience with snakes.

The method of graphical representation is a convenient one, with the independent variable plotted on the horizontal axis (the abscissa) and the dependent variable plotted on the vertical axis (the ordinate). The orderly relation between the variables can be stated more precisely by fitting a mathematical equation to the relationship. In other words, we are interested in more than the fact that when X increases, Y also increases; we want to know what the precise relationship is ?

Experimental and Control Group

Sometimes an experiment focuses only on the influence of a single condition, which can be either present or absent. (Such a condition is simply a variable with only two values, one representing its presence, the other its absence.) In this case, the experimental design commonly calls for an experimental group with the condition present and a control group with the condition absent. The results of such an experiment are presented in next page figure. Inspecting the figure, we see that the experimental group, which received computer-assisted instruction, scored higher on reading achievement tests than the control group, which did not receive such instruction.



EXPERIMENTAL AND CONTROL GROUPS.

Multivariate Design

In some instances it is necessary to investigate the simultaneous effects of several variables. Suppose, for example, that you are studying the effects of moisture, temperature, and illumination on plant growth. You could hold two of these variables constant and study the effect of the third. A little reflection will show how limited this design would be. Unless favourable levels of the other variables were chosen, the plant would not live; and the experiment could not be performed at all; but how favourable the other variables must be cannot be determined in advance. A better procedure would be to vary moisture, temperature, and illumination in different combinations. Then the effect of one variable would be studied not against a constant value of other variables, but against an array of other values. Many behavioural science problems have this multivariable character. School performance, for example, is affected by the child's native ability, diet, family background, school facilities, teacher skill, and go on. The statistical problems of such a multivariate design are more complex than those of a design involving changes in only one independent variable at a time, but the yield in information is often greater for the same amount of experimental effort.

Statistical Statements

The most common statistic is the mean, or arithmetic average, which is the sum of the measures divided by the number of these measures. In experiments with a control group there are often two means to be compared a mean for the sample of subjects run under the experimental condition and a mean for the sample of subjects run under the control condition. The difference between these two means is, of course, what interests us. If the difference is large, we may accept it at face value. But what if the difference is small? What if our measures are somewhat crude and subject to error? What if a few extreme cases are producing the difference? Statisticians solve these problems by producing tests of the significance of a difference. A psychologist who says that the difference between the experimental and the control group is "statistically significant" means that certain statistical tests have been applied to the data and that the observed difference is trustworthy. The psychologist is not commenting on the importance or practical significance of the results but is telling us that the statistical tests indicate that the difference observed is likely to occur again if the experiment is repeated. Many chance factors can influence the results of an experiment. By using statistical tests, psychologists can judge the likelihood that the observed difference is, in fact, due to the effect of the independent variable rather than an unhappy accident of chance factors.

Correlation (As an Alternative to Experimentation)

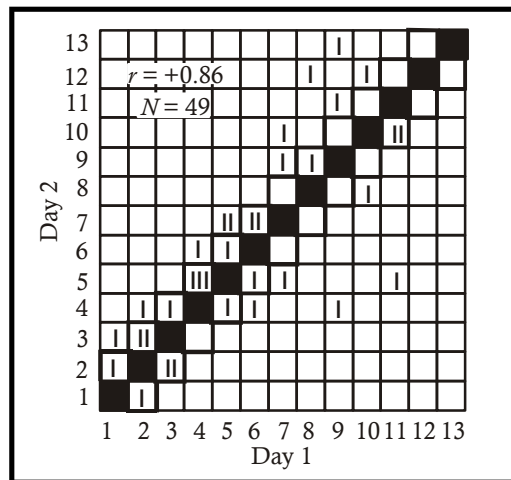
Sometimes strict experimental control is not possible. For example, the experimenter interested in the human brain is not free to remove portions at will, as he does with those of lower animals. But when brain damage occurs through disease, injury, or gunshot wound, the experimenter can study how parts of the brain are related to behaviour. For instance, a relationship may be found between the extent of damage at the back of the brain and the amount of difficulty in vision. This method of assembling correspondences, without experimental control over them, is known as correlation.

When large masses of data are available, correlation is often the best available method for discovering relationships. Suppose we have records of the high-school grades of stu-

dents entering college. The best way to find out the relation between high-school grades and freshmen grades in college is to correlate them, that is to find out if those who did well in high school generally do well in college, and *vice versa*. Measurement in correlation studies is provided by the coefficient of correlation, symbolized by the letter r which expresses the degree of relationship.

Coefficient of Correlation

The relationships expressed by a coefficient of correlation are made clearer by a diagram of actual test results. Forty-nine subjects were given a test of their susceptibility to hypnosis on two separated days. Each tally in the diagram represents the combined score of one subject on the two tests. Thus two subjects made scores of 1 on both days, and two other subjects made scores of 13 on both days. But one subject (see the lower right-hand portion of the diagram) made a score of 11 on the first test but only 5 on the second one.



A SCATTER DIAGRAM ILLUSTRATING CORRELATION.

If all subjects had repeated their original scores on the second day, all the tallies would have fallen in the diagonal shaded squares, and the correlation would have been $r = +1.00$. Enough tallies fall to either side, however, so in this case the correlation drops to $r = +0.86$. The interpretation of the coefficient of correlation requires a few words of explanation. Here are some rules of thumb:

1. A correlation of $r = +1.00$ means a perfect positive relationship between two variables. If weight corresponded exactly to height, so that you could precisely state a person's weight if you knew his height, then height and weight would be perfectly correlated. When the correlation is positive, the plus sign is often omitted.
2. A correlation of $r = -1.00$ means a perfect negative relationship. For example, if the price of a used car decreased as its age increased, so that one could precisely specify the price if the age were known, then the relation between the car's age and price would be expressed by a correlation of $r = -1.00$.
3. A correlation of $r = .00$ signifies no relation. Thus one would expect a zero correlation between the number of freckles on a person's face and his score on an

intelligence test. Knowing the value of one variable in no way helps predict the value of the other variable.

4. A correlation between $r = .00$ and either $+1.00$ or -1.00 indicates an imperfect relationship. The degree of relationship is specified by the extent to which the value of the correlation approaches 1.00, plus or minus. Negative and positive correlations of the same size represent the same degree of relationship.
5. A correlation is not a per cent, so a correlation of $r = .25$ cannot be interpreted as being half as great as one of $r = .50$. The relationship expressed by a correlation coefficient varies more nearly with its square. Thus a correlation of $r = .70$ (note that $.70^2 = .49$) expresses nearly double the relationship of $r = .50$ (note that $.50^2 = .25$). It helps to have some idea of the sizes of correlations commonly reported: (a) a correlation of $r = .50$ between the height of a parent and the adult height of a child of that parent; (b) correlations of about $r = .40$ between scholastic aptitude tests and freshman grades in college; (c) correlations of about $r = .75$ between grades in the first semester of the freshman year and those in the second semester.
6. Some supplementary information is needed to indicate whether or not a given correlation is statistically significant. This is the same problem met in determining whether a difference between two means is significant. For example, if too few cases have been studied, a few extreme cases might produce a high correlation although none actually exists in the total population. Hence statisticians have developed formulas for stating the significance of a correlation; the most dependable correlations are based on a large number of cases.



BIOLOGICAL BASIS OF BEHAVIOR

Receptors and Effectors

Physical energy excites receptors when they are sensitive to the given form and intensity of energy. The resulting sensations are elaborated by the CNS into perceptions whenever several sensation inputs and CNS trace from past experiences are integrated. Sensations are studied in man when a subject reports on those aspects of perception that depends on sensation. This is the method of introspection. In animals subjects, electrodes are used to detect responses of the receptors, nerves or brain in the method of evoked potentials. The inputs from stimuli are controlled by the experimenter in both methods.

Receptors

Stimuli are physical events to which receptors respond. Receptors are specialized to respond to specific forms of energy. Some specialized receptors excite those parts of the brain whose activity results in conscious sensation; some receptors cause of only somatic or arouse the brain in response to needs.

There are receptors specialized to respond to mechanical, thermal, chemical, acoustic or photic energy. When a receptor is activated by the form of energy for which it is specialized, it has received an adequate stimulus; if it is excited by another form of energy, it receives an inadequate stimulus. Either stimulus causes a generator potential that fires sensory neurons.

The modality of the resulting sensation depends on what part of the brain the nerve impulses reach. According to the law of specific nerve energies. Receptors transduce physical energy into nerve impulses. Receptors may be classified by their adequate stimulus. They may be classified by form as unspecialized nerve cells, specialized nerve cells

or specialized receptor cells. They may be classified by location in the head only, or in the whole body as special senses or general senses. The general senses are somesthesia (pressure, pain, warmth, and cold) and kinesthesia. The special senses are olfaction, vision, gustation, audition and vestibular sensitivity. Receptor can be classified by the functional location of their stimuli as exteroceptors, interoceptors or proprioceptors. Exteroceptors include cutaneous senses; interoceptors include organic senses. Proprioceptors respond to body position and movement.

Effectors

The muscles are called effectors because they produce effects, changing the individuals relations with the environment. The only other effector possessed by the human organism is certain glands such as the salivary gland, which moistens food in the mouth, (Food in the mouth or even in the stomach or intestine is still in the environment and the digestive juices poured out by the glands change the food before it is actually absorbed into the body.) The light-producing organ of the firefly and the electric shock-producing organ of certain fishes are effects.

Two basic classes of effectors can be distinguished. Muscle cells have specialized in the property of contractility; they respond to stimulation by initiating chemical processes which change the configuration of the cell. In vertebrates three varieties of muscle cells exist. The simplest smooth muscle has developed a special substance, the fibrillae, which changes in shape and produces the overall deformation of the cell. Smooth muscles are found primarily in the viscera. Striated muscles have developed two types of fibrillae, one dark and the other light in appearance. These are arranged in an orderly sequence, giving the muscle a striated appearance. Striated muscles make up all the somatic musculature of the vertebrate organism. The third type of muscle contains the same light and dark fibrillae organized in what appears to be a random network. This muscle is found exclusively in the heart and has therefore been called the cardiac muscle.

The second class of effector is the gland cell. This cell responds to irritating stimulation of the cell membrane by producing novel chemical substances which are eventually secreted through the cell membrane. Basically, there are two types of gland cells. Duct glands discharge their secretions into body cavities such as the stomach or intestine; the ductless glands empty directly into the bloodstream. The secretions of duct glands typically have a much localized effect than those of ductless glands.

Adjuster Mechanism (Homeostasis)

Human biological adjustments are basic in the sense that they are life functions and have their equivalents in animals. Needs continually arise and are met. Many physiological adaptations occur automatically. The blood remains slightly alkaline within a very narrow range of change; any larger variation in the direction of acidity or increased alkalinity would result in death. Body temperature is controlled by 'thermostat' in the brain. This thermostat regulates various processes such as combustion food, perspiring, and dilatation or constriction of the capillaries. Such adaptations are labeled as homeostasis. The hypothalamus plays a major role in the maintenance of homeostasis. It helps provide a constant body temperature and monitors the amount of nutrients stored in the cells.

In brief 'homeostasis', in physiology, is an overall term for the tendency of biological systems to maintain a state of equilibrium. This tendency ranges from systems of inter-

nal balance in individual organisms to ecological patterns of balance in a community of organisms, as between numbers of predators and prey. The concept was first advanced in the 19th century by the French physiologist Claude Bernard, and was given its name by the American physiologist Walter B. Cannon. Cannon made a distinction between hierarchical, homeostatic levels; reflex and instinctive behaviour, acquired habits and adaptations and willed actions, which serve the self-regulating system intent upon maintaining a 'fluid' equilibrium. Examples of homeostasis include the body's self-regulation of hormone and acid-base levels, the composition of body fluids and cell growth and body temperatures. On a broad scale, the world community of organisms tends to maintain some degree of evolving balance when not seriously disturbed. The so-called Gaia hypothesis of the earth as a living organism, which gained great popularity in the 1980s, may in some ways be regarded simply as an extension of homeostasis concepts.

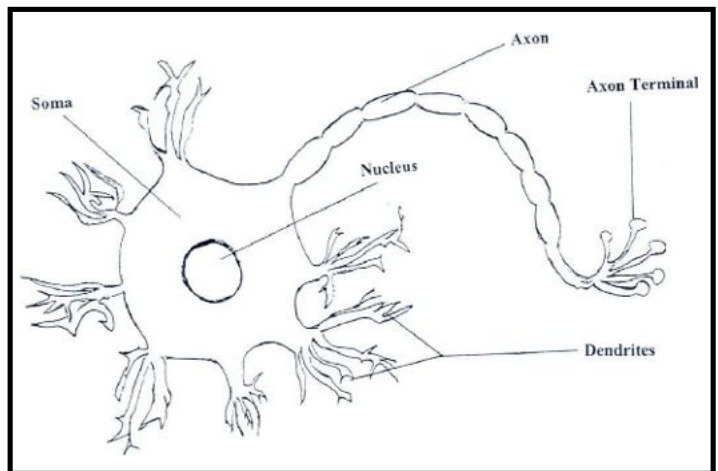
THE NEURAL IMPULSE

A neuron, or nerve cell, is the most basic component of the nervous system below figure.

To understand how neurons send messages, it is important to become familiar with their specialized structures.

The soma (or cell body) is the neuron's control centre. It contains the nucleus and other organelles which are necessary for the neuron's survival.

Dendrites are branched structures specialized to receive information from other neurons or cells. Therefore, the dendrites are the neuron's information receivers.



Axons are long, tail-like fibers that extend from the soma. Their function is to send information to other neurons. Axons can be thought of as the neuron's information senders.

The structure at the end of the axon is called the axon terminal. This is where neurotransmitters are released.

A neuron may have many dendrites, but it has only one axon.

Neurons' specialized structures allow them to transmit messages throughout the nervous system. The sending of a message from one neuron to the next is called the **neural impulse**.

Below figure is a series of simplified diagrams meant to familiarize you with the steps involved in the neural impulse. Refer to diagrams 2 and 3 as you read through the description of each step.

The transmission of a neural impulse occurs on the neuron's axon.

1. At rest, there is a different concentration of ions, or charged atoms, between the outside and the inside of the axon. Namely, there is a larger concentration of positive ions outside of the axon than there is inside of it.

As is well known, opposite charges are strongly attracted to each other. That is, the positive charges on the outside of the axon are strongly drawn to the axon's more negative interior.

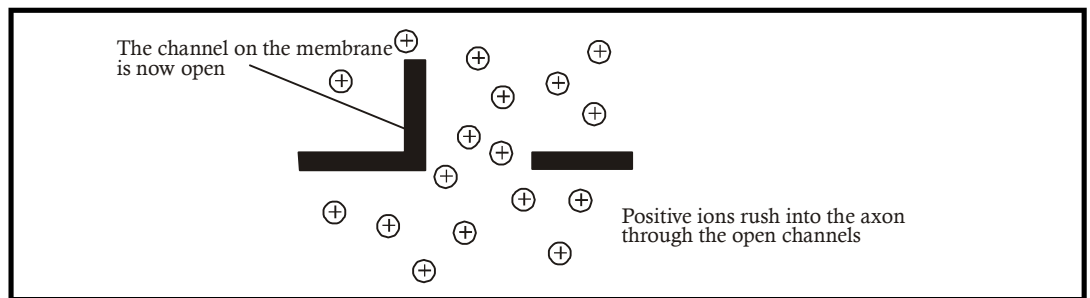
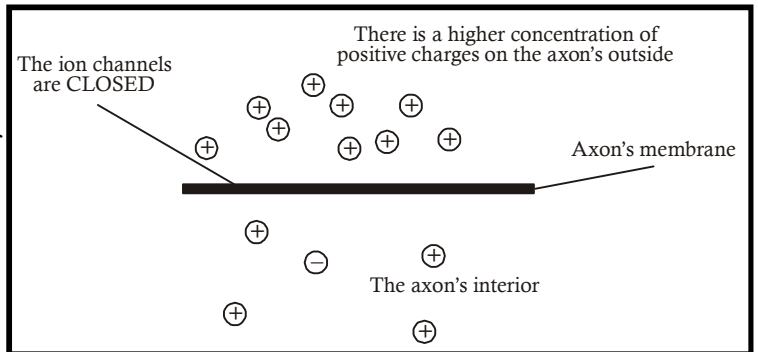
Keeping these charges apart is the **neuron's membrane**. The mem-

brane has small **channels** through which these ions can pass, but at rest, these channels are closed. This is called the resting potential.

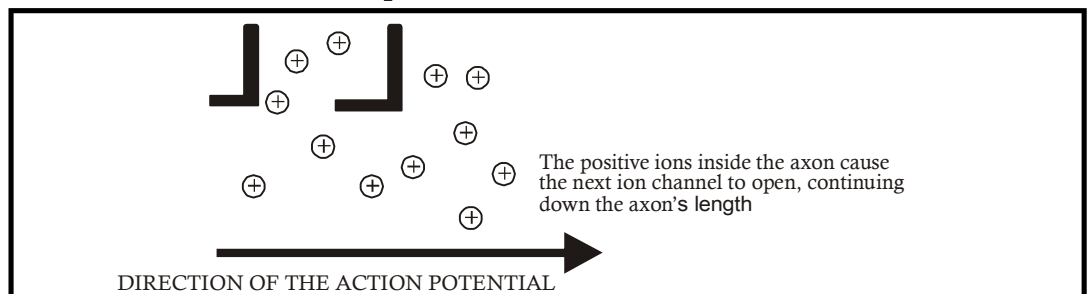
2. Eventually, a stimulus will excite a neuron's information receiver- the dendrite. This stimulus may come from an organism's external environment (such as touching a hot flame) or may originate from within the organism (for example, the release of hormones).

Once the stimulus reaches a certain strength or **the threshold**, the small channels on the axon's membrane open.

The opening of the channels on the axon's membrane allows the positive ions outside of the axon to rush in. The influx of positive ions changes the charge inside the axon, making it more positive. This process is called **depolarization**.



3. As ions rush in through one gate, their positive charge causes the next, nearby gate to open, letting in more positive ions. The gates on the axon continue to open, and in this way, transmit the message down its entire length. The message traveling down the axon is the **action potential**.



4. After transmitting a message, the axon returns to its resting state by pumping out positive ions through the channels on its membrane. This process is called the

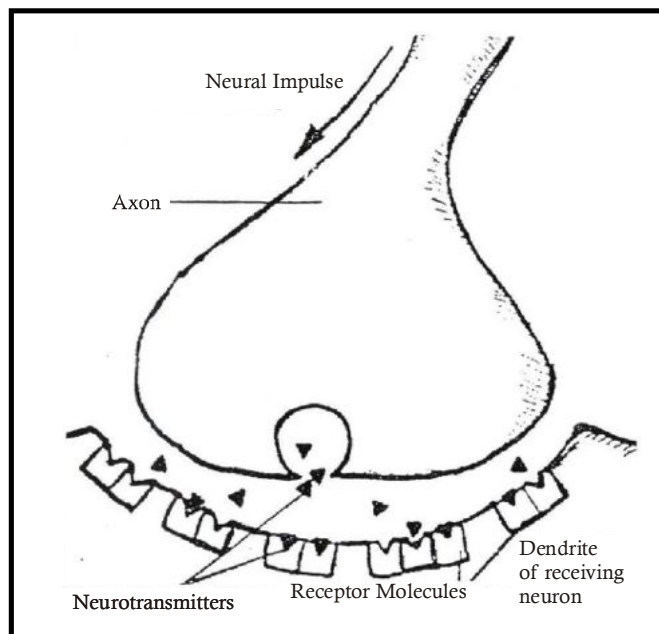
refractory period. Once the refractory period is over, the neuron is ready to transmit a message again.

How does the axon of one neuron transmit messages to other, nearby neurons?

In the nervous system, the axon terminals of one neuron are in very close contact with the dendrites of neighboring neurons. The signal is transmitted from one neuron to the next through the release of **neurotransmitters**, or chemical messengers.

More specifically, when the action potential reaches the axon terminal of the sending neuron, it causes neurotransmitters to be released. The neurotransmitters then attach to receptors located on the receiving dendrite. This starts another action potential, and the entire process is repeated.

After stimulating the receiving dendrite, neurotransmitters are taken back into the sending neuron. This recycling process is called **reuptake**.



SENSORY SYSTEM

We have a well organised sensory system which includes five external senses, sense of 'stability and balance' and some inner sense. These senses, work on mechanical basis, and, can be distinguished between higher and lower senses. Since auditory and visual senses play dominant role in our perception, they are known as higher senses. Rest of the other senses are lower senses. Though all these senses have been discussed in first chapter, we shall discuss, again, two major senses known as higher senses.

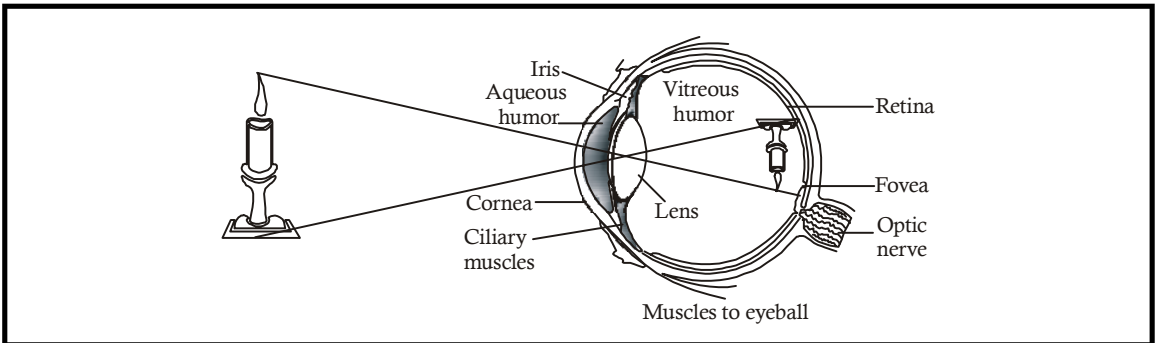
Vision

Because of this sensory system mind becomes able to make a particular shape, size and colour of an object. The visual sense is complex in structure and has a difficult job in the

process of perception. It is the place where the experience of the visible portion of electro-magnetic spectrum called light is produced.

The Human Eye (Structure)

Some main parts of human eye are cornea, pupil, lense and retina. Light enters the eye through the transparent cornea, the amount of light being regulated by the pupil; the lense then focuses the light on the sensitive surgace, the retina. Constriction and dilation of the pupil are controlled by the autonomic nervous system: the parasympathetic division controls the change in pupil size as a function of change in illumination (in much the same way as we increase the shutter opening of a camera to admit more light on a dark day and decrease the opening under conditions of bright illumination). The sympathetic division acts to dilate the pupil under conditions of strong emotion, either pleasant or unpleasant. Even conditions of mild emotional arousal or interest will result in systematic changes in pupil size.

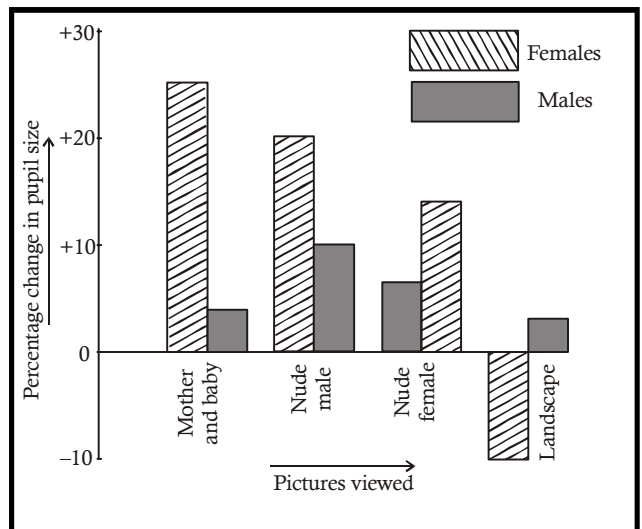


THE STRUCTURE OF EYE

The size of the pupil is regulated by the iris, the diaphragm that gives the eye its colour. The shape of the lens is regulated by the ciliary muscles. The external muscles that move the eye are not shown in given above figure.

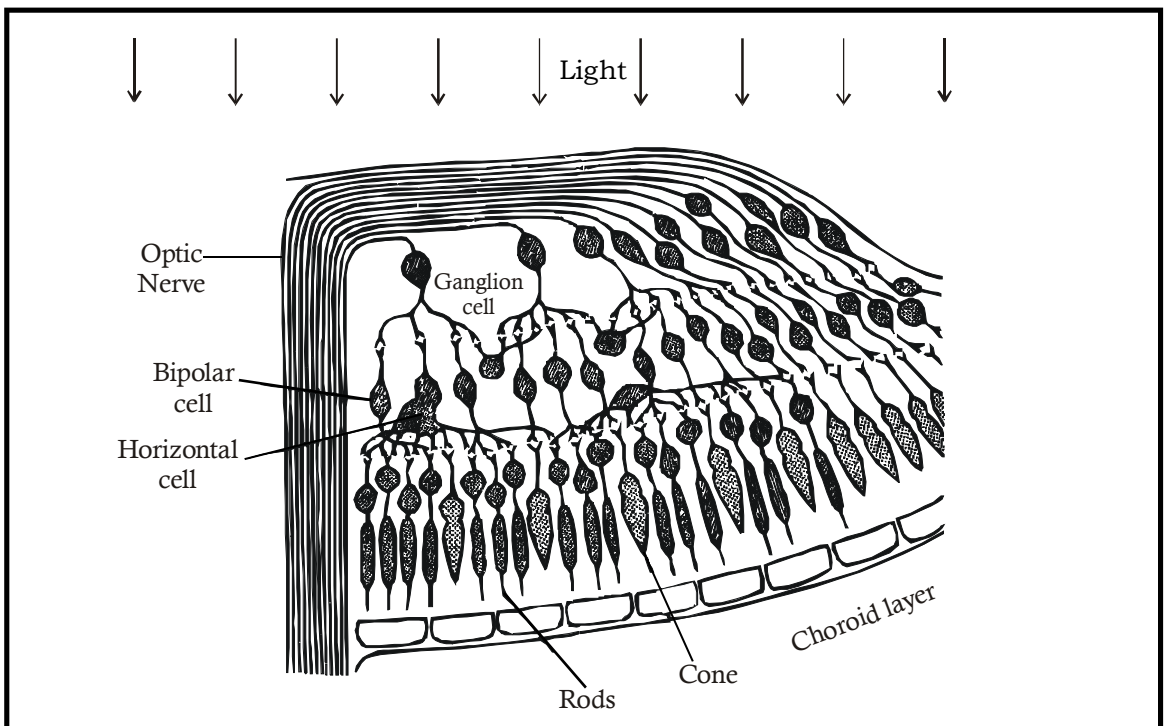
Changes in Pupil Size as a Response to Pictures

Changes in pupil size were recorded on film and later measured. The figure shows the percentage of increase or decrease in pupil area in response to various pictures for both male and female subjects. The amount of light entering the eye was constant for all pictures. The sexes differ quite markedly with regard to the interest value of these particular pictures. A later study showed that homosexuals could be distinguished from normal males on the basis of pupillary responses to pictures of female pinups.



Rods and Cons

The retina, the light-sensitive surface at the back of the eye, has three main layers: (1) the rods and cones, the photosensitive cells that convert light energy into nerve impulses; (2) the bipolar cells, which make synaptic connections with the rods and cones; (3) the ganglion cells the fibers of which form the optic nerve. Strangely enough, the rods and cones form the rear layer of the retina. The eye is a very imperfect optical system. The light waves not only have to pass through the lens and liquids that fill the eyeball, none of which is a perfect transmitter of light, but also have to penetrate the network of blood vessels and the bipolar and ganglion cells that lie on the inside of the eye before reaching the photoreceptors where light is converted into nervous impulses.



The retinal cells of special interest are the photoreceptors: the cylindrical rods and the more bulbous cones. The cones, active only in daylight vision, permit us to see both achromatic colours (white, black, and the intermediate grays) and chromatic colours (red, green, blue, etc.) The rods function mainly under reduced illumination (at twilight or night) and permit us to see only achromatic colours. The two types of photoreceptors differ in much the same way that colour film differs from black and white film. Black and white film (rods) is more sensitive than colour film and can produce a picture even under conditions of dim illumination. Colour film (cones) requires much more intense light for proper operation. More than 6 million cones and 100 million rods are distributed, somewhat unevenly, throughout the retina. The fovea contains only cones—some 50,000 of them packed together in an area smaller than a square millimeter. Outside the fovea are both rods and cones. With the cones decreasing in number from the centre of the retina to the periphery. The rods are connected in groups, and each group has one neuron running to the optic nerve. The cones in the periphery of the retina are grouped

together in units along with the rods, but each of the cones in the fovea has its own “private wire” to the brain. Our vision is more acute when light waves strike the fovea, because the non-converging “private line” does not mix signals. For this reason we turn our head to look directly at an object when we want to see it clearly. Under dim illumination, however, when the cones are not operative, we can more easily detect a faint stimulus, such as dim star, if we do not look directly at it but let its image fall just outside the foveal region where rod density is greatest.

Although the rods will respond to a much dimmer stimulus than will the cones, the image they give is less clear. At night, objects have indistinct outlines and lack much of the detail seen in daylight. If we remember that rods are grouped in their connections to bipolar and ganglion cells, we can realize why this would be the case. Because signals from many rods converge on a single optic nerve fiber, interpretation by the brain is less fine grained than it would be for the “private wire” cone cells in the fovea.

Blind Spots

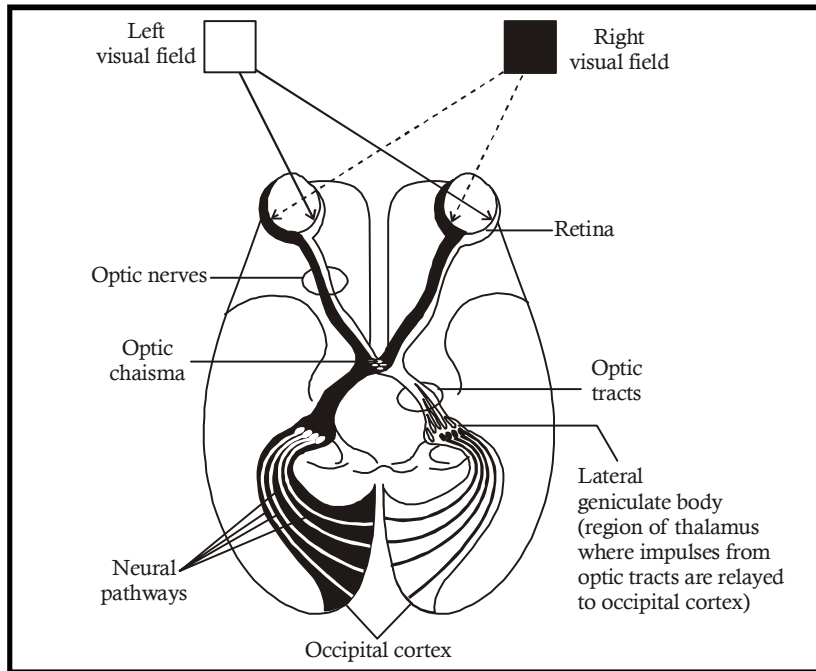
The most sensitive portion of the eye in normal daylight vision is a part of the retina called the fovea. This area plays a major role in perception and yet it is so small that the projection (or image) of a thumb-nail viewed at arms length will cover it. Not far from the fovea is an insensitive area called the blind spot, where the nerve fibers from the ganglion cells of the retina come together to form the optic nerve. Although we are not normally aware of the blind spot, its existence can easily be demonstrated. Follow the instructions given in below figure.



(A) With the right eye closed, stare at the upper right cross. Move the book back and forth about 1 foot from the eye. When the black circle on the left disappears, it is projected on the blind spot. (B) With the right eye closed, stare at the lower right cross. Move the book back and forth again. When the white space falls in the blind spot, the black line appears to be continuous. This phenomenon helps us to understand why we are not ordinarily aware of the blind spot.

Visual Pathways

Next page figure shows the optic nerve fibers leading from each eye to the cortical areas where vision is represented (the occipital lobes). Notice that some of the fibers go from the right eye to the right cerebral hemisphere and from the left eye to the left hemisphere, whereas other fibers cross over at a junction called the optic chiasma and go to the right hemisphere of the cerebral cortex, and fibers from the left sides of both eyes go to the left hemisphere. Consequently, damage to the occipital lobe of one hemisphere (say, the left) will result in blind areas in both eyes (the left sides of both eyes). This fact is sometimes helpful in pinpointing the location of a cerebral tumor or injury.



OPTICAL NERVE FIBRES.

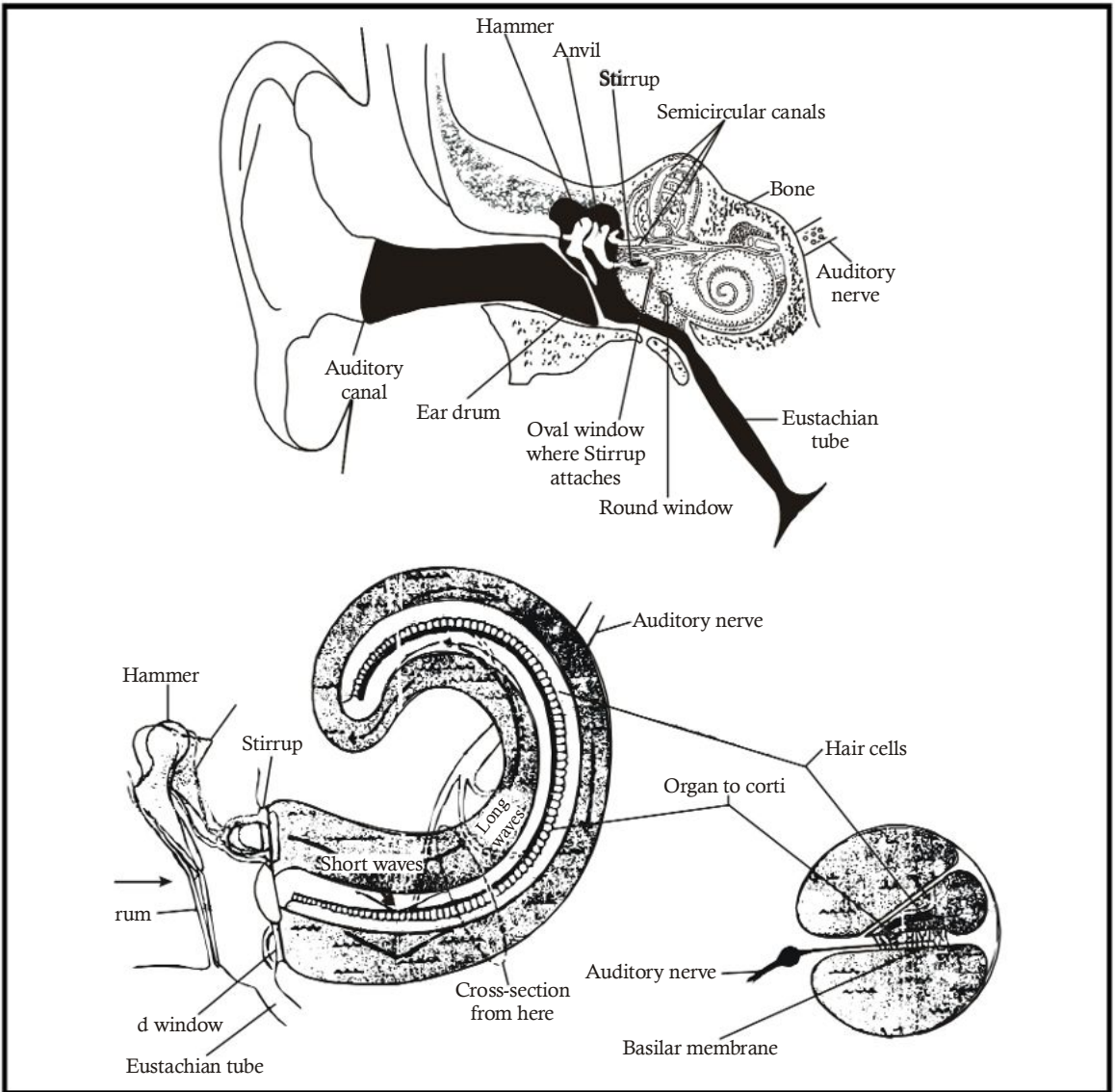
Audition

While the eye responds to electromagnetic energy, the ear is sensitive to mechanical energy – to pressure changes among the molecules in the atmosphere. A vibrating object, such as a tuning fork, causes successive waves of compression and expansion among the air molecules surrounding it. The sound waves generated by the vibration of molecules (in air, water, or some other medium) are the stimuli for hearing. Unlike light, sound cannot travel except through a medium a ringing bell suspended in a vacuum jar cannot be heard when the air is pumped out.

The Human Ear

The external ear connects with an auditory canal leading to the eardrum, a movable diaphragm activated by sound waves entering the ear. On the inner side of the diaphragm is a cavity housing the bony transmitters of the middle ear (three small bones called the hammer, anvil, and stirrup). The hammer is attached firmly to the eardrum, and the stirrup to another membrane, the oval window. The oval window conducts the sounds to the cochlea, the auditory portion of the inner ear. Because the oval window is much smaller than the eardrum, small movements at the eardrum are condensed into a magnified pressure on the oval window.

Pressure at the oval window sets into motion the fluid inside the cochlea. This pressure is relieved at the round window at the other end of the fluid-filled channel that runs through the cochlea. Pressure changes in the fluid displace the basilar membrane in the cochlea, upon which the organ of corti rests, and this displacement stimulates the sensitive elements in the hair cells of the organ of corti, which are connected with the auditory nerve.



Auditory Pathways

The pathways of the auditory nerves resemble those of the optic nerves in that nerve fibers from each ear travel to both cerebral hemispheres. Thus, destruction of one temporal lobe will not cause complete deafness in either ear.

NERVOUS SYSTEM

Basic Units

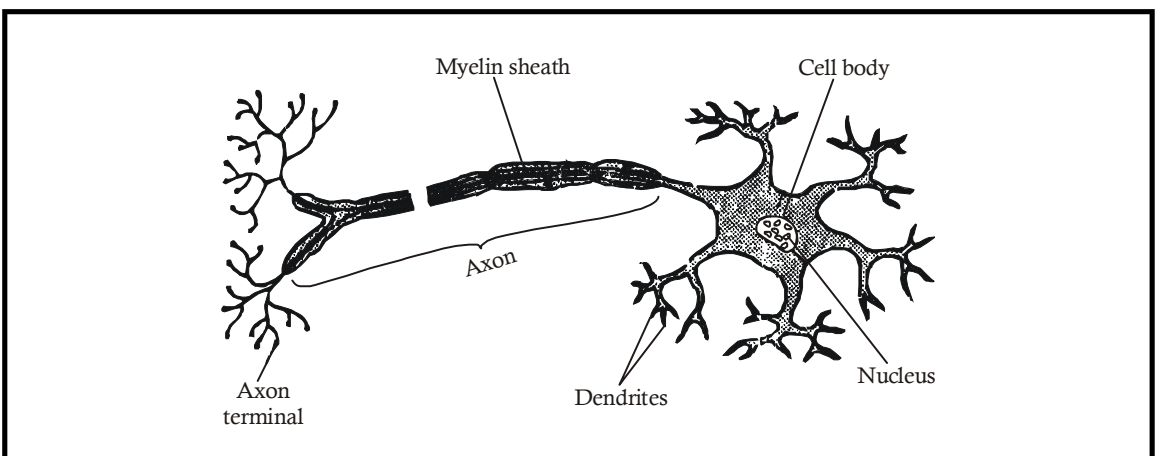
Some 10-12 billion specialized cells called neurons are the basic units of the nervous system. The human brain is composed of neurons. Neurons hold the secrets of learning and mental functioning. They play important role in the transmission and co-ordination

of nervous impulses. Each neuron is a living cell with a nucleus and other parts common to all cells. As a specialized structure the neuron consists of three main parts; the cell body, containing the nucleus; the dendrites, the many short fibres projecting from the cell body that receive activity from adjacent cells; and the axon, a single fibre extending away from one side of the cell body that transmits activity to other neurons or to muscles and glands. Nerve impulses normally move in one direction—from the dendrites through the cell body and along the axon to the dendrites or cell body of the next neuron, or to a muscle or gland. All neurons have these general features, but they vary greatly in dimension and construction. A neuron in the spinal cord may have an axon 2-3 to feet long running from the tip of the spine to the big toe; a neuron in the brain may cover only a few thousandths of an inch with all its parts. Closely interwoven among the neurons are a large number of cells, called glia cells. Until recently glia cells were thought to have only a nutritive and supportive function. New evidence suggests that they may be actively involved in the metabolic processes related to neural activity, but their exact role is as yet unknown.

A nerve is a bundle of elongated axons belonging to hundreds or thousands of neurons. The axons from a great many neurons (perhaps about 1,000) may connect with the dendrites and cell body of a single neuron. A junction between neurons is a synapse. The synapse is not a direct connection; a slight physical separation exists across which the nervous impulse is transmitted by a chemical intermediary.

Transmission of Nerve Impulses

A nerve impulse in the human body may travel at anywhere from 2-200 miles per hour, depending on the diameter of the axon and other factors. The analogy of a fuse has sometimes been used; when a fuse is lighted, one part of the fuse lights the next part, the impulse being regenerated along



A NERVE CELL.

The way. However, the details of neural transmission are much more complex than this. The process is electrochemical. The thin membrane that holds together the protoplasm of the cell is not equally permeable to the different types of electrically charged ions that normally float in the protoplasm of the cell and in the liquid surrounding it. In its resting

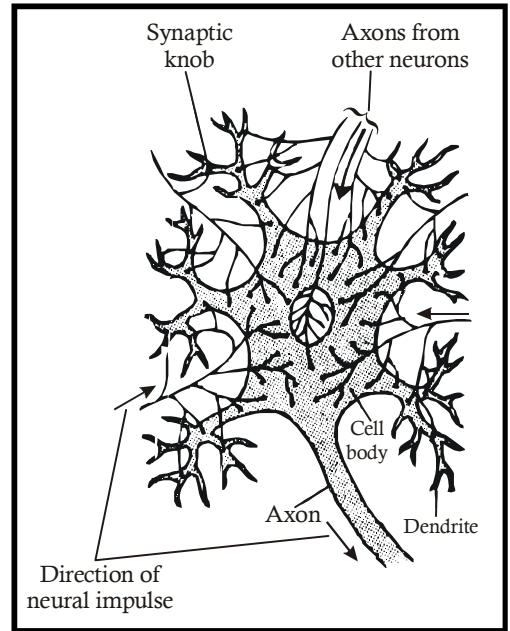
state the cell membrane keeps out positively charged sodium ions (Na^+) and allows in potassium ions (K^+) and chloride ions (Cl^-). As a result there is a small electrical potential, or difference, across the membrane. The inside of a nerve cell is slightly more negative than the outside; this is its resting potential.

When the axon of a nerve cell is stimulated, the electric potential across the membrane is reduced at the point of stimulation. If the reduction in potential is large enough, the permeability of the cell membrane suddenly changes, allowing the sodium ions to enter the cell. Now the outside of the cell membrane is negative with respect to the inside. This change affects the adjacent portion of the axon, causing its membrane to permit the inflow of sodium ions. This process, repeating itself down the length of the axon, is the nerve impulse. The nerve impulse is also known as the action potential, in contrast to the resting potential, of the neuron. Because the nerve impulse is generated anew at each stage along the axon, it does not diminish in size during transmission.

Some nerve fibres have an insulating sheath called myelin sheath; such fibres are known as myelinated fibers. The sheath is interrupted approximately every two millimeters by constrictions called nodes, where the myelin sheath is very thin or absent. Because conduction jumps along the fiber from node to node, it is much more rapid in myelinated fibers than in nonmyelinated fibers. The myelin sheath was a late development in evolution and is characteristic of the nervous systems of higher animals. The fact that the formation of the myelin sheath of many nerve fibers in the brain is not completed until some time after birth suggests that the slow maturation of some of the infant's sensory and motor functions may be related to the gradual process of myelination.

Synaptic Transmission

Switching of impulses occurs in the synaptic junction between neurons. Because of the switching, facilitation, inhibition, coordination and integration of impulses occurs through the way in which groups of neurons act together. A single neuron transmits an impulse, or "fires," when the impulses reaching it become strong enough. Because its axon does not transmit at all prior to this, the neuron has been said to follow an all-or-none principle of action. The neuron fires in a single brief, transient burst and is then temporarily inactive (in what is called a refractory phase) for a few thousandths of a second. The size of the action potential is constant, and once started it travels all the way down the axon to the synapses. But the decision of the neuron to fire or not depends on graded potentials (potentials that are not all-or-none but can be any size) in the dendrites or cell body. These graded potentials are induced by stimulation at the synapses by other neurons,

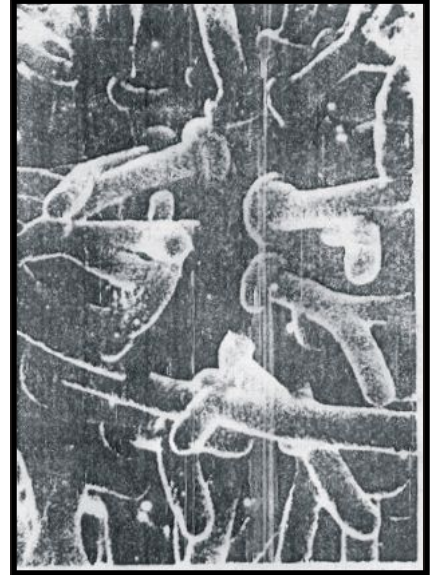


**SYNAPSES AT THE CELL BODY
OF A NEURON.**

and their size varies with the amount and kind of incoming activity. When the graded potential becomes sufficiently large, it generates enough depolarization in the cell body to trigger the all-or-none action potential that then travels down the axon. If the graded potential does not reach the discharge threshold of the action potential, no activity occurs. Thompson (1967) has likened the occurrence of the graded potentials in the nerve cell to a decision making process. The neuron considers all incoming activity and, depending on the amount and kind of this activity, decides to fire or not.

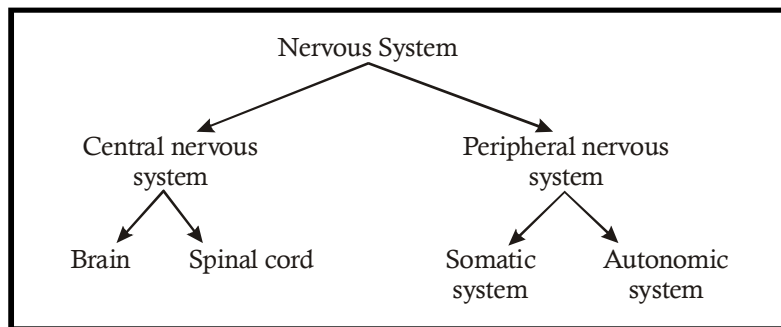
Organization of the Nervous System

Groups of neuron fibers bundled together form nerves which are spread out to every part of the body. Nerves sense receptors, skin, muscles and internal organs. Those nerves that carry sensory information from the periphery of the body and the internal organs to the spinal cord and brain are called afferent (sensory) nerves. Nerves transmitting impulses from the brain and spinal cord to the muscles and organs are called efferent (motor) nerves. The neuron cell bodies are not part of the long nerves but are grouped together into nuclei or ganglia which are found in or near the spinal cord, the brain, and some of the internal organs.



PHOTOGRAPH OF NERVE FIBERS AND SYNAPTIC KNOBS.

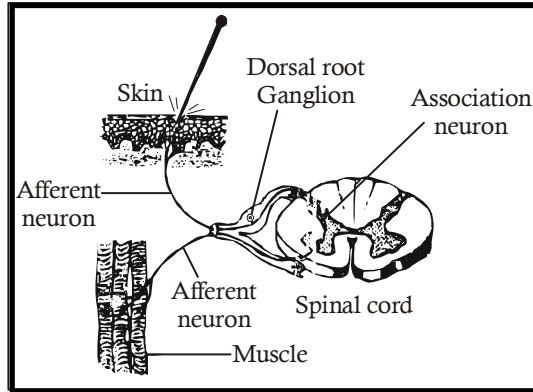
Divisions of Nervous System



Spinal Cord

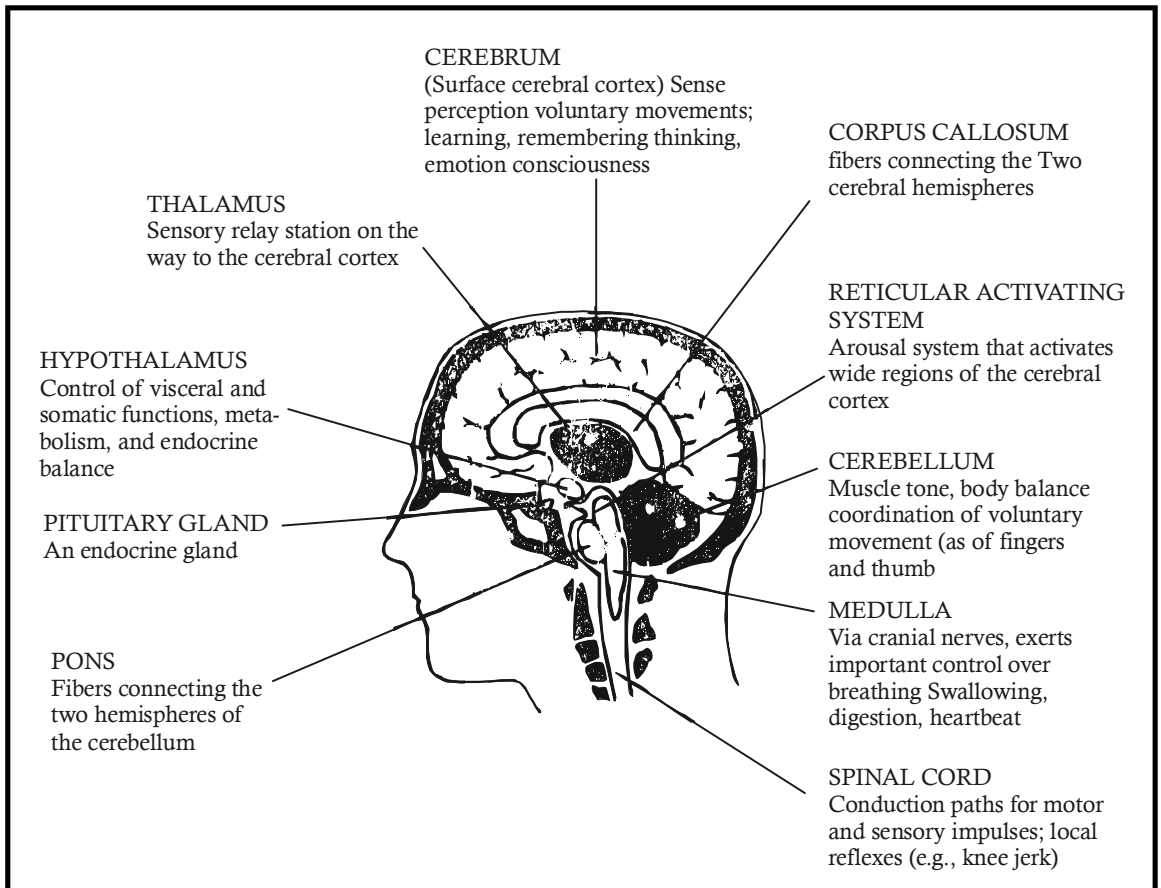
The nerve fibers running from various parts of the body to and from the brain are gathered together in the spinal cord, where they are protected by the bony spinal vertebrats. Some of the very simplest stimulus-response sequences are carried out within the spinal cord. One example is the knee jerk, the extension of the leg in response to a tap on the tendon that runs in front of the knee cap. Frequently a doctor makes this test to determine the efficiency of the spinal reflexes. The natural function of this reflex is to ensure that the leg will extend when the knee is bent by the force of gravity so that the organism remains standing. When the knee tendon is tapped, a message from sensory cells embedded in the attached muscle is transmitted through an afferent nerve to the

spinal cord where it synapses directly with many motor neurons whose axons run in the efferent nerve back to the same muscle and cause it to extend.

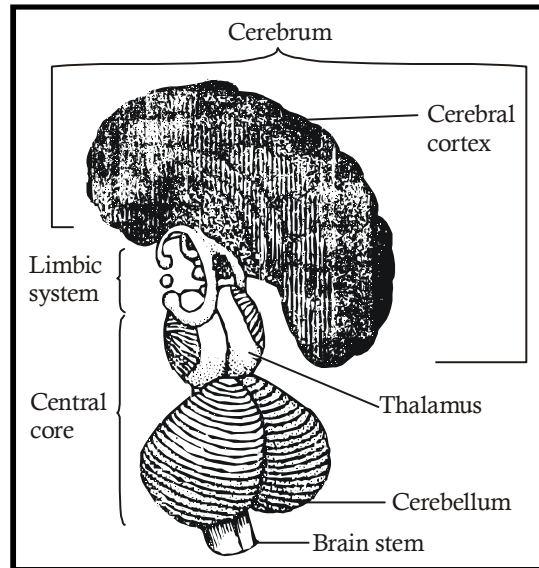


THREE-NEURON REFLEX ARC.

Structure of the Brain



THE HUMAN BRAIN.



THE THREE CONCENTRIC LAYERS OF THE HUMAN BRAIN.

Study of the Brain

The following four methods are used most often by physiological psychologists and neurophysiologists in studying the brain.

1. Injury or Surgical Ablation

Noting the kinds of symptoms produced when tumors or injuries damage certain parts of the brain may give clues about functions controlled by the area. Early observations that injury to the left side of the brain usually resulted in speech defects, whereas damage to the right side did not, led to localisation of a speech centre in the left cerebral hemisphere. Improved methods of locating the area of injury and assessing the kind of language functions disturbed have specified more exactly the areas involved in different linguistic abilities.

In experiments with animals it is possible to remove systematically parts of the brain (or destroy the tissue electrically) and observe the kinds of defects that result. Sometimes ablation operations are performed on human patients when the removal of abnormal brain tissue is essential to their well-being. Such patients are carefully studied to assess the effects of the operation on their abilities.

2. Electrical or Chemical Stimulation

Stimulating parts of the brain with mild electrical currents produces effects on behaviour. Brain surgery on human patients is often done under local anesthesia so the surgeon can tell (by the patient's responses when different points are stimulated electrically) which area to remove. From patient reports of sensations during stimulation fairly accurate maps of the cortex have been obtained.

Experimenters have used permanently implanted electrodes in animals to produce repeated stimulation of a local part of the brain. Studies with such electrodes help deter-

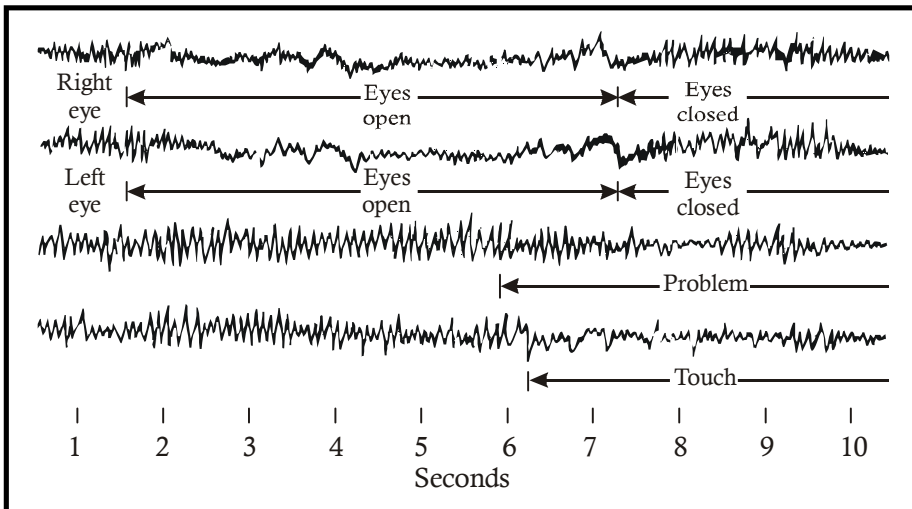
mine where sensory effects occur and where various types of muscular activity are controlled.

Chemical stimulation has also been used to affect behaviour. A small tube is inserted into the animal's brain so that its end touches the area of interest. A minute amount of some chemical, often one that resembles a neural transmitter, is delivered through the tube and the behavioral effects are observed.

3. Electrical Effects of Neural Activity

When neural action occurs, slight electrical currents are produced. By inserting at appropriate places electrodes connected to measuring devices, the experimenter can detect whether impulses starting at, say, the ear reach the part of the brain where the electrodes are inserted.

The brain as a whole also produces rhythmical electrical discharges. The record of these total brain discharges, known as an electroencephalogram (EEG), plays its part in the study of central nervous system activity. For example, if a particular kind of stimulation changes the rhythmic discharges picked up from one part of the brain and does not affect the discharges from another part, we can assume that the stimulation affected that particular region.



ELECTRICAL ACTION OF THE BRAIN.

4. Single Neuron Activity

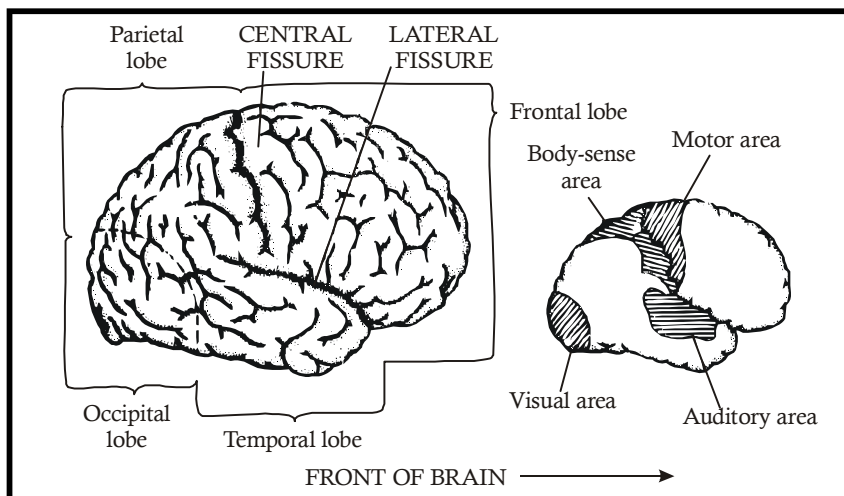
The development of extremely refined microelectrodes (about one thousandth of a millimeter in size) has made it possible to record the nerve impulse from a single neuron. This technique, which permits the investigator to study the activity of a single nerve cell while the organism is being exposed to different stimuli, yields information that cannot be obtained by recording massed electrical discharges.

Cerebral Cortex

The two large cerebral hemispheres that envelop the brain stem are the most recent part of the nervous system to evolve, and they are more highly developed in man than in any

other organism. The cerebral cortex is the thick layer of nerve cell bodies covering the cerebrum; in Latin the word cortex means "bark." The inside of the cerebrum beneath the cortex is composed mostly of myelinated axons, and appears white. (A live brain appears slightly pinkish because of the blood supply.) It is in the cerebral cortex that all complex mental activity takes place.

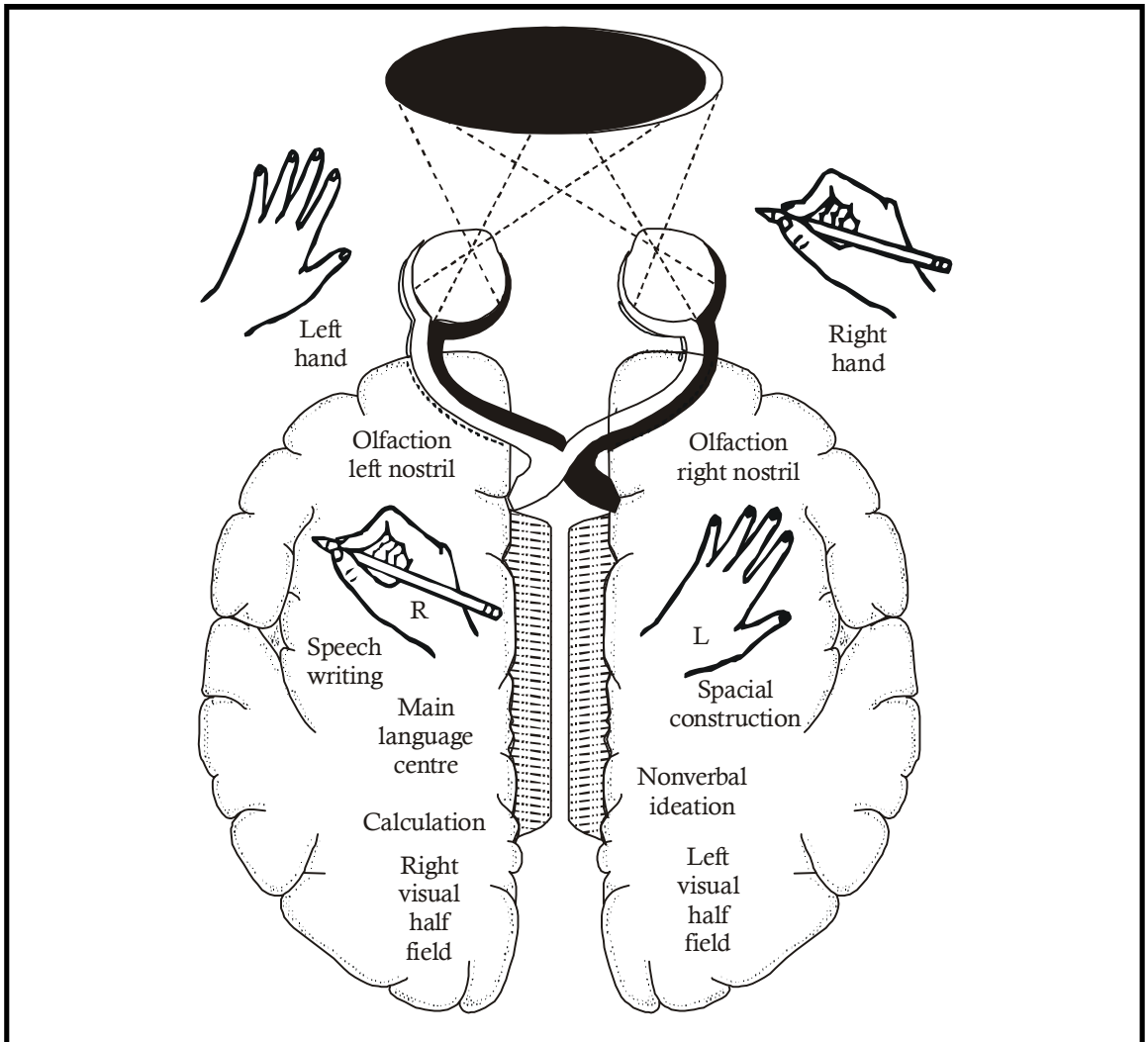
The cerebral cortex of a primitive mammal such as the rat is small and relatively smooth. As we ascend the phylogenetic scale toward the higher mammals, the amount of cortex relative to the amount of total brain tissue increases accordingly, and the cortex becomes progressively more wrinkled and convoluted so that its actual surfaces is far greater than it would be if it were a smooth covering over the surface of the cerebrum. There is a general correlation between the cortical development of a species, its position on the phylogenetic scale, and the complexity of its behaviour. Two-thirds of the cortex of the human brain is buried in the fissures. If the cortical tissue in the average human brain were spread flat, it would cover more than two square feet. The place where localized functions appear on a map of the cortex are called projection areas. Electrical stimulation of some areas of the cortex will produce specific kinds of motor responses (those involving motion or activity in parts of the body) or sensory effects (those involving sensation, feeling, awareness). Tumors that exert pressure on these projection areas disturb the responses. When these areas are destroyed by disease or injury the same functions are altered or obliterated. Yet we would be making an error in logic if we assumed that these functions are controlled by these areas alone. Even though an area is essential to a function, it may not be sufficient to control that function.



LOCALISATION OF FUNCTION IN THE HUMAN CORTX.

The two hemispheres are symmetrical, one on the right and one on the left, with a deep division between them, running from front to rear. So our first classification is the division into right and left hemispheres. For the most part, functions of the right side of the body are controlled by the left hemisphere and functions of the left side by the right hemisphere. Each hemisphere is divided into four lobes: the frontal, parietal, occipital, and temporal. The landmarks dividing these lobes are shown in figure. The frontal lobe is separated from the parietal lobe by the central fissure, running down from part of the

cerebrum near the top of the head sideways toward the ears. The division between the parietal lobe and the occipital lobe is not as clear-cut; it suffices for our purpose to know that the parietal lobe is at the top of the brain, behind the central fissure, while the occipital lobe is at the rear of the brain. The temporal lobe is well set off by a deep fissure at the side of the brain, the lateral fissure.



A DIVIDED BRAIN.

Sensory inputs to the two hemispheres are shown in the above figure. With the eyes fixated straight ahead, stimuli to the left of the fixation point go to the right cerebral hemisphere and stimuli to the right go to the left hemisphere. The left hemisphere controls movements of the right hand, and the right hemisphere controls the left hand. Hearing is largely crossed in its input, but some sound representation goes to the same hemisphere as the ear. Olfaction is received on the same side as the nostril. The left hemisphere is dominant for most people; it controls written and spoken language and

mathematical calculations. The minor, right hemisphere can understand only simple language, its main ability seems to involve spatial construction and pattern sense.

Autonomic Nervous System

We have discussed earlier that the peripheral nervous system includes those nerves connecting the brain and spinal cord with the outline regions of the body. The peripheral system consists of two divisions. The somatic system controls the skeletal muscles and receives sensory information from the skin, muscles, and various receptors. The autonomic system controls the glands and the smooth muscles that comprise the heart, blood vessels, and the lining of the stomach and intestines. These muscles are called smooth because microscopic examination shows that they lack the striated appearance characteristic of skeletal muscles. The somatic system then controls the striated muscles; the autonomic system controls the smooth muscles. The autonomic nervous system derives its name from the fact that many of the activities it controls are autonomous or self-regulating, such as digestion and circulation, which continue even when a person is asleep or unconscious. Again, autonomic nervous system has two divisions—which are often antagonistic in their action.

1. Sympathetic Division

On either side of the spinal column, closely connected with it through the spinal nerves, lie chains of nerve fibers and masses of cell bodies from which fibers extend to the various visceral organs. These chains are known as the sympathetic chains. The fibers coming from the spinal cord to the sympathetic chains originate in the thoracic and lumbar portions of the spine, between the cervical (neck) and the sacral (lower spine) regions. All the fibers and ganglia together constitute the sympathetic division of the autonomic system.

The sympathetic division tends to act as a unit. In emotional excitement it simultaneously speeds up the heart, dilates the arteries of the muscles and heart, and constricts the arteries of the skin and digestive organs; its action leads also to perspiration and to secretion of certain hormones that increase emotional arousal.

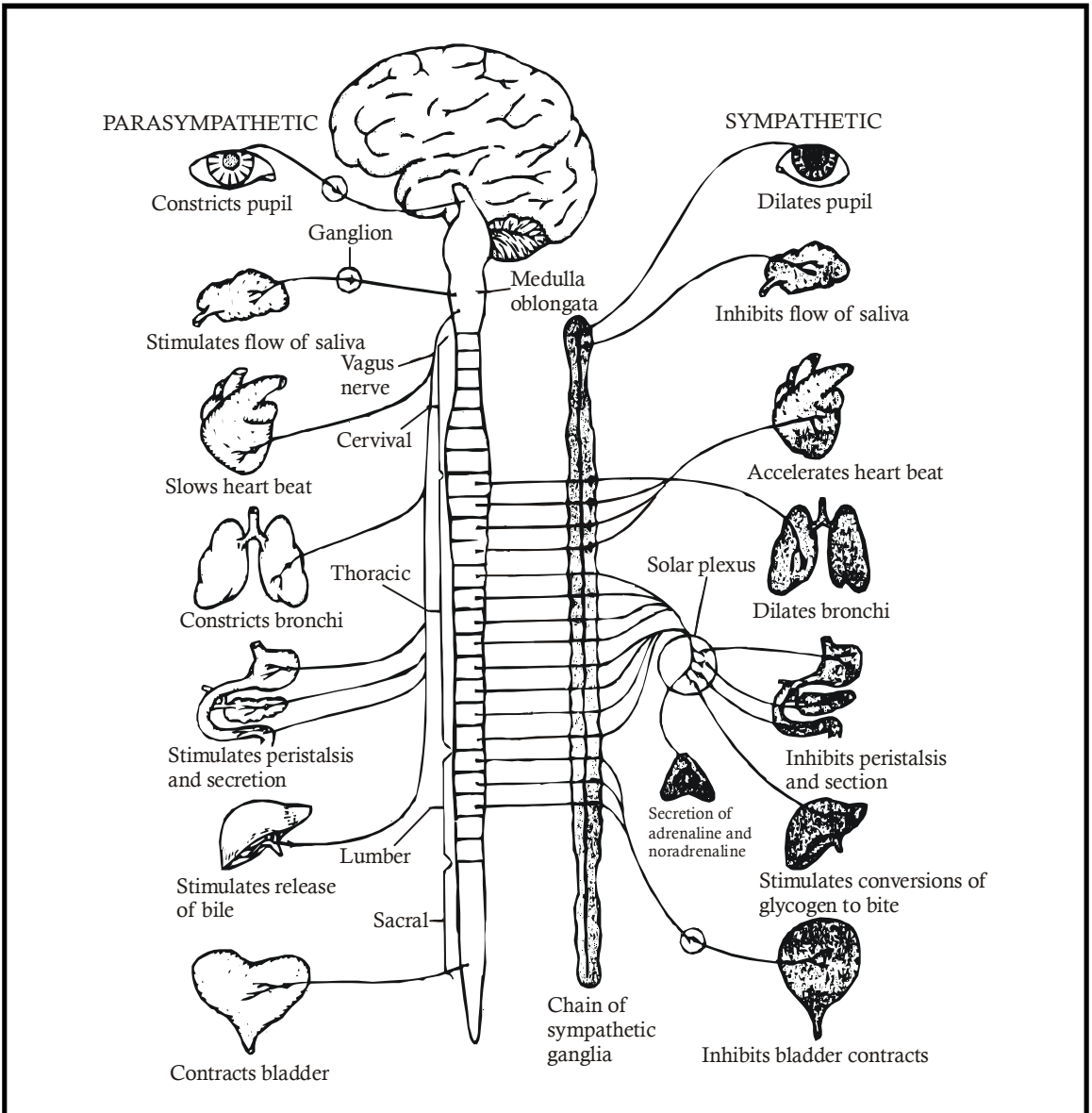
2. Parasympathetic Division

The parasympathetic division has two parts, some of its fibers originating in the cranial region (above those of the sympathetic system) and others originating in the sacral region (below those of the sympathetic system).

Unlike the sympathetic system, the parasympathetic system tends to act in a piecemeal fashion, affecting one organ at a time. If the sympathetic system is thought of as dominant in violent and excited activity, the parasympathetic system may be thought of as dominant in quiescence. It participates in digestion and, in general, maintains the functions that conserve and protect bodily resources.

Interaction Between the two Divisions

Both the sympathetic and parasympathetic parts act in opposite manners, when their fibers are connected to the same muscle or gland. Thus the sympathetic system speeds the heart rate, the parasympathetic system slows it; the sympathetic system inhibits digestive processes, the parasympathetic system facilitates them; the sympathetic system dilates the pupil of the eye, the parasympathetic system constricts it.



There are some exceptions to the principle that the two systems are antagonistic. Both divisions may be active at once, and in some cases they act together in sequence. Although the sympathetic system is usually dominant in fear and excitement, a not uncommon parasympathetic symptom in extreme emotion is the involuntary discharge of the bladder of bowels. Another example is the complete sex act in the male, which requires erection (parasympathetic) followed by ejaculation (sympathetic). Thus, while the two divisions are often antagonistic, they interact in complex ways, and their interaction is not yet fully understood.

METHODS OF PHYSIOLOGICAL PSYCHOLOGY

Degeneration Techniques

In the mid-1980s, Gyorgy Buzsáki was trying to get inside rats' heads. Working at the University of California, San Diego, he would anesthetize each animal with ether and hypothermia, cut through its scalp, and drill holes in its skull. Carefully, he'd screw 16 gold-plated stainless steel electrodes into the rat's brain. When he was done with the surgery, these tiny pieces of metal—just 0.5 mm in diameter—allowed him to measure voltage changes from individual neurons deep in the brain's folds, all while the rodent was awake and moving around. He could listen to the cells fire action potentials as the animal explored its environment, learning and remembering what it encountered.

In those days, recording from two cells simultaneously was the norm. The 16-site recording in Buzsáki's 1988 study "was the largest ever in a rat," he says. Nowadays, scientists can measure voltage changes from 1,000 neurons at the same time with silicon multielectrode arrays. But the basic techniques of using a probe to measure electrical activity within the brain (electrophysiology) or from outside it (electroencephalography, or EEG) are still workhorses of neural imaging labs. "The new tools don't replace the old ones," says Jessica Cardin, a neuroscientist at the Yale School of Medicine. "They add new layers of information."

Another decades-old neuroscientific technique that remains popular today is patch clamping. Developed in the late 1970s and early 1980s, it can detect changes in the electric potential of individual cells, or even single ion channels. With a tiny glass pipette suctioned against the cell's membrane, researchers can make a small tear, sealed by the pipette tip, and detect voltage changes inside the cell. With some improvements, the patch clamp, like electrophysiology and EEG, has remained a regular part of the neuroscientist's tool kit. Recently, researchers had a robot carry out the process.

Brain Control

Manipulating the brain can also shed light on how it works. In the late 1960s, Jose Delgado of Yale University placed electrodes in the brain of a chimpanzee named Paddy to alter the animal's emotional behavior. The transmitter produced an unpleasant sensation in response to a specific pattern of activity in Paddy's amygdala. After six days of repeated stimulation, she grew depressed, and the activity pattern decreased by 99 percent. Paddy recovered after two weeks, but when Delgado repeated the experiment, she became melancholic again. "What he did was incredible," says Buzsáki, now at New York University, because Delgado could listen to and precisely manipulate brain waves in an era when vacuum tubes were considered high tech.

Over the past 50 years, with ever more compact and precise electrodes, researchers have continued to use electromagnetic stimulation to manipulate neural activity with the goal of understanding brain function. Transcranial magnetic stimulation, or applying a non-invasive magnetic field outside the head, has been used to reveal the physiological underpinnings of neurological diseases and identify potential treatments in animal models, for example. And following in Delgado's footsteps, Buzsáki and his colleagues have obliterated ripples of neural activity responsible for consolidating the day's memories in sleeping rats. "The animal remembers nothing the next day, even though it had perfect sleep," he says.

In 2005, a powerful alternative to electrodes and magnetic fields emerged: controlling neurons with light. Boyden, along with Stanford's Karl Deisseroth and their colleagues, inserted algal ion channels that respond to light into mammalian neural cells. With photons, they could depolarize a cultured neuron's membrane at will, and researchers soon used the tool in vivo. Named science's Method of the Year by Nature in 2010, optogenetics allows researchers to target specific cell populations. The technique has been used in mice to alter or trigger a memory, shut down epileptic seizures, and inhibit aggressive behavior.

Lesion Techniques

In the lesion method, patients with brain damage are examined to determine which brain structures are damaged and how this influences the patient's behavior. Researchers attempt to correlate a specific brain area to an observed behavior by using reported experiences and research observations. Researchers may conclude that the loss of functionality in a brain region causes behavioral changes or deficits in task performance. For example, a patient with a lesion in the parietal-temporal-occipital association area will exhibit agraphia, a condition in which he/she is not able to write, despite having no deficits in motor ability. If damage to a particular brain region (structure X) is shown to correlate with a specific change in behavior (behavior Y), researchers may deduce that structure X has a relation to behavior Y.

In humans, lesion are most often caused by tumors or strokes. Through current brain imaging technologies, it is possible to determine which area was damaged during a stroke. Loss of function in the stroke victim may then be correlated with that damaged brain area. While lesion studies in humans have provided key insights into brain organization and function, lesions studies in animals offer many advantages.

First, animals used in research are reared in controlled environmental conditions that limit variability between subjects. Secondly, researchers are able to measure task performance in the same animal, before and after a lesion. This allows for within-subject comparison. And third the control groups can be watched who either did not undergo surgery or who did have surgery in another brain area. These benefits also increase the accuracy of the hypothesis being tested which is more difficult in human research because the before-after comparison and control experiments drop out.

To strengthen conclusions regarding a brain area and task performance, researchers may perform double dissociation. The goal of this method is to prove that two dissociations are independent. Through comparison of two patients with differing brain damage and contradictory disease patterns, researchers may localize different behaviors to each brain area. Broca's area is a region of the brain is responsible for language processing, comprehension and speech production. Patients with a lesion in Broca's area will exhibit Broca's aphasia or non-fluent aphasia. These patients are unable to speak fluently; a sentence produced by a patient with damage to the Broca's area may sound like: "I ... er ... wanted ... ah ... well ... I ... wanted to ... er ... go surfing ... and ..er ... well...". On the other hand, Wernicke's area is responsible for speech comprehension. A patient with a lesion in this area has Wernicke's aphasia. They may be able to produce language, but lack the ability to produce meaningful sentences. Patients may produce 'word salad': "I then did this chingo for some hours after my dazi went through meek and been sharko". Patients with Wernicke's aphasia are often unaware of speech deficits and may believe that they are speaking properly.

Certainly one of the famous "lesion" cases was that of Phineas Gage. On 13 September 1848 Gage, a railroad construction foreman, was using an iron rod to tamp an explosive charge into a body of rock when premature explosion of the charge blew the rod through his left jaw and out the top of his head. Miraculously, Gage survived, but reportedly underwent a dramatic personality change as a result of destruction of one or both of his frontal lobes. The uniqueness of Gage case (and the ethical impossibility of repeating the treatment in other patients) makes it difficult to draw generalizations from it, but it does illustrate the core idea behind the lesion method.

Non-invasive Methods – EEG, Scanning Methods

EEG, PET, MRI, and fMRI scan the brain through a variety of methods and have varying degrees of specificity and invasiveness.

Neuroimaging, or brain scanning, includes the use of various techniques to directly or indirectly image the structure, function, or pharmacology of the brain. It is a relatively new discipline within medicine, neuroscience, and psychology. Physicians who specialize in the performance and interpretation of neuroimaging in the clinical setting are known as neuroradiologists.

Neuroimaging falls into two broad categories:

1. Structural imaging, which deals with the structure of the brain and the diagnosis of large-scale intracranial disease (such as a tumor), as well as injury.
2. Functional imaging, which is used to diagnose metabolic diseases and lesions on a finer scale (such as Alzheimer's disease), and also for neurological and cognitive-psychology research. Functional imaging allows the brain's information processing to be visualized directly, because activity in the involved area of the brain increases metabolism and "lights up" on the scan.

Four of the most common types of brain scans are EEG, PET, MRI, and fMRI.

Electroencephalography (EEG)

Electroencephalography (EEG) is used to show brain activity in certain psychological states, such as alertness or drowsiness. It is useful in the diagnosis of seizures and other medical problems that involve an overabundance or lack of activity in certain parts of the brain.

To prepare for an EEG, electrodes are placed on the face and scalp. After placing each electrode in the right position, the electrical potential of each electrode can be measured. According to a person's state (waking, sleeping, etc.), both the frequency and the form of the EEG signal differ. Patients who suffer from epilepsy show an increase of the amplitude of firing visible on the EEG record. The disadvantage of EEG is that the electric conductivity—and therefore the measured electrical potentials—may vary widely from person to person and also over time, due to the natural conductivities of other tissues such as brain matter, blood, and bones. Because of this, it is sometimes unclear exactly which region of the brain is emitting a signal.

Positron Emission Tomography (PET)

Positron emission tomography (PET) scans measure levels of the sugar glucose in the brain in order to illustrate where neural firing is taking place. This works because active neurons use glucose as fuel. As part of the scan, a tracer substance attached to radioactive isotopes is injected into the blood. When parts of the brain become active, blood

(which contains the tracer) is sent to deliver oxygen. This creates visible spots, which are then picked up by detectors and used to create a video image of the brain while performing a particular task. However, with PET scans, we can only locate generalized areas of brain activity and not specific locations. In addition, PET scans are costly and invasive, making their use limited. However, they can be used in some forms of medical diagnosis, including for Alzheimer's.

Magnetic Resonance Imaging (MRI)

Magnetic resonance imaging (MRI) and functional magnetic resonance imaging (fMRI) scans are the form of neural imaging most directly useful to the field of psychology.

An MRI uses strong magnetic fields to align spinning atomic nuclei (usually hydrogen protons) within body tissues, then disturbs the axis of rotation of these nuclei and observes the radio frequency signal generated as the nuclei return to their baseline status. Through this process, an MRI creates an image of the brain structure. MRI scans are non-invasive, pose little health risk, and can be used on infants and in utero, providing a consistent mode of imaging across the development spectrum. One disadvantage is that the patient has to hold still for long periods of time in a noisy, cramped space while the imaging is performed.

The fMRI is a series of MRIs that measures both the structure and the functional activity of the brain through computer adaptation of multiple images. Specifically, the fMRI measures signal changes in the brain that are due to changing neural activity. In an fMRI, a patient can perform mental tasks and the area of action can be detected through blood flow from one part of the brain to another by taking pictures less than a second apart and showing where the brain "lights up." For example, when a person processes visual information, blood rushes to the back of the brain, which is where the occipital lobe is located. FMRI's make it possible to show when things happen, how brain areas change with experience, and which brain areas work together. They have been used to study a wide range of psychological phenomena, including (but by no means limited to) the neural activity of telling a lie, the differences between novices and experts when playing a musical instrument, and what happens inside our heads when we dream.

MUSCULAR AND GLANDULAR SYSTEM

Understanding human behaviour is a very complex task because of the different activities of both the body and mind. The bodily factors which have a bearing on our behaviour include sense organs, nervous system, muscles, endocrine glands, etc.

The role of sense organs and nervous system is already dealt. The muscles and glands influence our behaviour in many ways. Let us understand the role of muscles and glands in controlling our behaviour.

Muscles

The muscles are responsible for many types of movements and behaviours which are internal as well as external. The main characteristic of muscle tissue is its ability to contract and making movements possible.

There are three types of muscles in our body. They are:

1. **Skeletal muscles.** As the name implies skeletal muscle attaches to the skeleton and enables body movement. It is normally under voluntary (conscious) control.

2. **Cardiac muscle.** It is the muscle of heart and is responsible for pumping the blood. It is under involuntary control.
3. The smooth muscles form the walls of hollow organs (except heart) and also found in the skin and the eyes. These are responsible for a number of functions such as movement of food through the digestive tract and emptying of the urinary bladder. These are controlled involuntarily. The major functions of the muscular system are:
 - a. Overall movements of the body such as walking, running, and manipulation of objects with hands, maintenance of body posture, respiration, and production of body heat.
 - b. Muscles also help in communication functions such as speaking, writing, typing, gestures, and facial expressions.
 - c. They also help in constriction of organs and vessels which help propel and mix food and water in digestive tract, propel secretions from organs and regulate blood flow through vessels.

Muscle Tone

Another factor related to functioning of muscles is the muscle tone. This refers to the constant tension produced over long periods of time. Muscle tone is necessary for keeping the back and legs straight, the head held in an upright position and the abdomen from bulging. Muscle contraction is very important to enable the individual to perform long term activities like running, marathon, listening to long lecture, shooting and such other activities.

Energy for muscle contraction is supplied to the muscles in the form of adenosine triphosphate (ATP). Muscle fatigue results when ATP is used during muscle contraction faster than it can be produced in the muscle fibers. From the psychological point of view, the most common type of fatigue is 'psychological fatigue'.

This involves the central nervous system rather than the muscles themselves. The muscles are still capable of contracting, but the individual 'perceives' that additional muscle contraction is impossible. For example, a burst of activity in a tired athlete in response to spectator encouragement is an example of how psychological fatigue can be overcome.

It is important for the nurse to know these functions of muscles and help the patients to maintain rest, relaxation, and exercise to keep the health fit.

Glands

There are many glands in human beings situated in different parts of the body.

They are of two types: (1) Duct glands and (2) ductless or endocrine glands.

1. **Duct Glands.** The duct glands release their secretions through small ducts or tubes into the body cavities or on to the surface of the body. For example, salivary gland, sweat glands, lacrimal glands which produce tears, glands which secrete digestive juices, etc.

These glands are activated during emotional situations. For example, sweating increases during fear or anger, tears increase during grief, saliva decreases during fear and make our mouth become dry, digestion slows down during emotions, but

constant emotion leads to increased secretion of hydrochloric acid leading to peptic or duodenal ulcers.

2. **Endocrine Glands (Ductless Glands).** Endocrine glands are more important from the point of view of behaviour. Endocrine glands release their secretions called 'hormones' directly into blood stream.

The normal secretions of these hormones promote healthy and normal personality. But over or under secretion of these hormones affect the development of body, general metabolism, mental development and emotional behaviour. Some of the important glands which are directly related to the development of personality and behaviour are:

1. **Pituitary gland.** This is also called the master gland, because it controls the functions of many glands. There are two lobes in this gland. The anterior lobe secretes hormones called trophic hormones which influence the secretions of other glands. For example, thyrotrophic hormone, gonadotrophic hormone, etc.

The anterior lobe controls the growth of body, prolactin secretion in women, insulin secretion, metabolic activities, sexual activities, etc. The hyperactivity of this gland leads to gigantism and under activity leads to dwarfism.

2. **Thyroid gland.** It secretes thyroxin hormone. The normal secretion of thyroxin hormone regulates oxygen consumption and helps energy output. Hyperthyroidism causes increased nervous tension or excitement, insomnia and over activity.

Hypothyroidism causes sluggishness, forgetfulness, stupidity, dullness, etc. Hypothyroidism in childhood leads to a disease called cretinism and during adulthood it leads to myxoedema.

3. **Parathyroid gland.** These glands are four in number. They secrete a hormone called Parathyroxin. Hyposecretion leads to excitability, muscular tremors, spasms and cramps, complete decline in secretion leads to disease called tetany.

Mentally, the individual becomes highly sensitive to criticisms and unable to control emotions. Hyperactivity results in lassitude, lack of interest, physical weakness and softness of bones due to lack of calcium, and lethargy due nervous weakness.

4. **Adrenal glands.** These glands have two parts: Medulla and Cortex. The outer part is called cortex. It produces a hormone called cortin. Under secretion of cortin leads to lethargy, fatigue, lack of interest in sexual activities, irritability, depression, poor memory, sleep disturbances, indecisiveness, etc. over secretion results in over excitability, activeness, appearance of premature sexual characteristics, etc. Excess secretion in women leads to appearance of masculine characteristics such as moustache, hard voice, rough skin and growth of hairs on the chest.

Adrenal medulla secretes a hormone called Adrenaline and Noradrenalin (epinephrine and nor-epinephrine). This hormone plays a very important role during emotional experiences. Adrenaline mobilizes the person for emergencies.

It causes rapid heartbeat, high blood pressure, respiration, release of more energy, increasing muscular strength and decreases the function of digestive and excretory organs.

5. **Sex glands.** These glands secrete sex hormones. At least a few male and female traits are related to the balance existing between male and female hormones. Sex hormones are necessary for the development of interest in sex and personality traits.

The sex hormones in males are called Androgens and in females they are called Estrogens and Progesterone. Under or over secretion affects the personality development. So also, early and late maturation affects personality and behaviour in various ways.

In addition to these glands there are some other glands such as: the pancreas produces insulin. When the insulin quantity in blood goes very high, the sugar level comes down particularly in brain, resulting in giddiness, sweating, unconscious, lack of energy in limbs, etc. pineal gland which is very important in causing males and female characteristics. Thymus gland secretes the hormones which have a bearing on sex organs and sexual activities.

In this way various glands in our body have control over our behaviour. The knowledge of these glands and their functions help nurses to understand various signs and symptoms of diseases and also the behavioural deviations associated with the activities of these glands.

BIOLOGICAL BASIS OF MOTIVATION

Hunger

When someone is hungry, the need for food dominates everything else. It motivates people to obtain and consume food. Of course we must eat to live. But, what makes you feel hungry? Studies have indicated that many events inside and outside the body may trigger hunger or inhibit it. The stimuli for hunger include stomach contractions, which signify that the stomach is empty, a low concentration of glucose in the blood, a low level of protein and the amount of fats stored in the body. The liver also responds to the lack of bodily fuel by sending nerve impulses to the brain. The aroma, taste or appearance of food may also result in a desire to eat. It may be noted that none of these alone gives you the feeling that you are hungry. All in combination act with external factors (such as taste, colour, by observing others eating, and the smell of food, etc.) to help you understand that you are hungry. Thus, it can be said that our food intake is regulated by a complex feedingsatiety system located in the hypothalamus, liver, and other parts of the body as well as the external cues available in the environment.

Some physiologists hold that changes in the metabolic functions of the liver result in a feeling of hunger. The liver sends a signal to a part of the brain called hypothalamus. The two regions of hypothalamus involved in hunger are - the lateral hypothalamus (LH) and the ventro-medial hypothalamus (VMH). LH is considered to be the excitatory area. Animals eat when this area is stimulated. When it is damaged, animals stop eating and die of starvation. The VMH is located in the middle of the hypothalamus, which is otherwise known as hunger-controlling area which inhibits the hunger drive. Now can you guess about people who overeat and become obese, and people who eat very little or who are on a diet?

Thirst

What would happen to you, if you were deprived of water for a long time? What makes you feel thirsty? When we are deprived of water for a period of several hours, the mouth and throat become dry, which leads to dehydration of body tissues. Drinking water is necessary to wet a dry mouth. But a dry mouth does not always result in water drinking behaviour.

In fact processes within the body itself control thirst and drinking of water. Water must get into the tissues sufficiently to remove the dryness of mouth and throat.

Motivation to drink water is mainly triggered by the conditions of the body: loss of water from cells and reduction of blood volume. When water is lost by bodily fluids, water leaves the interior of the cells. The anterior hypothalamus contains nerve cells called 'osmoreceptors', which generate nerve impulses in case of cell dehydration. These nerve impulses act as a signal for thirst and drinking; when thirst is regulated by loss of water from the osmoreceptors, it is called cellular-dehydration thirst. But what mechanisms stop the drinking of water? Some researchers assume that the mechanism which explains the intake of water is also responsible for stopping the intake of water. Others have pointed out that the role of stimuli resulting from the intake of water in the stomach must have something to do with stopping of drinking water. However, the precise physiological mechanisms underlying the thirst drive are yet to be understood.

Sex

One of the most powerful drives in both animals and human beings is the sex drive. Motivation to engage in sexual activity is a very strong factor influencing human behaviour. However, sex is far more than a biological motive. It is different from other primary motives (hunger, thirst) in many ways like, (a) sexual activity is not necessary for an individual's survival; (b) homeostasis (the tendency of the organism as a whole to maintain constancy or to attempt to restore equilibrium if constancy is disturbed) is not the goal of sexual activity; and (c) sex drive develops with age, etc. In case of lower animals, it depends on many physiological conditions; in case of human beings, the sex drive is very closely regulated biologically, sometimes it is very difficult to classify sex purely as a biological drive.

Physiologists suggest that intensity of the sexual urge is dependent upon chemical substances circulating in the blood, known as sex hormones. Studies on animals as well as human beings have mentioned that sex hormones secreted by gonads, i.e. testes in males and the ovaries in females are responsible for sexual motivation. Sexual motivation is also influenced by other endocrine glands, such as adrenal and pituitary glands. Sexual drive in human beings is primarily stimulated by external stimuli and its expression depends upon cultural learning.

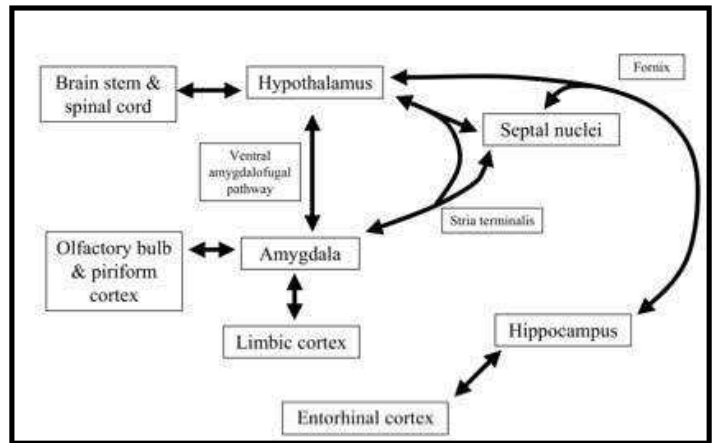
BIOLOGICAL BASIS OF EMOTION

Limbic System

The limbic system is a convenient way of describing several functionally and anatomically interconnected nuclei and cortical structures that are located in the telencephalon and diencephalon. These nuclei serve several functions, however most have to do with control of functions necessary for self preservation and species preservation. They regulate autonomic and endocrine function, particularly in response to emotional stimuli. They set the level of arousal and are involved in motivation and reinforcing behaviors. Additionally, many of these areas are critical to particular types of memory. Some of these regions are closely connected to the olfactory system, since this system is critical to survival of many species.

Areas that are typically included in the limbic system fall into two categories. Some of these are subcortical structures, while many are portions of the cerebral cortex. Cortical regions that are involved in the limbic system include the hippocampus as well as areas of neocortex including the insular cortex, orbital frontal cortex, subcallosal gyrus, cingulate gyrus and parahippocampal gyrus. This cortex has been termed the "limbic lobe" because it makes a rim surrounding the corpus callosum, following the lateral ventricle. Subcortical portions of the limbic system include the olfactory bulb, hypothalamus, amygdala, septal nuclei and some thalamic nuclei, including the anterior nucleus and possibly the dorsomedial nucleus.

One way in which the limbic system has been conceptualized is as the "feeling and reacting brain" that is interposed between the "thinking brain" and the output mechanisms of the nervous system. In this construct, the limbic system is usually under control of the "thinking brain" but obviously can react on its own. Additionally, the limbic system has its input and processing side (the limbic cortex, amygdala and hippocampus) and an output side (the septal nuclei and hypothalamus).



Hypothalamus

The hypothalamus, the primary output node for the limbic system, has many important connections. It is connected with the frontal lobes, septal nuclei and the brain stem reticular formation via the medial forebrain bundle. It also receives inputs from the hippocampus via the fornix and the amygdala via two pathways (ventral amygdalofugal pathway and stria terminalis). The hypothalamus has centers involved in sexual function, endocrine function, behavioral function and autonomic control.

In order to perform its essential functions, the hypothalamus requires several types of inputs. There are inputs from most of the body as well as from olfaction, the viscera and the retina. It also has internal sensors for temperature, osmolarity, glucose and sodium concentration. In addition, there are receptors for various internal signals, particularly hormones. These include steroid hormones, and other hormones as well as internal signals (such as hormones involved in appetite control such as leptin and orexin).

The hypothalamus strongly influences many functions including autonomic, endocrine functions and behaviors. Autonomic functions are controlled via projections to the brain stem and spinal cord. There are localized areas in the hypothalamus that will activate the sympathetic nervous system and some that will increase parasympathetic activity. Endocrine functions are controlled either by direct axonal connections to the posterior pituitary gland (vasopressin and oxytocin control) or via release of releasing factors into the hypothalamic-hypophyseal portal system (to influence anterior pituitary func-

tion). There are also projections to the reticular formation that are involved in certain behaviors, particularly emotional reactions.

Some functions are intrinsic to the hypothalamus. These are functions that require a direct input to the hypothalamus and where the response is generated directly via hypothalamic outputs. Included are such things as temperature and osmolarity regulation. There are many functions where the hypothalamus monitors the internal milieu and produces a regulatory response. These include the regulation of endocrine functions and appetite. For example, the ventromedial nucleus of the hypothalamus is considered a satiety area, while the lateral hypothalamic area is a feeding center.

Additionally, there are many complex behaviors that are patterned by the hypothalamus, including sexual responses. The preoptic area is one of the areas of greatest sexual dimorphism (i.e., difference in structure between the sexes) and, along with the septal nuclei, is an area of gonadotropin releasing hormone projections to the median eminence region of the hypothalamus. These sexual responses involve autonomic, endocrine and behavioral responses.

Finally, the suprachiasmatic nucleus receives direct retinal input. This nucleus is responsible for entraining circadian rhythms to the day-night cycle.

Amygdala

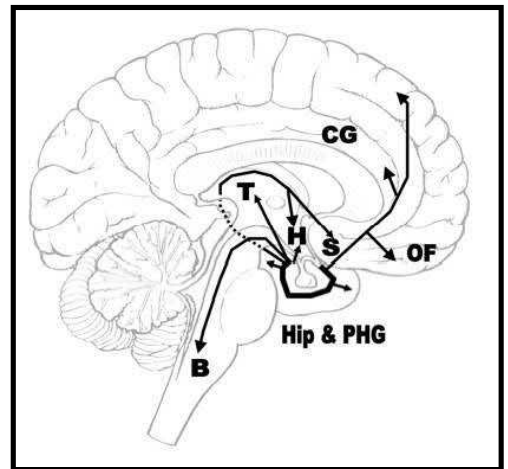
The amygdala is an important structure located in the anterior temporal lobe within the uncus. The amygdala makes reciprocal connections with many brain regions including the thalamus, hypothalamus, septal nuclei, orbital frontal cortex, cingulate gyrus, hippocampus, parahippocampal gyrus, and brain stem. The olfactory bulb is the only area that makes input to the amygdala and does not receive reciprocal projections from the amygdala.

The amygdala is a critical center for coordinating behavioral, autonomic and endocrine responses to environmental stimuli, especially those with emotional content. It is important to the coordinated responses to stress and integrates many behavioral reactions involved in the survival of the individual or of the species, particularly to stress and anxiety. Lesions of the amygdala reduce responses to stress, particularly conditioned emotional responses. Stimulation of the amygdala produces behavioral arousal and can produce directed rage reactions.

Various stimuli produce responses mediated by the amygdala. The convergence of inputs is important since it allows the generation of learned emotional responses to a variety of situations. The amygdala responds to a variety of emotional stimuli, but mostly those related to fear and anxiety.

Hippocampus

The hippocampus is an ancient area of cerebral cortex that has three layers. This is located in the medial aspect of the temporal lobe, forming the medial wall of the lateral

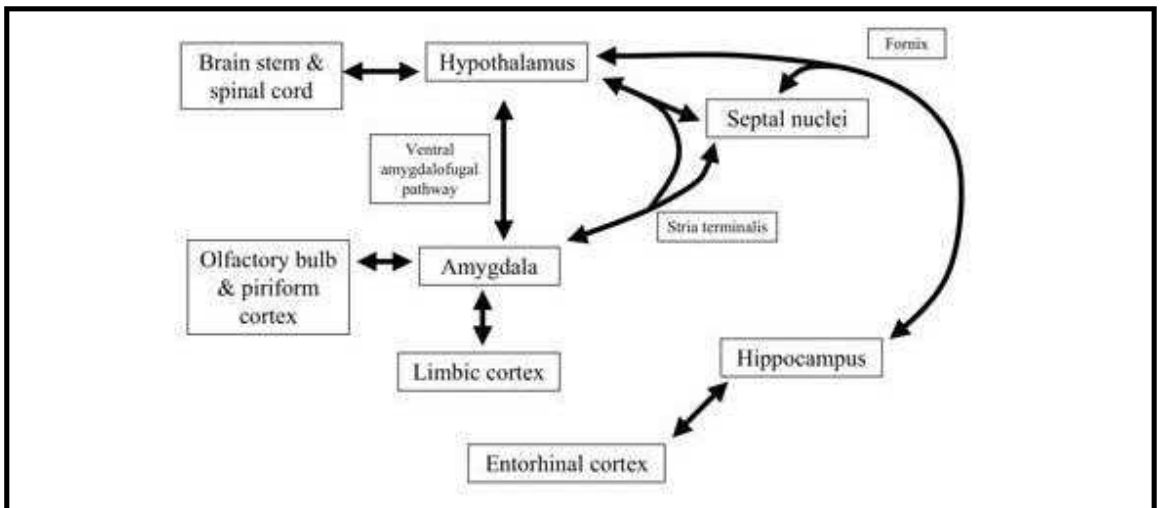


ventricle in this area. The hippocampus has several parts. The dentate gyrus contains densely packed granule cells. There is a curved area of cortex called the Cornu Ammonis (CA) that is divided into four regions called the CA fields. These are designated as CA1 to CA4. These contain prominent pyramidal cells. The CA fields blend into the adjacent subiculum, which, in turn, is connected to the entorhinal cortex on the parahippocampal gyrus of the temporal lobe.

There are several sources of hippocampal afferents. These are primarily from the septum and hypothalamus via the fornix and from the adjacent entorhinal cortex. This cortical region receives input from diffuse areas of the neocortex, especially the limbic cortex, and from the amygdala. The entorhinal cortex projects to the dentate gyrus of the hippocampus via the perforant pathway, synapsing on granule cells. These granule cells connect to pyramidal neurons in the CA3 region, which, in turn, project by Sheaffer collaterals to CA1 pyramidal cells. It is these latter cells that give rise primarily to the fornix. The physiology of these pathways has been studied extensively, particularly in terms of long-term physiological changes associated with memory. Hippocampal neurons have been studied extensively in terms of long-term potentiation. This requires activation of glutamate receptors and results in long-term changes in neuronal excitability by way of calcium mediated physiologic effects.

Outputs from the hippocampus pass primarily via two pathways. The first of these outputs is through the fornix. These fibers project to the mammillary bodies via the post-commissural fornix, to the septal nuclei, to the preoptic nucleus of the hypothalamus, to the ventral striatum and to portions of the frontal lobe through the pre-commissural fornix. There are large numbers of projections from the hippocampus back to the entorhinal cortex.

Figure demonstrates many of the important pathways within the limbic system. Note the hippocampus has reciprocal connections with the cortex as well as outputs along the fornix. Historically, the loop starting with the hippocampus projecting to the mammillary bodies, with relay to the anterior thalamic nucleus, then the cingulate gyrus, entorhinal cortex and back to the hippocampus was thought to be an important circuit. This received the name "Papez circuit". The circular nature of this connection, however, does not appear to be of functional significance.



The hippocampus has several functions. It helps control corticosteroid production. It also has significant contribution to understanding spatial relations within the environment. Additionally the hippocampus is critically involved in many declarative memory functions.

There are several types of memory. Explicit, or declarative memory refers to the memory of facts and events. Any memory that can be completely explained in words is of this type. Implicit or non-declarative memory, however, is also very important. The learning of skills as well as associative learning, such as conditioned and emotional responses are common examples of non-declarative or implicit memory. Explicit memory depends on the medial temporal lobe and the relationship between the hippocampus and entorhinal region of the parahippocampal gyrus.

There are several areas involved in explicit memory. The hippocampus plays a critical role in short-term memory, which is absolutely necessary if long-term memory patterns are to be established. Lesions of the hippocampus do not affect old, established memories. These lesions affect new declarative learning. Ultimately, memory storage is transferred to other areas of the cerebral cortex, and the location of encoding of these memories may be a function of the type of memory. Established memories involve association areas in the frontal lobe and parieto-temporo-occipital association cortex.

The hippocampus is not only active in encoding memories but also in retrieval of them. Activation of the hippocampus can be seen in this case of learning about new surroundings and of retrieving directions.

Limbic Cortex

The prefrontal cortex is anterior to the premotor cortex. The orbital frontal cortex is the portion over the orbits. This part of the cortex is extremely well-developed in humans and is critical to judgment, insight, motivation and mood. It is also important for conditioned emotional reactions. The prefrontal cortex receives input from the other areas of limbic cortex, from the amygdala and from septal nuclei and has reciprocal connections with each of these areas and with the dorsomedial nucleus of the thalamus.

Damage to the prefrontal area produces difficulties with abstract reasoning, judgment moods and puzzle solving. The effect of frontal lobe damage on mood depends on the specific part of the prefrontal cortex damaged. The patient's behavior is often described as tactless. Also, this part of the cortex can also be strongly affected by alcohol.

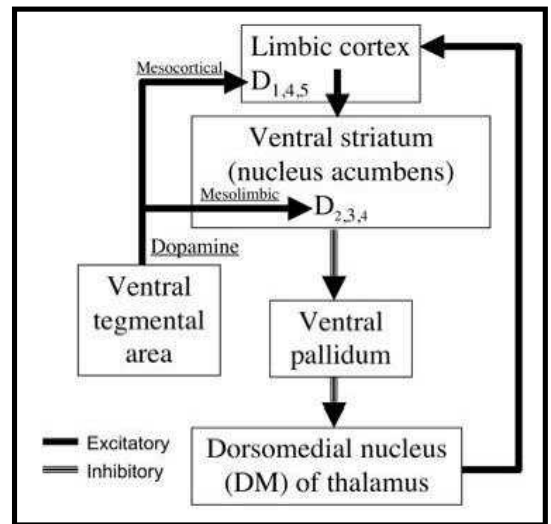
Prefrontal cortex function is abnormal in mood disorders. Depression is most often associated with increased activity in portions of the frontal lobe, especially the medial regions including the subgenual portion of the anterior cingulate cortex, and decreased activity in the posterior cingulate gyrus.

Olfaction makes strong connections with the anterior portions of the temporal lobe and the amygdala. The olfactory cortex is structurally simpler than other portions of the cerebral cortex and is termed allocortex. It includes the prepiriform and periamygdaloid cortex that comprises the anterior part of the parahippocampal gyrus covering the uncus. In some species, of course, olfaction is more important than in others. Olfactory filaments cross the cribriform plate and synapse with mitral cells in the olfactory bulbs. Axons from these cells make up the olfactory tract which extends to anterior temporal structures bilaterally as well as the basal forebrain.

Olfactory signals are relayed to several other brain regions after their initial termination in the olfactory cortex. The olfactory cortex affects the frontal lobe through connections with the dorsomedial nucleus of the thalamus. Olfactory cortex projections to the amygdala can influence emotional and endocrine reactions particularly via connections with the hypothalamus.

There are several interesting syndromes that elucidate aspects of limbic functions. The Kluver-Bucy syndrome occurs with bilateral lesions of the temporal lobes. It blocks emotional responses in animals, which become quite docile. They are unafraid of things that their species should react to, for example, in the case of an ape, a length of rope. Animals become hypersexual and engage in compulsive exploration behavior, especially with the mouth.

As previously described, there are pathways through the forebrain that are involved in reinforcement of behaviors and in "reward". Electrical stimulation of these sites is highly reinforcing for behavior. Many of these pathways involve dopamine and are commonly affected by drugs of addiction. Habituation in these pathways with chronic administration of addicting drugs is one of the most important targets of addiction research. Figure shows the ventral striatal projections to the ventral pallidum which, in turn, projects through the dorsomedial nucleus of the thalamus to limbic areas of cortex. The ventral striatum mostly consists of the nucleus accumbens, which is an important target of dopaminergic projections from the ventral tegmental area.



Various addictive compounds affect activity of the dopamine transmission in the nucleus accumbens (mesolimbic) and frontal cortical (mesocortical) systems. Additionally, these pathways appear to be functionally unbalanced in patients with schizophrenia. It appears that patients with schizophrenia have diminished dopamine effects through mesocortical systems to the prefrontal cortex. This could produce symptoms such as social withdrawal and diminished emotional responsiveness. Concurrently, there is a relative increase in dopamine effects via the mesolimbic system to the ventral striatal system, resulting in positive symptoms of delusions and hallucinations.

Hormonal Regulation of Behavior

'Divya is desperate to get a job. She has prepared well for the interview and feels confident. As she enters the room and the interview begins, she becomes extremely tense. Her feet go cold, her heart starts pounding, and she is unable to answer appropriately'.

Why did this happen? Try thinking about a similar situation that you have faced sometime in your life. Can you describe probable reasons for this? As we will see, a great deal of physiological changes happen when we experience emotion. When we are excited, afraid or angry, these bodily changes might be relatively easy to note. All of you must have noted the increase in heart rate, throbbing temples, increased perspiration, and

trembling in your limbs when you are angry or excited about something. Sophisticated equipment has made it possible to measure the exact physiological changes that accompany emotions. Both autonomic as well as somatic nervous system play important roles in the emotional process. The experience of emotions is a result of a series of neuro-physiological activations in which thalamus, hypothalamus, limbic system, and the cerebral cortex are involved significantly. Individuals with extensive injury in these brain areas have been known to demonstrate impaired emotional abilities. Selective activation of different brain areas has been experimentally shown to arouse different emotions in infants and adults.

One of the earliest physiological theories of emotion was given by James (1884) and supported by Lange, hence, it has been named the **James-Lange theory** of emotion. The theory suggests that environmental stimuli elicit physiological responses from viscera (the internal organs like heart and lungs), which in turn, are associated with muscle movement. For example, startling at an unexpected intense noise triggers activation in visceral and muscular organs followed by an emotional arousal. Put in other words, James-Lange theory argues that your perception about your bodily changes, like rapid breathing, a pounding heart, and running legs, following an event, brings forth emotional arousal. The main implication made by this theory is that particular events or stimuli provoke particular physiological changes and the individual's perception of these changes results in the emotion being experienced.

However, this theory faced a lot of criticism and fell in disuse. Another theory was proposed by Cannon (1927) and Bard (1934). **The Cannon-Bard theory** claims that the entire process of emotion is mediated by thalamus which after perception of the emotion-provoking stimulus, conveys this information simultaneously to the cerebral cortex and to the skeletal muscles and sympathetic nervous system. The cerebral cortex then determines the nature of the perceived stimulus by referring to past experiences. This determines the subjective experience of the emotion. At the same time the sympathetic nervous system and the muscles provide physiological arousal and prepare the individual to take action.

The ANS is divided into two systems, sympathetic and parasympathetic. These two manner. In a stressful situation the sympathetic system prepares the body to face the situation. It strengthens the internal environment of the individual by controlling the fall in heart rate, blood pressure, blood sugar, etc. It induces a state of physiological arousal that prepares the individual for fight or flight response in order to face the stressful situation. As the threat is removed the parasympathetic system gets active and restores the balance by calming the body. It restores and conserves energy and brings the individual back to a normal state. Though acting in an antagonistic manner, the sympathetic and parasympathetic systems are complementary to each other in completing the process of experience and expression of emotion.

GENETICS AND BEHAVIOR

Chromosomes and Genes

Chromosomes contain genetic material that can determine a person's characteristics.

Chromosomes are structures in the nucleus of a cell containing DNA, histone protein, and other structural proteins. Chromosomes also contain genes, most of which are made up of DNA and RNA.

DNA, or deoxyribonucleic acid, determines whether our eyes are blue or brown, how tall we will be, and even our preference for certain types of behavior. Known as our “genetic code,” it is shaped like a double helix, made of sequences of nucleic acids attached to a sugar phosphate backbone. Genes are subsections of DNA molecules linked together that create a particular characteristic.

Each chromosome is made up of a single DNA molecule coiled around histone proteins. Research dating back to the 1800s shows that every living creature has a specific set of chromosomes in the nucleus of each of its cells.

Human chromosomes are divided into two types—autosomes and sex chromosomes. Some genetic traits are linked to a person’s sex and therefore passed on by the sex chromosomes. The autosomes contain the remainder of a person’s genetic information. All human beings have 23 pairs of chromosomes by which genetic material is developed and characteristically demonstrated; 22 of these are autosomes, while the remaining pair (either XX, female, or XY, male) represents a person’s sex chromosomes. These 23 pairs of chromosomes work together to create the person we ultimately become.

Chromosomal abnormalities can occur during fetal development if something goes wrong during the replication of the cells. Common abnormalities include Down syndrome (caused by an extra chromosome #21), Klinefelter syndrome (caused by an extra X chromosome), and Turner syndrome (caused by a missing X chromosome). Genetic counseling is available for families in order to determine if any abnormalities exist that may be passed along to offspring. Many chromosomal abnormalities are of psychological importance, with substantial impacts on mental processes; for example, Down syndrome can cause mild to moderate intellectual disabilities.

As science advances, the ability to manipulate chromosomes is becoming more realized. Cloning is an example of taking chromosomal and genetic material and creating a new animal, and was first successfully achieved in the famous example of Dolly the sheep. There is much controversy surrounding the manipulation of chromosomes in human beings, with many people believing it to be unethical.

Gene-Environment Correlations: Nature or Nurture?

Genetic expression can be influenced by various social factors, as well as environmental factors, from light and temperature to exposure to chemicals.

Our genetic destiny is not necessarily written in stone; it can be influenced by several factors, such as social factors, as well as environmental influences among which we live, including anything from light and temperature to exposure to chemicals. The environment in which a person is raised can trigger the expression of behavior for which a person is genetically predisposed, while the same person raised in a different environment may exhibit different behavior.

Long-standing debates have taken place over the idea of which factor is more important, genes or environment. Is a person destined to have a particular outcome in life because of his or her genetic makeup, or can the environment (and the people in it) work to change what might be considered “bad” genes? Today, it is generally agreed upon that neither genes nor environment work alone; rather, the two work in tandem to create the people we ultimately become.

Environmental elements like light and temperature have been shown to induce certain changes in genetic expression; additionally, exposure to drugs and chemicals can significantly affect how genes are expressed. People often inherit sensitivity to the ef-

fects of various environmental risk factors, and different individuals may be differently affected by exposure to the same environment in medically significant ways. For example, sunlight exposure has a much stronger influence on skin cancer risk in fair-skinned humans than in individuals with an inherited tendency for darker skin. The color of a person's skin is largely genetic, but the influence of the environment will affect these genes in different ways.

Gene-environment correlations, known as rGE, can be explained in 3 particular ways—passive, evocative, or active.

Passive Gene-Environment Correlations

In passive gene-environment correlation, an association exists between a person's genetic makeup and the environment in which he or she is raised. In other words, the person's environment, particularly in the case of children, is largely determined by the parent's genetic characteristics. Parents create a home environment that is influenced by their own heritable characteristics. When the children's own genotype influences their behavior or cognitive outcomes, the result can be a misleading relationship between environment and outcome. For example, an intelligent parent is likely to create a home environment rich in educational materials and experience. Since intelligence is moderately heritable, it can be argued that intelligence in the child is inherited rather than a factor of the home environment created by the parents. It is relatively unclear whether the genetic or environmental factors had more to do with the child's development.

Evocative Gene-Environment Correlations

Evocative gene-environment correlation happens when an individual's (heritable) behavior evokes an environmental response. For example, the association between marital conflict and depression may reflect the tensions that arise when engaging with a depressed spouse rather than a causal effect of marital conflict on risk for depression.

Active Gene-Environment Correlations

In active gene-environment correlation, the person's genetic makeup may lead them to select particular environments. For example, a shy person is likely to choose quiet activities and less boisterous environments than an extroverted individual. He or she may be more likely to spend time at the library than at a dance club.

Adoption and Twin Studies in the Nature vs. Nurture Debate

Adoption and twin studies can help make sense of the influence of genes and the environment. Studies of adult twins are used to investigate which traits are heritable. Identical twins share the same genotype, meaning their genetic makeup is the same. Twins raised apart tend to be similar in intelligence and, in some cases, life events and circumstance, when studied years later, even when raised separately.

However, researchers have discovered that the phenotype (or the observable expression of a gene) of identical twins grows apart as they age. In adoption studies, identical twins raised by different families can give insight into the nature-versus-nurture debate. Since the child is being raised by parents who are genetically different from his or her biological parents, the influence of the environment shows in how similar the child is to his or her adoptive parents or siblings. Adoption studies make a strong case for the

influence of environment, whereas twin studies make a strong case for genetic influence.

The Influence of Genes on Behavior

Genetic makeup has a large role in determining human behavior.

The influence of genes on behavior has been well established in the scientific community. To a large extent, who we are and how we behave is a result of our genetic makeup. While genes do not determine behavior, they play a huge role in what we do and why we do it.

Behavioral Genetics

Behavioral genetics studies heritability of behavioral traits, and it overlaps with genetics, psychology, and ethology (the scientific study of human and animal behavior). Genetics plays a large role in when and how learning, growing, and development occurs. For example, although environment has an effect on the walking behavior of infants and toddlers, children are unable to walk at all before an age that is predetermined by their genome. However, while the genetic makeup of a child determines the age range for when he or she will begin walking, environmental influences determine how early or late within that range the event will actually occur.

Classical Genetics

Classical, or Mendelian, genetics examines how genes are passed from one generation to the next, as well as how the presence or absence of a gene can be determined via sexual reproduction. Gregor Mendel is known as the father of the field of genetics, and his work with plant hybridization (specifically pea plants) demonstrated that certain traits follow particular patterns. This is referred to as the law of Mendelian inheritance.

Genes can be manipulated by selective breeding, which can have an enormous impact on behavior. For example, some dogs are bred specifically to be obedient, like golden retrievers; others are bred to be protective, like German shepherds. In another example, Seymour Benzer discovered he could breed certain fruit flies with others to create distinct behavioral characteristics and change their circadian rhythms.

The Influence of Behavior on Genes

Behavior can influence genetic expression in humans and animals by activating or deactivating genes.

Behavior can have an impact on genetic makeup, even as early as the prenatal period. It is important to understand the implications of behavior on genetic makeup in order to reduce negative environmental and behavioral influences on genes.

EEG and PET scans have the ability to show psychologists how certain behaviors trigger reactions in the brain. This has led to the discovery of specific genes, such as those that influence addictive behaviors. A variety of behaviors have been shown to influence gene expression, including—but not limited to—drug use, exposure to the elements, and dietary habits.

Drugs and Alcohol

Prenatal exposure to certain substances, particularly drugs and alcohol, has detrimental effects on a growing fetus. The most serious consequences of prenatal drug or alcohol

exposure involve newborn addiction and fetal alcohol syndrome (FAS). Fetal alcohol syndrome affects both physical and mental development, damaging neurons within the brain and often leading to cognitive impairment and below-average weight. Exposure to drugs and alcohol can also influence the genes of children and adults. Addiction is thought to have a genetic component, which may or may not be caused by a genetic mutation resulting from drug or alcohol use.

Temperature

Temperature exposure can affect gene expression. For example, in Himalayan rabbits, the genetic expressions of fur, skin, and eyes are regulated by temperature. In the warm areas of the rabbits' bodies, the fur lacks pigment due to gene inactivity and turns white. On the extremities of the rabbits' bodies (nose, ears and feet) the gene is activated and therefore pigmented (usually black).

Light

Light exposure also influences genetic expression. Thomas Hunt Morgan performed an experiment in which he exposed some caterpillars to light and kept others in darkness. Those exposed to certain light frequencies had corresponding wing colors when they became butterflies (for example, red produced vibrant wing color, whereas blue led to pale wings). Darkness resulted in the palest wing color, leading him to conclude that light exposure influenced the genes of the butterflies. In this manner a caterpillar's behavior can directly affect gene expression; a caterpillar that actively seeks out light will appear different as a butterfly than one that avoids it.

Nutrition

Lack of proper nutrition in early childhood is yet another factor that can lead to the alteration of genetic makeup. Human children who lack proper nutrition in the first three years of life tend to have more genetic problems later in life, such as health issues and problems with school performance.



ATTENTION, PERCEPTION, LEARNING, MEMORY AND FORGETTING

ATTENTION

Attention is a limited resource used to selectively concentrate on some information while ignoring other perceivable information.

Attention is the behavioral and cognitive process of selectively concentrating on a discrete stimulus while ignoring other perceivable stimuli. It is a major area of investigation within education, psychology, and neuroscience. Attention can be thought of as the allocation of limited processing resources: your brain can only devote attention to a limited number of stimuli. Attention comes into play in many psychological topics, including memory (stimuli that are more attended to are better remembered), vision, and cognitive load.

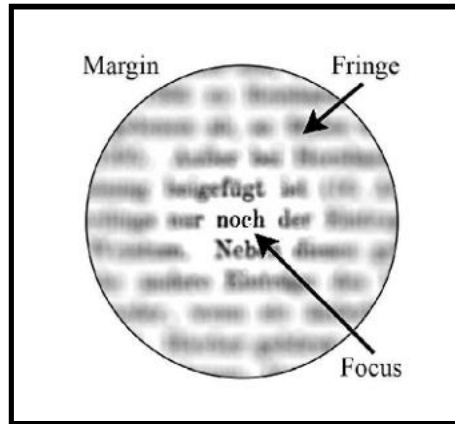
Visual Attention

Generally speaking, visual attention is thought to operate as a two-stage process. In the first stage, attention is distributed uniformly over the external visual scene and the processing of information. In the second stage, attention is concentrated to a specific area of the visual scene; it is focused on a specific stimulus. There are two major models for understanding how visual attention operates, both of which are loose metaphors for the actual neural processes occurring.

Spotlight Model

The term “spotlight” was inspired by the work of William James, who described attention as having a focus, a margin, and a fringe. The focus is the central area that extracts

“high-resolution” information from the visual scene where attention is directed. Surrounding the focus is the fringe of attention, which extracts information in a much more crude fashion. This fringe extends out to a specified area, and the cutoff is called the margin.



SPOTLIGHT MODEL: In the spotlight model, the focus is the central area of attention, which takes in highly detailed information. The fringe takes in less detailed information, and the margin is the cutoff for taking in any information.

Zoom-Lens Model

First introduced in 1986, this model inherits all the properties of the spotlight model, but it has the added property of changing in size. This size-change mechanism was inspired by the zoom lens one might find on a camera, and any change in size can be described by a trade-off in the efficiency of processing. The zoom-lens of attention can be described in terms of an inverse trade-off between the size of focus and the efficiency of processing. Because attentional resources are assumed to be fixed, the larger the focus is, the slower processing will be of that region of the visual scene, since this fixed resource will be distributed over a larger area.

Cognitive Load

Think of a computer with limited memory storage: you can only give it so many tasks before it is unable to process more. Brains work on a similar principle, called the cognitive load theory. “Cognitive load” refers to the total amount of mental effort being used in working memory. Attention requires working memory; therefore devoting attention to something increases cognitive load.

Multitasking and Divided Attention

Multitasking can be defined as the attempt to perform two or more tasks simultaneously; however, research shows that when multitasking, people make more mistakes or perform their tasks more slowly. Each task increases cognitive load; attention must be divided among all of the component tasks to perform them.

Older research involved looking at the limits of people performing simultaneous tasks like reading stories while listening to and writing something else, or listening to two separate messages through different ears (i.e., dichotic listening). The vast majority of

current research on human multitasking is based on performance of doing two tasks simultaneously, usually involving driving while performing another task such as texting, eating, and speaking to passengers in the vehicle or talking on a cell phone. This research reveals that the human attentional system has limits to what it can process: driving performance is worse while engaged in other tasks; drivers make more mistakes, brake harder and later, get into more accidents, veer into other lanes, and are less aware of their surroundings when engaged in the previously discussed tasks.

Selective Attention

Studies show that if there are many stimuli present (especially if they are task-related), it is much easier to ignore the non-task-related stimuli, but if there are few stimuli the mind will perceive the irrelevant stimuli as well as the relevant.

Some people can process multiple stimuli with practice. For example, trained Morse-code operators have been able to copy 100% of a message while carrying on a meaningful conversation. This relies on the reflexive response that emerges from “overlearning” the skill of Morse-code transcription so that it is an autonomous function requiring no specific attention to perform.

PERCEPTION

Perception is a highly complex process, which involves a complicated interaction of selection, organisation, and interpretation. Despite the fact that it relies upon senses for obtaining raw data, the process tends to amalgamate, improve and change these data because of the complexity of interaction. It adds as well as deducts from the sensory world.

“Perception is a process by which individuals organise and interpret their sensory impressions in order to give meaning to their environment.”

“The psychological processes that allow an individual to adjust his behaviour is called perception.

“Perception is a cognitive process which selects, organises and interprets the stimuli.”

“It is a cognitive process by which people attend to incoming stimuli, organise and interpret such stimuli into behaviour.”

“Perception is the selection and organisation of material which stems from the outside environment at some time or the other to provide meaning to what we experience.”

Elements of Perception

Perception is a process of sensory organs. The mind gets the information through the five sense organs, *i.e.*, eyes, nose, ears, tongue, and skin. The stimulation comes to the organs through action, written messages, oral communication, taste, touch, etc. The perception starts with the awareness of these stimuli. Recognising these stimuli takes place only after paying attention to them. These message are then translated into action.

Perception involves several elements (subprocesses) which are listed below:

Stimuli. The receipt of information is the stimulus which results in sensation. Knowledge and behaviour depend on senses and their stimulation. These senses are influenced by a larger number of stimuli. The family, social and the economic environment are important stimuli for the people. The physiological and psychological functions are

impact of these stimuli. The intensive and extensive forms of stimuli have a greater impact on the sensory organs. The physical work environment, socio-cultural environment and other factors have certain stimuli to influence the employee's perception.

In organisational settings, the supervisor may form the stimulus situation for the worker's perceptual process.

Attention. The stimuli that are paid attention depend purely on the people's selection capacity and the intensity of stimuli. Educated employees pay more attention to any stimuli, for example, announcement of bonus, appeal for efficiency, training, and motivation. The management has to find out suitable stimuli, which can appeal to the employees at the maximum level. An organisation should be aware of all those factors, which affect the attention of the employees. During the attention process, sensory and neural mechanisms are affected and the message receiver becomes involved in understanding the stimuli. Taking employees to the attention stage is essential in an organisation for making them behave in a systematic and required order.

Recognition. The messages or incoming stimuli are recognised before they are transmitted into behaviour. Perception is a two-phase activity, *i.e.*, receiving stimuli and translating the stimuli into action. The recognition process is dependent on mental acceptability. For example, if a car driver suddenly sees a child in front of his running car, he stops the car. He recognises the stimuli, *i.e.*, the life of the child is in danger. His mental process recognises the danger after paying attention to the stimuli. If he does not pay attention to the stimuli, he cannot recognise the danger. After recognising the stimuli, he translates the message into behaviour.

Translation. The management in an organisation has to consider the various processes of translating the message into action. The employees should be assisted to translate the stimuli into action. For example, the announcement of bonus should be recognised as a stimulus for increasing production. The employee should translate it into appropriate behaviour. In other words, they should be motivated by the management to increase productivity.

Behaviour. Behaviour is the outcome of the cognitive process. It is a response to change in sensory inputs, *i.e.*, stimuli. Perceptual behaviour is not influenced by reality, but is a result of the perception process of the individual, his learning and personality, environmental factors and other internal and external factors at the workplace. The psychological feedback that may influence the perception of an employee may be superior behaviour, his eye movement, raising of an eyebrow, the tone of voice, etc. The behaviour of employees depends on perception, which is visible in the form of action, reaction or other behaviour. The behavioural termination of perception may be overt or covert. The perception behaviour is the result of the cognitive process of the stimulus which may be a message or an action situation of management function. Perception is reflected in behaviour, which is visible in different forms of employees' action and motivation.

Performance. Proper behaviour leads to higher performance. High performers become a source of stimuli and motivation to other employees. A performance-reward relationship is established to motivate people.

Satisfaction high performance gives more satisfaction. The level of satisfaction is calculated with the difference in performance and expectation. If the performance is more than the expectation, people are delighted, but when performance is equal to expectation,

it results in satisfaction. On the other hand, if performance is less than the expectation, people become frustrated and this requires a more appealing form of stimulus of developing proper employee work behaviour and high performance. It is essential to understand the factors that influence the perception process and mould employees' behaviour towards the corporate objectives and self-satisfaction.

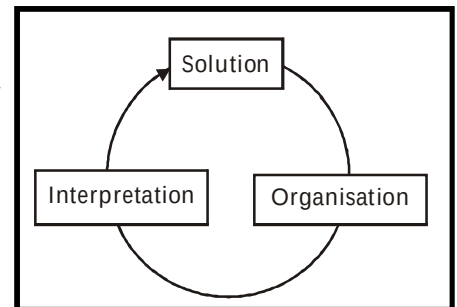
Perception Process

Perception is the process of selecting, organising, and interpreting information. This process, which is shown in below figure. "The Perception Process", includes the perception of select stimuli that pass through our perceptual filters, are organised into our existing structures and patterns, and are then interpreted based on previous experiences. Although perception is a largely cognitive and psychological process, how we perceive the people and objects around us affects our communication. We respond differently to an object or person that we perceive favorably than we do to something we find unfavorable. But how do we filter through the mass amounts of incoming information, organise it, and make meaning from what makes it through our perceptual filters and into our social realities?

Selecting Information. We take in information through all five of our senses, but our perceptual field (the world around us) includes so many stimuli that it is impossible for our brains to process and make sense of it all. So, as information comes in through our senses, various factors influence what actually continues on through the perception process. Selecting is the first part of the perception process, in which we focus our attention on certain incoming sensory information. Think about how, out of many other possible stimuli to pay attention to, you may hear a familiar voice in the hallway, see a pair of shoes you want to buy from across the mall, or smell something cooking for dinner when you get home from work. We quickly cut through and push to the background all kinds of sights, smells, sounds, and other stimuli, but how do we decide what to select and what to leave out?

We tend to pay attention to information that is salient. Salience is the degree to which something attracts our attention in a particular context. The thing attracting our attention can be abstract, like a concept, or concrete, like an object. For example, a person's identity as a Native American may become salient when they are protesting at the Columbus Day parade in Denver, Colorado. Or a bright flashlight shining in your face while camping at night is sure to be salient. We tend to find salient things that are visually or aurally stimulating and things that meet our needs or interests. Lastly, expectations affect what we find salient.

Visual and Aural Stimulation. It is probably not surprising to learn that visually and/or aurally stimulating things become salient in our perceptual field and get our attention. Creatures ranging from fish to humming birds are attracted to things like silver spinners on fishing poles or red and yellow bird feeders. Having our senses stimulated isn't always a positive thing though. Think about the couple that won't stop talking dur-



THE PERCEPTION PROCESS

ing the movie or the upstairs neighbour whose subwoofer shakes your ceiling at night. In short, stimuli can be attention-getting in a productive or distracting way. As communicators, we can use this knowledge to our benefit by minimising distractions when we have something important to say. It's probably better to have a serious conversation with a significant other in a quiet place rather than a crowded food court. Conversely, nonverbal adaptors, or nervous movements we do to relieve anxiety like pacing or twirling our hair, can be distracting. Aside from minimising distractions and delivering our messages enthusiastically, the content of our communication also affects salience.

Needs and Interests. We tend to pay attention to information that we perceive to meet our needs or interests in some way. This type of selective attention can help us meet instrumental needs and get things done. When you need to speak with a financial aid officer about your scholarships and loans, you sit in the waiting room and listen for your name to be called. Paying close attention to whose name is called means you can be ready to start your meeting and hopefully get your business handled. When we don't think certain messages meet our needs, stimuli that would normally get our attention may be completely lost. Imagine you are in the grocery store and you hear someone say your name. You turn around, only to hear that person say, "Finally! I said your name three times. I thought you forgot who I was!" A few seconds before, when you were focused on figuring out which kind of orange juice to get, you were attending to the various pulp options to the point that you tuned other stimuli out, even something as familiar as the sound of someone calling your name. Again, as communicators, especially in persuasive contexts, we can use this to our advantage by making it clear how our message or proposition meets the needs of our audience members. Whether a sign helps us find the nearest gas station, the sound of a ringtone helps us find our missing cell phone, or a speaker tells us how avoiding processed foods will improve our health, we select and attend to information that meets our needs.

We also find salient information that interests us. Of course, many times, stimuli that meet our needs are also interesting, but it's worth discussing these two items separately because sometimes we find things interesting that don't necessarily meet our needs. I'm sure we've all gotten sucked into a television show, video game, or random project and paid attention to that at the expense of something that actually meets our needs like cleaning or spending time with a significant other. Paying attention to things that interest us but don't meet specific needs seems like the basic formula for procrastination that we are all familiar with.

In many cases we know what interests us and we automatically gravitate toward stimuli that match up with that. For example, as you filter through radio stations, you likely already have an idea of what kind of music interests you and will stop on a station playing something in that genre while skipping right past stations playing something you aren't interested in. Because of this tendency, we often have to end up being forced into or accidentally experiencing something new in order to create or discover new interests. For example, you may not realise you are interested in Asian history until you are required to take such a course and have an engaging professor who sparks that interest in you. Or you may accidentally stumble on a new area of interest when you take a class you wouldn't otherwise because it fits into your schedule. As communicators, you can take advantage of this perceptual tendency by adapting your topic and content to the interests of your audience.

Expectations. The relationship between salience and expectations is a little more complex. Basically, we can find expected things salient and find things that are unexpected salient. While this may sound confusing, a couple examples should illustrate this point. If you are expecting a package to be delivered, you might pick up on the slightest noise of a truck engine or someone's footsteps approaching your front door. Since we expect something to happen, we may be extra tuned in to clues that it is coming. In terms of the unexpected, if you have a shy and soft-spoken friend who you overhear raising the volume and pitch of his voice while talking to another friend, you may pick up on that and assume that something out of the ordinary is going on. For something unexpected to become salient, it has to reach a certain threshold of difference. If you walked into your regular class and there were one or two more students there than normal, you may not even notice. If you walked into your class and there was someone dressed up as a wizard, you would probably notice. So, if we expect to experience something out of the routine, like a package delivery, we will find stimuli related to that expectation salient. If we experience something that we weren't expecting and that is significantly different from our routine experiences, then we will likely find it salient. We can also apply this concept to our communication. I always encourage my students to include supporting material in their speeches that defies our expectations. You can help keep your audience engaged by employing good research skills to find such information.

There is a middle area where slight deviations from routine experiences may go unnoticed because we aren't expecting them. To go back to the earlier example, if you aren't expecting a package, and you regularly hear vehicle engines and sidewalk foot traffic outside your house, those pretty routine sounds wouldn't be as likely to catch your attention, even if it were slightly more or less traffic than expected. This is because our expectations are often based on previous experience and patterns we have observed and internalised, which allows our brains to go on "autopilot" sometimes and fill in things that are missing or overlook extra things. Look at the following sentence and read it aloud: Perception is based on patterns, meaning we often reach a conclusion without considering each individual element. This example illustrates a test of our expectation and an annoyance to every college student. We have all had the experience of getting a paper back with typos and spelling errors circled. This can be frustrating, especially if we actually took the time to proofread. When we first learned to read and write, we learned letter by letter. A teacher or parent would show us a card with A-P-P-L-E written on it, and we would sound it out. Over time, we learned the patterns of letters and sounds and could see combinations of letters and pronounce the word quickly. Since we know what to expect when we see a certain pattern of letters, and know what comes next in a sentence since we wrote the paper, we don't take the time to look at each letter as we proofread. This can lead us to overlook common typos and spelling errors, even if we proofread something multiple times. As a side note, I'll share two tips to help you avoid proofreading errors: First, have a friend proofread your paper. Since they didn't write it, they have fewer expectations regarding the content. Second, read your papers backward. Since patterns of speech aren't the same in reverse you have to stop and focus on each word. Now that we know how we select stimuli, let's turn our attention to how we organise the information we receive.

Organising Information. Organising is the second part of the perception process, in which we sort and categorise information that we perceive based on innate and learned cognitive patterns. Three ways we sort things into patterns are by using proximity, simi-

larity, and difference. In terms of proximity, we tend to think that things that are close together go together. For example, have you ever been waiting to be helped in a business and the clerk assumes that you and the person standing beside you are together? The slightly awkward moment usually ends when you and the other person in line look at each other, then back at the clerk, and one of you explains that you are not together. Even though you may have never met that other person in your life, the clerk used a basic perceptual organising cue to group you together because you were standing in proximity to one another.

We also group things together based on similarity. We tend to think similar-looking or similar-acting things belong together. I have two friends that I occasionally go out with, and we are all three males, around the same age, of the same race, with short hair and glasses. Aside from that, we don't really look alike, but on more than one occasion a server at a restaurant has assumed that we're brothers. Despite the fact that many of our other features are different, the salient features are organised based on similarity and the three of us are suddenly related.

We also organise information that we take in based on difference. In this case, we assume that the item that looks or acts different from the rest doesn't belong with the group. Perceptual errors involving people and assumptions of difference can be especially awkward, if not offensive. My friend's mother, who is Vietnamese American, was attending a conference at which another attendee assumed she was a hotel worker and asked her to throw something away for her. In this case, my friend's mother was a person of colour at a convention with mostly white attendees, so an impression was formed based on the other person's perception of this difference.

These strategies for organising information are so common that they are built into how we teach our children basic skills and how we function in our daily lives. I'm sure we all had to look at pictures in grade school and determine which things went together and which thing didn't belong. If you think of the literal act of organising something, like your desk at home or work, we follow these same strategies. If you have a bunch of papers and mail on the top of your desk, you will likely sort papers into separate piles for separate classes or put bills in a separate place than personal mail. You may have one drawer for pens, pencils, and other supplies and another drawer for files. In this case you are grouping items based on similarities and differences. You may also group things based on proximity, for example, by putting financial items like your checkbook, a calculator, and your pay stubs in one area so you can update your budget efficiently. In summary, we simplify information and look for patterns to help us more efficiently communicate and get through life.

Simplification and categorising based on patterns isn't necessarily a bad thing. In fact, without this capability we would likely not have the ability to speak, read, or engage in other complex cognitive/behavioural functions. Our brain innately categorises and files information and experiences away for later retrieval, and different parts of the brain are responsible for different sensory experiences. In short, it is natural for things to group together in some ways. There are differences among people, and looking for patterns helps us in many practical ways. However, the judgments we place on various patterns and categories are not natural; they are learned and culturally and contextually relative. Our perceptual patterns do become unproductive and even unethical when the judgments we associate with certain patterns are based on stereotypical or prejudicial thinking.

We also organise interactions and interpersonal experiences based on our firsthand experiences. When two people experience the same encounter differently, misunderstandings and conflict may result. Punctuation refers to the structuring of information into a timeline to determine the cause (stimulus) and effect (response) of our communication interactions. Applying this concept to interpersonal conflict can help us see how the perception process extends beyond the individual to the interpersonal level. This concept also helps illustrate how organisation and interpretation can happen together and how interpretation can influence how we organise information and vice versa.

Where does a conflict begin and end? The answer to this question depends on how the people involved in the conflict punctuate, or structure, their conflict experience. Punctuation differences can often escalate conflict, which can lead to a variety of relationship problems. For example, Rose and Peter are on a project team at work and have a deadline approaching. Rose has been working on the project over the weekend in anticipation of her meeting with Peter first thing Monday morning. She has had some questions along the way and has e-mailed Peter for clarification and input, but he hasn't responded. On Monday morning, Rose walks into the meeting room, sees Peter, and says, "I've been working on this project all weekend and needed your help. I e-mailed you three times! What were you doing?" Peter responds, "I had no idea you e-mailed me. I was gone all weekend on a camping trip." In this instance, the conflict started for Rose two days ago and has just started for Peter. So, for the two of them to most effectively manage this conflict, they need to communicate so that their punctuation, or where the conflict started for each one, is clear and matches up. In this example, Rose made an impression about Peter's level of commitment to the project based on an interpretation she made after selecting and organising incoming information. Being aware of punctuation is an important part of perception checking, which we will discuss later. Let's now take a closer look at how interpretation plays into the perception process.

Interpreting Information. Although selecting and organising incoming stimuli happens very quickly, and sometimes without much conscious thought, interpretation can be a much more deliberate and conscious step in the perception process. Interpretation is the third part of the perception process, in which we assign meaning to our experiences using mental structures known as schemata. Schemata are like databases of stored, related information that we use to interpret new experiences. We all have fairly complicated schemata that have developed over time as small units of information combine to make more meaningful complexes of information.

We have an overall schema about education and how to interpret experiences with teachers and classmates. This schema started developing before we even went to pre-school based on things that parents, peers, and the media told us about school. For example, you learned that certain symbols and objects like an apple, a ruler, a calculator, and a notebook are associated with being a student or teacher. You learned new concepts like grades and recess, and you engaged in new practices like doing homework, studying, and taking tests. You also formed new relationships with teachers, administrators, and classmates. As you progressed through your education, your schema adapted to the changing environment. How smooth or troubling schema reevaluation and revision is varies from situation to situation and person to person. For example, some students adapt their schema relatively easily as they move from elementary, to middle, to high school, and on to college and are faced with new expectations for behaviour and academic engagement. Other students don't adapt as easily, and holding onto their old schema creates problems

as they try to interpret new information through old, incompatible schema. We've all been in a similar situation at some point in our lives, so we know that revising our schemata can be stressful and that such revision takes effort and usually involves some mistakes, disappointments, and frustrations. But being able to adapt our schemata is a sign of cognitive complexity, which is an important part of communication competence. So, even though the process may be challenging, it can also be a time for learning and growth.

It's important to be aware of schemata because our interpretations affect our behaviour. For example, if you are doing a group project for class and you perceive a group member to be shy based on your schema of how shy people communicate, you may avoid giving him presentation responsibilities in your group project because you do not think shy people make good public speakers. Schemata also guide our interactions, providing a script for our behaviours. We know, in general, how to act and communicate in a waiting room, in a classroom, on a first date, and on a game show. Even a person who has never been on a game show can develop a schema for how to act in that environment by watching *The Price Is Right*, for example. People go to great lengths to make shirts with clever sayings or act enthusiastically in hopes of being picked to be a part of the studio audience and hopefully become a contestant on the show.

As we have seen, schemata are used to interpret others' behaviour and form impressions about who they are as a person. To help this process along, we often solicit information from people to help us place them into a preexisting schema. In the United States and many other Western cultures, people's identities are often closely tied to what they do for a living. When we introduce others, or ourselves, occupation is usually one of the first things we mention. Think about how your communication with someone might differ if he or she were introduced to you as an artist versus a doctor. We make similar interpretations based on where people are from, their age, their race, and other social and cultural factors. We will learn more about how culture, gender, and other factors influence our perceptions as we continue through the chapter. In summary, we have schemata about individuals, groups, places, and things, and these schemata filter our perceptions before, during, and after interactions. As schemata are retrieved from memory, they are executed, like computer programs or apps on your smartphone, to help us interpret the world around us. Just like computer programs and apps must be regularly updated to improve their functioning, competent communicators update and adapt their schemata as they have new experiences.

External Factors Influencing the Perception Process

The external attention factors are:

- (a) **Intensity.** The intensity of stimulus implies that the more intense the stimulus audio or visual, the more is the likelihood it will be perceived. A loud noise, strong odour or bright light or bright colours will be more readily perceived than soft sound, weak odour or dim light. It is because of this advantage that advertisers employ intensity to draw the consumers' attention.
- (b) **Size.** The size of the object or stimulus has a greater impact on the perception process because the size influences attention and recognition in a more effective manner. A Great Den dog which is tall attracts the attention. At the same time a pocket dog also attracts attention because of its size. However, generally the larger the object the more likely it will be perceived. The amount of attention enhances

with the size of the newspaper advertisement exposed to the individuals although the increase in attention may not be directly proportional to the increase in size.

- (c) **Contrast.** Contrasting objects have more impact on behaviour. The contrast principle states that external stimuli, which stand out against the background or which, are not what the people expect will receive attention. Plant safety signs, which have black lettering on a yellow background or white lettering on a red background, are attentions getting.

Any change in the accustomed atmosphere attracts attraction. If one or more of the machines should come suddenly to a halt, the supervisor would immediately notice the difference in noise level. Also a person who has fallen asleep in a bus because of the drone of the engine wakes up immediately the engine stops.

- (d) **Repetition.** Repeated stimuli have more impact on performances than a single statement. Repetition has the advantage of being attention catching. Perhaps, it is because of this that supervisors tend to repeat directions regarding job instructions several times for even simple tasks to hold the attention of their workers. Advertisers while putting T.V. or radio advertisements repeat the brand name they are advertising.
- (e) **Motion.** The factor of motion implies that the individuals attend to changing objects in their field of vision than to static objects. It is because of this advantage that advertisers involve signs, which include moving objects in their campaigns. At an unconscious level the animals in the jungles make use of this principle. A tiger lying in wait is motionless until his prey is nearer him and then jumps at an appropriate moment.
- (f) **Novelty and familiarity.** A novel object in the familiar situation or a familiar object in a novel situation tends to attract attention. Thus a white person or a black person in India catches attention faster. Job rotation is an example of this principle. Recent research indicates that job rotation not only increased attention but also employees' acquisition of new skills.
- (g) **Situations.** Situations have a great influence on people's perceptions. A favourable work environment develops a positive attitude and work culture because the perception process is easily channelised and rightly directed.

Internal Factors Influencing the Perception Process

The internal set factors are as under:

- (a) **Habit.** Habits die hard and therefore individuals perceive objects, situations and conditions differently according to their habits. A Hindu will bow and do Namaskar when he sees a temple while walking on road, because of his well-established habit. These are several instances in life settings where individuals tend to react with the right response to the wrong signals. Thus a retired soldier may throw himself on the ground when he hears a sudden burst of car tyre.
- (b) **Motivation and interest.** Two examples of motivational factors are hunger and thirst. Motivational factors increase the individual's sensitivity to those stimuli which he considers as relevant to the satisfaction of his needs in view of his past experience with them. A thirsty individual has a perceptual set to seek a water fountain or a hotel to quench his thirst, which increases for him likelihood of perceiving restaurant signs and decreases the likelihood of visualising other objects

at that moment in time. A worker who has a strong need for affiliation, when walks into the lunchroom, the table where several coworkers are sitting tends to be perceived and the empty table or the table where only one person is sitting will attract no attention.

- (c) **Learning.** The state of learning influences and plays a crucial role in the perception process. However, it should be recognised that the role of learning is more pronounced in respect of complex forms of perception where the symbolic content creeps into the process. Although interrelated with motivation and personality, learning may play the single biggest role in developing perceptual set.

People perceive as per their levels of learning. It is therefore essential for the organisation to make its employees knowledgeable and educated for their effective performance and behaviour. The learning of managers and workers is a twin requirement.

- (d) **Organisational role and specialisation.** The modern organisations value specialisation. Consequently the specialty of a person that casts him in a particular organisational role predisposes him to select certain stimuli and to disregard others. Thus in a lengthy report a departmental head will first notice the text relating to his department.
- (e) **Economic and social background.** The employee perceptions are based on economic and social backgrounds. Socially and economically developed employees have a more positive attitude towards development rather than less developed employees.
- (f) **Personality.** The personality of the perceiver as well as the stimulator have an impact on the perception process. The age, sex, race, dress, etc. of both the persons have a direct influence on the perception process.

Perceptual Process

The perceptual process is a sequence of steps that begins with the environment and leads to our perception of a stimulus and an action in response to the stimulus. This process is continual, but you do not spend a great deal of time thinking about the actual process that occurs when you perceive the many stimuli that surround you at any given moment.

The process of transforming the light that falls on your retinas into an actual visual image happens unconsciously and automatically. The subtle changes in pressure against your skin that allow you to feel object occur without a single thought.

In order to fully understand how the perception process works, we'll start by breaking down each step.

The Steps in the perceptual process:

1. **The Environmental Stimulus.** The world is full of stimuli that can attract our attention through various senses. The environmental stimulus is everything in our environment that has the potential to be perceived.

This might include anything that can be seen, touched, tasted, smelled or heard. It might also involve the sense of proprioception, such as the movements of the arms and legs or the change in position of the body in relation to objects in the environment.

For example, imagine that you are out on a morning jog at your local park. As you perform your workout, there are a wide variety of environmental stimuli that might

capture your attention. The tree branches are swaying in the slight breeze; a man is out on the grass playing fetch with his Golden Retriever; a car drives past with the windows rolled down and the music blaring; a duck splashes in a nearby pond. All of these things represent the environmental stimuli, serving as a starting point for the perceptual process.

2. **The Attended Stimulus.** The attended stimulus is the specific object in the environment on which our attention is focused. In many cases, we might focus on stimuli that are familiar to us, such as the face of a friend in a crowd of strangers at the local coffee shop. In other instances, we are likely to attend to stimuli that have some degree of novelty.

From our earlier example, let's imagine that during your morning job you focus your attention on the duck floating in the nearby pond. The duck represents the attended stimulus. During the next step of the perceptual process, the visual process will progress.

3. **The Image on the Retina.** Next, the attended stimulus is formed as an image on the retina. The first part of this process involves the light actually passing through the cornea and pupil and onto the lens of the eye. The cornea helps focus the light as it enters the eye, and the iris of the eye controls the size of the pupils in order to determine how much light to let in. The cornea and lens act together to project an inverted image on the retina.

As you might already be aware, the image on the retina is actually upside down from the actual image in the environment. At this stage of the perceptual process, this is not terribly important. The image has still not been perceived, and this visual information will be changed even more dramatically in the next step of the process.

4. **Transduction.** The image on the retina is then transformed into electrical signals in a process known as transduction. This allows the visual messages to be transmitted to the brain to be interpreted.

The retina contains many photoreceptor cells. These cells contain proteins known as rods and cones. Rods are primarily for seeing things in low light, while cones are associated with detecting colour and shapes at normal light levels.

The rods and cones contain a molecule called retinal, which is responsible for transducing the light into visual signals that are then transmitted via nerve impulses.

5. **Neural Processing.** The electrical signals then undergo neural processing. The path followed by a particular signal depends on what type of signal it is (*i.e.* an auditory signal or a visual signal).

Through the series of interconnect neurons located throughout the body, electrical signals are propagated from the receptors cells to the brain. In our previous example, the image of duck floating in the pond is received as light on the retina, which is then transduced into an electrical signal and then processed through the neurons in the visual network.

In the next step of the perceptual process, you will actually perceive the stimuli and become aware of its presence in the environment.

6. **Perception.** In the next step of the perception process, we actually perceive the stimulus object in the environment. It is at this point that we become consciously aware of the stimulus.

Let's consider our previous example, in which we imagined that you were out for a morning jog in the park. At the perception stage, you have become aware of that there is something out on the pond to perceive.

Now, it is one thing to be aware of stimuli in the environment, and quite another to actually become fully consciously aware of what we have perceived. In the next stage of the perceptual process, we will sort the perceived information into meaningful categories.

7. **Recognition.** Perception doesn't just involve becoming consciously aware of the stimuli. It is also necessary for our brain to categorise and interpret what it is we are sensing. Our ability to interpret and give meaning to the object is the next step, known as recognition.

Continuing our example, it is at the recognition stage of the perceptual process that you realise that there is a duck floating on the water. The recognition stage is an essential part of perception since it allows us to make sense of the world around us. By placing objects in meaningful categories, we are able to understand and react to the world around us.

8. **Action.** The final step of the perceptual process involves some sort of action in response to the environmental stimulus. This could involve a variety of actions, such as turning your head for a closer look or turning away to look at something else.

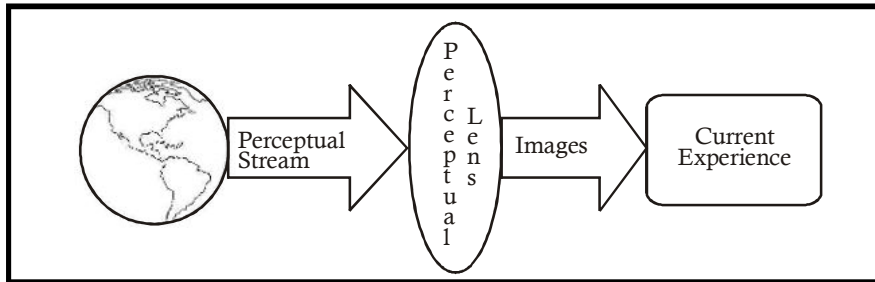
The action phase of perceptual development involves some type of motor action that occurs in response to the perceived and recognised stimulus. This might involve a major action, like running toward a person in distress, or something as subtle as blinking your eyes in response to a puff of dust blowing through the air.

Approaches to the Study of Perception

Schematically, we can summarise this way of looking at perception in what I'll call the Perceptual Process Model, shown here in a very simple form. This is essentially the model that I proposed in *The Reality Illusion*, though I used somewhat different terminology there.

Think of yourself as being continually immersed in an immense stream of information, which I'll call your perceptual stream. The globe represents the world around you, the source of that perceptual stream. In addition to the outside world it includes your body, itself a major source of the information. Your perceptual stream is processed through your perceptual lens, which we will treat conceptually as a "black box." ("Black box" is an engineering term used for a process described in terms of what the process accomplishes, without specifically addressing the internal detail of how that is actually done.) Within that lens, pieces of this overall perceptual stream are filtered out and assembled into a stream of perceptual images.

I'm using the term "image" here in a very general sense, to refer to more than just visual images. It encompasses all the perceptual representations you make for yourself. The words you hear when someone speaks to you and the melody you hear when you listen to music are auditory images; your sense of your body in space and of the movement of your arm when you pick up an object are kinesthetic images, etc. These images are not objective copies of the world "out there," as depicted in the eye-as-a-camera model. Rather, they are representations of things you know how to perceive and expect to encounter.



This model describes the way you produce the various individual dimension of perception – vision, hearing, touch, the kinesthetic sense of body position and movement, emotional feeling, intellectual understanding, etc., as well as how you produce your overall ongoing experience – the encompassing multidimensional stream of experience of which all the individual dimensions are components. I'll sometimes use the term current experience to refer to the ever changing present moment in that ongoing stream.

Looking at visual perception in isolation, the perceptual stream would be the flow of visual information coming into your eyes from the outside world, and the perceptual images would be the visual images you see as a result of that flow. More generally, your larger perceptual stream would include ambient sounds impinging on your eardrums, smells, the haptic (touch and pressure) sensations from your contact with the objects in the external world, etc. It would also include proprioceptive sensations from within your body, which we'll examine in greater depth in later postings. Your perceptual images would include your interpretations of what you hear, the sense of mass and volume of external objects derived from all your senses, and even your perceptions of opportunity or threat from external situations and your emotional responses to them.

This points up an important truth that it's easy to lose track of. Human beings are extremely complex organisms, living and functioning in extremely rich and complex environments. When we think and talk about who we are and how we function, we usually ignore most of that complexity and focus on much simpler abstractions as if that was all there is. That kind of simplification works pretty well, most of the time, but we need to keep in mind that there is always a lot more going on. Sometimes that difference can be important, and neglecting it can have serious consequences. I'll return to this point from time to time as we proceed.

Different parts of the flow produce different perceptual images.

Your perceptual image of an external situation or event is not uniquely determined by that situation or event. You compose that image by selecting and assembling pieces from your larger perceptual flow, so the parts of the flow you select will determine the resulting perception. Different parts of the perceptual flow will produce different perceptions.

Consider the familiar faces/vase figure, which comes up frequently in discussions of visual perception and visual illusion. What you see depends on where you look. The black figure in the middle looks like a vase, while the white areas on each side can be seen as two faces looking at each other. Objectively, it's neither; it's simply a pattern of black against a white background. This ambiguity is often described as stemming from figure/ground reversal. Which



you see depends on which part of the drawing you make the foreground (figure) and which part you relegate to the background (ground).

In terms of our Perceptual Process Model, we could say that the two different perceptions are composed using different portions of the incoming perceptual stream. You see the faces when you use the white portion of that stream, and the vase happens when you use the black. This illustrates a general perceptual principle that

Gestalt and Psychology Approaches

Gestalt psychology is a school of thought that looks at the human mind and behaviour as a whole. Originating in the work of Max Wertheimer, Gestalt psychology formed partially as a response to the structuralism of Wilhelm Wundt. The development of this area of psychology was influenced by a number of thinkers, including Immanuel Kant, Ernst Mach and Johann Wolfgang von Goethe.

"The fundamental "formula" of Gestalt theory might be expressed in this way," Max Wertheimer wrote. "There are wholes, the behaviour of which is not determined by that of their individual elements, but where the part-processes are themselves determined by the intrinsic nature of the whole. It is the hope of Gestalt theory to determine the nature of such wholes" (1924).

Absolute Threshold

Certain minimum stimulation of sense organs is required before any sensory experience will be evoked. The minimum physical energy necessary to activate a given sensory system is known as the absolute threshold. The absolute threshold can be determined by presenting the subject with a stimulus of given intensity and asking whether or not he detects it; on the next trial a different stimulus intensity is used, and so on, through a wide range of intensities. We find, instead of arriving at a single intensity value, a range of intensities over which the physical energy of the stimulus gradually moves from having no effect to having a partial effect (that is, it is sometimes detected and sometimes not) to having a complete effect.

SOME APPROXIMATE VALUES FOR ABSOLUTE THRESHOLDS.

Sense Modality	Threshold
Vision	A candle flame seen at 30 miles on a dark, clear night
Hearing	The tick of a watch under quiet conditions at 20 feet
Taste	One teaspoon of sugar in two gallons of water
Smell	One drop of perfume diffused into the entire volume of a six-room apartment
Touch	Wing of a fly falling on your cheek from a distance of 1 centimetre.

Psychophysical Function

The curve in below figure is called a psychophysical function because it expresses the relationship between a "psychological" variable (the experience of perceiving the stimulus) and a "physical" variable (the intensity of the stimulus). It plots the percentage of times the subject says, "Yes, I detect a stimulus," against a measure of the physical energy of the stimulus. In this case, the subject almost never reports the presence of a stimulus below an energy level equal to three units, whereas he almost always reports it

above nine units. The relative frequency of reporting the presence of a stimulus gradually increases between three and nine units.

Psychophysical functions can be obtained for any sensory modality; when vision is involved, the function is sometimes called the “frequency-of-seeing curve.”

On the basis of certain theoretical considerations, psychologists have agreed to define the absolute threshold as that value at which the stimulus is perceived 50 per cent of the time. Thus, for the data displayed in above figure, the absolute threshold would be six units.

Difference Threshold

Absolute threshold is identified at the transition between no experience and some experience. At the level of some experience we don't have recognition of any difference between two stimuli. The transition between no difference and some difference is identified as difference threshold. Thus, the minimum amount of stimulation necessary to tell two stimuli apart is known as the difference threshold. Two reds must differ by some finite amount before they can be discriminated from each other; two tones must differ in intensity by a measurable amount before one can be heard as louder than the other.

Like the absolute threshold, the difference threshold is defined as a statistical quantity. It is the amount of change in physical energy necessary for a subject to detect a difference between two stimuli 50 per cent of the time. Psychologists frequently use the term *just noticeable difference* (j.n.d.) to refer to this amount of change.

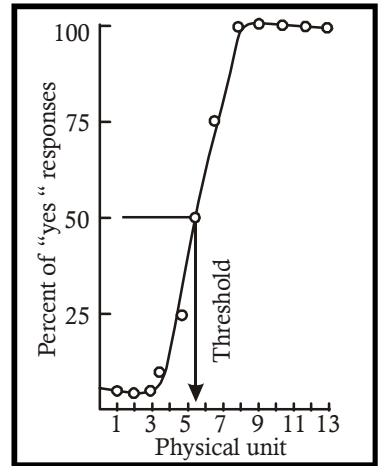
One remarkable feature of the human organism and, for that matter, of most animals is that the difference threshold tends to be a constant fraction of the stimulus intensity. To illustrate, suppose we estimate the difference threshold for a subject judging weights. If he is given a 100 gram weight, we note that his difference threshold is 2 grams; that is, the 100 gram weight must be compared to a weight of at least 102 grams in order for him to detect a j.n.d. about 50 per cent of the time. If we give him a 200 gram weight the difference threshold is 4 grams. For a 400 gram weight, the difference threshold is 8 grams; for an 800 gram weight, the difference threshold is 16 grams. Note that the difference threshold relative to the weight being judged is constant:

$$\frac{2}{100} = \frac{4}{200} = \frac{8}{400} = \frac{16}{800} = .02$$

The above relationship is known as Weber's law, *i.e.*, the ratio between the increase in stimulation necessary for a j.n.d. and the amount of stimulation taken as a referent is constant.

We may observe approximations of Weber's law operating in everyday experience. A twenty-minute increase in air-travel time from Los Angeles to San Francisco may be detected as a just noticeable difference, but a similar increase in travel time from San Francisco to London may not. An increase of five dollars in the cost of a shirt is quite noticeable, whereas a similar increase in the cost of a suit may be of little concern.

It is interesting to conjecture about the aspects of the sensory system that make Weber's law hold, and many theories have been proposed. We shall not discuss these



PSYCHOLOGICAL FUNCTION.

theories here, but the research initiated by them has made it clear that matters are more complex than indicated by Weber's law. Weber's law holds fairly well in the middle range of sensory dimensions but is somewhat in error at the extremes, particularly at very low levels of stimulation.

Weber's Law

We have seen earlier that the difference threshold relative to the weight being judged is constant. This relationship is known as Weber's law, named after Ernst Weber. Stated mathematically, if l is the amount of stimulation taken as a referent, and Δl is the increase in stimulation necessary for a j.n.d., then

$$\frac{\Delta l}{l} = k$$

where k is a constant that does not depend on l . The quantity k is called Weber's constant.

The Table given below presents values of Weber's constant for various sense modalities. The tremendous range in values reflects the fact that some sensory systems are much more responsive to changes in the physical environment than others.

Sense Modality	Weber's Constant
Pitch of a tone	$\frac{1}{33}$
Deep pressure from skin and subcutaneous tissue	$\frac{1}{80}$
Visual brightness	$\frac{1}{60}$
Lifted weights	$\frac{1}{50}$
Loudness of a tone	$\frac{1}{10}$
Cutaneous pressure	$\frac{1}{7}$
Taste for saline solution	$\frac{1}{5}$

Weber's Constant

Approximate values of Weber's constant for various sensory discriminations are given above, where the noticeable point is, the small the fraction, the greater the differential sensitivity.

Weber-Fechner Law

Fechner extended Weber's law (formulated for thresholds) to develop scales for measuring sensory experiences. By summing successive j.n.ds, he developed a scale for measuring how far a given stimulus was above threshold; this procedure in turn permitted him to measure the psychological distance between any two stimuli. His formulation, commonly called the Weber-Fechner law, holds that the sensory experience bears a logarithmic relationship to the intensity of the physical stimulus. This logarithmic equation, although approximately correct, has been modified by modern workers to fit results from a wide

variety of experiments. Equations relating sensory experience to the intensity of the physical stimulus have proved extremely helpful to engineers designing telephones, video displays, tape recorders, and other types of communication equipment. They tell the designer how intense a signal must be in order to be perceived accurately under varying conditions.

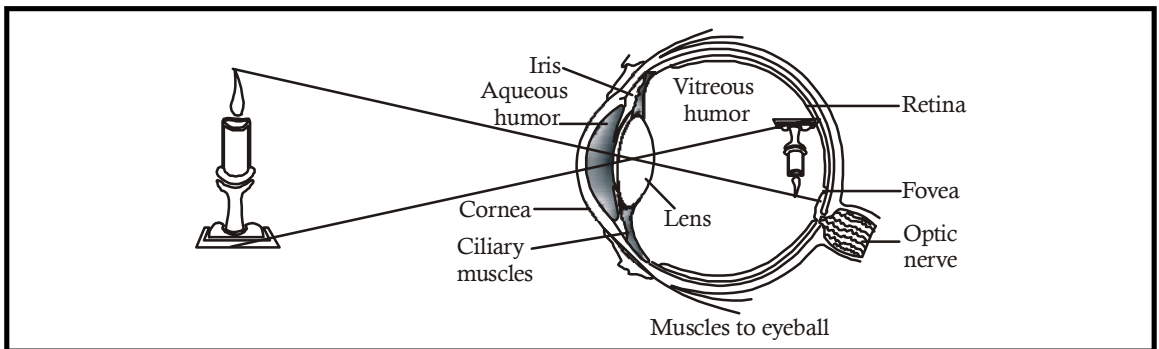
Our Senses

The Visual Sense

Each of our sense organs is sensitive to a particular type of physical energy. The eye responds to that portion of electromagnetic energy travelling through space that we call light. It is convenient to think of electromagnetic energy as travelling in waves, with wavelengths (the distance from one crest of a wave to the next) varying tremendously from the shortest cosmic rays (10 trillionths of an inch) to long radio waves that may measure many miles. The wavelengths that the human eye perceives as light extend only from about 380 nanometers (nm) to about 780 nm. Since a nanometer is one billionth millionth of a meter, it is clear that visible energy is but a very small section of the total electromagnetic spectrum.

The eye will respond to forms of stimulation other than light waves. Pressure on the eyeball or electrical stimulation of certain areas of the brain will produce the sensation of light. These observations point up the fact that the experience of light is a quality produced in the visual system. The visible portion of the electromagnetic spectrum is called light because it is what usually produces that sensation.

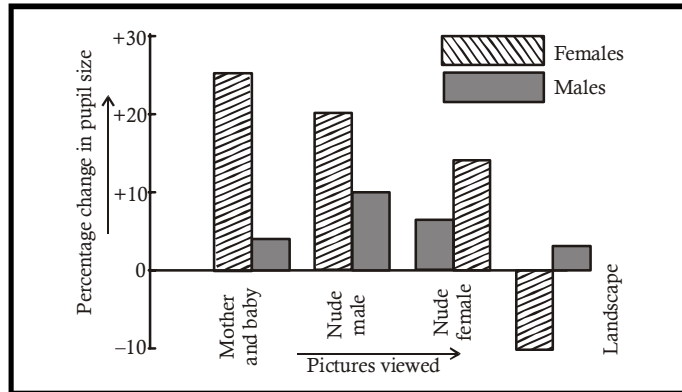
The Human Eye (Parts and Functions)



A CROSS SECTION OF THE LEFT HUMAN EYE.

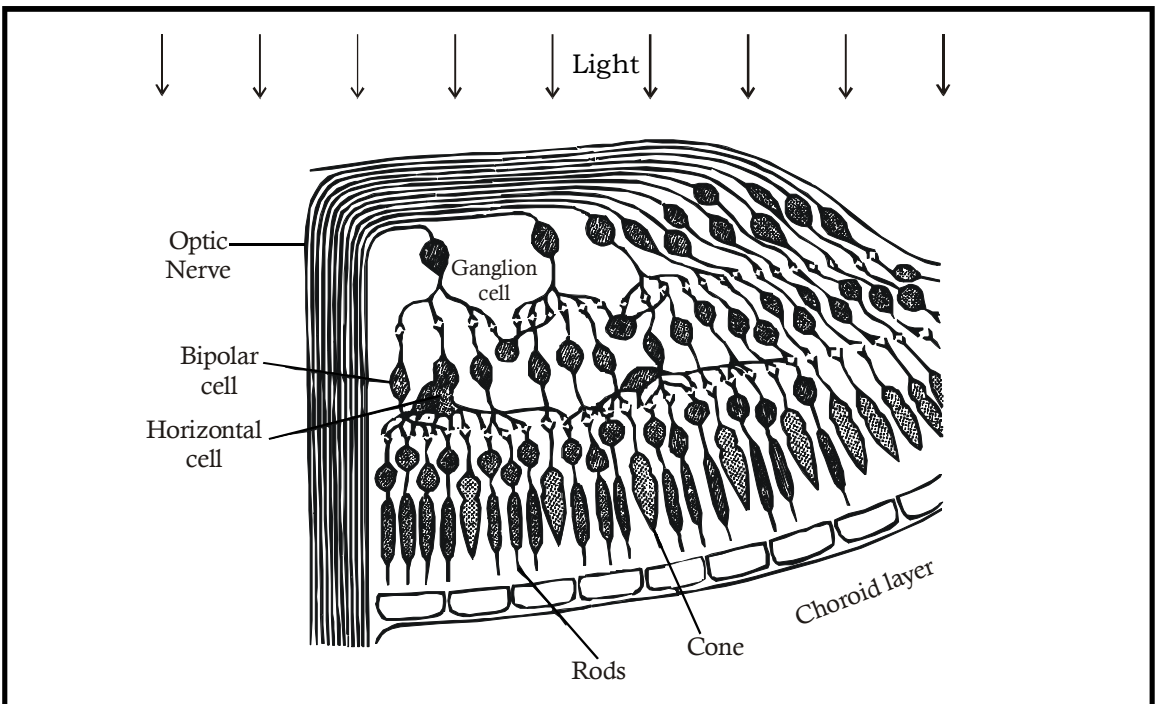
The main parts of human eye, shown in above figure, are as follows:

- (1) Light enters the eye through the transparent *cornea*.
- (2) The amount of light being regulated by the *pupil*. Construction and dilation of the pupil are controlled by the autonomic nervous system. The parasympathetic division controls the change in pupil size as a function of change in illumination. The sympathetic division acts to dilate the pupil under conditions of strong emotion, either pleasant or unpleasant. Even conditions of mild emotional arousal or interest will result in systematic changes in pupil size.



CHANGES IN PUPIL SIZE AS A RESPONSE TO PICTURES.

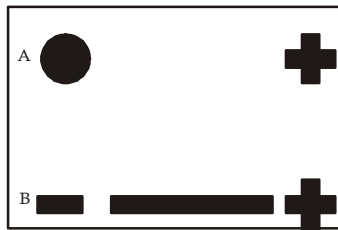
- (3) After pupil regulates the amount of light, the lens then focuses the light on the sensitive surface, the *retina*. This retina has three main layers.
- the **rods** and **cones**, the photosensitive cells that convert light energy into nerve impulses;
 - the **bipolar cells**, which make synaptic connections with the rods and cones; and
 - the **ganglion cells**, the fibres of which form the optic nerve. Strangely enough, the rods and cones form the rear layer of the retina. The eye is a very imperfect optical system. The light waves not only have to pass through the lens and liquids that fill the eyeball, none of which is a perfect transmitter of light, but also have



LAYERS OF RETINA.

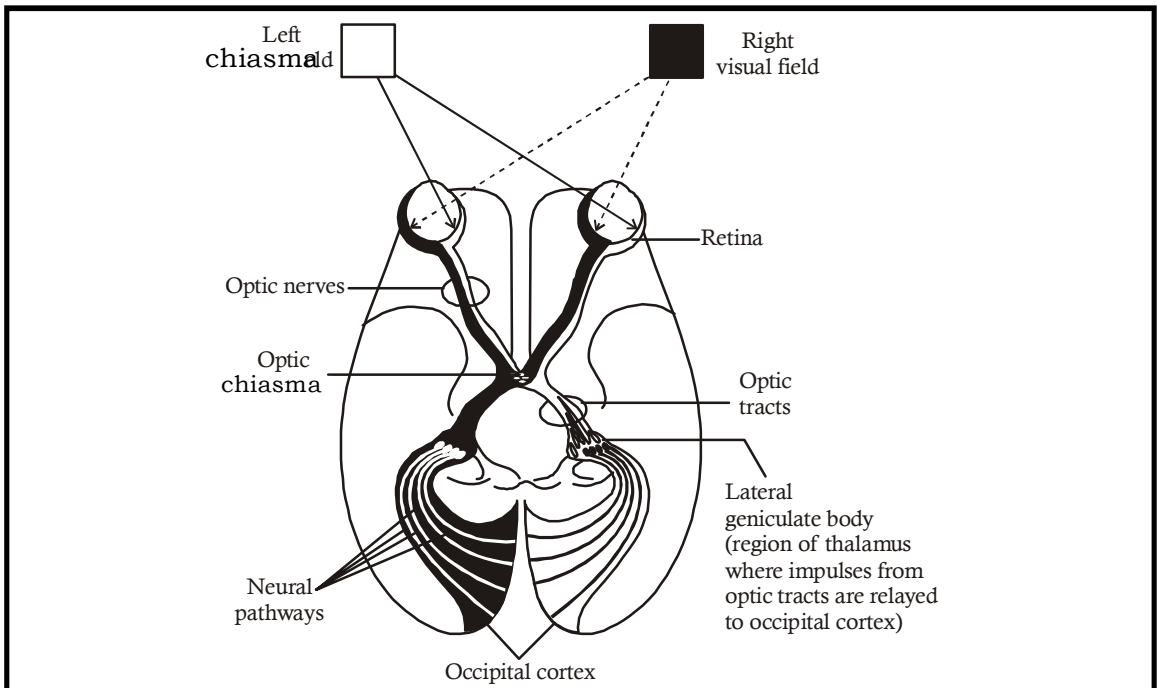
to penetrate the network of blood vessels and the bipolar and ganglion cells that lie on the inside of the eye before reaching the photoreceptors where light is converted into nervous impulses.

- (4) The most sensitive portion of the eye in normal daylight vision is a part of the retina called the *fovea*. This area plays a major role in perception and yet it is so small that the projection (or image) of a thumbnail viewed at arms length will cover it. Not far from the fovea is an insensitive area, called the blind spot, where the nerve fibres from the ganglion cells of the retina come together to form the optic nerve. Although we are not normally aware of the blind spot, its existence can easily be demonstrated.



Locating the blind spot where, A. with the right eye closed, stare at the upper right cross. Move the book back and forth about 1 foot from the eye. When the black circle on the left disappears, it is projected on the blind spot. B. with the right eye closed, stare at the lower right cross. Move the book back and forth again. When the white space falls in the blind spot, the black line appears to be continuous. This phenomenon helps us to understand why we are not ordinarily aware of the blind spot.

- (5) The below figure shows the *optic nerve fibres* leading from each eye to the cortical areas where vision is represented (the occipital lobes). Notice that some of the



VISUAL PATHWAYS

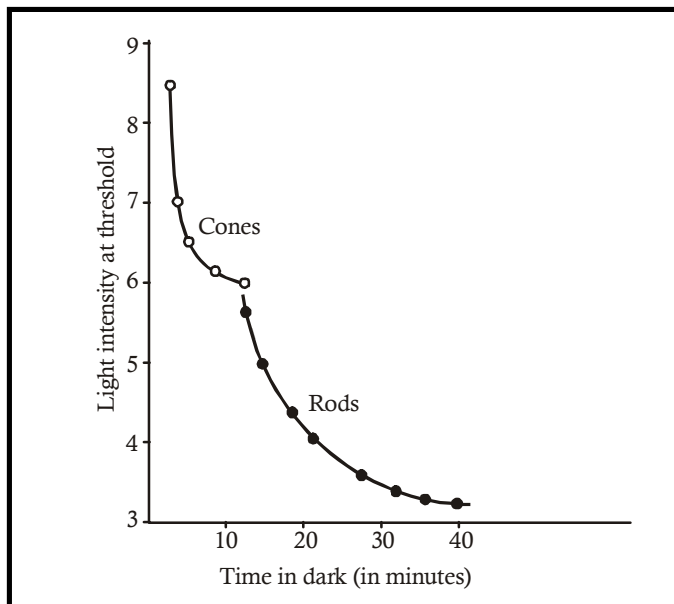
fibres go from the right eye to the right cerebral hemisphere and from the left eye to the left hemisphere, whereas other fibres cross over at a junction called the *optic chiasma* and go to the opposite hemisphere. Fibres from the right sides of both eyes go to the right hemisphere of the cerebral cortex, and fibres from the left sides of both eyes go to the left hemisphere. Consequently, damage to the occipital lobe of one hemisphere (say, the left) will result in blind areas in both eyes (the left sides of both eyes). This fact is sometimes helpful in pinpointing the location of a cerebral tumour or injury.

- (6) The size of the pupil is regulated by the *iris*.
- (7) The *diaphragm* gives the eye its colour.
- (8) The shape of the lens is regulated by the *ciliary muscles*.
- (9) The *external muscles* move the eye.

Dark Adaptation

We have all experienced the difficulty of finding our way to an empty seat when entering a dark theatre. After a few minutes our eyes become accustomed to the dark, and we are able to see the people around us even though the lighting has not changed. We have undergone dark adaptation.

The course of dark adaptation provides further evidence for the difference in action between the rods and cones. The first part of the curve shows that the cones gradually become sensitive to fainter lights, but after five minutes in the dark their sensitivity has increased as much as it will, as measured by the absolute threshold. The rods, however, continue to adapt and do not reach their maximum sensitivity for about a half-hour. The dark-adapted eye is much more sensitive to lights with wavelengths in the blue-green region than to the longer wavelengths in the red region.



THE COURSE OF DARK ADAPTATION

Colour Vision

An interesting relationship exists among colours. If the spectral colours are wrapped around the circumference of a circle, allowing room between the red and violet ends of the spectrum for the purples and reds not found on the spectrum, the colours opposite each other on the circle will be complementary. That is, if lights of these colours are mixed in proper proportions, they disappear to a neutral gray. The mixture of lights is an additive mixture, whereas the mixture of pigments is a subtractive mixture because of the way in which pigments selectively absorb some of the light. Light is the source of all colour, and pigments are simply reflectors and absorbers of colour. They achieve their colour by absorbing certain parts of the spectrum and reflecting the parts that remain.

Some of the colours on the colour circle appear more elementary than others; that is, they appear to be composed of a single hue. These elementary colours are called psychological primaries, and usually four primary colours are named—red, yellow, green, and blue. Between them are “secondary” colours, in which the primary components are still identifiable: orange between red and yellow, yellow-greens between yellow and green, blue-greens between green and blue, and purples and violets between blue and red. Another set of primaries is called colour-mixture primaries. Any three widely spaced colours on the spectrum can be used to produce all the other colours by additive mixture. The three colours usually chosen are red, green, and blue.

Psychological Dimensions of Colour

A person trying to describe what he sees must resort to three psychological dimensions: hue, brightness, and saturation. Hue refers to what we ordinarily think of as the “name” of the colour—for example, red, green, etc. The circumference of the colour circle provides the scale along which the hues can be placed in order. There the hues follow the order of shortest to the longest visible wavelength.

Another dimension along which colours can be scaled is brightness. The physical basis of brightness is primarily the energy of the light source, which corresponds to the amplitude of the wave. But brightness also depends to some extent upon wavelength. Yellow, for example, appears slightly brighter than certain red and blue wavelengths, even when all three have equal amplitudes.

A third dimension along which colours can be scaled is saturation, which refers to the apparent purity of the colour. Highly saturated colours appear to be pure hues, without any gray; colours of low saturation appear close to gray. The primary physical correlate of saturation is the complexity of the light wave. A light wave composed of only one or a few different wavelengths will produce the most highly saturated colour. Light waves composed of many components result in colours of low saturation. However, as colours of a single wavelength become brighter (merge into white) or darker (merge into black) they begin to lose the apparent purity of their hues; the change in brightness is accompanied by a reduction in saturation.

The Colour Solid

Colour solid describe the relationship between the three psychological dimensions of colour. The colour solid represents all the three dimensions of hue, brightness and saturation. The most highly saturated colours are of medium brightness.

Colourblindness

Normal eye, conveniently think, discriminates three systems of colour: light-dark, yellow-blue, and red-green. All other combinations can be derived from these. Colourblindness results from a deficiency in one or two of these systems, the light-dark system remaining intact if the person can see at all. The person with normal vision is called a trichromat. A person lacking one system but with use of the other two is called a dichromat. A dichromat is partially colour-blind. Finally, the person with only the light-dark system is a monochromat and totally colour-blind.

By far the most common form of colourblindness is red-green blindness, with the blue-yellow and light-dark systems intact. This deficiency affects some 7 per cent of men but less than 1 per cent of women. Total colourblindness, in which the person sees merely black, white and gray, is extremely rare; yellow-blue blindness, in which red-green discrimination is preserved, is rarer still.

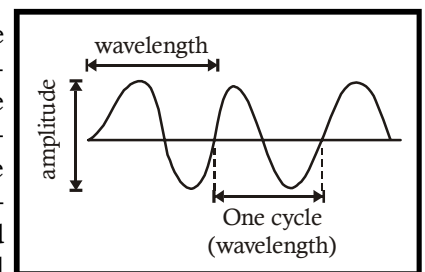
Afterimages and Colour Contrast

If you stare at a red circle and then look at a plain gray surface, you are likely to see a green circle on it; that is, you experience a negative afterimage. It is negative because green is the complementary colour of red. Not all afterimages are in the complementary colour however. After staring at a very bright light you are likely to see a whole succession of colours, but seeing the complementary colour is very common.

Complementary colours can also serve to enhance each other. When two complementary colours occur side by side, each colour appears more highly saturated than it would when placed next to a non-complementary colour. This effect, termed simultaneous contrast, is one reason for making pennants of such complementary pairs as red and green and yellow and blue.

The Auditory Sense

The ear is sensitive to mechanical energy—to pressure changes among the molecules in the atmosphere. A vibrating object, such as a tuning fork causes successive waves of compression and expansion among the air molecules surrounding it. The sound waves generated by the vibration of molecules (in air, water, or some other medium) are the stimuli for hearing. Unlike light, sound cannot travel except through a medium; a ringing bell suspended in a vacuum jar cannot be heard when the air is pumped out. Simple sound waves can be graphically represented as sine functions. The cycles of the sine wave represent the successive compression and expansion of the air as the sound wave moves along. The two main characteristics of such a wave are its frequency and its amplitude. Frequency is measured in number of vibrations per second, that is, the number of times per second that the complete cycle is repeated. The unit Hertz (abbreviated Hz) is used to denote cycles per second; that is, one cycle per second is one Hz². Amplitude refers to the amount of compression and expansion, as represented by the amount by which the curve is displaced above or below the baseline.

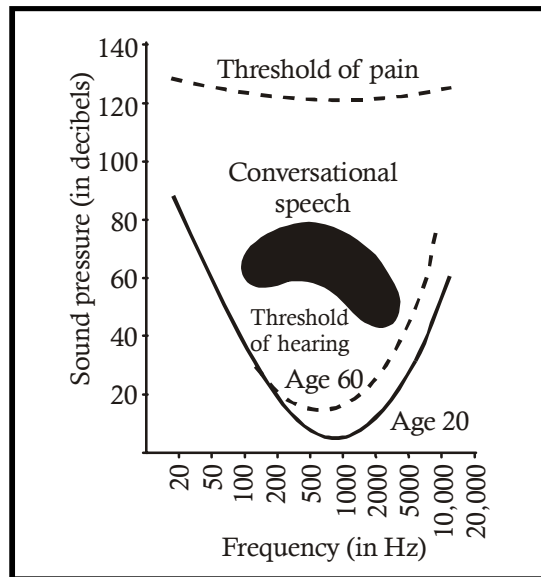


WAVE FUNCTION

Pitch and Loudness

The psychological correlate of frequency is pitch: the higher the vibrational frequency, the higher the perceived pitch. The amplitude of the sound wave determines the intensity with which sound pressure strikes the eardrum. The psychological correlate of intensity is loudness; the greater the intensity, the louder the tone (provided pitch remains constant).

The unit to measure the sound intensity is called decibel (one-tenth of a bel, named in honour of Alexander Graham Bell, and abbreviated db). The absolute threshold for hearing varies with the frequency of the source. Tones in the range from 800-6000 Hz require less than 10 db to reach threshold, whereas tones less than 100 Hz or greater than 15,000 Hz require 40 db or more to reach threshold.



THRESHOLD OF HEARING.

Complex Tones and Noise

The sounds we hear are seldom pure tones represented by a sound wave of a single frequency. Even the musical note produced by striking middle C on the piano has, in addition to its fundamental tone of 262 Hz, overtones, which are multiples of that frequency. The overtones occur because the piano wire vibrates not only as a whole, producing a fundamental tone of 262 Hz, but also in halves, thirds, quarters, fifths, and so on, with each partial vibration producing its own frequency. The sounds of one musical instrument differ from those of another in the number of overtones produced and in the way in which the construction of the instrument enhances (resonates) certain overtones and deadens others. This characteristic quality of a musical tone is called timbre. It is the timbre of a tone that tells us whether it is being produced by a piano or a clarinet. If all overtones are eliminated by the use of sound filters, it is difficult to determine what instrument is being played.

When two tones are sounded together, they may lead to a fusion that is heard as consonant (pleasant) or as dissonant (unpleasant). The two tones create a third tone based on

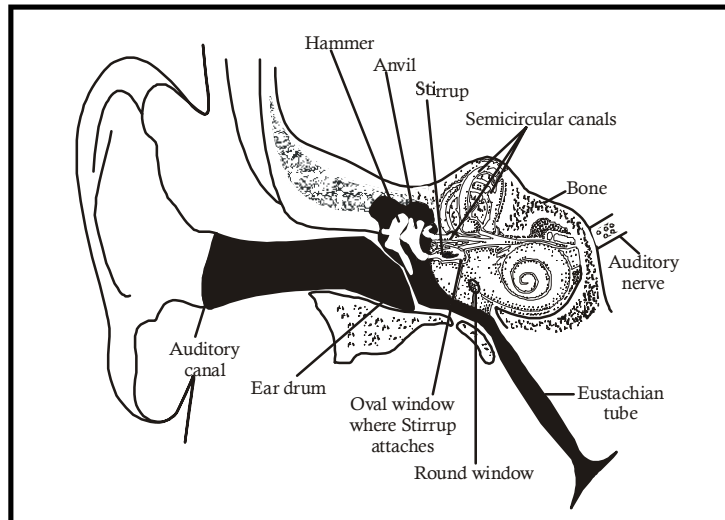
the difference in their frequencies. This difference tone may or may not harmonise with the fundamental tones sounded; for this reason some combinations of tones are preferred to others. Musical harmony depends in part on the interaction between fundamental tones, overtones, and difference tones, which combine to make up the complex tonal stimulus.

A noise is a sound composed of many frequencies not in harmonious relation to one another. Acoustical experts sometimes speak of white noise when referring to a noise composed of all frequencies in the sound spectrum at roughly the same energy level. White noise is analogous to white light, which is composed of all frequencies in the light spectrum. The sound of radio static or a bathroom shower approximates the sound of white noise.

A noise with energy concentrated in certain frequency bands may have a characteristic pitch. For example, we may legitimately use the musical term “bass” to characterise the sound of a drum, even though a drum is more noisy than tonal. Speech sounds make simultaneous use of tonal qualities and noise qualities: vowels are tonal, and consonants are noisy.

The Human Ear

The eardrum, a movable diaphragm activated by sound waves entering the ear, is connected to the external ear by an auditory canal.



A CROSS SECTION OF THE EAR

On the inner side of the diaphragm is a cavity housing the bony transmitters of the middle ear (three small bones called the hammer, anvil, and stirrup). The hammer is attached firmly to the eardrum, and the stirrup to another membrane, the oval window. The oval window conducts the sounds to the cochlea, the auditory portion of the inner ear. Because the oval window is much smaller than the eardrum, small movements at the eardrum are condensed into a magnified pressure on the oval window.

Pressure at the oval window sets into motion the fluid inside the cochlea. This pressure is relieved at the round window at the other end of the fluid-filled channel that runs through the cochlea. Pressure changes in the fluid displace the basilar membrane in the

cochlea, upon which the organ of corti rests, and this displacement stimulates the sensitive elements in the hair cells of the organ of corti, which are connected with the auditory nerve.

The pathways of the auditory nerves resemble those of the optic nerves in that nerve fibres from each ear travel to both cerebral hemispheres (terminating in the temporal lobes). Thus, destruction of one temporal lobe will not cause complete deafness in either ear.

The Sense of Smell

The sight and the hearing senses are known as “higher senses”. All other senses are lower senses because they, though important for survival, lack the richness of patterning and organisation. From an evolutionary viewpoint, smell is one of the most primitive and most important of the senses. The sense organ for smell has a position of prominence in the head appropriate to a sense intended to guide behaviour. Smell has a more direct route to the brain than any other sense. The receptors high in the nose, in the olfactory epithelium of each nasal cavity, are connected without synapse directly to the olfactory bulbs of the brain, lying just below the frontal lobes. The olfactory bulbs are in turn connected with the olfactory cortex on the inside of the temporal lobes and extend to the neighbouring cortex; the exact neural connections are still a matter of some uncertainty. In fish, the olfactory cortex makes up the entire cerebral hemispheres. In the dog, the olfactory cortex represents about one-third of the area of the side of the brain, as contrasted with one-twentieth of this area in man. For this reason both the U.S. Postal Service and the Bureau of Customs have dogs trained to check unopened packages for heroin and marihuana, and in several cases specially trained police dogs have sniffed out hidden explosives.

The Sense of Taste

We know that the primary taste qualities are sweet, sour, salt, and bitter. Every other taste experience is composed of fusions of these qualities with other senses. Smell, texture, temperature, and sometimes pain (judging from the pleasure some diners derive from highly spiced Mexican food) all contribute to the sensations we experience when we taste a food. When we drink a cup of coffee we enjoy its aroma and its warmth by means of senses other than taste; the taste sense provides only for the sweet-sour-bitter components.

The taste receptors are found in the taste buds on the edges and toward the back of the tongue; a few are located elsewhere in the soft palate, the pharynx, and the larynx. It is known that the number of taste buds decreases with age, so that older people are less sensitive to taste than children. Some taste buds at the tip of the tongue react only to sweet, salt, or sour; others react to some or all of these in combination. In general, sensitivity to sweet is greatest at the tip of the tongue, to salt on the tip and the sides, to sour on the sides, and to bitter on the back.

The Skin Sensations

The sense of touch can be felt as at least four sensations: touch, pain, warm, and cold—all of which are felt through distinct kinds of sensitive spots on the skin surface. All other skin sensations that we commonly describe, such as itch, tickle, quick-pricking pains, or dull, long-lasting pains, are variations of the four basic sensations. An itching sensa-

tion, for example, can be produced by stimulating pain spots by a light, repeated needle prick; tickle is experienced when adjacent touch spots on the skin are touched lightly in rapid succession.

The precise receptors for the various skin sensations have been the subject of much study and dispute. At one time, histologists identified a number of quite different nerve-end structures in the skin, each of which was thought to be the specific receptor for one of the four sensations. Subsequent studies, however, failed to substantiate such claims: when investigators “mapped” cold, warm, touch, and pain spots on their own skin, excised the underlying tissue, and examined it microscopically, there was no consistent relationship between the type of sensation experienced and the type of underlying nerve-end structures. Only two things can be stated with some degree of certainty:

1. Nerve fibres at the base of hair follicles serve as receptors for touch or light pressure (but they are not the only receptors, since the lips, which are hairless, are quite sensitive to pressure).
2. Free nerve-endings that terminate in the epidermis (as opposed to those nerves that end in certain encapsulated structures) are involved in pain reception.

Kinesthesia and Equilibratory Senses

There is another sensory system which informs us of the position and movement of parts of the body. In technical language this is kinesthesia—the muscle, tendon, and joint sense. Position and movement are detected by sense organs in the joints; sense organs in the muscles and tendons tell us whether a muscle is stretched or contracted and help to adjust muscular tension to the load upon it. Without kinesthesia we would have great difficulty in maintaining posture, walking, climbing, and in controlling voluntary movements such as reaching, grasping, and manipulating.

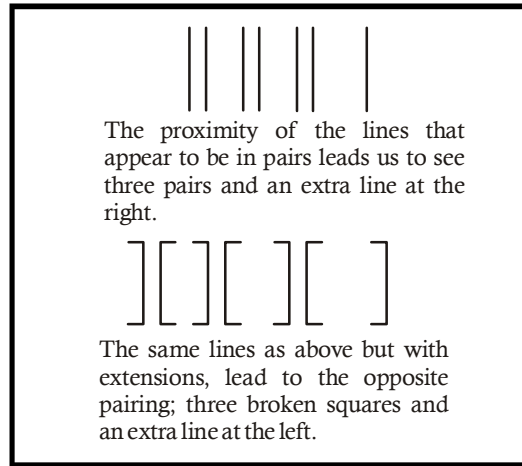
Cooperating with kinesthesia are the equilibratory senses, which deal with total body position in relation to gravity and with motion of the body as a whole. The relation of bodily parts to one another and to external objects is the responsibility of kinesthesia; the orientation of the body in space is the responsibility of the equilibratory senses.

Perceptual Organisation

Our daily experiences show that what we perceive always remains constant whether it is viewed from any direction, any angle, any side or at any moment. This quality of perception, perceptual constancy, is possible only when perception involves a proper organisation in itself. In a more analytical vein, let us identify some principles of perception that help to explain how objects and events are organised and perceived.

Gestalt View

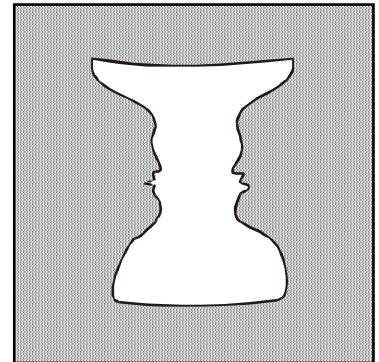
When we look at the simple patterns of lines and dots, we find that they fall into ordered relationships. In the top below Figure, we tend to see three pairs of lines, with an extra line at the right. But notice that we could have seen three pairs beginning at the right with an extra line at the left. The slight modification shown in the lower part of the figure causes us to do just that. This tendency to structure what we see is very compelling; what we see in figures seems to be forced on us by the patterns of stimulation. The properties of whole affect the ways in which parts are perceived. For that reason we may say—following the lead of Gestalt psychology—that the whole is different from the sum of its parts.



PATTERNING AND PERCEPTUAL STRUCTURING.

Figure and Ground

Figures are geometrical patterns against a background and thus appear to be like objects, with contours and boundaries. Figure-ground organisation is basic to stimulus patterning. Patterns do not have to contain identifiable objects to be structured as figure and ground. Patterns of black and white and many wallpaper designs are perceived as figure-ground relationships, and very often figure and ground are reversible. In below figure, note that the part that is seen as figure seems more solid and well defined and tends to appear slightly in front of the background, even though you know it is printed on the surface of the page. You seem to look through the spaces in and around the figure to a uniform background behind, whether the background is in white (or a light colour) or black (or a dark colour).



REVERSIBLE FIGURE AND GROUND.

We can perceive figure-ground relationships through senses other than vision. For example, we may hear the song of a bird against a background of outdoor noises or the melody played by the violin against the harmonies of the rest of the orchestra.

Gestalt Laws of Perceptual Organisation

Gestalt psychology was founded by German thinkers Max Wertheimer, Wolfgang Kohler and Kurt Koffka and focused on how people interpret the world. The Gestalt perspective formed partially as a response to the structuralism of Wilhelm Wundt, who focused on breaking down mental events and experiences to the smallest elements. Max Wertheimer noted that rapid sequences of perceptual events, such as rows of flashing lights, create the illusion of motion even when there is none. This is known as the phi phenomenon. Motion pictures are based upon this principle, with a series of still images appearing in rapid succession to form a seamless visual experience.

According to Gestalt psychology, the whole is different than the sum of its parts. Based upon this belief, Gestalt psychologists developed a set of principles to explain perceptual organisation, or how smaller objects are grouped to form larger ones. These principles are often referred to as the "laws of perceptual organisation."

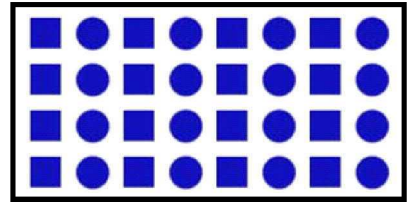
However, it is important to note that while Gestalt psychologists call these phenomena "laws," a more accurate term would be "principles of perceptual organisation." These principles are much like heuristics, which are mental shortcuts for solving problems.

Follow the links below to find more information and examples of the different Gestalt laws of perceptual organisation.

Law of Similarity

Items that are similar tend to be grouped together.

In the image below, most people see vertical columns of circles and squares.



Law of Pragnanz

Reality is organised or reduced to the simplest form possible.

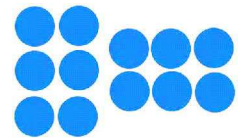
For example, we see the image above as a series of circles rather than as many much more complicated shapes.



Law of Proximity

Objects near each other tend to be grouped together.

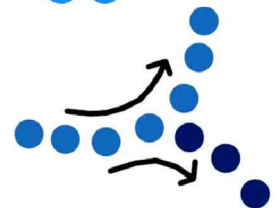
The circles on the left appear to be grouped in vertical columns, while those on the right appear to be grouped in horizontal rows.



Law of Continuity

Lines are seen as following the smoothest path.

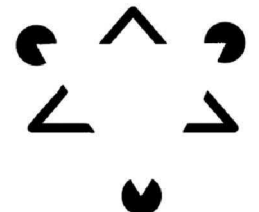
In the image above, the top branch is seen as continuing the first segment of the line. This allows us to see things as flowing smoothly without breaking lines up into multiple parts.



Law of Closure

Objects grouped together are seen as a whole.

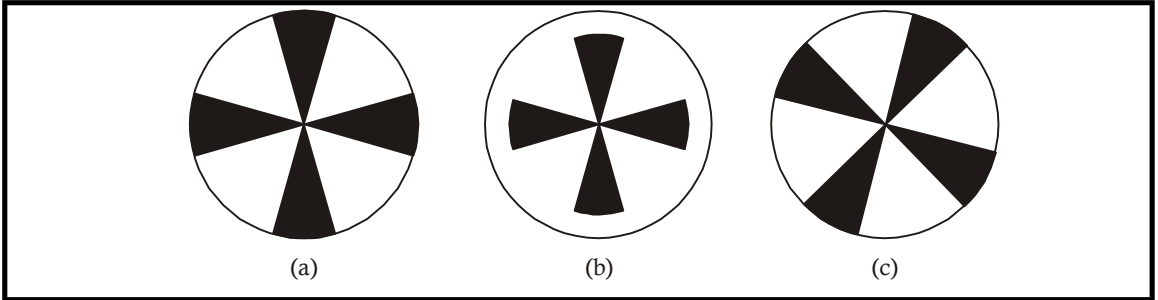
We tend to ignore gaps and complete contour lines. In the image above, there are no triangles or circles, but our minds fill in the missing information to create familiar shapes and images.



Relative Size, Surroundedness, Orientation, and Symmetry

Other things being equal, the smaller of two areas will be seen as a figure against a larger background. Thus given figure (a) will tend to be perceived as a black propellor shape against a white background because the black area is the smaller. This effect is enhanced if the white area actually surrounds the black as in figure (b), because surrounded areas tend to be seen as figures. However, if we orient the figure so that the white area is arranged around the horizontal and vertical axes then it is easier to see this larger area as a figure (c). There seems to be a preference for horizontally or verti-

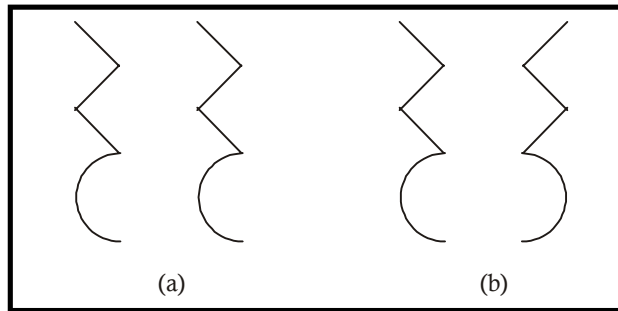
cally oriented regions to be seen as figures. Also note that both these sets of patterns are symmetrical. Symmetry is a powerful perceptual property, and may be more salient perceptually than non-reflected repetition. Examples of symmetry and repetition are shown in Figure.



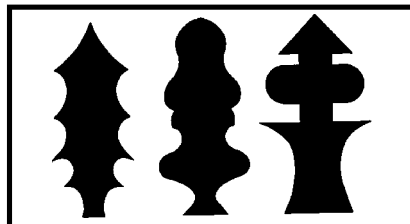
The preferred perception of (a) is a black propeller on a white background.

This preference is enhanced if the white area surrounds the black, as at (b). If the orientation of the forms is altered, so that the white area is oriented around the horizontal and vertical axes, as at (c), then it is easier to see the larger white area as a figure.

Symmetrical areas will tend to be perceived as figures, against asymmetrical backgrounds. Below figure how relative size, orientation, symmetry, and surroundedness may all operate together so that it is difficult if not impossible to see anything other than the black areas as the figures in the picture. The reader will note the perceptual stability of this picture compared with the ambiguity, where the relative sizes, surroundedness, and symmetries in the display favour neither the “faces” nor the “vase” particularly strongly.



At (a) one form is repeated without reflection around a vertical axis. This arrangement is not as perceptually salient as the arrangement shown at (b), where repetition with reflection around the vertical axis produces bilateral symmetry.



This picture clearly shows black shapes on a white background. The black shapes are vertically oriented, symmetrical, small (relative to the background), and surrounded by the background.

The Law of Pragnanz

For the Gestalt psychologists, many of these laws were held to be manifestations of the Law of Pragnanz, introduced by Wertheimer. Koffka described the law:

Of several geometrically possible organisations that one will actually occur which possesses the best, simplest and most stable shape.

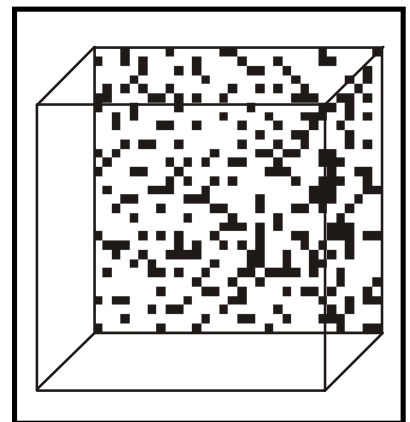
Thus an organisation of four dots arranged as though they were at the corners of a square will be seen as a “square” because this is a “better” arrangement than, say, a cross or a triangle plus an extra dot. The square is a closed, symmetrical form, which the Gestaltists maintained was the most stable.

Although the Gestaltists accepted that familiarity with objects in the world, and “objective set”, might influence perceptual organisation, they rejected an explanation solely in these terms. A major determinant of perceptual organisation for them was couched in terms of certain “field forces” that they thought operated within the brain. The Gestaltists maintained a *Doctrine of Isomorphism*, according to which there is, underlying every sensory experience, a brain event that is structurally similar to the experience. Thus when one perceives a circle, a “circular trace” is established, and so on. Field forces were held to operate to make the outcome as stable as possible, just as the forces operating on a soap bubble are such that its most stable state is a sphere.

Unfortunately, no evidence has been provided for such field forces, and the physiological theory of the Gestaltist has fallen by the wayside, leaving us with a set of descriptive principles, but without a model of perceptual processing. Indeed, some of their “laws” of perceptual organisation today sound vague and inadequate. What is meant by a “good” or a “simple” shape, for example? Recently workers have attempted to formalise at least some of the Gestalt perceptual principles.

Perceptual Hypotheses

Reversible figures like the Necker cube indicate that our perceptions are not a static mirroring of visual stimuli. Perceiving can be thought of as a search for the best interpretation of sensory information, based on our knowledge of object characteristics. From a cognitive viewpoint, a perceived object is a hypothesis suggested by the sensory data. The pattern of the Necker cube contains no clue as to which of two alternative hypothesis is correct, so the perceptual system entertains first one then the other hypothesis and never settles on an answer. The problem arises because the Necker cube is a three-dimensional object represented on a two-dimensional surface. If we were to see it in three-dimensional form, there would be many cues to tell us which hypothesis to choose (Gregory, 1970).



NECKER CUBE.

The notion of hypothesis testing emphasises the active nature of perception. The perceptual system does not passively sense inputs, but searches for the percept that is most consistent with the sensory data. In most situations there is only one reasonable interpretation of the sensory data, and the search for the correct percept proceeds so quickly and automatically that we are unaware of it. Only under unusual conditions, as when

viewing ambiguous figures, does the hypothesis-testing nature of perception become apparent.

Perceptual Constancy

The word “constancy” is an exaggeration, but it dramatizes our relatively stable perception of objects. A well-known object is perceived as permanent and stable regardless of the illumination on it, the position from which it is viewed, or the distance at which it appears. The tendency to see the colour of a familiar object as the same, regardless of the actual light conditions, is called colour constancy. The tendency to see an object’s shape as unchanging regardless of the viewing angle is called shape constancy. The tendency to see an object as the same size regardless of distance is called size constancy. Finally, the fact that an object appears to retain its “same” position, even as we move about, is known as location constancy.

Colour Constancy

We have already stated that the colour constancy is the tendency to see the colour of a familiar object as the same regardless of the actual light conditions. The owner of a blue car sees it as blue whether looking at it in bright sunlight, in dim illumination, or under a yellow street light. He is relying on his memory of the car’s colour, which is one factor contributing to colour constancy. Information about the nature of the illumination and the colour of surrounding objects are also clues to colour constancy. When these clues are eliminated, colour constancy diminishes or disappears. If you look at a ripe tomato through a narrow tube so that you do not know the nature of the object or the source and kind of illumination, the tomato will appear to be blue or brown or any of a number of other colours, depending on the wavelength of the light being reflected from it. Without colour constancy clues we see the colour of objects according to the wavelength of light being reflected to the eye.

Shape and Size Constancy

Shape constancy means constancy in shape whether the object is viewed at any angle. When a door swings open toward us, its shape as projected on the retina goes through a series of distortions. The door’s rectangular shape becomes a trapezoid, with the edge toward us looking wider than the hinged edge; then the trapezoid grows thinner, until all that is projected on the retina is a vertical line the thickness of the door. We can readily distinguish these changes, but the psychological experience is an unchanging door swinging on its hinges. The fact that the door does not seem to change its shape is an example of shape constancy. We see the top of a glass bottle as “circular” whether we view it from the side or from the top.

Size constancy refers to the fact that as an object is moved farther away we tend to see it as more or less invariant in size. Hold a quarter a foot in front of your eyes and then move it out to arm’s length. Does it appear to get smaller? Not noticeably so. Yet the retinal image of the quarter 12 inches away is half the size of the image of the quarter when it is 24 inches from the eye. It certainly does not appear to reduce to half its size as we move it to arm’s length.

Ways to Judge an Object Size

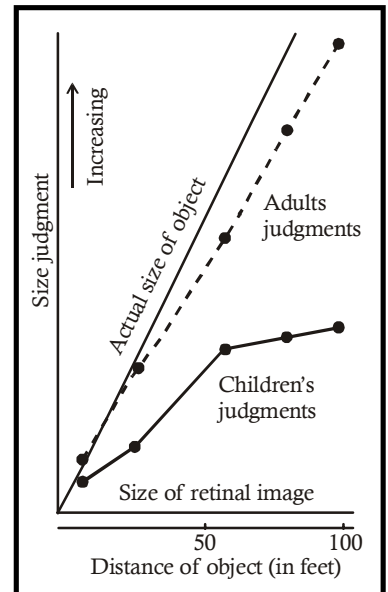
When we look at a distant object, we can judge its size in one of three ways:

1. *Perspective size.* We might judge it according to the geometry of perspective, seeing it as smaller the farther away it is. This size would correspond to the size of the image on the retina.
2. *Object size.* We might judge an object by its true size and hence see it as remaining constant in size regardless of its distance.
3. *Compromise* between perspective size and object size. We might compromise and see the object as smaller at a distance, but not as much smaller as the geometry of perspective indicates.

The last of the three alternatives is usually correct. Our size perceptions represent compromises between perspective size and object size. How well size constancy operates depends upon the presence of distance cues and upon our familiarity with the object. The more information available about the distance of the object, the more the perceived size approaches the actual size. As distance cues are eliminated, the perceived size approaches the size of the retinal image, unless the object is familiar. Familiarity with an object enables us to judge its appropriate size, even in the absence of depth cues.

Size Perception and Age

In an experiment below figure, adults and eight-year-old children viewed objects at distances ranging from 10 to 100 feet. The physical sizes of the objects were adjusted at the various distances so that the image projected on the retina was always the same size. The horizontal line indicates the size of the retinal image (which is constant for all distances), and the diagonal line the size of the physical object (which increases with distance). Note that adults make fairly accurate size judgments no matter how distant the objects. The judgments of children, however, appear to be increasingly influenced by the size of the retinal image as the object is placed farther away.



SIZE PERCEPTION AND AGE.

Location Constancy

Location constancy may be termed as perceptual stability of the things. Our world has perceptual stability for us because we perceive objects as enduring, as being the same as when we last looked. We also perceive these objects in a setting that remains essentially fixed, despite the fact that a myriad of changing impressions strike the retina as we move about. We take the stability of our perceptual world for granted, but unusual conditions demonstrate that it too depends upon past experience.

The role of learning in location constancy can be shown by experiments that use special glasses to rearrange the visual environment. In a classic study conducted more than 75 years ago, Stratton fitted himself with lenses that not only inverted the visual field so that he saw the world upside-down, but also reversed it so that objects perceived on the

left were actually on the right and vice versa. Stratton reports that at first the world seemed to lose its stability. After a few days this swinging or swirling sensation decreased, indicating some restoration of location constancy. Another sign of regained location constancy was that a fire was again heard to crackle in the fireplace where it was seen, a harmony of location that was at first lost because only the eyes, and not the ears, were perceiving in reverse. Although the distortion provided by the lenses made even the simplest task extremely difficult and laborious, Stratton found that as the experiment progressed he became more skilful in dealing with his mixed-up perceptual world. He bumped into objects less frequently and was able to perform such tasks as washing and eating, which initially had been very difficult. When the glasses were removed some adjustment was required before he regained his old visual-motor habits.

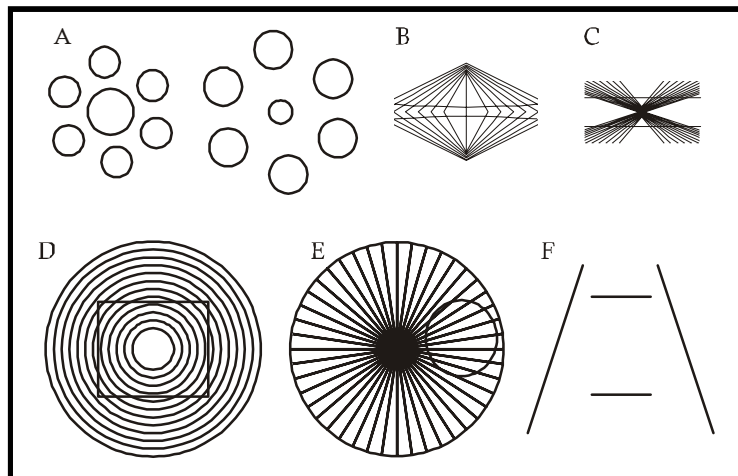
Brightness Constancy

The simplest example to understand is **brightness constancy** (more accurately, lightness constancy). A white piece of paper indoors reflects considerably less light than does a black lump of coal outside on a bright, sunny day. Yet the paper looks white, and the coal black.

Brightness constancy depends on extracting the *per cent of light* (albedo) that the coal and the paper reflect. This can be done by comparing the amount of light from the coal and from the paper to the average amount of light from their surroundings. The paper reflects more light than its surrounding; the coal reflects less. So, even though the total amount of light from white paper (or black coal) can vary greatly, the relation of light from white paper (or black coal) to light from the surroundings remains (relatively) unchanged – a higher order stability. So the visual system compares light from each object to the light from its neighbors. This comparison gives an accurate estimate of the per cent of light that each object reflects.

Visual Illusions

We experience an illusion when we select a perceptual hypothesis that is actually incorrect.

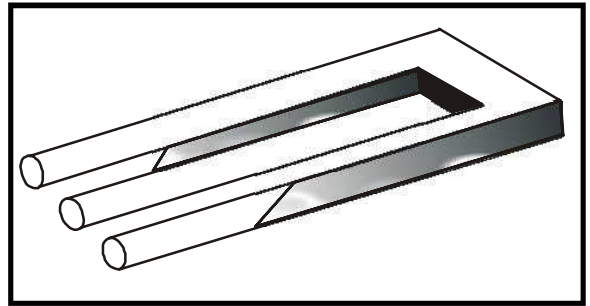


VISUAL ILLUSIONS.

Geometrical illusions have been studied for many years, but their explanations are still not fully agreed upon. Some illusions are based on relative size in contrast with surroundings (Fig. A). Others may be understood if we suppose the figures to be projected in the third dimension (Fig. B through D). If the lines in B and C were drawn on the surface of a solid double cone, or represented by a system of wires meeting at the horizon, they would have to be curved in order to be parallel as viewed. Because we tend to view these figures as though they were perspective drawings, we see the parallel lines as bent. In this case our tendency to “constancy” misleads us. Similarly, the backgrounds in illusions D and E can be viewed as three-dimensional (either concave or convex) thereby distorting the square and circle superimposed on them.

Below Figure F is known as ‘Ponzo illusion’. The illusion can be thought of as a flat projection of three-dimensional space, with the vertical lines converging in the distance, as in the picture of the railroad tracks. We know from experience that the distant railroad ties are the same size as the nearest ones, even though the retinal image they give is much smaller. If real objects were lying between the tracks, the upper rectangle in the picture would be perceived as more distant. Because the brain tries to compensate for the expected shrinkage of images with distance (even though in this case there is no shrinkage for which to compensate), we see the upper rectangle as larger. Such an illusion is an inevitable result of size constancy mechanisms.

The fact that the Ponzo illusion increases in magnitude between childhood and adulthood suggests that the illusion depends upon learning to use linear perspective cues in two-dimensional drawings (Parrish, Lundy, and Leibowitz, 1968). Some of the problems of seeing three-dimensional forms on a two-dimensional surface are illustrated by the “impossible figure”. Incompatible depth information is given to the eye, and the brain cannot decide how to interpret it.



AN IMPOSSIBLE FIGURE.

Perception of Depth and Movement

Depth Perception

The problems of perceiving distance and depth (the third dimension) have great importance in the study of perception. The retina is essentially a two-dimensional surface, then how it can be helpful in perceiving a space of three dimensions?

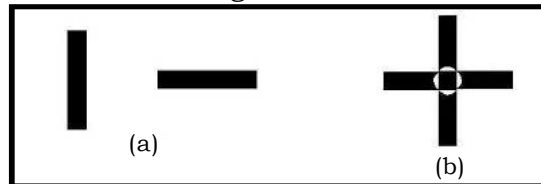
Stereopsis

Stereopsis refers to our ability to appreciate depth, that is the ability to distinguish the relative distance of objects with an apparent physical displacement between the objects. It is possible to appreciate the relative location of objects using one eye (monocular cues). However, it is the lateral displacement of the eyes that provides two slightly different views of the same object (disparate images) and allow acute stereoscopic depth discrimination.

Binocular Cues

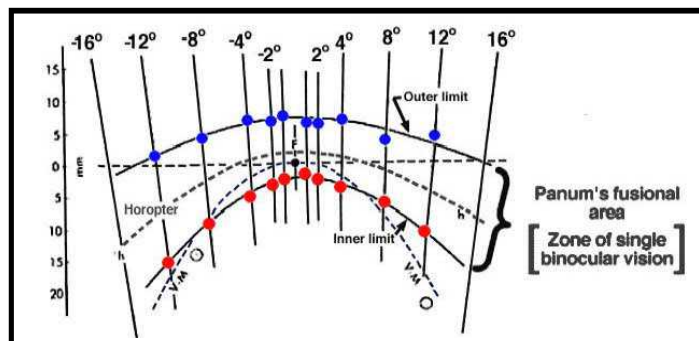
Stereopsis is an important binocular cue to depth perception. Stereopsis cannot occur monocularly and is due to binocular retinal disparity within Panum's fusional space. Stereopsis is the perception of depth produced by binocular retinal disparity. Therefore, two objects stimulates disparate (non-corresponding) retinal points within Panum's fusional area.

Fusion describes the neural process that brings the retinal images in the two eyes to form one single image. Fusion occurs to allow single binocular vision. Fusion takes place when the objects are the same. When the objects are different, suppression, superimposition or binocular ("retinal") rivalry may occurs. Suppression occurs to eliminate one image to prevent confusion. Superimposition results in one image presented on top of the other image. Binocular rivalry describes alternating suppression of the two eyes resulting in alternating perception of the two images. This usually occurs when lines are presented to the two eyes differ in orientation, length or thickness. An example of binocular rivalry occurs when one eye is presented with a horizontal line and the other eye is presented with a vertical line. Binocular rivalry occurs at the intersection of the lines and some suppression also exists below figure.



(a) Binocular rivalry can be demonstrated by placing a pen between yourself and the screen. Keep your eye on the tip of the pen and notice the two bars merge. You may need to slowly move the pen from the screen towards you. (b) Result of (a)

Panum's fusional area is the region of binocular single vision. Outside Panum's fusional area, physiological diplopia occurs. Using the haplopic method of determining the horopter, Panum's area can be determined below figure.



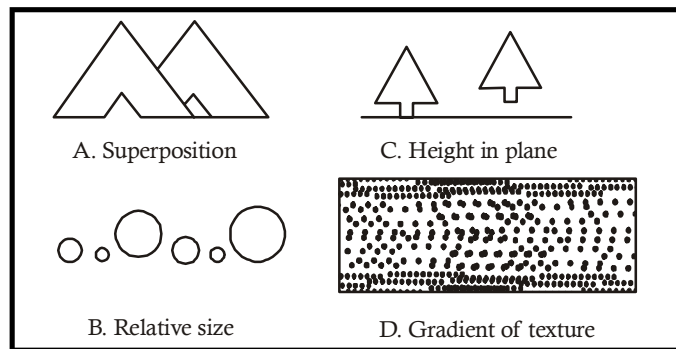
Haplopic method of determining the horopter involves locating the region of single binocular vision at a distance of 40 cm. Panum's fusional area lies between the outer and inner limits of the region of single binocular vision.

Retinal disparity. Retinal disparate points are retinal points that give rise to different principal visual direction and diplopia. However, retinal disparity within Panum's fusional area (zone of single binocular vision) can be fused resulting in single vision. Retinal disparity is essential for stereoscopic depth perception as stereoscopic depth perception

results from fusion of slightly dissimilar images. Due to the lateral displacement of our eyes, slightly dissimilar retinal images result from the different perception of the same object from each eye.

Monocular Cues

An artist is able to give depth to his picture because he can make use of the many monocular cues that tell us the distance of objects. Figure illustrates four types of cues that are used in the perception of depth. If one object appears to cut off the view of another, the presumption is strong that the first object is nearer (Fig. A). If there is an array of like objects of different sizes then the smaller ones are perceived as being in the distance. Even a series of scattered circles of different sizes may be viewed as spheres of the same size at varying distances (Fig. B). Another hint of perspective is height in the horizontal plane. As we look along a flat plane, objects farther away appear to be higher, so that we can create the impression of depth for objects of the same size by placing them at different heights (Fig. C). Even for irregular surfaces, such as a rocky desert or the waving surface of the ocean, there is a gradient of texture with distance, so that the “grain” becomes finer as distance becomes greater (Fig. D).

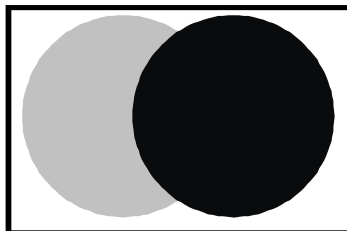


VISUAL DISTANCE PERCEPTION.

Several strong monocular cues allow relative distance and depth to be judged. These monocular cues include:

Relative Size. Retinal image size allow us to judge distance based on our past and present experience and familiarity with similar objects. As the car drives away, the retinal image becomes smaller and smaller. We interpret this as the car getting further and further away. This is referred to as size constancy. A retinal image of a small car is also interpreted as a distant car.

Interposition. Interposition cues occur when there is overlapping of objects. The overlapped object is considered further away.

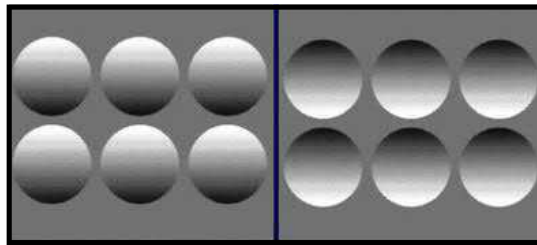


INTERPOSITION. THE LIGHT CIRCLE IS REPORTED TO BE CLOSER SINCE IT OVERLAPS THE DARK CIRCLE.

Linear Perspective. When objects of known distance subtend a smaller and smaller angle, it is interpreted as being further away. Parallel lines converge with increasing distance such as roads, railway lines, electric wires, etc.

Aerial Perspective. Relative colour of objects give us some clues to their distance. Due to the scattering of blue light in the atmosphere, creating “wall” of blue light, distance objects appear more blue. Thus distant mountains appear blue. Contrast of objects also provide clues to their distance. When the scattering of light blurs the outlines of objects, the object is perceived as distant. Mountains are perceived to be closer when the atmosphere is clear.

Light And Shade. Highlights and shadows can provide information about an object’s dimensions and depth. Because our visual system assumes the light comes from above, a totally different perception is obtained if the image is viewed upside down.



HIGHLIGHTS AND SHADOWS PROVIDE INFORMATION ABOUT DEPTH.

Monocular Movement Parallax. When our heads move from side to side, objects at different distances move at a different relative velocity. Closer objects move “against” the direction of head movement and farther objects move “with” the direction of head movement.

Movement Perception

Movement perception, process through which humans and other animals orient themselves to their own or others’ physical movements. Most animals, including humans, move in search of food that itself often moves; they move to avoid predators and to mate. Animals must perceive their own movements to balance themselves and to move effectively; without such perceptual functions the chances for survival would be sharply reduced.

Visual Cues to Movement

The eye is by far the most effective organ for sensing movement. Some animals are especially sensitive to visual stimuli that move in specific ways. For instance, electrical patterns from the eye of a frog show that some elements in the organ respond only when the stimulus is about the size of a fly moving in the insect’s range of speed. Generally the eyes of lower animals seem to respond selectively to what is of importance to survival. In these animals the eye’s retina does much of the visual processing. This is an economical arrangement since the animal tends to respond only to essential stimuli, the brain having little to do but relay signals to the motor system. It is an inflexible mechanism, however; higher animals process visual information in more elaborate ways, the brain being more heavily involved. Thus, some cells in the visual area of the cat’s brain respond only to moving stimuli, sets of movement-detector cells functioning specifically for

each direction across the field of view. Features of human visual experience also suggest that movement detectors exist in the human brain.

Each retina in most higher animals has a central (foveal) zone for detailed colour and pattern vision and a surrounding peripheral zone that effectively is sensitive only to the grosser features of the outer visual field. The peripheral retina is especially sensitive to movement (often a signal of danger), which induces a reflex turning of the eyes to project the image on the fovea and permit the moving object to be recognised.

Visual Stability

Mechanisms have evolved that yield stable, clear visual input despite swaying and other blurring factors. In a reflex mechanism called optokinetic nystagmus, the eyes pursue a moving scene to keep the image stationary on the retina. When they can move no farther, they snap back and pursue the scene again in a to-and-fro alternation of slow pursuit and quick return. These eye movements are readily observed in people who are looking at a moving pattern of stripes or turning their heads, this response being inhibited only when something stationary is visually fixated.

Similar nystagmic movements are triggered by impulses arising in the inner ear when the head moves. These persist even when the eyes are closed and may be felt by pressing the eyelids lightly as one rotates the whole body.

In a related stabilising activity the eyes scan in quick jerks (saccades) with short fixations; *e.g.*, in reading. Normally the eyes cannot move steadily over a stationary scene but make a series of stationary images (like still photographs); visual function tends to be suppressed when there is saccadic blurring. Yet the eyes can follow a steadily moving object smoothly.

When one looks from one point to another, movements of the retinal image are the same as those produced by a moving scene on a stationary eye. It might be thought that the sensory structures found in the eye muscles would provide the cues for judging whether it is the eye or the scene that has moved. Yet we see the scene as stationary only when we move our eyes voluntarily and not when they are moved passively by the finger. This suggests that motor-nerve signals inform us whether our eyes are moving, rather than the sensory structures in the eye muscles. When the eye is moved by pushing it with the finger there is no normal motor discharge to inform the brain, and changes in retinal image are perceived as movement of the scene. Indeed, people with paralyzed eye muscles experience the scene as moving when they try to move their eyes. When the motor discharge thus generated is not accompanied by the expected image motion, the person falsely perceives the scene and the eye to be moving together.

Relative Visual Movement

A visual field containing familiar objects provides a stable framework against which relative motion may be judged. People often report that an isolated point of light in a dark room is moving when it is not; the experience is known as autokinetic movement. It was observed in 1799 by Alexander von Humboldt while he was watching a star through a telescope, and he attributed it to movement of the star itself. Not until about 60 years later was the effect shown to be subjective, apparently arising from instability in the sense of eye position without a visual frame of reference.

Similarly, if a small object is presented in a frame with nothing else in view, movement usually is attributed to the object even when only the frame moves. This induced

movement effect reflects our tendency to use the larger surround as a stable frame of reference. Recall the illusion that your train is moving when it is really the moving train alongside that, seen through the window, is falsely accepted as the frame of reference.

People cannot perceive very slow movement; below a minimum speed (about that of the minute hand on a watch) movements become imperceptible and can only be inferred (as in remembering the previous position of the hour hand). At high speeds, one perceives a blurred streak rather than a definite object in motion.

Movement After Effect

When a parade is interrupted after some minutes, the pavement may seem to move in the opposite direction to the marchers who have passed. Phenomena similar to this movement aftereffect occur in other senses. For instance, after disembarking, a sailor feels the land to be rolling like a ship as the result of kinesthetic and vestibular aftereffects. The visual movement aftereffect probably arises when movement detectors in the brain that respond to the original direction of motion become fatigued, leaving predominant those detectors that respond to contrary movement.

Apparent Movement

Motion-picture film is a strip of discrete, still pictures but produces the visual impression of continuous movement. Stationary light bulbs coming on one after the other over the theatre entrance also produce an impression of steady movement. In part, such effects of apparent movement (called the visual phi phenomenon) depend on persistence of vision: visual response outlasts a stimulus by a fraction of a second. When the interval between successive flashes of a stationary light is less than this visual-persistence time, the flicker will appear to fuse into a continuous light. The flicker frequency at which this occurs is called the perceiver's flicker-fusion frequency (or critical flicker frequency) and represents the temporal resolving power of his visual system at the time. Another process on which apparent movement depends is a tendency (called visual closure or phi) to fill in the spaces between adjacent visual objects. This means that the movement detectors of the visual system are triggered as effectively by a closely spaced pair of lights alternately going on and off as by a single light moving back and forth. It would seem that two aspects of visual function (flicker fusion and phi) make the motion-picture industry possible.

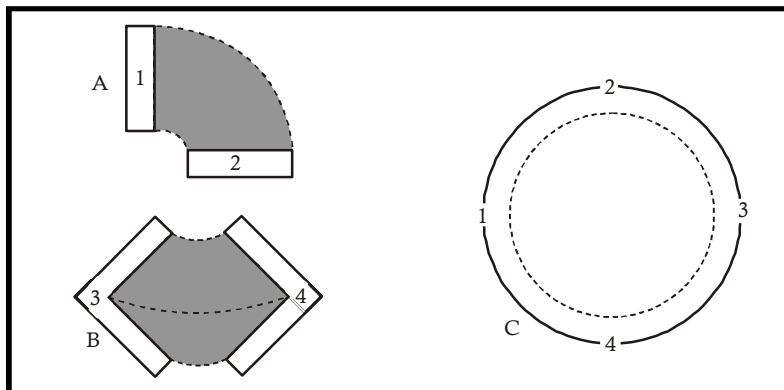
(i) Autokinetic Effect

Autokinetic effect is produced when perception of apparent motion of a stationary light takes place. If you stare for a few seconds at a single spot of light in a completely dark room, the light will appear to move about in an erratic manner—sometimes oscillating back and forth, sometimes swooping off in one direction. This apparent movement of a stationary light, known as the autokinetic effect, has been the subject of much experimentation, but there is still no certain explanation. What is clear though is that the autokinetic effect occurs only in a visually impoverished environment where there is no frame of reference against which to determine that the light spot is stationary. When other lights are introduced or the room is lightened, the effect disappears. Pilots during night flights are particularly susceptible to the autokinetic phenomenon; they sometimes line up a distant beacon with the edge of a windshield or some other frame of reference to minimise the effect.

(ii) Stroboscopic Motion

When separated stimuli, not in motion, are presented in succession, stroboscopic motion is perceived. Moving personality in a film is a best example of this kind of apparent motion.

A simpler form of stroboscopic motion, known as the phi phenomenon, has been studied extensively in the laboratory. One arrangement is diagrammed in Fig. C. The four lights can be turned on and off in any order. When one light blinks on and then off, followed shortly by another, there is the illusion of a single light moving from the position of the first to the position of the second, and so on. The apparent movement is seen as occurring through the empty space between the two lights. When the four lights of Fig. C flash on and off in proper sequence, you see a rotating circle, but the apparent diameter of the circle is less than that of a circle that would actually pass through the four lights. Whatever “attracts” the light to the position of the next light operates also to “attract” it toward the centre of the circle, thereby making the circle smaller. The two tendencies result in the compromise that is seen as a circle too small to pass through the actual position of the lights. Even though the phi phenomenon is illusory, it tends to preserve the perceptual structure that would be possible in real motion. For example, in Fig. A the perceived motion is through an arc but in the plane of the paper; in Fig. B the motion is seen in the third dimension, the figure turning over as it moves across.



SPECIAL CASES OF PHI MOVEMENT.

Stroboscopic Effect

When a rotating electric fan is illuminated by a flashing light source (called a stroboscope) so that a flash arrives whenever a fan blade passes a fixed position, the blades will seem to stand still. This is a useful way of observing fast-moving objects such as machinery or insect wings. If the flashes occur less frequently, the object will seem to move slowly in its actual direction; when the flashes arrive more frequently, the object will seem to move backward, as stagecoach wheels may do in the cinema.

Visual Movement in Depth

An object moving directly away from an observer provides fewer visual cues of movement than it would be moving across the field of view. However, changes in retinal-image size are produced that give a clue to its movement. Thus a stationary, but shrinking, lumi-

nous object in the dark is seen as if it were receding. Other clues to movement in depth are changes in the convergence angle of the two eyes, in the focussing mechanism, and in the haziness and brightness of the object.

Nonvisual Cues

Auditory

Sound waves travel well in water, and fish are accordingly able to rely heavily on acoustic cues to detect moving objects. Land animals, although typically more visually oriented, also make some use of such cues, including changes in intensity (loudness) and small differences in the time at which the wave reaches each ear. Some animals (*e.g.*, rabbit, horse) have mobile external ears that track moving sound sources. Bats vocalise high-frequency waves and are able to detect objects by sonic reflection (a technique similar to sonar).

Kinesthetic

Kinesthesia here refers to experiences that arise during movement from sense organs in the membranes lining the joints and from the sense of effort in voluntary movement; receptors in muscles seem to have little role in the perception of bodily movements. Depending on speed of motion and the joint involved, blindfolded people can detect a passive joint movement as small as a quarter of a degree. People vary widely in the accuracy with which they can actively produce movement of a given extent; this ability also varies with direction of movement and the opposing friction, mass, and springiness.

Kinesthetic perception may persist for a limb that has been amputated, giving rise to a hallucinatory experience known as the phantom limb. The patient may experience vividly the “movement” of the absent part; a recent amputee may attempt to stand on his missing leg or to grasp with his missing hand.

Vestibular System

Vestibular structures, enclosed in a fluid-filled cavity in the region of each inner ear, include the utricle, a small sac containing minute sensitive hairs associated with tiny sandlike granules called otoliths. The utricle functions as a linear accelerometer. When the head tilts relative to gravity or is accelerated, the relatively dense otoliths deflect the hair cells and nerve impulses are transmitted to the brain. At constant velocity the otoliths become stable, stimulation ceases, and a person must rely on other cues (*e.g.*, by observing the passing scene) to detect his motion.

Vestibular structures for each ear also include three fluid-filled semicircular canals, each in a different plane. Each canal has a swelling (ampulla) that contains the cupula, a cluster of sensitive hairs embedded in a jellylike mound. As the head moves in the plane of a given canal, motions of the fluid deflect the cupula to produce nerve impulses. These travel through the brainstem to other brain and spinal centres that mediate equilibrium or balance and that generate nystagmic eye movements.

Taken together, the semicircular canals serve as a rotary accelerometer. If a person is rotated at constant velocity and then is suddenly stopped, the cupula is redeflected to give a feeling of rotation in the opposite direction; this also gives rise to dizziness and post-rotational nystagmus. Dancers and skaters learn to overcome such effects by concentrating on some fixed, visible object; with their eyes closed they are as likely to fall as anyone.

Over-stimulation of the vestibular system (*e.g.*, on a ship or airplane) may induce motion sickness. A person with vestibular function totally destroyed is not subject to motion sickness; but if the vestibular mechanism is impaired only on one side, each movement of the head can be nauseating. Such a patient takes a long time to compensate for this imbalance.

In outer space there is no gravitation to mediate feelings of up or down, although these still may arise from visual cues. The utricles and vestibular canals still respond to movements of the head, however, and serve orientation within the spacecraft.

Role of Motivation in Perception

We cannot perceive all the things simultaneously. Our perceptions are selective. When we perceive things, we focus upon a few. This perceptual focusing is called attention. When a stimulus attracts our attention, we usually perform certain body movements that enhance our reception of the stimulation. If it is a visual stimulus we turn our head in the proper direction, our eyes turn so that the image falls on the fovea, our pupils dilate momentarily to allow more light to enter the eye, and the lens muscles work to bring the image clearly into focus. If the stimulus is auditory we may cup our hands behind our ears or turn one ear in the direction of the sound, keeping the rest of our movements very still so as to enhance the reception of a faint auditory stimulus. These body movements are accompanied by certain characteristic internal physiological changes. The physiological reactions that occur in response to stimulation changes in the environment form such a consistent pattern that they have been called the orienting reflex and have been studied extensively by psychologists.

The orienting reflex occurs in both man and animals in response to even minimal changes in the stimulus environment. The physiological accompaniments of attention, in addition to the body movements mentioned above, include dilation of the blood vessels in the head, constriction of the peripheral blood vessels, certain changes in the gross electrical responses of the brain (EEG), and changes in muscle tone, heart rate, and respiration. These responses serve the dual function of (1) facilitating the reception of stimulation and (2) preparing the organism to respond quickly in case action is needed. We can see why such a reflex is extremely valuable for self-preservation.

What a person perceives and how he perceives it may also be determined to some extent by his needs and personal values. The value an individual places on an object may affect such direct impressions as those of size. For example, it has been shown that children from poorer homes tend to overestimate the size of coins more than do children from well-to-do homes.

Certain internal variables, such as motives and expectations, are equally important in determining which stimulus attracts our attention. The advertiser counts on an appeal to the male sex drives when he uses pictures of scantily clad females to advertise anything from carpets to automobile tires. In a culture where hunger is a more generally unsatisfied drive than sex, pictures of food might prove to be a more powerful attention-getter.

The Role of Learning in Perception

One of the traditional problems of visual perception has been the question of whether our abilities to perceive the spatial aspects of our environment are learned or innate. This is

the familiar nature-nurture problem, and its investigation with relation to perception goes back to the philosophers of the seventeenth and eighteenth centuries. There are two theories concerned to this problem – Nativism and Empiricism.

One group, the nativists (Descartes, Kant), argued that we are born with the ability to perceive the way we do. In contrast, the empiricists (Berkeley, Locke) maintained that we learn our ways of perceiving through experience with objects in the world about us. Among the early sensory psychologists, Hering and Helmholtz held opposing views. Hering pointed to retinal disparity as evidence for the view that our eyes are innately designed to perceive depth; he developed a theory of distance vision based on the fact that each eye registers a different image. Helmholtz argued that visual perceptions were too variable to be explained on the basis of fixed receptor mechanisms and must therefore be learned.

Most contemporary psychologists believe that a fruitful integration of these two viewpoints is possible. No one today really doubts that practice and experience affect perception. The question is whether we are born with some ability to perceive objects and space in our environment or whether these abilities are completely learned.

Some conclusions, observed from several experiments for the role of learning in perception, are given below:

- (1) Our perceptions develop gradually from primitive visual experiences in which figure-ground relationships and colour predominate, becoming more accurate and more detailed with practice. They cannot, however, be taken as conclusive evidence of the innate visual ability of the infant. We do not know what deteriorative changes may have occurred over the years the adult subjects was blind, nor do we know what compensating skills this adult may have developed to overcome his handicap.
- (2) Gibson and Walk tested the response of infants, ranging in age from 6-14 months, when placed on the center board of the visual cliff. Since the infants could not be tested until they were old enough to crawl, the experiment does not prove that depth perception is present at birth. The results of studies with other organisms, however, indicate that depth perception is present at least as soon as the animal is able to locomote. Chickens tested when less than 24 hours old never made a mistake by stepping off on the deep side. Goats and lambs placed on the center board as soon as they could stand (some only one day old) always chose the shallow side. When placed on the deep side, such animals characteristically froze in a state of immobility.
- (3) Experiments designed to isolate the specific visual depth cues to which the organism responds on the visual cliff have yielded contradictory results; it seems, however, that monocular cues are sufficient. Infants who wore eye patches on one eye discriminated as well as those with binocular vision.
- (4) In one experiment, kittens were raised in an environment where they were exposed only to bright dots; they were never shown straight lines, contours, or edges. When these animals were subsequently tested using a specific procedure, their visual cortex contained neurons of an abnormal type responding well to small spots of light and showing little of the customary preference for lines and edges. Even at the level of cells in the visual cortex, the effects of prior experience are evident.
- (5) Visual acuity was tested with an apparatus that displayed a grating pattern in various orientation. The Euro-Canadians exhibited a much higher resolution for vertical and horizontal orientation than for diagonal orientations of the grating.

The Crees, however, showed no differences; they were equally good at all orientations. There is no evidence to suggest genetically determined differences in the visual system of the two groups. Rather, it appears that the acuity of the Crees is the result of their visual experience.

SIGNAL DETECTION THEORY

Detection theory or signal detection theory is a means to measure the ability to differentiate between information-bearing patterns (called stimulus in living organisms, signal in machines) and random patterns that distract from the information (called noise, consisting of background stimuli and random activity of the detection machine and of the nervous system of the operator). In the field of electronics, the separation of such patterns from a disguising background is referred to as signal recovery.

According to the theory, there are a number of determiners of how a detecting system will detect a signal, and where its threshold levels will be. The theory can explain how changing the threshold will affect the ability to discern, often exposing how adapted the system is to the task, purpose or goal at which it is aimed. When the detecting system is a human being, characteristics such as experience, expectations, physiological state (e.g., fatigue) and other factors can affect the threshold applied. For instance, a sentry in wartime might be likely to detect fainter stimuli than the same sentry in peacetime due to a lower criterion, however they might also be more likely to treat innocuous stimuli as a threat.

Much of the early work in detection theory was done by radar researchers. By 1954, the theory was fully developed on the theoretical side as described by Peterson, Birdsall and Fox and the foundation for the psychological theory was made by Wilson P. Tanner, David M. Green, and John A. Swets, also in 1954. Detection theory was used in 1966 by John A. Swets and David M. Green for psychophysics. Green and Swets criticized the traditional methods of psychophysics for their inability to discriminate between the real sensitivity of subjects and their (potential) response biases.

Detection theory has applications in many fields such as diagnostics of any kind, quality control, telecommunications, and psychology. The concept is similar to the signal to noise ratio used in the sciences and confusion matrices used in artificial intelligence. It is also usable in alarm management, where it is important to separate important events from background noise.

Applications

Signal Detection Theory has wide application, both in humans and animals. Topics include memory, stimulus characteristics of schedules of reinforcement, etc.

Sensitivity or Discriminability

Conceptually, sensitivity refers to how hard or easy it is to detect that a target stimulus is present from background events. For example, in a recognition memory paradigm, having longer to study to-be-remembered words makes it easier to recognize previously seen or heard words. In contrast, having to remember 30 words rather than 5 makes the discrimination harder. One of the most commonly used statistics for computing sensitivity is the so-called sensitivity index or d' . There are also non-parametric measures, such as the area under the ROC-curve.

Bias

Bias is the extent to which one response is more probable than another. That is, a receiver may be more likely to respond that a stimulus is present or more likely to respond that a stimulus is not present. Bias is independent of sensitivity. For example, if there is a penalty for either false alarms or misses, this may influence bias. If the stimulus is a bomber, then a miss (failing to detect the plane) may increase deaths, so a liberal bias is likely. In contrast, crying wolf (a false alarm) too often may make people less likely to respond, grounds for a conservative bias.

Compressed Sensing

Another field which is closely related to signal detection theory is called compressed sensing (or compressive sensing). The objective of compressed sensing is to recover high dimensional but with low complexity entities from only a few measurements. Thus, one of the most important applications of compressed sensing is in the recovery of high dimensional signals which are known to be sparse (or nearly sparse) with only a few linear measurements. The number of measurements needed in the recovery of signals is by far smaller than what Nyquist sampling theorem requires provided that the signal is sparse, meaning that it only contains a few non-zero elements. There are different methods of signal recovery in compressed sensing including basis pursuit, expander recovery algorithm, CoSaMP and also fast non-iterative algorithm. In all of the recovery methods mentioned above, choosing an appropriate measurement matrix using probabilistic constructions or deterministic constructions, is of great importance. In other words, measurement matrices must satisfy certain specific conditions such as RIP (Restricted Isometry Property) or Null-Space property in order to achieve robust sparse recovery.

SUBLIMINAL PERCEPTION

Subliminal is a two part word consisting of the prefix sub- and the root word limen (Latin origins). Sub-means below and limen means threshold. Thus, subliminal comes to mean below threshold. But what is a threshold?

Thresholds

According to Schmeidler, and the vast majority of psychologists studying subliminal phenomena, a threshold is the point at which a stimulus is perceived 50% of the time. For example, if my aural threshold were to be tested, I would be played numerous sounds in varying volumes. To signal that I heard the tone-I perceived the stimulus-I would raise my hand or press a button, whatever the researches had instructed me to do when I heard a sound. By controlling and tracking the frequency of the emitted sound, the researchers are able to find the volume at which I hear a sound-perceive the stimulus-half of the time. This point is my auditory threshold. A working definition of threshold is paramount. It acts as the scientific base for research in this area; a foundation on which all data are defined. For some, this is where the problems with subliminal research begin.

Visual Subliminal Perception

A series of nothings become something. When a person is flashed an image, the brain maintains a type of "footprint or "after image" of the stimulus. That is, despite the stimulus

as been shown and then taken away, for a short time that image is still seen by the brain. At first flash the subject may remember the shape of the beginning of a word. The next flash (very close together) the brain creates a footprint of the middle of the word. Continuing in such a way, a subject can piece together these footprints – these 'series of nothings' – and form/recognize the word as a whole.

How visual masking works. The masking of visual perception is generally the taking away of the "footprint" lingering on the subject's brain created by the target stimulus. A nonsense image (a series of letters that spell nothing, a line pattern, etc.) is flashed immediately after the target image. This leaves the footprint of the nonsense image lingering around as opposed to that of the target image.

Aural Subliminal Perception

Don't steal!!

Many department stores utilize subliminal messages to discourage theft. They use audio messages masked in the "store's music" (many stores play music over their intercom system). Masked audio messages are generally compressed or accelerated to a degree that renders them unintelligible, even if supraliminal. The message is then masked by the playing of, in this case, music. This music is the primary channel - it is the easiest to perceive. The hidden message becomes the secondary channel. (Moore) This tactic may not be futile, "numerous findings indicate not only analysis of secondary channel content at the level of individual words, but short persistence of memory for that content" (Greenwald). Perceived or not, there is still the controversy over whether or not it will influence one's behavior.

Difficulties with Research

Troubles with Thresholds. To begin with, psychologists have essentially massaged the theory of thresholds so that subliminal perception could "exist in a form that can be studied". In the 1800's, Herbart and Fechner used the term *limen* as a distinction between conscious and unconscious. Much of the research conducted during the 1950's and on is based on a distinctive breaking of "threshold" into two parts: sensory threshold and perceptual threshold: stimuli that are "sensed by the body, but not conscious" and stimuli that are "made conscious", respectively. It is the perceptual threshold that is utilized in the realm of subliminal perception research. Thus, what the researchers are studying are stimuli that are above the sensory threshold, but below the perceptual threshold (the point at which a subject can perceive a stimulus 50% of the time). This begs debate for it is sidestepping the entire concept of a dichotomy between conscious and unconscious, the exact concept of subliminal.

Already known from other areas of research is the fact that we do receive information in our lower brain that never makes it to the cortex. The cortex is where things are "made conscious". This is where sensory perception comes in to play. It is estimated that for every 1,000,000 stimuli that pass by the sensory threshold, one stimulus passes through the perceptual threshold. This is how the intricacies of human behavior are explained. Humans simply do too much to be conscious of it all. As McConnell points out, "We are a walking mass of thresholds."

Each person has their own unique set of thresholds. To measure the threshold of each subject prior to conducting the experiments is both time consuming and redundant, but does account for such idiosyncrasies. However, not only do thresholds vary from person

to person, but they also vary day by day within one individual. So what I can't hear today I may hear tomorrow and vice versa (McConnell, Moore). With no static threshold, or statistical foundation, research conducted on subliminal perception today becomes invalid tomorrow.

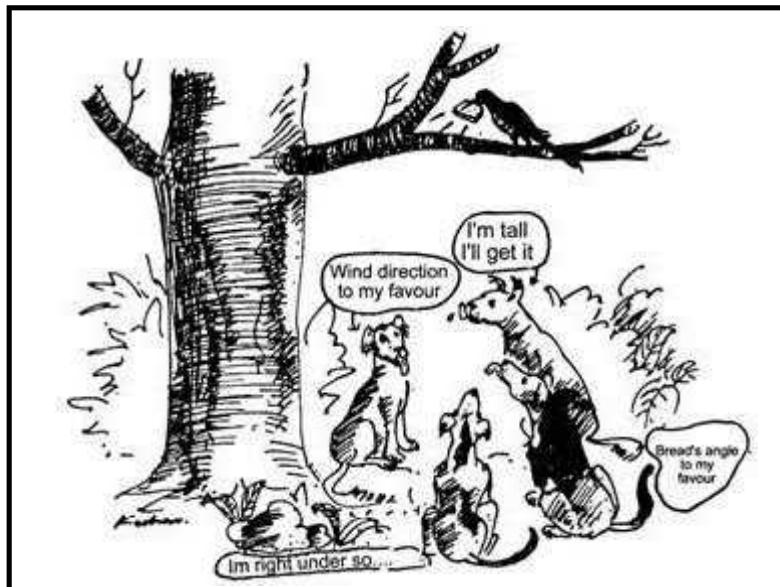
Erdelyi brings up an intriguing point. He asserts that these problems with thresholds are not methodological hindrances, but rather they are conceptual flaws. This reminds the community that indeed this research is not truly "subliminal".

Attention. Attention plays a role on the research of subliminal perception. It is capable of skewing results because the subject is instructed to listen for a sound or to look for a sight. When trying to ascertain if one can be influenced subliminally in day to day life, the person will not be actively looking or listening for the supposed stimuli/message. To counter this, study designs should keep the subject unaware of the target stimuli.

Information Processing Approach to Perception

During the 1960s, American psychologists investigating and exploring the principles of cognitive theories ultimately developed a new approach called cognitive psychology or information processing. Cognitive psychology included a spectra of processes like attention, perception, thinking, remembering, problem- solving, etc.

They fully gave up studying learning in isolation and this resulted in studying human learning as a whole rather than its different components. The term cognition refers to the processes through which information coming from the senses is transformed.



Information Processing

Scientific investigations show that there is a great resemblance between the brain and the computer. In view of this, it has been felt that the processes involved in the working of a computer could also explain those involved in human learning.

The aim of the information-processing approach is to isolate the mental stages of processing, their relationships, functions and performance. It declares that whatever mental processes act upon may best be described as information.

Later, we select, encode and retrieve information and use it to solve problems, make decisions and organise overt responses which may be called processing. Learning one chunk of information may depend upon the perceptual organisation brought into play by the individual. For example, if we give a subject a nonsense syllable 'lon' to remember, he may encode the attributes of this visual stimulus in terms of the first three letters of the word 'lonely' or may code it as the 11th, 14th and 13th letters of the alphabet, or may code it in terms of an abbreviation of London. The information that we pick up from the environment is said to be processed in stages through a succession of transformations from sensory input to response organisation and output.

The stages of processing are not observed directly and they are inferred from the evidence of responses to stimuli, particularly from differences in response patterns. We can only infer the operation of a memory system or a forgetting process by comparing the responses in a variety of stimulus situations.

All the information processing models assume that the central processor, the brain, has limited processing capacity and accept that all the information provided cannot be processed simultaneously. Not only does the brain have a limited amount of storage space for memories, but in addition, the flow of information through any mental stage will not be instant.

The transformation of information between stages is a process that takes time; the brain, like the other organs, cannot do everything at once. If this was not the case, there would be no delays in responding, no forgetting and the system would be prepared to accept as much information as is available.

Clearly, there are numerous everyday instances to observe this limited capacity instances where we suffer from 'information overload'. When too much information is presented, for the system to handle, a break occurs in the processing system and the result may be observable to an outsider in the form of an accident.

Suppose you are driving in heavy traffic and hoping to be on time for an important appointment that is on your mind. This occupies some of your available processing capacity and the passenger next to you is bothering you with tedious questions which are consuming some more capacity. At this point, we would agree that you are doing about as much as your capacity will permit.

You are on the brink of a processing limit with little spare capacity available. You certainly do not have enough processing capacity left to handle a sudden and unexpected task, such as the driver in front changing his mind and taking a turn and at the same time applying your brakes to avoid a collision. If capacity available is insufficient, you may not avoid a collision.

We are all familiar with instances of being asked to do too much at the same time. And it seems likely that extreme or continuous information overload could be a factor for the onset of a mental breakdown. There is a limit to the amount of information which we can respond to at any one time, but just where the limit is drawn continues to be a question asked by cognitive psychologists.

The limit may occur anywhere in the processing sequence. One group of workers insist that the limit applies to the amount of information perceived, whereas another

group argues that we perceive but respond selectively, and yet another group considers that selectivity occurs at many stages, and the breakdown can occur at any of these stages.

There is a limit to the amount of information we can process at any stage and at any one time. We need a process of selection to admit only required information at a particular stage. Such selection processes have been described as controlling processes, because they control the flow of information. Attention is, perhaps, the most noticeable of the control processes, in which the interaction takes place between the sense organs and the environment.

We attend first to one aspect of the environment and then to another as it becomes more important. For instance, in our earlier example, the car driver may be attending to the questions from his passenger. The new information, i.e. a car which is in the front is taking a sudden turn, calls for attention and then the conversation ceases to be important. Thought is given to the new problem about what is to be done to avoid smashed headlights!

The flow of information from one stage to another is said to be determined by control processes such as attention or by established pathways. In the performance of any well-practiced skills, e.g. driving a car, the performer may notice that he does not have to attend continuously to the information input in order to respond continuously.

When this absence of attention occurs, the driver may notice the 'time-gap phenomenon' whereby he may find himself at a point in the journey without knowing how he arrived there. Sometimes the time lapse may occur for a few seconds or it may last a minute or so, but during that time obstacles have been encountered and manoeuvres performed without the performer being aware of them.

A momentary diversion of attention from the environmental system and away from a guided response system does not always result in the breakdown of the information processing system. This suggests that processing can continue on 'automatic controls' under certain circumstances.

Analysts of information processing maintain that phenomena such as perception, memory, etc. cannot be investigated in isolation from one another. The system must be viewed as a whole rather than as a set of separate systems or processes.

The impact of information processing research has been tremendous. This impact can be traced as far away as below the seas and above the skies through the counterparts of the human beings like robots and computers. This system also explains, analyses and synthesizes behaviour that employs reasoning in a variety of situations like the selecting of stocks or horses or other securities in which an investment broker can wisely invest his or her client's capital, and a gambler's bet on horses.

Machines and robots designed to mimic organisms are built in such a way that they often exceed human capabilities in performance and have replaced man in exploring planets other than earth. All this is done through information that is well-programmed and executed to avoid and eliminate all chances of breakdown.

Investigations in information processing in the areas of sensation and perception extended themselves to reach out to the commonest sensory handicaps- blindness and deafness. Rehabilitation of handicapped persons involves a long process of education to enable them to learn skills that come naturally to the able.

The aim of all the schools of the deaf today is to teach speaking and the understanding of speech by lip-reading and sign language which has proved to be highly successful when planned and presented according to information processing models.

One of the most obvious effects of blindness is immobility. The blind could be made independently mobile by devices for obstacle detection and navigation. Their problem could be overcome to some extent by the use of obstacle detectors which may also locate some landmarks and by the use of special maps indicating landmarks which are of the greatest use to blind people.

One aid which has been devised is a long cane – a slightly longer version of the traditional stick, to be carried by the blind and which when tapped from side-to-side in a regular manner in front can detect obstacles such as foot-paths, walls, vehicles and so on. More technically sophisticated aids based on the reflection of inaudible ultrasonic rays have been developed based on the RADAR principle.

These devices have a narrow beam through which the direction and distance of the object can be gauged through sounds. The simplest way to do this is to hold the device in the hand and wave it. This device is called the sonic torch. The users are able to learn to process a great deal of information.

Some sonic torches are stereophonic and head mounted, and the user can detect widely-placed obstacles. In the use of this aid with blind children it has been found that there is a critical period during which the optimum amount of adjustment to the sonic sounds is made.

Insight

Scientists investigating this field realised that one of the main characteristics of thinking, rather creative thinking, is insight. The sudden appearance of a new idea is called insight. Detailed description provided by scientists and innovators who came out with spectacular discoveries showed similarities in the process described as creative thinking.

Analysis of these experiences showed the occurrence of sudden flashes of ideas or insight. Scientists are busy investigating into the origin, nature and functions of insight as a process. We should not be surprised if they evolve models and aids of insight which could be built into the human system of thinking. The author leave it to the readers' imaginations to visualize the consequences, good or bad.

Culture and Perception

While taking a break, we are happy to republish some of our favorite 'oldies but goodies'. This one was first put online in December of last year (2008). It was the first installment of a series of posts on Richard Nisbett's theory of culture and perception. Enjoy!

In a lively account published in *Trends in Cognitive Sciences*, Nisbett and Miyamoto (2005) made the case for "cultural" influences on perception. The crux of the argument is this : visual perception in Americans is more analytical, while in Asians it is more holistic. Americans pay attention to details, Asians to the larger picture. Americans examine objects in isolation, Asians are more sensitive to context. In the authors' own words:

"[...], we believe there is considerable evidence that shows that Asians are inclined to attend to, perceive and remember contexts and relationships whereas Westerners are more likely to attend to, perceive and remember the attributes of salient objects and their category memberships. It should be noted that the perceptual and attentional differences just described are in general quite large, sometimes even close to one standard deviation. Indeed, in the typical study, Asians and Westerners were found to behave in qualitatively different ways."

The evidence referred to above consists of psychological experiments that compared the behaviour of Westerners and Asians using mostly well-established paradigms. Change blindness, for example, is a popular staple of visual psychology: people often fail to detect large differences between two pictures shown in succession.

Masuda and Nisbett have found that Japanese subjects are more likely to detect changes to the background, while Americans are relatively more likely to detect changes in salient objects. From this and other findings, the authors argue that Asians and Westerners show qualitative differences in patterns of attention

This is meant as a challenge to a well-entrenched view in psychophysics (the part of psychology that deals with perception) that the main features of visual perception are defined by the common biological background of the human species and that issues of culture are, thankfully, someone else's headache. This allows for a convenient focus on exclusively running work colleagues and undergraduate students as subjects.

Psychophysicists like to cut up visual perception into an ill-defined hierarchy that runs from low-level stuff to high-level stuff. Examples of low-level stuff include contrast detection and orientation discrimination, whereas face perception is high-level stuff, and the perception of emotions is practically social psychology (ewww). It is understood that any topic that falls a level or two below your own favourite object of study is terribly technical and therefore boring, and anything above your own level is badly-controlled pop science, and boring. Hence, to someone who studies orientation discrimination, the properties of receptors in the retina is too low-level a topic and boring, and object recognition is a high-level nightmare, a philosophical quagmire, and boring.

It's probably fair to say that most people working in the field assume that low-level mechanisms are less susceptible to interindividual variation (pathologies aside), whereas high-level mechanisms are less constrained, people have better conscious control over them, etc. It is absolutely fair to question this assumption, but that is not really the point here. The point is that, given this assumption, evidence for inter-cultural variations in low-level mechanisms is more surprising and can be construed as going against the consensus. Color perception is thought to be low-level, and that is why there has been much interest in potential influences of language on the perception of colors ever since the original Berlin & Kay study. So when Nisbett & Masuda (2006) talk about cultural variation in "basic visual processes", what level in the hierarchy are they really talking about?

We would all be surprised to see intercultural variability in orientation selectivity in primary visual cortex cells, not because it is set in genetic stone (it isn't), but because we don't have any plausible reasons to expect variability. We all share a visual environment where all orientations are represented, it isn't like some people can get by without horizontal selectivity. What about visual attention, which is at the heart of Nisbett's theory?

The place of visual attention in the low-level/high-level hierarchy is hard to pin down. On the one hand some people argue that it influences vision all the way down to the appearance of visual objects, and the other hand others will argue that it's a catch-all term for what are actually several distinct phenomena. Some of it is thought to be an automatic response to certain stimuli, and therefore low-level, but to a large extent visual attention is modulated by conscious strategies and by context. In most contexts, your gaze will be automatically attracted to a large, looming object, but it is easy to learn to suppress that response.

Moreover, there are probably hundreds of studies that have trained people and monkeys to use specific patterns of visual attention. Anything related to the Posner paradigm probably falls under that description: subjects learn to allocate their attention flexibly given the constraints of the task. Therefore writing that:

"This research provides evidence that cultural variations are observable even in patterns of attention that one might assume to be governed by basic, invariant psychological processes" is a bit of a stretch, to say the least. I don't know of anyone who ever wrote that the allocation of attention is an invariant psychological process.

So: would it be surprising to find that attentional patterns vary from one individual to the next? Not that much. We know they are under relatively flexible, top-down control. The most surprising aspect in Nisbett's theory is the way different patterns of attention are supposed to arise in Asians and Westerners: through an internalisation in visual perception of broad cultural features. Western culture, in Nisbett's description, rewards individualistic thinking, while Asian culture favours the group over the individual. Therefore Asians need to "attend to the context", over attending to individual objects as Americans do.

There's quite a gap between having to attend more (as in having to pay more mind to) to social relations and interdependencies and attending more to global features of a visual scene. The potential cultural benefits of doing that are rather hard to discern – it's not as if important social information manifested itself "holistically" in a visual scene but not in its details. Clearly something is missing here.

To explain why holistic cultures should engender holistic perception, Nisbett and Miyamoto point to some observational data:

"When mothers and infants are observed playing with toys in their own home [...] it is found that American mothers label toys and point out their attributes more often than do Japanese mothers. By contrast, Japanese mothers tend to engage their infants in social routines more than do American mothers. American mothers' emphasis on labelling objects might lead infants to focus on the objects and their appropriate categorizations whereas Japanese mothers' emphasis on social practices might direct infants' attention to the relationship or to the context in which the object is located."

I'd be interested in an explanation of how exactly an emphasis on 'social practices' should affect an infant's attentional patterns, but let's assume the phenomenon is real, and that the experimental findings are solid. Here's a possible summary of what they tell us, along with a few caveats:

1. East Asians and Americans perform differently on average in a range of psychological tasks that are thought to be related to visual attention, whatever that is.
2. The general pattern of results is that Americans seem to have faster/more accurate access to more local aspects of a visual scene, while East Asians have better access to global and/or contextual aspects. The difference can be roughly described in terms of holistic vs. analytic perceptual strategies.
3. There is limited evidence that the two types of strategies can be elicited in bi-cultural individuals if cultural context is primed first. According to some studies, if you prime Chinese-Americans with their Chinese identity, for example by showing them items specific to one culture, they will behave more holistically, and vice-versa if primed with their American identity.

Now for the caveats, listed Asian-style from the more general to the more specific:

1. Very generally speaking, and aside from the priming experiments, all the data cited by Nisbett and Miyamoto are not experimental data. They should be treated with the same caution as survey data showing group differences, because that's what they are. They're subject to sample error and sample bias. There's a hundred potentially relevant variables that haven't been controlled for (more on that below). The causal mechanism evoked in the paper seems to me pretty weak, but others might disagree.
2. In some cases, there might be some non-perceptual confounds at play, especially when the tasks given to subjects are vague. In Masuda and Nisbett (2001), subjects were shown short animations and asked to "report what they had seen". It seems plausible that subjects reported what they felt were the most important aspects of the scene, but importance may not have anything to do with perceptual salience. The fact that Japanese gave more information about the context does not show that they hadn't perceived the features of the individual objects: it could equally well be that they felt they weren't worth reporting. Also, given that the Japanese participants hadn't reported much in way of individual features, it is not so surprising that they were less likely to recall them later on.

A related point is that subjects sometimes try to do what they think the experimenter wants them to do, sometimes they try to second-guess the experimenter's intention, and sometimes they just press response keys at random to get their 10 bucks or course credit. Different cultures may differ in their expectations about psychologists, and also in their view of appropriate ways to get course credit.

3. Visual attention is traditionally described as a filtering process: the brain receives more information than it really needs at any point in time, and needs to select information sources appropriate to the task at hand. The flip side of the coin is to view attention as a resource allocation problem: attention is about trading off accurate information about some aspects of the world for the sake of others. If I focus on one part of a visual scene, I have less information about other parts. In that respect the results of the first change blindness experiment reported in Masuda & Nisbett (2006) are surprising: if the Japanese are as fast as the Americans in detecting focal changes, and faster at detecting changes in the background, then where is the expected trade-off? There's no end of issues when analysing experimental data related to allocation of attention, and the review by Sperling & Doshier (1986) is a good place to start.

Although the results reported by Nisbett and Miyamoto are intriguing, one should be careful not to overinterpret them. Even if Americans and East Asians do indeed differ in how they allocate attention, that reflects a difference in default perceptual strategies – the data don't say anything about how general the finding is, nor do they imply that the strategies are inflexible. I doubt that Americans and East Asians differ much in their attentional strategies while driving or playing baseball. Not to belabour the point, but if someone were to show that East Asian boxing champions fight while focusing on the relationship between their opponents and its background, now that would be surprising.

Perceptual Styles

Perceptual Style is the way you take in information through your five senses and make that information meaningful to you.

Your Perceptual Style acts as a filter between sensation and understanding. It is at the core of who you are, and it impacts your values, your beliefs, your feelings, and your psychology.

Each of us has one of six unique Perceptual Styles that is innate. Our individual Perceptual Style is literally hard wired and has grown with us as we've aged and developed. The decisions you make, the actions you take, and the directions you choose, are all influenced by your Perceptual Style. This is because our Perceptual Style defines our reality.

We often assume perceive and respond to the same objective reality and that there is one absolute "right." Research implies that not only is that untrue, but perception is actually a filter applied to objective reality, resulting in natural differences between people.

It is because of our varying Perceptual Styles that we all experience a different reality.

Here is a brief look at the six Perceptual Styles (presented in alphabetical order):

Activity. People with the Activity Perceptual Style jump into life with both feet. They fully engage with the confidence that the details will sort themselves out. Direction, ideas, and pursuits emerge as the result of constant action and involvement with others and their surroundings. They engage until some new possibility or interest emerges to capture their attention. They cultivate extensive networks of friends and associates.

Adjustments. People with the Adjustments Perceptual Style see the world as an objective reality that can be known if they take the time to gather complete information about its intricacies and complexities. They pursue the acquisition and application of knowledge as the basis for their life experience. They enjoy sharing their knowledge with others and gathering new information from research or conversation. They have a strong sense of diplomacy and project a calm certainty.

Flow. People with the Flow Perceptual Style are instinctive advocates for the natural rhythms of life. They see the complex connectivity among seemingly unrelated people, environments, and situations. They intuitively integrate and harmonize their actions within a broadly defined community that provides them and others with a sense of belonging. They honor the continuity between past, present, and future.

Goals. People with the Goals Perceptual Style stride through life focused on the accomplishment of specific results and well-defined objectives. They experience a sense of urgency and clarity of purpose. They believe achievement is primary and method or process secondary – the end justifies the means. They evaluate all activities based on possible contribution towards the achievement of the results they expect. They thrive on competition and believe that life is a constant competition with winners and losers.

Methods. People with the Methods Perceptual Style approach life in a practical, matter-of-fact manner. They focus on how things need to be done. They believe that ordered processes, properly followed, will produce the desired results. They will discern the best process or technique to apply to any specific situation in order to produce reliable, repeatable outcomes. They impose order and they believe that everyone prefers to use well known and proven methods.

Vision. People with the Vision Perceptual Style approach life as a singular experience, a journey toward the future. They face the realities of a situation with

serious intent, an optimistic perspective that a solution will be found, and confidence that if one is not, there are always other alternatives to explore. They intuitively see new directions, and actions are taken or dropped opportunistically based on a sense of future possibilities and potential. They are highly persuasive and easily convince others to follow their vision.

All six Perceptual Styles provide distinctly different experiences of the world. These differences result in a profound psychological and perceptual diversity that is the most important diversity there is because it helps explain the differences between people.

The six Perceptual Styles together describe the total range of perceptual reality. However, individually each Perceptual Style has access to only 1/6 of the total. Everyone has one Perceptual Style that is innate and unchanging. All six Perceptual Styles are evenly distributed in the world, and research confirms there is no difference in regards to culture, race, gender, or age. All six have unique strengths and challenges.

Pattern Recognition

In psychology and cognitive neuroscience, pattern recognition describes a cognitive process that matches information from a stimulus with information retrieved from memory. Among others, the recognized patterns can be those perceived in facial features, units of music, components of language or characters and other symbols. One theory understands patterns as a set of characteristic features extracted from the stimulus, but it does not comprehensively describe the process or the role of context and there is a multitude of other theories with different approaches. Pattern recognition does not occur instantly, although it does happen automatically and spontaneously. Pattern recognition is an innate ability of animals.

Template Matching

The incoming sensory information is compared directly to copies (templates) stored in the long term memory. These copies are stored in the process of our past experiences and learning.

E.g. A A A are all recognized as the letter A but not B.

Note. This does not allow for variation in letters unless there are templates for each variation.

Prototype Matching

Prototype means a concept of average characteristics of a particular subject. It can be found throughout the world. For instance a concept of small animal with feathers, beak, two wings that can fly is a prototype concept of a crow, sparrow, hen, eagle, etc. Prototype matching, unlike template matching, does not emphasize a perfect match between the incoming stimuli and the stored concept in the brain.

Feature Analysis

According to this theory, the sensory system breaks down the incoming stimuli into its features and processes the information. Some features may be more important for recognition than others. All stimuli have a set of distinctive features. Feature analysis proceeds through four stages.

Detection
Pattern dissection
Feature comparison in memory
Recognition

Multiple Discrimination Scaling

Template and feature analysis approaches to recognition of objects (and situations) have been merged/reconciled/overtaken by multiple discrimination theory. This states that the amounts in a test stimulus of each salient feature of a template are recognized in any perceptual judgment as being at a distance in the universal unit of 50% discrimination from the amount of that feature in the template.

False Pattern Recognition

The human tendency to see patterns that do not actually exist is called apophenia. Examples of apophenia include the Man in the Moon, faces or figures in shadows, clouds and in patterns with no deliberate design, such as the swirls on a baked confection, and the perception of causal relationships between events which are, in fact, unrelated. Apophenia figures prominently in conspiracy theories, gambling, misinterpretation of statistics and scientific data, and some kinds of religious and paranormal experiences. Misperception of patterns in random data is called pareidolia.

Ecological Perspective on Perception

"Ecological psychology" is a term claimed by several schools of psychology with the main one involving the work of James J. Gibson and his associates, and another one the work of Roger G. Barker, Herb Wright and associates at the University of Kansas in Lawrence. Whereas Gibsonian psychology is always termed "ecological psychology", the work of Barker (and his followers) is also sometimes referred to as environmental psychology. There is some overlap between the two schools, although the Gibsonian approach is more philosophical and deeply reflective on its predecessors in the history of psychology.

Both schools emphasise 'real world' studies of behaviour as opposed to the artificial environment of the laboratory.

LEARNING PROCESS

Learning to read, to ride a bicycle, to act, to use a computer, to play synthesiser etc., each one of us learns something or the other intentionally or accidentally. Learning situations are most natural and common in our life. Every moment you learn something or the other because of the varied experience you have in life. In the educational process, the central idea is learning.

You are aware of a child learns right from his birth and goes on learning throughout his lifetime. An infant is quite helpless at birth, but slowly he learns to adopt himself to the environment around him. *E.g.* A child approaches a burning matchstick; the child burns his hand and withdraws. Another time when he comes to a burning matchstick, he takes no time to withdraw himself away. He learns to avoid not only the burning matchstick but also all burning things. When this happens, we say that the child has learned that if you touch a flame, you get burnt up. In this way, the change in the behaviour of an

individual occurs through direct or indirect experiences. This change in behaviour brought about by experience is called as learning. This is a very simple explanation of the term learning. Now let us understand the meaning and definitions of learning.

Concept of Learning

Learning, in psychology, is the process by which a relatively lasting change in potential behaviour occurs because of practice or experience. Learning is also a process of acquiring modifications in existing knowledge, skills, habits, or tendencies through experience, practice, or exercise.

Gates and others “Learning is the modification of behaviour through experience”

Henry, P smith “Learning is the acquisition of new behaviour or strengthening or weakening of old behaviour as a result of experience”.

Crow and Crow “Learning is the acquisition of habits, knowledge and attitudes. It involves new ways of doing things, and it operates in an individual’s attempt to overcome obstacles or to adjust to new situations.”

Skinner “Learning is the process of progressive behaviour adaptation.”

Munn “To learn is to modify behaviour and experience.”

M.L. Bigge “Learning may be considered as change in insights, behaviour, perception, motivation or a combination of these.”

The above definitions emphasise four attributes of learning...

As Process: the first is that learning is permanent change in behaviour.

It does not include change due to illness, fatigue, maturation and use of intoxicant.

The learning is not directly observable but manifests in the activities of the individual.

Learning depends on practice and experience.

Characteristics of Learning

Yoakum & Simpson have stated the following general characteristics of learning: Learning is growth, adjustment, organisation of experience, purposeful, both individual and social, product of the environment.

According to W.R. Mc law learning has the following characteristics.

1. Learning is a continuous modification of behaviour continues throughout life
2. Learning is pervasive. It reaches into all aspects of human life.
3. Learning involves the whole person, socially, emotionally & intellectually.
4. Learning is often a change in the organisation of behaviour.
5. Learning is developmental. Time is one of its dimensions.
6. Learning is responsive to incentives. In most cases positive incentives such as rewards are most effective than negative incentives such as punishments.
7. Learning is always concerned with goals. These goals can be expressed in terms of observable behaviour.
8. Interest & learning are positively related. The individual learns but those things, which he is interested in learning. Most boys find learning to play football easier than learning to add fractions.
9. Learning depends on maturation and motivation.

Types of Learning

Learning has been classified in many ways.

- I. **Informal, formal and non-formal learning.** Depending on the way of acquiring it learning may be informal, formal or non-formal.
 - Informal learning is incidental. It takes place throughout life. It is not planned.
 - Formal learning is intentional and organised. It takes place in formal educational institution.
 - Non-formal is also intentional & organised. It is flexible.
- II. **Individual or Group learning.** Learning is called either individual or group learning depending upon the number of individuals involved in the learning process.
- III. Another classification involves the types of activity involved
 - (a) **Motor learning.** When learning involves primarily the use of muscles it is called as motor learning. *e.g.*: learning to walk, to operate a typewriter
 - (b) **Discrimination learning.** Learning which involves the act of discrimination is called discrimination learning. *e.g.* infant discriminates between mother and aunt, milk and water.
 - (c) **Verbal learning.** When learning involves the use of words it is called as verbal learning.
 - (d) **Concept learning.** When learning involves the formation of concept it is called as concept learning.
 - (e) **Sensory learning.** When learning is concerned with perception and sense it is sensory learning.

Nature of Learning

- a. **Learning is adaptation or adjustment.** Friends, we all continuously interact with our environment. We often make adjustment and adapt to our social environment. Through a process of continuous learning, the individual prepares himself for necessary adjustment or adaptation. That is why learning is also described as a process of progressive adjustment to ever changing conditions, which one encounters.
- b. **Learning is improvement.** Learning is often considered as a process of improvement with practice or training. We learn many things, which help us to improve our performance.
- c. **Learning is organising experience.** Learning is not mere addition of knowledge. It is the reorganisation of experience.
- d. **Learning brings behavioural changes.** Whatever the direction of the changes may be, learning brings progressive changes in the behaviour of an individual. That is why he is able to adjust to changing situations.
- e. **Learning is active.** Learning does not take place without a purpose and self-activity. In any teaching learning process, the activity of the learner counts more than the activity of a teacher.
- f. **Learning is goal directed.** When the aim and purpose of learning is clear, an individual learns immediately. It is the purpose or goal, which determines what, the learner sees in the learning situations and how he acts. If there is no purpose or goal learning can hardly be seen.

- g. **Learning is universal and continuous.** All living creatures learn. Every moment the individual engages himself to learn more and more. Right from the birth of a child till the death learning continues.

Process of Learning

Learning is a process. It is carried out through steps. Learning process involves–

- (a) A motive or a drive.
- (b) An attractive goal.
- (c) A block to the attainment of the goal.

Let us see the steps one by one :

- (a) **A motive or a drive.** Motive is the dynamic force that energises behaviour and compels an individual to act. We do any activity because of our motives or our needs. When our need is strong, enough we are compelled to strive for its satisfaction. Learning takes place because of response to some stimulation. As long as our present behaviour, knowledge, skill and performance are adequate to satisfy all our needs, we do not feel any necessity to change our behaviour or acquire new knowledge and skills. It is this requirement, which initiates a learner to learn something.
- (b) **Goal.** Every individual has to set a definite goal for achievement. We should always have a definite goal for achieving anything. If a definite goal is set then learning becomes purposeful and interesting.
- (c) **Obstacle /block /barrier.** The obstacle or block or the barrier is equally important in the process of learning. The obstacle or the barriers keep us away from attaining the goal.

Now, you will think how the obstacle can be important in the process of learning. So let me tell you, if you face no difficulty of any kind in attaining the goal, you will not bring any change in your present behaviour or stock of knowledge or skills. Thus, the block or the barrier is an essential step in the learning process.

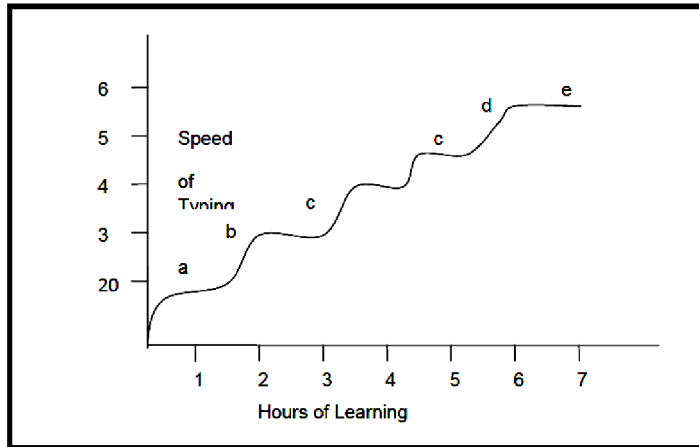
Let us clear the above views by taking an example. Suppose you wish to be included in your college hockey team. You want to have the esteem of your colleagues, your teachers. You are also motivated try the interesting experiences that you many enjoy. But you are blocked by your lack of skill in dodging, tackling and handling the ball. The obstacles in the path of goal achievement will set you make up your deficiency and acquire essential skill through sufficient practice and coaching.

Learning Curve

Learning curve is a graphic representation of how learning takes place in a particular situation. In all type of learning situations, the course of learning can be depicted and described graphically by drawing learning curves against x and y axis.

The figure shows a typical learning curve of many types of learning. The curve consists of a number of irregularities, as the progress is not constant. For the convenience, the curve is divided into 5 stages – a, b, c, d and e.

- (a) **Period of slow progress.** Generally, when a person has to start a learning of a given activity from a scratch, his early progress will be slow. *E.g.*, an infant's progress in learning to walk is very negligible in the beginning.



- (b) **Period of rapid progress.** In this stage, the learner's output raises rapidly. *E.g.* In typing once the learner has developed co-ordination of the movement of fingers he shows rapid progress.
- (c) **Period of no apparent progress.** Learning curves frequently display a period of no apparent progress. It is also known as plateau. A period of no visible learning progress, preceded and followed by improvement is called as plateaus. *E.g.* In typing, a person may after having made rather consistent progress for some time, reach a point where perhaps for weeks no further progress is made.

Causes of Plateau

- (i) The learner may be reorganising the previous learning into a new pattern before further progress is possible.
 - (ii) The learner may have hit upon bad habits
 - (iii) Lack of progress may be due to decrease in motivation.
 - (iv) The task may not be of uniform difficulty.
 - (v) Loss of interest.
 - (vi) The onset of fatigue is also one of the causes of a plateau.
- (d) **Period of sudden rise.** At the end of a plateau, there is generally a spurt in achievement. While on the plateau, the learner acquires better techniques, which help him later on to show rapid progress.
- (e) **Levelling.** All learning will finally slow down to such an extent that it will ultimately reach a period of no improvement. No one can continue to improve indefinitely in any given situation. The learning curve will eventually reach a limit, where no further improvement is possible. This limit is known as physiological limit.

Characteristics of Learning Curve

- (i) Slow initial progress.
- (ii) Spurt-like learning after some time.
- (iii) Declination in the rate of learning.
- (iv) Plateaus of learning.
- (v) Sudden increase in learning.
- (vi) Gradual levelling at the end.

Types of Learning Curve

We get different types of learning curves depending upon –

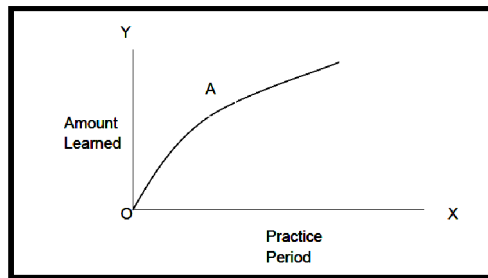
- (a) The nature of the learner
- (b) The nature of the task/learning material
- (c) Time available.
- (d) Conditions under which the learning takes place.

It is difficult to classify these learning curves. However, three common types of curves are there,

- (i) Negatively accelerated or the convex curve.
- (ii) Positively accelerated learning curve or the concave curve.
- (iii) Combination of convex-concave curve.

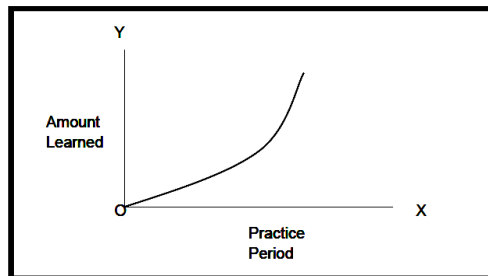
Convex curve

It depicts rapid initial improvement in learning that slows down with time. When the task is simple and the learner has previous practice on a similar task, we get this type of learning curve.



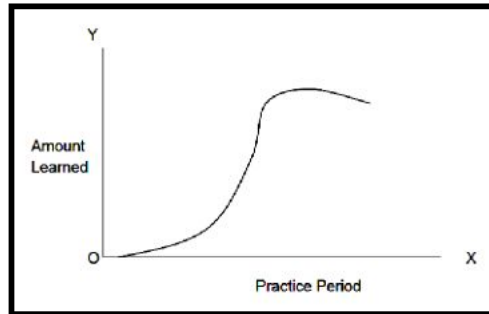
Concave Curve

There is slow initial improvement and learning increases with time. When the task is difficult we get such type of learning curve.



Combination of Convex Concave Curve

It looks like the capital letter 'S'. The curve takes concave or convex shape in the beginning depending upon the nature of the task.



In actual practical situations, such smooth curves are rarely found. Usually there are ups and downs (fluctuation).

Educational Importance of Learning Curve

1. In acquiring the basic skills in various subjects, the learner at times appears to show no progress. At such moments, the teacher can diagnose the reasons for the lack of progress.
2. A student's progress may be arrested because the work is too complex for him. The teacher can observe the student's work and detect the part that gives him trouble. The teacher should see if the student has developed any faulty study habits, which impede his progress.
3. The plateau may be due to the lack of motivation. The teacher should provide encouragement in order to maintain motivation at a high level.
4. The learning curves give a graphic evidence of one's progress, which is an effective motivational device for the learner.
5. Occurrence of plateaus can be minimised by using superior teaching methods.

LEARNING PROCESS

Learning as a process focuses on what happens when the learning takes place. Explanations of what happens constitute **learning theories**. A **learning theory** is an attempt to describe how people and animals learn, thereby helping us understand the inherently complex process of learning. **Learning theories** have two chief values according to Hill. One is in providing us with vocabulary and a conceptual framework for interpreting the examples of learning that we observe. The other is in suggesting where to look for solutions to practical problems. The theories do not give us solutions, but they do direct our attention to those variables that are crucial in finding solutions.

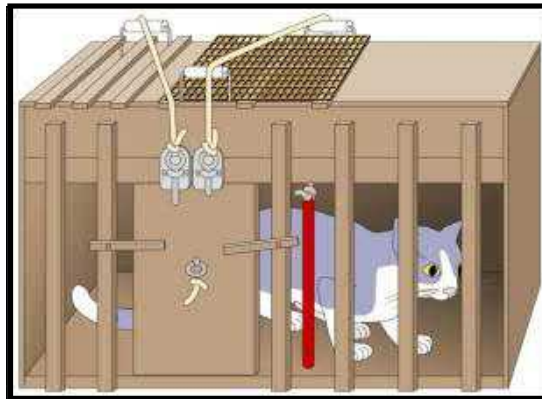
The three main categories or philosophical frameworks under which learning theories fall are behavioural, cognitive, and constructivism. Behaviourism focuses only on the objectively observable aspects of learning. Cognitive theories look beyond behaviour to explain brain-based learning. In addition, constructivism views learning as a process in which the learner actively constructs or builds new ideas or concepts.

We will discuss the behavioural theories under two broad categories: S-R theories.

S-R (Stimulus-Response) Theory with Reinforcement

(A) E.L. Thorndike – Trial and Error Theory of Learning

Edward Lee Thorndike (1874-1949) was the first American psychologist who put forward the Trial and Error Theory of learning. According to Thorndike, all learning takes place because of formation of bond or connection between stimulus and response. He further says that learning takes place through a process of approximation and correction. A person makes a number of trials, some responses do not give satisfaction to the individual but he goes on making further trials until he gets satisfactory responses.



Thorndike conducted a number of experiments on animals to explain the process of learning. His most widely quoted experiment is with a cat placed in a puzzle box. Thorndike put a hungry cat in a puzzle box. The box had one door, which could be opened by manipulating a latch of the door. A fish was placed outside the box. The cat being hungry had the motivation of eating fish outside the box. However, the obstacle was the latch on the door. The cat made random movements inside the box indicating trial and error type of behaviour biting at the box, scratching the box, walking around, pulling and jumping etc. to come out to get the food. Now in the course of her movements, the latch was manipulated accidentally and the cat came out to get the food. Over a series of successive trials, the cat took shorter and shorter time, committed less number of errors, and was in a position to manipulate the latch as soon as it was put in the box and learnt the art of opening the door.

Thorndike concluded that it was only after many random trials that the cat was able to hit upon the solutions. He named it as Trial and Error Learning. An analysis of the learning behaviour of the cat in the box shows that besides trial and error the principles of goal, motivation, explanation and reinforcement are involved in the process of learning by Trial and Error.

Laws of Learning

Based on Trial and Error Learning Theory, Thorndike gave certain laws of Learning. We shall discuss three fundamental Laws of Learning in this section. These laws are:

1. Law of Readiness

This law refers to the fact that learning takes place only when the learner is prepared to learn. No amount of efforts can make the child learn if the child is not ready to learn. The

dictum that 'you can lead a horse to the pond but you can't make it drink water unless it feels thirsty' goes very well with this law. In other words, if the child is ready to learn, he/she learns more quickly, effectively and with greater satisfaction than if he/she is not ready to learn. In the words of Thorndike the three stages of this Law of Readiness are :

For a conduction unit ready to conduct, to conduct is satisfying.

For a conduction unit ready to conduct, not to conduct is annoying.

For a conduction unit not ready to conduct, to conduct is annoying.

Thus, the Law of Readiness means mental preparation for action. It is not to force the child to learn if he is not ready. Learning failures are the result of forcing the learner to learn when he is not ready to learn something.

Educational Implications of Law of Readiness

The law draws the attention of teacher to the motivation of the child. The teacher must consider the psycho-biological readiness of the students to ensure successful learning experiences. Curriculum / Learning experiences should be according to the mental level of maturity of the child. If this is not so, there will be poor comprehension and readiness may vanish.

2. Law of Exercise

This law explains the role of practice in learning. According to this law, learning becomes efficient through practice or exercise. The dictum 'Practice makes a man perfect' goes very well with this law. This law is further split into two parts — Law of use and Law of disuse. The law of use means that a connection between a stimulus and response is strengthened by its occurrence, its exercise or its use. In other words, the use of any response strengthens it, and makes it more prompt, easy and certain. Regarding the law of disuse, it is said that when a modifiable connection is not made between a stimulus and a response over a length of time, the strength of that connection is decreased. This means that any act that is not practised for some time gradually decays. Anything that is not used exercised or practised for a certain period tends to be forgotten or becomes weak in strength, efficiency and promptness.

Educational Implications

Exercise occupies an important place in learning. Teacher must repeat, give sufficient drill in some subjects like mathematics, drawing, music or vocabulary for fixing material in the minds of the students. Thorndike later revised this law of exercise and accordingly it is accepted that practice does bring improvement in learning but it in itself is not sufficient.

Always practice must be followed by some reward or satisfaction to the learner. The learner must be motivated to learn.

3. Law of Effect

This is most important of Thorndike's laws, which state that when a connection between stimulus and response is accompanied by satisfying state, its strength is increased. On the other hand, when a connection is accompanied by an annoying state of affairs, its strength is reduced or weakened. The saying 'nothing succeeds like success' goes very well with this law. In other words, the responses that produce satisfaction or comfort for the learner are strengthened and responses that produce annoyance or discomfort for

the learner are weakened. Thorndike revised this law in 1930 and according to this revision, he stated that reward strengthened the response but punishment did not always weaken the response. Then he placed more emphasis on the reward aspect than on the punishment aspect of Law of Effect.

Educational Implications

This law signifies the use of reinforcement or feedback in learning. This implies that learning trials must be associated with satisfying consequences. The teacher can use rewards to strengthen certain responses and punishment to weaken others. However, the use of reward is more desirable than the use of punishment in school learning. The teacher for motivating the students for learning situations can exploit the use of reward.

(B) B.F. Skinners – Operant Conditioning

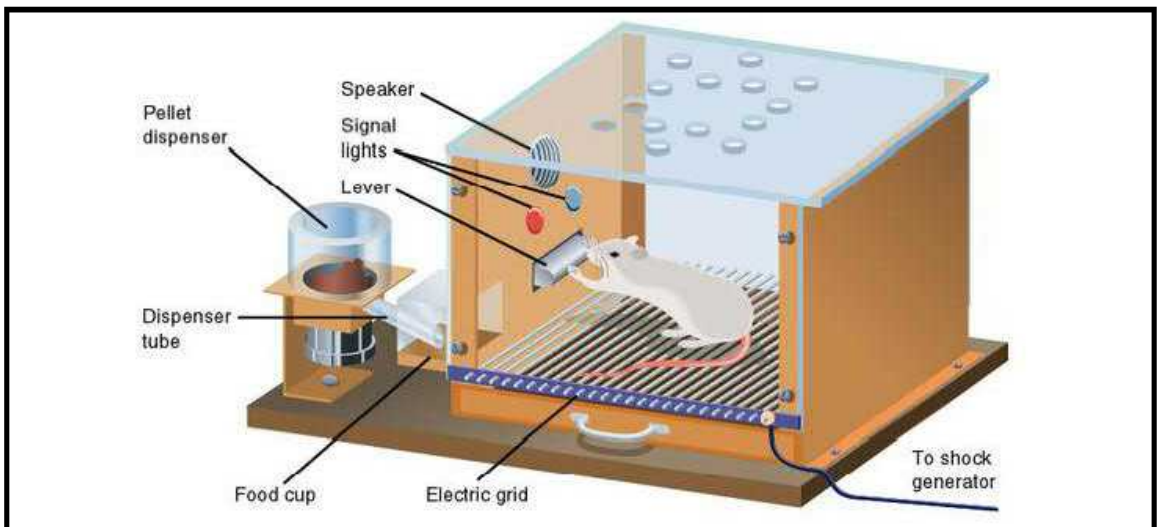
Operant conditioning (sometimes referred to as ***instrumental conditioning***) is a method of learning that occurs through rewards and punishments for behaviour. Through operant conditioning, an association is made between a behaviour and a consequence for that behaviour.

Behaviourist B.F. Skinner coined the term operant conditioning, which is why it is also referred as Skinnerian conditioning. As a behaviourist, Skinner believed that internal thoughts and motivations could not be used to explain behaviour. Instead, he suggested, we should look only at the external, observable causes of human behaviour.

Skinner used the term *operant* to refer to any “active behaviour that operates upon the environment to generate consequences”. In other words, Skinner’s theory explained how we acquire the range of learned behaviours we exhibit each and every day.

Skinner is regarded as the father of Operant Conditioning, but his work was based on Thorndike’s law of effect. Skinner introduced a new term into the Law of Effect - Reinforcement. Behaviour that is reinforced tends to be repeated (i.e. strengthened); behaviour that is not reinforced tends to die out-or be extinguished (i.e. weakened).

Skinner studied operant conditioning by conducting experiments using animals, which he placed in a “*Skinner Box*” which was similar to Thorndike’s puzzle box.



The Skinner box involved placing an animal (such as a rat or pigeon) into a sealed box with a lever that would release food when pressed. If food was released every time the rat pressed the lever, it would press it more and more because it learnt that doing so gives it food. Lever pressing is described as an operant behaviour, because it is an action that results in a consequence. In other words, it operates on the environment and changes it in some way. The food that is released as a result of pressing the lever is known as a reinforcer, because it causes the operant behaviour (lever pressing) to increase. Food could also be described as a conditioned stimulus because it causes an effect to occur.

Note. There is an important difference between a reward and a reinforcer in operant conditioning.

A reward is something, which has value to the person giving the reward, but may not necessarily be of value to the person receiving the reward.

A reinforcer is something, which benefits the person receiving it, and so results in an increase of a certain type of behaviour.

Skinner identified three types of responses or operant that can follow behaviour.

Neutral operants: Responses from the environment that neither increase nor decrease the probability of a behaviour being repeated.

Reinforcers are any event that strengthens or increases the behaviour it follows. There are two kinds of reinforcers.

1. **Positive reinforcers** are favorable events or outcomes that are presented after the behaviour. In situations that reflect positive reinforcement, a response or behaviour is strengthened by the addition of something, such as praise or a direct reward.
2. **Negative reinforcers** involve the removal of an unfavorable events or outcomes after the display of a behaviour. In these situations, a response is strengthened by the removal of something considered unpleasant.

In both of these cases of reinforcement, the behaviour **increases**.

Punishment is the presentation of an adverse event or outcome that causes a decrease in the behaviour it follows. Punishment weakens behaviour. There are two kinds of punishment:

1. **Positive punishment**, sometimes referred to as punishment by application, involves the presentation of an unfavorable event or outcome in order to weaken the response it follows.
2. **Negative punishment**, also known as punishment by removal, occurs when an favorable event or outcome is removed after a behaviour occurs.

In both of these cases of punishment, the behaviour **decreases**.

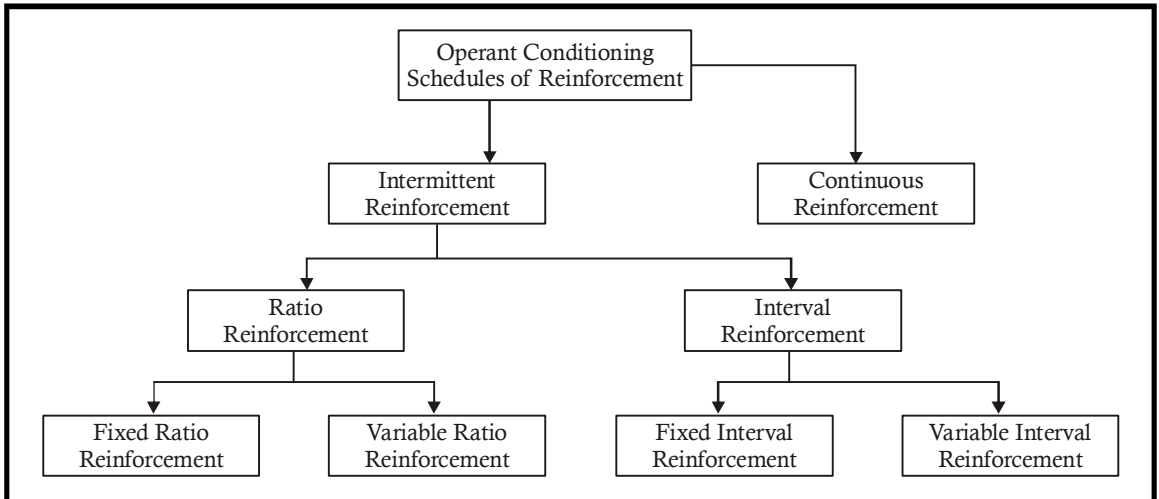
Schedules of Reinforcement

Intermittent reinforcement - reinforcement is given only part of the times the animal gives the desired response.

Continuous reinforcement - reinforcement is given every time the animal gives the desired response.

Ratio reinforcement - a pre-determined proportion of responses will be reinforced.

Fixed ratio reinforcement - reinforcement is given on a regular ratio, such as every fifth time the desired behaviour is produced.



Variable (random) fixed reinforcement- reinforcement is given for a predetermined proportion of responses, but randomly instead of on a fixed schedule.

Interval reinforcement- reinforcement is given after a predetermined period of time.

Fixed interval reinforcement - reinforcement is given on a regular schedule, such as every five minutes.

Variable interval reinforcement - reinforcement is given after random amounts of time have passed.

In animal studies, Skinner found that continuous reinforcement in the early stages of training seems to increase the rate of learning. Later, intermittent reinforcement keeps the response going longer and slows extinction.

Skinner specifically addressed the applications of behaviourism and operant conditioning to educational practice. He believed that the goal of education was to train learners in survival skills for self and society. The role of the teacher was to reinforce behaviours that contributed to survival skills, and extinguish behaviours that did not. Behaviourist views have shaped much of contemporary education in children and adult learning.

Implication of the Theory of Operant Conditioning

1. **Conditioning study behaviour.** Teaching is the arrangement of contingencies of reinforcement, which expedite learning. For effective teaching teacher should arranged effective contingencies of reinforcement. Example: For Self learning of a student teacher should reinforce student behaviour through variety of incentives such as prize, medal, smile, praise, affectionate patting on the back or by giving higher marks.
2. **Conditioning and classroom behaviour.** During learning process child acquire unpleasant experiences also. This unpleasantness becomes conditioned to the teacher, subject and the classroom and learner dislikes the subject and a teacher. Suitable behavioural contingencies, atmosphere of recognition, acceptance, affection and esteem helps child in approaching teacher and the subject. If student is not serious in study, teacher make use of negative reinforcement like showing

negligence, criticising student etc. but if student is serious in study, teacher make use of positive reinforcement like prize, medal, praise and smile.

3. **Managing Problem Behaviour:** Two types of behaviour is seen in the classroom viz. undesired behaviour and problematic behaviour. Operant conditioning is a behaviour therapy technique that shape students behaviour. For this teacher should admit positive contingencies like praise, encouragement etc. for learning. One should not admit negative contingencies. Example punishment (student will run away from the dull and dreary classes – escape stimulation).
4. **Dealing with anxieties through conditioning:** Through conditioning fear, anxieties, prejudices, attitudes, perceptual meaning develops. Examples of anxiety are signals on the road, siren blown during wartime, child receiving painful injection from a doctor. Anxiety is a generalised fear response. To break the habits of fear, a teacher should use desensitisation techniques. Initially teacher should provide very weak form of conditioned stimulus. Gradually the strength of stimulus should be increased.
5. **Conditioning group behaviour.** Conditioning makes entire group learn and complete change in behaviour is seen due to reinforcement. It breaks undesired and unsocial behaviour too. **Example.** Putting questions or telling lie to teachers will make teachers annoyed in such circumstances students learn to keep mum in the class. Asking questions, active participation in class discussion will make the teacher feel happy – interaction will increase and teaching learning process becomes more effective.
6. **Conditioning and Cognitive Processes.** Reinforcement is given in different form, for the progress of knowledge and in the feedback form. When response is correct, positive reinforcement is given. Example: A student who stands first in the class in the month of January is rewarded in the month of December. To overcome this Programme instruction is used. In this subject matter is broken down into steps. Organising in logical sequence helps in learning. Each step is build upon the preceding step. Progress is seen in the process of learning. Immediate reinforcement is given at each step.
7. **Shaping Complex Behaviour.** Complex behaviour exists in form of a chain of small behaviour. Control is required for such kind of behaviour. This extended form of learning is shaping technique. Smallest Behaviour is controlled at initial stage. On behalf of different contingencies, next order of chain of behaviours is controlled. Example: Vocabulary in English. Teaching spelling is mainly a process of shaping complex form of behaviour.

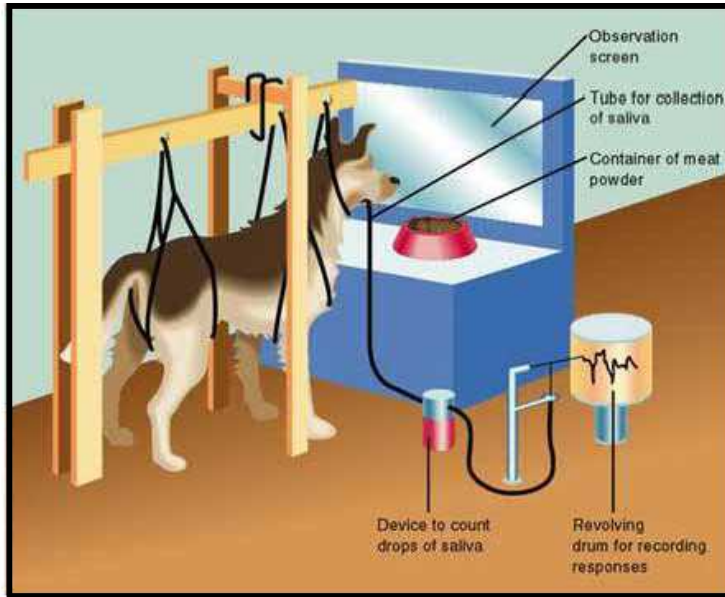
S-R (Stimulus-Response) Theory without Reinforcement

Pavlov – Classical Conditioning (1849-1936)

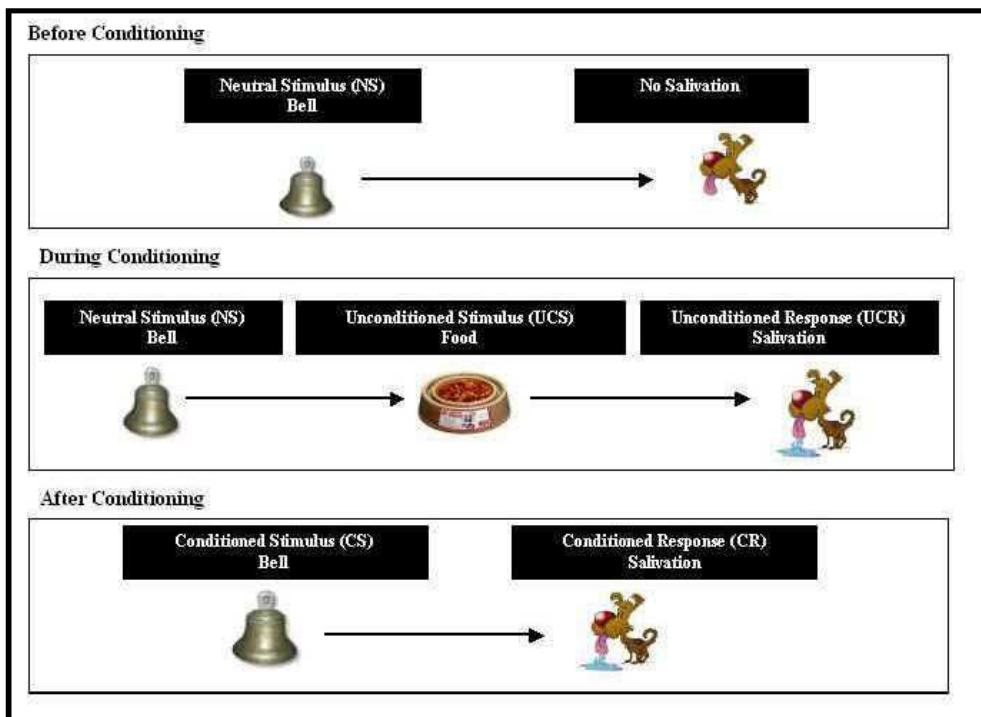
Classical conditioning is a term used to describe learning which has been acquired through experience. One of the best-known examples of classical conditioning can be found with the Russian psychologist Ivan Pavlov and his experiments on dogs.

In these experiments, Pavlov trained his dogs to salivate when they heard a bell ring. In order to do this he first showed them food, the sight of which caused them to salivate.

Later Pavlov would ring a bell every time he would bring the food out, until eventually, he could get the dogs to salivate just by ringing the bell and without giving the dogs any food.



In this simple but ingenious experiment, Pavlov showed how a reflex (salivation, a natural bodily response) could become conditioned (modified) to an external stimulus (the bell) thereby creating a conditioned reflex/response.



Components Involved in Classical Conditioning

We can gain a better understanding of classical conditioning by looking at the various components involved in his experiment;

- The unconditioned stimulus (UCS)
- The conditioned stimulus (CS)
- The unconditioned reflex/response (UCR)
- The conditioned reflex/response (CR)

So let's look at each of these classical conditioning components in more detail now.

Note. In its strictest definition classical conditioning is described as a previously neutral stimulus which causes a reflex (stimulus means something which causes a physical response).

The Unconditioned Stimulus (food) – (UCS). An unconditioned stimulus is anything, which can evoke a response without prior learning or conditioning.

For example, when a dog eats some food it causes his mouth to salivate. Therefore the food is an unconditioned stimulus, because it causes a reflex response (salivation) automatically and without the dog having to learn how to salivate.

Unconditioned Stimulus. This causes an automatic reflex response.

Conditioned Stimulus (bell) – (CS). The conditioned stimulus is created by learning, and therefore does not create a response without prior conditioning.

For example, when Pavlov rang a bell and caused the dogs to salivate, this was a conditioned stimulus because the dogs learnt to associate the bell with food. If they had not learnt to associate the bell with food, they would not have salivated when the bell was rung.

Conditioned Stimulus. You need to learn first before it creates a response. It is an acquired power to change something.

Unconditioned Reflex/Response (salivation) – (UCR) An unconditioned reflex is anything that happens automatically without you having to think about it, such as your mouth salivating when you eat. Unconditioned Reflex – Reflex that happens automatically and you did not have to learn how to do it.

Conditioned Reflex (salivation in response to bell) – (CR). A conditioned reflex is a response which you have learnt to associate with something.

For example, the dogs salivated when Pavlov rang a bell, when previously (without conditioning) the bell would not cause the dogs to salivate.

Conditioned Reflex. A reflex that can be evoked in response to a conditioned stimulus (a previously neutral stimulus).

Basic Concepts in Classical Conditioning

There are several principles that are associated with classical conditioning, some of these are:

Extinction a conditioned response will disappear over time when the conditioned stimulus is no longer presented.

Spontaneous recovery sometimes there is the weak appearance of a previously extinguished response.

Stimulus generalisation this is when individuals respond in this same way to experience stimuli. For example, all fuzzy animals scaring a young child instead of just a fuzzy cat.

Stimulus discrimination organisms can learn to discriminate between various stimuli.

Higher order conditioning this is when a neutral stimulus can cause the conditioned response sense if it had been associated with the conditioned stimulus.

Types of Classical Conditioning

Forward conditioning. Learning is fastest in forward conditioning. During forward conditioning the onset of the conditioned stimulus (CS) precedes the onset of the unconditioned stimulus (US). Two common forms of forward conditioning are delay and trace conditioning.

Delay conditioning. In delay, conditioning the conditioned stimulus (CS) is presented and is overlapped by the presentation of the unconditioned stimulus (US).

Trace conditioning. During trace conditioning, the conditioned stimulus (CS) and US do not overlap. Instead, the conditioned stimulus (CS) is presented, a period is allowed to elapse during which no stimuli are presented, and then the unconditioned stimulus (US) is presented. The stimulus-free period is called the *trace interval*. It may also be called the *conditioning interval*.

Simultaneous conditioning. During simultaneous conditioning, the conditioned stimulus (CS) and unconditioned stimulus (US) are presented and terminated at the same time.

Backward conditioning. Backward conditioning occurs when a conditional stimulus (CS) immediately follows an unconditional stimulus (US). Unlike traditional conditioning models, in which the conditional stimulus (CS) precedes the unconditional stimulus (US), the conditional response (CR) tends to be inhibitory. This is because the conditional stimulus (CS) serves as a signal that the unconditional stimulus (US) has ended, rather than a reliable method of predicting the future occurrence of the unconditional stimulus (US).

Temporal conditioning. The unconditioned stimulus (US) is presented at regularly timed intervals, and CR acquisition is dependent upon correct timing of the interval between unconditioned stimulus (US) presentations. The background, or context, can serve as the conditioned stimulus (CS) in this example.

Unpaired conditioning. The conditioned stimulus (CS) and unconditioned stimulus (US) are not presented together. Usually they are presented as independent trials that are separated by a variable, or pseudo-random, interval. This procedure is used to study non-associative behavioural responses, such as sensitisation.

CS-alone extinction. The conditioned stimulus (CS) is presented in the absence of the unconditioned stimulus (US). This procedure is usually done after the conditional response (CR) has been acquired through “forward conditioning” training. Eventually, the conditional response (CR) frequency is reduced to pre-training levels. Essentially, the stimulus is presented until habituation occurs.

Implications of Pavlov’s Theory to Classroom Situations

1. The theory believed that one must be able to practice and master a task effectively before embarking on another one. This means that a student needs to be able to

respond to a particular stimulus (information) before he/she can be associated with a new one.

2. Teachers should know how to motivate their students to learn. They should be versatile with various strategies that can enhance effective participation of the students in the teaching learning activities.
3. Most of the emotional responses can be learned through classical conditioning. A negative or positive response comes through the stimulus being paired with. For example, providing the necessary school material for primary school pupils will develop good feelings about school and learning in them, while, punishment will discourage them from attending the school.

Transfer of Training

The ability of the individual to apply the previous experience on the new related experience is what we call transfer of learning. Except students are able to transfer prior skills and knowledge on new ones, the continuity of learning will be difficult. This unit will explain how old learning can be transferred to a new one. You will know what the classroom teacher needs to do in order to facilitate transfer of experiences among his/her students.

The essence of learning is that a previously learnt fact should be linked with a present experience. This is because human being must be dynamic and that the prior experience will make them to develop the new skills and knowledge. The influence the past experience has on the succeeding experience is called transfer of learning.

Cormier and Hagman, define transfer of learning as the application of skills and knowledge learned in one context being applied in another context.

Oladele, defines transfer of learning as the effect of prior learning on the present. Learning is meaningful when the past learning smoothens the progress of something else. For example, if a learned experienced refuses to aid the new learning, the goal of training has seized to be accomplished. In the school, the teacher teaches some subjects in order that the experience gained in those subjects could be transferred into another.

Charham affirms that human and animal learning is normally affected by the past experience, and that the various subjects are included in the school curriculum because of their utility and wide application to real life situations. For instance, the teacher who has taught his/her students some skills in Mathematics would believe that such skills be transferred to related subjects like Physics or Accounting. If the students fail to apply these skills in their subsequent learning, it means that the students have not been successful in transferring the learning.

The above example gives us clues into the different types of transfer of learning that we have. These are explained below:

- (a) **Positive Transfer.** This is a situation whereby a previously learnt fact or information aids in the understanding of a new task. Aside from aiding the learners in their subsequent learning, it also helps the learners to learn better and effectively the new task.
- (b) **Negative Transfer.** This is a type of learning in which prior experience imparts negatively on the new one. In this case, the understanding of past skills inhibits the mastering of new ones. For example, if a student wrongly connects information, it can lead to negative transfer.

- (c) **Zero Transfer.** This type of learning reveals no link between the previously learnt task and the recent one. The evidence of zero transfer is hardly seen, it reveals no clear positive or negative effect.

Theories of Transfer of Learning

- a. **Theory of Mental Faculties.** This theory was propounded by the Greek Philosophers, notable among them was Aristotle. The basic tenet of the theory is that human mind is sub-divided into different powers of faculties like memory, judgment, reasoning or thinking. It is therefore believed that each of these faculties is reinforced and developed by cast and continuous memorisation of poetry/poem and similar works. This theory believes that exercises and regular practice will strengthen the mental faculties. The theory therefore dismisses the concept of transfer of learning, to it a well trained and disciplined mind is the ingredient needed for understanding of new information.
- b. **Theory of Identical Elements.** The theory which was developed by the duo of Thorndike and Woodworth (American Psychologists) indicates that it is possible for an individual to transfer the prior skills and knowledge to recent ones because both experiences are identical (share things in common). This theory suggests that a successful or effective learning will happen if there are connections or interrelatedness between the old and the new experiences. For example, it is expected that a student who has learnt about anatomical parts of human being in a Biology lesson, should be able to do well when he/she is asked to name anatomical parts of a goat during Agriculture lesson.
- c. **Theory of Generalisation.** This theory was advocated by a Psychologist named Charles Judd. The assumption of the theory is that general principles aid transfer of learning better than segregated facts. This theory believes in Gestalt, an assertion which views learning from a whole or complete form rather than in isolated form. For example, the theory of generalisation indicates that a learnt experience should be useful in other day-to-day related activities.

Classroom Implications of Transfer of Learning

1. The teacher should know that transfer of learning will not take place when both the old and new are unrelated. Hence, the teacher should endeavor to teach his/her subject-matter in a more meaningful and detailed way rather than by rote.
2. The teacher should provide the opportunity for his/her students to practice a subject-matter being discussed along with him/her. When the learners are allowed to take active part in teaching learning activities, they will be able to repeat the task at another time.
3. For a transfer of learning to take place, the teacher should always emphasise the relationship that exists between one subject matter and another.
4. The teacher should endeavor to develop positive attitudes towards a learning task so that the students can be motivated to like the task rather avoiding it.
5. It is believed that what students see, touch, feel or manipulate will be better remembered than the one they are not familiar with. Hence, for a meaningful transfer of learning to take place, the teacher should incorporate exercises that task the various senses of learners in the learning process.

INSTRUMENTAL LEARNING

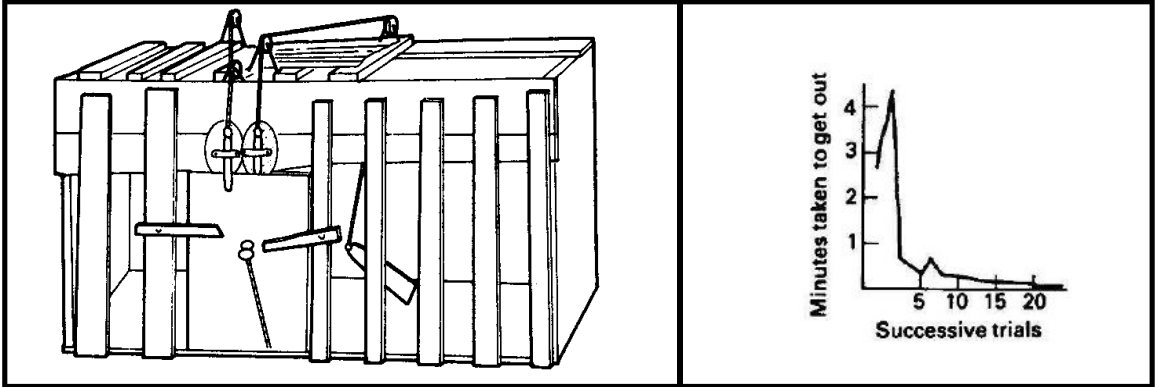
Although any mechanism which reduces responsiveness to a repetitious and insignificant series of events may have biologically useful functions, if only in energy-saving, and any mechanism which produces anticipatory reactions to the precursors of pain and pleasure has rather more obvious utilitarian possibilities, in neither of these cases is any account taken of on the-spot cost-benefit analyses, which might lead to either active seeking of goals or more passive abstention from possible courses of action, according to evaluations of the payoffs, positive or negative, thereby obtainable. The basics of habituation and classical conditioning do not therefore entail a truly purposive form of goal-oriented behaviour, even though Pavlov suggested that the arousal of food-seeking instincts by remote signs of the presence of food was part of the utility of the Pavlovian conditioning process. Thus, after considering theories which confine themselves to the effects of repeated stimuli, or of repeated stimulus pairings, we are entitled to ask whether a new theory is needed to encompass the learning of motor activities, especially those which are associated with wish and appetite. Various detailed ways in which the phenomena of classical conditioning may be contrasted with those of active and purposive, or, more conventionally, 'instrumental' learning, but clearly the question of motivation, in the form of drives, goals and incentives, is an initial reason for wanting to add something to theories of classical conditioning pure and simple.

A second obvious reason why one would want to add something to theories of learning concerned only with habituation and classical conditioning is that they say almost nothing about the learning of new response strategies or skills. Typically, in habituation an instinctive response disappears, and in classical conditioning an already present instinctive response is seen on new occasions, and thus there is no direct allowance for large increases in the repertoire of available behaviours. It is possible, though farfetched, to propose that anticipatory changes due to classical conditioning underlie new response patterns and skills. In a sense, all any of us can ever do is contract the muscles we were born with in different sequences and degrees of intensity, but the setting up of new sequences, whether in the human skills of carpentry and carpet-weaving or in the more limited performances of circus and laboratory animals, surely requires principles of response organisation which could never be made obvious from the study of salivation or withdrawal reflexes.

Thorndike's Stimulus-response Connections and the Law of Effect

A very ancient and very general principle of behavioural organisation is that of reward and punishment responses are directed so that the unpleasant is avoided and the pleasant pursued. However, there are many and various possible internal mechanisms which could produce behaviour which corresponds to this vague descriptive generalisation. Thorndike provided, at the same time as Pavlov's discoveries in stimulus-pairing experiments, both a more specific descriptive law and a theory of the mechanism of response selection, both of which were extremely influential. At the descriptive level, filtering out as much as possible of Thorndike's hypothetical mechanisms, his 'Law of Effect' says that individual responses are initially made at random, but are selected according to their effects, that is according to their consequences for the responding

animal. This is 'the method of trial and error with accumulated success', which has something in common with Darwinian natural selection, since particular behaviours which turn out to be advantageous to the animal are retained from a range of unorganised variation.



THORNDIKE'S PROBLEM BOX.

THORNDIKE'S RESULTS.

In the experiments which Thorndike performed, the change from variable reactions in the 'trial and error' phase to a single advantageous response was easily observed, and in some cases very gradual. The main experiments were performed by putting hungry cats inside small crates or 'problem boxes' from which they could escape: there were 15 boxes, each with a different mechanical system – a loop of string to be pulled, or a lever to be pressed, for instance – which allowed a cat to open a door if an appropriate response was made. A piece of fish was available outside the boxes, which a cat could eat if it got out, but the initial reactions of most of the cats he tested appeared to Thorndike to be directed at getting out of the box rather than reaching the food. In all the boxes these initial reactions were very similar in 11 of the 13 cats Thorndike tested (the other two being more sluggish). 'When put into the box the cat would show evident signs of discomfort and of an impulse to escape from confinement. It tries to squeeze through any opening; it claws and bites at the bars or wire; it thrusts its paws through any opening and claws at everything it reaches; it continues its efforts when it strikes anything loose and shaky; it may claw at things within the box'.

These various behaviours were counted by Thorndike as relatively undirected trial and error. But, in the course of these 'impulsive struggles' it was likely that the cat would, eventually and by accident, make the response which opened the door to the box it was confined in. For example, in the simplest box, Box A, there was a large wire loop hung 6 inches above the floor in the centre, in front of the door, which opened the box if pulled, and on the first occasion on which the cats were put in this, most of them had accidentally opened the door within three minutes.

But of course the important experimental result is that all the animals learned to do better than this, and escaped in less than a minute on the third and subsequent tests, on average, and in less than 10 seconds by the twentieth trial. This is the experimental demonstration of the Law of Effect – the effects or consequences of a response change the probability that it will be performed in the future. Learning curves such as that in above

figure (Thorndike's results) assess this in terms of the time taken to perform the response, or its latency, but the topography of the movements involved change as well, so that a well-trained cat, instead of struggling wildly, will, as soon as it is confined 'immediately claw the button or loop in a definite way'.

However, it was the theory which Thorndike used to explain the result, as much as the result itself, which attracted comment. Applying the rule of parsimony in explanation advocated in Lloyd Morgan's Canon, Thorndike attacked Lloyd Morgan's own view that animal learning of this kind was an example of the association of ideas, that is, that the cats should be assumed to have associated the response of pulling a loop of wire and the outcome of getting out of the box and eating fish. As an alternative,

Thorndike proposed that learned associations took a much simpler form, being always between the sense impression of an external stimulus situation and a direct impulse to act. The consequences of actions, in terms of perceived pleasure or pain, or in terms of 'satisfying' or 'annoying' states of affairs, were important in Thorndike's theory only because pleasurable results 'stamped-in' or burned in the situation-response connection which led to them, while, at least in the early versions of his theory, annoying results of an action stamped such connections out again. 'The impulse is the *sine qua non* of the association', and thus animals must have the impulse to act, and perform an act, before any associations are formed, and animals must always learn by doing and by actual experience of actions followed by reward.

Thus Thorndike did not expect, nor did he find, learning taking place by one animal imitating what another animal did, even in his experiments on cebus monkeys. For monkeys and other primates, and even eventually for people, Thorndike proposed that situation-response, or stimulus-response, connections were the basis of all learning. This proposal clearly bears some resemblance to Pavlov's idea of the formation of temporary connections in the nervous system, but the crucial addition made by Thorndike was immediate effect of reward and punishment in stamping-in or stamping-out connections between quite arbitrary preceding responses and their eliciting stimuli.

As we shall see, the basic results of Thorndike's experiments with cats, even though he replicated them with dogs, chicks and monkeys, are not sufficient to establish his theory that all the effects of reward and punishment take place via the formation of stimulus-response connections. It is therefore worth noting briefly some further details reported by Thorndike which, he admitted, did not seem consistent with the most extreme position which he considered – the possibility 'that animals may have *no images or memories at all, no ideas to associate*'. One such detail is the fact that cats were very slow to learn to lick or scratch themselves if these responses were rewarded by the opening of the door (by Thorndike). This would not be expected if the only principle at work was a mechanical tendency to repeated any rewarded act, but various additional explanations can be offered, such as 'an absence of preparation in the nervous system for connections between these particular acts and sense-impressions'.

The more elaborate alternative is to suppose that the idea of clawing a loop to get out is more readily formed than the idea of licking oneself to get out. Indeed, a general feature of Thorndike's discussion of his results is an appeal to the process of attention. A particular response such as pushing a lever was more likely to be learned if the animal 'pays attention to what it is doing'. This factor is also appealed to explain why it is that indi-

vidual cats learned more and more quickly as they had experience of more and more difficult problem boxes. Part of Thorndike's explanation for this was that 'the cat's general tendency to claw at loose objects within the box is strengthened, and its tendency to squeeze through holes and bite bars is weakened', but in addition he noted that 'its tendency to pay attention to what it is doing gets strengthened, and this is something which may properly be called a change in degree of intelligence'.

Finally, in ignorance of Pavlov's work, Thorndike performed a stimulus-pairing experiment, which led him to the view that his cats formed representations of sense- impressions. A hungry cat was kept in a large enclosure with a side of wire netting. Outside this Thorndike practised a routine of clapping his hands (and saying out loud, 'I must feed those cats'), ten seconds before going over and holding a piece of fish through the netting, 3 feet up from the floor. It will surprise few readers to learn that, after this had taken place 30 times, the cat climbed 3 feet up the wire netting immediately Thorndike clapped his hands, without waiting for the actual sight of the fish. Thorndike tentatively took the view that this was because the cat had some anticipatory idea or representation of the fish being presented in this case, without, however, acknowledging any anticipation of getting out and eating fish as a necessary factor in the string-pulling and lever-pressing experiments.

Hull's Stimulus-response Principles and Reward by Drive Reduction

It can be seen from the example above that Thorndike was less than completely rigorous in his application of the principle that learning takes place when motivationally significant events stamp in connections between sense- impressions and arbitrary impulses to action. His account of learning relies to a large extent on subjective factors, even though he minimised the importance of inference and reasoning in animal experiments, since the learned connections were supposedly between the internal perception of external stimuli, which could be significantly modified by attentional processes, and a subjective compulsion to perform a given response. Equally obvious is the fact that the 'satisfaction' or 'pleasure', which is responsible for burning in preceding stimulus response connections, is a hypothetical mental state of the subject, albeit one that can be inferred on the basis of behavioural evidence. Hull attempted to provide a far more rigorous and systematic theory of learning based on the principle of automatic stimulus-response connections, which had a far-reaching influence on many areas of both academic and applied psychology, but which stands today mainly as a monument to the limitations of this approach.

Hull attempted to strip away from Thorndike's Law of Effect all the more mentalistic or psychological extra processes involved in the perception of the environment, the organisation of response output, and the emotional assessment of reward and punishment. Learned connections were no longer between sense-impressions of what the animal was attending to and its impulse to perform a particular act, but directly between the neural impulses from activated sensory receptors and certain muscular reactions – the term Hull used was 'receptor-effector connections'. The complexities of attention and perception were thus side-stepped. Although this approach was initially plausible in terms of

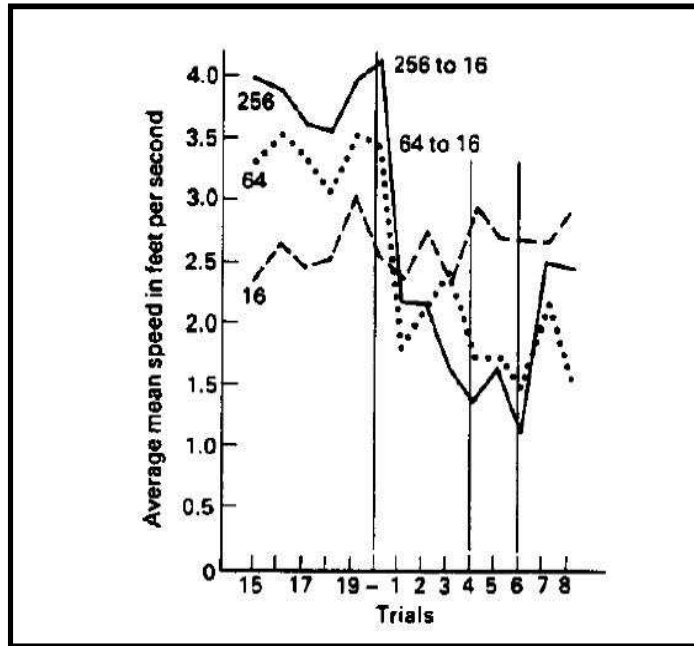
biological function, Hull also attempted what turned out to be an even more dangerous simplification by tying all motivation to states of physiological need, avoiding any reference to subjective states of pleasure or satisfaction. Animals have a biological need for food; lack of food was thus supposed to produce internal physiological stimuli described as the 'drive' of hunger. This would serve two purposes: the drive would in the first place be expected to activate and energise behaviours, but it also had an essential place in Hull's theory of reward, since eating food was supposed to affect the animal only in so far as it *reduced* the drive of hunger or the physiological deficit in nutrition.

Hull adopted Pavlov's (and Skinner's) term of 'reinforce-merit', for the event which strengthened stimulus-response connections, and thus Thorndike's Law of Effect became the *Law of primary reinforcement*, stated as follows:

Whenever an effector activity occurs in temporal contiguity with the afferent impulse, or the perseverative trace of such an impulse, resulting from the impact of stimulus energy upon a receptor, and this conjunction is closely associated in time with the diminution in the receptor discharge characteristic of a need, there will result an increment to the tendency for that stimulus on subsequent occasions to evoke that reaction.

It was clear even to Hull that there might be a digestive delay before eating food actually changed physiological need states, and therefore he proposed that, normally, eating food reinforced preceding stimulus response connections not by the 'primary reinforcement' of reducing the state of need, but by 'secondary reinforcement' caused because the stimuli involved in eating will have been 'closely and consistently associated' with eventual need satisfaction. There is no doubt that particular experiences of eating do indeed acquire motivational significance partly because of subsequent metabolic changes, and the appeal to drive reduction has the advantage of being directly applicable to the learning of responses which allow escape from external

imposed aversive states. However, the notion that the reduction of needs or drives is in any sense a *necessary* condition for learning has been almost universally abandoned. The most important reason for this is that the learning which occurs in the course of habituation, more active exploration of the environment and in several forms of Pavlovian conditioning, all lack the support of any physiological imbalance as obvious as those involved in hunger or lack of food. Further, learning under the influence of general social motivation, or that rewarded by specific sexual stimuli appears to be governed more by the properties of external events than by hypothetical internal needs. But, even in the prototypical case of the reduction of hunger by eating, motivational effects are considerably more complex than originally suggested by Hull. The tendency of human beings to eat things which are not medically good for them itself suggests that an abstract conception of biological need is not an adequate guide to motivational significance. A general alternative to drive reduction, indicated for instance by the apparent equivalence of low-calorie and high-calorie sweeteners as reinforcers for rats, is that the stimulus properties of food substance when ingested, as well as their post ingestional consequences, may have built-in rewarding effects.



THE EMOTIONAL EFFECTS OF RAPID INCENTIVE LEARNING.

Results of an experiment in which the amount of food obtained as reward by rats for running to the end of a straight alley was varied, using units of a 50th of a gram. Rats trained with larger rewards ran faster than rats trained with a small reward, but when all then received that small reward, a rapid reduction in speed of running took place in those rats which had formerly been given large rewards, so that they ran more slowly than rats already accustomed to receiving small rewards.

However, the experimental results which led Hull to modify his theory (though not to abandon drive reduction altogether) suggested that even the physical properties of reinforcing agents themselves do not adequately predict their behavioural effects, but rather, an internal evaluation of a given reward determines its effects on the learned performance of hungry rats. Crespi compared the performance of rats trained to run down an alley for food rewards of just less than a third of a gram with that of groups trained first with larger or smaller rewards, and then shifted to this reward amount. Above figure illustrates the case of rats which first experienced 5-gram rewards, and were then given a fraction of that amount for the same response, showing that their speed of running dropped sharply, and to a level well below that of animals which had received the smaller reward consistently. Crespi's own interpretation of this was that the reduction in reward size engendered an emotional depression in the animals. Hull retained the notion of a temporary emotional effect caused by reward reduction to account for the drop in response speed below the normal level, but also added a special process of 'incentive learning', detached from both drive reduction and the formation of stimulus-response connections, which he used to explain both the rapid change in performance in this experiment, and the immediate improvement in accuracy of maze-running when rewards are introduced following a period of unrewarded exploration. There are a number of other results from standard laboratory experiments, under the heading of 'contrast effects' or 'post-conditioning changes in the value of the instrumental reinforcer' which seem to indi-

cate that a reassessment of the emotional value to be put on a given reward can take place in separation from the response-learning process, and some of these will be discussed in a later section. It is usual now to conclude that 'instrumental learning cannot be reduced to an S-R association', and since the drive reduction factor in the reward process tends also to be discounted, Hull's reformulation of the Law of Effect has lost both its mainstays.

Skinnerian Pragmatics

The best-known modern protagonist of the Law of Effect is B. F. Skinner, who has gradually abandoned almost all theoretical speculation as to the role of drive, stimulus-response connections, or any other internal mechanism or process which might explain how or why learning takes place, in favour of empirical demonstration of the practical power of reward procedures in the animal laboratory, and rhetorical assertions about the ubiquity of strictly analogous effects in human life. I shall for the present confine myself to a discussion of the former, and ignore the latter

Thorndike's technique for studying learning in cats and dogs required him to replace the animal inside a box each time it made the successful response of getting out. In his experiments on cebus monkeys however, Thorndike found it expedient to change this procedure so that the animal remained inside an experimental enclosure, a piece of banana being dropped to it down a chute whenever it depressed a lever, Skinner used a similar set-up for laboratory rats, with automated devices for recording lever presses and delivering food pellets, and later developed a version which was used in very extensive experiments on pigeons thus inventing what is now a very widely used piece of apparatus in animal laboratories, known as the Skinner box. The isolation of the animal subject, and the automatising of the experimental procedures, may be partly responsible for the fact that a wide variety of extremely reliable (and replicable) behavioural results can be obtained by this method, and it is of course extensively used today by researchers with no commitment to Skinner's theoretical views. In fact, Skinnerian apparatus and behavioural techniques are used in so many theoretical contexts that it is quite inappropriate to gather together all the obtainable results here, but the basic phenomena associated with 'schedules of reinforcement' can usefully be reviewed at this point.

Schedules and Contingencies of Reinforcement

There is a good deal in common between certain kinds of Skinnerian reinforcement theory and general principles of economics. It is assumed in economics that certain global aspects of human business – savings, investment, employment and so on – may be ultimately determined by certain quantifiable variables – money supply or interest rates for instance, without too much detailed attention being given to the actual psychology of individual decision-making or the gradual change or growth of business enterprises. The success (or failure) of such assumptions rests partly on the degree to which people or companies do indeed act predictably in always acting to maximise their own economic advantage. In the same way it is a central assumption of reinforcement theory that animals will learn anything and everything which maximises their receipt of food rewards, with many subtle and global aspects of learned behaviour thus being ultimately determined by quantifiable variables concerned with the conditions of availability of food rewards (or 'contingencies of reinforcement'). The end relationship between the scheduling of food rewards and the exact patterns of behaviour which are eventually determined

by various conditions of reward can be and often is discussed without too much attention being given to gradual learned changes in behaviour (and the mechanisms which might be responsible for these). However, there can only be fixed relationships between contingencies of reinforcement and patterns of behaviour engendered by them if animals act predictably in accordance with the Law of Effect, and at the descriptive level always select behaviour which has beneficial consequences. This is not always the case – individual species are influenced by particular reinforcers in ways which reflect instinctive processes that are relatively immune from experienced payoffs but with the now standard laboratory procedures first used by Skinner, the predictability of relationships between contingencies of reinforcement and behaviours learned thereby is surprisingly high.

Continuous and Ratio Reinforcement

The simplest possible Skinner-box experiment, analogous to the Pavlovian case where a buzzer always signals food, is performed when an animal is able to deliver to itself a small amount of food each time it performs a single response such as pressing a lever or pushing a button. Left to itself, an individual of any common laboratory species will eventually discover this by trial and error, if the response is such as is made occasionally by accident; but it is usual to give prior adaptation to the working of the food delivery mechanism, and there are other methods of training, such as autoshaping, or ‘shaping’ the response by carefully watching the animal and delivering reward for successive approximations to the required behaviour (*e.g.* delivering rewards initially when a rat sniffs at a lever, then only when it touches it, then only when it presses the lever down). Once such a response has been learned, then if the animal is already hungry when it is placed in its experimental box, it will press the lever at regular intervals, gradually slowing down if it is allowed to become satiated. After such training, an additional learning process can be observed by disconnecting the food mechanism, since a hungry animal will at first continue vainly to make the response but will gradually cease to do so – the process of response extinction. Little can be concluded from this simple confirmation of trial and error learning, but the nature of the procedure allows for many extensions of the basic task. For instance, the animal may be required to make a fixed number of responses for each reward – a fixed-ratio schedule. This involves gradual learning, especially if the fixed number is large. Clearly the animal is not able to obtain rewards as frequently if it has to make 100 responses before each one, but additional standard results are that there is a pause in responding after each reward, roughly proportional to the size of the ratio, and that a run of responses is then made at a fast and gradually increasing rate. Obviously, the higher the ratio the higher in some sense is the cost in time and effort of obtaining rewards, and this is reflected in the variation both in vigour of performance and the size of the highest ratios that animals will continue to perform, which are systematically related to motivational factors such as degree of prior deprivation and the size and palatability of the rewards used. Ecologically speaking, the relationships between effort and food reward would be expected to be vastly different between species, depending on niche. For instance, herbivorous browsers, need to feed for long periods but carnivores (especially cold-blooded ones), eat relatively infrequently and difficult to obtain but richer meals. However, it appears that rats, being omnivorous, are extremely flexible in this respect, when they are left permanently in a Skinner box with various ratios of lever-pressing necessary to obtain food rewards. If allowed one small pellet for every press, then about once every hour they would go and press 20 times in succession, eating 20 pellets in less

than 10 minutes. When required to make 160 presses for each pellet, they were able to obtain almost as much food as before, by spending 14 hours a day making the necessary 70,000 responses. Even more flexibility was shown if the unusual procedure was used of rewarding rats by allowing them to enter a tunnel and eat for as long as they liked – but with the constraint that if they left the tunnel for more than 10 minutes then a fixed ratio of lever presses was necessary to make food available again. In this case rats required to make only one response to produce free access to food ate 10 separate meals a day, but if the ratio was increased they chose, reasonably enough, to eat fewer but longer meals, eating only five times a day when 40 responses were necessary to gain access to food, and eating only one large meal per day when just over 5,000 lever presses were required to gain access to food. In this experiment water was always available, but exactly analogous results were obtained when fixed ratios of responses were necessary to obtain water, with food always available, except that with indefinitely long access to water as the reward, a ratio of only 300 bar presses was required to persuade the rats to take just one (long) drink of water per day. This strongly suggests that frequent access to food is rather more rewarding for individual rats than frequent access to water. Lore and Flannelly deduced that access to water is ecologically important from their naturalistic study of the location of the burrows of wild rats on a large landfill, since these tended to be located within 50 metres of a stream but only within 100 metres of the food source, (a mound of refuse). Clearly many factors, notably in this case an obvious preference for sloping ground, may determine burrow location.

The general descriptive conclusion from experiments on fixed ratio schedules is simply that the size of the ratio significantly affects the pattern of responding for a given reward.

The theoretical implications of this are not always clear, but several experiments suggest that laboratory animals are capable of learning some representation of the size of the fixed ratio. That this is not strictly necessary for performance on ratio schedules is shown by the possibility of variable ratio schedules, in which a probabilistic device such as random number generator ensures that the exact point at which reward is given is always unpredictable, even though the average value of the ratio can be specified. Only a limited amount of evidence about the effects of this is available, but some rather complicated comparisons suggest that a variable ratio schedule produces somewhat faster responding than a fixed ratio of the same average value, with shorter pauses after reward is received, due to the effects of the occasional reinforcement of a single response or a very short run of responses. Given the choice, pigeons appear to prefer to work on the uncertain variable ratio than on a fixed ratio of equivalent average value.

Interval Schedules of Reinforcement

It also appears to be the case that variable ratio scheduling of reinforcements produces a higher rate of response than the same number of reinforcements (that is the equivalent frequency over time, and thus the same rate of gain of food) which are made available according to a different rule, based on the minimum interval of time between one reward and the next. For a variable interval schedule of reinforcement, a certain period of time, random about some mean, must elapse after one reward has been given before a *single* response produces the next. Since only one response per reward is necessary, it is not surprising that fewer are made than when multiple responses are mandatory on variable ratio schedules. Variable interval schedules are very widely used because they produce a

very consistent and steady rate of responding, which accelerates only slightly as time without a reward passes. For instance, if hungry pigeons can obtain reward for a single key peck on a variable interval of 1 minute, they typically begin pecking the key again within a few seconds of receiving food. Very occasionally they are rewarded at this point, but at other times they may continue to peck regularly, without anything else happening, for 4 or 5 minutes, until the next reward finally becomes available.

The theoretical analysis of performance on interval schedules is easier in the alternative case of a fixed interval, where the minimum period between successive rewards is always the same. Here the typical performance in a rat or pigeon, depending on the length of the interval, is that there is a distinct pause after a reinforcement has been received (as with fixed ratios) followed by a gradual increase in the rate of responding, such that the highest rates of response occur just before the next

reinforcement becomes due. This increase in response rate with time, along with other measurements, has been used to deduce that laboratory animals make use of an 'internal clock', which can be accurate to within a few seconds, to govern responses. According to Roberts, this clock is a linear measure of time, which can be reset by food rewards, stopped at a given point by appropriate external signals, and used to time intervals of several different lengths concurrently. This is rather hypothetical, but it is clear that one of the factors that may be involved in performance on interval schedules is the engagement of some sort of representation of the passage of time. Whether the internal clock is associated with expectancies of reward, or serves rather as a stimulus for more mechanical habits of response, is not usually obvious.

Choice in Reinforcement Schedules

The reinforcement schedules described so far have all concerned the relationship between a single response and the availability of reward. A strikingly more complex state of affairs arises if an animal is confronted with two available responses, each assigned its own schedule of reinforcement. A degree of orderliness prevails, however, if both available responses are rewarded on variable interval schedules. Suppose a pigeon is allowed for one hour every day to obtain grain in a box where pecking the left key is rewarded on average every 3 minutes (VI3min) and pecking the

right key is rewarded on average every 6 minutes (VI6min). What is the optimal way for it behave? The answer, of course, depends on what one means by 'optimal'. To gain the most food with the least amount of effort, physical or intellectual, it would be sufficient for the bird to peck the keys alternately, about once every 10 seconds. In fact, with the schedule as described, and a not-very-hungry pigeon, this is exactly what happens – the bird pecks at roughly the same rate on both keys, even though it gets 20 rewards per hour from the left key but only 10 from the right. The nature of the variable interval schedules means that the distribution and rate of responding has very little effect on the distribution of rewards, provided that both responses are made occasionally. But, by increasing the psychological separation between the two responses, the effects of rewarding one twice as often as the other can soon be observed: the standard procedure is to ensure that a left response is not rewarded if it is made just after a right response, or vice versa.

Once this is done, data of impressive regularity may be obtained, since the procedure of giving twice as many rewards for left pecks as for right pecks then means that pigeons peck left twice as often as they peck right. The interpretation of this unsurprising result has, however, proved to be difficult, and theories of increasing mathematical complexity

are now proposed to account for it. For present purposes it is sufficient merely to stress that the regularity does not arise only because a response is strengthened in proportion to the exact number of times it is rewarded.

There is a descriptive generality, called the 'matching law', which can be given either as

$$\frac{B_1}{B_1 + B_2} = \frac{R_1}{R_1 + R_2} \quad \dots(1)$$

or as

$$\frac{B_1}{B_2} = \frac{R_1}{R_2} \quad \dots(2)$$

where B_1 and B_2 are the absolute frequencies of two behaviours, and R_1 and R_2 are the absolute numbers of rewards each receives per hour. There are two general kinds of explanation for why this equation satisfactorily predicts behaviour, and therefore predicts in some sense the choice that a pigeon makes between two responses. The first kind says that the matching law arises because in some way or other animals are sensitive to the ratio of reinforcements for two responses and adjust their response choices accordingly. The second supposes that the matching relationship arises because the animals are attempting to respond optimally and that the observed allocation of responses results from their 'maximising' their momentary chances of reward. Although these theories sound very different, as it happens close examination of what the two theories predict about experimental results suggests that they are 'empirically indistinguishable'.

Relative Reward Value

There is, however, an extremely unambiguous empirical distinction between two possible rules which would both produce the matching law in practice. One rule implies that any response is performed according to the reinforcements it itself receives; the other says that a response is performed according to the *relative* value of the reinforcements it receives – most directly according to the proportion of the total rewards that are assigned to it in circumstances of choice. Thus

$$B_1 = kR_1 \quad \dots(3)$$

is the first case and

$$B_1 = \frac{kR_1}{R_1 + R_2} \quad \dots(4)$$

is the second. Either equation (3) or equation (4) would produce equations (1) and (2). But, 'the data unequivocally support' equation (4): This is because there is a very simple experimental test: a pigeon is rewarded at a standard level – say 20 times per hour, for pecks on the left key; while over a period of days or weeks, the frequency of rewards for right key pecks is varied between zero and 40 per hour. Even though the left key pecks are being rewarded at the same level, the rate of left-key pecking varies systematically, roughly in line with equation (4), although the fit of the data is even better when the total frequency of rewards in the denominator is raised to a fractional power. As with the matching law itself, the reason why equation (4) works is more obscure than might be expected. We can rule out the argument that left-key pecks are made less often if right-key pecks are rewarded a lot, because the animal is then too busy making right-key pecks, since it can be arranged, by giving a visual signal for exactly when they will be

rewarded, that it gets many rewards for right-key pecks without having to spend much time on that behaviour.

The descriptive alternative is that the effectiveness of a certain level of reward on the left key declines when an alternative source of reward is available. It is likely that this sometimes involves perceptual or emotional contrast effects between two levels of incentive experienced by the same animal, but a simpler possibility is that the satisfying or even satiating effects of one reward alternative detract from overall drive in general, or from the incentive properties of all alternatives. Hull found it necessary to distinguish between the drive-reducing and the incentive effects of rewards, and (at least) these two factors probably have to be taken into account. The satiating effect of rewards seems obvious, for instance, in the experiment of Rachlin and Baum, in which pigeons received 20 rewards per hour (V13 min) and 4 seconds access to grain, for pecks on their right key, while the same number of signalled rewards on the left key varied in duration from 1 to 16 seconds. When this alternative was 16 seconds, the birds gained 10 grams more weight in the experimental hour than when it was 1 second, and this would be sufficient to account for the fact that the rate of response for the standard reward was slower in the first case.

However, a fairly general finding is that the duration or magnitude of rewards given in concurrent schedules has much less effect on behaviour, and produces much less close matching of behaviour to relative reward properties, than does the variable of frequency of reinforcement. In other words, animals are much more likely to notice that they get an extra small reward from an alternative source, than that they get a reward which is twice as big. Frequent access to small amounts of food is generally much preferred to infrequent access to large amounts, and thus it is the details of the distribution of rewards over time, rather than the gross total of consumption, that has the strongest psychological effect in the experiments.

Skilled Performance and the Form of a Response

In the use of Skinnerian techniques for detailed laboratory work on 'the experimental analysis of behaviour', the emphasis is very often on the rate at which a given response is performed, with little attention given to the initial learning which determined its qualitative form. But reward procedures have just as much of an effect on 'how' as they do on 'how often' a response is performed. Skinner described the training of rats to press a lever with a certain force, or to press it down for a certain length of time. As a general rule, he found that 'Rats tend to adjust to a force which secures only slightly above the reinforcement of every other response'. Thus, if a lever needs to be pressed with a force equivalent to a dead weight of 20 grams, even experienced animals will make up to 40 per cent of their attempts at pushing it down with insufficient effort. They are strong enough to do better than this, since if the criterion is shifted, and a mechanical adjustment made so that a press of 60 grams or more is needed, then the animals quickly learn to make harder presses, but still make a third or more partial presses of less than 60 grams in force. Skinner found that one animal which weighed less than 200 grams itself could sustain presses averaging over 100 grams each, but even at much lower force requirements there seemed to be a principle of least effort bringing responses down below the minimum force necessary.

Normally for food rewards rats press a lever for only a fraction of a second. However, they may be gradually trained to hold it pressed down for 20 or 30 seconds at a time, and

various more complex techniques can be used to assess the accuracy of such timing, when there is no external feedback to signal when the lever has been pressed long enough. A further possible variation in the topography of a rat's lever pressing is the amount of angular movement:

Herrick succeeding in training his animals to depress a lever through an angular displacement of at least 20.35 degrees, but not more than 25.50 degrees. Rewards were delivered only when the lever returned to its home position after excursions which met this criterion, and therefore the judgment of a correct lever press required some muscular finesse on the part of the rats (and some technical ingenuity in the design of the apparatus). However Herrick's rats satisfied Skinner's expectation that they should adjust their behaviour so that at least every other response gained reward.

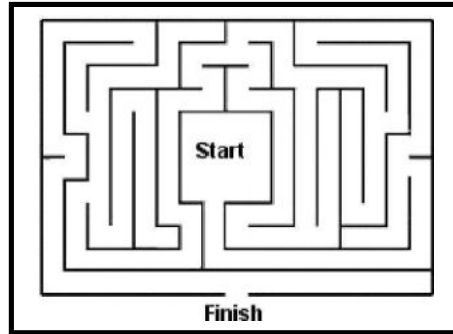
Although the acquisition of motor skills in animal learning is sometimes overlooked, it is clear that laboratory animals, starting with Thorndike's cats, are able to learn to operate with considerable accuracy mechanical apparatus that is quite different from anything members of the same species would ever encounter in a natural environment. Often natural and instinctive behaviour patterns are also elicited, but it is undeniable that new muscular patterns of some skill emerge under the influence of artificial training procedures.

Creativity in Response Selection

It is a Skinnerian dogma that changes in the form of a response always reflect the shaping effect of contingencies of reinforcement in the environment on a passive subject. This probably overstates the case even for laboratory rats and pigeons, but an explicit counter-example is available from systematic experiment on two rough-toothed porpoises, members of the only mammalian group, the whales, better endowed with brain tissue than the primates. Pryor *et al.* were initially attempting to demonstrate only the power of the Law of Effect in the shaping of new behaviours by reward. For five days a porpoise was, in the course of public performance at a sea life park in Hawaii, rewarded for a different particular behaviour each day, but the procedure had to be aborted because the animal began to perform a large number of response sequences which not only had not been specifically trained, but which had not previously been observed by the experimenters. These included various aerial flips and swimming with the tail out of the water. Pryor *et al.* assumed that 'novelty was an intrinsic factor' in the origin of the new responses, and that what had happened was that the procedure of giving rewards for something different each day had inadvertently rewarded novelty as a category of response. A second and more docile animal was therefore tested with a similar procedure with two observers and systematic recording of data.

For the first seven days only breaching, beaching and swimming upside down were observed and the animal tended to adopt a rigid pattern of these previously reinforced tricks. After a further seven sessions of training with particular new tricks, this second animal began to display the same varied behaviour as the first, emitting eight different kinds of behaviour in the sixteenth session, including a spin, a flip, an aerial spin, an upside-down tail slap, and a tail side-swipe, which were all seen for the first time. These were then rewarded one by one until, in five final sessions (28-33), the animal had to come up with a new response each time to obtain its fish rewards. On session 30, it performed 60 different patterns of movement, but all had been seen before, and thus it got no fish. On the next three sessions however, it managed to come up with a backwards

flip, an upside-down porpoise, and finally the response of standing on its tail and spitting water at the trainer. It may be stretching a point to refer to this as 'creativity', but clearly, the cumulative effects of the training procedure resulted in the animal learning something other than merely the mechanical repetition of stereotyped and unchanging patterns of response.



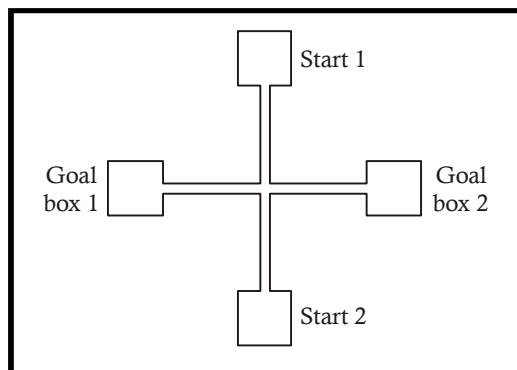
THE HAMPTON COURT MAZE FOR RATS.

Spatial learning and 'cognitive maps' The conclusion that instrumental conditioning typically consists of something more than the repetition of particular stamped-in stimulus response connections was of course reached earlier by Tolman on the basis of rather different evidence, namely, the behaviour of rats in mazes. Hardly anything about the behaviour of rats in mazes supports the existence of specific stimulus-response connections, since, in the first place, it is impossible to identify a specific stimulus, or a specific response. Small trained rats to follow the correct path to food in a modified form (suitably scaled down) of the maze built for human amusement at Hampton Court Palace. Yew hedges are replaced by high sided boards, which the animals cannot see over, and they must thus move in enclosed alleys. Clearly in this type of maze, detailed visual cues will be likely to be less important than they are in the other main type, the 'elevated maze', where the rats run on planks without sides fixed several feet above the ground. Both Small and later Watson who used the same maze with rats variously handicapped by blindness, deafness or lack of the sense of smell, speculated that the learning of the maze was accomplished as a motor pattern – by associations of 'motor images of turning', for Small, and 'a serially chained kinaesthetic reflex-arc system' for Watson.

An enormous amount of evidence suggests that associations between successive movements are rarely important in maze-learning. MacFarlane and Evans by flooding enclosed alley mazes with water, showed that rats which had learned a maze by swimming, running or wading could follow the connect path by an alternative form of locomotion, that is using very different kinaesthetic cues. More drastically, Lashley and Ball and Ingebritsen lesioned ascending proprioceptive nerve tracts in the spinal cord, producing in many cases aberrant posture or patterns of movement, but with relatively little effect on the rats' ability to learn or retain the correct pattern of turns in a maze. Since direct associations of series of movements does not thus seem important, and since also no individual sense modality appears crucial to maze-learning, the alternative theory is that maze performance has to be under the control of multiple stimuli, and feasible with multiple responses or, in what may be another way of saying the same thing, controlled by 'central' or 'symbolic' factors.

Tolman's contribution can be summarised as the designing of elegant behavioural experiments to demonstrate that rats use various cues to achieve a sense of geographic place, and as the providing of the long-lived term 'cognitive map' as part of his theory of 'purposive behaviourism'. The best examples of experiments used by Tolman to support his theory are thus those on the phenomenon of 'place learning'. The normal laboratory rat will give every appearance of expecting food at a particular familiar place in a maze it has learned: it will sniff and search for the food if it is absent, and attempt to burrow under or climb over obstacles placed in its normal path. Alternative routes to the same food may be used if several of equivalent length are available.

There are many anecdotal reports of rats finding short cuts. Tolman *et al.*, (1946) attempted a systematic study of this, by first training animals to take a right-angled dog's leg path to a goal box which was situated in a direction of 60 degrees from the starting point, and then providing them with 18 paths at 10-degree intervals. More than a third of Tolman's rats did indeed take the sixth path, going directly towards the site of the goal box, but another 17 per cent took the first path on the right, and other experimenters trying to replicate the result found rats apparently trying to retrace the dog's leg. However, a simpler experiment, which opposes turns in a particular direction against movement towards a particular place, produces easily replicable results. The procedure is to start rats from different sides of a cross-maze, and to compare animals always rewarded for turning in the same direction (and therefore rewarded in different places) with others always rewarded in the same place (and therefore rewarded after both left – and right-turns). Tolman *et al.* found that their animals were much better at always going to the same place, using an elevated maze with asymmetrical lighting and other obvious possible landmarks. It is not now doubted that this sort of place-learning comes relatively easily to rats, but, if a similar experiment is run in the dark, or under a homogeneous white dome, then, not surprisingly, it is easier for the animals to learn a consistent turning response (e.g. Blodgett and McCutchan).



PLACE VERSUS RESPONSE LEARNING.

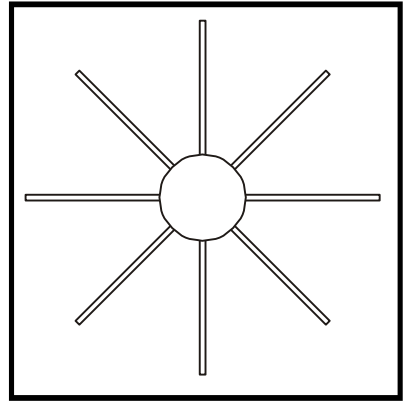
A maze in which rats may be trained to always make the same turn, to different places, or to turn either left or right, in order to get to the same place. See text.

Several more kinds of experiment have more recently confirmed the complicated nature of place learning. O'Keefe has identified not only particular external landmarks used by his rats (coatracks, etc., which when moved produced errors in a predictable direction), but also particular brain cells (in the hippocampus) which, among a large population, fire whenever the rat goes to a certain place in its apparatus. The prize for the most

Tolmanian experiment of the 1980s will probably go to R. G. M. Morris, since he combined the swimming which was used in Tolman's laboratory by MacFarlane with an appeal to remote landmarks. Rats were put in a large circular tank of water made opaque by the addition of milk (mud would have been more naturalistic). At a certain point in the tank was a cylinder with its top 1 centimetre below the surface – as soon as they found this rats would climb up on to it, and the test of learning was that, with experience, they would swim eventually to the invisible submerged platform from any point in the tank, thus demonstrating their degree of knowledge of its location.

If rats are placed on the central platform of a maze with a plan such as this, food rewards being available but out of sight at the end of each arm, experienced animals will visit each arm only once, thus demonstrating some form of memory for their activities in the recent past.

However, the most widely used spatial learning technique in modern animal laboratories is undoubtedly the radial maze, introduced by Olton and Samuelson. In this several arms (initially eight elevated planks of different widths) radiate out from a central point, where a hungry rat is placed to start with, and a single pellet of food is put, out of sight, at the end of some or all of these arms. If, as in the original experiment, all the arms are baited with food, then the rats' best strategy is to run down each arm only once, on a given daily test. After several days' training, rats on average choose about 7.5 novel arms in their first eight choices. Control experiments show that this statistically significant result is still obtained if odour cues are eliminated either by the crude method of drenching the apparatus with Old Spice after shave, or by the more revealing technique of confining the rat in the central area between choices, and surreptitiously exchanging an arm which had already been run down with another, or by rotating all the arms to different positions. Thus on any given day, the rats' cognitive map and associated memory is sufficiently detailed to record which seven places the animal has already been to; and even with a 17—arm maze, an average of 14 different arms were chosen in the first 17 choices. This is a matter of recent memory, since the procedure works without the animals having to visit arms in any fixed order, and therefore it works when the memory is of exactly what has happened on a particular day. A more traditional form of spatial learning is obtained if the radial maze is used with only the same one or two (or four) arms baited each day, so that the animal has to remember that food is always in some places but not in others.



THE RADIAL MAZE.

Multiple Levels of Representation in Instrumental Conditioning

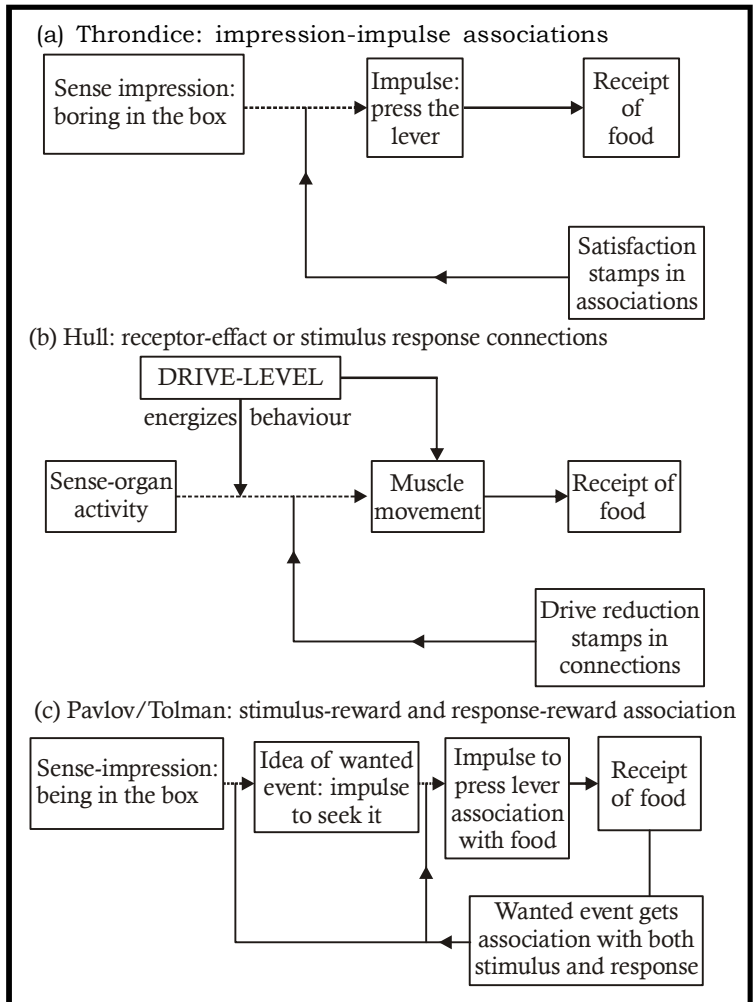
Tradition in the theoretical analysis of instrumental conditioning requires that we ask the questions of 'what is learned?' and 'what is the necessary and sufficient condition for learning?'. Modern research suggests the answers that there is no single kind of thing invariably learned in all cases of instrumental (or operant) conditioning; that there is no obvious essential ingredient; and that there are several limiting cases which are just sufficient to demonstrate one of the many phenomena which instrumental conditioning involves. However, in general the answers previously given to the questions of what is learned, and in what way, are not irrelevant to modern concerns, since it may be argued

that almost every proposed theoretical form of association should be retained as one of several possibilities, and almost every supposedly critical condition is a variable which may affect the course of learning, even if it is not critical.

Although the views of earlier theorists such as Thorndike and Hull should thus not be dismissed out of hand, it is undeniable that there has been a general shift of opinion away from the theory of learning-as-doing towards more cognitive theories of learning-as-knowing, even for laboratory animals. Part of this distinction corresponds in fact to a difference between two kinds of knowing, 'knowing how' and 'knowing that', which Dickinson has imported into learning theory from cognitive science as the contrast between 'procedural' and 'declarative' representations.

It is important that neither kind of representation necessarily has to be in verbal form; but the distinction can be illustrated by the example that an instruction in the form 'pull the gear lever back when the engine makes a loud high-pitched noise' is procedural; while instructions which are more explanatory, such as 'moving the lever changes the gear'; 'higher gears are used when the car goes faster', or 'the positions of the gears are as follows', are initially only declarative, but are well suited to the deriving of a wide range of procedural instructions, by processes of inference and integration. In human practice, the distinction between procedural and declarative representations usually maps onto the difference between automatic habits and more considered and controlled reflections. This is the case in most people's experience of learning to change gear, along with other aspects of learning to drive.

First attempts, and the early stages of learning, require a surprising amount of concentration and effort for activities which later on become second nature. When this has happened, the declarative aspects of early learning may become irrelevant, with eventual skills including some accomplished but fairly isolated habits. I hope I am not the only driver to have had the experience of hiring a



THEORIES OF INSTRUMENTAL LEARNING.

car on the Continent, and on the first occasions of reaching a suitable speed in first gear, putting in the clutch and making sweeping movements with my left hand, where the gear lever normally is (and, even more embarrassing, doing the same thing the opposite way around when first driving my own car on returning home). There is a wealth of anecdotal evidence, first emphasised by James that well-learned human skills produce automatic habits, which are independent of rational purpose; Reason and Mycielska have examined the degree to which this contributes to 'action slips' of mixing up habits.

In the context of learning theory, the contrast is between automatic habits and planned actions, but also between the level of representation, whether procedural or declarative. By and large, more habitual processes apply to muscle or limb movements, while purposive actions are coded in terms of goals, or sub-goals, but it is possible to speak of mental habits, and emotional reflexes, and relatively goal-directed muscle-twitches. As previously discussed, Thorndike's answer to the question of what is learned was very procedural, since it corresponded to the instruction 'when in the box, press the lever', but it was not at the lowest level of representation of the stimuli and responses involved, since it concerned learned connections between the sense-impression of being in the box, and an impulse to make a response, directed at a controlling feature of the environment, such as a lever or a loop of wire. By comparison, Hull's refinement of Thorndike's Law of Effect was equally procedural, but at a lower level of representation, since the learned connections were proposed to be directly between stimulus receptors and muscle movements. Schematic versions of both these theories are given in given Figure.

Opinion has decisively swung against them, and there is now much support for the Tolmanian alternative answer to the question of what is learned in instrumental conditioning, sketched at the bottom of Figure. It should be noted that the Tolmanian version differs from the others in at least two respects. First, it is at a higher level of stimulus and response representation than Hullian theory; it assumes internal ideas or representations of wanted or expected rewards, and coherent impulses for actions rather than direct conditioning of muscle movements. Second, it does not include a direct link between environmental input and response output at any level of representation: in Tolman's terms, what is learned is a 'means- end readiness', or a 'sign-Gestalt expectation'. In the sketch in above Figure, the sequence is that an animal first wants food, then searches for an action which, if performed, will produce the wanted food. The main learning in instrumental conditioning is thus the learning of what response is necessary to gain desired consequences, or, in its simplest form, an association between a response and the reward which follows it. In freer language, it is now said that in instrumental conditioning an animal is required to 'reach certain conclusions' about what its behaviour should be, or to 'track causal relationships' between its behaviour and reward. This kind of theory, that is (c) in above Figure, is certainly considerably more complicated than the other two which it has supplanted, and we should thus briefly review the evidence in its favour.

The evidence in favour of the Tolmanian two-stage theory of instrumental learning, in which responses are selected according to whether they are associated with a desired goal, is in part indirect, in the form of evidence *against* stimulus-response principles, and in part more positive, in the form of evidence that an evaluation of goals determines instrumental performance and that responses become associated with consequent rewards.

Spatial Learning as Opposed to Stimulus-response Connections

The difficulty of accounting for an animal's learned movements in space in terms of specific stimulus-response connections was a niggling problem for S-R theory of the Watsonian type from its inception, and Tolman's criticisms of S-R theory, with recent developments, are discussed above. Only two further points need be added. First, the recent use of the radial maze, and similar techniques in which animals learn not to return to a depleted food source, emphasises the failings of the Thorndike principle that rewards stamp in only immediately preceding behaviours. Theories about cognitive maps, or working memory, are vague, but the vagueness is necessary to account for the behavioural results. As a second point, we may resurrect an old anecdotal result which may have considerable ecological validity. Helson argued against stimulus-response theories on the grounds that reward procedures could induce relatively novel behaviours. He had attempted to train rats to distinguish between two shades of grey card that indicated which of two parallel routes they should take over a raised barrier which was placed across the end of a long deep box, with a partition down the middle dividing these routes and two food compartments on the other side of the barrier. The barrier was covered with wires, the side with the incorrect card being electrified. The animals showed no sign of detecting the difference between shades of grey. However they quickly acquired a distaste for wires which delivered shock. One small rat of the four tested refused to climb over either barrier, while two others circumvented the barriers by climbing up the wall of the experimental enclosure, and walking round on top of it to the food boxes. Helson argued that this was not an accidental response which was stamped in. There is very little doubt that laboratory rats, and most wild animals, will attempt alternative methods of locomotion, via alternative routes, to get to a desired location, if this becomes necessary. A strict interpretation of the Law of Effect would require that animals be permanently halted if a tree falls across their accustomed route to a water hole. Stimulus-response theorists have always made attempts to account for the emergence of relatively novel goal-directed behaviour, in terms of generalisation of previously acquired responses, 'habit-family hierarchies and the like. These attempts have never been particularly convincing, but are now less often attempted because of an accumulation of other evidence against stimulus-response theory, and against the explanatory version of the Law of Effect.

Re-evaluation of Goals and Response Consequences

When 'stimulus-response connections' was the answer to the question of 'what is learned?', 'reward' or 'drive reduction' was the force which made the connections, and hence was once regarded as a necessary condition for all learning. This view can also now be speedily dismissed, and has already been bypassed by the previous chapters in which learning is presumed to take place in the course of habituation and classical conditioning, without the operation of a reward mechanism being necessary. Recent evidence has however been directed at the proposition that expectations of a goal or specific reward is one of the mechanisms responsible for the results of instrumental conditioning procedures.

Latent Learning

A phenomenon which weakened the assumption that drive reduction was necessary for learning, and which suggests that performance in mazes is influenced by expectancies of rewards, was investigated in Tolman's laboratory by Blodgett and Tolman and Honzik.

Rats allowed to find food in the goal box of fairly complicated mazes gradually improve their speed of getting through, and gradually make fewer errors, over a matter of weeks. This could be interpreted, albeit implausibly in view of the number of possible wrong alternatives, as the gradual strengthening of accidental correct responses, by the experience of reward. However, if other animals are allowed an equivalent amount of experience of wandering about the maze without getting food at the end, and therefore without very much change in behaviour, it can be shown that a form of 'behaviourally silent' learning has taken place, since the sudden introduction of reward in the goal box leads to a sudden change in performance, animals with unrewarded experience immediately catching up with those always rewarded after their first reward in the experiment of Tolman and Honzik. Rats allowed to explore a maze thoroughly will return efficiently on the next opportunity to any point in the maze where food is first given and thus the evidence is very strong that geographical information is learned for its own sake.

The independence of expectations with regard to goals, and methods of response which achieve goals, is best illustrated by experiments in which these two factors are separately manipulated with monkeys or chimpanzees, showing an animal where food is being hidden will induce strategies of response designed to retrieve it, and there is little argument about the role of internal representations of the hidden food in these cases. With laboratory rats, it appears to be possible to change expectations of what will be found in the goal box of a maze simply by placing the rat in the goal box, and thus 'placement studies' supply evidence that expectations can in some circumstances dictate choice and intensity of response. For instance Seward allowed 64 rats to explore a simple T-maze with goal boxes of differing texture and colour, these differences not being visible from the choice point. He then placed the rats in one of the goal boxes by hand, where they were allowed to eat.

When they were next put in the start box of the T-maze, 54 of them turned towards the goal box in which they had been previously fed. Since they had not made the turning response before being fed, the assumption must be that they had previously learned that turning in a given direction brought them to a given goal box, and that when a particular goal box was made attractive by an association with feeding, this new evaluation of the goal could be applied to the old spatial knowledge. Similarly, if rats have already learned to go in a particular direction to food in a T-maze, then if they are placed in the usually correct goal box with no food present, this is sufficient to increase subsequent turns in the opposite direction. A slightly different form of latent learning occurs if changes in motivational states occur after an explanatory phase. Rats may remember where food was in a maze when they are made hungry, even if they were not hungry at the time when they saw it. Rats allowed to drink salty water for lever-pressing when just thirsty, will press that lever, but not one which has been previously rewarded with plain water, if they become seriously salt deficient. 'Latent extinction' of a food rewarded lever-pressing response is obtained if the animals are preexposed to the functioning of the reward-delivery mechanism without the usual rewards.

Devaluation of Reward

In all the above cases, learning about the consequences of a response took place without performance of the response itself, and subsequent effects on behaviour are therefore probably attributable to the prior association between the response and its consequences. In latent learning, it can be said that expectations as to the value of responding in a

certain way are increased, while in latent extinction, expectations with regard to the value of responding are lowered. A further example of changes in the value of a goal is provided by the taste-aversion technique. Animals poisoned after eating food of a certain flavour or smell eat very much less of that particular food subsequently. Under certain circumstances, rats poisoned after eating the particular flavour of food which they have previously received as rewards for bar-pressing in a Skinner box will thereafter show a greatly reduced tendency to perform this response. In these circumstances, it is arguable that the lever-pressing should have the status of 'a true action that is under the control of the current value of the reinforcer or goal'.

Specific Associations between Responses and Rewards

In the ideal form of the expectancy theory of instrumental learning, all responses are made for specific consequences. If a rat knows that food is available at one end of its cage but water at the other, it will go to the appropriate place when hungry or thirsty, and sometimes similar behaviour can be demonstrated in less familiar T-mazes, with water on one side and food on the other. It is unlikely, however, that an exact representation of reward is necessary, in all cases of instrumental conditioning. For instance, rats rewarded unpredictably with 10 or zero units of food, or with 8 or 2, learn to run up an alley more vigorously, if anything, than rats always rewarded with 5 units. Though not strictly necessary, it has always been suspected that very specific expectancies of reward are sometimes formed in experiments with animal subjects. The best-known study was performed with rhesus monkeys by Tinkelpaugh, under Tolman's supervision. The main purpose was to demonstrate what would now be termed 'representational' factors in monkeys by allowing them to watch items of food being put in particular places or containers at some time before they were permitted to retrieve the food for themselves. Overnight delays of up to 20 hours proved to be possible. However, the study is now remembered for the checks made on memory for the identity as well as the location of hidden food. Films were taken of the animals' facial expressions and searching activities aroused when a piece of banana had been hidden, but a piece of lettuce (which the animals liked much less than banana) substituted during the delay interval. Reactions to lettuce were sufficiently pronounced to convince Tinkelpaugh that the monkey had specific expectations of finding 'banana', rather than 'some food object', in the container, although an expectation of 'highly desired food object' would account for the results. It is difficult to obtain any more direct evidence of the specificity of representations of the reinforcer in instrumental conditioning.

Loge and Sells claimed to have repeated Tinkelpaugh's finding when rats were trained in Thorndikean problem boxes. Nine rats had already been trained with two trials per day in each of three different boxes. The responses required for escape were designated as 'face-washing', 'stand-up' and 'begging' which after training were essentially three different stereotyped postural responses. Throughout training adoption of the correct posture for a particular box resulted in a door being opened so that the animals could exit from the box to a small cup containing bread and milk. To examine 'representative factors' the nine well-trained rats were simply run as usual, but with sunflower seeds in the reward cup instead of the bread and milk. Although rats' facial expressions are not as revealing as those of monkeys, their general behaviour was changed considerably by the substitution of sunflower seeds. For the two days that they were observed, all rats on all occasions, on making the appropriate response and getting to the food cup, picked up one

of the seeds, threw it aside, then ran back into the box and repeated the response. Although the animals had previously eaten sunflower seeds, this experiment would have been better controlled if some of the animals had received sunflower seeds throughout the experiment. However, precisely this comparison was possible in the earlier experiment of Elliot, who trained rats on a multi-unit maze for either sunflower seed or branmash as reward, and found that animals switched from mash to seeds began to make many errors, whereas animals always rewarded with seeds performed accurately.

Although there is very little doubt that chimpanzees may form specific expectancies about particular food stuffs, which then determine both direct goal-seeking and other forms of behaviour which make the expectancies obvious to an observer, evidence supporting a role for specific expectancies of reward in instrumental learning by laboratory rats and pigeons is understandably less strong. The experiments in which rewards are changed, such as those of Elliot, just mentioned, and Crespi can be interpreted as indicating an overall motivational value of reward associated with instrumental behaviour, with reductions in motivational value having various emotional effects. However, a different kind of experiment, first suggested by Trapold, provides an additional source of evidence for the theory that the qualitative nature of rewarding events, other than just their degree of goodness or badness, is usually noticed by rats and pigeons, and occasionally assists in the selection of appropriate responses. In Trapold's experiment the task for rats was to press one lever, on the left, when a clicker was sounding, but another lever, on the right, when they heard a tone. Providing the two stimuli, and the two separate responses to be made to them, are easily distinguishable, rats are perfectly capable of learning such tasks, by trial and error, although they may need thousands of trials to get things completely right. Trapold's finding was that the task was made much easier if two different rewards were used, food pellets for the left lever and sugar solution for the right, even if both rewards were presented at the same place. Furthermore, in a separate experiment, he showed that associating particular rewards with the clicker and tone signals, before these were used as cues for the bar-pressing task, also made the eventual learning of the task faster if the rewards were kept the same way around, but slower if they were reversed. This supports the expectancy theory of instrumental learning illustrated in above Figure c (Pavlov/Tolman: stimulus-reward and response-reward association), in which the animal is supposed to think of the reward first, and then decide which response must be made to obtain the reward. Clearly this decision should be easier if separate rewards need separate responses.

Various elaborations of this type of experiment have confirmed that what is known as the 'differential outcome procedure' usually enhances learning or has other effects predictable in terms of differential expectancies of specific rewards, with both rats and pigeons. For pigeons a rather more difficult task has been used, in which one signal, which may be either green or red, say, determines which of two stimuli, say black or white lines, should be chosen next. If the first green signal means that the bird should now choose black to get a reward of food, but a red signal means it should choose white, to get a reward of water, then learning is quicker, and performance more accurate, than when all correct choices are reinforced by either food or water at random. With this procedure, if the task is made even more difficult by a delay between the red/green signal and the opportunity to make the next choice, between horizontal or vertical lines, then differential reward outcomes of food and water for the two correct choices allowed 95 per cent correct performance with a 10-second delay, while the use of only a single reward re-

duced accuracy to chance levels. It was thus argued that it was the expectancy of a particular reward, retained over the delay interval, which was responsible for directing the eventual choice. If green and vertical are two visual stimuli both associated with food, and red and horizontal both associated with water, then representations of food or water (or peripheral responses specific to these) will be sufficient to bridge the delay, and that should certainly make the task easier by comparison with that of remembering whether the first signal has just been green or red, and knowing that green means horizontal and red means vertical.

Various Alternative Sorts of Instrumental Learning

The evidence in the section above might tempt one to conclude that the Tolmanian explanation for what is learned is right, whereas Hull's and Thorndike's ideas were wrong. But not only are there some reasons for hesitation in interpreting this favourable evidence, there are also many results which argue quite unequivocally against specific expectations of rewards being always a necessary factor in instrumental learning. For instance, some attempts to demonstrate latent learning fail. More important, there are many cases where laboratory animals appear to learn rigid and mechanical automatic habits which become more or less completely independent of the desirability of the goals used to establish the habits. For instance Holman trained rats to press a lever for a reward of saccharin solution, and then made the rats averse to saccharin by injections of lithium chloride. This stopped the rats drinking saccharin solution, but had no effect on their habit of pressing the lever. Adams and Dickinson *et al.*, having previously shown that in some circumstances a conditioned aversion to previous rewards will immediately reduce the strength of habits they reinforced, agree that with long-established habits, especially those conditioned on fixed interval schedules of reinforcement, the learned performance is very little affected, if at all, by manifest devaluation of the ostensible goals.

The relative independence of habits from goals was stressed theoretically by Thorndike but with the empirical support of the fact that cats which had learned to press a latch to get out of a box often (but not always) continued to do so when he cut a hole in the roof. More dramatic apparent obliviousness to the final purpose of their activities is easy to demonstrate in rats, which, after being thoroughly trained to run down an alley for food pellets, may then run quickly over identical food pellets strewn in their path. Similarly, it is a reliable result that rats trained in Skinner boxes to press a lever for food pellets may ignore similar rewards presented in a bowl, preferring to obtain the same objects in the more usual way. This is partly a trade-off between accustomed habits and degree of effort, since, understandably, rats required to press many times for a single food pellet are relatively eager to accept free pellets from a new place. However, there is sufficient rigidity in rats' learned behaviour to support a good measure of Thorndikean or Hullian habits – it must be remembered that human behaviour, and not only that classed as neurotic or abnormal, is composed of established routines performed unthinkingly, as much as of rational purposes.

There is thus good reason to believe that rats and people, though occasionally and fleetingly directing their behaviour towards expected goals, make extensive use of the less taxing mechanism of automatic habit for behaviours which are often repeated. Morgan, just before Thorndike, had clearly stated the proposition that goal-seeking is always responsible for the early stages of learning, and for dealing with new problems, but that

However, for the general theory of learning, the more important cases are those at the opposite end of the scale of intellectual capacity. Although claims for instrumental learning in *Aplysia* are at present muted, the basic feature of a tendency to repeat rewarded acts is claimed for creatures as diverse as earthworms and flatworms and not only for advanced arthropods like honey-bees but also for less advanced insects like cockroaches, and even for the decapitated cockroach, and legs of cockroaches detached from the rest of the body. There are ample physiological reasons for doubting whether these neural systems are capable of sustaining, for instance, specific expectations of reinforcers which are as rich in information as those claimed for rats and pigeons. Further, for rats and pigeons, and other laboratory animals; it is possible to demonstrate forms of instrumental conditioning after radical interference with normal brain functions such as complete removal of the neocortex of the cerebral hemispheres (decortication) or removal of most of the hemispheres themselves. Oakley has reviewed such results, obtained by himself and others. Decorticate rabbits will eventually, after slightly slower learning, press levers for food in Skinner boxes, on basic fixed interval and fixed ratio schedules. There must therefore exist subcortical mechanisms which are sufficient for the demonstration of instrumental learning. This raises the possibility, at least, that there are subcortical methods of stimulus-response association, which differ in some way from the full range of psychological processes available to the normal animal.



Evidence from common laboratory animals, from cross-species comparisons and from physiological investigations, all suggests that there may be more than one answer to the question of what is learned. As a general rule of thumb, any mechanism proposed in the last 100 years as the universal principle of learning may be accepted as *a* form of learning in some or other species or preparation. Instead of choosing between the alternatives presented in above Figure, a more accurate and comprehensive answer to 'what is learned?' is obtained by combining all three processes illustrated in above Figure (Theories of instrumental learning), with some additions suggested by the results discussed above, to give a diagram such as that shown in above Figure. The purpose of this is to show that more than one process may be involved in instrumental learning, and there are many more possibilities than can conveniently be drawn in.

For instance, the example chosen is the pressing of a lever for food rewards by a hungry animal in a Skinner box. Instrumental learning to escape from dangerous or unpleasant external stimuli has not yet been considered, and would require some changes, though not wholesale ones. Also not immediately obvious is the fact that learning of some kinds is possible without any strong motivational events occurring; latent learning of the kind which commonly takes place when rats explore mazes is subsumed along with much else in the box which refers to responses which are 'appropriately associated with the wanted event'. Although this covers a multitude of possible forms of association, including any method of getting to the known location of a wanted event, it does not seem unreasonable to assume that getting a food reward after a particular response may contribute to the association between the reward and that response. This may work in both directions, so that wanting a reward will in future produce an impulse for that response (1), and when the response is made, a firmer expectation of specific impending reward might very well be justified (2). In the situation in which rewards are regularly obtained, it is assumed that some representation of these goals is conditioned to the context (3); but this needs to be modifiable by the effects of specific signals, which often have stronger conditioned effect than background or contextual stimuli. Clearly, ordinary classical conditioning should provide representation of the reinforcer associated with particular signals, as at (3), and similar effects are frequently now supposed to play a role in instrumental conditioning. But as Jenkins pointed out, and as Mackintosh has cogently emphasised, this cannot be the only role for what is known as a discriminative stimulus in operant conditioning for, to put it loosely, this does not merely inform the animal that rewards may be forthcoming (as in classical conditioning); it also instructs the animal that if it wants a reward, then a particular response is necessary. Thus, laboratory animals learn easily, if they have to, that rewards are sometimes delivered free but that when a signal is on they must work for them. The signal in this case does not indicate any increase in expectations of rewards, but rather that a given response is now appropriately associated with reward, and thus there is an input from the discriminative stimulus at (1).

Now, in order to accommodate the evidence for Thorndikean habits, we can further assume that the pure frequency of occurrence (not illustrated), or the effects of reward (4), will increase an automatic impulse to make the behavioural act normally rewarded, either in response to the background cues, as in Thorndike's original experiment, or connected with specific experimental signals (or discriminative stimuli). This is a fairly high level of representation of the response, since a variety of muscular coordinations, some undoubtedly unlearned, will be necessary to actually do something. However, both in animals lacking that level of representation of the response and for detailed stimulus-

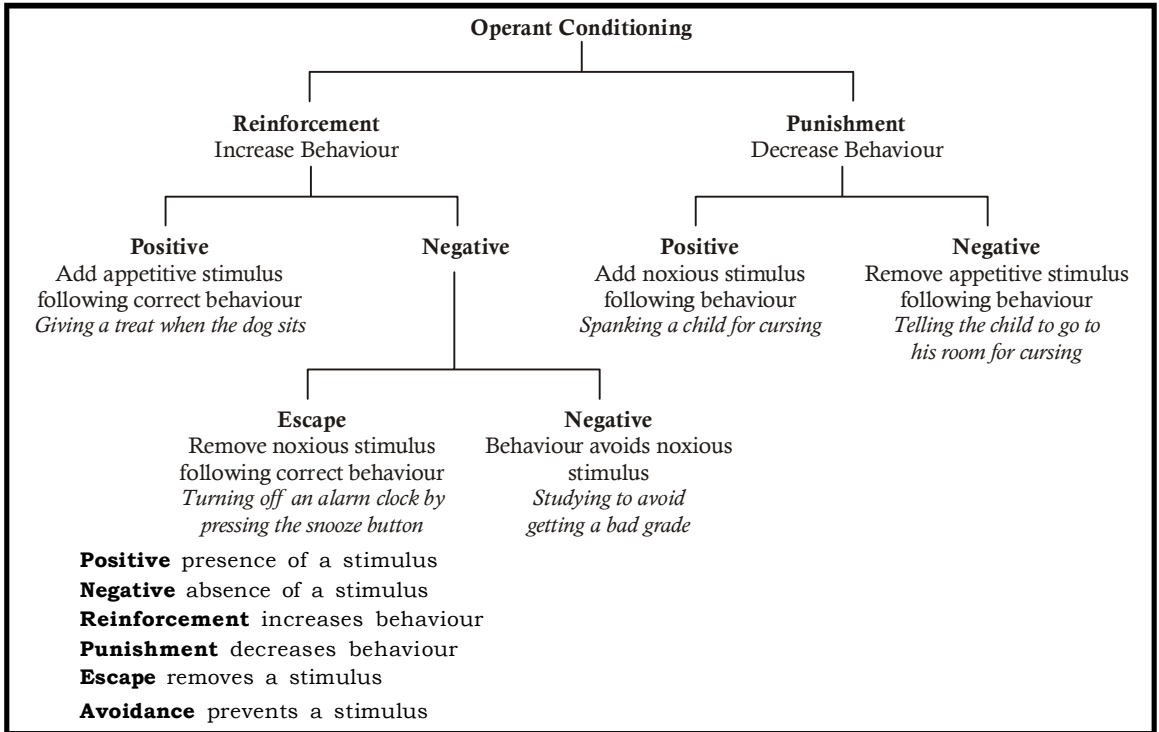
response co-ordination, such as is involved in gear-changing in higher animals, then the muscular coordinations themselves will be one of the things learned in instrumental conditioning procedures (5). We might expect that these will be mainly internal (some kinesthetic) associations and less tied either to the external cues or to the motivating effects of reward, but we have Locke's example of dance steps learned in a particular room to remind us that skills may sometimes need familiar external conditions, and the rewards are likely to be involved both as feedback to confirm correctness and in various kinds of motivation.

Finally, although considerable attention is currently given to expectations of reward, representations of the reinforcer and such like, concern with the 'wanted' character of reward representation and the role of specific drive states is probably due for a comeback. Drive states such as hunger, thirst or salt deficiency presumably have an innate multiplying effect on internal representations of food, water and salt: there may be many other in-built drive states, and there can certainly be alterations of these by which cravings for cigarettes, chocolates, morphine and numerous other ingestible substances are controlled metabolically. Therefore the level of specific drive is shown as modulating the degree to which external cues arouse ideas of reward and the intensity with which any such ideas are followed by the various response-producing processes encompassed by the 'perform response' box. In the variation of Hullian theory adopted by Spence and perpetuated by Bindra and others, the effect of rewards was interpreted as 'incentive motivation', which appeared to consist in part of general activation and energy available, with which to perform conditioned responses. Whether or not such a process is of any great importance in conditioning experiments has been disputed but, depending on the exact reward, it is possible that some or other form of arousal or emotion can become conditioned to external cues (6). The restlessness exhibited by caged animals in anticipation of regular feeding times would seem to be partly attributable to general arousal, and since a similar effect is observed in general movements of decorticated rabbits, and in measures of heartbeat in goldfish, it seems wisest to allow associations of arousal with external cues which are separable from those of more concrete and informative expectations. However, the distinction is not intended as a hard and fast one: some kinds of conditioned arousal may be reward specific – possibly including salivation, licking of lips and so on in the case of food, and hormonal and autonomic changes in anticipation of sexual contacts – while what are regarded as more cognitive expectancies, as of a goal in a lucky dip, or a prize behind the curtain, may have general motivating effects, while being altogether vague, nebulous, or mistaken.

REINFORCEMENT

Inspired by behaviourist psychology, **reinforcement learning** is an area of machine learning in computer science, concerned with how software agents ought to take *actions* in an *environment* so as to maximize some notion of cumulative *reward*. The problem, due to its generality, is studied in many other disciplines, such as game theory, control theory, operations research, information theory, simulation-based optimisation, statistics, and genetic algorithms. In the operations research and control literature, the field where reinforcement learning methods are studied is called *approximate dynamic programming*. The problem has been studied in the theory of optimal control, though most studies there are concerned with existence of optimal solutions and their characterisation, and not

with the learning or approximation aspects. In economics and game theory, reinforcement learning may be used to explain how equilibrium may arise under bounded rationality.



In machine learning, the environment is typically formulated as a Markov decision process (MDP), and many reinforcement learning algorithms for this context are highly related to dynamic programming techniques. The main difference between the classical techniques and reinforcement learning algorithms is that the latter do not need knowledge about the MDP and they target large MDPs where exact methods become infeasible.

Reinforcement learning differs from standard supervised learning in that correct input/output pairs are never presented, nor sub-optimal actions explicitly corrected. Further, there is a focus on on-line performance, which involves finding a balance between exploration (of uncharted territory) and exploitation (of current knowledge). The exploration vs. exploitation trade-off in reinforcement learning has been most thoroughly studied through the multi-armed bandit problem and in finite MDPs.

In behavioural psychology, **reinforcement** is a strengthening of a specific behaviour due to its association with a stimulus. Reinforcement is an important part of operant or instrumental conditioning. A *reinforcer* is the stimulus that strengthens the behaviour, in contrast to punishment that weakens the behaviour. The effect of reinforcement may be measured as an increase in the frequency of its expression (e.g., pulling a lever more frequently), duration (e.g., pulling a lever for longer periods of time), magnitude (e.g., pulling a lever with greater force), or decrease in latency (e.g., pulling a lever more quickly following the onset of an environmental event).

Although in many cases a reinforcing stimulus is a rewarding stimulus which is “valued” or “liked” by the individual (e.g., money received from a slot machine, the taste of

the treat, the euphoria produced by an addictive drug), this is not a requirement. Indeed, reinforcement does not even require an individual to consciously perceive an effect elicited by the stimulus. Furthermore, stimuli that are “rewarding” or “liked” are not always reinforcing: if an individual eats at McDonald’s (response) and likes the taste of the food (stimulus), but believes it is bad for their health, they may not eat it again and thus it was not reinforcing in that condition. Thus, reinforcement can only be assumed if there is a causal link (positive or negative) between the stimulus and the enhancement of behaviour.

In most cases reinforcement refers to an enhancement of behaviour but this term may also refer to an enhancement of memory. One example of this effect is called post-training reinforcement where a stimulus (*e.g.* food) given shortly after a training session enhances the learning. This stimulus can also be an emotional one. A good example is that many people can explain in detail where they were when they found out the World Trade Center was attacked.

B.F. Skinner was a high profile researcher that articulated many of the theoretical constructs of reinforcement and behaviourism. Skinner defined reinforcers according to the change in response strength rather than to more subjective criteria, such as what is pleasurable or valuable to someone. Accordingly, activities, foods or items considered pleasant or enjoyable may not necessarily be reinforcing (because they produce no increase in the response preceding them). Stimuli, settings, and activities only fit the definition of reinforcers if the behaviour that immediately precedes the potential reinforcer increases in similar situations in the future, for example, a child who receives a cookie when he or she asks for one. If the frequency of “cookie-requesting behaviour” increases, the cookie can be seen as reinforcing “cookie-requesting behaviour”. If however, “cookie-requesting behaviour” does not increase the cookie cannot be considered reinforcing.

The sole criteria that determines if an item, activity, or food is reinforcing is the change in probability of a behaviour after administration of that potential reinforcer. Other theories may focus on additional factors such as whether the person expected the strategy to work at some point, but in the behavioural theory, reinforcement is descriptive of an increased probability of a response.

The study of reinforcement has produced an enormous body of reproducible experimental results. Reinforcement is the central concept and procedure in special education, applied behaviour analysis, and the experimental analysis of behaviour.

Much of the work regarding reinforcement began with behavioural psychologists such as Edward Thorndike, J. B. Watson and B.F. Skinner and their use of animal experiments. B.F. Skinner is famous for his work on reinforcement and believed that positive reinforcement is superior to punishment in shaping behaviour. At first glance, punishment can seem like just the opposite of reinforcement, yet Skinner argued that they differ immensely; he claimed that positive reinforcement results in lasting behavioural modification (long-term) whereas punishment changes behaviour only temporarily (short-term) and has many detrimental side-effects. Skinner defined reinforcement as creating situations that a person likes or removing a situation he doesn’t like, and punishment as removing a situation a person likes or setting up one he doesn’t like. Thus, the distinction was based mainly on the pleasant or aversive (unpleasant) nature of the stimulus.

Two other researchers, Azrin and Holz, expanded upon operant conditioning by focusing on the definition of punishment in their chapter to Honig’s volume on operant

behaviour, and they defined it as a “consequence of behaviour that reduces the future probability of that behaviour.” Skinner’s assumptions regarding reinforcement and punishment were thus challenged throughout the 1960s, and some studies have shown that positive reinforcement and punishment are equally effective in modifying behaviour; that debate, however, continues in studies today as to whether or not reinforcement is more or equally as effective as punishment. Edward Thorndike also did some work regarding reinforcement in learning theory and believed that learning could occur unconsciously; that is, reinforcements or punishments could have an effect upon learning even if the person or organism is unaware of it. The research on the effects of positive and negative reinforcement alongside punishment continue today as those concepts apply directly to many forms of learning and behaviour.

Types of Operant Conditioning

The basic definition is that a positive reinforcer *adds* a stimulus to increase or maintain frequency of a behaviour while a negative reinforcer *removes* a stimulus to increase or maintain the frequency of the behaviour. As mentioned above, positive and negative reinforcement are components of operant conditioning, along with positive punishment and negative punishment, all explained below:

Positive Reinforcement

Occurs when a stimulus is presented as a result of operant behaviour and that behaviour increases.

Example. If a dog “sits” on command and this behaviour is followed by the reward of a dog treat, then the dog treat serves to positively reinforce the behaviour of “sitting.”

Example. A father gives candy to his daughter when she picks up her toys. If the frequency of picking up the toys increases or stays the same, the candy is a positive reinforcer (to reinforce the behaviour of cleaning up).

Negative Reinforcement

Occurs when an aversive (unpleasant) stimulus is removed as a result of operant behaviour and the rate of the behaviour increases.

Example. A child cleans his or her room, and this behaviour is followed by the parent stopping “nagging” or asking the child repeatedly to do so. Here, the nagging serves to negatively reinforce the behaviour of cleaning because the child wants to remove that aversive stimulus of nagging.

Example. A person puts ointment on a bug bite to soothe an itch. If the ointment works, the person will likely increase the usage of the ointment because it resulted in removing the itch, which is the negative reinforcer.

Punishment

Positive punishment is the adding of an aversive (unpleasant) stimulus to decrease a behaviour or response.

Example. A mother yells at a child when he or she runs into the street. If the child stops running into the street, the yelling acts as positive punishment because the mother presents (adds) an unpleasant stimulus in the form of yelling.

Negative punishment is the removal of a pleasant stimulus to decrease a behaviour or response.

Example. A teenager comes home after curfew and the parents take away a privilege, such as cell phone usage. If the frequency of the child coming home late decreases, the removal of the phone is negative punishment because the parents are taking away a pleasant stimulus (the phone) and motivating the child to return home earlier.

Simply put, reinforcers serve to increase behaviours whereas punishments serve to decrease behaviours; thus, positive reinforcers are stimuli that the subject will work to attain, and negative reinforcers are stimuli that the subject will work to be rid of or to end. The table below illustrates the adding and subtracting of stimuli (pleasant or aversive) in relation to reinforcement vs. punishment.

	Pleasant Stimulus	Aversive (unpleasant) Stimulus
Adding/Presenting	Positive Reinforcement	Positive Punishment
Removing/Taking Away	Negative Punishment	Negative Reinforcement

Further Ideas and Important Concepts to Remember:

Distinguishing between positive and negative can be difficult and may not always be necessary; focusing on *what* is being removed or added and *how* it is being removed or added will determine the nature of the reinforcement.

A negative reinforcer is not a punishment. The two, as explained above, differ in the addition or removal of a stimulus.

The increase in behaviour is independent of (i.e. not related to) whether or not the organism finds the reinforcer to be pleasant or aversive. Example: A child is given detention for acting up in school, but the frequency of the bad behaviour increases. Thus, the detention is a positive reinforcer even if the detention is not a pleasant stimuli, perhaps because the child now feels like a “rebel” or sees it as an opportunity to get out of class.

Some reinforcement can be simultaneously positive and negative, such as a drug addict taking drugs for the added euphoria (a positive feeling) and eliminating withdrawal symptoms (which would be a negative feeling). Or, in a warm room, a current of external air serves as positive reinforcement because it is pleasantly cool and as negative reinforcement because it removes uncomfortable hot air.

Both positive and negative reinforcement *increase* behaviour. Most people, especially children, will learn to follow instruction by a mix of positive and negative reinforcement.

Primary Reinforcers

A primary reinforcer, sometimes called an *unconditioned reinforcer*, is a stimulus that does not require pairing to function as a reinforcer and most likely has obtained this function through the evolution and its role in species’ survival. Examples of primary reinforcers include sleep, food, air, water, and sex. Some primary reinforcers, such as certain drugs, may mimic the effects of other primary reinforcers. While these primary reinforcers are fairly stable through life and across individuals, the reinforcing value of different primary reinforcers varies due to multiple factors (*e.g.*, genetics, experience). Thus, one person may prefer one type of food while another abhors it. Or one person may

eat lots of food while another eats very little. So even though food is a primary reinforcer for both individuals, the value of food as a reinforcer differs between them.

Secondary Reinforcers

A secondary reinforcer, sometimes called a *conditioned reinforcer*, is a stimulus or situation that has acquired its function as a reinforcer after pairing with a stimulus that functions as a reinforcer. This stimulus may be a primary reinforcer or another conditioned reinforcer (such as money). An example of a secondary reinforcer would be the sound from a clicker, as used in clicker training. The sound of the clicker has been associated with praise or treats, and subsequently, the sound of the clicker may function as a reinforcer. As with primary reinforcers, an organism can experience satiation and deprivation with secondary reinforcers.

Other Reinforcement Terms

A generalised reinforcer is a conditioned reinforcer that has obtained the reinforcing function by pairing with many other reinforcers (One example of this is money because it is paired with many other reinforcers).

In reinforcer sampling, a potentially reinforcing but unfamiliar stimulus is presented to an organism without regard to any prior behaviour.

Socially-mediated reinforcement (direct reinforcement) involves the delivery of reinforcement that requires the behaviour of another organism.

The Premack principle is a special case of reinforcement elaborated by David Premack, which states that a highly-preferred activity can be used effectively as a reinforcer for a less-preferred activity.

Reinforcement hierarchy is a list of actions, rank-ordering the most desirable to least desirable consequences that may serve as a reinforcer. A reinforcement hierarchy can be used to determine the relative frequency and desirability of different activities, and is often employed when applying the Premack principle.

Contingent outcomes are more likely to reinforce behaviour than non-contingent responses. Contingent outcomes are those directly linked to a causal behaviour, such a light turning on being contingent on flipping a switch. Note that contingent outcomes are not necessary to demonstrate reinforcement, but perceived contingency may increase learning.

Contiguous stimuli are stimuli closely associated by time and space with specific behaviours. They reduce the amount of time needed to learn a behaviour while increasing its resistance to extinction. Giving a dog a piece of food immediately after sitting is more contiguous with (and therefore more likely to reinforce) the behaviour than a several minute delay in food delivery following the behaviour.

Non-contingent reinforcement refers to response-independent delivery of stimuli identified as reinforcers for some behaviours of that organism. However, this typically entails time-based delivery of stimuli identified as maintaining aberrant behaviour, which decreases the rate of the target behaviour. As no measured behaviour is identified as being strengthened, there is controversy surrounding the use of the term noncontingent "reinforcement".

Natural and Artificial

In his 1967 paper, *Arbitrary and Natural Reinforcement*, Charles Ferster proposed classifying reinforcement into events that increase frequency of an operant as a natural consequence of the behaviour itself, and events that are presumed to affect frequency by their requirement of human mediation, such as in a token economy where subjects are “rewarded” for certain behaviour with an arbitrary token of a negotiable value. In 1970, Baer and Wolf created a name for the use of natural reinforcers called “behaviour traps”. A behaviour trap requires only a simple response to enter the trap, yet once entered, the trap cannot be resisted in creating general behaviour change. It is the use of a behavioural trap that increases a person’s repertoire, by exposing them to the naturally occurring reinforcement of that behaviour. Behaviour traps have four characteristics:

They are “baited” with virtually irresistible reinforcers that “lure” the student to the trap.

Only a low-effort response already in the repertoire is necessary to enter the trap. Interrelated contingencies of reinforcement inside the trap motivate the person to acquire, extend, and maintain targeted academic/social skills.

They can remain effective for long periods of time because the person shows few, if any, satiation effects.

As can be seen from the above, artificial reinforcement is in fact created to build or develop skills, and to generalise, it is important that either a behaviour trap is introduced to “capture” the skill and utilise naturally occurring reinforcement to maintain or increase it. This behaviour trap may simply be a social situation that will generally result from a specific behaviour once it has met a certain criterion (*e.g.*, if you use edible reinforcers to train a person to say hello and smile at people when they meet them, after that skill has been built up, the natural reinforcer of other people smiling, and having more friendly interactions will naturally reinforce the skill and the edibles can be faded).

Intermittent Reinforcements

Pigeons experimented on in a scientific study were more responsive to **intermittent reinforcements**, than **positive reinforcements**. In other words, pigeons were more prone to act when they only *sometimes could get what they wanted*. This effect was such that behavioural responses were maximised when the reward rate was at 50% (in other words, when the uncertainty was maximised), and would gradually decline toward values on either side of 50%. R.B. Sparkman, a journalist specialised on what motivates human behaviour, claims this is also true for humans, and may in part explain human tendencies such as gambling addiction.

The Principle of Reinforcement

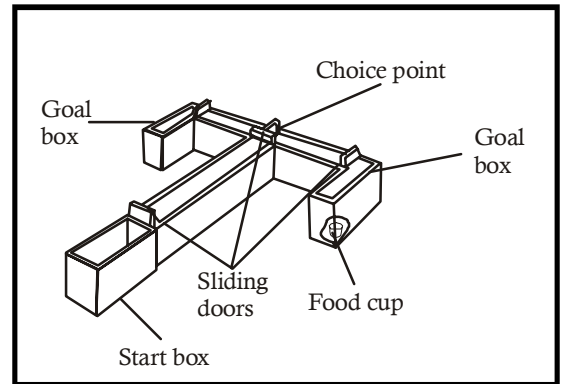
The term reinforcement used in the discussion of classical conditioning is referred to the paired presentation of the unconditioned stimulus and the conditioned stimulus. In operant conditioning, reinforcement referred to the occurrence of an event, like the giving of food or water, following the desired response. But in other terms, in classical conditioning reinforcement elicits the response, and in operant conditioning reinforcement follows the response. Although the reinforcement is quite different in the two situations, the result in both cases is an increase in the likelihood of the desired response. We can therefore, define reinforcement as any event whose occurrence increases the probab-

ity that a stimulus will on subsequent occasions evoke a response. We customarily distinguish between two types of reinforcers: positive reinforcers (such as food), which on being presented increase the probability of a response, and negative reinforcers (such as shock), which on being terminated increase response probability.

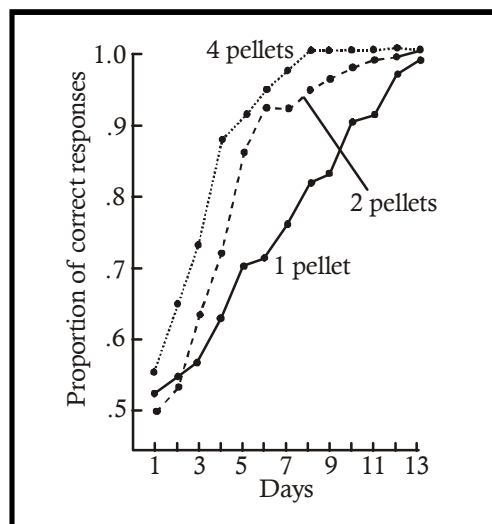
Basic Variables

Variables affecting Reinforcement

On the course of learning the effect of a number of reinforcement variables have systematically been investigated. The amount of reinforcement has been found to be an important parameter. Within limits the greater the amount of reinforcement, the more rapid the rate of learning. This relationship is illustrated in an experiment using a T-maze (Fig. A). After the rat was placed in the start box, it ran to the choice point, where it had to decide between a right or left turn to reach the food placed in one of the goal boxes. In this experiment there were three groups of rats, each group differing in the amount of food received for a correct turn: one group received four food pellets, another group two pellets, and a third group one pellet. Each rat ran four trials a day. The results in terms of proportion of correct responses per day are shown in Fig. (B). Note that the group with the largest amount of reinforcement learned at the fastest rate, while the other two groups learned more slowly. The curves start at about .5 since there are only two choices; the animal should make the correct turn 50 per cent of the time by chance alone.

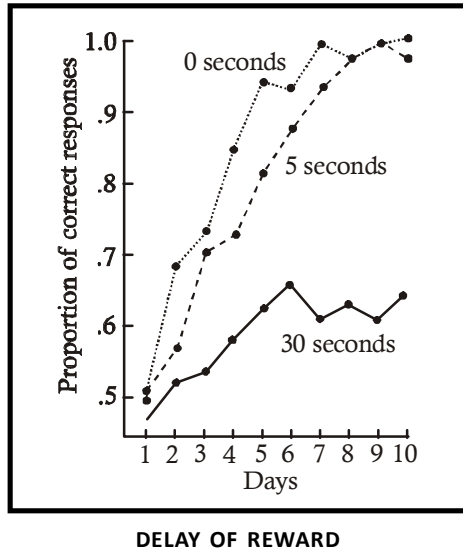


(A) T-MAZE



(B) AMOUNT OF REINFORCEMENT

The delay of reinforcement is another important parameter of reinforcement. A common assumption in training animals or young children has been that it is most effective to reward or punish the organism immediately after it responds.



The effectiveness of immediate reinforcement in a laboratory learning situation is demonstrated by the following experiment. The apparatus used was a T-maze having goal boxes equipped with food dispensers that could be set to delay the presentation of food pellets. One group of rats received their food immediately upon entering the correct goal box (zero-second delay); another group was fed following a five-second delay; and a third group was delayed thirty seconds before receiving their food. Above figure shows the learning curves for each of the three groups. The zero-second group and the five-second group both reached near perfect responding by the ninth day, but the zero-second group learned at a faster rate. The thirty-second-delay group was markedly inferior and never achieved a very high level of performance.

When reinforcement is applied on an irregular basis, they are called variable schedules.

Variable Ratio. This refers to applying a reinforcer after a variable number of responses. Variable ratio schedules have been found to work best under many circumstances and knowing an example will explain why. Imagine walking into a casino and heading for the slot machines. After the third coin you put in, you get two back. Two more and you get three back. Another five coins and you receive two more back. How difficult is it to stop playing?

Variable Interval. Reinforcing someone after a variable amount of time is the final schedule. If you have a boss who checks your work periodically, you understand the power of this schedule. Because you don't know when the next 'check-up' might come, you have to be working hard at all times in order to be ready.

In this sense, the variable schedules are more powerful and result in more consistent behaviours. This may not be as true for punishment since consistency in the application is so important, but for all other types of reinforcement they tend to result in stronger responses.

Schedules

When an animal's surroundings are controlled, its behaviour patterns after reinforcement become predictable, even for very complex behaviour patterns. A schedule of reinforcement is a rule or program that determines how and when the occurrence of a response will be followed by the delivery of the reinforcer, and extinction, in which no response is reinforced. Schedules of reinforcement influence how an instrumental response is learned and how it is maintained by reinforcement. Between these extremes is *intermittent* or *partial reinforcement* where only some responses are reinforced.

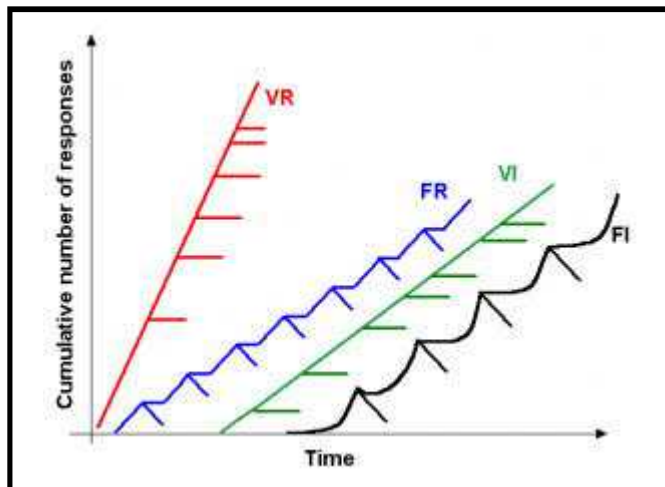
Specific variations of intermittent reinforcement reliably induce specific patterns of response, irrespective of the species being investigated (including humans in some conditions). The orderliness and predictability of behaviour under schedules of reinforcement was evidence for B.F. Skinner's claim that by using operant conditioning he could obtain "control over behaviour", in a way that rendered the theoretical disputes of contemporary comparative psychology obsolete. The reliability of schedule control supported the idea that a radical behaviourist experimental analysis of behaviour could be the foundation for a psychology that did not refer to mental or cognitive processes. The reliability of schedules also led to the development of applied behaviour analysis as a means of controlling or altering behaviour.

Many of the simpler possibilities, and some of the more complex ones, were investigated at great length by Skinner using pigeons, but new schedules continue to be defined and investigated.

Simple Schedules

A chart demonstrating the different response rate of the four simple schedules of reinforcement, each hatch mark designates a reinforcer being given

Ratio schedule – the reinforcement depends only on the number of responses the organism has performed.



Continuous reinforcement (CRF) – a schedule of reinforcement in which every occurrence of the instrumental response (desired response) is followed by the reinforcer.

Lab example: each time a rat presses a bar it gets a pellet of food.

Real world example: each time a dog defecates outside its owner gives it a treat; each time a person puts \$1 in a candy machine and presses the buttons he receives a candy bar.

Simple schedules have a single rule to determine when a single type of reinforcer is delivered for specific response.

Fixed ratio (FR) – schedules deliver reinforcement after every n th response.

Example: FR2" = every second desired response the subject makes is reinforced.

Lab example: FR5" = rat reinforced with food after every 5 bar-presses in a Skinner box.

Real-world example: FR10" = Used car dealer gets a \$1000 bonus for each 10 cars sold on the lot.

Variable ratio schedule (VR) – a reinforcement schedule in which the number of responses necessary to produce reinforcement varies from trial to trial. A VR schedule of VR10 means that if one averaged the number of reinforcers, on the average every tenth desired response was reinforced.

Lab example: VR4" = first pellet delivered on 2 bar presses, second pellet delivered on 6 bar presses, third pellet 4 bar presses ($2 + 6 + 4 = 12$; $12/3 = 4$ bar presses to receive pellet).

Real-world example: slot machines (because, though the probability of hitting the jackpot is constant, the number of lever presses needed to hit the jackpot is variable).

Fixed interval (FI) – reinforced after every n th amount of time.

Example: FI1" = reinforcement provided for the first response after 1 second.

Lab example: FI15" = rat is reinforced for the first bar press after 15 seconds passes since the last reinforcement.

Real world example: washing machine cycle.

Variable interval (VI) – reinforced on an average every n th amount of time. The 'n' is an average.

Example: VI4" = first pellet delivered after 2 minutes, second delivered after 6 minutes, third is delivered after 4 minutes ($2 + 6 + 4 = 12$; $12/3 = 4$). Reinforcement is delivered on the average after 4 minutes.

Lab example: VI10" = a rat is reinforced for the first bar press after an average of 10 seconds passes since the last reinforcement.

Real world example: checking your e-mail or pop quizzes. Going fishing—you might catch a fish after 10 minutes, then have to wait an hour, then have to wait 18 minutes.

Other simple schedules include:

Differential reinforcement of incompatible behaviour – Used to reduce a frequent behaviour without punishing it by reinforcing an incompatible response. An example would be reinforcing clapping to reduce nose picking.

Differential reinforcement of other behaviour (DRO) – Also known as omission training procedures, an instrumental conditioning procedure in which a positive reinforcer is periodically delivered only if the participant does something other than the target response. An example would be reinforcing any hand action other than nose picking.

Differential reinforcement of low response rate (DRL) – Used to encourage low rates of responding. It is like an interval schedule, except that premature responses reset the time required between behaviour.

Lab example: DRL10" = a rat is reinforced for the first response after 10 seconds, but if the rat responds earlier than 10 seconds there is no reinforcement and the rat has to wait 10 seconds from that premature response without another response before bar pressing will lead to reinforcement.

Real world example: "If you ask me for a potato chip no more than once every 10 minutes, I will give it to you. If you ask more often, I will give you none."

Differential reinforcement of high rate (DRH) – Used to increase high rates of responding. It is like an interval schedule, except that a minimum number of responses are required in the interval in order to receive reinforcement.

Lab example: DRH10"/15 responses = a rat must press a bar 15 times within a 10 second increment to get reinforced.

Real world example: "If Lance Armstrong is going to win the Tour de France he has to pedal x number of times during the y -hour race."

Fixed time (FT) – Provides reinforcement at a fixed time since the last reinforcement, irrespective of whether the subject has responded or not. In other words, it is a non-contingent schedule.

Lab example: FT5" = rat gets food every 5" regardless of the behaviour.

Real world example: a person gets an annuity check every month regardless of behaviour between checks

Variable time (VT) – Provides reinforcement at an average variable time since last reinforcement, regardless of whether the subject has responded or not.

Effects of different types of simple schedules. Fixed ratio: activity slows after reinforcer and then picks up.

Variable ratio: high rate of responding greatest activity of all schedules, responding rate is high and stable.

Fixed interval: activity increases as deadline nears, can cause fast extinction.

Variable interval: steady activity results, good resistance to extinction.

Ratio schedules produce higher rates of responding than interval schedules, when the rates of reinforcement are otherwise similar.

Variable schedules produce higher rates and greater resistance to extinction than most fixed schedules. This is also known as the Partial Reinforcement Extinction Effect (PREE).

The variable ratio schedule produces both the highest rate of responding and the greatest resistance to extinction (for example, the behaviour of gamblers at slot machines).

Fixed schedules produce "post-reinforcement pauses" (PRP), where responses will briefly cease immediately following reinforcement, though the pause is a function of the upcoming response requirement rather than the prior reinforcement.

The PRP of a fixed interval schedule is frequently followed by a "scallop-shaped" accelerating rate of response, while fixed ratio schedules produce a more "angular" response.

- fixed interval scallop: the pattern of responding that develops with fixed interval reinforcement schedule, performance on a fixed interval reflects subject's accuracy in telling time.

Organisms whose schedules of reinforcement are “thinned” (that is, requiring more responses or a greater wait before reinforcement) may experience “ratio strain” if thinned too quickly. This produces behaviour similar to that seen during extinction.

Ratio strain: the disruption of responding that occurs when a fixed ratio response requirement is increased too rapidly.

Ratio run: high and steady rate of responding that completes each ratio requirement. Usually higher ratio requirement causes longer post-reinforcement pauses to occur.

Partial reinforcement schedules are more resistant to extinction than continuous reinforcement schedules.

Ratio schedules are more resistant than interval schedules and variable schedules more resistant than fixed ones.

Momentary changes in reinforcement value lead to dynamic changes in behaviour.

Compound Schedules

Compound schedules combine two or more different simple schedules in some way using the same reinforcer for the same behaviour. There are many possibilities; among those most often used are:

Alternative schedules. A type of compound schedule where two or more simple schedules are in effect and whichever schedule is completed first results in reinforcement.

Conjunctive schedules. A complex schedule of reinforcement where two or more simple schedules are in effect independently of each other, and requirements on all of the simple schedules must be met for reinforcement.

Multiple schedules. Two or more schedules alternate over time, with a stimulus indicating which is in force. Reinforcement is delivered if the response requirement is met while a schedule is in effect.

Example. FR4 when given a whistle and FI6 when given a bell ring.

Mixed schedules. Either of two, or more, schedules may occur with no stimulus indicating which is in force. Reinforcement is delivered if the response requirement is met while a schedule is in effect.

Example. FI6 and then VR3 without any stimulus warning of the change in schedule.

Concurrent schedules. A complex reinforcement procedure in which the participant can choose any one of two or more simple reinforcement schedules that are available simultaneously. Organisms are free to change back and forth between the response alternatives at any time.

Real world example. changing channels on a television.

Concurrent-chain schedule of reinforcement. A complex reinforcement procedure in which the participant is permitted to choose during the first link which of several simple reinforcement schedules will be in effect in the second link. Once a choice

has been made, the rejected alternatives become unavailable until the start of the next trial.

Interlocking schedules. A single schedule with two components where progress in one component affects progress in the other component. An interlocking FR60–FI120, for example, each response subtracts time from the interval component such that each response is “equal” to removing two seconds from the FI.

Chained schedules. Reinforcement occurs after two or more successive schedules have been completed, with a stimulus indicating when one schedule has been completed and the next has started

Example. FR10 in a green light when completed it goes to a yellow light to indicate FR3, after it is completed it goes into red light to indicate VI6, etc. At the end of the chain, a reinforcer is given.

Tandem schedules. Reinforcement occurs when two or more successive schedule requirements have been completed, with no stimulus indicating when a schedule has been completed and the next has started.

Example. VR10, after it is completed the schedule is changed without warning to FR10, after that it is changed without warning to FR16, etc. At the end of the series of schedules, a reinforcer is finally given.

Higher-order schedules completion of one schedule is reinforced according to a second schedule; *e.g.* in FR2 (FI10 secs), two successive fixed interval schedules require completion before a response is reinforced.

Superimposed Schedules

The psychology term, *superimposed schedules of reinforcement*, refers to a structure of rewards where two or more simple schedules of reinforcement operate simultaneously. Reinforcers can be positive, negative, or both. An example is a person who comes home after a long day at work. The behaviour of opening the front door is rewarded by a big kiss on the lips by the person’s spouse and a rip in the pants from the family dog jumping enthusiastically. Another example of superimposed schedules of reinforcement is a pigeon in an experimental cage pecking at a button. The pecks deliver a hopper of grain every 20th peck, and access to water after every 200 pecks.

Superimposed schedules of reinforcement are a type of compound schedule that evolved from the initial work on simple schedules of reinforcement by B.F. Skinner and his colleagues. They demonstrated that reinforcers could be delivered on schedules, and further that organisms behaved differently under different schedules. Rather than a reinforcer, such as food or water, being delivered every time as a consequence of some behaviour, a reinforcer could be delivered after more than one instance of the behaviour. For example, a pigeon may be required to peck a button switch ten times before food appears. This is a “ratio schedule”. Also, a reinforcer could be delivered after an interval of time passed following a target behaviour. An example is a rat that is given a food pellet immediately following the first response that occurs after two minutes has elapsed since the last lever press. This is called an “interval schedule”.

In addition, ratio schedules can deliver reinforcement following fixed or variable number of behaviours by the individual organism. Likewise, interval schedules can deliver reinforcement following fixed or variable intervals of time following a single response by the organism. Individual behaviours tend to generate response rates that differ based

upon how the reinforcement schedule is created. Much subsequent research in many labs examined the effects on behaviours of scheduling reinforcers.

If an organism is offered the opportunity to choose between or among two or more simple schedules of reinforcement at the same time, the reinforcement structure is called a “concurrent schedule of reinforcement”. Brechner (1974, 1977) introduced the concept of superimposed schedules of reinforcement in an attempt to create a laboratory analogy of social traps, such as when humans overharvest their fisheries or tear down their rainforests. Brechner created a situation where simple reinforcement schedules were superimposed upon each other. In other words, a single response or group of responses by an organism led to multiple consequences. Concurrent schedules of reinforcement can be thought of as “or” schedules, and superimposed schedules of reinforcement can be thought of as “and” schedules. Brechner and Linder and Brechner expanded the concept to describe how superimposed schedules and the social trap analogy could be used to analyze the way energy flows through systems.

Superimposed schedules of reinforcement have many real-world applications in addition to generating social traps. Many different human individual and social situations can be created by superimposing simple reinforcement schedules. For example a human being could have simultaneous tobacco and alcohol addictions. Even more complex situations can be created or simulated by superimposing two or more concurrent schedules. For example, a high school senior could have a choice between going to Stanford University or UCLA, and at the same time have the choice of going into the Army or the Air Force, and simultaneously the choice of taking a job with an internet company or a job with a software company. That is a reinforcement structure of three superimposed concurrent schedules of reinforcement.

Superimposed schedules of reinforcement can create the three classic conflict situations (approach–approach conflict, approach–avoidance conflict, and avoidance–avoidance conflict) described by Kurt Lewin and can operationalise other Lewinian situations analyzed by his force field analysis. Other examples of the use of superimposed schedules of reinforcement as an analytical tool are its application to the contingencies of rent control (Brechner, 2003) and problem of toxic waste dumping in the Los Angeles County storm drain system (Brechner, 2010).

Concurrent Schedules

In operant conditioning, concurrent schedules of reinforcement are schedules of reinforcement that are simultaneously available to an animal subject or human participant, so that the subject or participant can respond on either schedule. For example, in a two-alternative forced choice task, a pigeon in a Skinner box is faced with two pecking keys; pecking responses can be made on either, and food reinforcement might follow a peck on either. The schedules of reinforcement arranged for pecks on the two keys can be different. They may be independent, or they may be linked so that behaviour on one key affects the likelihood of reinforcement on the other.

It is not necessary for responses on the two schedules to be physically distinct. In an alternate way of arranging concurrent schedules, introduced by Findley in 1958, both schedules are arranged on a single key or other response device, and the subject can respond on a second key to change between the schedules. In such a “Findley concurrent” procedure, a stimulus (*e.g.*, the colour of the main key) signals which schedule is in effect.

Concurrent schedules often induce rapid alternation between the keys. To prevent this, a “change over delay” is commonly introduced: each schedule is inactivated for a brief period after the subject switches to it.

When both the concurrent schedules are variable intervals, a quantitative relationship known as the matching law is found between relative response rates in the two schedules and the relative reinforcement rates they deliver; this was first observed by R.J. Herrnstein in 1961. Matching law is a rule for instrumental behaviour which states that the relative rate of responding on a particular response alternative equals the relative rate of reinforcement for that response (rate of behaviour = rate of reinforcement). Animals and humans have a tendency to prefer choice in schedules.

Persuasive Communication & The Reinforcement Theory

Persuasive Communication

Persuasion influences any person the way they think, act and feel. Persuasive skill tells about how people understand the concern, position and needs of the people. Persuasion can be classified into informal persuasion and formal persuasion.

Informal Persuasion

This tells about the way in which a person interacts with his/her colleagues and customers. The informal persuasion can be used in team, memos as well as e-mails.

Formal Persuasion

This type of persuasion is used in writing customer letter, proposal and also for formal presentation to any customer or colleagues.

Process of Persuasion

Persuasion relates how you influence people with your skills, experience, knowledge, leadership, qualities and team capabilities. Persuasion is an interactive process while getting the work done by others. Here are examples for which you can use persuasion skills in real time. Interview: you can prove your best talents, skills and expertise. Clients: to guide your clients for the achievement of the goals or targets. Memos: to express your ideas and views to coworkers for the improvement in the operations. Resistance identification and positive attitude are the vital roles of persuasion.

Persuasion is a form of human interaction. It takes place when one individual expects some particular response from one or more other individuals and deliberately sets out to secure the response through the use of communication. The communicator must realise that different groups have different values.

In instrumental learning situations, which involve operant behaviour, the persuasive communicator will present his message and then wait for the receiver to make a correct response. As soon as the receiver makes the response, the communicator will attempt to fix the response by some appropriate reward or reinforcement.

In conditional learning situations, where there is respondent behaviour, the communicator presents his message so as to elicit the response he wants from the receiver, and the stimulus that originally served to elicit the response then becomes the reinforcing or rewarding element in conditioning.

VERBAL LEARNING

Materials Used in Verbal Learning Experiments

Prose, poetry, passage, lists of words, paired words, lists of numbers are some of the materials which are used in the experiments on verbal learning. It is better to think of verbal materials in terms of units and their characteristics or properties. A word or a nonsense syllable like PEN OR ZEG are examples of word units and nonsense syllable units. The property of a verbal unit may be its meaning or length or difficulty. German psychologist Ebbinghaus was the first to use verbal units known as nonsense syllables in his experiments as early as 1885. Since his time, nonsense syllables have been used increasingly in verbal learning experiments. Besides, in recent years, new techniques have been developed to investigate verbal learning.

Experiments have also been performed, in which single letters have constituted a verbal unit, while others have employed bigrams (two letter syllables); but the use of triagram, or three letter sequence, is very common. Nonsense syllables can be constructed with vowels in between, or only with consonants. The common practice is to form verbal units into lists ranging from 7 to 100 items, depending upon the method used for verbal learning.

Methods of Verbal Learning

Since the use of standard methods for learning makes comparisons of results possible, the methods of verbal learning has great importance. Besides, each method makes the learner use only particular ways of responding in which the experimenter is interested. The four methods which are presently used in verbal learning research are outlined below.

1. Method of Serial Learning

In serial learning used by Ebbinghaus in his experiments, units are presented in the same order from trial to trial, and S must produce them in the order presented. Recitation of alphabets or numbers one to ten or hundred illustrates serial learning. The memory drum is a most convenient apparatus for studies on serial learning because it can expose each verbal unit to S for the desired length of time (varying between one to five seconds). When a memory drum is used learning is carried on usually by a technique known as anticipation method, in which the entire list is once exposed to S and then he is asked to anticipate each item in the list before it is exposed on the memory drum. Thus, when a mark indicating the beginning of the list comes in the window, S has to anticipate the first item of the list; when the first comes he has to anticipate the second, and so on to keep one step ahead. The list is presented again and again and a trial wise record of Ss right and wrong anticipation is kept until he reaches a predetermined criterion of learning—like getting all items correct on two successive trials, or 80 per cent items correct, and so on. The serial method is useful in investigating what are known as chained association (often called vertical associations), because each item in the list serves as a stimulus for the next one, and a response to earlier one.

2. Methods of Paired Association

Paires of verbal units are generally used in a paired association, but sometimes geometrical forms or figures are also used as one of the units. S acquires an association

between the units of each pair so that the second member of the pair can be given when the first one is presented alone. The paired associates learning is an exact counterpart of learning the foreign language equivalents of native language words, where the first member, called stimulus unit, is a native language word and the second word, called response unit, is its equivalent in a foreign language.

3. Free Learning Method

It is the simplest learning task we can present to S, therefore, it is also known as free learning task method. In this, S is presented with a list of 40 to 50 verbal units and is asked to learn them for some specific period of time. After that time, he is asked to write them down. The list is exposed for the second time, the third time and so on, till a desired criterion of learning is achieved by S. This method is specially used in testing long-term and short-term retention of memory, and also in investigating the methods of organising information in memory. The materials can be presented on a memory drum and learned to a desired criterion.

4. Recognition Method

The materials to be learned are given to S in the same way as in the free recall method; but the measures of learning differ. S's job is to recognise the items when they are presented later along with some other items which were not there in the original list. The new items in the list are called distractors. This method can be useful only if the distractors are similar to the originally learned materials. For instance, if the original list is of word units, the distractors should also be word units and not nonsense syllables.

5. Verbal Discrimination Method

This is an unusual method, but it has certain special features which makes it a valuable tool in studying verbal learning. A series of paired words or nonsense syllables are presented to S one by one. In each pair, only one word is arbitrarily decided as correct by the experimenter. It can be either a stimulus word or a response word. S tries to guess the correct one. E tells whether his guess is correct or not, then proceeds to the next pair. The learning is carried on till S is able to guess all the correct items in the list. Note the similarity and differences in this and paired associates learning. In paired associates learning the correct response is presented to S, in this method he has to find it out.

Organisation in Verbal Learning

From verbal learning it has been observed that meaningfulness of lists sometimes comes from relationships existing among the items in the list. Such meaningfulness is called organisation, which may be provided by the materials themselves or may develop often by active thinking on the part of the learner about the inherent relationship between the items or between the items and the learner's own experiences. Two general kinds of organisations may be demonstrated. (1) conceptual, and (2) hierarchical.

A conceptual organisation is a class that includes a number of verbal units. If the materials to be learned are chosen in such a way that its units can fall into a few well defined categories, learning is greatly facilitated. For instance, while learning, a subject is likely to organise his units in some conceptual categories and remember them. This may actually speed up his learning. For example, the following list from Underwood's

study proves that the lists which facilitate conceptual categories may be learned faster than those which do not facilitate such an organisation. The words in the list were: Bob, Rabbi, Cow, Rumbu, Bill, Priest, Horse, Frustrat, Joe, Bishop, Dog, Tango, John, Minister, Cat, Waltz. Most Ss recalled the lists category by category. For instance, Bob, Bill, Joe, John, Cow, Horse, Dog, Cat and so on. One can see that the first four words are clustered because they are proper nouns, and the remaining because they are the names of animals. The organisation, resulting from a clustering of the verbal unit on the basis of identifiable cues, may thus help speed up learning.

Memory is our ability to encode, store, retain and subsequently recall information and past experiences in the human brain.

MEMORY AND FORGETTING

Memory is our ability to encode, store, retain and subsequently recall information and past experiences in the human brain.

It is the sum total of what we remember, and gives us the capability to learn and adapt from previous experiences as well as to build relationships. It is the ability to remember past experiences, and the power or process of recalling to mind previously learned facts, experiences, impressions, skills and habits. It is the store of things learned and retained from our activity or experience, as evidenced by modification of structure or behaviour, or by recall and recognition.

Etymologically, the modern English word "memory" comes to us from the Middle English *memorie*, which in turn comes from the Anglo-French *memoire* or *memorie*, and ultimately from the Latin *memoria* and *memor*, meaning "mindful" or "remembering".

In more physiological or neurological terms, memory is, at its simplest, a set of encoded neural connections in the brain. It is the re-creation or reconstruction of past experiences by the synchronous firing of neurons that were involved in the original experience. As we will see, though, because of the way in which memory is encoded, it is perhaps better thought of as a kind of collage or jigsaw puzzle, rather than in the traditional manner as a collection of recordings or pictures or video clips, stored as discrete wholes. Our memories are not stored in our brains like books on library shelves, but are actually on-the-fly reconstructions from elements scattered throughout various areas of our brains.

DID YOU KNOW

Recent studies suggest that repeated bouts of jet lag may cause harm to the temporal lobe, an area of the brain important to memory, causing it to shrink in size, and compromising performance on spatial memory tests.

It is thought that stress hormones, such as cortisol, released by the body during times of stress (such as the sleep disturbance, general stress and fatigue caused by long flights) are responsible for this impairment of memory and other mental skills.

Memory is related to but distinct from learning, which is the process by which we acquire knowledge of the world and modify our subsequent behaviour. During learning, neurons that fire together to produce a particular experience are altered so that they have a tendency to fire together again. For example, we learn a new language by studying it, but we then speak it by using our memory to retrieve the words that we have learned.

Thus, memory depends on learning because it lets us store and retrieve learned information. But learning also depends to some extent on memory, in that the knowledge stored in our memory provides the framework to which new knowledge is linked by association and inference. This ability of humans to call on past memories in order to imagine the future and to plan future courses of action is a hugely advantageous attribute in our survival and development as a species.

Since the development of the computer in the 1940s, memory is also used to describe the capacity of a computer to store information subject to recall, as well as the physical components of the computer in which such information is stored. Although there are indeed some parallels between the memory of a computer and the memory of a human being, there are also some fundamental and crucial differences, principally that the human brain is organised as a distributed network in which each brain cell makes thousands of connections, rather than as an addressable collection of discrete files.

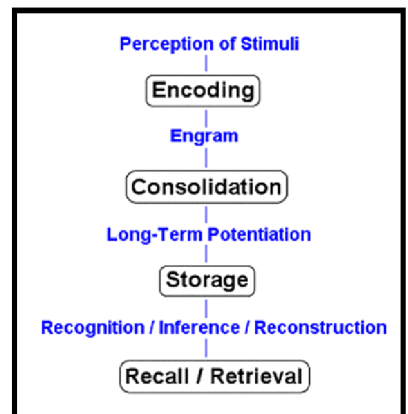
The sociological concept of collective memory plays an essential role in the establishment of human societies. Every social group perpetuates itself through the knowledge that it transmits down the generations, either through oral tradition or through writing. The invention of writing made it possible for the first time for human beings to preserve precise records of their knowledge outside of their brains. Writing, audiovisual media and computer records can be considered a kind of external memory for humans.

MEMORY PROCESSES

We have already looked at the different stages of memory formation (from perception to sensory memory to short-term memory to long-term memory) in the section on Types of Memory. This section, however, looks at the overall processes involved.

Memory is the ability to encode, store and recall information. The three main processes involved in human memory are therefore encoding, storage and recall (retrieval). Additionally, the process of memory consolidation (which can be considered to be either part of the encoding process or the storage process) is treated here as a separate process in its own right.

Some of the physiology and neurology involved in these processes is highly complex and technical (and some of it still not completely understood), and lies largely outside the remit of this entry level guide, although at least a general introduction is given here. More information on the architecture of the human brain, and the neurological processes by which memory is encoded, stored and recalled can be found in the section on Memory and the Brain.



**MEMORY PROCESSES: DIAGRAM
BY LUKE MASTIN**

Memory Encoding

Encoding is the crucial first step to creating a new memory. It allows the perceived item of interest to be converted into a construct that can be stored within the brain, and then recalled later from short-term or long-term memory.

Encoding is a biological event beginning with perception through the senses. The process of laying down a memory begins with attention (regulated by the thalamus and the frontal lobe), in which a memorable event causes neurons to fire more frequently, making the experience more intense and increasing the likelihood that the event is encoded as a memory. Emotion tends to increase attention, and the emotional element of an event is processed on an unconscious pathway in the brain leading to the amygdala. Only then are the actual sensations derived from an event processed.

The perceived sensations are decoded in the various sensory areas of the cortex, and then combined in the brain's hippocampus into one single experience. The hippocampus is then responsible for analyzing these inputs and ultimately deciding if they will be committed to long-term memory. It acts as a kind of sorting centre where the new sensations are compared and associated with previously recorded ones. The various threads of information are then stored in various different parts of the brain, although the exact way in which these pieces are identified and recalled later remains largely unknown. The key role that the hippocampus plays in memory encoding has been highlighted by examples of individuals who have had their hippocampus damaged or removed and can no longer create new memories. It is also one of the few areas of the brain where completely new neurons can grow.

DID YOU KNOW

Studies suggest that characteristics of the environment are encoded as part of the memory trace, and can be used to enhance retrieval of the other information in the trace.

In other words, you can recall more when the environments are similar in both the learning (encoding) and recall phases.

Thus, deep-sea divers tend to remember their training more effectively when trained underwater rather than on land, and students perform better on exams by studying in silence, because exams are usually done in silence.

Although the exact mechanism is not completely understood, encoding occurs on different levels, the first step being the formation of short-term memory from the ultra-short term sensory memory, followed by the conversion to a long-term memory by a process of memory consolidation. The process begins with the creation of a memory trace or engram in response to the external stimuli. An engram is a hypothetical biophysical or biochemical change in the neurons of the brain, hypothetical in the respect that no-one has ever actually seen, or even proved the existence of, such a construct.

An organ called the hippocampus, deep within the medial temporal lobe of the brain, receives connections from the primary sensory areas of the cortex, as well as from associative areas and the rhinal and entorhinal cortexes. While these anterograde connections converge at the hippocampus, other retrograde pathways emerge from it, returning to the primary cortexes. A neural network of cortical synapses effectively records the various associations which are linked to the individual memory.

There are three or four main types of encoding:

1. Acoustic encoding is the processing and encoding of sound, words and other auditory input for storage and later retrieval. This is aided by the concept of the phonological loop, which allows input within our echoic memory to be sub-vocally rehearsed in order to facilitate remembering.

2. Visual encoding is the process of encoding images and visual sensory information. Visual sensory information is temporarily stored within the iconic memory before being encoded into long-term storage. The amygdala (within the medial temporal lobe of the brain which has a primary role in the processing of emotional reactions) fulfills an important role in visual encoding, as it accepts visual input in addition to input from other systems and encodes the positive or negative values of conditioned stimuli.
3. Tactile encoding is the encoding of how something feels, normally through the sense of touch. Physiologically, neurons in the primary somatosensory cortex of the brain react to vibrotactile stimuli caused by the feel of an object.
4. Semantic encoding is the process of encoding sensory input that has particular meaning or can be applied to a particular context, rather than deriving from a particular sense.

It is believed that, in general, encoding for short-term memory storage in the brain relies primarily on acoustic encoding, while encoding for long-term storage is more reliant (although not exclusively) on semantic encoding.

DID YOU KNOW

In a positive example of disfluency (the subjective feeling of difficulty associated with any mental task), a recent study at Princeton University has shown that students learning new material printed in a difficult-to-read font or typeface were able to recall significantly more than those learning the same material in a font considered easy to read.

It is believed that presenting information in a way that is hard to digest means that a person has to concentrate more, leading to deeper processing and therefore better retrieval afterwards.

Human memory is fundamentally associative, meaning that a new piece of information is remembered better if it can be associated with previously acquired knowledge that is already firmly anchored in memory. The more personally meaningful the association, the more effective the encoding and consolidation. Elaborate processing that emphasises meaning and associations that are familiar tends to lead to improved recall. On the other hand, information that a person finds difficult to understand cannot be readily associated with already acquired knowledge, and so will usually be poorly remembered, and may even be remembered in a distorted form due to the effort to comprehend its meaning and associations. For example, given a list of words like "thread", "sewing", "haystack", "sharp", "point", "syringe", "pin", "pierce", "injection" and "knitting", people often also (incorrectly) remember the word "needle" through a process of association.

Because of the associative nature of memory, encoding can be improved by a strategy of organisation of memory called elaboration, in which new pieces of information are associated with other information already recorded in long-term memory, thus incorporating them into a broader, coherent narrative which is already familiar. An example of this kind of elaboration is the use of mnemonics, which are verbal, visual or auditory associations with other, easy-to-remember constructs, which can then be related back to the data that is to be remembered. Rhymes, acronymns, acrostics and codes can all be used in this way. Common examples are "Roy G. Biv" to remember the order of the colours of the rainbow, or "Every Good Boy Deserves Favour" for the musical notes on the lines of the treble clef, which most people find easier to remember than the original list of colours

or letters. When we use mnemonic devices, we are effectively passing facts through the hippocampus several times, so that it can keep strengthening the associations, and therefore improve the likelihood of subsequent memory recall.

In the same way, associating words with images is another commonly used mnemonic device, providing two alternative methods of remembering, and creating additional associations in the mind. Taking this to a higher level, another method of improving memory encoding and consolidation is the use of a so-called memory palace (also known as the method of loci), a mnemonic techniques that relies on memorised spatial relationships to establish, order and recollect other memories. The method is to assign objects or facts to different rooms in an imaginary house or palace, so that recall of the facts can be cued by mentally "walking through" the palace until it is found. Many top memorisers today use the memory palace method to a greater or lesser degree. Similar techniques involve placing the items at different landmarks on a favourite hike or trip (known as the journey method), or weaving them into a story.

The old and popular notion of the brain as a kind of "muscle" which strengthens with repeated use (also known as faculty theory) is now largely discredited. Research, dating back to William James towards the end of the 19th Century, shows that long hours spent memorising does not build up the powers of memory at all, and, on the contrary, may even diminish it. This is not to say that individual memories cannot be strengthened by repetition, but that, as James found, daily training in the memorisation of a poetry of one author, for example, does not improve a person's ability to learn the poetry of another author, or poetry in general.

Many studies have shown that the most vivid autobiographical memories tend to be of emotional events, which are likely to be recalled more often and with more clarity and detail than neutral events. One theory suggests that high levels of emotional arousal lead to attention narrowing, where the range of sensitive cues from the stimulus and its environment is decreased, so that information central to the source of the emotional arousal is strongly encoded while peripheral details are not (e.g. the so-called "weapon focus effect", in which witnesses to a crime tend to remember the gun or knife in great detail, but not other more peripheral details such as the perpetrator's clothing or vehicle).

Memory Consolidation

Consolidation is the processes of stabilising a memory trace after the initial acquisition. It may perhaps be thought of part of the process of encoding or of storage, or it may be considered as a memory process in its own right. It is usually considered to consist of two specific processes, synaptic consolidation (which occurs within the first few hours after learning or encoding) and system consolidation (where hippocampus-dependent memories become independent of the hippocampus over a period of weeks to years).

Neurologically, the process of consolidation utilises a phenomenon called long-term potentiation, which allows a synapse to increase in strength as increasing numbers of signals are transmitted between the two neurons. Potentiation is the process by which synchronous firing of neurons makes those neurons more inclined to fire together in the future. Long-term potentiation occurs when the same group of neurons fire together so often that they become permanently sensitised to each other. As new experiences accumulate, the brain creates more and more connections and pathways, and may "re-wire" itself by re-routing connections and re-arranging its organisation.

As such a neuronal pathway, or neural network, is traversed over and over again, an enduring pattern is engraved and neural messages are more likely to flow along such familiar paths of least resistance. For example, if a piece of music is played over and over, the repeated firing of certain cells in a certain order in your brain makes it easier to repeat this firing later on, with the result that the musician becomes better at playing the music, and can play it faster, with fewer mistakes.

In this way, the brain organises and reorganises itself in response to experiences, creating new memories prompted by experience, education or training. The ability of the connection, or synapse, between two neurons to change in strength, and for lasting changes to occur in the efficiency of synaptic transmission, is known as synaptic plasticity or neural plasticity, and it is one of the important neurochemical foundations of memory and learning.

DID YOU KNOW

Studies have shown that we often construct our memories after the fact, and that we are susceptible to suggestions from others that help us fill in the gaps in our memories.

This malleability of memory is why, for example, a police officer investigating a crime should not show a picture of a single individual to a victim and ask if the victim recognises the assailant.

If the victim is then presented with a line-up and picks out the individual whose picture the victim had been shown, there is no real way of knowing whether the victim is actually remembering the assailant or just the picture.

It should be remembered that each neuron makes thousands of connections with other neurons, and memories and neural connections are mutually interconnected in extremely complex ways. Unlike the functioning of a computer, each memory is embedded in many connections, and each connection is involved in several memories. Thus, multiple memories may be encoded within a single neural network, by different patterns of synaptic connections. Conversely, a single memory may involve simultaneously activating several different groups of neurons in completely different parts of the brain.

The inverse of long-term potentiation, known as long-term depression, can also take place, whereby the neural networks involved in erroneous movements are inhibited by the silencing of their synaptic connections. This can occur in the cerebellum, which is located towards the back of the brain, in order to correct our motor procedures when learning how to perform a task (procedural memory), but also in the synapses of the cortex, the hippocampus, the striatum and other memory-related structures.

Contrary to long-term potentiation, which is triggered by high-frequency stimulation of the synapses, long-term depression is produced by nerve impulses reaching the synapses at very low frequencies, leading them to undergo the reverse transformation from long-term potentiation, and, instead of becoming more efficient, the synaptic connections are weakened. It is still not clear whether long-term depression contributes directly to the storage of memories in some way, or whether it simply makes us forget the traces of some things learned long ago so that new things can be learned.

Sleep (particularly slow-wave, or deep, sleep, during the first few hours) is also thought to be important in improving the consolidation of information in memory, and activation patterns in the sleeping brain, which mirror those recorded during the learning of tasks

from the previous day, suggest that new memories may be solidified through such reactivation and rehearsal.

Memory re-consolidation is the process of previously consolidated memories being recalled and then actively consolidated all over again, in order to maintain, strengthen and modify memories that are already stored in the long-term memory. Several retrievals of memory (either naturally through reflection, or through deliberate recall) may be needed for long-term memories to last for many years, depending on the depth of the initial processing. However, these individual retrievals can take place at increasing intervals, in accordance with the principle of spaced repetition (this is familiar to us in the way that "cramming" the night before an exam is not as effective as studying at intervals over a much longer span of time).

The very act of re-consolidation, though, may change the initial memory. As a particular memory trace is reactivated, the strengths of the neural connections may change, the memory may become associated with new emotional or environmental conditions or subsequently acquired knowledge, expectations rather than actual events may become incorporated into the memory, etc.

Research into a cognitive disorder known as Korsakoff's syndrome shows that the retrograde amnesia of sufferers follows a distinct temporal curve, in that the more remote the event in the past, the better it is preserved. This suggests that the more recent memories are not fully consolidated and therefore more vulnerable to loss, indicating that the process of consolidation may continue for much longer than initially thought, perhaps for many years.

Memory Storage

Storage is the more or less passive process of retaining information in the brain, whether in the sensory memory, the short-term memory or the more permanent long-term memory. Each of these different stages of human memory function as a sort of filter that helps to protect us from the flood of information that confront us on a daily basis, avoiding an overload of information and helping to keep us sane. The more the information is repeated or used, the more likely it is to be retained in long-term memory (which is why, for example, studying helps people to perform better on tests). This process of consolidation, the stabilising of a memory trace after its initial acquisition, is treated in more detail in a separate section.

Since the early neurological work of Karl Lashley and Wilder Penfield in the 1950s and 1960s, it has become clear that long-term memories are not stored in just one part of the brain, but are widely distributed throughout the cortex. After consolidation, long-term memories are stored throughout the brain as groups of neurons that are primed to fire together in the same pattern that created the original experience, and each component of a memory is stored in the brain area that initiated it (e.g. groups of neurons in the visual cortex store a sight, neurons in the amygdala store the associated emotion, etc.). Indeed, it seems that they may even be encoded redundantly, several times, in various parts of the cortex, so that, if one engram (or memory trace) is wiped out, there are duplicates, or alternative pathways, elsewhere, through which the memory may still be retrieved.

Therefore, contrary to the popular notion, memories are not stored in our brains like books on library shelves, but must be actively reconstructed from elements scattered

throughout various areas of the brain by the encoding process. Memory storage is therefore an ongoing process of reclassification resulting from continuous changes in our neural pathways, and parallel processing of information in our brains.

The indications are that, in the absence of disorders due to trauma or neurological disease, the human brain has the capacity to store almost unlimited amounts of information indefinitely. Forgetting, therefore, is more likely to be result from incorrectly or incompletely encoded memories, and/or problems with the recall/retrieval process. It is a common experience that we may try to remember something one time and fail, but then remember that same item later. The information is therefore clearly still there in storage, but there may have been some kind of a mismatch between retrieval cues and the original encoding of the information. "Lost" memories recalled with the aid of psychotherapy or hypnosis are other examples supporting this idea, although it is difficult to be sure that such memories are real and not implanted by the treatment.

DID YOU KNOW

Genuine eidetic or photographic memory (an "unprocessed" sensory memory of sensory events that is as accurate as if the person were still experiencing the original objects or events) is extremely rare, although not unheard of.

Most extraordinary memory skills which make claims of photographic memory, however, result from a combination of innate skills, learned tactics, mnemonic devices and extraordinary knowledge bases, rather than eidetic memory as such.

Young children, with their lack of verbal and conceptual systems, are more likely to have real eidetic memories than adults.

A classic case from the 1970s was a woman named Elizabeth, who was able to write out poetry in a foreign language (of which she had no prior knowledge) several years after seeing the original text.

Having said that, though, it seems unlikely that, as Richard Schiffrin and others have claimed, ALL memories are stored somewhere in the brain, and that it is only in the retrieval process that irrelevant details are "fast-forwarded" over or expurgated. It seems more likely that the memories which are stored are in some way edited and sorted, and that some of the more peripheral details are never stored.

Forgetting, then, is perhaps better thought of as the temporary or permanent inability to retrieve a piece of information or a memory that had previously been recorded in the brain. Forgetting typically follows a logarithmic curve, so that information loss is quite rapid at the start, but becomes slower as time goes on. In particular, information that has been learned very well (e.g. names, facts, foreign-language vocabulary, etc.), will usually be very resistant to forgetting, especially after the first three years. Unlike amnesia, forgetting is usually regarded as a normal phenomenon involving specific pieces of content, rather than relatively broad categories of memories or even entire segments of memory.

Theorists disagree over exactly what becomes of material that is forgotten. Some hold that long-term memories do actually decay and disappear completely over time; others hold that the memory trace remains intact as long as we live, but the bonds or cues that allow us to retrieve the trace become broken, due to changes in the organisation of the neural network, new experiences, etc., in the same way as a misplaced book in a library is "lost" even though it still exists somewhere in the library.

Interestingly, it appears not to be possible to deliberately delete memories at will, which can have negative consequences, for example if we experience traumatic events we would actually prefer to forget. In fact, such memories tend to be imprinted even more strongly than normal due to their emotional content, although recent research involving the use of beta blockers (such as propranolol) suggests that it may be possible to tone down the emotional aspects of such memories, even if the memories themselves cannot be erased. The way this works is that the act of recalling stored memories makes them "malleable" once more, as they were during the initial encoding phase, and their re-storage can then be blocked by drugs which inhibit the proteins that enable the emotional memory to be re-saved.

Memory Recall/Retrieval

Recall or retrieval of memory refers to the subsequent re-accessing of events or information from the past, which have been previously encoded and stored in the brain. In common parlance, it is known as remembering. During recall, the brain "replays" a pattern of neural activity that was originally generated in response to a particular event, echoing the brain's perception of the real event. In fact, there is no real solid distinction between the act of remembering and the act of thinking.

These replays are not quite identical to the original, though - otherwise we would not know the difference between the genuine experience and the memory - but are mixed with an awareness of the current situation. One corollary of this is that memories are not frozen in time, and new information and suggestions may become incorporated into old memories over time. Thus, remembering can be thought of as an act of creative reimagination.

Because of the way memories are encoded and stored, memory recall is effectively an on-the-fly reconstruction of elements scattered throughout various areas of our brains. Memories are not stored in our brains like books on library shelves, or even as a collection of self-contained recordings or pictures or video clips, but may be better thought of as a kind of collage or a jigsaw puzzle, involving different elements stored in disparate parts of the brain linked together by associations and neural networks. Memory retrieval therefore requires re-visiting the nerve pathways the brain formed when encoding the memory, and the strength of those pathways determines how quickly the memory can be recalled. Recall effectively returns a memory from long-term storage to short-term or working memory, where it can be accessed, in a kind of mirror image of the encoding process. It is then re-stored back in long-term memory, thus re-consolidating and strengthening it.

DID YOU KNOW

Memory recall appears to be state-dependent, at least to some extent.

Studies have shown that, when material is learned under the influence of a drug or alcohol, for example, it is subsequently recalled better when in the same drug state than when sober.

Similarly, individuals tend to retrieve information more easily when it has the same emotional content as their current emotional state, and when the emotional state at the time of retrieval is similar to the emotional state at the time of encoding.

The efficiency of human memory recall is astounding. Most of what we remember is by direct retrieval, where items of information are linked directly to a question or cue, rather than by the kind of sequential scan a computer might use (which would require a systematic search through the entire contents of memory until a match is found). Other memories are retrieved quickly and efficiently by hierarchical inference, where a specific question is linked to a class or subset of information about which certain facts are known. Also, the brain is usually able to determine in advance whether there is any point in searching memory for a particular fact (e.g. it instantly recognises a question like "What is Socrates' telephone number?" as absurd in that no search could ever produce an answer).

There are two main methods of accessing memory: recognition and recall. Recognition is the association of an event or physical object with one previously experienced or encountered, and involves a process of comparison of information with memory, e.g. recognising a known face, true/false or multiple choice questions, etc. Recognition is a largely unconscious process, and the brain even has a dedicated face-recognition area, which passes information directly through the limbic areas to generate a sense of familiarity, before linking up with the cortical path, where data about the person's movements and intentions are processed. Recall involves remembering a fact, event or object that is not currently physically present (in the sense of retrieving a representation, mental image or concept), and requires the direct uncovering of information from memory, e.g. remembering the name of a recognised person, fill-in the blank questions, etc.

Recognition is usually considered to be "superior" to recall (in the sense of being more effective), in that it requires just a single process rather than two processes. Recognition requires only a simple familiarity decision, whereas a full recall of an item from memory requires a two-stage process (indeed, this is often referred to as the two-stage theory of memory) in which the search and retrieval of candidate items from memory is followed by a familiarity decision where the correct information is chosen from the candidates retrieved. Thus, recall involves actively reconstructing the information and requires the activation of all the neurons involved in the memory in question, whereas recognition only requires a relatively simple decision as to whether one thing among others has been encountered before. Sometimes, however, even if a part of an object initially activates only a part of the neural network concerned, recognition may then suffice to activate the entire network.

In the 1980s, Endel Tulving proposed an alternative to the two-stage theory, which he called the theory of encoding specificity. This theory states that memory utilises information both from the specific memory trace as well as from the environment in which it is retrieved. Because of its focus on the retrieval environment or state, encoding specificity takes into account context cues, and it also has some advantages over the two-stage theory as it accounts for the fact that, in practice, recognition is not actually always superior to recall. Typically, recall is better when the environments are similar in both the learning (encoding) and recall phases, suggesting that context cues are important. In the same way, emotional material is remembered more reliably in moods that match the emotional content of these memories (e.g. happy people will remember more happy than sad information, whereas sad people will better remember sad than happy information).

According to the levels-of-processing effect theory, another alternative theory of memory suggested by Fergus Craik and Robert Lockhart, memory recall of stimuli is also a function of the depth of mental processing, which is in turn determined by connections with

pre-existing memory, time spent processing the stimulus, cognitive effort and sensory input mode. Thus, shallow processing (such as, typically, that based on sound or writing) leads to a relatively fragile memory trace that is susceptible to rapid decay, whereas deep processing (such as that based on semantics and meanings) results in a more durable memory trace. This theory suggests, then, that memory strength is continuously variable, as opposed to the earlier Atkinson-Shiffrin, or multi-store, memory model, which just involves a sequence of three discrete stages, from sensory to short-term to long-term memory.

The evidence suggests that memory retrieval is a more or less automatic process. Thus, although distraction or divided attention at the time of recall tends to slow down the retrieval process to some extent, it typically has little to no effect on the accuracy of retrieved memories. Distraction at the time of encoding, on the other hand, can severely impair subsequent retrieval success.

The efficiency of memory recall can be increased to some extent by making inferences from our personal stockpile of world knowledge, and by our use of schema (plural: schemata). A schema is an organised mental structure or framework of pre-conceived ideas about the world and how it works, which we can use to make realistic inferences and assumptions about how to interpret and process information. Thus, our everyday communication consists not just of words and their meanings, but also of what is left out and mutually understood (e.g. if someone says "it is 3 o'clock", our knowledge of the world usually allows us to know automatically whether it is 3 am or 3 pm). Such schemata are also applied to recalled memories, so that we can often flesh out details of a memory from just a skeleton memory of a central event or object. However, the use of schemata may also lead to memory errors as assumed or expected associated events are added that did not actually occur.

There are three main types of recall:

Free recall is the process in which a person is given a list of items to remember and then is asked to recall them in any order (hence the name "free"). This type of recall often displays evidence of either the primacy effect (when the person recalls items presented at the beginning of the list earlier and more often) or the recency effect (when the person recalls items presented at the end of the list earlier and more often), and also of the contiguity effect (the marked tendency for items from neighbouring positions in the list to be recalled successively).

Cued recall is the process in which a person is given a list of items to remember and is then tested with the use of cues or guides. When cues are provided to a person, they tend to remember items on the list that they did not originally recall without a cue, and which were thought to be lost to memory. This can also take the form of stimulus-response recall, as when words, pictures and numbers are presented together in a pair, and the resulting associations between the two items cues the recall of the second item in the pair.

Serial recall refers to our ability to recall items or events in the order in which they occurred, whether chronological events in our autobiographical memories, or the order of the different parts of a sentence (or phonemes in a word) in order to make sense of them. Serial recall in long-term memory appears to differ from serial recall in short-term memory, in that a sequence in long-term memory is represented in memory as a whole, rather than as a series of discrete items. Testing of serial recall by psychologists have yielded several general rules:

more recent events are more easily remembered in order (especially with auditory stimuli);
recall decreases as the length of the list or sequence increases;
there is a tendency to remember the correct items, but in the wrong order;
where errors are made, there is a tendency to respond with an item that resembles the original item in some way (e.g. "dog" instead of "fog", or perhaps an item physically close to the original item);
repetition errors do occur, but they are relatively rare;

DID YOU KNOW

Several recent studies in the growing area of neuro-education have shown the value of the "testing effect" (or "retrieval effect"), where quizzes a short time after initial learning significantly improves subsequent retrieval of facts and ideas, as well as overall understanding of topics and the ability to solve related problems.

Testing helps protect against "proactive interference" (the familiar feeling of being overwhelmed by too much information), and the studies suggest that a quick test is much more effective than an extra hour of study or re-reading.

if an item is recalled earlier in the list than it should be, the missed item tends to be inserted immediately after it;

if an item from a previous trial is recalled in a current trial, it is likely to be recalled at its position from the original trial.

If we assume that the "purpose" of human memory is to use past events to guide future actions, then keeping a perfect and complete record of every past event is not necessarily a useful or efficient way of achieving this. So, in most people, some specific memories may be given up or converted into general knowledge (i.e. converted from episodic to semantic memories) as part of the ongoing recall/re-consolidation process, so that we are able to generalise from experience. It is also possible that false or wrongly interpreted memories may be created during recall, and carried forward thereafter. One can also, up to a point, choose to forget, by blocking out unwanted memories during recall.

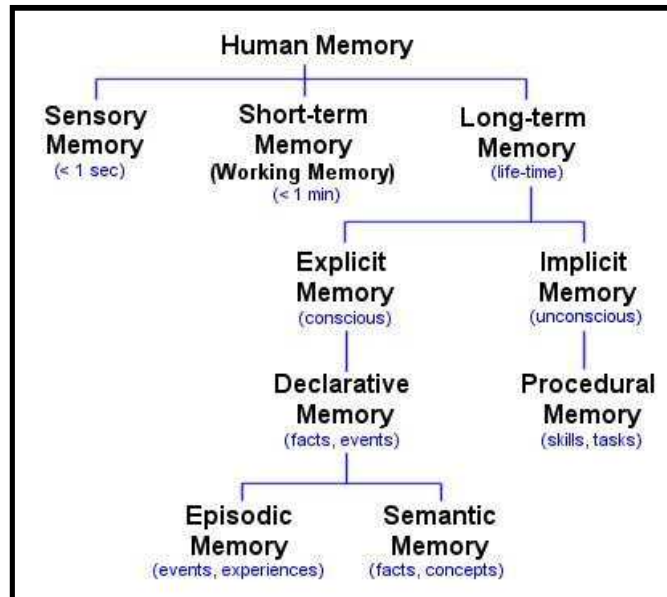
However, there is a rare condition called hyperthymesia (also known as hypermnnesia or superior autobiographical memory) in which a few people show an extraordinary capacity to recall detailed specific events from a person's personal past, without relying on practised mnemonic strategies. Although only a handful of cases of hyperthymesia have ever been definitively confirmed, some of these cases are quite startling, such as a California woman who could recall every day in complete detail from the age of 14 onwards, a young English girl with an IQ of 191 who had a perfect photographic memory spanning almost 18 years, and a Russian man known simply as "S." who was only able to forget anything by a deliberate act of will. One of the most famous cases, known as "A.J.", described it as a burden rather than a gift, but others seem to be able to organise and compartmentalise their prodigious memories and do not appear to feel that their brains are "cluttered" with excess information. There is a good "60 Minutes" documentary on the subject at. Interestingly, recent research has shown that such individuals tend to have significantly larger than average temporal lobes and caudate nuclei, and many exhibit mild Obsessive Compulsive Disorder-like behaviour (the caudate nucleus is also associated with OCD).

STAGE OF MEMORY

What we usually think of as "memory" in day-to-day usage is actually long-term memory, but there are also important short-term and sensory memory processes, which must be worked through before a long-term memory can be established. The different types of memory each have their own particular mode of operation, but they all cooperate in the process of memorisation, and can be seen as three necessary steps in forming a lasting memory.

This model of memory as a sequence of three stages, from sensory to short-term to long-term memory, rather than as a unitary process, is known as the modal or multi-store or Atkinson-Shiffrin model, after Richard Atkinson and Richard Shiffrin who developed it in 1968, and it remains the most popular model for studying memory. It is often also described as the process of memory, but I have used this description for the processes of encoding, consolidation, storage and recall in the separate Memory Processes section.

It should be noted that an alternative model, known as the levels-of-processing model was proposed by Fergus Craik and Robert Lockhart in 1972, and posits that memory recall, and the extent to which something is memorised, is a function of the depth of mental processing, on a continuous scale from shallow (perceptual) to deep (semantic). Under this model, there is no real structure to memory and no distinction between short-term and long-term memory.



TYPES OF HUMAN MEMORY: DIAGRAM BY LUKE MASTIN

Sensory Memory

Sensory memory is the shortest-term element of memory. It is the ability to retain impressions of sensory information after the original stimuli have ended. It acts as a kind of buffer for stimuli received through the five senses of sight, hearing, smell, taste and touch, which are retained accurately, but very briefly. For example, the ability to look at

something and remember what it looked like with just a second of observation is an example of sensory memory.

The stimuli detected by our senses can be either deliberately ignored, in which case they disappear almost instantaneously, or perceived, in which case they enter our sensory memory. This does not require any conscious attention and, indeed, is usually considered to be totally outside of conscious control. The brain is designed to only process information that will be useful at a later date, and to allow the rest to pass by unnoted. As information is perceived, it is therefore stored in sensory memory automatically and unbidden. Unlike other types of memory, the sensory memory cannot be prolonged via rehearsal.

DID YOU KNOW

Studies have shown that attention significantly affects memory during the encoding phase, but hardly at all during recall.

Thus, distractions or divided attention during initial learning may severely impair subsequent retrieval success, whereas distractions at the time of recall may slow down the process a little, but has little to no effect on its accuracy.

Sensory memory is an ultra-short-term memory and decays or degrades very quickly, typically in the region of 200 - 500 milliseconds ($1/5$ - $1/2$ second) after the perception of an item, and certainly less than a second (although echoic memory is now thought to last a little longer, up to perhaps three or four seconds). Indeed, it lasts for such a short time that it is often considered part of the process of perception, but it nevertheless represents an essential step for storing information in short-term memory.

The sensory memory for visual stimuli is sometimes known as the iconic memory, the memory for aural stimuli is known as the echoic memory, and that for touch as the haptic memory. Smell may actually be even more closely linked to memory than the other senses, possibly because the olfactory bulb and olfactory cortex (where smell sensations are processed) are physically very close - separated by just 2 or 3 synapses - to the hippocampus and amygdala (which are involved in memory processes). Thus, smells may be more quickly and more strongly associated with memories and their associated emotions than the other senses, and memories of a smell may persist for longer, even without constant re-consolidation.

Experiments by George Sperling in the early 1960s involving the flashing of a grid of letters for a very short period of time (50 milliseconds) suggest that the upper limit of sensory memory (as distinct from short-term memory) is approximately 12 items, although participants often reported that they seemed to "see" more than they could actually report.

Information is passed from the sensory memory into short-term memory via the process of attention (the cognitive process of selectively concentrating on one aspect of the environment while ignoring other things), which effectively filters the stimuli to only those which are of interest at any given time.

Short-term (Working) Memory

People with a good working memory, on the other hand, are more likely to be optimistic and self-assured, and more likely to lead a happy and successful life.

Short-term memory acts as a kind of "scratch-pad" for temporary recall of the information which is being processed at any point in time, and has been referred to as "the brain's Post-it note". It can be thought of as the ability to remember and process information at the same time. It holds a small amount of information (typically around 7 items or even less) in mind in an active, readily available state for a short period of time (typically from 10 to 15 seconds, or sometimes up to a minute).

For example, in order to understand this sentence, the beginning of the sentence needs to be held in mind while the rest is read, a task which is carried out by the short-term memory. Other common examples of short-term memory in action are the holding on to a piece of information temporarily in order to complete a task and simultaneous translation (where the interpreter must store information in one language while orally translating it into another). What is actually held in short-term memory, though, is not complete concepts, but rather links or pointers (such as words, for example) which the brain can flesh out from its other accumulated knowledge.

However, this information will quickly disappear forever unless we make a conscious effort to retain it, and short-term memory is a necessary step toward the next stage of retention, long-term memory. The transfer of information to long-term memory for more permanent storage can be facilitated or improved by mental repetition of the information or, even more effectively, by giving it a meaning and associating it with other previously acquired knowledge. Motivation is also a consideration, in that information relating to a subject of strong interest to a person, is more likely to be retained in long-term memory.

The term working memory is often used interchangeably with short-term memory, although technically working memory refers more to the whole theoretical framework of structures and processes used for the temporary storage and manipulation of information, of which short-term memory is just one component.

The central executive part of the prefrontal cortex at the front of the brain appears to play a fundamental role in short-term and working memory. It both serves as a temporary store for short-term memory, where information is kept available while it is needed for current reasoning processes, but it also "calls up" information from elsewhere in the brain. The central executive controls two neural loops, one for visual data (which activates areas near the visual cortex of the brain and acts as a visual scratch pad), and one for language (the "phonological loop", which uses Broca's area as a kind of "inner voice" that repeats word sounds to keep them in mind). These two scratch pads temporarily hold data until it is erased by the next job.

DID YOU KNOW

A recent study at the University of Michigan suggests that attention and short-term memory processing are directly affected by a person's surroundings and environment.

Two groups of individuals were tested on their attention and working memory performance, one group after a relaxed walk in a quiet park and the other group after navigating busy city streets.

Those who had been walking the city streets scored far lower on the tests.

Although the prefrontal cortex is not the only part of the brain involved – it must also cooperate with other parts of the cortex from which it extracts information for brief periods – it is the most important, and Carlyle Jacobsen reported, as early as 1935, that damage to the prefrontal cortex in primates caused short-term memory deficits.

The short-term memory has a limited capacity, which can be readily illustrated by the simple expedient of trying to remember a list of random items (without allowing repetition or reinforcement) and seeing when errors begin to creep in. The often-cited experiments by George Miller in 1956 suggest that the number of objects an average human can hold in working memory (known as memory span) is between 5 and 9 (7 ± 2 , which Miller described as the "magical number", and which is sometimes referred to as Miller's Law). However, although this may be approximately true for a population of college students, for example, memory span varies widely with populations tested, and modern estimates are typically lower, of the order of just 4 or 5 items.

DID YOU KNOW

Short-term working memory appears to operate phonologically.

For instance, whereas English speakers can typically hold seven digits in short-term memory, Chinese speakers can typically remember ten digits.

This is because Chinese number words are all single syllables, whereas English are not.

The type or characteristics of the information also affects the number of items which can be retained in short-term memory. For instance, more words can be recalled if they are shorter or more commonly used words, or if they are phonologically similar in sound, or if they are taken from a single semantic category (such as sports, for example) rather than from different categories, etc. There is also some evidence that short-term memory capacity and duration is increased if the words or digits are articulated aloud instead of being read sub-vocally (in the head).

The relatively small capacity of the short-term memory, compared to the huge capacity of long-term memory, has been attributed by some to the evolutionary survival advantage in paying attention to a relatively small number of important things and not to a plethora of other peripheral details which would only interfere with rapid decision-making.

"Chunking" of information can lead to an increase in the short-term memory capacity. Chunking is the organisation of material into shorter meaningful groups to make them more manageable. For example, a hyphenated phone number, split into groups of 3 or 4 digits, tends to be easier to remember than a single long number. Experiments by Herbert Simon have shown that the ideal size for chunking of letters and numbers, whether meaningful or not, is three. However, meaningful groups may be longer (such as four numbers that make up a date within a longer list of numbers, for example). With chunking, each chunk represents just one of the 5 - 9 items that can be stored in short-term memory, thus extending the total number of items that can be held.

It is usually assumed that the short-term memory spontaneously decays over time, typically in the region of 10 - 15 seconds, but items may be retained for up to a minute, depending on the content. However, it can be extended by repetition or rehearsal (either by reading items out loud, or by mental simulation), so that the information re-enters the short-term store and is retained for a further period. When several elements (such as digits, words or pictures) are held in short-term memory simultaneously, they effectively compete with each other for recall. New content, therefore, gradually pushes out older content (known as displacement), unless the older content is actively protected against interference by rehearsal or by directing attention to it. Any outside interference tends to cause disturbances in short-term memory retention, and for this reason people often feel a distinct desire to complete the tasks held in short-term memory as soon as possible.

DID YOU KNOW

The use of mnemonic devices can significantly increase memory, particularly the recall of long lists of names, numbers, etc.

One case, known as "S.F.", was able to increase his digit span (the longest list of number that a person can repeat back in correct order) from 7 to 79 with the use of mnemonic strategies.

Akira Haraguchi and Lu Chao's record-breaking recitations of the digits of the number Pi (100,000 and 67,890 digits respectively) also make use of mnemonic systems.

The forgetting of short-term memories involves a different process to the forgetting of long-term memories. When something in short-term memory is forgotten, it means that a nerve impulse has merely ceased being transmitted through a particular neural network. In general, unless an impulse is reactivated, it stops flowing through a network after just a few seconds.

Typically, information is transferred from the short-term or working memory to the long-term memory within just a few seconds, although the exact mechanisms by which this transfer takes place, and whether all or only some memories are retained permanently, remain controversial topics among experts. Richard Schiffrin, in particular, is well known for his work in the 1960s suggesting that all memories automatically pass from a short-term to a long-term store after a short time (known as the modal or multi-store or Anderson-Schiffrin model).

However, this is disputed, and it now seems increasingly likely that some kind of vetting or editing procedure takes place. Some researchers (e.g. Eugen Tarnow) have proposed that there is no real distinction between short-term and long-term memory at all, and certainly it is difficult to demarcate a clear boundary between them. However, the evidence of patients with some kinds of anterograde amnesia, and experiments on the way distraction affect the short-term recall of lists, suggest that there are in fact two more or less separate systems.

Long-term Memory

Long-term memory is, obviously enough, intended for storage of information over a long period of time. Despite our everyday impressions of forgetting, it seems likely that long-term memory actually decays very little over time, and can store a seemingly unlimited amount of information almost indefinitely. Indeed, there is some debate as to whether we actually ever "forget" anything at all, or whether it just becomes increasingly difficult to access or retrieve certain items from memory.

Short-term memories can become long-term memory through the process of consolidation, involving rehearsal and meaningful association. Unlike short-term memory (which relies mostly on an acoustic, and to a lesser extent a visual, code for storing information), long-term memory encodes information for storage semantically (i.e. based on meaning and association). However, there is also some evidence that long-term memory does also encode to some extent by sound. For example, when we cannot quite remember a word but it is "on the tip of the tongue", this is usually based on the sound of a word, not its meaning.

Physiologically, the establishment of long-term memory involves a process of physical changes in the structure of neurons (or nerve cells) in the brain, a process known as

long-term potentiation, although there is still much that is not completely understood about the process. At its simplest, whenever something is learned, circuits of neurons in the brain, known as neural networks, are created, altered or strengthened. These neural circuits are composed of a number of neurons that communicate with one another through special junctions called synapses. Through a process involving the creation of new proteins within the body of neurons, and the electrochemical transfer of neurotransmitters across synapse gaps to receptors, the communicative strength of certain circuits of neurons in the brain is reinforced. With repeated use, the efficiency of these synapse connections increases, facilitating the passage of nerve impulses along particular neural circuits, which may involve many connections to the visual cortex, the auditory cortex, the associative regions of the cortex, etc.

This process differs both structurally and functionally from the creation of working or short-term memory. Although the short-term memory is supported by transient patterns of neuronal communication in the regions of the frontal, prefrontal and parietal lobes of the brain, long-term memories are maintained by more stable and permanent changes in neural connections widely spread throughout the brain. The hippocampus area of the brain essentially acts as a kind of temporary transit point for long-term memories, and is not itself used to store information. However, it is essential to the consolidation of information from short-term to long-term memory, and is thought to be involved in changing neural connections for a period of three months or more after the initial learning.

Unlike with short-term memory, forgetting occurs in long-term memory when the formerly strengthened synaptic connections among the neurons in a neural network become weakened, or when the activation of a new network is superimposed over an older one, thus causing interference in the older memory.

Over the years, several different types of long-term memory have been distinguished, including explicit and implicit memory, declarative and procedural memory (with a further sub-division of declarative memory into episodic and semantic memory) and retrospective and prospective memory.

DID YOU KNOW

Several studies have shown that both episodic and semantic long-term memories can be better recalled when the same language is used for both encoding and retrieval.

For example, bilingual Russian immigrants to the United States can recall more autobiographical details of their early life when the questions and cues are presented in Russian than when they are questioned in English.

Declarative (Explicit) & Procedural (Implicit) Memory

Long-term memory is often divided into two further main types: explicit (or declarative) memory and implicit (or procedural) memory.

Declarative memory ("knowing what") is memory of facts and events, and refers to those memories that can be consciously recalled. It is sometimes called explicit memory, since it consists of information that is explicitly stored and retrieved, although it is more properly a subset of explicit memory. Declarative memory can be further sub-divided into episodic memory and semantic memory.

Procedural memory ("knowing how") is the unconscious memory of skills and how to do things, particularly the use of objects or movements of the body, such as playing a guitar or riding a bike. It is composed of automatic sensorimotor behaviours that are so deeply embedded that we are no longer aware of them, and, once learned, these "body memories" allow us to carry out ordinary motor actions automatically. Procedural memory is sometimes referred to as implicit memory, because previous experiences aid in the performance of a task without explicit and conscious awareness of these previous experiences, although it is more properly a subset of implicit memory.

These different types of long-term memory are stored in different regions of the brain and undergo quite different processes. Declarative memories are encoded by the hippocampus, entorhinal cortex and perirhinal cortex (all within the medial temporal lobe of the brain), but are consolidated and stored in the temporal cortex and elsewhere. Procedural memories, on the other hand, do not appear to involve the hippocampus at all, and are encoded and stored by the cerebellum, putamen, caudate nucleus and the motor cortex, all of which are involved in motor control. Learned skills such as riding a bike are stored in the putamen; instinctive actions such as grooming are stored in the caudate nucleus; and the cerebellum is involved with timing and coordination of body skills. Thus, without the medial temporal lobe (the structure that includes the hippocampus), a person is still able to form new procedural memories (such as playing the piano, for example), but cannot remember the events during which they happened or were learned.

Perhaps the most famous study demonstrating the separation of the declarative and procedural memories is that of a patient known as "H.M.", who had parts of his medial temporal lobe, hippocampus and amygdala removed in 1953 in an attempt to cure his intractable epilepsy. After the surgery, H.M. could still form new procedural memories and short-term memories, but long-lasting declarative memories could no longer be formed. The nature of the exact brain surgery he underwent, and the types of amnesia he experienced, allowed a good understanding of how particular areas of the brain are linked to specific processes in memory formation. In particular, his ability to recall memories from well before his surgery, but his inability to create new long-term memories, suggests that encoding and retrieval of long-term memory information is mediated by distinct systems within the medial temporal lobe, particularly the hippocampus. The fact that he was able to learn hand-eye coordination skills such as mirror drawing, despite having absolutely no memory of having learned or practised the task before, also suggested the existence of different types of long-term memory, which are now known as declarative and procedural memories.

DID YOU KNOW

Children under the age of about seven pick up new languages easily without giving it much conscious thought, using procedural (or implicit) memory.

Adults, on the other hand, actively learn the rules and vocabulary of a new language using declarative (or explicit) memory.

There is strong evidence, notably by studying amnesic patients and the effect of priming, to suggest that implicit memory is largely distinct from explicit memory, and operates through a different process in the brain. Studies of the effects of amnesia have shown that it is quite possible to have an intact implicit memory despite a severely impaired explicit memory. Priming is the effect in which exposure to a stimulus influ-

ences response to a subsequent stimulus, so that, for instance, if a person reads a list of words including the word "concert", and is later asked to complete a word starting with "con", there is a higher probability that they will answer "concert" than, say, "contact", "connect", etc. Studies from amnesic patients indicate that priming is controlled by a brain system separate from the medial temporal system that supports explicit memory.

Episodic Memory

Declarative memory can be further sub-divided into episodic memory and semantic memory.

Episodic memory represents our memory of experiences and specific events in time in a serial form, from which we can reconstruct the actual events that took place at any given point in our lives. It is the memory of autobiographical events (times, places, associated emotions and other contextual knowledge) that can be explicitly stated. Individuals tend to see themselves as actors in these events, and the emotional charge and the entire context surrounding an event is usually part of the memory, not just the bare facts of the event itself.

Semantic memory, on the other hand, is a more structured record of facts, meanings, concepts and knowledge about the external world that we have acquired. It refers to general factual knowledge, shared with others and independent of personal experience and of the spatial/temporal context in which it was acquired. Semantic memories may once have had a personal context, but now stand alone as simple knowledge. It therefore includes such things as types of food, capital cities, social customs, functions of objects, vocabulary, understanding of mathematics, etc. Much of semantic memory is abstract and relational and is associated with the meaning of verbal symbols.

DID YOU KNOW

Experiments on rats in the 1970s showed that there are over a million "place cells" in a rat's hippocampus, each of which only becomes active when the rat is located in a very specific part of its environment.

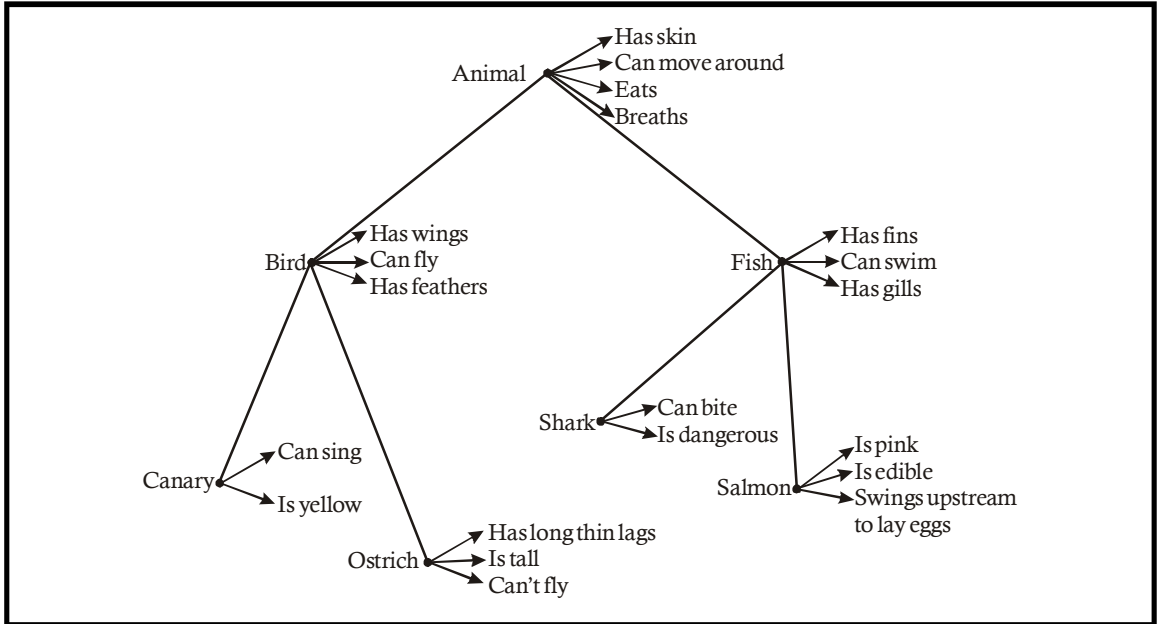
All together they can form a very precise cognitive map that tells the animal where it is at any given time.

When the rat explores a new environment, it forms a new cognitive map of place cells that can be very stable, lasting weeks or months.

Semantic Memory

Information stored in memory must have some organisation, otherwise our ability to reconstruct events and reason in terms of remembered facts would be hopelessly haphazard. Memory necessary for the use of language is called semantic memory. Collins and Quillian (1972) have proposed that semantic memory might be organised as an interconnected network. A part of such a semantic network is illustrated in below Figure. In the network, each word is stored with a set of pointers to other words in the network. The word concepts are the nodes, and the pointers are the connections in the network. Basic to this network is the assumption that information about a particular class of things is stored only at the level of the hierarchy for that class. Information generally true of birds (for example, wings, feathers) is not stored at the node for each separate type of bird.

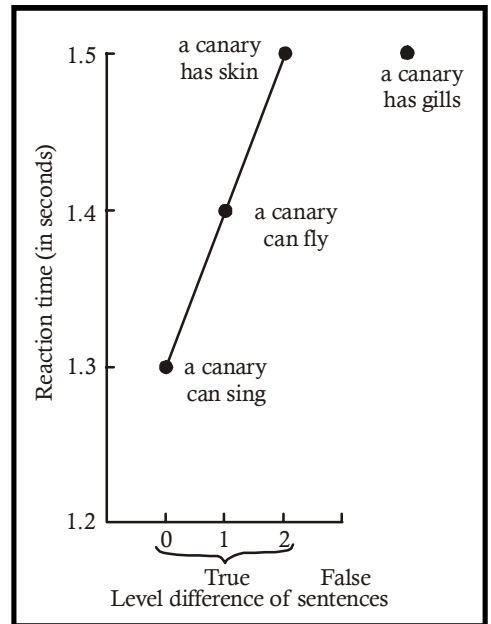
Rather, the fact that “A canary has wings” can be inferred by finding that a canary is a bird and that birds have wings. Storing information in this way economises on the amount of space required to maintain necessary facts in memory. In the case of exceptions, such as an ostrich, the fact that it cannot fly is stored at the ostrich node.



SEMANTIC NETWORK.

According to the model, a sentence is comprehended as being true if a path can be found through the semantic network that interrelates the words in the sentence. By referring to above figure, the sentence “A shark can move around” would be comprehended as true by finding the path: a shark is a fish, a fish is an animal, and an animal can move around.

If we assume that it requires time to move from one node in the hierarchy to the next, this time should be reflected in the speed with which the subject decides true or false. Above figure presents data on the time needed to make a true response. Points on the graph are labeled with an example of a typical sentence at each of three levels; the information to judge “A canary is yellow” is all at one level of the semantic network, whereas “A canary eats” required the subject to move from the canary node to the animal node, two levels away. As predicted, the time to decide that a sentence is true increases with the number of levels that have to be consid-



SEMANTIC MEMORY AND SENTENCE COMPREHENSION.

ered to answer the question. The experiment confirms the representation of semantic memory given in semantic network figure. There is no absolute requirement that human memory should be organised in this way, but the experiment supports this hierarchical view.

Our spatial memory in particular appears to be much more confined to the hippocampus, particularly the right hippocampus, which seems to be able to create a mental map of space, thanks to certain cells called "place cells". Episodic memory does also trigger activity in the temporal lobe, but mainly in order to ensure that these personal memories are not mistaken for real life. This difference in the neurological processing of episodic and semantic memory is illustrated by cases of anterograde amnesia cases (a good example being a case known as "C.L.") in which episodic memory is almost completely lost while semantic memory is retained.

A further category of declarative memory, referred to as autobiographical memory, is sometimes distinguished, although really it is just one area of episodic memory. Autobiographical memory refers to a memory system consisting of episodes recollected from an individual's own life, often based on a combination of episodic memory (personal experiences and specific objects, people and events experienced at particular times and places) and semantic memory (general knowledge and facts about the world).

One specific type of autobiographical memory is known as a "flashbulb memory", a highly detailed, exceptionally vivid "snapshot" of a moment or circumstances in which surprising and consequential (or emotionally arousing) news was heard, famous examples being the assassination of John Kennedy, the terrorist bombings on 9/11, etc. Such memories are believed by some to be highly resistant to forgetting, possibly due to the strong emotions that are typically associated with them. However, a number of studies also suggest that flashbulb memories are actually not especially accurate, despite apparently being experienced with great vividness and confidence.

Retrospective & Prospective Memory

An important alternative classification of long-term memory used by some researchers is based on the temporal direction of the memories.

Retrospective memory is where the content to be remembered (people, words, events, etc.) is in the past, i.e. the recollection of past episodes. It includes semantic, episodic and autobiographical memory, and declarative memory in general, although it can be either explicit or implicit.

DID YOU KNOW

MRI studies have shown that the same parts of the brain are used when remembering the past as when imagining a similar event in the future, which shows that past memories are also accessed and drawn on when projections are made into the future.

This is sometimes referred to as "mental time travel" as it allows us to project ourselves at will either backwards or forwards in time within our personal lives.

Prospective memory is where the content is to be remembered in the future, and may be defined as "remembering to remember" or remembering to perform an intended action. It may be either event-based or time-based, often triggered by a cue, such as going to the doctor (action) at 4 pm (cue), or remembering to post a letter (action) after seeing a mailbox (cue).

Clearly, though, retrospective and prospective memory are not entirely independent entities, and certain aspects of retrospective memory are usually required for prospective memory. Thus, there have been case studies where an impaired retrospective memory has caused a definite impact on prospective memory. However, there have also been studies where patients with an impaired prospective memory had an intact retrospective memory, suggesting that to some extent the two types of memory involve separate processes.

THEORIES OF FORGETTING

Rather surprisingly, research into forgetting in psychology has been rather limited in recent years. Perhaps this has something to do with the dominance of the computer metaphor in cognitive psychology and the fact the computers tend not to forget.

On the other hand, people have a tendency to forget lots of things. We forget appointments, have difficulty remembering someone's name, lose our keys and can't remember the precise details of books we read last week. Numerous different explanations have been proposed to try and explain why we forget.

Why do we forget? There are two simple answers to this question. First, the memory has disappeared - it is no longer available. Second, the memory is still stored in the memory system but, for some reason, it cannot be retrieved.

These two answers summaries the main theories of forgetting developed by psychologists. The first answer is more likely to be applied to forgetting in short term memory, the second to forgetting in long term memory.

Forgetting information from short term memory (STM) can be explained using the theories of trace decay and displacement.

Forgetting from long term memory (LTM) can be explained using the theories of interference and lack of consolidation.

Trace Decay Theory of Forgetting

This theory relates to both short term memory and long term memory, and also relates to lack of availability.

This theory suggests short term memory can only hold information for between 15 and 30 seconds unless it is rehearsed. After this time the information decays (fades away).

This explanation of forgetting in short term memory assumes that memories leave a trace in the brain. A trace is some form of physical and/or chemical change in the nervous system. Trace decay theory states that forgetting occurs as a result of the automatic decay or fading of the memory trace. Trace decay theory focuses on time and the limited duration of short term memory.

No one disputes the fact that memory tends to get worse the longer the delay between learning and recall, but there is disagreement about the explanation for this effect. According to the trace decay theory of forgetting, the events between learning and recall have no affect whatsoever on recall. It is the length of time the information has to be retained that is important. The longer the time, the more the memory trace decays and as a consequence more information is forgotten.

There are a number of methodological problems confronting researchers trying to investigate the trace decay theory. One of the major problems is controlling for the events

that occur between learning and recall. Clearly, in any real-life situation, the time between learning something and recalling it will be filled with all kinds of different events. This makes it very difficult to be sure that any forgetting which takes place is the result of decay rather than a consequence of the intervening events.

Support for the idea that forgetting from short-term memory might be the result of decay over time came from research carried out by Brown (1958) in the United Kingdom, and Peterson and Peterson (1959) in the United States. The technique they developed has become known as the Brown-Peterson task.

Evaluation

There is very little direct support for decay theory as an explanation for the loss of information from short-term and long-term memory. One of the problems with decay theory is that it is more or less impossible to test it. In practice, it is not possible to create a situation in which there is a blank period of time between presentation of material and recall. Having presented information participants will rehearse it. If you prevent rehearsal by introducing a distracter task, it results in interference.

Decay theory has difficulty explaining the observation that many people can remember events that happened several years previously with great clarity, even though they haven't thought about them during the intervening period. If our memories gradually decayed over time, then people should not have clear memories of distant events which have lain dormant for several years. However, there is evidence to suggest that information is lost from sensory memory through the process of decay (Sperling, 1960).

Displacement from Short Term Memory

Displacement seeks to explain forgetting in short term memory, and suggests it's due to a lack of availability.

The short-term memory store in Atkinson & Shiffrin's (1968) model of memory was assumed to have certain characteristics, including a limited capacity, and if information was not rehearsed, it would be forgotten. In the late 1960s and early 1970s, when Atkinson and Shiffrin's model was at its most popular, it was thought that the loss of information could be explained through the concept of displacement.

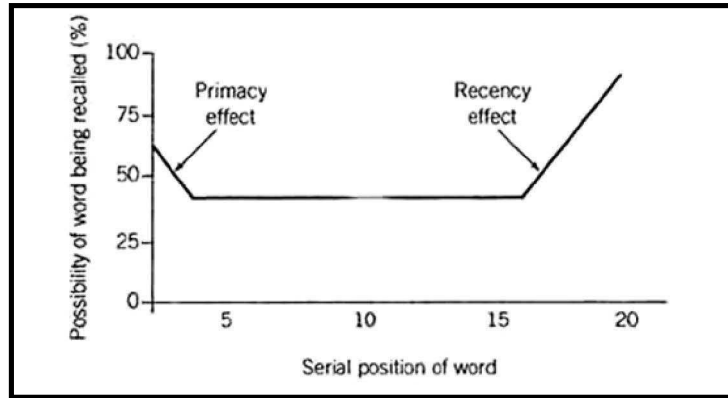
Displacement theory provides a very simple explanation of forgetting. Because of its limited capacity, suggested by Miller to be 7 ± 2 items, STM can only hold small amounts of information. When STM is 'full', new information displaces or 'pushes out' old information and takes its place. The old information which is displaced is forgotten in STM.

It was also assumed that the information that had been in the short-term store for the longest was the first to be displaced by new information, similar to the way in which boxes might fall off the end of a conveyor belt – as new boxes are put on one end, the boxes which have been on the conveyor belt the longest drop off the end.

Support for the view that displacement was responsible for the loss of information from short-term memory came from studies using the 'free-recall' method. A typical study would use the following procedure: participants listen to a list of words read out at a steady rate, usually two seconds per word; they are then asked to recall as many of words as possible. They are free to recall the words in any order, hence the term 'free recall'.

The findings from studies using free recall are fairly reliable and they produce similar results on each occasion. If you take each item in the list and calculate the probability of

participants recalling it (by averaging recall of the word over all participants) and plot this against the item's position in the list, it results in the serial position curve.



SIMPLIFIED REPRESENTATION OF THE SERIAL POSITION CURVE FOR IMMEDIATE RECALL.

Good recall of items at the beginning of the list is referred to as the primacy effect and good recall of items at the end of the list are referred to as the recency effect. The displacement theory of forgetting from short-term memory can explain the recency effect quite easily. The last few words that were presented in the list have not yet been displaced from short-term memory and so are available for recall.

The primacy effect can be explained using Atkinson & Shiffrin's (1968) multi-store model which proposes that information is transferred into long-term memory by means of rehearsal. The first words in the list are rehearsed more frequently because at the time they are presented they do not have to compete with other words for the limited capacity of the short-term store. This means that words early in the list are more likely to be transferred to long-term memory. So the primacy effect reflects items that are available for recall from long-term memory. However, words in the middle of the list used to be in short term memory until they were pushed out - or displaced by the words at the end of the list.

Evaluation

Displacement theory provided a good account of how forgetting might take place in Atkinson & Shiffrin's (1968) model of short-term memory. However, it became clear that the short-term memory store is much more complex than proposed in Atkinson and Shiffrin's model (re: working memory).

Murdock's (1962) serial position experiment supports the idea of forgetting due to displacement from short term memory, although it could be due to decay. Forgetting from short term memory can occur due to displacement or due to decay, but it is often very difficult to tell which one it is.

Interference Theory

If you had asked psychologists during the 1930s, 1940s, or 1950s what caused forgetting you would probably have received the answer "Interference". It was assumed that memory can be disrupted or interfered with by what we have previously learned or by what we will

learn in the future. This idea suggests that information in long term memory may become confused or combined with other information during encoding thus distorting or disrupting memories.

Interference theory states that forgetting occurs because memories interfere with and disrupt one another, in other words forgetting occurs because of interference from other memories (Baddeley, 1999). There are two ways in which interference can cause forgetting:

1. Proactive interference (pro=forward) occurs when you cannot learn a new task because of an old task that had been learnt. When what we already know interferes with what we are currently learning - where old memories disrupt new memories.
2. Retroactive interference (retro=backward) occurs when you forget a previously learnt task due to the learning of a new task. In other words, later learning interferes with earlier learning - where new memories disrupt old memories.

Proactive and retroactive Interference is thought to be more likely to occur where the memories are similar, for example: confusing old and new telephone numbers. Chandler (1989) stated that students who study similar subjects at the same time often experience interference.

Previous learning can sometimes interfere with new learning (e.g. difficulties we have with foreign currency when travelling abroad). Also new learning can sometimes cause confusion with previous learning. (Starting French may affect our memory of previously learned Spanish vocabulary). In the short term memory interference can occur in the form of distractions so that we don't get the chance to process the information properly in the first place. (e.g. someone using a loud drill just outside the door of the classroom.)

Key Study: Postman (1960)

Aim. To investigate how retroactive interference affects learning. In other words, to investigate whether information you have recently received interferes with the ability to recall something you learned earlier.

Method. A lab experiment was used. Participants were split into two groups. Both groups had to remember a list of paired words - e.g. cat - tree, jelly - moss, book - tractor. The experimental group also had to learn another list of words where the second paired word if different - e.g. cat - glass, jelly- time, book - revolver. The control group were not given the second list. All participants were asked to recall the words on the first list.

Results. The recall of the control group was more accurate than that of the experimental group.

Conclusion: This suggests that learning items in the second list interfered with participants' ability to recall the list. This is an example of retroactive interference.

Evaluation

Although proactive and retroactive interference are reliable and robust effects, there are a number of problems with interference theory as an explanation of forgetting.

First, interference theory tells us little about the cognitive processes involved in forgetting. Secondly, the majority of research into the role of interference in forgetting has been carried out in a laboratory using lists of words, a situation which is likely to occur fairly infrequently in everyday life (i.e. low ecological validity). As a result, it may not be possible to generalise from the findings.

Baddeley states that the tasks given to subjects are too close to each other and, in real life; these kinds of events are more spaced out. Nevertheless, recent research has attempted to address this by investigating 'real-life' events and has provided support for interference theory. However, there is no doubt that interference plays a role in forgetting, but how much forgetting can be attributed to interference remains unclear (Anderson, 2000).

Lack of Consolidation

The previous accounts of forgetting have focused primarily on psychological evidence, but memory also relies on biological processes. For example, we can define a memory trace as:

'some permanent alteration of the brain substrate in order to represent some aspect of a past experience'.

When we take in new information, a certain amount of time is necessary for changes to the nervous system to take place – the consolidation process – so that it is properly recorded. During this period information is moved from short term memory to the more permanent long term memory.

The brain consists of a vast number of cells called neurons, connected to each other by synapses. Synapses enable chemicals to be passed from one neuron to another. These chemicals, called neurotransmitters, can either inhibit or stimulate the performance of neurons. So if you can imagine a network of neurons all connected via synapses, there will be a pattern of stimulation and inhibition. It has been suggested that this pattern of inhibition and stimulation can be used as a basis for storing information. This process of modifying neurons in order form new permanent memories is referred to as consolidation.

There is evidence that the consolidation process is impaired if there is damage to the hippocampus. In 1953, HM had brain surgery to treat his epilepsy, which had become extremely severe. The surgery removed parts of his brain and destroyed the hippocampus, and although it relieved his epilepsy, it left him with a range of memory problems. Although his STM functioned well, he was unable to process information into LTM.

The main problem experienced by HM is his inability to remember and learn new things. This inability to form new memories is referred to as anterograde amnesia. However, of interest in our understanding of the duration of the process of consolidation is HM's memory for events before his surgery. In general, his memory for events before the surgery remains intact, but he does have some memory loss for events which occurred in the two years leading up to surgery.

Pinel (1993) suggests that this challenges Hebb's (1949) idea that the process of consolidation takes approximately 30 minutes. The fact that HM's memory is disrupted for the two-year period leading up to the surgery indicates that the process of consolidation continues for a number of years.

Finally, aging can also impair our ability to consolidate information.

Evaluation

The research into the processes involved in consolidation reminds us that memory relies on biological processes, although the exact manner by which neurons are altered

during the formation of new memories has not yet been fully explained. However, there is no doubt that investigating the role of neurons and neurotransmitters will provide new and important insights into memory and forgetting.

Retrieval Failure Theory

Retrieval failure is where the information is in long term memory, but cannot be accessed. Such information is said to be available (i.e. it is still stored) but not accessible (i.e. it cannot be retrieved). It cannot be accessed because the retrieval cues are not present.

When we store a new memory we also store information about the situation and these are known as retrieval cues. When we come into the same situation again, these retrieval cues can trigger the memory of the situation. Retrieval cues can be:

External/Context - in the environment, e.g. smell, place etc.

Internal/State- inside of us, e.g. physical, emotional, mood, drunk etc.

There is considerable evidence that information is more likely to be retrieved from long-term memory if appropriate retrieval cues are present. This evidence comes from both laboratory experiments and everyday experience. A retrieval cue is a hint or clue that can help retrieval.

Tulving (1974) argued that information would be more readily retrieved if the cues present when the information was encoded were also present when its retrieval is required. For example, if you proposed to your partner when a certain song was playing on the radio, you will be more likely to remember the details of the proposal when you hear the same song again. The song is a retrieval cue - it was present when the information was encoded and retrieved.

Tulving suggested that information about the physical surroundings (external context) and about the physical or psychological state of the learner (internal context) is stored at the same time as information is learned. Reinstating the state or context makes recall easier by providing relevant information, while retrieval failure occurs when appropriate cues are not present. For example, when we are in a different context (i.e. situation) or state.

Context (External) Dependent Forgetting

Retrieval cues may be based on context-the setting or situation in which information is encoded and retrieved. Examples include a particular room, driving along a motorway, a certain group of people, a rainy day and so on.

Context also refers to the way information is presented. For example, words may be printed, spoken or sung, they may be presented in meaningful groups - in categories such as lists of animals or furniture - or as a random collection without any link between them. Evidence indicates that retrieval is more likely when the context at encoding matches the context at retrieval.

You may have experienced the effect of context on memory if you have ever visited a place where you once lived (or an old school). Often such a visit helps people recall lots of experiences about the time they spent there which they did not realise were stored in their memory.

A number of experiments have indicated the importance of context-based cues for retrieval. An experiment conducted by Tulving and Pearlstone (1966) asked participants to learn lists of words belonging to different categories, for example names of animals, clothing and sports.

Participants were then asked to recall the words. Those who were given the category names recalled substantially more words than those who were not. The categories provided a context, and naming the categories provided retrieval cues. Tulving and Pearlstone argued that cue-dependent forgetting explains the difference between the two groups of participants. Those who recalled fewer words lacked appropriate retrieval cues.

An interesting experiment conducted by Baddeley indicates the importance of setting for retrieval. Baddeley asked deep-sea divers to memorise a list of words. One group did this on the beach and the other group underwater. When they were asked to remember the words half of the beach learners remained on the beach, the rest had to recall underwater.

Half of the underwater group remained there and the others had to recall on the beach. The results show that those who had recalled in the same environment (i.e. context) which that had learned recalled 40% more words than those recalling in a different environment. This suggests that the retrieval of information is improved if it occurs in the context in which it was learned.

State (Internal) Dependent Cues

The basic idea behind state-dependent retrieval is that memory will be best when a person's physical or psychological state is similar at encoding and retrieval. For example, if someone tells you a joke on Saturday night after a few drinks, you'll be more likely to remember it when you're in a similar state - at a later date after a few more drinks. Stone cold sober on Monday morning, you'll be more likely to forget the joke.

State Retrieval clues may be based on state-the physical or psychological state of the person when information is encoded and retrieved. For example, a person may be alert, tired, happy, sad, drunk or sober when the information was encoded. They will be more likely to retrieve the information when they are in a similar state.

Tulving and Pearlstone's study involved external cues (e.g. presenting category names). However, cue-dependent forgetting has also been shown with internal cues (e.g. mood state). Information about current mood state is often stored in the memory trace, and there is more forgetting if the mood state at the time of retrieval is different. The notion that there should be less forgetting when the mood state at learning and at retrieval is the same is generally known as mood-state-dependent memory.

A study by Goodwin et al. investigated the effect of alcohol on state-dependent retrieval. They found that when people encoded information when drunk, they were more likely to recall it in the same state. For example, when they hid money and alcohol when drunk, they were unlikely to find them when sober. However, when they were drunk again, they often discovered the hiding place. Other studies found similar state-dependent effects when participants were given drugs such as marijuana.

People tend to remember material better when there is a match between their mood at learning and at retrieval. The effects are stronger when the participants are in a positive mood than a negative mood. They are also greater when people try to remember events having personal relevance.

Evaluation

According to retrieval-failure theory, forgetting occurs when information is available in LTM but is not accessible. Accessibility depends in large part on retrieval cues. Forgetting is greatest when context and state are very different at encoding and retrieval. In this situation, retrieval cues are absent and the likely result is cue-dependent forgetting.

There is considerable evidence to support this theory of forgetting from laboratory experiments. The ecological validity of these experiments can be questioned, but their findings are supported by evidence from outside the laboratory. For example, many people say they can't remember much about their childhood or their schooldays. But returning to the house in which they spent their childhood or attending a school reunion often provides retrieval cues which trigger a flood of memories



THINKING, INTELLIGENCE AND CREATIVITY

THINKING

Human thinking, and in particular, the human ability to solve complex, real-life problems contributes more than any other human ability to the development of human culture and the growth and development of human life on earth. However, the human ability to solve complex problems is still not well understood, partly because it has for a long time been largely ignored by traditional problem-solving research in the field of psychology.

In this article, we present a definition of complex problem solving and describe a theoretical framework that accommodates the theoretical and empirical strides that have been made in understanding complex problem solving thus far and may serve as a guide for future research. We discuss the dominant methodological approaches that have been employed to study complex problem solving, and offer our own recommendations on which of the various approaches might be the most promising.

The breadth of human thinking is truly striking. As is evident in the examples provided above, human thinking can lead to catastrophic disasters. Yet, on the other hand, human thinking also makes possible the most wondrous achievements. Whatever “thinking” is, and scientists disagree heartily on the proper meaning of the term, there can be little doubt that human thinking determines human culture to an extent that is unrivalled by any other human ability. Of course, culture determines the way we think as well—as many scientists have pointed out in the past.

In this article, we describe some of the main properties of human thinking. To stay with the general theme of this encyclopedia, after providing a definition of thinking and briefly discussing the various categories or types thinking may be divided into, we concentrate on one particular type of human thinking that contributes more than any other to the development of human culture and the growth and development of human life on

earth, namely, problem solving. Our discussion will focus on complex, real-life problem solving rather than on solving small and artificial laboratory-type problems, and will concentrate on both empirical research and attempts to explain the phenomenon theoretically.

Many of our daily activities involve thinking of some sort. For example, we decide what to wear in the morning, which route to take to get to our office, which job-related duties to perform in which sequence once we arrive at our office, what to have for lunch, and so on. Of course, not all thinking is alike. There is thinking that involves only a few mental steps, and there is thinking that requires many steps. Some thinking involves situations we have never encountered before, and other thinking involves familiar situations. Sometimes thinking is tied to clear goals, and sometimes it is not. "Thinking," then, can be distinguished on any number of meaningful dimensions, and the mental processes—that is, the steps we engage in when thinking, as well as the mental representations the processes operate on—may differ widely for different types of thinking.

Given the multidimensionality of thinking, it may come as no surprise that different researchers, all claiming to study the phenomenon of thinking, have in the past differed widely in their definitions of the term thinking. On the one hand, thinking has often been defined in terms of its possible functions. For example, Aebli likens thinking to "bringing order into one's doing," and Johnson, Drner, and many others define thinking in terms of problem solving.

On the other hand, some researchers have attempted to define thinking in a manner that is independent of its assumed functions. We will follow the latter approach and for the purpose of this article define thinking as the "cognitive processing of internal memory representations that may occur both consciously and subconsciously and may not always follow the laws of logic."

Given this definition of human thinking, we can distinguish between various subtypes, or categories, of thinking. A taxonomy of thinking that has been developed by Johnson-Laird. According to Johnson-Laird, five different types of thinking can be distinguished. For example, thinking that is goal-oriented, does not follow a predetermined sequence of mental steps, and has no precise starting point is usually labeled "creative."

Alternatively, thinking that is goal-oriented, does not follow a predetermined sequence of mental steps, has a precise starting point, and leads to an increase in semantic information in the human memory system is called "inductive." Inductive and deductive thinking together constitute the category of problem solving..

Readers should note that the taxonomy has more than just intuitive appeal. The questions used divide the area of thinking into truly distinct subtypes that differ from each other in terms of both the mental processes conducted and the mental representations involved. Thus, "association-based" thinking, for instance, differs from "calculation" both in terms of the underlying processes (i.e. mental steps) as well as in the mental representations that the processes operate on.

In the rest of the article, we focus primarily on one particular type of thinking that is inherent in the taxonomy, namely, problem solving (subsuming both inductive and deductive thinking). We will do so because we are convinced that this type of thinking is the most important in supporting and ensuring lasting local, regional, and global welfare. In the next section, we summarize the currently dominant empirical approaches to studying problem solving, after first providing a brief historical background within which the development of the dominant approaches can be understood. Notice that we will concen-

trate almost exclusively on what is called complex problem solving (CPS), that is, problem solving that occurs in the context of real-world problems. We present a definition of CPS, and describe a theoretical framework that accommodates the theoretical and empirical strides that have been made in understanding CPS thus far, and serves as a guide for future research. Last but not least, we discuss the dominant methodological approaches that have been employed to study CPS, and offer our own recommendations on which approach might turn out to be the most promising.

Problem Solving

Beginning with the early experimental work of the Gestaltists in Germany, and continuing through the 1960s and early 1970s, research on problem solving was typically conducted with relatively simple laboratory tasks that were novel to research participants. Simple novel tasks were used for a variety of reasons: they had clearly defined optimal solutions, they were solvable in a relatively short time, research participants' problem-solving steps could be traced, and so on. The underlying assumption was, of course, that simple tasks, such as the Tower of Hanoi, capture the main properties of "real" problems, and that the cognitive processes underlying participants' solution attempts on simple problems were representative of the processes engaged in when solving real problems. Thus, simple problems were used for reasons of convenience, and generalizations to more complex problems were thought possible. Perhaps the best-known and most impressive example of this line of research is the work by Newell and Simon.

However, beginning in the 1970s, researchers became increasingly convinced that empirical findings and theoretical concepts derived from simple laboratory tasks were not generalizable to more complex, real-life problems. Even worse, it appeared that the processes underlying CPS in different domains were different from each other. These realizations have led to rather different responses in North America and Europe.

In North America, initiated by the work of Herbert Simon on learning by doing in semantically rich domains, researchers began to investigate problem solving separately in different natural knowledge domains (*e.g.* physics, writing, chess playing) thus relinquishing their attempts to extract a global theory of problem solving. Instead, these researchers frequently focused on the development of problem solving within a certain domain, that is, on the development of expertise. Areas that have attracted rather intense attention in North America include such diverse fields as reading, writing, calculation, political decision making, managerial problem solving, lawyers' reasoning, mechanical problem solving, problem solving in electronics, computer skills, game playing, and personal problem solving.

In Europe, two main approaches have surfaced, one initiated by Donald Broadbent in Great Britain and the other by Dietrich Drner in Germany. The two approaches have in common an emphasis on relatively complex, semantically rich, computerized laboratory tasks that are constructed to be similar to real-life problems. The approaches differ somewhat in their theoretical goals and methodology. The tradition initiated by Broadbent emphasizes the distinction between cognitive problem-solving processes that operate under awareness versus outside of awareness, and typically employs mathematically well-defined computerized systems. The tradition initiated by Drner, on the other hand, is interested in the interplay of cognitive, motivational, and social components of problem solving, and utilizes very complex computerized scenarios that contain up to 2000 highly interconnected variables.

Complex Problem Solving

With the above considerations in mind, it is not surprising that there exist a wide variety of definitions of CPS that have little in common. Indeed, researchers in the area of problem solving have long been troubled by the absence of agreement on the exact meaning of many of the basic terms in the area. Any general conclusion regarding CPS, however, and any theoretical model of CPS can be meaningful only if all agree on what constitutes a problem and what constitutes CPS. For the rest of this article we define CPS as follows: CPS occurs to overcome barriers between a given state and a desired goal state by means of behavioral and/or cognitive, multi-step activities. The given state, goal state, and barriers between given state and goal state are complex, change dynamically during problem solving, and are non-transparent. The exact properties of the given state, goal state, and barriers are unknown to solvers at the outset. CPS implies the efficient interaction between solvers' and the situational requirements of the task, and involves solvers cognitive, emotional, personal, and social abilities and knowledge.

Readers should notice that this definition differs rather substantially from definitions that feature prominently in the North American tradition. John Anderson, as an example of the North American approach, has defined problem solving as "any goal-directed sequence of cognitive operations" regardless of whether the task is novel or familiar to the solvers, regardless of whether the task is complex, and regardless of whether a single barrier or multiple barriers exist between given state and goal state. Our definition, in contrast, constrains potential problems by requiring that they be (a) novel tasks that problem solvers are unfamiliar with, (b) complex, (c) dynamically changing over time, and (d) non-transparent. In order to solve these problems, solvers have to be able to anticipate what will happen over time, and have to consider side effects of potential actions.

In addition and in contrast to earlier, often implicit, views, CPS is not viewed as deterministic in the sense that any problem-solving activity will always lead to the solution of a problem. Rather, CPS may lead to an approximate solution that may advance the solvers but may not lead to actually solving the problem. For example, research participants performing the duties of the mayor of a computer-simulated town may, even after some practice, still not be able to generate the best possible solution to a given problem. In fact, many, often computerized, tasks exist for which—due to the complex non-linear relations among the task variables—the optimal solution is unknown. Of course, the absence of an optimal solution, while theoretically reasonable and even desirable, poses a problem to experimenters who want to determine the quality of problem solvers' performances, and to those who use micro worlds for personnel selection purposes.

A Theoretical Framework for Complex Problem Solving

Adopting any definition has consequences not only for how an area is studied but also for how empirical findings are theoretically interpreted. For example, if problem solving is defined in terms of cognitive rather than neurophysiological, biological, or behavioral processes, then it makes little sense to construct a theory of problem solving at a neurophysiological, biological, or behavioral level. In the following, we present our thoughts on how a general theoretical framework for understanding problem solving that is based on the above definition of CPS might look. The framework is based on the assumptions that: (a) the theoretical goal of CPS research is to understand the interplay among cognitive, motivational, personal, and social factors when complex, novel, dynamic, non-transparent

ent tasks are solved, and (b) the interplay among the various components can best be understood within an information-processing model. The framework is constrained, of course, by what is known already about CPS. We therefore first present a brief, non-exhaustive list of the main empirical phenomena that have been demonstrated for the area of CPS in recent years.

Studies on Person Factors

Studies exploring the effect of person factors on CPS tend to focus on test intelligence as one dominant and important person variable. In addition, comparisons between experts and novices and analyses of clinical groups and of strategic preferences belong in this category. Other person variables that have been explored theoretically as well as experimentally but are not discussed here because of space limitations include self-reflection, emotions, and language.

1. Test Intelligence

Strohschneider has compared the predictive value of test intelligence for CPS performance under two different experimental conditions. All research participants operated an abstract system called VEKTOR first, and were then confronted with a semantically rich Peace Corps worker simulation system, MORO. The Berlin Intelligence Structure test (BIS) was used to assess participants' intelligence. The BIS differentiates between two factors: a content-oriented component representing knowledge in three different modalities, and a process-oriented component representing four operative abilities. For the MORO system, all of the seven subtest scales correlated significantly with a general measure of control performance; the same was found for the VEKTOR system with respect to six out of the seven subscales. Strohschneider thus concluded that test intelligence was indeed a significant predictor of CPS performance. Comparing the two systems, Strohschneider found that performance on the two systems was not correlated significantly. This indicates that a single CPS trait may not be responsible for performance under all conditions.

S, Oberauer, and Kersting also assessed the value of test intelligence for predicting CPS performance measures. In their study, the authors used the well-known system TAILORSHOP under non-transparency conditions. All research participants were also asked to complete the BIS intelligence test. Using traditional measures of performance for the TAILORSHOP (total assets at the end of simulation and the number of simulated months with a revenue), the authors found no significant correlations with any of the BIS scales. Based on the assumption that the (unknown and therefore perhaps low) reliability of the CPS measures could have caused this null-effect, the authors tried other ways of operationalizing the TAILORSHOP performance. In a task analysis, they discovered that two subgoals were utilized by smart problem solvers that unfortunately conflicted with each other: shirt sales and profit margin. Due to system-inherent characteristics, all research participants had a negative profit margin; lucky problem solvers, however, decreased this negative value and at the same time increased shirt sales. Because revenues are the product of sales and profit margin per shirt, good problem solvers increased their losses despite the fact that they were using efficient strategies. When S, Oberauer, and Kersting constructed a new dependent measure—the sum of the increases in shirt sales and in profit margin—then the new measure of CPS quality correlated significantly with the BIS scale “capacity of information processing.”

Taken together, the studies demonstrate that intellectual abilities have predictive value for CPS results when certain conditions are met: (1) instead of using a global I.Q. measure, one needs to separate different components of intelligence of which the capacity of information processing appears to be the most promising predictor; and, (2) CPS quality has to be measured reliably – a condition that is rarely met.

2. Expert-Novice Comparisons

Reither was the first to validate the developing-world scenario DAGU by comparing the control performances of experienced technical advisers who had about 10 years practice in developing countries (i.e. experts), and of postgraduate students who had just begun their first mission as development aid volunteers (i.e. novices). Both experts and novices worked on the DAGU system in groups of three. The main result of Reither's study was that experts showed a broader range of actions and a greater willingness to make decisions from the start, but also that experts used only standard strategies and were not able to adapt to changing task conditions. Reither calls this behavior the "blindness of the specialists." Despite these strategic differences between novices and experts, both groups performed terribly on the system, however. In both groups, the number of inhabitants of the simulated country had decreased dramatically after 20 simulated years, due to starvation. This finding leads to the (as yet unanswered) question whether either the experts did not acquire any usable knowledge during their 10 years of practice, or the simulated system does not capture reality in a valid manner.

Schaub and Strohschneider examined whether managers and students act differently when dealing with the MORO scenario. In MORO, research participants have to take on the role of a Peace Corps worker in Africa. The authors reported that the managers' problem-solving behavior was characterized by a more intense exploration of the scenario and by a more cautious and adaptive way of adjusting to the demands of the task. On average, the managers achieved better results than the students. One potentially confounding factor in this comparative study was the age of the research participants, however. The managers were 25 years older than the students, on average, and thus had more life experience. Similar studies comparing students of economy with professors of that discipline on their performances on the TAILORSHOP scenario have been conducted by Putz-Osterloh.

Repeated exposure to a problem, of course, also produces a certain degree of expertise and should therefore lead to better problem performance and representation. This could indeed be demonstrated in a number of studies where research participants had to work on the same scenario for more than one simulation period. In general, it seems fair to argue that knowledge is an important predictor of control performance, although dissociations between the two variables in CPS situations have also been reported.

3. Clinical Groups

In a prospective longitudinal study, Fritz and Funke compared the quality of CPS in pupils who had minimal cerebral dysfunctions (MCD) and in matched controls (CON). Working with the dynamic system OEKOSYSTEM, all participants were asked first to explore and then to control the system. In terms of the quality of participants' knowledge acquisition (as revealed by causal diagrams of the assumed structural relations), the authors found that the MCD group did not acquire significantly less knowledge than did the CON group. However, the strategies used differed markedly for the two groups. Participants in the

CON group used single variation interventions that tend to reveal the causal relations among variables three times more often than did subjects in the MCD group. With respect to the quality of system control, there was no significant group difference, although 20% of the subjects in the CON group reached the required goal state at least once, whereas almost none of the MCD participants did.

4. Strategies

Schmuck used a self-constructed instrument to assess the degree to which problem solvers spontaneously exert executive control and compared solvers with high and low efficiency of executive control on their performances in the FIRE scenario. Solvers classified as highly efficient showed better performance from the start but also needed more time and made more interventions. These solvers also showed a greater variability in behavior than did participants classified as low in efficiency. Schmuck argued that these strategic differences explain the low stability scores found in many CPS studies. The explanation relies strongly on the assumption, however, that Schmuck's instrument allows a reliable differentiation between people who differ in efficiency, a fact that has yet to be demonstrated.

Vollmeyer, Burns, and Holyoak have analyzed the strategies problem solvers use when exploring, controlling, and predicting an unknown complex dynamic system called BIOLOGY LAB. Participants were categorized according to their exploration behavior as using: (1) a scientific strategy, (2) systematic variations of a strategy, or (3) unsystematic variations of a strategy. As expected, strategies (1) and (2) led to a better representation of the system and to a better prediction of system states than did strategy (3). Surprisingly however, no group differences were found for participants' control performance. The authors interpret their result as indicating that different types of knowledge are necessary for the three different tasks.

Putz-Osterloh also strongly recommends strategy analyses for the explanation of individual differences. Using the DYNAMIS micro world, she found significant improvements in structural system knowledge for participants using efficient strategies for intervention. Results from Schoppek demonstrate the differential influence of two kinds of knowledge (specific rules, general operations) on strategies and the necessity to consider processes of intention regulation in the control of dynamic systems. This influential role of knowledge could also be demonstrated in experiments from Preuler.

Strategic differences may show up also in cross-cultural studies. Strohschneider and Gss could demonstrate, for example, differences between Brazilian and German students in planning and problem solving. The participants completed two instruments, a planning inventory that consisted of everyday problem scenarios, and the cold-storage problem, a computer-simulated dynamic problem. Results demonstrated that in the planning inventory, Brazilian participants accepted the given descriptions and were more optimistic about the results of their efforts. German participants tended to inquire more about the background of the problems and took a more active approach but were less optimistic. In the cold-storage simulation, the Brazilian participants had more difficulties controlling the system. This was attributed to their orientation toward the present time, making it difficult to detect developmental patterns.

Strategic differences were also found when comparing Indian and German samples working on the MORO scenario. Results show that German students dealt with the computer-simulated problem more actively, whereas Indian participants committed more strategic and tactical errors; experimental variation had minimal effect on participants' behavior. It was concluded that German students used more control-oriented strategies than the Indian students and were more competent in solving the problem.

Even within one country, marked differences may exist, as was found by Strohschneider who compared East and West Germans after unification with respect to their problem-solving strategies and abilities. Whereas East Germans analyzed problems very carefully and made clear plans, the West Germans tried to work as quickly as possible and did not develop extensive plans.

Studies on Situation Factors

In studies exploring the role of situation factors on CPS, several variables have been experimentally manipulated, including the type of task, the effects of noise-induced stress, individual versus group problem solving, the transparency of system variables, and the type of the system presentation.

1. Type of Task

Funke and Miller conducted an experiment with the SINUS system in which the problem solvers' task was first to explore an unknown dynamic system for a given number of trials through either passive observation of another person interacting with the system or active intervention. Later, all participants were asked to control the system such that given goal states were reached. The dependent variables in this study were the quality of knowledge acquisition and the quality of control performance. Results showed that active intervention led to better control performance but reduced the amount of verbalizable knowledge. Surprisingly, the observers, who were poor in control performance, constructed better causal diagrams showing the system variables; thus, they appeared to have acquired knowledge about the variables and their interrelations but not about how to control the system.

Berry performed a similar study using the SUGAR PRODUCTION and the PERSONAL INTERACTION tasks. In her Experiment 1, participants first had to watch an experimenter interacting with the system and then to control the system by themselves. It seemed as if participants did not learn anything through pure observation, neither on the control scores nor on the post-task questionnaires. In a second experiment, Berry found that learning by observation was possible when the task was changed from a task with non-salient relations to a task with salient relations among the system variables. The effect of this modification was apparent on both diagnostic measures, on the quality of control performance as well as on the system knowledge as measured by a questionnaire.

2. Stress

Drner and Pfeifer tested the effects of noise-induced stress on CPS. Although stress can be considered a person variable, this study is subsumed under situation factors because the experimental conditions manipulated situational aspects. The authors used a version of the FIRE scenario developed by Brehmer and his associates. The participants' task was to manage five different fires either under conditions of a stressful white noise or under quiet conditions. Time pressure was present in both conditions. At a global level,

there was no difference between the two conditions with respect to their success and failure rates. A more fine-grained analysis—looking at participants' tactical decisions—revealed, however, that although stress did not affect the number of errors made, it did affect the types of errors made. For example, an incorrect dosage of fire fighting interventions and a more reactive type of behavior was characteristic of the stressed participants.

3. Individual versus Group Complex Problem Solving

Kller, Dauenheimer, and Strau compared group CPS to individual CPS. In a first session, all participants worked individually on the scenario FUEL OIL DISTRIBUTION. Performance on the system was then used to classify participants as either good or poor problem solvers. Then, in a second session, participants worked on a very similar scenario called TEXTILESHOP, either individually or in a dyad consisting of either two poor or two good problem solvers. It turned out that the individual problem solvers' performances were worse than the performances achieved by the dyads. For the latter, it did not seem to matter whether they were composed of good or poor problem solvers.

Leutner worked with pupils who had to deal with a derivative of TAILORSHOP either as individuals or in groups of three. In contrast to the previously reported work it turned out here that knowledge acquisition was significantly higher for individuals than for groups but with respect to control performance there was no difference.

Badke-Schaub analyzed problem-solving strategies of individuals and groups dealing with a model for the epidemic of AIDS. Participants had to propose interventions to prevent the spread of the disease. Badke-Schaub found that groups have problems defining a common goal but have advantages in finding problem-relevant information. Groups also produced more proposals for solutions but found it difficult to select one or more of these proposals.

4. Transparency

Putz-Osterloh and Ler investigated the effect of transparency on problem-solving quality in the scenario TAILORSHOP. One group received the system under conditions of non-transparency; here, participants were told only which interventions were possible but did not receive further information. The second group received a graphical representation of the relations among (almost all of) the system variables. After 15 simulated time cycles (i.e. months), participants in the transparency condition had achieved better scores on the performance measure. In addition, the correlation between system performance and test intelligence was also moderated by transparency. Only under transparent conditions was a small but significant rank correlation between the two variables observed; under non-transparency conditions, the correlation was zero. Putz-Osterloh and Ler argued that the equivalence of the two tasks—the intelligence test and the CPS task—might have been much higher under transparency than under non-transparency conditions. In the former case, both tasks shared the attribute that information was given to participants who had to analyze it. In the latter case, CPS required additional information search procedures that were not necessary to complete the intelligence test. Although it may have been true for the Putz-Osterloh and Ler study, Funke has shown empirically that this assumption does not generally hold: the moderating effect of transparency on the I.Q.-CPS relationship is lost in favor of a main effect of test intelligence if one selects a larger range of I.Q. values than those shown normally by student participants.

In a recent study, Putz-Osterloh again manipulated the transparency of a system by presenting, or not presenting, a structural diagram of the DYNAMIS system. In this study, the experimental group that received the diagram was not superior to a control group without a diagram on measures of task performance and strategy selection. But on a follow-up transfer task with a modified system, the experimental group outperformed the control group on both types of indicators. Putz-Osterloh concluded from these results that knowledge acquisition is not necessarily a prerequisite for good control. The strategies that are applied may be more important for predicting the quality of performance.

5. Information Presentation

Hbner performed an experiment in which 20 participants had to control a simulated GAS ABSORBER. The system state was displayed either in an analog or a numerical format. With respect to the dependent variable quality of control, it turned out that the analog group was significantly better and also needed less time than the group with numeric presentation.

Studies on System Factors

Experimental research manipulating system attributes has concentrated on the effects of the variables *Eigendynamik*, feedback delay, and semantic embedding.

1. Eigendynamik

In a series of experiments, Funke systematically varied several system factors, one of which was the *Eigendynamik* of the system. *Eigendynamik* occurs when a system changes its state without intervention from outside, due to its inherent dynamics. In more technical terms, *Eigendynamik* is present when a system changes its state at time t due to the values of some variables at time $t-1$ but does so independently of any input by the operator. In the extreme case, *Eigendynamik* means that a system changes over time despite the fact that no active intervention has occurred. Many natural systems show this property requiring an operator to anticipate the system's inherent changes due to the *Eigendynamik*. Funke has used the SINUS system, an artificial system simulating the growth of living creatures from a distant planet with three exogenous and three endogenous variables, to study the effect of *Eigendynamik*. There were three different conditions, a control condition with no *Eigendynamik*, and two conditions with different degrees of *Eigendynamik*. The results demonstrated that increased *Eigendynamik* yielded a decrease in the quality of system control although the quality of system identification remained unaffected by the manipulation. This pattern of findings suggests that the two dependent variables may tap different processes that are differentially affected by *Eigendynamik*.

2. Feedback Delays

Heineken et al. tested the effects of feedback delay on CPS by using a simple system called TEMPERATURE in which participants had to control the temperature of an artificial system for 1200 simulation cycles. Feedback concerning the quality of the intervention was either immediate or after little or much delay. In addition, half of the participants were informed in advance which delay condition would be realized. Heineken et al. reported that: (a) the quality of system control decreased with increasing delay and (b) *a priori* information about the delay was not effective. Interestingly enough, even in the

much-delay condition, participants were – after a long period of time – able to control the system. This indicates that although feedback delay may influence the rate of learning, it does not appear to block completely the ability to master a time-delayed system.

3. Semantic Embeddedness

Hesse has compared two different semantic embeddings for the same underlying system. EPIDEMIC simulates the spread of a disease in a small community. In one condition, participants, as the managers of a local health service, were asked to care for people who had influenza. In the second condition, the disease was changed to a life-threatening smallpox epidemic. The change in semantics changed participants' behavior drastically; in the more "dangerous" situation, participants tended to be, among other things, much more involved and to take more time making decisions.

Another interesting study on the effects of semantic embedding has been reported by Beckmann. The author compared two semantic embeddings (CHERRY TREE vs. MACHINE) of the same system structure with respect to participants' knowledge acquisition and control performances. In this experiment, the semantically rich embedding seemed to prevent problem solvers using efficient analytic knowledge acquisition strategies.

Studies on Interaction Effects

The interactions between person, situation, and system factors have been researched less frequently than have the individual factors. One selected area concerns the interaction between person and situation variables.

Person and Situation

Rost and Strau analyzed the interaction between type of information presentation (numerically vs. graphically) and type of induced mental model (propositional vs. analog) using a simple simulation system called SHOP. Their study demonstrates the usefulness of interaction analysis in CPS research. The authors started with the assumption that the advantages of a certain presentation format (presentation of system information in numeric or in graphic form) would affect performance only if it corresponded to the format in which knowledge about the system was internally represented. The internal representation format was induced in this study in a short training session that either stimulated thinking about the system in terms of propositions (if-then statements) or in terms of a graphical network in which the nodes represented the variables connected by causal links, and the diameter of the nodes indicated the quantitative state of the variables. Rost and Strau assumed that a propositional representation of system knowledge would best fit a numerical presentation and that the analog representation would best fit the graphic presentation. The central system variable in their scenario was money. For each of the 25 simulation cycles, the dependent variable problem-solving quality was set to +1 if an increase in money had occurred, -1 in case of a decrease, and 0 in case of no change.

The Components of a Theory of Complex Problem-solving

The empirical findings summarized above lead to a simple theoretical framework for understanding CPS that is depicted in the summarizes the basic components of the framework and the interrelations among the components. As can be seen, the framework contains three separate components: problem solver, task, and environment.

In the problem solver, static memory content and dynamic information processing are distinguished. Memory is divided further into domain-general and domain-specific knowledge both of which affect CPS performance. Information processing includes the task strategies that are selected and the processes of task monitoring and progress evaluation. In addition, non-cognitive problem-solver variables such as motivation and personality also factor into CPS performance.

The task itself is depicted in terms of the barriers that exist between a given state and a goal state. As noted above, the barriers are assumed to be complex, dynamically changing, and non-transparent; the transition from given to goal state is constrained by the problem solvers' memory content and information processing, and by the tools that are available to solvers.

The environment includes the resources available for problem solving, as well as feedback, expectations, cooperation, peer pressure, disturbances, etc. The environment affects both problem solvers and the task. It affects problem solvers by constraining the information processes that can be used and by influencing which knowledge is accessible. The environment affects the task by offering additional information, constraining which tools may be used, and so on. In addition, the environment can be changed actively by the problem solvers but not by the task.

From this relatively simple view of CPS, it should become clear that two of the main questions that will need to be addressed by future research are: (a) which components within the problem solver, task, and environment affect CPS in which way and (b) how do the various components (person, task, and environment) interact in affecting CPS performance? Clearly, much more research will need to be conducted before one can attempt to answer these questions.

Methodological Approaches to Studying Complex Problem-solving

Of the many different methods that could be used to study CPS, two approaches have become favorites in recent years: the experimental method and single case studies. Although some researchers have argued that CPS cannot be fruitfully approached with classical experimental techniques, others have demonstrated rather convincingly that experimental techniques can be used to further our understanding of CPS. It now appears that the controversy between proponents and opponents of the use of experimental methodology may, at least in part, be based on a false conception of what experimental methods really are. The experimental method is a set of techniques that, at least in our view, can be employed not only to test a set of static assumptions, but also to test dynamic process models. Proponents of the experimentally oriented research strategy argue that false theoretical models can be identified only with the experimental method; that is, only experiments allow one to make ultimate decisions about scientific hypotheses. Thus, the experimental method may be most fruitfully employed for the purpose of theory testing.

The phenomenologically oriented research strategy, on the other hand, relies strongly on the precise reconstruction of single cases of CPS. Cognitive modeling is then used to reconstruct certain aspects of individual behavior. In general, it appears to be rather difficult with this approach to identify false theoretical assumptions. Proponents of the single-case methodology therefore frequently argue that single cases are useful primarily for explorative purposes (i.e. for the development of scientific hypotheses).

We consider the two approaches, the experimental method and single case studies, to be complementary. Single case studies may be employed most fruitfully during theory development; the strength of the experimental method, in contrast, is that it provides a strong test of proposed assumptions. While both of these approaches are helpful and both have the potential to advance our understanding of CPS, the effectiveness of the two approaches is critically affected by some *general* problems of the research domain. Addressing these problems is of critical importance before both the experimental method and the single case studies can be employed. We list four of these general problems below:

1. The first general problem concerns the *measurement of CPS knowledge and performance*. The adequate measurement of the problem solvers' knowledge and performance in CPS situations represents a major hurdle that needs to be addressed and resolved before we can make any real progress toward understanding CPS. To this end, Hbner, for instance, has proposed mathematical procedures for the operationalization of certain aspects of task performance. Kolb, Petzing, and Stumpf propose the use of operations research methods for the same purpose. We believe that real progress will not come from these propositions but will come only from theoretical advances. Any good theory of CPS must prescribe the dependent variables and must outline how these variables can be measured. Additionally, a theory of the formal system itself may help to select important and reliable indicators of system performance.
2. The second general problem concerns *generalizability and external validity*. Although the artificial systems currently used in our labs are much more complex than they were in the early 1980s, we cannot necessarily assume that increased complexity has also led to improved generalizability. Dörner's attempt to bring complexity into the labs of the scholars of thinking and problem solving was successful – but has the situation really changed with respect to our understanding of real-world phenomena? We agree with Hunt, who argues “Geneticists have a theory that explains how one generalizes from inheritance in the fruit fly to inheritance in human beings. Cognitive psychology does not have a theory to explain how we move from game behaviors to behaviors in other situations.”
3. The third general problem concerns the *analysis of problem solving processes*. It is critical that we try to understand the process of CPS, rather than the product. The experimental method has not been specifically designed for process analyses, although experimental treatments can help in testing assumptions about parameters and their assumed dependence on external factors. Thus, process models and experiments are not contradictory; they are complementary tools that help us understand CPS. For this reason, modeling approaches of problem-solving processes may offer interesting insights but need to be enriched by empirical data.
4. And finally, the *development of problem-solving theories* is in a rather desolate condition. Developing a theory, or multiple theories, is the most difficult job – and yet at the same time it is the prerequisite for additional experimental research. A good theory prescribes and determines experimental research. Theoretical assumptions can be derived from everyday experiences, from cognitive modeling, or from single case or field studies. Most, if not all, of the assumptions can be tested

experimentally – but neither the experimental method nor the single case approach prescribes the development of theories.

It is our view that the experimental method will remain one central method of choice for studying human CPS simply because no other method is as capable of providing decisive answers to clearly formulated questions. At the same time, however, it remains clear that progress in this difficult research area can be achieved only if different approaches work together to achieve insights into how people deal with complex problems.

THEORETICAL PERSPECTIVES ON THOUGHT PROCESSES

Most Important Theories of Thinking are: 1. Associationism, 2. Gestalt, 3. Information-processing models, 4. The Central Theory, 5. The Peripheral Theory.

Associationism

Originally this was a philosophical theory which sought to reduce the contents of the mind to a set of elementary sensations or ideas from which more complex ideas were built up by associations between them. Associationism theory has always been reductive, in the sense that it sought to analyse complex ideas into their simpler components.

According to this approach, events or ideas will be associated if they have frequently or consistently occurred together. This can be expressed as one law or can be analysed into at least three separate notions, temporal contiguity, spatial contiguity, and constant conjunction.

One idea will be associated with and hence give rise to another, if the events producing those ideas have occurred at the same time as one another, or in the same place as one another, or if they have consistently occurred together. To these fairly generally accepted laws some theorists would add such subsidiary factors as similarity: one idea will give rise to another to the extent that they resemble one another.

Gestalt

Gestalt psychologists disputed that the mind could be understood as a passive collection of simple ideas joined together by the inexorable operation of a few “laws of association”. The slogan of Gestalt psychology that the whole is greater than the sum of its parts, is best understood as a reaction against associationism.

Gestalt theory worked chiefly in the field of the psychology of thought and studied how patterns of behaviour which have already been firmly acquired are reorganized into relevant behaviour for supplying a solution.

The construct of insight was intended to explain this reorganization. Gestalt theorists approached problems in the psychology of thought from the angle of that of perception. Successful thinking involves a perception or an understanding of the structural and functional relationships within a problem context.

To solve a problem one must grasp the “inner relationships” that bind that context into an organized whole.

As a process, problem solving begins with centering of attention on problem elements and their connection with the basic difficulty. A goal or solution is assumed to exist, and the subject is drawn to it by the most direct path.

Solution attempts involve some active reorganization of elements by the individual with a resulting recentering of attention. Each recentering sets-up a pattern of tension towards the goal within the organism, which he seeks to resolve.

These tensions vary according to a number of laws including those which pertain to figural qualities of the goal object and to subject's geographical location relative to that object. Resolution or closure is achieved only by the attainment of the correct recentering, i.e., of a solution which is perceived as satisfactory.

The resulting behavior change is often immediate, leading Gestalters to speak of insight as one of the fundamental phenomena of thinking (and learning).

Insight, as suggested by Miller, can be thought of as a creator of a metaprogram (probably linguistic) or solution plan which controls the application of subordinate programs (patterns of behavior) already established and available.

Thus, in essence, the presence of a problem motivates us to think. Through thinking we search means for the solution of a problem. In this effort we make use of all the tools and previously learnt concepts. We discover some new concepts in this effort. We can solve a problem by analysing its different aspects with the knowledge of the whole situation.

Opposition to Associationism

The origin of the gestalt theory is connected with the difficulties encountered by the traditional associationism in regard to perception and thinking. The associationists believed that perception is a copy of objects, or a mental image of what has been perceived, and thinking is a mechanical combination of those images. Gestalt psychology challenged both parts of this belief.

Origin

In 1890, Christian Von Ehrenfels introduced the idea of gestalt or form, shape and structure. The name gestalt psychology was originally applied to two schools of thought. Max Wertheimer and his students, Kurt Koffka and Wolfgang Kohler cantered their activities in Germany another, similarly oriented theory developed in Austria.

Both sought to explain what might underline sensory experience; both addressed themselves to the problem of discovering the nature of perceptual organization. Their explanatory hypothesis distinguish them from each other. The Austrian Gestaltists believed that perception was the result of processes within the brain actively to combine a number of separate sensations into a meaningful experience. Example the melody one heard when listening to a symphony was constructed by the brain from a series of individual neural impulses; one for each note.

The German Gestaltists held that organisation of experience occurred automatically as a result of the physiological functioning of brain, particularly the sensory areas. For them the melody of a symphony was imposed on the brain by the prior working of the nervous system.

From their interest in perceptual organisation the German Gestaltists gradually moved to a consideration of behaviour in general. It was their interest in these more general principles which enabled German Gestaltists to flourish while Austrian group dried out.

Sources of Gestalt Psychology

The Gestalt psychology of Wertheimer, Koffka and Kohler arose from two sources. Firstly, a rejection of the atomistic approach of other theories lent negative impetus. Secondly, while Wertheimer's work with phi phenomenon suggested an alternative.

More Oppositions

The negative pressures which fostered their work includes the following. Gestaltists objected structuralism because it reduced mind to elements and in doing so destroyed something very important. The patterning or organization of sensory events. Their attempt was mechanistic, artificial and misleading. According to Gestaltists, the whole thing was something more than its parts taken separately and could only be understood by laws focussing on gestalts (organised wholes).

A second disagreement between the structuralists and the founders of the gestalt school was the constancy hypothesis. The structuralists assumed that things were forever and always, what they seemed to be. According to them, behavioural or perceptual environment was identical. By concentrating on the raw feel of experience and disregarding the perception of its meaning or connotations, the structuralists were led to exaggerate the validity of sensory processes and to neglect the importance of situational determinants in experience. Gestaltist, who recognised constancy as a naturally occurring event, the study of which could supply valuable information about perception.

The Gestaltists had equally strong criticism to make of their contemporaries, the behaviourists. It was commendable for the behaviourists to be so objective, but in their objectivity they left out everything that intervened between stimulus and response, neglecting many important aspects of psychology purely because they could find no scientific way to investigate such things. The Gestaltists in contrast to behaviourist's S-R theory, they maintained an S-O-R.

Patterns of stimuli Organism's perceptual organization response determined by perception.

Accusations were also levelled at Thorndike's whose concept of trial and error learning was not acceptable to the Gestaltists. The whole concept was wrong, the errors which an animal made in a puzzle box were not blind errors at all. Errors did not occur accidentally, but were produced by the organism's perception of the field in which he was operating. The law of effect was also mechanistic.

The Gestaltists concentrated on insight learning and were convinced that learning was discontinuous, subject to abrupt changes. It also seemed erroneous to the Gestaltists to propose that reinforcement stamped associations in or out. Learning was a matter of appreciating perceptual relationships and reinforcement was an event to be appreciated, not an instrument to aid in the appreciation of events.

Phi Phenomenon

The phi phenomenon is produced when two small slits, at right angles to each other on an otherwise opaque screen, are lighted successively so that the bar of light appears to move from one position to another. Wertheimer, Koffka and Kohler were impressed with the effect of different patterns of stimuli on perception and concluded that such patterns must be the result of interaction among the neural processes with the brain. Having arrived at this conclusion, they developed themselves to testing it and to generalize upon it.

Definition of Psychology

According to Gestaltists “psychology is the study of the immediate experience of the whole organism”.

Basic Unit of Study (Methodological Consideration)

The basic unit of study is the physical gestalt, its totality of experience and consciousness. They distinguish between the physical and phenomenological reality. The shapes and figures human beings perceive are established by the action of forces that tend towards equilibrium. These forces do not lie outside but are within the mind. They are phenomenologically real, but besides phenomenology, there is physical reality. This includes the nervous system and its interaction with the environment.

They do not totally reject the physical reality. They relate physical and phenomenological reality by the concept of isomorphism. They are the set of laws that govern physical and phenomenological reality. Gestalt existed both in mind and nature. Thus, according to Gestaltists order, regularity and gestalt are the universal principles that bind psychology and physiology.

Gestalt psychologists had to seek this order, the gestalt in nature or accept the idea that all gestalts were Apriori ideas. Kohler accepted Mach's hypotheses, which implied that macroscopic physical states develop in the direction of equilibrium, stability, regularity and simplicity. This backs the assumption of unity within the universe. Thus, it becomes a reductionist concept.

Kohler found a regularity in gestalt with law of pragnaz and he explained this by talking about electrostatic changes that work to produce some equilibrium. Gestalt psychologists introduced the concept of organization between the stimulus and response of the behaviouristic theory. Between the stimulus and the response processes of organization take place which build the elements into a complex unit.

For the Gestaltists the only plausible approach to scientific investigation was a molar one, considering the complex act or experience as the unit of significance, and not as some simple S-R sequence which might compose it.

Basic Postulates

The primary postulate of the Gestaltists is related to the whole-part attitude. According to them the whole dominates the parts and constitutes the primary reality, the primary datum for psychology, or any simple function of its parts, but a field whose character depends upon all of itself. The secondary postulates include the isomorphism principle and laws of organization.

Isomorphism

Kohler's theory of isomorphism is based on the assumption of unity of the universe. They are same set of laws that governs phenomenological and physical reality. According to Kohler, the principle of isomorphism covers practically the whole field of psychology. It says that experienced order in time is always structurally identical with a functional order in the sequence of correlated brain processes.

Psychological Field

Since the functions of the brain represent a dynamic field, the overt behavior of humans is a dynamic field too. This is the essence of the principle of isomorphism. The dynamic

field of psychological experience is the individual and his environment, and their interaction within the field forms the centre of the behavior. This field is a field of forces which tend to establish the best possible balance, the most symmetrical, stable and simple configuration. The laws of Gestalt, especially of pragnaz, apply to this field.

Human actions are usually related to physical reality not as it is but as it appears to the perceiving subject. The environment as perceived together with the perceiving subject form the psychological field. Koffka's psychological field called the behavioral environment corresponds to the phenomenological environment.

The idea of a psychological field isomorphic to the physical field enabled the gestalt psychologists to develop a theory of volition and emotions. Disequilibrium between organism and environment causes tension and activates forces moving in the direction of restoration of balance. A hungry organism seeks food, an angry man starts a fight. Personality is often called ego by the Gestaltists.

Each psychological field contains an important subfield, personality or the ego. Koffka says emotions will arise when an object with powerful positive valence appears in the field, and the subject is barred from achieving it. The impenetrable barriers cause tension, involving the ego area, and explosive emotional behavioral flows.

This approach to the problem enabled the gestalt psychologists to minimize the anamnestic factors and to emphasize the influence of the present environment. The present environment and the person are the two poles of a dynamic field with forces acting between them. The present action of the forces and their striving toward an equilibrium are the determinants of behavior.

Neurological Theory

Gestaltists have rejected the models of central nervous system given by earlier psychologists like Hebb, Lshley etc. They are against atomism. As per kohler mental events have been explained in terms of either inherited machine or secondary acquired constraints.

Gestaltists are also against the step by step process (chain of S-R units). They conceive of personal whole based on dynamic distribution and organization in an analogy with an electric circuit. The theory of electric circuit has been used by the gestalt psychologists in their studies of brain mechanism. The brain activity is guided by the principle of equilibrium or atleast reduced tension. Whenever there is a gap in the current, tensions build up on both the sides of the gap and the electric current tends to close the gap. Thus, the principle of closure is the most important principle of both the physical and phenomenological worlds.

Kohler assumed that in the cerebral cortex a chemical substance is produced depending upon the intensity of stimulation, and the figure ground phenomenon should be interpreted by a huge concentration of ions inside the figure in distinction from the low concentration in the cortex. The cortex is the field, and in it as in other dynamic field, the process itself and its conditions should be considered. The brain functions as a dynamic field, and its forces can be self distributed and self regulated always in direction of equilibrium.

Opposition to Quantification

Gestalt psychologists could not believe in the usefulness of statistical computation for various reasons. Measurements are based on the assumption of commensurability and

such assumption was feasible for atomistic psychology (Gestaltists were against atomistic psychology). Gestalt psychologists found that the same number of the same elements could form various configurations and present various experiences. Example what's the use of adding the number of dots if the same number of dots can form a circle, a square, a rectangle or a triangle?

Gestalt: Perception, Learning and Thinking

Learning By Insight

Gestalt psychology introduced a new theory of learning associated with the name Wolfgang Kohler and the term insight. He conducted experiments on chimpanzees. In that chimpanzees, grasped the new situation; they displayed orientation in learning and a great deal of intelligence. They did not try blindly all the possible responses; their behavior showed that learning included a considerable amount of goal directed activity. Kohler's chimpanzees were put in the cage and a banana was placed at a distance from the cage. Strings, sticks and boxes were used by the experimental animals to get the banana; they apparently somehow perceived the situation and displayed insight in solving the problem.

Koffka concluded that all learning was perceptual reorganization. What the animal has developed during its latent learning is a trace of a maze; this trace being in communication with the present activity, regulates it more or less as the perception of the maze world. Koffka applied the laws of perception to the learning theory. Instead of reward or reinforcement, he introduced the concept of goal. So long as a activity is incomplete, every new situation created by it is still to the animal a transitional situation; whereas when the animal has attained his goal, he has arrived at a situation which is to him an end situation.

The main factor of the gestalt theory is insight. Since the individual and his environment form a psychological field, perception of the field and gradual restructuring of it is insight. A description of what takes place in insightful learning was given by Yerkes after a series of experiments with primates. Insight, in different organisms may reveal common characteristics.

1. Survey, inspection, or persistent examination of problematic situation;
2. Hesitation, pause, attitude of concentrated attention;
3. Trial of more or less adequate mode of response;
4. Incase initial mode of response proves inadequate, trial of some other mode of response, the transition from one method to the other being sharp and often sudden;
5. Persistent or frequently recurrent attention to the objective or goal and motivation thereby;
6. Appearance of critical point at which the organism suddenly, directly and definitely performs the required adaptive act;
7. Ready repetition of adaptive response after once performed;
8. Notable ability to discover and attend to the essential aspect or relation in the problematic situation and to neglect, relatively, variations in non-essentials.

Learning and Perception

Since learning is interrelated with perception, the law of perception apply to learning. The most general principle of learning is *Pragnanz* or the goal directed tendency to re-

store the equilibrium. Learning takes place when there is a tension or disequilibrium of forces in the psychological field; the learning process removes the tension and is therefore, guided by the principle of *Pragnanz*.

The additional laws of learning are similarity; proximity; closure and good continuation. When an organism learns a certain material which contains similar or dissimilar elements, the similar ones are learned more readily than the dissimilar ones. This is the law of similarity.

The law of proximity in perception becomes a law of temporal contiguity in the learning theory. The elements are grouped together both by factors of physical proximity (to form pattern in space) and by proximity in time.

The law of closure reflects the idea of striving for completion. Koffka applied the law of closure as applied to learning as follows: so long as activity is incomplete, every new situation created by it is still to the animal a transitional situation; whereas when the animal has attained his goal, he has arrived at a situation which is to him an end situation.

The law of good continuation implied that we tend to learn better those elements which show consistency in their configuration.

The gestalt psychologists seemed to indicate that learning is improvement in gestalt; the learned figures become more symmetric, better organized, in accordance with law of *pragnanz*. Thus, the field becomes restructured and becomes a better gestalt.

The problems of recall and retention presented considerable difficulties for the gestalt theory. Koffka and Kohler solved this problem of presenting learning processes as cognitive fields. Fields can be structured and restructured. Learning processes change the structure of the fields; accordingly, they leave traces. In each repetition the former trace system is modified by the new trace system.

Several gestalt experiments indicated that three processes take place in learning. The first is leveling or changing in the direction of symmetry and good distribution. The second, sharpening, consists in the accentuation of the essential elements of a figure, which makes it easily distinguishable. The third process is related to the clarity and simplicity of the perceived figure and is called normalizing. These three processes correspond to the general law of *pragnanz* or moving toward a good gestalt. It is the productive, goal directed insight versus the reproductive conditioning which has been emphasized in the gestalt theory of learning.

Productive Thinking

The ground for productive thinking was prepared by Wurzburg school. Wertheimer undertook the study of productive thinking and problem solving using a great many subjects. He distinguished three types of processes.

Type *a* processes are concerned with decisive issues pertaining to structural problems and include such operations as grouping, reorganization, and discovery of essential features. The best thinking is productive thinking. It relates the means to the tasks and the goals, and to the total situation. Whenever an organism faces a problem, the tensions will lead to an activity called problem solving or productive thinking.

Productive thinking is assured by type *a* processes and almost eliminated by type *y* processes. Type *y* processes are blind and result in premature conclusions, miscentering or a restless pursuit of a hypotheses without a sense of direction and subject to the

influence of external factors. This type corresponds to learning by drill, associations, conditioning and blind trial and error. Type b processes are partially productive and partially mechanized.

Analysis is futile when it is only a mere division into parts. In combination or association, things cannot be thrown together or related at random with utter disregard of structural aspects and interrelations. This does not lead to problem solving. Productive thinking deals with whole-qualities and part-whole relationships rather than with analysis into particles and association of them. Productive thinking looks for the structural truth rather than for a piecemeal truth, said Wertheimer.

Wertheimer in his theory also states centering and recentering. In centering, there is a transition from the subjectively or personally centered view to a detached view with an objective grasp of the whole situation and of structural and functional requirements. Recentering is the obtaining of a new and penetrating perspective. It provides a new angle from which to view the question of multiple interests and achievement of creative persons, and the ability of scientists and engineers trained in certain fields at high levels of productivity.

Repetition is useful, but continuous use of mechanical repetition also has harmful effects. It is dangerous because it easily induces habits of sheer mechanical action, blindness, tendencies to perform slavishly instead of thinking, instead of facing a problem freely.

Wertheimer denied any application of trial and error in thinking. Thinking is goal directed and insightful and creates new gestalts. It changes the structure of the perceptual field. Apparently the gestalt theory went a long way beyond the studies of Wurzburg school. The school proposed the set and goal. The Gestaltists ascribed purposeful activity to nature and men, activity leading toward a perfect gestalt.

Information Processing

At the very heart of cognitive psychology is the idea of **information processing**. Cognitive psychology sees the individual as a processor of information, in much the same way that a computer takes in information and follows a program to produce an output.

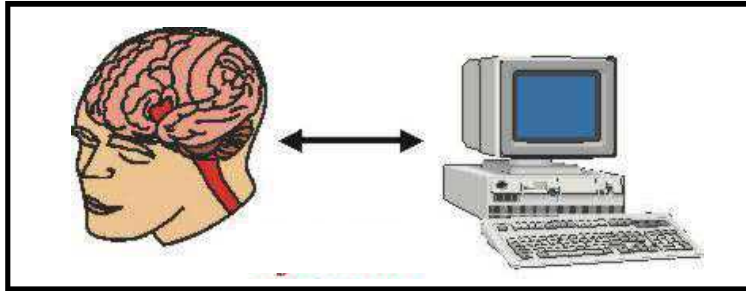
Cognitive psychology compares the human mind to a computer, suggesting that we too are information processors and that it is possible and desirable to study the internal mental/mediational processes that lie between the stimuli (in our environment) and the response we make.

Basic Assumptions

The information processing approach is based on a number of assumptions, including:

1. information made available by the environment is processed by a series of processing systems (*e.g.* attention, perception, short-term memory);
2. these processing systems transform or alter the information in systematic ways;
3. the aim of research is to specify the processes and structures that underlie cognitive performance;
4. information processing in humans resembles that in computers.

Computer - Mind Analogy



INFORMATION PROCESSING PARADIGM.

The development of the computer in the 1950s and 1960s had an important influence on psychology and was, in part, responsible for the cognitive approach becoming the dominant approach in modern psychology (taking over from behaviorism). The computer gave cognitive psychologists a metaphor, or analogy, to which they could compare human mental processing. The use of the computer as a tool for thinking how the human mind handles information is known as the computer analogy.

Essentially, a computer codes (i.e. changes) information, stores information, uses information, and produces an output (retrieves info). The idea of information processing was adopted by cognitive psychologists as a model of how human thought works.

For example, the eye receives visual information and codes information into electric neural activity which is fed back to the brain where it is “stored” and “coded”. This information can be used by other parts of the brain relating to mental activities such as memory, perception and attention. The output (i.e. behavior) might be, for example, to read what you can see on a printed page.

Hence the **information processing approach** characterizes thinking as the environment providing input of data, which is then transformed by our senses. The information can be stored, retrieved and transformed using “mental programs”, with the results being behavioral responses.

Cognitive psychology has influenced and integrated with many other approaches and areas of study to produce, for example, social learning theory, cognitive neuropsychology and artificial intelligence (AI).

Information Processing & Attention

When we are selectively attending to one activity, we tend to ignore other stimulation, although our attention can be distracted by something else, like the telephone ringing or someone using our name.

Psychologists are interested in what makes us attend to one thing rather than another (**selective attention**); why we sometimes switch our attention to something that was previously unattended (e.g. Cocktail Party Syndrome), and how many things we can attend to at the same time (**attentional capacity**).

One way of conceptualizing attention is to think of **humans as information processors** who can only process a limited amount of information at a time without becoming overloaded. Broadbent and others in the 1950’s adopted a **model of the brain** as a **limited capacity information processing system**, through which external input is transmitted.

The Information Processing System



Information processing models consist of a series of stages, or boxes, which represent stages of processing. Arrows indicate the flow of information from one stage to the next.

Input processes are concerned with the analysis of the stimuli.

Storage processes cover everything that happens to stimuli internally in the brain and can include coding and manipulation of the stimuli.

Output processes are responsible for preparing an appropriate response to a stimulus.

Critical Evaluation

A number of models of attention within the Information Processing framework have been proposed including:

Broadbent's Filter Model (1958), Treisman's Attenuation Model (1964) and Deutsch and **Deutsch's Late Selection Model** (1963).

However, there are a number of evaluative points to bear in mind when studying these models, and the information processing approach in general.

These include:

1. The information processing models assume serial processing of stimulus inputs.

Serial processing effectively means one process has to be completed before the next starts.

Parallel processing assumes some or all processes involved in a cognitive task(s) occur at the same time.

There is evidence from dual-task experiments that parallel processing is possible. It is difficult to determine whether a particular task is processed in a serial or parallel fashion as it probably depends (a) on the processes required to solve a task, and (b) the amount of practice on a task. Parallel processing is probably more frequent when someone is highly skilled; for example a skilled typist thinks several letters ahead, a novice focuses on just 1 letter at a time.

2. **The analogy** between human cognition and computer functioning adopted by the information processing approach is limited. Computers can be regarded as information processing systems insofar as they:
 - (i) combine information presented with stored information to provide solutions to a variety of problems, and
 - (ii) most computers have a central processor of limited capacity and it is usually assumed that capacity limitations affect the human attentional system.

BUT

- (i) the human brain has the capacity for extensive parallel processing and computers often rely on serial processing;
- (ii) humans are influenced in their cognitions by a number of conflicting emotional and motivational factors.

3. **The evidence** for the theories/models of attention which come under the information processing approach is largely based on experiments under controlled, scientific conditions. Most laboratory studies are artificial and could be said to **lack ecological validity**.

In everyday life, cognitive processes are often linked to a goal (e.g. you pay attention in class because you want to pass the examination), whereas in the laboratory the experiments are carried out in isolation from other cognitive and motivational factors. Although these laboratory experiments are easy to interpret, the data may not be applicable to the real world outside the laboratory. More recent ecologically valid approaches to cognition have been proposed (e.g. the Perceptual Cycle, Neisser, 1976).

Attention has been studied largely in isolation from other cognitive processes, although clearly it operates as an interdependent system with the related cognitive processes of perception and memory. The more successful we become at examining part of the cognitive system in isolation, the less our data are likely to tell us about cognition in everyday life.

4. **The Models proposed by Broadbent and Treisman are ‘bottom-up’ or ‘stimulus driven’ models of attention.** Although it is agreed that stimulus driven information in cognition is important, what the individual brings to the task in terms of expectations/past experiences are also important. These influences are known as ‘top-down’ or ‘conceptually-driven’ processes. For example, read the triangle below:



Expectation (top-down processing) often overrides information actually available in the stimulus (bottom-up) which we are, supposedly, attending to. How did you read the text in the triangle above?

The Central Theory

According to this theory thinking and reasoning is taking place only in the brain and the brain is solely responsible for all kinds of thinking and problem solving activities. When we think with our brains it's often said that the brain secretes thoughts.

The Peripheral Theory

The peripheral nervous system is mainly involved in this. The peripheral theory is also called motor theory. According to this theory the brain is one of the parts of thinking

process. Since all our learning are both mental and physical, we also think with our whole parts, brain and the nervous system-the whole body.

There is a continuous chain of S-R relations from brain to body and from body to the brain. This goes on till the process of thinking takes place. The founder of behaviourism Watson said that thinking is silent speaking as there were muscular movements in mouth and in throat and speaking is louder thinking.

There is strong evidence that muscular activity is present during thinking. Because of this we can-not reject the central theory totally. According to them, thinking is going on in the brain without the co-operation of the peripheral or motor system. But because of the end product of verbal and muscular activities the peripheral theory seems to be more agreeable than the central theory.

CONCEPT FORMATION

A concept is the mental model of an object. The mental model encompasses the object's definition, blueprint and examples. For example, the concept of "boat" includes linguistic description (definition), drawing (blueprint), and picture (example). We develop a concept through a prototyping process, where tentative models are tested until a polished model is derived.

Converting concept into emergent behavior often takes "critical mass." To propagate a concept, either it appeals to a group of capable people or catches the attention of the mass media.

Concepts serve multiple functions. We recognize objects by matching the sensory data with known spatial concepts. We describe causal mechanisms with a causal model. Finally, we develop concepts to solve problems.

Naming of a Concept

We name a concept (mental model) so that the concept can be identified and shared. There are two categories of names. The general name is a symbolic representation of a class of objects. The particular name is a symbol that represents a particular object.

Not every distinct object merits a name. For instance, a cloud does not need a particular name, rather it falls into the general name of "cloud." On the other hand, a pet deserves to a unique name since there is a need to distinguish it from others.

General or Particular Concept

A vague pattern which appears on the imagery layer is not a useful concept since we do not know how to distinguish or reproduce it. A concept should have a clear and identifiable mental model such as definition, blueprint, or example.

A concept can be general or particular. The general concept is a generic form with variable parameters such as a mathematical function. The particular concept is the identification of a particular object. It is a substantiation of general concept.

The Definition of a Concept

We can define a concept with special features, or traits. For instance, "chair" has four legs, a flat seat, a back, and it can be sat on. Some concepts have formal definitions, which are based on the universal conceptual system. The basic concepts are defined in

the dictionary while the advanced concepts are defined by specialized texts. Some concepts have a commonly accepted definition, and some are more open to debate. All the traits in an outstanding *trait set* are defined recursively. For instance, “pleasure, smile” is a trait set. Thus we may define “pleasure” as a state that causes us to “smile,” and recursively, define “smile” as an expression of “pleasure.” Ideally, we should present all the outstanding trait sets that contain this trait in order to fully define a concept.

Schemas

Our collections of concepts are our *schemas*. We use our schemas to interpret familiar experiences, and to incorporate or assimilate new experiences. Our schemas consists of names and categories.

Rule of Concept Formation

The brain develops many conceptual sectors through learning. A conceptual sector is composed of templates and scripts that can manipulate coherent mental models stored in the explicit memory.

Each person’s conceptual activity is supported by his or her distinct cognitive styles, including systematic method, heuristic strategy, insight, intuition, and trial and error. In a conceptual sector, information is processed in input stage, analysis stage, and composition stage.

Input Rule

The meaning of information is defined by its mental models, which can be innate or learned. There are two types of concept that can be extracted in the perceptual process. We identify sensory concept of physical objects with their distinct traits and we identify symbolic concept of abstract matter through the deliberate learning of symbolic rules. Sensory information can be described with lower level distinct fundamental prototypes where an object is defined with traits (object={trait1, trait2, ...}). When the traits match, an object is identified.

Traits

A concept can be defined with traits. A *trait* is a simple pattern that can be analyzed autonomously with feature space. The conceptual input module identifies patterns based on predefined traits. For instance, a triangle is a trait that is made up of three lines connecting to each other. The cognitive system handles distinct patterns better than random images because patterns can be stored into memory and recalled without losing significant details.

There are several types of traits. A spatial trait can be reduced into lower resolution without losing its resemblance to the original image. An auditory trait is the pattern of sound that is associated with an object or event. A linguistic trait is the pattern of words that corresponds to an event. A mathematical trait describes the quantitative nature of matter. Some of these traits have no immediate spatial implication and are abstract. A non-measurable trait cannot be quantified and is referred to as a secondary quality of matter. For instance, the quality of “kindness” is a non-measurable secondary trait. Other traits that do not belong to either one of these categories often utilize other prototyping tools as needed.

We can learn traits through formal or informal education. We can also extract traits ourselves by observing patterns of event.

Common Traits

We can often develop a *set of traits* for multiple applications. For example, all sports require muscle strength, agility, competitiveness, and training. The fundamental traits for a human being are self-confidence, patience, and integrity.

Analysis Stage

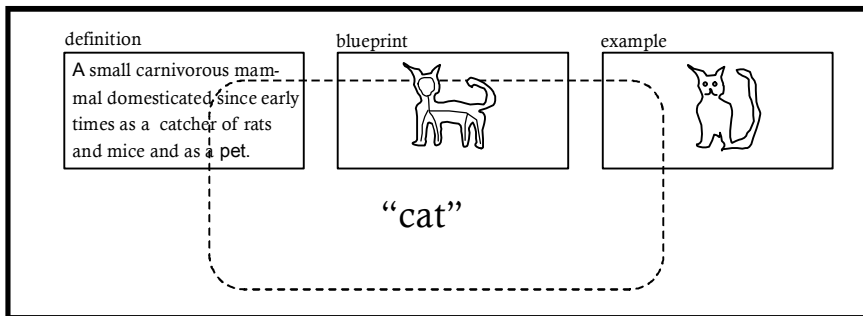
The analytical modules in our brain analyze data based on predefined mental models. These modules conduct several analytical processes in tandem, and they present several potential prototypes to later stages. Each person has a distinct style of analysis, which is either visual or verbal. Visual-type people translate words into images when they think, while verbal-type people translate images into words.

Category

Concept can be divided into several categories according to their relationships with reality. Empirical concept is a mental model that has correspondence in reality. Imaginative concept, in contrast, may have no correspondence in reality. Impressional concept generates outstanding impressions on thoughts.

Composition Rule

In the composition stage we derive definition, blueprint, or examples of a concept through a prototyping process. Since we can fit a given phenomenon to different mental models, we cannot claim that a model is the only one that can explain a given phenomenon.



A WORKING MEMORY FOR THE CONCEPT FORMATION OF "CAT".

Prototyping Process

We create concepts with a prototyping process by using language, mathematics, logics, charts, computers, and physical materials.

There are two levels of prototyping. We manipulate primitive objects in a working memory with moving, sizing, and rotation. We also use symbols such as language or logic to create symbolic models. Symbolic prototyping ability is unique to higher animal, including non-human primates and human beings.

Innate Prototyping Capacity

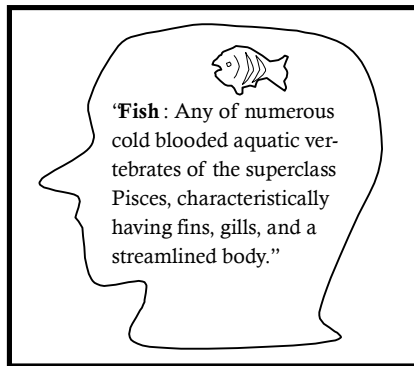
The brain has many innate capacities to conduct prototyping. The cognitive system can manipulate the imagery layer by rotating, shifting, and zoom into objects in the visual imagery layer of the working memory. The language sector represents objects with words. The auditory sector perceives frequency and rhythm with auditory prototypes.

Learned Prototyping Capacity

Advanced prototyping capacities are acquired through learning. Mathematics uses quantitative models to analyze objects. Logic uses models of validity and domain to validate transformations. Language uses linguistic models to represent all things in universe.

Definition With Words

We can describe or define objects or abstract ideas with words.



DEFINING AN OBJECT WITH WORDS.

Prototyping with Basic Objects

We can move, enlarge or shrink, and rotate an object in the imagery layer of the working memory.

Association

The concepts stored in the explicit memory can be linked to form associations. When we recall something, we also recall its associated information.

Connecting the Dots

At any given moment, the cognitive system maintains a large number of objects in mind and arranges these objects to form information. For example, when we listen to music, we put pieces of sound together to recreate the music in mind.

During the conceptual prototyping, we first collect elements, or handles that are related. Then like connecting dots into a figure, we put these elements together into a meaningful structure. The result is a prototype that may develop into a mental model (concept).

Simplicity of Prototypes

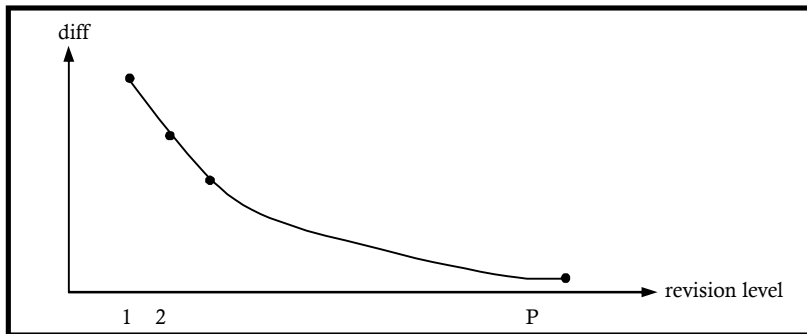
Simple prototypes are easier to learn. Civilizations have been built upon simple prototypes. As society progresses, many complex prototypes are developed, many of them have

no particular benefits. These prototypes sap the energy of society and indirectly cause the decline of a civilization.

Revision of Prototypes

A prototype is developed from the first to the final revision. Every revision in-between is created by modifying the previous one: revision $n+1 = \text{revision } n + \text{diff}$, where *diff* is the difference between two revisions. A stabilized prototype has the smallest possible amount of modification between revisions.

Most systems can be described as a *state machine* with many states (revisions). Each state can be transformed into next state by making modifications. We make changes to it and generate the next state (revision). We repeat this process many times until a desirable final state is reached.



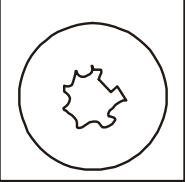
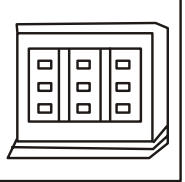
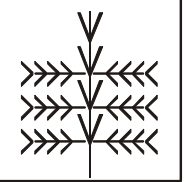
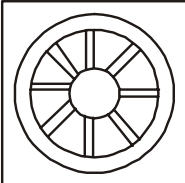
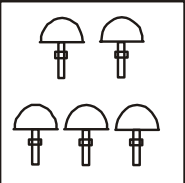
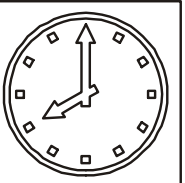
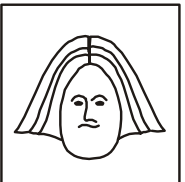
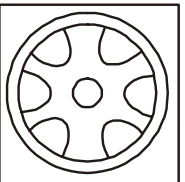
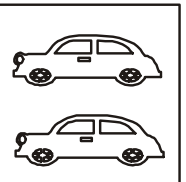
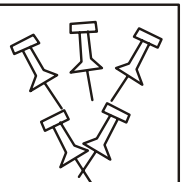
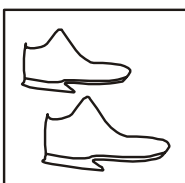
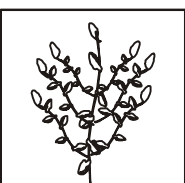
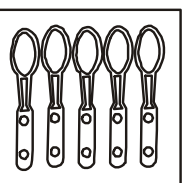
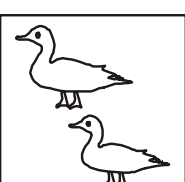
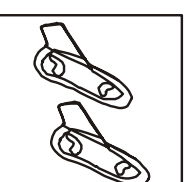
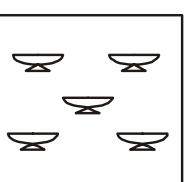
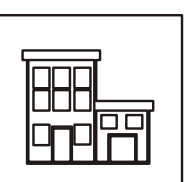
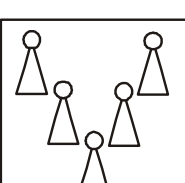
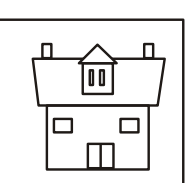
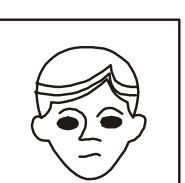
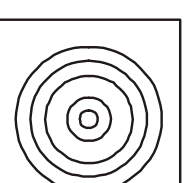
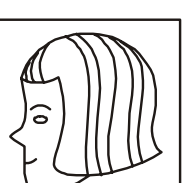
A PROTOTYPE'S REVISION LEVEL VERSUS THE DIFFERENCE BETWEEN REVISION N AND $N+1$. THE *DIFF* CAN BE MEASURED BY THE NUMBER OF WORDS IN AN ARTICLE OR BY THE LINES OF CODE IN A PROGRAM.

Strategy

Concept learning utilizes the psychological processes of generalization and discrimination. In learning the concept of triangularity, a rat generalizes the response initially to other geometrical forms, but since these responses go unrewarded they are extinguished, and it eventually narrows the discrimination to triangles. A child learning the concept dog may generalize the term initially to include all small animals. But from parental corrections and personal observations the child learns to make finer discrimination until his concept approximates the conventional conception of the dog. The child may refine the concept further and distinguish between friendly dogs, whose wagging tails indicate that approach is safe, and unfriendly dogs, whose growls signify that avoidance is the best response. Eventually the child will learn to distinguish among various breeds.

Characteristics

Studies by Heidebreder indicated that concrete characteristics are generally easier to conceive than the more abstract relationships of form and number. These studies used a paired-associate technique with the type of material illustrated in next figure. On each trial several pictures were presented one at a time, and the subject was required to anticipate the response paired with each one. The experimenter arbitrarily assigned a different nonsense word as the response for each concept. Thus, MULD might refer to the object concept tree, FARD to the spatial concept of circular patterns, and DILT to the number concept of five objects. On the second trial new pictures representing the same

Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
				
RELK	FARD	LETH	MULP	LING
				
FARD	DILT	MULP	LETH	FARD
				
LETH	RELK	FARD	LING	DILT
				
LING	MULP	DILT	RELK	MULP
				
MULP	LING	LING	DILT	LETH
				
DILT	LETH	RELK	FARD	RELK

CONCEPT FORMATION.

concepts were paired with the original responses. For example, shown in next page figure shows six pictures that might be presented on trial 1, a new set of six pictures for trial 2, and so on. The experiment continued until the subject responded correctly to all stimuli on a trial.

A series of such experiments showed that object concepts (shoe, book, bird) were the easiest to learn, spatial forms the next easiest, and then numbers. Our thinking apparently tends to run to objects rather than to abstractions. Piaget's stages of intellectual development indicate that the child first learns object concepts and develops more abstract concepts only as he grows older. Interestingly enough, with certain types of brain damage an individual may lose the ability to deal with abstract concepts and respond only in terms of concrete ideas. For example, he may be able to throw balls accurately into three boxes located at different distances from him but not be able to state which box is nearest and which farthest or explain his procedure in aiming. This inability to think abstractly is so marked in certain types of brain damage that performance on concept-learning tasks is sometimes used as a basis for diagnosis.

TYPES OF REASONING

Reasoning is the process by which we make decisions. There are two ways of reasoning. One of these two ways is inductive while the other one is deductive. We shall discuss these two ways of reasoning one by one.

Inductive

Inductive is a process by which we reach up to universal conclusion from particular observations. Or in other words, inductive is a process of reasoning whereby we arrive at universal generalisations from particular facts. Inductive gives rise to empirical generalisations, and is opposite to deductive. Inductive involves a passage from observed to unobserved. Inductive involves two processes—observation and generalization. If, in a number of cases, it is observed that educated girls have got expensive habits, one may conclude that all educated girls have got expensive habits. This is the simplest kind of inductive, and is called inductive by enumeration. But scientific inductive is based on known causal connection. Inductive by enumeration gives us only probable conclusion, but scientific inductive gives us certain conclusions.

A perfect inductive is the process of establishing a universal proposition by an exhaustive enumeration of all the instances of the type covered by the universal proposition. Perfect inductive is not opposite to deductive; but on the other hand, it is an example of deductive. The conclusion in perfect inductive is arrived at by strict syllogistic reasoning. But since in perfect inductive there is really no generalisation and there is no passage from observed to unobserved, it is not sometimes regarded as inductive in the true sense of the term.

Merits of Inductive

1. Inductive generalizations are based on observed facts and realistic foundation, and as such, they are precise and accurate.
2. Inductive method is scientific in character. Some of the important theorems of physical and social sciences have been developed through this method.

3. Inductive method underlines the relativity of laws or generalisations. It points out that a particular generalisation is valid under particular circumstances.
4. Inductive supplies the universal premise and is helpful in finding out material truth.

Demerits of Inductive

1. It is a time-consuming, expensive and complicated method.
2. It often leads to hurried and wrong generalisation.
3. The collection of data for inductive is a complex job. An investigator using this method requires a high degree of competence and training, besides sophisticated tools of analysis.

Deductive

Deductive is the process through which we reach up to particular conclusion with the help of universal facts. Or in other words, deductive is the process of drawing generalisation, through a process of reasoning on the basis of certain assumptions which are either self-evident or based on observation. In deductive, we deduce generalisations from universal to particular. Deductive can give conclusive evidence. Depending upon the premises, every deductive reasoning is either valid or invalid. The main task of deductive logic is to clarify the nature of relation between premises and conclusions in valid arguments. It is concerned with the working out of logical implications between propositions. Let us consider an example:

All men are mortal.

John is a man.

John is mortal.

Merits of Deductive

1. It is a simple and easy method which is not time-consuming and expensive.
2. This method leads to accuracy and precision in generalisation because it makes use of logic, and mathematical tools of analysis. If the premises are true, we can easily get a true conclusion.
3. In a social science, like economics, where there is limited scope for experimentation, this method becomes the only available method for the development of the subject.

Demerits of Deductive

1. It is harmful, when universal validity is claimed for the generalisation arrived at by deductive, particularly, when the premises are incorrect or partially correct. If policy prescriptions are based on deductive, the consequences may be dangerous. The deductive “armchair” analysis should be taken with caution and care.
2. If the assumptions upon which deductive reasoning is based are untrue or partially true, the inferences drawn become automatically beyond truth, therefore, having no operational validity.
3. Deductive method is abstract. If a large dose of abstraction is used in theorising the result is the creation of “intellectual toys” and useless “implicit theorising”.

Distinction between Deductive and Inductive

1. In deductive, we deduce generalisation from universal to particular but in inductive we arrive at universal generalisations from particular facts. Therefore, sometimes deductive is thought to be opposite to inductive.
2. The propositions from which deductives are made are assumed. But in inductive this is not the case. Inductive is concerned with discovering facts and relations between them. Observed facts provide the basis of inductive, but they are not relevant for deductive.
3. Deductive is not concerned with the material truth of the premises; but inductive is concerned with the establishment of the material truth of universal propositions.
4. In deductive, the conclusion only seeks to unfold what is in the premises. In does not go beyond premises. The conclusion in deductive, in other words, is never more general than the premises. But in inductive, the conclusion goes beyond the premises or, what is in the data. Therefore, in inductive, the conclusion is more general than the premises.
5. Deductive method gives us conclusions which are certain; but the conclusions of the inductive method are only probable and not always certain. This is so because the conclusion in deductive reasoning follows from the premises logically, or it is implied in the premises; but in inductive method conclusion is not implied in the premises. Thus, the conclusion is certain, if we say that since all men are mortal, and Ram is a man. Ram is mortal. But conclusion is only probable or uncertain, if we say that since some educated girls have expensive habits, all educated girls have expensive habits.

PROBLEM SOLVING

We have already been introduced with information processing models which are dealt with problem solving. The ability to simulate complex cognitive processes is a major accomplishment in itself. But what is more significant is that information-processing models developed for quite different problems (for example, model to prove theorems in geometry versus a model to describe neurotic behaviour) have many common component processes. This commonality of processes among models suggests that a general theory of complex cognitive processes may not be too far away. Simon and Newell (1964) were the first to isolate some of the components common to many information-processing models and pull them together into a single model, which they dubbed the General Problem Solver (GPS). They propose that we should be able to combine the special information of any particular task (chess playing, theorem providing, music composition) with GPS and come up with a composite program that can solve the task, using strategies and tactics of the type employed by human beings.

The GPS simulates in a formal manner what the individual is believed to do when attacking a problem. The programs are very complex and can be characterized here only in the barest outline. The actual program for GPS is built around two basic processes that follow each other in repeated cycles until the problem is solved, or abandoned as too difficult or insoluble. The first process, part of what is called the problem-solving organization, is to set subgoals that might be appropriate to the solution of the problem. These

subgoals are then evaluated, and a promising one is selected for exploration. Note that this is a kind of “executive,” or “decision-making,” function, including both a search and an evaluation phase. A subgoal, for example, might be the solution of a simplified version of the more general problem. Once the executive routine selects a subgoal, the process known as the means-end analysis applies relevant heuristics to reach the subgoal. This requires that the information-processing mechanism begin with data that are given and follow permissible transformations as in ordinary problem solving. Because the heuristic approach does not guarantee a solution, if the initial approach fails, the executive routine then searches for subgoals that appear more productive.

The approach to complex cognitive processes exemplified by GPS is quite promising. If it turns out that information-processing models based on only a few basic methods of symbolic representation and a small number of elementary information processes can simulate complex human behaviour, then we have truly advanced our understanding.

Types of Problems

In school and in everyday life, we all have to solve a wide variety of problems. In school, these problems might be how to complete an algebraic equation or remembering the order of U.S. Presidents. In everyday life, problems might be how to maintain a long-distance relationship or how to pay bills on a limited income. Either way, in order to be successful, we must have the ability to solve different types of problems using different types of solution strategies.

Problem solving is the application of ideas, skills, or factual information to achieve the solution to a problem or to reach a desired outcome. Let’s talk about different types of problems and different types of solutions.

Educational psychology has broken down problems in two different ways. The first way is to make a distinction between well-defined and poorly-defined problems. A **well-defined problem** is one that has a clear goal or solution, and problem solving strategies are easily developed. In contrast, a **poorly-defined problem** is the opposite. It’s one that is unclear, abstract, or confusing, and that does not have a clear problem solving strategy.

The second way that educational psychology has broken down different types of problems is by making a distinction between routine and non-routine problems. Just like the name indicates, a **routine problem** is one that is typical and has a simple solution. In contrast, a **non-routine problem** is more abstract or subjective and requires a strategy to solve. Routine problems are what most people do in school: memorizing simple facts, how to do addition and subtraction, how to spell words, and so on. However, in more advanced years or in more advanced subjects in school, teachers might present students with non-routine problems that require **critical thinking** skills and subjective solutions. For example, the ethics of social issues such as the death penalty, or the role of civil rights in laws, or themes in famous literature, might be considered non-routine problems. Non-routine problems require more complicated or creative problem solving strategies. Let’s talk about problem solving strategies now and go over several possible options.

Problem Solving Strategies

Depending on the type of problem, you have a lot of options regarding possible solution strategies. In this lesson, let’s start with solutions that are common strategies for well-defined or routine problems.

The first strategy you might try when solving a routine problem is called an **algorithm**. Algorithms are step-by-step strategies or processes for how to solve a problem or achieve a goal. The most common example you might think of for using algorithms might be in math class. When presented with an algebraic equation, you might have learned about how to solve for x using certain well-defined steps. But algorithms can be used in other subjects as well. For example, when you are learning about how to take apart and clean a car engine, you will want to approach this problem using a set series of steps, making sure that you don't misplace or mix up any of the parts.

Another solution that many people use to solve problems is called **heuristics**. Heuristics are general strategies used to make quick, short-cut solutions to problems that sometimes lead to solutions but sometimes lead to errors. Heuristics are sometimes referred to as mental short-cuts, and we often form them based on past experiences. You have probably used heuristics all the time in your daily life, maybe without knowing what they are called.

So, in summary, heuristics are a common problem solving strategy for everyday life types of problems, and usually they lead to good decisions but they can sometimes lead to mistakes.

A third type of problem solving that you might have used in the past is to use **graphic representations**. Graphic representations are visual-based illustrations of a problem that might lead to clarification of a problem or creative solutions. Examples of graphic representations are flow charts, diagrams, outlines or mind maps. With any of these options, you can draw the problem out, and this might help you see the problem in a new way.

Let's talk about one more problem solving strategy that comes from educational psychology. This final strategy is called **The IDEAL Strategy**, where 'IDEAL' is an acronym, meaning each letter stands for one idea. This strategy includes five steps, each corresponding to one letter.

The 'I' stands for **identify**, which is the first step in the IDEAL strategy. Here, you identify what the problem is, as clearly as possible.

The 'D' stands for **define**. Here, you define what the possible final goal or solution might be.

Next, the 'E' stands for **explore**. This is the step where you explore possible ways to reach the solution or goal. You could use other strategies we've discussed here, such as a graphic representation, heuristics, or brainstorming.

The fourth letter is 'A,' which stands for **anticipate**. In this step, you look forward to possible outcomes of different solutions you've created, and try to see which solution will work the best. You then choose one of the solutions and act on it. While the 'A' in IDEAL stands for anticipate, not act, you can remember that the 'A' could include both anticipation and action if it's helpful.

The final step in the IDEAL solution is the 'L,' which stands for **look**. After you have chosen one solution and acted on it, you now look back and learn from what you did. So again, you could think of the 'L' as standing for both look and learn.

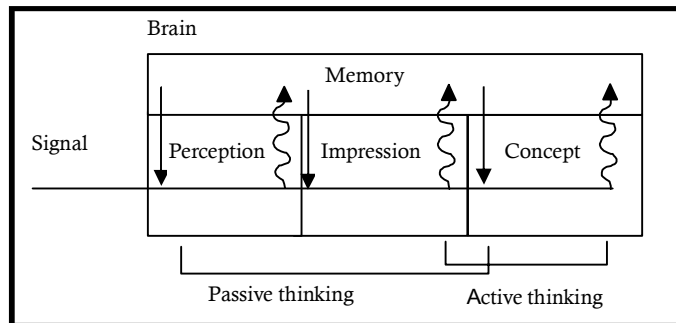
ROLE OF CONCEPTS IN THINKING

Human intelligence are developed through thinking. There are several modes of thinking. When we just thoughts floating in our brain without searching for any goal in particular is called aimless thinking (or passive thinking). Examples are the state of mind

when we are in holiday. Problem solving is to resolve problems in life, from puzzlement of some symptoms, curiosity of the mysteries in the universe, failure of some system, to just unhappiness of any kind. Even if there is no problem at hand, people often think just to generalize an pattern they observed. This kind of thinking is pure conceptual. For example, the first human who look up at the stars and try to figure out the mystery behind the outer space. Finally, we often need to make a decision or form a policy under certain time constraint.

States of Thinking

Thinking is the combination of focus switching and composition that generates a stream of thought in the working memories. It includes the bottom-up involuntary focus switching (triggered by sensation) and topdown voluntary focus switching (conducted by the cognitive system). One can think with either *primitive elements* or *symbolic elements*, or both. The style of thinking varies from person to person; some people think with primitive form, and others with symbolic forms. In most cases, there is a translation between the two. We translate scenes into words, and we evaluate the meaning of words with its corresponding reality.



THE THINKING PROCESS OF THE BRAIN.

The thinking process consists of perception, impression, and concept.

The perception process is part of the passive thinking process. The brain retrieves involuntary memory automatically during the process of perception to help recognize predefined objects.

The impression process is partly passive thinking and partly active thinking. The impressional system continuously gauges the impression of current imageries. It is mostly passive thinking, but once the process gets enough attention, it can trigger active thinking and enter the concept state.

In a stimulus-free environment, sensation and perception are kept to a minimum. The mind enters the active thinking state with impression in the background and concept in the foreground. The cognitive system conducts memory recall to support this active process. We can engage in intensive conceptual activity without engaging much in the impression stage. Logical thinking, for instance, is purely conceptual.

Convergent and Divergent Thinking

Convergent thinking looks for sameness in different things or related things. It sorts out many information and derive a general rules. We first collect many related information and then compare them and found a general rules. We generalize particulars.

Divergent thinking looks for variation after learning a knowledge or rules. This kind of thinking is often required in creativity.

Stream of Thought

Each person's consciousness is presented to him or her in a stream of thought, which is a mixture of primitive and symbolic elements. Each thought is presented in a working memory that consists of background and objects in the foreground. Each working memory consists of several objects that can be shifted, zoomed, or activate associated information.

The information appearing in the imagery layers of active working memory is conscious thought, which is of central format. The 40 Hz synchronization wave links all the imagery layers and provides a continuous stream of thought to the mind through focus switching.

The construction of thoughts is carried out in two phases. Similar to the selection of views in a TV broadcasting room, the focus switching mechanism selects the most significant information presented in the imagery layers. The basic or executive cognitive kernel composes the information that is chosen.

There are two levels of filtering of thoughts that generate public speech. The selection of thoughts is conducted on all the imageries. Should nothing in the imagery layer stand out, all of them are meshed into one. The stream of thought in the consciousness must undergo additional conscious filtering before it can be presented to the public. It would not be proper for some private thoughts to come out in the speech.

Some of the activities in creating a stream of thought are verbal activities (such as speaking), multimodal activities (such as the making of movies), symbolic activities (such as reading, researching, and thinking), memory recall, and travel.

Symbols and Thinking

The cognitive system uses *symbol* to represent whole set of objects to facilitate thinking and communication. Concepts can be derived from either the primitive or symbolic thinking process. A concept derived from a primitive thinking process tends to become a primitive concept, lacking a broad foundation for a wider application. The ability to reason relies on the ability of symbolic thinking – each symbol can bring in massive information through the zoom-in zoom-out mechanism.

Imagination

Human imagination are mainly based on existing memory. We change the forms in our mind with rotation, mutation, or combination, and rarely think of something radically different from what is already in our memory. *Creativity* happens when something radically different from our memories pops into our mind.

The greatest contrast between sensation and imagination is that sensation can generate vivid memory while imagination cannot, because the imagery layers have lower resolution than the sensory layers.

Thinking Activities

The major thinking activities are analysis, synthesis, and composition.

Analysis

The analysis includes the following acts of think. Categorization is the act that put things into proper category. Generalization is the act to find sameness in different things and to find pattern from similar things. Abstraction is using simple pattern to represent complex matter. Using one notion (a symbol) to represent another object.

Synthesis and Composition

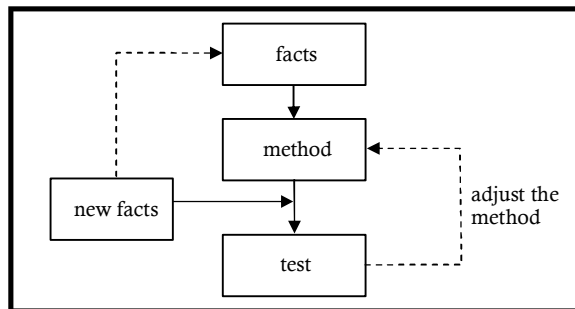
Compose elements into a whole and putting small parts into a hierarchy. Systemization is the act of defining the rules in a system consisting of parts.

Thinking Elements

The elements of thoughts can be divided into several categories. The descriptive elements describe physical or abstract things. Physical things are perceived as sensation while abstract things are perceived as abstract concept. For example, “cat” is a physical concept while “freedom” is an abstract concept. The explanatory elements explain the reason with logic. For example, “The tire inflates under the sun because the volume of gas increases with the rise in temperature.” Logic elements can be causal logic or formal. *Causal logic* includes physical causation and cognitive causation. *Formal logic* includes domain logic, conditional logic, and predicate logic

Thinking Methods

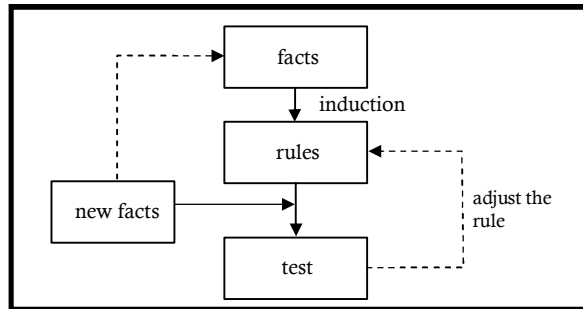
We can use one of five thinking methods to develop a concept or to solve a problem: systematic method, heuristic strategy, insight, intuition, and trial and error. Figure depicts the generic scheme of these cognitive methods.



THE GENERIC SCHEME OF COGNITIVE METHOD.

Systematic Method

The systematic method (algorithm) defines rules in a system or in a procedure. The two major principles of systematic method are inductive and deductive. In inductive, we collect information and generate rules. Then we apply these rules to new information and draw a conclusion.



THE SYSTEMATIC METHOD OF INDUCTIVE AND DEDUCTIVE.

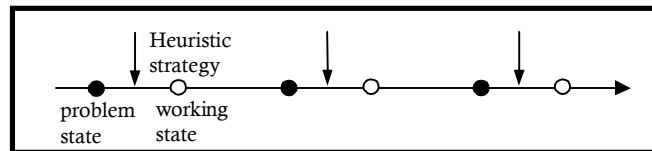
The systematic method requires two phases. Original creativity often occurs in the inductive phase, where we generate *rules* from random facts. This phase takes intuition, insight, and trial and error, and is difficult to teach. Education often teaches students the rules and how to apply them in the deductive phase. Deductive is governed by rules and is easier to learn.

The lateralization of the brain may facilitate the processes of inductive and deductive. The right hemisphere handles inductive with intuition, memory search, and matching patterns. Rules formulated by inductive are transferred to the left hemisphere. The left hemisphere specializes in applying rules in the deductive phase.

Heuristic Strategy

The systematic method is not always feasible. We may not have enough time or ability to study the system thoroughly, or a complete knowledge of a subject has not been developed. When this happens, we can try the heuristic strategy. The heuristic strategy uses a simple procedure or algorithm to convert a non-working state to a working state. Deriving method based on experience, it uses successful examples in the past for guidance. There is no guarantee that an improvised solution will work. Probability, however, is on our side if we use short-cuts that have previously succeeded.

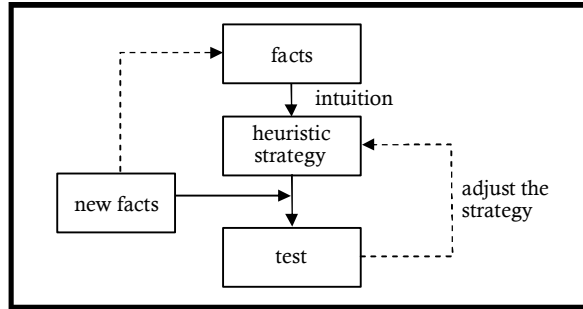
In a complex environment, a systematic method or insight is not always available. Trial and error can be time consuming, and intuition is unreliable. We can use heuristic strategies in such an environment to perform routine tasks that require quick and effective response.



USING HEURISTIC STRATEGY TO DEAL WITH REPETITIVE SITUATIONS.

We develop a heuristic strategy through a combination of intuition and experimentation. Experience gives us intuition, which in turn enables us to develop a heuristic strategy.

When developed this way, the heuristic strategy differs from the systematic method. We know *how* to use a heuristic strategy but we do not know *why* it works. We can integrate systematic rules into a coherent framework because the rules are based on the causal mechanisms that govern the universe. Heuristic strategies, on the other hand, do not have an underlying framework.



THE DEVELOPMENT OF HEURISTIC STRATEGY.

The following are examples of general heuristic strategies for problem solving. We should keep things simple since simplicity goes a long way. We can analyze complex ideas with its block structure. Another way to analyze complex ideas is to reduce the level of abstraction. We can divide a complex task into steps – trying to solve a difficult problem all at once proves overwhelming.

Intuition

When facing problems, we often rely on intuition, which is based on association and underlying schema. Accumulated experiences build up the underlying schema, or personal preferences. Like insight, intuition has a mysterious quality. The difference is that insight is based on systematic knowledge while intuition is based on a mixture of personal experience.

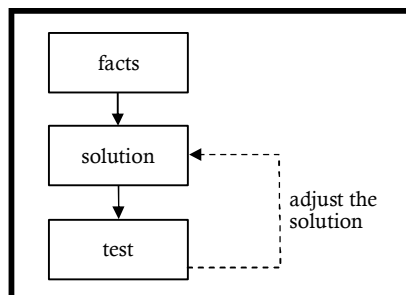
Intuition is also involved in the generation of attempted solutions, such as theories, hypotheses, and prototypes.

Insight

Insight is a special type of intuition that is acquired by over-learning a systematic knowledge. Insight enables us to single out the key of a problem and provide an elegant solution. This is the most gratifying kind of achievement. Whenever we apply this insight to a problem, we add concreteness to ambiguity.

In general, we have insight into selective disciplines. When a problem arises, we can first try to match our stock of insight to this problem. If we fail to solve it this way, then we can try a systematic method. Finally, we may settle for trial and error.

Trial and Error



THE TRIAL AND ERROR METHOD.

Trial and error is a common technique in problem solving. We all know that if we try several things, we eventually can hit upon a solution. For instance, when we design landscaping, we can move plants around until we are satisfied with the arrangement. Other mental activities use similar techniques.

Most new forms are created through trial and error – from impressional forms like paintings and dancing to conceptual forms like industrial design and computer programming. Trial and error provides a basic and simple means of exploration.

Even though we rely on trial and error to solve many problems, we do not feel comfortable with this technique, because experimentation does not necessarily help us to understand the system. There is no conceptual frame-work behind the solution.

Procedural Problems

For a system of precise procedures, problem solving should focus on checking the order of steps, finding missing steps, and examining the specifications of each step.

Replicable Concepts

Replicable concepts (called memes) are the concepts that have a life of their own. Not every concept has a life of its own, most concepts faded into wasteland. To be replicable, a concept must be concise and easily replicable. For example, a melody is replicable only if it is easy to repeat and sing.

Like gene, meme is not by default “self replicating.” It takes effort to keep it alive. The exception is some famous thought that are engraved into human knowledge.

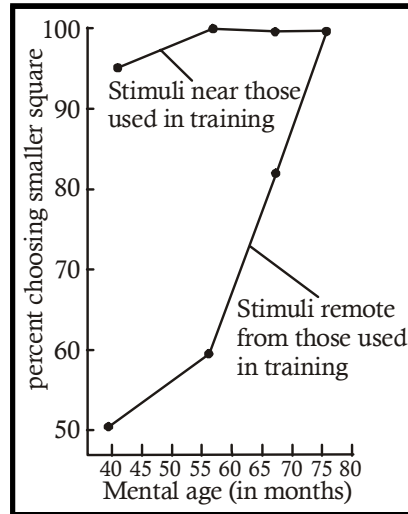
Similar memes attract each other. The togetherness increases their appeals and encourages replication. Some memes have short-term success (bribe, cheat, lies), some have long-term success (honesty, integrity, reputation). In the long run, some meme persevere and become the focus of admiration.

Various ideas pass from one generation to the next. The idea that works the best tend to be propagated and caught most attention for replication in the form of formula or recipe.

Language in Children's Thinking

A child's ability to use language corresponds closely to his ability to deal with concepts and relationships. This correspondence is illustrated by an experiment in which pre-school children were taught to select the smaller of a pair of squares, each mounted on the lid of a box. If the child chose the smaller square, he found the box open and an attractive toy inside. If he mistakenly chose the larger square, he found the box locked. The child began by learning to choose a six-inch square in preference to an eight-inch one. When he had learned to choose the six-inch square regularly, he was ready for the crucial tests with smaller squares.

He was now confronted with two test pairs, all pairs being smaller than the original ones. Of these smaller pairs, one (known as the “near pair”) was close to the original squares in size (4.5 inches and 6 inches), the other (known as the “remote pair”) was much smaller (1.4 and 1.9 inches). If the child had learned to “transpose,” that is, to choose the smaller square regardless of absolute size, he should succeed in choosing the smaller square for both the near and the remote pairs. The results are plotted in given figure. Although all children did well on the near pair, they showed a striking increase in success with age on the remote pair.



LANGUAGE AND THE PERCEPTION OF RELATIONSHIP.

The older children's success with the remote pair was clearly related to their use of language. If a child could say the equivalent of "The smaller one is always right," he could succeed with the remote pair. None of the children who failed to express in words the principle of correct choice during initial learning transposed the correct response on the test with the remote pair; 73 per cent of the children who stated the correct solution in words, either spontaneously or upon questioning, successfully transposed what they had learned.

Of course, the older children were more developed as problem solvers, regardless of whether they relied on language; the younger children were able to use language to some extent, but not well enough to serve as a tool for thinking in the transposition. We can recognize that language and thought are related, without assuming they are identical. Studies comparing the performances of deaf-mute children and those with normal hearing indicate that language may aid in solving problems of relationships and concept formation, but it is by no means essential for the development of such cognitive abilities.

Linguistic-Relativity Hypothesis

Whorf's thesis (known as the linguistic-relativity hypothesis because it proposes that thought is relative to the language in which it is conducted) has been the subject of much debate among psychologists and anthropologists. Most of them accept a correspondence between the language and the ways of conceiving the world, but they tend to turn things around and try to show that those experiences significant to people affect the way things are expressed in language. Thus Eskimos have different words for different kinds of snow that we would scarcely be able to tell apart, and the Hanunoo of the Philippine islands have names for ninety-two varieties of rice. For us, the important part of Whorf's conjecture is that there is a close correspondence between language and thinking.

Researchers interested in his theory have tested it in a study of two groups of Navajo children. Both groups lived on a reservation, but one spoke only English and the other

only Navajo. A special characteristic of the Navajo language is that certain verbs of handling – the Navajo equivalents of to pick up, to drop, to hold in the hand, and so on – require special forms depending upon the nature of the object being handled. There are 11 different forms, one for round spherical objects, one for round thin objects, one for long flexible objects, and so forth. Even the very young Navajo-speaking children knew and used these forms correctly. These children were compared to English-speaking Navajos of the same age, with respect to how often they used shape, form, or material rather than colour as a basis for sorting objects. The objects given were those usually sorted by young children on the basis of colour. The Navajo-speaking children tended to sort on the basis of form at significantly younger ages than did the English-speaking children. The fact that the Navajo language required attention to shapes, forms, and materials of objects presumably made the Navajo-speaking child pay more attention to this aspect of his environment.

DECISION-MAKING

You have to make decisions both large and small throughout every single day of your life. What do you want to have for breakfast? What time should you meet a friend for dinner? What college should you go to? How many children do you want to have?

When faced with some decisions, you might be tempted to just flip a coin and let chance determine your fate. In most cases, we follow a certain strategy or series of strategies in order to arrive at a decision. For many of the relatively minor decisions that we make each and every day, flipping a coin wouldn't be such a terrible approach. For some of the complex and important decisions, we are more likely to invest a lot of time, research, effort, and mental energy into coming to the right conclusion.

So how exactly does this process work? The following are some of the major decision-making strategies that you might use.

The Single-Feature Model

This approach involves hinging your decision solely on a single feature. For example, imagine that you are buying soap. Faced with a wide variety of options at your local superstore, you decide to base your decision on price and buy the cheapest type of soap available. In this case, you ignored other variables (such as scent, brand, reputation, and effectiveness) and focused on just a single feature.

The single-feature approach can be effective in situations where the decision is relatively simple and you are pressed for time. However, it is generally not the best strategy when dealing with more complex decisions.

The Additive Feature Model

This method involves taking into account all the important features of the possible choices and then systematically evaluating each option. This approach tends to be a better method when making more complex decisions.

For example, imagine that you are interested in buying a new camera. You create a list of important features that you want the camera to have, then you rate each possible option on a scale of -5 to +5. Cameras that have important advantages might get a +5 rating for that factor, while those that have major drawbacks might get a -5 rating for that

factor. Once you have looked at each option, you can then tally up the results to determine which option has the highest rating.

The additive feature model can be a great way to determine the best option for a variety of choices. As you can imagine, however, it can be quite time-consuming and is probably not the best decision-making strategy to use if you are pressed for time.

The Elimination by Aspects Model

The elimination by aspects model was first proposed by psychologist Amos Tversky in 1972. In this approach, you evaluate each option one characteristic at a time beginning with whatever feature you believe is the most important. When an item fails to meet the criteria you have established, you cross the item off your list of options. Your list of possible choices gets smaller and smaller as you cross items off the list until you eventually arrive at just one alternative.

Making Decisions in the Face of Uncertainty

The previous three processes are often used in cases where decisions are pretty straightforward, but what happens when there is a certain amount of risk, ambiguity, or uncertainty involved? For example, imagine that you are running late for your psychology class. Should you drive above the speed limit in order to get there on time, but risk getting a speeding ticket? Or should you drive the speed limit, risk being late, and possibly get docked points for missing a scheduled pop quiz? In this case, you have to weigh the possibility that you might be late for your appointment against the probability that you will get a speeding ticket.

When making a decision in such a situation, people tend to employ two different decision-making strategies: the availability heuristic and the representativeness heuristic. Remember, a heuristic is a rule-of-thumb mental short-cut that allows people to make decisions and judgments quickly.

The Availability Heuristic. When we are trying to determine how likely something is, we often base such estimates on how easily we can remember similar events happening in the past. For example, if you are trying to determine if you should drive over the speed limit and risk getting a ticket, you might think of how many times you have seen people getting pulled over by a police officer on a particular stretch of highway. If you cannot immediately think of any examples, you might decide to go ahead and take a chance, since the availability heuristic has led to you judge that few people get pulled over for speeding on your particular route. If you can think of numerous examples of people getting pulled over, you might decide to just play it safe and drive the suggested speed limit.

The Representativeness Heuristic. This mental shortcut involves comparing our current situation to our prototype of a particular event or behavior. For example, when trying to determine whether you should speed to get to your class on time, you might compare yourself to your image a person who is most likely to get a speeding ticket. If your prototype is that of a careless teen that drives a hot-rod car and you are a young businesswoman who drives a sedan, you might estimate that the probability of getting a speeding ticket is quite low.

The decision-making process can be both simple (such as randomly picking out of our available options) or complex (such as systematically rating different aspects of the existing choices). The strategy we use depends on various factors, including how much time

we have to make the decision, the overall complexity of the decision, and the amount of ambiguity that is involved.

METACOGNITION

In the late 1970's, John Flavell originally coined the word "metacognition." He defined the word as "cognition about cognitive phenomenon", or basically thinking about thinking. Subsequent studies on metacognition described the term comparative to Flavell's meaning. Cross and Paris (1988) defined it as "the knowledge and control children have over their own thinking and learning activities." For Hennessey (1999), it is the "awareness of one's own thinking, awareness of the content of one's conception, as active monitoring of one's cognitive processes, an attempt to regulate one's cognitive processes in relationship to further learning, and an application of a set of heuristics as an effective device for helping people organize their methods of attacks in general."

Metacognition denotes in-depth thinking in which cognitive processes involved in learning are actively controlled. This includes planning how to accomplish a given learning task, monitoring understanding, and estimating progress toward the completion of a task. It was said that students can have greater ability to control goals, dispositions, and attention when they are more aware of their thinking processes as they learn. This means that self-regulation is a result of self-awareness. If a student is aware of how dedicated he is to reaching his goals, of how strong his trait of perseverance is, and of how focused his attention is to a task that involves thinking or writing, he can regulate his dedication, disposition, and attention. For instance, when a student is aware of his lack of commitment to write his thesis, and bears the knowledge that he is procrastinating, delaying, and allowing himself to be distracted by other less important things, then he could take action to get started on doing the task. This is possible only if the student becomes aware of his procrastination and takes control in planning on how to approach his thesis completion.

Although the term has been known for a long time, especially in the field of educational psychology, defining metacognition can be difficult. Until now, there are still debates as to what the term exactly means. This confusion can be rooted from the idea that there are terms presently used to characterize the same basic phenomenon, such as self-regulation, executive control, or an aspect of that phenomenon such as meta-memory. These terms are commonly used interchangeably in literature. While there are some differences between explanations of the term, the role of executive processes in monitoring and regulation of cognitive methods is emphasized.

Components of Metacognition

Metacognition is classified as having three components: (1) Metacognitive knowledge, (2) Metacognitive regulation, and (3) Metacognitive experiences. Metacognitive knowledge refers to the awareness individuals possess about themselves and other people as cognitive processors. Metacognitive regulation, on the other hand, has to do with people's control over cognition and learning experiences through a set of methods that help people regulate their learning while metacognitive experiences involve cognitive efforts that are currently taking place.

Metacognitive Knowledge

Aside from these three components, metacognition also has three different types of metacognitive knowledge: (1) Declarative knowledge, (2) Procedural knowledge, and (3) Conditional knowledge. Declarative knowledge refers to the factual information that one knows, and can both be spoken or written. This is also the knowledge about oneself as a learner and about what factors can influence one's performance. Procedural knowledge comprises information on how to do something or how to perform the procedural steps that make up a task. A high degree of procedural knowledge allows individuals to perform tasks more automatically through a variety of strategies. Conditional knowledge, on the other hand, refers to the knowledge about when to use a procedure, skill, or strategy or when not to, information as to why a procedure works and under what conditions it works, in addition to why one procedure is better than another. Such knowledge affords the individual a chance to assign resources when using strategies.

Skills in Metacognitive Regulation

There are three important skills in metacognitive regulation: (1) Planning, (2) Monitoring, and (3) Evaluating. Planning involves suitable selection of strategies and the right assignment of resources. Monitoring includes awareness of understanding and task performance, while evaluating refers to the assessment of the final result of a task and the efficiency carried out during task performance.

Metacognitive Strategies

Research has shown that teaching students metacognitive strategies can improve learning. Studies reveal, however, that there are only a few instructors who explicitly teach study strategies because most instructors assume that students have already acquired these skills in high school. Among the different learning strategies that are commonly used when studying or doing homework are rote memorization, goal setting, monitoring, self-assessing, and regulating during thinking and writing processes.

INTELLIGENCE

What exactly is intelligence? While intelligence is one of the most talked about subjects in psychology, there is no standard definition of what exactly constitutes intelligence. Some researchers have suggested that intelligence is a single, general ability, while others believe that intelligence encompasses a range of aptitudes, skills, and talents.

Intelligence has been an important and controversial topic throughout psychology's history. Despite the substantial interest in the subject, there is still considerable disagreement about what components make up intelligence. In addition to questions of exactly how to define intelligence, the debate continues today about whether accurate measurements are even possible.

At various points throughout recent history, researchers have proposed some different definitions of intelligence. While these definitions can vary considerably from one theorist to the next, current conceptualizations tend to suggest that intelligence involves the level of ability to do the following:

1. **Learn.** The acquisition, retention, and use of knowledge is an important component of intelligence.

2. **Recognize problems.** To put knowledge to use, people must be able to identify possible problems in the environment that need to be addressed.
3. **Solve problems.** People must then be able to take what they have learned to come up with a useful solution to a problem they have noticed in the world around them.

Intelligence involves some different mental abilities including logic, reasoning, problem-solving, and planning. While the subject of intelligence is one of the largest and most heavily researched, it is also one of the topics that generates the greatest controversy.

While psychologists often disagree about the definition and causes of intelligence, research on intelligence plays a significant role in many areas. These areas include decisions regarding how much funding should be given to educational programs, the use of testing to screen job applicants, and the use of testing to identify children who need additional academic help.

The term "intelligence quotient," or IQ, was first coined in the early 20th century by a German psychologist named William Stern. Psychologist Alfred Binet developed the very first intelligence tests to help the French government identify schoolchildren who needed extra academic assistance. Binet was the first to introduce the concept of mental age, or a set of abilities that children of a certain age possess.

Since that time, intelligence testing has emerged as a widely used tool that has led to developing many other tests of skill and aptitude. However, it continues to spur debate and controversy over the use of such testing, cultural biases that may be involved, influences on intelligence, and even the very way we define intelligence.

Theories of Intelligence

Different researchers have proposed a variety of theories to explain the nature of intelligence. Here are some of the major theories of intelligence that have emerged during the last 100 years:

Charles Spearman

British psychologist Charles Spearman (1863-1945) described a concept he referred to as general intelligence, or the g factor. After using a technique known as factor analysis to examine some mental aptitude tests, Spearman concluded that scores on these tests were remarkably similar. People who performed well on one cognitive test tended to perform well on other tests, while those who scored badly on one test tended to score badly on others. He concluded that intelligence is a general cognitive ability that can be measured and numerically expressed.

Louis L. Thurstone

Psychologist Louis L. Thurstone (1887-1955) offered a differing theory of intelligence. Instead of viewing intelligence as a single, general ability, Thurstone's theory focused on seven different primary mental abilities. The abilities that he described include:

- Verbal comprehension
- Reasoning
- Perceptual speed
- Numerical ability
- Word fluency

Associative memory

Spatial visualization

Jensen

Jensen (1969) proposed two level-theories of mental abilities, which sparked a debate on the association of heredity versus environment with intelligence after his studies, got published in the "Harvard Educational Review" in 1969. He demonstrated clear-cut genetic differences in the average intelligence of races and social classes. Jensen suggested that two genetically based levels of intelligence exist.

In other word intelligence consists of two levels of abilities. The Level I ability is associative learning, which consists of Short-Term Memory (STM), rote learning, attention and simple associative skills. The Level II is called cognitive learning, which consists of abstract thinking, symbolic thought, conceptual learning, and use of language in problem solving.

He argued that Level I ability, i.e. associative learning is equally distributed across all racial and national groups but on the contrary, Level II, i.e., cognitive learning is concentrated more the middle class Anglo-American populations than in the lower class black populations.

According to him, genetic differences in intelligence exist among people coming from different races, nationalities, and social classes.

Jensen conducted studies of intelligence comparing identical fraternal twins. Identical twins have identical genetic dispositions for that their IQs are assumed to be similar.

Fraternal twins like ordinary siblings genetically less similar, which results in the assumption that their IQs less similar. Jensen found support for his views in these findings. Studies with identical twins produced an average correlation of .86, whereas studies of ordinary siblings produced an average correlation of .47.

Jensen on basis of his studies believed that genetic factors are more important than environmental factors for one's intelligence.

Cattell

In the 1930s and early 1940s, both Raymond B. Cattell and Donald O. Hebb were critical of the ability of the currently available tests to accurately measure intelligence and both began to develop improved tests of intelligence. Cattell was in search of a "culture-free" test of intelligence while Hebb was concerned with the development of tests of intelligence which could measure the effects of localized brain damage at different ages in patients with little schooling.

At that time, intelligence testing was fuelled by a scientific interest to quantify the intelligence of different groups of people, with a focus on special classes of individuals, such as the insane, feeble-minded, deaf, and delinquent. The intelligence scales used were influenced by the work of Alfred Binet and Theodore Simon with the publication of the first Binet-Simon scale in 1905 (Binet and Simon, 1905). This "measuring scale of intelligence" was originally developed for use with school children in Paris and consisted of a series of 30 brief cognitive tests. The Binet-Simon intelligence scale and its modifications (such as the Stanford-Binet scale) were the most commonly used tests of intelligence in the 1930's (Pintner, 1931), but the Wechsler-Bellevue intelligence scale, published in 1939 (Wechsler, 1939) introduced a number of innovations in intelligence test-

ing including a test more suitable for measuring intelligence in adults rather than children; the inclusion of both verbal and performance measures of intelligence; and the omission of memory tests.

The state of intelligence testing in America in the 1930's was reviewed by Pintner (1931), who described the intelligence tests then available and the "various fields in which mental tests have been successfully used". As pointed out by Pintner, "The beginnings of the intelligence testing movement were closely bound up with the study of mental deficiency and abnormality" and "much of the early work had to do with the selection of mentally deficient children." By the 1930's intelligence testing was also used to test army recruits, employees, the deaf, the blind and the foreign-born. They were also used to examine sex and racial differences in intelligence; to detect children of superior intelligence; and to study the inheritance of intelligence.

Cattell's Work on Intelligence Testing

Cattell (1940a) gave a detailed critique of the Binet test and was concerned that differences in scores on intelligence tests were due to differences in social status, special abilities distinct from intelligence, or other environmental factors. The problem that he identified was that tests of intelligence focused on acquired skills and verbal ability rather than on intelligence per se. Cattell was interested in developing "culture free" intelligence tests based on non-verbal skills that would define a person's general intelligence "g" irrespective of their social status, race, verbal ability or environmental experience. Cattell thus developed a culture-free test with seven sub-tests. The tests were designed to be non-verbal and instructions were given using a series of worked examples rather than verbal instructions.

Cattell et al. (1941) outlined seven variables which influenced intelligence or "general ability" and then analyzed the effects of environmental variables such as cultural knowledge and training (practice) on each test. He then compared the scores of American-born and immigrant children on his culture-free test and three other tests of intelligence: the Terman-Merrill Revision of the Binet Test, the arithmetic section of the American Council of Education (A.C.E.) Test, and the Arthur Performance Test. Based on these studies, Cattell et al. (1941) found that the Arthur test was the most culture-free, the culture-free test was second, the Terman-Merrill-Binet test was third and the A.C.E. test fourth, and concluded that culture-free tests should be administered using people from a wide range of cultures, "primitive and otherwise."

Howard Gardner

One of the more recent ideas to emerge is Howard Gardner's theory of multiple intelligences. Instead of focusing on the analysis of test scores, Gardner proposed that numerical expressions of human intelligence, such as in the IQ test, are not a full and accurate depiction of people's abilities. His theory describes eight distinct types of intelligence based on skills and abilities that are valued in different cultures.

The eight kinds of intelligence Gardner described are:

- Visual-spatial intelligence
- Verbal-linguistic intelligence
- Bodily-kinesthetic intelligence
- Logical-mathematical intelligence

Interpersonal intelligence
Musical intelligence
Intrapersonal intelligence
Naturalistic intelligence

Robert Sternberg

Psychologist Robert Sternberg defined intelligence as "mental activity directed toward purposive adaptation to, selection, and shaping of real-world environments relevant to one's life." While he agreed with Gardner that intelligence is much broader than a single, general ability, he instead suggested that some of Gardner's types of intelligence are better viewed as individual talents. Sternberg proposed what he referred to as "successful intelligence," which involves three different factors:

Analytical intelligence. Your problem-solving abilities.

Creative intelligence. Your capacity to deal with new situations using past experiences and current skills.

Practical intelligence. Your ability to adapt to a changing environment.

Goleman

In 1990, in my role as a science reporter at The New York Times, I chanced upon an article in a small academic journal by two psychologists, John Mayer, now at the University of New Hampshire, and Yale's Peter Salovey. Mayer and Salovey offered the first formulation of a concept they called "emotional intelligence."

Those were days when the preeminence of IQ as the standard of excellence in life was unquestioned; a debate raged over whether it was set in our genes or due to experience. But here, suddenly, was a new way of thinking about the ingredients of life success. I was electrified by the notion, which I made the title of this book in 1995. Like Mayer and Salovey, I used the phrase to synthesize a broad range of scientific findings, drawing together what had been separate strands of research – reviewing not only their theory but a wide variety of other exciting scientific developments, such as the first fruits of the nascent field of affective neuroscience, which explores how emotions are regulated in the brain.

I remember having the thought, just before this book was published ten years ago, that if one day I overheard a conversation in which two strangers used the phrase emotional intelligence and both understood what it meant, I would have succeeded in spreading the concept more widely into the culture. Little did I know.

The phrase emotional intelligence, or its casual shorthand EQ, has become ubiquitous, showing up in settings as unlikely as the cartoon strips Dilbert and Zippy the Pinhead and in Roz Chast's sequential art in The New Yorker. I've seen boxes of toys that claim to boost a child's EQ; lovelorn personal ads sometimes trumpet it in those seeking prospective mates. I once found a quip about EQ printed on a shampoo bottle in my hotel room.

And the concept has spread to the far corners of our planet. EQ has become a word recognized, I'm told, in languages as diverse as German and Portuguese, Chinese, Korean, and Malay. (Even so, I prefer EI as the English abbreviation for emotional intelligence.) My e-mail inbox often contains queries, from, for example, a doctoral student in Bulgaria, a school teacher in Poland, a college student in Indonesia, a business consult-

ant in South Africa, a management expert in the Sultanate of Oman, an executive in Shanghai. Business students in India read about EI and leadership; a CEO in Argentina recommends the book I later wrote on the topic. I've also heard from religious scholars within Christianity, Judaism, Islam, Hinduism, and Buddhism that the concept of EI resonates with outlooks in their own faith.

Most gratifying for me has been how ardently the concept has been embraced by educators, in the form of programs in "social and emotional learning or SEL. Back in 1995 I was able to find only a handful of such programs teaching emotional intelligence skills to children. Now, a decade later, tens of thousands of schools worldwide offer children SEL. In the United States many districts and even entire states currently make SEL curriculum requirement, mandating that just as students must attain a certain level of competence in math and language, so too should they master these essential skills for living.

In Illinois, for instance, specific learning standards in SEL abilities have been established for every grade from kindergarten through the last year of high school. To give just one example of a remarkably detailed and comprehensive curriculum, in the early elementary years students should learn to recognize and accurately label their emotions and how they lead them to act. By the late elementary years lessons in empathy should make children able to identify the nonverbal clues to how someone else feels; in junior high they should be able to analyze what creates stress for them or what motivates their best performance. And in high school the SEL skills include listening and talking in ways that resolve conflicts instead of escalating them and negotiating for win-win solutions.

Around the world Singapore has undertaken an active initiative in SEL, as have some schools in Malaysia, Hong Kong, Japan, and Korea. In Europe, the U.K. Has led the way, but more than a dozen other countries have schools that embrace EI, as do Australia and New Zealand, and here and there countries in Latin America and Africa. In 2002 UNESCO began a worldwide initiative to promote SEL, sending a statement of ten basic principles for implementing SEL to the ministries of education in 140 countries.

In some states and nations, SEL has become the organizing umbrella under which are gathered programs in character education, violence prevention, antibullying, drug prevention and school discipline. The goal is not just to reduce these problems among school-children but to enhance the school climate and, ultimately, students' academic performance.

In 1995, I outlined the preliminary evidence suggesting that SEL was the active ingredient in programs that enhance children's learning while preventing problems such as violence. Now the case can be made scientifically: helping children improve their self-awareness and confidence, manage their disturbing emotions and impulses, and increase their empathy pays off not just in improved behavior but in measurable academic achievement.

This is the big news contained in a recently completed meta-analysis of 668 evaluation studies of SEL programs for children from preschoolers through high school. The massive survey was conducted by Roger Weissberg, who directs the Collaborative for Academic, Social and Emotional Learning at the University of Illinois at Chicago – the organization that has led the way in bringing SEL into schools worldwide.

The data show that SEL programs yielded a strong benefit in academic accomplishment, as demonstrated in achievement test results and grade-point averages. In participating schools, up to 50 percent of children showed improved achievement scores and up to 38 percent improved their grade-point averages. SEL programs also made schools safer:

incidents of misbehavior dropped by an average of 28 percent; suspensions by 44 percent; and other disciplinary actions by 27 percent. At the same time, attendance rates rose, while 63 percent of students demonstrated significantly more positive behavior. In the world of social science research, these remarkable results for any program promoting behavioral change, SEL had delivered on its promise.

In 1995 I also proposed that a good part of the effectiveness of SEL came from its impact in shaping children's developing neural circuitry, particularly the executive functions of the prefrontal cortex, which manage working memory – what we hold in mind as we learn – and inhibit disruptive emotional impulses. Now the first preliminary scientific evidence for that notion has arrived. Mark Greenberg of Pennsylvania State University. A codeveloper of the PATHS curriculum in SEL, reports not only that the program for elementary school students boasts academic achievement but, even more significantly, that much of the increased learning can be attributed to improvements in attention and working memory, key functions of the prefrontal cortex. This strongly suggests that neuroplasticity, the shaping of the brain through repeated experience, plays a key role in the benefits from SEL.

Perhaps the biggest surprise for me has been the impact of EI in the world of business, particularly in the areas of leadership and employee development (a form of adult education). The Harvard Business Review has hailed emotional intelligence as “a ground-breaking, paradigm-shattering idea,” one of the most influential business ideas of the decade.

Such claims in the business world too often prove to be fads, with no real underlying substance. But here a far-flung network of researchers has been at work, ensuring that the application of EI will be grounded in solid data. The Rutgers University-based Consortium for Research on Emotional Intelligence in Organizations (CREIO) has led the way in catalyzing this scientific work, collaborating with organizations that range from the Office of Personnel Management in the federal government to American Express.

Today companies worldwide routinely look through the lens of EI in hiring, promoting, and developing their employees. For instance, Johnson and Johnson (another CREIO member) found that in divisions around the world, those identified at mid career as having high leadership potential were far stronger in EI competencies than were their less-promising peers. CREIO continues to foster such research, which can offer evidence-based guidelines for organizations seeking to enhance their ability to achieve their business goals or fulfill a mission.

Das, Kar & Parrila

The Planning, Attention-Arousal, Simultaneous and Successive (PASS) theory of intelligence, first proposed in 1975, and later elaborated by Das, Naglieri & Kirby (1994) and Das, Kar & Parrila, (1996) challenges g-theory on the grounds that the brain is made up of interdependent, but separate, functional systems. Neuroimaging studies and clinical studies of individuals with brain lesions make it clear that the brain is modularized; for example, damage to a very specific area of the left temporal lobe will impair the production (but not the comprehension) of spoken and written language. Damage to an adjacent area will have the opposite impact, preserving the individual's ability to produce, but not understand speech and text.

CREATIVITY

Torrance

Built on J.P. Guilford's work and created by Ellis Paul Torrance, the Torrance Tests of Creative Thinking (TTCT), a test of creativity, originally involved simple tests of divergent thinking and other problem-solving skills, which were scored on four scales:

Fluency. The total number of interpretable, meaningful, and relevant ideas generated in response to the stimulus.

Flexibility. The number of different categories of relevant responses.

Originality. The statistical rarity of the responses.

Elaboration. The amount of detail in the responses.

The third edition of the TTCT in 1984 eliminated the Flexibility scale from the figural test, but added Resistance to Premature Closure (based on Gestalt Psychology) and Abstractness of Titles as two new criterion-referenced scores on the figural. Torrance called the new scoring procedure Streamlined Scoring. With the five norm-referenced measures that he now had (fluency, originality, abstractness of titles, elaboration and resistance to premature closure), he added 13 criterion-referenced measures which include: emotional expressiveness, story-telling articulateness, movement or actions, expressiveness of titles, syntheses of incomplete figures, synthesis of lines, of circles, unusual visualization, extending or breaking boundaries, humor, richness of imagery, colourfulness of imagery, and fantasy.

According to Arasteh and Arasteh (1976) the most systematic assessment of creativity in elementary school children has been conducted by Torrance and his associates, who have developed and administered the Minnesota Tests of Creative Thinking (MTCT), which was later renamed as the TTCT, to several thousands of school children. Although they have used many of Guilford's concepts in their test construction, the Minnesota group, in contrast to Guilford, has devised tasks which can be scored for several factors, involving both verbal and non-verbal aspects and relying on senses other than vision. These tests represent a fairly sharp departure from the factor type tests developed by Guilford and his associates, and they also differ from the battery developed by Wallach and Kogan (1965), which contains measures representing creative tendencies that are similar in nature.

To date, several longitudinal studies have been conducted to follow up the elementary school-aged students who were first administered the Torrance Tests in 1958 in Minnesota. There was a 22-year follow-up, a 40-year follow-up, and a 50-year follow-up.

Torrance (1962) grouped the different subtests of the Minnesota Tests of Creative Thinking (MTCT) into three categories.

Verbal tasks using verbal stimuli

Verbal tasks using non-verbal stimuli

Non-verbal tasks

Getzels & Jackson

Mr. Getzels spent a half-century on the faculty of the University of Chicago, helping establish educational administration as a field of scholarship and conducting groundbreaking

studies in the 1960's that challenged traditional notions of what makes a student gifted and the reliability of I.Q. tests as a measure of intelligence.

He wrote four books and dozens of articles and was a mentor for generations of graduate students.

His work helped lead selective colleges to seek well-rounded applicants rather than just those with top grades and test scores, and it continues to influence how critics evaluate artists and teachers assess children.

"Getzels was one of the founders of creativity research," said Robert Sternberg, a professor of psychology and education at Yale University. "He was the person who recognized the importance of finding the right problem, not just seeking the right solution."

The 1962 book "Creativity and Intelligence," which Mr. Getzels wrote with his Chicago colleague Philip Jackson, challenged accepted theory by showing that some children who did poorly on I.Q. tests often earned good grades.

He identified a group of highly creative students who excelled on open-ended questions like, "How many uses can you think of for the following objects?" – a key approach of the standardized tests now in broad use in the nation's schools.

Both those with high I.Qs and those who scored well on Mr. Getzels' creativity tests outperformed their peers in class. The high I.Q. students showed more energy and were more goal-oriented, he found, while the creative children were more emotionally stable and had better senses of humor.

Successful artists, entrepreneurs, even scientists, Mr. Getzels found, were often not the best at answering straightforward questions posed by teachers or routine exams.

"Most schools, all you learn is solving problems; then you get out in the real world, you feel lost because nobody's telling you what to solve," said Mihaly Csikszentmihalyi, one of his protégés and co-authors.

"In real life, most of us are not confronted with clearly stated problems with a simple solution. Usually, we have to first say what the problem is."

Mr. Getzels was born in 1912 in what is now Bialystok, Poland, and came to the United States as a child. He graduated from Brooklyn College and received a master's degree in comparative literature from Columbia University. He then joined the Army, where he switched fields to study psychology. He received a Ph.D. in social relations from Harvard in 1951 and immediately went to the University of Chicago.

Though he retired from the faculty in 1982, Mr. Getzels continued to pass his days reading and writing in his office in the university library. In addition to his research, he read widely in physics and math and, lately, novels.

Mr. Getzels served on many government panels, including a 1967 presidential task force on education and a cultural mission to the Soviet Union in 1960.

Guilford

Before I am able to explain several of the leading theories of creativity, one must understand a confusing intricacy surrounding this topic of study. What is different about the study of creativity is that in other fields of research the newest theory often replaces the older theories. The newest theory is often deemed as being "more correct" than the previously held school of thought. However when researching creativity, one might find that there are a multitude of theories that are seen as "correct". The theories themselves

each argue that creativity is caused by different factors. They each argue that the components of creativity are different. Interestingly enough, they each are correct in their own respect. Now this may seem a bad confusing, but because creativity is a such a dynamic concept many theories are only able to define creativity in one particular field or career at one specific point in time (i.e. creativity among engineers, creativity among artists, etc.) That being said, it is important to view each model of creativity only as a single perspective among many perspectives.

Guilford's Structure of Intellect Model and Model of Creativity

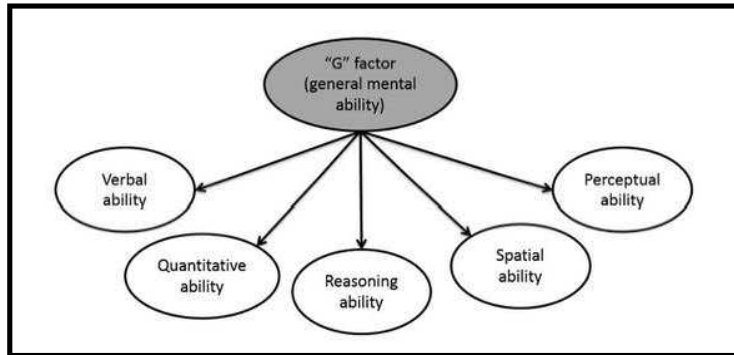
One of the earliest models of creativity was created by a man named J. P. Guilford. Originally, Guilford was trying to create a model for intellect as a whole, but in doing so also created a model for creativity. Guilford hypothesized that every mental task was made up of three separate parts: an operation, a content, and a product. He stated that there were five types of operations, four types of content, and six types of products adding to a total of one hundred and twenty different types of possible mental tasks (he would later expand this number to one hundred and eighty different mental tasks, but for now let's focus on the original one hundred and twenty.) Of these one hundred and twenty different mental tasks Guilford identified one specific operation as "divergent production" and marked it as being a vital component of creativity. This divergent production coupled with a content and a product created twenty four possibilities that Guilford labeled collectively as "divergent thinking".

By labeling this group of mental tasks as divergent thinking Guilford made an important assumption for creative research: creativity isn't one abstract concept. The idea that creativity is a category rather than one single concept opened up the ability for other researchers to look at creativity with a whole new perspective.

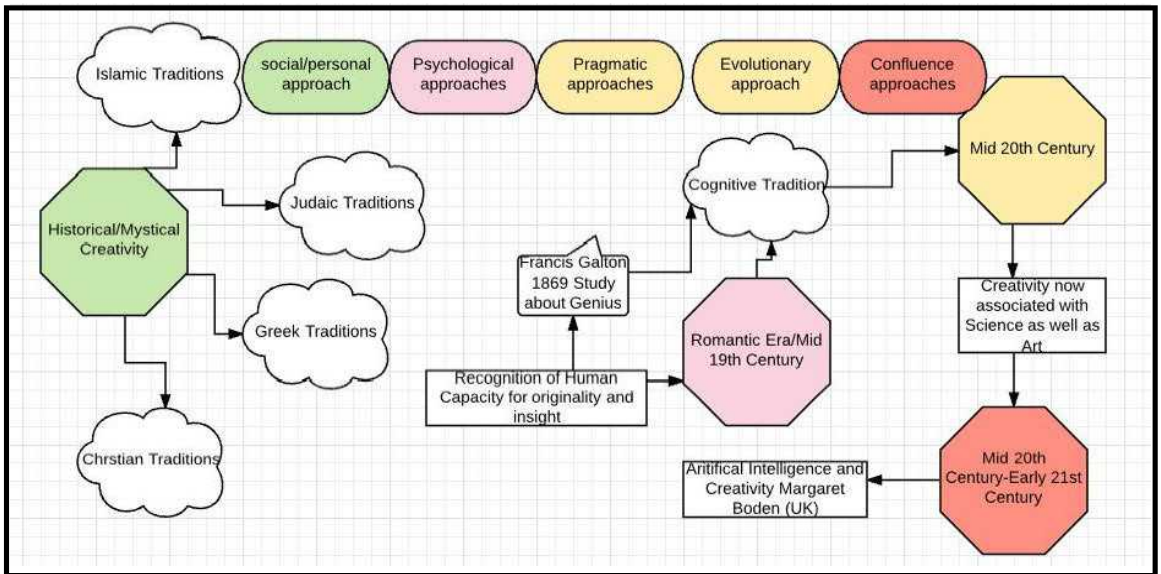
Additionally, Guilford hypothesized one of the first models for the components of creativity. He explained that creativity was a result of having:

1. Sensitivity to problems, or the ability to recognize problems.
2. Fluency, which encompasses
 - a. Ideational fluency, or the ability rapidly to produce a variety of ideas that fulfill stated requirements.
 - b. Associational fluency, or the ability to generate a list of words, each of which is associated with a given word.
 - c. Expressional fluency, or the ability to organize words into larger units, such as phrases, sentences, and paragraphs.
3. Flexibility, which encompasses
 - a. Spontaneous flexibility, or the ability to demonstrate flexibility.
 - b. Adaptive flexibility, or the ability to produce responses that are novel and high in quality

This represents the base model by which several researchers would take and alter to produce their new theories of creativity years later. This base theory stated that creativity was a result of cognitive ability (intellect) alone. As Guilford was originally only studying intellect, it is not strange that he categorizes creativity as just a specific subset of cognitive ability.



After Guilford's initial model, researchers would begin to categorize and re-categorize mental ability into subsets that would further hypothesize how mental ability was broken up (as shown in the image to the left). These models would help others understand and put into place one specific part of creativity that Guilford brought to light: the intellectual requirement.



Wallach & Kogan

In Wallach and Kogan's (1965) assessment of creativity examinees are asked to come up with many possible items that contain a specific component, such as with wheels, round things, or things that make noise.

Example. Name things with wheels:

- a car
- a truck
- a train
- a go cart
- your mind

Scoring

Scoring is comprised of four components:

Originality each response it compared to the total amount of responses from all of the people you gave the test to. Responses that were given by only 5% of your group are unusual (1 point), responses that were given by only 1% of your group are unique - 2 points). Total all the point.

Fluency or total. Just add up all the responses. In this example it is 5.

Flexibility or different category, in this case 2 (the first four are all methods of transportation but the "wheel turning in your mind is a different category from transportation).

Elaboration amount of detail (for Example "a car" = 0 whereas "a car racing down the street" = 1 (another point is given for the detail of where the car is going).

Administration

Any one can administer Wallas and Kogan's (1965) test. No training required. Materials can be created by the examiner and individually administered to the examinees or shown on an overhead projector to a group of examinees.

Typically the test is administered in a classroom setting. However, the test can also been an unlimited time "take home": since time is an issue. The majority of responses given by the examinees in the first few minutes tend to be their least creative.

RELATIONSHIP BETWEEN INTELLIGENCE AND CREATIVITY

Intelligence is classically defined as "the ability to acquire and utilize knowledge." In testing circumstances, an Intelligence Quotient (IQ) is gauged by one's ability to utilize information gained historically.

Creativity is the ability to come up with new ideas through a mental process of connecting existing concepts. The ideas don't have to be revolutionary (which is a common misconception many people have about creative thinking), they just have to be new for the thinker.

Intelligence certainly plays a part in creative thinking, but not how you might expect.

Your IQ is generally gauged by an ability to interpret information and provide solutions, no matter the circumstance. In mathematics and basic sciences IQ is immensely important, because it demonstrates your ability to memorize concepts and repeat their results on similar problems. If I tell you that two plus two equals four, you should (ideally) be able to intelligently conclude that four plus four equals twice the original answer.

This fact alone demonstrates intelligence's relation to creativity, one that is vital for not only understanding creative thinking, but for improving it.

Another important aspect of intelligence is the ability to filter solutions efficiently.

If you're great at acquiring knowledge (say, through reading or lectures or watching videos on YouTube) and you have the ability to put that knowledge to use effectively, but lack the ability to efficiently filter through solutions, you may come up with effective ideas, but it's going to take you a long time. As opposed to those with high intelligence levels who can filter through ideas quickly.

Of course – and this is the real kicker – intelligence only gets you so far when it comes to creativity.

To be creative is to pull existing knowledge into a new situation and quickly sort through potential outcomes. Of course: existing knowledge is something that anyone above a certain threshold on the IQ scale can amass. That intelligence number, it seems, is right around 100 (right in the middle of the average range for IQ test-takers in the United States). If you're reading this, you have the creative potential of anyone with an IQ of 100 or above.

Being able to come up with creative ideas isn't something you need an overly-high IQ to accomplish. Once you've got a level of knowledge gathering and utilization that's about average, you're well on your way to having the creative potential of Albert Einstein, Bill Gates, or Steve Jobs. Mr. Jobs even stated this himself while he was alive by saying;

“Creativity is just connecting things. When you ask creative people how they did something, they feel a little guilty because they didn't really do it, they just saw something. It seemed obvious to them after a while. That's because they were able to connect experiences they've had and synthesize new things. And the reason they were able to do that was that they've had more experiences or they have thought more about their experiences than other people.”

So intelligence matters, it demonstrates your ability to gather knowledge and effectively use it. Creativity is the ability to go beyond the intelligence frame and capitalize on seemingly random connections of concepts.



PERSONALITY, MOTIVATION, EMOTION, STRESS AND COPING

Almost everyday we describe and assess the personalities of the people around us. Whether we realize it or not, these daily musings on how and why people behave as they do are similar to what personality psychologists do.

While our informal assessments of personality tend to focus more on individuals, personality psychologists instead use conceptions of personality that can apply to everyone. Personality research has led to the development of a number of theories that help explain how and why certain personality traits develop.

Definitions of Personality

While there are many different theories of personality, the first step is to understand exactly what is meant by the term *personality*. The word personality itself stems from the Latin word *persona*, which referred to a theatrical mask worn by performers in order to either project different roles or disguise their identities.

A brief definition would be that personality is made up of the characteristic patterns of thoughts, feelings and behaviors that make a person unique. In addition to this, personality arises from within the individual and remains fairly consistent throughout life.

Some other definitions of personality:

- (i) "Personality refers to individuals' characteristic patterns of thought, emotion, and behavior, together with the psychological mechanisms – hidden or not – behind those patterns. This definition means that among their colleagues in other subfields of psychology, those psychologists who study personality have a unique mandate: to explain whole persons."
- (ii) "Although no single definition is acceptable to all personality theorists, we can say that personality is a pattern of relatively permanent traits and unique

characteristics that give both consistency and individuality to a person's behaviour."

Components of Personality

So what exactly makes up a personality? As described in the definitions above, you would expect that traits and patterns of thought and emotion make up an important part. Some of the other fundamental characteristics of personality include:

- (i) **Consistency.** There is generally a recognizable order and regularity to behaviors. Essentially, people act in the same ways or similar ways in a variety of situations.
- (ii) **Psychological and physiological.** Personality is a psychological construct, but research suggests that it is also influenced by biological processes and needs.
- (iii) **It impacts behaviors and actions.** Personality does not just influence how we move and respond in our environment; it also *causes* us to act in certain ways.
- (iv) **Multiple expressions.** Personality is displayed in more than just behavior. It can also be seen in our thoughts, feelings, close relationships and other social interactions.

DETERMINANTS OF PERSONALITY

Personality Traits

Personality is the collection of characteristic thoughts, feelings, and behaviors that are associated with a person. Personality **traits** are characteristic behaviors and feelings that are consistent and long lasting.

Traits vs. States

Unlike traits, which are stable characteristics, states are temporary behaviors or feelings that depend on a person's situation and motives at a particular time. The difference between traits and states is analogous to the difference between climate and weather. Los Angeles has a warm climate, but on some days it may have cool weather. In the same way, a person who has the trait of calmness may experience a state of anxiety on a day when he or she faces a difficult challenge.

Ancient Greek Ideas

The ancient Greeks believed that people's personalities depended on the kind of **humor**, or fluid, most prevalent in their bodies. The ancient Greeks identified four humors – blood, phlegm, black bile, and yellow bile – and categorized people's personalities to correspond as follows:

- (i) **Sanguine.** Blood. Cheerful and passionate.
- (ii) **Phlegmatic.** Phlegm. Dull and unemotional.
- (iii) **Melancholic.** Black bile. Unhappy and depressed.
- (iv) **Choleric.** Yellow bile. Angry and hot-tempered.

The Greek theory of personality remained influential well into the eighteenth century.

Cattell's Sixteen Traits

Like the ancient Greeks, modern researchers believe in the existence of a few basic personality traits. Combinations of these basic traits, they believe, form other traits. Psychologist Raymond Cattell used a statistical procedure called **factor analysis** to identify basic personality traits from a very long list of English words that identified traits. Factor analysis allowed Cattell to cluster these traits into groups according to their similarities. He found that personality is made up of sixteen basic dimensions.

The Big Five Traits

Other researchers have since clustered personality traits into even fewer categories. Today, many psychologists believe that all personality traits derive from five basic personality traits, which are commonly referred to as the Big Five:

1. Neuroticism
2. Extraversion
3. Openness to experience
4. Agreeableness
5. Conscientiousness

The big five traits remain quite stable over the life span, particularly after the age of thirty. Although researchers identified the big five traits by using a list of English words, these traits seem to be applicable in many countries.

Criticisms of the Big Five Model

Critics of the big five have various arguments against the model:

- (i) Some critics think that more than five traits are needed to account for the wide personality differences among people.
- (ii) Other critics argue that five traits are too many. For example, they point out that openness correlates positively with extraversion. These critics argue that just three traits – neuroticism, extraversion, and agreeableness – should be enough to fully describe personality.
- (iii) Still other critics argue that the big five are somewhat arbitrary because they depend on the words used in the statistical analysis that produced them. A different list of words may have yielded different basic traits.
- (iv) Some psychologists have questioned the research supporting the stability of the Big Five traits across cultures. They argue that the research could be biased because the use of Western tests is more likely to uncover cultural similarities than differences.

Biological Determinants of Personality

The human brain with its twelve billion nerve cells and almost infinite number of inter-connections and pathways may well be the most complex structure in the universe. In principle, all psychological events are represented in some manner by the activity of the brain and nervous system in conjunction with the other body systems. One approach to the study of man attempts to relate his actions to events taking place inside his body, particularly within the brain and nervous system. This approach tries to reduce observable (overt) behaviour and mental events, such as thoughts and emotions, to neurobio-

logical processes. For example, a psychologist studying learning from the biological approach is interested in the changes that take place in the nervous system as the result of learning how to perform a new task. Visual perception can be studied by recording the activity of nerve cells in the brain as the eye is exposed to simple patterns of lines.

Recent discoveries have made it dramatically clear that there is an intimate relationship between brain activity and behaviour and experience. Emotional reactions, such as fear and rage have been produced in animals and humans by mild electrical stimulation of specific areas deep in the brain. Electrical stimulation of certain areas in the human brain will produce sensations of pleasure and pain, and even vivid memories of past events.

Ultimately, it may be possible to specify the neurobiological mechanisms underlying even the most complex human actions. However, a comprehensive neurobiological theory of man is at present only a remote possibility.

Psychologists agree that environmental factors interact with genetic factors to form personality. Some psychologists have proposed theories that emphasize these genetic influences on personality.

Hans Eysenck's Theory

Psychologist **Hans Eysenck** believes that genetics are the primary determinate of personality, although he thinks conditioning also plays a role. According to Eysenck, personality traits are hierarchical, with a few basic traits giving rise to a large array of more superficial traits. Genetically determined differences in physiological functioning make some people more vulnerable to behavioral conditioning. Eysenck suggests that introverted people have higher levels of physiological arousal, which allows them to be conditioned by environmental stimuli more easily. Because of this, such people develop more inhibitions, which make them more shy and uneasy in social situations.

Empirical evidence for genetic contributions to personality comes mainly from two kinds of studies: studies of children's temperaments and heritability studies.

Studies of Temperament

Temperament refers to innate personality features or dispositions. Babies show particular temperaments soon after birth. Temperaments that researchers have studied include reactivity, which refers to a baby's excitability or responsiveness, and soothability, which refers to the ease or difficulty of calming an upset baby.

Researchers have studied children from infancy to adolescence and found that temperaments remain fairly stable over time. However, temperaments can also be modified over time by environmental factors.

Heritability Studies

Heritability studies also provide evidence for genetic contributions to personality. **Heritability** is a mathematical estimate that indicates how much of a trait's variation in a population can be attributed to genes. For more information about heritability.

Twin studies help researchers to determine heritability, "Evolution and Genes." Researchers have shown that identical twins raised together are more similar than fraternal twins raised together in traits such as positive emotionality, negative emotionality, and constraint. Identical twins separated early in life and raised apart are more similar

in these traits than are fraternal twins raised together. Both of these research findings suggest the existence of a genetic component to personality.

Behavioral geneticists have shown, after doing studies in many different countries, that the heritability of personality traits is around .5, which means that 50 percent of the variation in personality traits in a group of people can be attributed to genetic differences among those people.

The Influence of Family Environment

Surprisingly, research shows that sharing a family environment doesn't lead to many similarities in personality. There is no or little correlation between the personality traits of adopted children and their adoptive parents. Researchers think this is because parents don't act the same way with all their children. Children's temperaments influence how a parent behaves toward them, and a child's gender and place in a birth order can also affect how that child is treated.

Environmental Influences

The environment also has important influences on personality. These include peer relationships and the kinds of situations a child encounters. "Walter Mischel's Ideas," the interactions between innate characteristics and environmental factors are two-way. Children's temperaments are likely to influence their peer relationships and the situations they encounter. Similarly, peers and situations can modify children's personality characteristics.

Evolutionary Approaches

Evolutionary theorists explain personality in terms of its adaptive value. Theorists such as David Buss have argued that the Big Five personality traits are universally important because these traits have given humans a reproductive advantage.

Socio-Cultural Determinants of Personality

There are also wide individual differences that are strongly correlated with everyday behaviour. For example, academic self-esteem predicts children's school achievement as well as their willingness to try hard at challenging tasks. Children with high social self-esteem are consistently better liked by their peers. And boys come to believe they have more athletic talent than do girls, and they are also more advanced in a variety of physical skills.

Because self-esteem is so powerfully related to behaviour, researchers have been intensely interested in finding out which social influences cause it to be high for some children and low for others. If ways can be found to improve children's sense of self-worth, then many aspects of child development might be enhanced as well.

Cultural forces have a profound impact on self-esteem. During adolescence early maturing girls and late maturing boys tend to feel poorly about themselves – outcomes influenced by cultural standards of physical beauty. Gender-stereotyped expectations for physical attractiveness and achievement have an especially detrimental effect on the self-esteem of girls. Beginning in adolescence, they score substantially lower than boys in overall sense of self-worth – partly because girls worry more about their appearance and partly because they feel more insecure about their abilities.

Furthermore, the strong role of social comparison in self-esteem does not characterize children everywhere. Puerto Rican fishing village children, discussed earlier in this chapter, almost never make statements referring to social comparison. Yet among American children, such statements are common, both in self descriptions and in everyday conversations. An even greater emphasis on social comparison may underlie the consistent finding that Taiwanese children score lower in self-esteem than American children, despite their higher academic achievement. In Taiwanese classrooms, competition is tough and achievement pressure is high. Because so much stress is placed on school performance, Taiwanese children have fewer opportunities than American children to feel successful in other ways.

Challenges for Cultural Psychology

Cultural psychologists face the difficult challenge of studying and describing differences among cultures without stereotyping any particular culture. Ideally, cultural psychologists acknowledge that all members of a culture don't behave similarly. Variation exists within every culture, in terms of both individuals and subcultures. Cultural psychologists also try not to exaggerate differences among cultures.

APPROACHES TO THE STUDY OF PERSONALITY

The most intensive topic in the world of philosophy and psychology is personality and the self. The major task of philosophy is to seek the self and to shape personality. While psychology rests on measuring and shaping the personality. Psychologically, most personality theories can be grouped into one of four classes: trait, psychoanalytic, social learning, and humanistic. These theoretical approaches differ markedly in the constructs they propose as forming the structure of personality (e.g. traits, id-ego-superego, learned habits or self-concept) and the way they relate these constructs to behaviour. They also differ in the methods they use to assess or measure personality. In the remainder of this chapter we will summarize the main constructs of each theoretical approach, note how these constructs are related to behaviour, and give examples of the methods used to assess personality. Personality cannot be studied scientifically unless there are satisfactory ways of measuring personality variables.

Psychoanalytic

Sigmund Freud was the founder of psychoanalytic and the psychodynamic approach to psychology. This school of thought emphasized the influence of the unconscious mind on behavior. Freud believed that the human mind was composed of three elements: the id, the ego, and the superego.

Freud's theories of psychosexual stages, the unconscious, and dream symbolism remain a popular topic among both psychologists and laypersons, despite the fact that his work is viewed with skepticism by many today.

Many of Freud's observations and theories were based on clinical cases and case studies, making his findings difficult to generalize to a larger population. Regardless, Freud's theories changed how we think about the human mind and behaviour and left a lasting mark on psychology and culture.

Another theorist associated with psychoanalytic is Erik Erikson. Erikson expanded upon Freud's theories and stressed the importance of growth throughout the life-span. Erikson's psychosocial stage theory of personality remains influential today in our understanding of human development.

Major Thinkers Associated With Psychoanalysis

Sigmund Freud

Erich Fromm

Otto Rank

Anna Freud

Carl Jung

Sabina Spielrein

Erik Erikson

Karl Abraham

Key Psychoanalytic Terms

Free Association

In Sigmund Freud's **Outline of Psychoanalytic** (1940), he compared the human mind to an iceberg in order to describe the structure of personality. The small portion of the iceberg that lies above the water represents the conscious mind, or all of the thoughts, feelings, and desires that you are fully aware of. The massive chunk below the surface represents your unconscious mind where all of the dreams, impulses, and repressed memories that are outside of your conscious awareness are housed. In order to begin to understand these unconscious mysteries, Freud used free association tactics to get a glimpse at what lay beneath. This exercise required the patient to talk about whatever came to mind no matter how relevant, nonsensical, or foreign the thoughts sounded. This technique, along with dream analysis and childhood memory evaluation, attempted to help patients understand themselves and their actions.

Stream of consciousness is a term applied to a writing exercise utilized by James Joyce, Virginia Woolf, and Beat writer James Kerouac, to name a few, as a means of recording the conscious experience of a continuous flow of ideas, feelings, and images running through the mind at the exact moment they occur.

In addition to free association, Freud's idea that personality is constructed of three components (the id, ego, and superego) is one of the most widely recognized theories in psychology, and is often the subject of cartoons and comics

Case Study

A case study is an in-depth study of one person. Much of Freud's work and theories were developed through individual case studies.

In a case study, nearly every aspect of the subject's life and history is analyzed to seek patterns and causes for behavior. The hope is that learning gained from studying one case can be generalized to many others. Unfortunately, case studies tend to be highly subjective and it is difficult to generalize results to a larger population.

Types of Case Studies

- (i) **Explanatory.** Used to do causal investigations.
- (ii) **Exploratory.** A case study that is sometimes used as a prelude to further, more in-depth research. This allows researchers to gather more information before developing their research questions and hypotheses.
- (iii) **Descriptive.** Involves starting with a descriptive theory. The subjects are then observed and the information gathered is compared to the pre-existing theory.

- (iv) **Intrinsic.** A type of case study in which the researcher has a personal interest in the case.
- (v) **Collective.** Involves studying a group of individuals.
- (vi) **Instrumental.** Occurs when the individual or group allows researchers to understand more than what is initially obvious to observers.

Case Study Methods

- (i) **Prospective.** A type of case study in which an individual or group of people is observed in order to determine outcomes. For example, a group of individuals might be watched over an extended period of time to observe the progression of a particular disease.
- (ii) **Retrospective.** A type of case study that involves looking at historical information. For example, researchers might start with an outcome, such as a disease, and then backwards at information about the individuals life to determine risk factors that may have contributed to the onset of the illness.

Sources of Information Used in a Case Study

There are a number of different sources and methods that researchers can use to gather information about an individual or group. The six major sources that have been identified by researchers are:

1. **Direct observation.** This strategy involves observing the subject, often in a natural setting. While an individual observer is sometimes used, it is more common to utilize a group of observers.
2. **Interviews.** One of the most important methods for gathering information in case studies. An interview can involve structured survey-type questions, or more open-ended questions.
3. **Documents.** Letters, newspaper articles, administrative records, etc.
4. **Archival records.** Census records, survey records, name lists, etc.
5. **Physical artifacts.** Tools, objects, instruments and other artifacts often observed during a direct observation of the subject.
6. **Participant observation.** Involves the researcher actually serving as a participant in events and observing the actions and outcomes.

Conscious

In Sigmund Freud's psychoanalytic theory of personality, the conscious mind consists of everything inside of our awareness. This is the aspect of our mental processing that we can think and talk about in a rational way.

The conscious mind includes such things as the sensations, perceptions, memories, feeling and fantasies inside of our current awareness. Closely allied with the conscious mind is the preconscious, which includes the things that we are not thinking of at the moment but which we can easily draw into conscious awareness.

Things that the conscious mind wants to keep hidden from awareness are repressed into the unconscious mind. While we are unaware of these feelings, thoughts, urges and emotions, Freud believed that the unconscious mind could still have an influence on our behavior.

Freud often used the metaphor of an iceberg to describe the two major aspects of human personality. The tip of the iceberg that extends above the water represents the

conscious mind. As you can see in the image at the right, the conscious mind is just the “tip of the iceberg.” Beneath the water is the much larger bulk of the iceberg, which represents the unconscious.

While the conscious and preconscious are important, Freud believed that they were far less vital than the unconscious. The things that are hidden from awareness, he believed, exerted the greatest influence over our personalities and behaviors.

Defense Mechanism

According to Anna Freud, daughter of the famous Sigmund, you employ several types of defense mechanisms in order to get yourself through tough times until you are capable of handling the situation from a more realistic approach. Repression, rationalization, reaction formation, projection, intellectualization, denial, and displacement are all ways in which you engage in some form of artifice. Let’s take a closer look at these.

Repression is the most basic and most important mechanism in which painful and frightful memories that incite guilt, shame, or self-worthlessness are kept out of your conscious reach and stored in the unconscious region of the mind. Rationalization is best illustrated by Aesop’s fable where the fox decides the grapes are sour only after he can’t reach them. You follow the reasoning that sounds good at the time, rather than the reasoning that best reflects the situation, taking on an “I didn’t really want it anyway” attitude.

Reaction formation is hiding a motive from yourself by performing acts that display the extreme opposite of that motive. For example, you may hate your mother-in-law, but are ashamed that you feel that way, so you go out of your way to be extra nice to her. Projection is when you assign your own offensive traits to other people in an attempt to avoid disliking yourself. Intellectualization is coping with stressful situations in purely abstract and intellectual terms. People who work in a field where suffering and death are constant must attempt to detach themselves emotionally in order to perform the job successfully.

Denial occurs when the reality of the situation is too hard to accept, and therefore you refuse to acknowledge it. Displacement is the defense mechanism in which motives that cannot be fulfilled in the intended way are expressed in other ways such as art, music or poetry.

Ego

According to Freud, the ego is part of personality that mediates the demands of the id, the superego and reality. The ego prevents us from acting on our basic urges (created by the id), but also works to achieve a balance with our moral and idealistic standards (created by the superego). While the ego operates in both the preconscious and conscious, its strong ties to the id means that it also operates in the unconscious.

The ego operates based on the reality principle, which works to satisfy the id’s desires in a manner that is realistic and socially appropriate. For example, if a person cuts you off in traffic, the ego prevents you from chasing down the car and physically attacking the offending driver. The ego allows us to see that this response would be socially unacceptable, but it also allows us to know that there are other more appropriate means of venting our frustration.

Observations

According to **Sigmund Freud**, “One might compare the relation of the ego to the id with that between a rider and his horse. The horse provides the locomotor energy, and the rider has the prerogative of determining the goal and of guiding the movements of his powerful mount towards it. But all too often in the relations between the ego and the id we find a picture of the less ideal situation in which the rider is obliged to guide his horse in the direction in which it itself wants to go.”

According to **Anna Freud**, “All the defensive measures of the ego against the id are carried out silently and invisibly. The most we can ever do is to reconstruct them in retrospect: we can never really witness them in operation. This statement applies, for instance, to successful repression. The ego knows nothing of it; we are aware of it only subsequently, when it becomes apparent that something is missing.”

Quotations About the Ego

According to **Sigmund Freud**, “The ego is not master in its own house.”

According to **Sigmund Freud**, “It is easy to see that the ego is that part of the id which has been modified by the direct influence of the external world.”

According to **Sigmund Freud**, “The ego represents what we call reason and sanity, in contrast to the id which contains the passions.”

According to **Sigmund Freud**, “Towards the outside, at any rate, the ego seems to maintain clear and sharp lines of demarcation. There is only one state – admittedly an unusual state, but not one that can be stigmatized as pathological – in which it does not do this. At the height of being in love the boundary between ego and object threatens to melt away. Against all the evidence of his senses, a man who is in love declares that “I” and “you” are one, and is prepared to behave as if it were a fact.”

Id

According to Sigmund Freud’s psychoanalytic theory of personality, the id is the personality component made up of unconscious psychic energy that works to satisfy basic urges, needs, and desires. The id operates based on the pleasure principle, which demands immediate gratification of needs.

The id is the only part of personality that is present at birth. Freud also suggested that this primitive component of personality existed completely within the unconscious. The id acts as the driving force behind personality. It not only strives to fulfill our most basic urges, many of which are tied directly to survival, it also provides *all* of the energy necessary to drive personality.

During infancy, before the other components of personality begin to form, children are ruled entirely by the id. Satisfying basic needs for food, drink, and comfort are of the utmost importance. As we grow older, it would obviously be quite problematic if we acted out to satisfy the needs of the id whenever we felt an urge, need, or desire. Fortunately, the other components of personality develop as we age, allowing us to control the demands of the id and behave in socially acceptable ways.

As mentioned earlier, the id acts according to the pleasure principle, which is the idea that needs should be met immediately. When you are hungry, the pleasure principle directs you to eat. When you are thirsty, it motivates you to drink. But of course, we can’t

always satisfy our urges right away. Sometimes we need to wait until the right moment or until we actually have access to the things that will fulfill our needs.

When we are unable to immediately satisfy a need, tension results. The id relies on the primary process to temporarily relieve the tension. The primary process involves creating a mental image either through daydreaming, fantasizing, hallucinating, or some other process. For example, when you are really thirsty, you might start fantasizing about a tall, cold glass of ice water.

Remember, however, that the id is just one of the three major components of personality. You can learn more about the ego and superego and how these elements of personality interact in this overview of the id, ego, and superego.

Observations

According to **Sigmund Freud**, “It is the dark, inaccessible part of our personality, what little we know of it we have learned from our study of the Dreamwork and of the construction of neurotic symptoms, and most of that is of a negative character and can be described only as a contrast to the ego. We approach the id with analogies: we call it a chaos, a cauldron full of seething excitations... It is filled with energy reaching it from the instincts, but it has no organization, produces no collective will, but only a striving to bring about the satisfaction of the instinctual needs subject to the observance of the pleasure principle.”

“One might compare the relation of the ego to the id with that between a rider and his horse. The horse provides the locomotor energy, and the rider has the prerogative of determining the goal and of guiding the movements of his powerful mount towards it. But all too often in the relations between the ego and the id we find a picture of the less ideal situation in which the rider is obliged to guide his horse in the direction in which it itself wants to go.”

Superego

According to Freud’s psychoanalytic theory of personality, the superego is the component of personality composed of our internalized ideals that we have acquired from our parents and from society. The superego works to suppress the urges of the id and tries to make the ego behave morally, rather than realistically.

In Freud’s theory of psychosexual development, the superego is the last component of personality to develop. The id is the basic, primal part of personality that is present from birth. Next, the ego begins to develop during the first three years of a child’s life. Finally, the superego starts to emerge around the age of five.

The ideals that contribute to the formation of the superego include not just the morals and values that we have learned from our parents, but also the ideas about right and wrong that we acquire from society and the culture in which we live.

The Two Parts of the Superego

The superego can be further divided into two components: the ego ideal and the conscience. The ego ideal is made up of all of our rules for good behavior. The conscience is composed of the rules for which behaviors are considered bad. When we engage in actions that conform to the ego ideal, we feel good about ourselves or proud of our accomplishments. When we do things that our conscience considers bad, we experience feelings of guilt.

The Goals of the Superego

The primary action of the superego is to entirely suppress any urges or desires of the id that are considered wrong or socially unacceptable. It also tries to force the ego to act morally rather than realistically. Finally, the superego strives for moral perfections, without taking reality into account.

As you can see in the image at the right, the superego is present in all three levels of consciousness. Because of this, we can sometimes experience guilt without understanding exactly why we feel that way. When the superego acts in the conscious mind, we are aware of our resulting feelings. If, however, the superego acts unconsciously to punish or suppress the id, we might end up with feelings of guilt and no real understanding of why we feel that way.

Observations

“The superego’s contents are for the most part conscious and so can be directly arrived at by endopsychic perception. Nevertheless, our picture of the superego always tends to become hazy when harmonious relations exist between it and the ego. We then say that the two coincide, i.e. at such moments the superego is no perceptible as a separate institution either to the subject himself or to an outside observer. Its outlines become clear only when it confronts the ego with hostility or at least with criticism. The superego, like the id, become perceptible in the state which it produces within the ego: for instance, when its criticism evokes a sense of guilt.”

Unconscious

In Freud’s psychoanalytic theory of personality, the unconscious mind is a reservoir of feelings, thoughts, urges, and memories that outside of our conscious awareness. Most of the contents of the unconscious are unacceptable or unpleasant, such as feelings of pain, anxiety, or conflict. According to Freud, the unconscious continues to influence our behavior and experience, even though we are unaware of these underlying influences.

Criticisms of Psychoanalysis

- (i) Freud’s theories overemphasized the unconscious mind, sex, aggression and childhood experiences.
- (ii) Many of the concepts proposed by psychoanalytic theorists are difficult to measure and quantify.
- (iii) Most of Freud’s ideas were based on case studies and clinical observations rather than empirical, scientific research.

Strengths of Psychoanalysis

- (i) While most psychodynamic theories did not rely on experimental research, the methods and theories of psychoanalytic thinking contributed to experimental psychology.
- (ii) Many of the theories of personality developed by psychodynamic thinkers are still influential today, including Erikson’s theory of psychosocial stages and Freud’s psychosexual stage theory.
- (iii) Psychoanalysis opened up a new view on mental illness, suggesting that talking about problems with a professional could help relieve symptoms of psychological distress.

Neofreudian

Many psychologists who studied with Freud eventually broke away from the Vienna group to develop own theories of personality and establish own schools of psychology.

Collectively, these theorists are known as the **neo-Freudians**. They retained many basic Freudian concepts and assumptions.

Limits of Freud's theory. Failure to recognize personality change after the first few years of life; emphasis on instinctual over social influences; & generally negative picture Freud painted of human nature.

Alfred Adler. Introduced concept of striving for superiority to account for most human motivation – we are motivated to overcome feelings of helplessness that begin in infancy. The primary struggle in personality development was the effort to overcome feelings of inferiority.

Adler's Individual Psychology

Individual psychology views people as motivated by purposes and goals.

People take an active role in creating their own lives.

Social factors are more important than sexual factors in shaping personality.

Striving for superiority is the human motivation to adapt to, improve, and master the environment.

In striving for mastery, a person may compensate or overcompensate for weaknesses.

Overcompensation is his term for attempts to deny real situations or control a weakness, which may result in the development of an inferiority complex or a superiority complex.

Differences between Freud and Adler. Adler saw motivation from social influences and striving for success. (not sex and aggression)

Adler is optimistic; people responsible for who they are; present behavior shaped by own view of the future; psychologically healthy people usually aware of what doing and why.

Adler suggested social interest and compassion could be central to motivation.

Born weak—leads to feelings of inferiority and dependence on people feeling of unity with others (social interest, ultimate standard for psychological health).

Striving for superiority or success one dynamic force behind behaviour – way of compensating for feelings of inferiority or weakness; need to overcome inferiority feelings.

People's subjective perceptions shape their behavior and personality.

Personality is unified and self-consistent person's thoughts, feelings and actions all directed toward single goal and serve single purpose. Behavior leads to neurotic or healthy style of life depends on degree of social interest developed early life.

Value of all human activity must be seen from viewpoint of social interest – strive for perfection for all people in an ideal community – not for personal superiority. Feelings of oneness with humanity, social community. Develops from mother-child relationship.

Immature people lack social interest – self-centered, strive for personal power and superiority over others.

Psychologically Healthy – genuinely concerned about others and have goal of success including well-being of others. Move beyond striving for personal gain; Concerned with goals beyond themselves.

The self-consistent personality structure develops into a person's style of life – flavor of life –goals, self concept, attitude toward the world. Express social interest through action. Creative power, freedom to create own style of life, in control of own life.

Adler identified parental pampering and neglect as two sources of later personality problems.

Birth order – argued that middle born children were the most achieving and were less likely to experience psychological disorders than were first-borns or last-borns.

Carl Jung's Analytical Psychology

Carl Jung contributed the concepts of introversion, extroversion, personal unconscious, and collective unconscious to the study of personality.

Jung stated that the roots of personality go back to the dawn of human existence and are contained in a collective unconscious. Jung proposed the existence of a collective unconscious that houses primordial images he called archetypes.

The collective unconscious contains material each of us inherited from past generations and is basically the same for all people. Among the most important of the archetypes are the anima, the animus, and the shadow. Jung pointed to the recurrent surfacing of archetypal symbols in folklore, art, dreams, and psychotic patients as evidence for their existence.

Erik Erikson: Ego Psychology

Ego performs constructive function – as opposed to Freud's mediator function. Goals of ego – establish identity; satisfy need for mastery over environment. Powerful and independent part of personality.

Identity: complex inner state includes sense of individuality, uniqueness and continuity with past and present. Term from Erikson – **identity crisis** – confusion and despair from lacking sense of identity. Uncertain about who you are, your values, lacking direction.

Erikson outlined eight stages of personality development that we pass through during our lifetimes. At each stage we are faced with a crisis and two means to resolve the crisis.

Ego helps adapt to various conflicts and crises of life.

Stages are an interaction of opposites – people have harmonious and disruptive experiences. Conflict or clash between elements produces a basic ego strength.

Too little basic strength at one stage results in core pathology (lack hope from infancy – withdrawal) Ego identity shaped by multiplicity of conflicts and events – past, present and anticipated

Identity crisis – characterized personality development from adolescence on. Not catastrophic event – but opportunity for either adaptive or maladaptive adjustment

Stages in Life-span

Karen Horney – anxious insecurity stems from inadequate child rearing experiences are the source of all personality conflicts.

The need for security is a prime motive in human existence.

Freud's views of women bothered her – instead cultural and social factors responsible for differences between gender – not biology emphasized the importance of sociocultural factors in personality development.

Person's reaction to real or imagined dangers or threats – termed anxiety – stronger motivating force than sexual drive or libido. Neurotic people – trapped in a self-defeating life-style. Neurotic trends or strategies used to cope with emotional problems.

Individuals cope with anxiety using three interaction styles or strategies

Moving toward people – compliant type, submission. Helpless, dependent. Attach selves to whomever is available – any relationship better than loneliness and feeling unwanted.

Moving against people – aggressive type of personality; aggression and hostility compensation for feelings of insecurity and inadequacy. Take advantage of others in adulthood. Externalization (like projection) so respond to others as if others will take advantage of them.

Moving away from people – detached type. Tune out the world, instead of interacting with others. Desire privacy and self-sufficiency, avoid affection, love and friendship. Avoid anxiety by avoiding involvement.

Womb envy – men's envy of women's ability to bear and nurse children. (Freud's ideas of penis envy). Each gender has attributes that the other admires, and men compensate for inability to have kids through achievement.

Freud was highly critical of organized religion, calling it wish fulfillment and a type of neurosis.

Jung explained humankind's persistent need for religion in terms of a God archetype modern psychotherapists take the place of religious leaders when patients become disenchanted with the answers provided by their religion.

Well known neo-Freudian personality assessment instruments is the Myers-Briggs Type Indicator. Test measures psychological types, as outlined by Jung. Test scores divide people into types along four dimensions: extraversion-introversion, sensing-intuitive, thinking-feeling, and judgment-perception. Researchers have challenged the way the test divides people into categories.

Strengths of the neo-Freudian theories in contributions made to psychoanalytic theory. Corrected some of the limitations they found in Freud's work, many theorists introduced important concepts to the field of psychology, influencing many later approaches to personality.

Criticisms of the neo-Freudians include use of biased and questionable data to support the theories. In addition, critics have charged that some of the theories are oversimplified and incomplete.

Social Learning Approaches

Social learning theorists share many assumptions with behaviourists, particularly the belief that people are shaped in fundamental ways by their **environment** through **learning** processes. Social learning theorists also acknowledge that classical and operant **conditioning** are an important influence on human behaviour. However, they add to these learning processes a third: **observational learning**. They believe people learn by observing others and therefore that other people (the **social environment**) are particularly important as an influence on behaviour. With the emphasis on observational learning comes

a belief that it is impossible to explain human behavior without considering the role of internal, **mental processes** in human behaviour, something that behaviourists reject.

Observational learning involves a number of cognitive and behavioural processes. In order to learn the behaviour of another, the person must first pay **attention** to what the other person (called a model) is doing. They must then **encode** and form a memory of the behaviour the model performs. At a later time, this memory must be translated back into a behaviour so that the observer may **imitate** it. In order to imitate the behaviour effectively the observer may need to **practise** it. Whether or not the observer actually makes use of the behaviour they have learned depends on whether they are motivated to do so. The observer's **motivation** may be affected by several factors, principally, whether they believe that reinforcement is available if they imitate. Their beliefs may be influenced by the consequences of their past behaviour (as in behaviourism) but also by the **observed consequences** of the model's behaviour. If the model is observed to be reinforced for their actions, then imitation becomes more likely, assuming that the observer values the reinforce that the model was observed to get. Conversely, if the model is observed to be punished then imitation becomes less likely (these processes are called **vicarious reinforcement** and punishment).

A number of factors besides reinforcement and punishment influence whether a particular person is likely to be selected as a model and imitated. Models are more likely to be imitated if the observer perceives them to be **similar** to themselves. Imitation is also more likely when the model has **high status** and is **attractive** and when the observer has **low self-esteem**.

Social learning theorists share with behaviourists a commitment to the **scientific method**. They favour objective, **quantitative** approaches to research and use the **experimental method** if possible. Unlike behaviourists they do not study animal learning. They conduct research in any context in which learning occurs, particularly favouring research using **children**, whose behaviour is more obviously influenced by the people around them than adults' (although adults also learn through observation). Social learning research therefore may include **observation** of children and adults in school and family settings. However, the **laboratory** setting is favoured by many researchers because of the opportunities it gives to isolate and examine the causal effects of different influences on observational learning.

An example of this approach comes from the studies carried out by **Albert Bandura** on the social learning of aggressive behaviours. Bandura (1965) showed children a film of an adult behaving aggressively towards an inflatable toy called a bobo doll. Some children saw the aggressive adult being reinforced by another adult, others saw them being punished and a control group saw the model behaving aggressively with no positive or negative consequences. Afterwards the children were given the opportunity to play with a range of toys including a bobo doll. It was found that those who had seen the model getting punished were much less likely to imitate the aggressive actions, showing the influence of vicarious punishment on imitation.

Its commitment to scientific methods is a strength of the social learning approach as its research studies are **reliable** and allow inferences about **cause and effect** to be drawn. However, this does give rise to the criticism that it relies heavily on research conducted in rather **artificial** settings. For example, some critics argue that the behaviour of the children in Bandura's studies was significantly influenced by **demand characteristics**

and that they were, in effect, deliberately producing the behaviour they thought the experimenters wanted to see.

Because the social learning approach takes into account the **cognitive factors** that mediate between stimuli and responses, it addresses one of the most important criticisms of behaviourism: its neglect of thinking processes. Consequently, social learning theory can explain a range of processes that behaviourism has difficulties with, especially behaviours that emerge without the person that produces them being directly conditioned. As such it has contributed significantly to our understanding of processes like **aggression** and gender development and has also formed the basis of a range of **treatments** for problems like phobias. Modelling-based therapies are particularly effective with children, who may find behaviour therapies using direct conditioning hard to deal with.

Some criticisms of social learning theory arise from their commitment to the environment as the chief influence on behaviour. For example, there are several gender differences that appear to be universal, such as preferences for particular characteristics in a potential heterosexual partner (men prioritise youth and fertility, women prioritise status and resources) and differences in the gender thinking of boys and girls (boys tend to be more rigid) that social learning theory cannot account for. It may be that differences like these reflect **genetic influences** on behaviour that social learning theory takes no account of.

Although the approach appears less mechanistic than behaviourism, social learning theorists generally do not believe in free will, and take a **deterministic** view of human behaviour. Their emphasis on the role of cognitive and motivational factors may appear to give freedom of choice a role in behaviour, but it should be remembered that a person's motivation is the product of learning that took place at an earlier time and that therefore their 'choices' are not free. They share with the behaviourists a tendency to favour the **nurture** side of the nature-nurture debate, although it is generally assumed that the capacity for observational learning must be innate. Like behaviorism and other scientific approaches to psychology, social learning theory is **nomological** in its outlook and holds the similarities between people to be more important than the differences. Along with the nomological outlook comes a **reductionist** approach to explaining behaviour, whereby even the most complex things that people do are explained in terms of relatively simple fundamental processes of learning.

Person-Situation Interaction

Social learning theorists regard the situation as an important determinant of behaviour. A person's actions in a given situation depend upon the specific characteristics of the situation, the individual's appraisal of the situation, and past reinforcement for behaviour in similar situations (or observations of others in similar situations). People behave consistently insofar as the situations they encounter and the roles they are expected to play remain relatively stable.

Some of the person variables that determine what an individual will do in a particular situation include the following:

- (i) **Competencies** – intellectual abilities, social skills, and other abilities.
- (ii) **Cognitive strategies** – habitual ways of selectively attending to information and organizing it into meaningful units.
- (iii) **Outcome expectancies** – expectations about the consequences of different behaviours and the meaning of certain stimuli (e.g. “shifty eyes” may mean not

trustworthy). Both types of expectancies depend upon inferences about the intentions motivating the behaviour of others.

- (iv) **Subjective value of outcome** – even if individuals have similar expectancies, they may choose to behave differently because of differences in the subjective values of the outcomes they expect. Two students may expect that a certain behaviour will please the professor, but for one this outcome is important, while for the other it is not.
- (v) **Self-regulatory systems and plans** – individual differences in self-imposed goals, rules guiding behaviour, self-imposed rewards for success or punishment for failure, and ability to plan and executed steps leading to a goal will lead to differences in behaviour.

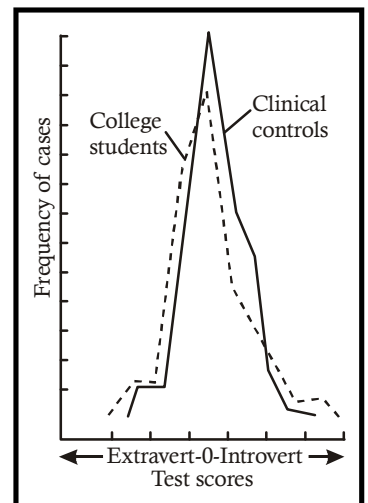
Assessment and Behaviour Changes

Social learning theorists are interested in personality assessment mainly for the purpose of behaviour of behaviour change. They define a specific behaviour pattern and attempt to discover the stimulus conditions that control that pattern. They are also interested in assessing the individual's response capabilities; a particular behaviour may fail to appear not because the proper reinforcing stimuli are lacking but because that response is not in the individual's repertoire. In addition to assessing stimulus conditions and response capabilities, the social learning theorist may want to measure the degree and type of self-reinforcement the individual uses. All these variables are relevant to behaviour and the possibilities for modifying that behaviour.

The Trait Approach of Personality

One of the earliest "personality theories" attempted to classify individuals into personality types on the basis of body build. A short, plump person was said to be sociable, relaxed, and even-tempered; a tall, thin person (ectomorph) was characterized as restrained, self-conscious, and fond of solitude; a heavy-set muscular individual was described as noisy, callous, and fond of physical activity. Although a person's physique may well have some influence on personality, the relationship is much more subtle than this sort of classification implies, and research has shown little correlation between body build and specific personality characteristics.

Personality type theories have also been based on purely psychological characteristics. One of Freud's pupils, the Swiss psychiatrist Carl Jung, divided all personalities into introverts and extroverts. The introvert tends to withdraw into himself, particularly in times of emotional stress and conflict; he tends to be shy and prefers to work alone. The introvert may take to the speaking platform in support of some movement to which he is strongly committed, but even then he is impelled from within. The extrovert under stress seeks the company of others. He is likely to be very sociable, a "hail fellow well met," and tends to choose occupations that permit him to deal directly with people, such as sales or promotional work.



INTROVERSION-EXTROVERSION.

You probably know a “typical introvert” and a “typical extrovert.” But it is also likely that most of your friends fall somewhere between the two extremes. This is one of the major problems of type theories. Most typologies, whether they are based on physical or psychological characteristics, involve a continuum of individual differences rather than discrete types. Type theories are appealing because they provide a simple way of looking at personality, but, in actuality, personality is far more complex.

Traits vs Types

A trait represents each of the positions on a number of continuous dimensions or scales describes a personality. While type theory tries to sort people into types, trait theory assume traits as its base.

Thus, we could rate an individual on a scale of intelligence, emotional stability, aggressiveness, creativeness, or any of a number of other dimensions. A trait, then, refers to any characteristic in which one individual differs from another in a relatively permanent and consistent way.

Psychologists working in the area of trait theory are concerned with:

- (1) determining the basic traits that provide a meaningful description of personality and,
- (2) finding some way to measure them.

FIVE TRAIT DIMENSIONS AND SOME OF THEIR COMPONENTS.

Trait Dimension	Descriptive Adjective Pairs
Extraversion	Talkative-Silent Open-Secretive Adventurous-Cautious
Agreeableness	Good natured-irritable Gentle-Headstrong Cooperative-Negativistic
Conscientiousness	Tidy-Careless Responsible-Undependable Persevering-Quitting
Emotional Stability	Calm-Anxious Poised-Nervous Not Hypochondriacal-Hypochondriacal
Culture	Artistically Sensitive-Artistically Insensitive Refined-Boorish Intellectual-Unreflective

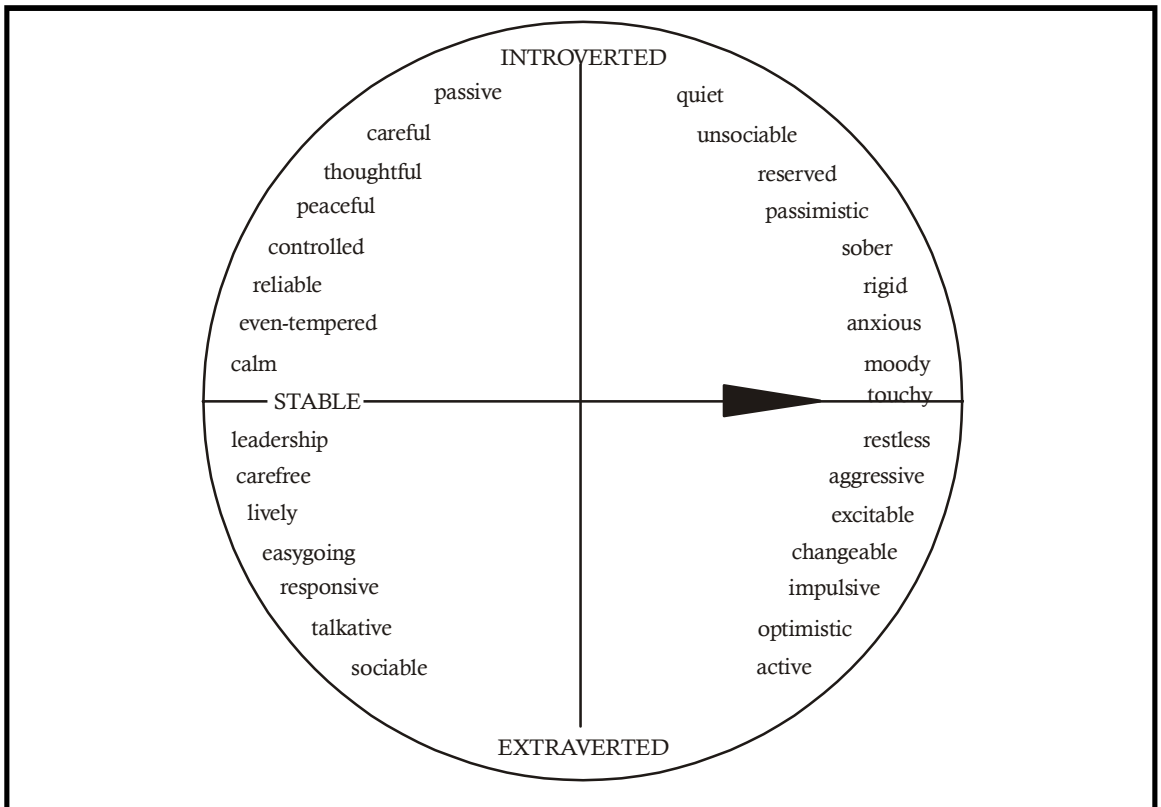
One approach uses factor analysis. Factor analysis is a complex statistical technique for reducing a large number of measures to a smaller number of independent dimensions. With this technique several hundred test responses might be reduced to a few underlying dimensions or factors that account for all of the response data.

The five trait dimensions listed in above Table were found in one study of this kind.

Factor analysis appears to provide a fairly straightforward way of arriving at a meaningful group of basic traits. But, unfortunately, because of differences in statistical techniques and in the type of data selected for analysis studies do not always find the same

basic factors. A factor found in one study may not show up in another. While some Investigators have found 5 factors that they believe to be the basic dimensions of personality, others have discovered as many as 20.

Despite the lack of agreement on basic traits, some overlap does occur. Two dimensions found consistently in factor-analytic studies of personality are introversion-extraversion and stability-instability. Introversion-extraversion refers to the degree to which one's basic orientation is turned toward the self or outward toward the external world. It is essentially the same distinction made by Carl Jung. Stability-instability is dimension of emotionality varying from calm, well-adjusted, reliable individuals at the stable end to those who are moody, anxious, temperamental, and unreliable at the other end. A British psychologist, Hans Eysenck, has developed a theory of personality based on these two dimensions. Given below figure shows the dimensions as related to various personality traits.



EYSENCK'S DIMENSIONS RELATED TO PERSONALITY TRAITS.

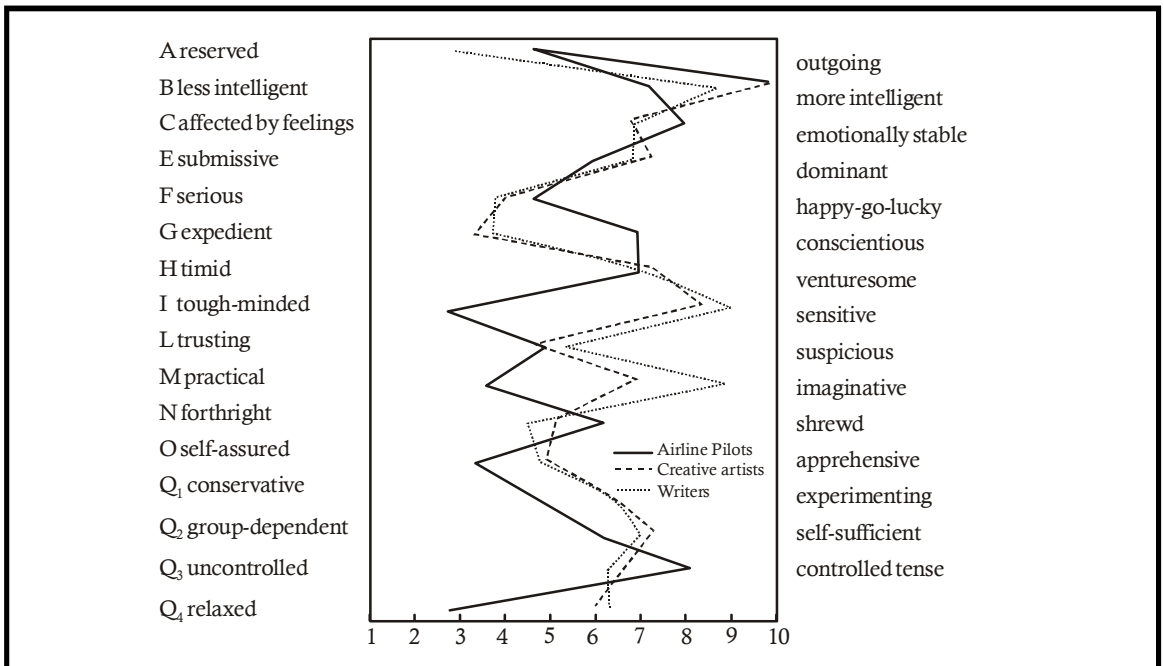
Assessing Personality Traits

Personality traits can be assessed by two methods: (1) the person describes himself by answering questions about his attitudes, feelings, and behaviours; (2) someone else evaluates the person's traits either from what he knows about the individual or from direct

observations of behaviour. With the first method a personality inventory is most often used, whereas the second usually involves a rating scale.

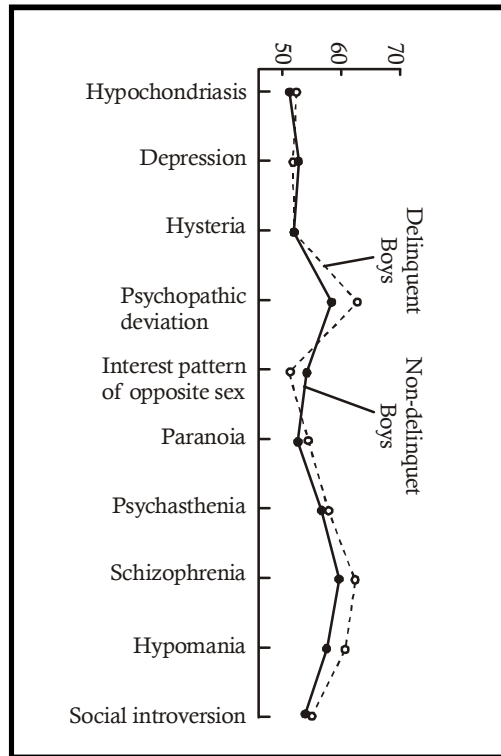
The Minnesota Multiphasic Personality Inventory, is based on the method of empirical construction rather than factor analysis. Instead of assuming specific personality traits and designing questions to measure them, many items were given to well-defined groups of individuals who were known to differ from the norm on some external criterion. Only those questions that discriminated significantly between groups were retained to form the inventory.

1. I like mechanics magazines.
2. I have a good appetite.
3. I wake up fresh and rested most mornings.
4. I think I would like the work of a librarian.
5. I am easily awakened by noise.
6. I like to read newspaper articles on crime.
7. My hands and feet are usually warm enough.
8. My daily life is full of things that keep me interested.



PERSONALITY PROFILES.

9. I am about as able to work as I ever was.
10. There seems to be a lump in my throat much of the time.
11. A person should try to understand his dreams and be guided by or take warning from them.



PERSONALITY PROFILES OF MALE DELINQUENTS AND NONDELINQUENTS.

SOME EXAMPLES OF RATING SCALES.

How would you describe the individual's self-confidence ?	Considers himself incapable of much success	Underestimates his own abilities	Knows just what he is capable of	Exaggerates his abilities	Judges himself capable of anything
Place a check at the point that describes the individual's poise.	Nervous and ill at ease	Somewhat tense easily upset	Average poise and security	Sure of himself	Composed; adapts well to crises
How would you rate the subject's emotional control	Very low	Below average	Average	Above average	Very high
Does the individual antagonize others ?	Never	Rarely	Sometimes	Often	Always
How would you rate the parent's behaviour toward the child	No control of emotion in response to child's behaviour	Controlled more by emotion than reason in dealing with child	Emotion freely expressed but behaviour controlled	Remains calm and objective toward child despite provocation	Never shows any signs of emotion always controlled
Does the individual need constant prodding or proceed with his work without being told ?	Needs much prodding in doing ordinary assignments	Needs occasional prodding	Does ordinary assignments of own accord	Completes suggested supplementary work	Seeks and sets for himself additional tasks

12. I enjoy detective or mystery stories.
13. I work under a great deal of tension.
14. I have diarrhoea once a month or more.
15. Once in a while I think of things too bad to talk about.

Gordon Allport's Trait Theory

In 1936, psychologist Gordon Allport found that one English-language dictionary alone contained more than 4,000 words describing different personality traits. He categorized these traits into three levels:

- (i) **Cardinal Traits.** Traits that dominate an individual's whole life, often to the point that the person becomes known specifically for these traits. People with such personalities often become so known for these traits that their names are often synonymous with these qualities. Consider the origin and meaning of the following descriptive terms: Freudian, Machiavellian, narcissism, Don Juan, Christ-like, etc. Allport suggested that cardinal traits are rare and tend to develop later in life.
- (ii) **Central Traits.** These are the general characteristics that form the basic foundations of personality. These central traits, while not as dominating as cardinal traits, are the major characteristics you might use to describe another person. Terms such as *intelligent*, *honest*, *shy* and *anxious* are considered central traits.
- (iii) **Secondary Traits.** These are the traits that are sometimes related to attitudes or preferences and often appear only in certain situations or under specific circumstances. Some examples would be getting anxious when speaking to a group or impatient while waiting in line.

Raymond Cattell's Sixteen Personality Factor Questionnaire

Trait theorist Raymond Cattell reduced the number of main personality traits from Allport's initial list of over 4,000 down to 171, mostly by eliminating uncommon traits and combining common characteristics. Next, Cattell rated a large sample of individuals for these 171 different traits. Then, using a statistical technique known as factor analysis, he identified closely related terms and eventually reduced his list to just 16 key personality traits. According to Cattell, these 16 traits are the source of all human personality. He also developed one of the most widely used personality assessments known as the Sixteen Personality Factor Questionnaire (16PF).

Eysenck's Three Dimensions of Personality

British psychologist Hans Eysenck developed a model of personality based upon just three universal traits:

1. **Introversion/Extraversion.** Introversion involves directing attention on inner experiences, while extraversion relates to focusing attention outward on other people and the environment. So, a person high in introversion might be quiet and reserved, while an individual high in extraversion might be sociable and outgoing.
2. **Neuroticism/Emotional Stability.** This dimension of Eysenck's trait theory is related to moodiness versus even-temperedness. Neuroticism refers to an individual's tendency to become upset or emotional, while stability refers to the tendency to remain emotionally constant.

3. **Psychoticism.** Later, after studying individuals suffering from mental illness, Eysenck added a personality dimension he called psychoticism to his trait theory. Individuals who are high on this trait tend to have difficulty dealing with reality and may be antisocial, hostile, non-empathetic and manipulative.

The Five-Factor Theory of Personality

Both Cattell's and Eysenck's theory have been the subject of considerable research, which has led some theorists to believe that Cattell focused on too many traits, while Eysenck focused on too few. As a result, a new trait theory often referred to as the "Big Five" theory emerged. This five-factor model of personality represents five core traits that interact to form human personality. While researchers often disagree about the exact labels for each dimension, the following are described most commonly:

- | | | |
|-----------------|------------------|----------------------|
| 1. Extraversion | 2. Agreeableness | 3. Conscientiousness |
| 4. Neuroticism | 5. Openness | |

Assessing the Trait Approach to Personality

While most agree that people can be described based upon their personality traits, theorists continue to debate the number of basic traits that make up human personality. While trait theory has objectivity that some personality theories lack, it also has weaknesses. Some of the most common criticisms of trait theory center on the fact that traits are often poor predictors of behavior. While an individual may score high on assessments of a specific trait, he or she may not always behave that way in every situation. Another problem is that trait theories do not address how or why individual differences in personality develop or emerge.

The Cognitive Approach

This theory is humanistic in nature, because according to this theory man has potential of self-direction and freedom of choice. Therefore, this theory is also known as humanistic theory.

Humanistic theories reject both the psychoanalytic and the behaviouristic conceptions of human nature as too mechanistic, portraying people as creatures helplessly buffeted about by internal instincts or external stimuli. They are less concerned with motivational constructs as explanations of behaviour than with the individual's perception of himself, immediate experiences, and his personal view of the world.

Most humanistic theories stress our positive nature—our push toward growth and self-actualization. Their emphasis is also on the "here and now" rather than on events in early childhood that may have shaped the individual's personality.

The main feature of the humanistic approach to personality will become clearer as we consider the ideas of two of its leading spokesmen, Carl Rogers and Abraham Maslow.

Comparison to Other Approaches

Cognitive psychologists attempt to create rules and explanations of human behavior and eventually generalise them to everyone's behaviour. The Humanist approach opposes this, taking into account individual differences that make us each behave differently. The cognitive approach attempts to apply a scientific approach to human behaviour, which is reductionist in that it doesn't necessarily take into account such differences. How-

ever, popular case studies of individual behaviour such as HM have lead cognitive psychology to take into account ideosyncracies of our behaviour. On the other hand, cognitive psychology acknowledges the thought process that goes into our behaviour, and the different moods that we experience that can impact on the way we respond to circumstances.

Roger's Self Theory

Rogers' theory of personality developed from his experiences with a specific therapeutic method for helping troubled individuals. His "nondirective," or "client-centered," therapy assumes that each person has the motivation and the ability to change; the therapist's task is simply to facilitate progress toward this change. In psychoanalysis therapy the therapist "analysed" the patient's history to arrive at the problem. The assumption of nondirective therapy is that we are the best experts on ourselves. The patient, not the therapist, knows the factors that shape his personality and the best procedures for modifying them. The therapist's role is to act as a sounding board while the patient explores and analyses problems.

Self Concept

The self consists of all the ideas, perceptions, and values that characterize "I" or "me"; it includes the awareness of "what I am" and "what I can do." This perceived self (the self-concept) in turn influences both the person's perception of the world and his behaviour. An individual with a strong, positive self-concept views the world quite differently from one whose self-concept is weak. The self-concept does not necessarily reflect reality; a person may be highly successful and respected yet view himself as a failure.

There is another self in Rogers' theory, and that is the ideal self. We all have an ideal self, a conception of the kind of person we would like to be. This concept is similar to Freud's ego-ideal. The closer the ideal self is to the real self, the more fulfilled and happy the individual. A large discrepancy between the two results in an unhappy, dissatisfied individual.

Thus, two kinds of incongruence can develop: one between the self and the experiences of reality, and the other between the self and ideal self.

Development of the Self

A child learns to discriminate between those thought and actions that are considered worthy and those that are not, because his behaviour is continuously being evaluated by parents and others.

The unworthy experiences become excluded from the self-concept, even though they may be quite valid or natural experiences. For example, relieving physiological tension in the bowel or bladder is experienced by the child as pleasurable. However, unless he urinates or defecates privately and in the proper place, parents usually condemn such activities as "bad" or "naughty." The child must then, because he needs to retain his parents' positive regard, deny his own experience. He must deny to himself that he receives satisfaction from defecating or urinating.

Sometimes this learning may be negative as a girl child feels herself neglected and inferior. Even a boy learns to beat his sister without any guilt feeling.

There must be certain restrictions on behaviour. Considerations of household efficiency and sanitation require some restraints on elimination. And children cannot be permitted to beat their siblings. Rogers suggests that the best approach is for the parents to recognize the child's feelings as valid, while explaining their feelings and the reasons for restraint.

Self-Actualization

According to Roger self actualization is "a tendency toward fulfilment, toward actualization, toward the maintenance and enhancement of the organism." As the organism grows, it seeks to fulfil its potential within the limits of its heredity.

The innate tendency toward self-actualization often runs into conflict with two learned needs; the need for positive regard and the need for self-regard. Rogers assumes that the need for positive regard is universal; we all want to be accepted and loved by others. Sometimes the regard we receive from others is unconditional: "I love you no matter how badly you may behave at times." But more often it is conditional, that is, it depends on specific behaviour.

The need for self-regard develops later; it is essentially the internalization of those actions and values that others approve. In this sense it is similar to Freud's superego.

The need for self-actualization directs the individual to seek or to avoid activities depending upon whether the experience is seen as one that will enhance the person. The need for positive regard directs behaviour according to whether the resulting experience will elicit favourable reactions from others – from those important in the individual's life. Obviously, there are situations where the two needs conflict.

Ideally, the more completely the individual is given positive regard – acceptance that is not conditional to specific behaviours – the more congruence there will be between his self-concept and his actual experiences, as well as between his self-concept and ideal self.

Assessment of Humanistic Personalities

Some psychologists at the time disliked psychodynamic and behaviourist explanations of personality. They felt that these theories ignored the qualities that make humans unique among animals, such as striving for self-determination and self-realization. In the 1950s, some of these psychologists began a school of psychology called **humanism**.

Humanistic psychologists try to see people's lives as those people would see them. They tend to have an optimistic perspective on human nature. They focus on the ability of human beings to think consciously and rationally, to control their biological urges, and to achieve their full potential. In the humanistic view, people are responsible for their lives and actions and have the freedom and will to change their attitudes and behavior.

Two psychologists, Abraham Maslow and Carl Rogers, became well known for their humanistic theories.

Rotter's Locus of Control

Locus of Control as a principle was originated by Julian Rotter in 1954. It considers the tendency of people to believe that control resides internally within them, or externally, with others or the situation.

Note that, like other preferences, this is a spectrum. Some people have a wholly internal or external locus of control, but many will have some balance both views, perhaps varying with situation. For example some may be more internal at home but more external at work.

Internal

People with a high internal locus of control believe in their own ability to control themselves and influence the world around them. They see their future as being in their own hands and that their own choices lead to success or failure.

Rotter (1990) describes the internal locus of control as:

'the degree to which persons expect that a reinforcement or an outcome of their behavior is contingent on their own behavior or personal characteristics'

Their belief in their ability to change things may well make them more confident and they will hence seek information that will help them influence people and situations. They will also likely be more motivated and success-oriented. These beliefs may even lead them to be more politically active.

They are more likely to have expectancy shifts, where a sequence of similar events are expected to have different outcomes. They tend to be more specific, generalizing less and considering each situation as unique. People in middle age tend to have the highest internal locus of control.

A downside of an internal locus of control is that, in accepting responsibility, the person has to also accept blame for failures.

External

People with a high external locus of control believe that control over events and what other people do is outside them, and that they personally have little or no control over such things. They may even believe that others have control over them and that they can do nothing but obey.

Rotter (1990) describes the external locus of control as:

'the degree to which persons expect that the reinforcement or outcome is a function of chance, luck, or fate, is under the control of powerful others, or is simply unpredictable.'

With such beliefs, people with an external locus of control tend to be fatalistic, seeing things as happening to them and that there is little they can do about it. This tends to make them more passive and accepting. When they succeed, they are more likely to attribute this to luck than their own efforts.

They are less likely to have expectancy shifts, seeing similar events as likely to have similar outcomes. They hence step back from events, assuming they cannot make a difference. Younger and older people tend to have higher external locus of control than people in middle age.

Seligman's Explanatory styles

Within positive psychology, optimism has been conceptualised in several ways. This discussion will focus on the 'explanatory style' view (Buchanan & Seligman, 1995) of optimism and will show that whilst the theory has been well researched – with valid, reliable, flexible and complimentary measurements – its application in, for example, education

has led to interventions primarily concerned with the treatment or prevention of depression rather than an extensive promotion of optimism for its own sake.

Given that the aim of positive psychology is to address the imbalance between the focus of psychology on dysfunction and optimal functioning, it is clear that the potential contribution of explanatory style of optimism has not yet been fully realised within positive psychology, particularly in schools and early years programmes.

This essay will first critically outline optimism as conceptualised by explanatory style in terms of its historical development, methods of measurement, evolutionary significance, origins, some key research findings and focus of interventions within education. It will then consider the implications for further research to heighten its contribution within positive psychology of education for children and young adults.

Explanatory Style and Dispositional Optimism

It is perhaps useful at this point to distinguish explanatory style from dispositional optimism, which sees optimism as a broad personality trait characterised by a global expectation that more good things will happen than bad.

In contrast, the explanatory style view of optimism taps into the immediate reactive optimistic/pessimistic tendencies, which explain events and contribute to a person's general optimistic/pessimistic coping response going forward.

The Learned Helpless Model

The learned helpless model which predated, and is said to have perhaps pre-empted the empirical birth of positive psychology, found that when unable to respond in a controllable way to electric shocks, rats became helpless.

When the rats learnt that response-outcome was independent they became passive because there was no point in them doing anything. This model, however, did not account for individual differences when it was applied to people and did not consider the possibility of learned optimism.

The question arose as to why, in situations where there is no control over the outcome, some people give up more easily and succumb to depression while others do not. In the reformulation of the learned helplessness model Abramson, Seligman, & Teasdale (1978) applied the attribution theory, which described the pattern of causes people attribute to good or bad events in their lives.

This reformulated cognitive-behavioural approach believes that it is often the person's habitual way of looking at experience – a person's explanatory style – which determines how they explain an event they have experienced and influences their cognitive (optimistic/pessimistic) expectation of future events.

The Learned Helpless Model

Explanatory style consists of three dimensions: internal/external; stable/unstable and global/specific.

1. Internal/external refers to whether or not a person believes that they have control or influence over events.
2. The stable/unstable dimension represents whether a person believes a repeated event will be the same or subject to change.

3. Global versus specific refers to whether or not a person's explanation generalises the event to others beyond the specific event in hand.

For example the attributional style model believes that if a person has an explanatory style that tends to consider bad events as internal, stable and global they are said to have a pessimistic explanatory style because they see the bad experience or event as one that was their fault, will not change in the future and is generally problematic rather than specific to that particular event.

This contrasts with a person with an optimistic explanatory style who is said to explain the causes of negative events as external, unstable and specific. The optimistic person places no blame on themselves, believes there is room for change and that the bad experience was specific to that particular event and is not to be generalised to all others.

If a pleasing event is experienced an optimist would exhibit an internal, stable and global explanatory style, whereas a pessimist would be showing external, unstable and specific explanatory style. One's explanatory style is believed to influence a person's view of the future and as a result their projected perceptions and subsequent behaviour.

Historical Origins of Explanatory Style

It is perhaps the theory's historical origins in traditional psychology which has meant that much of the research into explanatory style has focussed on preventing depression rather than the development of optimistic explanatory style from a very early age. I believe that this model has huge potential for application in education and schools.

It could provide the key to cognitions that lead to optimism and the benefits that this tendency potentially promotes such as academic achievement and self-actualisation – and many of the outcomes for the current Every Child Matters agenda. A particular strength of the model is the methods of measurement.

Kohlberg's Theory of Moral Development

Kohlberg identified three levels of moral reasoning: pre-conventional, conventional, and post-conventional. Each level is associated with increasingly complex stages of moral development.

Level 1: Preconventional

Throughout the preconventional level, a child's sense of morality is externally controlled. Children accept and believe the rules of authority figures, such as parents and teachers. A child with pre-conventional morality has not yet adopted or internalized society's conventions regarding what is right or wrong, but instead focuses largely on external consequences that certain actions may bring.

Stage 1: Obedience-and-Punishment Orientation

Stage 1 focuses on the child's desire to obey rules and avoid being punished. For example, an action is perceived as morally wrong because the perpetrator is punished; the worse the punishment for the act is, the more "bad" the act is perceived to be.

Stage 2: Instrumental Orientation

Stage 2 expresses the "what's in it for me?" position, in which right behavior is defined by whatever the individual believes to be in their best interest. Stage two reasoning shows a limited interest in the needs of others, only to the point where it might further the

individual's own interests. As a result, concern for others is not based on loyalty or intrinsic respect, but rather a "you scratch my back, and I'll scratch yours" mentality. An example would be when a child is asked by his parents to do a chore. The child asks "what's in it for me?" and the parents offer the child an incentive by giving him an allowance.

Level 2: Conventional

Throughout the conventional level, a child's sense of morality is tied to personal and societal relationships. Children continue to accept the rules of authority figures, but this is now due to their belief that this is necessary to ensure positive relationships and societal order. Adherence to rules and conventions is somewhat rigid during these stages, and a rule's appropriateness or fairness is seldom questioned.

Stage 3: Good Boy, Nice Girl Orientation

In stage 3, children want the approval of others and act in ways to avoid disapproval. Emphasis is placed on good behavior and people being "nice" to others.

Stage 4: Law-and-Order Orientation

In stage 4, the child blindly accepts rules and convention because of their importance in maintaining a functioning society. Rules are seen as being the same for everyone, and obeying rules by doing what one is "supposed" to do is seen as valuable and important. Moral reasoning in stage four is beyond the need for individual approval exhibited in stage three. If one person violates a law, perhaps everyone would—thus there is an obligation and a duty to uphold laws and rules. Most active members of society remain at stage four, where morality is still predominantly dictated by an outside force.

Level 3: Postconventional

Throughout the postconventional level, a person's sense of morality is defined in terms of more abstract principles and values. People now believe that some laws are unjust and should be changed or eliminated. This level is marked by a growing realization that individuals are separate entities from society and that individuals may disobey rules inconsistent with their own principles. Post-conventional moralists live by their own ethical principles – principles that typically include such basic human rights as life, liberty, and justice – and view rules as useful but changeable mechanisms, rather than absolute dictates that must be obeyed without question. Because post-conventional individuals elevate their own moral evaluation of a situation over social conventions, their behavior, especially at stage six, can sometimes be confused with that of those at the pre-conventional level. Some theorists have speculated that many people may never reach this level of abstract moral reasoning.

Stage 5: Social-Contract Orientation

In stage 5, the world is viewed as holding different opinions, rights, and values. Such perspectives should be mutually respected as unique to each person or community. Laws are regarded as social contracts rather than rigid edicts. Those that do not promote the general welfare should be changed when necessary to meet the greatest good for the greatest number of people. This is achieved through majority decision and inevitable compromise. Democratic government is theoretically based on stage five reasoning.

Stage 6: Universal-Ethical-Principal Orientation

In stage 6, moral reasoning is based on abstract reasoning using universal ethical principles. Generally, the chosen principles are abstract rather than concrete and focus on ideas such as equality, dignity, or respect. Laws are valid only insofar as they are grounded in justice, and a commitment to justice carries with it an obligation to disobey unjust laws. People choose the ethical principles they want to follow, and if they violate those principles, they feel guilty. In this way, the individual acts because it is morally right to do so (and not because he or she wants to avoid punishment), it is in their best interest, it is expected, it is legal, or it is previously agreed upon. Although Kohlberg insisted that stage six exists, he found it difficult to identify individuals who consistently operated at that level.

Critiques of Kohlberg's Theory

Kohlberg has been criticized for his assertion that women seem to be deficient in their moral reasoning abilities when compared to men. Carol Gilligan (1982), a research assistant of Kohlberg, criticized her former mentor's theory because it was based so narrowly on research using white, upper-class men and boys. She argued that women are not deficient in their moral reasoning and instead proposed that males and females reason differently: girls and women focus more on staying connected and maintaining interpersonal relationships.

Kohlberg's theory has been criticized for emphasizing justice to the exclusion of other values, with the result that it may not adequately address the arguments of those who value other moral aspects of actions. Similarly, critics argue that Kohlberg's stages are culturally biased – that the highest stages in particular reflect a westernized ideal of justice based on individualistic thought. This is biased against those that live in non-Western societies that place less emphasis on individualism.

Another criticism of Kohlberg's theory is that people frequently demonstrate significant inconsistency in their moral judgements. This often occurs in moral dilemmas involving drinking and driving or business situations where participants have been shown to reason at a lower developmental stage, typically using more self-interest driven reasoning (i.e., stage two) than authority and social order obedience driven reasoning (i.e., stage four). Critics argue that Kohlberg's theory cannot account for such inconsistencies.

BASIC MOTIVATIONAL CONCEPTS

Rationalistic concept of the human being was the predominant view of philosophers and theologians for hundreds of years in beginning. Reason was supposed to be the main determinant of what a person did. A person was free to choose, and choices were good or bad depending upon one's intelligence and education. It was assumed that the good choice, if known, would automatically be selected. Within this viewpoint a person is very much responsible for his own behaviour. If people are viewed as rational beings whose intellects are free to choose goals and decide on courses of action, then a concept of motivation is unnecessary. Against this view, the philosophers of seventeenth and eighteenth century (such as Hobbes, Locke and Hume) suggested a more mechanistic view, according to which some actions arise from internal or external forces over which we have no control. Hobbes, for example, held that we behave in such a way as to achieve pleasure and avoid pain. No matter what reasons we may give for our conduct, these two tenden-

cies are the underlying causes of all behaviour. This doctrine of hedonism is still very influential in some motivation theories.

Instincts

An innate biological force that predisposes the organism to act in a certain way, is known as instinct. The theory of instinct is extreme of the mechanistic view. Since animals had no soul or intellect and could not operate on the basis of reason, their behaviour had long been attributed to instincts.

Darwin suggested that there was no sharp distinction between humans and animals, the door was opened for the use of instincts in explaining human behaviour. The strongest advocate of instinct theory was the psychologist William McDougall who maintained that all of our thoughts and behaviour were the result of instincts. In his book *Social Psychology*, published in 1908, McDougall mentioned the following instincts:

flight	self-assertion
repulsion	reproduction
curiosity	gregariousness
pugnacity	acquisition
self-abasement	construction

McDougall thought these instincts were inherited and compelling source of conduct, but modifiable by learning and experience. He later expanded his list to 18 instincts, including some that related to specific bodily needs. By modifying and combining these instincts he attempted to explain all human behaviour.

Instinct theory is diametrically opposed to a rationalistic view of human beings. Instead of choosing goals and actions a person is at the mercy of innate forces which determine or motivate his course of action. Psychoanalytic theory also attributed behaviour to powerful innate forces. Freud believed that behaviour was determined by two basic energies; the life instincts which found expression in sexual behaviour and the death instincts which underlie aggressive acts. These instincts, though unconscious, were powerful motivational forces. Both psychoanalytic theory and instinct theory were influential in turning interest away from a rationalistic conception of people toward a motivational view that saw behaviour as the result of unconscious, irrational forces within the individual.

Needs and Drives

It is found that most of the instincts listed by psychologists is only an imaginable behaviour and not a common feature belonging to human class. To say that a man fought because he had an instinct of pugnacity does not offer much more than a description of his behaviour. Anthropologists noted that many instincts were not found in all cultures, Pugnacity, for example was not typical of all primitive societies. There were peaceful societies in which men found no need to fight.

Because of these and other difficulties, the notion of instincts was replaced during the 1920s by the concept of drives. A drive is an aroused state resulting from some bodily or tissue need. Such as a need for food, water, oxygen, or avoidance of painful stimuli. This aroused condition motivates the organism to initiate behaviour to remedy the need. For example, a lack of food produces certain chemical changes in the blood, indicating a need

for food, which in turn create a drive state of arousal or tension. The organism seeks to reduce this drive by doing something (in this case, finding food) to satisfy the need.

Sometimes the terms need and drive are used interchangeably, but more often need refers to the physiological state of tissue deprivation, whereas drive refers to the psychological consequences of a need. Need and drive are parallel but not identical. Drive does not necessarily get stronger as need gets stronger. A starved organism may be so weakened by its great need for food that drive (the motivation to get it) is weakened. People who have fasted for long periods report that their feelings of hunger (drive level) come and go even though their need for food persists.

Like instinct, drive is a hypothetical construct. It is not something we can observe but is inferred from behaviour. If we deprive an animal of food for a specified period of time, we note that it runs more rapidly down a runway to obtain food than a non-deprived animal and eats more than its counterpart. We infer from this behaviour that the animal's hunger drive has been activated. Although drive is a hypothetical construct, it proved to be more amenable to experimental investigation than instinct. Needs could be defined objectively and conditions could be specified for creating and eliminating them.

Homeostasis and Drive Theory

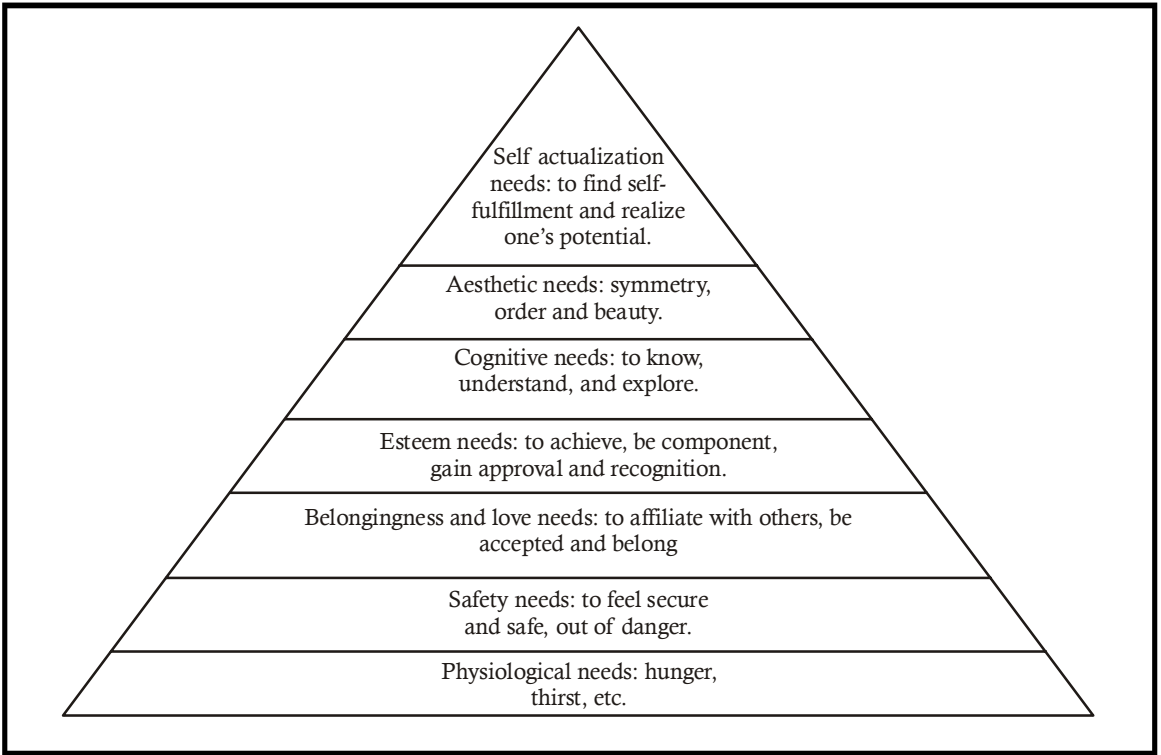
Homeostasis, which refers to the body's tendency to maintain a constant internal environment, is a principle basic to the drive concept. Numerous physiological states must be maintained within fairly narrow limits: the concentration of blood sugar, the levels of oxygen and carbon dioxide in the blood, water balance in the cells, to mention but a few. Various mechanisms within the body operate to keep these conditions stable. Presumably, sensors in the body detect changes from the optimal level and activate mechanisms designed to correct the imbalance. The principle is the same as a furnace thermostat, which turns the heat on when the temperature falls below a certain level and off when the temperature rises.

Hunger and thirst can be viewed as homeostatic mechanisms because they initiate behaviour designed to restore the balance of certain substances in the blood. Within the framework of homeostasis, a need is any physiological imbalance or departure from the optimal state; its psychological counterpart is drive. When the physiological imbalance is restored, drive is reduced and motivated activity ceases. Many physiological imbalances can be corrected automatically. The pancreas releases sugar stored in the liver to maintain the proper balance of sugar in the blood. But when automatic mechanisms can no longer maintain a balanced state, the organism becomes aroused (drive is activated) and is motivated to restore the balance. A person goes in search of food on feeling the symptoms of low blood-sugar level.

The principle of homeostasis have been extended to cover more than just tissue needs or physiological balances. This broader sense of this principle contains the theory that any physiological or psychological imbalance will motivate behaviour designed to restore equilibrium. A hungry, anxious, uncomfortable, or fearful person will be motivated to do something to reduce the tension. As we shall see later, there has been criticism of this view that motivation is solely tension arousal and ceases with tension reduction.

Maslow's Hierarchy of Needs

The needs that are low in the hierarchy must be at least partially satisfied before those that are higher can become important sources of motivation.



MASLOW'S HIERARCHY OF NEEDS.

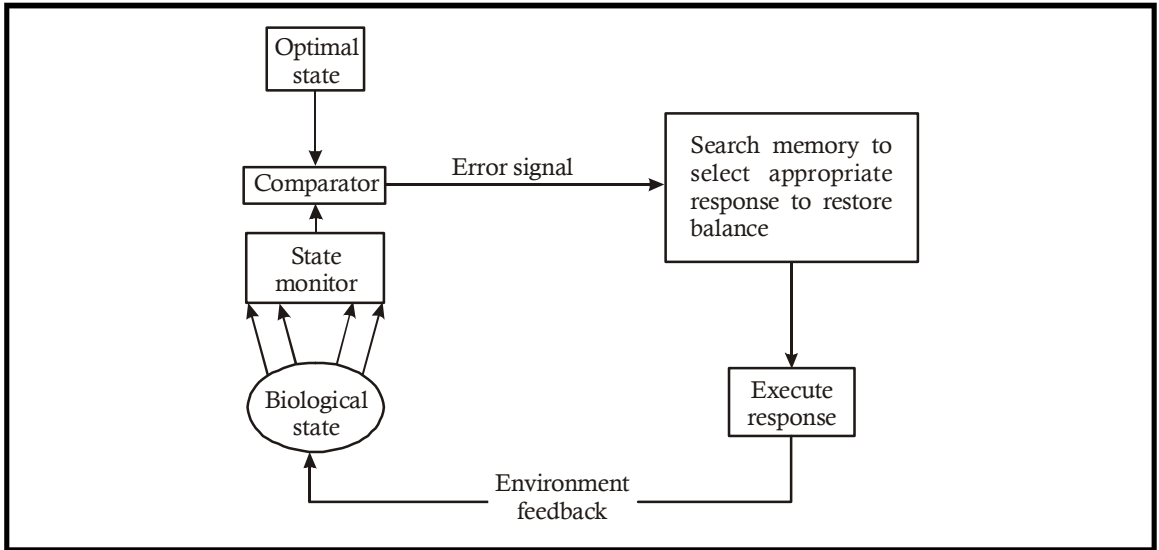
Incentive Theory

During the 1950s, external stimuli were being overlooked as instigators of behaviour. During this period, psychologists began to be disenchanted with the drive-reduction theory of motivation as an explanation of all types of behaviour.

The organism was not solely pushed into activity by internal drives. Certain objects in the environment (called incentives) were important in arousing behaviour. Motivation could be better understood as an interaction between stimulus objects in the environment and a particular physiological state of the organism.

It became clear, too, that the homeostatic concept of drive as reducing tension and returning the organism to a quiescent state did not agree with all the evidence. Human beings often seek tension-arousing experiences such as roller-coaster rides, car racing, white-water canoeing, or horror movies. These activities certainly increase tension rather than reduce it.

More recent approaches to a theory of motivation have focused on the role of incentives. Incentives refer to objects or conditions in the environment. The organism will approach positive incentives; it will avoid negative incentives. For a thirsty animal water would be a positive incentive; for a sexually aroused animal, a mate. An object or situation that has caused pain would be a negative incentive. Incentives have two functions – they arouse the organism and they direct behaviour either toward or away from themselves.



MOTIVATIONAL CONTROL SYSTEM.

The dual function of incentives can be demonstrated experimentally. A hungry rat will run through a maze to a goal box that it knows contains food; the positive incentive directs behaviour. If the rat is given a morsel of food at the start of the maze, it will run even faster toward the goal box; the incentive also serves to arouse behaviour. As with drive theory, arousal is an important aspect of motivation, but here arousal is evoked by an external incentive rather than by conditions of deprivation.

Motivational Cycle

Most of our motives have a cyclical nature- they are aroused, they trigger behaviour, which leads to a goal and finally, after the goal is reached, they are shut off. The behaviour that is related to drive state is called “Instrumental behaviour” because it is instrumental in bringing about the goal or the satisfaction of need.

Stages of Motivational Cycle

The first stage of the motivational cycle is the Driving State, simply called as drive. This term is often used when the motive state has a biological or physiological basis. It is regarded as impelling a person or animal to action.

Drives can originate when an organism lacks something. Drives can also be aroused by environmental stimuli, such as advertisements from newspaper magazines and television etc.

For example, when someone sees a sexy picture in a magazine, his sex drive is aroused. Besides all these factors, some cognitive factors like thoughts and memory can also trigger driving states. Very often term “need” is used to refer to the driving state. Drives can originate when an organism lacks something in its needs. This is one meaning of the word need. The need creates a drive state in the organism. It acts as if it pushes the organism to behave in a certain fashion to satisfy its need.

The second stage of the motivational cycle is the instrumental behaviour. It is instigated by the driving state. Thirst, for example, drives an animal to explore water. Sooner or later this behaviour succeeds, thereby reducing the need and the drive state, in other words, the animal's exploratory behaviour is instrumental in finding water and reducing the thirst drive.

When instrumental behaviour achieves its goal, the need is satisfied there follows a temporary period of relief. This is the third stage of motivational cycle. For example, in thirst, lack of water in the body produce need and a drive state (first stage); the organism has shown some explore behaviour (second stage) and finally he quenches his thirst by drinking water *i.e.*, reaching the goal (third stage).

When the thirst is satisfied, motivational cycle is over. But when the need for water will build up again, the organism will once more go through motivational cycle.

Goals may be positive or negative. Food, sexual satisfaction and championship are some of the examples of positive goals. These are the goals, which the individual approaches or tries to reach.

On the other hand, negative goals are those, which the organism tries to escape from or avoid. Dangerous, unpleasant and embarrassing situations are some of the examples of negative goals. However, goals depend upon the drives or needs, which are active.

APPROACHES TO THE STUDY OF MOTIVATION

There are two main approaches to the study of motivation. According to Freud's psychoanalytic theory our actions are determined by inner forces and impulses, often operating below the level of consciousness. In contrast, social learning theory, as formulated by Albert Bandura at Stanford University, sees our behaviour as learned through interaction with, and observation of the environment.

The Psychoanalytic Theory of Motivation

One of the paradigms for the present study of human motivation is Psychoanalytic theory. Before going specifically into the details of the motivational theory, we shall attempt a survey of Psychoanalytic theory.

Psychoanalysis was one of the most influential theories of 20th century, but it was no less controversial. Both chance and self-determinism played their role in the development of the theory. Chance, because Sigmund Freud, the father of psychoanalysis, had been dragged into the field of psychiatry for personal reasons. Determinism because, Freud's obsessive nature and "penchant for rejecting people who did not totally agree with him" largely determined the history of psychoanalysis (Boeree). Freud owed influences both to his contemporaries in the field of psychiatry and to the scientific tradition of his time.

Freud was born on the 6th of May 1856 in Freiberg in a small town in Moravia- a part of Austria- Hungary. Hailing from a middleclass Jewish family, he graduated in medicine and started his research career involving the dissection of nervous system of eels. Following this, he worked for the next six years in the Psychological Laboratory under Brucke, from where he acquired the outlines of physical sciences in general. Here he worked on the anatomy of central nervous system. Unable to support his family with the earnings from laboratory studies, in 1882 he left Brucke and began to work at Vienna General

Hospital. In 1885 he started work at Salpêtrière, the famous Parris Hospital for nervous diseases, with Charcot, the then reigning figure in psychiatry. Freud's career with Charcot influenced much of his later theories. In 1886 Freud returned to Vienna and got married. He set up a private clinic there to practise as a consultant for nervous diseases.

Charles Darwin in *On the Origin of Species* rejected the special divine status of man, and henceforth mind became a legitimate subject for scientific investigation. The concept of psychic energy governed much of Freud's theories. The channelling out, displacement and sublimation of the energy played a key role in 'defence mechanisms' and neurotic behaviour in Freudian psychoanalysis. These concepts were much influenced by the then emerged notion of 'law of conservation of energy', according to which energy can neither be created nor be destroyed. Freud's application of these principles into his concept of psychic energy led him to the postulation of sublimation and displacement of that energy.

Freud's ideas of the human mind are shot through with this metaphor – whether blocked instinctual drives or repressed memories, he believed our greatest destructive and creative achievements stemmed from forces denied their natural release.

Freud was a thoroughgoing determinist. Being influenced by the nature of physical sciences, the then predominant trend in psychology was concerned with conscious contents of mind. Freud deviated much from this standpoint and postulated the role of unconscious mental activities in interconnecting the isolated conscious psychic activities.

The Freudian psychoanalysis was developed out of the observation and the analyses of data derived from his clinical practice. However, there are three major practitioners who influenced Freud's theories. They are Anton Mesmer (1734-1815), Jean Charcot (1825-1893) and Joseph Breuer (1842-1925).

Franz Anton Mesmer, a distinguished German neurologist, was the one who suggested the influence of gravitational force on animal spirit. The notion of gravitational force was later replaced with the magnetism and his theory is known as animal magnetism. He treated his patients by putting them into trance using magnetised bar over them. The method known as mesmerism is essentially one of suggestion. Later, mesmerism came to be known as hypnotism.

Jean Charcot, who is considered as the father of modern neurology, was more concerned with the neurology of motor disorders, resulting diseases, aneurysms and localisation of brain functions. He used hypnosis in tracing women with hysteria. He associated hysteria with weakness in the nervous system and found that hypnosis brings the patients in a state similar to that of hysteria. Charcot however "was only interested studying hysteria, not in curing it" (Boeree).

Joseph Breuer experimented with his patients using hypnosis in a slightly different way. During the hypnotic trance, Breuer asked the patients about their problems and conflicts, and discussed them in depth. The result was a marked relief from the symptoms and the development of insight into their problems. Freud worked in association with Breuer and they together proposed the influence of some unconscious elements in the overt behaviour of human beings. This was one of the important developments in the history of psychopathology.

Breuer and Freud also found that "it is therapeutic to recall and relieve emotional trauma that has been made unconscious (in a process called repression) and release some of the emotional material became known as catharsis". The understanding of the

inner material causing the obvert symptoms was referred to as insight. Breuer's systematic treatment of hysterical symptoms of Anna O provided scientific considerations about the effect of catharsis in treating neurotic illness.

The method of using catharsis during hypnotic trance was replaced by Freud by the method of free-association. In free association, "the patient repeats all ideas and impulses regardless of their significance". This is regarded as the therapist's gateway to the patients' unconscious mind. The method of treatment consisted in patient's gaining insight into his/her own repressed memories and conflicts.

An Outline of the Psychoanalytic Theory

The term psychoanalysis has three distinct meanings. Firstly it is a school of psychology, which emphasises psychic determinism and dynamics. As a school of psychology it also emphasises the importance of childhood experiences in moulding one's adult personality and behaviour. Secondly, psychoanalysis, with its emphasis on the role of unconscious in determining human behaviour, is a specialised method for investigating the unconscious mental activities. Finally, psychoanalysis is a therapeutic method for the investigation and treatment of mental disorders, especially the neurotic disorders.

Our concern here is the theoretical one, taking psychoanalysis as a study of psychic determinants of human behaviour. Psychoanalysis as a psychological theory has undergone constant revisions both during Freud's time and afterwards. Psychoanalysis is considered to be a High-Level theory containing various sub theories such as 'levels of consciousness', 'psychic structure of personality', 'psychosexual development', 'defence mechanisms' and 'theory of instincts', and it serves to unify them to some extent.

It is to be noted that much of psychoanalytic theory was derived from Freud's clinical experience in treating neurotic patients. So the theory focused more on the origin of abnormal behaviour in the formulation of the concept of mind. It is also to be considered that the distinction between sanity and insanity is not that of type but that of degree. The defence mechanisms that serve to compensate for the frustrations of the sane human, in exaggeration lead to the abnormality, of both neurotic and psychotic. Hence it can be seen that clinical data may not be insufficient for the formulation of a theory of mind.

The Theory of Instincts

Freud recognized two fundamental motivating forces. The first one is the constructive one called the Eros or life urges and the other is the destructive one called the Thanatos or the death urges (Freud, BPP). Eros finds its output through drives known as Self-preservation drive or ego drive and sex drive which play a significant role in Freudian psychoanalysis.

The sex drive in the organism is based on a special form of energy called libido. This has its seat in that part of personality called id, which is essentially unconscious. The libido or sex energy is best released through amorous activities. However, "society can see no more menace to its culture than would arise from the liberation of sexual impulses and a return to their original goal".

The motivating principle in life is the pleasure principle, which according to psychoanalysis, is the tendency to avoid pain and to seek pleasure. This dominates in sex drive. As one attains maturity, this is supplemented by the reality principle, which is the voice of reason that aims at rational acts to avoid future pains. The Nirvana principle

is expressed in death drive, which is aimed at the final return of living matters to the inorganic state.

Levels of Consciousness

Deviating from the view held during his time, Freud introduced the threefold division of mind into conscious mind, pre-conscious mind and unconscious mind. Of these three divisions of mind, it is the conscious mind of which we generally are aware. The conscious mind is constituted by events, memories, fantasies and the sensations from sense organs along with the feelings emotions and the like, of which one is aware at the moment. In the next moment when one is not aware of these mental images they become latent. Some of the latent memories can come back to conscious mind again to form the content of the moment and thus it becomes conscious. The activity of conscious mind is not governed by its constituents alone. The unconscious mental processes, much different from those observed in conscious mental states, act behind the latter. These two the unconscious and the conscious- functionally differentiate into two levels of mind, the conscious mind and the unconscious mind. Between these two levels is the third level called the pre-conscious mind.

The pre-conscious mind consists of the past psychic experiences and desires which are readily recallable and is the storehouse for conscious mind.

The unconscious proper – excluding the preconscious from the whole realm of unconscious –consists of the buried memories, thoughts, emotions and impulses for which conscious mind has no direct voluntary access. Neither the social or moral rules nor the categories of space and time have any relevance in this psychic division. While conscious and preconscious mind are characterised by internal consistency, temporal arrangement and adaptability to outer world, the unconscious ‘is timeless, chaotic, infantile and primitive’.

The unconscious has two sources for its contents -of inheritance and of experience. The inherited contents are primitive, pleasure dominated, and brutal instincts. The contents formed out of individual’s experience are repressed in nature. These inherited instinctual drives and repressed memories have no direct access to conscious mind. However, they exert a profound influence on the conscious mental processes.

Freud, basically being a neurologist always “believed that physical treatments (medicine) would eventually be the cure for mental illness” (Burns 44). This is because he thoroughly believed in the neurological foundation of mind and the possibility of reducing psychology to neurology. Freud belongs to the materialistic tradition in the history of psychology (Stokes 139).

The Freudian concept of conscious mind is analogous to the modern notion of qualia or subjectivity of conscious experience. His notion of pre-conscious mind and unconscious mind may be compared with those neural processes that never give rise to consciousness. In this way it can be seen that Freud’s triple division of conscious, preconscious and unconscious mind are not ontological entities, but only the functional levels of consciousness. It is significant here to note that Freud never mentioned the notion of self-consciousness in his description of the levels of consciousness.

The Psychic Structure of Personality

Freud distinguished three parts or functional principles within the mind; the id, the ego and the superego. The dynamics of interplay between these functions determines one’s

actions and overt behaviours. These principles of psychic dynamism are central to the interpretation of abnormal behaviour in psychoanalysis.

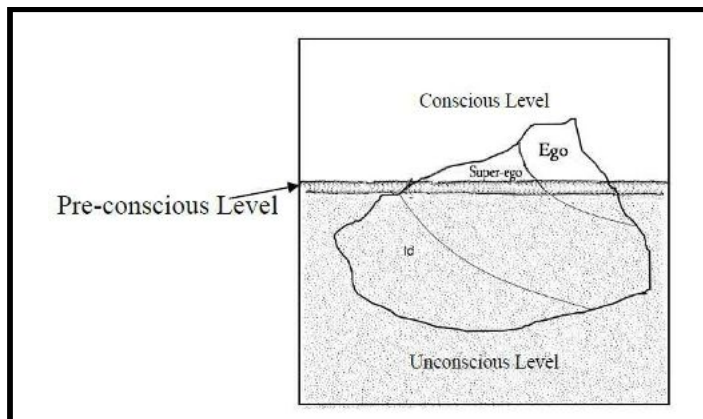
Id is the prime-mover; it is the source of all psychic energy. The psychic energy or drive within it called the libido, if left totally unchecked, will lead to amorous activities and if fully checked will lead to damming up of libido. Both are dangerous, to the society and to the individual respectively.

Id is governed by the pleasure principle, with an overriding goal maximising pleasure and eliminating tensions and conflicts associated with achieving pleasure. The characterising way of processing information or thinking by id is referred to by Freud as primary process. The primary process is primarily emotional, irrational, illogical “and filled with fantasies and preoccupations of sex, aggression, selfishness and envy”. This dark and inaccessible part of personality functionally belongs to the unconscious mind.

The instincts or drives are dominated by the pleasure principle (the chief characteristic of psychic energy) and are inherited. And id is the part of personality from where the fundamental instincts seek their first outlet.

Fortunately, according to Freud, the dangerous and selfish drive does not go unchecked. The instinct of id, which is primitive and somewhat brutal, seeking its direct expression in activities, is faced with dangers from external world. So a portion of id- that has been expediently modified by the proximity of the external world with its threats of dangers- called ego, modifies the instinctual needs of id to match for the needs of society. The ego forms the second part of personality, which is partially conscious and partially unconscious.

The ego is governed by reality principle instead of pleasure principle that governs the id. The information-processing or thinking of ego is characterised by logic and reason and is referred to as the secondary process. Ego is drawing power from the id while controlling it as a rider on a horse. The horse in this metaphor is id; the primitive and animal like source of energy. The rider is the ego which may be weak or strong, clumsy or skilful. The rider can direct the energy (if skilfully and well-controlled) towards positive aims.



THE LEVELS OF CONSCIOUSNESS AND THE STRUCTURE OF PERSONALITY (MODIFIED FROM BOEREE).

The third part of personality called the super-ego is the moral censor, which is identified with the voice of conscience. The superego is partially unconscious and partially conscious. So, the man in psychoanalysis is a primitive being, driven by pleasure domi-

nating principles, being exposed to external world. Ego meets the demands of id by channelizing it to the activities acceptable to external world that are being censored by the super-ego. About the formation of superego.

The role of ego is to mediate the conflict between the id and the superego. Ego must find some outlet for the instincts of id and at the same time it has to restrict them within the demands of superego. According to Freud, ego is often caught between the id and the superego and also it has to compensate for the demands of external world.

The relationship between the structure of personality and the levels of consciousness is compared by Freud himself to an iceberg floating on water. Only one third of the iceberg is visible and the visible part (conscious mind) is largely controlled by the invisible (unconscious mind).

Anxiety and Defence Mechanisms

The immoral, anti social, brutal and primitive instincts of the id are rarely given a direct outlet to the external world. Ego regulates and transforms them to meet the demands of the external world and the super ego. As noted earlier, the aim of ego is to harmonize the demands of the three tyrannical masters it serves. In the process of harmonizing the needs of id, superego and the external world, ego often becomes the battleground of conscious and unconscious conflicts.

We have noted earlier that ego essentially is the part of id that is functionally modified to meet the requirements of external world. Any split in this constitution shown by the inability of ego to satisfy the demands of the id, indicates the weakness of the ego. On the other hand, if the ego is to satisfy id's demands then it shows the strength of the ego.

Anxiety. In the earlier analogy of the horse and its rider, the rider's privilege to guide the powerful animal to meet his aim is not always warranted; at times the horse takes the route at its will. In doing so, the weakened ego falls into the state of anxiety; realistic, moral or neurotic.

Ego need not always to fall into neurotic anxiety. The instinctual needs of the id may be given indirect outlet in satisfying the id. The Ego achieves this with various dynamic processes. These processes include defence mechanisms such as fantasy, identification, sublimation and displacement, and other mental activities such as dreams. In many of these processes the thirst of id is satisfied through symbolized acts. The symbolization enables the ego to meet the needs of id and repressed unconscious contents without committing the individual to the dangers from external world, which would have been imposed on him/ her if a direct outlet to these instincts was given.

Defence Mechanisms. Defence mechanisms are unconscious activities; they are not directly known to the person. They are individualistic too; different individuals use different sets of them. They are considered to be normal, and only if exaggerated beyond limit they give rise to abnormality. It is to be noted that the primary intention of them is to meet the demands of id without being in conflict with the external world. Some of them are given below.

- a. **Denial.** It is the non-perception of reality that is anxiety generating as in the case of going for a second diagnosis by denying the credibility of the first one, when a critical illness is diagnosed.
- b. **Fantasy.** It is the conjuring of an imagined scenario to replace a real one. It is done in all day dreaming.

- c. **Compensation.** A deficit in a particular area is compensated using the skill in a different area as in the case of student who is poor in his studies performs well in sport.
- d. **Projection.** Placing one's own unacceptable impulses on others as in the case of blaming the question setter for low grade in the examination.
- e. **Displacement.** Redirecting the impulses on a low risk target as done in slamming the door instead of hitting a person.
- f. **Sublimation.** Redirecting the impulses through socially approved channels. This is obvious in one's interest in expelling his aggression by working as a butcher.
- g. **Reaction Formation.** This is to do actions opposite to one's actual motivation. This happens when an alcoholic works in anti-liquor movement.
- h. **Regression.** It is the instance of one's return to earlier stage of development. An adult crying on listening to a bad news is an example.
- i. **Repression.** It is the defence mechanism that is most important in psychoanalysis. In this the unfavourable events are pulled back to unconscious. Even though it can never become explicit, it has profound influence on the later behaviour of the individual. This is done when a girl, for example, is sexually harassed in her childhood. The memory becomes latent and is never recollected in her life, but may be expressed as a fear to be alone in house or workplace. Most of the psychoanalytic clinical techniques are intended to investigate into the repressed unconscious memories that give rise to the neurotic symptoms. In doing so, the patient gains insight into his/her mental dynamics and this reduces the symptoms.

Dreams

Dream is the process through which the ego maintains its strength without having a dissatisfied id. The interpretation of dreams plays a significant role in psychoanalysis as a therapeutic technique. Dream-analysis is significant also in psychoanalytic theory as a mode of revealing the unconscious; other processes being slips of tongue, fantasies, inner conflicts, mental symptoms and the like. Freud finds that sleep fulfils one of the necessary conditions for psychosis in the sense that it is turning away from reality. In psychosis, the mental symptoms like hallucination and delusion are repressed experiences and unconscious conflicts that find expression through actions. Similarly dream – as a wish-fulfilling activity (Freud, Dreams) – fulfils the expressions of the day's residues in terms of stressful thoughts and repressed experiences.

Dream as a wish fulfilling activity is not accomplished through any direct means. It is one of the ways that provide an outlet for the repressed content in the unconscious. However, the contents of dreams are not the direct expressions of the repressed contents but are censored by the ego. The censoring makes it necessary to interpret the dream in order to reveal the unconscious. Often the contents of a dream as observed in the process of dreaming, which Freud calls the manifest dream, are symbolized acts. They express the conflicts and stress, the meaning of which is to be found in latent-dream. The practical task of transforming the manifest-dream to latent-dream and explaining how the latter has become the former is known as dream interpretation or dream work.

The latent-dream, which is the actual meaning of dream, is censored through various processes and it becomes necessary to unroll these processes before the manifest contents are interpreted. The processes through which the latent content is transformed to

the manifest content are condensation, displacement, representation and secondary revision.

Condensation is the means through which many elements of the latent dream are represented through a single manifest dream. So “condensation results in the dream’s multiple layers of meaning; the censorship is served by the apparently superficial association through which the composite figures are formed”.

Displacement is the shifting of accent from one dream element to another. So displacement is the principal means of dream distortion and hence it becomes necessary to identify the way the distortion is done here in order to interpret it.

Representation is the way of forming symbols in dream. Through this, complex and vague concepts are converted into dream image. Dream symbols are mostly sexual in meaning and are disguised verbally.

Secondary revision is the final stage of dream work. Freud used the method of free association to discover the latent dream content. Secondary revision covers up the contradiction between the dream and the dreamer’s everyday life.

So in psychoanalysis, dreams, the royal road to the unconscious (Lear 88), can be interpreted as the ego’s endeavour to satisfy the essential needs of the id without imposing any harm either from the superego or from the external world.

Psychosexual Development

The psychic energy supplied by id takes a child through three stages of development in becoming an adult. They are (a) infantile sexuality (b) latent period and (c) genital stage. The psychic energy – specifically called the libido- is sexual in nature. It manifests in its biological, social and psychological form during these stages of development. During these development processes, the sexual instinct may undergo more or less serious failures to pass through certain stages of development called the fixation of the instinct. The fixation will have serious consequences in later life as it determines certain characteristic traits of the individual.

- i. **Infantile Sexuality.** The period of infantile sexuality stretches from the birth of the child till the age of six years. There are three specific stages of development in this.
 - a. **Oral state.** Dominated by pleasure principle, oral stage last for first two years of infancy. Mouth becomes the source of pleasure and sucking, biting and related activities that bring pleasure are reckoned to be sexual in nature. In Freud’s own words,
The first organ to make appearance as an erotogenic zone and to make libidinal demands upon mind is, from the time of birth onwards, is mouth. ... The baby’s obstinate persistence in sucking gives evidence at an early stage of a need for satisfaction which, although it originates from and is stimulated by the taking of nourishment, nevertheless seeks to obtain pleasure independently of nourishment and for that reason may and should be described as ‘sexual’.
At this stage, the child knows the world through mouth. Character fixation at this stage is marked by the interest in the activities like eating, gum chewing, smoking and talkativeness in later life.
 - b. **Anal Stage.** During the age of two to four years the child becomes aware of himself as an independent individual, who can control the emotions of his parents at

will, with toilet habits. Pleasure is derived from expulsion and retention. The psychic force -the libido- is turned towards oneself to form what is called self-love or narcissism. The reality principle starts ruling over the pleasure principle. Fixation at this stage is characterized by liking of yellow colour, orderliness, rigidity etc. Further,

Sadistic impulses already begin to occur sporadically during the oral phase along with the appearance of the teeth. Their extent increases greatly during the second phase, which we describe as the sadistic-anal phase, because satisfaction is then sought in aggression and in the excretory function.

- c. **Phallic stage.** During the age of four to six, little boys fall in love with their mother and hate their father and little girls like their father and hate their mother. This phenomenon is called Oedipus complex. The girl-father relation is specifically known as Electra complex. During this stage, the instinctual sex energy gets localized in genital organs. Narcissism gives way to libido to make parent of opposite sex to become its object. Infantile masturbation is common among children at this stage.
- ii. **Latent Stage.** From the end of phallic stage to puberty, there is an apparent renouncement in the sexual interest of the child. During this period the child becomes interested in both the parents. Narcissistic interests are being reduced. With the beginning of education, moral and intellectual growth predominates other interests.
- iii. **Genital Stage.** Genital stage lasts from puberty to maturity. Firstly, the oral, anal and phallic stages of infantile sexuality are revived at the onset of puberty. The first phase of this stage called homoerotic stage is marked with the libido concentration on children of the same sex. By this, the child overlooks his heterosexual interests, which the society discourages at this early life. The child feels more comfortable with peer group of same sex and is even afraid of the opposite sex. The homoerotic period is followed by the period of heterosexuality. The narcissistic drive still persists. It is expressed in the early adolescent love affairs as this intends to demonstrate the ability to attract members of opposite sex. This allows the individual to attain self-confidence in sexual maturity. The genital stage starts with a dominating pleasure principle that in turn gives way to reality principle in attaining sexual maturity.

Criticisms of Psychoanalytic Theory

The criticisms of Psychoanalytic theory arise mainly from two quarters. The first one is from within the realm of psychoanalysis itself, directed mainly against the theory of sexuality. The other comes from outside the sphere of psychoanalysis and is intended mainly to question the scientific status of the theory and its methodology.

Around 1912, Alfred Adler (1870-1937) and Carl Gustav Jung (1875-1961), both pupils of Freud departed from the mainstream of psychoanalysis with their dissent to the libido theory. However, they had not much in common. Both departed from the Freudian orientation and moved in their own way, establishing their own schools of psychology- Individual Psychology and Analytical Psychology respectively.

Alfred Adler

Adler attacked the sexual aetiology of neurosis. He replaced the libidinal theory with the theory of inferiority feeling. He finds the feeling of inferiority in human being to be universal as it has its root in the very childhood, since the child by virtue of his being helpless and smaller before the adults. This inborn feeling of inferiority is supplemented by the attitudes of parents during the development period of the child.

This feeling of inferiority necessitates the compensation for the feeling of inferiority that the individual ego is subjected to. In general, this is achieved in two ways. In the first way, through the flight into illness, one becomes able to manipulate the factors affecting the significance of the inferiority and also to gain the attention of others. The child's control over these factors helps him to assume a sense of superiority. The second way is that of direct compensation. In this, the lack of ability to get attention because of the child's inferiority is overcome through a direct struggle to master his inferiorities.

Carl G. Jung

Jung had acquired a wide knowledge of symbolism, literature and philosophy of many cultures. This influenced very much the theoretical and therapeutic aspects of his version of psychoanalysis. Without an open conflict with Freud and without depending upon the sexual theory of Freud, Jung developed his theory side-by-side with that of Freud until he departed from mainstream psychoanalysis in 1912. Jung stresses the effect of parental neurotic difficulties on the development of the difficulties in the child. Since the child's mind is more sensitive and mouldable, during the developmental period, it is subjected to a deeper impression from the child's interaction with his parents. Jung emphasizes the significant role of mother in the development of child. This maternal influence is strong in the child even before the development of Oedipus complex and is regressive in nature.

For Jung, regression is an important neurotic craving and is in its final analysis the desire to return to mother's womb as the symbol of security or rebirth. Jung finds that symbols are not limited to sexual interpretation. In this respect he presents a wider interpretation of the symbols than that of Freud.

In *Psychology of the Unconscious*, Jung introduced the term primal libido, which essentially is a life force. This is the undifferentiated energy and a major part of this is sexual in nature. The sexual origin of libido is denied. According to Jung's theory of collective unconscious, the significant memories of the human race are a part of every one's heritage. This heritage includes not only human traits but also residues from the animal past. This collective unconscious represented the wisdom of the ages and hence it is viewed as superior to the individual values. This view influenced Jung's therapeutic approaches, which involves, in part, bringing the patient in contact with his collective unconscious. The interpretation of dream is used for this purpose.

Jung's approach to repression is different from that of Freud. For Freud, repression contains memories and instincts, which the individual's ego is unable to tolerate because of their negative nature. Jung's idea of repression contains positive aspects of personality as well. He views education as a factor that introduces conflict in the individual's life. Education forces a person to divert from his individual line of life developing spontaneously. Finding out this individual line of life or self-realization then forms a major aspect of his system of therapy. Jung's goal of therapy is different from that of

Freud. Freud's therapy is a retrospective analytic understanding of the past. In Jung's analytical psychology the functional analysis looks at the future in order to find the meaning in the present and future. So self-realization forms an important guiding force in Jung's therapeutic technique.

This approach to therapy has been subjected to severe criticism. In this the patient is taken away from reality, which is substituted with mystical and semi-religious fantasies of life. This substitution does not root out the patient's problem, but an obsession of something else rules over the patient's problem. This is the classical mechanism of obsessional neurosis (Thompson 169). Jung's contribution includes the emphasis on parent-child relationship, viewing therapy as a patient-analyst mutual interaction and an attitude of respect for the patient and his neurosis. His mode of therapy also includes the revealing of the repressed positive potentialities. This helps in developing the undeveloped aspect of patient's personality.

B.F. Skinner

B.F. Skinner, the great exponent of Behaviourism, in his essay 'Critique of Psychoanalytic Theory' has levelled serious criticisms against psychoanalysis. These criticisms, by and large attack the methodology of psychoanalysis. According to Skinner, Freud took his mental apparatus as real rather than a scientific construct. By mental apparatus, Freud meant the conscious, pre-conscious and unconscious mind and the dissection of personality into id, ego and superego. The dynamism includes the interplay of forces, which are necessarily derived from psychic energy.

No matter what logicians may eventually make of this mental apparatus, there is little doubt that Freud accepted it as a real rather than a scientific construct or theory. One does not at the age of 70 define the goal of one's life as the exploration of an explanatory fiction. Freud did not use his "mental apparatus" as a postulate system from which he deduced theorems to be submitted to empirical check. If there was any interaction between the mental apparatus and empirical observations, such interaction took the form of modifying the apparatus to account for newly discovered facts.

Another criticism of Skinner is about the use of analogies for explanation in theoretical psychology. Keeping in mind Freud's view that analogies, it is true, decide nothing, but we can make one feel more at home.

From a behaviourist point of view, Skinner criticises Freud on the use of introspection as a means for observing mental life. Further the manipulation of this mental life as a means for treatment is also questioned. This is done from a standpoint observing Freud as a thoroughgoing determinist and dualist. One further criticism on psychoanalysis is made on Freud's use of terms such as forces, processes, and mechanism from other disciplines.

Karl Popper

Now we come to the criticism against the scientific status of psychoanalysis. The main exponent in this field is the Austrian-born British philosopher of science, Sir Karl Popper (1902-1994). Popper's attacks on what he called pseudo-sciences were mainly directed against three theories. They are Marx's theory of history, Freud's psychoanalysis and Adler's 'Individual Psychology'. It is the possibility of being refuted, and not that of confirmation, which he says, determines the scientific status of a theory. And in this sense both versions of psychoanalytic theories (that of Freud and Adler) are not refutable. The

logical problem of induction requires infinite number of confirmatory evidence in order to prove a theory to be true. This is obviously impossible. Both the theories mentioned earlier, Popper says, are able to explain the whole realm of human behaviour. They are even potent enough to explain the explicitly conflicting behaviour with the same strength. He observes that these theories “always fitted... were always confirmed-which in the eyes of their admirers constituted the strongest argument in favour of these theories...this apparent strength was in fact their weakness”.

Popper denies the scientific status of these theories on the ground that they are non-testable and irrefutable.

Popper's attitude to psychoanalysis may be viewed from two perspectives. Firstly, we can question the very status of the scientific, and sideline it to be one among the various ways of looking into facts, and thus deny scientific status to psychoanalysis without undermining its meaning and significance.

Adolf Grunbaum

The second way of looking into Popper's view is to deny the argument that psychoanalysis is unscientific. This is the position that Adolf Grunbaum had held. Grunbaum disagreed with popper. He, while maintaining the falsification criteria of Popper shows that psychoanalysis is scientific. Grunbaum explains this with what he calls Freud's Master Proposition, also known as the Necessary Condition Thesis (NCT). NCT explains that, ONLY psychoanalysis can produce a durable cure of psychoneurosis. This, Grunbaum says, is a strong statement that could be falsified if, for example, another form of therapy such as behaviour therapy cured someone of a neurosis, or even if spontaneous remission occurred. This falsifiability makes psychoanalysis scientific. Grunbaum further argues that NCT is in fact falsifiable, and falsifiable in a number of ways. This, according to him makes psychoanalysis a bad science.

Ned Block and Gabriel Segal evaluate Grunbaum's argument and put forward a modified NCT. Some of Grunbaum's arguments against NCT were the spontaneous remission of neurosis, the outcome of psychoanalytic treatment as that of placebo effect and Freud's own position that psychoanalysis often offers only temporary cure. Block and Segal offer their version of NCT like this: (NCT) other things being equal, a patient under analysis would get better unless some of the analyst's interpretations were correct.

Motivation as Cause

As we have seen earlier, Skinner describes Freud as a thoroughgoing determinist. Determinism in psychology means the denial of the concept of free will. This is to assert that every psychological event is necessarily subjected to psychological laws. Or, in other words, it is the theory of psychic causation.

The term causation when described under the subject of psychology requires a further analysis. This is necessitated by the fact that the word causation is primarily used to describe the interconnection between the material objects or phenomena. When we work with psychological events, the elements that contributed from outside the human mind, we call stimuli; and that from within the mind, motivation. Here it is worth to remember Schopenhauer's words that “Motivation is causality seen from within”. That is to say, when the causal elements under consideration are psychological in nature, it is appropriate to use the word motivation to describe their casual aspect. When the phenomena

under consideration are physical objects, the cause is attributed to an agent external to it.

Even when we replace the word cause with motivation, in the psychic realm, we are confronted with two major problems that are necessarily involved with the concept of causation. Firstly, whether the concept of motivation alone is sufficient to explain the whole psychic phenomena? This obviously raises the problem of plurality of causes of the causal theories. The second problem is the status of intentions and volitions in the mental causation; whether they stand external to the causal chain and still influence the causal process? If so, it guarantees the possibility of free will.

Among these, the first one compels us to examine the concepts of necessary and sufficient cause. J.S. Mill explains cause as always a sufficient condition.

This sense of cause is much different from the one we normally have. We normally attribute cause to the single event or factor. In saying that the spark caused explosion in a laboratory, the condition under which the inflammable gas is formed in the laboratory is not taken into account. It is obvious that without the inflammable gas formed, the explosion could not have taken place. In Mill's sense it is the whole of the situation that led to the explosion and he describes this as cause or the sufficient condition. But the sufficient condition may also include factors that are irrelevant.

The necessary cause on the other hand is the condition in the causal process in the absence of which the effect fails to occur. We may say that oxygen is a necessary cause of fire we mean that in the absence of oxygen there cannot be an instance of fire. Our endeavour to explain cause as necessary or sufficient condition is determined by our specific intention towards the effect (Copi 450). We search for cause as a necessary condition when our aim is to cease the effect. We can remove the necessary element from the sum total of existing conditions and hence remove the effect. On the other hand where the effect is desirable or we want to produce the effect, we may view cause as a sufficient condition. Now it may also be seen that all necessary conditions put together form the sufficient condition.

If Mill's sufficient condition is taken as cause, the lack of any one element among the sufficient condition will not yield the effect. So, cause may be viewed as the element which when introduced to an already existing condition makes it sufficient to produce the effect. This concept of cause is essentially that of necessary condition. Our search for this element which is the one introduced mainly depends on our specific interest.

The kind of elements or changes for which we look depends on the structure of the order in which we are interested. The answer to question, "Who killed the Archduke Ferdinand at Sarajevo?" must be of the form: "The person A,B,C and so on, are the assassins of the Archduke." On the other hand, the question, "What killed the Archduke?" must be answered according to the kind of specific order for which we are in search, and according to the purpose of inquiry... What is an adequate answer to one question will not, in general, be adequate to another (248-49).

So, finally we see that the attribution of cause to a single event among others mainly depends on our specific interest. Then, in analyzing the psychic phenomena, we are apt to consider the external factors as more or less insignificant (though not irrelevant) and attribute their cause to psychic elements. This does not deny the importance of external or of physical factors. It may be noted that the external factors sometimes become irrelevant in the sense that the subject does not take it seriously. Whether a persisting noise

of a machine will distract me from my reading or not largely depends on the way I take it. Hence, it only means that we are now considering the significance of psychic elements in determining an action. This particular element, which leads to an action, we call motive. And in the realm of psychology, motivation may be viewed as causation. Obviously a motive cannot be the cause in the sense of sufficient condition but may be seen as necessary cause and hence it is the cause that we intend to investigate.

The second problem we stated viz. the status of intentions and volitions as factors of psychological causation may be examined specifically within the context of psychoanalytical motivational theory.

In the Outline of Psychoanalysis, Freud called the force that acts behind the needs of id (which we saw earlier, as the prime mover) the instincts. They represent the somatic demands upon mental life. These instincts or drives the satisfaction of which is the prime intention of mental activity are essentially somatic needs. The aim of Eros – one among the two fundamental motivating forces that Freud recognized (will be described later) is self-preservation and preservation of the species. Such instincts, which are basically connected with the preservation of self and of species, may be considered as needs. These needs are not psychologically manifested enough for a direct observation. They are rather inferred from psychic activities. Intentions and volitions may be seen as manifestations of underlying instinctual needs. So they may be reckoned cause in a secondary sense only. To state it otherwise, intentions and volitions are the secondary elements in the chain of psychological causation. The first element, which is instinct, is necessarily unconscious. According to Freud, instincts act not only behind intentions and volitions but also behind all psychic activities.

Motivational Theory of Psychoanalysis

The motivational theory of psychoanalysis has developed through two major phases. In the first phase, with his most important essay on motivation, *Instincts and their Vicissitudes*, Freud described the fundamental motivation factors as instincts and examined their characteristics. Here, the motivating factors were taken to be sexual in their nature and were based on the pleasure principle alone. Later in 1922, with the publication of *Beyond the Pleasure Principle*, Freud introduced the concept of death instinct. This concept, finding its expression in aggressive and destructive activities, has profoundly changed Freud's view on motivation. This, according to Cofer and

Appley may be attributed to three factors:

- (1) dissatisfaction with the earlier explanation of aggression as a manifestation of the sexual instincts (The widespread occurrence of cruelty and destruction of the First World War profoundly impressed Freud);
- (2) the conviction that the repetition – compulsion principle was more fundamental and all-pervasive than the more limited sexual and self-preserved instincts could explain; and, most important;
- (3) the need to find a force to counteract the then monistic life instincts.

In *Instincts and their Vicissitudes*, Freud distinguished between instincts and stimuli. Instincts, according to him, are principal motivating factor of behaviour. Stimuli, on the other hand refer to the external factors affecting the organism. Freud originally used the German word *Trieb* to express the motivating factor. "Literally the word *Trieb* means the mechanical provocation to action" (Bolles 61). Its translation into English as instinct

created confusion since the original English meaning of instinct varies marginally from that of Trieb.

Freud's theory of motivation is essentially a stimulus intensity reduction theory, according to which the intention of every motive is to keep the nervous system to the lowest level of stimulation. The stimulus reduction approach to the theory of motivation is arrived at from Freud's initial concept of equilibrium (Bolles 58) according to which the tendency of nervous system is to discharge any increase in excitation. But the energy can be discharged through pre-established channels. The psychic structure of personality, we have seen earlier, does not permit the direct outlet to all instincts necessitate the instincts to be reduced through channelization through the personality structure. So in stimulus intensity reduction, the excitation created by the instincts is reduced without having a direct discharge.

According to Freud, instincts are characterized by their impetus, aim, object and source. Impetus is the overt expression of the instinct. The aim is to reduce the intensity of the instinctual drive. Object is the means through which the organism establishes its aim. In determining the object, dream and various defence mechanisms like fantasy, sublimation displacement etc. play their role. This is because the direct expression of the instinct to the motivated object might not be warranted either by super-ego or by the external world.

Here it may be noted that in the description of the source of instinct, Freud maintains a biological explanation and leaves it outside the realm of psychology. Then it becomes implicit that he allows a biological explanation of organic behaviour (the contemporary psychiatry) distinct from the psychological account. This differentiates between the neurobiology of consciousness and folk psychology (the contemporary name for the Freudian tradition).

The Basic Instincts

As stated earlier, Freud recognised two fundamental motivating factors; the life instincts (Eros) and the death instincts (Thanatos). Some reasons for Freud's postulation of the death instinct are to be discussed in the last section. Further, Freud noticed the inevitable significance of death as the final state of all organisms. He also based this postulation on the idea of the returning of all things to their origin.

Further the introduction of death instinct is influenced by the stimulus reduction nature of Freud's theory. The struggle for reduction of intensity and maintaining it at a constant level is the removal of inner stimulus tension (the nirvana principle), "a struggle which comes to the expression in the pleasure principle" is one strong reason for Freud's belief in death instinct.

The aim of life instinct is the preservation of the individual and the species. The former is expressed through the drives such as thirst and hunger and the latter through sexual urges. During the development through psychosexual stages, the objects and the aim of sexual instincts become varied and the failure to pass through certain stages fixes the development at that stage. As seen earlier this determines some of the characteristic traits of the individual. Even though Eros is explained as consisting of two drives – the ego drive and sex drive – Freud recognized the principal energy behind both of them to be the same. He called this energy libido or sex energy. Libido is the principle of unification. This aims at preservation and maintenance.

The death instinct on the other hand is the principle of alienation. Here lies the contrast between Eros and death instinct. The death instinct seeks return to an earlier state of the organism- the physical constituent states; the life instinct aims at reunion.

The basic instinct, Freud says, cannot be localized to any region of the mind. The whole available energies of Eros- libido, he says, "is present as yet undifferentiated ego-id and serve to neutralize the destructive impulses which are simultaneously present". And Freud illustrates such interaction between life and death instincts in biological functions such as eating and sexual act. "Eating is a destructive activity with the object of incorporating it finally and sexual act is an act of aggression having as its purpose in the most intimate union". However it is difficult to identify the source of death instinct.

Some Observations on the Psychoanalytic Theory of Motivation

Noted below are some of the key observations on the Psychoanalytic theory of motivation that will be useful in formulating the final conclusion of this work.

Stimulus Intensity Reduction Nature

This is a version of tension reduction theory of motivation. This model, according to Coffey and Appley, is perhaps the most widely met motivational hypothesis in biology and psychology (601) and is the key to the psychoanalytic theory of motivation. Freud correlates an increase in mental stimulation with pain and its decrease with pleasure.

The Dual Principle

The dual principle on which the Psychoanalytic theory of motivation is based is life instinct and death instinct. This enables the theory to explain psychic phenomena on the basis of one or both the principles.

This is an outcome of what Freud calls the necessary postulate regarding the functioning of the nervous system, which in Freud's conception is to master stimuli (Atkinson and Birch 209).

Arousal Aspect

Charms explains the tension arousal aspect of Freudian motivational theory. According to him,

"Freud makes it particularly clear that he feels it necessary to account for the apparent pleasurable aspects of increasing stimulation in his account of sexual forepleasure. He depends primarily on the ultimate reduction of tension for his explanation, but it is clear that the contradiction of the theory implicit in evidence that some tension is sought out and apparently pleasurable bothered him. In dealing with these phenomena, he comes close to an arousal jag type position".

The Origin of Instincts

From the interaction between life and death instincts and their counter balancing as suggested by Freud, it will be reasonable to search for a common source for both these instincts. To state it otherwise, the possibility of both the instinct to have arisen out of a single psychic principle or characteristic cannot be ruled out. Such an origin if discovered will be of immense value as it will enable us to bring the whole of psychic phenom-

ena, which Freud brought under his theory of motivation, under one principle that unifies both the motivating principles of Freud.

Dialectical Interplay of the Basic Instincts

Freud called the death instinct ego instinct and explained that “it springs from the vitalising of inanimate matter, and have as their aim the reinstatement of lifelessness”. In *An Outline of Psychoanalysis*, Freud states that “We shall be justified in saying that there arises at birth an instinct to return to the intra-uterine life that has been abandoned-and instinct to sleep”. And in *New Introductory Lectures on Psychoanalysis*, Freud speaks of the “phantasy of returning into the mother’s womb”. All these show the importance of death instinct in Freud’s theory.

It is significant here to note that birth (the acquiring of separate identity- alienation) is the primary realisation of the death instinct, which is governed by the nirvana principle. Death (the physical decomposition-identification) is the final realisation of the life instinct, which is governed by the pleasure principle. And life is a striving for the dialectical synthesis of birth and death, and is governed by reality principle.

Freud’s shift from a single motivating principle to the dual one can be seen necessitated by the insufficiency of a single principle to explain the dynamics of psychic life. The dual principle fulfils it in the way mentioned above. Hence life is a process of dialectical evolution aimed at the realisation of an ultimate motivating factor. Once such a stage is attained all principles (pleasure, nirvana and reality) becomes insignificant.

Relation to Indian Theories of Mind

Freud’s theory of ‘levels of consciousness’, has got a striking similarity with the Upanisadic concept of the levels of mind described in the last chapter. The levels of consciousness described in *Upanisads*, *jagrat*, *swapna* and *susupti* correspond to Freud’s notion of conscious, preconscious and unconscious minds respectively. However, there is no parallel in Freud’s theory for the concept of *turiya*- the level of transcendental reality. Another similarity can be seen between Freud’s theory of ‘structure of personality’ and the *Samkhya*-Yoga theory of *gunas*.

It may be seen that the similarity between Freudian theories and the Indian theories of mind are not necessarily the one of accidental coincidence. Freud was aware of the Upanisadic tradition as he had referred to the comparison of the dual nature of motivation with division of the primal reality into two as described in *Brihadaranyakopanishad*. His remark here that Plato might have been influenced by this notion of Upanisads in his formulation of the concept of Zeus shows Freud’s regard for the traditional Indian theories of mind.

Social Learning Theory

In social learning theory instinctual drives have been neglected and the emphasis is on the reciprocal interaction between behaviour and environment neither driven by internal forces nor are we passive reactors to external stimulation. The type of behaviour we exhibit partly determines the reward or punishment we receive, and these in turn influence our behaviour.

Patterns of behaviour can be acquired through direct experience or by observing the behaviour of others. Some responses may be successful; others may produce unfavourable results. Through this process of differential reinforcement, the person eventually selects

the successful behaviour patterns and discards the others. There are some theories based on social learning given below:

1. Ethological Processes

Social learning theory also stresses the importance of vicarious learning, that is, learning by observation. Many patterns of behaviour are learned by watching the behaviour of others and observing its consequences for them. Emotions can also be learned vicariously by watching the emotional responses of others as they undergo painful or pleasant experiences. A young boy who observes the pained expressions of his older brother in the dentist's chair will probably be fearful when the time comes for his first dental appointment. Social learning theorists emphasize the role of models in transmitting both specific behaviours and emotional responses. And they have exerted much of their research efforts toward discovering how modelled behaviour is transmitted—what types of model are most effective, and what factors determine whether the modelled behaviour that is learned will actually be performed.

2. Self-Regulatory Process

A third emphasis of social learning theory is the importance of self-regulatory processes. A specific behaviour produces an external outcome, but it also produces a self-evaluative reaction. People set their own standards of conduct or performance and respond to their behaviour in self-satisfied or self-critical ways, depending upon how the behaviour relates to their standards. Thus reinforcement has two sources: external and self-evaluative. Sometimes the two coincide, and sometimes they are contradictory. A person may be rewarded socially, or materially, for behaviour that is not acceptable according to self-standards. Indeed, self-reproach is an important influence in motivating people to adhere to accepted standards of conduct in the face of opposing influences. One person is tempted to falsify information on an income tax return. The chances of getting caught (external punishment) are slim and the financial gain is substantial. But the anticipation of personal feelings of self-contempt prevent the individual from doing so: The behaviour is not in accord with self-standards.

3. Cognitive Process

Social learning theory departs from a strict behaviourist position by stressing the importance of cognitive processes. Because we can think and represent situations symbolically, we are able to foresee the probable consequences of our actions and alter our behaviour accordingly. Our actions are governed to a large extent by anticipated consequences. We don't wait until we have experienced frostbite to decide to wear warm gloves in subfreezing weather. Future consequences, represented symbolically in one's thoughts, can motivate behaviour in much the same way as actual consequences.

Humanistic Approach

The humanistic approach in psychology developed in the 1960s and 70s in the United States as a response to the continual struggle between behavioural theorists and cognitive psychologists. The humanists brought in a new perspective, believing that the study of psychology should focus not just on the purely mechanistic aspects of cognition, nor purely on the impact of environment on behaviour, but rather felt that the emphasis of psychological study should be on the particulars of human experience. For the humanist

psychologist, the goal of psychological study isn't so much the total understanding of human behaviour, but rather the ability to help people deal with life more successfully. In this sense, it is a highly therapeutic approach, as opposed to a purely theoretical one.

The humanistic approach has its roots in existential philosophy. Throughout the latter half of the nineteenth century and the first half of the twentieth century, philosophers such as Kierkegaard, Camus, and Sartre began to view the condition of man's existence on earth as somewhat "absurd". In an increasingly post-Christian European society, they felt that man had finally reached a state in which an honest appraisal of reality would give one the impression that the universe was neither benevolent nor malevolent, but rather totally indifferent to the trials and tribulations of humankind. Instead of reacting to this news depressively, however, the existentialists felt empowered with a sense of freedom and personal responsibility. Humanistic psychology takes its approach from this philosophy, attempting to help people come to terms with their often painful existence in the world by accepting the responsibility of their existential freedom.

Maslow's Hierarchy of Needs

One of the first major proponents of humanistic psychology was Abraham Maslow. Coming from a background of developmental psychology, Maslow attempted to develop a theory by which to explain the diversity of human needs and motivation. In the course of his studies, he was able to develop a model to express these needs that has since become quite popular across the world, being referred to as either the "hierarchy of needs" or sometimes "Maslow's Pyramid".

For Maslow, the goal of any human being is to reach a state of "self-actualization" in which all their needs are met and a state of contented happiness is achieved. However, these needs must be met in a particular order. The base of the pyramid shows the physical needs that all humans share, such as the need to eat, drink, have shelter from the elements, and so on. According to Maslow, higher forms of happiness are not achievable so long as these needs are not met.

Moving up the pyramid from there, one must have safety needs met, then one must feel love towards and from his friends and family. After this, one must receive self-validation and a feeling of self-worth from one's self. Lastly, a state of self-actualization and creative work is possible. However, if any of the levels of the pyramid goes unfulfilled at any time, the subject must return to that level. In other words, self-actualization isn't a permanent state, but rather a continuous battle. The primary idea to retain from this model is that human motivation is complex, and while working towards the primary goal of self-actualization, their particular needs and pursuits might change dramatically.

Rollo May's Existential Psychology

For other thinkers like Rollo May, the problem was less with deciphering human behaviour and motivation and more with simply helping humans exist in a state of existential crisis. Like many philosophers and psychologists with existential leanings, May felt that the absence of an innate moral structure in the universe usually led to people experience anxiety and distress because their expectations of the universe were simply not being met. His school of existential psychology is intended to deal with exactly this, the problems that arise from human consciousness working against the problems inherent in existence.

Generally speaking, the problem for May is that most people feel as if the universe is either against them or on their side, and they approach life accordingly. Since this is not reality, May's therapeutic approach was to attempt to help people accept the coincidental and "meaningless" state of their existence by demonstrating that freedom was the natural consequence of this meaningless state, and that the lack of a prescribed fate was a cause to be celebrated rather than feared.

Carl Rogers and Unconditional Positive Regard

Carl Rogers was another of the important figures in early humanistic psychology. He was the first to advocate for the importance of "unconditional positive regard", a way of viewing one's self in which one is unconditionally positive and accepting, maintaining the right to be critical without being judgmental or overly harsh. For Rogers, this state of positive regard towards one's self was essential to psychological development, and in maintaining strong personal relationships.

Therapeutically, Rogers believed in a client-directed approach to matters. In a therapy session, he would simply listen non-judgmentally and allows his clients to speak about anything they wished, reasoning that with a little bit of guidance, people had the intelligence and self-awareness to direct their own progress towards wellness. This degree of "patient control" in clinical practice was revolutionary in the realm of psychology, and in the realm of humanistic thought in general.

Evaluation of the Humanistic Approach

Humanistic psychology is the psychological approach which states that the human is the most important thing, more important than the complex, the disorder, the behaviour, or the environment. It seeks to help people live better lives through practical therapeutic measures.

Humanistic psychology has its origins in existential philosophy. Believing that the universe is inherently meaningless, existentialists see themselves as ultimately free creatures able to define their own personally meaningful destinies.

Abraham Maslow attempted to explain human motivation from the standpoint that all people are looking to achieve "self-actualization". There is an entire hierarchy of needs that have to be fulfilled in order to reach that stage however.

Rollo May's existential psychology attempted to help people come to terms with the "meaninglessness" of life, reframing this "problem" as a profound state of individual freedom, free from the constraints of fate.

Carl Rogers' approach to therapy was to allow clients to direct their own recovery through the principle of "unconditional positive regard".

Sources of Conflict

There are many causes or reasons for conflict in any work setting. Some of the primary causes are:

Poor Communication different communication styles can lead to misunderstandings between employees or between employee and manager. Lack of communication drives conflict 'underground'.

Different Values any workplace is made up of individuals who see the world differently. Conflict occurs when there is a lack of acceptance and understanding of these differences.

Differing Interests conflict occurs when individual workers 'fight' for their personal goals, ignoring organizational goals and organizational well-being.

Scarce Resources too often, employees feel they have to compete for available resources in order to do their job. In a resource scarce environment, this causes conflicts – despite awareness of how scarce resources may be.

Personality Clashes all work environments are made up of differing personalities. Unless colleagues understand and accept each other's approach to work and problem-solving, conflict will occur.

Poor Performance when one or more individuals within a work unit are not performing – not working up to potential – and this is not addressed, conflict is inevitable.

BIOLOGICAL MOTIVES

Not only mental functions but body and its requirements are also powerful motivators of human behaviour. These physical requirements are hunger, thirst, sleep and sex. Human life is a simple equation of these four terms. Now, we shall describe these biological motives one by one.

Hunger

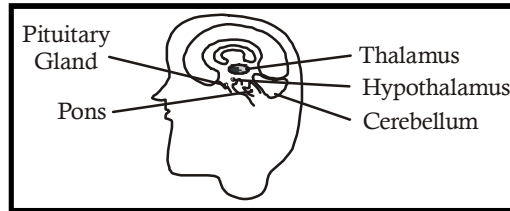
The body needs an adequate supply of the essential nutrients in order to function efficiently. Depletion of these nutrients activates homeostatic mechanisms to release food stored in the body. For example, the liver releases stored sugar into the blood stream. Replenishment from body stores enables a person to continue functioning even though he has missed several meals. When the body stores are diminished to a certain point, however, the automatic homeostatic mechanisms are no longer adequate, and the entire organism becomes mobilized to seek food. Therefore, hunger can be a powerful motivator, as anyone who has gone without food for any length of time knows. Those who have subsisted on semistarvation diets report that thoughts of food and eating occupy much of their waking hours as well as their dreams.

The feelings most people describe as hunger – an empty or aching sensation in the region of the stomach, sometimes accompanied by a feeling of weakness – give us some clues. But stimuli outside the organism can influence feelings of hunger and eating behaviour. These stimulus (outsiders) may be of two kinds:

1. The odour or sight of food (external stimuli) can arouse hunger even when there is no physiological need. You walk past a bakery only an hour after your last meal and find your mouth watering at the sight of a piece of cake in the window. Your signal for hunger in this case is not related to the depletion of nutrients in the body or to stomach pangs or a feeling of weakness.
2. Habits and social customs can also influence eating behaviour. You are accustomed to eating at certain times of day and may suddenly feel hungry when you notice that it is noon. You may consume more when having dinner with a group of friends who are all eating voraciously than you would when driving alone.

Regulatory Centres in the Hypothalamus

The various control systems that regulate feeding are integrated in a region of the brain called the hypothalamus.



THE HUMAN BRAIN (SHOWING HYPOTHALAMUS).

The hypothalamus, a small collection of cell nuclei located at the base of the brain has numerous connections with other brain parts and with the pituitary gland. It also has a greater density of blood vessels than any other area of the brain, and thus can be readily influenced by the chemical state of the blood.

The hypothalamus plays an important role in the regulation of food intake. Physicians have observed that patients with tumours or injury in the region of the hypothalamus overeat (*hyperphagia*) and become obese. The development of precise instruments for exploring the brain has made it possible to specify more exactly the regulatory centres in the hypothalamus. We now know that two areas regulate food intake the lateral *hypothalamus* (LH) initiates eating – it is a “start” or “feeding center”, the ventromedial hypothalamus (VMH) inhibits eating – it is a “stop” or “satiety center.”

Regulation of Food Intake

Regulation of food intake in brain can be studied by following two ways:

1. One way to study the function of a specific brain area is to insert a microelectrode and stimulate the spot with a weak electric current. Stimulation of LH cells causes a satiated animal to eat. Electrical stimulation of VMH cells inhibits eating; with a weak current the animal slows its feeding behaviour, and with a stronger current it will stop entirely (Hoebel and Teitelbaum, 1962).
2. Another way to study the function of a brain area is to destroy cells in the region (also with a microelectrode but using direct instead of alternating current) and observe how the animal behaves when the area no longer exerts control. On destruction of cells in the LH the animal refuses to eat or drink and will die unless kept alive through artificial feeding and watering. Damage to cells in the VMH produces overeating an obesity in every species investigated, from rat and chicken to monkey and human. No major metabolic disturbance has been found. The animals become fat simply because of overeating. If restricted to the same amount of food that a normal animal consumes, they do not gain weight.

Studies of the sort described above have made it clear that the VMH area (satiety center) and the LH area (feeding center) act in opposite ways to regulate food intake. Moreover, two kinds of control systems appear to be integrated in the hypothalamus. One system, responsive to the immediate nutritive needs of the organism, tells the brain when to start a meal and when to stop. A second, long-term control system, attempts to

maintain a stable body weight over a period of time, regardless of how much the organism may or may not eat in any one meal.

(a) Short-term Control of Food Intake

Studies indicate that the hypothalamus contains “glucoreceptors” cells sensitive to the rate at which glucose passes through them. The glucoreceptors in the VMH respond differently to glucose level than do those in the LH. Microelectrodes were implanted in the hypothalamus of dogs and cats so that the activity of single neurons in the VMH and LH could be recorded. Records of neural activity were obtained before and after injections of glucose or insulin. The results showed that after glucose injections, cells in the VMH (satiety center) become more active, whereas cells in the LH (feeding center) show decreased activity. The reverse is true after insulin injections. Cells in other parts of the brain monitored as controls showed no changes.

But digestion is a slow process, and we stop eating long before the food we have consumed can be transformed into enough blood sugar to make up a deficit in the blood stream. A more immediate signal is needed to let the brain know that food is on its way. One such signal is provided by a full stomach. If food is injected directly into the stomach of a hungry animal (without passing through the mouth and throat) it eats much less than it would otherwise. Experiments suggest that cells in the VMH respond to distention of the stomach to inhibit further eating.

An empty stomach produces the periodic contractions of muscles in the stomach wall that we identify as “hunger pangs.” This increased movement of the stomach wall has been shown to activate cells in the LH. Thus, an empty stomach signals the LH to start the organism eating; a full stomach signals the VMH that it is time to stop eating.

A third short-term control mechanism is temperature. In a warm environment most animals and humans eat less than in a cold environment. Cooling the brain has a similar effect on food intake. The nature of these “thermoreceptors” is not clear, but evidence shows that the LH responds to decreased brain temperature and the VMH responds to increased brain temperature.

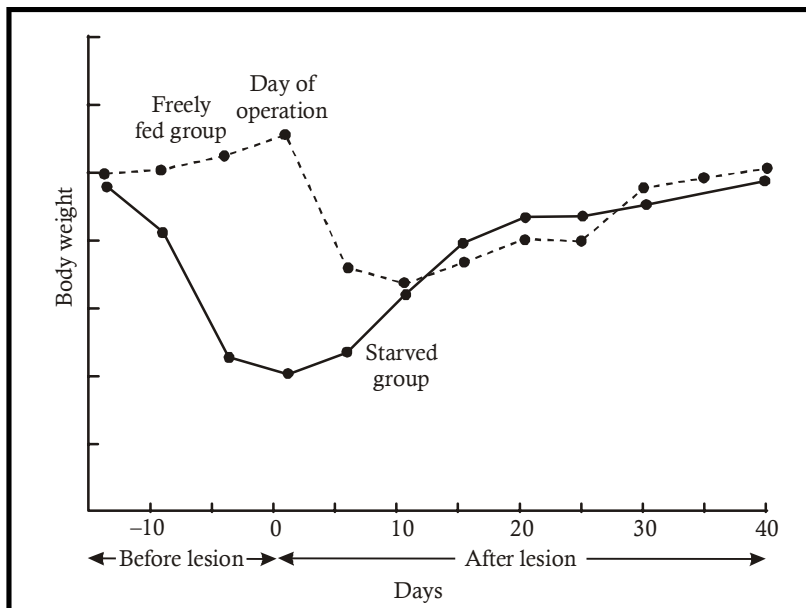
(b) Long-Term Control of Food Intake

Most wild animals maintain their weight at about the same level throughout their lifetime, despite the fact that food may be plentiful one week and scarce the next. Human beings have greater difficulty maintaining a constant weight because their eating behaviour is more strongly influenced by emotional and social factors. Even so, most people stay at about the same weight from year to year. The hypothalamus appears to provide, in addition to its short-term controls of food intake, a delicate system for ensuring that the organism’s weight remains at a stable level.

We noted that rats with damage to the VMH overeat and become obese. Originally, this gluttonous behaviour was attributed to an increased appetite resulting from destruction of part of the short-term control system. But once the rat reaches a certain level of obesity (usually two or three times its normal weight), it no longer overreacts. It reduces its food intake to only slightly more than normal and maintains itself at its new obese weight. If the animal’s diet is restricted, it will drop to its original normal weight. If, once again, it is allowed to eat freely, it will overeat as before until it is back up to its obese

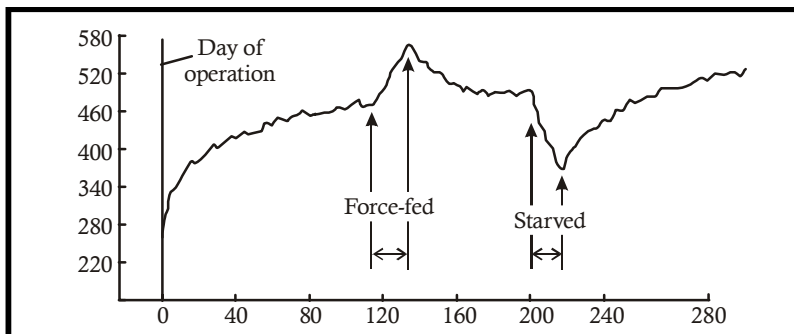
state. It looks as if damage to the VMH disturbs the animal's long-term weight control system so that weight is regulated at a higher level.

If obese rats are force-fed after their weight has levelled off, so that they become "super-obese," they reduce their food intake until their weight returns to its "normal-obese" level. Some correlate of body weight must act on the VMH to influence food intake. Since autopsies have shown the surplus weight of animals with VMH lesions to be composed almost entirely of fat, a likely candidate is the amount of free fatty acids in the blood stream.



BODY WEIGHT AND THE LATERAL HYPOTHALAMUS.

Various experiments indicate that the hypothalamus provides a fairly delicate system for maintaining the organism's weight at a stable level. The VMH and LH have reciprocal effects on the "set point" for body weight. If the VMH is damaged, the set point is raised; damage to the LH lowers the set point. If very careful lesions are made in both areas so that an equivalent amount of tissue is destroyed, animals neither overeat nor undereat but maintain their weight at the presurgery level.



EFFECTS OF FORCE FEEDING AND STARVATION ON BODY WEIGHT OF RAT WITH VMH LESIONS.

Thirst

There are two ways by which an organism can replenish its water deficit. First by drinking and second by recovering water from the kidneys before it can be excreted as urine. A water deficit motivates the organism to drink. It also sets off a homeostatic mechanism by stimulating the release of the antidiuretic hormone (ADH) from the pituitary gland. ADH acts on the kidneys so that water is reabsorbed into the blood stream and only very concentrated urine is formed. After a night's sleep you may notice that your urine is much more concentrated (darker colour and stronger odour) than at other times of the day. The body has recovered water from the kidneys to compensate for the fact that you have not consumed fluids while sleeping. This homeostatic mechanism can maintain the body's water balance only up to a certain point. When the water deficit is too great, thirst becomes intense and the organism is impelled to find water.

Peripheral Cues

Receptors in the mouth and throat must somehow measure water intake, because the animal stops drinking after swallowing its usual amount. After a few minutes, however, the body realizes it has been duped and that although sufficient water entered the mouth, it never reached the stomach. There must be additional signals regulating water intake. Experiments with dogs provide fairly clear evidence of the role receptors in the mouth and throat play in the regulation of water intake. Dogs were fitted with a plastic tube leading from the oesophagus to outside the body. They could drink normally, but the water drained out, never reaching the stomach. A thirsty dog equipped with this kind of oesophagus bypass will drink its usual amount of water and then stop, apparently satisfied by fluid that never reaches the stomach. After about 10 minutes, however, it begins to drink again and consumes the same amount of water. This cycle continues with the dog gradually increasing its intake as the water deficit builds up.

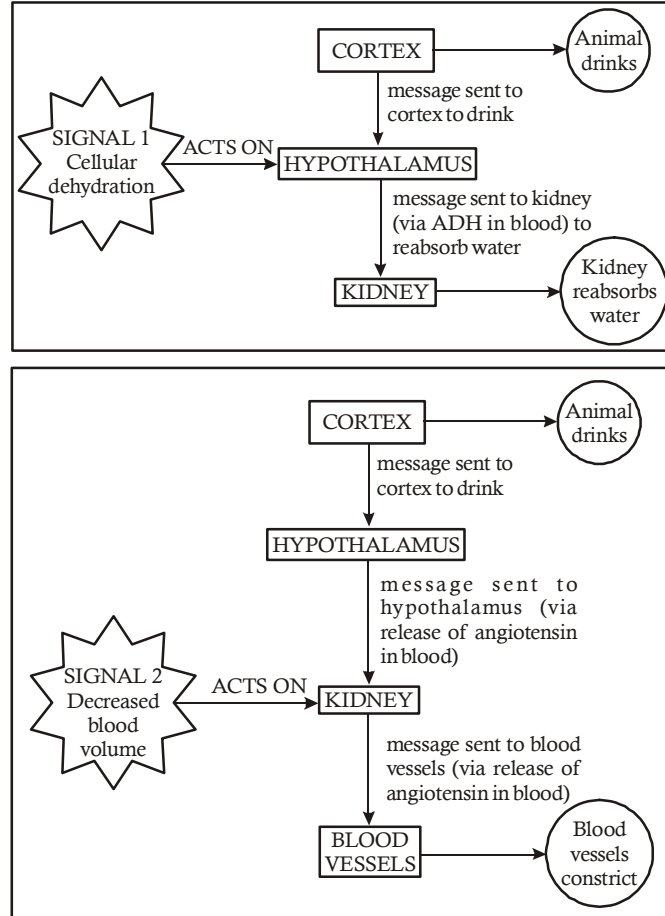
If a dog is allowed immediate access to water, it will drink the usual amount. If there is a 50-, 10, or 15-minute delay, the amount consumed decreases as the delay increases. After a 20-minute interval the dog does not drink at all. These results indicate that a certain amount of water must be absorbed through the stomach wall into the blood stream before the mechanism that responds to water intake is activated.

Hypothalamic Regulation of Water Intake

Water deficit produces two changes in the blood and fluids surrounding the body cells: (I) decreases their volume and (II) increases the concentration of certain chemicals, primarily sodium. When the body fluids surrounding tissue cells become too concentrated, water passes out of the cells by osmosis, leaving them dehydrated. Current theories of thirst postulate two brain mechanisms for control of water intake: osmoreceptors that are sensitive to the chemical concentration of the body fluids and volumetric receptors that respond to the total volume of body fluids.

Osmoreceptors. Although cellular dehydration occurs in all body cells, certain nerve cells in the hypothalamus respond specifically to dehydration. Dehydration causes these cells to become slightly deformed or shrivelled, and it is probably this mechanical change that triggers neural activity. The osmoreceptors are located in the hypothalamus just above the pituitary gland and they stimulate the release of ADH from the pituitary. This hor-

more signals the kidneys to reabsorb water from urine into the bloodstream where it dilutes the chemical concentration of the blood and body fluids.



CONTROL MECHANISM FOR REGULATION OF WATER INTAKE.

Dehydration of the osmoreceptors can be produced by depriving the organism of water or by injecting a salt solution directly into the brain. Even a minute amount of salt solution injected into the preoptic region of a goat's hypothalamus will cause the animal to drink several gallons of water. A thirsty animal prefers a drink of cool water, but other fluids are accepted provided they are not more concentrated than normal body fluids. Lemonade is an acceptable thirst quencher; salty bouillon is not.

Volumetric Receptors. Two physiological states stimulate the hypothalamus to elicit drinking: cellular dehydration and decreased blood volume. Cellular dehydration acts directly on the hypothalamus to start the organisms drinking and to stimulate the release of ADH from the pituitary, which tells the kidneys to reabsorb water into the bloodstream; when angiotensin reaches the hypothalamus it stimulates drinking. When the body fluids become too concentrated, the brain signals the kidneys to retrieve water from the urine and send it back into the bloodstream; when the blood volume is low, the kidneys

signal the brain to start the organism drinking. In both instances a hormone carries the message.

It is not certain what neural mechanisms tell the organism to stop drinking. Cellular hydration, the reverse of dehydration, cannot be the main stimulus for thirst satiation because the organisms stops drinking long before fluid has time to enter the cells in any significant amount. Signals from the mouth, as it meters the amount of incoming water, and from the stomach, as it is distended by water, may operate to depress the activity of the hypothalamic cells responsible for drinking. But the exact process is far from clear.

Sex

Another powerful motivator is sex which differs in many respects from hunger and thirst. Though sex is not vital to the survival of the organism, it is essential to the survival of the species. Eating and drinking serve to reduced tissue deficits. With sex, however there is no deficit, and sexual behaviour uses energy rather than restores it. Freud proposed that sexual energy or libido builds up within the organism and must find some outlet. But there is no physiological evidence to suggest a buildup of some substance which is reduced by sexual activity.

Sexual behaviour depends upon a combination of internal factors (hormones and brain mechanisms) and external ones (learned and unlearned environmental stimuli). We will look first at the internal or physiological variables, and then discuss the function of external variables in sexual behaviour.

Physiological Basis of Sexual Behaviour

Hormones are responsible for the development and functioning of the reproductive organs. These hormones are controlled by the pituitary gland. Pituitary hormones in the female stimulate the ovaries to manufacture the sex hormones, estrogen and progesterone. In the male, pituitary hormones stimulate the cells of the testes to manufacture and secrete the sex hormone testosterone. The marked increase of these hormones at puberty produces changes in the primary and secondary sex characteristics at adolescence.

Although hormones influence sexual motivation, they do not elicit specific patterns of sexual behaviour. Rather, they produce a state of readiness to respond appropriately to certain classes of stimuli. Administration of hormones of the opposite sex has been shown to reverse sexual response patterns in a number of different species. Female dogs given testosterone show mounting behaviour; male rats given estrogen show the typical female response pattern. But often the administration of hormones of the opposite sex produces bisexual behaviour, the direction of which is largely controlled by environmental stimuli. Female rats treated with male hormones will show male sexual behaviour in the presence of normal females, but will revert to the female pattern when confronted with a sexually aggressive male. Similarly, estrogen treated males will adjust their sex role in accordance with the sex of their partner. Hormones prime the animal to act, but the behaviour displayed is largely determined by environmental stimuli.

Neural Mechanisms

Neural control of sexual behaviour is complex, and the mechanisms that influence sexual behaviour vary considerably among different species. Some of the basic reflexes (such as

erection, pelvic movements, and ejaculation in the male) are controlled at the level of the spinal cord and do not involve the brain. Men whose spinal cord has been severed from the brain by injury (paraplegics) are still capable of these movements. Much of the regulation of sexual arousal and behaviour takes place in the hypothalamus, although the extra area varies across species. Electrical stimulation of the posterior hypothalamus of the rat produces not only copulation but the entire repertoire of sexual behaviour. Male rats stimulated in the area do not mount indiscriminately but court the female by nibbling her ears and nipping the back of her neck until she responds. Intromission and ejaculation follow unless the electrical stimulation is terminated. Even a sexually satiated male rat will respond to stimulation by pressing a bar to open the door leading to the female, courting and mating her.

Role of Experience

Mating behaviour in the lower mammals is less affected by experiences. As we go from the lower to higher mammals experience and learning play increasingly important roles in sexual behaviour. Young monkeys raised together show in their play with each other many of the postural responses necessary for later copulation. In wrestling with their peers infant male monkeys display hindquarter grasping and thrusting responses that later become components of the adult sexual pattern. Infant female monkeys retreat when treated by an aggressive-male infant, and stand steadfastly in a posture that is a forerunner of the stance later required to support the weight of the male during copulation. These presexual responses appear as early as 60 days of age and increases in frequency and refinement as the monkeys mature. Their early appearance suggests that they are innate responses to specific stimuli; the modification and refinement that occurs with experience indicates the role of learning in the development of the adult sexual pattern.

But humans shows a little difference in sexual experiences. Human infants first develop their feelings of trust and affection through a warm and loving relationship with the mother. This basic trust is necessary for satisfactory interactions with one's peers. And affectionate relationships with other youngsters of both sexes, prior to and during adolescence, lay the groundwork for the intimacy required for heterosexual relationships among young adults.

Cultural Influences

Cultural influences are strong determiners of human sexual behaviour. Every society, even the most primitive, places some restrictions on sexual behaviour. Incest (sexual relations within the immediate family) is prohibited by almost all cultures. Other aspects of sexual behaviour – sexual activities among children, homosexuality, masturbation, and premarital sex – are viewed with varying degrees of tolerance by different societies. Among preliterate cultures studied by anthropologists the amount and type of sexual activity considered acceptable vary widely. Some very permissive societies encourage autoerotic activities and sex play among children of both sexes. The children may be instructed in sex or allowed to observe adult sexual activity. The Chewa of Africa, for example, believe that unless children exercise themselves sexually they will be unable to produce offspring later.

PERCENTAGE OF MARRIED INDIVIDUALS EVER HAVING HAD PREMARITAL COITUS.

Age	Under 25	25-34	35-44	45-54	55-
Males	95	92	86	89	84
Females	81	65	41	36	31

ORAL-GENITAL FOREPLAY USED PREMARITALLY.

	1940s	1970s
Fellatio	33%	72%
Cunnilingus	14%	69%

SOCIAL MOTIVES

Social motives are otherwise known as acquired or learned motives. These some complex forms of motives, which result mainly from man's interaction with his social environment. These motives are called social because they learned in social groups. These peculiar human motives can be looked upon as general states that lead to particular behaviours. Social motives are general characteristics of a person and since they are learned motives, their strength differs greatly from one individual to other.

The social motive that will activate an individual is dependent upon an individual's own social experience. It is unique to himself and depends upon ways of perceiving things. For this reason, the psychologists have always found it difficult to arrive at a commonly agreed list of social motives as they have done in case of biological motives.

Since these motives are inferred from behaviour, it is very difficult to measure these motives. This type of difficulty stems from the problem that we do not know which particular type of behaviour is associated with which particular motive.

Need for Achievement

Need for achievement (N-Ach) refers to an individual's desire for significant accomplishment, mastering of skills, control, or high standards. The term was first used by Henry Murray and associated with a range of actions. These include: "intense, prolonged and repeated efforts to accomplish something difficult. To work with singleness of purpose towards a high and distant goal. To have the determination to win". The concept of N-Ach was subsequently popularised by the psychologist David McClelland.

This personality trait is characterized by an enduring and consistent concern with setting and meeting high standards of achievement. This need is influenced by internal drive for action (intrinsic motivation), and the pressure exerted by the expectations of others (extrinsic motivation). Measured by thematic appreciation tests, need for achievement motivates an individual to succeed in competition, and to excel in activities important to him or her.

Need for Achievement is related to the difficulty of tasks people choose to undertake. Those with low N-Ach may choose very easy tasks, in order to minimise risk of failure, or highly difficult tasks, such that a failure would not be embarrassing. Those with high N-Ach tend to choose moderately difficult tasks, feeling that they are challenging, but within reach.

People high in N-Ach are characterised by a tendency to seek challenges and a high degree of independence. Their most satisfying reward is the recognition of their achievements. Sources of high N-Ach include:

1. Parents who encouraged independence in childhood
2. Praise and rewards for success
3. Association of achievement with positive feelings
4. Association of achievement with one's own competence and effort, not luck
5. A desire to be effective or challenged
6. Intrapersonal Strength
7. Desirability
8. Feasibility
9. Goal Setting Abilities

The pioneering research work of the Harvard Psychological Clinic in the 1930s, summarised in *Explorations in Personality*, provided the start point for future studies of personality, especially those relating to needs and motives. David C. McClelland's and his associates' investigations of achievement motivation have particular relevance to the emergence of leadership. McClelland was interested in the possibility of deliberately arousing a motive to achieve in an attempt to explain how individuals express their preferences for particular outcomes – a general problem of motivation. In this connection, the need for achievement refers to an individual's preference for success under conditions of competition. The vehicle McClelland employed to establish the presence of an achievement motive was the type of fantasy a person expressed on the Thematic Apperception Test (TAT), developed by Christiana Morgan and Henry Murray, who note in *Explorations in Personality* that "...when a person interprets an ambiguous social situation he is apt to expose his own personality as much as the phenomenon to which he is attending... Each picture should suggest some critical situation and be effective in evoking a fantasy relating to it". The test is composed of a series of pictures that subjects are asked to interpret and describe to the psychologist. The TAT has been widely used to support assessment of needs and motives.

In 1961 McClelland published *The Achieving Society*, which articulated his model of human motivation. McClelland contended that three dominant needs -for achievement, for power, and for affiliation- underpin human motivation. McClelland believed that the relative importance of each need varies among individuals and cultures. Arguing that commonly used hiring tests using IQ and personality assessments were poor predictors of competency, McClelland proposed that companies should base hiring decisions on demonstrated competency in relevant fields, rather than on standardized test scores. Iconoclastic in their time, McClelland's ideas have become standard practice in many corporations.

The procedure in McClelland's initial investigation was to arouse in the test audience a concern with their achievement. A control group was used in which arousal was omitted. In the course of this experiment, McClelland discovered through analyzing the stories on the TAT that initial arousal was not necessary. Instead, members of the control group – individuals who had no prior arousal – demonstrated significant differences in their stories, some writing stories with a high achievement content and some submitting stories with a low achievement content. Using results based on the Thematic Apper-

ception Test, McClelland demonstrated that individuals in a society can be grouped into high achievers and low achievers based on their scores on what he called “N-Ach”.

McClelland and his associates have since extended their work in fantasy analysis to include different age groups, occupational groups, and nationalities in their investigations of the strength of need for achievement. These investigations have indicated that the N-Ach score increases with a rise in occupational level. Invariably, businessmen, managers, and entrepreneurs are high scorers. Other investigations into the characteristics of the high achievers have revealed that accomplishment on the job represents an end in itself; monetary rewards serve as an index of this accomplishment. In addition, these other studies found that the high achievers, though identified as managers, businessmen, and entrepreneurs, are not gamblers. A high emotional intelligence calls for a high need for achievement while a low emotional intelligence calls for a lower need for achievement. They will accept risk only to the degree they believe their personal contributions will make a difference in the final outcome.

An experiment realized to entry level managers of AT&T from 1956 to 1960, studied the level of achievement attained during a period of 8 to 16 years, showing that High n-Achievement was associated with managerial success at lower levels of management jobs, in which promotion depends more on individual contributions than it does at higher levels. At the higher levels, in which promotion depends on demonstrated ability to manage others, a high n-Achievement is not associated with success; by contrast, the leadership motive pattern is so associated, in all likelihood because it involves a high n-Power, emerging as a concern for influencing people.

These explorations into the achievement motive seem to turn naturally into the investigation of national differences based on Max Weber’s thesis that the industrialization and economic development of the Western nations were related to the Protestant ethic and its corresponding values supporting work and achievement. McClelland and his associates have satisfied themselves that such a relationship, viewed historically through an index of national power consumption, indeed exists. Differences related to individual, as well as to national, accomplishments depend on the presence or absence of an achievement motive in addition to economic resources or the infusion of financial assistance. High achievers can be viewed as satisfying a need for self-actualization through accomplishments in their job assignments as a result of their particular knowledge, their particular experiences, and the particular environments in which they have lived.

Measurement

The techniques McClelland and his collaborators developed to measure N-Ach, N-Affil and N-Pow can be viewed as a radical break with the dominant psychometric tradition. However, it should be recognised that McClelland’s thinking was strongly influenced by the pioneering work of Henry Murray, both in terms of Murray’s model of human needs and motivational processes (1938) and his work with the OSS during World War Two. It was during this period that Murray introduced the idea of “situation tests” and multi-rater/multi-method assessments. It was Murray who first identified the significance of Need for Achievement, Power and Affiliation and placed these in the context of an integrated motivational model.

Whilst trait-based personality theory assume that high-level competencies like initiative, creativity, and leadership can be assessed using “internally consistent” measures (see psychometrics), the McClelland measures recognize that such competencies are

difficult and demanding activities which will neither be developed nor displayed unless people are undertaking activities they care about (*i.e.* are strongly motivated to undertake). Furthermore, it is the cumulative number of independent, but cumulative and substitutable, components of competence they bring to bear while seeking to carry out these activities that will determine their success. Accordingly, the N-Ach, N-Aff and N-Pow scoring systems simply count how many components of competence people bring to bear whilst carrying out activities they have a strong personal inclination (or motivation) to undertake.

McClelland's research led him to formulate psychological characteristics of persons with strong need for achievement. According to McClelland and David Winter (*Motivating Economic Achievement*), the following features accompany high level of achievement motivation: - Moderate risk propensity; - Undertaking innovative and engaging tasks; - Internal locus of control and responsibility for own decisions and behaviours; - Need for precise goal setting.

An important corollary is that there is no point in trying to assess people's abilities without first finding out what they care about. So one cannot (as some psychometricians try to do) assess such things as "creativity" in any general sense. One has always to ask "creativity in relation to what?" So McClelland's measures, originally presented as means of assessing "personality", are best understood as means of measuring competence in ways which break radically with traditional psychometric approaches. Everybody has different goals and needs for achievement based off many factors like past experiences, education, the risks, and the feedback they will receive.

Need for Affiliation

The **Need for affiliation (N-Affil)** is a term that was popularized by David McClelland and describes a person's need to feel a sense of involvement and "belonging" within a social group; McClelland's thinking was strongly influenced by the pioneering work of Henry Murray who first identified underlying psychological human needs and motivational processes. It was Murray who set out a taxonomy of needs, including achievement, power and affiliation – and placed these in the context of an integrated motivational model. People with a high need for affiliation require warm interpersonal relationships and approval from those with whom they have regular contact. People who place high emphasis on affiliation tend to be supportive team members, but may be less effective in leadership positions.

Affiliation is a positive, sometimes intimate, personal relationship. Affiliation can include "concern over establishing, maintaining, or restoring a positive affective relationship with another person or persons."

Situations

There are many situations in which people feel a need for affiliation. One situation that causes a greater need for affiliation is during a stressful situation. An example where there was an increase in the need for affiliation among individuals was right after the September 11 terrorist attack on the World Trade Center. This event led to Americans' putting their differences aside and coming together. The increase in an individual's need for affiliation allowed individuals responding to the same stressor to come together and find security in one another. Situations that include fear often lead people to want to

be together and trigger a need for affiliation. Research done by Schacter shows that fear that comes from anxiety increases the need for the person to affiliate with others who are going through the same situation or that could help them through the stressful event. The strength of this need changes from one person to the next, there are moments that people just want to be together.

The need for affiliation for an individual can vary over short amounts of time; there are times when individuals wish to be with others and other times to be alone. In one study, completed by Shawn O'Connor and Lorne Rosenblood, beepers were distributed to the students. The students were then asked to record, when their beepers went off, whether or not they wanted to be alone or if they wanted to be with others at that particular moment. This study was done to observe how frequently college students were in the presence of others and how frequently they were alone. The next step in this study asked for the students to record whether, at the time their beeper went off, they wanted to be alone or in the company of others. This response that they gave usually reflected which of the two situations they were experiencing the next time their beepers went off. The information retained from this study helped to show the strength of an individual's need for affiliation. By showing how frequently they obtained the presence of others when they felt that it was what they wanted at that moment it showed how strong their need for affiliation was at that particular moment.

Depending on the specific circumstances, an individual's level of need for affiliation can become increased or decreased. Yacov Rofo suggested that the need for affiliation depended on whether being with others would be useful for the situation or not. When the presence of other people was seen as being helpful in relieving an individual from some of the negative aspects of the stressor, an individual's desire to affiliate increases. However, if being with others may increase the negative aspects such as adding the possibility of embarrassment to the already present stressor, the individual's desire to affiliate with others decreases. Individuals are motivated to find and create a specific amount of social interactions. Each individual desires a different amount of a need for affiliation and they desire an optimal balance of time to their self and time spent with others.

Need for Power

As an independent motive, the need for power expresses itself behaviours, which tend to control and influence the course of events including the behaviours of others. Traditionally it was believed that power was desired by people as an instrument to satisfy other motives like greed, aggression affiliation etc.

In the past, we have seen that mankind has always struggled for power. Cities have been destroyed and wars waged by people in reckless manner to increase their power. However, in recent years, emphasis has been placed on power motive as independent in itself and not derivable from other motives. Such a view was expressed by McClelland.

David McClelland has also described four general ways in which power motivation is expressed:

People do things to gain feelings of power and strength from sources outside themselves. For example, we express power motivation by read stories about sports or we may attach ourselves to a leader from who can draw strength.

People do things to gain feelings of power and strength from sources within themselves. For example, an individual may express power motivation building up the

body and by mastering urges and impulses. A person may express power by trying to gain control over things- collecting pistols, fancy, cars, and credit cards and so on. People do things as individuals to have impact on others. For example person may argue with another person or may have competitive attitude another individual in order to have impact or influence on that person.

People do things as members of organizations to have an impact on others. For example, the leader of a political party may use the party apparatus to influence others or an army officer may express the need for power through the chain of command.

For any individual, one of these ways of expressing power motivation may dominate. But a combination of power motives cannot be ruled out. However, ever dominant mode of expression often changes with age and life experiences a person.

Different studies have shown that women seem to have less strong needs for power than men. Generally women are said to express their power motivation by being counsellors, advisors and resource persons for other people. In this way, they have indirect impact and influence.

Dependency Motive

Dependency refers to interpersonal relationships where an individual behaves in a way in order to gain attention, reassurance, assistance, comfort, and support from fellow men. People use to spend more time with parents, or intimate friends when in difficult situations. They seek advice and mutually spend upon one another in working on the solution to various problems. In her words, individuals appear to be more dependent on social interactions and approval. Research studies indicate that the girls tend to be more dependent and more affiliative than boys.

At times of stress people want to resort to dependency, *i.e.* they are in need of someone's care, and thus rely on the other (or others) for emotional, economic, and other support like love and security etc.

Cooperation Motive

Cooperation is an acquired motive. It is a condition manifested when two or more individuals or groups work together to achieve a common goal, cooperation signifies lack of mutual disagreement and opposition among fellow group-members, and absence of rivalry. Citizens of Zuni of New Mexico, Pacific found to be extremely cooperative. Golman reports that being wealthy in Zuni brings no status.

On the other hand status is derived not from power or landed property but from friendships. A happy and successful Zuni is who has the most friends.

Studies of altruism among children provide evidence that helping-behaviour can be fostered through the use of models. Peulson compared 36 children who had watched all programmes in the Sesame Street Series with those who had not been exposed to the series. He found significantly more cooperative behaviour among the experimental group.

Conformity Motive

Conformity generally refers to the tendency to allow one's opinions, attitudes, actions, and even perceptions to be affected by prevailing opinion; attitudes, actions and perceptions. There is behavioural conformity, *i.e.*, the tendency to "go along with the group" – to act in ways consistent with the majority. When changes in attitude, and belief take place

because of pressure from significant others, attitudinal conformity takes place. There is also conformity of personality traits, *i.e.*, underlying characteristics of a person undergo changes according to the norms of society.

F.H. Allport described, the conformity motive phenomena with the help of a conformity curve, which reflects that most people exhibit complete conformity to social norms with fewer and fewer people having deviation. Norms are rules or laws or implicit moral standards of a society. Norms refer to behaviour that is usual, or expected, acceptable and socially prescribe. Conformity to social norms takes many forms such as “Honor your mother or “Love your country”, which are very encouraging. Norms are discouraging, *i.e.*, “Do not steal”, “Do not lie”. They take widely held belief and opinions, and are manifested in forms of dress, speech, manners, and fashions. Our submissiveness to social influences is due to conformity motives to the norms of society in which we live.

EXPLORATION BEHAVIOUR AND CURIOSITY

What makes people curious? Why do individuals explore the unknown? The research in these areas is inconclusive and often contradictory. Are curiosity and exploration motivations or drives? Can curiosity and exploration be operationally defined independent of one another? Motivation is defined as the arousal, direction and persistence of behaviour (Franken, 1994); an internal state or condition that activates behaviour and gives it direction; desire or want, that energizes and directs goal-oriented behaviour; the influence of needs and desires on the intensity and direction of behaviour. Drive is defined as a basic or instinctive need; a vigorous effort toward a goal; to cause and guide the movement.

Curiosity is defined as a need, thirst or desire for knowledge. The concept of curiosity is central to motivation. The term can be used as both a description of a specific behaviour as well as a hypothetical construct to explain the same behaviour. Berlyne (1960) believes that curiosity is a motivational prerequisite for exploratory behaviour. The term curiosity is used both as a description of a specific behaviour as well as a hypothetical construct to explain the same behaviour. Exploration refers to all activities concerned with gathering information about the environment. This leads to the conflict and question of whether exploratory behaviour should be defined in terms of the movements that an animal or human performs while exploring or in terms of the goal or purpose of the behaviour observed. A clear distinction between these two may not always be possible.

What exactly are curiosity and exploration? Loewenstein points out four central issues of curiosity: definition and dimensionally, cause, voluntary exposure to curiosity, and situational determinants. He adds a fifth issue of superficiality and intensity since he states that curiosity can arise, change focus or end abruptly. Loewenstein believes that despite its transience, curiosity can be a powerful motivational force. “Curiosity often produces impulse behaviour and attempts at self control”. Just look at the stories of Pandora and Eve, in which curiosity causes people to expose themselves knowingly to terrible consequences. Langevin (1971) has conducted research in the area of curiosity and classifies measures of curiosity into two categories. First, curiosity is viewed as a motivational state and measured with behavioural indices. Second, he conceptualizes curiosity as a personality trait that is assessed by personality measures. It has been suggested that curiosity is not a unitary construct. At the conceptual level there are

numerous definitions of curiosity which tend to encompass a broad range of characteristics. For Fowler, boredom is one prerequisite or motivation for curiosity (exploration).

Curiosity and exploration are difficult to define independently when looking at them from a psychological perspective, the concepts motivation and drive come into play and become intertwined. The underlying problem in using and defining (scientifically and socially) is the circularity of the terms.

Roget's Thesaurus says that the absence of curiosity is boredom, ennui, satiety, take no interest, mind one's own business, uninquisitive.

The earliest discussions of curiosity were conducted by philosophers and religious thinkers and centered on the question of curiosity's moral status rather than on its psychological underpinnings. Cicero referred to curiosity as a "passion for learning" and argued that the story of Ulysses and the Sirens was really a parable about curiosity. Several forms of curiosity related behaviour such as search behaviour, movement toward an unknown object and asking questions are included in the area of motivational psychology, however, curiosity does not fit well into the conceptual framework developed along the traditional pathways of behavioural sciences. Firstly, the conception of an intrinsically motivated behavioural system, which cannot be linked to a reducible drive raises serious questions about motivational psychology since the 1950's. The idea of curiosity was rediscovered when laboratory researchers wondered about the maze activities of the lab rat when none of the drive states such as thirst or hunger were aroused. Secondly, Wohlwill states that the curiosity phenomena cannot be investigated without reference to the natural environment of an individual. Before 1950 curiosity was seen in the light of its social function, for example: the eagerness or greed to get to know something new for the sake of newness, and in early psychological literature the term curiosity had a negative connotation. The scientific term "curiosity" is more neutral.

The most basic problem that has occupied curiosity researchers and theorists is the underlying cause of curiosity. Is curiosity a primary or secondary drive? A primary drive is inborn or innate whereas a secondary drive is learned or acquired. The research is inconclusive. If secondary, from what more basic drive or motive does it derive? Older theories oriented toward instinct and drive concepts. The defining feature is that curiosity produces an unpleasant sensation (usually labeled arousal) that is reduced by exploratory behaviour.

William James (1890) pointed out two kinds of curiosity. He emphasized the biological function of curiosity as a mechanism of instinct driven behaviour that serves in approaching new objects. Approach and exploration are described as being characteristic forms of behaviour. The second kind of curiosity pointed out by James is "scientific curiosity" and "metaphysical wonder" with which "the practical instinctive root has probably nothing to do" rather "the philosophical brain responds to an inconsistency or a gap in its knowledge".

In the psychoanalytical literature Freud views curiosity as a derivative of the sex drive. The partial impulse of looking motivates the child's great interest in all things and all events that have to do with sexuality. Whereas the looking impulse and curiosity are primarily sexual in origin, the child's exploratory interest and desire for knowledge can be considered to be a by product of cognitive development. Due to social pressure, sexual exploration is later abandoned.

Blarer (1951) states that the inhibition of curiosity may result in different forms of pathological behaviour, such as depression, and higher levels of sensation-seeking or

thrill seeking behaviour. Blarer proposed curiosity to be intrinsic to the individuals perceptions and world experiences and thus Blarer is the basis for the intrinsic motivation viewpoint in curiosity theory.

Theoretical Conceptions of Curiosity & Exploration

The postulation of an independent exploratory drive is based on the observation of the exploratory activities of animals in situations where there were no external stimuli to satisfy homeostatic drives (water or food). Nissen experimented with rats and defines exploratory behaviour in two ways 1. as an inborn exploratory drive, 2. that curiosity is a secondary or learned drive acquired through classical conditioning. Originally, Berlyne thought that the aversive and drive-reducing effect of deviations of the arousal potential from the individual's optimum level as the underlying mechanism of curiosity. Since then, Berlyne has come to believe that curiosity is externally stimulated, and that the curiosity drive is aroused by external stimuli specifically stimulus conflict. This encompasses complexity, novelty and surprise. Berlyne believed that in the short term, stimulus change and novelty is accompanied by physiological change. However, over longer periods of time, investigating behaviours are not accompanied by readily identifiable physiological changes. Berlyne also holds that exploratory behaviour serves to maintain or attain a medium to optimal activation level for the organism. In all cases where exploration takes place, arousal or desire is reduced.

Fiske and Maddi hold a medium arousal level model and differentiate between the terms arousal and activation. They define arousal as "diverse manifestations of activation, such as muscle tone, heart rate, and increased sensitivity for stimuli". Their definition of activation is "the state of a catalytic and energizing mechanism in the central nervous system".

McReynolds bases his theories of exploratory behaviour on animal experiments. Motivational aspects of exploratory behaviour, for example: a living being is active in order to receive new perceptual information from its environment, as well as adaptive aspects, for example: a living being is in a situation of stimuli that it must regulate and adapt to. Fowler's boredom-based perspective interprets curiosity as a homeostatic drive (internally stimulated) since the curiosity drive seems to be both evoked and satisfied by the same stimuli. He attacked Berlyne by noting the inherent contradiction in the view that the curiosity drive was both evoked and satisfied by the same stimuli. Theorists that believe that curiosity is externally stimulated were "forced to ascribe both drive-eliciting and reinforcing properties to the same stimuli- namely the novel stimuli for which the animal responded". Fowler observed animals producing the exploration- initiating response before, rather than after, exposure to the stimulus.

Hunt (1963) states that curiosity refers to a "motivation inherent in information processing" this means that curiosity is a mixture of cognition and motivation. The main principle, which is equally as important as the drive reduction hypothesis is the establishment and maintenance of an optimal amount of incongruence. Incongruence determines the strength, direction and affective qualities of behaviour.

Drive Theories

Drive theories differ on whether they view curiosity as a primary or secondary drive. Some research has shown that unsatisfied curiosity tends to intensify over some interval as do other drives such as hunger and thirst.

Curiosity has a motivated force that is stimulated internally (boredom) or by external stimuli. However, Hebb believes that curiosity seeking behaviour poses a paradox for drive based accounts of curiosity. "Drive is not simply a state the decrease of which is rewarding. At high levels the reduction of drive is rewarding, but at low levels, an increase may be rewarding" (Hebb, 1955).

What is the role of homeostatic drives to curiosity? Harlow states that exploration is an example of human motivation that is independent from homeostatic drives. Harlow's non-homeostatic intrinsic drive theory has been attacked by drive and learning theorists. Kreitler and Kreitler (1976) have changed positions from the basic assumption of drive theory to a more cognitive process in the development of exploratory behaviour.

Curiosity & Culture

There is evidence for cross-cultural similarities in exploratory behaviour. However, cultures generally vary both in attitudes towards exploration and information seeking as well as in the range of situations allowing the expression of the various manifestations of exploration and curiosity, this is especially true for the sensation-seeking motive. Zuckerman defines sensation seeking as "the seeking of varied, novel, complex and intense sensations and experiences, and the willingness to take physical, social, legal, and financial risks for the sake of such experiences."

Berlyne also conducted research on cross cultural comparisons in the area of curiosity. His findings conclude that there is a high similarity of demand characteristics of stimuli in two cultures of widely differing historical antecedents and technological development. Also, different cultures from various geographical regions show evidence for cross-cultural similarities in exploratory behaviour. More research is needed to study curiosity behaviour in its own cultural context to gain a better understanding of the functional relationships between various environmental and social facilitators and inhibitors of curiosity in a given society.

There are varying points of view when it comes to theoretically conceptualizing curiosity and exploration. Research findings seem to be dependent on the theoretical orientation of the theorist and the emphasis lies on internal or external stimulus conditions, primary-inborn or secondary-acquired drives, and homeostatic-biogenetic versus nonhomeostatic-psychological motivations.

This is a difficult topic to conceptualize because of the circular nature of the terms and the contradiction and inconclusiveness of the research. Curiosity, exploration, motivation and drive are defined, described, explained and operationally defined in terms of one another, and thus become embedded and intertwined.

PHYSIOLOGICAL CORRELATES OF EMOTION

During intense emotion, we become aware of a number of physiological changes resulted from activation of the sympathetic division of the autonomic nervous system as it prepares the body for emergency action. The sympathetic system gears the organism for energy output. As the emotion subsides, the parasympathetic system, the energy-conserving system, takes over and returns the organism to its normal state.

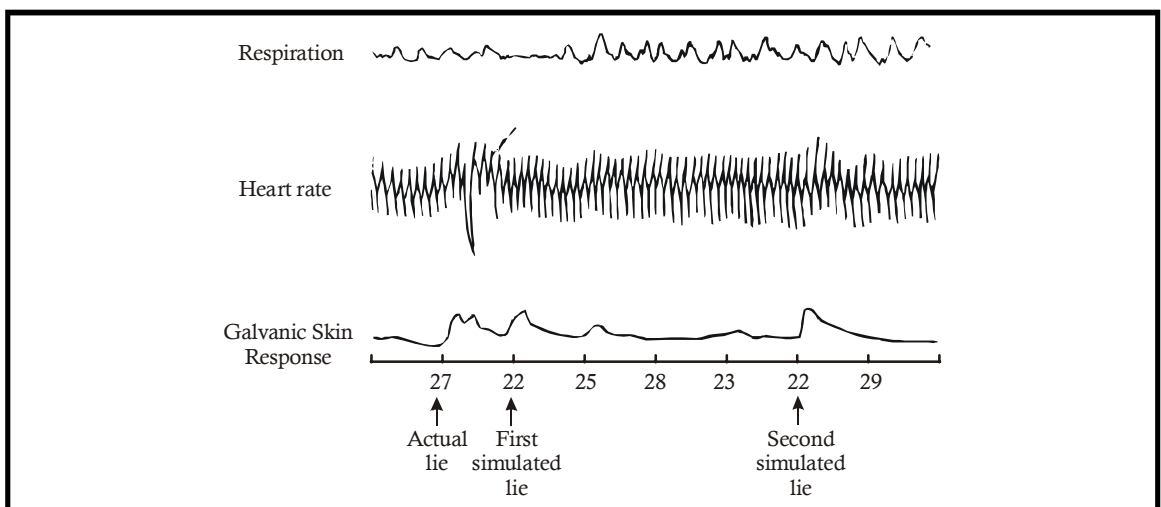
The sympathetic system is responsible for the following changes:

1. Blood pressure and heart rate increase.

2. Respiration becomes more rapid.
3. The pupils of the eyes dilate.
4. Electrical resistance of the skin decreases.
5. Blood sugar level increases to provide more energy.
6. The blood becomes able to clot more quickly in case of wounds.
7. Motility of the gastrointestinal tract decreases or stops entirely; blood is diverted from the stomach and intestines and sent to the brain and skeletal muscles.
8. The hairs on the skin erect, causing "goose pimples."

SYMPTOMS OF FEAR IN COMBAT FLYING.

During Cobat Missions did you Feel	Often	Sometimes	Total
A pounding heart and rapid pulse	30%	56%	86%
That your muscles were very tense	30	53	83
Easily irritated, angry, or "sore"	22	58	80
Dryness of the throat or mouth	30	50	80
"Nervous perspiration" or "cold sweat"	26	53	79
"Butterflies" in the stomach	23	53	76
Sense of unreality, that this couldn't be happening to you	20	49	69
Need to urinate very frequently	25	40	65
Trembling	11	53	64
Confused or rattled	3	50	53
Weak or faint	4	37	41
Right after a mission, unable to remember details of what happened	5	34	39
Sick to the stomach	5	33	38
Unable to concentrate	3	32	35
That you had wet or soiled your pants	1	4	5



THE POLYGRAPH.

These physiological changes accompanying intense emotions are the basis for use of the polygraph, commonly known as the “lie detector,” in checking the reliability of an individual’s statements. The term lie detector is actually incorrect. The polygraph does not detect lies; it simply measures some of the physiological accompaniments of emotion. The measures most frequently recorded are alternations in heart rate, blood pressure, respiration, and the galvanic skin response, or GSR (the GSR is a change in the electrical conductivity of the skin).

The standard procedure in operating a polygraph is to first take a recording while the subject is relaxed; this recording serves as a base line for evaluating subsequent responses. The examiner then asks a series of carefully worded questions that are answered “yes” or “no” “Critical questions” are interspersed among “neutral questions,” and sufficient time is allowed between questions (usually a minute) for the measures to return to normal. Presumably, the subject’s guilt is revealed by heightened physiological responses to the critical questions.

But a practiced liar may show little emotional response when lying. And a knowledgeable subject may be able to “beat” the machine by thinking about something exciting or by tensing muscles during neutral questions, thus creating a baseline comparable to reactions to the critical questions. The recording in the right half of above figure shows the responses to an actual lie and a simulated lie. In this experiment the subject picked a number and then tried to conceal its identity from the examiner. The number was 27, and a marked change in heart rate and GSR can be seen where the subjects denies number 27. The subject simulates lying to number 22 by tensing his toes, producing noticeable reactions in heart rate and GSR.

Because of these and other problems, lie detector results have been ruled inadmissible as court evidence. Such tests are frequently used, however in preliminary criminal investigations and by prospective employers hiring personnel for trusted positions. The polygraph also serves as a research tool in studies measuring emotional responses to stress or observing the effects of drugs on physiological responses.

THEORIES OF EMOTIONS

1. James-Lange Theory

This theory states that the perception of the physiological changes is the emotion. William James, a famous psychologist at Harvard during the late 1800s, believed that the important factor in our felt emotion is the feedback from the bodily changes that occur in response to a frightening or upsetting situation.



JAMES-LANGE THEORY.

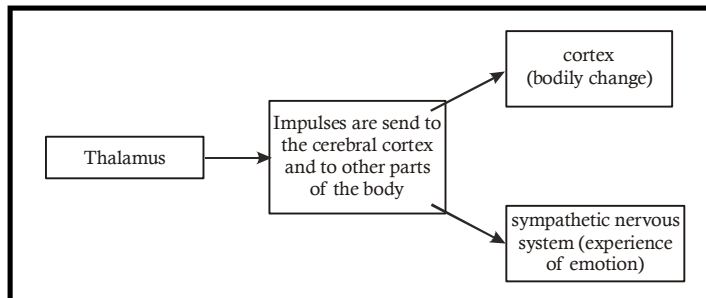
He stated this theory in a form that seems to put the cart before the horse: “We are afraid because we run.” “We are angry because we strike.” A Danish physiologist named

Carl Lange arrived at a similar proposal about the same time, and the theory is referred to as the James-Lange theory.

We can think of instances where the recognition of emotion does follow bodily responses. If you stumble on the stairs, you automatically grab for the handrail before you have time to recognize a state of fear. Your felt emotion, after the crisis is over, includes the perception of a pounding heart, rapid breathing, and a feeling of weakness or trembling in the arms and legs. Because the feeling of fear follows the bodily responses, this situation gives some plausibility to the James-Lange theory.

2. Cannon-Bard Theory

The major objections to the James-Lange theory came from Walter Cannon, a physiologist at the University of Chicago, who pointed out that : (1) bodily changes do not seem to differ very much from one emotional state to another despite the fact that we as individuals are usually pretty clear about which emotion we are experiencing; (2) the internal organs are relatively insensitive structures not well supplied with nerves, and internal changes occur too slowly to be a source of emotional feeling; (3) artificially inducing the bodily changes associated with an emotion (*e.g.*, injecting a person with adrenaline) does not produce the experience of the true emotion.



CANNON'S THEORY.

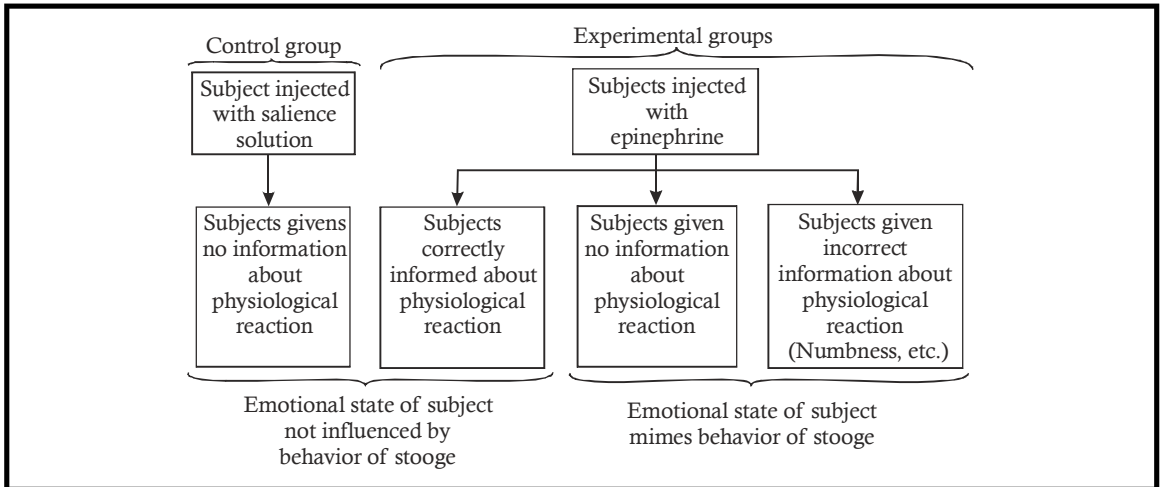
Cannon assigned the central role in emotion to the thalamus. He suggested that the thalamus responded to an emotion-producing stimulus by sending impulses simultaneously to the cerebral cortex and to other parts of the body; emotional feelings were the result of joint arousal of the cortex and the sympathetic nervous system. According to this view the bodily changes and the experience of emotion occur at the same time.

3. Schachter's Theory

Schachter (1959) proposed that emotional states are a function of the interaction of cognitive factors and a state of physiological arousal. He suggested that if a person were injected with epinephrine without his knowledge, he would be under pressure to understand his bodily feelings – and would do so in terms of his knowledge of the immediate situation. If he were with a beautiful woman, he might decide he was in love or sexually excited. If he were in a confrontation with his wife, he might explode in anger. If he were at a party where everyone around him was acting euphoric, he might decide he was very happy. If the situation provided him with no reasonable explanation for his bodily feelings, he might simply decide he was sick. In any event Schachter proposed that an emo-

tion depends both on the state of physiological arousal and on a cognition appropriated to the state of arousal; the cognition determines the label of the emotion. He designed an experiment to test this hypothesis.

To produce physiological arousal, he injected some subjects with epinephrine, telling them it was a vitamin injection and that he was studying the effects of vitamin C on vision. Below figure illustrates the main experimental groups and summarizes the effect each condition had on the emotional state of the subject.



EPINEPHRINE INJECTION STUDY.

CONFLICTS

Conflict occurs between people in all kinds of human relationships and in all social settings. Because of the wide range of potential differences among people, the absence of conflict usually signals the absence of meaningful interaction. Conflict by itself is neither good nor bad. However, the manner in which conflict is handled determines whether it is constructive or destructive.

Conflict is defined as an incompatibility of goals or values between two or more parties in a relationship, combined with attempts to control each other and antagonistic feelings toward each other. The incompatibility or difference may exist in reality or may only be perceived by the parties involved. Nonetheless, the opposing actions and the hostile emotions are very real hallmarks of human conflict.

Conflict has the potential for either a great deal of destruction or much creativity and positive social change. Therefore, it is essential to understand the basic processes of conflict so that we can work to maximize productive outcomes and minimize destructive ones. This paper will briefly describe some common sources of conflict, the levels of social interaction at which conflict occurs, and the general strategies of approaching conflict that are available.

Sources of Conflict

Early reviews in the field of conflict resolution identified a large number of schemes for describing sources or types of conflict. One of the early theorists on conflict, Daniel Katz,

created a typology that distinguishes three main sources of conflict: economic, value, and power.

1. **Economic conflict** involves competing motives to attain scarce resources. Each party wants to get the most that it can, and the behaviour and emotions of each party are directed toward maximizing its gain. Union and management conflict often has as one of its sources the incompatible goals of how to slice up the “economic pie”.
2. **Value conflict** involves incompatibility in ways of life, ideologies – the preferences, principles and practices that people believe in. International conflict (*e.g.*, the Cold War) often has a strong value component, wherein each side asserts the rightness and superiority of its way of life and its political-economic system.
3. **Power conflict** occurs when each party wishes to maintain or maximize the amount of influence that it exerts in the relationship and the social setting. It is impossible for one party to be stronger without the other being weaker, at least in terms of direct influence over each other. Thus, a power struggle ensues which usually ends in a victory and defeat, or in a “stand-off” with a continuing state of tension. Power conflicts can occur between individuals, between groups or between nations, whenever one or both parties choose to take a power approach to the relationship. Power also enters into all conflict since the parties are attempting to control each other.

It must be noted that most conflicts are not of a pure type, but involve a mixture of sources. For example, union-management conflict typically involves economic competition, but may also take the form of a power struggle and often involves different ideologies or political values. The more sources that are involved, the more intense and intractable the conflict usually is.

Another important source of conflict is ineffective communication. Miscommunication and misunderstanding can create conflict even where there are no basic incompatibilities. In addition, parties may have different perceptions as to what are the facts in a situation, and until they share information and clarify their perceptions, resolution is impossible. Self-centeredness, selective perception, emotional bias, prejudices, etc., are all forces that lead us to perceive situations very differently from the other party. Lack of skill in communicating what we really mean in a clear and respectful fashion often results in confusion, hurt and anger, all of which simply feed the conflict process. Whether the conflict has objective sources or is due only to perceptual or communication problems, it is experienced as very real by the parties involved.

Escalation of Conflict

A final source of conflict is more additional than basic, that is, it comes in after the conflict has started. Conflicts have a definite tendency to escalate, *i.e.*, to become more intense and hostile, and to develop more issues, *i.e.*, what the parties say the conflict is about. Therefore, escalating conflicts become more difficult to manage. The process of escalation feeds on fear and defensiveness. Threat leads to counterthreat, usually with higher stakes at each go-round. Selective and distorted perception justifies a competitive and cautious approach as opposed to a trusting and cooperative one. Through Deutsch's crude law of social relations, competition breeds competition, rather than cooperation. The self-fulfilling prophecy comes into play. Each party believes in the evil intentions of the other and the inevitability of disagreement, and therefore takes precautionary ac-

tions which signal mistrust and competitiveness. When the other party then responds with a counteraction, this is perceived as justifying the initial precautionary measure, and a new spiral of action and counteraction begins. Through the norm of reciprocity, stronger attempts to control are met not only with stronger resistance, but more contentious attempts to gain the upper hand.

With each succeeding spiral of conflict, polarization grows and the parties become more adamant and intransigent in their approach to the situation. Even though the intensity of the conflict may moderate for periods of time, the issues remain, and a triggering event induces conflictual behaviour with negative consequences, and the conflict has moved one more step up the escalation staircase. When parties become “locked in” to a conflict they are usually unable to get out by themselves, and the intervention of a third party in the role of arbitrator, mediator or consultant may be required.

Levels of Conflict

Conflict can occur at a number of levels of human functioning. Conflict in your head between opposing motives or ideas is shown by your “internal dialogue” and is at the intrapersonal level. Beyond that, the primary concern here is with social conflict, *i.e.*, conflict between people whether they are acting as individuals, as members of groups, or as representatives of organizations or nations.

Interpersonal conflict occurs when two people have incompatible needs, goals, or approaches in their relationship. Communication breakdown is often an important source of interpersonal conflict and learning communication skills is valuable in preventing and resolving such difficulties. At the same time, very real differences occur between people that cannot be resolved by any amount of improved communication. “Personality conflict” refers to very strong differences in motives, values or styles in dealing with people that are not resolvable. For example, if both parties in a relationship have a high need for power and both want to be dominant in the relationship, there is no way for both to be satisfied, and a power struggle ensues. Common tactics used in interpersonal power struggles include the exaggerated use of rewards and punishments, deception and evasion, threats and emotional blackmail, and flattery or ingratiation. Unresolved power conflict usually recycles and escalates to the point of relationship breakdown and termination.

Role conflict involves very real differences in role definitions, expectations or responsibilities between individuals who are interdependent in a social system. If there are ambiguities in role definitions in an organization or unclear boundaries of responsibilities, then the stage is set for interpersonal friction between the persons involved. Unfortunately, the conflict is often misdiagnosed as interpersonal conflict rather than role conflict, and resolution is then complicated and misdirected. The emotional intensity is often quite high in role conflict since people are directly involved as individuals and there is a strong tendency to personalize the conflict.

Intergroup conflict occurs between collections of people such as ethnic or racial groups, departments or levels of decision making in the same organization, and union and management. Competition for scarce resources is a common source of intergroup conflict, and societies have developed numerous regulatory mechanisms, such as collective bargaining and mediation, for dealing with intergroup conflict in less disruptive ways. Social-psychological processes are very important in intergroup conflict. Group members tend to develop stereotypes (oversimplified negative beliefs) of the opposing group, tend to

blame them for their own problems (scapegoating), and practice discrimination against them. These classic symptoms of intergroup conflict can be just as evident in organizations as in race relations in community settings. Intergroup conflict is especially tense and prone to escalation and intractability when group identities are threatened. The costs of destructive intergroup conflict can be extremely high for a society in both economic and social terms.

Multi-party conflict occurs in societies when different interest groups and organizations have varying priorities over resource management and policy development. These complex conflicts typically involve a combination of economic, value and power sources. This complexity is often beyond the reach of traditional authoritative or adversarial procedures, and more collaborative approaches to building consensus are required for resolution.

International conflict occurs between states at the global level. Competition for resources certainly plays a part, but value and power conflict are often intertwined and sometimes predominate. The differences are articulated through the channels of diplomacy in a constant game of give and take, or threat and counterthreat, sometimes for the highest of stakes. Mechanisms of propaganda can lead to many of the same social-psychological distortions that characterize interpersonal and intergroup conflict.

Methods of Conflict Resolution

Regardless of the level of conflict, there are differing approaches to deal with the incompatibilities that exist. Conflict can result in destructive outcomes or creative ones depending on the approach that is taken. If we can manage conflict creatively, we can often find new solutions that are mutually satisfactory to both parties. Sometimes this will involve a distribution of resources or power that is more equitable than before, or in creating a larger pool of resources or forms of influence than before. Creative outcomes are more probable when the parties are interdependent, *i.e.*, each having some degree of independence and autonomy from which to influence the other, rather than one party being primarily dependent on the other. Given interdependence, three general strategies have been identified that the parties may take toward dealing with their conflict; win-lose, lose-lose, and win-win.

The **win-lose approach** is all too common. People learn the behaviours of destructive conflict early in life – competition, dominance, aggression and defense permeate many of our social relationships from the family to the school playground. The “fixed pie” assumption is made, often incorrectly, that what one party gains, the other loses. The strategy is thus to force the other side to capitulate. Sometimes, this is done through socially acceptable mechanisms such as majority vote, the authority of the leader, or the determination of a judge. Sometimes, it involves secret strategies, threat, innuendo – whatever works is acceptable, *i.e.*, the ends justify the means. There is often a strong we-they distinction accompanied by the classic symptoms of intergroup conflict. The valued outcome is to have a victor who is superior, and a vanquished who withdraws in shame, but who prepares very carefully for the next round. In the long run, everyone loses.

The **lose-lose strategy** is exemplified by smoothing over conflict or by reaching the simplest of compromises. In neither case is the creative potential of productive conflict resolution realized or explored. Disagreement is seen as inevitable, so therefore why not split the difference or smooth over difficulties in as painless a way as possible? Some-

times, this is indeed the reality of the situation, and the costs are less than in the win-lose approach, at least for the loser. Each party gets some of what it wants, and resigns itself to partial satisfaction. Neither side is aware that by confronting the conflict fully and cooperatively they might have created a more satisfying solution. Or the parties may realistically use this approach to divide limited resources or to forestall a win-lose escalation and outcome.

The **win-win approach** is a conscious and systematic attempt to maximize the goals of both parties through collaborative problem solving. The conflict is seen as a problem to be solved rather than a war to be won. The important distinction is we (both parties) versus the problem, rather than we (one party) versus they (the other party). This method focuses on the needs and constraints of both parties rather than emphasizing strategies designed to conquer. Full problem definition and analysis and development of alternatives precedes consensus decisions on mutually agreeable solutions. The parties work toward common and superordinate goals, *i.e.*, ones that can only be attained by both parties pulling together. There is an emphasis on the quality of the long term relationships between the parties, rather than short term accommodations. Communication is open and direct rather than secretive and calculating.

Threat and coercion are proscribed. The assumption is made that integrative agreements are possible given the full range of resources existing in the relationship. Attitudes and behaviours are directed toward an increase of trust and acceptance rather than an escalation of suspicion and hostility. The win-win approach requires a very high degree of patience and skill in human relations and problem solving.

Conflict is an inevitable fact of human existence. If we work to understand and manage it effectively, we can improve both the satisfaction and productivity of our social relationships.

Types of Conflicts

Sometimes conflict arises between a motive and a person's internal standards, rather than between two external goals. For example, sexual desires may conflict with one's standards of what constitutes acceptable social behaviour. A woman's motive for achievement may conflict with her standards for appropriate feminine behaviour. Conflicts between motives and internal standards can often be more difficult to resolve than conflicts between external goals.

Most conflicts involve goals that are simultaneously desirable and undesirable – both positive and negative. Candy is delicious, but fattening. Going off for a weekend of skiing is fun, but the consequences of lost study time can be anxiety-producing. The attitude toward a goal at once wanted and not wanted, liked and disliked, is called an ambivalent attitude. Ambivalent attitudes are very common: a young woman leaves home to escape parental domination, only to come back to receive parental protection her attitude toward her parents is ambivalent.

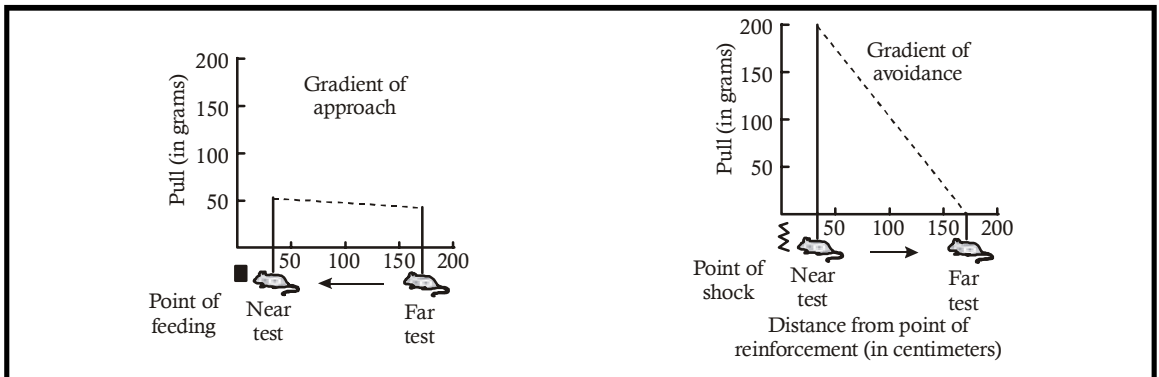
A person confronted by a goal that is at once attractive and dangerous vacillates while trying to decide what to do. The dangers seem less real when the goal is at a distance, so that its inviting aspects lead to approach reactions. But the sense of danger increases as the goal is approached so that nearer to the incentive one has a tendency to withdraw. This simultaneous tendency “to” and “from” leads to vacillation at some point near enough to the goal for one to be aware of the dangers but distant enough to be safe from them.

When a shy adolescent boy is about to call a girl for a date, he is drawn to the telephone by the possibility of success but his anxiety about possible rebuff mounts as he approaches. As a result he may make several false starts before he either carries through his plan or abandons it.

This type of conflict – called approach avoidance conflict – is illustrated by an experiment in which hungry rats were taught to run the length of an alley to obtain food in a goal box at the end of the alley; this training established approach reactions. Then the rats were given a brief electric shock while eating. The shock added avoidance tendencies to the approach tendencies and hence produced an approach-avoidance conflict. To test the resulting conflict behaviour, the rats were placed at the start of the alley. The characteristic behaviour corresponded to that predicted: the rat started in the direction of the goal box but came to a stop before reaching it. The place of stopping could be controlled by modifying the strength of either hunger or shock.

Gradients of Approach and Avoidance

The behaviour of the rats in the preceding experiment can be better understood by using the concept of gradients of approach and avoidance. By a gradient we mean a change in response strength as a function of the distance from the goal-object. The pull of a magnet upon a piece of iron is an analogous gradient. The pull (or strength of the gradient) increases as the distance between the piece of iron and the magnet is shortened.



GRADIENTS OF APPROACH AND AVOIDANCE.

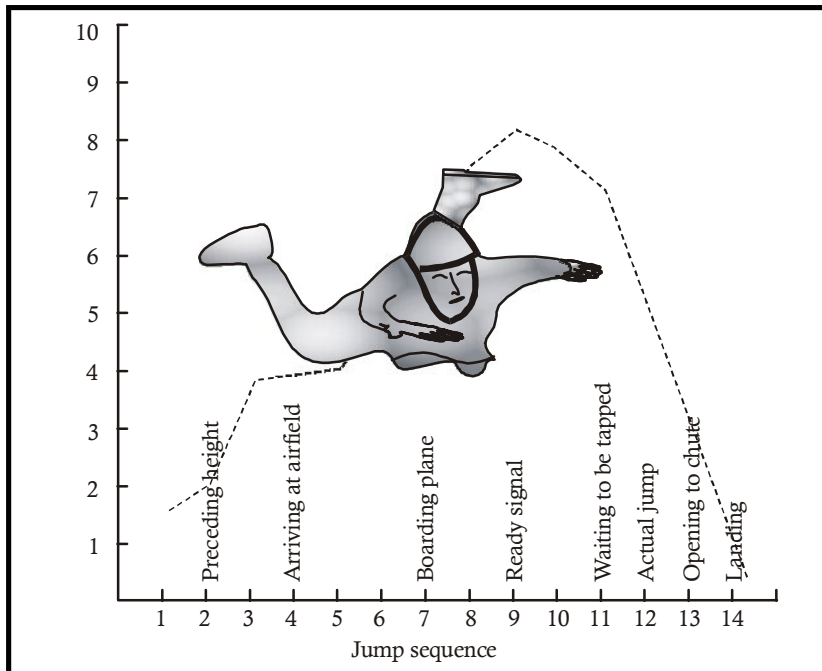
The difference between a test at a point near the goal box and that far from the goal box is shown in the gradient of avoidance in the bottom panel of above figure. Note that the slope of the two lines differs: the gradient for avoidance is much steeper than the one for approach.

The studies on gradients illustrate several principles important in the understanding of ambivalent behaviour.

1. The tendency to approach a positive incentive is stronger the nearer the subject is to it.
2. The tendency to go away from a negative incentive is stronger the nearer the subject is to it.
3. In some situations the strength of avoidance increases more rapidly with proximity than does that of approach. In other words, the avoidance gradient tends to be steeper than the approach gradient.

The third principle helps explain how a person may be repeatedly drawn back into an old conflict situation. At a distance the positive aspects seem more inviting than the negative ones appear forbidding. Everyone knows of couples who go steady, break up, make up again, only to break up once more. Away from each other, their mutual attraction takes precedence because negative feelings are reduced; close to each other, the negative feelings drive them apart. To an outsider it appears irrational that people who get along so poorly attempt reconciliation. Once the ambivalence of their attitudes is recognized, however, their attempts at reconciliation become understandable—even if not reasonable.

Studies of the reactions of skydivers to their first parachute jump show how negative feelings (fear of death or injury) become stronger as the moment of danger becomes more imminent.



REACTIONS OF PARACHUTISTS.

Interestingly enough, the peak of fear occurs not at the moment of greatest danger (in the free fall before the parachute opens) but at the point of final commitment after which it would be difficult for the parachutist to change his mind. Once the decision is made, avoidance feelings begin to decrease.

In our society the approach-avoidance conflicts that are most pervasive and difficult to resolve generally take place in the following areas:

1. **Independence versus dependence.** We may in times of stress want to resort to the dependency characteristic of childhood, to have someone take care of us and solve our problems for us. But we are taught that the ability to stand on our own and assume responsibilities is mark of maturity.
2. **Cooperation versus competition.** In American life much emphasis is placed on competition and success. Competition begins in early childhood among siblings,

continues through school and college, and culminates in business and professional rivalry. At the same time we are urged to cooperate and help others. The concept of “team spirit” is as American as the success story. Such contradictory expectations are potentially conflict-producing.

3. **Impulse expression versus moral standards.** All societies have to place some degree of regulation upon impulse control. We noted that much of childhood learning involves imposing cultural restrictions upon innate impulses. Sex and aggression are two areas in which our impulses most frequently conflict with moral standards, and violation of these standards may generate strong feelings of guilt.

STRESS AND COPING

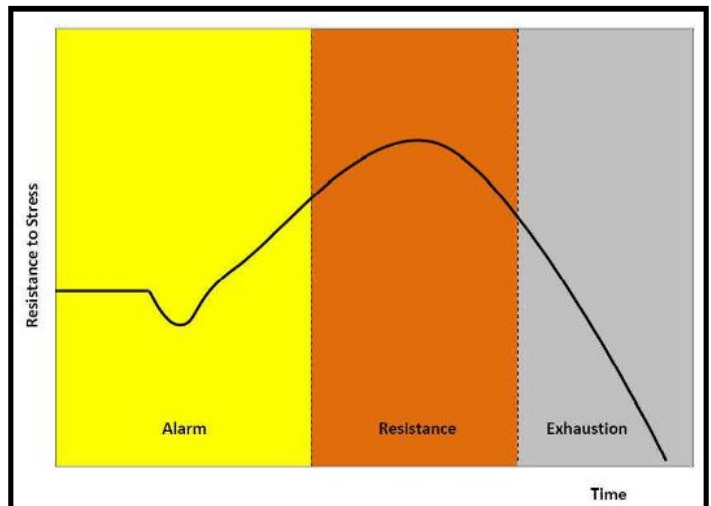
In order to understand how people learn to cope with stress, it is important to first reflect on the different conceptualizations of stress and how the coping research has emerged alongside distinct approaches to stress. Stress has been viewed as a response, a stimulus, and a transaction. How an individual conceptualizes stress determines his or her response, adaptation, or coping strategies.

Stress as a Response

Stress as a response model, initially introduced by Hans Selye (1956), describes stress as a physiological response pattern and was captured within his general adaptation syndrome (GAS) model. This model describes stress as a dependent variable and includes three concepts:

1. Stress is a defensive mechanism.
2. Stress follows the three stages of alarm, resistance, and exhaustion.
3. If the stress is prolonged or severe, it could result in diseases of adaptation or even death.

Later, in *The Stress Concept: Past, Present and Future* (1983), Selye introduced the idea that the stress response could result in positive or negative outcomes based on cognitive interpretations of the physical symptoms or physiological experience (Figure, “The General Adaptation to Stress Model”). In this way, stress could be experienced as eustress (positive) or distress (negative). However, Selye always considered stress to be a physiologically based construct or response. Gradually, other researchers expanded the thinking on stress to include and involve psychological concepts earlier in the stress model.



GENERAL ADAPTATION TO STRESS (GAS) MODEL.

The response model of stress incorporates coping within the model itself. The idea of adaptation or coping is inherent to the GAS model at both the alarm and resistance stages. When confronted with a negative stimulus, the alarm response initiates the sympathetic nervous system to combat or avoid the stressor (i.e., increased heart rate, temperature, adrenaline, and glucose levels). The resistance response then initiates physiological systems with a fight or flight reaction to the stressor, returning the system to homeostasis, reducing harm, or more generally accommodating the stressor, which can lead to adaptive diseases such as sleep deprivation, mental illness, hypertension, or heart disease. Thus, along with the early conceptualization of stress as a physiological response, early research on coping was also born. As early as 1932, Walter Cannon described the notion of self-regulation in his work *The Wisdom of the Body*.

Stress as a Stimulus

The theory of stress as a stimulus was introduced in the 1960s, and viewed stress as a significant life event or change that demands response, adjustment, or adaptation. Holmes and Rahe (1967) created the Social Readjustment Rating Scale (SRRS) consisting of 42 life events scored according to the estimated degree of adjustment they would each demand of the person experiencing them (e.g., marriage, divorce, relocation, change or loss of job, loss of loved one). Holmes and Rahe theorized that stress was an independent variable in the health-stress-coping equation — the cause of an experience rather than the experience itself. While some correlations emerged between SRRS scores and illness (Rahe, Mahan, & Arthur, 1970; Johnson & Sarason, 1979), there were problems with the stress as stimulus theory. The stress as stimulus theory assumes:

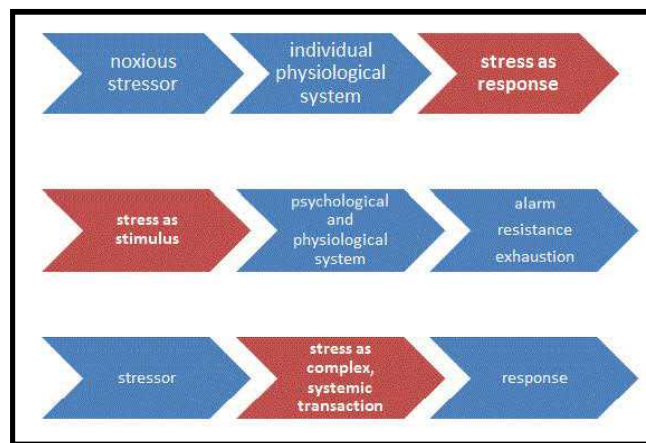
1. Change is inherently stressful.
2. Life events demand the same levels of adjustment across the population.
3. There is a common threshold of adjustment beyond which illness will result.

Rahe and Holmes initially viewed the human subject as a passive recipient of stress, one who played no role in determining the degree, intensity, or valence of the stressor. Later, Rahe introduced the concept of interpretation into his research (Rahe & Arthur, 1978), suggesting that a change or life event could be interpreted as a positive or negative experience based on cognitive and emotional factors. However, the stress as stimulus model still ignored important variables such as prior learning, environment, support networks, personality, and life experience.

Stress as a Transaction

In attempting to explain stress as more of a dynamic process, Richard Lazarus developed the transactional theory of stress and coping (TTSC) (Lazarus, 1966; Lazarus & Folkman, 1984), which presents stress as a product of a transaction between a person (including multiple systems: cognitive, physiological, affective, psychological, neurological) and his or her complex environment. Stress as a transaction was introduced with the most impact when Dr. Susan Kobasa first used the concept of hardiness (Kobasa, 1979). Hardiness refers to a pattern of personality characteristics that distinguishes people who remain healthy under life stress compared with those who develop health problems. In the late 1970s, the concept of hardiness was further developed by Salvatore Maddi, Kobasa, and their graduate students at the University of Chicago (Kobasa, 1982; Kobasa & Maddi, 1981; Kobasa, Maddi, & Kahn, 1982; Kobasa, Maddi, Puccetti, & Zola, 1985; Maddi & Kobasa,

1984). Hardiness has some notable similarities with other personality constructs in psychology, including locus of control (Rotter, 1966), sense of coherence (Antonovsky, 1987), self-efficacy (Bandura, 1997), and dispositional optimism (Scheier & Carver, 1985), all of which will be discussed in the next section. Researchers introduced multiple variables to the stress-as-transaction model, expanding and categorizing various factors to account for the complex systems involved in experiencing a stressor (Werner, 1993). The nature of stress was described in multiple ways: acute, episodic or intermittent, and chronic. Different types of stressors emerged, such as event, situation, cue, and condition, which then fell into categories based on locus of control, predictability, tone, impact, and duration. Figure illustrates theories of stress as a response, stimulus, and transaction.



THEORIES OF STRESS AS RESPONSE, STIMULUS, AND TRANSACTION.

In his book *Psychological Stress and the Coping Process* (1966), Lazarus presented an elegant integration of previous research on stress, health, and coping that placed a person's appraisal of a stressor at the centre of the stress experience. How an individual appraises a stressor determines how he or she copes with or responds to the stressor. Whether or not a stressor is experienced as discomforting is influenced by a variety of personal and contextual factors including capacities, skills and abilities, constraints, resources, and norms (Mechanic, 1978). Lazarus and Folkman (1984) unpacked the concept of interpretation further in their model of stress appraisal, which includes primary, secondary, and reappraisal components (see Figure 16.5, "The Transactional Theory of Stress and Coping"). Primary appraisal involves determining whether the stressor poses a threat. Secondary appraisal involves the individual's evaluation of the resources or coping strategies at his or her disposal for addressing any perceived threats. The process of reappraisal is ongoing and involves continually reappraising both the nature of the stressor and the resources available for responding to the stressor.

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Coping with Stress

There are many ways that people strive to cope with stressors and feelings of stress in their lives. A host of literature, both popular and academic, extols the practice of stress management and whole industries are devoted to it. Many techniques are available to help individuals cope with the stresses that life brings. Some of the techniques listed in figure, "Stress Management Techniques," induce a lower than usual stress level temporarily to compensate the biological tissues involved; others face the stressor at a higher level of abstraction. Stress management techniques are more general and range from cognitive (mindfulness, cognitive therapy, meditation) to physical (yoga, art, natural medicine, deep breathing) to environmental (spa visits, music, pets, nature).

Cognitive	Physical	Environmental	Other
Therapy	Artistic expression	Music	Conflict resolution
Hobbies	Deep breathing	Nature	Prayer
Meditation	Natural medicine	Pets	
Mindfulness	Physical exercise	Spa visits	
Planning	Relaxation		
Reading	Yoga		
Time management			

STRESS MANAGEMENT TECHNIQUES.

Stress coping, as described by researchers such as Lazarus and Folkman, implies a more specific process of cognitive appraisal to determine whether an individual believes he or she has the resources to respond effectively to the challenges of a stressor or change. The appraisal literature explains the response or coping process in terms of problem-focused coping or emotion-focused coping, also referred to as active and passive coping styles. As well, approach and avoidance-style measures of coping exist involving assertiveness or withdrawal. When faced with a challenge, an individual primarily appraises the challenge as either threatening or non-threatening, and secondarily in terms of whether he or she has the resources to respond to or cope with the challenge effectively. If the individual does not believe he or she has the capacity to respond to the challenge or feels a lack of control, he or she is most likely to turn to an emotion-focused coping response such as wishful thinking (e.g., I wish that I could change what is happening or how I feel), distancing (e.g., I'll try to forget the whole thing), or emphasizing the positive (e.g., I'll just look for the silver lining). If the person has the resources to manage the challenge, he or she will usually develop a problem-focused coping response such as analysis (e.g., I try to analyze the problem in order to understand it better; I'm making a

plan of action and following it). It is theorized and empirically demonstrated that a person's secondary appraisal then determines coping strategies. Coping strategies vary from positive thinking to denial and are measured and tested using a variety of instruments and scales such as the COPE inventory.

Positive reinterpretation and growth
Mental disengagement
Focus on and venting of emotions
Use of instrumental social support
Active coping
Denial
Religious coping
Humour
Behavioral disengagement
Restraint
Use of emotional social support
Substance use
Acceptance
Suppression of competing activities
Planning

COPE INVENTORY. THE COPE INVENTORY SCALE OF COPING TECHNIQUES.

Related concepts to stress coping include locus of control, sense of coherence, self-efficacy, and stress-related growth. Rotter posited that a person with an internal locus of control believes that their achievements and outcomes are determined by their own decisions and efforts. If they do not succeed, they believe it is due to their own lack of effort. Whereas, a person with an external locus of control believes that achievements and outcomes are determined by fate, luck, or other. If the person does not succeed, he or she believes it is due to external forces outside of the person's control. Aaron Antonovsky (1987) defined sense of coherence as:

a global orientation that expresses the extent to which one has a pervasive, enduring though dynamic feeling of confidence that (1) the stimuli deriving from one's internal and external environments in the course of living are structured, predictable and explicable; (2) the resources are available to one to meet the demands posed by these stimuli; and (3) these demands are challenges, worthy of investment and engagement.

Self-efficacy is often confused with self-confidence, but in fact confidence is merely one of the many factors that make up a strong sense of self-efficacy. Albert Bandura (1997) defined self-efficacy as the extent or strength of one's belief in one's own ability to complete tasks and reach goals. Self-confidence is a trait measure (a quality that is built over time) whereas self-efficacy is a state measure (a capacity experienced at a specific point in time and concerning a specific task). Stress-related growth or thriving is a dispositional response to stress that enables the individual to see opportunities for growth as opposed to threat or debilitation. Spreitzer and colleagues (2005) offered a preliminary definition of thriving as a "psychological state in which individuals experience both a sense of vitality and a sense of learning at work". Carver (1998) described thriving as being "better off after adversity". There are many examples of individuals surpassing previous performances when faced with particularly stressful scenarios, showing increased growth and strength in the face of adversity.

Type A, B, C, D Behaviors

Have you ever heard someone say that they were a Type A personality? Ever wondered what they meant by that? There are actually 4 different personality types that have been classified into major categories. They have been designated as types A, B, C, and D. Each personality type has certain traits and characteristics which set it apart from the others.

It is important to note that people can have traits and characteristics from all for classifications. The type of personality that they are assigned when tested is based on the majority of characteristics and traits that are on display.

Type A Personality Traits

People who have a Type A personality tend to focus on competition. They love to achieve greatness and they don't like it when it takes a lot of time to experience success. People with this personality are often working on multiple projects at the same time because they have an almost desperate need to be seen as a winner. They sometimes feel insecure, especially if they aren't seen as being successful, and push themselves to the breaking point on a regular basis for the chance to reach their goals quickly.

Type B Personality Traits

People who have a Type B personality tend to be more relaxed. They don't take stressful situations personally. They can still be highly competitive, but they don't have the same urgency to see a successful outcome happen. You'll find that this personality trait tends to procrastinate a lot, waiting until the last possible second to submit work – and if there isn't a firm deadline in place, they'll just submit the work on their own time. If given the choice between working or chilling with their friends, they'll likely choose the latter.

Type C Personality Traits

People who have a Type C personality like to focus on the fine details of any given situation. They like to get lost in whatever information is available to them to glean out new opinions from the data. Instead of being assertive about what they want to do, this personality type tends to let other people follow their own path and set their desires aside for a later time. This tends to cause higher levels of stress in these folks and often regret: "If I had only been a little more assertive..."

Type D Personality Traits

People who have a Type D personality tend to see the world in a very negative way. Although there may be situations that most people would find to be optimistic, this personality type will always be pessimistic about any potential outcome. Their outlook toward life itself can be severely negative as well. There is a constant fear of rejection in some way, which often causes this personality type to live a fairly isolated life. This is the personality type that has the highest risk of suffering from depression at some point during their lifetime.

Stress Management Strategies

Biofeedback

Biofeedback is a way to measure the body's physiological responses in real time, and a tool to learn to control them. Biofeedback generally relies on machines that measure

heart rate, muscle tension or even brain waves, and usually requires a therapist or other health professional to operate the machine, explain what the readings mean, and work with clients to incorporate the information into lifestyle changes. However, there are some exceptions to this, which we'll discuss later on in this article.

How Does Biofeedback Work?

Most often, biofeedback helps people control their stress response, by realizing when it's underway and employing relaxation techniques like deep breathing, visualizations, and meditation to calm their physiological arousal. Many of the benefits of biofeedback simply come from the increased relaxation in your body and the lack of a chronically triggered fight-or-flight response. Because chronic stress can be a trigger for many negative health symptoms, this can offer a significant and palpable improvement in the way people feel and how their bodies function.

By helping you learn how your body is currently functioning, biofeedback can help you to know what to change. Also, by showing you in 'real time' which relaxation techniques are working and which aren't, you're able to more easily grasp effective ways to relax your body's physiology and incorporate healthier habits into your lifestyle.

What Does It Treat?

Because biofeedback can with stress-related conditions by helping to reverse the stress response (and help you keep from letting it be triggered in the first place!), it can be useful with a variety of conditions.

Biofeedback has been popularly used for anxiety, tension headaches, pain, IBS, general stress, and several other conditions.

How To Get Started With Biofeedback

There are some ways in which you're probably already using a very simple form of biofeedback:

1. **Physical Awareness.** Some physical responses can be easily sensed without equipment, of course. When your body becomes tense, that's a natural message that you're stressed. Shallow breathing can be another. Becoming aware of your body's stress signals can arguably be a sort of natural form of 'biofeedback'.
2. **Household Tools.** Scales, thermometers, and even mirrors can tell you about some of your body's functioning by pointing out excess weight, fevers, and visible signs of stress like acne.

Both of these methods tell you about your physiology and help you to know that healthy changes are required, but are only the tip of the iceberg. Traditional biofeedback usually involves much more sophisticated measurements that can change more rapidly and are more difficult for the layperson to detect on their own. Usually, standard biofeedback necessitates the use of a biofeedback technician – a therapist or health professional – but recently, home biofeedback equipment has become readily available to consumers. Wild Divine is a pioneer in this field, and has created a home biofeedback system that hooks up to your home computer, takes measurements from three fingers, and tells you in real time how your physiology is responding to the meditation and stress management techniques it teaches you with the help of health gurus like Deepak Chopra, Dr. Dean Ornish, and Dr. Andrew Weil. I've used their system and personally think it's great.

As mentioned, there are also many biofeedback practitioners out there who can help you understand how your physiology is responding to stress, and help you to perfect strategies to calm your body and your mind, and stay healthier in the process. One of the best ways to find a good practitioner is to ask your doctor for a referral.

Music Therapy

The effects of anxiety and stress on human behaviour have been widely recognized. Experts estimate that between 50% and 80% of all medically related disorders are directly attributable to stress related etiologies. Hanser's literature review (1985) found that psychosocial stressors are considered to be major catalysts in the development or progression of hypertension and associated illnesses, gastrointestinal problems, skin disorders, headaches, insomnia, coronary artery disease, and other potentially life-threatening diseases.

Strategies for coping with stress seem to follow two themes (Hanser, 1985): first, direct attempts to change environmental demands or the capacity of the self; and second, regulating emotions associated with stress, managing tension, and facilitating relaxation. In practice, the emotional and behavioural elements are combined to form an integrated coping methodology.

Music therapy, with its potential to influence both psychological and physiological processes, then becomes an important alternative in the practice of stress management. Recognition of the inherent qualities of music in association with medicine has its roots in antiquity. Throughout history there are references to the use of music as therapy for a variety of illnesses. Particularly in the past two decades, a resurgence of interest in this field has focused attention on clinical and experimental research evaluating the physiological and psychological effects of music on the workings of the body, both in health and in disease (for a comprehensive review see Standley, 1986).

The bulk of this research in the area of stress reduction has examined the effects of music on the physiological processes of relaxation. Numerous studies have reported significant decreases in, for example, heart rate, skin temperature, and muscle activity when music has been used in the relaxation process. Recent studies of psychological responses to music in the relaxation process have been overwhelmingly positive. Anxiety measures and self-reports find people indicating greatly increased feelings of relaxation and calmness. Other researchers have studied the positive contribution of music on the effectiveness of biofeedback, guided imagery, and other established techniques.

In light of recent research in music therapy and stress reduction, it seems evident that music can play a significant role in the treatment of persons suffering from stress-related disorders. It has been demonstrated that music can dramatically influence physiological and psychological processes, and that it also provides, for most individuals, a non-threatening, pleasurable experience. Further, the dramatic effects that music can have, as well as the variability of individual responses to music, should underscore the need for a trained music therapist, knowledgeable in this area, to ensure a more positive and successful experience when using music in stress reduction treatment and relaxation training.

For persons with stress and anxiety related disorders, music therapy can:

Provide a stimulus for relaxation. A variety of studies have demonstrated the effectiveness of music and/or music therapy techniques in inducing a relaxation response, both physiologically and psychologically. Jellison (1975), for example, has shown that back-

ground music can reduce stress responses and help induce relaxation under stress. Music has also been successfully combined with biofeedback techniques (Scartelli, 1984; Scartelli & Borling, 1986) to reduce tension and facilitate relaxation responses. Hanser (1985), in a comprehensive review of the literature, reported on numerous studies wherein music was found to be effective in eliciting perceptions of improved states of relaxation as measured by A-State Anxiety Inventory scores and verbal reports. Throughout much of the literature, subjects have overwhelmingly conveyed a feeling of deep calm and relaxation. In many cases (Scartelli & Borling, 1986), it was clearly the music which induced and maintained an enhanced state of relaxation, both psychologically and physiologically.

Provide a reinforcing stimulus for relaxation. The strength of music therapy in eliciting and reinforcing relaxation lies, perhaps, in the fact that music elicits responses more compatible with relaxation behaviours than do other modes of reinforcement. Saperston (1989), found that verbal stimuli, through activating verbally based cognitive processes, could distract from the relaxation training. Music, being processed predominantly by nonverbal areas of the brain, could more consistently produce desired responses. This relates also to the findings of Scartelli & Borling in their 1986 study. Subjects in this study reported that the music listening helped to prevent their minds from wandering, and reinforced their ability to focus on relaxation. These studies support many researchers' contention that a more efficient relaxation process results from enhanced nondominant hemispheric activity, since music listening is a nondominant hemisphere function. Music is also an effective reinforcer because, as an auditory stimulus, it pervades the environment and is difficult to ignore. Individuals can also learn to actively make use of musical experience, a pleasurable one for most people, for their own reinforcement. By cognitively engaging in the music, individuals can create their own distraction and/or relaxation reinforcement during stressful feelings or events.

Promote recognition of bodily responses to stress and enhance the, physiological processes of relaxation. Music therapy techniques can be effectively employed to educate the individual regarding his or her own bodily responses to stress. Hanser (1987) successfully employed a music-based exercise programme designed to promote recognition, and eventual alleviation, of body tension. Participants became much more adept at identifying and recognizing their own tension responses. This ability to better identify one's own responses can ultimately enhance the ability to relax. As Scartelli & Borling (1986) point out, by bridging the gap between the conscious and normally unconscious activities of the body, individuals can develop a new understanding and increasing sense of control over their internal states.

To further facilitate the physiological processes of relaxation, music used in conjunction with biofeedback measures has also been shown to be effective. Scartelli (1984), in studying the effects of biofeedback and/or sedative music and biofeedback alone, found that it was subjects in biofeedback with music and music alone conditions who experienced significant decreases in frontalis muscle activity. Scartelli & Borling (1986), conducting a similar experiment, also found significant decreases in muscle tension at targeted sites. Davis & Thaut (1989) found physiological patterns that could actually be indicative of a variety of pleasant or unpleasant experiences. When participants reported that the experience was, subjectively relaxing and anxiety reducing, they speculated that the moderate increases in physiological arousal they observed simply accompanied the enjoyment of the music and thus facilitated feelings of reduced anxiety and increased relaxation.

Most researchers also agree on the importance of individual attitudes and preferences to ward music and its effectiveness in relaxation processes. While individual physiological responses to musical stimuli will likely vary, musical stimuli perceived by the individual as pleasant and relaxing may best enhance the physiological and psychological processes of relaxation (Davis & Thaut, 1989). At the same time, however, the client's preferred relaxation music may not elicit the best physiological response. In such a case, Saperston (1989) successfully modified a client's music preference to facilitate more relaxed responding. He observed that clients who initially achieved lower EMG levels with preferred 'stimulative' music, eventually attained considerably lower EMG levels with 'sedative' music.

Facilitate identification and expression of feelings associated with stress. Much of the research base in music therapy and stress reduction literature has focused on the effects of music on, specifically, physiological processes, of relaxation. Many of these studies have employed music in a somewhat "passive" role, providing carefully selected music listening, stimuli for relaxation, reinforcement, and so on. Often concurrent with the enhanced physiological processes of relaxation, subjects report feeling more relaxed on a psychological level as well. Music therapy, however, can play a much more significant active role in enhancing the psychological processes of relaxation.

Music therapy can be effective in helping the individual to learn to recognize feelings and behaviours associated with stress and to express these emotions in positive, appropriate ways. Stress and stress-related disorders can be terribly disruptive to normal functioning and interrelating. The music therapy session can heighten an individual's awareness of the emotional toll stress is taking on his or her life. If stress-related feelings and emotions are not identified and dealt with, they may simply exacerbate the situation. For the individual who has so successfully internalized these feelings, musical experience, being both personal and internal, may provide an important cathartic release.

Lienhard (as cited in Peters, 1987), also discusses the integral part that music therapy can play as an adjunct to group therapy in stress treatment programmes. According to Lienhard, instrumental improvisation, music listening, their reactions. Song lyrics, for example, can be used as a catalyst for discussion, helping clients to identify areas of stress, identify and express feelings associated with this stress, discuss emotions, and share constructively with the group and receive valuable feedback.

Provide alternative coping mechanisms for dealing more successfully with stress. While it is critically important for persons with stress-related disorders to increase or enhance their ability to relax, it is just as important that they learn alternate ways of dealing with the stress in their lives. Music therapy can be effective in this role. Wolfe (1978) successfully employed music therapy discussion groups designed to encourage group members to talk to one another as well as to stimulate constructive discussion concerning ways of improving unpleasant and stressful situations. Learning to play instruments while in music therapy has the added benefit of providing the client with a leisure skill that serves as a means of tension release.

Further, in using the music setting as a model for the environment in which stress-evoking interactions and/or events may occur, problems may be identified and solved within the confines of the therapy session (Hanser, 1985). Stress-related behaviours and/or patterns can be observed within the music therapy context and dealt with or changed within this controlled environment. Also, beyond simply learning relaxation techniques, Saperston (1989), for example, uses the music therapy setting to help indi-

viduals learn to recognize when they would benefit from performing their relaxation induction, providing another effective means of coping with stress.

Breathing Exercises

Have you ever noticed how you breathe when you feel relaxed? The next time you are relaxed, take a moment to notice how your body feels. Or think about how you breathe when you first wake up in the morning or just before you fall asleep. Breathing exercises can help you relax, because they make your body feel like it does when you are already relaxed.

Deep breathing is one of the best ways to lower stress in the body. This is because when you breathe deeply, it sends a message to your brain to calm down and relax. The brain then sends this message to your body. Those things that happen when you are stressed, such as increased heart rate, fast breathing, and high blood pressure, all decrease as you breathe deeply to relax.

The way you breathe affects your whole body. Breathing exercises are a good way to relax, reduce tension, and relieve stress.

Breathing exercises are easy to learn. You can do them whenever you want, and you don't need any special tools or equipment to do them.

You can do different exercises to see which work best for you.

How do you do breathing exercises?

There are lots of breathing exercises you can do to help relax. The first exercise below—belly breathing—is simple to learn and easy to do. It's best to start there if you have never done breathing exercises before. The other exercises are more advanced. All of these exercises can help you relax and relieve stress.

Belly Breathing. Belly breathing is easy to do and very relaxing. Try this basic exercise anytime you need to relax or relieve stress.

Sit or lie flat in a comfortable position.

Put one hand on your belly just below your ribs and the other hand on your chest.

Take a deep breath in through your nose, and let your belly push your hand out. Your chest should not move.

Breathe out through pursed lips as if you were whistling. Feel the hand on your belly go in, and use it to push all the air out.

Do this breathing 3 to 10 times. Take your time with each breath.

Notice how you feel at the end of the exercise.

Next steps. After you have mastered belly breathing, you may want to try one of these more advanced breathing exercises. Try all three, and see which one works best for you:

4-7-8 breathing. This exercise also uses belly breathing to help you relax. You can do this exercise either sitting or lying down.

1. To start, put one hand on your belly and the other on your chest as in the belly breathing exercise.
2. Take a deep, slow breath from your belly, and silently count to 4 as you breathe in.
3. Hold your breath, and silently count from 1 to 7.
4. Breathe out completely as you silently count from 1 to 8. Try to get all the air out of your lungs by the time you count to 8.

5. Repeat 3 to 7 times or until you feel calm.
6. Notice how you feel at the end of the exercise.

Roll breathing. Roll breathing helps you to develop full use of your lungs and to focus on the rhythm of your breathing. You can do it in any position. But while you are learning, it is best to lie on your back with your knees bent.

1. Put your left hand on your belly and your right hand on your chest. Notice how your hands move as you breathe in and out.
2. Practice filling your lower lungs by breathing so that your "belly" (left) hand goes up when you inhale and your "chest" (right) hand remains still. Always breathe in through your nose and breathe out through your mouth. Do this 8 to 10 times.
3. When you have filled and emptied your lower lungs 8 to 10 times, add the second step to your breathing: inhale first into your lower lungs as before, and then continue inhaling into your upper chest. Breathe slowly and regularly. As you do so, your right hand will rise and your left hand will fall a little as your belly falls.
4. As you exhale slowly through your mouth, make a quiet, whooshing sound as first your left hand and then your right hand fall. As you exhale, feel the tension leaving your body as you become more and more relaxed.
5. Practice breathing in and out in this way for 3 to 5 minutes. Notice that the movement of your belly and chest rises and falls like the motion of rolling waves.
6. Notice how you feel at the end of the exercise.

Practice roll breathing daily for several weeks until you can do it almost anywhere. You can use it as an instant relaxation tool anytime you need one.

Caution. Some people get dizzy the first few times they try roll breathing. If you begin to breathe too fast or feel lightheaded, slow your breathing. Get up slowly.

Morning breathing. Try this exercise when you first get up in the morning to relieve muscle stiffness and clear clogged breathing passages. Then use it throughout the day to relieve back tension.

1. From a standing position, bend forward from the waist with your knees slightly bent, letting your arms dangle close to the floor.
2. As you inhale slowly and deeply, return to a standing position by rolling up slowly, lifting your head last.
3. Hold your breath for just a few seconds in this standing position.
4. Exhale slowly as you return to the original position, bending forward from the waist.
5. Notice how you feel at the end of the exercise.

Progressive Muscular Relaxation

Progressive muscle relaxation (PMR) is a deep relaxation technique that has been effectively used to control stress and anxiety, relieve insomnia, and reduce symptoms of certain types of chronic pain. Progressive muscle relaxation is based upon the simple practice of tensing, or tightening, one muscle group at a time followed by a relaxation phase with release of the tension. Doctors have used progressive muscle relaxation in combination with standard treatments for symptom relief in a number of conditions, including headaches, cancer pain, high blood pressure, and digestive disturbances.

The technique of progressive muscle relaxation was described by Edmund Jacobson in the 1930s and is based upon his premise that mental calmness is a natural result of physical relaxation. Progressive muscle relaxation can be learned by nearly anyone and requires only 10 minutes to 20 minutes per day to practice.

Most practitioners recommend tensing and relaxing the muscle groups one at a time in a specific order, generally beginning with the lower extremities and ending with the face, abdomen, and chest. You can practice this technique seated or lying down, and you should try to practice with comfortable clothing on, and in a quiet place free of all distractions.

Here is how it works;

1. While inhaling, contract one muscle group (for example your upper thighs) for 5 seconds to 10 seconds, then exhale and suddenly release the tension in that muscle group.
2. Give yourself 10 seconds to 20 seconds to relax, and then move on to the next muscle group (for example your buttocks).
3. While releasing the tension, try to focus on the changes you feel when the muscle group is relaxed. Imagery may be helpful in conjunction with the release of tension, such as imagining that stressful feelings are flowing out of your body as you relax each muscle group.
4. Gradually work your way up the body contracting and relaxing muscle groups.
5. People who suffer from insomnia often report that practicing progressive muscle relaxation at night helps them fall asleep. Progressive muscle relaxation is also an excellent tool to help learn about the body and the signals it may be telling you. With practice and time, you can learn to accurately identify and diminish the signs and signals of stress and tension in your body.

Guided Imagery

You're nervous and stressed, and when you try to rehearse your opening lines, your mind goes completely blank. The more you try to practice your material, the more stressed you feel!

So, you take a break, you close your eyes, and you remember the last vacation that you took in the mountains. You think of the gentle stream where you stopped to rest. You can hear the birdsong, smell the clean air, and feel the sun's warmth on your skin. You slowly begin to relax as you imagine this peaceful scene, and your heart rate and breathing slow down. When you open your eyes a few minutes later, you feel relaxed and in control, and you have no trouble remembering your opening lines.

Have you ever used your imagination to escape, or cope with a stressful situation? If so, you were using "guided imagery" to relax. In this article, we'll look at how to use imagery to manage stress, and we'll discuss how you can use this technique to cope with difficult situations.

Guided imagery is a stress management technique, where you use your imagination to picture a person, place, or time that makes you feel relaxed, peaceful and happy. Imagery is slightly different from other stress management techniques, in that it relies on the use of all of your senses.

For instance, in your imagination you hear the sound of birds chirping, you see the drops of dew on the grass, you feel the breeze on your skin, you smell the wildflowers, and

you taste the cold drink. In imagery, using all of your senses is what creates such a powerfully relaxing experience, and this is why it's so useful in managing stress and coping with difficult situations.

There are several other ways that you can use imagery to help you relax. For example, you could create mental pictures of stress flowing out of your body, or of your problems, your distractions, and your everyday concerns being folded away and stashed in a pad-locked chest.

Some people are skeptical about the effectiveness of using imagery. However, research suggests that it can be incredibly effective in lowering your stress levels.

For instance, one study found that using stress management techniques alongside relaxation imagery, and even just using imagery alone, significantly reduced participants' blood pressure. Another study, which researched the effectiveness of imagery on breast cancer patients, found similar benefits: patients who used imagery to cope with their disease experienced less stress, more vigor, and a higher quality of life than those who didn't use the technique.

As well as these examples, many other studies have successfully used imagery to lower stress in patients suffering from post-traumatic stress syndrome, abuse, depression, and other conditions, including occupational stress.

Mindfulness

Would you like to reduce your stress? Very few people would say no to that offer! But what exactly does it mean?

We all have a sense of how stress manifests in our personal life and what we want to change. Many people would say they are looking for relief from the worries that plague them, while others might point to their stiff neck or tight jaw, or mention their headaches. However we experience it personally, stress impacts both our minds and bodies. It is a state of hyper-arousal where our minds and bodies are on alert, our adrenaline is flowing, and we feel a need to do something to protect ourselves.

What are some of the effects of stress?

In an aroused state, your body produces an adrenaline rush that increases your heart rate, elevates your respiration, tenses your muscles, and changes your blood flow. While this might be just what you need to complete a task or escape a dangerous situation, it might be not be a useful reaction to the current situation and could be counter-productive.

Under stress, you might find your thoughts racing in cycles you can't escape. You might feel restless and anxious, unable to concentrate. You might become irritable, impulsive, angry, or aggressive. Over time, you might experience insomnia, reduced pain tolerance, and hyper-vigilance.

So it is helpful to know how to shift out of this aroused state to a relaxed state where your mind and body are at rest. Practicing mindfulness can provide that shift.

Meditation

If stress has you anxious, tense and worried, consider trying meditation. Spending even a few minutes in meditation can restore your calm and inner peace.

Anyone can practice meditation. It's simple and inexpensive, and it doesn't require any special equipment.

And you can practice meditation wherever you are — whether you're out for a walk, riding the bus, waiting at the doctor's office or even in the middle of a difficult business meeting.

Understanding Meditation

Meditation has been practiced for thousands of years. Meditation originally was meant to help deepen understanding of the sacred and mystical forces of life. These days, meditation is commonly used for relaxation and stress reduction.

Meditation is considered a type of mind-body complementary medicine. Meditation can produce a deep state of relaxation and a tranquil mind.

During meditation, you focus your attention and eliminate the stream of jumbled thoughts that may be crowding your mind and causing stress. This process may result in enhanced physical and emotional well-being.

Benefits of Meditation

Meditation can give you a sense of calm, peace and balance that can benefit both your emotional well-being and your overall health.

And these benefits don't end when your meditation session ends. Meditation can help carry you more calmly through your day and may help you manage symptoms of certain medical conditions.

Meditation and emotional well-being

When you meditate, you may clear away the information overload that builds up every day and contributes to your stress.

The emotional benefits of meditation can include:

- Gaining a new perspective on stressful situations
- Building skills to manage your stress
- Increasing self-awareness
- Focusing on the present
- Reducing negative emotions
- Increasing imagination and creativity
- Increasing patience and tolerance

Meditation and illness

Meditation might also be useful if you have a medical condition, especially one that may be worsened by stress.

While a growing body of scientific research supports the health benefits of meditation, some researchers believe it's not yet possible to draw conclusions about the possible benefits of meditation.

With that in mind, some research suggests that meditation may help people manage symptoms of conditions such as:

- Anxiety
- Cancer

- Asthma
- Chronic pain

Depression

Heart disease

High blood pressure

Irritable bowel syndrome

Sleep problems

Tension headaches

Be sure to talk to your health care provider about the pros and cons of using meditation if you have any of these conditions or other health problems. In some cases, meditation can worsen symptoms associated with certain mental and physical health conditions.

Meditation isn't a replacement for traditional medical treatment. But it may be a useful addition to your other treatment.

Types of Meditation

Meditation is an umbrella term for the many ways to a relaxed state of being. There are many types of meditation and relaxation techniques that have meditation components. All share the same goal of achieving inner peace.

Ways to meditate can include:

Guided meditation. Sometimes called guided imagery or visualization, with this method of meditation you form mental images of places or situations you find relaxing.

You try to use as many senses as possible, such as smells, sights, sounds and textures. You may be led through this process by a guide or teacher.

Mantra meditation. In this type of meditation, you silently repeat a calming word, thought or phrase to prevent distracting thoughts.

Mindfulness meditation. This type of meditation is based on being mindful, or having an increased awareness and acceptance of living in the present moment.

In mindfulness meditation, you broaden your conscious awareness. You focus on what you experience during meditation, such as the flow of your breath. You can observe your thoughts and emotions, but let them pass without judgment.

Qi gong. This practice generally combines meditation, relaxation, physical movement and breathing exercises to restore and maintain balance. Qi gong (CHEE-gung) is part of traditional Chinese medicine.

Tai chi. This is a form of gentle Chinese martial arts. In tai chi (TIE-CHEE), you perform a self-paced series of postures or movements in a slow, graceful manner while practicing deep breathing.

Transcendental Meditation. Transcendental Meditation is a simple, natural technique. In Transcendental Meditation, you silently repeat a personally assigned mantra, such as a word, sound or phrase, in a specific way.

This form of meditation may allow your body to settle into a state of profound rest and relaxation and your mind to achieve a state of inner peace, without needing to use concentration or effort.

Yoga. You perform a series of postures and controlled breathing exercises to promote a more flexible body and a calm mind. As you move through poses that require balance and concentration, you're encouraged to focus less on your busy day and more on the moment.

Elements of Meditation

Different types of meditation may include different features to help you meditate. These may vary depending on whose guidance you follow or who's teaching a class. Some of the most common features in meditation include:

Focused attention. Focusing your attention is generally one of the most important elements of meditation.

Focusing your attention is what helps free your mind from the many distractions that cause stress and worry. You can focus your attention on such things as a specific object, an image, a mantra, or even your breathing.

Relaxed breathing. This technique involves deep, even-paced breathing using the diaphragm muscle to expand your lungs. The purpose is to slow your breathing, take in more oxygen, and reduce the use of shoulder, neck and upper chest muscles while breathing so that you breathe more efficiently.

A quiet setting. If you're a beginner, practicing meditation may be easier if you're in a quiet spot with few distractions, including no television, radios or cellphones. As you get more skilled at meditation, you may be able to do it anywhere, especially in high-stress situations where you benefit the most from meditation, such as a traffic jam, a stressful work meeting or a long line at the grocery store.

A comfortable position. You can practice meditation whether you're sitting, lying down, walking, or in other positions or activities. Just try to be comfortable so that you can get the most out of your meditation. Aim to keep good posture during meditation.

Open attitude. Let thoughts pass through your mind without judgment.

Everyday ways to Practice Meditation

Don't let the thought of meditating the "right" way add to your stress. If you choose to, you can attend special meditation centers or group classes led by trained instructors. But you can also practice meditation easily on your own.

And you can make meditation as formal or informal as you like, however it suits your lifestyle and situation. Some people build meditation into their daily routine. For example, they may start and end each day with an hour of meditation. But all you really need is a few minutes of quality time for meditation.

Here are some ways you can practice meditation on your own, whenever you choose:

Breathe deeply. This technique is good for beginners because breathing is a natural function.

Focus all your attention on your breathing. Concentrate on feeling and listening as you inhale and exhale through your nostrils. Breathe deeply and slowly. When your attention wanders, gently return your focus to your breathing.

Scan your body. When using this technique, focus attention on different parts of your body. Become aware of your body's various sensations, whether that's pain, tension, warmth or relaxation.

Combine body scanning with breathing exercises and imagine breathing heat or relaxation into and out of different parts of your body.

Repeat a mantra. You can create your own mantra, whether it's religious or secular. Examples of religious mantras include the Jesus Prayer in the Christian tradition,

the holy name of God in Judaism, or the om mantra of Hinduism, Buddhism and other Eastern religions.

Walk and meditate. Combining a walk with meditation is an efficient and healthy way to relax. You can use this technique anywhere you're walking, such as in a tranquil forest, on a city sidewalk or at the mall.

When you use this method, slow down your walking pace so that you can focus on each movement of your legs or feet. Don't focus on a particular destination. Concentrate on your legs and feet, repeating action words in your mind such as "lifting," "moving" and "placing" as you lift each foot, move your leg forward and place your foot on the ground.

Engage in prayer. Prayer is the best known and most widely practiced example of meditation. Spoken and written prayers are found in most faith traditions.

You can pray using your own words or read prayers written by others. Check the self-help section of your local bookstore for examples. Talk with your rabbi, priest, pastor or other spiritual leader about possible resources.

Read and reflect. Many people report that they benefit from reading poems or sacred texts, and taking a few moments to quietly reflect on their meaning.

You can also listen to sacred music, spoken words, or any music you find relaxing or inspiring. You may want to write your reflections in a journal or discuss them with a friend or spiritual leader.

Focus your love and gratitude. In this type of meditation, you focus your attention on a sacred image or being, weaving feelings of love, compassion and gratitude into your thoughts. You can also close your eyes and use your imagination or gaze at representations of the image.

Building your Meditation Skills

Don't judge your meditation skills, which may only increase your stress. Meditation takes practice.

Keep in mind, for instance, that it's common for your mind to wander during meditation, no matter how long you've been practicing meditation. If you're meditating to calm your mind and your attention wanders, slowly return to the object, sensation or movement you're focusing on.

Experiment, and you'll likely find out what types of meditation work best for you and what you enjoy doing. Adapt meditation to your needs at the moment. Remember, there's no right way or wrong way to meditate. What matters is that meditation helps you reduce your stress and feel better overall.

Yogasana

Dating back over 5000 years, yoga is considered by many to be the oldest defined practice of self-development. The methods of classical yoga include ethical disciplines, physical postures, breathing control and meditation. Traditionally an Eastern practice, it's now becoming popular in the West. In fact, many companies, especially in Britain, are seeing the benefit of yoga, recognizing that relaxed workers are healthier and more creative, and are sponsoring yoga fitness programs. Many studies have found that a little yoga in the morning, at night, or even on a lunch break, can minimize stress and increase productivity. It is believed that yoga is so effective for stress relief because, aside from

the physical benefits that yoga brings, it encourages a good mood, an increase in mindfulness, and a healthy dose of self-compassion.

Overview of Yoga

Many of the popular techniques found to reduce stress derive from yoga:

- Controlled breathing
- Meditation
- Physical movement
- Mental imagery
- Stretching

Yoga, which derives its name from the word, “yoke”—to bring together—does just that, bringing together the mind, body and spirit. But whether you use yoga for spiritual transformation or for stress management and physical well-being, the benefits are numerous.

Effects on the Body

The following is only a partial list of yoga’s benefits:

- Reduced stress
- Sound sleep
- Reduced cortisol levels
- Improvement of many medical conditions
- Allergy and asthma symptom relief
- Lower blood pressure
- Smoking cessation help
- Lower heart rate
- Spiritual growth
- Sense of well-being
- Reduced anxiety and muscle tension
- Increased strength and flexibility
- Slowed aging process

Yoga’s benefits are so numerous, it gives a high payoff for the amount of effort involved.

What’s Involved With Yoga?

The practice of yoga involves stretching the body and forming different poses while keeping breathing slow and controlled. The body becomes relaxed and energized at the same time. There are various styles of yoga, some moving through the poses more quickly, almost like an aerobic workout, and other styles relaxing deeply into each pose. Some have a more spiritual angle, while others are used purely as a form of exercise.

Benefits. Virtually everyone can see physical benefits from yoga, and its practice can also give psychological benefits, such as stress reduction and a sense of well-being, and spiritual benefits, such as a feeling of connectedness with God or Spirit, or a feeling of transcendence. Certain poses can be done just about anywhere and a yoga program can go for hours or minutes, depending on one’s schedule.

There are several mechanisms in yoga that have an effect on stress levels, meaning there are multiple ways that yoga can minimize your stress levels. Studies show that the most effective ways in which yoga targets stress are by lifting your mood (or positive

effect), by allowing for increased mindfulness, and by increasing self-compassion. By simultaneously getting us into better moods, enabling us to be more focused on the present moment, and by encouraging us to give ourselves a break, yoga is a very effective stress reliever.

Drawbacks. Yoga does require some commitment of time and is more difficult for people with certain physical limitations. Some people feel self-conscious doing some of the poses. Also, yoga classes can be expensive although it is possible, albeit perhaps more challenging, to learn from a book or video.

Comparison to Stress Reduction Methods. As yoga combines several techniques used for stress reduction, it can be said to provide the combined benefits of breathing exercises, stretching exercises, fitness programs, meditation practice, and guided imagery, in one technique. However, for those with great physical limitations, simple breathing exercises, meditation or guided imagery might be a preferable option and provide similar benefits. Yoga also requires more effort and commitment than taking pills or herbs for stress reduction.

How to Get Started. Yoga classes can be wonderful for beginners – you will be surrounded by others of all levels of expertise and ability, and you will have someone talking you through the process and helping you to learn the correct forms. Aside from taking a class, there are several useful apps that can help you with yoga, and this site contains many useful videos and articles to help you learn everything you need to know to make the practice of yoga a regular part of your life.

Stress Inoculation Training

Stress Inoculation Therapy (SIT) is a psychotherapy method intended to help patients prepare themselves in advance to handle stressful events successfully and with a minimum of upset. The use of the term "inoculation" in SIT is based on the idea that a therapist is inoculating or preparing patients to become resistant to the effects of stressors in a manner similar to how a vaccination works to make patients resistant to the effects of particular diseases.

Stress Inoculation Therapy is conceptually similar to Relapse Prevention methods used in addictions therapy. In SIT, patients are educated about stressful situations and the general nature of stress, the negative outcomes they may be vulnerable to experiencing when confronted with stress, and steps they can take to avoid those negative outcomes. At the conclusion of stress inoculation efforts, patients should feel like they can anticipate pitfalls that may occur during an event, and have a workable and practical plan in place for helping themselves avoid those pitfalls.

Stress inoculation has three phases:

In the **initial conceptualization** phase, the therapist educates the patient about the general nature of stress (offering much the same information as contained in the earlier sections of this document), and explains important concepts such as appraisal and cognitive distortion that play a key role in shaping stress reactions. The idea that people often and quite inadvertently make their stress worse through the unconscious operation of bad coping habits is conveyed. Finally, the therapist works to develop a clear understanding of the nature of the stressors the patient is facing.

A key part of what needs to be communicated in the SIT conceptualization stage is the idea that stressors are creative opportunities and puzzles to be solved, rather than mere

obstacles. Patients are helped to differentiate between aspects of their stressors and their stress-induced reactions that are changeable and aspects that cannot change, so that coping efforts can be adjusted accordingly. Acceptance-based coping is appropriate for aspects of situations that cannot be altered, while more active interventions are appropriate for more changeable stressors.

The second phase of SIT focuses on **skills acquisition and rehearsal**. The particular choice of skills taught is important, and must be individually tailored to the needs of individual patients and their particular strengths and vulnerabilities if the procedure is to be effective. A variety of emotion regulation, relaxation, cognitive appraisal, problem-solving, communication and socialization skills may be selected and taught on the basis of the patient's unique needs.

In the final SIT phase, **application and follow through**, the therapist provides the patient with opportunities to practice coping skills. The patient may be encouraged to use a variety of simulation methods to help increase the realism of coping practice, including visualization exercises, modeling and vicarious learning, role playing of feared or stressful situations, and simple repetitious behavioral practice of coping routines until they become over-learned and easy to act out.

SIT has been conducted with individuals, couples, and groups (both small and large). The length of intervention can be as short as 20 minutes or as long as 40 one hour weekly and biweekly sessions. In most instances, SIT consists of 8 to 15 sessions, plus booster and follow-up sessions, conducted over a 3-to-12-month period.



SOCIAL PSYCHOLOGY

NATURE, SCOPE AND HISTORY OF SOCIAL PSYCHOLOGY

Social psychology is the scientific study of how people's thoughts, feelings, and behaviors are influenced by the actual, imagined, or implied presence of others (Allport 1998). By this definition, scientific refers to the empirical method of investigation. The terms thoughts, feelings, and behaviors include all of the psychological variables that are measurable in a human being. The statement that others may be imagined or implied suggests that we are prone to social influence even when no other people are present, such as when watching television, or following internalized cultural norms.

Social psychology is an empirical science that attempts to answer a variety of questions about human behavior by testing hypotheses, both in the laboratory and in the field. Such approach to the field focuses on the individual, and attempts to explain how the thoughts, feelings, and behaviors of individuals are influenced by other people.

A relatively recent field, social psychology has nonetheless had a significant impact not only on the academic worlds of psychology, sociology, and the social sciences in general, but has also influenced public understanding and expectation of human social behavior. By studying how people behave under extreme social influences, or lack thereof, great advances have been made in understanding human nature. Human beings are essentially social beings, and thus, social interaction is vital to the health of each person. Through investigating the factors that affect social life and how social interactions affect individual psychological development and mental health, a greater understanding of how humankind as a whole can live together in harmony is emerging.

Links Between Social Psychology and Sociology

Social psychology is a branch of psychology that studies cognitive, affective, and behavioral processes of individuals as influenced by their group membership and interactions,

and other factors that affect social life, such as social status, role, and social class. Social psychology examines the effects of social contacts on the development of attitudes, stereotypes, discrimination, group dynamics, conformity, social cognition and influence, self-concept, persuasion, interpersonal perception and attraction, cognitive dissonance, and human relationships.

A significant number of social psychologists are sociologists. Their work has a greater focus on the behavior of the group, and thus examines such phenomena as interactions and social exchanges at the micro-level, and group dynamics and crowd psychology at the macro-level. Sociologists are interested in the individual, but primarily within the context of social structures and processes, such as social roles, race and class, and socialization. They tend to use both qualitative and quantitative research designs. Sociologists in this area are interested in a variety of demographic, social, and cultural phenomena. Some of their major research areas are social inequality, group dynamics, social change, socialization, social identity, and symbolic interactionism.

Social psychology bridges the interest of psychology (with its emphasis on the individual) with sociology (with its emphasis on social structures). Most social psychologists are trained within the discipline of psychology. Psychologically oriented researchers place a great deal of emphasis on the immediate social situation, and the interaction between person and situation variables. Their research tends to be highly empirical and is often centered round lab experiments. Psychologists who study social psychology are interested in such topics as attitudes, social cognition, cognitive dissonance, social influence, and interpersonal behavior. Two influential journals for the publication of research in this area are *The Journal of Personality and Social Psychology* and *The Journal of Experimental Social Psychology*. Read more about Sociological Social Psychology.

History of Social Psychology

The discipline of social psychology began in the United States at the dawn of the twentieth century. The first published study in this area was an experiment by Norman Triplett (1898) on the phenomenon of social facilitation. During the 1930s, many Gestalt psychologists, particularly Kurt Lewin, fled to the United States from Nazi Germany. They were instrumental in developing the field as something separate from the behavioral and psychoanalytic schools that were dominant during that time, and social psychology has always maintained the legacy of their interests in perception and cognition. Attitudes and a variety of small group phenomena were the most commonly studied topics in this era.

During World War II, social psychologists studied persuasion and propaganda for the US military. After the war, researchers became interested in a variety of social problems, including gender issues and racial prejudice. In the 1960s, there was growing interest in a variety of new topics, such as cognitive dissonance, bystander intervention, and aggression. By the 1970s, however, social psychology in America had reached a crisis. There was heated debate over the ethics of laboratory experimentation, whether or not attitudes really predicted behavior, and how much science could be done in a cultural context (Gergen 1973). This was also the time when a radical situationist approach challenged the relevance of self and personality in psychology.

During the years immediately following World War II, there was frequent collaboration between psychologists and sociologists (Sewell 1989). However, the two disciplines have become increasingly specialized and isolated from each other in recent years, with soci-

ologists focusing on macro variables (such as social structure) to a much greater extent. Nevertheless, sociological approaches to social psychology remain an important counterpart to psychological research in this area.

Social psychology reached maturity in both theory and method during the 1980s and 1990s. Careful ethical standards now regulate research, and greater pluralism and multicultural perspectives have emerged. Modern researchers are interested in a variety of phenomena, but attribution, social cognition, and self-concept are perhaps the greatest areas of growth. Social psychologists have also maintained their applied interests, with contributions in health and environmental psychology, as well as the psychology of the legal system.

Social psychology is the study of how social conditions affect human beings. Scholars in this field are generally either psychologists or sociologists, though all social psychologists employ both the individual and the group as their units of analysis. Despite their similarity, the disciplines tend to differ in their respective goals, approaches, methods, and terminology. They also favor separate academic journals and professional societies.

Social Psychology Experiments

In its simplest form, experimentation is a method of determining the presence or absence of a causal relationship between two variables by systematically manipulating one variable (called the independent variable) and assessing its effect on another variable (called the dependent variable).

Some scholars have questioned the utility of experimentation, noting that the experiments which researchers design sometimes do not resemble the circumstances that people encounter in their everyday lives. However, experimentation is the only research method that allows one to definitively establish the existence of a causal relationship between two or more variables.

Social Psychology Research Methods

Social Psychology Social psychologists typically explain human behavior as a result of the interaction of mental states and immediate, social situations. In Kurt Lewin's (1951) Heuristic behavior can be viewed as a function of the person and the environment, $B = f(P, E)$. Experimental methods involve the researcher altering a variable in the environment and measuring the effect on another variable. An example would be allowing two groups of children to play violent or nonviolent videogames, and then observing their subsequent level of aggression during free-play period. A valid experiment is controlled and uses random assignment.

Co-relational methods examine the statistical association between two naturally occurring variables. For example, one could correlate the amount of violent television children watch at home with the number of violent incidents the children participate in at school. Note that this study would not prove that violent TV causes aggression in children. It is quite possible that aggressive children choose to watch more violent TV programs.

Observational methods are purely descriptive and include naturalistic observation, contrived observation, participant observation, and archival analysis. These are less common in social psychology but are sometimes used when first investigating a phenomenon. An example would be to unobtrusively observe children on a playground (with a video camera, perhaps) and record the number and types of aggressive actions displayed.

Whenever possible, social psychologists rely on controlled experimentation. Controlled experiments require the manipulation of one or more independent variables in order to examine the effect on a dependent variable. Experiments are useful in social psychology because they are high in internal validity, meaning that they are free from the influence of confounding or extraneous variables, and so are more likely to accurately indicate a causal relationship. However, the small samples used in controlled experiments are typically low in external validity, or the degree to which the results can be generalized to the larger population. There is usually a trade-off between experimental control (internal validity) and being able to generalize to the population (external validity).

Because it is usually impossible to test everyone, research tends to be conducted on a sample of persons from the wider population. Social psychologists frequently use survey research when they are interested in results that are high in external validity. Surveys use various forms of random sampling to obtain a sample of respondents that are representative of a population. This type of research is usually descriptive or co-relational because there is no experimental control over variables. However, new statistical methods, like structural equation modeling, are being used to test for potential causal relationships in this type of data.

Regardless of which method is used, it is important to evaluate the research hypothesis in the light of the results, either confirming or rejecting the original prediction. Social psychologists use statistics and probability testing to judge their results, which define a significant finding as less than 5% likely to be due to chance. Replications are important to ensure that the result is valid and not due to chance or some feature of a particular sample.

Social Psychology Ethics

Social Psychology. The goal of social psychology is to understand cognition and behavior as they naturally occur in a social context, but the very act of observing people can influence and alter their behavior. For this reason, many social psychology experiments utilize deception to conceal or distort certain aspects of the study. Deception may include false cover stories, false participants (known as confederates or stooges), false feedback given to the participants, and so on.

The practice of deception has been challenged by some psychologists who maintain that deception under any circumstances is unethical, and that other research strategies (such as role-playing) should be used instead. Unfortunately, research has shown that role-playing studies do not produce the same results as deception studies and this has cast doubt on their validity. In addition to deception, experimenters have at times put people into potentially uncomfortable or embarrassing situations (for example, Milgram's Obedience to Authorities experiments, Zimbardo's Stanford Prison experiment), and this has also been criticized for ethical reasons.

To protect the rights and well-being of research participants, and at the same time discover meaningful results and insights into human behavior, virtually all social psychology research must pass an ethical review process. At most colleges and universities, this is conducted by an ethics committee or institutional review board. This group examines the proposed research to make sure that no harm is done to the participants, and that the benefits of the study outweigh any possible risks or discomforts to people taking part in the study.

Furthermore, a process of informed consent is often used to make sure that volunteers know what will happen in the experiment and understand that they are allowed to quit the experiment at any time. A debriefing is typically done at the conclusion of the experiment in order to reveal any deceptions used and generally make sure that the participants are unharmed by the procedures. Today, most research in social psychology involves no more risk of harm than can be expected from routine psychological testing or normal daily activities.

TRADITIONAL THEORETICAL PERSPECTIVES

Field Theory

Field theory is a psychological theory (more precisely: Topological and vector psychology) which examines patterns of interaction between the individual and the total field, or environment. The concept first made its appearance in psychology with roots to the holistic perspective of Gestalt theories. It was developed by Kurt Lewin, a Gestalt psychologist, in the 1940s.

Lewin's field theory can be expressed by a formula: $B = f(p, e)$, meaning that behavior (B) is a function of the person (p) and their environment (e).

The Life Space

The idea that an individual's behavior, at any time, is manifested only within the coexisting factors of the current "life space" or "psychological field." So a life space is the combination of all the factors that influences a person's behavior at any time. Therefore, behavior can be expressed as a function of the life space $B = f(LS)$. Furthermore, the interaction of the person (P), and the environment (E) produces this life space. In symbolic expression, $B = f(LS) = F(P, E)$. An example of a more complex life-space concept is the idea that two people's experience of a situation can become one when they converse together. This does not happen if the two people do not interact with each other, such as being in the same room but not talking to each other. This combined space can be "built" up as the two people share more ideas and create a more complex life-space together.

Environment

The environment as demonstrated in the life space, refers to the objective situation in which the person perceives and acts. The life space environment (E) is completely subjective within each context as it depends not only on the objective situation, but also on the characteristics of the person (P). It is necessary to consider all aspects of a person's conscious and unconscious environment in order to map out the person's life space. The combined state, influenced by the environment as well as the person's perspective, conscious, and unconscious, must be viewed as a whole. While each part can be viewed as a separate entity, to observe the totality of the situation one must take all inputs into consideration.

Person

Lewin applied the term person in three different ways.

1. Properties/characteristics of the individual. (needs, beliefs, values, abilities).

2. A way of representing essentially the same psychological facts of "life space" itself.
3. "The behaving self".

"The behaving self may be seen as the individual's perception of his relations to the environment he perceives."

The development of the person inevitably affects the life space. As a person undergoes changes with their body or their image of themselves changes, this can cause an instability in the region of life space. Additionally, an instability in the psychological environment or life space can lead to the instability of the person.

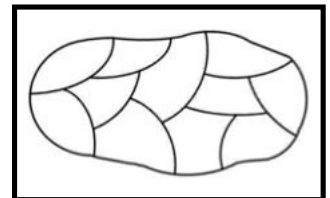
Behavior

Any change within the life space subject to psychological laws. Accordingly, an action of the person (P) or a change in the environment (E) resulting from said action, can be considered behavior (B). These behaviors can make large or small influences on the totality of the life space. Regardless, they must be taken into consideration. Field theory holds that behavior must be derived from a totality of coexisting facts. These coexisting facts make up a "dynamic field", which means that the state of any part of the field depends on every other part of it. This not only includes both mental and physical fields, but also unseen forces such as magnetism and gravity. This can be elaborated by imagining the difference that a force can make by acting from a distance. When considering something such as the Moon's influence on the Earth, it is clear that there is an effect even though it acts from a large distance away. Behavior depends on the present field rather than on the past or the future.

Development also plays a major role in life space behavior. From the beginning of one's life behavior is molded in all respects to his or her social situation. This of course brings up the sociological discussion of nature versus nurture. Experimental psychology studies have shown the formation of aspiration, the driving factor of actions and expressions (behavior), is directly influenced by the presence or absence of certain individuals within one's life space. A child's development naturally leads to an opening up of new unknown life space regions. Transitional periods such as adolescence are characterized by a greater effect of these new regions. Therefore, an adolescent entering a new social group or life space can be seen psychologically as entering a cognitively unstructured field. This new field makes it difficult for the individual to know what behavior is appropriate within the field. This is believed to be a possibility for changes in child and adolescent behavior.

Theory and Experimental Evidence

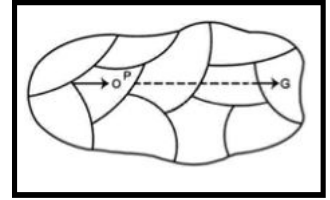
According to field theory, a person's life is made up of multiple distinct spaces. Image 1 is an example of the total field, or environment. Image 2 is showing a person, and a goal they have. This image shows that there are forces pushing a person toward their goal. The dotted line is everything one must go through to reach their goal, and how one must go through many different spaces. Individuals may have the same goal, but the field to get there may be different. One's field may be adjusted in order to gain the most in life. Some fields may be deleted, and some added, all depending on certain events that occur in a person's lifetime.



FIELD THEORY IMAGE 1.

Field theory also includes the idea that every person holds a different experience for a situation. This is not to say that two people's experience of an event will not be similar,

but that there will be some difference. This leads to the idea that no two experiences are the same for a person either, as the dynamic field is constantly changing. This is to say that the dynamic field is like a stream, constantly flowing while changing slightly. Another piece of field theory is the idea that no part of a person's field can be viewed as being pointless. Every part of a total field must be viewed as having possible meaning and importance. This must be done regardless of how pointless or non-important the part of the field may seem, it should still be accounted for. The totality of an individual's field seems to have no bounds, as research has shown that even an infant's experience of World War II could possibly affect life later on, due to the change in field. This is a good example of how broad field theory can span, as a person's preconsciousness may be altered due to field changes that occurred before any major development.



FIELD THEORY IMAGE 2.

Cognitive Dissonance

Psychologist Leon Festinger first proposed a theory of cognitive dissonance centered on how people try to reach internal consistency. He suggested that people have an inner need to ensure that their beliefs and behaviors are consistent. Inconsistent or conflicting beliefs lead to disharmony, which people strive to avoid.

In his 1957 book, *A Theory of Cognitive Dissonance*, Festinger explained, "Cognitive dissonance can be seen as an antecedent condition which leads to activity oriented toward dissonance reduction just as hunger leads toward activity oriented toward hunger reduction. It is a very different motivation from what psychologists are used to dealing with but, as we shall see, nonetheless powerful."

Influential Factors

The degree of dissonance people experience can depend on a few different factors, including how highly they value a particular belief and the degree to which their beliefs are inconsistent.

The overall strength of the dissonance can also be influenced by several factors:

- Cognitions that are more personal, such as beliefs about the self, tend to result in greater dissonance.

- The importance of the cognitions; things that involve beliefs that are highly valued typically result in stronger dissonance.

- The ratio between dissonant (clashing) thoughts and consonant (harmonious) thoughts

- The greater the strength of the dissonance, the more pressure there is to relieve the feelings of discomfort.

Cognitive dissonance can often have a powerful influence on our behaviors and actions.

Examples

Cognitive dissonance can occur in many areas of life, but it is particularly evident in situations where an individual's behavior conflicts with beliefs that are integral to his or her self-identity. A common example of cognitive dissonance occurs in the purchasing

decisions we make on a regular basis. Consider a situation in which a man who places a value on being environmentally responsible just purchased a new car that he later discovers does not get great gas mileage.

The conflict:

It is important for the man to take care of the environment.

He is driving a car that is not environmentally friendly.

In order to reduce this dissonance between belief and behavior, he can sell the car and purchase another one that gets better gas mileage, or he can reduce his emphasis on environmental responsibility. In the case of the second option, his dissonance could be further minimized by engaging in actions that reduce the impact of driving a gas-guzzling vehicle, such as utilizing public transportation more frequently or riding his bike to work.

Most people want to hold the belief that they make good choices. When a purchase turns out badly, it conflicts with their previously-existing belief about their decision-making abilities.

In A Theory of Cognitive Dissonance, Festinger offers an example of how an individual might deal with dissonance related to a health behavior by discussing individuals who continue to smoke, even though they know it is bad for their health.

According to Festinger, a person might decide that they value smoking more than health, deeming the behavior "worth it" in terms of risks versus rewards.

Another way to deal with this dissonance is to minimize the potential drawbacks. The smoker might convince himself that the negative health effects have been overstated. He might also assuage his health concerns by telling himself that he cannot avoid every possible risk out there.

Festinger suggested that the smoker might try to convince himself that if he does stop smoking then he will gain weight, which also presents health risks. By using such explanations, the smoker is able to reduce the dissonance and continue the behavior.

Sociobiology

Sociobiology is based on the premise that some behaviors are at least partly inherited and can be affected by natural selection. It begins with the idea that behaviors have evolved over time, similar to the way that physical traits are thought to have evolved. Animals will, therefore, act in ways that have proven to be evolutionarily successful over time, which can result in the formation of complex social processes, among other things.?

According to sociobiologists, many social behaviors have been shaped by natural selection. Sociobiology investigates social behaviors such as mating patterns, territorial fights, and pack hunting. It argues that just as selection pressure led to animals evolving useful ways of interacting with the natural environment, it also led to the genetic evolution of advantageous social behavior. Behavior is therefore seen as an effort to preserve one's genes in the population and certain genes or gene combinations are thought to influence particular behavioral traits from generation to generation.

Charles Darwin's theory of evolution by natural selection explains that traits less adapted to particular conditions of life will not endure in a population because organisms with those traits tend to have lower rates of survival and reproduction. Sociobiologists model the evolution of human behaviors in much the same way, using various behaviors as the relevant traits. In addition, they add several other theoretical components to their theory.

Sociobiologists believe that evolution includes not just genes, but also psychological, social, and cultural features. When humans reproduce, offspring inherit the genes of their parents, and when parents and children share genetic, developmental, physical, and social environments, the children inherit the gene-effects of their parents. Sociobiologists also believe that the different rates of reproductive success are related to different levels of wealth, social status, and power within that culture.

Example of Sociobiology in Practice

One example of how sociobiologists use their theory in practice is through the study of sex-role stereotypes. Traditional social science assumes that humans are born with no innate predispositions or mental contents and that sex differences in children's behavior is explained by the differential treatment of parents who hold sex-role stereotypes. For example, giving girls baby dolls to play with while giving boys toy trucks, or dressing little girls in only pink and purple while dressing boys in blue and red.

Sociobiologists, however, argue that babies do have innate behavioral differences, which trigger the reaction by parents to treat boys one way and girls another way. Further, females with low status and less access to resources tend to have more female offspring while females with high status and more access to resources tend to have more male offspring. This is because a woman's physiology adjusts to her social status in a way that affects both the sex of her child and her parenting style. That is, socially dominant women tend to have higher testosterone levels than others and their chemistry makes them more active, assertive, and independent than other women. This makes them more likely to have male children and also to have a more assertive, dominant parenting style.

Psychodynamic Approaches

Sigmund Freud – psychiatrist, neurologist and key pioneer of the psychodynamic approach (including his psychoanalytic theory of personality) – theorised that our mental activity is mostly unconscious, and it is this unconscious activity that causes our behaviour. He explained that traumatic childhood experiences pushed into the unconscious mind can later lead to mental disorders, and developed 'talking cures' (psychoanalysis, or more generally termed psychotherapy) to help release problematic repressed memories and relieve symptoms.

Key Assumptions of the Psychodynamic Approach

Unconscious activity is the key determinate of how we behave.

We possess innate 'drives' (or 'instincts') that 'energize' our minds to motivate behaviour as we develop through our lives.

Our [three-part] personality – the psyche – is comprised of the ID, ego and superego. Childhood experiences have significant importance in determining our personality when we reach adulthood.

The Role of the Unconscious

The psyche, forming the structure of personality, has three parts:

ID driving us to satisfy selfish urges (i.e. acts according to the 'pleasure principle') (exists from birth).

Ego acts rationally, balancing the ID and the superego (i.e. acts according to the 'reality principle') (develops years 2-4).

Superego concerned with keeping to moral norms (i.e. acts according to the 'morality principle'), and attempts to control a powerful ID with feelings of guilt (develops years 4-5).

Psychosexual Stages

Freud also thought that humans progress through 'psychosexual stages', during the development of the psyche. He named five stages, each with a particular characteristic behaviour:

Oral – sucking behaviour (0-18 months)

Anal – holding or discarding faeces (18 months – 3.5 years)

Phallic – fixation on genitals (3.5 – 6 years)

Latency – repressed sexual urges (6 years - puberty)

Genital – awakened sexual urges (puberty onwards)

Freud claimed that, during development, becoming fixated on one of these stages would restrict full development result in displaying specific personality symptoms. An 'anally retentive' personality is one such symptom – he proposed that when conflict occurs over potty training, a person could become fixated on cleanliness and orderliness to an extreme.

Ego Defence Mechanisms

The ego balances potential conflict between the ID and superego, and tries to reduce anxiety. In areas of significant conflict, the ego can redirect psychic energy using 'defence mechanisms'. Three key mechanisms Freud proposed are:

Repression burying an unpleasant thought or desire in the unconscious (e.g. traumatic childhood experiences may be repressed and so forgotten).

Displacement emotions are directed away from their source or target, towards other things (e.g. wringing a dishcloth in anger, which would have otherwise been directed at the cat scratching the furniture).

Denial a threatening thought is ignored or treated as if it were not true (e.g. a wife might find evidence that her husband is cheating on her, but explain it away using other reasons).

Research Methods used by the Approach

Freud's psychoanalytical theory was based on case studies, which gather large amounts of detailed information about individuals or small groups. These cases were of patients with which he used psychoanalysis ('talking cures'), the aim of which to bring unconscious mental activity to the conscious to release anxiety. Techniques to do this include:

Free association expressing immediate [unconscious] thoughts, as they happen

Dream interpretation analysing the latent content (i.e. underlying meaning) of manifest content (i.e. what was remembered from the dream).

Current psychologists using the psychodynamic approach still utilize case study evidence (e.g. recording psychoanalytic interviews) to provide proof of their explanations.

Evaluation of the Psychodynamic Approach/Methods

Strengths

Freud highlighted a widely accepted link between childhood experience and adult characteristics.

Case study methodology embraces our complex behaviour by gathering rich information, and on an individual basis – an idiographic approach – when conducting research.

Some evidence supports the existence of ego defence mechanisms such as repression, e.g. adults can forget traumatic child sexual abuse (Williams, 1994).

Modern day psychiatry still utilizes Freudian psychoanalytic techniques.

Weaknesses

It could be argued that Freud's approach over-emphasises childhood experience as the source of abnormality (although modern psychodynamic theories give more recognition to the adult problems of everyday life, such as the effects of negative interpersonal relationships).

By using case studies to support theories, the approach does not use controlled experiments to collect empirical evidence, so is considered far less scientific than other approaches.

Case study evidence is difficult to generalise to wider populations.

Many of Freud's ideas are considered non-falsifiable – theories may appear to reflect evidence, but you cannot observe the relevant constructs directly (namely the unconscious mind) to test them scientifically, such that they could be proved wrong. Philosopher of science Karl Popper famously argued that a theory is not scientific if it is not falsifiable.

Social Cognition

Social cognition is a sub-topic of social psychology that focuses on how people process, store, and apply information about other people and social situations. It focuses on the role that cognitive processes play in our social interactions. The way we think about others plays a major role in how we think, feel, and interact with the world around us.

Defining Social Cognition

How exactly do psychologists define social cognition? While there is no single definition, there are some common factors that many experts have identified as being important.

Social Cognition Involves

The processes involved in perceiving other people and how we come to know about the people in the world around us.

It involves the study of these mental processes that involved in perceiving, remembering, thinking about, and attending to the other people in our social world.

Psychologists are interested in why we attend to certain information about the social world. How this information is stored in memory, and how it is then used to interact with other people.

It is not simply a topic within social psychology - it is an approach to studying any subject with social psychology. Using a social-cognitive perspective, researchers can study a wide range of topics including attitudes, person-perception, prejudice, stereotypes, self-concept, discrimination, persuasion, decision-making, and other areas.

The Role of Social Cognition

For example, imagine that you are getting ready to go on a blind date. Not only do you worry about the impression and signals that you are sending to the other person, you are also concerned with interpreting the signals given by the other individual. How do you form an impression of this person? What meaning do you read into the other person's behavior?

This is just one example of how social cognition influences a single social interaction, but you can probably think of many more examples from your daily life. We spend a considerable portion of every day interacting with others, which is why an entire branch of psychology formed to help understand how we feel, think and behave in social situations.

The Development of Social Cognition

Developmental psychologists also study how social cognition develops over the course of childhood and adolescence. As children grow, they become more aware not only of their own feelings, thoughts, and motives but also of the emotions and mental states of others. As this awareness increases, children become more adept at understanding how others are feeling, knowing how to respond in social situations, engaging in prosocial behaviors, and taking the perspective of others.

While there are many different theories that look at how social cognition develops, one of the most popular focuses on the work of the psychologist Jean Piaget. According to Piaget, a child's cognitive development goes through a series of stages.

During the earliest stages of development, children are very egocentric. They see the world from their own perspective and struggle to think about how other people may view the world.

As children grow older, children become increasingly adept at perspective-taking and have an increased ability to think about how and why people act the way they do in social situations.

More recently, research has provided evidence that children develop the ability to think about the perspectives of other people at an earlier age than Piaget previously believed. Even young preschoolers exhibit some ability to think about how other people might view a situation.

One of the most important developments in the early emergence of social cognition is the growth of a theory of mind. A theory of mind refers to a person's ability to understand and think about the mental states of other people. It is the emergence of a theory of mind that is critical to being able to consider the thoughts, motives, desires, needs, feelings, and experiences that other people may have. Being able to think about how these mental states can influence how people act is critical to forming social impressions and explaining how and why people do the things that they do.

Questions about Social Cognition

So what are some of the different questions related to social cognition that researchers are interested in understanding? Our perceptions of others play such an important role in how we forge relationships, how we interact with others, how we treat others, and how others treat us. Some of the topics that psychologists are interested in when it comes to social cognition include:

How do we interpret other people's feelings and emotions? How do we figure out what they are thinking or feeling? What cues or indicators do we use to make these assumptions?

What influence do our thoughts have on our feelings?

How do we develop attitudes? What role do these attitudes play in our social lives?

How is self-concept formed and how does it influence our relationships with others?

What mental processes influence person perception, or how we form impressions of other people?

Cultural Differences

Social psychologists have also found that there are often important cultural differences in social cognition. When looking at a social situation, any two people may have wildly different interpretations. Each person brings a unique background of experiences, knowledge, social influences, feelings, and cultural variations.

Some researchers have found that there are also collective, cultural influences that can affect how people interpret social situations. The same social behavior in one cultural setting may have a very different meaning and interpretation if it was to take place in another culture. As people interpret behavior, extract meaning from the interaction, and then act based upon their beliefs about the situation, they are then further reinforcing and reproducing the cultural norms that influence their social cognitions.

Possible Shortcomings

One criticism of some of the research on social cognition suggests that it is too focused on individualistic behavior. Because the topic itself is so social, some suggest that many of the information-processing models that have traditionally been used to understand the cognitive processes behind social cognition are too limited. Focusing more on the collective and interactive aspects of human thought may provide a better understanding of how people think about and understand social behavior.

SOCIAL PERCEPTION

Social perception theories and investigations deal with the nature, causes, and consequences of perceptions of social entities, including one's self, other individuals, social categories, and aggregates or groups to which one may or may not belong. The content of a perception can be virtually any property. Individual attributes may include personality traits, behavioral dispositions, physical characteristics, and ability evaluations. Group attributes can include properties such as size, cohesiveness, cultural traits, stratification patterns, network patterns, legitimacy, and historical elements. With some notable exceptions, however, the field of social perception traditionally has emphasized the micro side, focusing on individual inferences regarding one individual or a very small number of other individuals.

Social perception is best viewed as an umbrella label that covers a range of loosely related and usually loosely formulated theoretical conjectures and associated research. Today “social cognition” may be the more popular label, subsuming theory and research indexed under numerous other headings: person perception, social judgment, social representation, schema theory, reference group theory, impression formation, attribution theory, and more. Little of this work will be discussed here, although most of it is easy to access. Reference lists in books, and articles under the various headings tend to intersect rather than be isolated. Review articles have appeared with regularity, and so it is relatively easy to locate the seminal, general, or esoteric references one seeks.

Perception

Social perception is only one manifestation of the general phenomenon of human perception. All perceptions begin with energy-producing events, either inside people or from the environment. Each of the senses operates as a “transducer,” encoding a particular form of energy (e.g., radiant, kinetic, chemical) into neurological signals that are carried to the brain as complex, parallel streams of bioelectrical impulses. In the brain, these streams of information are filtered and transformed through several stages, producing dynamic neural representations almost instantaneously. Depending on anatomic factors, prior experiences, and the nature of the signals, these representations may or may not reach the level of conscious awareness.

When sensations survive this elaborate preprocessing and exceed sensory thresholds, however, they break through into a person’s conscious awareness, appearing as coherent and meaningful perceptions: hunger pangs, one’s reflection in the mirror, a smile from a friend. These perceptions seem to capture all the essential properties of the events that instigated them.

Roughly speaking, then, people acquire energetic impulses from internal and external environments that, in turn, impinge on the sensory apparatuses as sensations and are transformed by the brain into perceptions. This suggests a close relationship between the perceptions that are formed and the subsequent actions taken on their behalf. For instance, on the basis of perceptions of the personal qualities of others (and perceptions of others’ perceptions of those qualities), people make judgments about those qualities (e.g., good or bad); on the basis of those judgments, people formulate intentions about how they will behave toward others (e.g., plan to engage with them or avoid them); on the basis of those intentions, the actions of others, the prevailing context, and so on, people enact their impressions and intentions in social interactions.

Three qualities of perceptions bear further elaboration: Structure, stability, and meaning are definitive subjective properties. A fourth quality—accuracy—is best understood as an objective property of perceptions, at least in principle. Accuracy and bias in social perception are addressed later in this article.

Structure. Humans experience the world as structured. Rather than seeming chaotic and unpredictable, elements and events generally appear to correspond to one another in patterned ways. Things seem to happen for reasons. Much of this patterning is imposed, however, and one person’s perceptions may be very different from those of others, even under identical conditions. This is especially relevant in regard to the interpretations people impose on complex phenomena. For instance, people tend not to be aware of how differences in the expectations they bring to a situation color their perceptions. People

cannot take in information on everything around them and those expectations direct a person to attend selectively to the available stimuli in a situation. This biasing of attention can have a tremendous influence on interpretations of the situation. Moreover, each person may impose a unique subjective structure on the same objective reality. Every sports fan has experienced the perception that compared to the opponent, his or her favored team is consistently the victim of “bad calls” by the officials. Supporters of the opponent generally disagree, and rarely does one perceive the opposition to have been treated unfairly by officials. What fans actually “see” are slices of reality unconsciously chosen to conform with their beliefs and expectations.

Stability. Different sports fans may see different things, but if pressed, they probably could identify broad areas of the game they agree about and take for granted. They would agree that there were no sudden changes in the sport they were watching; they would profess not to have seen players on the field dematerializing in one place and rematerializing elsewhere; the ball appears to stay the same size and shape even as it moves nearer to and farther from the fans’ points of view. In general, while observers may disagree on some points, most of what is observed has an underlying sense of continuity. Indeed, among the myriad sensations to which one might attend in a given situation, the bias is toward those which engender a sense of stability—a sense of the temporal endurance of these patterned sensations.

Meaning. If structure and stability were the only properties of experience, the world would appear merely as successions of discrete, insignificant objects and events, each with no particular import transcending the moment. In contrast, most perceptions seem meaningful. That is, perceptions are conceived of as threads in a larger fabric. Through their interconnections and the patterns they form, they seem to have significance, purpose, causes, and consequences beyond their own existence. With cognitive development comes the ability to recognize and select impressions and events that are significant in terms of the information they convey. As will be discussed below, meaningfulness and significance do not imply accuracy. Perceptions—especially social perceptions—are imperfect representations that can be highly misleading.

Among the variety of ways in which one could organize the social perception literature, one of the simplest and most useful proceeds from the individual perceiver, to perceptions of other individuals, and finally to group perceptions. These categories define the three sections below, which are followed by some remarks about future directions.

Attitude

In psychology, attitude is a psychological construct, a mental and emotional entity that inheres in, or characterizes a person. They are complex and an acquired state through experiences. It is an individual's predisposed state of mind regarding a value and it is precipitated through a responsive expression towards a person, place, thing, or event (the attitude object) which in turn influences the individual's thought and action. Prominent psychologist Gordon Allport described this latent psychological construct as “the most distinctive and indispensable concept in contemporary social psychology.” Attitude can be formed from a person's past and present. Key topics in the study of attitudes include attitude strength, attitude change, consumer behavior, and attitude-behavior relationships.

Prosocial Behavior

Prosocial behaviors are those intended to help other people. Prosocial behavior is characterized by a concern for the rights, feelings, and welfare of other people. Behaviors that can be described as prosocial include feeling empathy and concern for others and behaving in ways to help or benefit other people.

In *The Handbook of Social Psychology*, C. Daniel Batson explains that prosocial behaviors refer to "a broad range of actions intended to benefit one or more people other than oneself—behaviors such as helping, comforting, sharing and cooperation."

The term prosocial behavior originated during the 1970s and was introduced by social scientists as an antonym for the term antisocial behavior.

Motivates Prosocial Behavior

Prosocial behavior has long posed a challenge to social scientists seeking to understand why people engage in helping behaviors that are beneficial to others, but costly to the individual performing the action. In some cases, people will even put their own lives at risk in order to help other people, even those that are complete strangers. Why would people do something that benefits someone else but offers no immediate benefit to the doer?

Psychologists suggest that there are a number of reasons why people engage in prosocial behavior. In many cases, such behaviors are fostered during childhood and adolescence as adults encourage children to share, act kindly, and help others.

Evolutionary psychologists often explain prosocial behaviors in terms of the principles of natural selection. Obviously, putting your own safety in danger makes it less likely that you will survive to pass on your own genes. However, the idea of kin selection suggests that helping members of your own genetic family makes it more likely that your kin will survive and pass on genes to the future generations. Researchers have been able to produce some evidence that people are often more likely to help those to whom they are closely related.

The norm of reciprocity suggests that when people do something helpful for someone else, that person feels compelled to help out in return. Essentially, helping others means that they might help us in return. This norm developed, evolutionary psychologists suggest because people who understood that helping others might lead to reciprocal kindness were more likely to survive and reproduce.

Prosocial behaviors are often seen as being compelled by a number of factors including egoistic reasons (doing things to improve one's self-image), reciprocal benefits (doing something nice for someone so that they may one day return the favor), and more altruistic reasons (performing actions purely out of empathy for another individual).

Situational Influences on Prosocial Behavior

Characteristics of the situation can also have a powerful impact on whether or not people engage in prosocial actions. The bystander effect is one of the most notable examples of how the situation can impact helping behaviors. The bystander effect refers to the tendency for people to become less likely to assist a person in distress when there are a number of other people also present.

For example, if you drop your purse and several items fall out on the ground, the likelihood that someone will stop and help you decreases if there are many other people present. This same sort of thing can happen in cases where someone is in serious danger, such as when someone is involved in a car accident. In some cases, witnesses might assume that since there are so many other people present, someone else will have surely already called for help.

The tragic murder of a young woman named Kitty Genovese was what spurred much of the interest and research on the bystander effect. In 1964, Genovese was attacked as neared her apartment on her way home from work late one night. She was stabbed and left lying on the sidewalk. She called for help and reports later indicated that many of her neighbors heard her cries yet did not call for help or attempt to interfere with the attack that lasted approximately 30 minutes. A neighbor eventually called police, but Genovese died before reaching the hospital.

The story generated considerable interest in the bystander effect and in understanding why people help in some situations but not in others, and experts have discovered a number of different situational variables that contribute to (and sometimes interfere with) prosocial behaviors.

First, the more people that are present decreases the amount of personal responsibility people feel in a situation. This is known as the diffusion of responsibility.

People also tend to look to others for how to respond in such situations, particularly if the event contains some level of ambiguity. If no one else seems to be reacting, then individuals become less likely to respond as well.

Fear of being judged by other members of the group also play a role. People sometimes fear leaping to assistance, only to discover that their help was unwanted or unwarranted. In order to avoid being judged by other bystanders, people simply take no action.

Lantane and Darley have suggested that five key things must happen in order for a person to take action. An individual must:

- Notice what is happening
- Interpret the event as an emergency
- Experience feelings of responsibility
- Believe that they have the skills to help
- Make a conscious choice to offer assistance

Other factors that can help people overcome the bystander effect including having a personal relationship with the individual in need, having the skills and knowledge to provide assistance, and having empathy for those in need.

Prosocial Behavior Versus Altruism

Altruism is sometimes seen as a form of prosocial behavior, but some experts suggest that there are actually different concepts. While prosocial behavior is seen as a type of helping behavior that ultimately confers some benefits to the self, altruism is viewed as a pure form of helping motivated purely out of concern for the individual in need.

Others argue, however, that reciprocity actually does underlie many examples of altruism or that people engage in such seemingly selfless behaviors for selfish reasons, such as to gain the acclaim of others or to feel good about themselves.

GROUP AND SOCIAL INFLUENCE

Social Facilitation

Not all intergroup interactions lead to the negative outcomes we have described. Sometimes being in a group situation can improve performance. Social facilitation occurs when an individual performs better when an audience is watching than when the individual performs the behavior alone. This typically occurs when people are performing a task for which they are skilled. Can you think of an example in which having an audience could improve performance? One common example is sports. Skilled basketball players will be more likely to make a free throw basket when surrounded by a cheering audience than when playing alone in the gym. However, there are instances when even skilled athletes can have difficulty under pressure. For example, if an athlete is less skilled or nervous about making a free throw, having an audience may actually hinder rather than help. In sum, social facilitation is likely to occur for easy tasks, or tasks at which we are skilled, but worse performance may occur when performing in front of others, depending on the task.

Social Loafing

Another way in which a group presence can affect our performance is social loafing. Social loafing is the exertion of less effort by a person working together with a group. Social loafing occurs when our individual performance cannot be evaluated separately from the group. Thus, group performance declines on easy tasks (Karau & Williams, 1993). Essentially individual group members loaf and let other group members pick up the slack. Because each individual's efforts cannot be evaluated, individuals become less motivated to perform well. For example, consider a group of people cooperating to clean litter from the roadside. Some people will exert a great amount of effort, while others will exert little effort. Yet the entire job gets done, and it may not be obvious who worked hard and who didn't.

As a college student you may have experienced social loafing while working on a group project. Have you ever had to contribute more than your fair share because your fellow group members weren't putting in the work? This may happen when a professor assigns a group grade instead of individual grades. If the professor doesn't know how much effort each student contributed to a project, some students may be inclined to let more conscientious students do more of the work. The chance of social loafing in student work groups increases as the size of the group increases.

Interestingly, the opposite of social loafing occurs when the task is complex and difficult. Remember the previous discussion of choking under pressure? This happens when you perform a difficult task and your individual performance can be evaluated. In a group setting, such as the student work group, if your individual performance cannot be evaluated, there is less pressure for you to do well, and thus less anxiety or physiological arousal (Latané, Williams, & Harkens, 1979). This puts you in a relaxed state in which you can perform your best, if you choose (Zajonc, 1965). If the task is a difficult one, many people feel motivated and believe that their group needs their input to do well on a challenging project (Jackson & Williams, 1985). Given what you learned about social loafing, what advice would you give a new professor about how to design group projects? If you suggested that individuals' efforts should not be evaluated, to prevent the anxiety of chok-

ing under pressure, but that the task must be challenging, you have a good understanding of the concepts discussed in this section. Alternatively, you can suggest that individuals' efforts should be evaluated, but the task should be easy so as to facilitate performance. Good luck trying to convince your professor to only assign easy projects!

The table below summarizes the types of social influence you have learned about in this module.

TYPES OF SOCIAL INFLUENCE

Type of Social Influence	Description
Conformity	Changing your behavior to go along with the group even if you do not agree with the group
Compliance	Going along with a request or demand
Normative social influence	Conformity to a group norm to fit in, feel good, and be accepted by the group
Informational social influence	Conformity to a group norm prompted by the belief that the group is competent and has the correct information
Obedience	Changing your behavior to please an authority figure or to avoid aversive consequences
Groupthink	Group members modify their opinions to match what they believe is the group consensus
Group polarization	Strengthening of the original group attitude after discussing views within a group
Social facilitation	Improved performance when an audience is watching versus when the individual performs the behavior alone
Social loafing	Exertion of less effort by a person working in a group because individual performance cannot be evaluated separately from the group, thus causing performance decline on easy tasks.

Social Influence

Social influence is described as the change in person's behaviour, thoughts, feelings and attitudes that results from interaction with another individual in society. It can be intentionally or unintentionally, as a result of the way the changed person perceives themselves in relationship to the influencer. It is different from conformity, power and authority. Many psychologists affirmed that social influence is the process by which individuals make real changes to their outlooks and behaviours as a result of communication with others who are perceived to be similar, desirable, or expert. People adjust their views with respect to others to whom they feel similar in accordance with psychological principles such as balance. Persons are also influenced by the majority: when a large portion of an individual's referent social group holds a particular attitude, it is possible that the individual will adopt it as well. Additionally, individuals may change a belief under the influence of another who is perceived to be an expert in the matter at hand.

Social influence has numerous connotations in psychology field. It is usually used to summarise the field of social psychology. It studies, "how thoughts, feelings and behaviour of individuals are influenced by actual, imagined or implied presence of others" (Allport, 1968).

French and Raven (1959) presented an early formalization of the notion of social influence in their dialogue of the bases of social power. According to French and Raven, agents

of change included not just individuals and groups, but also norms and roles. They viewed social influence as the outcome of the application of social power from one of five bases such as reward power, coercive power, legitimate power, expert power, or referent power. A change in reported opinion or attitude (conformity) was considered as an instance of social influence whether or not it represented a true private change.

Human social life is characterised by social influences; influences, they are aware of and some they are not. As individuals, they rarely give way to social influence to fit in and at times they do it because they are not sure of the proper way to feel or act and so use others as a resource of information. Social life of individual is also governed by social norms, which are usually accepted ways of thinking, feeling and behaving that are shared among others in a social group. When a social group has well recognised norms, there is a huge pressure for people to maintain such norm.

It is apparent that social influence is a phenomenon that involves a change in behaviour, actions or perspective as a result of a stimulus in environment. It is generally evident in everyday world, from how we adhere to unwritten social norms that systemize human lives to obeying instructions given by an authority figure. Effect of Social influence is especially visible in the long history of humankind, particularly World War II, where individuals were observed to commit immoral acts because they scuffle to act in sync with their personal judgments when faced with external pressure. The changes that precede social influences can be intended or unintended, instant or delayed, explicit or implicit.

Since 1959, researchers have distinguished true social influence from forced public acceptance and from changes based on reward or coercive power. Social scholars are still concerned with public compliance, reward power, and coercive power, but those concerns are differentiated from social influence studies. Current research on social influence generally uses experimental methodology in several areas such as minority influence in group settings, research on persuasion, dynamic social impact theory, a structural approach to social influence, and social influence in expectation states theory.

Minority Influence in Group Settings

Several studies regarding social influence have concentrated on the power of the majority within groups. Moscovici, Lage, and Naffrechoux (1969) recommended that characteristics such as consistency, autonomy, and rigidity empower minorities to act as influential agents. Additionally, evidence of the power of minority influence was established by Moscovici (1980), who suggested the conversion theory. Evidence exists that firmly establishes the minority as a strong source of influence.

Dynamic Social Impact Theory

Social impact theory was originated by Bibb Latane in 1981. This theory has three basic rules which consider how individuals can be sources or targets of social influence. Social impact is the result of social forces including the strength of the source of impact, the immediacy of the event, and the number of sources exerting the impact. Social impact theory is described as the idea that conforming to social influence depends on the strength of the group's importance, its immediacy, and the number of people in the group. According to Aronson et al. (2007) social impact theory envisages that conformity will increase as the strength and proximity increase. The more important the group is to us the more likely we are to conform to it. Argo and colleagues (2005) described social impact theory as people are impacted by the real, implied, or imagined presence or action of a social

presence (i.e. another person or group of people). This impact results from three “social forces”: number, immediacy and social source strength.

Conformity

It is changing how people behave to be more like others. This element plays to belonging and esteem needs as we seek the approval and friendship of others. Conformity can run very deep, as people will even change their beliefs and values to be like those of peers and admired superiors. Conformity refers to the act of changing a particular belief or behaviour to fit in with one's social environment. Main factor that influences conformity is social norms. Social norms are the expected behaviour within a specific culture or society. Once a particular way of doing things is established as a norm, people will start conforming to it as it gives the impression of being the 'right' thing to do. Person who conform to social group have low self-esteem, high need for social support or approval, need for self-control, low IQ, high anxiety, feeling of self blames and insecurity in the group, and lastly, feeling of inferiority.

Numerous researches demonstrate that when a person is confronted with social norms, one will often adjust their behaviour to closer approximation of the perceived norm (Bond & Smith, 1996). Dissimilar to popular belief, conformity is not personality-driven but highly situational (Goldberg 1952). In his experiment, Goldberg observed that conformity usually occurs in the initial stages of exposure and any additional exposure thereafter does not affect the influence. The results from his experiment also demonstrated that the more displeasing the subject initially was to the particular social norm in question, the greater the conformity, as the compromise in this case will be larger.

Factors that affect conformity. Conformity categorized into two parts that include normative conformity and informational conformity. Normative conformity is usually prompted by a need to fit in while informational conformity usually occurs when a person is looking for guidance in a vague situation. While a person involved in normative conformity usually conforms for fear of being rejected by a group. A person involved in informative conformity usually conforms because he is uncertain of the situation, and thus, do not have his own viewpoint in that particular situation to begin with. Finally, while normative conformity usually ends in compliance where the changes are evident in overt behaviour and actions (explicit), the influence of informational conformity usually results in internalization (implicit), where a person adopts the views and opinion of the group for his own.

Sherif (1935) conducted an experiment to measure conformity. The aim of the experiment was to test for informational conformity by placing participants in an ambiguous situation. They were first individually tested before being tested in groups of threes. Results demonstrated that the answers given in a group coincide with the rest even though the answer initially given in the first test was greatly different. Asch (1951) did not agree with the results of the auto-kinetic experiment as there was no correct answer in the experiment. Hence, he conducted another experiment that has an obvious answer to investigate the extent to which an individual would conform to the social pressure from a majority group. Participants in this experiment were placed with other confederates who gave the wrong answer on purpose. The results obtained from this experiment displayed that 75% of participants conformed at least once while only 25% of them did not conform. Interviews held after the experiment revealed that participants went along with the rest of the group for fear of being ridiculed. Some of them said that they genuinely believe the

group's answers were correct. This experiment exemplifies both normative influence and informational influence.

Other Studies have revealed that conformity varies across different cultures. People from Western cultures are categorized as individualist while people who are from Asian cultures are classified as collectivities. While the former places greater emphasis on self-development, the latter usually put the needs of family and other social groups above their own. Because of this difference, people who are from Asian cultures tend to conform more (Smith and Bond, 1998).

Peer Pressure

Peers play a large role in the social and emotional development of children and adolescents. Their influence begins at an early age and increases through the teenage years. It is natural, healthy and important for children to have and rely on friends as they grow and mature.

Peers can be positive and supportive. They can help each other develop new skills, or stimulate interest in books, music or extracurricular activities.

However, peers can also have a negative influence. They can encourage each other to skip classes, steal, cheat, use drugs or alcohol, share inappropriate material online, or become involve in other risky behaviors. The majority of teens with substance abuse problems began using drugs or alcohol as a result of peer pressure. This pressure can happen in person or on social media.

Kids often give in to peer pressure because they want to fit in. They want to be liked and they worry that they may be left out or made fun of if they don't go along with the group.

The following are tips about peer pressure to share with your kids:

Stay away from peers who pressure you to do things that seem wrong or dangerous.

Learn how to say "no," and practice how to avoid or get out of situations which feel unsafe or uncomfortable.

Spend time with other kids who resist peer pressure. It helps to have at least one friend who is also willing to say "no."

If you have problems with peer pressure, talk to a grown up you trust, like a parent, teacher, or school counselor.

Parents can also help by recognizing when their child is having a problem with peer pressure. The following are tips for parents to help your child deal with peer pressure:

Encourage open and honest communication. Let kids know they can come to you if they're feeling pressure to do things that seem wrong or risky.

Teach your child to be assertive and to resist getting involved in dangerous or inappropriate situations or activities.

Get to know your child's friends. If issues or problems arise, share your concerns with their parents.

Get to know how your child interacts with friends and others online. Communicate openly about safe internet and social media use.

Help your child develop self-confidence. Kids who feel good about themselves are less vulnerable to peer pressure.

Develop backup plans to help kids get out of uncomfortable or dangerous situations. For example, let them know you'll always come get them, no questions asked, if they feel worried or unsafe.

If your child has ongoing difficulties with peer pressure, talk to his or her teacher, principal, school counselor or family doctor. If you have questions or concerns about your child's mood, self-esteem or behavior, consider a consultation with a trained and qualified mental health professional.

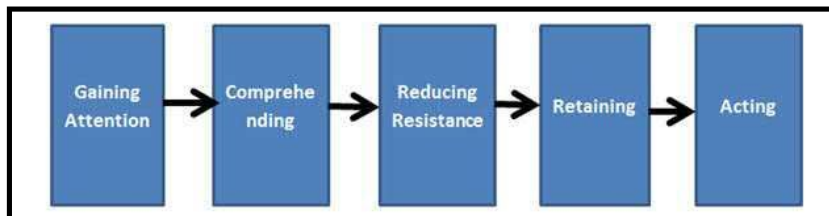
Persuasion

Generally, persuasion is defined as communicative activities that are mediated. According to the communication scholar Gerald R. Miller, any message that is aimed to shape, reinforce or change the perceptions, emotions, beliefs, behavioural intentions and behaviours is to be considered as persuasive communication (2002). When tracking historical record, the study of persuasive communication started in Ancient Greece and has a history of more than two eras. Ancient intellectuals like Plato and Aristotle framed rhetoric as a technique for oral persuasion, which was followed by the notable Roman scholars Buntillian and Cicero. Cicero described rhetoric as a speech designed to persuade (Quoted in Burke, 1969,). After many decades, the development of mass media facilitated the broadening of the concept of rhetoric beyond oratory. The philosopher Kenneth Burke was the first to recognize the persuasive potential of nonverbal domains. Burke's work gave rise to the study of persuasiveness in other domains, increasing interest in visual rhetoric, understood as the art of using imagery and visuals persuasively. It is established in literature that persuasion is long term endeavour. It is a way of seeing and being in the world.

In management literature, persuasion is the process of changing or reinforcing attitudes, beliefs or behaviour of a person. People respond to persuasive messages in two ways: thoughtfully and mindlessly. When people are in thoughtful mode, the persuasiveness of the message is determined by merits of the message. When people respond to messages mindlessly, their brains are locked on automatic. Persuasion is mainly dependent upon the attractiveness of the speakers and reaction of the listeners. Persuasion is exclusively related with communication, learning, awareness and thought.

Process of Persuasion

In the process of persuasion, both the persuader and the receiver of the persuasive message are wilfully active. As Bettinghaus writes, "perception of a persuasive message is not a passive process. The receiver is as active in the receiving process as is the source in the transmitting process. The attitudes and beliefs of the receiver mediate the way in which the message will be received and responded to"



The process of persuasion involves a series of consecutive steps:

The communication is presented; the person pays attention to it; he comprehends the contents of the message and also the basic conclusion being urged. However, for persuasion to be effected the individual must agree with or yield to the point being urged and then finally act on it or in other words carry out the behaviour implied due to the new change in his attitude.

Components of Persuasion. The components in the communication process are source, message, the context of the message channel, receiver and the audience.

1. “Source” factors include the perceived sender of the communication. The “message” denotes to what person says and includes style, content and organization, while “Channel” designates the medium (e.g. press, radio, television) through which the message is communicated. The source of a persuasive message is the communicator who is presenting it. A source is more persuasive if he or she is seen as credible (believable) and attractive.
There are two ways for a source to be trustworthy: (a) claiming to be an expert, and (b) appearing to be trustworthy.
2. **The Message.** Persuasive messages can involve emotional appeals or rational opinions. When time is limited, short emotional appeals may be more effective than rational arguments. There is also evidence that more intelligent audiences are persuaded better by two-sided messages, probably because they more readily recognize that there are two sides to the issue.
3. **The Context.** Promoters often have difficulty in capacitating the internal arguments that compete with their persuasive messages. When people listen to or read a persuasive message, they are usually free to limit our attention or silently counter argue with its arguments. Research has revealed that when subjects are distracted, they are more likely to accept a persuasive message than when they have been allowed to concentrate on their counterarguments.
4. **The Audience.** Numerous research concentrated on the recipients of persuasive messages, the audience, to discover when some people are more persuadable than others. Many audience characteristics interact with message variables, such as involvement or intelligence. Intellectual recipients are more persuaded by complex messages, while unintelligent recipients are more persuaded by simple emotional messages.

Persuasive communication can be directed at:

1. **Cognition.** Persuasion can be used to change individuals’ beliefs about an object or an issue, which includes attributes, interpretation, definition, outcome, etc.
2. **Attitude.** Persuasion can be used to change individuals’ attitude toward an object or an issue, which refers to the categorization of an object or an issue along an evaluative dimension.
3. **Behaviour.** Persuasion can be used to alter individuals’ behaviour, which is the overt actions regarding an object or an issue.
4. Famous theorist, Perloff (2003) explained persuasion a symbolic process in which communicators try to persuade other people to change their attitudes or behaviours regarding an issue through the transmission of a message in an atmosphere of free choice.

Compliance

Compliance is described as a particular kind of response, acquiescence to a particular kind of communication, a request. The request may be obvious, as in the direct solicitation of funds in a door-to-door campaign for charitable donations, or it may be implicit, as in a political advertisement that sellers the qualities of a candidate without directly asking for a vote (Cialdini & Goldstein, 2004). But in all cases, the target recognizes that he or she is being urged to respond in a desired way. In general term, compliance is where a person does something that they are asked to do by another. They may choose to comply or not to comply, although the thoughts of social reward and punishment may lead them to compliance when they really do not want to comply.

Compliance greatly impact everyday behaviour, especially in social relations. Social psychologists interpret compliance as way of social influence used to realize goals or accomplish social or personal gains. To study compliance, social psychologists investigate overt and subtle social influences and their relationship to compliance.

Factors that impact compliance are as under:

Group strength. The more important the group is to an individual, the more likely the individual is to obey with social influence.

Immediacy. The closeness of the group makes an individual more likely to comply with group pressures. Pressure to comply is sturdiest when the group is closer to the individual and made up of people the individual cares about.

Number. Compliance increases as the number of people in a group increases. Importantly, the influence of adding people starts to decrease as the group gets larger.

Similarity. Perceived shared characteristics cause an individual to be more likely to comply with a request, particularly when the shared feature is perceived as unplanned and rare.

Obedience

It is different from compliance, in that it complies with an order from someone that people accept as an authority personality. In compliance, there is some choice. In obedience, there is no choice. Basically, obedience refers to a social influence in which a person follows explicit instructions that were given by an authority figure. In the long history of humankind, the effect of obedience is bold. It is deduced that the most conspicuous feature of obedience is the presence of an authority figure.

Milgram (1963) stated that there are various factors that affect the extent of obedience. Firstly, it was observed that prestige and obedience are positively related. When the experiment was moved to somewhere less prestigious as opposed to the original location (a university), obedience level dropped. Secondly, surveillance was also observed to affect level of obedience. Buffers that prevent the participant from being fully aware of the impact of their actions also increase the level of obedience. It was also observed that authority amplifies level of obedience.

An individual is said to conform if they choose a course of action that is preferred or considered socially acceptable by the majority. Because the individual is influenced by how the majority thinks or behaves is referred to as majority influence. The fact that an individual conforms along with the majority in public, does not necessarily mean they

have changed their private outlook or beliefs. Therefore, most majority influence is characterised by public compliance rather than private acceptance.

Factors that affect Obedience

Milgram and Zimbardo concluded through their experiments that there are many factors affect obedience:

Proximity to the authority figure. Proximity designates physical closeness; the closer the authority figure is, the more obedience is demonstrated. In the Milgram experiment, the experimenter was in the same room as the participant, likely eliciting a more obedient response.

Prestige of the experimenter. Both researchers have advocated that the prestige associated with Yale and Stanford respectively may have influenced obedience in their experiments.

Expertise. A subject who has neither the ability nor the expertise to make decisions, especially in a crisis, will leave decision making to the group and its hierarchy.

Deindividuation. In obedience, people come to view themselves not as individuals but as instruments for carrying out others' desires, and thus no longer perceive themselves as responsible for their actions.

It can be said that Social influence is the ways in which external factors produce change in an individual. It guides the way person form his/her thoughts and organize overt behaviour and actions. Social influence is a phenomenon people experience every day. Because of some noticeable differences, social influence has been grouped into three different categories as conformity, compliance and obedience. While they share various resemblances when compared against each other, and even among each other, they have some distinct qualities. Compliance involves an explicit request that is less aggressive as opposed to that of obedience. It can be concluded that though the notions of compliance and obedience are analogous, conformity stands distinctly apart because of its unique stimulus and result.

Social Power

Social power is the potential for social influence. The available tools one has to exert influence over another can lead to a change in that person. Social power and social influence are separate and distinct concepts. Although social power is potential (which may or may not be used), social influence is an effect, an actual change (or deliberate maintenance) in the beliefs, attitudes, behavior, emotions, and so on, of someone because of the actions or presence of another. The person or group that is the source of influence is commonly known as the influencing agent, whereas the object of the attempted or successful influence attempt is commonly known as the target (of the influence). Thus, influencing agents have social power, which are the means they may use to influence targets.

The ability of one person (or group) to get others to do his or her will, also known as social power, has long been of interest to social psychologists. Perhaps this is because so much of human interaction involves the change or the attempt to change the beliefs, attitudes, or behaviors of another. Because of the long-standing interest in the topic, several different investigations have used different definitions of social power and differ-

ent ways of measuring power. However, an extensive survey has found that the approach most commonly used originally identified five distinct potential tactics one could use in an influence attempt. This approach was updated some years ago and now includes six distinct tactics that can be subdivided into eleven varieties. Of course, in an influence attempt multiple types of influence are often used at the same time. The types of social power are as follows:

Informational. This type is the ability to rationally persuade someone.

Expert. This social power is similar to informational power except that arguments are not necessary because the target trusts the influencing agent.

Referent. The referent type is based on the target's identifying and liking the influencing agent and, because of this, wanting to comply with his or her requests.

Coercive power. This type involves threat of punishment. These can be things such as monetary fines (impersonal) or simply personal disapproval (personal).

Reward power. This social power type stems from the ability of the influencing agent to grant some kind of reward, either impersonal or personal.

Legitimate power. Based on what general society typically expects of us, this includes (a) the formal legitimate (or position) norm, which is the right to ask for something based simply on position or job title; (b) the reciprocity norm, whereby if someone does something for you, you owe him or her the favor in return; (c) the equity norm, the idea that one is expected to help others receive what they deserve, for example, if you work hard, you should get rewarded; and (d) social responsibility (or dependence), whereby people are obligated to help those who depend on them.

The type of social power used in an influence attempt often depends on a person's motivations. Sometimes people are consciously aware of their motivations, and sometimes they are not. Clever influencing agents often choose the kind of influence they use based on considerations of potential effectiveness and other factors. These factors can be quite varied. For example, some people are motivated by the desire to appear powerful. To feel powerful, an influencing agent may choose a type of influence strategy that makes him or her feel as though he or she is in control of the target of influence. If so, the influencing agent may choose to use coercion or reward in the influence attempt. Similarly, a desire to enhance one's sense of power in the eyes of others, status, security, role requirements, the desire to harm a target of influence, and self-esteem considerations might lead one to choose the more controlling, stronger, or harsher types of influence tactics (such as coercion). Others may wish to maintain a friendship or appear humble. In that case, they would rely more on information.

When these stronger or harsher types of power are used effectively, they enhance the influencing agent's view of himself or herself because the influencing agent can attribute subsequent change in the target to himself or herself. When this occurs, the power-holder then tends to think less of the target of influence. It has been argued that simply through the continual exercise of successful influence, the power-holder's view of others and himself or herself changes in a harmful way. The power-holder begins to view himself or herself as superior to the person that over which he or she is exercising power, and because of this feeling of superiority, may treat the target of influence in a demeaning manner. This effect would be consistent with the common belief that power is a corrupting influence.

Social Power Evidence

Hundreds of studies published in respected scientific journals involving social power as described earlier have been conducted in several diverse fields, including health and medicine, family relations, gender relations, education, marketing and consumer psychology, social and organizational psychology, and examinations of confrontation between political figures. Studies have been conducted in the context simulations and questionnaires, strictly controlled laboratory settings using traditional experimental methods, real-world settings such as hospitals and other large organizations, and through historical case study analysis.

Importance of Social Power

Much of what humans do as individuals and society involves influencing others. People want and need things from others, things such as affection, money, opportunity, work, and justice. How they get those things often depends on their abilities to influence others to grant their desires. In addition, people are also the constant targets of the influence attempts of others. Thus, it is important to understand what causes people to comply with others' wishes, and how the exercise of power affects both targets and influencing agents. The study of social power provides that knowledge.

Reactance

Broadly, reactance refers to the idea that people become upset when their freedom is threatened or eliminated, so much so that they attempt to reassert their lost freedom. The theory is relevant to the idea that humans are motivated to possess and preserve as many options and choices as possible. When people's options are restricted, they experience aversive emotional consequences. Reactance is very similar to a layperson's idea of reverse psychology: Humans will tend to do the opposite of what they are told to. Being ordered to do something by an external person or source implies that someone is trying to reduce one's freedom. Reactance also refers to the idea that people will want something more if they are told they cannot have it. As a result, humans may act in a manner that will oppose a resistance presented to their freedom.

Psychological reactance theory was first proposed by the social psychologist Jack Brehm in 1966. Reactance theory is still considered to be one of the basic psychological theories; it has withstood decades of testing and can be applied to many aspects of human behavior.

Reactance theory is important because it highlights people's need for control, freedom of action and choice, as well as people's desire to preserve as many options as possible. Indeed, the theory was devised during a decade when people were constantly advocating and rallying about freedom of choice and action. Brehm observed that humans react strongly to having options taken away by external forces; they become quite upset and will take action to preserve or regain their lost options.

Many psychologists have noted that humans have a very strong aversion to loss, both in options and choices. Essentially, humans value freedom greatly. They like having options so much that they will incur costs to their own self just to maintain options, even if the options they keep open aren't that important or profitable. Think about what would happen if you woke up one day and heard on the news that you no longer have the right to vote; most likely you would become very upset—people value the ability to vote in a demo-

cratic society. Though this may seem like an extreme example, even people who do not exercise the right to vote would be upset. Indeed, many people would immediately revolt because someone else is trying to infringe on one of their basic freedoms.

Reactance theory highlights the simple, but important, fact that people value their freedom: When this freedom of behavior and choice is threatened, people will engage in motivated behavior, designed to take steps that will reassert and regain that freedom. In the former example, citizens will rally, petition, they may even become aggressive, if necessary, to try and regain freedom or options they feel are jeopardized.

A subtler example can be demonstrated by one of the original studies on reactance. Participants were asked to rate a series of records and then list the three they desired the most. Importantly, participants were promised that they could keep one of the records. After ranking their top three choices, participants were told that their third choice was unavailable. The researchers found that when participants were asked to rate the records again, the choice that was no longer available (their third choice) would then be rated as more attractive than it originally was. Simply because the option was no longer available, people actually valued it more.

Consequences of Reactance

When people react, they become aroused. That is, they become upset, distressed, angry, or emotionally charged. Over the decades, researchers have been able to identify three main ways that people direct this arousal. These are known as the main consequences of psychological reactance.

First, an object, action, or freedom becomes more attractive after it has been eliminated or threatened. That is, the desire for that behavior or object will increase, as seen in the previous example. This consequence also applies to things such as people and behaviors, not just objects. For example, teenagers who are told by their parents that they cannot attend a party on the weekend want to go to that party more than before their parents restricted the teenagers' behavior. Even if the teenagers originally had no intention of attending the party, once they are told they cannot, they will desire going to the party more than before.

Second, people will engage in behavioral attempts to reassert the threatened or eliminated freedom. That is, a person will try to regain his or her freedom or options. According to reactance theory, when parents forbid teenagers to attend the party, the teenagers will engage in behaviors that they think will increase their chances of regaining their options. For example, they may begin arguing with their parents about the benefits (e.g., social acceptance) and costs (e.g., exclusion, being the only one in the class not attending) of attending the party. Hence, the teenagers will try to regain the ability to attend the party.

Often people will even engage exactly in the same behavior that was threatened or eliminated. Thus, if the teenagers cannot convince their parents to let them go, they may go anyway, either by sneaking out of the house or pretending to do something else, such as going to a respected friend's house.

Finally, reactance may lead people to feel or act aggressively toward the person who is attempting to restrict their freedom. For example, in times of war, citizens whose country is being occupied may feel intense hatred toward the enemy (occupiers) such that they have aggressive thoughts, and sometimes even aggressive actions, toward the enemy.

Influences on Degree of Reactance

The magnitude of reactance is not exactly the same for each person, nor for each situation. Rather, it depends on several key factors. First, the importance of the action or choice determines the degree of reactance to the loss. That is, when something that is very important to a person is in jeopardy, that person will probably experience stronger reactance (i.e., more arousal, increased attempt to regain). For example, students wishing to enroll in a course would probably value enrolling in it more if it is required to graduate than if it is only an elective. Consequently, if it is required to graduate and they are unable to enroll in it because the course is full, they will react more strongly than if they had wanted to take it simply as an elective. Moreover, the students who value it more will probably try and reassert their ability to take that course by pleading their case to the professor or department, whereas those students who wanted to take it as an elective might just attempt to enroll in the course next semester (though, to be sure, they will probably want to take the course more than before).

If an option or behavior has not been taken away, but has only been threatened to be taken away, the perceived magnitude of the threat (that is, if only a threat exists) will determine the strength of the reactance experienced by the person. If the threat is blatantly strong, then the person will experience stronger psychological reactance in response to the threat.

Importance of Reactance

Having control over their actions and behavior is one of human beings' most important and valued needs. Indeed, people become distressed, angry, and even aggressive to actual loss of freedom, even perceived infringement on freedom. For example, after a couple breaks up, the person who initiated the end of the relationship is better able to cope and often feels a maintained sense of control. The person who did not have control over the termination of the relationship, however, will typically want his or her ex-partner back even more. That person also tends to feel a lack of control over the situation, which can be accompanied by wanting the ex-partner back more, being unable to think about anything else, and taking extreme steps to try and win that person back.

Men who are refused by women they believe they should have the opportunity to sleep with may become angry and coercive, even to the point of raping her. Moreover, sometimes reactance will produce behavior that is opposite of what was intended. This could be one reason why restrictions on violent video games and movies, pornographic material, or unhealthy behaviors such as smoking or drinking underage leads to the opposite of the intended effect. Humans will even use this basic knowledge to their advantage. For example, some parents may try to have their children cooperate by using reverse psychology on them.

Aggression

In psychology, the term aggression refers to a range of behaviors that can result in both physical and psychological harm to yourself, others, or objects in the environment. This type of behavior centers on harming another person either physically or mentally. It can be a sign of an underlying mental health disorder, a substance use disorder, or a medical disorder.

Forms

Aggression can take a variety of forms, including:

- Physical
- Verbal
- Mental
- Emotional

While we often think of aggression as purely in physical forms such as hitting or pushing, psychological aggression can also be very damaging. Intimidating or verbally berating another person, for example, are examples of verbal, mental, and emotional aggression.

Purposes

Aggression can serve a number of different purposes, including:

- To express anger or hostility
- To assert dominance
- To intimidate or threaten
- To achieve a goal
- To express possession
- A response to fear
- A reaction to pain
- To compete with others

Types

Psychologists distinguish between two different types of aggression:

Impulsive aggression. Also known as affective aggression, impulsive aggression is characterized by strong emotions, usually anger. This form of aggression is not planned and often takes place in the heat of the moment. When another car cuts you off in traffic and you begin yelling and berating the other driver, you're experiencing impulsive aggression. Research suggests that impulsive aggression, especially when it's caused by anger, triggers the acute threat response system in the brain, involving the amygdala, hypothalamus, and periaqueductal gray (PAG).

Instrumental aggression. Also known as predatory aggression, instrumental aggression is marked by behaviors that are intended to achieve a larger goal. Instrumental aggression is often carefully planned and usually exists as a means to an end. Hurting another person in a robbery or car-jacking is an example of this type of aggression. The aggressor's goal is to obtain money or a vehicle, and harming another individual is the means to achieve that aim.

Factors That Can Influence Aggression

A number of different factors can influence the expression of aggression, including:

Biological factors. Men are more likely than women to engage in physical aggression. While researchers have found that women are less likely to engage in physical aggression, they also suggest that women do use non-physical forms, such as verbal aggression, relational aggression, and social rejection.

Environmental factors. How you were raised may play a role. People who grow up witnessing more forms of aggression are more likely to believe that such violence and hostility are socially acceptable. Bandura's famous Bobo doll experiment demonstrated that observation can also play a role in how aggression is learned. Children who watched a video clip where an adult model behaved aggressively toward a Bobo doll were more likely to imitate those actions when given the opportunity.

Physical factors. Epilepsy, dementia, psychosis, alcohol abuse, drug use, and brain injuries or abnormalities can also influence aggression.

Group Dynamics

Group dynamics deals with the attitudes and behavioral patterns of a group. Group dynamics concern how groups are formed, what is their structure and which processes are followed in their functioning. Thus, it is concerned with the interactions and forces operating between groups.

Group dynamics is relevant to groups of all kinds – both formal and informal. If the UPA government has set up Group of Ministers for every governance issue, the Supreme Court of India has 27 Group of Judges committees overseeing all manner of non-judicial work in the apex court. In an organizational setting, the term groups are a very common and the study of groups and group dynamics is an important area of study.

Leadership Styles

Leaders' navigation through different leadership styles is important, as there is "no best style" that is appropriate for all situations. Goleman (2014a) identified six leadership styles such as authoritative, coaching, affiliative, democratic, pacesetter and coercive. He argued that a leader use a combination of the six styles. According to Goleman (2014a), a leader is authoritative when he/she is able to promote organisational vision alongside specific actions for its attainment. While the coaching style of leadership requires leader's expertise and time but is rewarding in the end. The affiliative style of leadership creates happy ambiance, and gives positive feedback. Goleman (2014) suggested the use of this style alongside authoritative style is best in keeping the balance between serious work targets and harmonious work environment. Democratic style is consultative in nature, this makes subordinates feel they are involved in decision-making and therefore lessen oppositions. The last two styles are pacesetting and coercive styles. He suggested the use of these styles sparingly or only when the need arise as these styles bring negative setback to the organisation for it kills employees' motivation. What is interesting is that Goleman's (2014a) paper on leadership styles is in accord with Blanchard and Hersey's, "Situational Leadership" theory. What these experts and researchers in the field are saying is that leaders use a variation of leadership styles depending on the situation confronting them.

Team Effectiveness

It is worth noting that certain leadership styles facilitate team effectiveness. Recent studies reported the positive effect of empowering leadership style on team members' task performance and cohesiveness, respectively, which led to greater team performance. Whilst effective leadership is important, perfect blend of people in teams is also a factor in achieving organisational targets. It is therefore important for today's leaders to know

the impact of team composition to organisational performance. Recent study reports that it is best to put together senior employees and new employees in teams for they are more likely to succeed in reaching optimal productivity. This is in contrast to the notion that workers familiarity in teams deliver higher level of productivity. However, one study observed that diverse orientation of team members create negative effect of overall team performance.

Aside from mere knowledge about team composition, leaders should have an in-depth knowledge on how to blend people in teams to boost team performance. For example, one study found that impact of team membership change, team internal and external communication, team conflict, trust and psychological safety all impact on team performance. This study gives valuable insights for leaders. For example, a change in team membership could lower overall team performance as the new comer might not be adept in the new assignment and therefore could negate productivity. Whilst internal communication foster team innovation and creativity. Likewise, the presence of conflicting views among team members in accomplishing team tasks contributes to creativity. In similar vein, external communication also help team to build network of support. Open communication, trust, collaboration and interdependence promote positive team outcomes. These are relevant inputs for modern organisational leaders.

Theories of Intergroup Relations

Minimal Group Experiment

Prior to Tajfel's work, it was believed that group bias (favoritism and/or prejudice) arose from personal interests of group members or from conflict. Tajfel demonstrated that the minimal condition needed for group favoritism is simply categorization into a group, no matter how arbitrary the criteria for categorization.

In 1971, Henri Tajfel conducted experiments to find out what the minimal conditions were for intergroup bias. In order to conduct proper experiments, he followed several criteria:

- No face-to-face interaction between subjects (in- or out-group).

- No subjects could know to which group other subjects belonged.

- The responses of the subjects that would demonstrate bias could not in any way be justified by the group membership inclusion criteria.

- The responses of the subject should not benefit them in any utilitarian way.

- A strategy of intergroup differentiation should be in competition with a more utilitarian or rational strategy that benefits the in-group in absolute terms.

- The response should be made important and real to the subject

Two experiments were conducted. The first of which consisted of 64 male subjects aged 14 to 15 who estimated the number of dots projected on a screen. They were divided into four groups, under two different experimental conditions. In the "neutral" condition, subjects were told that some overestimate and others underestimate, but this did not reflect accuracy. In the "value" condition, the subjects were told that some people are more accurate than others.

The subjects were then randomly assigned to one of the four groups (over estimators, under estimators, accurate, inaccurate) regardless of their actual performance (which

was irrelevant for this experiment). Now each subject had the opportunity to give cash awards to different subjects based on group membership. Did group membership matter?

In the second experiment, 48 of the same boys were divided into three groups this time. They were shown slides of reproductions of paintings by two artists, Klee and Kandinsky, with slides identified by number and painting by a letter. They were asked which ones they preferred, although the artist remained anonymous to them. Again, each of the groups were randomly assigned to either the "Klee" or the "Kandinsky" group, and the subjects were presented with matrices where they could assign monetary rewards to their own group and the other group.

Results. In the first experiment, the different conditions resulted in no significant differences in the results. All the groups shown significant favoritism for their in-group, and there was striking evidence for discriminatory in-group behavior. Similarly, experiment 2 resulted in significant in-group favoritism.

Social Identity Theory

Tajfel and Turner's social identity theory explains that part of a person's concept of self comes from the groups to which that person belongs. An individual does not just have a personal selfhood, but multiple selves and identities associated with their affiliated groups. A person might act differently in varying social contexts according to the groups they belong to, which might include a sports team they follow, their family, their country of nationality, and the neighborhood they live in, among many other possibilities.

When a person perceives themselves as part of a group, that is an ingroup for them. Other comparable groups that person does not identify with are called outgroups. We have an "us" vs. "them" mentality when it comes to our ingroups and their respective outgroups.

There are three processes that create this ingroup/outgroup mentality:

Social Categorization. First, we categorize people in order to understand and identify them. Some examples of social categories include black, white, professor, student, Republican, and Democrat. By knowing what categories we belong to, we can understand things about ourselves, and we can define appropriate behavior according to the groups that we and others belong to. An individual can belong to several groups at the same time.

Social Identification. We adopt the identity of the group that we belong to, and we act in ways that we perceive members of that group act. For example, if you identify as a Democrat, you will most likely behave within the norms of that group. As a consequence of your identification with that group, you will develop emotional significance to that identification, and your self-esteem will be dependent on it.

Social Comparison. After we categorize ourselves within a group and identify ourselves as being members of that group, we tend to compare our group (the ingroup) against another group (an outgroup). To maintain your self-esteem, you and your group members will compare your group favorably against other ones. This helps explain prejudice and discrimination, since a group will tend to view members of competing groups negatively to increase self-esteem.

Relative Deprivation Theory

Relative deprivation is the lack of resources to sustain the diet, lifestyle, activities and amenities that an individual or group are accustomed to or that are widely encouraged or

approved in the society to which they belong. Measuring relative deprivation allows an objective comparison between the situation of the individual or group compared to the rest of society. Relative deprivation may also emphasise the individual experience of discontent when being deprived of something to which one believes oneself to be entitled, however emphasizing the perspective of the individual makes objective measurement problematic.

It is a term used in social sciences to describe feelings or measures of economic, political, or social deprivation that are relative rather than absolute. The term is inextricably linked to the similar terms poverty and social exclusion. The concept of relative deprivation has important consequences for both behavior and attitudes, including feelings of stress, political attitudes, and participation in collective action. It is relevant to researchers studying multiple fields in social sciences. It has sometimes been related to the biological concept of relative fitness, where an organism that successfully out-produces its competitors leaves more copies in the gene pool.

Social scientists, particularly political scientists and sociologists, have cited 'relative deprivation' (especially temporal relative deprivation) as a potential cause of social movements and deviance, leading in extreme situations to political violence such as rioting, terrorism, civil wars and other instances of social deviance such as crime. For example, some scholars of social movements explain their rise by citing grievances of people who feel deprived of what they perceive as values to which they are entitled. Similarly, individuals engage in deviant behaviors when their means do not match their goals.

Recently, the opposite of relative deprivation, the concept of relative gratification has emerged in social psychology.

Realistic Conflict Theory

Realistic conflict theory (initialized RCT), also known as realistic group conflict theory (initialized RGCT) is a social psychological model of intergroup conflict. The theory explains how intergroup hostility can arise as a result of conflicting goals and competition over limited resources, and it also offers an explanation for the feelings of prejudice and discrimination toward the outgroup that accompany the intergroup hostility. Groups may be in competition for a real or perceived scarcity of resources such as money, political power, military protection, or social status.

Feelings of resentment can arise in the situation that the groups see the competition over resources as having a zero-sums fate, in which only one group is the winner (obtained the needed or wanted resources) and the other loses (unable to obtain the limited resource due to the "winning" group achieving the limited resource first). The length and severity of the conflict is based upon the perceived value and shortage of the given resource. According to RCT, positive relations can only be restored if superordinate goals are in place.

Balance Theories

In the psychology of motivation, balance theory is a theory of attitude change, proposed by Fritz Heider. It conceptualizes the cognitive consistency motive as a drive toward psychological balance. The consistency motive is the urge to maintain one's values and beliefs over time. Heider proposed that "sentiment" or liking relationships are balanced if the affect valence in a system multiplies out to a positive result.

In social network analysis, balance theory is the extension proposed by Frank Harary and Dorwin Cartwright. It was the framework for the discussion at a Dartmouth College symposium in September 1975.

Equity Theory

Equity theory focuses on determining whether the distribution of resources is fair to both relational partners. Equity is measured by comparing the ratio of contributions (or costs) and benefits (or rewards) for each person. Considered one of the justice theories, equity theory was first developed in the 1960s by J. Stacy Adams, a workplace and behavioral psychologist, who asserted that employees seek to maintain equity between the inputs that they bring to a job and the outcomes that they receive from it against the perceived inputs and outcomes of others (Adams, 1963). According to Equity Theory, in order to maximize individuals' rewards, we tend to create systems where resources can be fairly divided amongst members of a group. Inequalities in relationships will cause those within it to be unhappy to a degree proportional to the amount of inequality. The belief is that people value fair treatment which causes them to be motivated to keep the fairness maintained within the relationships of their co-workers and the organization. The structure of equity in the workplace is based on the ratio of inputs to outcomes. Inputs are the contributions made by the employee for the organization.

Social Exchange Theory

Social exchange theory proposes that social behavior is the result of an exchange process. The purpose of this exchange is to maximize benefits and minimize costs. According to this theory, developed by sociologist George Homans, people weigh the potential benefits and risks of social relationships. When the risks outweigh the rewards, people will terminate or abandon that relationship.

How Social Exchange Theory Works

Most relationships are made up of a certain amount of give-and-take, but this does not mean that they are always equal. Social exchange suggests that it is the valuing of the benefits and costs of each relationship that determine whether or not we choose to continue a social association.

Costs vs. Benefits

Costs involve things that you see as negatives such as having to put money, time, and effort into a relationship. For example, if you have a friend that always has to borrow money from you, then this would be seen as a high cost.

The benefits are things that you get out of the relationship such as fun, friendship, companionship, and social support. Your friend might be a bit of a freeloader, but he brings a lot of fun and excitement to your life. As you are determining the value of the friendship, you might decide that the benefits outweigh the potential costs.

Social exchange theory suggests that we essentially take the benefits and subtract the costs in order to determine how much a relationship is worth. Positive relationships are those in which the benefits outweigh the costs while negative relationships occur when the costs are greater than the benefits.

Expectations and Comparison Levels

Cost-benefits analysis plays a major role in the social exchange process, but so do expectations. As people weigh the benefits of a relationship against the costs of the relationship, they do so by establishing a comparison level that is often influenced by social expectations and past experiences. If you have always had poor friendships, your comparison levels at the start of a relationship will be much lower than a person who has always had a close-knit circle of supportive and caring friends.

For example, if your previous romantic partner showered you with displays of affection, your comparison level for your next relationship is going to be quite high when it comes to levels of affection. If your next romantic partner tends to be more reserved and less emotional, that person might not measure up to your expectations.

Evaluating the Alternatives

Another aspect of the social exchange process involves looking at the possible alternatives. After analyzing the costs and benefits and contrasting these against your comparison levels, you might start to look at the possible alternatives. The relationship might not measure up to your comparison levels, but as you survey the potential alternatives, you might determine that the relationship is still better than anything else that is available. As a result, you might go back and reassess the relationship in terms of what may be now a somewhat lower comparison level.

The Honeymoon Phase

The length of a friendship or romance can also play a role in the social exchange process. During the early weeks or months of a relationship, often referred to as the "honeymoon phase," people are more likely to ignore the social exchange balance. Things that would normally be viewed as high cost are dismissed, ignored, or minimized while the potential benefits are often exaggerated.

So what happens when this honeymoon period finally comes to an end? In many cases, there will be a gradual evaluation of the exchange balance. Downsides will become more apparent and benefits will start to be seen more realistically. This recalibration of the exchange balance might also lead to the termination of the relationship if the balance is tipped too far toward the negative side.

APPLIED SOCIAL PSYCHOLOGY

Applied social psychology can be defined as using social psychological theories, principles, research findings, and experimental methods to understand social issues and to offer real-world solutions for a variety of social problems. As a discipline, applied social psychology functions on the premise that social problems are, at their heart, caused by human behavior. To understand and change these problem behaviors, applied social psychologists conduct a scientific examination of individuals' thoughts, feelings, and behaviors as they pertain to a variety of social influences. Through their research, applied social psychologists hope to offer practical suggestions for improving human social behavior in areas ranging from workplace productivity to safer sexual activity.

While an earlier generation of psychologists had been predominantly interested in the structure and measurement of mental tasks in the laboratory, the 20th century saw a substantial increase in researchers advocating the application of theories outside the

laboratory. In 1903, experimental psychologist Walter Dill Scott wrote *The Theory and Practice of Advertising*, suggesting that consumer habits could be influenced by emotional suggestions. In 1908, psychologist Hugo Munsterberg defined applied psychology as research adjusted to fit the problems encountered in everyday life. In works on industrial psychology, advertising, and education, Munsterberg, Scott, and others began exploring the possibilities of an applied psychology. In 1917, psychologist G. S. Hall founded the *Journal of Applied Psychology* to further explore the potential of this new field.

By the 1920s, there was an undeniable enthusiasm for applied research in the psychological community, despite its reputation as an “undignified” pursuit. In addition to the new challenges it presented, applied research was also attractive because private corporations often provided better salaries than did academic institutions. John B. Watson, former American Psychological Association president and one of the founders of modern behaviorism, began a successful career at the J. Walter Thompson advertising firm after being fired from Johns Hopkins University in 1920. Watson was able to put psychological principles into practice with such advertising industry standards as expert and celebrity testimonials and focus group research. He was promoted to vice president in only 4 years and was eventually honored for his achievements by the American Psychological Association. Even those psychologists who stayed in traditional laboratories found that it was easier to justify lab materials and costs if their research had the potential for application.

The work of Kurt Lewin (1840-1947) marks the beginning of modern applied social psychology. Lewin, best known for his field theory suggesting that behavior is a function of an individual’s personality and his or her environment, proposed that social psychologists should engage in what he called action research. A social activist himself, Lewin believed that social issues should inspire social psychological research. This research could then be used to provide solutions for social problems. Lewin’s action research sought to define a social problem, recommend countermeasures, and test the effectiveness of those countermeasures through community involvement, surveys, case studies, and controlled experiments. To this end, Lewin formed several organizations to engage in action research, including the Society for the Psychological Study of Social Issues, the Research Center for Group Dynamics, the Commission on Community Interrelations, and the National Training Laboratories. These groups studied interracial housing, ingroup loyalty, and leadership styles, establishing Lewin as one of the foremost proponents of combined applied and theoretical social psychology in the history of psychology.

As social psychology sought greater acceptance as a science in the 1950s and 1960s, action research in the field became less popular and was replaced by basic academic social psychology. It was not until the late 1960s and 1970s, while American society was undergoing radical transformations as a result of the civil rights movement, the Vietnam War, and the Watergate scandal, that social psychologists returned to the field in an attempt to understand and explain what was happening around them. Through the 1970s and 1980s, social psychologists debated if applied research could inform public policy. If so, many felt that the researchers themselves must present and explain their findings to policymakers or risk dangerous misinterpretation of their work.

Applied social psychology has been growing in prestige since the 1980s. With applications ranging from improving the criminal justice system to informing education and health issues, the last few decades have seen substantial increases in nontraditional funding sources. While applied social psychological research continues within academic institutions, private and government grants, as well as full-time research positions within

large corporations, have allowed researchers the flexibility and opportunity to study a diverse array of social phenomena.

Unique Features

While research in basic social psychology often begins with scientific curiosity, work in applied social psychology typically starts with the identification of a specific social problem, such as teen pregnancy or hate crimes. Applied social psychologists seek to understand and treat these social maladies through the application of the theories and methods of social psychology. While basic social psychologists attempt to isolate the causal relationships between a small number of specific variables that can be carefully controlled in the lab, applied social psychologists work to identify and predict large-scale effects that can be used to design and implement social programs. Real social issues rarely involve only one or two psychological variables. Therefore, it is often useful for applied social psychologists to consider broad combinations of psychological principles when attempting to understand a social issue. It is also common for applied social psychologists to adopt an interdisciplinary approach to their work, incorporating economic, sociological, and political perspectives.

Within the laboratory, social psychologists are generally able to conduct precise and carefully controlled experimental manipulations to test their hypotheses. Within communities and businesses, applied researchers often work in unpredictable and unrestricted environments, relying on less-precise research techniques such as surveys, self-reports, and rough “before and after” evaluations. Often at the mercy of corporate or government sponsors, applied social psychologists work under program deadlines, funding restrictions, and political pressures.

While laboratory effects need to reach statistical significance to support the hypothesis of the experimenter, implemented social programs need to show effects that are not only statistically significant but also large enough to have real-world consequences for the program’s sponsor. Applied social psychologists must also make quantitative estimates when designing an experimental social program to show that the program’s benefits will ultimately outweigh the initial costs.

Applied in Working

To list all the areas to which social psychology is currently being applied would be almost impossible. Broadly, applied social psychologists are active in studying and improving educational programs, industrial and organizational productivity, environmental and health care issues, justice system reform, and all types of mass communication, including advertising, public relations, and politics.

Some applied social psychologists conduct research for academic institutions, some for private foundations and corporations, and some for government organizations. Some evaluate the success or failure of a specific experimental social program while others work as internal consultants to government agencies and businesses, providing feedback on a variety of projects. Some applied social psychologists give policy advice to corporate or government managers from outside an organization while still others become managers themselves. Finally, some applied social psychologists become full-time advocates for social change, working with activist groups rather than from inside a government or corporate body.

Criticisms of applied social psychology come from psychologists as well as from policymakers. Laboratory psychologists question the methodology behind much of the applied social research. With no control groups, few experimental manipulations, and a heavy reliance on self-report and correlation, can applied social psychology really contribute to social psychological theories? After all, a successful social program will need to be tailored differently to every community in which it is implemented. Policymakers question whether the effects of experimental social programs are large enough (or cheap enough) to be widely implemented. Others argue that the large-scale behavior modification implied in some social psychological programs is unethical. Ultimately, there is a genuine tradeoff between conducting basic and applied social psychological research. In basic studies, researchers seek general principles that may not directly apply to more complicated real-world problems; in applied studies, researchers address specific problems, yet their conclusions may fail to generalize to other situations.

While both groups have valid critiques of an admittedly murky field, it is important to emphasize that applied social psychologists are actively engaged in the scientific study of social issues. While this problem-oriented approach does take some emphasis away from theory building, the plethora of field observations and unique program implementations essential to applied research will only serve to strengthen and refine current and future social psychological theories. Although the oversimplification of research findings could lead to the improper implementation of social programs, applied social psychologists seek to prevent such use by taking an active role in designing socially responsible policy.

A final issue to note is that the implied divide between applied and basic research ignores the contributions that basic social psychologists often make to areas of application. While applied social psychology refers to research with a problem-oriented focus, it is much more difficult to define who meets the definition of an applied social psychologist. Many psychologists primarily focused on basic laboratory research are also actively involved in the Society for the Psychological Study of Social Issues and other organizations committed to the application of social psychological principles.



HUMAN DEVELOPMENT AND INTERVENTIONS

DEVELOPMENTAL PROCESSES

The role of nature-nurture must be reconsidered in light of the Human Genome Project's surprising results. Conventional biology emphasizes that human expression is controlled by genes, and is under the influence of nature. Since 95% of the population possess "fit" genes, dysfunctions in this population are attributable to environmental influences (nurture). Nurture experiences, initiated in utero, provide for "learned perceptions." Along with genetic instincts, these perceptions constitute the life-shaping subconscious mind. The conscious mind, which functions around age six, operates independently of the subconscious. Conscious mind can observe and criticize behavioral tapes, yet can not "force" a change in subconscious.

One of the perennial controversies that tends to evoke rancor among biomedical scientists concerns the role of nature versus nurture in the unfoldment of life [Lipton, 1998a]. Those polarized on the side of nature invoke the concept of genetic determinism as the mechanism responsible for "controlling" the expression of an organism's physical and behavioral traits. Genetic determinism refers to an internal control mechanism resembling a genetically-coded "computer" program. At conception, it is believed that the differential activation of selected maternal and paternal genes collectively "download" an individual's physiologic and behavioral character, in other words, their biological destiny.

In contrast, those endorsing "control" by nurture argue that the environment is instrumental in "controlling" biological expression. Rather than attributing biological fate to gene control, nurturists contend that environmental experiences provide an essential role in shaping the character of an individual's life. The polarity between these philosophies simply reflects the fact that those endorsing nature believe in an internal control

mechanism (genes) while those supporting nurture mechanisms ascribe to an external control (environment).

The resolution of the nature and nurture controversy is profoundly important in regard to defining the role of parenting in human development. If those endorsing nature as the source of “control” are correct, the fundamental character and attributes of a child are genetically predetermined at conception. Genes, presumed to be self-actualizing, would control organismal structure and function. Since development would be programmed and executed by the internalized genes, the basic role of the parent would be to provide nutrition and protection for their growing fetus or child.

In such a model, developmental characters that deviate from the norm imply that the individual expresses defective genes. The belief that nature “controls” biology fosters the notion of victimization and irresponsibility in the unfoldment of one’s life. “Don’t blame me for this condition, I got it in my genes. Since I can’t control my genes, I am not responsible for the consequences.” Modern medical science perceives of a dysfunctional individual as one possessing a defective “mechanism.” Dysfunctional “mechanisms” are currently treated with drugs, though pharmaceutical companies are already touting a future in which genetic engineering will permanently eliminate all deviant or undesirable characters and behaviors. Consequently, we relinquish personal control over our lives to the “magic bullets” proffered by pharmaceutical companies.

The alternative perspective, supported by a large number of lay people and a growing contingency of scientists, expands upon the role of parents in human development. Those endorsing nurture as life’s “control” mechanism contend that parents have a fundamental impact on the developmental expression of their offspring. In a nurture-controlled system, gene activity would be dynamically-linked to an ever changing environment. Some environments enhance the potential of the child, while other environments may induce dysfunction and disease. In contrast to the fixed-fate mechanism envisioned by naturists, nurture mechanisms offer an opportunity to shape an individual’s biological expression by regulating or “controlling” their environment.

In reviewing the nature-nurture controversy over the years, it is apparent that at times, support for nature mechanisms predominates over the concept of nurture, while at other times the reverse is true. Since the revelation of the DNA genetic code by Watson and Crick in 1953, the concept of self-regulated genes controlling our physiology and behavior has prevailed over the perceived influence of environmental signals. Removing personal responsibility in the unfolding of one’s life leaves us with the belief that almost all negative or defective human traits represent a mechanical failure of the human molecular mechanism. By the early 1980’s, biologists were fully convinced that genes “control” biology. It was further assumed that a map of the completed human genome would provide science with all the necessary information to not only “cure” all of mankind’s ills, but also create a Mozart or another Einstein. The resulting Human Genome Project was designed as a global effort dedicated to deciphering the human genetic code.

The primary function of genes is to serve as biochemical blueprints that encode the complex chemical structure of proteins, the molecular “parts” from which cells are constructed. Conventional thought held that there was one gene to code for each of the 70,000 to 90,000 different proteins that make up our bodies. In addition to protein-coding genes, the cell also contains regulatory genes that “control” the expression of other genes. Regulatory genes presumably orchestrate the activity of a large number of structural

genes whose actions collectively contribute the complex physical patterns providing each species with its specific anatomy. It is further presumed that other regulatory genes control the expression of such traits as awareness, emotion, and intelligence.

Before the project got off the ground, scientists had already estimated that human complexity would necessitate a genome (the total collection of genes) in excess of 100,000 genes. This was based upon a conservative estimate that there were in excess of 30,000 regulatory genes and over 70,000 protein-coding genes stored in the human genome. When the results of the human genome project were reported this year, the conclusion presented itself as a "cosmic joke." Just when science thought it had life all figured out, the universe threw a biological curve ball. In all the hoopla over the sequencing of the human genetic code and being got caught up in the brilliant technological feat, we have not focused on the actual "meaning" of the results. These results overturn a foundational core belief embraced by conventional science.

The Genome project's cosmic joke concerns the fact that the whole human genome consists of only 34,000 genes. Two thirds of the anticipated and presumed necessary genes do not exist! How can we account for the complexity of a genetically-controlled human when there are not even enough genes to code just for the proteins?

The "failure" of the genome to confirm our expectations reveals that our perception of how biology "works" is based upon incorrect assumptions or information. Our "belief" in the concept of genetic determinism is apparently fundamentally flawed. We can not attribute the character of our lives solely to the consequence of inherent genetic "programming." The genome results force us to reconsider the question: "From whence do we acquire our biological complexity?" In a commentary on the surprising results of the Human Genome study, David Baltimore (2001), one of the world's most prominent geneticists and Nobel prize winner, addressed this issue of complexity:

"But unless the human genome contains a lot of genes that are opaque to our computers, it is clear that we do not gain our undoubted complexity over worms and plants by using more genes.

Understanding what does give us our complexity – our enormous behavioral repertoire, ability to produce conscious action, remarkable physical coordination, precisely tuned alterations in response to external variations of the environment, learning, memory...need I go on?- remains a challenge for the future.

Of course the most interesting consequence of the project's results is that we must now face that "challenge for the future" alluded to by Baltimore. What does "control" our biology, if not the genes? In the heat of the genome frenzy, emphasis on the project overshadowed the brilliant work of many biologists who were revealing a radically different understanding of organismal "control" mechanisms. Emerging at the cutting edge of cell science is the recognition that the environment, and more specifically, our perception of the environment, directly controls our behavior and gene activity .

Conventional biology has built its knowledge upon what is referred to as the "Central Dogma." This inviolable belief claims that the flow of information in biological organisms is from DNA to RNA and then to Protein. Since DNA (genes) is the top rung of this information flow, science adopted the notion of the Primacy of DNA, with "primacy" in this case meaning first cause. The argument for genetic determinacy is based upon the premise that DNA is in "control." But is it?

Almost all of the cell's genes are stored in its largest organelle, the nucleus. Conventional science maintains that the nucleus represents the "command center of the cell," a notion based upon the assumption that genes "control" (determine) the expression of the cell. As the cell's "command center," it is implied that the nucleus represents the equivalent of the cell's "brain."

If the brain is removed from any living organism, the necessary consequence of that action is immediate death of the organism. However, if the nucleus is removed from a cell, the cell does not necessarily die. Some enucleated cells can survive for two or months without possessing any genes. Enucleated cells are routinely used as "feeder layers" that support the growth of other specialized cell types. In the absence of a nucleus, cells maintain their metabolism, digest food, excrete waste, breathe, move through their environment recognizing and appropriately responding to other cells, predators or toxins. Ultimately these cells die, for without their genome, enucleated cells are unable to replace worn-out or defective proteins required for life functions.

The fact that cells maintain a successful and integrated life in the absence of genes, reveals that genes are not the "brain" of the cell. The primary reason why genes can not "control" biology is that they are not self emergent. This means that genes can not self-actualize, they are chemically unable to turn themselves on or off. Gene expression is under the regulatory control of environmental signals that act through epigenetic mechanisms.

However, genes are fundamental to the normal expression of life. Rather than serving in the capacity of "control," genes represent molecular blueprints necessary in manufacturing the complex proteins that provide for the cell's structure and functions. Defects in the gene programs, mutations, may profoundly impair the quality of life in those possessing them. It is important to note that the lives of less than 5% of the population are impacted by defective genes. These individuals express genetically-propagated birth defects, whether they are manifest at birth or appear later in life.

The significance of this data is that more than 95% of the population came into this world with an intact genome, one that would code for a healthy and fit existence. While science has focused its efforts at assessing the role of genes by studying the 5% of the population with defective genes, it has not made much progress as to why the majority of the population, which possess a fit genome, acquire dysfunction and disease. We simply can not "blame" their reality on the genes (nature).

Scientific attention as to what "controls" biology is shifting from the DNA to the cell's membrane. In the economy of the cell, the membrane is the equivalent of our "skin." The membrane provides an interface between the ever-changing environment (not-self) and the enclosed controlled environment of the cytoplasm (self). The embryonic "skin" (ectoderm) provides for two organ systems in the human body: the integument and the nervous system. In cells, these two functions are integrated within the simple layer that envelopes the cytoplasm.

Protein molecules in the cell membrane interface the demands of the internal physiologic mechanisms with existing environmental exigencies (Lipton, 1999). These membrane "control" molecules are comprised of couplets consisting of receptor proteins and effector proteins. Protein receptors recognize environmental signals (information) in the same way our receptors (e.g., eyes, ears, nose, taste, etc.) read our environment. Specific receptor proteins are chemically "activated" upon receiving a recognizable environmental signal (stimulus). In its activated state, the receptor protein couples with, and in turn,

activates specific effector proteins. The “activated” effector proteins selectively “control” the cell’s biology in coordinating a response to the initiating environmental signal.

Receptor-effector protein complexes serve as “switches” that integrate the function of the organism within its environment. The receptor component of the switch provides “awareness of the environment” and the effector component generates a “physical sensation” in response to that awareness. By structural and functional definition, the receptor-effector switches represent molecular units of perception, which is defined as “awareness of the environment through physical sensation.” Perception protein complexes “control” cell behavior, regulate gene expression and have been implicated in the rewriting of the genetic code (Lipton, 1999).

Every cell is innately intelligent in that it generally possesses genetic “blueprints” to create all of the necessary perception complexes that enable it to survive and thrive in its normal environmental niche. The DNA coding for these perceptual protein complexes have been acquired and accumulated by cells during four billion years of evolution. Perception coding genes are stored in the cell’s nucleus and are duplicated prior to cell division, providing each daughter cell with a set of life sustaining perception complexes.

However, environments are not static. Changes in the environments generate a need for “new” perceptions on the part of organisms inhabiting those environments. It is now evident that cells create new perception complexes through their interaction with novel environment stimuli. Utilizing a newly discovered group of genes, collectively referred to as “genetic engineering genes,” cells are able to create new perception proteins in a process representing cellular learning and memory.

This evolutionarily advanced gene-writing mechanism enables our immune cells to respond to foreign antigens by creating life-saving antibodies. Antibodies are specifically-shaped proteins that the cell manufactures to physically complement the invasive antigens. As proteins, antibodies require a gene (“blueprint”) for their assembly. Interestingly, the specifically tailored antibody genes that are derived from the immune response did not exist before the cell was exposed to the antigen. The immune response, which takes about three days from the initial exposure to the antigen till the appearance of specific antibodies, results in the “learning” of a new perception protein (the antibody) whose DNA “blueprint” (“memory”) can be genetically passed on to all daughter cells.

In creating a life conserving perception, the cell must couple a signal-receiving receptor with an effector protein the “controls” the appropriate behavioral response. The character of a perception can be scored by the type of response the environmental stimulus evokes. Positive perceptions produce a growth response, while negative perceptions activate the cell’s protection response.

Although perception proteins are manufactured through molecular genetic mechanisms, activation of the perception process is “controlled” or initiated by environmental signals. The expression of the cell is primarily molded by its perception of the environment and not by its genetic code, a fact that emphasizes the role of nurture in biological control. The controlling influence of environment is underscored in recent studies on stem cells (Vogel, 2000). Stem cells, found in different organs and tissues of the adult body, are similar to embryonic cells in that they are undifferentiated, though they have the potential to express a wide variety of mature cell types. Stem cells do not control their own fate. The differentiation of stem cells is based upon the environment the cell finds itself in. For example, three different tissue culture environments can be created. If a

stem cell is placed in culture number one, it may become a bone cell. If the same stem cell was put into culture two, it will become a nerve cell or if placed into culture dish number three, the cell matures as a liver cell. The cell's fate is "controlled" by its interaction with the environment and not by a self-contained genetic program.

While every cell is capable of behaving as a free-living entity, late in evolution cells began to assemble into interactive communities. Social organizations of cells resulted from an evolutionary drive to enhance survival. The more "awareness" an organism possesses, the more capable it is of surviving. Consider that a single cell has X amount of awareness. Then a colony of 25 cells would have a collective awareness of 25X. Since each cell in the community has an opportunity of sharing awareness with the rest of the group, then every single cell effectively possesses a collective awareness of 25X. Which is more capable of surviving, a cell with 1X awareness or one with 25X awareness? Nature favors the assembly of cells into communities as a means of expanding awareness.

The evolutionary transition from unicellular life forms to multicellular (communal) life forms represented an intellectually and technically profound high point in the creation of the biosphere. In the world of unicellular protozoa, each cell is an innately intelligent, independent being, adjusting its biology to its own perception of the environment. However, when cells join together to form multicellular "communities," it required that the cells establish a complex social intercourse. Within a community, individual cells can not behave independently, otherwise the community would cease to exist. By definition, the members of a community must follow a single "collective" voice. The "collective" voice controlling the community's expression represents the sum of all of the perceptions of every cell in the group.

Original cellular communities consisted of from tens to hundreds of cells. The evolutionary advantage to living in community soon led to organizations comprised of millions, billions or even trillions, of socially interactive single cells. In order to survive at such high densities, the amazing technologies evolved by the cells led to highly structured environments that would boggle the minds and imagination of human engineers. Within these environments, cell communities subdivide the workload among themselves, leading to the creation of hundreds of specialized cell types. The structural plans to create these interactive communities and differentiated cells are written into the genome of each cell within the community.

Though each individual cell is of microscopic dimensions, the size of multicellular communities may range from the barely visible to the monolithic in proportion. At our level of perspective, we do not observe individual cells but we do recognize the different structural forms cell communities acquire. We perceive these macroscopic structured communities as plants and animals, which includes ourselves among them. While you might consider yourself as a single entity, in truth you are the sum of a community of approximately 50 trillion single cells.

The effectiveness of such large communities is enhanced by the subdivision of labor among the component cells. Cytological specialization's enable the cells to form the specific tissues and organs of the body. In larger organisms, only a small percent of the cells function in perceiving the community's external environment. Groups of specialized "perception cells" form the tissues and organs of the nervous system. The function of the nervous system is to perceive the environment and coordinate the cellular community's biological response to the impinging environmental stimuli.

Multicellular organisms, like the cells they are comprised of, are genetically endowed with fundamental protein perception complexes that enable the organism to effectively survive in their environment. Genetically programmed perceptions are referred to as instincts. Similar to cells, organisms are also capable of interacting with the environment and creating new perceptual pathways. This process provides for learned behavior.

As one ascends the tree of evolution, moving from more primitive to more advanced multicellular organisms, there is a profound shift from the predominant use of genetically programmed perceptions (instinct) to the use of learned behavior. Primitive organisms primarily rely upon instincts for the greater proportion of their behavioral repertoire. In higher organisms, especially humans, brain evolution offers a great opportunity for creating a large database of learned perceptions, which reduces dependence upon instincts. Humans are endowed with an abundance of genetically propagated vital instincts. Most of them are not evident to us, for they operate below our level of consciousness, providing for the function and maintenance of cells, tissues and organs. However, some basic instincts generate overt and observable behavior. For example, the suckling response of the neonate, or the retraction of a hand when a finger gets burned in a flame.

"Human beings are more dependent on learning for survival than other species. We have no instincts that automatically protect us and find us food and shelter, for example." (Schultz and Lavenda, 1987) As important as instincts are to our survival, our learned perceptions are more important, especially in light of the fact that they can over-ride genetically programmed instincts. Since perceptions direct gene activity and engage behavior, the learned perceptions we acquire are instrumental in "controlling" the physiologic and behavioral character of our lives. The sum of our instincts and learned perceptions collectively form the subconscious mind, which in turn, is the source of the "collective" voice that our cell's "agreed" to follow.

Although we are endowed at conception with innate perceptions (instincts) we only begin to acquire learned perceptions at the time that our nervous systems become functional. Until recently, conventional thought held that the human brain was not functional until some time after birth, in that many of its structures are not fully differentiated (developed) until that time. However, this assumption has been invalidated by the pioneering work of Thomas Verny (1981) and David Chamberlain (1988), among others, who have revealed the vast sensory and learning capabilities expressed by the fetal nervous system.

The significance of this understanding is that perceptions experienced by the fetus would have a profound effect upon its physiology and development. Essentially, the perceptions experienced by the fetus are the same as those experienced by the mother. Fetal blood is in direct contact with the mother's blood via the placenta. Blood is one of the most important components of the connective tissue, through it pass most of the organizing factors (e.g., hormones, growth factors, cytokines) that coordinate the function of the body's systems. As the mother responds to her perceptions of the environment, her nervous system activates the release of behavior-coordinating signals into her bloodstream. These regulatory signals control the function, and even gene activity, of the tissues and organs needed by her to engage in the required behavioral response.

For example, if a mother is under environmental stress, she will activate her adrenal system, a protection system that provides for fight or flight. These stress hormones released into the blood prepare the body to engage a protection response. In this process, blood vessels in the viscera constrict forcing blood to nourish the peripheral muscles and

bones that provide protection. Fight-or-flight responses depend upon reflex behavior (hind-brain) rather than conscious reasoning (forebrain). To facilitate this process, the stress hormones constrict the forebrain's blood vessels forcing more blood to go to the hind brain in support of reflex behavior functions. Constriction of blood vessels in the gut and fore-brain during a stress response respectively repress growth and conscious reasoning (intelligence).

It is now recognized that, along with nutrients, stress signals and other coordinating factors in the mother's blood cross the placenta and enter into the fetal system (Christensen 2000). Once these maternal regulatory signals enter the fetal blood stream, they affect the same target systems in the fetus as they did in the mother. The fetus simultaneously experiences what the mother is perceiving in regard to her environmental stimuli. In stressful environments, fetal blood preferentially flows to the muscles and hind brain, while shorting the flow to the viscera and the forebrain. The development of fetal tissues and organs is proportional to the amount of blood they receive. Consequently, a mother experiencing chronic stress will profoundly alter the development of her child's physiologic systems that provide for growth and protection.

The learned perceptions acquired by an individual begin to arise in utero and can be subdivided into two broad categories. One set of outward-directed learned perceptions "control" how we respond to environmental stimuli. Nature has created a mechanism to facilitate this early learning process. Upon encountering a novel environmental stimulus, the neonate is programmed to first observe how the mother or father responds to the signal. Infants are particularly adept at interpreting parental facial characters in discriminating the positive or negative nature of a new stimulus. When an infant encounters new environmental features, it generally focuses first on the parent's expression in learning how to respond. Once the new environmental feature is recognized, it is coupled with an appropriate behavioral response. The coupled input (environmental stimulus) and output (behavioral response) program is stored in the subconscious as a learned perception. If the stimulus ever reappears, the "programmed" behavior encoded by the subconscious perception is immediately engaged. Behavior is based upon a simple stimulus-response mechanism.

Outwardly-directed learned perceptions are created in response to everything from simple objects to complex social interactions. Collectively, these learned perceptions contribute to an individual's enculturation. Parental "programming" of a child's subconscious behavior enables that child to conform with the "collective" voice, or beliefs, of the community.

In addition to the outward-directed perceptions, humans also acquire inward-directed perceptions which provide us with beliefs about our "self-identity." In order to know more about ourselves, we learn to see ourselves as others see us. If a parent provides a child with a positive or negative self image, that perception is recorded in the child's subconscious. The image acquired of self becomes the subconscious "collective" voice which shapes our physiology (e.g., health characteristics, weight) and behavior. Though every cell is innately intelligent, by communal agreement, it will give its allegiance to the collective voice, even if that voice engages in self destructive activities. For example, if a child is given a perception of itself that it can succeed, it will continuously strive to do just that. However, if the same child was provided with a belief that it was "not good enough," the body must conform to that perception, even by using self-sabotage if necessary, in order to thwart success.

Human biology is so dependent upon learned perceptions, that it is not surprising evolution has provided us with a mechanism that encourages rapid learning. Brain activity and states of awareness can be measured electronically using electroencephalography (EEG). There are four fundamental states of awareness distinguished by the frequency of electromagnetic activity in the brain. The time that an individual spends in each of these EEG states is related to a patterned sequential expressed during child development (Laibow, 1999).

DELTA waves (0.5-4 Hz), the lowest level of activity, are primarily expressed between birth and two years of age. When a person is in DELTA, they are in an unconscious (sleep-like) state. Between two years and six years of age, the child begins to spend more of its time in a higher level of EEG activity characterized as THETA (4-8 Hz). THETA activity is the state we experience upon just arising, when we are half asleep and half awake. Children are in this very imaginative state when they play, creating delicious pies made out of mud or gallant steeds from old brooms.

The child begins to preferentially express a still higher level of EEG activity called ALPHA waves around the age of six. ALPHA (8-12 HZ) is associated with states of calm consciousness. At around 12 years, the child's EEG spectrum may express sustained periods of BETA (12-35 HZ) waves, the highest level of brain activity characterized as "active or focused consciousness."

The significance of this developmental spectrum is that an individual does not generally sustain active consciousness (ALPHA activity) until after five years of age. Before birth and through the first five years of life, the infant is primarily in DELTA and THETA, which represents a hypnogogic state. In order to hypnotize an individual it is necessary to lower their brain function to these levels of activity. Consequently, the child is essentially in a hypnotic "trance" through the first five years of its life. During this time it is down-loading biology-controlling perceptions without even the benefit, or interference, of conscious discrimination. The potential of a child is "programmed" into its subconscious mind during this phase of development.

Learned perceptions are "hard-wired" as synaptic pathways in the subconscious, which essentially represents what we recognize as the brain. Consciousness, which functionally expresses itself with the appearance of ALPHA waves at around six years of life, is associated with the most recent addition to the brain, the prefrontal cortex. Human consciousness is characterized by an awareness of "self." While most of our senses, such as eyes, ears and nose, observe the outer world, consciousness resembles a "sense" that observes the inner workings of its own cellular community. Consciousness feels the sensations and emotions generated by the body and has access to the stored data base comprising our perceptual library.

To understand the difference between subconscious and consciousness, consider this instructive relationship: The subconscious mind represents the brain's hard drive (ROM), and the conscious mind is the equivalent of the "desktop" (RAM). Like a hard disk, the subconscious can store an unimaginable quantity of perceptual data. It can be programmed to be "on line," meaning that incoming signals go directly to the data base and are processed without the necessity of conscious intervention.

By the time consciousness evolves to a functional state, most of the fundamental perceptions about life have been programmed into the hard drive. Consciousness can access this data base and open up for review a formerly learned perception, such as a behavioral script. This would be the same as opening up a document from the hard drive on to the

desk top. In consciousness, we have the ability review the script and edit the program as we see fit, just as we do with open documents on our computers. However, the editing process in no way changes the original perception which is still hardwired in the subconscious. No amount of yelling or cajoling by the consciousness can change the subconscious program. For some reason we think there is an entity in the subconscious that listens and responds to our thoughts. In reality the subconscious is a cold, emotionless database of stored programs. Its function is strictly concerned with reading environmental signals and engaging the hard wired behavior programs, no questions asked, no judgments made.

Through sheer will power and intent, consciousness can attempt to over-ride a subconscious tape. Usually such efforts are met with varying degrees of resistance, since the cells are obligated to adhere to the subconscious program. In some cases the tensions between conscious will power and subconscious programs can result in serious neurological disorders. For example, consider the fate of Australian concert pianist David Helfgott whose story was presented in the film *Shine*. David was programmed by his father, a survivor of the holocaust, to not succeed, for success would make him vulnerable in that he would stand out from others. In spite of the relentlessness of his father's programming, David was consciously aware that he was a world class pianist. In order to prove himself, Helfgott purposely chose one of the most difficult piano compositions, a piece by Rachmaninoff, to play in the national competition. As the film reveals, in the final stage of his amazing performance, a major conflict occurred between his conscious will to succeed and the subconscious program to fail. When he successfully played the last note he passed out, upon awakening he was irreparably insane. The fact that his conscious will power forced his body mechanism to violate the programmed "collective" voice led to a neurological melt down.

The conflicts we generally experience in life are frequently related to our conscious efforts of trying to "force" changes upon our subconscious programming. However, through a variety of new energy psychology modalities (e.g., Psych-K, EMDR, Avatar, etc.) the content of subconscious beliefs can be assessed and using specific protocols, consciousness can facilitate a rapid "reprogramming" of limiting core beliefs.

THEORIES OF DEVELOPMENT

Psychoanalytical

According to Freud's psychoanalytic theory, personality develops through a series of stages, each characterized by a certain internal psychological conflict.

Sigmund Freud's psychoanalytic theory of personality argues that human behavior is the result of the interactions among three component parts of the mind: the id, ego, and superego. This theory, known as Freud's structural theory of personality, places great emphasis on the role of unconscious psychological conflicts in shaping behavior and personality. Dynamic interactions among these fundamental parts of the mind are thought to progress through five distinct psychosexual stages of development. Over the last century, however, Freud's ideas have since been met with criticism, in part because of his singular focus on sexuality as the main driver of human personality development.

Freud's Structure of the Human Mind

According to Freud, our personality develops from the interactions among what he proposed as the three fundamental structures of the human mind: the id, ego, and superego. Conflicts among these three structures, and our efforts to find balance among what each of them “desires,” determines how we behave and approach the world. What balance we strike in any given situation determines how we will resolve the conflict between two overarching behavioral tendencies: our biological aggressive and pleasure-seeking drives vs. our socialized internal control over those drives.

The Id

The id, the most primitive of the three structures, is concerned with instant gratification of basic physical needs and urges. It operates entirely unconsciously (outside of conscious thought). For example, if your id walked past a stranger eating ice cream, it would most likely take the ice cream for itself. It doesn't know, or care, that it is rude to take something belonging to someone else; it would care only that you wanted the ice cream.

The Superego

The superego is concerned with social rules and morals – similar to what many people call their “conscience” or their “moral compass.” It develops as a child learns what their culture considers right and wrong. If your superego walked past the same stranger, it would not take their ice cream because it would know that would be rude. However, if both your id and your superego were involved, and your id was strong enough to override your superego's concern, you would still take the ice cream, but afterward you would most likely feel guilt and shame over your actions.

The Ego

In contrast to the instinctual id and the moral superego, the ego is the rational, pragmatic part of our personality. It is less primitive than the id and is partly conscious and partly unconscious. It's what Freud considered to be the “self,” and its job is to balance the demands of the id and superego in the practical context of reality. So, if you walked past the stranger with ice cream one more time, your ego would mediate the conflict between your id (“I want that ice cream right now”) and superego (“It's wrong to take someone else's ice cream”) and decide to go buy your own ice cream. While this may mean you have to wait 10 more minutes, which would frustrate your id, your ego decides to make that sacrifice as part of the compromise– satisfying your desire for ice cream while also avoiding an unpleasant social situation and potential feelings of shame.

Freud believed that the id, ego, and superego are in constant conflict and that adult personality and behavior are rooted in the results of these internal struggles throughout childhood. He believed that a person who has a strong ego has a healthy personality and that imbalances in this system can lead to neurosis (what we now think of as anxiety and depression) and unhealthy behaviors.

Psychosexual Stages of Development

Freud believed that the nature of the conflicts among the id, ego, and superego change over time as a person grows from child to adult. Specifically, he maintained that these conflicts progress through a series of five basic stages, each with a different focus: oral, anal, phallic, latency, and genital. He called his idea the psychosexual theory of develop-

ment, with each psychosexual stage directly related to a different physical center of pleasure.

Across these five stages, the child is presented with different conflicts between their biological drives (id) and their social and moral conscience (superego) because their biological pleasure-seeking urges focus on different areas of the body (what Freud called “erogenous zones”). The child’s ability to resolve these internal conflicts determines their future ability to cope and function as an adult. Failure to resolve a stage can lead one to become fixated in that stage, leading to unhealthy personality traits; successful resolution of the stages leads to a healthy adult.

Criticism of Freud’s Theories

Although Freud’s theories have many advantages that helped to expand our psychological understanding of personality, they are not without limits.

Narrow Focus

In his singular emphasis on the structure of the human mind, Freud paid little to no attention to the impact of environment, sociology, or culture. His theories were highly focused on pathology and largely ignored “normal,” healthy functioning. He has also been criticized for his myopic view of human sexuality to the exclusion of other important factors.

No Scientific Basis

Many critics point out that Freud’s theories are not supported by any empirical (experimental) data. In fact, as researchers began to take a more scientific look at his ideas, they found that several were unable to be supported: in order for a theory to be scientifically valid, it must be possible to disprove (“falsify”) it with experimental evidence, and many of Freud’s notions are not falsifiable.

Misogyny

Feminists and modern critics have been particularly critical of many of Freud’s theories, pointing out that the assumptions and approaches of psychoanalytic theory are profoundly patriarchal (male-dominated), anti-feminist, and misogynistic (anti-woman). Karen Horney, a psychologist who followed Freud, saw the mainstream Freudian approach as having a foundation of “masculine narcissism.” Feminist Betty Friedan referred to Freud’s concept of “penis envy” as a purely social bias typical of the Victorian era and showed how the concept played a key role in discrediting alternative notions of femininity in the early to mid-twentieth century.

Behavioristic

Behaviorism equates learning with changes in either the form or frequency of observable performance. Learning is accomplished when a proper response is demonstrated following the presentation of a specific environmental stimulus. For example, when presented with a math flashcard showing the equation “ $2 + 4 = ?$ ” the learner replies with the answer of “6.” The equation is the stimulus and the proper answer is the associated response. The key elements are the stimulus, the response, and the association between the two. Of primary concern is how the association between the stimulus and response is made, strengthened, and maintained.

Behaviorism focuses on the importance of the consequences of those performances and contends that responses that are followed by reinforcement are more likely to recur in the future. No attempt is made to determine the structure of a student's knowledge nor to assess which mental processes it is necessary for them to use (Winn, 1990). The learner is characterized as being reactive to conditions in the environment as opposed to taking an active role in discovering the environment.

Although both learner and environmental factors are considered important by behaviorists, environmental conditions receive the greatest emphasis. Behaviorists assess the learners to determine at what point to begin instruction as well as to determine which reinforcers are most effective for a particular student. The most critical factor, however, is the arrangement of stimuli and consequences within the environment.

Memory, as commonly defined by the layman, is not typically addressed by behaviorists. Although the acquisition of "habits" is discussed, little attention is given as to how these habits are stored or recalled for future use. Forgetting is attributed to the "nonuse" of a response over time. The use of periodic practice or review serves to maintain a learner's readiness to respond (Schunk, 1991).

Transfer refers to the application of learned knowledge in new ways or situations, as well as to how prior learning affects new learning. In behavioral learning theories, transfer is a result of generalization. Situations involving identical or similar features allow behaviors to transfer across common elements. For example, the student who has learned to recognize and classify elm trees demonstrates transfer when (s)he classifies maple trees using the same process. The similarities between the elm and maple trees allow the learner to apply the previous elm tree classification learning experience to the maple tree classification task.

Behaviorists attempt to prescribe strategies that are most useful for building and strengthening stimulus-response associations (Winn, 1990), including the use of instructional cues, practice, and reinforcement. These prescriptions have generally been proven reliable and effective in facilitating learning that involves discriminations (recalling facts), generalizations (defining and illustrating concepts), associations (applying explanations), and chaining (automatically performing a specified procedure). However, it is generally agreed that behavioral principles cannot adequately explain the acquisition of higher level skills or those that require a greater depth of processing (e.g., language development, problem solving, inference generating, critical thinking) (Schunk, 1991).

Many of the basic assumptions and characteristics of behaviorism are embedded in current instructional design practices. Behaviorism was used as the basis for designing many of the early audio-visual materials and gave rise to many related teaching strategies, such as Skinner's teaching machines and programmed texts. More recent examples include principles utilized within computer-assisted instruction (CAI) and mastery learning.

Specific assumptions or principles that have direct relevance to instructional design include the following (possible current ID applications are listed in *italics* and brackets following the listed principle):

An emphasis on producing observable and measurable outcomes in students [behavioral objectives, task analysis, criterion-referenced assessment].

Pre-assessment of students to determine where instruction should begin [learner analysis].

Emphasis on mastering early steps before progressing to more complex levels of performance [sequencing of instructional presentation, mastery learning].

Use of reinforcement to impact performance [tangible rewards, informative feedback].

Use of cues, shaping and practice to ensure a strong stimulus-response association [simple to complex sequencing of practice, use of prompts].

The goal of instruction for the behaviorist is to elicit the desired response from the learner who is presented with a target stimulus. To accomplish this, the learner must know how to execute the proper response, as well as the conditions under which that response should be made. Therefore, instruction is structured around the presentation of the target stimulus and the provision of opportunities for the learner to practice making the proper response. To facilitate the linking of stimulus-response pairs, instruction frequently uses cues (to initially prompt the delivery of the response) and reinforcement (to strengthen correct responding in the presence of the target stimulus).

Behavioral theories imply that the job of the teacher/designer is to (1) determine which cues can elicit the desired responses; (2) arrange practice situations in which prompts are paired with the target stimuli that initially have no eliciting power but which will be expected to elicit the responses in the “natural” (performance) setting; and (3) arrange environmental conditions so that students can make the correct responses in the presence of those target stimuli and receive reinforcement for those responses.

For example, a newly-hired manager of human resources may be expected to organize a meeting agenda according to the company’s specific format. The target stimulus (the verbal command “to format a meeting agenda”) does not initially elicit the correct response nor does the new manager have the capability to make the correct response. However, with the repeated presentation of cues (e.g., completed templates of past agendas, blank templates arranged in standard format) paired with the verbal command stimulus, the manager begins to make the appropriate responses. Although the initial responses may not be in the final proper form, repeated practice and reinforcement shape the response until it is correctly executed. Finally, learning is demonstrated when, upon the command to format a meeting agenda, the manager reliably organizes the agenda according to company standards and does so without the use of previous examples or models.

Cognitive

In the late 1950’s, learning theory began to make a shift away from the use of behavioral models to an approach that relied on learning theories and models from the cognitive sciences. Psychologists and educators began to de-emphasize a concern with overt, observable behavior and stressed instead more complex cognitive processes such as thinking, problem solving, language, concept formation and information processing (Snelbecker, 1983). Within the past decade, a number of authors in the field of instructional design have openly and consciously rejected many of ID’s traditional behavioristic assumptions in favor of a new set of psychological assumptions about learning drawn from the cognitive sciences. Whether viewed as an open revolution or simply a gradual evolutionary process, there seems to be the general acknowledgment that cognitive theory has moved to the forefront of current learning theories (Bednar et al., 1991). This shift from a behavioral orientation (where the emphasis is on promoting a student’s overt performance by

the manipulation of stimulus material) to a cognitive orientation (where the emphasis is on promoting mental processing) has created a similar shift from procedures for manipulating the materials to be presented by an instructional system to procedures for directing student processing and interaction with the instructional design system.

Cognitive theories stress the acquisition of knowledge and internal mental structures and, as such, are closer to the rationalist end of the epistemology continuum. Learning is equated with discrete changes between states of knowledge rather than with changes in the probability of response. Cognitive theories focus on the conceptualization of students' learning processes and address the issues of how information is received, organized, stored, and retrieved by the mind. Learning is concerned not so much with what learners do but with what they know and how they come to acquire it. Knowledge acquisition is described as a mental activity that entails internal coding and structuring by the learner. The learner is viewed as a very active participant in the learning process.

Cognitivism, like behaviorism, emphasizes the role that environmental conditions play in facilitating learning. Instructional explanations, demonstrations, illustrative examples and matched non-examples are all considered to be instrumental in guiding student learning. Similarly, emphasis is placed on the role of practice with corrective feedback. Up to this point, little difference can be detected between these two theories. However, the "active" nature of the learner is perceived quite differently. The cognitive approach focuses on the mental activities of the learner that lead up to a response and acknowledges the processes of mental planning, goal-setting, and organizational strategies (Shuell, 1986). Cognitive theories contend that environmental "cues" and instructional components alone cannot account for all the learning that results from an instructional situation. Additional key elements include the way that learners attend to, code, transform, rehearse, store and retrieve information. Learners' thoughts, beliefs, attitudes, and values are also considered to be influential in the learning process. The real focus of the cognitive approach is on changing the learner by encouraging him/her to use appropriate learning strategies.

As indicated above, memory is given a prominent role in the learning process. Learning results when information is stored in memory in an organized, meaningful manner. Teachers/designers are responsible for assisting learners in organizing that information in some optimal way. Designers use techniques such as advance organizers, analogies, hierarchical relationships, and matrices to help learners relate new information to prior knowledge. Forgetting is the inability to retrieve information from memory because of interference, memory loss, or missing or inadequate cues needed to access information.

According to cognitive theories, transfer is a function of how information is stored in memory. When a learner understands how to apply knowledge in different contexts, then transfer has occurred. Understanding is seen as being composed of a knowledge base in the form of rules, concepts, and discriminations. Prior knowledge is used to establish boundary constraints for identifying the similarities and differences of novel information. Not only must the knowledge itself be stored in memory but the uses of that knowledge as well. Specific instructional or real-world events will trigger particular responses, but the learner must believe that the knowledge is useful in a given situation before he will activate it.

Because of the emphasis on mental structures, cognitive theories are usually considered more appropriate for explaining complex forms of learning (reasoning, problem-solving, information-processing) than are those of a more behavioral perspective. However, it is important to indicate at this point that the actual goal of instruction for both of these viewpoints is often the same: to communicate or transfer knowledge to the students in the most efficient, effective manner possible. Two techniques used by both camps in achieving this effectiveness and efficiency of knowledge transfer are simplification and standardization. That is, knowledge can be analyzed, decomposed, and simplified into basic building blocks. Knowledge transfer is expedited if irrelevant information is eliminated. For example, trainees attending a workshop on effective management skills would be presented with information that is “sized” and “chunked” in such a way that they can assimilate and/or accommodate the new information as quickly and as easily as possible. Behaviorists would focus on the design of the environment to optimize that transfer, while cognitivists would stress efficient processing strategies.

Many of the instructional strategies advocated and utilized by cognitivists are also emphasized by behaviorists, yet usually for different reasons. An obvious commonality is the use of feedback. A behaviorist uses feedback (reinforcement) to modify behavior in the desired direction, while cognitivists make use of feedback (knowledge of results) to guide and support accurate mental connections.

Learner and task analyses are also critical to both cognitivists and behaviorists, but once again, for different reasons. Cognitivists look at the learner to determine his/her predisposition to learning (i.e., How does the learner activate, maintain, and direct his/her learning?). Additionally, cognitivists examine the learner to determine how to design instruction so that it can be readily assimilated (i.e., What are the learner’s existing mental structures?). In contrast, the behaviorists look at learners to determine where the lesson should begin (i.e., At what level are they currently performing successfully?) and which reinforcers should be most effective (i.e., What consequences are most desired by the learner?).

Specific assumptions or principles that have direct relevance to instructional design include the following (possible current ID applications are listed in *italics* and brackets following the listed principle):

Emphasis on the active involvement of the learner in the learning process [*learner control, metacognitive training (e.g., self-planning, monitoring, and revising techniques)*].

Use of hierarchical analyses to identify and illustrate prerequisite relationships [*cognitive task analysis procedures*].

Emphasis on structuring, organizing, and sequencing information to facilitate optimal processing [*use of cognitive strategies such as outlining, summaries, synthesizers, advance organizers, etc.*].

Creation of learning environments that allow and encourage students to make connections with previously learned material [*recall of prerequisite skills; use of relevant examples, analogies*].

Behavioral theories imply that teachers ought to arrange environmental conditions so that students respond properly to presented stimuli. Cognitive theories emphasize making knowledge meaningful and helping learners organize and relate new information to existing knowledge in memory. Instruction must be based on a student’s existing mental

structures, or schema, to be effective. It should organize information in such a manner that learners are able to connect new information with existing knowledge in some meaningful way. Analogies and metaphors are examples of this type of cognitive strategy. For example, instructional design textbooks frequently draw an analogy between the familiar architect's profession and the unfamiliar instructional design profession to help the novice learner conceptualize, organize and retain the major duties and functions of an instructional designer. Other cognitive strategies may include the use of framing, outlining, mnemonics, concept mapping, advance organizers and so forth.

Such cognitive emphases imply that major tasks of the teacher/designer include: (1) understanding that individuals bring various learning experiences to the learning situation which can impact learning outcomes; (2) determining the most effective manner in which to organize and structure new information to tap the learners' previously acquired knowledge, abilities, and experiences; and (3) arranging practice with feedback so that the new information is effectively and efficiently assimilated and/or accommodated within the learner's cognitive structure.

Consider the following example of a learning situation utilizing a cognitive approach: A manager in the training department of a large corporation had been asked to teach a new intern to complete a cost-benefit analysis for an upcoming development project. In this case, it is assumed that the intern has no previous experience with cost-benefit analysis in a business setting. However, by relating this new task to highly similar procedures with which the intern has had more experience, the manager can facilitate a smooth and efficient assimilation of this new procedure into memory. These familiar procedures may include the process by which the individual allocates his monthly paycheck, how (s)he makes a buy/no-buy decision regarding the purchase of a luxury item, or even how one's weekend spending activities might be determined and prioritized. The procedures for such activities may not exactly match those of the cost-benefit analysis, but the similarity between the activities allows for the unfamiliar information to be put within a familiar context. Thus processing requirements are reduced and the potential effectiveness of recall cues is increased.

Various Aspects of Development

Sensory Motor

The foundation of growth, development, and learning in a child starts with sensory and motor interaction with the world. The brain is built from the bottom up and this starts with movement and sensory exploration. Sensory stimulation and feedback drive the brain, but the motor system drives sensory stimulation—you can't have one without the other. This is at the core of everything we do at Brain Balance Achievement Centers. It is crucial to improve motor skills, sensory detection, and processing before any higher learning, behavioral or academic changes can truly happen.

Motor Skills

In almost every case, children who have been assessed with a developmental deficit in one or more areas are the kids who struggle with learning or behavioral issues. Invariably, these kids are the ones with inadequately developed sensory and motor systems. We often see they are uncoordinated, clumsy, and awkward with their gait. Motor activities involve a number of different skills, including:

Muscle tone, strength, and coordination
Rhythm and timing
Bilateral coordination
Dominance
Gross and fine-motor skills
Primitive and postural reflexes
Eye-muscle balance and coordination
Vestibular balance and posture

Sensory Skills

Much like underdeveloped motor skills, the same applies when there are atypical sensory deficits — smell, touch, vision, hearing, and balance. No sensory function works in isolation. All the senses are dependent on other sensory functions, which are dependent on a baseline and acceptable level of development. Brain Balance sensory activities address any issues regarding:

Hearing	Vision
Smell	Taste
Touch	Vestibular (inner ear)
Proprioception (the ability to know where one's body is in space)	
Balance and spatial perception	

At Brain Balance Achievement Centers, we assess all of these skills with standardized and accepted testing and we quantify them by grade, age, or functional level. Our program then works to integrate sensory input and strengthen motor skills through the frequency and duration of activities that are designed to improve balance and stability and to increase muscle tone so a child can react more quickly.

Some of these exercises include:

Vision exercises	Vestibular activities
Balance exercises	Primitive reflex training
Gait training	And so much more!

These activities coupled with academic skill training, confidence building, and easy-to-follow dietary guidelines work to establish proper balance, while improving rhythm and timing. Our unique integrated, whole child approach helps your child achieve optimum body and mind balance, which leads to improved learning—academically, socially, and emotionally. You'll be amazed at the physical and academic advances your child will make during our program.

Cognitive

Cognitive development is the study of childhood neurological and psychological development. Specifically, cognitive development is assessed based on the level of conception, perception, information processing, and language as an indicator of brain development. It is generally recognized that cognitive development progresses with age, as human awareness and understanding of the world increases from infancy to childhood, and then again into adolescence. The process of cognitive development was first described by Jean Piaget, in his Theory of Cognitive Development.

Language

One of the best-studied examples of cognitive development is language development. While some theories propose that language development is a genetically inherited skill common to all humans, others argue that social interactions are essential to language development. Most scientists recognize that language is influenced by the complex interaction between genetics and the environment. Language development can be further characterized into distinct process of learning, including the development of language sounds, organizing these sounds, forming the basic linguistic units (e.g., root words, tone, etc.), syntax (e.g., grammatically correct sentence patterns), the meaning of certain words or phrases, and the relationship between statements. Another aspect of cognitive development is bilingualism. Recent research indicates that bilingualism acts on the executive function of the brain because the selection of a particular language is an active process.

Social and emotional development is the change over time in children's ability to react to and interact with their social environment.

Emotional and Social

Social and emotional development is complex and includes many different areas of growth. Each is described in more detail below:

temperament the way a young child acts and responds to different situations, caregivers, and strangers.

attachment the emotional bond between a child and caregiver.

social skills or social competence the ability to get along with other people.

emotion regulation the ability of a child to control his or her emotions and reactions to the environment.

Moral

Morality is our ability to learn the difference between right or wrong and understand how to make the right choices. As with other facets of development, morality doesn't form independently from the previous areas we have been discussing. Children's experiences at home, the environment around them, and their physical, cognitive, emotional, and social skills influence their developing sense of right vs. wrong.

Between the ages of 2 and 5, many children start to show morally-based behaviors and beliefs. For example, Tasha may see Juan take the blocks out of Tyler's hands and say, "Juan! You're gonna get in trouble!" At this point, many young children also start to show empathy-based guilt when they break the rules. For example, if Juan from the above example sees Tyler cry because his blocks were stolen, Juan might start feeling somewhat bad that he hurt Tyler's feelings. As a younger child, however, Juan would feel badly only if he was punished for taking the blocks rather than making someone else sad.

According to Piaget, children between the ages of 5 and 10 see the world through a Heteronomous Morality. In other words, children think that authority figures such as parents and teachers have rules that young people must follow absolutely. Rules are thought of as real, unchangeable guidelines rather than evolving, negotiable, or situational. As they grow older, develop more abstract thinking, and become less self-focused, children become capable of forming more flexible rules and applying them selectively for the sake of shared objectives and a desire to co-operate.

Developmental psychologist Lawrence Kohlberg built on Piaget's work to create his theory of the Stages of Moral Understanding. According to Kohlberg, young children at this age base their morality on a punishment and obedience orientation. Much like Piaget, Kohlberg believed that young children behave morally because they fear authority and try to avoid punishment. In other words, little kids follow the rules because they don't want to get in trouble. It's too much to expect preschool-aged children to automatically "do the right thing". However, most young children can understand the difference between "good" and "bad" behavior, and this understanding provides the basis for more complicated moral thinking in the future.

Contemporary research has provided us with additional information about how young children understand morals. Children between the ages 5 and 6 typically think in terms of distributive justice, or the idea that material goods or "stuff" should be fairly shared. In other words, everyone should get his or her exact "fair share." For example, Sally may think that it's only fair if each child gets exactly 2 cookies and the same amount of milk in their glass. Other factors, such as need or effort, are not considered. Sally wouldn't think that Susie should get an additional cookie because her lunch fell on the floor. By age 6 or 7, children begin to consider what people have earned or worked for when thinking about distributive justice. Children can also reason that some people should get more because they worked harder. For example, Jane begins to understand that Jill should earn a bigger prize because she sold more Girl Scout cookies.

PSYCHOPATHOLOGY

Psychopathology is the scientific study of mental disorders, including efforts to understand their genetic, biological, psychological, and social causes; develop classification schemes (nosology) which can improve treatment planning and treatment outcomes; understand the course of psychiatric illnesses across all stages of development; more fully understand the manifestations of mental disorders; and investigate potentially effective treatments.

At least conceptually, psychopathology is a subset of pathology, which is the "... scientific study of the nature of disease and its causes, processes, development, and consequences." Psychopathology is distinct from psychiatry by virtue of being a theoretical field of scientific research rather than a specialty of medical practice.

Early explanations for mental illnesses were influenced by religious belief and superstition. Psychological conditions that are now classified as mental disorders were initially attributed to possessions by evil spirits, demons, and the devil. This idea was widely accepted up until the sixteenth and seventeenth centuries. Individuals who suffered from these so-called "possessions" were tortured as treatment. Doctors used this technique in hoping to bring their patients back to sanity. Those who failed to return to sanity after torture were executed.

The Greek physician Hippocrates was one of the first to reject the idea that mental disorders were caused by possession of demons or the devil. He firmly believed the symptoms of mental disorders were due to diseases originating in the brain. Hippocrates suspected that these states of insanity were due to imbalances of fluids in the body. He identified these fluids to be four in particular: blood, black bile, yellow bile, and phlegm.

Furthermore, not far from Hippocrates, the philosopher Plato would come to argue the mind, body, and spirit worked as a unit. Any imbalance brought to these compositions of

the individual could bring distress or lack of harmony within the individual. This philosophical idea would remain in perspective [vague] until the seventeenth century.

In the eighteenth century's Romantic Movement, the idea that healthy parent-child relationships provided sanity became a prominent idea. Philosopher Jean-Jacques Rousseau introduced the notion that trauma in childhood could have negative implications later in adulthood.

In the nineteenth century, greatly influenced by Rousseau's ideas and philosophy, Austrian psychoanalyst Sigmund Freud would bring about psychotherapy and become the father of psychoanalysis, a clinical method for treating psychopathology through dialogue between a patient and a psychoanalyst. Talking therapy would originate from his ideas on the individual's experiences and the natural human efforts to make sense of the world and life.

As the Study of Psychiatric Disorders

The scientific discipline of psychopathology was founded by Karl Jaspers in 1913. It was referred to as "static understanding" and its purpose was to graphically recreate the "mental phenomenon" experienced by the client.

The study of psychopathology is interdisciplinary, with contributions coming from clinical, social, and developmental psychology, as well as neuropsychology and other psychology subdisciplines; psychiatry; neuroscience generally; criminology; social work; sociology; epidemiology; statistics; and more. Practitioners in clinical and academic fields are referred to as psychopathologists.

How do scientists (and people in general) distinguish between unusual or odd behavior on one hand, and a mental disorder on the other? One strategy is to assess a person along four dimensions: deviance, distress, dysfunction and danger, known collectively as the Four D's.

The four Ds

A description of the four Ds when defining abnormality:

Deviance. This term describes the idea that specific thoughts, behaviours and emotions are considered deviant when they are unacceptable or not common in society. Clinicians must, however, remember that minority groups are not always deemed deviant just because they may not have anything in common with other groups. Therefore, we define an individual's actions as deviant or abnormal when their behaviour is deemed unacceptable by the culture they belong to. However, many disorders have a relation between patterns of deviance and therefore need to be evaluated in a differential diagnostic model.

Distress. This term accounts for negative feelings by the individual with the disorder. They may feel deeply troubled and affected by their illness. Behaviors and feelings that cause distress to the individual or to others around him or her are considered abnormal, if the condition is upsetting to the person experiencing it. Distress is related to dysfunction by being a useful asset in accurately perceiving dysfunction in an individual's life. These two are not always related because an individual can be highly dysfunctional and at the same time experiencing minimal stress. One should know the important characteristic of distress is not involved with dysfunction, but rather the limit to which an individual is stressed by an issue.

Dysfunction. This term involves maladaptive behaviour that impairs the individual's ability to perform normal daily functions, such as getting ready for work in the morning, or driving a car. This maladaptive behaviour has to be a problem large enough to be considered a diagnosis. It's highly noted to look for dysfunction across an individual's life experience because there is a chance the dysfunction may appear in clear observable view and in places where it is less likely to appear. Such maladaptive behaviours prevent the individual from living a normal, healthy lifestyle. However, dysfunctional behaviour is not always caused by a disorder; it may be voluntary, such as engaging in a hunger strike.

Danger. This term involves dangerous or violent behaviour directed at the individual, or others in the environment. The two important characteristics of danger is, danger to self and danger to others. When diagnosing, there is a large vulnerability of danger in which there is some danger in each diagnosis and within these diagnoses there is a continuum of severity. An example of dangerous behaviour that may suggest a psychological disorder is engaging in suicidal activity. Behaviors and feelings that are potentially harmful to an individual or the individuals around them are seen as abnormal.

The p factor

Instead of conceptualizing psychopathology as consisting of several discrete categories of mental disorders, groups of psychological and psychiatric scientists have proposed a "general psychopathology" construct, named the p-factor, because of its conceptual similarity with the g factor of general intelligence. Although researchers initially conceived a tripartite (three factor) explanation for psychopathology generally, subsequent study provided more evidence for a unitary, factor that is sequentially comorbid, recurrent/chronic, and exists on a continuum of severity and chronicity. Thus, the p factor is a dimensional, as opposed to a categorical, construct.

Higher scores on the p factor dimension have been found to be correlated with higher levels of functional impairment, greater incidence of problems in developmental history, and more diminished early-life brain function. In addition, those with higher levels of the p factor are more likely to have inherited a genetic predisposition to mental illness. The existence of the p factor may explain why it has been "... challenging to find causes, consequences, biomarkers, and treatments with specificity to individual mental disorders."

The p factor has been likened to the g factor of general intelligence, which is also a dimensional system by which overall cognitive ability can be defined. As psychopathology has typically been studied and implemented as a categorical system, like the Diagnostic and Statistical Manual system developed for clinicians, the dimensional system of the p factor provides an alternative conceptualization of mental disorders that might improve our understanding of psychopathology in general; lead to more precise diagnoses; and facilitate more effective treatment approaches.

Benjamin Lahey and colleagues first proposed a general psychopathology factor in 2012.

As Mental Symptoms

The term psychopathology may also be used to denote behaviors or experiences which are indicative of mental illness, even if they do not constitute a formal diagnosis. For

example, the presence of a hallucination may be considered as a psychopathological sign, even if there are not enough symptoms present to fulfill the criteria for one of the disorders listed in the DSM or ICD.

In a more general sense, any behaviour or experience which causes impairment, distress or disability, particularly if it is thought to arise from a functional breakdown in either the cognitive or neurocognitive systems in the brain, may be classified as psychopathology. It remains unclear how strong the distinction between maladaptive traits and mental disorders actually is, e.g. neuroticism is often described as the personal level of minor psychiatric symptoms.

Diagnostic and Statistical Manual of Mental Disorders

The Diagnostic and Statistical Manual of Mental Disorders (DSM) is an guideline for the diagnosis and understanding of mental disorders. It serves as reference for a range of professionals in medicine and mental health in the United States particularly. These professionals include psychologists, counselors, physicians, social workers, psychiatric nurses and nurse practitioners, marriage and family therapists, and more.

Examples of mental disorders classified within the DSM include:

Major depressive disorder is a mood disorder defined by symptoms of loss of motivation, decreased mood, lack of energy and thoughts of suicide.

Bipolar disorders are mood disorders characterized by depressive and manic episodes of varying lengths and degrees.

Dysthymia is a mood disorder similar to depression. Characterized by a persistent low mood, dysthymia is a less debilitating form of depression with no break in ordinary functioning.

Schizophrenia is characterized by altered perception of reality, including delusional thoughts, hallucinations, and disorganized speech and behaviour. Most cases arise in patients in their late teens or early adulthood, but can also appear later on in life.

Borderline personality disorder occurs in early adulthood for most patients; specific symptoms include patterns of unstable and intense relationships, chronic feelings of emptiness, emotional instability, paranoid thoughts, intense episodes of anger, and suicidal behavior.

Bulimia nervosa "binge and purge", an eating disorder specified as reoccurring episodes of uncontrollable binge eating followed by a need to vomit, take laxatives, or exercise excessively. Usually begins occurring at adolescence but most individuals do not seek help until later in life when it can be harder to change their eating habits.

Phobias Found in people of all ages. Characterized by an abnormal response to fear or danger. Persons diagnosed with Phobias suffer from feelings of terror and uncontrollable fear, exaggerated reactions to danger that in reality is not life-threatening, and is usually accompanied by physical reactions related to extreme fear: panic, rapid heartbeat, and/or shortened breathing.

Pyromania this disorder is indicated by fascination, curiosity, or attraction to purposely setting things on fire. Pyromaniacs find pleasure and/or relief by watching things burn. Can occur due to delusional thinking, impaired judgement due to other mental disorders, or simply as aggressive behavior to express anger.

PSYCHOTHERAPIES

Psychoanalysis

Psychoanalytic or psychodynamic psychotherapy draws on theories and practices of analytical psychology and psychoanalysis. It is a therapeutic process which helps patients understand and resolve their problems by increasing awareness of their inner world and its influence over relationships both past and present. It differs from most other therapies in aiming for deep seated change in personality and emotional development.

Psychoanalytic and psychodynamic psychotherapy aim to help people with serious psychological disorders to understand and change complex, deep-seated and often unconsciously based emotional and relationship problems thereby reducing symptoms and alleviating distress. However, their role is not limited only to those with mental health problems. Many people who experience a loss of meaning in their lives or who are seeking a greater sense of fulfilment may be helped by psychoanalytic or psychodynamic psychotherapy.

Sometimes people seek help for specific reasons such as eating disorders, psychosomatic conditions, obsessional behaviour, or phobic anxieties. At other times help is sought because of more general underlying feelings of depression or anxiety, difficulties in concentrating, dissatisfaction in work or inability to form satisfactory relationships. It may benefit adults, children, and adolescents. It can help children who have emotional and behavioural difficulties which are evident at home or school. These can include personality problems, depression, learning difficulties, school phobias, eating or sleeping disorders.

Psychoanalytic or psychodynamic psychotherapy provides an effective treatment for a range of psychological disorders, both as a treatment in its own right and as an adjunct to other forms of treatment. It can contribute significantly to patient's mental and physical health, to their sense of well-being and to their ability to manage their lives more effectively.

Whether psychoanalytic or psychodynamic psychotherapy is the treatment of choice for a particular individual depends on a variety of factors. It is often helpful to have one or more preliminary consultations with an experienced psychotherapist before deciding whether psychoanalytic psychotherapy is an appropriate treatment for the person concerned. Occasionally, the treatment might be of short duration but generally speaking psychoanalytic psychotherapy is best considered as a long-term treatment involving considerable commitment for both patient and therapist.

The relationship with the therapist is a crucial element in the therapy. The therapist offers a confidential and private setting which facilitates a process where unconscious patterns of the patient's inner world become reflected in the patient's relationship with the therapist (transference). This process helps patients gradually to identify these patterns and, in becoming conscious of them, to develop the capacity to understand and change them.

Person-centered

Person-centered therapy uses a non-authoritative approach that allows clients to take more of a lead in discussions so that, in the process, they will discover their own solu-

tions. The therapist acts as a compassionate facilitator, listening without judgment and acknowledging the client's experience without moving the conversation in another direction. The therapist is there to encourage and support the client and to guide the therapeutic process without interrupting or interfering with the client's process of self-discovery.

Anyone who would be better off gaining more self-confidence, a stronger sense of identity, and the ability to build healthy interpersonal relationships and to trust his or her own decisions could benefit from person-centered therapy. This approach, alone or in combination with other types of therapy, can also be helpful for those who suffer from grief, depression, anxiety, stress, abuse, or other mental health conditions. Person-centered therapists work with both individuals and groups. Since the client must do a lot of the work in person-centered therapy, those who are more motivated are likely to be more successful.

Person-centered therapy is talk therapy wherein the client does most of the talking. Your therapist will not judge or try to interpret what you say, but may restate your words in an attempt to fully understand your thoughts and feelings. When you hear your own words repeated back to you, you may then wish to self-edit and clarify your meaning. This may happen several times until you decide that you have expressed exactly what you are thinking and how you feel. There may be moments of silence to allow your thoughts to sink in. This client-focused process facilitates your self-discovery, self-acceptance, and a provides a means toward healing and positive growth.

Person-centered therapy, also known as Rogerian therapy, originated in the work of the American psychologist, Carol Rogers, who believed that everyone is different and, therefore, everyone's view of his or her own world, and ability to manage it, should be trusted. Rogers believed that all of us have the power to find the best solutions for ourselves and make appropriate changes in our lives. Person-centered therapy was a movement away from the therapist's traditional role—as an expert and leader—toward a process that allows clients to use their own understanding of their experiences as a platform for healing. The success of person-centered therapy relies on three conditions:

Unconditional positive regard, which means therapists must be empathetic and non-judgmental to convey their feelings of understanding, trust, and confidence that encourage their clients to make their own decisions and choices

Empathetic understanding, which means therapists completely understand and accept their clients' thoughts and feelings

Congruence, which means therapists carry no air of authority or professional superiority but, instead, present a true and accessible self that clients can see is honest and transparent.

Licensed mental health professionals from a variety of disciplines who have training and experience in the Rogerian approach can use person-centered interventions in therapy. In addition to finding someone with the educational background and relevant experience, look for a therapist or counselor who is especially empathetic and with whom you feel comfortable discussing personal issues.

Gestalt

Gestalt therapy is a client-centered approach to psychotherapy that helps clients focus on the present and understand what is really happening in their lives right now, rather than

what they may perceive to be happening based on past experience. Instead of simply talking about past situations, clients are encouraged to experience them, perhaps through re-enactment. Through the gestalt process, clients learn to become more aware of how their own negative thought patterns and behaviors are blocking true self-awareness and making them unhappy.

Gestalt therapy can help clients with issues such as anxiety, depression, self-esteem, relationship difficulties, and even physical ones like migraine headaches, ulcerative colitis, and back spasms. Good candidates for gestalt therapy are those who are interested in working on their self-awareness but may or may not understand the role they play in their own unhappiness and discomfort. Gestalt techniques are often used in combination with body work, dance, art, drama, and other therapies.

A gestalt therapist focuses on what is happening in the moment and finding solutions in the present time. For examples, rather than discuss why something happened in the past, the therapist will encourage you to re-enact the moment and discuss how it feels right now. In other words, you will be asked to experience your feelings, rather than simply talk about them. The therapist will ask you questions like, “What’s going on in this moment?” Or “How does this make you feel now?” Your therapist may experiment with dream work, guided fantasy, role-playing, confrontation, and other techniques that can help bring past and current struggles to life in the therapeutic setting. Your goal, as you become more aware of yourself and your senses, is to take more responsibility for yourself, accept the consequences of your behavior, and learn to satisfy your own needs while still respecting the needs of others.

The word “gestalt” means whole. Gestalt therapy was developed by psychotherapist Fritz Perls on the principle that humans are best viewed as a whole entity consisting of body, mind, and soul, and best understood when viewed through their own eyes, not by looking back into the past but by bringing the past into the present. Gestalt therapy emphasizes that to alleviate unresolved anger, pain, anxiety, resentment, and other negative feelings, these emotions cannot just be discussed, but must be actively expressed in the present time. If that doesn’t happen, both psychological and physical symptoms can arise. Perls believed that we are not in this world to live up to others’ expectations, nor should we expect others to live up to ours. By building self-awareness, gestalt therapy helps clients better understand themselves and how the choices they make affect their health and their relationships. With this self-knowledge, clients begin to understand how their emotional and physical selves are connected and develop more self-confidence to start living a fuller life and more effectively deal with problems.

Look for a licensed, experienced psychotherapist with a gestalt approach toward therapy. In addition to fulfilling their general education and licensing requirements, some therapists may take continuing education courses and training in gestalt therapy techniques. Once you have established that a therapist has the credentials and experience you are looking for, be sure you understand and are comfortable with the process as explained by the therapist.

Existential

Existential therapy focuses on free will, self-determination, and the search for meaning—often centering on you rather than on the symptom. The approach emphasizes your capacity to make rational choices and to develop to your maximum potential.

The existential approach stresses that:

All people have the capacity for self-awareness.

Each person has a unique identity that can be known only through relationships with others.

People must continually re-create themselves because life's meaning constantly changes.

Anxiety is part of the human condition.

What else is existential therapy recommended for? Psychological problems—like substance abuse—result from an inhibited ability to make authentic, meaningful, and self-directed choices about how to live, according to the existential approach. Interventions often aim to increase self-awareness and self-understanding. Existential psychotherapists try to comprehend and alleviate a variety of symptoms, including excessive anxiety, apathy, alienation, nihilism, avoidance, shame, addiction, despair, depression, guilt, anger, rage, resentment, embitterment, purposelessness, psychosis, and violence. They also focus on life-enhancing experiences like relationships, love, caring, commitment, courage, creativity, power, will, presence, spirituality, individuation, self-actualization, authenticity, acceptance, transcendence, and awe.

Here's what you can expect from a course of therapy. Existential psychotherapies use a range of approaches, but major themes focus on your responsibility and freedom. Therapists help you find meaning in the face of anxiety by choosing to think and act responsibly and by confronting negative internal thoughts rather than external forces like societal pressures or luck. Fostering creativity, love, authenticity, and free will are common avenues that help move you toward transformation. Similarly, when treating addiction disorders, the existential therapist coaches you to face the anxiety that tempts you to abuse substances and guides you to take responsibility. The goal: You learn to make more willful decisions about how to live, drawing on creativity and love, instead of letting outside events determine your behavior.

This practice—due to its focus on existence and purpose—is sometimes perceived as pessimistic, but it's meant to be a positive and flexible approach. At its best, according to 20th-century philosopher Paul Tillich, existential psychotherapy fairly and honestly confronts life's "ultimate concerns," including loneliness, suffering, and meaninglessness. Specific concerns are rooted in each individual's experience, but contemporary existential psychotherapist Irvin Yalom says that the universal ones are death, isolation, freedom, and emptiness. Existential therapy focuses on the anxiety that occurs when you confront these inherent conflicts, and the therapist's role is to foster personal responsibility for making decisions. Yalom, for example, perceives the therapist as a "fellow traveler" through life, and he uses empathy and support to elicit insight and choices. And because people exist in the presence of others, the relational context of group therapy is an effective approach, he says. The core question addressed in this kind of therapy is "how do I exist in the face of uncertainty, conflict, or death?"

In addition to their mental health training, existential therapists often have a background in philosophy. Licensure varies state by state, but many existential therapists complete graduate degrees in psychology or counseling, for example. They also complete additional supervised fieldwork in existential therapy.

Acceptance Commitment Therapy

Acceptance and commitment therapy (ACT) is an action-oriented approach to psychotherapy that stems from traditional behavior therapy and cognitive behavioral therapy. Clients learn to stop avoiding, denying, and struggling with their inner emotions and, instead, accept that these deeper feelings are appropriate responses to certain situations that should not prevent them from moving forward in their lives. With this understanding, clients begin to accept their issues and hardships and commit to making necessary changes in their behavior, regardless of what is going on in their lives, and how they feel about it.

ACT has been used effectively to help treat workplace stress, test anxiety, social anxiety disorder, depression, obsessive-compulsive disorder, and psychosis. It has also been used to help treat medical conditions such as chronic pain, substance abuse, and diabetes.

Working with a therapist, you will learn to listen to your own self-talk, or the way you talk to yourself specifically about traumatic events, problematic relationships, physical limitations, or other issues. You can then decide if an issue requires immediate action and change or if it can—or must—be accepted for what it is while you learn to make behavioral changes that can affect the situation. You may look at what hasn't worked for you in the past, so that the therapist can help you stop repeating thought patterns and behaviors that are causing you more problems in the long run. Once you have faced and accepted your current issues, you make a commitment to stop fighting your past and your emotions and, instead, start practicing more confident and optimistic behavior, based on your personal values and goals.

The theory behind ACT is that it is not only ineffective, but often counterproductive, to try to control painful emotions or psychological experiences, because suppression of these feelings ultimately leads to more distress. ACT adopts the view that there are valid alternatives to trying to change the way you think, and these include mindful behavior, attention to personal values, and commitment to action. By taking steps to change their behavior while, at the same time, learning to accept their psychological experiences, clients can eventually change their attitude and emotional state.

Look for a licensed, experienced therapist, social worker, professional counselor or other mental-health professional with additional training in ACT. There is no special certification for ACT practitioners. Skills are acquired through peer counseling, workshops, and other training programs. In addition to these credentials, it is important to find a therapist with whom you feel comfortable working.

Behavior Therapy

In behavioral therapy, the goal is to reinforce desirable behaviors and eliminate unwanted or maladaptive ones. Behavioral therapy is rooted in the principles of behaviorism, a school of thought focused on the idea that we learn from our environment. The techniques used in this type of treatment are based on the theories of classical conditioning and operant conditioning.

One important thing to note about the various behavioral therapies is that unlike some other types of therapy that are rooted in insight (such as psychoanalytic and humanistic therapies), behavioral therapy is action-based. Behavioral therapists are

focused on using the same learning strategies that led to the formation of unwanted behaviors.

Because of this, behavioral therapy tends to be highly focused. The behavior itself is the problem and the goal is to teach clients new behaviors to minimize or eliminate the issue. Old learning led to the development of a problem and so the idea is that new learning can fix it.

There are also three major areas that also draw on the strategies of behavioral therapy:

Cognitive-behavioral therapy relies on behavioral techniques but adds a cognitive element, focusing on the problematic thoughts that lie behind behaviors.

Applied behavior analysis utilizes operant conditioning to shape and modify problematic behaviors.

Social learning theory centers on how people learn through observation. Observing others being rewarded or punished for their actions can lead to learning and behavior change.

Edward Thorndike was one of the first to refer to the idea of modifying behavior. Other early pioneers of behavior therapy included psychologists Joseph Wolpe and Hans Eysenck.

Behaviorist B.F. Skinner's work had a major influence on the development of behavior therapy and his work introduced many of the concepts and techniques that are still in use today.

Later on, psychologists such as Aaron Beck and Albert Ellis began adding a cognitive element to behavioral strategies to form a treatment approach known as cognitive-behavioral therapy (CBT).

The Foundation of Behavioral Therapy

In order to understand how behavioral therapy works, let's start by exploring the two basic principles that contribute to behavioral therapy: classical and operant conditioning.

Classical conditioning involves forming associations between stimuli. Previously neutral stimuli are paired with a stimulus that naturally and automatically evokes a response. After repeated pairings, an association is formed and the previously neutral stimulus will come to evoke the response on its own.

Operant conditioning focuses on how reinforcement and punishment can be utilized to either increase or decrease the frequency of a behavior. Behaviors followed by desirable consequences are more likely to occur again in the future, while those followed by negative consequences become less likely to occur.

Behavior Therapy Based on Classical Conditioning

Classical conditioning is one way to alter behavior, and a number of techniques exist that can produce such change. Originally known as behavior modification, this type of therapy is often referred to today as applied behavior analysis.

Some of the techniques and strategies used in this approach to therapy include:

Flooding. This process involves exposing people to fear-invoking objects or situations intensely and rapidly. It is often used to treat phobias, anxiety and other stress-related disorders. During the process, the individual is prevented from escaping or avoiding the situation.

For example, flooding might be used to help a client who is suffering from an intense fear of dogs. At first, the client might be exposed to a small friendly dog for an extended

period of time during which he or she cannot leave. After repeated exposures to the dog during which nothing bad happens, the fear response begins to fade.

Systematic Desensitization. This technique involves having a client make a list of fears and then teaching the individual to relax while concentrating on these fears. The use of this process began with psychologist John B. Watson and his famous Little Albert experiment in which he conditioned a young child to fear a white rat. Later, Mary Cover Jones replicated Watson's results and utilized counterconditioning techniques to desensitize and eliminate the fear response.

Systematic desensitization is often used to treat phobias. The process follows three basic steps.

First, the client is taught relaxation techniques.

Next, the individual creates a ranked list of fear-invoking situations.

Starting with the least fear-inducing item and working their way up to the most fear-inducing item, the client confronts these fears under the guidance of the therapist while maintaining a relaxed state.

For example, an individual with a fear of the dark might start by looking at an image of a dark room before moving on to thinking about being in a dark room and then actually confronting his fear by sitting in a dark room. By pairing the old fear-producing stimulus with the newly learned relaxation behavior, the phobic response can be reduced or even eliminated.

Aversion Therapy. This process involves pairing an undesirable behavior with an aversive stimulus in the hope that the unwanted behavior will eventually be reduced. For example, someone suffering from alcoholism might utilize a drug known as disulfiram, which causes severe symptoms such as headaches, nausea, anxiety and vomiting when combined with alcohol. Because the person becomes extremely ill when they drink, the drinking behavior may be eliminated.

Behavior Therapy Based on Operant Conditioning

Many behavior techniques rely on the principles of operant conditioning, which means that they utilize reinforcement, punishment, shaping, modeling and related techniques to alter behavior. These methods have the benefit of being highly focused, which means that they can produce fast and effective results.

Some of the techniques and strategies used in this approach to behavioral therapy include:

Token Economies. This type of behavioral strategy relies on reinforcement to modify behavior. Clients are allowed to earn tokens that can be exchanged for special privileges or desired items. Parents and teachers often use token economies to reinforce good behavior. Kids earn tokens for engaging in preferred behaviors and may even lose tokens for displaying undesirable behaviors. These tokens can then be traded for things such as candy, toys, or extra time playing with a favorite toy.

Contingency Management. This approach utilizes a formal written contract between the client and the therapist that outlines the behavior change goals, reinforcements and rewards that will be given and the penalties for failing to meet the demands of the agreement. These types of agreements aren't just used by therapists – teachers and parents also often use them with students and children

in the form of behavior contracts. Contingency contracts can be very effective in producing behavior changes since the rules are spelled out clearly in black-and-white, preventing both parties from backing down on their promises.

Modeling. This technique involves learning through observation and modeling the behavior of others. The process is based on Albert Bandura's social learning theory, which emphasizes the social components of the learning process. Rather than relying simply on reinforcement or punishment, modeling allows individuals to learn new skills or acceptable behaviors by watching someone else perform those desired skills. In some cases, the therapist might model the desired behavior. In other instances, watching peers engage in the sought after behaviors can also be helpful.

Extinction. Another way to produce behavior change is to stop reinforcing a behavior in order to eliminate the response. Time-outs are a perfect example of the extinction process. During a time-out, a person is removed from a situation that provides reinforcement. For example, a child who starts yelling or striking other children would be removed from the play activity and required to sit quietly in a corner or another room where there are no opportunities for attention and reinforcement. By taking away the attention that the child found rewarding, the unwanted behavior is eventually extinguished.

REBT

Rational Emotive Behavior Therapy (REBT) is a short-term form of psychotherapy that helps you identify self-defeating thoughts and feelings, challenge the rationality of those feelings, and replace them with healthier, more productive beliefs. REBT focuses mostly on the present time to help you understand how unhealthy thoughts and beliefs create emotional distress which, in turn, leads to unhealthy actions and behaviors that interfere with your current life goals. Once identified and understood, negative thoughts and actions can be changed and replaced with more positive and productive behavior, allowing you to develop more successful personal and professional relationships.

REBT can help you with negative emotions such as anxiety, depression, guilt, and extreme or inappropriate anger. This approach is also used to help change stressful and self-defeating behaviors, such as aggression, unhealthy eating, and procrastination that get in the way of your quality of life and reaching your goals.

To help you manage and overcome difficulties or achieve life goals, the therapist will work with you to identify the beliefs and rigid thought patterns that are holding you hostage. The therapist will help you see how irrational these thoughts are and how they harm you. Through a variety of mental exercises, you will then learn how to reduce your negative thoughts and responses, and replace them with healthier, more constructive and self-accepting thoughts. REBT makes use of a variety of methods and tools, including positive visualization, reframing your thinking, and the use of self-help books and audio-visual guides, as well as assigned homework for reinforcement between sessions.

REBT evolved from the work of psychiatrist Albert Ellis in the mid-1950s as the original form of cognitive-behavioral therapy. Ellis believed that most people are not aware that many of their thoughts about themselves are irrational and negatively affect the way they behave in important relationships and situations. According to Ellis, it is these thoughts that lead people to suffer negative emotions and engage in self-destructive be-

havior. At the same time, humans are capable of challenging and changing their irrational beliefs, if they are willing to do the work. While specific life events may contribute to mental health issues, REBT therapists believe that it is an individual's own faulty and irrational belief system that causes the most problems. By letting go of negative thoughts and replacing them with positive beliefs, one is better able to accept one's self and others and, in turn, live a happier life.

CBT

Cognitive-behavioral therapy (CBT) is a form of psychotherapy that treats problems and boosts happiness by modifying dysfunctional emotions, behaviors, and thoughts. Unlike traditional Freudian psychoanalysis, which probes childhood wounds to get at the root causes of conflict, CBT focuses on solutions, encouraging patients to challenge distorted cognitions and change destructive patterns of behavior.

CBT rests on the idea that thoughts and perceptions influence behavior. Feeling distressed, in some cases, may distort one's perception of reality. CBT aims to identify harmful thoughts, assess whether they are an accurate depiction of reality, and if they are not, employ strategies to challenge and overcome them.

CBT is appropriate for people of all ages, including children, adolescents, and adults. Evidence has mounted that CBT can benefit numerous conditions, such as major depressive disorder, anxiety disorders, post-traumatic stress disorder, eating disorders, obsessive-compulsive disorders, and many others. Research also indicates that CBT can be delivered effectively online, in addition to face-to-face therapy sessions.

The tools deployed in CBT—which include learning to identify and dispute unrealistic or unhelpful thoughts and developing problem-solving skills—have been used to treat a broad range of mental health challenges. CBT is now considered among the most efficacious forms of talk therapy, especially when clients incorporate strategies into their daily life. This effort to gain insight into one's cognitive and behavioral processes and modify them in a constructive way often involves ongoing practice, but is favored by many clients as it can require fewer therapy sessions than other modalities.

MBCT

Mindfulness-Based Cognitive Therapy (MBCT) is a modified form of cognitive therapy that incorporates mindfulness practices such as meditation and breathing exercises. Using these tools, MBCT therapists teach clients how to break away from negative thought patterns that can cause a downward spiral into a depressed state so they will be able to fight off depression before it takes hold.

MBCT was developed for people with recurring episodes of depression or unhappiness, to prevent relapse. It has been proven effective in patients with major depressive disorder who have experienced at least three episodes of depression. Mindfulness-based relapse prevention may also be helpful for treating generalized anxiety disorders and addictions. MBCT has also been shown to improve symptoms of depression in some people with physical health conditions, such as vascular disease and traumatic brain injury.

MBCT is group therapy, with once-a-week, two-hour sessions, led by your therapist, as part of an eight-week program. You will learn meditation techniques as well as basic principles of cognition, such as the relationship between the way you think and how you feel. You will also have the opportunity to learn more about your depressive condition. On

the days when there is no session, there is homework, which includes practicing breathing exercises and mindful meditation.

Sometimes normal sadness is a powerful trigger for someone who has recovered from a depressive state to relapse into another bout of depression. Rather than try to avoid or eliminate sadness or other negative emotions, one learns to change their relationship with these emotions by practicing meditation and other mindfulness exercises. These activities rebalance neural networks, allowing the client to move away from automatic negative responses toward an understanding that there are other ways to respond to situations. By developing a routine meditation practice, clients can use the technique whenever they start to feel overwhelmed by negative emotions. When sadness occurs and starts to bring up the usual negative associations that trigger relapse of depression, the client is equipped with tools that will help them replace negative thought patterns with positive.

An MBCT therapist is a mental health professional who has additional training in mindfulness-based practices and techniques and is skilled at teaching these techniques to others. Institutions, like the UC San Diego Center for Mindfulness, trains and certifies MBCT teachers around the U.S. There is no formal referral service for MBCT programs or therapists. In addition to checking credentials, it is important to find an MBCT therapist with whom you feel comfortable working.

Play Therapy

Although sometimes used with adults, play therapy is a psychotherapeutic approach primarily used to help children ages 3 to 12 explore their lives and freely express repressed thoughts and emotions through play. Therapeutic play normally takes place in a safe, comfortable playroom, where very few rules or limits are imposed on the child, encouraging free expression and allowing the therapist to observe the child's choices, decisions, and play style. The goal is to help children learn to express themselves in healthier ways, become more respectful and empathetic, and discover new and more positive ways to solve problems.

Therapeutic play helps children with social or emotional deficits learn to communicate better, change their behavior, develop problem-solving skills, and relate to others in positive ways. It is appropriate for children undergoing or witnessing stressful events in their lives, such as a serious illness or hospitalization, domestic violence, abuse, trauma, a family crisis, or an upsetting change in their environment. Play therapy can help children with academic and social problems, learning disabilities, behavioral disorders, anxiety, depression, grief, or anger, as well as those with attention deficit disorders or who are on the autism spectrum.

The parent or caregiver plays an important role in play therapy for children. After conducting an initial intake interview with the parent, when the therapist collects information about the child, and, often, a separate interview with the child, the therapist can make an assessment prior to beginning treatment. An assessment allows the therapist to decide the best treatment approach for the child. In the playroom, the child is encouraged to play with very specific types of toys that encourage self-expression and facilitate the learning of positive behaviors. Arts and crafts, music, dancing, storytelling, and other tools may also be incorporated into play therapy. Play therapy usually occurs in weekly sessions for an average of 20 sessions lasting 30 to 45 minutes each.

Play therapy responds to the unique developmental needs of young children, who often express themselves better through play activities than through verbal communication. The therapist uses play and other creative activities to communicate with the child and observe how the child uses these activities to express thoughts and feelings that are not expressed in words. There are two approaches to play therapy:

Nondirective play therapy is based on the principle that children can resolve their own issues given the right conditions and the freedom to play with limited instruction and supervision.

Directive play therapy uses more input from the therapist to help speed up results. Play therapists use both approaches, depending on the circumstances.

Play therapists are well-trained in child development, attachment, and the use of play as a way to communicate with children. The play therapist should also be trained in a recognized therapeutic approach, such as child-centered, cognitive-behavioral, Adlerian, or Gestalt therapy. In addition to finding someone with the appropriate educational background and relevant experience, look for a therapist with whom you feel comfortable working on personal and family issues.

Positive Psychotherapy

The term psychotherapy typically elicits thoughts of treating a psychiatric disorder – such as depression, OCD, or social phobia, helping couples work through marital conflict, or helping individuals overcome grief or work through a major life challenge. In other words, the goal is to eliminate or reduce something negative.

Positive psychotherapy – often referred to as PPT for short – is a relatively new and unique therapeutic approach that moves away from focusing on what’s wrong or negative to what’s good and positive. It strives to help you explore, identify, and build up your strengths – and what’s going right in your life – rather than dissect and “fix” your weaknesses and whatever’s wrong or broken in your world. That’s not to say problems and weaknesses aren’t discussed and explored; rather, building upon your strengths and helping you experience happiness in the present are viewed as the key to improved well-being and psychological health.

Positive psychotherapy is based on the idea that people find happiness in a variety of ways. Unfortunately, many people are robbed from joy and well-being in the present moment because they see happy times only in retrospect. Positive psychotherapy helps them identify present moment happiness.

Positive psychotherapy was originally developed with the intent of promoting optimal functioning and a greater sense of well-being in people who were already psychologically healthy. However, it is now being used by many therapists – not as a replacement for traditional psychotherapy – but as a complementary adjunct to it. According to psychologist Martin Seligman – one of the leading authorities on positive psychology – PPT can enhance traditional therapy’s goal of fixing what’s wrong by also building what’s strong. This approach is highly beneficial to therapy clients in many ways.

Unfortunately, skeptics of positive psychology and PPT tend to disregard it. Focusing on positive, happy things may seem a bit too Pollyannaish rather than genuinely therapeutic. However, clinicians who implement PPT into their regular practice argue that it adds a healthy balance to the usual focus on psychological disorders and emotional suffering.

Positive psychotherapy is based on the tenets of positive psychology. Humanistic in nature, positive psychotherapy and psychology stem from the work of several prominent psychologists and psychoanalysts over the years. These include Carl Rogers (client-centered therapy), Abraham Maslow (self-actualization and the “hierarchy of needs”), Albert Bandura (social learning theory), Erich Fromm (author of *The Art of Loving*), Christopher Peterson (author of *A Primer in Positive Psychology*), and Martin Seligman (known for his work on learned optimism). Seligman and Peterson are frequently credited with co-founding the field of positive psychology. Another prominent figure in positive psychotherapy is University of Toronto psychologist Tayyab Rashid.

It should be noted that the positive psychotherapy being discussed here is not quite the same as European positive psychotherapy, which was developed in the late 1960s by psychiatrist Nossrat Peseschkian and his colleagues.

Three Key Assumptions of Positive Psychotherapy

According to Seligman and Rashid, there are three key assumptions that form the basis of PPT:

1. People have an innate desire for happiness, fulfillment, and personal growth. They’re not just looking to avoid distress or negative feelings. Psychological problems develop when they’re unable to grow.
2. A person’s strengths are just as real and valid as any symptoms they’re experiencing or disorders they may have.
3. A beneficial therapeutic alliance doesn’t require a focus on weaknesses or psychopathology in order to develop; it can be formed by talking about a client’s strengths and resources as well.

Five Essentials of Happiness – Seligman’s PERMA Model

Throughout the course of his research on what makes people happy, Seligman identified 5 key elements which have come to be known as the PERMA model of well-being. This is based on the 5 things that play a key role in people’s happiness. They include:

Positive emotion

Engagement

Relationships

Meaning

Accomplishment

People who feel fulfilled in each of these key areas of life are much less vulnerable to depression and experience a greater overall sense of well-being.

Positive emotions bring many benefits, including a more positive mood, increased creativity, greater willingness to take risks and try new things, better overall health, and happier relationships. They make it easier to let go of past regrets, appreciate and enjoy the present, and feel optimistic about what the future holds.

In PPT, therapists help you to develop more positive emotions. This helps you feel more hopeful (rather than expecting something bad to happen) and enables you to look at your life from a more positive perspective.

Engaging with life is essential to well-being. Sitting on the sidelines keeps you bored, depressed, and unmotivated – and reinforces the tendency to dwell on the past and fret

about the future. When you engage in life, you gain momentum and are able to focus on – and enjoy – the moment.

As you identify your strengths in PPT, you can begin to find more ways to use them in meaningful and fulfilling ways – and in ways that bring value to others through your work, relationships, and other endeavors.

Relationships that are meaningful and happy bring great joy into our lives. They also help us keep a more balanced and realistic perspective than we have when we're isolated. Positive relationships with others also make us feel supported, loved, and valued. Positive psychotherapy helps you identify and nurture positive relationships in your life, and recognize those relationships that are unhealthy or even toxic.

Meaning and purpose in life make it rich and rewarding – and worth living. People find meaning in many different ways, including family, faith, a worthy cause, and their work.

Positive psychotherapy helps you identify the things that fulfill you and truly give your life meaning. Your therapist will encourage you to do things that align with your values and find others who care about the things that matter to you.

Accomplishing a goal or challenging task is very rewarding – whether it's finishing college, mastering a skill like tennis or playing the piano, or shedding those unwanted pounds. People who are happy can look at their life and feel a genuine sense of accomplishment. This doesn't mean they haven't made mistakes or experienced failures along the way. Focusing on their successes – their accomplishments – inspires them to work towards future goals, and stick with them even when it's tough to do so.

In PPT, your therapists will encourage you to define your goals and enhance the positive traits required to attain them. Your therapist will also help you stay focused on your goals and encourage you to celebrate all your successes – even if they're small. A strong sense of personal accomplishment boosts your resilience, and enables you to stay the course no matter how tedious or grueling – even when obstacles present themselves.

Values in Action Inventory

One of the key tools used in positive psychotherapy is the Values in Action Inventory of Strengths (VIA-IS). This tool, developed by Seligman and Peterson, consists of a questionnaire that enables you to determine your five greatest strengths. Because it focuses on your strengths, it can help offset the stigma of a mental health diagnosis. The latter, by its very nature, emphasizes areas of weakness.

The VIA-IS can serve as a powerful reminder that you are not your depression, bipolar disorder, OCD, or any other derogatory diagnostic label. A perceived flaw, personal failure, or psychiatric disorder doesn't define you. This is one of the most empowering aspects of positive psychotherapy. It enables you to see yourself as a whole person, rather than as a "broken" or "messed up" individual who needs to be "fixed".

Identifying your key strengths can be incredibly empowering. It reinforces the shift from what's wrong to what's strong. Research has shown that people who are aware of their strengths and use them regularly have a greater sense of well-being and fewer depressive symptoms.

Three Phases of Positive Psychotherapy

Positive psychotherapy can be divided into three distinct phases. The first phase is focused on helping therapy clients examine and identify their personal strengths.

The second phase of PPT is geared towards building and strengthening clients' positive emotions, while helping them let go of negative thought patterns and emotions that have been causing problems in their life.

The third and last phase of PPT is focused on the positive relationships in clients' lives, including ways in which those can be strengthened. This phase also focuses on helping clients gain a greater understanding of what gives their life purpose and meaning.

Therapist Techniques

Therapists use various techniques to incorporate the tenets of positive psychology into their work with clients. These include helping clients to:

Encourage strength-based language. Most clients come to therapy expecting to talk about all the negative things – hurts, resentments, conflicts with others, sadness, and losses. As the saying goes, what you focus on expands. Shifting the language to incorporate more positive words helps shift the focus to more positive things in a person's life. It can improve clients' ability to recognize their strengths and identify what's working in their lives.

Focus on the positive instead of the negative. In therapy (and often in life) people tend to focus on the negative rather than the positive. People who battle depression are especially prone to dwelling on the negative aspects of their lives, as well as all the negative things that have happened in their past. In PPT, the therapist strives to move the clients' focus to the positive – e.g. the good things that happened during the day or week.

This shift helps clients develop a more balanced – and realistic – perspective (i.e., there's almost always some good with the bad). It also enables them to become more adept at observing the positive things in the present, as well as identifying anything positive that came out of a difficult situation or experience in the past.

Reinforce feelings of hopefulness. Instilling a sense of hope is a common therapeutic goal. In PPT, therapists strive to find ways to build and strengthen their clients' sense of hopefulness and optimism. Helping clients recognize their strengths can contribute to this goal, as it facilitates greater confidence in their ability to handle problems and challenges in the future.

Send reminders. Some therapists will use a pager, text message, or other means to send the client a reminder to record what they're experiencing in that moment, with an emphasis on the positive. The idea is that these records (which the client can expand upon later in the day) provide a sample of sorts. At some point later on in therapy, these can be discussed to see how the client is progressing.

Transactional Analysis

Transactional analysis (TA) is a widely recognised form of modern psychology, and one of the most accessible theories of psychology at that. In simple terms, TA is designed to promote personal growth and change. It is considered a fundamental therapy for well-being and for helping individuals to reach their full potential in all aspects of life.

Founded by Eric Berne in the late 1950s, TA therapy is based on the theory that each person has three ego-states: parent, adult and child. These are used along with other key transactional analysis concepts, tools and models to analyse how individuals communicate and to identify what interaction is needed for a better outcome.

Throughout therapy, the TA therapist will work directly on problem-solving behaviours, whilst helping clients to develop day-to-day tools for finding constructive, creative solutions. The ultimate goal is to ensure clients regain absolute autonomy over their lives. Eric Berne defines this autonomy as the recovery of three vital human capacities - spontaneity, awareness and intimacy.

Transactional analysis is a talking therapy and sessions are designed to explore an individual's personality and how this has been shaped by experience - particularly those stemming from childhood. This is achieved through skilful questioning and the utilisation of various models, techniques and tools.

Sessions can be carried out in the form of one-on-one counselling, or with families, couples or groups. And, although it is commonly recognised as a brief and solution-focused approach, transactional analysis can also be applied as an effective long-term, in-depth therapy.

In counselling, TA therapy is very versatile, for it can be used in a wide range of areas and incorporates key themes from humanistic, integrative and psychoanalytical and psychodynamic therapies.

The atmosphere that supports transactional analysis is non-judgemental, secure and respectful, ensuring that a positive relationship is forged between the therapist and client(s) in order to provide a model for subsequent relationships and communication that are developed outside of therapy.

In this setting, the therapist works collaboratively with the individual. Together, they will identify what has gone wrong in the client's communication and provide opportunities for them to change repetitive patterns that limit their potential.

Key Concepts of Transactional Analysis

Below is an exploration of some of the key concepts of transactional analysis that a therapist will use in their work.

Ego-states

Ego-states refer to the three major parts of an individual's personality, and they each reflect an entire system of thought, feeling and behaviour.

These ego-states are:

Parent. Rooted in the past; a set of thoughts, feelings and behaviours learnt from our parents and other important people. This part of our personality can be supportive or critical.

Adult. Rooted in the present; relates to direct responses in the 'here and now' that are not influenced by our past. This tends to be the most rational part of our personality.

Child. Rooted in the past; a set of thoughts, feelings and behaviours learnt from our childhood. These can be free and natural or strongly adapted to parental influences.

Our ego-states determine how we express ourselves as individuals, interact with each other and form relationships. The simplicity of the terminology used in TA makes the model very accessible.

Unconscious Scripts

TA therapists recognise that we all have the potential to live the life we want, rather than the life we are programmed to live. Sometimes, however, this potential is hindered by repetitive patterns or 'unconscious scripts' that stem from childhood decisions and teachings.

TA therapists use script theory to identify these unconscious scripts. These will be analysed using the ego-state model, and their identification is crucial to helping clients realise how certain permissions and prohibitions they received as a child are impacting their lives and how they communicate.

These unconscious scripts often exist as repetitive patterns of behaviour, thoughts and feelings - characteristics that suggest the child ego-state is overbearing and tainting other parts of an individual's personality.

Transactions

When individuals communicate, their ego-states interact to create transactions. If the ego-states interact and blend in a healthy way, transactions tend to be healthier. But, sometimes ego-states can contaminate each other to create a distorted view of the world. Transactions may be classified as straightforward, crossed-up, or ulterior, and understanding these is key to conflict resolution.

Strokes

Strokes refer to compliments, acceptance and recognition, which are influential in how people lead their lives. TA therapy recognises that we are greatly motivated by the reinforcement we get as children, and if this was dysfunctional, we are likely to adopt dysfunctional patterns of living as we get older.

Intimacy

Another motivation recognised in transactional analysis is intimacy. Similarly to strokes, if the intimacy a child experiences is dysfunctional, it can lead to problems. Children may learn that this type of intimacy is the best that he or she can do to meet basic needs and communicate with others. This can lead to the development of repetitive patterns of behaviour that can hinder a person's potential.

Redecision

This refers to an individual's capacity to 'redecide' and make changes to certain decisions that were made as a child - those stemming from unconscious scripts. Redecision reflects the assumption of TA therapy that individuals have the potential to lead their lives as they choose. This power is released after a redecision is made while a client is in their child-ego state.

Ultimately, therapists will encourage clients to challenge their current beliefs and the way he or she uses their life script. This will help them to better understand the direction and patterns of their life for themselves, and this awareness can help them to make the decision to change their behaviour.

Dialectic Behavior Therapy

Dialectical behavior therapy (DBT) provides clients with new skills to manage painful emotions and decrease conflict in relationships. DBT specifically focuses on providing

therapeutic skills in four key areas. First, mindfulness focuses on improving an individual's ability to accept and be present in the current moment. Second, distress tolerance is geared toward increasing a person's tolerance of negative emotion, rather than trying to escape from it. Third, emotion regulation covers strategies to manage and change intense emotions that are causing problems in a person's life. Fourth, interpersonal effectiveness consists of techniques that allow a person to communicate with others in a way that is assertive, maintains self-respect, and strengthens relationships.

DBT was originally developed to treat borderline personality disorder. However, research shows that DBT has also been used successfully to treat people experiencing depression, bulimia, binge-eating, bipolar disorder, post-traumatic-stress disorder, and substance abuse. DBT skills are thought to have the capability of helping those who wish to improve their ability to regulate emotions, tolerate distress and negative emotion, be mindful and present in the given moment, and communicate and interact effectively with others.

DBT treatment typically consists of individual therapy sessions and DBT skills groups. Individual therapy sessions consist of one-on-one contact with a trained therapist, ensuring that all therapeutic needs are being addressed. The individual therapist will help the patient stay motivated, apply the DBT skills within daily life, and address obstacles that might arise over the course of treatment.

DBT skills group participants learn and practice skills alongside others. Members of the group are encouraged to share their experiences and provide mutual support. Groups are led by one trained therapist teaching skills and leading exercises. The group members are then assigned homework, such as practicing mindfulness exercises. Each group session lasts approximately two hours, and groups typically meet weekly for six months. Groups can be shorter or longer, depending on the needs of the group members. DBT can be delivered by therapists in many ways. For instance, some people complete the one-on-one therapy sessions without attending the weekly skills group. Others might choose the group without regular one-on-one sessions.

DBT is a cognitive-behavioral treatment developed by Marsha Linehan, Ph.D., in the 1980s to treat people with borderline personality disorder. Those diagnosed with BPD often experience extremely intense negative emotions that are difficult to manage. These intense and seemingly uncontrollable negative emotions are often experienced when the individual is interacting with others—friends, romantic partners, family members. People with borderline often experience a great deal of conflict in their relationships.

As its name suggests, DBT is influenced by the philosophical perspective of dialectics: balancing opposites. The therapist consistently works with the individual to find ways to hold two seemingly opposite perspectives at once, promoting balance and avoiding black and white—the all-or-nothing styles of thinking. In service of this balance, DBT promotes a both-and rather than an either-or outlook. The dialectic at the heart of DBT is acceptance and change.

DBT assumes that effective treatment, including group skills training, must pay equal attention to the behavior and experience of providers working with clients as it does to clients' behavior and experience. Thus, treatment of the providers is an important part of any DBT program, and therapists should practice the skills themselves. They need to know basic behavior therapy techniques and DBT treatment strategies. Look for a mental health professional with specialized training and experience in DBT. The Linehan Board of Certification, a non-profit organization, has developed certification standards for

clinicians. In addition, it is important to find a therapist with whom you feel comfortable working.

Art Therapy

Art therapy involves the use of creative techniques such as drawing, painting, collage, coloring, or sculpting to help people express themselves artistically and examine the psychological and emotional undertones in their art. With the guidance of a credentialed art therapist, clients can "decode" the nonverbal messages, symbols, and metaphors often found in these art forms, which should lead to a better understanding of their feelings and behavior so they can move on to resolve deeper issues.

Art therapy helps children, adolescents, and adults explore their emotions, improve self-esteem, manage addictions, relieve stress, improve symptoms of anxiety and depression, and cope with a physical illness or disability. Art therapists work with individuals, couples, and groups in a variety of settings, including private counseling, hospitals, wellness centers, correctional institutions, senior centers, and other community organizations. No artistic talent is necessary for art therapy to succeed, because the therapeutic process is not about the artistic value of the work, but rather about finding associations between the creative choices made and a client's inner life. The artwork can be used as a springboard for reawakening memories and telling stories that may reveal messages and beliefs from the unconscious mind.

As with any form of therapy, your first session will consist of your talking to the therapist about why you want to find help and learning what the therapist has to offer. Together, you will come up with a treatment plan that involves creating some form of artwork. Once you begin creating, the therapist may, at times, simply observe your process as you work, without interference or judgment. When you have finished a piece of artwork—and sometimes while you are still working on it—the therapist will ask you questions along the lines of how you feel about the artistic process, what was easy or difficult about creating your artwork, and what thoughts or memories you may have had while you were working. Generally, the therapist will ask about your experience and feelings before providing any observations.

Art therapy is founded on the belief that self-expression through artistic creation has therapeutic value for those who are healing or seeking deeper understanding of themselves and their personalities. According to the American Art Therapy Association, art therapists are trained to understand the roles that color, texture, and various art media can play in the therapeutic process and how these tools can help reveal one's thoughts, feelings, and psychological disposition. Art therapy integrates psychotherapy and some form of visual arts as a specific, stand-alone form of therapy, but it is also used in combination with other types of therapy.

An art therapist has the minimum of a master's degree, generally from an integrated program in psychotherapy and visual arts at an educational institution accredited by the Council for Higher Education Accreditation (CHEA). The initials ATR after a therapist's name means he or she is registered with the Art Therapy Credentials Board (ATCB). The initials ATR-BC means the therapist is not only registered but has passed an examination to become board-certified by the ATCB.

Performing Art Therapy

Creative interventions have been formalized through the disciplines of art therapy, music therapy, dance/movement therapy, drama therapy or psychodrama, poetry therapy, and play therapy, including sandtray therapy. Each discipline has been applied in psychotherapy and counseling with individuals of all ages, particularly children, for more than 70 years. Art, music, dance, drama, and poetry therapies are referred to as “creative arts therapies” because of their roots in the arts and theories of creativity. These therapies and others that utilize self-expression in treatment are also called “expressive therapies”. Expressive arts therapies are defined as the use of art, music, drama, dance/movement, poetry/creative writing, bibliotherapy, play, and sandplay within the context of psychotherapy, counseling, rehabilitation, or medicine. Additionally, expressive therapies are sometimes referred to as “integrative” when various arts are purposively used in combination in treatment.

Individual approaches to creative arts therapy are defined as follows:

Art therapy is the purposeful use of visual arts materials and media in intervention, counseling, psychotherapy, and rehabilitation; it is use with individuals of all ages, families, and groups.

Music therapy is the prescribed use of music to effect positive changes in the psychological, physical, cognitive, or social functioning of individuals with health or educational problems.

Drama therapy is the systematic and intentional use of drama/theater processes, products, and associations to achieve the therapeutic goals of symptom relief, emotional and physical integration, and personal growth. It is an active approach that helps the client tell his or her story to solve a problem, achieve catharsis, extend the depth and breadth of his or her inner experience, understand the meaning of images, and strengthen his or her ability to observe personal roles while increasing flexibility between roles.

Dance/movement therapy is based on the assumption that body and mind are interrelated and is defined as the psychotherapeutic use of movement as a process that furthers the emotional, cognitive, and physical integration of the individual. Dance/movement therapy effects changes in feelings, cognition, physical functioning, and behavior.

Poetry therapy and bibliotherapy are terms used synonymously to describe the intentional use of poetry and other forms of literature for healing and personal growth.

Play therapy is the systematic use of a theoretical model to establish an interpersonal process wherein trained play therapists use the therapeutic powers of play to help clients prevent or resolve psychosocial difficulties and achieve optimal growth and development.

Sandplay therapy is a creative form of psychotherapy that uses a sandbox and a large collection of miniatures to enable a client to explore the deeper layers of his or her psyche in a totally new format; by constructing a series of “sand pictures,” a client is helped to illustrate and integrate his or her psychological condition.

Integrative approaches involve two or more expressive therapies to foster awareness, encourage emotional growth, and enhance relationships with others. This approach distinguishes itself through combining modalities within a therapy

session. Integrative approaches are based on a variety of orientations, including arts as therapy, arts psychotherapy, and the use of arts for traditional healing.

While some practitioners define art, dance/movement, music, or drama therapies as play therapies, creative arts therapies and expressive therapies are not merely subsets of play therapy and have a long history in mental health with distinct approaches. While the arts may sometimes be a form of play, encouraging individuals to express themselves through a painting, music, or dance involves an understanding of the media beyond the scope of play. In brief, the arts therapies are different from play therapy because they integrate knowledge of art with principles of psychotherapy and related fields.

In addition to the disciplines and approaches mentioned above, many therapists integrate activities that enhance relaxation as part of trauma intervention. Relaxation techniques often include creative components such as music, movement, or art making. For example, guided imagery or visualization, meditation, yoga, and other methods of stress reduction are also used with individuals who have experienced trauma or loss.

Finally, art, music, and dance/movement therapies and other creative interventions such as play have sometimes been incorrectly labeled as “nonverbal” therapies. They are both verbal and nonverbal because verbal communication of thoughts and feelings is a central part of therapy in most situations. In fact, most therapists who use these methods integrate them within a psychotherapy approach, including but not limited to psychodynamic, humanistic, cognitive, developmental, systems, narrative, solution-focused, and others. There are also creative interventions that specifically focus on verbal communication and self-expression as part of treatment, such as drama therapy, creative writing and poetry therapy, and bibliotherapy. In all cases, these approaches are “brain-wise” interventions that stimulate whole-brain responses to help individuals of all ages experience reparation, recovery and well-being.

Family Therapy

Family therapy or family counseling is designed to address specific issues that affect the psychological health of the family, such as major life transitions or mental health conditions. It may be used as the primary mode of treatment or as a complementary approach.

The Benefits of Family Counseling

Families can benefit from therapy when they experience any stressful event that may strain family relationships, such as financial hardship, divorce, or the death of a loved one. In addition, it can be effective in treating mental health concerns that impact the family as a whole, such as depression, substance abuse, chronic illness, and food issues, or everyday concerns, like communication problems, interpersonal conflict, or behavioral problems in children and adolescents.

Family counseling aims to promote understanding and collaboration among family members in order to solve the problems of one or more individuals. For example, if a child is having social and academic problems, therapy will focus on the family patterns that may contribute to the child's acting out, rather than evaluating the child's behavior alone. As the family uncovers the source of the problem, they can learn to support the child and other family members and work proactively on minimizing or altering the conditions that contribute to the child's unwanted behavior.

Meeting with a Family Therapist

Family counseling is provided by licensed marriage and family therapists (LMFT). Other mental health professionals—such as professional counselors, social workers, and psychologists—who have received formal training in family therapy approaches may incorporate those principles into their own work.

This mode of therapy is solution-focused and short-term, with as few as nine sessions required, on average. Meetings are often held once per week and typically last for 50 minutes. The number of family members who attend each session may vary, depending on therapy goals, and often a therapist will offer individual sessions to supplement the family sessions. Family counseling is conducted in a variety of settings including family counseling services, community agencies, and residential treatment centers.

Family Therapy Approaches

Most forms of family counseling borrow heavily from systems theory, though there are others based on psychological approaches such as experiential, cognitive-behavioral, and psychodynamic. Family systems therapy contends that the internal dynamics of the family system can produce and sustain problematic behaviors in family members. All family therapy approaches are designed to help families improve communication, problem-solving, and coping skills, and enhance their sense of connection to one another.

History of Family Therapy

father and daughter with man seated in background observing Elements of family therapy can be found in the family-focused social work of the early 20th century and the work of child psychoanalysts, such as Nathan Ackerman, in the 1930s. But it wasn't until the late 1950s that family therapy was developed in earnest, as many psychiatrists and psychoanalysts—including Virginia Satir, Murray Bowen, Milton Erickson, and Jay Haley—were seeking a novel, alternative approach to traditional individual therapy. Family therapy was formally accepted in the psychotherapeutic community in the 1960s and has continued to evolve over the years.

APPLICATIONS OF THEORIES OF MOTIVATION AND LEARNING IN SCHOOL

Cognitive Dissonance

Cognitive dissonance occurs when there is an unresolved conflict in our mind between two beliefs, thoughts or perceptions we hold about a given subject. The level of tension resulting from such conflict will be a function of:

- How strong the conflict is between the two dissonant thoughts;
- How important the issue they relate to are to a specific individual or group;
- How difficult it is to rationalize (justify through logical or pseudo-logical reasoning) the dissonance.

Cognitive dissonance is a very powerful motivator which, as I shall discuss in a future post, is often used in transformational change programs both in the business and educational world. The reason why it is so powerful is because, when used effectively, Cogni-

tive dissonance creates a sense of discomfort in an individual which in order to be resolved results in one of two outcomes:

The individual changes behavior (possibly replacing the existing behavior with the newly modelled one);

The individual does not adopt the new behavior and justifies his/her behaviour by changing the conflicting cognition created by the new information, instead.

Drive Reduction Theory

This theory is centred on the notion that we all have needs that we attempt to satisfy in order to reduce the tension or arousal they cause. The internal stimuli these needs produce are our main drives in life. There are Primary drives which refer to basic needs (food, sleep, procreation, etc.) and Secondary drives which refer to social identity and personal fulfillment.

As we act on our needs we are conditioned and acquire habits and subconscious responses. So, for example: when a child needs to feel good about himself, he may recite a poem, sing a song, perform a dance or other 'feats' to his parents knowing he is going to get some recognition. Whenever he needs recognition in other contexts, this individual will possibly use the same tactics in order to get the same response from any other figure of authority – including teachers.

When the driven action does not satisfy his needs or the enacting of drives is frustrated, negative emotions (e.g. anxiety) arise. To go back to the previous example: if the boy is looking for a chance to show off to an authority figure his 'skills' but he is not given the opportunity to do so, he will feel frustrated, angry and/or unappreciated – a very common scenario in school, often dismissed as the child being 'naughty' or 'unruly'.

Attribution Theory

When we make a mistake or 'fail' at something we tend to go through a two-step process. We first experience an automatic response involving internal attribution (i.e. the error is our fault); then a conscious, slower reaction which seeks to find an alternative external attribution (e.g. the error is due to an external factor). This is because we all have a vested interest in 'looking good' in our own eyes – a sort of survival mechanism. This type of response, however, is unlikely to lead to self-improvement, as it results in an individual not addressing the real cause of their error/bad performance in the future. Roesch and Amirkham (1997) found that more experienced and successful athletes made more self-serving attributions which lead to identifying and addressing the internal causes of their performance errors.

Endowed progress effect

When people feel they have made some progress towards a goal, they will feel more committed towards its achievement. Conversely, people who are making little or no progress are more likely to give up early in the process. In my work with very low achieving 'difficult' students when I operated in challenging inner-city-area schools, sitting with them at the beginning of a task and guiding them through open questioning often 'did the trick' where threatening them with sanctions had failed miserably.

Cognitive Evaluation Theory

When looking at a task, we assess it in terms of how well it meets our need to feel competent and in control. We will be intrinsically motivated by tasks we believe fall in our current level of competency and 'put off' by those which we deem we will do poorly at. This issue is often more about self-perception of one's levels of competency than objective truth.

Self-determination Theory; Intrinsic and Extrinsic Motivation

Individuals differ from one another in terms of PLOC (personal locus of causality). Some will feel that their behavior is self-determined; they are the initiators and sustainers of their actions and their PLOC will be internal. Others will see external forces as determinants of their lives; coercing them into actions. These people's PLOC will be external. The internal locus is connected with intrinsic motivation, whilst the external locus is connected with extrinsic motivation.

Extrinsic motivation is when one is motivated by external factors, such as rewards, social recognition or fear of punishment. This kind of motivation focuses people on rewards rather than action.

Intrinsic motivation, on the other hand, refers to the desire to do things because we enjoy doing them, hence it is a stronger motivator than Extrinsic motivation. Three needs lead to intrinsic motivation:

Being successful at what we do (i.e. I enjoy French because I am good at it).

Being connected with others (i.e. I love my French class because I have bonded well with the rest of the class).

Having autonomy.

An important factor leading to Intrinsic Motivation creation is providing learners with the opportunity to develop effectance. Effectance refers to one of the 3 points made above (being successful at what we do) and is given rise to when the learner accomplishes success at something that they perceive challenging and falling in what Deci (1997) terms 'Optimal zone of development' – i.e.: a task that it is perceived as difficult enough to be challenging but within the stretch of the learner's ability. Effectance does not arise when we simply give students 'easy' work; that is why the 'easy wins' strategy often fails with students with poor intrinsic motivation; students are not stupid, they know you are dumbing down the work to make them feel good and the ensuing praise will not affect their self-regard as learners of your subject.

Self-determination theory assumes that there are individuals for whom a feeling of being in control of their life and responsible for their actions is very important for their personal fulfillment and, consequently, for their motivation.

There is a myth circulating amongst some educators these days (including some of my colleagues) that Extrinsic motivation should not be tapped into as a strategy to encourage students to improve. However, there is no sufficient credible evidence that Extrinsic Motivation is detrimental to learning to do away with it; on the contrary, research shows consistently that extrinsic motivation, when not overused and deployed in synergy with some of the other strategies discussed in this post, can eventually bring about Intrinsic motivation. Example: a student that does not enjoy French may, through experiencing a sense of effectance and obtaining consistent (thoroughly deserved and proportionate) praise and rewards become more appreciative of the subject, especially if she is experiencing

steady growth in her mastery of the language and feels connected and supported by his peers.

It is self-evident that using Extrinsic motivation will work with certain individuals rather than others; hence, as already mentioned, identifying the orientation of their Personal Locus Of Causality (PLOC) is fundamental, prior to carrying out any intervention.

Valence- Instrumentality- Expectancy (VIE) Theory

In this paradigm, motivation refers to three factors:

Valence what we think we will get out of a given action/behaviour (what's in it for me?).

Instrumentality the belief that if I perform a specific course of action I will succeed (clear path?).

Expectancy the belief that I will be definitely able to succeed (self-efficacy).

Goal-related Theory

In order to direct ourselves in our personal, educational and professional life we set ourselves goals. These can be:

Clear (so we know what to do and what not to do)

Challenging (so we get some stimulation)

Achievable (so we do not fail)

If we set goals ourselves, rather than having them imposed on us, we are more likely to work harder in order to achieve them. Moreover, Locke and Kristoff (1996) identified specific and challenging goals as those which are more likely to lead to higher achievement.

I picked the above theories as they are the ones that I have been using more successfully over 25 years as a classroom teacher and subject leader. It goes without saying that in order to apply them effectively one has to ensure first and foremost that they know and listen to the learners they are dealing with. Cognitive and emotional empathy are a must for the success of any of the above motivational strategies.

These strategies work best in synergy rather than in isolation. In a future post, for instance, I will endeavour to show how attitudinal change may be brought about by using a combination of the above principles to achieve the desired outcomes.

FACTORS IN EDUCATIONAL ACHIEVEMENT

An academic classroom is one where the primary goal is to promote proficiency of academic standards. Everything else, while appreciated and winked at, comes after. The class, curriculum, and instruction, by design, are built to move students in their academic achievement.

If that's your goal, what might your focuses be? What's important? What matters? Let's take a look at six important factors at work here.

Instructional Design Quality

While there are countless factors in the academic achievement of a student, few are more powerful than instructional design.

This is a topic for an entire book or a post of its own. But just consider that the ‘understanding’ is in large part a product of design and that starts with the most macro elements of all: what is being studied (curriculum), how it is sequenced (curriculum maps, for example), and how each are designed to maximize learning opportunities.

Accuracy and Alignment

Accurately unpacking the standard – not oversimplifying it, nor making it more complex than it has to be. Making sense of what the standard says, and understanding exactly what the student needs to know and be able to do to get there.

Accurately acknowledging the rigor of the standard—accurately, which means what it says, not what you think it should be. The standards describe a minimal level of proficiency—feel free to push students above and beyond, but not until they’ve mastered the language described in the standard.

Consider Using. Your PLN. Collaboration with other teachers is critical, and not necessarily through a formal 9-step PLC process, but conversationally, over time trying to make better sense of what the standards both say and imply.

Alignment between student practice with the standard. The work students do throughout the year shouldn’t merely ‘engage’ students or ‘push them to think, make, or create.’

Those things are great, but if there isn’t alignment between the practice and the standard, the mastery isn’t going to show because what the students aren’t mastering the standard, but rather thinking, making, and creating. Understanding by Design’s Backwards Planning makes a lot of sense here. As in, do it.

Alignment between student readiness and the academic standard. Think the Zone of Proximal Development. It does no good to create ‘rigorous’ work for the student who lacks the background knowledge, content knowledge, or literacy to perform the work while you browbeat them to ‘meet your expectation.’

Meet the student where they are implies exactly that. Meet them, then move them.

Consider Using. Understand by Design, especially Backward Planning thinking.

A climate of assessment and ways to measure understanding provide usable data that teachers can grab-and-use to revise planned instruction. Fresh, trustworthy, and relevant data teachers can use and students can understand to let them know what should come next. If it doesn’t answer that question—what now?—it’s an assessment of narrow value.

Data as in, units, lessons, and activities are designed to flexibly respond to that data in authentic ways without leaving it all on the teacher’s shoulders to do so on the fly—or, worse, at the local “data team” meeting.

Consider Using: Exit-Slip Teaching (we’re working on a model that might help here), technology that can help aggregate and report usable data consistently.

Whole Literacy

Literacy in terms of being able to read a variety of texts both critically and holistically. To be able to decode grade level texts of appropriate complexity, and to then be able to take that text apart in terms of ideas, evidence, theme, and craft.

And further, ‘whole’ literacy, as in being able to write clearly and eloquently about what is read, learned, and thought across physical and digital forms, for a variety of audiences

and purposes. In short, to be able to seamlessly use the writing process to create and refine arguments and narrate experiences for both academic and authentic purposes.

Student Motivation and Engagement

Student effort as in, helping students developing intrinsic motivation.

Student effort as in, supporting students emotionally, metacognitively, and intellectually to give themselves to the process of mastering academic content.

Student effort is everything. Teaching with engagement is like tweeting constantly with zero followers. Singing to an empty stadium. Dancing alone in the dark. We often use words and phrases like student engagement, student-centeredness, curiosity, and so on—and all of these are great. But a motivated student with zero technology, limited resources, and even a novice teacher will perform light years better than an apathetic student in a ‘21st-century classroom.’

Consider Using. Consistent learning feedback that makes sense to students; positive reinforcement; clear indicators of progress for students (think about video games, which constantly clarify yes/no, good/bad, warmer/colder to students); also, think damnification—points, levels, badges, unlocks, etc.; lots of ‘voice’ and choice; also, work within their ZPD that feels designed for them specifically will carry more weight than one-size-fits-all-do-it-because-I-said so; “Teaching What Matters Most” by Silver, Strong, and Perini is a great resource as well.

TEACHER EFFECTIVENESS

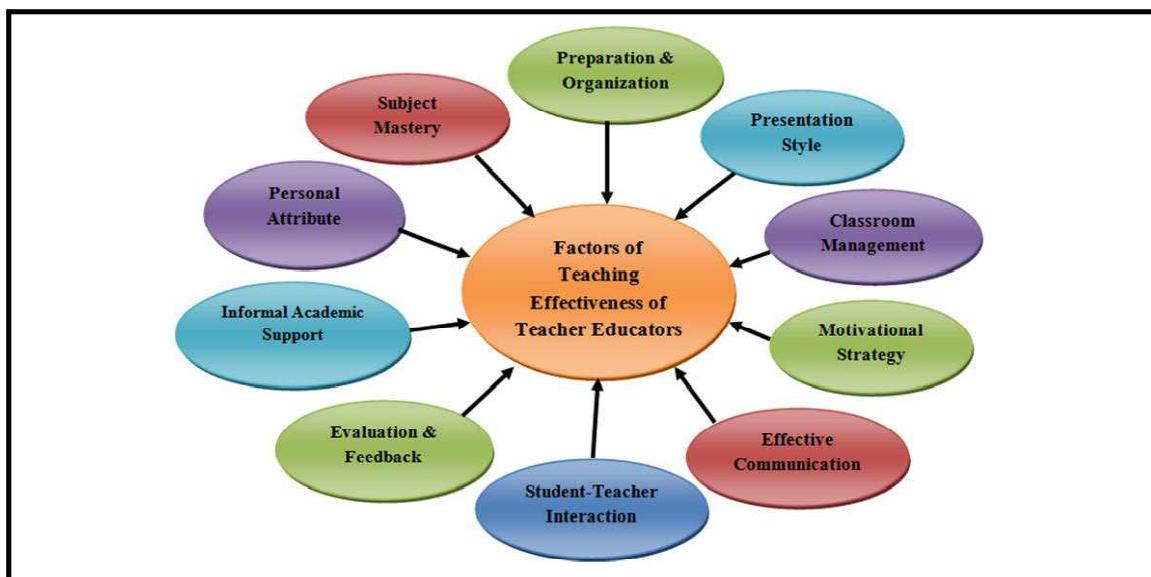
Teaching effectiveness is important because effective teaching helps student learning. It has become even more important as the emphasis on quality in higher education has increased. Effective teaching does not occur by chance. Effective teachers have become good at what they do because they evaluate their practice. James (n/d) suggests that “educational evaluation is a professional responsibility for academic staff, arising from a commitment to understanding the effects of teaching on students and to enhance student learning.” There are numerous ways of evaluating teaching or monitoring its effectiveness.

Beck (2005) identifies "twelve potential sources of evidence of teaching effectiveness." These include:

- Student ratings (such as Student Evaluations of Teaching);
- Peer reviews;
- Self-reviews;
- Videos of practice;
- Interviews with student;
- Alumni, employer and administrator ratings;
- Teaching awards and scholarship;
- Learning outcome measures; and
- Maintenance of teaching portfolios.

The sources identified above provide a diverse range of measures of teaching effectiveness. Institutions, departments and schools encourage a broad range of sources to evidence good teaching practice. The source that is used depends on why teaching effectiveness is being measured. For example, if the intention is promotion then a review

may be performed by a supervisor using a specific set of criteria which aids in making a summative decision on the academic's effectiveness. If the objective is to improve teaching practice and to modify the teaching plan or structure then a different set of criteria is applied. For example, a number of student evaluations may be used to determine which aspects of teaching are effective.



Evaluations to improve teaching practice and design are referred to as formative evaluation, while evaluations used in making decisions (for example, for purposes of promotion) are referred to as summative evaluations of teaching effectiveness.

A number of evaluation and assessment tools are available to measure teaching effectiveness. At Flinders University these are outlined in the Policy on Course and Topic Evaluation, Monitoring and Review. The evaluation tools in use at Flinders include the Student Evaluation of Teaching, peer evaluations and supervisor assessment.

GUIDANCE IN SCHOOLS

This article throws light upon the top three types of guidance in school which helps individual to tackle the problems better taking clear cut decisions in case of educational career, adjustment to peers, teachers, parents, decision about future occupation, etc. The top three types of guidance in school are: 1. Educational Guidance 2. Vocational Guidance 3. Personal Guidance.

1. Educational Guidance

Educational guidance has secured an important place in the field of guidance meant for the students. It is not a recent attempt by the modern guidance workers. It is concerned with every aspect of education especially meant for the students. It is an intellectual attempt which is concerned mainly with different problem relating to choose suitable courses for students, to complete it smoothly, to prepare students for future vocations etc.

It is designed to help students to achieve success in different, phases of education starting from primary education to its end including vocational preparation. Therefore much more care should be seriously and sincerely viewed by the guidance experts. Before highlighting on different relating facts regarding educational guidance it is the first and foremost task to define educational guidance at a glance. Several authors have given their own definitions regarding educational guidance which are stated here for the further discussions.

Brewer (1932). "Educational guidance is concerned strictly with the pupils success in his educational career".

Dunsmoor and Miller (1949). "Educational guidance is primarily concerned with the student's success in his educational career. It relates to the student's adjustment to school and to the preparation and carrying out of suitable educational plans in keeping with his educational needs, abilities and career interests."

Myers (1954). "Educational guidance is a process concerned with bringing about, between an individual pupil with his distinctive characteristics on the one hand, and differing groups of opportunities and requirements on the other, a favourable setting for the individual's development or education."

Hamrin and Erikson (1939). "Guidance in the secondary school refers to that aspect of the education programme which is concerned especially with helping the pupil to become adjusted to his present situation and to plan his future in the line with his interests, abilities and social needs."

Jone (1951). "Educational guidance is concerned with assistance given to pupils in their choices and adjustments with relation to schools, curriculums, courses and school life."

Ruth Strang. "Educational guidance is intended to aid the individual in choosing an appropriate programme and in making progress in it."

The above cited definitions of different experts reveal that educational guidance is a continuous and conscious effort made by guidance workers to assist students those who are facing problems pertaining to choosing a suitable institution for study, selecting courses and subjects of their own interests and objectives, improving study habits, gaining scholarships, achieving satisfactory progress in studies, carrying examinations properly, continuing higher education successfully including college and university education and managing school and college life with due adjustments in day to day affairs.

However, educational guidance is mainly concerned with such problems of education as are faced by pupils engaged in the study of different courses and subjects for their vocational preparation. The main interest of educational guidance is to see the students on successful floors of life making them better in the educational career and life. It is a fact that several problems in the school or educational institutions are faced by the young students which check the development of the pupils so far educational prosperity is concerned.

Therefore the prosperity in educational career is arrested as the faced problems are not tackled properly by the students. For the solution of the educational problems much more careful attention should be paid by teachers or guidance workers to show the real path for the progress of the students.

Now a number of questions arise in the mind of guidance workers or teachers about the problems relating educational guidance. To be more clear about the questions certain hints are given.

In this connection, Brewer, (1932) the eminent guidance expert has given the following problematic questions pertaining to classroom morale and situation:

- a. When the subject of the year's work has begun, did the pupils see how it was connected to their past work and preceding studies?
- b. Do the pupils understand the aims and values of the study? Do they understand how what their learning will be useful to them, now and in the future, in educational, civic and other activities?
- c. Do the pupils find the work interesting do they attack it with happiness and success?
- d. Do they all find the work right as to difficulty? If not, are the lesson tasks appropriately differentiated, and are slower pupils given the extra help they need?
- e. Are all the pupils showing reasonably satisfactory results?
- f. If not, are the causes being discovered and are the necessary corrective adjustments being applied?
- g. Do all the pupils know how to attack each lesson? Do they know how to use appropriate methods of study?
- h. Do the pupils understand the outside assignment, and are these recorded so that there is no misunderstanding about them?
- i. Do pupils understand how to make up for absence or other irregularities in the work?
- j. Do they know the best methods to use in answering questions in the recitation?
- k. Do they know how to prepare for and take the examinations?
- l. Are they shown, at the end of the course what additional studies and courses are ahead of them, and how this course is related to these?

These above questions give clue about the broad field of educational guidance. Therefore creative, suggestive and satisfactory educational guidance would have been offered to the needy students if these problems are tackled properly with due emphasis by the teachers or guidance workers.

Objectives of Educational Guidance

It is obvious that the objectives of educational guidance can be easily determined the broad sketch of objectives of education. Keeping same thing in the mind the physical, mental, social, emotional and spiritual development of the individual can be considered as the most important objectives of educational guidance. In this context the proper and due care must be taken for the all round development of the students.

In the state like U.S.A. the commission Secondary Education of National Education Association (1918) gave recommendations on the following seven areas as the goal of education which may state in favour of objectives of educational guidance:

- a. Health.
- b. Command of fundamental processes.
- c. Worthy of home membership.
- d. Vocations.
- e. Civic education.
- f. Worthy use of leisure.
- g. Ethical character.

Later on four general objectives in regard to education were recommended by educational policies commission of the National Educational Association, U.S.A. (1938).

The following objectives may be kept in the mind to determine the objectives of educational guidance:

- a. The objective of self realization.
- b. The objective of human relationship.
- c. The objective of economic efficiency.
- d. The objective of civic responsibility.

The above four objectives especially give much emphasis on the personal and social development of the students as a result of which they may be able to achieve healthy social relationship and economic prosperity in the society. So that one is recognized as a good citizen of the nation. Besides above related discussions it is note worthy to cite aims and purposes of educational guidance given by Crow and Crow (1962) so far high school level is concerned.

These are as follows:

- a. Select the curriculum that best fits his abilities, interests, and future needs.
- b. Develop work and study habits that enable him to achieve satisfactory success in his studies.
- c. Gain some experience in learning areas outside the particular field of his special interests and talents.
- d. Understand the purpose and the function of the school in relation to his needs.
- e. Discover all that he has to offer and plan a programme of studies accordingly.
- f. Learn about the purpose and function of the college or school he may wish to attend later.
- g. Elect, try out courses or exploratory courses in order to gain insight in to learning areas that still lie ahead.
- h. Participate in out-of-class activities in which he can develop potential leadership qualities.
- i. Appraise his fitness for continued study in a college or other school or in a particular vocation.
- j. Develop an attitude which will stimulate him to continue his education in a school selected for its worth to him in relation to his talents and training.
- k. Adjust to the curriculum and the life of the school.

After a careful and intensive study of the above cited objectives, certain important objectives of guidance are cited here.

- (i) To help the student to choose suitable and appropriate courses for study according to his interest, abilities, needs and goals.
- (ii) To enable the student to know about various types of courses available for higher and technical education.
- (iii) To make cautious the student to know about new aims of education as the aims of education is changed from time to time.
- (iv) To enable the student to adjust and cooperate with the curricular activities of the school.

- (v) To inspire the student to participate in different co-curricular activities organised by the school.
- (vi) To enable the student to improve proper study habits for the purpose of attaining better learning.
- (vii) To assist student for better utilization of leisure time available in school and home situations.
- (viii) To help student to overcome his day to day problems pertaining to his study.
- (ix) To enable the student personally evaluate his strengths, weaknesses, needs, abilities and interests.
- (x) To help student to know and cooperate teachers, classmates, schoolmates and other official members of the school.
- (xi) To assist student about admission procedure and prospectus for different available courses and institutions of new higher study.
- (xii) To enable student go through courses both intensively and extensively with proper acceptance and motivation.
- (xiii) To provide student adequate information's about sources of scholarships and stipends for the purpose of financial help meant for them.
- (xiv) To help student to adjust with hostel situations and friend in case of boarder of the hostel.
- (xv) To assist adolescent boys and girls to adjust and cooperate with the members of opposite sex with positive attitudes.
- (xvi) To enable the student to attend examinations properly with due preparation for same.
- (xvii) To acquaint the student to proceed in his self direction, self expression and self development in accordance with his best possible capacities.

Need of Educational Guidance

With best possible hopes educational guidance is highly needed at present in all most all the schools owing to the following reasons:

1. In almost all the cases it is found that an individual student differs from his classmates so far ability to learn, rate of learning, and motivation and interest for learning is concerned. Because the concept like individual difference is firmly believed as it is based on thousands of experiments. But in case of classroom teaching learning situations equal instruction is provided to the students which do not take individual difference into account.
Generally the gifted students and slow learners are neglected in normal class teaching as teacher does not get time to take initiative for them. So the difference between gifted students and slow learners creates a great gap between them which is not expected. Keeping this thing in the mind guidance worker and teachers welcome educational guidance just to overcome the problem of individual difference in the classroom situation.
2. In most of the cases it is seen that students are choosing their own subjects and courses not knowing their abilities and interests. Also some of the students like to obey their parent's word in relation to choose the subject on courses. Due to both the reasons students do not be able to achieve success in the examinations which brings frustration in their minds.

So in this context to arrest academic frustration, reduce the number of failures and check stagnation and dropouts from the educational situations, the educational guidance is highly preferred by the guidance workers and teachers in the school.

3. Sometimes students select certain subject or courses aim-lessly not judging their abilities, interests and own liking field as a result of which they do not get suitable employment for future. They are put in to such employment difficulties as they select subject or courses not thinking its future perspectives and vocational implication. To eradicate this problem, educational guidance came forward to help students in the school life.
4. In some of the cases it is found that students having abilities and interests for a particular subjects or courses do not perform better in the examinations. Of course there are several factors which are responsible for it. In educational situations factors like poor study habit of the students, ineffective instructions, lack of teaching aids etc. should be taken into account at the time of teaching and learning. Therefore the educational guidance is needed by teachers and guidance workers to improve standard of teaching as well as standard of students.
5. Generally students from different family background and socio-economic status come to the school for the purpose of study. Sometimes some of the students do not adjust with their classmates, other peer groups, teachers and other concerning persons of the school as they face a new situation. Besides this problem of truancy and deliquesce also comes to the picture. All problems of concerning adjustment of students can be easily tackled by educational guidance with best possible manner.
6. Always it is not possible in the part of students to know about available new subjects or courses, admission procedures for it, financial help for same and occupational scope of subject etc. So the most important job of guidance worker and teachers to assist students in this regard.
7. It is a fact that time and tide wait for none. So much more emphasis should be given on time and its utilization. Generally students get leisure hour in the school as well as the home. Therefore teachers and guidance workers suggest students how to use leisure time in best possible manner in order to get maximum benefits so far their academic career and vocational career are concerned.

Functions of Educational Guidance

1. **Elementary stage.** Educational guidance carries out the following specific functions for the pupils at the elementary stage of schooling:
 - (i) Educational guidance helps pupils to make a better beginning in their educational career by which they can be able to have a better start and stay in a preferable type of education.
 - (ii) Educational guidance assists pupils to plan intelligently for suitable subject or courses which should have vocational implication for creating positive relationship between academic achievement and employment.
 - (iii) Educational guidance enables pupils to achieve best according to their abilities and interests overcoming the learning difficulties which come to arrest the development of students.
 - (iv) Educational guidance prepares pupils for successful entry into the secondary stage of schooling as a result of which the pupils achieve their goal of education through better start for the same.

2. **Secondary stage.** Educational guidance performs the following specific functions at the secondary stage of schooling:
 - (i) Educational guidance makes acquainted the students with the new purposes of education in relation to nature, type, role and scope of education considering suitable employment, good leadership and better individual as well as social life.
 - (ii) Educational guidance helps students to choose suitable courses and activities according to their abilities and interests keeping in view to their admission for higher secondary schools, junior colleges, vocational institutions in order to receive employment and better social and cultural life.
 - (iii) Educational guidance enables students to make an appraisal of their abilities, interests, aptitudes and skills, and to tag them to curricular courses and co-curricular activities.
 - (iv) Educational guidance helps students to overcome difficulties in learning some subjects and to make them fit future progress in educational career.
 - (v) Educational guidance assists students to develop positive attitudes towards study and to motivate them for the same aiming at better learning.
3. **Higher Secondary Stage.** Educational guidance has the following specific functions at the higher secondary stage:
 - (i) Educational guidance helps students in the selection of suitable courses for further study and vocations for future in accordance with their abilities and interests.
 - (ii) Educational guidance enables students to understand the clear-cut objectives of higher education as a result of which they will be able to decide about their further study.

Basic Principles of Educational Guidance

With due emphasis Crow and Crow uttered that educational guidance can be offered successfully for all students if the following principles are observed:

1. Upon entrance into the high school or shortly thereafter the pupils should have administered them standardized tests that will be prognostic of success in one or another curriculum.
2. Guidance towards the selection of a curriculum should be decided in the light of test results, degree of achievement on the previous school level, and pupil and parent interest.
3. A pupil's achievement during each term or year in each subject area of study should be followed closely by his counsellor and help should be offered whenever it is needed.
4. If a pupil appears to be having difficulty in his chosen programme of studies, this should be recognized early by the counsellor, the pupil and his parents. Steps should be taken promptly to make whatever programme changes are necessary in order that the pupil does not lose too much time or become so discouraged that he is tempted to dropout high school.
5. A pupil should not be required to repeat more than once with the same teacher any course in which he fails. Sometimes personality differences between the teacher and the pupil interfere with learning progress.

After knowing the principles of educational guidance programmes given by Crow and Crow, certain questions come to the minds of guidance workers and teachers regarding principles which are commonly accepted by different experts of the concerning field.

The followings are the some of the principles on which educational guidance should be based and initiated:

- a. Educational guidance is meant for all not for few only.
- b. Educational guidance should be pupil oriented and centred.
- c. Educational guidance must be objective oriented and should be given without any personal biasness.
- d. Individual difference must be taken in to consideration at the time of offering educational guidance to the students.
- e. Educational guidance should be based on a principle that each and every student is capable of gaining and achieving the best of his own ability, interest, aptitude and skill.
- f. Educational guidance should be evaluated considering the resources and facilities available to the students.
- g. Educational guidance must be based on time to time appraisal of student's performance and achievements.
- h. Educational guidance is not job of compulsion or imposition so far student's will is concerned.

Vocational Guidance

With due importance vocational guidance came to the picture first in the guidance movement. As the Director of Vocational Bureau of Boston Frank Parsons did a lot of useful work in the concerning field. Perhaps later on vocational guidance and its different related facts are explored by various experts of the same field.

Really vocational guidance is a some sort of help given to the students in choosing and preparing for a suitable and satisfactory vocation in the world of works. It also assists students to solve the problems pertaining to vocational choices, plans, developments, adjustments and maturity with utmost satisfaction. Vocational guidance is more careful for the development of vocational career of students.

After completion of diplomas or degrees one searches for job to satisfy bread and butter aim of education as well as life. Therefore vocational guidance is highly needed by the students to make himself dependent. Vocational guidance helps students for better career in the occupational world handling different programmes such as career talk, career corner, career conference, industrial visit, simulated interview and vocational counselling etc.

However vocational guidance has become an appreciable attempt by the guidance workers and teachers to take proper care for vocational career of students. To be more clear about the concept of vocational guidance certain definitions are cited here.

National Vocational Guidance Association (U.S.A., 1937). "Vocational guidance is the process of assisting the individual to choose an occupation, prepare for it, enter upon and progress in it. It is concerned primarily with helping individuals make decisions and choices involved in planning a future and building a career decisions and choices necessary in effecting satisfactory vocational adjustment."

International Labour Organisation (I.L.O.). “Vocational guidance is the assistance rendered by an individual to another in the latter’s solving of problems related to his progress and vocational selection keeping in mind the individual’s peculiarities or special abilities and their relations with his occupational opportunity”.

Myers (1941). “Vocational guidance is fundamentally an effort to conserve the price-less native capacities of youth and costly training provided for youth in the schools..... It assists the individual to invest and use his capacities in those vocations which will bring greatest satisfaction and success to himself and greatest benefit to society.”

Super (1957). “Vocational guidance is the process of helping a person to develop and accept an integrated and adequate picture of himself and of his role in the world of work, to test this concept against reality and to convert it in to a reality with satisfaction to himself and benefit to society.”

Analysing the gist’s of above cited definitions it can be said that vocational guidance plays a major role in the life of students. Also it is a process of helping the student or individual to take right decision in case of occupational choice, prepares himself for a suitable job and achieve vocational development and maturity with a satisfactory progress and adjustments after completion of his study or related training.

Mere joining by individual in a particular employment should not be only mission of life. But gaining satisfaction and achieving expectable progress in an occupation should be taken into consideration. Even if adjustment in vocational life cannot be ignored. Therefore vocational guidance takes potent role in solving all those problems which generally rush in vocational life of the individual or student.

On the same line the concept of vocational guidance can be understood easily that it deals with various problematic questions of individual’s life such as what he is, what he wants to be, what does he best fitted for, to what extent he achieves satisfactory progress “in occupation, etc. In other words, it can be envisaged ornamentally as the process of fitting-round pegs in round holes and square pegs in square holes.”

However, vocational guidance is some sort of assistance meant for student in order to take right decision in placement of job, adjust well in occupational centre, acquire satisfactory progress and achieve maximum satisfaction in job.

Aims & Objectives of Vocational Guidance

Jone has pointed out certain important aims of vocational guidance which are as follows:

- a. To assist the student to acquire such knowledge of the characteristics and functions, the duties and rewards of the group of occupation within which his choice will probably lie as he may need for intelligent choice.
- b. To enable him to find what general and specific abilities, skills, etc., are required for the group of occupations under considerations and what are the qualifications of age, preparation, sex, etc., for entering them.
- c. To give opportunity for experiences in school (try out courses) and out of school (after school and vacation jobs) that will give such information about conditions of work as will assist the individual to discover his own abilities and help in the development of wider interests.
- d. To help the individual develop the point of view that all honest labour is worthy and that the most important bases for choice of an occupation are:
 - (i) The peculiar service that the individual can render to society,

- (ii) Personal satisfaction in the occupation, and
- (iii) Aptitude for the work required.
- e. To assist the individual to acquire a technique of analysis of occupational information and to develop the habit of analyzing such information before making a final choice.
- f. To assist him secure such information about himself, his abilities, general and specific, his interests, and his powers as he may need for choice.
- g. To assist economically handicapped children who are above the compulsory attendance age to secure, through public or private funds, scholarships or other financial assistance so that they may have opportunities for further education in accordance with their vocational plans.
- h. To assist the student to secure a knowledge of the facilities offered by various educational institutions for vocational training and the requirements for admission to them, the length of training offered, and the cost of attendance.
- i. To help the workers to adjust himself to the occupation in which he is engaged ; to assist him to understand, his relationships to workers in his own and related occupations and to society as a whole.
- j. To enable the student to secure reliable information about the danger of alluring short cuts to fortune through short training courses and setting propositions ; and of such unscientific methods as phrenology, physiognomy, astronomy, astrology, numerology or graphology and to compare these methods with that of securing really trustworthy information and frank discussion.

According to Crow and Crow the specific aims of vocational guidance are the following:

- a. Assisting a pupil to acquire knowledge of the functions, duties, responsibilities, and rewards of occupations that lie within the range of his choice.
- b. Assisting a pupil to discover his own abilities and skills and to fit them in to the general requirements of the occupations under consideration.
- c. Assisting the pupil to evaluate his own capacities and interests with regard to their greatest worth to him and to society.
- d. Helping the individual to develop an attitude towards work that will dignify whatever type of occupation he may wish to enter. The important bases for choice should be personally achieved satisfactions and the service that can be rendered.
- e. Giving exploratory opportunities in different areas of school learning and vocational exploration that will enable the learner to get the feel of several types of activities.
- f. Assisting the individual to think critically about various types of occupations and to learn a technique for analysing information about vocations.
- g. Assisting the mentally handicapped, the physically handicapped, or the economically handicapped to make the adjustments that will be best for them in their struggle for a fuller life and for personal and social welfare.
- h. Instilling in the pupil a confidence in the teachers and other guidance personnel that will encourage him when he confers with them on personal and vocational problems.
- i. Assisting the pupil to secure the necessary information about the facilities offered by various educational institutions engaging in vocational training.
- j. Providing information for the learner about admission requirements, the length of training, and the cost of attending and institution of higher learning to which

he may wish to go after graduation from high school in order to continue his vocational preparation.

- k. Giving assistance during school years so that the individual will be able to adjust on the job to work conditions and to other workers.
- l. Assisting each pupil to appreciate his rightful place in a group of workers and to become a functional member of the team.
- m. Alerting the pupil to the long range training needed to become proficient in most lines of endeavor.
- n. Cautioning each learner concerning fads and pseudo scientific short cuts to vocational competency.
- o. Helping the learner to realize that the success is purchased at the price of effort, and that satisfaction on the job derives from doing his work conscientiously and competently.

The following are the various objectives of vocational guidance:

- a. To enable students to gain knowledge of the features, functions, scope, nature and duty requirement of employment in which they want to be engaged.
- b. To assist students to receive up-to-date and useful information about abilities and skill as they possess in the context of related qualifications and competencies required to accept a preferable job.
- c. To help students to know their potentials, abilities and interests related to identified vocation which they want to secure.
- d. To enable students to develop capacities for analysing available vocational information's to have a better choice in relation to available information's in this regard.
- e. To help students to avail information regarding scope and prospectus of various educational training, apprenticeship schemes and different vocational training.
- f. To assist students to choose the right type of employment according to their liking and satisfaction.
- g. To enable students to develop entrepreneurship abilities within them for keeping feet in the kingdom of self employment.
- h. To assist students to develop abilities and skills for achieving successful progress and satisfactory performance in the occupation.
- i. To enable students to gain maximum satisfaction out of his world of works.

Need of Vocational Guidance

In this complex and competitive world vocational guidance is highly needed for youngsters owing to various following reasons:

1. In course of time the present world has received enormous changes such as rapid industrialisation, excess population explosion and scientific advancement etc. Among that rapid industrialization implies a higher man power requirement as it gives scope, hope and inspiration for both public and private sector to establish varieties of industry to satisfy the present demands of the people.

So that to challenge this man power requirements everybody has to be cautious from very beginning as it prefers to right type of personnel. Keeping this in the mind vocational guidance should proceed in the school to assist students to choose

a right type of job for them in future. Then the question like “what shall I do in life” can be easily answered.

Otherwise an improper selection for a student so far job placement is concerned invites unhappiness, harassment, failure and frustration in life which is not desired. Because an occupation is not mere appointment for earning money to maintain life. But every student should remember this fact that an occupation is something more than that and it can be understood as a way of life.

2. Just mere placement in a particular job is not the mission of life. Therefore before entering into a job, a student should realize the question such as “What am I best suited for”. At present due to family and financial problems; many youngsters join in a job not possessing that much of required abilities or capacities. Naturally this situation leads to failure, dissatisfactions and discontent in the occupational life. Due to this reason vocational guidance is highly needed.
3. In some cases it is found that young men and women are attracted to a particular type of employment due to handsome salary and other ample facilities offered by service authorities. They do not take abilities or capacities into account in relation to their fitness for the job. So that they do not achieve success and satisfaction in the service life which compels them to change from one occupation to another. So that this situation invites failure, frustration and discontent in the occupational life as well as in social life of the students.
4. The choice and preference for the job should be made by students according to their abilities, limitations; interests and aptitudes judging the nature of the occupation; its demand, future prospective and requirement of occupation. One should not feel happy as just he got an appointment somewhere in the private or public sector. Because self development, successful adjustment and occupational maturity are different missions of service life which should not be ignored in no ways so far occupational life of the students are concerned.
5. Vocational guidance is needed to set a right person for the right job taking individual difference into account so far the abilities, interests, aptitudes, training experiences and vocational capacities are concerned.
6. Vocational guidance plays a vital role providing useful information's and data regarding different occupations to the students. Because the variety of occupations are more than three thousands in number in the world of work which are to be noted for the students. At the same time students would be able to know the nature, characteristics, demands of the different job as a result of which they can go on choosing right job for them.
7. Vocational guidance is also helpful for the students that it places round pegs in round holes and square pegs in square holes. In other words it assists students to choose right job according to their abilities judging what can be done by one or what cannot be done by one.
8. Vocational guidance should be offered to high school students, higher secondary school students and also college and university students as they leave education in a particular stage of education. After completion of education they need a job to survive. Therefore they should be accustomed with vocational guidance before leaving education.

9. Vocational guidance also assists students how to adjust in occupational life, how to gain satisfactory progress in job and so on. For this reason vocational guidance is highly needed.

Functions of Vocational Guidance

1. **Elementary Stage.** Vocational guidance has the following specific functions at the elementary stage.
 - (i) Vocational guidance assists students to develop their favourable attitude towards better habits, skills in relation to their own liking field of work.
 - (ii) Vocational guidance prepares students for better courses for a job those who leave educational campus after primary education.
 - (iii) Vocational guidance prepares students to participate in curricular and co-curricular activities which are carried on in the school to develop the pupil's skills and attitudes for successful work in future life.
 - (iv) Vocational guidance assists pupils to know their own abilities and limitations in regards to their preferable job.
 - (v) Vocational guidance assists to students providing a lot of occupational information's for the future of the students.
 - (vi) Vocational guidance helps students preparing them for secondary school courses so far vocational aspect is concerned.
2. **Secondary Stage.** The followings are the different functions of vocational guidance at the secondary school level:
 - (i) Vocational guidance assists students to appraise their own abilities, interests, aptitudes, skills and other qualities as a result of which they can be able to know themselves and choose best so far their occupational choice is concerned.
 - (ii) Vocational guidance enables students to know vocational implications of various subjects and courses by which they can be able to choose suitable subject and courses for the purpose of study.
 - (iii) Vocational guidance provides all the information's and required data about employment situation, job trends, nature and conditions of job and its different benefits by which students make up their mind for a suitable job.
 - (iv) Vocational guidance helps-students to prepare them for a bright future so far their choice of job and further vocational training course are concerned.
3. **Higher Secondary Stage.** Vocational guidance has the following specific functions at the higher secondary stage:
 - (i) Vocational guidance assists students to know about different opportunities given by private or public sectors so far job is concerned.
 - (ii) Vocational guidance enables students to understand the vocational implications of their subjects or courses which are to be studied by them.
 - (iii) Vocational guidance assists students to acquaint with different available scholarships, stipends, grants and fellowships.
 - (iv) Vocational guidance gives comprehensive outlook to the students about different types of vocations and careers.
 - (v) Vocational guidance is helpful for students from contact point of view so far different related agencies, institutions and programmes are concerned.

Principles of Vocational Guidance

Vocational guidance strategies must be based the following principles:

- a. Occupation should be felt by everybody as a source of income.
- b. Job is to be looked at as important source of satisfying needs, abilities, interests and aptitudes.
- c. The pupil needs to know the total perspective of the job for which he has taken interest to join.
- d. Different social and personal factors should be considered at time of selection of a particular occupation.
- e. The proper selection of a job is a time consuming affair and it does not believe in single or fixed decisions.
- f. The principle of individual difference should be taken into account at the time of offering vocational guidance programme.
- g. Vocational guidance service must satisfy the occupational needs of the students.

Basic Assumptions of Vocational Guidance

No two individuals are alike in the world so far abilities, interest and personalities are concerned. It cannot be said that exceptional children do not possess occupational potentialities.

3. Personal Guidance

In-fact personal guidance occupies an important place in the kingdom of guidance as well as in the life of individual. Each and every moment of life individual faces thousands of problems besides educational and vocational problems of life. However, personal guidance is concerned with the problems of health, emotional adjustment, social adjustment including recreation and leisure's time activities etc.

Especially personal guidance is a type of assistance offered to an individual to overcome his emotional problems and to help him to control his emotions which do occur in the individual's life. A sound and satisfactory personality of an individual can be developed if he becomes able to check and control different powerful emotions such as anger, fear, anxiety, jealousy, nervousness, joy and tensions in different context of life.

In individual's life anxiety, failure, frustration, dissatisfaction and disappointment come to picture if he does not check different emotions. Therefore from beginning to end of life personal guidance does assist individual a lot in the different required situations. Personal guidance may be defined as the assistance given to the individual to solve his emotional, social, ethical and moral as well as health problems. Hence personal guidance deals with all those problems of life which are not covered under educational and vocational guidance.

Personal guidance may be expressed as the help to the individuals to divert his emotional powers and feelings into a positive direction in relation to his progress of life. It is also meant to solve the emotional problems which generally arise in the family and different situations of different organisations or institutions. It is found that problems of pre childhood age which come due to family also lead to dissatisfactory performance in the school career. On the same line low achievement of school does affect directly vocational adjustment and vocational career of one's life.

In this way the entire life span of individual does not achieve satisfaction as well as happiness. The same state of mind and situation does hamper seriously the progress of the individual. Of course several factors are responsible for individual's maladjustment in family and some other situations.

In the life of individual factors like jealousy among siblings, domination of elders, lack of love and affection, maltreatment of children, lack of a sense of belongingness, parental authoritarianism, conflict among family members, socio-economic status, parental dissatisfaction, (educational level of the family members, attitude of parents towards education play dominant role in context of maladjustment of the children.

Also problems like lack of friends, loneliness, failure, feelings of inadequacy, inferiority complex, maladjustment with girl friends, negative attitude towards girl friend and other typical problems do hamper the students' progress in academic life well as social life.

Purpose of Personal Guidance

An organised personal guidance programme in institutions would serve the following purposes:

- a. Personal guidance assists students to know emotional problems which occur in day to day life of the students.
- b. Personal guidance helps students to resolve their emotional problems of life.
- c. Personal guidance enables students to explore different adjustment's mechanisms.
- d. Personal guidance helps students to check the emotions which are not desired for the development of the individual student.
- e. Personal guidance assists students.
- f. Personal guidance enables students to carry out social and civic activities properly.
- g. Personal guidance assists students to develop awareness about personal health and physical activities.
- h. Personal guidance enables students to well use of leisure time.
- i. Personal guidance helps students to carry out character building activities.
- j. Personal guidance assists students to understand family situations and adjust accordingly.
- k. Personal guidance helps students to understand different social setup and situations and deal effectively.
- l. Personal guidance enables students to derive maximum satisfaction and pleasure out of different social activities and various institutions.
- m. Personal guidance helps students to understand various emotional characteristics of adolescents such as hostility, fear, anxiety, jealousy, etc. and divert it in a positive and right ways.
- n. Personal guidance enables students to study different types of emotional problems such as frustration, anxiety, nervousness, stress, neurosis etc. and find out solutions for it for happy life.
- o. Personal guidance assists students to adjust and co-operate with the friends of opposite sex effectively in a positive direction.

Steps of Personal Guidance

The process of personal guidance develops through certain following steps:

- (1) Personal guidance needs various possible useful related data or information's regarding students touching their physical, mental, social, emotional, interest, aptitude and attitudes, aspects of life which are to be collected carefully.
- (2) Adequate diagnosis of causes of problems should be made on the basis of collection of data regarding students.
- (3) The guidance personnel should think rightly about the remedial measures after the step like diagnosis of the causes of problems.
- (4) The guidance personnel renders guidance service to help the students studying their problems and its causes by means of techniques like required counselling, psychological analysis, imitation, advice with affection and situational feelings.
- (5) Lastly follow-up service should be made to evaluate the effectiveness of given guidance service. To fulfil this purpose, the guidance personnel may take help of the techniques like interview or question answer methods or the like.

Functions of Personal Guidance

1. **Pre Primary Stage.** Personal guidance has the following specific functions at the pre primary stage:
 - (i) It assists tiny children to get with others by sharing experiences as well as toys and models, being courteous, experience to control anger, experience to be a group leader, learning to be a follower and playing fairly with pleasure.
 - (ii) It enables children to express themselves by working with his hands, learning many rhythms, listening to others and dramatizing stories.
 - (iii) It assist children to assume responsibilities by putting away toys, caring for pets, passing out materials and taking care for own food habits and clothing.
2. **Elementary Stage.** Personal guidance has the following functions at this stage:
 - (i) It assists children to fulfil some of the basic needs such as good health, knowledge of fundamental skills, feeling of security and assurance, desire for friends and social acceptance, discipline, leisure time activities and vocational skills concerning general knowledge of the world of work.
 - (ii) It enables children to develop ability of self discipline at home and schools.
 - (iii) It helps children to make them learn right from wrong in different situations.
3. **Secondary Stage.** Personal guidance has the certain broad and important functions at this stage which includes both junior high school stage and high school stage:
 - (i) It enables students to adjust in their new school situations and environments.
 - (ii) It assists students to develop a feeling of belongingness and fraternity.
 - (iii) It helps students to inspire them for active leadership and group life.
 - (iv) It enables students to fulfil educational, vocational and personal needs and interests of life.
 - (v) It helps students to overcome the problems of adjustment in relation to adolescent age and other personal adjustment problems.
 - (vi) It enables students to know useful information's pertaining to sex life.
 - (vii) It assists students to develop leadership abilities and good citizenship.
 - (viii) It enables students to achieve social and moral development in life.

- (ix) It helps students to carry out certain recreational activities for themselves.
4. **College and University Stage.** Personal guidance has following functions at college and university stage:
- (i) Personal guidance helps students to enable them have a satisfactory personal and social adjustment in their new environment.
 - (ii) It assists students to develop in young adults a sense of social service, social responsibility, patriotism and tolerance in the context of ethical and moral development.
 - (iii) It enables students to develop good relationship with community and college to overcome financial problems and some other related problems.
 - (iv) It assists adult students to appreciate the importance of religious and moral values in life.
 - (v) It enables adult students to perform duty rightly in relation to educational, vocational, social, moral and personal life.

COUNSELLING: PROCESS, SKILLS, AND TECHNIQUES

School counseling takes place in public and private school settings in grades K-12. Counseling is designed to facilitate student achievement, improve student behavior and attendance, and help students develop socially. Mental health professionals with master's degrees or beyond, school counselors both provide counseling and serve an educational role in and around schools. Many schools have full-time counselors on staff in order to support students who are experiencing personal or academic challenges, help students choose careers and plan for college, and intervene when students face behavioral, physical, or mental health challenges.

In the early 20th century, as industrial centers grew throughout the United States, secondary schools began to increase their focus on courses that would help prepare students to enter the workforce. Some teachers doubled as vocational counselors in order to assist in these efforts.

1917. Specific legislation is drafted to provide funding for vocational guidance programs. Following this, the school guidance counseling profession grows.

1920s. New York becomes the first state to develop certification requirements for school counselors.

1930s. Urban elementary schools begin to offer school counseling services.

1950s. The humanistic psychology and person-centered psychology movements gaining traction lead the focus of school counseling to undergo a significant shift.

1970s. School counseling incorporates wider goals of helping students develop socially, personally, and academically.

1990s. A nationwide shift toward standards-based education and the adoption of legislation such as the No Child Left Behind Act make it necessary for school counselors to find increasingly innovative ways to serve students.

What do School Counselors Do?

School counselors, also known as guidance counselors, were first primarily responsible for facilitating career development. Today, the role of the school counselor is multifaceted and may vary greatly, depending on the requirements of both the state and each

individual school.

The duties of school counselors may include:

Providing instruction on psychological and social issues. School counselors might teach sex education classes, provide information to students about bullying, or offer seminars on study skills.

Vocational guidance. Many school counselors help students prepare for college or select careers.

Counseling. School counselors often help students mediate conflicts with their peers, teachers, or parents. Many school counselors also provide therapy and counseling services to students during school hours.

Early intervention. School counselors receive training about learning difficulties and psychological concerns that commonly manifest in children and adolescents. They may also provide referrals, recommendations, and education to parents about mental health concerns.

Special needs services. Counselors often help special needs students integrate into classrooms and may oversee programs that address requirements for students with special needs or learning difficulties.

Further, counselors often help students:

Maintain academic standards and set goals for academic success.

Develop skills to improve organization, study habits, and time management.

Work through personal problems that may affect academics or relationships.

Improve social skills.

Cope with school or community-related violence, accidents, and trauma.

Identify interests, strengths, and aptitudes through assessment.

School counselors offer individual counseling to help students resolve personal or interpersonal problems. They may also offer small group counseling to help students enhance listening and social skills, learn to empathize with others, and find social support through healthy peer relationships. For students who are otherwise unable to access mental health services, school counselors provide support at no cost. School counselors also provide support to school staff by assisting with classroom management techniques and the development of programs to improve mental health or school safety. When necessary, counselors may also intervene in a disrupted learning environment.

Required Training for School Counselors

School counselors must complete a master's degree, at minimum, in school counseling, psychology, or social work and obtain the relevant state certification, endorsement, or licensure to gain employment. This may involve taking a comprehensive exam and logging hours in a supervised counseling setting. In many cases, counselors will need to complete an internship or practicum, and some states also require previous teaching experience.

School counselors are required to renew their licensure every three to five years. This timeline depends on the requirements of the state in which they are employed. In order to renew licensure, continuing education classes or professional development courses are generally necessary.

Many states require public schools to provide school counseling services, and these programs are funded at the state or local level. The American School Counselor Association (ASCA) recommends a student-to-school-counselor ratio of 250:1, although the average ratio is currently 471:1.

Do School Counselors Diagnose Mental Health Concerns?

Serious diagnosable mental health conditions affect 21% of U.S. children between the ages of 9 and 17, but only 20% of these children obtain a diagnosis and receive treatment in any given year. While school counselors may suspect the presence of learning difficulties or other conditions such as ADHD, they are not licensed to diagnose or prescribe medication. Some schools do have school psychiatrists, however, and these professionals are able to prescribe medication to students, though parental permission is typically necessary.

When a school counselor suspects the presence of a learning, behavioral, or mental health concern, they will typically provide a referral to a specialist in the community. Learning difficulties can be diagnosed by school or educational psychologists or neuropsychologists, and ADHD is generally diagnosed by psychiatrists, physicians, or clinical psychologists in private practice.



EMERGING AREAS

ISSUES OF GENDER, POVERTY, DISABILITY AND MIGRATION

Cultural bias and Discrimination

Despite recognition in the Millennium Declaration of the importance of human rights, equality, and non-discrimination for development, the Millennium Development Goals (MDGs) largely bypassed these key principles. The fundamental human rights guarantees of equality and non-discrimination are legally binding obligations and do not need instrumental justifications. That said there is a growing body of evidence that human rights-based approaches, and these key guarantees in particular, can lead to more sustainable and inclusive development results.

Discrimination can both cause poverty and be a hurdle in alleviating poverty. Even in countries where there have been significant gains toward achieving the MDGs, inequalities have grown. The MDGs have supported aggregate progress – often without acknowledging the importance of investing in the most marginalized and excluded, or giving due credit to governments and institutions which do ensure that development benefits these populations. Recognition of this shortcoming in the MDGs has brought an increasing awareness of the importance of working to reverse growing economic inequalities through the post-2015 framework, and a key element of this must be actively working to dismantle discrimination.

The post-2015 framework should be grounded in a fundamental guarantee of equality and non-discrimination. Under international law, this requires states to identify and eliminate discrimination and ensure equality. This may require legislative or administrative reform to repeal discriminatory provisions or address discriminatory practices by the government or private actors, a change in resource allocation, or educational measures. The post-2015 framework should embody the responsibility of states, when acting

together or alone, to take proactive measures to identify and address entrenched discrimination, both direct and indirect. It should embody the responsibility of states, international institutions, and corporations to avoid and remedy discrimination for which they are directly or indirectly responsible. The framework should go some way toward achieving this by including goals, targets, and indicators directed at reducing discrimination and ensuring that the social and economic needs of the most marginalized communities are being addressed fairly, and at reducing wealth inequalities more broadly.

Recommendations

Targets and indicators in the post-2015 framework should be designed to reduce existing inequalities, address entrenched discrimination, and realize the social and economic rights of the most in need while also remaining feasible, affordable, and implementable.

The post-2015 framework should emphasize the importance of development reaching the most marginalized populations, including indigenous peoples. It can go some way toward achieving this by including:

A specific target of addressing the social and economic needs of the most marginalized or discriminated against groups in each country. The framework should establish the methodology for identifying marginalized or disadvantaged groups, but the groups identified would vary country to country.

Indicators should look to identify structural discrimination, including consideration of discriminatory laws and discrimination by private actors. Considerations should include whether governments have non-discrimination laws that bind public and private entities (with a definition of discrimination consistent with international human rights law), require public and private institutions to develop non-discrimination action plans, and fully implement such laws and policies.

Indicators should measure realization of urgent social and economic needs of the most marginalized populations.

Indicators should measure respect of indigenous peoples' rights, including land and cultural rights, and recognize free, prior, and informed consent.

Indicators which measure the achievement of each target for the most marginalized or discriminated against groups in each country.

In order to achieve this, disaggregated data will be essential. It may not be feasible to disaggregate data by all potential grounds of discrimination. At a minimum, states and international institutions should collect disaggregate by gender, demographic group (i.e. ethnic background, language, religion), locale (rural/urban/slum household, state/territory), age, and disability. States and international institutions should also analyze all existing disaggregated data.

The post-2015 framework should emphasize the importance of development reaching the poorest. It can go some way toward achieving this by including:

A specific target of addressing the social and economic needs of the poorest two wealth quintiles in each country.

Indicators which measure the achievement of each target by wealth quintiles. This will require the collection of data along wealth quintiles.

A specific target on reducing income inequalities within countries, with an emphasis on reducing the gap between the richest and poorest quintiles (20:20 gap) or between the top quintile and the bottom two quintiles (10:40 gap).

Populations at Risk of Discrimination

Human rights advocates have long observed the close link between discrimination and poverty. While disaggregated data is not available with respect to each marginalized group, recently published data suggests that more than two thirds of extremely poor people in low income countries and lower-middle income countries live in households where the head of household is from an ethnic minority group. It also tells us that more than three quarters of extremely poor people live in rural areas. Further, more than 80 percent of people with disabilities live in developing countries, illustrating both the confluence of poverty and disability and the importance of proactively addressing the needs of people with disabilities in development strategies.

Human rights law prohibits discrimination on the basis of a wide range of prohibited grounds. The Universal Declaration of Human Rights (UDHR), the International Covenant on Civil and Political Rights (ICCPR), and the International Covenant on Economic, Social and Cultural Rights (ICESCR) prohibit discrimination on the grounds of race, colour, sex, language, religion, political or other opinion, national or social origin, property, birth or 'other status.' The ICCPR and ICESCR include further an undertaking to "ensure the equal right of men and women to the enjoyment of all" rights in their respective covenants. Children are to be protected against discrimination on the basis of the status, activities, expressed opinions, or beliefs of the child's parents, legal guardians, or family members. These lists are far from exhaustive. Other human rights treaties have since prohibited discrimination on the grounds of marital status, descent or ethnic origin, disability, nationality, age, and economic position. Treaty bodies have interpreted treaties to prohibit discrimination on the basis of geographical residence, health status, and sexual orientation. Human rights law also protects against discrimination on the basis of gender identity, family status, health status (e.g. HIV status), homelessness, or because they engage in sex work.

In one of the most comprehensive statements of the meaning of discrimination and state economic, social, and cultural rights obligations, the Committee on Economic, Social, and Cultural Rights has stated that "discrimination constitutes any distinction, exclusion, restriction or preference or other differential treatment that is directly or indirectly based on the prohibited grounds of discrimination and which has the intention or effect of nullifying or impairing the recognition, enjoyment or exercise, on an equal footing, of [human] rights. Discrimination also includes incitement to discriminate and harassment." Both direct and indirect forms discrimination are prohibited. "Direct discrimination occurs when an individual is treated less favourably than another person in a similar situation for a reason related to a prohibited ground.... Indirect discrimination refers to laws, policies or practices which appear neutral at face value, but have a disproportionate impact on the exercise of [human] rights as distinguished by prohibited grounds of discrimination."

The Need to Dismantle Entrenched Discrimination and Inequality

Discrimination is a key underlying cause of inequality and needs to be addressed by the post-2015 framework if it is to successfully reduce inequality more broadly. By supporting aggregate progress, the MDGs have rewarded strategies directed at "low-hanging fruit" as much as strategies directed at the most deprived populations. The post-2015 framework should support development strategies which are designed to reach and benefit the most marginalized, excluded and in need populations. This includes addressing the urgent

social needs of such populations as well as assessing difficulties that marginalized and excluded groups experience in enjoying economic, social and cultural rights, and taking the necessary steps to address these difficulties.

Despite significant progress on the MDG on safe drinking water, the UN Special Rapporteur on the human rights to safe drinking water and sanitation, the European Committee of Social Rights (ECSR), the WHO, and others have documented discrimination in access to water and sanitation on the basis of race and ethnicity, including indigenous peoples in Costa Rica and Rwanda; Dalits in Bangladesh; Roma in Slovenia and Portugal; and poor communities of Korean descent in Japan.

The relationship between discrimination and poverty is overwhelmingly evident in developed as well as in developing countries. As the UN Committee on Racial Discrimination has recognized, in the US, “racial, ethnic, and national minorities, especially Latino and African American persons, are disproportionately concentrated in poor residential areas characterised by sub-standard housing conditions, limited employment opportunities, inadequate access to health care facilities, under-resourced schools, and high exposure to crime and violence.” The US Census reported that in 2009, 25 percent of black people and people of Hispanic origin live below poverty level, compared to 14 percent of people of all races.

Racial and ethnic minorities have long been disproportionately represented in the US criminal justice system. While accounting for only 13 percent of the US population, African Americans represent 28.4 percent of all arrests. According to Bureau of Justice Statistics approximately 3.1 percent of African American men, 1.3 percent of Latino men, and 0.5 percent of white men are in prison. Because they are disproportionately likely to have criminal records, members of racial and ethnic minorities are more likely than whites to experience stigma and legal discrimination in employment, housing, education, public benefits, jury service, and the right to vote.

In Europe, Roma are among the poorest as well as being among the most discriminated against. In Bosnia and Herzegovina for instance, Human Rights Watch documented pervasive discrimination against Roma and other national minorities in the constitution, national laws, and public institutions. This research shows the wider impact of discrimination on the daily lives of Roma in accessing housing, education, health care, and employment.

The relationship between discrimination, inequality, and poverty can also be seen, for instance, in Burma. Discrimination against certain ethnic minorities is widespread in Burma and addressing this discrimination should be a key element of poverty reduction strategies. The Rohingya, a Muslim minority group, are among the most marginalized in Burma. The Burmese government has long denied Rohingya the right to obtain citizenship in Burma, which has facilitated human rights abuses against them and rendered them stateless, posing a serious obstacle to achieving a durable solution to the sectarian violence in Arakan State and resolving the situation of Rohingya refugees. The Rohingya face restrictions on movement, employment and access to opportunity, education, marriage, and on other aspects of their everyday lives. For decades they have faced violent abuses by state security forces such as abusive forced labor, rape, arbitrary detention and torture, and killings, based on ethnic and religious grounds. This discrimination has played a role in pushing Rohingya into increased poverty and is a hurdle in the realization of social and economic rights. Discrimination also limits peoples’ ability to participate in the development of poverty reduction strategies or government policies and limit

access to justice, compounding the problem. As international donors enhance engagement with the Burmese government in response to significant governance and human rights reforms, donors and the government should work to address the urgent social needs of the most in need.

As set out below, dismantling discrimination and addressing inequality requires a range of fully implemented laws, policies and programmes, and may include temporary special measures. It may also require a change in resource allocation, with devotion of greater resources to marginalized groups.

Enact Laws against Discrimination, Amend Discriminatory Laws

States have a clear obligation to end discrimination under the law and to enact effective laws against both direct and indirect discrimination by both government and private actors, including in access to development assistance. Failure to do so may lead to unequal outcomes of development and poverty alleviation programs. Maintenance of state-sponsored discrimination can undermine development activities for specific sectors of the population, even if overall a country is meeting its goals and indicators.

For example, in Bangladesh discriminatory family laws on marriage, separation, and divorce push some women further into poverty—even as the country is seen as a positive example of successful gains related towards meeting the Millennium Development Goals and in poverty alleviation. In 20 years Bangladesh has raised life expectancy by 10 years. It has doubled female primary school enrollment in 10 years, and more than halved infant mortality. According to some studies, both the rich and the poor have benefitted from these gains. However, Human Rights Watch research demonstrated that certain women and families may not benefit from these societal gains due to discriminatory family laws that push them deeper into poverty.

Human Rights Watch found that family laws in Bangladesh establish greater barriers to divorce for women than men, provide vague guidance on maintenance claims, ignore women's contributions to households, lack provisions for division of marital property upon divorce, and are incompatible with international human rights law. The United Nations country team in Bangladesh has identified “marital instability” as a key cause of poverty among female-headed households and the Bangladesh Planning Commission has said that women are more susceptible to poverty after abandonment or divorce. Yet, current indicators demonstrating Bangladesh's success towards poverty alleviation obfuscate the impact of these discriminatory laws on poor women.

Dismantle Discriminatory Practices

The post-2015 framework should recognize the importance of identifying, prohibiting, and dismantling discriminatory practices—both of both a direct and indirect nature—for poverty alleviation. Governments should implement the necessary strategies, policies, and action plans, which may include temporary special measures to accelerate the achievement of equality, to address discrimination.

As the MDGs recognize, access to primary education is a core development goal. Yet millions of children suffer from discriminatory barriers to education and either never attend school or are compelled to leave school early. While MDG 2 highlighted equal education for girls, it did not highlight other discriminatory barriers to education. Human Rights Watch research has also found that migrant children, children from rural areas, ethnic or religious minorities, internally displaced and refugee children, indigenous chil-

dren, and low-caste children were often denied equal access to education, or in some cases, access to any education at all.

Human Rights Watch's research on education in Nepal has documented widespread discrimination experienced by children with disabilities, who are prevented from getting access to school and realising their right to education by a range of barriers. These include inadequately trained teachers, a lack of appropriate materials, no transport provision, and negative attitudes towards children with disabilities. As a result, children with disabilities represent a significant proportion of the 330,000 primary school age children who remain out of school in Nepal. These patterns are replicated elsewhere—globally, children with disabilities are less likely to start school and have significantly lower rates of school completion than non-disabled children. Development strategies should be directed at ensuring the full agency of persons with disabilities. The government, donors, and international institutions need to take the necessary steps to address the barriers that keep children with disabilities from attending school.

Human Rights Watch research has also documented how discrimination and abuse against women and children has impacted their ability to access quality health care. In the area of maternal health, research carried out by Human Rights Watch in South Africa has documented a strong link between the neglect and abuse of women's rights—especially during pregnancy and labor—and poor maternal health outcomes. The way in which the health MDGs have been framed has encouraged a focus on rolling out proven maternal, newborn, and child health interventions, but they have been less effective at promoting action to tackle more entrenched issues of discrimination that impact maternal and child health outcomes, such as the practice of child marriage.

Similarly, Human Rights Watch found in a 2008 report that efforts to roll out antiretroviral treatment in Kenya had limited impact as they failed to address the discrimination, stigma, abuse, and neglect that many people living with HIV experience. Human Rights Watch identified a range of barriers and forms of discrimination which children faced in accessing HIV testing and treatment. HIV-positive mothers who were victims of violence and property rights abuses were unable to access treatment for themselves or their children because they could not afford transport to health centers or enough food to avoid serious side effects from the drugs. Parents or caregivers lacked accurate information about medical care for children, or avoided testing and treatment because of stigma and discrimination. Orphans also faced neglect and abuse. Human Rights Watch found that as a result, Kenya's HIV care programs had failed to deliver lifesaving drugs to the majority of children who needed them. HIV testing and treatment programs should work to remove discriminatory barriers to care.

Address Discrimination by Private Actors

Discrimination by private actors in workplaces, the provision of services, or other sectors of society may prevent the enjoyment of human rights. States have an obligation to adopt measures “to ensure that individuals and entities in the private sphere do not discriminate on prohibited grounds.” The post-2015 framework should recognize the importance and legal obligations of identifying, prohibiting and addressing discrimination by private actors for poverty reduction and sustainable development.

Gender-based violence is a form of private discrimination that may prevent some women equal access to the benefits of state or donor-led development programs. Human Rights Watch found evidence of that in HIV-treatment programs in Zambia.

Zambia is one of many countries setting ambitious targets for rapidly scaling up antiretroviral treatment for HIV/AIDS and is making impressive progress. It is currently on track to meet the MDG on HIV by addressing a range of obstacles to treatment and receiving substantial donor support to overcome them. However, in 2007 Human Rights Watch found that women's unequal status in Zambian society gravely undermined their ability to access and adhere to antiretroviral treatment (ART). Women in Zambia reported that gender-based human rights abuses presented very real barriers to accessing and adhering to treatment. Treatment policies and programs did not take into account the connection between domestic violence or women's insecure property rights, both forms of entrenched discrimination, and women's ability to seek, access, and adhere to HIV treatment – leading to unequal health outcomes for these women.

Nondiscrimination in the workplace is enshrined as a core labor right and key to addressing inequality and poverty. Human Rights Watch has documented a state-owned gold company in DRC favoring workers from one ethnic group over another in compensation and promotions, employers in the Dominican Republic discriminating against workers based on their actual or perceived HIV status, and Saudi employers forcibly confining low-paid women migrant workers, including those who work in hospitals and dress shops. The mistreatment in Saudi Arabia of migrant workers from Asia and Africa also has been aggravated by deeply rooted religious and racial discrimination.

Dismantle Economic Barriers

The post-2015 framework should support the dismantling of economic barriers to the progressive realization of social and economic rights. As the Committee on Economic, Social and Cultural Rights has emphasized, “Equity demands that poorer households should not be disproportionately burdened with health expenses as compared to richer households.” The same can be said for other social, economic, and cultural rights. Human Rights Watch research has shown how user fees and transport costs present a barrier for poorer families are burdened in accessing health care or education. User fees are prohibited in human rights law for primary education, which also requires they be progressively eliminated in secondary and tertiary education.

The Committee has further recognized that eliminating systemic discrimination and realizing economic, social and cultural rights without discrimination frequently requires greater devotion of resources to traditionally neglected groups.

The post-2015 framework should prioritize investment in the poorest and most marginalized populations. It can go some way toward achieving this by including indicators which measure the achievement of targets by wealth quintiles, with a specific target of addressing the social and economic needs of the poorest two quintiles. Several NGOs have suggested also including a specific target on reducing income inequalities within countries, with an emphasis on reducing the gap between the richest and poorest quintiles (20:20 gap) or between the top quintile and the bottom two quintiles (10:40 gap).

Ensure Development Initiatives Do Not Discriminate or Otherwise Reinforce Discrimination

In addition to proactively addressing discrimination and inequality, governments and international institutions must ensure that their policies do not discriminate—directly or indirectly—against at-risk populations. The post-2015 framework should recognize the risk of discrimination in development policies and actively prevent against it. Hu-

man Rights Watch has documented discrimination in the distribution of aid, disproportionate negative impacts of development policies, and failure in project design to consider at-risk groups. We have also documented how development initiatives have violated the rights of indigenous peoples rather than seeking to realize their social and economic rights while respecting their cultural rights.

There is no reference in the MDGs to people being hurt or disadvantaged by development efforts. Nor do many developing country governments, bilateral donors, or the international financial institutions use a human rights framework for understanding, mitigating, and remedying the harm that may result from development efforts. An explicit focus within the MDGs and in development strategies on human rights and associated principles of transparency, consultation, participation, and accountability could have helped to correct this.

Avoid Discrimination in the Distribution of Aid

Too often, government funds or development aid are misused for political gain rather than investing in the realization of social and economic rights. This can come in the form of outright corruption, in discrimination on the basis of political opinion in the distribution of aid, or in investing in areas where politicians will most benefit rather than investing in the areas of greatest need.

Human Rights Watch has documented discrimination in development projects and policies in Ethiopia on the basis of political opinion. The government has used donor-supported programs, salaries, and training opportunities as political weapons to control the population, punish dissent, and undermine perceived political opponents. Local officials denied these people access to seeds and fertilizer, agricultural land, credit, food aid, and other resources for development on the basis of their political opinion. For example, local officials offered to “forgive” opposition members in need of food and give them access to a cash-for-work program if they wrote a letter of regret to the administration for aligning with the opposition.

People have also been excluded from development programs, deliberately or inadvertently, because of their religion, ethnic background, gender, sexual orientation, disability amongst other grounds. The post-2015 framework should acknowledge this risk and work to prevent it.

Design Projects to Consider At-Risk Groups and Avoid Adverse Impacts

Governments and donors should ensure that their development strategies and projects are designed to consider at-risk groups and avoid any adverse rights impacts and develop accessible, effective accountability mechanisms for those discriminated against.

For example, Human Rights Watch’s research in the Gambella region of Ethiopia has documented serious human rights abuses against indigenous peoples and other ethnic minorities in the carrying out of the government’s so-called “villagization” programme. This programme is being carried out in the name of development—transferring tens of thousands of people from their existing homes to new model villages, where they were promised to be provided with improved infrastructure and better services. The scheme is also described as a voluntary one. It is anything but. Our research shows that people are being forced to move against their will and government soldiers have beaten and abused those who have objected to the move. Fear and intimidation are widespread amongst affected populations. Despite government pledges, the land near the new villages still

needs to be cleared, while food and agricultural assistance have not been provided. As a result, some of the relocated populations have faced hunger and even starvation.

Indigenous peoples' social and economic rights must be realized while respecting their cultural and other human rights. The post-2015 framework should expressly commit to the protection of indigenous peoples rights and prevent against violations of these rights in the name of development. This includes recognition of the specific rights of indigenous people concerning their ancestral lands, and their right to free, prior and informed consent concerning projects affecting them or their lands. Governments, donors, and international institutions should invest in providing education, health care, water and sanitation, and other social and economic rights in ways that are consistent with the cultural practices and values of indigenous peoples.

Also in Ethiopia, Human Rights Watch has documented forced relocations of agro-pastoralist indigenous peoples linked to the creation of 245,000 hectares of state-run sugar plantations along the Omo River. State security forces used intimidation, assaults and arbitrary arrests when people questioned the relocations or refused to move. The ongoing cost of this development to indigenous groups is massive: their farms are being cleared, prime grazing land is being lost, and traditional livelihoods are being decimated. The Ethiopian government has failed to meaningfully consult, compensate, or discuss with these communities alternative means of livelihoods.

The Legal Basis

Development grounded in a human rights-based approach draws upon the principles and legal framework of human rights and requires that respect for the human rights of those affected by aid or development programs is central to planning and operationalizing that activity. It recognizes beneficiaries of aid as rights-holders with legal entitlements and identifies governments and their partners, including international institutions, as duty bearers with correlating obligations to meet those entitlements.

The Principle of Equality and Non-Discrimination

First set out in the Universal Declaration of Human Rights (UDHR), equality and non-discrimination are foundational principles of the international human rights legal framework. All of the core human rights treaties adopted since the UDHR contain legal obligations related to equality and non-discrimination.

The human rights legal framework requires states not only to refrain from discrimination, but also obliges them to take appropriate measures to end discrimination by state and private actors and promote equality. Discrimination in law or in practice may prevent the realization and enjoyment of other rights; therefore, a development approach firmly established in human rights obligates states to confront the systemic and root-causes of discrimination or inequality, while actively working towards the fulfillment of rights without discrimination. This requires states to establish equality under the law and address policies, programs, or even stereotypes that create or perpetuate discrimination.

Development based in human rights needs to emphasize and ensure equality and non-discrimination in both process and outcomes. It requires particular attention to the needs of protected groups, the impact of programs on their respective rights, and the establishment of procedures to ensure accountability and participation in development that affects them. It also requires that human rights standards guide all stages of programming.

Millennium Declaration and Millennium Development Goals

While the Millennium Declaration grounded poverty alleviation objectives in principles of human rights, equality, and non-discrimination, these principles were not embodied in the Millennium Development Goals.

In the declaration, states rededicated themselves to “respect for human rights and fundamental freedoms, [and] respect for the equal rights of all without distinction as to race, sex, language or religion,” recognized a “collective responsibility to uphold the principles of human dignity, equality and equity” and recognized that democratic and participatory governance based on the will of the people best assures the right to live life in dignity, free from hunger and from fear of violence, oppression or injustice. In pledging to the principle of equality, the declaration stated, “No individual and no nation must be denied the opportunity to benefit from development.”

Despite states’ recognition of the centrality of human rights, equality, and non-discrimination to development and poverty eradication in the declaration, the goals largely overlooked them.

Duty-Bearers

The commitment to equality and non-discrimination should extend not only directly from states, but also from donors and multilateral organizations. In compliance with their international obligations, states should respect the enjoyment of human rights in other countries and prevent third parties, through political or legal means, from interfering with the enjoyment of rights. In furtherance of this obligation, UN treaty and charter bodies have found that states’ membership in international organizations should take due account of human rights. State members of international organizations, including multilateral development banks or other institutions, cannot set aside their obligations to respect, protect, and fulfill human rights. All member states of the United Nations, and UN institutions themselves, carry a fundamental duty to act consistently with the principles of the UN Charter, which requires “universal respect for, and observance of, human rights and fundamental freedoms for all...”

Further, while non-governmental organizations, when operating as non-state actors, do not bear the same international obligations under human rights law as governments do, they do not function in a human rights void and should operate in a manner that promotes human rights. Governments are also required to ensure they enforce human rights on all within their jurisdiction. Thus, all actors engaged in development should adopt a rights-based approach to their work, which requires a specific action to ensure equality and non-discrimination in process and outcome.

Stigma

The ability of social groups to assign a stigma (a discrediting mark or negative judgment) to people, attributes and/or behaviors and to sanction people who are so stigmatized is a fundamental mechanism of informal social control in all societies. Socially defined stigmas are applied through a process that begins when a difference is labeled, then linked to negative stereotypes, leading to a separation of “us” from “them,” and finally to status loss and discrimination for those carrying the trait. Stigmas persist because they are reinforced and replicated by a range of interlocking social norms and institutions. Organizational and national policies reflect social values and norms. They may replicate and

reinforce social stigma against certain behaviors or groups actively (e.g., through criminalization, or disqualification from services), or passively (e.g., through neglect and lack of investment), according to structural factors ranging from economic and geographic distinctions to differences in ethnicity, occupation, or sexual orientation. Thus, stigma and discrimination can be profoundly political.

Marginalization

Marginalization describes the position of individuals, groups or populations outside of 'mainstream society', living at the margins of those in the centre of power, of cultural dominance and economical and social welfare. It is defined as, "a process by which a group or individual is denied access to important positions and symbols of economic, religious, or political power within any society...a marginal group may actually constitute a numerical majority...and should perhaps be distinguished from a minority group, which may be small in numbers, but has access to political or economic power".

"To be marginalized, ..., is to be distanced from power and resources that enable self-determination in economic, political, and social settings...It is an inherent characteristic of 'those in the margin', that they have poor access to economical and other recourses like education and social services, meanwhile participation and self determination are on a low level. However, definitions of what is regarded as marginalised are highly depending on the historical and socio-economical context of a society."

In the last decennia, globalisation, migration, economical developments – e.g. in the area of communication technologies – have had a major influence on the economical level of millions of people and consequently, on their place in the society. Gender, culture, language, race, sexual orientation, lifestyle and the socio-economic position or class are factors, which influence the position of an individual or a group in the society. Vulnerable groups like migrants and ethnic minorities, homeless people, drug addicts, sex workers, youth with risk behaviour, isolated older people or people with disabilities face higher risks of social exclusion and marginalisation. Their problems can be related to homelessness, unemployment, poor access to social and health services, low health status and poor living conditions.

Social Suffering

'Social inclusion' refers to the position, in which someone can access and benefit from the full range of opportunities available to members of a society. The term is not a fixed theory or concept, but continuously under development, regarding the challenges and problems in different situations and circumstances. Social inclusion has a value as process and as a goal, and should be approached pro-actively, not as a passive defending of rights, but as active interventions, rooted in community (self) organization and leading to real policy changes by transforming given structures.

"Social inclusion is the political response to exclusion. Most analyses of racism and sexism, for example, focus on the removal of systemic barriers to effective participation and focus on equality of opportunity. These analyses tend to be essentialist and consequently are unable to develop a comprehensive vision that cuts across all the areas of injustice.

Social inclusion is about more than the removal of barriers, it is about a comprehensive vision that includes all. Social inclusion, by virtue of the fact that it is both process and outcome, can hold governments and institutions accountable for their policies. The

yardstick by which to measure good government becomes the extent to which it advances the well-being of the most vulnerable and most marginalized in society.

Social inclusion is about advocacy and transformation. It is about the political struggle and political will to remove barriers to full and equitable participation in society by all."

Even if there is still a gap between policy statements and what is happening in practice, in the last decennia more and more awareness and attention has been given to improve social inclusion of disadvantaged groups and people, also by international bodies as the United Nations (UN Millennium Goals) and the European Union.

National governments and NGO's installed measures and projects to improve the situation on different levels. Usually, the overall aim is to "ensure the marginalised and those living in poverty have greater participation in decision making which affects their lives, allowing them to improve their living standards and their overall wellbeing" and "to remove barriers for people or for areas that experience a combination of linked problems (...)".

"Programmes often are set up with multidisciplinary approaches, addressing problems at several levels of society. For instance, the British Social Inclusion Programme 2007 addressed aspects like stigma and discrimination, the role of health and social care in tackling social exclusion, implementing evidence-based practice in vocational services and enabling reintegration into the community, employment and taking part in the local community.

Child Abuse

The term 'Child Abuse' may have different connotations in different cultural milieu and socio-economic situations. A universal definition of child abuse in the Indian context does not exist and has yet to be defined. According to WHO² :

Physical Abuse. Physical abuse is the inflicting of physical injury upon a child. This may include burning, hitting, punching, shaking, kicking, beating or otherwise harming a child. The parent or caretaker may not have intended to hurt the child. It may, however, be the result of over-discipline or physical punishment that is inappropriate to the child's age.

Sexual Abuse. Sexual abuse is inappropriate sexual behaviour with a child. It includes fondling a child's genitals, making the child fondle the adult's genitals, intercourse, incest, rape, sodomy, exhibitionism and sexual exploitation. To be considered 'child abuse', these acts have to be committed by a person responsible for the care of a child (for example a baby-sitter, a parent, or a daycare provider), or related to the child. If a stranger commits these acts, it would be considered sexual assault and handled solely by the police and criminal courts.

Emotional Abuse. Emotional abuse is also known as verbal abuse, mental abuse, and psychological maltreatment. It includes acts or the failures to act by parents or caretakers that have caused or could cause, serious behavioural, cognitive, emotional, or mental trauma. This can include parents/caretakers using extreme and/or bizarre forms of punishment, such as confinement in a closet or dark room or being tied to a chair for long periods of time or threatening or terrorizing a child. Less severe acts, but no less damaging, are belittling or rejecting treatment, using derogatory terms to describe the child, habitual tendency to blame the child or make him/her a scapegoat.

Neglect. It is the failure to provide for the child's basic needs. Neglect can be physical, educational, or emotional. Physical neglect can include not providing adequate food or clothing, appropriate medical care, supervision, or proper weather protection (heat or cold). It may include abandonment. Educational neglect includes failure to provide appropriate schooling or special educational needs, allowing excessive truancies. Psychological neglect includes the lack of any emotional support and love, never attending to the child, substance abuse including allowing the child to participate in drug and alcohol use.

Domestic Violence

Domestic violence is destructive for both the battered and the batterer. Its tendency to be passed down over generations makes it all the more important that we develop effective methods for combating abusive behaviors. Domestic violence can be physical or psychological, and it can affect anyone of any age, gender, race, or sexual orientation. It may include behaviors meant to scare, physically harm, or control a partner. While every relationship is different, domestic violence generally involves an unequal power dynamic in which one partner tries to assert control over the other in a variety of ways. Examples include insults and threats, emotional abuse, and sexual coercion. Some perpetrators may even use children, pets, or other family members as emotional leverage to get the victim to do what they want. Victims experience diminished self-worth, anxiety, depression, and a general sense of helplessness that can take time and often professional help to overcome.

PEACE PSYCHOLOGY

Peace psychology is a subfield of psychology and peace research that deals with the psychological aspects of peace, conflict, violence, and war. Peace psychology can be characterized by four interconnected pillars: (1) research, (2) education, (3) practice, and (4) advocacy. The first pillar, research, is documented most extensively in this article.

Peace psychological activities are based on psychological models (theories) and methods; they are usually normatively bound in their means and objectives by working towards the ideal of sustainable peace using (as far as possible) non-violent means. Violence and peace are usually defined in terms of Johan Galtung's extended conceptualization of peace, according to which peace is not merely the absence of personal (direct) violence and war (= negative peace), but also the absence of structural (indirect) and cultural violence (= positive peace). The ideal of peace can also be conceptualized as the comprehensive implementation of human rights (civil, political, economic, social, and cultural rights); this should, among other purposes, ensure the satisfaction of basic human needs, such as positive personal and social identity, sense of control, security, (social) justice, well-being, a safe environment, and access to adequate food and shelter.

Organizations that focus on peace psychology include, for example, in the United States the Society for the Study of Peace, Conflict, and Violence (Peace Psychology Division [Division 48] of the American Psychological Association) and Psychologists for Social Responsibility, a nongovernmental organization based in Washington, DC. Germany has the Forum Friedenspsychologie, and the Australian Psychological Society has an Interest Group called Psychologists for Peace. On the international level, there is the Committee for the Psychological Study of Peace as well as the International Network of Psycholo-

gists for Social Responsibility, which links organizations from (among other countries) Germany, Finland, the United States, Australia, Costa Rica, India, and Italy.

Peace psychological research can be analytically (research on peace) or normatively (research for peace) oriented. Regardless of its analytical or normative orientation, peace psychological research mainly deals with the psychological aspects of the formation, escalation, reduction, and resolution of conflicts (including war), the psychosocial conditions conducive or detrimental to a sustainable peace, and the psychosocial effects of war and violence. In each case, different levels of analysis and explanation are relevant: from the level of individuals to groups, social organizations and institutions, states and state systems (e.g., the European Union), military alliances (e.g., NATO), and collective security systems (e.g., the United Nations and the Organization for Security and Cooperation in Europe [OSCE]).

Formation and Escalation of Conflict

Peace psychology focuses on the psychological aspects of the formation, escalation, reduction, and resolution of conflicts. A conflict exists when the expectations, interests, needs, or actions of at least two parties to the conflict are perceived by at least one of the parties to be incompatible. Peace psychology is mainly concerned with conflicts between social groups (intergroup conflicts, such as between ethnic groups, clans, religious groups, states etc.), in terms of domains like power, wealth, access to raw materials and markets, cultural or religious values, honor, dignity, or recognition. In conflicts one has to distinguish between (overt) positions (e.g., "we don't negotiate with X") and underlying interests (e.g., power, spheres of influence and wealth) as well as between current triggers (e.g., violence at a political protest) and systematic, enduring, structural causes (e.g., deprivation of a group's political participation or access to professional employment). Although conflicts are inevitable and can lead to positive change when dealt with constructively, the escalation of conflicts and in particular the occurrence of violence are preventable. Psychological processes of information processing (attention, perception, memory, thinking, judgment), emotion, and motivation influence substantially how a conflict is handled, and in particular whether conflicts escalate to violent episodes. An important factor is the different points of view of the conflict parties, such as when behavior that is based on positive intentions is perceived by the opponent as aggressive and therefore contributes to escalation. Conflicts can easily escalate. A cycle of violence can arise in which both parties are involved, and original victims can become perpetrators, without realizing it ("victim myth").

Conflicts can be intensified specifically through the construction of enemy images, psychological warfare, and propaganda promulgated by the media, political elite, educational systems, socialization, cultural symbols and other means. Enemy images may have a kernel of truth, but overstate the negative sides of the opponent. The core features of a strong enemy image include: (1) a negative evaluation of the opponent (e.g., aggressive, immoral, but also inferior), (2) a one-sided blame for negative events, and (3) a different evaluation of similar actions of one's own side than the enemy ("double standard"; e.g., build-up of arms on one's own side is self-defense, on the enemy's it is aggression). These constructions can cause dehumanization of the opponent, so that moral standards no longer apply. In extreme cases, it may seem acceptable, even desirable, for the opponent to suffer and be killed. The construction of the enemy image has

the central function of justifying armament, violence, and war. In addition, it enhances the individual and collective self-image.

Psychological warfare includes methods of generating or strengthening war support among the civilian population and the military. These methods include disinformation using the media (war propaganda), but also sabotage, displacement, murder, and terror. War propaganda consists of two complementary strategies: (1) repeating, highlighting, and embellishing with detail information that functions to intensify the enemy image or threat perceptions, and (2) ignoring and devaluing information that may lead to de-escalation. In addition, negative behavior of the adversary may be provoked (e.g., by maneuvers at the state borders) or charges that the enemy engaged in heinous acts may be entirely invented (e.g., the Nayirah testimony).

Conflict Reduction and Resolution (Peace Psychological Strategies)

Different peace psychological strategies for non-violent conflict resolution are discussed (conflict de-escalation, conflict resolution, conflict transformation). One can distinguish between strategies on the official level (e.g., measures of tension reduction and trust build such as Charles E. Osgood's "Graduated and Reciprocated Initiatives in Tension Reduction" [GRIT], negotiations, mediation), approaches of unofficial diplomacy (interactive problem-solving workshops), and strategies at the level of peace and conflict civil society (e.g., peace journalism, contact between social groups).

Official Level

Osgood's GRIT model was designed as a counter-concept to the arms race in the East-West conflict, in which the former superpowers, USA and USSR, constantly increased the quantity and quality of their arms so that the destruction of humankind by a nuclear war seemed increasingly possible. The GRIT model, in contrast, aimed to de-escalate and create an atmosphere of mutual trust. One party publicly announces and performs a verifiable, concrete step to reduce tension, and asks or invites the other side to do something similar (developing a spiral of trust). Care is taken so that each step does not endanger the safety of one's own side. GRIT was designed to reverse the tension involved in the nuclear arms race by having each side engage in graduated and reciprocal initiatives. While there is no firm evidence, it has been suggested that U.S. President Kennedy and the Soviet leader Khrushchev based their negotiations after the Cuban missile crisis on this concept.

When conflicted parties are engaged in long-lasting, severe conflicts, it can be difficult to have constructive bilateral negotiations. In this case, a third party (e.g., a social scientist or reputed politician) can serve as a mediator in order to facilitate conflict management. Mediators must be well aware of the conflict and its history, should have the confidence of both conflict parties, and need to be familiar with conflict analysis and communication strategies. Important strategies include establishing trust, working out the essential elements of conflict, and possibly dividing the problem so that at least partial solutions can be achieved and violence can be prevented or stopped. Problems arise when mediators are biased and have strong individual interests. Mediation success is more likely if the conflict is moderately intense, the power difference between the parties is low, and the mediators have high prestige (as a person or because of organizational affiliation).

Unofficial Level

In severe, long-lasting conflicts, it may be advisable to intervene at a level below official diplomacy. Interactive problem solving is such an informal approach to bring members of the conflict parties together. These can include citizens who are well-respected from different areas of society, such as media, business, politics, education, or religion. A team of social scientists (e.g., psychologists) initiate and promote a problem-solving process with the elements of conflict diagnosis, generation of alternatives, and development of nonviolent solutions that results in outcomes that are satisfactory to all parties involved. There is the expectation or hope that the participants influence their governments and public opinion so that official negotiations can follow. Psychologically important components of the process are that the respective self and enemy images are corrected. Interactive problem solving was used in particular in the Israel-Palestine conflict by the U.S. psychologist Herbert Kelman and his team.

Civil Society Level

Media are often involved in the formation of enemy images and escalation of conflict. Peace journalism, in contrast, has the objective of investigating and using the influence of the media as a means of encouraging the constructive, non-violent resolution of conflict. Key strategies include representing the conflicting parties as well as the conflict and its history appropriately, identifying propaganda, and articulating the suffering of the people.

The collective action and peaceful demonstrations of the population toward peaceful and socially just ends can have an influence on the decisions of those in power – particularly in democracies. Citizens' commitment depends, among other factors, on the existence of opportunities in society, individual value orientations (e.g., valuing non-violence, social justice), the presence of role models, and the perceived probability of success of one's actions.

Contacts between opposing groups (e.g., on the level of municipalities, associations, universities, trade unions) can contribute to building positive relationships and the reduction of prejudice (see contact hypothesis). Conditions associated with the improvement of intergroup relations when groups come in contact with one another include: The actors involved have similar social status; there are common goals that can be achieved through cooperation; and the contacts are supported by authorities in society.

In asymmetric conflicts, where one conflict party is politically, economically, and/or militarily clearly superior, the stronger party may not be interested in a truly sustainable conflict resolution. Under asymmetric conditions, when the root causes of the conflict cannot be sufficiently addressed, structural violence persists. For such situations, approaches have been developed such as nonviolent resistance and liberation psychology, which originated in Latin America and is related to liberation theology.

Nonviolent resistance refers to public, non-violent behavior directed against injustice; it involves publicly explicating one's own intentions, committing to communicate with the other side, and the willingness to endure negative consequences of one's own actions. Methods of nonviolent resistance range from protests (e.g., demonstrations) to non-cooperation (e.g., strikes, boycotts) to civil resistance (e.g., traffic blockades). Particularly well known are the actions, speeches, and writings of Mahatma Gandhi and Martin Luther King.

Effects of War and Violence

Peace psychology examines war and violence between groups also with the aim of illustrating the psychological and social costs of war and violence and to document the human suffering caused. The psychological consequences include, in particular, traumatization (mainly of the civilian population, but also members of the military), cognitive and emotional damage, and the destruction of trustful social relationships. Wars often do not resolve the underlying problems; they often provoke new violence and wars. For example, in post-war societies an increased level of family and community violence can be observed. In addition, resources necessary to deal with civilian issues (e.g., education, health, social welfare) are lost. There is still little comprehensive and objective research on the consequences and costs of war.

Psychosocial Conditions of Sustainable Peace

Even when violence has been stopped or a Peace Treaty reached, to prevent the risk of a renewed escalation, physical and economic reconstruction as well as socio-political and psychosocial interventions are required. These interventions aim to cure psychosocial wounds of war, build trust, develop a common collective memory, recognize past wrongdoing, and achieve reconciliation and/or forgiveness. Examples are trauma therapy and Truth and Reconciliation Commissions.

Also, irrespective of any specific conflict and violence, peace psychological research looks at the psychosocial conditions that hamper or promote sustainable peace. The basic aim is to transform cultures of violence into cultures of peace.

The following cultural characteristics are obstacles to the development of sustainable peace: the view of one's own group (ethnicity, religion, nation, etc.) to be superior and more valuable and others as inferior and of little value (or in the extreme case: no value); the development of enemy images, dehumanization of others, legitimization of violence and damage; underlying beliefs (ideologies) such as ethnocentrism, social dominance orientation, authoritarianism, nationalism, militarism, and an education system that promotes these ideologies; power differentials that are defended or increased by the powerful and that create unequal conditions in areas such as wealth, health, education, and political participation (structural violence).

Among factors conducive to the development of sustainable peace are: the fundamental belief that conflicts are frequent, but that they can be solved without violence and for the benefit of the various conflict parties; the concept of humanism with the features of human dignity, pacifism, empathy, respect, tolerance and solidarity, and respect for all people or for humanity as a whole; critical proximity to one's own group that – in addition to positive identification – also integrates own weaknesses, mistakes, and committed wrongdoings in the collective self-concept.

In the transformation of cultures of violence into cultures of peace the focus on human rights is of high importance. Human rights are inalienable rights that apply to all human beings, without distinction as to sex, color, ethnicity, language, religion, political opinion, or social origin (prohibition of discrimination). The UN Human Rights Charter contains the essential documents of the Universal Declaration of Human Rights (UDHR, 1948) and the Twin Covenants (1966, International Covenant on Economic, Social and Cultural Rights and International Covenant on Civil and Political Rights). The UDHR consists of 30 articles with more than 100 individual rights, including civil and political

rights (e.g., right to life, prohibition of torture, right to fair and public trial, right to asylum, freedom of speech, regular elections), but also social, economic, and cultural rights (including the right to work, rest, holidays with pay, protection from unemployment, the right to food, clothing, housing, medical care, and free primary education). Of particular importance in the UN's human rights concept is that all human rights are significant (indivisibility) and that they apply to all people (universality). Psychological research on human rights has mainly examined knowledge, attitudes, and readiness to act in support of human rights. Representative surveys in Germany show that the realization of human rights is considered to be very important, but at the same time knowledge of human rights is low and inaccurate. The results show a "halving" of human rights: Some civil rights are known, while economic and social rights are hardly considered human rights. Of importance in peace psychology are also analyses of whether human rights are used in the sense of peace or whether they are abused for the construction of enemy images or to prepare wars.

WELLBEING AND SELF-GROWTH

Types of Wellbeing

The concept of happiness is the corner stone of the assumptions of positive psychology. Happiness is characterised by the experience of more frequent positive affective states than negative ones as well as a perception that one is progressing toward important life goals. Identifying factors that contribute to happiness has proven to be challenging. Interestingly though, one thing that does stand out in the research to date is that the attainment and pursuit of pleasure may not always lead to happiness.

Certain kinds of environmental factors or conditions have been found to be associated with happiness and include such things as: individual income, labour market status, health, family, social relationships, moral values and many others. Ultimately, in the pursuit of understanding happiness, there are two main theoretical perspectives which focus on addressing the question of what makes people feel good and happy. These are the hedonic and eudaimonic approaches to happiness.

Hedonic wellbeing is based on the notion that increased pleasure and decreased pain leads to happiness. Hedonic concepts are based on the notion of subjective wellbeing. Subjective well-being is a scientific term that is commonly used to denote the 'happy or good life'. It comprises of an affective component (high positive affect and low negative affect) and a cognitive component (satisfaction with life). It is proposed that an individual experiences happiness when positive affect and satisfaction with life are both high.

Eudaimonic wellbeing, on the other hand, is strongly reliant on Maslow's ideas of self actualisation and Roger's concept of the fully functioning person and their subjective well being. Eudaimonic happiness is therefore based on the premise that people feel happy if they experience life purpose, challenges and growth. This approach adopts Self-Determination Theory to conceptualise happiness. Self determination theory suggests that happiness is related to fulfilment in the areas of autonomy and competence.

From this perspective, by engaging in eudaimonic pursuits, subjective well being (happiness) will occur as a by product. Thus, life purpose and higher order meaning are believed to produce happiness. It appears that the general consensus is that happiness does not result from the pursuit of pleasure but from the development of individual strengths

and virtues which ties in with the concept of positive psychology. The differences between eudaimonic and hedonic happiness are listed below.

Hedonic (Subjective Wellbeing)

- Presence of positive mood
- Absence of negative mood
- Satisfaction with various domains of life (e.g. work, leisure)
- Global life satisfaction

Eudaimonic (Psychological Wellbeing)

- Sense of control or autonomy
- Feeling of meaning and purpose
- Personal expressiveness
- Feelings of belongingness
- Social contribution
- Competence
- Personal growth
- Self acceptance

views happiness from both the hedonistic and eudaimonic view in which they define happiness in terms of the pleasant life, the good life and the meaningful life. Peterson et al., identified three pathways to happiness from the positive psychological view:

Pleasure is the process of maximising positive emotion and minimising negative emotion and is referred to as the pleasant life which involves enjoyable and positive experiences.

Engagement is the process of being immersed and absorbed in the task at hand and is referred to as the good life which involves being actively involved in life and all that it requires and demands. Thus the good life is considered to result from the individual cultivating and investing their signature strengths and virtues into their relationships, work and leisure (Seligman, 2002) thus applying the best of self during challenging activities that results in growth and a feeling of competence and satisfaction that brings about happiness.

Meaning is the process of having a higher purpose in life than our selves and is referred to as the meaningful life which involves using our strengths and personal qualities to serve this higher purpose. The meaningful life, like the good life, involves the individual applying their signature strengths in activities, but the difference is that these activities are perceived to contribute to the greater good in the meaningful life.

Character Strengths

The study was designed to explore the relationships among character strengths, strengths use, future self-continuity and subjective well-being. A total of 225 undergraduates completed paper-and-pencil questionnaires assessing character strengths, strengths use, future self-continuity, and subjective well-being. Results suggested several character strengths were correlated with subjective well-being and the strongest correlations were found for hope, curiosity, zest, perseverance and love. All character strengths were sig-

nificantly correlated with strengths use. Strengths use and future self-continuity were robustly correlated with subjective well-being. The mediation analysis showed that strengths use mediates the relationship between character strengths and subjective well-being, and specifically, the indirect effects of strengths use varies from different character strengths. The moderated mediator suggested that future self-continuity moderated the mediation of strengths use because future self-continuity moderates the effect of strengths use on subjective well-being. Furthermore, the indirect effect of strengths use was stronger with high level of future self-continuity than those with low level of future self-continuity. The present findings make a contribution to understand the underlying mechanisms involving in character strengths are associated with higher level of well-being. Additionally, the findings expand knowledge about future self-continuity and its relation to strengths use and subjective well-being among undergraduates, having significant implications in the educational context.

Resilience

Wellbeing and resilience are important in preventing the onset of mental health problems as well as potentially lessening the severity of existing mental health problems. Wellbeing and resilience are vital to developing efficient problem solving skills, building and maintaining interpersonal relationships and realistic goal setting, all of which greatly enhance an individual's ability to perform and contribute meaningfully in daily life.

When to focus on wellbeing and resilience:

Resilience is always important, but it can particularly equip people for times of change and stress such as:

life transitions and changes, such as starting at a new school.

after the breakdown of a relationship, or a family.

if diagnosed with, or worried about a mental health problem (such as anxiety, or an eating disorder).

Wellbeing and Resilience

Good health is about more than just the absence of sickness. Focusing on wellbeing and building resiliency is important in establishing a holistic approach to health, addressing both physical and psychological states.

The World Health Organisation defines wellbeing as “the state in which an individual realises his or her own abilities, can cope with normal stresses of life, can work productively, and is able to make a contribution to his or her own community”. Wellbeing involves having positive self image and esteem.

Resilience, which is directly related to wellbeing, is about having the ability to cope with and adapt to new situations. Having a sense of resilience and positive wellbeing enables a person to approach other people and situations with confidence and optimism, which is especially important for young people given the enormous changes that occur with the transition into adolescence and adulthood.

Signs of resilience and wellbeing:

confidence to approach new situations and approach new people

realistic optimism

avoiding constant self-blame

ability to set goals
positive self image and self esteem.

What Young People can do to Focus on Wellbeing and Resilience

In a practical sense developing and improving on setting realistic and achievable goals, problem solving and social skills all contribute to wellbeing and resiliency. Other important skills include identifying and becoming aware of one's own strengths and weaknesses.

Setting realistic, achievable and measurable goals is a great way to promote self-efficacy. Setting goals should be motivating. One way to increase motivation (and minimise feeling pressure or 'failure') is to set sub-goals, smaller and achievable related targets.

Understanding and avoiding negative self-talk as well as actively practicing positive self-talk, is an important tool for building self-esteem. Self-talk, which is essentially internal reflections on personal ability and/or image, can greatly influence self esteem and perceptions about personal ability.

Developing and focusing on interpersonal skills, especially learning how to engage with people from different backgrounds, is a valuable tool for young people that will enhance self esteem and ability to maintain personal and fulfilling relationships.

Post-Traumatic Growth

You've probably heard of post-traumatic growth. It's been touted in the New York Times ("the surprisingly positive flip side of PTSD"), studied in countless empirical papers, and sermonized in TED talks, books, and academic research centers. It's one of those zeitgeist ideas upon which entire empires are built: First, we had PTSD, but now we have post-traumatic growth.

The basic principle is this: Something horrendous happens (a mugging, a serious car crash, a cancer diagnosis), and in the weeks and months that follow, we have trouble not thinking about the experience. We avoid reminders of it, and we have intrusive memories that pop into our heads and ratchet up our anxiety. We are debilitated by these symptoms. But at some unspecified time later—perhaps years after—we grow from that trauma, and we become a better person. We become wiser, stronger, more accepting of others; we develop closer relationships and have more compassion for others; we change our philosophy of living, re-evaluate our priorities, and become more spiritual. All in all, we are better, because we were traumatized.

It's an immensely appealing idea. Who wouldn't want it to be true?

The only problem is: There's very little evidence for it. Wait, didn't you just say that countless empirical papers have been published on post-traumatic growth?

Yes, but here's what almost all of those studies do. They ask people who have experienced a potentially traumatic event whether they are better off because of it. And what do you know, many people report that they are indeed a better person. They have grown. They see new possibilities in life. They are able to find in their suffering a bright kernel of redemption.

But does this mean they actually are better? In other words, are they actually wiser, more compassionate, and closer to others? Or do they just perceive that they are?

Perceived Growth and Actual Growth are two Different Things

It turns out that it is very difficult to separate these two things: the perception that we are better and the actuality that we are. In fact, almost no studies separate perceived and actual growth for a very good reason. It is extraordinarily difficult to know how someone is doing before a trauma occurs.

One remarkable study did just that: They measured a large sample of undergraduates ($N = 1,528$) at the beginning and the end of a semester. They then identified a group of participants who experienced a traumatic event during the semester that caused considerable distress ($n = 122$). They asked whether they had grown from the trauma (including "I have a greater feeling of self-reliance" or "I am able to do better things with my life"). In theory, among this trauma-exposed group, if you perceived that you grew from the trauma, you should actually show improvements in your overall functioning. In other words, when you say you think you are better (perceived growth), you are in fact better (actual growth). Well, the researchers found that the perception of growth was unrelated to actual growth. Moreover, the perception of growth was linked to more distress at the semester's end.

In short, just because someone believes that they are better off does not mean that they are. In fact, it may very well mean that they are not.

Perceived Growth Is Likely a "Positive Illusion"

Why would someone perceive that they have grown when they have not? One explanation is that perceiving growth is a way of coping with the event itself. In this framework, post-traumatic growth isn't growth at all. It's a "motivated positive illusion" whose purpose is to protect us from the possibility that we may have been damaged. In fact, one unusually rigorous experimental study found that when an event threatens our sense of self, we are more likely to believe that the event made us better in some way.

Alas, this coping strategy is ineffective. People who perceive growth tend to do worse over the short and the long term compared to people who don't. How do we know? One recent study looked at soldiers returning from a deployment to Iraq. They found that soldiers who reported posttraumatic growth five months after returning home saw an increase in PTSD symptoms at 15 months. Another study published this year on survivors of the Oslo bombings, the horrific 2011 massacre in Norway, found the exact same result—early posttraumatic growth = later PTSD. In short, perceiving growth portends worse functioning, not better.

HEALTH

Health Promoting and Health Compromising Behaviors

Health promotion is, as stated in the 1986 World Health Organization (WHO) Ottawa Charter for Health Promotion, "the process of enabling people to increase control over, and to improve, their health. To reach a state of complete physical, mental and social well-being, an individual or group must be able to identify and to realize aspirations, to satisfy needs, and to change or cope with the environment. Health is, therefore, seen as a resource for everyday life, not the objective of living. Health is a positive concept emphasizing social and personal resources, as well as physical capacities. Therefore, health pro-

motion is not just the responsibility of the health sector, but goes beyond healthy lifestyles to well-being".

Life Style and Chronic Diseases

Diabetes

Keeping your blood sugar levels within the range recommended by your doctor can be challenging. That's because many things make your blood sugar levels change, sometimes unexpectedly. Following are some factors that can affect your blood sugar levels.

Food

Healthy eating is a cornerstone of healthy living — with or without diabetes. But if you have diabetes, you need to know how foods affect your blood sugar levels. It's not only the type of food you eat but also how much you eat and the combinations of food types you eat.

What to do:

Learn about carbohydrate counting and portion sizes. A key to many diabetes management plans is learning how to count carbohydrates. Carbohydrates are the foods that often have the biggest impact on your blood sugar levels. And for people taking mealtime insulin, it's crucial to know the amount of carbohydrates in your food, so you get the proper insulin dose.

Learn what portion size is appropriate for each type of food. Simplify your meal planning by writing down portions for the foods you eat often. Use measuring cups or a scale to ensure proper portion size and an accurate carbohydrate count.

Make every meal well-balanced. As much as possible, plan for every meal to have a good mix of starches, fruits and vegetables, proteins and fats. It's especially important to pay attention to the types of carbohydrates you choose. Some carbohydrates, such as fruits, vegetables and whole grains, are better for you than are others. These foods are low in carbohydrates and contain fiber that helps keep your blood sugar levels more stable. Talk to your doctor, nurse or dietitian about the best food choices and the appropriate balance of food types.

Coordinate your meals and medications. Too little food in proportion to your diabetes medications – especially insulin – may result in dangerously low blood sugar (hypoglycemia). Too much food may cause your blood sugar level to climb too high (hyperglycemia). Talk to your diabetes health care team about how to best coordinate meal and medication schedules.

Avoid sugar-sweetened beverages. Sugar-sweetened beverages – including those sweetened with high fructose corn syrup or sucrose – tend to be high in calories and offer little in the way of nutrition. And because they cause blood sugar to rise quickly, it's best to avoid these types of drinks if you have diabetes.

The exception is if you are experiencing a low blood sugar level. Sugar-sweetened beverages, such as soda, juice and sports drinks, can be used as an effective treatment for quickly raising blood sugar that is too low.

Exercise

Physical activity is another important part of your diabetes management plan. When you exercise, your muscles use sugar (glucose) for energy. Regular physical activity also helps your body use insulin more efficiently.

These factors work together to lower your blood sugar level. The more strenuous your workout, the longer the effect lasts. But even light activities — such as housework, gardening or being on your feet for extended periods — can improve your blood sugar.

What to do:

Talk to your doctor about an exercise plan. Ask your doctor about what type of exercise is appropriate for you. In general, most adults should exercise at least 30 minutes a day on most days of the week. If you've been inactive for a long time, your doctor may want to check your overall health before advising you. He or she can recommend the right balance of aerobic and muscle-strengthening exercise.

Keep an exercise schedule. Talk to your doctor about the best time of day for you to exercise so that your workout routine is coordinated with your meal and medication schedules.

Know your numbers. Talk to your doctor about what blood sugar levels are appropriate for you before you begin exercise.

Check your blood sugar level. Check your blood sugar level before, during and after exercise, especially if you take insulin or medications that lower blood sugar. Exercise can lower your blood sugar levels even a day later, especially if the activity is new to you, or if you're exercising at a more intensive level. Be aware of warning signs of low blood sugar, such as feeling shaky, weak, tired, hungry, lightheaded, irritable, anxious or confused.

If you use insulin and your blood sugar level is below 100 milligrams per deciliter (mg/dL), or 5.6 millimoles per liter (mmol/L), have a small snack before you start exercising to prevent a low blood sugar level.

Stay hydrated. Drink plenty of water or other fluids while exercising because dehydration can affect blood sugar levels.

Be prepared. Always have a small snack or glucose tablets with you during exercise in case your blood sugar level drops too low. Wear a medical identification bracelet when you're exercising.

Adjust your diabetes treatment plan as needed. If you take insulin, you may need to reduce your insulin dose before exercising, or wait awhile after exercise to inject insulin. Your doctor can advise you on appropriate changes in your medication. You may also need to adjust treatment if you've increased your exercise routine.

Medication

Insulin and other diabetes medications are designed to lower your blood sugar levels when diet and exercise alone aren't sufficient for managing diabetes. But the effectiveness of these medications depends on the timing and size of the dose. Medications you take for conditions other than diabetes also can affect your blood sugar levels.

What to do:

Store insulin properly. Insulin that's improperly stored or past its expiration date may not be effective. Insulin is especially sensitive to extremes in temperature.

Report problems to your doctor. If your diabetes medications cause your blood sugar level to drop too low or if it's consistently too high, the dosage or timing may need to be adjusted.

Be cautious with new medications. If you're considering an over-the-counter medication or your doctor prescribes a new drug to treat another condition — such as high blood pressure or high cholesterol — ask your doctor or pharmacist if the medication may affect your blood sugar levels. Liquid medications may be sweetened with sugar to cover their taste. Sometimes an alternate medication may be recommended. Always check with your doctor before taking any new over-the-counter medication, so you know how it may impact your blood sugar level.

Illness

When you're sick, your body produces stress-related hormones that help your body fight the illness, but they also can raise your blood sugar level. Changes in your appetite and normal activity also may complicate diabetes management.

What to do:

Plan ahead. Work with your health care team to create a sick-day plan. Include instructions on what medications to take, how often to measure your blood sugar and urine ketone levels, how to adjust your medication dosages, and when to call your doctor.

Continue to take your diabetes medication. However, if you're unable to eat because of nausea or vomiting, contact your doctor. In these situations, you may need to adjust your insulin dose or temporarily stop taking your medication because of a risk of hypoglycemia.

Stick to your diabetes meal plan. If you can, eating as usual will help you control your blood sugar levels. Keep a supply of foods that are easy on your stomach, such as gelatin, crackers, soups and applesauce. Drink lots of water or other fluids that don't add calories, such as tea, to make sure you stay hydrated. If you're taking insulin, you may need to sip sugar-sweetened beverages, such as juice or a sports drink, to keep your blood sugar level from dropping too low.

Alcohol

The liver normally releases stored sugar to counteract falling blood sugar levels. But if your liver is busy metabolizing alcohol, your blood sugar level may not get the boost it needs from your liver. Alcohol can result in low blood sugar shortly after you drink it and for as many as 24 hours more.

What to do:

Get your doctor's OK to drink alcohol. Alcohol can aggravate diabetes complications, such as nerve damage and eye disease. But if your diabetes is under control and your doctor agrees, an occasional alcoholic drink is fine. Moderate alcohol consumption is defined as no more than one drink a day for women of any age and men over 65 years old and two drinks a day for men under 65. One drink equals a 12-ounce beer, 5 ounces of wine or 1.5 ounces of distilled spirits.

Don't drink alcoholic beverages on an empty stomach. If you take insulin or other diabetes medications, be sure to eat before you drink or drink with a meal to prevent low blood sugar.

Choose your drinks carefully. Light beer and dry wines have fewer calories and carbohydrates than do other alcoholic drinks. If you prefer mixed drinks, sugar-free mixers — such as diet soda, diet tonic, club soda or seltzer — won't raise your blood sugar.

Tally your calories. Remember to include the calories from any alcohol you drink in your daily calorie count. Ask your doctor or dietitian how to incorporate calories and carbohydrates from alcoholic drinks into your diet plan.

Check your blood sugar level before bed. Because alcohol can lower blood sugar levels long after you've had your last drink, check your blood sugar level before you go to sleep. If your blood sugar isn't between 100 and 140 mg/dL (5.6 and 7.8 mmol/L), have a snack before bed to counter a drop in your blood sugar level.

Menstruation and Menopause

Changes in hormone levels the week before and during menstruation can result in significant fluctuations in blood sugar levels. And in the few years before and during menopause, hormone changes may result in unpredictable variations in blood sugar levels that complicate diabetes management.

What to do:

Look for patterns. Keep careful track of your blood sugar readings from month to month. You may be able to predict fluctuations related to your menstrual cycle.

Adjust your diabetes treatment plan as needed. Your doctor may recommend changes in your meal plan, activity level or diabetes medications to make up for blood sugar variation.

Check blood sugar more frequently. If you're likely approaching menopause or experiencing menopause, talk to your doctor about whether you need to monitor your blood sugar level more often. Symptoms of menopause can sometimes be confused with symptoms of low blood sugar, so whenever possible, check your blood sugar before treating a suspected low to confirm the low blood sugar level.

Most forms of birth control can be used by women with diabetes without a problem. However, oral contraceptives may raise blood sugar levels in some women.

Stress

If you're stressed, the hormones your body produces in response to prolonged stress may cause a rise in your blood sugar level. Additionally, it may be harder to closely follow your usual diabetes management routine if you're under a lot of extra pressure.

What to do:

Look for patterns. Log your stress level on a scale of 1 to 10 each time you log your blood sugar level. A pattern may soon emerge.

Take control. Once you know how stress affects your blood sugar level, fight back. Learn relaxation techniques, prioritize your tasks and set limits. Whenever possible, avoid common stressors. Exercise can often help relieve stress and lower your blood sugar level.

Get help. Learn new strategies for coping with stress. You may find that working with a psychologist or clinical social worker can help you identify stressors, solve stressful problems or learn new coping skills.

The more you know about factors that influence your blood sugar level, the more you can anticipate fluctuations — and plan accordingly. If you're having trouble keeping your blood sugar level in your target range, ask your diabetes health care team for help.

Hypertension

Hypertension (HTN or HT), also known as high blood pressure (HBP), is a long-term medical condition in which the blood pressure in the arteries is persistently elevated. High blood pressure typically does not cause symptoms. Long-term high blood pressure, however, is a major risk factor for coronary artery disease, stroke, heart failure, atrial fibrillation, peripheral vascular disease, vision loss, chronic kidney disease, and dementia.

High blood pressure is classified as either primary (essential) high blood pressure or secondary high blood pressure. About 90–95% of cases are primary, defined as high blood pressure due to nonspecific lifestyle and genetic factors. Lifestyle factors that increase the risk include excess salt in the diet, excess body weight, smoking, and alcohol use. The remaining 5–10% of cases are categorized as secondary high blood pressure, defined as high blood pressure due to an identifiable cause, such as chronic kidney disease, narrowing of the kidney arteries, an endocrine disorder, or the use of birth control pills.

Blood pressure is expressed by two measurements, the systolic and diastolic pressures, which are the maximum and minimum pressures, respectively. For most adults, normal blood pressure at rest is within the range of 100–130 millimeters mercury (mmHg) systolic and 60–80 mmHg diastolic. For most adults, high blood pressure is present if the resting blood pressure is persistently at or above 130/80 or 140/90 mmHg. Different numbers apply to children. Ambulatory blood pressure monitoring over a 24-hour period appears more accurate than office-based blood pressure measurement.

Lifestyle changes and medications can lower blood pressure and decrease the risk of health complications. Lifestyle changes include weight loss, physical exercise, decreased salt intake, reducing alcohol intake, and a healthy diet. If lifestyle changes are not sufficient then blood pressure medications are used. Up to three medications can control blood pressure in 90% of people. The treatment of moderately high arterial blood pressure (defined as >160/100 mmHg) with medications is associated with an improved life expectancy. The effect of treatment of blood pressure between 130/80 mmHg and 160/100 mmHg is less clear, with some reviews finding benefit and others finding unclear benefit. High blood pressure affects between 16 and 37% of the population globally. In 2010 hypertension was believed to have been a factor in 18% of all deaths (9.4 million globally).

Coronary Heart Disease

Coronary artery disease develops when the major blood vessels that supply your heart with blood, oxygen and nutrients (coronary arteries) become damaged or diseased. Cholesterol-containing deposits (plaque) in your arteries and inflammation are usually to blame for coronary artery disease.

When plaque builds up, it narrows your coronary arteries, decreasing blood flow to your heart. Eventually, the decreased blood flow may cause chest pain (angina), shortness of breath, or other coronary artery disease signs and symptoms. A complete blockage can cause a heart attack.

Because coronary artery disease often develops over decades, you might not notice a problem until you have a significant blockage or a heart attack. But there's plenty you can do to prevent and treat coronary artery disease. A healthy lifestyle can make a big impact.

Symptoms

If your coronary arteries narrow, they can't supply enough oxygen-rich blood to your heart — especially when it's beating hard, such as during exercise. At first, the decreased blood flow may not cause any coronary artery disease symptoms. As plaque continues to build up in your coronary arteries, however, you may develop coronary artery disease signs and symptoms, including:

Chest pain (angina). You may feel pressure or tightness in your chest, as if someone were standing on your chest. This pain, referred to as angina, usually occurs on the middle or left side of the chest. Angina is generally triggered by physical or emotional stress.

The pain usually goes away within minutes after stopping the stressful activity. In some people, especially women, this pain may be fleeting or sharp and felt in the neck, arm or back.

Shortness of breath. If your heart can't pump enough blood to meet your body's needs, you may develop shortness of breath or extreme fatigue with exertion.

Heart attack. A completely blocked coronary artery will cause a heart attack. The classic signs and symptoms of a heart attack include crushing pressure in your chest and pain in your shoulder or arm, sometimes with shortness of breath and sweating.

Women are somewhat more likely than men are to experience less typical signs and symptoms of a heart attack, such as neck or jaw pain. Sometimes a heart attack occurs without any apparent signs or symptoms.

When to see a doctor

If you suspect you're having a heart attack, immediately call 911 or your local emergency number. If you don't have access to emergency medical services, have someone drive you to the nearest hospital. Drive yourself only as a last resort.

If you have risk factors for coronary artery disease — such as high blood pressure, high cholesterol, tobacco use, diabetes, a strong family history of heart disease or obesity — talk to your doctor. He or she may want to test you for the condition, especially if you have signs or symptoms of narrowed arteries.

Causes

Coronary artery disease is thought to begin with damage or injury to the inner layer of a coronary artery, sometimes as early as childhood. The damage may be caused by various factors, including:

- Smoking
- High blood pressure
- High cholesterol
- Diabetes or insulin resistance
- Sedentary lifestyle

Once the inner wall of an artery is damaged, fatty deposits (plaque) made of cholesterol and other cellular waste products tend to accumulate at the site of injury in a process called atherosclerosis. If the surface of the plaque breaks or ruptures, blood cells called platelets will clump at the site to try to repair the artery. This clump can block the artery, leading to a heart attack.

Risk Factors

Risk factors for coronary artery disease include:

Age. Simply getting older increases your risk of damaged and narrowed arteries.

Sex. Men are generally at greater risk of coronary artery disease. However, the risk for women increases after menopause.

Family history. A family history of heart disease is associated with a higher risk of coronary artery disease, especially if a close relative developed heart disease at an early age. Your risk is highest if your father or a brother was diagnosed with heart disease before age 55 or if your mother or a sister developed it before age 65.

Smoking. People who smoke have a significantly increased risk of heart disease. Exposing others to your secondhand smoke also increases their risk of coronary artery disease.

High blood pressure. Uncontrolled high blood pressure can result in hardening and thickening of your arteries, narrowing the channel through which blood can flow.

High blood cholesterol levels. High levels of cholesterol in your blood can increase the risk of formation of plaque and atherosclerosis. High cholesterol can be caused by a high level of low-density lipoprotein (LDL) cholesterol, known as the "bad" cholesterol. A low level of high-density lipoprotein (HDL) cholesterol, known as the "good" cholesterol, can also contribute to the development of atherosclerosis.

Diabetes. Diabetes is associated with an increased risk of coronary artery disease. Type 2 diabetes and coronary artery disease share similar risk factors, such as obesity and high blood pressure.

Overweight or obesity. Excess weight typically worsens other risk factors.

Physical inactivity. Lack of exercise also is associated with coronary artery disease and some of its risk factors, as well.

High stress. Unrelieved stress in your life may damage your arteries as well as worsen other risk factors for coronary artery disease.

Unhealthy diet. Eating too much food that has high amounts of saturated fat, trans fat, salt and sugar can increase your risk of coronary artery disease.

Risk factors often occur in clusters and may build on one another, such as obesity leading to type 2 diabetes and high blood pressure. When grouped together, certain risk factors put you at an even greater risk of coronary artery disease. For example, metabolic syndrome — a cluster of conditions that includes elevated blood pressure, high triglycerides, low HDL, or "good," cholesterol, elevated insulin levels and excess body fat around the waist — increases the risk of coronary artery disease.

Sometimes coronary artery disease develops without any classic risk factors. Researchers are studying other possible factors, including:

Sleep apnea. This disorder causes you to repeatedly stop and start breathing while you're sleeping. Sudden drops in blood oxygen levels that occur during sleep apnea increase blood pressure and strain the cardiovascular system, possibly leading to coronary artery disease.

High sensitivity C-reactive protein. High sensitivity C-reactive protein (hs-CRP) is a normal protein that appears in higher amounts when there's inflammation somewhere in your body. High hs-CRP levels may be a risk factor for heart disease.

It's thought that as coronary arteries narrow, you'll have more hs-CRP in your blood.

High triglycerides. This is a type of fat (lipid) in your blood. High levels may raise the risk of coronary artery disease, especially for women.

Homocysteine. Homocysteine is an amino acid your body uses to make protein and to build and maintain tissue. But high levels of homocysteine may increase your risk of coronary artery disease.

Preeclampsia. This condition that can develop in women during pregnancy causes high blood pressure and a higher amount of protein in urine. It can lead to a higher risk of heart disease later in life.

Alcohol use. Heavy alcohol use can lead to heart muscle damage. It can also worsen other risk factors of coronary artery disease.

Autoimmune diseases. Conditions such as rheumatoid arthritis and lupus (and other inflammatory rheumatologic conditions) have an increased risk of atherosclerosis.

Complications

Coronary artery disease can lead to:

Chest pain (angina). When your coronary arteries narrow, your heart may not receive enough blood when demand is greatest — particularly during physical activity. This can cause chest pain (angina) or shortness of breath.

Heart attack. If a cholesterol plaque ruptures and a blood clot forms, complete blockage of your heart artery may trigger a heart attack. The lack of blood flow to your heart may damage your heart muscle. The amount of damage depends in part on how quickly you receive treatment.

Heart failure. If some areas of your heart are chronically deprived of oxygen and nutrients because of reduced blood flow, or if your heart has been damaged by a heart attack, your heart may become too weak to pump enough blood to meet your body's needs. This condition is known as heart failure.

Abnormal heart rhythm (arrhythmia). Inadequate blood supply to the heart or damage to heart tissue can interfere with your heart's electrical impulses, causing abnormal heart rhythms.

Prevention

The same lifestyle habits that can help treat coronary artery disease can also help prevent it from developing in the first place. Leading a healthy lifestyle can help keep your arteries strong and clear of plaque. To improve your heart health, you can:

Quit smoking.

Control conditions such as high blood pressure, high cholesterol and diabetes.

Stay physically active.

Eat a low-fat, low-salt diet that's rich in fruits, vegetables and whole grains.

Maintain a healthy weight.

Reduce and manage stress.

Psychoneuroimmunology [Cancer, HIV/AIDS]

Psychoneuroimmunology (PNI), also referred to as psychoendoneuroimmunology (PENI) or psychoneuroendocrinoimmunology (PNEI), is the study of the interaction between psychological processes and the nervous and immune systems of the human body. PNI takes an interdisciplinary approach, incorporating psychology, neuroscience, immunology, physiology, genetics, pharmacology, molecular biology, psychiatry, behavioral medicine, infectious diseases, endocrinology, and rheumatology.

The main interests of PNI are the interactions between the nervous and immune systems and the relationships between mental processes and health. PNI studies, among other things, the physiological functioning of the neuroimmune system in health and disease; disorders of the neuroimmune system (autoimmune diseases; hypersensitivities; immune deficiency); and the physical, chemical and physiological characteristics of the components of the neuroimmune system *in vitro*, *in situ*, and *in vivo*.

Cancer

Cancer is a group of diseases involving abnormal cell growth with the potential to invade or spread to other parts of the body. These contrast with benign tumors, which do not spread. Possible signs and symptoms include a lump, abnormal bleeding, prolonged cough, unexplained weight loss, and a change in bowel movements. While these symptoms may indicate cancer, they can also have other causes. Over 100 types of cancers affect humans.

Tobacco use is the cause of about 22% of cancer deaths. Another 10% are due to obesity, poor diet, lack of physical activity or excessive drinking of alcohol. Other factors include certain infections, exposure to ionizing radiation and environmental pollutants. In the developing world, 15% of cancers are due to infections such as *Helicobacter pylori*, hepatitis B, hepatitis C, human papillomavirus infection, Epstein–Barr virus and human immunodeficiency virus (HIV). These factors act, at least partly, by changing the genes of a cell. Typically, many genetic changes are required before cancer develops. Approximately 5–10% of cancers are due to inherited genetic defects from a person's parents. Cancer can be detected by certain signs and symptoms or screening tests. It is then typically further investigated by medical imaging and confirmed by biopsy.

Many cancers can be prevented by not smoking, maintaining a healthy weight, not drinking too much alcohol, eating plenty of vegetables, fruits and whole grains, vaccination against certain infectious diseases, not eating too much processed and red meat and avoiding too much sunlight exposure. Early detection through screening is useful for cervical and colorectal cancer. The benefits of screening in breast cancer are controversial. Cancer is often treated with some combination of radiation therapy, surgery, chemotherapy and targeted therapy. Pain and symptom management are an important part of care. Palliative care is particularly important in people with advanced disease. The chance of survival depends on the type of cancer and extent of disease at the start of treatment. In children under 15 at diagnosis, the five-year survival rate in the developed world is on average 80%. For cancer in the United States, the average five-year survival rate is 66%.

In 2015, about 90.5 million people had cancer. About 14.1 million new cases occur a year (not including skin cancer other than melanoma). It caused about 8.8 million deaths (15.7% of deaths). The most common types of cancer in males are lung cancer, prostate cancer, colorectal cancer and stomach cancer. In females, the most common types are

breast cancer, colorectal cancer, lung cancer and cervical cancer. If skin cancer other than melanoma were included in total new cancer cases each year, it would account for around 40% of cases. In children, acute lymphoblastic leukemia and brain tumors are most common, except in Africa where non-Hodgkin lymphoma occurs more often. In 2012, about 165,000 children under 15 years of age were diagnosed with cancer. The risk of cancer increases significantly with age, and many cancers occur more commonly in developed countries. Rates are increasing as more people live to an old age and as lifestyle changes occur in the developing world. The financial costs of cancer were estimated at \$1.16 trillion USD per year as of 2010.

HIV/AIDS

Human immunodeficiency virus infection and acquired immune deficiency syndrome (HIV/AIDS) is a spectrum of conditions caused by infection with the human immunodeficiency virus (HIV). Following initial infection, a person may not notice any symptoms or may experience a brief period of influenza-like illness. Typically, this is followed by a prolonged period with no symptoms. As the infection progresses, it interferes more with the immune system, increasing the risk of developing common infections such as tuberculosis, as well as other opportunistic infections, and tumors that rarely affect people who have uncompromised immune systems. These late symptoms of infection are referred to as acquired immunodeficiency syndrome (AIDS). This stage is often also associated with unintended weight loss.

HIV is spread primarily by unprotected sex (including anal and oral sex), contaminated blood transfusions, hypodermic needles, and from mother to child during pregnancy, delivery, or breast-feeding. Some bodily fluids, such as saliva and tears, do not transmit HIV. Methods of prevention include safe sex, needle exchange programs, treating those who are infected, pre- and post-exposure prophylaxis, and male circumcision. Disease in a baby can often be prevented by giving both the mother and child antiretroviral medication. There is no cure or vaccine; however, antiretroviral treatment can slow the course of the disease and may lead to a near-normal life expectancy. Treatment is recommended as soon as the diagnosis is made. Without treatment, the average survival time after infection is 11 years.

In 2016, about 36.7 million people were living with HIV and it resulted in 1 million deaths. There were 300,000 fewer new HIV cases in 2016 than in 2015. Most of those infected live in sub-Saharan Africa. From the time AIDS was identified in the early 1980s to 2017, the disease has caused an estimated 35 million deaths worldwide. HIV/AIDS is considered a pandemic—a disease outbreak which is present over a large area and is actively spreading. HIV originated in west-central Africa during the late 19th or early 20th century. AIDS was first recognized by the United States Centers for Disease Control and Prevention (CDC) in 1981 and its cause—HIV infection—was identified in the early part of the decade.

HIV/AIDS has had a large impact on society, both as an illness and as a source of discrimination. The disease also has large economic impacts. There are many misconceptions about HIV/AIDS such as the belief that it can be transmitted by casual non-sexual contact. The disease has become subject to many controversies involving religion including the Catholic Church's position not to support condom use as prevention. It has attracted international medical and political attention as well as large-scale funding since it was identified in the 1980s.

PSYCHOLOGY AND TECHNOLOGY INTERFACE

Digital Learning

Digital Learning is "learning facilitated by technology that gives students some element of control over time, place, path and/or pace."

Time. Learning is no longer restricted to the school day or the school year. The Internet and a proliferation of Internet access devices have given students the ability to learn anytime.

Place. Learning is no longer restricted within the walls of a classroom. The Internet and a proliferation of Internet access devices have given students the ability to learn anywhere and everywhere.

Path. Learning is no longer restricted to the pedagogy used by the teacher. Interactive and adaptive software allows students to learn in their own style, making learning personal and engaging. New learning technologies provide real-time data that gives teachers the information they need to adjust instruction to meet the unique needs of each student.

Pace. Learning is no longer restricted to the pace of an entire classroom of students. Interactive and adaptive software allows students to learn at their own pace, spending more or less time on lessons or subjects to achieve the same level of learning.

Digital learning is more than just providing students with a laptop. Digital learning requires a combination of technology, digital content and instruction.

Technology. Technology is the mechanism that delivers content. It facilitates how students receive content. It includes Internet access and hardware, which can be any Internet access device – from a desktop to a laptop to an iPad to a smartphone. Technology is the tool, not the instruction.

Digital Content. Digital content is the high quality academic material which is delivered through technology. It is what students learn. It ranges from new engaging, interactive and adaptive software to classic literature to video lectures to games. It isn't simply a PDF of text or a PowerPoint presentation.

Instruction. Educators are essential to digital learning. Technology may change the role of the teacher but it will never eliminate the need for a teacher. With digital learning, teachers will be able to provide the personalized guidance and assistance to ensure students learn and stay on track – throughout the year and year after year – to graduate from high school. Teachers may be the guide on the side, not the sage on the stage.

Digital Etiquette

Digital Etiquette is integrity, or sense of self management, and general responsibility for your actions as well as yourself on the internet and treating people like you would in real life. Some basic digital etiquette would be like respecting others, following the rules of the website you're using, or just being kind to others. This probably sounds familiar and if you haven't caught on yet these are the ways you probably treat people in real life, or at least how you're supposed to treat them. Also, it's about how you write messages online; even if you didn't intend it to be mean and was only joking, some people can take it the

wrong way because the empathy or feelings can't be transferred into the messages. Like when you write in caps lock, people assume that you are yelling at them or are angry.

Cyber Bullying

Cyberbullying is bullying that takes place over digital devices like cell phones, computers, and tablets. Cyberbullying can occur through SMS, Text, and apps, or online in social media, forums, or gaming where people can view, participate in, or share content. Cyberbullying includes sending, posting, or sharing negative, harmful, false, or mean content about someone else. It can include sharing personal or private information about someone else causing embarrassment or humiliation. Some cyberbullying crosses the line into unlawful or criminal behavior.

The most common places where cyberbullying occurs are:

Social Media, such as Facebook, Instagram, Snapchat, and Twitter.

SMS (Short Message Service) also known as Text Message sent through devices.

Instant Message (via devices, email provider services, apps, and social media messaging features).

Email

Special Concerns

With the prevalence of social media and digital forums, comments, photos, posts, and content shared by individuals can often be viewed by strangers as well as acquaintances. The content an individual shares online – both their personal content as well as any negative, mean, or hurtful content – creates a kind of permanent public record of their views, activities, and behavior. This public record can be thought of as an online reputation, which may be accessible to schools, employers, colleges, clubs, and others who may be researching an individual now or in the future. Cyberbullying can harm the online reputations of everyone involved – not just the person being bullied, but those doing the bullying or participating in it. Cyberbullying has unique concerns in that it can be:

Persistent. Digital devices offer an ability to immediately and continuously communicate 24 hours a day, so it can be difficult for children experiencing cyberbullying to find relief.

Permanent. Most information communicated electronically is permanent and public, if not reported and removed. A negative online reputation, including for those who bully, can impact college admissions, employment, and other areas of life.

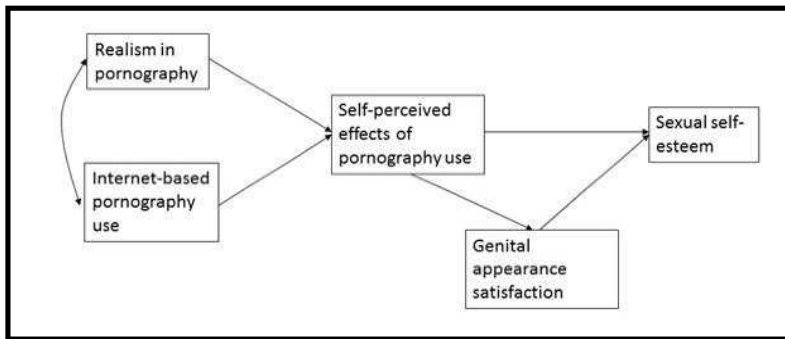
Hard to Notice. Because teachers and parents may not overhear or see cyberbullying taking place, it is harder to recognize.

Laws and Sanctions

Although all states have laws requiring schools to respond to bullying, many states do not include cyberbullying under these laws or specify the role schools should play in responding to bullying that takes place outside of school. Schools may take action either as required by law, or with local or school policies that allow them to discipline or take other action. Some states also have provisions to address bullying if it affects school performance. You can learn about the laws and policies in each state, including if they cover cyberbullying.

Cyber Pornography: Consumption, Implications

During the past two decades, there seems to have been a mainstreaming of pornography in the mass culture, for instance, in music videos and women's literature (Comella, 2013). Continuous advancements in technology have made Internet pornography increasingly more accessible to young adults worldwide. In the past decade, Swedish and Norwegian researchers have concluded that the majority of young adults have been exposed to pornography and report positive effects of using it. Nevertheless, research on the impact of Internet pornography use continues to be deficient in several areas. Stulhofer et al. (2007) suggested that a new research agenda should have a greater focus on individuals' well-being and avoid ideological framing as much as possible. In addition, the role of the genre (type of pornographic content) should be included in the design of research on pornography. This recommendation involves a shift from the traditional focus on the relationship between watching pornography and sexual risk behavior to the potential for positive outcomes of using mainstream and non-mainstream pornography. Wright, Bae, and Funk (2013) also recommended that future research on pornography broaden its scope to include women in general, and more specifically, to include individuals' perceptions of pornography's realism and the role of pornography in sexual learning.



THE CONCEPTUAL MODEL TO BE TESTED.

This study proposed a theoretical model that addresses the relationships between Internet pornography use, beliefs about the realism portrayed in pornography, the self-perceived effects of one's own pornography use, genital appearance satisfaction, and positive sexual self-esteem in young adults in Scandinavia.

Scripted Lusts

Gagnon and Simon (2005) claimed that people's sexuality develops through a social process of continuous interaction with significant and generalized others. Through social processes, sexuality is conferred through the expressions and meanings that society imposes. Sexuality is "scripted" on three distinct levels: cultural scenarios, interpersonal scripts, and intrapsychic scripts. Cultural scenarios are historical, culturally shared guidelines for sexuality that "...specify appropriate objects, aims, and desirable qualities of self-other relations,..." sequencing of events, and what the actors are assumed to be feeling. In addition to learning what is viewed as appropriate sexual conduct, the individual also learns what is considered as deviant patterns of behavior. The interpersonal script deals with the organization of context-specific behavior with shared social conventions and expectations that allow the involved actors to negotiate and participate in a

complex dyadic process. Thus, the interpersonal script enables individuals to find mutuality in sexuality. The intrapsychic script comprises factors that motivate the individual to become sexually aroused, and to commit to, and engage in, sexual activities. The individual adaption and transformation of local cultural scenarios can serve as “private sexual cultures” within the shared public sexual culture, and thus can be the basis for sexual diversity in a society. The prevailing sexual script in the Nordic context is one that connects sexual activity to love. “The Love Ideology” forms an important theme that legitimizes sex in a normative fashion, particularly when it concerns legitimizing partner sexuality among young people. However, this does not rule out at least a partial acceptance of recreational sex among consenting adults.

Sexual conduct seldom is learned by observing parents or other persons having sex in real life, and adults do not teach the young how to have sex. In this context, sexually explicit media (SEM) represent an exception, as they show the young individual possible sexual acts and arousal stimuli. Much of the existing SEM, particularly the media targeting men, deviate to some extent from the prevailing sexual script and show a hedonistic sexuality detached from the love script. In SEM, sexual acts are the center of attention. Few studies on Internet pornography have operationalized the form, content, or genre of pornography. When operationalized, the definition of Internet pornography has a range of broader descriptions, such as soft-core vs. hard-core, mainstream vs. paraphilia/deviant, types of sexual behavior (e.g., intercourse, anal, group sex), sexual orientation preferences, and media format (e.g., amateur, animation). What is considered common or mainstream will be highly subjective on the individual level, but also changes relative to the social acceptability and accessibility of pornography. However, there is support for making a distinction between a mainstream and non-mainstream genre. Previous research has found that groups with a preference for paraphilia or deviant pornography had higher sexual activity and an earlier age of onset for viewing pornography. Although the SEM industry has an offering for nearly every sexual preference, people tend to watch mainstream pornography. Thus, it appears that most people prefer watching a pornography genre that mirrors their prevailing intrapsychic script.

Consumption and Self-Perceived Positive Effects of Pornography

Changes in accessibility and privatization of Internet pornography use may change the sexual scripts that guide sexual behavior, both interpersonally and privately. Cultural and sub-cultural differences in the consumption of Internet pornography in the younger generation may reflect variations in their accessibility to technology, and in the degree of normalization of pornography and thus the sexual scripts.

In 2013, young men reported using SEM more frequently; and as compared to women, they regarded the effects of their SEM use as more positive and uncomplicated. Most of them reported that SEM had motivated them to try out new sexual positions and acts, increased their understanding of their sexual orientation, and made them more aware of what they like to do sexually. A recent study of men who have sex with men (MSM) in Norway, also concluded that the frequent consumption of gay SEM seemed to play a positive role in MSM’s sexuality in a similar way. In a Swedish study of women’s pornography use, 65% of the women reported generally positive effects and increased lust, and felt that pornography influenced their sexual behavior by giving them tips on new positions. However, it is possible that frequent users of SEM also assess the effects more positively to avoid cognitive dissonance between their attitudes and behaviors. This tendency may

be more common among women because of a need to justify their use of SEM in a society where the majority of young women have ambivalent or negative attitudes towards pornography.

Viewing Pornography as Realistic and Self-Perceived Positive Effects of Pornography

The sexual communication model advocates that, on a psychological level, the average pornography user is cognitively and perceptually capable of recognizing that a large portion of pornography content is composed of fictional or exaggerated representations of sexuality. This claim may be true if the consumer has similar sexual interaction experiences in real life as those seen on film. In this context, the recognition of exaggerated or fictional representations may be more difficult if the viewed pornography has non-mainstream rather than mainstream content. Young adults with non-mainstream preferences most likely will have fewer real life arenas in which to experiment with their desires. Therefore, they will not have as many opportunities to adjust their beliefs based on their own experiences, and SEM may become the central arena within which they live out their desires. Those who prefer mainstream pornography may have adjusted their opinions about the realism of SEM because they have more opportunities to experiment with their desires in real life. Therefore, it is reasonable to assume that a higher frequency of pornography use, type of pornographic content, and the degree to which one thinks pornography realistically portrays sex may influence an individual's self-evaluation of the effects of pornography.

Self-Perceived Positive Effects of Pornography and Sexual Self-Esteem

Stulhofer et al. (2007) found that the exposure to pornography affected sexual scripts as well as sexual experiences, which had a positive effect on sexual satisfaction, particularly in young men. This finding reflects the positive outcomes of pornography use, as described by young Norwegian adults, and generates a new hypothesis involving a link between the use of pornography and sexual self-esteem. Sexual self-esteem is associated with sexual satisfaction, and represents "one's affective reactions to one's sexual thoughts, feelings, and behaviors". Positive sexual self-esteem includes feelings of pride, satisfaction, and security associated with sexual conduct. If people believe that the use of pornography makes them try out new sexual positions and acts and helps them to find out what they desire sexually, the self-perceived effects of pornography use could transfer to a positive evaluation of oneself as a good lover. However, some individuals perceive their pornography use negatively, which may be related to negative sexual self-esteem, including disappointment, dissatisfaction, or a sense of vulnerability or insecurity.

The Role of Genital Appearance Satisfaction

The portrayal of idealized bodies by the media is often regarded as one of the most important sources of an individual's body image. Body dissatisfaction, specifically, genital appearance dissatisfaction among women, has been associated with lower sexual self-esteem. This association is often due to increased body self-consciousness. Negative beliefs about penis size for men are associated with shame, lack of masculinity, and lack of sexual prowess. A common belief in the popular media is that the SEM industry has influenced opinion regarding the ideal body in general and the ideal genital appearance in particular. The increase in the removal of pubic hair among women has been linked to

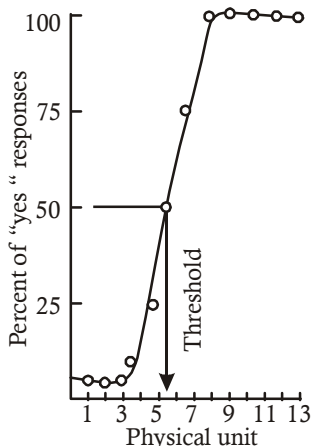
the media effects of pornography. The low variability in the appearance of genitals in traditional pornography has been reported to be a source of many men's opinions that their penis is too small. However, the belief that the frequent use of SEM leads to genital appearance dissatisfaction has not been confirmed in the research literature, which is sparse. In fact, a study of MSM found no association between frequent pornography use and dissatisfaction with their own genital appearance. One can also argue that the diversity of SEM on the Internet can provide users with the chance to see a larger variety of genitals, which may lead to an increased sense of normality and acceptance of one's own genitals. This feature could be important for women in particular, because they cannot see their own genitals directly without using a mirror. Female sexuality is subject to social restraint, and in general, women are not socialized to perceive their own genitals as acceptable. Therefore, SEM may contribute to women's perceptions of the appearance of their genitals as more acceptable.

From the perspective of social comparison theory, the context in which the idealized media image is presented can influence the body image of the perceiver. The self-perceived effects of one's own pornography use may be related to the degree of satisfaction one experiences with the appearance of one's own genitals. A person watching SEM with a focus on the similarities between themselves and the actor, or on the positive aspects of the situation depicted, may develop a more positive self-image and alleviate the pressure exerted by idealized standards of appearance in SEM. In addition to contextual differences, there are individual differences in the tendency to compare oneself with others. Doubts about oneself, a higher level of self-consciousness, and an orientation towards others are related to a greater tendency to engage in social comparisons; for some MSM, these comparisons result in body dissatisfaction and lowered self-esteem.

In previous research differences have been identified between men and women in relation to consumption patterns as well as in relation to social norms concerning the use of pornography. In the present article additional differences between the sexes, such as use of mainstream or non-mainstream is discussed as being expressions not only of gender differences but also related to differences in social context and individual differences in sexual experience. Use of sexually explicit material available over the Internet among young adults in Scandinavia will also be discussed in relation to perceived realism of pornography, self-perceived effects of own use of sexually explicit material as well as satisfaction with the appearance of one's own genitals and general sexual self-esteem. It is, based on the findings reported here, suggested that future research should take into account how gender differences as well as variations in preferences for different types of pornography interact in order to understand the mechanisms related to use of sexually explicit material.

MULTIPLE CHOICE QUESTIONS

1. The minimum physical energy necessary to activate a given sensory system is known as
 - (A) perception
 - (B) sense capacity
 - (C) absolute threshold
 - (D) none of the above.
2. The psychophysical function curve expresses the relationship between
 - (A) a psychological variable and a physical variable
 - (B) a psychological variable and an absolute threshold variable
 - (C) sense modality and threshold
 - (D) none of the above.
3. The graph, given below, shows



- (A) a sense modality curve
 - (B) physical sufficiency curve
 - (C) a psychophysical function curve
 - (D) an absolute threshold function curve.
4. Psychophysical functions can be obtained for
 - (A) any sensory modality
 - (B) some sensory modalities
 - (C) vision only
 - (D) all sensory modalities except vision.
5. Psychologists define the absolute threshold as that value at which the stimulus is perceived
 - (A) 75 per cent of the time
 - (B) 60 per cent of the time
 - (C) 50 per cent of the time
 - (D) 45 per cent of the time.
6. The approximate absolute threshold for the sense modality vision is
 - (A) a candle flame seen at 30 miles on a dark, clear night
 - (B) a candle flame seen at 25 miles on a dark, clear night
 - (C) a black sphere of radius 1 cm, seen at 30 miles, on a plain white paper
 - (D) a black sphere of radius 1 cm, seen at 25 miles, on a plain white paper.
7. The approximate absolute threshold for the sense modality, hearing, is
 - (A) sound of air having slow pace in quiet conditions
 - (B) the tick of a watch under quiet conditions at 20 feet
 - (C) the tick of a watch under quiet conditions at 30 feet
 - (D) sound of air of a fan in a closed sound-proof room.
8. The approximate absolute threshold for the sense modality, taste, is
 - (A) one teaspoon of sugar in five gallons of water
 - (B) one teaspoon of sugar in three gallons of water
 - (C) one teaspoon of sugar in one gallons of water
 - (D) one teaspoon of sugar in two gallons of water.
9. The approximate absolute threshold for the sense modality smell is one drop of

perfume diffused into the entire volume of a

- (A) one room apartment
- (B) three room apartment
- (C) six room apartment
- (D) nine room apartment.

10. The approximate absolute threshold for the sense modality touch is wing of a fly falling on your cheek from a distance of

- (A) 0.01 centimeter
- (B) 0.25 centimeter
- (C) 0.50 centimeter
- (D) 1.00 centimeter.

11. The minimum amount of stimulation necessary to tell two stimuli apart is known as

- (A) the difference threshold
- (B) the absolute threshold
- (C) Weber's constant
- (D) ROC values.

12. If l is the amount of stimulation taken as a referent, and Δl is the increase in stimulation necessary for a just noticeable difference, then which of the following relationships is true?

- (A) $\frac{l}{\Delta l} = k$
- (B) $l = k \cdot \Delta l$
- (C) $\frac{\Delta l}{l} = k$
- (D) None of these.

13. The relationship

$$\frac{\Delta l (\text{increase in stimulation necessary for a j.n.d.})}{l (\text{amount of stimulation taken as a referent})} = K (\text{constant})$$

is known as

- (A) threshold law
- (B) law of perception
- (C) Thilly's law
- (D) Weber's law.

14. The constant k , used in relationship

$$\frac{\Delta l}{l} = k, \text{ is known as}$$

- (A) threshold constant
- (B) Weber's constant
- (C) amount of sense capacity
- (D) a point of sense limit.

15. Match List I (Sense modality) with List II (Weber's constant) and select your answer from the codes given below:

List I (sense modality)	List II (Weber's constant)
--------------------------------	-----------------------------------

(a) Pitch of a tone	(1) $\frac{1}{50}$
---------------------	--------------------

(b) Deep pressure from skin and subcutaneous tissue	(2) $\frac{1}{60}$
---	--------------------

(c) Visual brightness	(3) $\frac{1}{80}$
-----------------------	--------------------

(d) Lifted weights	(4) $\frac{1}{333}$
--------------------	---------------------

Codes:

a	b	c	d	a	b	c	d
(A) 1	2	3	4	(B) 4	3	2	1
(C) 3	2	1	4	(D) 2	1	4	3

16. Match List I with List II and select your answer from the codes given below:

List I (Sense modality)	List II (Weber's constant)
--------------------------------	-----------------------------------

(a) Lifted weights	1. $\frac{1}{50}$
--------------------	-------------------

(b) Loudness of a tone	2. $\frac{1}{10}$
------------------------	-------------------

(c) Cutaneous pressure	3. $\frac{1}{7}$
------------------------	------------------

(d) Taste for saline solution	4. $\frac{1}{5}$
-------------------------------	------------------

Codes:

a	b	c	d	a	b	c	d
(A) 1	2	3	4	(B) 2	3	4	1
(C) 3	4	1	2	(D) 4	1	2	3

17. Weber's law was formulated for

- (A) measuring sensory experiences
- (B) measuring Weber's constant
- (C) threshold
- (D) none of the above

18. Fechner extended the Weber's law to develop scales for measuring

- (A) threshold
- (B) increase in stimulation
- (C) sense experiences
- (D) none of the above.

19. Which of the following laws holds that the sensory experience bears a logarithmic relationship to the intensity of the physical stimulus ?
 (A) Weber's law (B) Nativists' law
 (C) Empiricists' law
 (D) Weber-Fechner law.
20. Who amongst the following discovered that sunlight passing through a prism breaks into a band of varicolored light such as we see in a rainbow ?
 (A) Newton (B) Weber
 (C) Fechner (D) b and c both.
21. The wavelengths that the human eye perceives as light extend only from
 (A) about 150 nm to about 300 nm
 (B) about 380 nm to about 780 nm
 (C) about 880 nm to about 1080 nm
 (D) about 1080 nm to about 1580 nm.
22. Which of the following terms describes the fact that the curve measures the operating or sensitivity characteristics of a person receiving signals ?
 (A) BOSS (B) CROSS
 (C) ROC (D) CONE.
23. If a curve sketched in a manner that the hit and false alarm probabilities would differ from those in the table but would fall somewhere on the curve, then this curve is called
 (A) BOSS (B) CROSS
 (C) ROC (D) CONE.
24. Signal detectability theory is given by
 (A) Weber
 (B) Fechner
 (C) Newton
 (D) Green and Swets.
25. The rods and cones are
 (A) which make synaptic connections with the ganglion cells
 (B) the fibers of which form the optic nerve
 (C) the photosensitive cells that convert light energy into nerve impulses
 (D) none of the above.
26. Match List I with List II and choose the correct answer from the codes given below.

List I
(Parts of eye)

- (a) Cornea
 (b) Pupil
 cons-
 dila-
 (c) Retina
 (d) Autonomic

List II
(Functions)

1. On which the light is focused
 2. By which the truction and tion of the pupil are controlled
 3. Through which light enters the eye
 4. By which the nervous system amount of light being regulated

Codes:

- | | a | b | c | d | | a | b | c | d |
|-----|---|---|---|---|-----|---|---|---|---|
| (A) | 3 | 4 | 1 | 2 | (B) | 1 | 2 | 3 | 4 |
| (C) | 4 | 3 | 2 | 1 | (D) | 2 | 1 | 4 | 3 |

27. Which of the following controls the change in pupil size as a function of change in illumination ?
 (A) Retina
 (B) The sympathetic division
 (C) The parasympathetic division
 (D) None of the above
28. The sympathetic division (in eye) acts
 (A) to control the pupil size as a function of change in illumination
 (B) to dilate the pupil under conditions of strong emotion
 (C) to form the rear layer of the retina
 (D) none of the above.
29. Match List I with List II and select the correct answer from the codes given below:
- | List I | List II |
|-----------------------|--|
| (a) Rods and cones | 1. Which form the optic nerve |
| (b) The bipolar cells | 2. Make synaptic connections with the rods and cones |
| (c) The ganglion | 3. Convert light cells energy into nerve impulses |

Codes:

a	b	c	a	b	c
(A) 1	2	3	(B) 3	2	1
(C) 2	3	1	(D) 1	3	2

30. Which of the following form the rear layer of the retina ?
 (A) Rods and cones
 (B) Bipolar cells
 (C) Ganglion cells
 (D) None of the above.
31. Which of the following statements is not true ?
 (A) The cones will respond to a much dimmer stimulus than will the rods
 (B) The rods will respond to a much dimmer stimulus than will the cones
 (C) The image rods give less clear
 (D) The fovea contains only cones.
32. Cones are most responsive to wave-lengths in which part of the spectrum ?
 (A) Violet-blue (B) Blue-green
 (C) Green-yellow (D) Orange-red.
33. Rods are more sensitive to wavelength in which of the following parts of the spectrum ?
 (A) Violet-blue (B) Blue-green
 (C) Green-yellow (D) Orange-red.
34. Most motorists find driving at dusk hazardous. This is because
 (A) the transition from day to night vision takes place gradually as daylight diminishes
 (B) the optic nerves stop working as daylight diminishes
 (C) the wavelengths of blue-green or green-yellow spectrum begin to be small as daylight diminishes
 (D) none of the above.
35. Complete the statement given below:
 At twilight both the rods and cones are operating
 (A) with full effectiveness
 (B) on opposite objects
 (C) but not with full effectiveness
 (D) on similar objects.
36. The course of dark adaptation provides further evidence for the difference in action between

- (A) rods and cones
 (B) cornea and retina
 (C) pupil and nerves
 (D) lens and fovea.

37. Match List I with List II and select the correct answer from the codes given below:

List I
(Psychological dimensions of color)

List II
(Meaning)

- | | |
|----------------|--|
| (a) Hue | (1) Name of the color |
| (b) Brightness | (2) Scaling of color |
| (c) Saturation | (3) Scaling the apparent purity of the color |

Codes:

a	b	c	a	b	c
(A) 1	2	3	(B) 2	3	1
(C) 3	1	2	(D) 1	3	2

38. Which of the following represents all the three dimensions of color simultaneously ?
 (A) Color-solid
 (B) Colorblindness
 (C) Color contrast
 (D) Color mixture.
39. In a color circle, the dimension of hue is represented by points
 (A) along the radius
 (B) along the vertical axis
 (C) around the circumference
 (D) at the centre.
40. The color solid helps in our understanding of the relationship between
 (A) hue and brightness
 (B) brightness and saturation
 (C) hue and saturation
 (D) none of the above.
41. Match List I with List II and select the correct answer from the codes given below:

List I

List II

- | | |
|------------------|----------------------------|
| (a) Trichromate | (1) Partially colour-blind |
| (b) Dichromate | (2) Totally colour-blind |
| (c) Monochromate | (3) Normal vision |

Codes:

	a	b	c		a	b	c
(A)	1	2	3	(B)	2	3	1
(C)	3	1	2	(D)	3	2	1

- 42.** The ear is sensitive to
(A) electromagnetic energy
(B) mechanical energy
(C) electrical energy
(D) none of the above.
- 43.** The psychological correlate of frequency is
(A) pitch (B) loudness
(C) tones (D) none of these.
- 44.** The psychological correlate of intensity is
(A) pitch (B) loudness
(C) tones (D) none of these
- 45.** Select the true statements from the codes given below
(1) The higher the vibration frequency, the higher the perceived pitch
(2) The higher the vibration frequency, the lower the perceived pitch
(3) The greater the intensity, the louder the tone
(4) The greater the intensity, the lower the tone.

Codes:

- (A) 1 & 4 (B) 2 & 4
(C) 1 & 3 (D) 2 & 3.

- 46.** We can hear frequencies that range from about
(A) 200 to 2000 Hz
(B) 20 to 20,000 Hz
(C) 2000 to 20,000 Hz
(D) 20,000 to 80,000 Hz.
- 47.** The absolute threshold for hearing depends on the
(A) wavelength of the tone
(B) frequency of the tone
(C) amplitude of the tone
(D) phase of the tone.
- 48.** A quality that depends on the overtones and other impurities differing from one instrument to another is known as
(A) pitch (B) loudness
(C) timbre (D) none of these.
- 49.** Complex sounds composed of many frequencies not in harmonious relation to one another are called
(A) pitch (B) loudness
(C) timbre (D) noise.
- 50.** Theories attempting to give a physiological explanation of pitch are
(1) frequency theory
(2) sound theory
(3) place theory
(4) value theory
Select the correct answer from the codes given below:
- Codes:**
(A) 1 & 2 (B) 2 & 3
(C) 1 & 3 (D) 3 & 4.
- 51.** The theory which emphasizes the place on the basilar membrane where a particular frequency produces its maximum effect is known as
(A) frequency theory (B) sound theory
(C) place theory (D) value theory
- 52.** The theory which assumes that pitch is determined by the frequency of impulses travelling up the auditory nerve is called
(A) sound theory
(B) place theory
(C) frequency theory
(D) value theory.
- 53.** The place theory applies to
(A) negligible frequencies
(B) lower frequencies
(C) high frequencies
(D) none of the above.
- 54.** Synchronous discharge is important for the
(A) negligible frequencies
(B) lower frequencies
(C) higher frequencies
(D) none of the above.
- 55.** Select the lower senses from the following options and choose your answer from the codes given below:
(1) Skin sensations
(2) Smell and taste
(3) Visual and auditory
(4) Kinesthesia (5) Equilibratory

Codes:

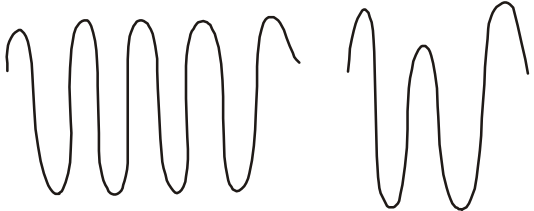
- (A) 1, 2 and 3 (B) 2, 3 and 4
(C) 1, 2, 4 and 5 (D) 3, 4 and 5.
- 56.** The stability of perception depends upon the
(A) Constancies of colour and shape
(B) Constancies of size
(C) Constancies of location
(D) All the above.
- 57.** Size perception usually represents a compromise between
(A) perspective size and object size
(B) perspective size and environment
(C) object size and environment
(D) distant perspective size and nearest perspective size
- 58.** How well size constancy operates depends upon the
(A) presence of distance cues
(B) presence of air
(C) our familiarity with the object
(D) a and c both.
- 59.** When we look at a distant object, we can judge its size in one of three ways (given in options). Which of the three alternatives is usually correct ?
(A) Perspective size
(B) Object size
(C) Compromise between perspective size and object size
(D) To be familiar with the object.
- 60.** Perceiving motion without a successive pattern of stimulation is known as
(A) apparent motion
(B) stroboscopic motion
(C) autokinetic motion
(D) none of these.
- 61.** Which of the following tells us the distance of objects ?
(A) Hue (B) Monocular cues
(C) Visual cortex (D) None of these.
- 62.** An artist is able to give depth to his picture because he can make use of the
(A) brush (B) colours
(C) paper (D) monocular cues.
- 63.** 'Our ability to perceive the spatial aspects of our environment is innate' is the theory accepted by
(A) Locke (B) Berkeley
(C) Descartes (D) Hume.
- 64.** The motion of hypothesis testing emphasizes the active nature of
(A) perception (B) learning
(C) memory (D) forgetting.
- 65.** Horizontal lines in the figure of railway tracks show
(A) illusion based on relative size
(B) illusion based on intersecting lines
(C) ponzo illusion
(D) none of the above.
- 66.** Running of movie is possible because of
(A) real motion
(B) stroboscopic motion
(C) autokinetic motion
(D) electrical motion.
- 67.** Who amongst the following accepts the view that we learn our ways of learning through experiences ?
(A) Descartes (B) Spinoza
(C) Kant (D) Locke.
- 68.** Who amongst the following is an empiricist ?
(A) Descartes (B) Spinoza
(C) Berkeley (D) Kant.
- 69.** Who amongst the following is nativist ?
(A) Descartes (B) Kant
(C) Berkeley (D) Both (A) and (B)
- 70.** Locke supports which of the following views ?
(A) Nativism (B) Empiricism
(C) Both (A) and (B)
(D) None of the above.
- 71.** The advertiser counts on an appeal to the male sex drives when he uses pictures of scantily clad females to advertise anything from carpets to automobile tyres. This is an example of
(A) role of motivation in perception
(B) role of learning in perception
(C) role of need in perception
(D) role of attention in perception.
- 72.** The minimum physical energy necessary to activate a given sensory system is
(A) absolute threshold
(B) differential threshold

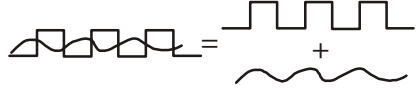
- (C) physical limit
(D) just noticeable difference.
- 73.** The difference threshold tends to be
(A) independent of the stimulus intensity
(B) a constant fraction of the stimulus intensity
(C) an integral multiple of the stimulus intensity
(D) a constant.
- 74.** The scale to measure the psychological distance between any two stimuli was developed by
(A) Weber (B) Fechner
(C) Titchner (D) Binet.
- 75.** Signal detectability theory
(A) permits one to separate response and stimulus
(B) does not permit one to separate the sensory and the non-sensory factors
(C) permits one to separate the sensory and the non-sensory factors
(D) permits one to measure the response bias.
- 76.** The absolute threshold depends upon the individual's
(A) physical condition
(B) motivation state
(C) both the above
(D) none of the above.
- 77.** According to Weber's law, if l is the amount of stimulus taken as a referent and Δl is the increase in stimulation necessary for a j.n.d.
(A) $\Delta l = \text{constant}$
(B) $\Delta l = \text{constant}$
(C) $\Delta l \times l = \text{constant}$
(D) $\Delta l/l = \text{constant}$.
- 78.** Which of the following is not true ?
(A) The cones are active only in day-light vision
(B) The rods permit us to see only achromatic colours
(C) The cones permit us to see both achromatic and chromatic colours
(D) The cones permit us to see only chromatic colours (red, green, blue, etc.).
- 79.** Damage to the occipital lobe of one hemisphere (say, the left) will result in blind areas in
(A) the left sides of both eyes
(B) the right sides of both eyes
(C) the left eye only
(D) the right eye only.
- 80.** Through signal detection experiments on rats, the drug lysergic acid (LSD) is found to affect
(A) response bias
(B) sensory sensibility
(C) both response bias and sensory sensitivity
(D) neither response bias nor sensory sensitivity.
- 81.** The most sensitive portion of the eye in normal daylight vision is a part of the retina called
(A) cornea (B) blind spot
(C) fovea (D) optic chiasma
- 82.** Which of the following statements is false? The colour-mixture primaries are
(A) psychological primaries
(B) any three widely-spaced colours on the spectrum that can be used to produce all the other colours by additive mixture
(C) red, green, blue
(D) both (B) and (C) are false.
- 83.** The most highly saturated colours are of
(A) lowest brightness
(B) medium brightness
(C) highest brightness
(D) there is no relationship between saturation and brightness.
- 84.** Which of the following statements is false? A totally colourblind person is
(A) a monochromat
(B) has only the light-dark system
(C) also night-blind
(D) has neither the yellow-blue system nor the red-green system.
- 85.** The fovea is
(A) insensitive during day time
(B) sensitive only in the day time

- (C) sensitive only in the night time
(D) both (A) and (B) are correct.
- 86.** When blue light and yellow light are mixed, the result is
(A) blue-green (B) green
(C) purple (D) violet.
- 87.** When blue pigment and yellow pigment are mixed, the result is
(A) green (B) grey
(C) blue-green (D) violet.
- 88.** The absolute threshold for learning
(A) varies with frequency of the source
(B) is independent of the frequency of the source
(C) is the same for all individuals
(D) does not vary from time to time for a particular individual.
- 89.** Which of the following statements is false ?
(A) The mixture of light is an additive mixture
(B) The mixture of lights as well as pigments is an additive mixture
(C) The mixture of pigments is a subtractive mixture
(D) The pigments are simply reflectors and absorber of colours, and are not the source of colours.
- 90.** The psychological primaries are
(A) red, yellow, green
(B) red, green, blue
(C) red, yellow, green and blue
(D) grey, black and white.
- 91.** The auditory sense is sensitive to
(A) electromagnetic waves
(B) electromagnetic energy
(C) mechanical energy
(D) chemical energy.
- 92.** The psychological correlate of frequency (of a sound wave) is
(A) timbre (B) loudness
(C) quality (D) pitch
- 93.** Theories of colour vision have not taken as its starting point the facts about
(A) colour mixture
(B) colour contrast and after image
(C) colour blindness
(D) colour constancy.
- 94.** If you stare at a red circle and then look at a plain grey surface, you are likely to see
(A) a red circle on it
(B) a green circle on it
(C) a dark grey circle on it
(D) a white circle on it.
- 95.** If one compares the dimensions of tone with those of colour, the following correspondence holds approximately
(A) hue <-> timbre
(B) saturation <-> pitch
(C) saturation <-> timbre
(D) brightness <-> pitch.
- 96.** The basic skin sensations are
(A) touch, pain, warm and cold
(B) touch, warm and cold
(C) itch, tickle, warm and cold
(D) itch, pain, itch, warm and cold
- 97.** When cold water (0-5 °C) is circulated through one coil and warm water (40-44 °C) through another intertwining coil, the subject on grasping the coils experiences the following sensation.
(A) An ambivalent sensation of hot and cold
(B) Hot, burning sensation
(C) Cold, freezing sensation
(D) Warm, comfortable sensation.
- 98.** Perception are not affected by
(A) behaviour
(B) tension and anxiety
(C) suggestion
(D) hunger.
- 99.** The number of separate and distinct impressions, which can be grasped after limited display, is called
(A) perceptual limit
(B) apprehension span
(C) perceptual span
(D) perceptual threshold.
- 100.** The laws of perception were formulated by the
(A) behaviourist (B) psychologists
(C) gestaltists (D) existentialists.

- 101.** Sudden perceptual reorganisation of a stimulus situation is best described as
(A) concept formation
(B) imagination
(C) recognition (D) insight.
- 102.** The following is not an external factor that directs attention
(A) intensity (B) illusion
(C) contrast (D) repetition.
- 103.** Contours are formed whenever there is a
(A) gradual change in the brightness of background
(B) change in the colour of the background
(C) discrete change in the brightness or colour of the background
(D) all the above.
- 104.** Perception is a
(A) sensory process
(B) nervous mechanism
(C) sensory stimulation
(D) process of interpreting sensory experiences.
- 105.** Perceiving involves
(A) adjustment to the stimuli in and around
(B) understanding whatever is going
(C) organization of stimuli
(D) awareness of somebody or something.
- 106.** As compared to noises, tunes and melodies are better perceived because
(A) they are more pleasing in effect
(B) they have order and form
(C) they appeal to our interests
(D) they are capable of arousing emotions.
- 107.** The position, orientation and movement of parts of the body in relation to gravity are informed by the
(A) kinesthesia
(B) equilibrium senses
(C) kenesthesia and equilibratory senses
(D) Auditory senses and kenesthesia.
- 108.** If you mark off three equally distant points on the back of person's hand, touch a pencil to the first two points, and pause slightly before you touch the third, the person will report that the first two points were closer together than the second and third. This perception illustrates grouping of tactile stimuli according to
(A) nearness in shape
(B) nearness in time
(C) continuation in time
(D) closure in time.
- 109.** The principles of organisation hold good for
(A) visual sense only
(B) visual and auditory senses
(C) visual and tactile senses
(D) visual, auditory and tactile senses.
- 110.** Objects nearer to you than the spot at which you are looking (fixation point) move in a direction
(A) opposite to the direction in which your head moved
(B) same as the direction of your head movement
(C) do not move at all
(D) cannot be seen as they are out of focus.
- 111.** The stability of the environment as we experience is termed as
(A) sensory bias
(B) perceptual constancy
(C) continuity in perception
(D) sensory constancy
- 112.** How well size constancy operates does not depend upon the presence of
(A) distance cues
(B) familiarity with the object
(C) clues about illumination
(D) past experience.
- 113.** Experiments using special glasses (or prism) in which the subjects adapted his visual—motor habits in a visually rearranged environment indicated the role of learning in
(A) location constancy
(B) size constancy
(C) shape constancy
(D) all the above.
- 114.** The following are the internal factors that direct attention

- (A) motive (B) preparatory set
(C) needs (D) all of these
- 115.** Which of the following statements is false ?
(A) Contours have shape
(B) In camouflage, contours are broken
(C) Objects with different shapes can be formed by the same contour
(D) Contour can be seen without any energy difference at all on the two sides of the contour
- 116.** The perceptual constancies imply
(A) role of thinking
(B) organisation within perception
(C) role of emotions
(D) role of memory.
- 117.** When we look at a figure for some time and then look at another figure, the perception of second figure is distorted. This phenomenon is referred to as
(A) after image (B) illusion
(C) the figure after-effect
(D) distorted perception.
- 118.** Figure is different from the ground in that the figure appears to be
(A) darker than the ground
(B) longer than the ground
(C) farther than the ground
(D) closer than the ground.
- 119.** Phi-phenomenon is a form of
(A) autokinetic effect
(B) stroboscopic motion
(C) real motion
(D) induced motion.
- 120.** Two vertical bars of light are arranged at a certain distance inside a dark room. The lights are alternately turned on and off. When the time interval is less than 30 milliseconds, the lights appear simultaneously. When the interval is 60 milliseconds, the light is seen to move across the open space between the two stimulus lights. This is an experiment on
(A) visual movement
(B) autokinetic effect
(C) phi-phenomenon
(D) induced movement.
- 121.** The moon is often seen moving behind the clouds. This is an example of
(A) real motion
(B) phi- phenomenon
(C) autokinetic
(D) induced motion.
- 122.** Reversible figure—ground relationships
(A) cannot exist
(B) illustrate that objects with different shapes are formed only by different contours
(C) illustrate the multistability of perceptual organization
(D) illustrate the instability of perceptual organization.
- 123.** Which of the following is not a law of perceptual organization ?
(A) Proximity (B) Similarity
(C) Pregnanz (D) Constant.
- 124.** The statement 'we perceive things as we are and not as they are' means
(A) our perceptions are more subjective than objectives
(B) our perceptions are unreal
(C) our perceptions are influenced by our motives
(D) our perceptions are distorted.
- 125.** The neurons in the visual cortex that respond only to stimuli of particular orientation and specific length and width are called
(A) simple cells
(B) complex cells
(C) hypercomplex cells
(D) hypocomplex cells.
- 126.** Perception (*i.e.* perceiving things as we do) is
(A) innate
(B) learned through experience
(C) effected by practice
(D) all of the above.
- 127.** Stereoscopic experiences are due to
(A) retinal disparity
(B) monocular cues
(C) superposition effect
(D) relativity in size.
- 128.** Experiments using 'visual cliff' indicate that depth perception in infants is

- (A) innate
(B) present at least as soon as the animal is able to locomote
(C) learned only through experience and practice
(D) influenced by rearing practices.
- 129.** Plasticity (Modifiability) of perception is demonstrated by experiments with
(A) prisms
(B) sensory deprivation
(C) perceptual learning
(D) all the above.
- 130.** Which of the following statements about perceptual learning is not correct ?
(A) It can be learned from book
(B) It is observed when we study the increased auditory and tactile sensitivity of the blind people
(C) It has practical and adaptive value
(D) It illustrates the plasticity of perception.
- 131.** Which of the following is not one of the basic skin senses discussed in the text?
(A) Warmth (B) Sharpness
(C) Cold (D) Pain.
- 132.** Which of the following is not the basic categories of taste ?
(A) Sour (B) Salty
(C) Bland (D) Bitter.
- 133.** The phenomenon of perceiving patterns according to closeness in position is called
(A) simplicity (B) grouping
(C) proximity (D) continuity.
- 134.** Which of the following is not a binocular for depth perception ?
(A) Retinal disparity
(B) Movement
(C) Convergence
(D) All the above are binocular cues for depth perception .
- 135.** Which of the following statements about orienting reflex is false?
(A) It occurs in response to stimulation changes in the environment
(B) It has survival value
(C) It does not habituate over time
(D) It prepares the organism to respond quickly in case action is needed.
- 136.** Most dreams occur during which of the following stage of sleep ?
(A) REM stage (B) Stage 1
(C) Stage 2 (D) Stage 3.
- 137.** Which of the following statements regarding hypnosis is false ?
(A) Under hypnosis, attention becomes highly selective
(B) The EEG of the hypnotic state is that of sleeping
(C) The hypnotised subject readily enacts unusual roles
(D) The ability to be hypnotised may have a hereditary component.
- 138.** Alcohol is
(A) stimulant (B) hallucinogen
(C) depressant (D) all of these
- 139.** Decibels are measurement of
(A) loudness (B) pitch
(C) brightness (D) frequency.
- 140.** Suppose the lines in the figure below represent two sound waves. Their frequencies would be ____ and their intensities would be ____ .
- 
- (A) equal; equal
(B) different; different
(C) different; equal
(D) equal; different.
- 141.** The actual structure that converts sound to neural impulse in the ear is the
(A) tectorial membrane
(B) cochlea
(C) hair cell
(D) basilar membrane.
- 142.** Cochlear implants are an experimental treatment for
(A) conduction deafness
(B) a broken eardrum

- (C) neural deafness
(D) colour blindness.
- 143.** According to the principle of sensory adaptation, our sensory systems are particularly well designated to detect
(A) duration (B) intensity
(C) quality (D) change.
- 144.** Weber's Law describes the relationship between initial stimulus and
(A) stimulus duration
(B) magnitude of sensation
(C) the just noticeable difference
(D) the sensory threshold.
- 145.** Which of the following parts of the eye are closest to the back of the eyeball?
(A) bipolar cells (B) ciliary muscles
(C) ganglion cells (D) rods, cones.
- 146.** In young people, far-sightedness and near-sightedness are generally caused by
(A) defective lenses
(B) weak ciliary muscles
(C) unresponsive
(D) misshapen eyeballs.
- 147.** Which of the following statements regarding the rods and cones is false?
(A) They are sense light
(B) There are more types of cones than rods
(C) Rods are concentrated in the fovea
(D) Cones are involved in high-intensity vision.
- 148.** Drug use should be discouraged. Which of the following is not a primary reason for discouraging drug use? Drug use is
(A) dangerous to health and even to life
(B) is costly and likely to lead to illegal activities to maintain a supply
(C) may lead to drug dependency
(D) may result in unpleasant experiences.
- 149.** Sound waves, light waves, heat, and pressure are example of
(A) receptor (B) sensations
(C) stimuli (D) perceptions.
- 150.** Which of the following is a function of the quality of stimulus?
(A) Pitch (B) Loudness
(C) Brightness (D) Saltiness.
- 151.** The figure given below illustrates which of the following organizing principle?
- 
- (A) Continuation (B) Common fate
(C) Closer (D) Proximity.
- 152.** Which of the following cues regarding depth perception requires the use of both of your eyes?
(A) Binocular disparity
(B) Motion parallax
(C) Linear perspective
(D) Relative size.
- 153.** In one experiment by having a different eye covered each day, laboratory kittens lost their proficiency at depth perception. This demonstrates the impact on perception of
(A) Motivation
(B) Experience
(C) Genetics
(D) Disease and malnutrition.
- 154.** Perceptual set refers to
(A) the level at which our thresholds are set
(B) the tendency to group objects in a certain way
(C) the figure-ground phenomenon
(D) the tendency to perceive what we think should.
- 155.** Suppose Mary wants to know if she can learn Spanish by playing some language tapes while she sleeps. Her question concerns the general topic of _____ and, according to the text, the answer would most likely be _____.
(A) subliminal perception; may be
(B) subliminal perception; no
(C) signal detection; may be
(D) signal detection; no.
- 156.** Suppose you find that when lifting a fifty pound barbell you can just notice the difference when one pound is added. According to Weber's Law, if you were lifting a twenty-five-pound barbell, the just noticeable difference would be
(A) one-half pound

- (B) one pound
(C) two pound (D) four pound.
- 157.** Some people compare the human eye to a camera. Using this analogy, the shutter which regulates how much light gets in, corresponds to the
(A) retina (B) lens
(C) iris (D) cornea.
- 158.** Suppose I believe that vision is best described as a series of on-off switches and that the pattern of ons and offs determines the colours I see. I would subscribe to
(A) place theory
(B) trichromatic theory
(C) volley theory
(D) opponent-process theory.
- 159.** Which of the following has/have a function that differs from the rest ?
(A) Auditory canal
(B) Hammer, anvil, and stirrup
(C) Eardrum
(D) Hair cells.
- 160.** Which of the following statements most accurately reflects the text's conclusion about subliminal advertisement ?
(A) It is unethical
(B) It is illegal
(C) It is ineffective
(D) It should be used to prevent shop-lifting.
- 161.** Janet believes that everything you sense is conditioned and affected by your own unique experiences. Thus, no two people can perceive something the same way. Which of following ideas is most closely associated with Janet's view ?
(A) Nativist view
(B) Conceptually driven (top-down) processing
(C) Subliminal perception
(D) Signal detection theory
- 162.** You suspect you have a hearing loss, so you go to a specialist who fits you ear-phones and tells you to raise hand whenever you hear a tone. This specialist is testing your
(A) just noticeable difference
(B) sensory threshold
(C) Weber's ratio
(D) adaptation.
- 163.** Suppose you believe that hearing is set up same way as a harmonica. The note you hear depends on where on the instrument you blow. You probably subscribe to one of these theories.
(A) frequency theory
(B) volley theory
(C) opponent-process theory
(D) place theory.
- 164.** Which of the following technically does not change physical energy to sensation?
(A) Meissner's corpuscles
(B) taste buds
(C) basilar membrane
(D) rods and cones.
- 165.** The diagram given below demonstrates the principle of
- 
- (A) continuity
(B) figure-ground
(C) signal detection
(D) perceptual constancy.
- 166.** Most people see the following display as three groups, one of 3 Bs, and one of 3 Cs: AAABBBCCC. This is consistent with the Gestalt principle of
(A) similarity (B) proximity
(C) continuity
(D) subjective contour.
- 167.** The major difference between a green light and a blue light is the
(A) wave frequency (B) wave purity
(C) wavelength
(D) wave saturation.
- 168.** If the eye is compared to a camera, the role of retina would most closely resemble the role of the
(A) lens (B) film
(C) shutter (D) flash cube.
- 169.** Which theory of colour vision best explains why the colour of an after image is the complement of the original colour?

- (A) The trichromatic theory
(B) The opponent process theory
(C) Both theories explain this phenomenon equally well
(D) Neither theory explain this phenomenon.
- 170.** When watching a wild car chase scene in a movie we can be thankful for
(A) chunking
(B) lateral processing
(C) bottom-up processing
(D) the phi-phenomenon.
- 171.** Which of the following is one of the binocular distance cues ?
(A) Convergence
(B) Linear perspective
(C) Relative height
(D) Texture gradients
- 172.** Which of the following is an example of what is meant by perceptual constancy?
(A) Moths are always attracted to light
(B) A round pie tin always appear to as round
(C) Proximal and distal stimuli are always identical
(D) All of the above.
- 173.** Middle C sounded on a piano sounds different than middle C on a violin because of the difference in
(A) Himber (B) Purity
(C) Amplitude (D) All of these
- 174.** Research has shown that the perception of pitch depends on
(A) the area stimulated on the basilar membrane
(B) the frequency at which the basilar membranes vibrates
(C) both the above
(D) none of the above.
- 175.** Which of the following is not considered to be one of the four fundamental tastes?
(A) Sour (B) Sweet
(C) Burnt (D) Bitter.
- 176.** The fact that we are generally much more aware of changes in our sensory environments rather than the constants is the general idea behind
(A) signal detection theory
(B) sensory adaptation
(C) the method of constant stimuli
(D) sensory equalization
- 177.** The finding that identical twins tend to see illusion in the same way more frequently than best friends do demonstrates the impact on perception of
(A) heredity (B) experience
(C) motivation
(D) signal detection.
- 178.** A blindfolded subject is touched on the arm with a metal bar that the subject has been told is very hot. Even though the bar is cool, the subject winces and complains of being burned. This is an example of
(A) form perception
(B) perceptual set
(C) perceptual illusion
(D) subliminal perception.
- 179.** Stimuli cannot be perceived when they are presented below the absolute threshold. This statement is
(A) true
(B) false
(C) could not be interpreted
(D) none of the above.
- 180.** According to Weber's Law, which of the following changes would be most difficult to detect ?
(A) A 36-inch line lengthened by 1 inch
(B) A 60-inch line lengthened by 2 inches
(C) A 90-inch line lengthened by 3 inches
(D) All the above would be equally difficult to detect.
- 181.** Which law or theory of detecting stimuli places a major emphasis on subjective factors ?
(A) Weber's law
(B) Fechner's law
(C) Steven's law
(D) Signal detection theory
- 182.** The receiving area of a retina is called as
(A) cone (B) foveal field
(C) rod (D) receptive field.

183. Our sense of balance depends upon

- (A) the semi-circular canals
(B) the kinesthetic senses
(C) visual cues
(D) all the above

184. Which of the following terms perhaps best describes human perception ?

- (A) Accurate (B) Objective
(C) Subjective (D) Unknowable

Direction: match List I with List II and select the correct answer from the codes given below each question.

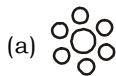
185. List I List II
(Philosophers) (Views)

- (a) Descartes (1) Nativism
(b) Kant (2) Empiricism
(c) Locke
(d) Berkeley

Codes:

- | a | b | c | d | a | b | c | d |
|-------|---|---|---|-------|---|---|----|
| (A) 1 | 1 | 2 | 2 | (B) 1 | 2 | 1 | 2 |
| (C) 2 | 1 | 2 | 1 | (D) 2 | 1 | 1 | 2. |

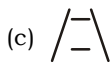
186. List I List II
(Figures) (Illusion)



(1) Panzo illusion



(2) Illusion based on intersecting lines



(3) Illusion based on relative size

Codes:

- | a | b | c | a | b | c |
|-------|---|---|-------|---|---|
| (A) 1 | 2 | 3 | (B) 3 | 2 | 1 |
| (C) 2 | 3 | 1 | (D) 1 | 3 | 2 |

187. List I List II
(Sense Modality) (Weber's Constant)

- (a) Pitch of a tone (1) $\frac{1}{10}$
(b) Visual brightness (2) $\frac{1}{50}$

- (c) Lifted weights (3) $\frac{1}{60}$
(d) Loudness of a tone (4) $\frac{1}{333}$

Codes:

- | a | b | c | d | a | b | c | d |
|-------|---|---|---|-------|---|---|----|
| (A) 1 | 2 | 3 | 4 | (B) 2 | 3 | 4 | 1 |
| (C) 4 | 3 | 2 | 1 | (D) 3 | 2 | 4 | 1. |

188. List I List II

- (a) Lower sense (1) Balance
(b) Spin sensation (2) Pain
(c) Kinesthesia (3) Joint sense
(d) Equilibratory senses (4) Tongue

Codes:

- | a | b | c | d | a | b | c | d |
|-------|---|---|---|-------|---|---|----|
| (A) 1 | 2 | 3 | 4 | (B) 2 | 3 | 4 | 1 |
| (C) 3 | 4 | 1 | 2 | (D) 4 | 2 | 3 | 1. |

189. List I List II

- (a) Rods and cones (1) From the optic nerve
(b) Bipolar cells (2) Make synaptic connection with rods and cones
(c) Ganglion cells (3) Convert light energy into nerve impulses

Codes:

- | a | b | c | a | b | c |
|-------|---|---|-------|---|---|
| (A) 3 | 2 | 1 | (B) 2 | 3 | 1 |
| (C) 2 | 1 | 3 | (D) 3 | 1 | 2 |

190. List I List II
(Dimensions of colour) (Dimensions of colour)

- (a) Hue (1) Timbre
(b) Brightness (2) Pitch
(c) Saturation (3) Loudness

Codes:

- | a | b | c | a | b | c |
|-------|---|---|-------|---|---|
| (A) 1 | 2 | 3 | (B) 2 | 3 | 1 |
| (C) 3 | 1 | 2 | (D) 1 | 3 | 2 |

191. List I List II

- (a) Eye (1) Lower sense
(b) Ear (2) Higher sense
(c) Skin
(d) Nose

Codes:

	a	b	c	d		a	b	c	d
(A)	1	2	1	2	(B)	2	1	2	1
(C)	2	2	1	1	(D)	1	1	2	2

Direction: Following questions contain two statements 'Assertion (A) and Reason (R)'. Select the correct answer for each question according to the codes given below:

Codes:

- (A) Both A and R are true and R is correct explanation of A.
- (B) Both A and R are true but R is not a correct explanation of A.
- (C) A is true and R is false
- (D) A is false but R is true.

192. Assertion (A): Something we stare at over a period of time does not disappear while the steady, unchanging stimulation is ignored by the retinal receptors.

Reason (R): The eyes continually make minute Oscillatory movements that cause light from the stimulus to strike different retinal receptors from one moment to the next.

193. Assertion (A): Stereoscopic experience differs from the experience of the third dimension in single flat pictures.

Reason (R): Retinal disparity occurs during experiences of single flat pictures.

194. Assertion (A): An artist is able to give depth to his pictures.

Reason (R): He can make use of the many monocular cues that tell us the distance of objects.

195. Assertion (A): The mixture of light is an additive mixture, whereas the mixture of pigments is a subtractive mixture.

Reason (R): Pigments have quality to distract or absorb the lights of different colors.

196. Assertion (A): Stereoscopic combination cannot take place in the eyes

Reason (R): Nerve fibers from the eyes are separated in passing to the brain.

197. Assertion (A): Descartes was a nativist.

Reason (R): He believed in innateness of ideas.

198. Assertion (A): Locke was an empiricist philosopher.

Reason (R): According to Locke, mind is a clean slate.

199. Assertion (A): Some certain body movements enhance our reception of the stimulation.

Reason (R): The orienting reflex occurs in response to even minimal changes in the stimulus environment.

200. Assertion (A): All the senses except hearing and vision are lower senses.

Reason (R): All the senses except hearing and visual do not enter as much into man's symbolic behaviour.

201. The term 'learning' includes

- (A) all changes in behaviour
- (B) changes in behaviour resulting from practice
- (C) all permanent changes in behaviour
- (D) improvement in behaviour pattern.

202. An essential aspect of all learning is

- (A) emotion
- (B) imitation
- (C) motivation
- (D) association.

203. The fundamental property of learning is that it

- (A) modifies some stimulus—response relations
- (B) promotes physical maturation
- (C) requires perceptual skills
- (D) modifies human behaviour.

204. An animal learns by trial and error when

- (A) there is a strong motivation
- (B) the goal is clear
- (C) the problem cannot be solved
- (D) all the above conditions are fulfilled.

205. Much learning in human being takes place by

- (A) trial and error
- (B) imitation and insight
- (C) insight and conditioning
- (D) conditioning and imitation.

206. In his theory of learning, Thorndike stressed most, the importance of

- (A) law of effect
- (B) law of exercise
- (C) law of frequency
- (D) law of readiness.

- 207.** Learning, involving a change in the way information, is processed as a result of experience that an organism has had, and which has not been rewarded explicitly is
(A) foresight learning
(B) instrumental learning
(C) cognitive learning
(D) motivated learning.
- 208.** The principle of learning that an act which has a satisfying effect will be learned more quickly than one which has a classifying effect was put forth by
(A) Watson (B) Thorndike
(C) Skinner (D) Gulthrie.
- 209.** For efficient learning the learning task should be
(A) rewarding (B) interesting
(C) adjusted to the learner's level of maturity
(D) lower than the learner's level of maturity.
- 210.** In an experiment the conditioned stimulus (CS) was the word 'good' pronounced aloud by the experimenter. After a conditioning had been established, sentences like 'The pioneer helps his comrade' also produced salivation. This is known as
(A) semantic conditioning
(B) stimulus conditioning
(C) operant conditioning
(D) response conditioning.
- 211.** Mirror drawing is an example of
(A) sensory skill
(B) sensory motor skill
(C) motor skill
(D) visual skill.
- 212.** Reinforcement can be
(A) negative only
(B) positive only
(C) both negative and positive
(D) material only.
- 213.** The following reinforcement schedule is more resistant to extinction
(A) fixed and variable interval schedule
(B) fixed and variable ratio schedule
(C) continuous reinforcement schedule
(D) both (A) and (B)
- 214.** It has been found that
(A) massed practice is more efficient than spaced practice
(B) massed practice is less efficient than spaced practice
(C) massed practice is as efficient as spaced practice
(D) only spaced practice is a variable way to learning.
- 215.** When a conditioned response to a stimulus has been acquired other similar stimuli will evoke the same response. This is due to
(A) gradient of generalisation
(B) generalisation
(C) discrimination
(D) extinction.
- 216.** In operant training the reinforced behaviour and the behaviour normally elicited by the reinforcing stimulus bear the following relationship. They are
(A) same
(B) similar though they generally differ in some details
(C) totally different
(D) extinction.
- 217.** Which of the following statements is true?
(A) In classical conditioning, the response is elicited whereas in operant conditioning, the response is emitted
(B) In both classical and operant conditioning, the response is elicited
(C) In classical conditioning, the response is emitted whereas in operant conditioning the response is emitted
(D) None of the above.
- 218.** The classical and operant conditioning do not differ in the following respect.
(A) In classical conditioning the animal is passive, whereas in operant conditioning the animal is active
(B) The conditioned behaviour in classical conditioning is called respondent behaviour whereas that in operant

- conditioning is called operant behaviour.
- (C) In classical training the reinforced behaviour is similar to the behaviour normally elicited by the reinforcing stimulus whereas in operant training, it is not
- (D) By classical conditioning, a subject can be made to alter his autonomic and muscular responses, whereas by operant conditioning this is not possible.
- 219.** Social approval work as a
(A) primary reinforcer
(B) secondary reinforcer
(C) non-contingent reinforcer
(D) discriminative stimulus.
- 220.** The rate of responding is rapid in
(A) variable interval schedule
(B) fixed ratio schedule
(C) variable-ratio schedule
(D) both fixed and variable-ratio schedule.
- 221.** Learning may be best defined as change in
(A) behaviour (B) personality
(C) habits (D) aptitudes.
- 222.** Punishment results in the response being
(A) weakened
(B) suppressed
(C) unlearned
(D) substituted always by a more desirable response.
- 223.** Punishing a child for wetting the bed often
(A) substitutes the behaviour by some other behaviour
(B) decreases the frequency of the behaviour
(C) increases the frequency of the behaviour
(D) does not have any effect.
- 224.** The learning curve is one of
(A) constant gains
(B) increasing gains
(C) decreasing gains
(D) increases gains linearly up to a point and then becomes constant.
- 225.** In studying learning how to type, it has been found that there is a shift from a letter habit. This occasionally results in
(A) a linear increment in the learning curve
(B) an abrupt increase in the learning curve
(C) an abrupt decrease followed by a gradual increase
(D) a plateau in the learning curve.
- 226.** In memorising poetry, the following procedures are used.
(A) Serial memorisation
(B) Paired-associate memorisation
(C) Substance memorisation
(D) Syntactic memorisation.
- 227.** Latent learning gives evidence to
(A) trial and error-learning
(B) paired-associate learning
(C) cognitive learning
(D) learning by insight.
- 228.** Insight learning supports
(A) Psychoanalytic approach to learning
(B) Behaviourist approach to learning
(C) Neurobiological approach to learning
(D) Gestalt-approach to learning.
- 229.** Latent learning goes on under
(A) low levels of drive
(B) high levels of drive
(C) strong motivation to learn
(D) stimulating conditions of rewarded punishment that are perceived subconscious level
- 230.** Punishment differs from negative reinforcement in that punishment produces
(A) punishment and negative reinforcement do not differ
(B) active avoidance learning
(C) escape learning
(D) passive avoidance learning.
- 231.** In association stage of skill learning, the skill
(A) is learned
(B) is perfect

- (C) becomes automatic and the person no longer need to think about performing the task
(D) all the above.
- 232.** The following factor does not influence the rate at which a person learns a skill
(A) feedback about one's progress
(B) distribution of practice
(C) use of memory aids like the method of loci
(D) transfer of a previous training.
- 233.** Shifting from right-hand driving (in America) to a left-hand driving (in Britain) is an example of
(A) positive transfer of learning
(B) negative transfer of learning
(C) no transfer of learning
(D) learning by past experience.
- 234.** A boy who paints because he derives satisfaction by painting is motivated by
(A) selective reward
(B) extrinsic reward
(C) mixed reward
(D) intrinsic reward.
- 235.** Which of the following is not true about extrinsic rewards? Giving extrinsic rewards may lead to
(A) originality and self-initiated activity
(B) decrease in interest in an activity which is intrinsically satisfying
(C) docility and reference to authority
(D) competitive behaviour.
- 236.** Which of the following is not true for low and high anxious subjects ?
(A) Pressure to finish a task results in a substantially improved score for high anxious subjects but not for low anxious subjects
(B) Ego-involving instructions raises scores for low anxious subjects
(C) High-anxious subjects learn a simple classically conditioned response (e.g. eye blink to airpuff) more rapidly than low-anxious subjects
(D) High-anxious subjects act on complex tasks, the low-anxious subjects do better.
- 237.** In skill learning (e.g. learning to drive a car), the following stage(s) is (are) involved.
(A) Cognitive stage
(B) Association stage
(C) Automation stage
(D) All the above.
- 238.** Which of the following statements is true regarding transfer of learning ?
(A) The direction of transfer depends upon the degree of response similarity and the amount of transfer depends upon the degree of stimulus similarity
(B) The direction and amount of transfer depends on stimulus similarity
(C) The direction of transfer depends on response similarity
(D) The direction to transfer depends upon the degree of stimulus similarity
(e) The amount of transfer depends upon the degree of response similarity
- 239.** The grade point average is least affected by the degree anxiety at
(A) the middle range of ability
(B) the extremes of ability
(C) the effect is approximately the same at all levels of ability
(D) none of the above.
- 240.** Which of the following statements is not true ?
(A) Punishment is more effective when it is administered consistently
(B) Punishment is more effective when it is administered closer in time and place to the behaviour being punished
(C) Punishment is more effective when it is administered for stronger response tendencies
(D) Punishment is more effective when it is administered to suppress one behaviour, while at the same time positive reinforcement is used to make another behaviour more likely to occur.
- 241.** According to Gestalt psychologists, learning involves
(A) insight (B) trial and error

- (C) imitation
(D) conditioned response.
- 242.** Retention is improved when the time assigned to a learning assignment is devoted
(A) equally to reading and writing
(B) partly to reading and partly to recitation
(C) entirely to reading
(D) entirely to recitation.
- 243.** In Pavlov's original demonstration of classical conditioning, salivation to the bell was the
(A) conditioned stimulus
(B) conditioned response
(C) unconditioned stimulus
(D) unconditioned response.
- 244.** General reading ability
(A) increases regularly with age
(B) increase up to forty five years and then declines very slowly
(C) increases up to middle twenties, declines very slowly to about age of forty five, and more sharply thereafter
(D) decreases regularly but the rate of decrease is imperceptible up to twenty five, very slow up to forty five, and faster thereafter
- 245.** Verbal skill are acquired on the basis of
(A) trial and error
(B) imitation
(C) reflex vocalisation
(D) all the above.
- 246.** Sally developed a fear of balconies after falling from balcony on a couple of occasions. What was the conditioned response?
(A) The balcony
(B) Fear of the balcony
(C) Almost falling
(D) Fear resulting from almost falling.
- 247.** When the UCS is removed and the CS is presented alone for a period of time what will occur?
(A) Classical conditioning
(B) Generalization
(C) Acquisition (D) Extinction.
- 248.** Which of the following statements about 'learning by imitation' is wrong ?
(A) Imitation does not involve any reasoning
(B) There is no trial and error before imitation
(C) Imitation as a form of learning is more important for the animals of higher species, including man
(D) Man learns more by insight than by imitation.
- 249.** The loud ticking of a clock does not distract a student preparing for an examination. This phenomenon is referred to as
(A) negative adaptation
(B) adaptation
(C) conditioning (D) assimilation.
- 250.** John says, "Please pass the salt.", Ralph passes the salt. "Thank you," says John. John's request precedes a behaviour (salt passing) which is reinforced ("Thank you"); thus, the request is analogous to a
(A) discriminative stimulus
(B) response
(C) positive reinforcer
(D) conditioned stimulus (CS).
- 251.** A rat is reinforced for the first level-pressing response that occurs, on the average, after 60 seconds. Which schedule is the rat on?
(A) FR (B) VR
(C) FI (D) VI.
- 252.** When the rat presses a lever, the mild electric shock on the cage floor is turned off. What procedure is being used ?
(A) Punishment
(B) Escape
(C) Discrimination training
(D) Avoidance.
- 253.** In terms of traditional view of conditioning, research on conditioned taste aversion was surprising because
(A) there was a very long delay between CS and UCS
(B) the dislike of a particular taste was operantly conditioned

- (C) conditioning occurred to all stimuli present when the food was consumed
- (D) the sense of taste seem to be relatively weak.
- 254.** Animal trainers (the Brelands) trained pigs to put coins in a piggy bank for a food reward. The animals learned the response but, instead of depositing the coins immediately in the bank, the pigs began to toss them in the air, drop them, push them on the ground, and so on. What had occurred that interfered with conditioning ?
- (A) Conditioned taste aversion
- (B) Blocking
- (C) Instinctive drift
- (D) S & L scandal.
- 255.** A cue light comes on in dog's shuttle box. It jumps the hurdle to the other side. What procedure is being used ?
- (A) Punishment
- (B) Escape
- (C) Discrimination training
- (D) Avoidance.
- 256.** In Mowrer's explanation of avoidance, the cue stimulus acquires the capacity to elicit fear through the process of
- (A) operant conditioning
- (B) classical conditioning
- (C) generalization
- (D) discrimination.
- 257.** The contingencies are as follows: if the response does not occur, the stimulus is not presented. Under this procedure the strength of the response decrease. What procedure is being used ?
- (A) Positive reinforcement
- (B) Negative reinforcement
- (C) Punishment
- (D) Avoidance training.
- 258.** Periods of dead-level performance in the learning curves are called
- (A) plateaus
- (B) negative adaptation
- (C) latent learning period
- (D) extinction.
- 259.** Children learn new words for specific objects mostly by
- (A) imitation
- (B) conditioning process
- (C) insight
- (D) trial and error.
- 260.** The correlation between retention and learning is
- (A) zero
- (B) low and positive
- (C) low and negative
- (D) high and negative.
- 261.** Learning by insight involves
- (A) establishment of bond between stimuli and appropriate responses
- (B) retention of satisfying responses and elimination of dissatisfying responses
- (C) sudden awareness of the relationship between the goal and means
- (D) the use of intelligence.
- 262.** Sally developed a fear of balconies from almost falling. Although she has had no dangerous experiences on bridges, cliffs and the view from tall building, she now fear these stimuli as well. Which of the following is likely to have produced a fear of these other stimuli ?
- (A) Instinctive drift
- (B) Spontaneous recovery
- (C) Generalization
- (D) Discrimination.
- 263.** A researcher reinforces closer and closer approximations to a target behaviour. What is the name of the procedures she is using ?
- (A) Shaping
- (B) Classical conditioning
- (C) Discrimination training
- (D) Extinction.
- 264.** The major difference between stimulants and alcohol is that stimulants
- (A) are stronger
- (B) are more dangerous
- (C) are pleasant to use
- (D) produce central nervous system arousal.

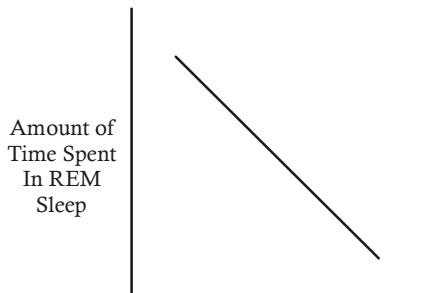
- 265.** According to the text, which of the following hallucinations is especially common after taking large doses of cocaine ?
(A) "I am dead"
(B) "There are bugs crawling under my skin"
(C) "I see wonderful shapes and colours"
(D) "I can't control myself".
- 266.** Amphetamines commonly have been used by people for all of the following except
(A) losing weight (B) staying awake
(C) reducing anxiety
(D) getting "high".
- 267.** A stimulus paired with a UCS fails to establish a CR because it is redundant with an already established CS. What is involved ?
(A) Spontaneous recovery
(B) Stimulus generalization
(C) Instinctive drift
(D) Blocking.
- 268.** Earlier learning viewpoint considered classical and operant conditioning to be automatic process involving environmental events that did not depend at all on biological or cognitive factors. Research in which of the following areas casts doubt on this point of view ?
(A) Blocking and signal relations
(B) Instinctive drift and conditioning taste aversion
(C) Response-outcome relations
(D) All the above.
- 269.** According to Freud, the deeper hidden meaning of a dream is called its
(A) manifest content
(B) latent content
(C) defense mechanism
(D) libido.
- 270.** According to text, how often does a normal adult day dream ?
(A) About once every 90 minutes
(B) About 3 to 4 times a day
(C) About 3 to 4 times a week
(D) It varies widely from one person to another.
- 271.** According to the text, most widely used mind-altered drug in the United States is
(A) cocaine (B) marijuana
(C) thorazine (chlorpromazine)
(D) alcohol.
- 272.** EEG patterns during REM sleep are most similar to those of a person who is
(A) awake (B) sleep-walking
(C) in theta sleep (D) in delta sleep.
- 273.** The average person goes through about how many sleep cycles in a typical night's sleep ?
(A) One (B) Five
(C) Eight (D) Hundreds.
- 274.** Which of the following statements regarding REM sleep is true ?
(A) REM deprivation enhances memory consolidation.
(B) As you get older, you spend more time in REM sleep
(C) If we are continuously deprived of REM sleep, our bodies get used to functioning with less of it.
(D) One half of an infant's sleep is REM sleep.
- 275.** Marijuana, like alcohol, seem to produce the greatest impairment in
(A) our ability to remember names of people and common objects
(B) our ability to remember events from the distant past
(C) our ability to store new information in a long-term store
(D) our ability to speak.
- 276.** According to test, what is the relationship between drug use and creativity ?
(A) Drugs usually produce creativity
(B) Drugs almost never produce creativity
(C) The relationship is unclear
(D) Some drugs produce creativity; other inhibits it.
- 277.** In some cases, hypnosis is used to help witness remember the details of a crime. The major objection to using this kind of evidence to support the idea that hypnosis improves memory is that

- (A) the hypnotists are not usually too frightened to be hypnotized effectively
- (B) the witness are usually too frightened to be hypnotized effectively
- (C) the courts will not allow the evidence to be studied outside the courtroom
- (D) there is no adequate control group with which to compare.

278. According to Hilgard's dissociation view, hypnosis is

- (A) an illusion
- (B) an altered state of consciousness
- (C) a hoax
- (D) an extreme form of persuasion.

279. The figure below shows a decreasing function between the amount of time a person spends in REM sleep and some other variable. What is that other variable?



- (A) the person's age
 - (B) how much total sleep the person gets
 - (C) the person's intelligence
 - (D) fatigue.
- 280.** Which of the following physiological changes does not occur during meditation?
- (A) Breathing slows
 - (B) Skin resistance to electrical conductivity rises
 - (C) Blood pressure falls
 - (D) Oxygen consumption increases.
- 281.** The text describe an experiment in which subjects had an image steadily cast on to one spot on their retinas. What was the result ?
- (A) They started daydreaming
 - (B) The image gradually disappeared and their brain produced alpha waves
 - (C) They became hypnotized

- (D) They started hallucinating much as if they had taken LSD.

282. About what percentage of the population can be hypnotized ?

- (A) 10 per cent (B) 50 per cent
- (C) 75 per cent (D) 95 per cent.

283. You observe that a sleeper's eyes move up and down repetitively. You wake the sleeper and find that she has been dreaming about watching cliff divers. This experience would support

- (A) neodissociation theory
- (B) the scanning hypothesis
- (C) cognitive role enactment theory
- (D) psychoanalysis.

284. Max dreams of being chased by a snake, and he is sure this dream represents insecurity with women. Max is interpreting the dream's

- (A) irregular content
- (B) hypothetical content
- (C) manifest content
- (D) latent content.

285. According to role enhancement theory, hypnotism is most similar to

- (A) being awake (B) delta sleep
- (C) narcolepsy (D) REM sleep.

286. Suppose you discover that your attention span varies as a function of the time, day being longest when you awaken, and growing increasingly shorter throughout the day. The concept which is most closely related to your discovery would be

- (A) dissociation
- (B) narcolepsy
- (C) role enactment
- (D) circadian rhythm.

287. Suppose you observe a person whose EEG shows a very high-level of alpha waves. You should guess that person was

- (A) relaxed but not asleep
- (B) in delta sleep
- (C) walking or talking in his or her sleep
- (D) daydreaming.

288. Darcy has no trouble falling asleep but wakes up every morning feeling tired. His wife complaints that she often gets him

- gasping for breath when he is asleep, as if he forgot to breathe, and then remembered. Darcy's problem is most likely
- (A) Korsakoff's syndrome
 - (B) insomnia
 - (C) sleep apnea
 - (D) narcolepsy.
- 289.** You are working in a lab and you analyze a blood sample that contains .06 per cent alcohol. You conclude that the person it was drawn from probably
- (A) had a drink before a routine physical exercise
 - (B) got picked up for drunk driving
 - (C) passed out from drinking too much alcohol
 - (D) died from drinking too much alcohol.
- 290.** Suppose you drink 15 ounces of alcohol per week. According to presented in the text, to stay the healthiest
- (A) you should drink about 2.5 ounces per day
 - (B) you should drink only on weekdays
 - (C) you should drink it all in one day
 - (D) your drinking pattern does not matter; and total amount you drink does.
- 291.** All instances of classical conditioning are based on
- (A) the orienting reflex
 - (B) naturally occurring stimulus response patterns
 - (C) habituation
 - (D) operant conditioning.
- 292.** In Pavlov's example, the unconditioned stimulus (UCS) was
- (A) a tone or bell
 - (B) salivation
 - (C) hunger
 - (D) food placed in the mouth.
- 293.** In most cases of classical conditioning, the
- (A) CS occurs just before the UCS
 - (B) CS occurs just after the UCS
 - (C) CR occurs before the CS
 - (D) CS and UCS occur randomly.
- 294.** In classical conditioning, when the UCS occurs before the CS, this is called
- (A) spontaneous recovery
 - (B) chaining
 - (C) forward conditioning
 - (D) backward conditioning.
- 295.** You teach a child to be afraid of the dog next door because it bites children, but now the child becomes afraid of all dogs. This is an example of
- (A) discrimination
 - (B) generalization
 - (C) stimulus control
 - (D) a discriminative stimulus.
- 296.** The tendency for an extinguisher response to recur after a rest from the experimental procedures is called
- (A) superstitious behaviour
 - (B) discrimination
 - (C) spontaneous recovery
 - (D) behaviour modification.
- 297.** The device used by Thorndike in an experiment with cats who learned to escape for a food reward is called a
- (A) maze
 - (B) Skinner box
 - (C) Pavlovian apparatus
 - (D) puzzle box.
- 298.** In his explanation of behaviour, Edward Thorndike had most in common with
- (A) Ivan Pavlov
 - (B) Edward Tolman
 - (C) Albert Bandura
 - (D) B. F. Skinner.
- 299.** Generalizing from an experiment described in the text, if you hypnotized a person and suggest to the person that he was blind in his eye, you would expect him to
- (A) be truly blind in that eye
 - (B) be able to see with that eye only when his right eye was closed
 - (C) be able to see with that eye but deny the sight
 - (D) admit to being able to see if confronted with evidence that the eye was working.
- 300.** Dr. Rogers believes that all hypnotists are "fake" and that hypnosis is just a matter of steady concentration. Her view would fit most closely with which of the following?

- (A) The scanning hypothesis
(B) Latent content
(C) Cognitive role enactment
(D) Dissociation.
- 301.** Which of the following is a characteristic of learning but not of performance ?
(A) Observable
(B) Measurable
(C) A mental activity
(D) It changes.
- 302.** Every time you get drunk, you are outgoing and introduced yourself to people. When you are sober, you don't remember any of them. This is an example of
(A) inhibitory conflict
(B) consolidation memory
(C) alcoholic blackout
(D) Korsakoff's syndrome.
- 303.** Linda has taken a drug and now is panicking because she cannot control the intense hallucinations she is experiencing. The drug, she has taken, is most likely
(A) marijuana (B) cocaine
(C) LSD
(D) an amphetamine.
- 304.** The normal sleep-wake cycle for humans spans a total of about how many hours ?
(A) 12 hours (B) 23 hours
(C) 24 hours (D) 25 hours.
- 305.** In the figure below, the shaded portions represent
- 
- (A) sleep apnea
(B) delta sleep
(C) REM sleep
(D) wakeful periods during a night's sleep.
- 306.** _____ is also known as 'Instrumental condition'.
(A) Skinner experiment
(B) Operant experiment
(C) Pavlov experiment
(D) None of the above
- 307.** A rat learns that he will reinforced only a red light in his cage is on. This is an example of
(A) generalization
(B) stimulus control
(C) continuous reinforcement
(D) backward conditioning.
- 308.** Which of the following schedule exists when a reward is delivered after ten responses ?
(A) Fixed interval schedule
(B) Variable interval schedule
(C) Fixed ratio schedule
(D) Variable ratio schedule.
- 309.** New operant behaviour is usually best established by using which of the following reinforcement schedule ?
(A) Fixed-interval
(B) Variable-interval
(C) Continuous
(D) Variable-ratio.
- 310.** Decreasing the frequency of a response by following the response with a negative consequence is
(A) systematic desensitization
(B) punishment
(C) negative reinforcement
(D) behaviour modification.
- 311.** Which of the following is most likely to be a primary reinforcer ?
(A) Praise (B) Money
(C) Attention (D) Water.
- 312.** How would you extinguish an operantly conditioned response ?
(A) Stop reinforcement
(B) Prevent the response
(C) Punish the response
(D) Remove the stimulus.
- 313.** Which of the following reinforcement schedules provides a reward at the end of a particular amount of time, regardless of the number of responses made during that time ?
(A) Fixed ratio
(B) Fixed interval
(C) Variable ratio
(D) Variable interval.

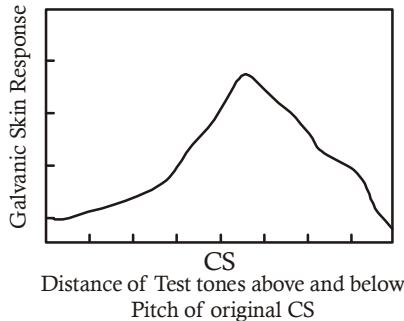
- 314.** When an individual mistakenly assumes a contingency between a response and reinforcement when no such contingency exists, the behaviour that produced is called
(A) superstitious behaviour
(B) forward conditioning
(C) backward conditioning
(D) negative reinforcement.
- 315.** According to Garcia's studies, rats are especially likely to attribute the occurrence of shock to
(A) the taste of the food they were eating when they were shocked
(B) the taste of the food they were eating about two hours before they were shocked
(C) the auditory and visual characteristic of the situation in which they ate food.
(D) the severity of shock.
- 316.** The process of a pig discontinuing dripping coins in a slot and instead rooting them around the pen is called
(A) instinctive drift
(B) discrimination
(C) generalization
(D) prepared learning.
- 317.** Observational learning is based predominantly upon
(A) backward conditioning
(B) vicarious reinforcement
(C) classical conditioning
(D) operant conditioning.
- 318.** You have conditioned your dog to bark every time you say the word 'Friday'. Now, however, you are no longer reinforcing the dog for barking. This second circumstance should produce
(A) spontaneous recovery
(B) extinction
(C) superstitious behaviour
(D) a phobia.
- 319.** You are teaching a mentally retarded child to dress herself. When she does something correctly, you give her a poker chip, which she can later exchange for food or privileges. This is an example of
(A) programmed instruction
(B) discrimination training
(C) a token economy
(D) systematic desensitization.
- 320.** You work on an assembly line, and you know that every day your output is checked by a computer. However, you do not know when it is checked, since the time changes from day-to-day. Which of the following reinforcement schedule are you on ?
(A) Fixed ratio (B) Fixed interval
(C) Variable ratio (D) Variable interval
- 321.** Latent learning refers to process of learning based on
(A) reinforcement
(B) observation
(C) operant conditioning
(D) classical conditioning.
- 322.** You teach your dog to sit by giving him a biscuit every time he sits on command. This is an example of
(A) backward conditioning
(B) observational learning
(C) classical learning
(D) operant conditioning.
- 323.** The principles of reinforcement are most similar to the ideas expressed in
(A) the law of effect
(B) discrimination training
(C) latent learning
(D) vicarious learning.
- 324.** A fourth-grade teacher promises that if Billy does well on the spelling test, he will not have to stay in and study during recess. She is using the technique of
(A) positive reinforcement
(B) negative reinforcement
(C) punishment (D) generalization.
- 325.** Teaching a tiger to jump through a flaming hoop (something it would never do on its own) almost certain involves the process of
(A) shaping
(B) systematic desensitization
(C) discrimination training
(D) classical conditioning.
- 326.** Recent research has indicated that different people have different cognitive

styles, or ways of thinking about problems. This research would be most consistent with the beliefs of

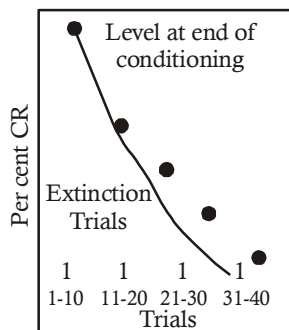
- (A) Edward Thorndike
- (B) B. F. Skinner
- (C) Edward Tolman
- (D) John B. Watson.

327. You drive around all day in a strange city looking for a particular building. You finally find it. Now your friend wants you to go to another building. Even though you weren't looking for it, you can drive directly to the second building, since you remember it from earlier in the day. This example is similar to
- (A) latent learning
 - (B) stimulus control
 - (C) modelling
 - (D) discrimination training.

328. The following figure is a graphic representation of the result of



- (A) reinforcement
 - (B) latent learning
 - (C) discrimination training
 - (D) extinction.
329. Which of the following classical conditioning contingencies would produce the result graphed in the figure below ?



- (A) Presenting the UCS just before the CS
- (B) No longer pairing the CS and the UCS
- (C) Presenting the CR just after the CS
- (D) Presenting the CS and the USC together.

Direction: There are questions given below. Each question contains two lists (I and II) related in some sense. Match List I with List II and select the correct answer from the codes given below each question.

330. List I (Experiments) List II (Conditioning)
- (a) Pavlov's (1) Classical
 - (b) Skinner (2) Operant

Codes:

- | | a | b | | a | b |
|-----|---|---|-----|---|----|
| (A) | 1 | 2 | (B) | 2 | 1 |
| (C) | 1 | 1 | (D) | 2 | 2. |

331. List I (Process) List II (Meaning)
- (a) Generalization (1) Reaction to differences
 - (b) Discrimination (2) Repetition of the conditioned stimulus without reinforcement
 - (c) Acquisition (3) Association between the CS and the US
 - (d) Extinction (4) Reaction to similarities

Codes:

- | | a | b | c | d | | a | b | c | d |
|-----|---|---|---|---|-----|---|---|---|----|
| (A) | 1 | 2 | 3 | 4 | (B) | 4 | 1 | 3 | 2 |
| (C) | 4 | 2 | 3 | 1 | (D) | 1 | 2 | 3 | 4. |

332. Learning process involves
- (i) A motive or drive
 - (ii) An attractive goal
 - (iii) A block to the attainment of the goal
- Code:**
- (A) Only (i)
 - (B) Only (i) and (ii)
 - (C) Only (ii) and (iii)
 - (D) All of the above

333. List I List II
- (a) T-maze (1) Apparent motion

- (b) Phi-phenomena (2) Visual Illusion
(c) Ponzo-effect (3) Operant conditioning
(d) Cumulative recorder (4) Simple choice learning

Codes:

	a	b	c	d		a	b	c	d
(A)	1	4	2	3	(B)	4	2	1	3
(C)	1	2	3	4	(D)	4	1	2	3.

334. List I List II
(Reinforcement (Condition) schedules)

- (a) FR (1) Occurs after a specified period of time non-reinforced responses
(b) FI (2) Occurs after a fixed number of nonreinforced responses
(c) VR
(d) VI

Codes:

	a	b	c	d		a	b	c	d
(A)	1	2	1	2	(B)	2	1	2	1
(C)	2	2	1	1	(D)	1	1	2	2.

Direction: In questions are two statements 'Assertion (A)' and 'Reason (R)'. You are to check whether these two statements are true or not; if a true, whether R is the correct explanation of A or not. Select your answer from the codes given below:

Codes:

- (A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true, R is false.
(D) A is false but R is true.

335. Assertion (A): Classical conditioning has been regarded by many psychologists as an appropriate starting point for the investigation of the learning process.

Reason (R): Classical conditioning represents an extremely simple form of learning.

336. Assertion (A): Generalization and discrimination appear in ordinary behaviour.

Reason (R): The young child who has learned to say "bow-wow" to a dog may understandably respond in like manner to similar stimulus, such as a sheep.

337. Assertion (A): The rat can respond to the bar in a skinner box as frequently or infrequently as it chooses.

Reason (R): The bar is always present in the skinner box.

338. Assertion (A): Secondary reinforcement has important practical implications.

Reason (R): Secondary reinforcement has wide degree of generalization.

339. Assertion (A): The procedure of pairing the CS and US is called reinforcement.

Reason (R): Any tendency for the CR to appear is facilitated by the presence of the US and the response to it.

340. Assertion (A): Information is transferred to LTM from STM.

Reason (R): Information maintained in the rehearsal buffer is prevented from decaying in STM.

341. 'Saving Score' is used as a measure of

- (A) recall (B) relearning
(C) redintegration (D) recognition.

342. Which of the following is correct ?

- (A) $\frac{\text{Right recognition}}{\text{Wrong recognition}} \times 100$
(B) $\frac{\text{Right recognition}}{\text{Total recognition}} \times 100$
(C) $\frac{(\text{Right} - \text{Wrong}) \text{ recognition}}{\text{Total recognition}} \times 100$
(D) $1 - \frac{\text{Wrong recognition}}{\text{Total recognition}} \times 100$

343. Which of the following is a more sensitive measure of memory ?

- (A) Recall (B) Redintegration
(C) Relearning (D) Recognition.

344. The most essential feature of memory is

- (A) recall (b) recognition
(C) retention (d) fixation.

- 345.** The amount of retention depends on
(A) degree of learning
(B) meaningful subject matter
(C) nature of interference
(D) all the above.
- 346.** Recollection of childhood memories on the basis of partial cues is a method of remembering called
(A) redintegration (B) recall
(C) recognition (D) relearning.
- 347.** Altered state of living tissue responsible for memory is called
(A) axon (B) engram
(C) mneme (D) phoneme.
- 348.** Among the following situations, the one that would be easiest to recall is
(A) a small beautiful passage
(B) series of events in a logical order
(C) material committed to memory
(D) an event causing intense emotional disturbance.
- 349.** The phenomenon of a tune 'running in the head' is called
(A) involuntary recall
(B) reinforcement
(C) reinstatement
(D) preservation.
- 350.** Eidetic imagery is more frequent among
(A) children (B) adolescents
(C) adults (D) old people.
- 351.** The most convincing proof of retention is
(A) relearning (B) recognition
(C) recollection (D) recall.
- 352.** Recognition is impossible in the absence of
(A) attitude (B) interest
(C) distinction (D) aptitude.
- 353.** Very often, we readily locate in a group someone we met earlier but cannot get to his name. This is an aspect of retention called
(A) memory of form
(B) recognition
(C) delayed response
(D) recalls.
- 354.** Recall is facilitated by
(A) recalling the facts of the relevant phenomenon
(B) special drugs
(C) an incentive, such as a reward
(D) repression.
- 355.** Which of the following statements is true?
(A) There is no memory before language develops
(B) Ordinarily fast learners retain more than slow learners
(C) Forgetting necessarily involves obliteration of memory traces
(D) The amount of forgetting after inactivity and sleep is more than after inter-polation of activities between original learning and tests of retention.
- 356.** Mnemonic devices
(A) improve memory
(B) improve retention
(C) aid recall
(D) improve recognition.
- 357.** Which of the following is the most scientific and experimentally tested explanation of forgetting? Forgetting is through
(A) decay through disuse
(B) interference effects
(C) motivated forgetting
(D) disintegration of memory traces.
- 358.** When prior learning interferes with recall of a later learning, it is called
(A) retroactive inhibition
(B) reverse inhibition
(C) proactive inhibition
(D) suppression.
- 359.** Aphasia's pertains to
(A) memory defects
(B) personality defects
(C) behavioural disorder
(D) emotional disorders.
- 360.** In cases of amnesia; usually the following memories are lost.
(A) The undated memories
(B) The personal memories
(C) The impersonal memories
(D) None of the above.

- 361.** TOT (Tip of the Tongue) state is an example of
(A) loss of information
(B) trace-dependent forgetting
(C) cue dependent forgetting
(D) loss of cue.
- 362.** Amnesia may be an extreme case of
(A) senile dementia
(B) retroactive inhibition
(C) proactive inhibition
(D) repression.
- 363.** Which of the following statements is false ?
(A) Episodic memory is less highly organised than semantic memory
(B) Episodic memory is part of the long-term memory organisation
(C) Episodic memory is more susceptible to forgetting than semantic memory
(D) Episodic memory may become a part of semantic memory.
- 364.** At the time of input, the material is elaborated, simplified and changed; these modifications at the time of input are called
(A) redintegration
(B) reconstructive processes
(C) constructive process
(D) confabulation.
- 365.** Information processed in the rehearsal buffer may be
(A) passed to the sensory register
(B) passed along to the short-term store
(C) lost
(D) passed along to the long-term store.
- 366.** Transfer of information from the sensory register to the short-term store depends on
(A) attention (B) rehearsal buffer
(C) sensation (D) active retrieval.
- 367.** TOT phenomenon indicates that information is
(A) organised in semantic memory
(B) stored in short-term memory
(C) organized in form of episodic memory
(D) organised in short-term memory.
- 368.** In an experiment subjects learned lists in which the stimuli and responses of paired associates varied in their image—arousing capability. C items evoked concrete imagery, while A, or abstract items evoked little imagery. The mean total recall would be highest for
(A) A—C (B) C—A
(C) C—C (D) A—A.
- 369.** The duration upto which the short-term store holds information is about
(A) 3 seconds (B) 3 milli-seconds
(C) 30 minutes (D) 30 seconds.
- 370.** The efficiency of immediate memory is determined by testing
(A) intelligence (B) memory span
(C) vocabulary (D) attention span.
- 371.** The most popular memory span material consists of
(A) words (B) figures
(C) numbers (D) objects.
- 372.** The capacity of sensory register as compared to the short-term store is
(A) more (B) less
(C) same (D) insignificant.
- 373.** The levels of processing in order to increasing depth of processing are
(A) perception, structural analysis, elaboration
(B) perception, structural analysis, semantic analysis
(C) perception, elaboration, semantic analysis
(D) perception, semantic analysis, structure analysis.
- 374.** Long-term memory is more likely to result from
(A) passive process of repetition
(B) elaborative rehearsal
(C) maintenance rehearsal
(D) maintenance rehearsal.
- 375.** The sensory storage seems to be (in vision at least) on the form of a
(A) abstract image
(B) polar image
(C) concrete image
(D) iconic image.
- 376.** In contrast to the short-term store, the capacity of the long-term store is
(A) huge and has no known limit

- (B) small and limited
(C) large but limited
(D) the capacities of short and long-term stores are comparable.
- 377.** Which of the following is not a major reason for information loss in long-term memory?
(A) Faculty organisation
(B) Displacement of old information by coming information
(C) Interference
(D) Inappropriate retrieval strategy.
- 378.** The major reason for information loss in the sensory register is
(A) interference
(B) displacement of old information by incoming information
(C) decay of trace
(D) both (A) and (B).
- 379.** Memory Span is affected by
(A) sex
(B) age
(C) neither by age nor sex
(D) both by sex and age.
- 380.** The saving method of testing memory
(A) makes use of the lists of non-sense syllables
(B) compares learning and relearning scores
(C) measures the amount of recall
(D) uses non-verbal material.
- 381.** With intermittent reinforcement
(A) learning is slow
(B) both learning and forgetting are slow
(C) forgetting is slow
(D) both learning and forgetting are fast.
- 382.** Forgetting is primarily a function of
(A) time interval between presentation and recall of an item
(B) amount of interpolated activity
(C) rate of interpolated activity
(D) all the above.
- 383.** Interference in recalling a previously learnt material due to an interpolated similar material learnt later is an example of
(A) reproductive inhibition
(B) amnesia
(C) reactive inhibition
(D) retroactive inhibition.
- 384.** The number of items a person can grasp in a single act of attention is called
(A) immediate memory span
(B) primary effect
(C) long-term memory
(D) retained number.
- 385.** Complete loss of memory is termed
(A) trauma (B) amnesia
(C) coma (D) paramnesia.
- 386.** Information is primarily transferred from short-term memory to long-term memory through the process of
(A) elaboration
(B) verbal rehearsal
(C) clustering (D) sensory-coding.
- 387.** In learning a list of 20 new Spanish words you are likely to experience
(A) a primacy effect
(B) a recent effect
(C) a serial position effect
(D) all the above.
- 388.** Interference is to forgetting as clustering is to
(A) categories (B) decay
(C) rehearsal (D) recognition.
- 389.** When you attempt to recall the name of a high school classmate by imagining yourself back in the English class with her, you are making use of
(A) retrieval cues (B) context cues
(C) schemas (D) recognition cues
- 390.** Taking this particular self-test measures yours
(A) constructive errors
(B) reconstructive errors
(C) recall (D) recognition.
- 391.** Ineffective coding of information may result in
(A) the primary effect
(B) the recency effect
(C) pseudo forgetting
(D) chunking.
- 392.** Decay theory is best able to explain the loss of memory in

- (A) sensory store
(B) long-term memory
(C) short-term memory
(D) both short-term and long-term memory
- 393.** Which of the following is not one of the three basic human memory processes ?
(A) Storage (B) Retrieval
(C) Decoding (D) Encoding.
- 394.** Which one of the three levels of processing would probability be employed when attempting to memorize the following three-letter sequences WAB WAC WAD?
(A) Structural (B) Semantic
(C) Phonemic (D) Chunking.
- 395.** Retrieval from long-term memory is usually best when the information has been stored at which level of processing ?
(A) Structural (B) Semantic
(C) Phonemic (D) Chunking.
- 396.** According to Pavo's dual-coding theory
(A) words are easier to encode than images
(B) abstract words are easier to encode than concrete words
(C) visual imagery may hinder the retrieval of words
(D) none of the above.
- 397.** Which of the memory stores can store the least amount of information ?
(A) Sensory store
(B) Short-term memory
(C) Long-term memory
(D) None of the above.
- 398.** Hypnotized subjects are generally able to achieve highly accurate retrieval of information from long-term memory. This statement is
(A) True
(B) False
(C) could not be interpreted
(D) none of the above.
- 399.** Forgetting is
(A) an active mental process
(B) a passive mental process
(C) neither an active nor a passive mental process
(D) an active mental process in some cases and a passive mental process in others
- 400.** Forgetting may be described as
(A) negative retention
(B) negative learning
(C) negative recall
(D) negative recognition.
- 401.** The curve of forgetting was first drawn by
(A) Thorndike (B) Ebbinghaus
(C) Skinner (D) Guthrie.
- 402.** With the number of repetitions, the asymptom of the forgetting curve
(A) There is no direct relationship
(B) Decreases
(C) Remain constant
(D) Increases.
- 403.** Each repetition of the verbal response can be viewed as producing an increment in
(A) background conditioning and in cue learning
(B) background conditioning only
(C) cue learning only
(D) none of the above.
- 404.** When you violate encoding specificity principle you are likely to experience an inability to
(A) encode information
(B) store information
(C) retrieve information
(D) none of the above.
- 405.** Evidence continues to grow that RNA provides the chemical code for memory. This statement is
(A) true (B) false
(C) could not be interpreted
(D) none of the above.
- 406.** It is very easy to recall the name of your high school because it has been subject to extensive
(A) de processing (B) clustering
(C) chunking (D) overlearning.
- 407.** Memories are permanently stored in
(A) LTM (B) Sensory
(C) STM (D) Mnemonic devices.

408. In the following diagram, where is short-term memory ?



- (A) Box A
(B) Box B
(C) Box C
(D) Between box A and box B.
409. In his research on sensory memory, Sperling flashed a twelve-item pattern on a screen. He found that subjects could report
- (A) only the top row of four items
(B) only the bottom row of four items
(C) any one of the rows
(D) all the three rows.
410. One way to interfere with information held in short-term memory is
- (A) to engage in semantic rehearsal
(B) to engage in acoustic rehearsal
(C) to count backward from one hundred by threes
(D) to use a mnemonic device.
411. The idea that memories which are stored when the person was in a particular frame of mind are best recalled when the person is again in that same frame of mind is called
- (A) mood-confabulation memory
(B) confabulation
(C) mood-dependent memory
(D) connectionist framework.
412. The feeling that what you are presently experiencing has happened before, even though you know it has not, is called
- (A) Deju vu
(B) Peg word system repression
(C) Eidetic imagery
(D) None of the above.
413. The memory that lasts about twenty seconds and is maintained by acoustic rehearsal is
- (A) sensory memory
(B) long-term memory
(C) short-term memory
(D) procedural memory.
414. If you were asked to “shadow” a message, you would
- (A) ignore it
(B) repeat it aloud
(C) focus attention on it
(D) try to remember it for later recall.
415. Which of the following is not a type of long-term memory ?
- (A) Procedural memory
(B) Acoustic memory
(C) Episodic memory
(D) Semantic memory.
416. Mental representations of objects, states, and qualities of the world and also of how these things are related to each other are called
- (A) episodic memories
(B) procedural memories
(C) eidetic images
(D) semantic memories.
417. Essay tests require____; multiple choice tests require____ .
- (A) recall; recognition
(B) recognition; recall
(C) recall; recall
(D) recognition: recognition.
418. Shallow processing memory is distinguished from deep processing memory by
- (A) part of the brain in which storage occurs
(B) kind of material being remembered
(C) amount of material being remembered
(D) kind or rehearsal used to remember.
419. Techniques we can use to improve our memories are called
- (A) eidetic techniques
(B) confabulation techniques
(C) consolidation techniques
(D) mnemonic devices.
420. Retrograde amnesia involves the loss of part of
- (A) long-term memory
(B) sensory memory
(C) short-term memory
(D) eidetic images.
421. Neurophysiologists believe that the temporary circulation of electrical impulses

- around complex loops of interconnected neurons is responsible for
- (A) confabulation
 - (B) retrograde amnesia
 - (C) long-term memory
 - (D) short-term memory.
- 422.** A memory expert remembers long lists of words by imagining that he is walking down a familiar street and picks up objects. This technique is an example of
- (A) déjà vu
 - (B) method of loci
 - (C) peg word system
 - (D) eidetic imagery
- 423.** In an effort to remember a list of complicated words for a psychology examination, you associate each item with another item on a sequential list. This technique is called
- (A) peg word system
 - (B) eidetic memory
 - (C) method of loci
 - (D) consolidation.
- 424.** You participate in four word-recall experiments. In the first, you rehearse by looking at the word lists. In the second, you concentrate on words that rhyme with the list words. In the third, you study the meaning of the words. In the fourth, you count backward while being presented with the list will you probably recall best?
- (A) First
 - (B) Second
 - (C) Third
 - (D) Fourth.
- 425.** Suppose you conceptualize learning as building a complex system of roads in your brain. Forgetting involves taking the wrong road. This view is similar to the concept of
- (A) repression
 - (B) decay
 - (C) motivated forgetting
 - (D) network model.
- 426.** Which of the following pieces of experimental evidence indicates that memories are not permanent and unchangeable?
- (A) Penfield's brain stimulation studies
 - (B) Improved recall under hypnosis
 - (C) Improved recall in psychoanalysis
 - (D) Loftus' studies of eyewitness testimony
- 427.** Which of the following groups of letters is within the limits of short-term memory?
- (A) P-B-G-Q-X-R-C-L-D-S
 - (B) Q-X-L-T-M-A-R-C-F-P-A-Z-T
 - (C) D-O-G-C-A-T-C-O-W-H-O-R-S-E
 - (D) C-T-U-M-A-I-P-R-Q-Q-T-W-W.
- 428.** In an experiment, you are asked to repeat the message you hear in your right ear verbally while you listen to a different message in each ear. This is called
- (A) confabulation
 - (B) selective attention
 - (C) proactive interference
 - (D) shadowing.
- 429.** If you delivered electro-convulsive shock to a depressed patient, you would expect the resulting memory loss to be for that period of time
- (A) In the long distant past
 - (B) Just after the patient woke up after the shock treatment
 - (C) About one to three days after the shock
 - (D) Just before the shock was given.
- 430.** Jack is eighty-seven year old and has good recall of events before his sixty-fifth birthday. However, he remembers virtually nothing since then. He probably suffers from
- (A) anterograde amnesia
 - (B) retrograde amnesia
 - (C) repression
 - (D) consolidation.
- 431.** The occurrence of anterograde amnesia supports the idea of
- (A) confabulation
 - (B) consolidation
 - (C) influences of learning in human memory
 - (D) genetic influences in human memory.
- 432.** Marlene is raped as a child but has no memory of this traumatic event. What theory best accounts for this loss?
- (A) Motivated forgetting

- (B) Interference
(C) Consolidation (D) Decay.
- 433.** Loftus's research on eyewitness testimony indicates
(A) eyewitnesses often lie to protect themselves
(B) eyewitnesses can distort the truth without realizing it
(C) eyewitnesses are typically accurate in their recall.
(D) eyewitnesses are entirely unreliable.
- 434.** Mood-dependent memory may be best explained by the
(A) power of mood on states of mind
(B) networks of relationship between ideas
(C) schemas involved in memory
(D) inferences caused by mood.
- 435.** Anterograde amnesia refers to the loss of
(A) sensory memory
(B) things stored in long-term memory
(C) memory for new information
(D) recognition and recall
- 436.** The notion that storage into long-term memory requires time is called
(A) consolidation
(B) synaptic rehearsal
(C) enrichment
(D) confabulation.
- 437.** You are talking to a friend at a large party when you hear someone else mention your name. You immediately start paying attention to the other conversation. This demonstrates
(A) dichotic listening
(B) selective attention
(C) the tip-of-the-tongue phenomenon
(D) shadowing.
- 438.** You are taking in an experiment in which each eye sees a different slide projected for a fraction of a second. This experiment is similar to experiment in
(A) dichotic listening
(B) selective attention
(C) eidetic imaging
(D) retroactive interference.

Direction: Match List (I) with List (II) in the questions are select the correct answer from the codes given below the lists.

439. Match the following:

List I (Kinds of memory)

- (a) Redintegrative memory
(b) Recall
(c) Recognition (d) Relearning

List II (Related issues)

- (1) It is familiar that we have met it before
(2) There is some residue from the past to learn easily
(3) Activity of some performance learned in the past
(4) To check details of the recovery of events in the personal past

Codes:

- | | a | b | c | d | | a | b | c | d |
|-----|---|---|---|---|-----|---|---|---|----|
| (A) | 4 | 3 | 1 | 2 | (B) | 1 | 2 | 4 | 3 |
| (C) | 1 | 4 | 2 | 3 | (D) | 4 | 1 | 3 | 2. |

440. List I

(Kinds of forgetting)

- (a) Decay through disuse
(b) Retroactive inhibition
(c) Proactive inhibition
(d) Repression

List II (Related issues)

- (1) Forgetting takes place simply through the passage of time
(2) The new learning may interfere with old one
(3) The old learning may interfere with new one
(4) Retrieval, in some way, is unacceptable to the person.

Codes:

- | | a | b | c | d | | a | b | c | d |
|-----|---|---|---|---|-----|---|---|---|----|
| (A) | 4 | 3 | 2 | 1 | (B) | 3 | 2 | 1 | 4 |
| (C) | 2 | 1 | 4 | 3 | (D) | 1 | 2 | 3 | 4. |

441. List I

- (a) Encoding
(b) Storage

List II

- (1) Transformation of sensory information.
(2) Transformation of the encoded information

- (c) Retrieval (3) Information is located in the memory when it is needed

Codes:

a	b	c	a	b	c
(A) 3	2	1	(B) 1	2	3
(C) 2	3	4	(D) 3	1	2

442. List I

- (a) Cue-dependent forgetting
(b) Trace-dependent forgetting
(c) STM
(d) LTM

List II

- (1) Actual decay of the memory trace
(2) Tip-of-the-tongue state
(3) Trace-dependent forgetting
(4) Cue-dependent forgetting

Codes:

a	b	c	d	a	b	c	d
(A) 1	2	3	4	(B) 2	1	4	3
(C) 2	1	3	4	(D) 1	2	4	3.

Direction: Following questions contain two statements 'Assertion (A) and Reason (R)'. Select the correct answer for each question according to the codes given below:

Codes:

- (A) Both A and R are true and R is correct explanation of A.
(B) Both A and R are true but R is not a correct explanation of A.
(C) A is true and R is false
(D) A is false but R is true.

- 443. Assertion (A):** Retention of earlier learning may be detected by one method and not by another.

Reason (R): Each kind of memory makes a somewhat different demand on the subject.

- 444. Assertion (A):** There has been a move away from a behaviouristic or stimulus-response approach to memory.

Reason (R): The approach has not proved fruitful in dealing with practical problems, and certain newly discovered phenomena appear to be inconsistent with its basic assumptions.

- 446. Assertion (R):** Psychologists have adopted a more cognitive view of memory rather than they have behaviouristic.

Reason (R): They want to isolate some of the processes that act between the input of a stimulus and the response output.

- 447. Assertion (A):** Learning leaves a "trace" in the brain.

Reason (R): Forgetting takes place simply through the passage of time.

- 448. Assertion (A):** In free-recall experiments, a primacy effect and a recency effect are apparent.

Reason (R): Probability of recall varies with an item's serial position in a list.

- 449. Assertion (A):** Recall reflects retrieval from LTM only.

Reason (R): The arithmetic task causes the loss of all the words in STM.

- 450. Assertion (A):** Cue-dependent forgetting is most obvious in the tip-of-the tongue state.

Reason (R): A word is stored in memory, but the appropriate cues that would allow retrieval are lacking.

- 451.** In behaviouristic school, thinking is explained as

- (A) subvocal speech
(B) a conditioned reflex
(C) an unconscious activity
(D) symbolism.

- 452.** In autistic thinking, there is

- (A) due regard to reality
(B) little regard to reality
(C) free association of idea
(D) manifestation of behaviour disorder.

- 453.** The first step in the process of reasoning is

- (A) abstract thinking
(B) identification of the goal
(C) assembling ideas pertaining to the problem
(D) organizing ideas pertaining to the problem.

- 454.** Delayed response is a test of

- (A) ideation capacity

- (B) short-term memory
(C) reasoning capacity
(D) none of the above.
- 455.** Thinking is carried on by
(A) imperceptible muscular movements
(B) mental images representing a variety of symbols
(C) both of these processes
(D) neither of these processes.
- 456.** Reasoning involves
(A) free association of ideas
(B) intensive thinking of the problem
(C) recall of ideas
(D) controlled sequence of association.
- 457.** The inference 'a shifty-eyed man who looks like a fox is very cunning' illustrates
(A) deductive thinking
(B) deductive reasoning
(C) inductive reasoning
(D) reasoning by analogy.
- 458.** Mental activity which is controlled by the wishes of the individual is called
(A) autistic thinking
(B) pleasure thinking
(C) creative thinking
(D) no thinking.
- 459.** In addition to preparation, creative thinking necessarily involves
(A) a period of relative inactivity
(B) sudden illumination
(C) both the above two stages
(D) neither of the above two stages.
- 460.** Which of the following statements is correct?
(A) There is no evidence for the existence of reasoning in animals
(B) Thinking stops during paralysis
(C) Recall in reasoning is random
(D) Reasoning can occur without language
- 461.** A complex concept defined by the joint presence of two or more features of objects or events is called
(A) disjunctive concept
(B) conjunctive concept
(C) relational concept
(D) an algorithm.
- 462.** The rules used in problem-solving are called
(A) algorithm (B) heuristics
(C) both (A) and (B)
(D) theorems.
- 463.** A condition which is not favourable in reasoning is
(A) interest (B) attention
(C) motivation (D) suggestion.
- 464.** Which of the following acts as an incentive in the reasoning process?
(A) The solution of the problem
(B) The problem
(C) The ideas
(D) The concept.
- 465.** The most important mental activity in the reasoning process is
(A) Recognition
(B) Thinking
(C) Assimilation
(D) Concept formation.
- 466.** When a concept similar to the one being learned is already known, the new concept is learned very easily. This is due to
(A) distinctiveness (B) mental set
(C) discrimination (D) transfer.
- 467.** The starting point of thinking is
(A) goal (B) conflict
(C) drive (D) tension.
- 468.** Which of the following statements is true?
(A) Algorithm and heuristics are sets of rules which if followed correctly will guarantee a solution to a problem
(B) Algorithm are strategies usually based on past experience with problems
(C) Heuristics are strategies which do not guarantee success
(D) Algorithm help to solve problems but do not guarantee success.
- 469.** Dreams are an example of
(A) autistic thinking
(B) cognitive thinking
(C) directed thinking
(D) no thinking.

- 470.** A class of objects or properties common to a class of objects is called
(A) a symbol (B) a concept
(C) a percept (D) an inventory.
- 471.** The reasoning process is akin to
(A) uncontrolled flight of thoughts
(B) random thinking
(C) learning by trial and error
(D) learning by sudden insight.
- 472.** The thinking process does not involve
(A) edictic images
(B) images
(C) verbal symbols
(D) non-verbal concepts.
- 473.** The human ability to form concepts enables us to divide things into
(A) symbols (B) classes
(C) operations (D) units.
- 474.** When the muscles necessary for speaking are paralysed,
(A) thinking cannot go on
(B) verbal thinking cannot go on
(C) verbal thinking can go on
(D) problem solving ability is hampered.
- 475.** The symbols used in thinking are
(A) images (B) language
(C) concepts (D) all the above.
- 476.** Functional fixedness is
(A) concentrating on a fixed goal
(B) adjusting functions to arrive at a particular solution
(C) finding alternative solutions to a problem
(D) adopting a fixed approach to solve a problem.
- 477.** A personality sketch might be given and it be required to judge whether it is a librarian or a farmer who is being described. A person described as meek, shy, helpful and tidy would probably be judged to be a librarian. Here the judgement is on the basis of
(A) representativeness
(B) size of the sample
(C) adjustment (D) algorithms.
- 478.** False belief which cannot be shaken by reason is termed
(A) fantasy (B) delusion
(C) illusion (D) hallucination.
- 479.** Which of the following statements is false? A set in thinking
(A) biases thinkers at the start of the problem
(B) acts as an implied assumption
(C) may be induced by instructions which revive old habits
(D) always acts as negative in its effect.
- 480.** People make decisions which will maximize
(A) subjectively expected utility
(B) objectively calculated utility
(C) objective utility
(D) the estimated profit.
- 481.** A symbol has an/a
(A) universal meaning
(B) innate meaning
(C) natural meaning
(D) arbitrary meaning.
- 482.** In an experiment, one group of subjects was asked to estimate the product of $8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$ in 5 seconds. The other group was asked to estimate the product of $1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8$ in the same time. The former group attained a larger estimate than the latter group. The biasing effect involved in this case is
(A) availability of information
(B) adjustment
(C) representativeness
(D) gambler's fallacy.
- 483.** If a baseball manager has evidence to know that the chances of success of a player are nearly 50-50. If, further this player has failed the last three times on a row, and the events are independent of each other, then the odds are
(A) less than 50-50
(B) more than 50-50
(C) still 50-50
(D) it is almost certain that the player will succeed in the next chance.
- 484.** An important feature of creative thinking is
(A) divergent thinking
(B) convergent thinking

- (C) operational thinking
(D) autistic thinking.
- 485.** At higher intelligence levels, the correlation between creativity and intelligence is
(A) high (B) low
(C) nil (D) very high.
- 486.** The following is not true of the personality traits of a creative thinker. The relative thinker
(A) prefers complexity and some degree of apparent imbalance in phenomena
(B) are more independent in their judgements
(C) are less self-assertive and dominant
(D) are disposed to entertain impulses and ideas that are commonly taboo.
- 487.** According to the generative or transformation theory (of language), it is possible to
(A) have different meanings to the same sentence
(B) have sentences with different deep structures that have a common surface structure
(C) have sentences with different surface structures that have a common deep structure
(D) have different connotations for the same surface structure.
- 488.** A major difficulty we encounter in problem-solving is the cognitive limitations imposed by
(A) short-term memory
(B) long-term memory
(C) both short-term and long-term memory
(D) our perceptual systems.
- 489.** Animals
(A) can communicate
(B) cannot communicate
(C) have natural languages
(D) can communicate and have natural languages.
- 490.** Spoken language is
(A) an ability found among all animals
(B) a species specific ability limited to *Homo sapiens*
(C) an ability confined only to the primates
(D) an innately determined mechanism in all animals.
- 491.** The factors involved in the dynamics of speech are
(A) loudness of the voice
(B) emotions communicated by speech
(C) fundamental pitch of the voice
(D) all the above.
- 492.** The smallest units of meaning in speech perception are
(A) morphemes (B) phonemes
(C) words (D) syllables.
- 493.** Intelligibility of speech is rendered more by the
(A) vowels
(B) high frequencies
(C) intermediate frequencies
(D) low frequencies.
- 494.** Much of creative thinking is
(A) problem-solving
(B) conscious reasoning
(C) logic and deductive
(D) unconscious rearrangement of symbols.
- 495.** Creative thinking involves the phenomenon of
(A) deductive
(B) insight
(C) inductive
(D) decision-making.
- 496.** Creative thinking proceeds in the following stages
(A) evaluation, preparation, revision, incubation, illumination
(B) preparation, evaluation, illumination, incubation, revision
(C) preparation, incubation, illumination, evaluation, and revision
(D) preparation, illumination, revision, incubation, evaluation.
- 497.** Language involves the use of
(A) signs (B) innate signs
(C) acquired signs (D) symbols.
- 498.** A thunder is often the sign of rain to come. This is an/a

- (A) acquired sign (B) innate sign
(C) symbol (D) phoneure.
- 499.** Most of creative thinking activity occurs during the stage of
(A) preparation (B) illumination
(C) incubation (D) evaluation.
- 500.** According to the text, the mental constructions involved in making classifications are
(A) heuristics (B) constituents
(C) algorithms (D) concepts.
- 501.** Categories that we use every day, such as furniture of dog, are called
(A) constructs
(B) arranged constructs
(C) predetermined concepts
(D) natural concepts.
- 502.** Rosch proposes that in everyday usage, concepts are encoded according to
(A) representing the problem
(B) deciding when a satisfactory answer has been found
(C) searching for solutions
(D) establishing a mental set.
- 503.** The inclination to repeat a solution that has worked in the past, even though it is not helpful, is called
(A) mental set
(B) functional fixedness
(C) the availability heuristic
(D) incubation.
- 504.** In another name for a rule-of-thumb problem-solving strategy is
(A) algorithm (B) heuristic
(C) mental set
(D) subgoal analysis.
- 505.** The problem-solving strategy which involves comparing one's current position with a desired end point and then trying to find a means of closing the gap between the two is called
(A) subgoal analysis
(B) prototype matching
(C) backward search
(D) means-end analysis.
- 506.** The tendency to view an object as being usable only for its customary use is called
(A) mental set
(B) natural construct
(C) an algorithm
(D) functional fixedness.
- 507.** Which of the following is not one of the basic levels of structural analysis for language?
(A) Morphology
(B) Syntax
(C) Linguistic relativity
(D) Phonology.
- 508.** When a person remembers a sentence, he or she generally remembers
(A) the entire sentence verbatim
(B) the propositions
(C) the nouns but not the adjectives
(D) the prepositions.
- 509.** The approach to understanding sentences that depends on our understanding of the social rules which apply is called
(A) utility (B) semantics
(C) syntax (D) pragmatics.
- 510.** SOAR is a(n)
(A) language used by chimps to talk on a computer
(B) algorithm used to estimate the probability of an event
(C) linguistic universal
(D) computer program that mimics.
- 511.** The ability to generate many different answers to questions is called
(A) divergent thinking
(B) syntax
(C) the availability
(D) the representativeness heuristic.
- 512.** The two factors that affect a rational decision-making process are the probability of each potential outcome and each outcome's
(A) length (B) utility
(C) certainty (D) consequences.
- 513.** Frequent events are generally better recalled than infrequent events. This is the assumptions we make when using
(A) means-end analysis
(B) subgoal analysis

- (C) the availability heuristic
(D) the representativeness heuristic.
- 514.** Which of the following would most likely be a characteristic rather than a defining feature of the category sheep ?
(A) Has hair (B) Bears live young
(C) Feeds its young with milk
(D) Is white.
- 515.** Suppose that in working on a problem, you cannot reach the correct answer because you fail to consider the possibility that it can be viewed from an entirely different perspective. Your tendency to view the problem from only one perspective would be an example of
(A) availability heuristic
(B) subgoal analysis
(C) representativeness heuristic
(D) functional fixedness.
- 516.** Suppose I tell you that to calculate the mean of a distribution, you must always add the scores and divide the total by the 4 number of scores. This is an example of
(A) a subgoal analysis
(B) a heuristic strategy
(C) an algorithmic strategy
(D) a means-end analysis.
- 517.** Suppose you are constructing a suit for an astronaut to use on the moon. You tackle the problem by dividing it into smaller problems, such as fabric, life support, elimination, etc. This is an example of
(A) subgoal analysis
(B) means-end analysis
(C) utility analysis
(D) global hypothesis.
- 518.** The linguistic relativity hypothesis holds that
(A) language determines thought
(B) thought determines language
(C) intelligence determines thought
(D) culture determines language.
- 519.** Features found in all languages as a result of shared characteristics of thought are called
(A) culture prepositions
(B) cultural constituents
(C) linguistic universal
(D) whorfian units.
- 520.** Why are chimpanzees unable to learn spoken language ?
(A) They are not trainable
(B) They lack the necessary vocal apparatus
(C) They are culturally deprived.
(D) English is too difficult.
- 521.** _____ occurs to overcome barriers between a given state and a desired goal state by means of behavioral and/or cognitive, multi-step activities.
(A) ASL (B) SOAR
(C) GPS (D) CPS
- 522.** You see a birch tree and declare that all trees have white bark, drop leaves in the fall, and have catkin seeds. You have identified
(A) a prototype
(B) a set of defining features
(C) an availability
(D) a representativeness heuristic.
- 523.** Suppose you are working on a word puzzle in which you are to arrange the following letters into a word: SEVOARI. You repeatedly try to solve the problem by starting the word with an R, although you know that this response will not produce a correct answer. (Note: the word is ovaries) Your tendency is called
(A) utility
(B) backward search
(C) subgoal analysis
(D) mental set.
- 524.** Suppose you are receiving help in making a difficult decision. Your consultant, first, describes all possible outcomes and then asks you to evaluate how desirable each outcome would be. This second step is called establishing
(A) probability
(B) a rationalization
(C) utility
(D) a mean-end analysis.
- 525.** You meet John, who is forty-five years old, intelligent, dedicated, verbal, concerned about other people's welfare, and wealthy. You conclude that he is more

likely to be a doctor than a truck driver. You are using the heuristic called

- (A) probabilistic inference
- (B) utility analysis
- (C) availability
- (D) representativeness.

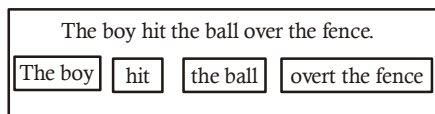
526. Which of the following represents more than one English phoneme ?

- (A) i (B) th
- (C) sh (D) do.

527. You analyse *anticommunism* by breaking it into *anti* plus *commun* plus *ism*. You are analysing this word according to its

- (A) syntax (B) grammar
- (C) morphology (D) phonology.

528. The diagram below is an example of breaking a sentence into its



- (A) morphemes
- (B) constituents
- (C) linguistic universal
- (D) underlying representation.

529. Suppose you stumble onto a culture that has only three basic colour terms. According to research, which of the following will not be one of them ?

- (A) black (B) blue
- (C) red (D) white.

530. Which of following explanations best explains the success of the cognitive revolution in psychology ?

- (A) The refining of introspection as a research method
- (B) The development of empirical methods
- (C) The development of empirical analogies for human thinking
- (D) Both (A) and (C).

531. Which of the following represents the top of the hierarchy in the structure of language ?

- (A) meaningful units
- (B) sentences
- (C) phrases (D) words.

532. Which of the following is not one of basic properties of language ?

- (A) generative (B) symbolic
- (C) structured (D) alphabetical.

533. Which of the following heuristics would you probably employ if assigned the task of carrying out a school election ?

- (A) Work backwards
- (B) Representativeness
- (C) Search for analogies
- (D) Form subgoal.

534. Which one of Greeno's problems is exemplified by the anagram ?

- (A) Arrangement
- (B) Inducing structure
- (C) Transformation
- (D) Chunking.

535. Experts differ from novices in their problem-solving in that they are more likely to make use of

- (A) planning (B) chunking
- (C) focusing on the deeper structures
- (D) all the above.

536. The word FAIRLY would be an example of a

- (A) morpheme (B) phoneme
- (C) syntactical unit
- (D) none of the above.

537. Children and chimpanzees do not appear to learn language in the same manner. This statement is

- (A) true (B) false
- (C) could not be interpreted
- (D) none of the above.

538. When a child says that TUB and BUT are constructed of the same three letters she is showing an awareness of

- (A) morphemes (B) phonemes
- (C) metalinguistics (D) syntax.

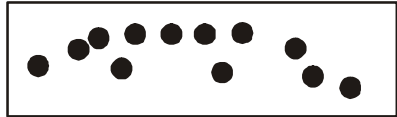
539. The fact that children appear to learn rules, rather than specific word combinations, when acquiring language skills argues most strongly against which theory of language development ?

- (A) Cognitive (B) Behaviourist
- (C) Nativist (D) None of these.

540. People generally prefer a choice that provides an 80 per cent chance of success

- over one that provides a 20 per cent chance of failure. This illustrates the effect of
- (A) the availability heuristic
 - (B) the representativeness heuristic
 - (C) framing
 - (D) confirmation bias.
- 541.** When faced with having to choose among numerous alternatives, most persons will opt for
- (A) a non-compensatory decision
 - (B) a compensatory decision model
 - (C) a means/ends analysis
 - (D) a subjective-utility model.
- 542.** What is the most you can know when making a risky decision ?
- (A) The amount you can win
 - (B) The amount you can lose
 - (C) Both the above
 - (D) None of the above.
- 543.** Most persons mistakenly believe that more people die from tornadoes than from asthma. This is because they mistakenly apply
- (A) a mean/ends analysis
 - (B) a compensatory decision model
 - (C) an availability heuristic
 - (D) a representativeness heuristic.
- 544.** Failure to actively seek out contrary evidence may lead to
- (A) overestimating the improbable
 - (B) the conjunction fallacy
 - (C) the gambler's fallacy
 - (D) confirmation bias.
- 545.** Which of the following persons is known for his nativist theory of language development ?
- (A) Noam Chomsky
 - (B) B. F. Skinner
 - (C) Herbert Simon
 - (D) Amos Tversky.
- 546.** Development psychologists believe that human development progresses
- (A) sequentially
 - (B) unpredictably
 - (C) with little individual variation
 - (D) quantitatively.
- 547.** A substance that can interrupt prenatal development is called a(n)
- (A) teratogen
 - (B) executive control structure
 - (C) syndrome
 - (D) nativist factor.
- 548.** The finding that Down's Syndrome children can improve their performance on sensorimotor tasks emphasizes the importance of this factor in development
- (A) the environment
 - (B) heredity
 - (C) metacognition
 - (D) over-regularization.
- 549.** All normal humans have twenty-three pairs of
- (A) genes
 - (B) chromosomes
 - (C) DNA molecules
 - (D) executive control structures.
- 550.** Which of the following is a newborn not able to do ?
- (A) See with 20/20 vision
 - (B) Locate the direction from which a sound comes
 - (C) discriminate different sounds
 - (D) discriminate different smells.
- 551.** According to an experiment reported in the text, if you tied a string from a baby's arm to mobile hanging above her crib, what outcome could you expect ?
- (A) She would cry and be frightened
 - (B) She would ignore the mobile
 - (C) She would exhibit pleasurable response when she learned she could control the mobile
 - (D) She would not learn the relationship between her movement and that of the mobile, but she would enjoy its movement.
- 552.** Which of the following is not one of Piaget's stage of development ?
- (A) Concrete operations
 - (B) Proportional period
 - (C) Sensorimotor period
 - (D) Hypothetical period.
- 553.** The process of changing preexisting schemes to conform to new information is called

- (A) assimilation
(B) accommodation
(C) egocentrism
(D) over-regularization.
- 554.** A child applies the learned *ed* ending to all words to express past tense. This is called
(A) overintoning
(B) underextension
(C) over-regulation
(D) under-regulation.
- 555.** According to studies reported in the text, parents most often reinforce their children for
(A) correct grammar
(B) correct pronunciation
(C) truthfulness
(D) over-regularization.
- 556.** The study of appropriate use of language in social context is called
(A) pragmatics
(B) semantics
(C) rhetoric
(D) information processing theory.
- 557.** Sensorimotor intelligences is best characterized by
(A) action
(B) concept construction
(C) thought
(D) conservation.
- 558.** A young infant often acts as if objects stop existing when they are moved out or the infant's perceptual range. This example would indicate that the infant has not yet developed
(A) metacognition
(B) representational thought
(C) sensorimotor intelligence
(D) the object concept.
- 559.** The inability to understand that others have a different point of view than you have is the basis for
(A) habituation (B) egocentrism
(C) metacognition
(D) over-regularization
- 560.** Robbie Case argues that we increase short-term storage space by
(A) assimilation (B) accommodation
(C) equilibration (D) chunking.
- 561.** The child's current ways of representing the world coupled with the child's current strategies for dealing with specific kinds of problems are called
(A) executive control structure
(B) metacognitions
(C) short-term storage spaces
(D) nativist representations.
- 562.** Which of the following develop first ?
(A) Babbling phonemes
(B) Intonation patterns
(C) Simple words for objects
(D) Simple words for commands.
- 563.** The increased memory capacity older children have is probably related to all of the following except
(A) greater familiarity with commonly used items
(B) more highly developed short-term memories
(C) better learning strategies
(D) increased use of memory cues.
- 564.** According to the text, people are intellectually most flexible during this decade of life
(A) thirty to forty (B) forty to fifty
(C) fifty to sixty (D) sixty to seventy.
- 565.** Some decline in old age is typically associated with which of the following memory systems ?
(A) Short-term memory
(B) Long-term memory
(C) Sensory memory
(D) Motor memory.
- 566.** Women are warned not to smoke cigarettes during pregnancy because nicotine can harm their unborn babies. Here, nicotine is an example of
(A) a metacognitive element
(B) a teratogen
(C) a genetic marker
(D) Down's Syndrome.
- 567.** Julie is in an experiment where every 30 seconds a loud tone is sounded. For the first several tone bursts, Julie jumped when she heard the tone. Gradu-

- ally, though, she came to ignore the tone and hardly noticed it when it sounded. This is best thought of as an example of
(A) assimilation (B) metacognition
(C) accommodation (D) habituation.
- 568.** Bob has almost no memory for any event of his life that occurred before his fourth birthday. This example is best thought of as an example of
(A) in fantile amnesia
(B) egocentrism
(C) metacognition
(D) habituation.
- 569.** Bobby is shown two balls of clay that he believes are the same size. As he watches, one ball is flattened. He now claims the flatter ball contains more clay. Which of Piaget's stages is Bobby in ?
(A) Formal operations
(B) Preoperational
(C) Concrete operations
(D) Sensorimotor.
- 570.** Jack is given an experiment to conduct. He approaches it by first specifying all possible solutions and then systematically testing each one until he arrives at the correct answer. Which of Piaget's stages is he in?
(A) Proportional
(B) Concrete operations
(C) Metacognitive
(D) Formal operations.
- 571.** Barbara complains that she can't solve a particular problem because she doesn't have enough room in her memory to be able to think about all the details of the problem at the same time. This situation is best thought of as relating most closely to the concept of
(A) assimilation
(B) accommodation
(C) equilibration
(D) short-term storage space.
- 572.** Which of the following is an example of telegraphic speech ?
(A) I must have it now (accompanied by a gesture)
(B) I'd like that
(C) Doggy bye-bye
(D) We go to the store.
- 573.** Which of the following is an example of over regularization ?
(A) Me, too
(B) I goed, too
(C) Mommy likes me
(D) Daddy bye-bye.
- 574.** Typing is primarily a speed-related skill. Which age group could generally be expected to be best at typing ?
(A) Twenty-to-thirty-year-olds
(B) Thirty-to-forty-year-olds
(C) Forty-to-fifty-year-olds
(D) Fifty-to-sixty-year olds.
- 575.** Jenny calls all animals with fur and four legs "doggie." Now she learns that some of these animals are "bears" and some are "bunnies." Her learning here is best thought of as an example of:
(A) habituation
(B) metacognition
(C) accommodation
(D) assimilation.
- 576.** Refer to the marbles represented in the figure below. According to Piaget's work, how would a typical four-year-old respond to the question ? "Which row has more marbles ?"
- 
- (A) "The top row."
(B) "The bottom row."
(C) "Both rows have the same number of marbles."
(D) "I don't know."
- 577.** Suppose you tested a group of four and five-year-old children with a classic Piagetian conservation task that Rochel Gelman tested which used stuffed mice on plates. If the children behaved as predicted, they _____ be able to do the Piagetian task and _____ be able to do Gelman's task.
(A) would; would
(B) would not; would not

- (C) would not; would
(D) would; would not.

Direction: match List I with List II and select the correct answer from the codes given below each question.

578. List I (Dimensions)	List II (Meanings)
(a) Evaluative dimension	(1) Strong-weak
(b) Potency dimension	(2) Sharp-dull
(c) Activity dimension	(3) Clean-dirty

Codes:

a	b	c	a	b	c
(A) 1	2	3	(B) 2	3	1
(C) 3	1	2	(D) 1	3	2

579. List I (Kinds of thinking)	List II (Examples)
(a) Autistic thinking	(1) Planning
(b) Creative thinking	(2) Fantasy
(c) Deterministic thinking	(3) Illusion
(d) Cognitive thinking	(4) Dreams

Codes:

a	b	c	d	a	b	c	d
(A) 1	2	3	4	(B) 4	3	2	1
(C) 4	1	2	3	(D) 1	4	3	2

580. List I	List II
(a) Deductive reasoning	(1) Universal to universal
(b) Inductive reasoning	(2) Universal to particular
	(3) Particular to particular
	(4) Particular to universal

Codes:

a	b	a	b
(A) 1	2	(B) 3	1
(C) 2	4	(D) 4	3.

Direction: Following questions contain two statements 'Assertion (A) and Reason (R)'. Select the correct answer for each question according to the codes given below:

Codes:

- (A) Both A and R are true and R is correct explanation of A.
(B) Both A and R are true but R is not a correct explanation of A.
(C) A is true and R is false
(D) A is false but R is true.

- 581.** For the human beings the most important triggers of sexual drive are
(A) adrenaline level
(B) hormone levels
(C) sensory stimuli
(D) all the above.
- 582.** Sensory stimuli can arouse the following drive states
(A) hunger (B) pain
(C) sexual drive state
(D) all the above.
- 583.** Motivation results in
(A) goal-directed behaviour
(B) undirected behaviour
(C) exited behaviour
(D) problem-solving behaviour.
- 584.** Motives cannot be
(A) inferred from behaviour
(B) observed directly
(C) used to explain behaviour
(D) used to predict behaviour.
- 585.** Which of the following statements is false?
(A) Incentives are external stimuli acting on the organism
(B) Incentives are not motivators of behaviour
(C) Incentives are direct behaviour
(D) Incentives arouse the organism.
- 586.** Which of the following does not represent motives ?
(A) Organic needs (B) Attitudes
(C) Aptitudes (D) Interests.
- 587.** The drive-reduction theory of motivation explained behaviour in terms of
(A) storation of physiological balance in the body
(B) internal drives pushing the organism into activity

- (C) external stimuli acting as motivators of behaviour
(D) both (A) and (B) but not (C).
- 588.** The aroused condition of an organism that results from some bodily or tissue deficit is called
(A) drive (B) motive
(C) goal (D) incentive.
- 589.** A motive or drive rises to behaviour that is
(A) goal-oriented
(B) pleasure seeking
(C) pain avoiding
(D) instinctive in nature.
- 590.** The strongest among the following motives is
(A) self-assertion
(B) self-preservation
(C) social approval
(D) love.
- 591.** The body's tendency to maintain a balance among its internal physiological states and the environment is termed
(A) ambivalence (B) heterostasis
(C) homeostasis
(D) cognitive dissonance.
- 592.** Motivation could be described as
(A) as instinctual response to a need
(B) a modifier of behaviour
(C) an acquired response to a need
(D) an energizer of behaviour.
- 593.** The contractions of muscles in the stomach wall do not
(A) active cells in lateral hypothalamus
(B) influence hypothalamic control of immediate appetite
(C) result in hunger pangs
(D) activate thermoreceptors in the brain.
- 594.** Injections of insulin will
(A) increase the food intake
(B) decrease the food intake
(C) not affect the eating behaviour
(D) result in increased activity of ventromedial cells of hypothalamus.
- 595.** Obesity in normal animals can be induced by
(A) social factors
(B) lesions in the ventromedial hypothalamus
(C) external stimuli
(D) all the above.
- 596.** Need for affiliation is not
(A) a social need
(B) a learned need
(C) derived from experience
(D) innate.
- 597.** Need for achievement can be measured by using
(A) TAT (B) Binnet's scale
(C) Thurstone's scale
(D) Semantic differential scale.
- 598.** People with high need for achievement prefer tasks that are
(A) not difficult
(B) very difficult
(C) moderately difficult
(D) impossible.
- 599.** Correlation of achievement scores and performance in college (college grade) is found to be
(A) low and negative
(B) low and positive
(C) high and positive
(D) erratic.
- 600.** Food intake in obese people is markedly affected by
(A) fear
(B) stomach contraction cues
(C) taste and smell cues
(D) all the above.
- 601.** Among primates, maternal behaviour is primarily influenced by
(A) experience and learning
(B) the hormone prolactin
(C) the hormone estrogen
(D) external stimuli.
- 602.** In human beings, sexual arousal is controlled at the level of
(A) spinal cord (B) cerebral cortex
(C) cerebellum (D) medulla.
- 603.** All behaviour is directed to
(A) reduce tissue deficit (restore energy)
(B) increase tissue deficit (use energy)

- (C) use energy or restore it depending on the kind of behaviour
(D) none of the above.
- 604.** Specific hungers result from
(A) nutritional deficits
(B) ethnic and cultural influences
(C) past experiences
(D) all the above.
- 605.** Thirst and drinking behaviour result from
(A) cellular dehydration
(B) cellular dehydration and hypervolemia
(C) cellular hydration and hypervolemia
(D) increase in volume of the blood.
- 606.** The things that we learn to fear are
(A) primary goals
(B) primary negative goals
(C) secondary negative goals
(D) none of the above.
- 607.** The changes in achievement need come about
(A) 50 years before changes in economic activity
(B) 50 years after changes in economic activity
(C) at the same as the changes in economic activity
(D) 200 years after changes in economic activity.
- 608.** The relationship between fear and affiliation motive is
(A) negative (B) positive
(C) zero
(D) only slightly negative.
- 609.** Lack of sensory stimulation in human adults for long periods of time may cause
(A) hallucinations
(B) disorientation in time and space
(C) inability to concentrate
(D) all the above.
- 610.** An instinct is characterized by
(A) cognition (B) conation
(C) affection (D) all the above.
- 611.** The pecking order among barnyard fowls pertains to motive called
(A) self-assertion (B) aggression
(C) pugnacity (D) gregariousness.
- 612.** A fairly complex pattern of behaviour emerging without prior training found in all members of a species is termed
(A) imitative behaviour
(B) reflex action
(C) instinct
(D) habit.
- 613.** An individual experiences a conflict when
(A) his strong needs remain unfilled
(B) he wants to do many things all at the same time
(C) he has to continue with the work despite loss of interest
(D) he has to choose between two strongest and contradictory motives.
- 614.** When a person is hungry and sleepy at the same time, he experiences
(A) approach-approach conflict
(B) approach-avoidance conflict
(C) avoidance-avoidance conflict
(D) double approach-avoidance conflict.
- 615.** Which of the following is not one of the main causes of frustration ?
(A) Environmental obstacles
(B) Aggression
(C) Conflict between and among motives
(D) Personal inadequacies
- 616.** Which of the following statements is false? Exploration, competence, and self-actualization motives are
(A) powerful and persistent even in lower animal species
(B) difficult to satisfy
(C) part of the human and animal species heritage
(D) innate.
- 617.** Competence motivation is largely
(A) extrinsic
(B) intrinsic
(C) passive
(D) intrinsic and passive
- 618.** Going from the lowest needs to higher order needs, the needs in Maslow's hierarchy are
(A) N. physiological, n. safety, n. belongingness, n. understand, n. self actualization

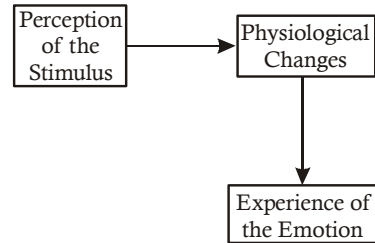
- (B) N. physiological, n. belongingness, n. safety, n. esteem, n. self-actualization
- (C) n. physiological, n/safety, n. belongingness, n. esteem, n. self-actualization
- (D) n. safety, n. esteem, n. understand, n. control, n. self-actualization.
- 619.** When two negative goals are involved, the resulting behaviour may be
- (A) vacillation
- (B) attempt to leave the conflict situation
- (C) day-dreaming
- (D) any of the above.
- 620.** If marrying a man gives security to a woman but requires her to leave an attractive job offer in another city, the woman experiences
- (A) approach-avoidance conflict
- (B) approach-approach conflict
- (C) externalized obstacles
- (D) behavioural disorders.
- 621.** According to James-Lange, theory of emotions, felt emotions depend on the
- (A) activity of lower brain areas
- (B) perception of bodily changes
- (C) activity of higher brain processes
- (D) interpretation of bodily arousal.
- 622.** Schacter-Singer theory is
- (A) not a theory of emotions
- (B) neurobiological theory of emotions
- (C) cognitive theory of emotions
- (D) psychoanalytic theory of emotions.
- 623.** Facial expressions of certain primary emotions are
- (A) innate
- (B) learned
- (C) derived from experience
- (D) all the above.
- 624.** Pleasure is a reaction to
- (A) removal of pain stimuli
- (B) satisfaction of a motive
- (C) environmental stimuli
- (D) change in the level of adrenaline.
- 625.** The following are important in dealing with stress.
- (A) Adrenal glands
- (B) Sympathetic system
- (C) Pituitary glands
- (D) All the above.
- 626.** The following disorder does not have a psychosomatic basis.
- (A) Serum hepatitis
- (B) Asthma
- (C) Dermatitis
- (D) Obesity
- 627.** Most primary emotions are closely related to
- (A) learning
- (B) attention
- (C) degree of arousal
- (D) perception.
- 628.** Anger and hostility are frequently related to
- (A) frustration of motives
- (B) injuries
- (C) threats
- (D) all the above.
- 629.** An innate mechanism that ensures that the young animal will follow and remain close to the mother (normally the first moving object it sees) is called
- (A) imprinting
- (B) intrinsic motivation
- (C) instinct
- (D) influence.
- 630.** Social learning theory was formulated by
- (A) Tolman
- (B) Bandura
- (C) Skinner
- (D) Erikson.
- 631.** If some one battles to defend the rights of an 'underdog', he is manifesting
- (A) hostile aggression
- (B) power motive
- (C) instrumental aggression
- (D) affiliation motive.
- 632.** Social learning theory stresses on the importance of
- (A) models
- (B) self-regulatory processes
- (C) cognitive processes
- (D) all the above.
- 633.** Which of the following statements about instincts and reflex action is wrong ?

- (A) Instinct is a response to the total situation; reflex is a response to some particular stimulus
(B) Instinctive activity defies modification
(C) Reflex action cannot be modified
(D) Instinctive activities take much longer time than reflex activity.
- 634.** The desire to be with one's own kind is termed as
(A) sympathy
(B) self-preservation
(C) gregariousness
(D) identification.
- 635.** In contrast to feeling like pressure, fear, etc., mood is
(A) episodic emotional feeling
(B) relatively short-lived emotional feeling
(C) relatively long-lasting
(D) none of these.
- 636.** Specific fears may be learned through
(A) conditioning (B) modelling
(C) by symbolic means
(D) all the above.
- 637.** Phobias are
(A) learned through modelling
(B) intense fears
(C) vague fears experienced without our knowing exactly what the source is
(D) all the above.
- 638.** A grief-stricken person gets a feeling of
(A) worthlessness (B) inadequacy
(C) external loss
(D) personal inadequacy.
- 639.** The cause of anxiety may be
(A) an unconscious memory of a fear stimulus
(B) a response to a specific fear stimulus
(C) a conscious memory of a fear stimulus
(D) specific fear experience when source of fear is known.
- 640.** A son who has a harsh father may feel generally hostile to all superiors. This is an example of
(A) generalized anxiety
(B) generalized anger
(C) generalized fear
(D) direct aggression.
- 641.** When both sympathetic and parasympathetic fibres are connected to the same muscle or gland
(A) they usually act in opposite manners
(B) they are mutually exclusive
(C) they may be active at once, and in some cases they act together in sequence
(D) both (A) and (C).
- 642.** The sympathetic system
(A) acts as a unit
(B) acts in a piecemeal manner, affecting one organ at a time
(C) is dominant in quiescence
(D) both (a) and (c).
- 643.** The parasympathetic system
(A) leads to secretion of certain hormones that increase emotional arousal
(B) maintains the functions that conserve and protect bodily resources
(C) accelerates heart beat
(D) inhibits bladder contracts.
- 644.** In emotional excitement, the sympathetic division of the autonomic nervous system
(A) constricts the arteries of the muscles and heart
(B) dilates the arteries of the skin and the digestive organs
(C) speeds up the heart
(D) all the above.
- 645.** Polygraphs rely on all of the following measures except
(A) blood pressure (B) EDA
(C) urinalysis (D) EEG.
- 646.** The results of a study conducted on the emotionality of men with severed spinal cords indicated that
(A) emotions are unrelated to the semantic nervous system
(B) emotions are unrelated to the autonomic nervous system
(C) a fully functioning nervous system contributes to emotion

- (D) spinal cord injuries intensify negative emotions and reduce positive emotions.
- 647.** An EDA (electrodermal activity) measure works by recording
- (A) amount of sweat on the skin
 - (B) electrical activity in the brain
 - (C) heart functions, such as blood pressure
 - (D) muscle movement.
- 648.** According to research reported in the text, an emotional display of rage can be initiated by stimulating what part of a cat's brain ?
- (A) The thalamus
 - (B) The hypothalamus
 - (C) The right frontal hemisphere
 - (D) The left frontal hemisphere.
- 649.** According to recent evidence, which part of your brain is probably not directly involved in emotion ?
- (A) Hypothalamus
 - (B) Cerebral cortex
 - (C) Amygdala
 - (D) Cerebellum.
- 650.** According to research presented in the text, activation in the left cerebral hemisphere is related to
- (A) positive emotions
 - (B) depression
 - (C) periods of uncontrollable crying
 - (D) violent behaviour.
- 651.** According to Charles Darwin, non-verbal communication should
- (A) vary according to one's culture
 - (B) vary according to the specific way in which one is raised
 - (C) vary according to one's personality typed
 - (D) be consistent across cultures and individuals.
- 652.** Suppose you want to measure how sensitive a group of subjects is to non-verbal cues. One test you could give would be the
- (A) PONS
 - (B) EDA
 - (C) OQ
 - (D) ASL.
- 653.** Which of the following is one of the dimensions on the circle of emotion ?
- (A) Inherited-to-learned emotions
 - (B) High-to-low activation
 - (C) Internal-to-external sources
 - (D) Verbal-to-non-verbal display.
- 654.** The comparison theory of happiness says that you are happiest when
- (A) you are less fortunate than others around you
 - (B) you are as fortunate as others around you
 - (C) you are more fortunate than others around you
 - (D) you have what you need
- 655.** The adaptation theory of happiness says that you are happiest when
- (A) you have what you need
 - (B) you are completely absorbed in an activity
 - (C) your circumstances are constantly improving
 - (D) you are an extrovert and experience little anxiety.
- 656.** According to the text, which of the following beliefs about crisis reactions is probably true ?
- (A) There are universal reactions to crisis
 - (B) People go through stages as they respond to crisis
 - (C) The response to crisis varies widely from person to person
 - (D) Crises are ultimately resolved.
- 657.** Suppose you are driving in a car when a truck coming towards you crosses the center line and seems to come at you. The part of your nervous system immediately activates is the
- (A) sympathetic division
 - (B) para-sympathetic division
 - (C) central nervous system
 - (D) sensory nervous system.
- 658.** Extrapolating from evidence presented in the text, if your best friend suddenly developed violent outbursts of temper, you might suspect the presence of a tumor in his
- (A) sympathetic nervous system
 - (B) parasympathetic nervous system

- (C) amygdala
(D) thalamus.

659. Dr. Madras is very interested in studying how the immune system is related to emotional disorders, such as temper outbursts and depression. The field she probably worked in is
(A) behavioural genetics
(B) immunosuppressive psychiatry
(C) psychoneuroimmunology
(D) pragmatic psychotherapeutic medicine
(e) pragmatic psychotherapeutic medicine.
660. Jason believes that many emotions share the same physical arousal patterns, but they differ according to the behaviours they trigger. These behaviours are called
(A) action tendencies
(B) cognitive appraisals
(C) cognitive arousals
(D) circles of emotion.
661. Suppose you volunteer to be in an experiment in which the experimenter intentionally embarrasses you and tells you that you are stupid. You should guess that the experimenter is really attempting to
(A) initiate non-verbal behaviour in you
(B) test the facial feedback hypothesis
(C) induce in you in negative mood
(D) activate your para-sympathetic nervous system.
662. According to the text, which of the following traits is most typically associated with happiness ?
(A) Winning a large amount of money
(B) Having a satisfying marriage and family life
(C) Having money
(D) Lucky.
663. The stages of the emotional process displayed in the figure below correspond to which of the following theories of emotion ?



- (A) Cannon-Bard (B) James-Lange
(C) Facial feedback
(D) Cognitive appraisal.
664. Which of the following was not one of Cannon's criticisms of the James-Lange theory ?
(A) No evidence existed for different physiological responses
(B) Physiological change occurs too slowly
(C) Injections of epinephrine alone do not produce emotion
(D) There are only a few basic emotions all others are modifications of these.
665. In Schachter's and Singer's experiment with the "vitamin" injections, which group of subjects imitated the Cohor's behaviour ?
(A) Informed and misinformed
(B) Misinformed only
(C) Placebo and informed
(D) Ignorant and misinformed.
666. The focus of the cognitive appraisal theory of emotion is on
(A) how we judge situations
(B) how accurately we judge our won physiological states
(C) how well we interpret the nonverbal communication of other
(D) how much we pay attention to our own emotional states.
667. Our evaluation of how we might handle a problem is called
(A) primary appraisal
(B) secondary appraisal
(C) our circle of emotions
(D) an action tendency.
668. Which of the following people who display the following trait patterns is most

- likely to report the greatest overall level of happiness ?
- (A) Mary, who experiences episodes of intense happiness
 - (B) Laura, who experiences episodes of intense unhappiness
 - (C) Linda, who seldom experiences happy emotions
 - (D) Rita, who frequently experiences happy emotions.
- 669.** Suppose you volunteer to be in an experiment in which the experimenter intentionally embarrasses you and tells you that you are stupid. You should guess that the experimenter is really attempting to
- (A) Initiate non-verbal behaviour in you
 - (B) Test the facial feedback hypothesis
 - (C) Induce in you a negative mood
 - (D) Activate your para-sympathetic nervous system.
- 670.** What test is generally used to measure need for achievement ?
- (A) TAT
 - (B) GSR
 - (C) Rorschach
 - (D) MMPI.
- 671.** John believes that happiness is achieved when your circumstances change frequently and for the better. He would probably subscribe to which perspective?
- (A) Comparison theory
 - (B) Adaptation theory
 - (C) Trait theory
 - (D) Absorption theory.
- 672.** Mathilde has just been raped. Her recovery will probably be most enhanced if
- (A) she doesn't report the rape
 - (B) she lies about the rape
 - (C) she has a supportive group of friends to talk to
 - (D) she never gets married.
- 673.** Which of the following early theorists proposed a long list of human instincts ?
- (A) Sigmund Freud
 - (B) William McDougall
 - (C) Clark Hull
 - (D) Richard Nisbett
- 674.** Freud categorized human instincts into two basic groups, the instincts of
- (A) primary drives and secondary drives
 - (B) drives and incentives
 - (C) emotion and motivation
 - (D) life and death.
- 675.** Bob believes different situations produce different physiological response patterns. These, in turn, arouse their corresponding emotions. Bob's theory most closely corresponds to
- (A) cognitive arousal theory
 - (B) cognitive appraisal theory
 - (C) Cannon-Bard theory
 - (D) James-Lange theory.
- 676.** Marlene believes that emotion is produced by the combination of physiological factors and situational cues. Her theory most closely corresponds to
- (A) cognitive arousal theory
 - (B) Cannon-Bard theory
 - (C) James-Lange theory
 - (D) facial feedback theory.
- 677.** Matthew believes that if he smiles a lot, he is more likely to feel happy. His view is most consistent with the
- (A) adaptation theory of happiness
 - (B) comparison theory of happiness
 - (C) trait theory of emotion
 - (D) facial feedback hypothesis.
- 678.** Suppose you are coping with a crisis but are not following the stages that health care professionals believe to be beneficial. According to the text, your response should be considered
- (A) normal
 - (B) reterdec
 - (C) deviant
 - (D) neurotic.
- 679.** An incentive differs from a drive in that it involves
- (A) an expectation
 - (B) humans, not animals
 - (C) animals, not humans
 - (D) biology but not learning.
- 680.** A lesion in which part of the brain will suppress (stop) eating behaviour ?
- (A) Ventromedial hypothalamus
 - (B) Reticular formation
 - (C) Lateral hypothalamus
 - (D) Amygdala

- 681.** Anorexia nervosa is an eating disorder that is most common in
(A) the elderly (B) young women
(C) athletes (D) children.
- 682.** Schachter believes that people are overweight primarily because
(A) they have more fat cells
(B) they have a slower metabolism
(C) they are more responsive to external cues
(D) they constantly feel hungry.
- 683.** In an experiment reported in the text, obese and normal-weight subjects were allowed to eat as many crackers as they wanted. The number of crackers eaten by obese subjects_____influenced by whether they has eaten a sandwich just before; the number of crackers of eaten by normal weight subjects_____ influenced by eating the sandwich.
(A) was; was
(B) was not; was not
(C) was not; was
(D) was; was not.
- 684.** Some obese people and people who are starving behave in a somewhat similar manner. Which of the following is not one of their common characteristics ?
(A) They are highly sensitive to food cues
(B) They are easily frustrated and upset
(C) They have very high energy levels
(D) They have high levels of free fatty acids in their blood.
- 685.** According to the text, the most successful long-term weight loss technique is
(A) drugs
(B) diet and exercise combined
(C) behaviour modification
(D) fat diets, especially the Stillman diet.
- 686.** A female rat is sexually responsive
(A) any time an interested male is present
(B) only during ovulation
(C) when her testosterone levels are high
(D) when her estrogen levels are low.
- 687.** Which of the following is not directly involved in human sexual behaviour ?
(A) Hypothalamus
(B) Thyroid gland
(C) Ovaries and testes
(D) Pituitary gland.
- 688.** Compared to that of animals, the human sexual response is more controlled by
(A) hormones
(B) preoptic region of the hypothalamus
(C) reticular formation
(D) cerebral cortex.
- 689.** Which of the following made scientific study of the sexual response cycle ?
(A) Clark Hull
(B) Sigmund Freud
(C) Stanley Schachter and Richard Nisbett
(D) William Masters and Virginia Johnson
- 690.** The phrase that refers to cases in which a female experiences orgasms, but not with her primary sexual partner, is
(A) primary orgasmic dysfunction
(B) vaginismus
(C) secondary orgasmic dysfunction
(D) primary erectile failure
- 691.** Martina Horner's explanation for women's tendency to set less challenging goals is their fear of
(A) failure (B) success
(C) their own sexuality
(D) competitions.
- 692.** Mary was raped by her father when she was seven years old. Now at age fourteen, she has no memory of this traumatic event. Her reaction could be an example of
(A) repression (B) sublimation
(C) an incentive
(D) a death instinct.
- 693.** Which of the following is an example of a secondary rather than a primary drives?
(A) Hunger
(B) Need for warm clothing (in a cold climate)
(C) Creativity (D) Thirst.
- 694.** Although of normal weight, Janine occasionally goes on eating binges in which she consumes several days' worth of food

- and then makes herself vomit. The term that best describes this behaviour is
- (A) abnormal setpoint
 - (B) opponent process
 - (C) anorexia nervosa
 - (D) bulimia.
- 695.** Bob always seems to be on a diet, and he constantly feels hungry when he diets. He fits the pattern of a person with a
- (A) low setpoint
 - (B) high setpoint
 - (C) mid-range setpoint
 - (D) highly variable setpoint.
- 696.** Kelly is very obese. He is forced in an experiment to eat a large piece of chocolate cream pie, after which he can eat as many pieces of other kinds of pie as he wishes. If he is a restrained eater, you would expect him to
- (A) eat no more pie
 - (B) eat a little more pie
 - (C) eat about as much more pie as a normal eater
 - (D) eat much more pie than a normal eater.
- 697.** Maria gets married, but is terribly distressed on her wedding night to find that she is unable to engage in sexual intercourse because of intense and painful muscle spasms around her vagina. If this condition is serious and persists, it would best be thought of as an example of
- (A) vaginismus
 - (B) primary orgasmic dysfunction
 - (C) secondary orgasmic dysfunction
 - (D) premature orgasmic dysfunction.
- 698.** The optimal level of arousal concept is based on the finding that all of the following parts of the brain are directly involved in arousal except the
- (A) hypothalamus
 - (B) thalamus
 - (C) cerebral cortex
 - (D) reticular formation.
- 699.** The Hebb experiment on sensory deprivation reported in the text was_____ drive-reduction theories and_____ optimal level theories.
- (A) supportive of, supportive of
 - (B) supportive of, contradictory to
 - (C) contradictory to, supportive of
 - (D) unrelated to; supportive of.
- 700.** According to Eysenck, compared to introverts, extroverts have_____ level of internal stimulation and seek_____ levels of external stimulation.
- (A) higher; the same
 - (B) lower; the same
 - (C) higher; lower
 - (D) lower; higher
- 701.** According to Solomon and Corbin's model, an opponent process is
- (A) similar to sensory overload
 - (B) the desire to reduce the emotion to a normal level
 - (C) an opposite response
 - (D) an increased level of arousal.
- 702.** In playing a ringtoss game, people with a high need for achievement generally
- (A) stand close to the target
 - (B) stand at an intermediate distance from the target
 - (C) stand far from the target
 - (D) refuse to play.
- 703.** Suppose a fire begins in your classroom everyone just stands paralyzed, watching it grow. This situation is_____ optimal level theory and_____ opponent process theory.
- (A) supportive of; unrelated to
 - (B) contradictory to; unrelated to
 - (C) unrelated to; supportive of
 - (D) supportive of; supportive of.
- 704.** Suppose you are asked by a psychologist to look at some pictures and write a story about each one. The test you are probably taking is the
- (A) sensation-seeking scale
 - (B) TAT
 - (C) MMPI
 - (D) Schacheter scale of motivation responses.
- 705.** Even though Lynn is very intelligent and has excellent grades in college, she is not interested in going to medical school because "it would probably be too difficult." Which of the following constructs would probably describe Lynn?
- (A) High fear of success
 - (B) High fear of failure

- (C) Low need for achievement
(D) High need for achievement

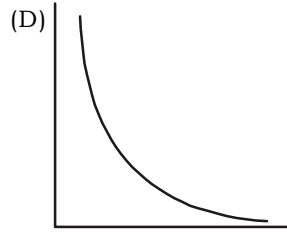
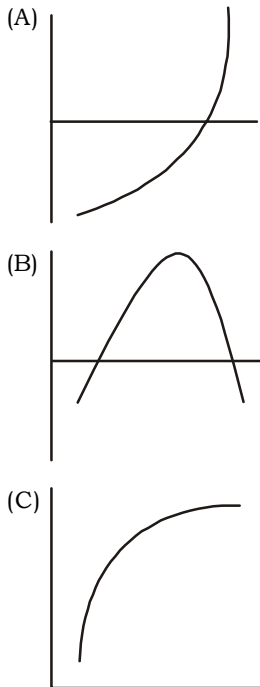
706. Which of the following behaviour is most easily explained by a drive-reduction theory of motivation ?

- (A) A hungry person gets food from a refrigerator
(B) A religious person fasts for weeks before eating
(C) Students stay up all night watching old movies on cable television
(D) A widow grieves over her husband's death for three years.

707. You are going to take a very important test and want to achieve the level of arousal that will maximize your performance. You conclude the level of arousal you should strive for is

- (A) zero
(B) a low level (to reduce tension)
(C) a moderate amount
(D) a high level.

708. Which of the following graphs best represents the relationship between performance and level of arousal ?



709. Evidence regarding facial expression in different cultures and observation of the blind suggest that

- (A) Schachter's two factor theory is correct
(B) facial expression of emotion is to a large extent innate
(C) emotions originate in the cortex
(D) learning is the major factor in explaining basic facial expressions.

710. According to Schachter's theory of emotion

- (A) Different emotions are represented by different autonomic reactions
(B) Emotions originate in subcortical brain structures
(C) Fear produces a desire to avoid affiliation
(D) Both arousal and cognition are needed to produce emotion.

711. Whether or not achievement behaviour will occur depends on one's need for achievement, fear of failure, and the probabilities and incentive values of success and failure on a specific task. Whose theory does this statement represent ?

- (A) McClelland (B) Atkinson
(C) Cannon (D) James.

712. Which of the following is true ?

- (A) Fantasizing during sexual encounters is generally evidence of repressed conflicts.
(B) Talking rather than acting diminishes sexual satisfaction
(C) Premature ejaculation is usually caused by unconscious hostility
(D) Orgasmic difficulties often derive from negative attitudes about sex.

713. Which of the following is unlearned, uniform in expression, and universal within a particular species ?

- (A) Incentive (B) Drive
(C) Instinct (D) Motivation.
- 714.** In Maslow's hierarchy of needs, which of the following needs would have to be satisfied before the need for self-actualization became activated ?
(A) Physiological needs
(B) Safety and security needs
(C) Esteem needs (D) All the above.
- 715.** What happens when a rat's lateral hypothalamus is lesioned ?
(A) It starts eating
(B) It looks for a sexual partner
(C) It stops eating
(D) It loses bladder and bowel control.
- 716.** What is the effect of insulin on blood glucose ?
(A) Glucose level increases
(B) Glucose level decreases
(C) Glucose changes to free fatty acids
(D) CCK increases.
- 717.** Which of the following is thought to be a major determinant of set point ?
(A) Number of fat cells
(B) Size of fat cells
(C) Amount of exercise
(D) Skill at tennis.
- 718.** Estrogens are found in
(A) males
(B) females
(C) both males and females
(D) androids
- 719.** A chemical secreted by one animal that affects the behaviour of another animal is known as a(n)
(A) androgen (B) affiliator
(C) hormone (D) pheromone.
- 720.** What is the name of the sexual phase during which orgasm occurs ?
(A) excitement (B) plateau
(C) orgasm (D) resolution.
- 721.** Psychologists recognise three kinds of intelligence activities. These do not include
(A) psychological (B) mechanical
(C) abstract (D) social.
- 722.** Animals do not possess
(A) mechanical intelligence
(B) abstract intelligence
(C) social intelligence
(D) intelligence of any kind.
- 723.** The concept of mental age was developed by
(A) Binet and Stanford
(B) Stanford and Simon
(C) Simon and Binet
(D) Binet.
- 724.** The power to perform an act, physical or mental, either before or after training is called
(A) trait (B) aptitude
(C) attitude (D) ability.
- 725.** To predict the outcome of training following test(s) are most useful
(A) aptitude tests
(B) achievement tests
(C) intelligence tests
(D) none of the above.
- 726.** Ability includes
(A) aptitude
(B) achievement
(C) both aptitude and achievement
(D) none of the above.
- 727.** Test scores are reliable when they are
(A) dependable (B) reproducible
(C) consistent (D) all the above.
- 728.** By giving tests in two different but equivalent forms, one can ascertain the
(A) reliability (B) validity
(C) predictive (D) validity.
- 729.** Which of the following statements is true?
(A) A test can be valid without being reliable
(B) A test can be reliable without being valid
(C) If a test measures what it is intended to measure then by definition it is reliable
(D) If a test measures what it is intended to measure then by definition it is reliable

- 730.** Wechsler tests consist of
(A) verbal items
(B) performance items
(C) both verbal items and performance items
(D) either verbal items or performance items.
- 731.** The first tests designed to measure intelligence were developed by
(A) Alfred Binet
(B) Theidore Simon
(C) Francis Galton
(D) Louis Thurstone.
- 732.** Intelligence is the ability of
(A) adjusting in new situations
(B) availing of past experiences
(C) abstract thinking
(D) all the above.
- 733.** Galton developed tests of intelligence by measuring tasks requiring
(A) reasoning
(B) perceptual motor skills
(C) problem-solving
(D) all the above.
- 734.** According to Binet, intelligence is the characteristics of an individual thought process that enable him
(A) to take and maintain a direction without becoming distracted
(B) to adapt means to ends
(C) to criticize his own attempts problem solution
(D) all the above.
- 735.** Intelligence quotient is
(A) based on the ratio between mental age and the chronological age
(B) equal to the mental age of the person
(C) equal to the chronological age divided by the mental age
(D) both (B) and (C) correct.
- 736.** General intelligence tests are
(A) achievement tests
(B) aptitude tests
(C) performance tests
(D) success tests.
- 737.** If an adult has IQ less than 25, he is described as
(A) an imbecile (B) a moron
(C) an idiot-servant (D) an idiot.
- 738.** The I.Q. range for average intelligence is
(A) 70-120 (B) 85-115
(C) 80-110 (D) 70-110.
- 739.** The study of gifted children was initiated by
(A) Spearman (B) Thorndike
(C) Terman (D) Thurstone.
- 740.** The I.Q.'s above 130 are found in about
(A) 1 per cent of the total population
(B) 4 per cent of the total population
(C) 3 per cent of the total population
(D) 2 per cent of the total population.
- 741.** Thurstone felt that intelligence could be broken down into a number of primary abilities. To find these abilities he applied the method of
(A) factor analysis
(B) content analysis
(C) context analysis
(D) synthesis.
- 742.** The first group test of intelligence was
(A) the Army Alpha
(B) the T.A.T.
(C) The Ctandard Binet
(D) AGCT.
- 743.** In an intelligence test, a 10-year-old boy was found to have mental age of 12 years
(A) 80 (B) 120
(C) 102 (D) 98.
- 744.** Mental deficiency is characterized by the I.Q. range
(A) 0-25 (B) 0-50
(C) 0-70 (D) 0-100.
- 745.** 'Culture-free' tests of intelligence were constructed by
(A) Binet (B) Davis
(C) Cattell (D) Kohs.
- 746.** Although I.Q.'s are relatively stable for most people, I.Q. may show large shifts because of
(A) environmental condition
(B) emotional factors
(C) motivational factors
(D) all the above.

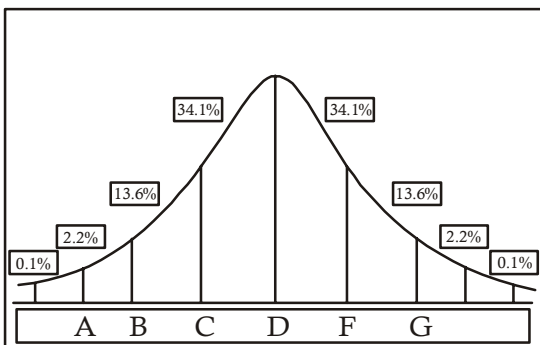
- 747.** As compared to dull children, mental age for bright children would generally grow
(A) more rapidly
(B) less rapidly
(C) at the same rate
(D) less rapidly to start with and then more rapidly.
- 748.** With age, the following shows little decline
(A) speed of intellectual functioning
(B) general knowledge
(C) short-term memory
(D) efficiency of intellectual functioning.
- 749.** The intelligence of a person
(A) does not change with age
(B) grows continuously
(C) grows up to a certain level and then decreases
(D) grows up to a certain level and then remains constant.
- 750.** Which of the following statements is wrong?
(A) Those with low intelligence reach their upper limit of mental growth quite late and those with high intelligence reach theirs at a relatively early age
(B) Mental growth continues even beyond the time of physical maturity
(C) The I.Q.s of very bright children invariably drop with age
(D) I.Q. usually changes very little with age, if at all.
- 751.** The general intelligence factor (called *g*) is the major determinant of performance on the intelligence test items according to
(A) Thurstone (B) Spearman
(C) Guttman (D) Binet.
- 752.** High intelligence (as measured by conventional tests)
(A) does not guarantee creativity
(B) guarantees creativity
(C) is negatively correlated to creativity
(D) is positively and strongly correlated to creativity.
- 753.** Conventional tests of intelligence measure
(A) divergent thinking
(B) convergent thinking
(C) both (A) and (B)
(D) neither (A) nor (B).
- 754.** Which do you think that it seems the most appropriate statement?
(A) There are innate differences in intelligence between the black and white population in United States
(B) There is no evidence for a genetic factor in I.Q. test differences
(C) It is not possible to draw valid inferences about innate racial differences in intelligence from available I.Q. data.
(D) All the above.
- 755.** The following may be a cause of mental deficiency.
(A) Phenylketonuria
(B) Inadequate infant nutrition
(C) Inadequate medical care
(D) Lack of intellectual stimulation.
- 756.** Ordinarily, the I.Q.
(A) does not appreciably change during growing years
(B) increases with frequent regarded training
(C) does not remain constant
(D) fluctuates with health conditions.
- 757.** For fraternal twins reared together the coefficient of correlation of intelligences scores is about
(A) 0.00 (B) 0.25
(C) 0.50 (D) 0.75.
- 758.** In the above example, the product is
(A) units (B) classes
(C) relations (D) transformations.
- 759.** In example 37, operation is
(A) cognition
(B) convergent production
(C) memory (D) evaluation.
- 760.** Consider an anagrams test. The subject is asked to rearrange four-letter combinations (e.g.: IRAP, PANL, EKAM) to form familiar words (pair, plan, make). According to Guilford's structure of intelligence model, the content is

- (A) behavioural (B) semantic
(C) figural (D) symbolic.
- 761.** In question 760, the OPERATION is
(A) cognition
(B) divergent production
(C) evaluation
(D) memory.
- 762.** The group-factor theory of intelligence was proposed by
(A) Brown (B) Thorndike
(C) Kelley (D) Spearman.
- 763.** Thurstone's Primary Mental Abilities Test (PMA) are based on the following theory of intelligence
(A) Group factor theory
(B) G-factor theory
(C) Structure of intellect model
(D) The concept of crystallized intelligence.
- 764.** A generalization-perceiving capacity which represents one's potential intelligence some-what independent of socialization and education is termed by Cattell as:
(A) crystallized intelligence
(B) fluid intelligence
(C) fluctuating intelligence
(D) the general intelligence factor.
- 765.** Which of the followings statements is wrong ? According to Cattell
(A) fluid intelligence includes the flexibility to organize and reorganized concepts
(B) fluid intelligence peaks early and begins to decline shortly after age 20
(C) fluid intelligence includes scholastic ability and is determined by the culture
(D) crystallized intelligence may continue to increase for many years after the end of formal schooling.
- 766.** According to Cattell, the crystallized intelligence
(A) reflects one's cultural exposure
(B) reflects one's formal education
(C) is innate
(D) both (A) and (B) are true.
- 767.** Thurstone's primary abilities
(A) are completely independent
(B) have significant inter-correlation
(C) do not give support to Spearman's idea of a general intelligence factor
(D) are insignificantly inter-correlated.
- 768.** Guilford's structure of intellect model suggests
(A) lumping together of items to form test
(B) 7 primary abilities
(C) 120 unique intellectual factors
(D) the general intelligence factor 'g'.
- 769.** Assume that a subject is given a long list of unrelated words to study and is asked to recall the words at a later time. Then according to Guilford's structure of intellect model, the content of this test is
(A) figural (B) symbolic
(C) behavioural (D) semantic.
- 770.** In question 769, PRODUCT is
(A) transformation (B) unit
(C) system (D) class.
- 771.** Highly creative persons tend frequently
(A) low scorers in intelligence tests
(B) moderately intelligent
(C) of average intelligence
(D) highly intelligent.
- 772.** For identical (Mz) twins reared apart the coefficient of correlation of intelligence scores is about
(A) 0.90 (B) 0.25
(C) 0.50 (D) 0.75.
- 773.** Nearly three-fourths of the total variance of I.Q.s scores is associated with
(A) environmental factors
(B) genetic factors
(C) genetic plus environmental factors
(D) maturational factors.
- 774.** An innate difference in intelligence between the black and white population was suggested by
(A) Lindzey (B) Jenkins
(C) Jensen (D) none of these
- 775.** This self-test you are now taking is an example of
(A) an aptitude test

- (B) an achievement test
(C) an intelligence test
(D) a criterion-related test.
- 776.** Which of the following statistics is generally used to estimate reliability and validity ?
(A) The correlation coefficient
(B) The standard deviation
(C) The percentile score
(D) The median.
- 777.** What kind of validity do tests such as the SAT and ACT particularly strive for?
(A) Content validity
(B) Construct validity
(C) Absolute validity
(D) Criterion-related validity.
- 778.** Spearman's *g* infers that
(A) most kinds of intelligence are highly related
(B) most kind of intelligence are not highly related
(C) intelligence is highly correlated with personality characteristics
(D) intelligence is primarily inherited.
- 779.** If the heritability ratio for intelligence is 80%, this means that for you as an individual 80% of your intelligence is determined by heredity and 20% is determined by your environment. This statement is
(A) true
(b) false
(C) could not be interpreted
(D) none of the above.
- 780.** According to Raymond Cattell, the trait which was largely the result of heredity and which corresponds to our ability to reason is
(A) *g*
(B) *gf* (fluid intelligence)
(C) *gh* (heritability factor)
(D) *ge* (crystallized intelligence).
- 781.** Binet's original intelligence test was developed to
(A) measure psychological adjustment to stress
(B) predict successes in school
(C) discriminate between racial groups
(D) prove the superiority of the upper socioeconomic group's children.
- 782.** Which of the following tests measures intelligence individually and reports a single I.Q. score ?
(A) Stanford-Binet
(B) Scholastic Aptitude Test
(C) Wechsler Intelligence Scale for Children
(D) Thematic Apperception Test.
- 783.** The following report form is representative of that used for which of the following tests ?
- | | Scaled score | IQ |
|-------------------|--------------|-----|
| Verbal score | 55 | 106 |
| Performance score | 49 | 90 |
| Full-scale score | 104 | 103 |
- (A) MMPI (B) WISC
(C) Stanford-Binet (D) SCII.
- 784.** According to the text, which of the following is the best predictor of success in college ?
(A) High school grades and letters of reference
(B) SAT scores and IQ score
(C) SAT scores and high school grades
(D) Student self-reports of motivation.
- 785.** About 95 per cent of the American population scores between _____ and _____ on a standardized intelligence test.
(A) 85 and 115 (B) 70 and 130
(C) 55 and 145 (D) 30 and 170.
- 786.** "Mild Retardation" has been associated with and intelligence test score of about
(A) 85 (B) 75
(C) 65 (D) 45.
- 787.** Perhaps the strongest evidence for heredity factor for intelligence comes from studies of
(A) unrelated children reared together
(B) deprived home environments
(C) enriched home environments
(D) none of the above.
- 788.** If the reaction range concept of human intelligence is correct, then a child with

- exactly normal intelligence will probably not exceed an I.Q. of
(A) 100 (B) 110
(C) 120 (D) 130.
- 789.** Which of the following retarded groups can often pass for normal as adults ?
(A) Mild (B) Moderate
(C) Profound
(D) Both (A) and (B).
- 790.** What percentage of mental retardation cases have been definitely linked to organic causes ?
(A) Approximately 25%
(B) Approximately 50%
(C) Approximately 75%
(D) Approximately 90%.
- 791.** Terman's long-term study of gifted children found that they tended to excel in
(A) physical development
(B) social development
(C) emotional development
(D) all the above.
- 792.** Which of the following groups shows the highest correlation with respect to intelligence ?
(A) Fraternal twins reared together
(B) Fraternal twins reared apart
(C) Identical twins reared apart
(D) Both (A) and (C).
- 793.** The search for biological correlates of intelligence are beginning to prove quite fruitful. This statement is
(A) true (B) false
(C) could not be interpreted.
(D) all the above.
- 794.** What is the role of metacomponents in Struberg's triarchic theory of intelligence
(A) learning and storing information
(B) carrying out instructions
(C) giving instructions
(D) all the above.
- 795.** What form of mental illness has been frequently found to be associated with outstanding creative ability ?
(A) Anxiety disorders
(B) Mood disorders
(C) Schizophrenia
(D) Anti-social personality.
- 796.** Most tests of creativity emphasize
(A) convergent thinking
(B) divergent thinking
(C) both the above
(D) none of the above.
- 797.** The more _____ the environments of a group of people being studied, the _____ the genetic contribution will be to variations in intelligence.
(A) dissimilar, more average
(B) similar; lower
(C) dissimilar; higher
(D) similar; higher.
- 798.** Arthur Jensen argued that individual and racial differences in IQ come predominantly from
(A) cultural differences
(B) impoverished environments
(C) test biases
(D) genetic differences.
- 799.** The major problem with Arthur Jensen's conclusion about the heritability of intelligence is
(A) he failed to consider that black Americans have not been in this country as long as white Americans
(B) he overemphasized the crucial role of environment as a contributor to intelligence
(C) he suggested that the factors responsible for variation of intelligence within groups could also explain variation between groups
(D) he based his findings on old measures and tests of intelligence which have very little validity in today's world.
- 800.** SOMPA (System of Multi-cultural Pluralistic Assessment) is an intelligence test that attempts to control for
(A) cultural biases in intelligence testing
(B) genetic differences in intelligences
(C) environmental differences in intelligences
(D) gender bias.
- 801.** According to the text, which of the following tests would most typically be used by a vocational counsellor ?
(A) TAT (B) MMPI
(C) Stanford-Binet (D) SCII.

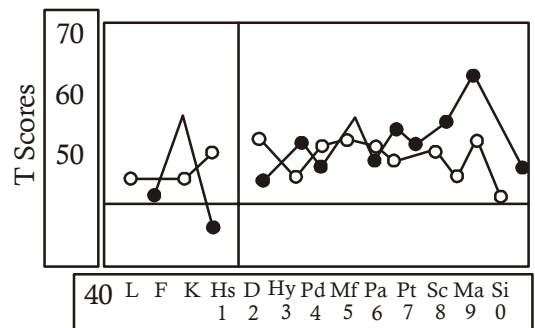
- 802.** The authors of the text believe that the use of tests
 (A) is generally good
 (B) is generally bad
 (C) depends on the test
 (D) depends on how the test is used.
- 803.** Suppose you have just taken the SAT. Your test administrator now asks you to answer twenty more questions. They will not be used in scoring your test, but your responses will help to determine whether these questions will be included in next year's test. You have been asked to be in
 (A) a control group
 (B) an experimental group
 (C) a validity group
 (D) a standardization group.
- 804.** A person with an IQ score of 28 and whose functioning is at this same level would be considered
 (A) severely retarded
 (B) mildly retarded
 (C) profoundly retarded
 (D) moderately retarded.
- 805.** In the figure below, people within the normal range of intelligence are represented by the area under the curve between point ____ and ____.



- (A) A and G (B) B and F
 (C) C and E (D) D and G.
- 806.** According to Turman's study of genius, children with exceptionally high IQs are more likely to
 (A) become alcoholics
 (B) turn to a life of crime

- (C) attain a superior income level
 (D) get divorced.

- 807.** According to the text, one of the problems extremely gifted children sometimes face is
 (A) becoming emotionally maladjusted
 (B) having a typical social relation
 (C) being physically unhealthy
 (D) never being able to get established in a career.
- 808.** The figure below shows scores for which of the following tests ?



- (a) MMPI (B) SAT
 (C) WISC (D) SCII.
- 809.** Selecting test items that are answered differently by "normal" and "mental ill" people and using them to test for possible mental illness is the rationale behind which of the following tests ?
 (A) Rorschach Inkblot Test
 (B) MMPI
 (C) K-ABC (D) TAT.
- 810.** Suppose you wish to know whether the test you have developed to place children in an advanced math. Class is consistently measuring the same characteristic. You would want to assess the test's
 (A) norms (B) validity
 (C) standardization (D) reliability.
- 811.** If time for development of a multiple choice test is limited and the answers on the test are easily remembered, the preferred measure of consistency would be
 (A) predictive validity

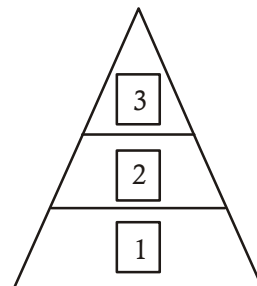
- (B) test-retest reliability
(C) split-halves reliability
(D) concurrent validity.
- 812.** A personnel director wants to develop a test to use for placing job applicants in jobs where they will do well. She would be most interested in
(A) predictive validity
(B) criterion-related validity
(C) face validity
(D) concurrent validity.
- 813.** Janet is 10 years old. Her IQ is 120. According to Binet's original IQ formula, what is Janet's mental age ?
(A) 6 (B) 8
(C) 10 (D) 12.
- 814.** Dr. Anderson believes that traditional intelligence tests are too narrowly focused and should include measure of such abilities as ability to adapt, ability to have sudden insights, and the ability to do different tasks at the same time. The person whose philosophy of testing she would most agree with is
(A) Francis Galton
(B) Arthur Jensen
(C) Alfred Binet
(D) Robert Sternberg.
- 815.** Suppose you determine that genes are responsible for 90 per cent of the individual variation of a particular trait, such as eye colour. You are here considering eye colour's
(A) trait distribution
(B) heritability factor
(C) normative distribution
(D) standard score.
- 816.** You are taking a personality test and are asked to respond to the words the tester says by saying the first related word that comes into your mind. You are probably taking
(A) the MMPI
(B) an intelligence test
(C) a projective test
(D) an interest test.
- 817.** You takes the MMPI and are given several assertions, such as "I never think bad thoughts about other people." You answer all these statements with "applies to me" response. You would probably score unusually high on
(A) the L scale
(B) the depression scale
(C) the hypochondria scale
(D) all the clinical scales.
- 818.** You are taking a test that asks you whether you like, dislike, or are indifferent to activities such as playing with children, working with your hands, and so on. You conclude you are taking the
(A) Strong-Camphbell Interest Inventory
(B) Wechsler Adult Intelligence Scale
(C) Thematic Apperception Test
(D) System of Multi-cultural Pluralistic Assessment.
- 819.** Probably the most severe problem of the personnel tests concerns the assessment of the tests
(A) standardization
(B) predictive validity
(C) test-retest reliability
(D) split-halves reliability.
- 820.** Suppose I had the desire and potential to be a violinist but became, instead, a psychologist. What is wrong with me, according to Maslow ?
(A) I suffer from incongruence
(B) I suffer from irony
(C) I have not achieved self-actualization
(D) I am directed towards overcoming inferiority.
- 821.** Which of the following is not personality trait ?
(A) Intelligence (B) Interest
(C) Emotionality (D) Honest.
- 822.** A personality trait is a
(A) particular ability
(B) distinct way of behaving in a specific situation
(C) specific motive
(D) specific activity.
- 823.** The 'types' of personality were discussed by
(A) Freud (B) Jung
(C) Adler (D) Pavlov.

- 824.** The most predominant aspect of the personality picture is the
(A) cultural aspect
(B) social aspect
(C) physical aspect
(D) intellectual aspect.
- 825.** Introversion, in contrast to extroversion, is highly correlated to
(A) intelligence (B) sociability
(C) creativity
(D) abstract thinking
- 826.** Personality tests are designed to
(A) identify personality traits
(B) measure personality trait
(C) identify and measure personality traits
(d) none of the above.
- 827.** The term personality refers to
(A) the modes of behaviour
(B) the various personality traits
(C) the physical and intellectual abilities
(D) the whole individual.
- 828.** Reference to individuals as 'introverts' and 'extroverts' was first made by
(A) Freud (B) Adler
(C) Jung (D) Allport.
- 829.** People with personality pattern type which is competitive, hard-driving, alert and efficient
(A) Suffer more from heart disease
(B) Suppress their fatigue
(C) Perform near their maximum capacity no matter what the situation calls for
(D) All the above.
- 830.** A light-hearted, carefree, sociable individual will show a high score in
(A) Anxiety (B) Repression
(C) Aggression (D) Extroversion.
- 831.** The dimension of body built characterized by predominance of fat in the abdominal region is called
(A) Endomorphy (B) Ectomorphy
(C) Mesomorphy (D) Pyknic.
- 832.** The distribution of a randomized sample of men along any 'type' dimension (e.g. 'leader' and 'follower', 'introvert' and 'extrovert') is
(A) normal, bell shaped
(B) humped at the two extremes
(C) Maswellian
(D) Gaussian.
- 833.** According to Adler, personality development is greatly influenced by reactions to
(A) sensory stimuli
(B) inferiority feelings
(C) superiority feelings
(D) all types of strong feelings.
- 834.** A person is sometimes extremely boastful and sometimes self-effacing; sometimes sociable and sometimes exclusive. He belongs to the personality type
(A) introvert (B) extrovert
(C) ambivert (D) mesomorph.
- 835.** The fundamental units of personality are referred to as
(A) interests (B) abilities
(C) motives (D) traits.
- 836.** Which of the following is not an important factor in determining personality traits?
(A) Home (B) Religion
(C) Sex instruction (D) Birth order.
- 837.** Personality theories concentrate on
(A) Situational factors
(B) Traits
(C) Interaction of situations with traits
(D) Separate influences of situations and traits.
- 838.** Almost all definitions of personality emphasize
(A) individual personality traits
(B) integration of personality traits
(C) moral traits
(D) traits pertaining to character.
- 839.** A person who does not show marked emotional responses belongs to the personality type known as
(A) effeminate (B) aggressive
(C) phlegmatic (D) stable.
- 840.** The gland which plays a key role in the development of personality is the
(A) pituitary (B) thyroid
(C) adrenal (D) thymus.

- 841.** Personality can be studied in terms of the unique organization of a particular individual specimen. This approach is known as
(A) nomothetic (B) idiographic
(C) omnibus (D) hierarchical.
- 842.** Personality is determined by
(A) non-verbally communicated attitudes between the parent and the child
(B) the behaviour of the parents
(C) the relationship between the child and the parent
(D) all the above.
- 843.** Freud's psychoanalytic theory of personality stresses on
(A) biological determinants of personality
(B) social determinants
(C) cultural determinants
(D) situational determinants.
- 844.** According to Freud's psychoanalytic theory the structure of personality consists of
(A) id, preconscious and conscious
(B) unconscious, preconscious and conscious
(C) id, ego and superego
(D) unconscious, ego and superego.
- 845.** Which of the following statements is correct? Out of id, ego and superego
(A) only id is unconscious
(B) only id and superego are unconscious
(C) all of id and superego and much of ego are unconscious
(D) all of id and much of ego and superego are unconscious.
- 846.** Slips of tongue reveal
(A) id urges
(B) ego ideals
(C) superego urges
(D) controls of the conscience.
- 847.** The defence that normally emerges during the phallic stage in psychosexual development is
(A) repression (B) identification
(C) suppression (D) individuation
- 848.** According to Freud, fixation at a particular stage of psychosexual development would take place if the needs during that stage were
(A) under-satisfied
(B) over-satisfied
(C) under-satisfied or over-satisfied
(D) satisfied.
- 849.** The central theme of Roger's personality theory is the
(A) self (B) ego
(C) experience
(D) external reality.
- 850.** Personality tests are not helpful answering such questions as
(A) How does personality change with age?
(B) Do identical twins differ in personality or not ?
(C) How can one personality type changed into another ?
(D) How are personality traits related socioeconomic status ?
- 851.** The 'executive' of personality according to psychoanalysis is
(A) id (B) superego
(C) conscience (D) ego.
- 852.** The prohibitions learned from parents and other authorities comprise the
(A) conscience (B) ego ideal
(C) libido (D) ego.
- 853.** Individual often act in ways which seem to be irrational. This was explained by Freud in terms of
(A) preconscious motivation
(B) unconscious motivation
(C) predominance of superego over id and ego
(D) predominance of ego over id and superego
- 854.** The concept of collective unconscious was formulated by
(A) Jung (B) Sigmund Freud
(C) Anna Freud (D) Adler.
- 855.** Allport formulated the idea of
(A) functional fixedness
(B) functional autonomy
(C) functional psychoses
(D) functionalism.

- 856.** The social learning theorists portray
(A) learner as passive recipient of environmental influences
(B) no individual differences with explaining behaviour
(C) learner as having internal cognitive processes to explain imitation, etc.
(D) both (A) and (C)
- 857.** Jung formulated the concepts of
(A) anima and animus
(B) extroversion and introversion
(C) personal unconscious
(D) all the above
- 858.** According to Roger's anxiety an defensive behaviour results due to
(A) inadequate childhood training
(B) incongruence between the organism (the total person) and the self
(C) faulty learning
(D) inadequate experiences.
- 859.** Roger's personality theory falls in the regime of
(A) existential psychology
(B) psychoanalytic psychology
(C) humanistic psychology
(D) behaviourism.
- 860.** The subjects studied by Maslow were
(A) neurotics (B) psychotics
(C) normal people (D) self actualise.
- 861.** The search for primary traits of personality uses the techniques of
(A) holistic assessment
(B) projective tests
(C) content analysis
(D) factor analysis.
- 862.** Attribution of one's own motive to others is called
(A) identification (B) projection
(C) association
(D) reaction formation.
- 863.** The drawback of Rorschach test is that it is
(A) impersonal
(B) not reliable
(C) not easy to administer
(D) difficult to score and to interpret.
- 864.** The most convenient kind of measures to use for almost any psychological purpose is
(A) pencil - and - tests
(B) sentence completion tests
(C) thematic apperception tests
(D) interview tests.
- 865.** In a projective test of personality, the stimulus must be
(A) simple (B) ambiguous
(C) complicated (D) pictorial.
- 866.** Holistic assessment involves
(A) the Rorschach test
(B) T.A.T.
(C) interviews
(D) all the above.
- 867.** Humanistic approaches to assessment of personality include techniques for measuring one's
(A) self-concept
(B) degrees of self-disclosure
(C) self concept and degree of self disclosure
(D) social desirability.
- 868.** Psychoanalysts assess personality through
(A) questionnaire methods
(B) interview methods
(C) MMPI
(D) projective techniques.
- 869.** Variables that influence the correlations found in trait research have been called
(A) moderator variables
(B) interfering variable
(C) independent variables
(D) intervening variables.
- 870.** The Minnesota Multiphasic Personality Inventory
(A) has empirical validity
(B) is easy to interpret
(C) is infallible
(D) all the above.
- 871.** When the content of the responses is ignored, but the pattern of a person's responses are matched with the patterns of answers given by groups of people with known characteristics, we get a test which has

- (A) content validity
(B) empirical validity
(C) face validity
(D) construct validity.
- 872.** Which of the following is a projective test ?
(A) WAIS (B) DAT
(C) TAT (D) MMPI.
- 873.** Sentence completion test is a
(A) test used in California Psychological inventory
(B) test for creativity in writing
(C) pencil and paper test
(D) projective test.
- 874.** The sixteen Factor Personality Questionnaire was developed by
(A) Thurstone (B) Cattell
(C) Gough (D) Murray.
- 875.** Most of the psychoanalytic theorists who broke away from Freud came to place more emphasis than Freud on the
(A) ego (B) id
(C) unconscious (D) superego.
- 876.** According to the text, a major problem with psychoanalytic theories is that they are
(A) not descriptive
(B) too research-driven
(C) untestable
(D) based on sexuality
- 877.** Which of the following was one of Horney's objections to Freud's theory ?
(A) His notion of unconscious desires
(B) The emphasis he placed on the ego
(C) His bias about female inferiority
(D) His emphasis on genital sex.
- 878.** Which of the following does not describe Freud's view of women as compared to men ?
(A) Morally and culturally inferior
(B) Envious of the male penis
(C) Having no id
(D) Less productive than males.
- 879.** Which of the following theories is most concerned with explaining the consistency of behaviour ?
(A) Freudian theory
(B) Humanistic theory
(C) Trait theory
(D) Cognitive theory.
- 880.** Eysenck concluded that personality structure varies on three dimensions: neuroticism versus emotional stability, introversion versus extroversion, and psychoticism. Underlying these traits are differences in
(A) drive
(B) neurological activity levels
(C) early experiences in the environment
(D) schemas and problem solving strategies
- 881.** According to the text, the factor which is most directly linked to a higher risk for coronary heart disease is
(A) Type A personality
(B) Type B personality
(C) self-efficacy
(D) hostility.
- 882.** According to social cognitive theorist, an organized set of knowledge about the person that guides perception and interpretation of information in social situations is called
(A) a strategy
(B) a self-schema
(C) a task (D) self-efficacy.
- 883.** Which of the following theorists has done his work primarily with normal adult humans ?
(A) Rogers (B) Jung
(C) Freud (D) Maslow.
- 884.** Maslow's theory proposes a hierarchy of needs. In the figure given below, which need corresponds to level 3, the highest-level need ?



- (A) Safety
(B) Love and belongingness
(C) Self-esteem
(D) Self-actualization.
- 885.** Bob is outgoing and sociable, seeks excitement, and loves a good time. He also has a good job and a stable family life. According to Eysenck, he would be especially high on which of the following dimensions?
(A) Neuroticism (B) Introversion
(C) Psychoticism (D) Extroversion.
- 886.** As presented in this chapter, Mischel's research on the consistency of personality would be most threatening to which one of the following theoretical positions?
(A) Psychoanalytic theory
(B) Humanistic theory
(C) Trait theory
(D) Social-cognitive theory.
- 887.** According to Freud, which of the following is the major motivating force in personality?
(A) Unconscious (B) conscious
(C) ego (D) conscience.
- 888.** To Freud, a slip of the tongue represented
(A) a neurological disorder
(B) an unhealthy childhood
(C) an unconscious conflict
(D) a simple mistake.
- 889.** According to the discussion of Freud's theory in the text, the id, ego and superego are
(A) physical divisions in the brain
(B) different ways of processing information
(C) various biochemical processes in the brain
(D) strong psychological forces.
- 890.** Which of the following is present at birth?
(A) superego (B) id
(C) ego
(D) reality principle.
- 891.** Which of the following defence mechanisms underlines all others?
(A) Projection (B) Fixation
(C) Regression (D) Repression.
- 892.** Sublimation is a special kind of
(A) repression (B) regression
(C) displacement (D) projection.
- 893.** The intense desire to take the place of the same-sex parent in the affections of the parent of the opposite sex is called
(A) the Oedipus conflict
(B) reaction formation
(C) sublimation
(D) the pleasure principle.
- 894.** According to McAdams' theory of psychobiography, the major human drive is one for
(A) self-coherence
(B) self-actualization
(C) superiority
(D) reproductive success.
- 895.** According to Gangestad and Simpson a female who takes a relatively long time to select a mate and requires from him strong attachment is using
(A) a nomothetic approach
(B) an ideographic approach
(C) a restricted strategy
(D) an unrestricted strategy.
- 896.** According to the text, which of the following is not one of the major themes which all personality theories address?
(A) Self-fulfilment
(B) Continuity and consistency
(C) The importance of external influences
(D) The meaning of life.
- 897.** Maria is continually worrying about whether her actions are socially acceptable. A Freudian psychologist might say her personality was being controlled by her
(A) ego (B) id
(C) superego
(D) defense mechanisms.
- 898.** Jack finds any mention of male femininity offensive. His therapist suggests that, in actuality, this may be Jack's way of reducing anxiety from his true fear of being a homosexual. If so, Jack is using the defense mechanism of

- (A) reaction formation
(B) displacement
(C) projection
(D) regression.
- 899.** Suppose fixation occurs during the first year of Tom's life. You might expect that, as an adult, Tom would be
(A) a homosexual
(B) excessively next
(C) passive and overtly dependent
(D) compulsive.
- 900.** James believes that the major motivation in a person's life is to strive for perfection. In this respect, he is most like
(A) Sigmund Freud (B) Alfred Adler
(C) Carl Jung (D) Karen Honey.
- 901.** Sally's roommate is continually doing favours for people—to a point where she gets frustrated because she has no time of her own. According to Horney, she is moving
(A) against people
(B) towards people
(C) away from people
(D) with people.
- 902.** Jeremy believes that the major purpose of a personality theory is to explain the consistency of human behaviour. Your best guess as to which type of personality theory describes his views is
(A) humanistic theory
(B) Psychoanalytic theory
(C) social cognitive theory
(D) trait theory.
- 903.** The gradual cutting back of rewards for appropriate behaviour is called
(A) self-monitoring
(B) self-reinforcement
(C) stimulus control
(D) shaping.
- 904.** According to the text, placebo drinking involves
(A) acting drunk even though you are not feeling the effects of alcohol
(B) substituting non-alcoholic beverages for alcoholic ones when you are drinking
(C) imagining how it feels to become progressively drunker so that you can negatively condition yourself to dislike alcohol
(D) substituting another activity, like smoking, for drinking.
- 905.** Ray seems to have no self-confidence whatsoever. He seldom undertakes challenging tasks, and he usually does not participate in a group because he thinks others would tease him. The concept which best describes Ray's problem is
(A) self-efficacy (B) sublimation
(C) type A personality
(D) homothetic.
- 906.** Joe's wife always treats him with respect and is warm, accepting, and sympathetic. Further more, she encourages Joe to speak his mind on whatever is troubling him. According to Rogers, she is offering Joe
(A) unconditional positive regard
(B) conditional positive regard
(C) self-actualization
(D) humanistic worth.
- 907.** Dr. Barnes states that personality is the result of one's genetic background in combination with one's environmental situation. Dr. Barnes most likely subscribes to
(A) evolutionary theory
(B) the behavioural genetic approach
(C) the psycho-physiological approach
(D) the ideographic view point.
- 908.** The theory which argues that sickness is the result of biological malfunction is called the
(A) general adaptation syndrome
(B) contingency model
(C) biomedical model
(D) bio psychosocial model.
- 909.** According to the text, about _____ per cent of American women will get breast cancer, about _____ per cent of the women who are treated are cured.
(A) 9; 35; 85 (B) 1; 9; 35
(C) 35; 9; 85 (D) 1; 85; 35.

- 910.** Most studies of stop smoking programmes find a relapse rate of about
(A) 5 per cent (B) 20 per cent
(C) 70 per cent (D) 95 per cent.
- 911.** The cells which attack bacteria and coat them or neutralize their toxins are called
(A) macrophages
(B) natural killer cells
(C) T-lymphocytes (D) B-lymphocytes.
- 912.** The SRRS assigns stress to events in one's life that involve
(A) only extreme changes
(B) negative changes
(C) both positive and negative changes
(D) both changes and unchanging conditions
- 913.** Surgeon General Koop's conclusion found that _____ women experience psychological distress after an abortion; the APA's conclusion was that _____ women experience significant psychological distress.
(A) many; many (B) few, few
(C) many; few (D) few; many.
- 914.** Research reported in the text found that, following the termination of an unwanted pregnancy _____ per cent of the women felt relief and _____ per cent felt guilty.
(A) 55; 49 (B) 9; 28
(C) 76; 71 (D) 17; 76.
- 915.** The text reports a study in which one group of nursing home residents was told that they were responsible for their arrangements and a second group were told that they would be taken care of
(A) the group who took care of themselves were happier and healthier
(B) the group who were taken care of were happier and healthier
(C) the first group were happier but less healthy
(D) the second group were happier but less healthy.
- 916.** The component of Type A behaviour which seems to be most directly linked to coronary heart disease is
(A) hostility
(B) detachment
(C) aggressiveness
(D) inability to express emotion.
- 917.** Generally speaking how are optimism related to stress ?
(A) Optimist experience more stress than pessimists
(B) Pessimists experience more stress than optimists
(C) Optimists and pessimists experience the same amount of stress, but pessimists report less conflict as a result
(D) None of the above
- 918.** Standing your ground and fighting for what you want is a strategy of coping called
(A) planful problem-solving
(B) distancing
(C) positive reappraisal
(D) confrontation coping.
- 919.** The coping strategy of turning to others for encouragement, sympathy, and advice forms the basis for which of the following coping strategies ?
(A) Planful problem-solving
(B) Positive reappraisal
(C) Accepting responsibility
(D) Social support.
- 920.** According to the text, which of the following support tactics is most likely to be helpful to the person needing support?
(A) Giving advice
(B) Listening
(C) Minimizing the loss
(D) Trying to identify with the bereaved's feeling of loss ("I know how you feel").
- 921.** Jung wants to lose weight so she makes a deal with her husband that for every five pounds she loses, he will contribute \$100 towards a vacation for the two of them. This is an example of
(A) a multimodal approach
(B) contingency contracting
(C) planful problem-solving
(D) stimulus control.
- 922.** You have just found out you are going to be called on in class to give a twenty-

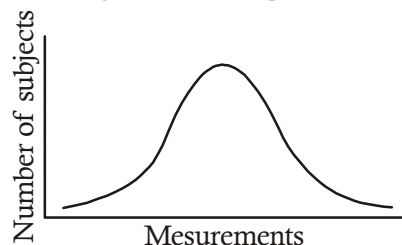
- minute lecture on material you hardly understand. You are in which stage of Selye's general adaptation syndrome ?
- (A) panic
(B) alarm resistance
(C) resistance (D) exhaustion.
- 923.** Marty has just undergone chemotherapy to suppress the activity of his immune system. The term which best describes how well his immune system is working is
- (A) immunocompetence
(B) SRRS score
(C) psychoimmunology
(D) catharsis.
- 924.** Mary was just divorced. John just got fired. Bob just borrowed some money. Judy just started working the night shift. According to the SRRS, who is under the most stress ?
- (A) Mary (B) John
(C) Bob (D) Judy.
- 925.** In general, the research on how terminating an unwanted pregnancy in the first trimester affects the psychological health of the mother can best be summarized as
- (A) it causes few short-term problems but frequently problems develop over the long-term.
(B) most women experience major problems in coping at the time, but nearly all work them out satisfactorily with time.
(C) abortion causes significant short-term and long-term psychological problems
(D) abortion does not pose a psychological hazard for most women
- 926.** Which of the following is not one of the stages of a stress management program ?
- (A) Training in techniques to reduce stress
(B) Practice in applying the skills
(C) Education about stress and its effects
(D) Development of a positive self-attitude.
- 927.** As an adolescent, Martha was kidnapped, beaten, and raped. Now as an adult, she still has flashbacks of the assault, has trouble sleeping, and has difficulty in her relationship with men. The term which most directly applies to her situation is
- (A) general adaptation syndrome
(B) hostility
(C) post-traumatic stress disorder
(D) catharsis.
- 928.** Jim has just been fired, and he is coping by trying to look at all the positive things that will result, such as being able to go out and find an even better job. His style of coping is called
- (A) distancing
(B) planful problem-solving
(C) positive reappraisal
(D) confrontational coping.
- 929.** Whenever Roger gets angry, he goes out for a long run to "let-off some steam". Roger's behaviour is best thought of an example of
- (A) catharsis
(B) contingency contracting
(C) stimulus control
(D) shaping.
- 930.** Suppose you are being treated for stress by being hooked up to electronic recording devices that tell you when you are changing your internal psychological processes. This technique is called
- (A) imagery therapy
(B) attribution therapy
(C) social readjustment
(D) biofeedback.
- 931.** Lonnie's therapist teaches him to tense and then relax various muscle groups in his as a means of reducing stress. Lonnie's therapist is using a technique called
- (A) body as a means of relaxation therapy
(B) biofeedback
(C) catharsis
(D) self-monitoring.
- 932.** The fact that exercise relieves stress is most consistent with which of the following theoretical models of stress ?
- (A) Contingency contracting
(B) General adaptation syndrome

- (C) Fight-or-flight
(D) The distancing model.
- 933.** Personality traits are characterized by
(A) consistency and distinctiveness
(B) charm and wit
(C) changes as a function of the situation
(D) lack of individual differences.
- 934.** Which of the following are among the basic issues around which personality theories have been constructed ?
(A) Determinism versus free will
(B) Consciousness versus the unconscious
(C) Nature versus nurture
(D) All the above
- 935.** Which of the following statements best reflects the text's conclusion about the effectiveness of biofeedback ?
(A) It is much effective than other therapies but is also considerably more expensive
(B) It works with animals but not with people
(C) It works for disorders associated with addiction but not for stress-related disorders
(D) Other therapies are just as effective and easier and cheaper to implement.
- 936.** Regular vigorous exercise seems to have what effect on stress ?
(A) It increases it
(B) It decreases it
(C) The effect it has varies from person to person
(D) It has no effect on stress.
- 937.** According to the text, healthy adults should exercise at least _____times a week for at least _____minutes per session.
(A) 2; 120 (B) 3; 15
(C) 5; 30 (D) 7; 20.
- 938.** The test reports a 1990 study of physical fitness which found that _____per cent of school age children could successfully complete physical fitness tests. This percentage is _____than the percentage of children who could perform these tasks ten years ago.
(A) 32; higher (B) 32; lower
(C) 82; higher (D) 82; lower.
- 939.** Suppose you try to quit smoking and the programme you enrol it includes self-monitoring, stimulus control, self-reinforcement, and shaping. It would best be considered an example of which of the following ?
(A) A biomedical model
(B) Biofeedback
(C) A multi-modal approach
(D) Guided relaxation.
- 940.** Which of the following is entirely unconscious, according the Freud ?
(A) The id (B) The ego
(C) The superego (D) The archetype.
- 941.** Although Osmo at an unconscious level has great hatred for Cosmo, he says he likes Cosmo and, to the outside world, gives all the appearance of liking him which defense mechanism is Osmo using ?
(A) Regression
(B) Reaction formation
(C) Projection
(D) Rationalization.
- 942.** The Oedipal complex occurs during the
(A) oral stage (B) anal stage
(C) phallic stage (D) genital stage.
- 943.** Which of the following concepts did Carl Jung develop ?
(A) Archetypes
(B) The collective unconscious
(C) Introversion-extraversion
(D) All the above.
- 944.** According to Eysenck differences in personality result in large part from differences in
(A) introversion-extroversion
(B) genetic inheritance
(C) condition ability
(D) all the above.
- 945.** Recent studies of the personalities of identical and fraternal twins are surprising and important because they indicate that
(A) personality is in large part genetically determined

- (B) environment is more important than heredity
(C) birth order is critical in personality formation
(D) monads and archetypes can be accounted for genetically.
- 946.** Sally believes that whatever happens to her, be it good or bad, is largely a matter of luck of fate. Sally has
(A) an internal locus of control
(B) an external locus of control
(C) high self-monitoring
(D) a sensation seeking tendency.
- 947.** Amanda is extraverted, impulsive, and risk-taking, and she likes fast cars, spicy food, and adventurous men. She is also nonconforming, especially in the sense that she will not adapt her personality to the person she happens to be with. Amanda is high on
(A) external locus of control
(B) self-monitoring
(C) sensation-seeking
(D) illegal substances.
- 948.** Which of the following has Adler emphasized in his theory of personality ?
(A) Striving for superiority
(B) Castration anxiety
(C) Introversion-extraversion
(D) All the above.
- 949.** Much of the behaviour that we call personality results from reinforcement and observational learning according to
(A) Jung (B) Skinner
(C) Bandura (D) Adler.
- 950.** Which of the following accepts a deterministic view of human behaviour ?
(A) The psychoanalytic approach
(B) The biological approach
(C) The behavioural approach
(D) All the above.
- 951.** An attempt to estimate the opinions, characteristics, or behaviours of a particular population by investigating a representative sample is a
(A) correlational study
(B) correlational coefficient
(C) case study
(D) survey.
- 952.** Research conducted by watching other people's behaviour as they go about their normal routine is called
(A) naturalistic observation
(B) participant observation
(C) survey
(D) case study.
- 953.** An experiment does an in-depth study of a particular individual. This is called a(n)
(A) experiment (B) survey
(C) case study (D) correlation.
- 954.** A sample is generally used when the group being studied is
(A) located in the same geographic area
(B) very large
(C) widely variant in the opinions they express
(D) predominantly male or female.
- 955.** The method that will allow cause-and-effect inferences to be made is the
(A) correctional study
(B) survey
(C) experiment (D) case study.
- 956.** A study that divides a population into subgroup according to certain criteria (such as age) and then makes comparisons between the subgroups is a
(A) time-series design
(B) correlation study
(C) longitudinal study
(D) cross-sectional study.
- 957.** Which of the following is used to determine whether a given set of conclusions can be arrived at from a particular set of data ?
(A) descriptive statistics
(B) standard deviations
(C) inferential statistics
(D) or relations coefficient.
- 958.** Which of the following research methods allows the combination of the findings of several studies so that a general conclusion can be reached ?
(A) Meta-analysis
(B) Time-series design
(C) Correlational study
(D) Inferential analysis.

- 959.** The type of research that determines the strength of a relationship between two variables is the
(A) experiment
(B) correlational study
(C) survey
(D) quasi-experiment.
- 960.** The first step in conducting an experiment is to formulate the
(A) hypothesis
(B) independent variable
(C) dependent variable
(D) control condition.
- 961.** The variable the experimenter measures is the
(A) experimental condition
(B) independent variable
(C) placebo
(D) dependent variable.
- 962.** The variable the experimenter intentionally manipulates in an experiment is called the
(A) placebo
(B) dependent variable
(C) experimental condition
(D) independent variable.
- 963.** In an unbalanced, or skewed, distribution, which measure of central tendency is least biased is
(A) mean (B) median
(C) mode (D) range.
- 964.** The average extent to which all scores in a particular set vary from the mean is the
(A) standard derivation
(B) central tendency
(C) probability statistic
(D) statistical significance.
- 965.** Suppose you are tossing a fair coin and it comes up heads on the first five tosses. The probability of it coming up heads again on the sixth toss is
(A) 1 : 2 (B) 1 : 6
(C) 1 : 36 (D) 1 : 180.
- 966.** When neither experimenter nor subject knows who is assigned to the experimental and control groups, the result is called a
(A) placebo effect
(B) single-blind technique
(C) double-blind technique
(D) self-fulfilling prophecy.
- 967.** Most experimental subjects try to help the researcher by doing their best to perform as they think they should. This problem is called
(A) the self-fulfilling prophecy
(B) demand characteristics
(C) the placebo effect
(D) the Hawthorne effect.
- 968.** Which of the following is not a way to reduce or control demand characteristics?
(A) Video-tape all instructions to subjects
(B) Interview subjects about their participation after the experiment
(C) be particularly friendly to the subject
(D) use deception as to the real intent of the experiment.
- 969.** To replicate an experiment, you would
(A) reanalyze the data using inferential rather than descriptive statistics
(B) calculate a correlation coefficient
(C) perform a meta-analysis
(D) read the experiment much as it was originally performed.
- 970.** According to the American Psychological Association, a researcher must not
(A) subject a human to physical discomfort
(B) subject a human to psychological discomfort
(C) refuse to let a human quit in the middle of an experiment
(D) refuse to let a human subject join an experiment.
- 971.** Suppose the population you are studying is 51 per cent female and 49 per cent male. Which of the following methods is used to get a sample that is also 51 per cent female and 49 per cent male?
(A) Public opinion poll
(B) Survey
(C) Random sample
(D) Representative sample.

- 972.** You decide that the best way to study the Ku Klux Klan is to become a member yourself and observe what really happens from the inside. You are using a technique called
 (A) participant observation
 (B) naturalistic observation
 (C) representative sampling
 (D) case study.
- 973.** Jack's parents have always expected him to be a lawyer, and now that he is twenty-one, he is applying to law school. This is best considered to be an example of
 (A) demand characteristics
 (B) the self-fulfilling prophecy
 (C) the Hawthorne effect
 (D) a double-blind situation.
- 974.** According to your text, psychology is a(n)
 (A) science
 (B) extension of common sense
 (A) humanity
 (B) art.
- 975.** Supposed you wish to study the effects of room temperature on job productivity on an assembly line. The independent variable this study would be
 (A) the temperature in the room
 (B) the number of people present
 (C) the number of goods assembled
 (D) how hot or cold people said they felt.
- 976.** Suppose you wish to study the effect of eating breakfast on test performance you find a group of subjects who never eat breakfast and ask half of them to eat breakfast every morning for a week. This would be an example of establishing a
 (A) control group
 (B) placebo
 (C) dependent variable
 (D) experimental group.
- 977.** Suppose you go into a laboratory and are asked to taste several differently foods and indicate which you like most and least. The experimenter is using a
 (A) behavioural measure
 (B) physiological measure
 (C) self-report
 (D) double blind technique.
- 978.** A group of four-year-olds is given an IQ test, then retested at the age of eight, twelve, sixteen, twenty, thirty and fifty. This is an example of a
 (A) time-series design
 (B) longitudinal study
 (C) cross-sectional study
 (D) replication.
- 979.** In 1934, LaPiere travelled across the United States with two Chinese friends. He later wrote to the establishments they visited to determine the strength of anti-Oriental prejudice. The major finding of his study was this
 (A) In 1934 there was a great deal of anti-Oriental prejudice
 (B) In 1934 there was almost no anti-Oriental prejudice; it developed after World War II
 (C) People admitted they were prejudiced but did not show it in their behaviour
 (D) People behaved in a prejudiced way but would not admit to being prejudiced.
- 980.** The text describes a study by Carol Dweck in which groups of children experimented for 25 days either easy problems or a combination of easy and difficult problems. Commonsense would have suggested that when these children were tested later on difficult problems, the _____ group would perform better. Dweck's results showed that the _____ group performed better.
 (A) easy; easy (B) easy; mixed
 (C) mixed; easy (D) mixed; mixed.
- 981.** "A quest for knowledge purely for its own sake" is one definition of
 (A) science (B) basic science
 (C) applied science
 (D) research.
- 982.** The diagram below represents a



- (A) normal curve
(B) description of meta-analysis
(C) skewed distribution
(D) result produced by an experiment that did not have an appropriate control group.
- 983.** Given the following set of scores--- 3, 4, 4, 4, 5, 7, 7, 8, 9, -4 is the
(A) mean (B) median
(C) mode (D) range.
- 984.** Given the following set of scores— 3, 4, 5, 5, 6, 6, 6, 8, 10 —what is the range ?
(A) 5 (B) 5.9
(C) 6 (D) 7.
- 985.** Suppose that, generally speaking, the taller people are, the more they weigh. Which of the following is the best estimate of the correlation coefficient between height and weight ?
(A) +63 (B) -11
(C) 0.00 (D) -77.
- 986.** Suppose I discover that as I eat more vegetables, my body weight decreases. I have demonstrated
(A) a positive correlation
(B) a negative correlation
(C) a cause-and-effect relationship
(D) random sampling.
- 987.** Which of the following people does not fit into the same theoretical perspective as the other three ?
(A) Ivan Pavlov (B) B. I. Skinner
(C) John B. Watson
(D) Abraham Mamslow.
- 988.** Interpretations of events and experiences are what cognitive psychologists call
(A) schemas
(B) information-processing systems
(C) mental strategies
(D) free associations.
- 989.** Which of the following is not one of the three goals of modern psychology ?
(A) Prediction (B) Translation
(C) Explanation (D) Description.
- 990.** Which of the following people established the first psychology laboratory ?
(A) Sigmund Freud
(B) William James
(C) Edward Thorndike
(D) Wilhem Wundt.
- 991.** Which of the following is most intimately tied to Darwin's Theory of biological evolution ?
(A) Behaviourism
(B) Functionalism
(C) Structuralism
(D) The humanistic view.
- 992.** For behaviourists, the primary determinant of whether a particular phenomenon was fit for study by psychologists was if it
(A) was uniquely human
(B) had applicability
(C) could be predicted on the basis of common sense
(D) was observable and therefore measurable.
- 993.** Which of the following psychologists took the strongest view that behaviour is controlled entirely by events in the environment ?
(A) B. F. Skinner
(B) Wilhelm Wundt
(C) William James
(D) John B. Watson.
- 994.** Freud asked patients to talk about their dreams in order to
(A) provide symbolic clues to the nature of their unconscious conflicts
(B) put them to sleep
(C) relax them so they would talk about what was really bothering them
(D) observe their natural behaviour.
- 995.** Which of the following terms does not fit Freud's theory ?
(A) Unconscious
(B) Free association
(C) Psychoanalysis
(D) Mental strategies.
- 996.** Which of the following groups of psychologists most stress the positive side of human nature ?
(A) Behaviourists
(B) Neurobiologists

- (C) Humanists
(D) Psychoanalysts.
- 997.** Suppose Dr. White is studying how a particular drug acts on the brain to produce an overeating response. Dr. White would most likely be considered a
(A) personality psychologist
(B) clinical psychologist
(C) health psychologist
(D) neuropsychologist.
- 998.** Which of the following types of psychologists is most likely to work directly in the school system, advising students and teachers and applying knowledge about teaching and learning ?
(A) educational psychologists
(B) experimental psychologists
(C) school psychologists
(D) developmental psychologist.
- 999.** The field of psychology that tries to explain difference among individuals is
(A) developmental psychology
(B) personality psychology
(C) industrial psychology
(D) psychopharmacology.
- 1000.** Researchers in which of the following perspectives would most likely to be interested in new developments in computer processing ?
(A) Psychoanalytic perspective
(B) Behavioural perspective
(C) Cognitive perspective
(D) Neurobiological perspective.
- 1001.** Behaviourists view learning as the development of
(A) memory traces in the brain
(B) consciousness
(C) association between events
(D) self-actualization.
- 1002.** Which of the following terms does not fit with the other three ?
(A) Stimulus (B) Response
(C) Reward
(D) Free association.
- 1003.** The goal of psychoanalysis is
(A) to make you forget the things that are troubling you
(B) to make you conform to society's expectations
(C) to show you what things you need to do make your life happier
(D) to help you consciously recognized the roots of your problems.
- 1004.** Which of the following approaches most strongly rejects the idea that human nature is determined by past and present events?
(A) Freud's psychoanalysis
(B) Behaviourism
(C) Structuralism (D) Humanism.
- 1005.** The task of determining how to lay out the keys on a computer keyboard would probably be given to which type of psychologist ?
(A) Social psychologist
(B) Developmental psychologist
(C) Industrial psychologist
(D) Personality psychologist.
- 1006.** Psychologist involved in the field of health psychology are probably particularly interested in
(A) psychological problems of doctors
(B) diagnosis of mental illness
(C) treatment of mental illness
(D) the effect of stress on disease.
- 1007.** Ms Jones is concerned that the production workers in her building are becoming fatigued because the lighting is bad and they must strain to see their detail work. To determine whether her hunch is valid, she would most likely call in a(n)
(A) environmental psychologist
(B) personnel psychologist
(C) social psychologist
(D) artificial intelligence specialist.
- 1008.** The most likely advantage for a student who takes only this course in psychology is
(A) ability to analyse people
(B) greater self-understanding
(C) better chance at a career
(D) ability to manipulate others.
- 1009.** You meet a woman who tells you she is a psychologist. From this information,

- you infer that professionally she is probably most interested in
- (A) what you do and the how you think
 - (B) your personality
 - (C) your emotional stability
 - (D) animal (not human) behaviour.
- 1010.** The methods of study used in psychology are most closely related to those used in
- (A) sociology
 - (B) religion
 - (C) physics
 - (D) philosophy.
- 1011.** The clearest example of applied research would be a study examining the relationship between
- (A) brightness of street lights and amount of crime
 - (B) colours of light and feelings of stresses
 - (C) order of words on a list and ability to remember the list
 - (D) age and problem-solving ability.
- 1012.** Dr. Smith believes that the human mind cannot be analysed into its constituent parts, but rather must be investigated by studying how it adapts to solve problems. Dr. Smith's view is most similar to that proposed by the
- (A) behaviourists
 - (B) structuralists
 - (C) psychoanalysts
 - (D) functionalists.
- 1013.** Suppose you are gravely concerned because you think you are losing your mind. Which kind of psychologist would you probably go to see ?
- (A) An applied organizational psychologist
 - (B) A social psychologist
 - (C) A personality psychologist
 - (D) A clinical psychologist.
- 1014.** A persons who works with fairly common social problems, such as drug addiction, marriage and divorce therapy, and family life therapy, would typically be trained as a
- (A) psychoanalyst
 - (B) counselling psychologist
 - (C) psychiatrist
 - (D) clinical psychologist.
- 1015.** An experimenter tests the hypothesis that physical exercise helps people's mood (makes them happier). Subjects in the experimental group participate on Monday and Tuesday and those in the control on Wednesday and Thursday. What is the extraneous (confounding) variable ?
- (A) The hypothesis
 - (B) Day of the week
 - (C) The exercise
 - (D) The mood (degree of happiness).
- 1016.** What is the mode of the following data?
2, 3, 3, 3, 5, 5, 7, 12.
- (A) 3
 - (B) 4
 - (C) 5
 - (D) 6.
- 1017.** What is the median of the following data? 1, 3, 4, 4, 5, 6, 9.
- (A) 3
 - (B) 4
 - (C) 4.57
 - (D) 6.
- 1018.** Researchers tend to find an inverse relationship between alcohol consumption and speed of response. Which of the following fictitious statistics could possibly represent that relationship ?
- (A) -4.57
 - (B) -.87
 - (C) .91
 - (D) .05.
- 1019.** The term statistical significance refers to
- (A) how important the data are for future research on the topic
 - (B) the conclusion that there are no reasonable alternative explanations
 - (C) the inference that the observed effects are unlikely to be due to chance
 - (D) the representativeness of the sample.
- 1020.** The major advantage of the experimental method over the correlation approach is that the experimental method
- (A) permits one to make causal conclusions
 - (B) allows for prediction
 - (C) is generally less artificial than correlational procedures
 - (D) permits the study of people in groups.

- 1021.** In looking through some medical records you find that there is a strong relationship between depression and chronic pain: the stronger the physical pain that people report, the higher their scores on an inventory that measures depression. Which of the following conclusions are justified ?
- (A) Depression tends to produce chronic pain
 - (B) Chronic pain tends to produce depression
 - (C) Both chronic pain and depression result from some unknown third factor
 - (D) None of the above.
- 1022.** An instructor wishes to find out whether a new teaching method is superior to his usual procedures, so he conducts an experiment. Everyone in his classes is quite excited about the prospect of learning under the new procedure, but of course he cannot administer the new teaching method to everyone. A random half of the students receive the new method and the remaining half receive the old. What is the most obvious flaw in this experiment ?
- (A) Subjects should have been systematically assigned to groups
 - (B) The sample is not representative of the population
 - (C) Placebo effects or experimenter bias will affect results
 - (D) Distortions in self-report will affect results.
- 1023.** What procedure helps correct for experimenter bias ?
- (A) Extraneous or confounding variables
 - (B) Sleep learning or hypnosis
 - (C) A higher standard for statistical significance
 - (D) Use of the double-blind procedure.
- 1024.** With regard to the topic of deception in research with man subjects, which of the following is true ?
- (A) Researchers are careful to avoid deceiving subjects
 - (B) Some topics could not be investigated unless deception were used
 - (C) It has been empirically demonstrated that deception causes severe distress
 - (D) All psychological research must involve some deception
- 1025.** Which of the following is among the six standard parts of a psychological journal article ?
- (A) Conclusions
 - (B) Bibliography
 - (C) Data summary
 - (D) Results.
- 1026.** The Author Index in the *Psychological Abstracts* provides
- (A) names of current APA members
 - (B) registration and biographical information about frequent authors
 - (C) index numbers that locate article summaries
 - (D) names of authors who specialize in abstractions.
- 1027.** According to Darley and Larane one reason that bystanders may fail to help the victim in an emergency is
- (A) diffusion of responsibility
 - (B) burnout
 - (C) groupthink
 - (D) emergencies are so obvious they cannot be mistaken for ordinary events.
- 1028.** You make a commitment to enrol in a summer program that involves training and work experience. You understand that room and board will be covered by your work contribution. After making the commitment, you discover an exorbitant travel charge and processing fee. That you will follow through is expected by the
- (A) foot-in-the-door phenomenon
 - (B) door-in-the-face technique
 - (C) low-ball phenomenon
 - (D) that's-not-all technique.
- 1029.** "A quest for knowledge purely for its own sake" is one definition of
- (A) Science
 - (B) Basic science
 - (C) Applied science
 - (D) Research.
- 1030.** For behaviourists, the primary determinant of whether a particular phenom-

- enon was fit for study by psychologists was if it
- (A) was uniquely human
 - (B) had applicability
 - (C) could be predicted on the basis of commonsense
 - (D) was observable and therefore immeasurable.
- 1031.** Freud asked patients to talk about their dreams in order to
- (A) provide symbolic clues to the nature of their unconscious conflicts
 - (B) put them sleep
 - (C) Relax them so they would talk about what was really bothering them
 - (D) Observe their nature behaviour.
- 1032.** Psychologists involved in the field of health psychology are probably particularly interested in
- (A) Psychological problem of doctors
 - (B) Diagnosis of mental illness
 - (C) Treatment of mental illness
 - (D) The effect of stress on disease.
- 1033.** Which of the following is most clearly responsible for speeding the conduction of a neural impulse ?
- (A) Myelin
 - (B) Glial cells
 - (C) Interneurons
 - (D) Hormones.
- 1034.** The gland that controls such diverse activities as the level of sugar in the blood and the secretion of the sex hormone androgen is called
- (A) Thyroid
 - (B) Gonads
 - (C) Adrenal cortex
 - (D) Adrenal medulla.
- 1035.** Bem, in his self-perception theory, believes we arrive at our attitudes by
- (A) A dissonance reduction
 - (B) Trying to explain our behaviour
 - (C) Being reinforced for them
 - (D) Learning them from our parents.
- 1036.** Jones and Nisbett argue that we tend to attribute our own behaviour to environmental causes, whereas we attribute others' behaviours to some enduring personality trait. This is called
- (A) the actor observer bias
 - (B) the fundamental attribution error
 - (C) causation correlation confounding
 - (D) the salience effect.
- 1037.** According to Sternberg, a relationship high in commitment but low in intimacy and passion is called
- (A) compassionate love
 - (B) empty love
 - (C) consummate love
 - (D) infatuated love.
- 1038.** Which of the following is a dispositional attribution regarding why Rex failed his first psychology test ?
- (a) "The test was too hard."
 - (B) "The teacher did not cover this material in class."
 - (C) I have no aptitude for psychology
 - (D) My roommate would not let my study.
- 1039.** John has heard that Mary is shy. When he meets her, he treats her as if she were shy, and she responds as if she were shy, even though she is not. This is an example of
- (A) The primacy effect
 - (B) The self-serving bias
 - (C) Cognitive dissonance
 - (D) A self-fulfilling prophecy.
- 1040.** The most long-lasting kind of conformity is
- (A) identification
 - (B) compliance
 - (C) obedience
 - (D) internalisation.
- 1041.** When Milgram asked psychology majors and psychiatrists what percentage of subjects would deliver. In his experiment, he found that _____ per cent in fact delivered all the shocks.
- (A) 50; 50
 - (B) 10; 10
 - (C) 1; 65
 - (D) 65; 1.
- 1042.** Zanna's resolution of the social facilitation contradiction described in the test was to suggest that the presence of other serves to
- (A) increase drive
 - (B) provide reinforcement
 - (C) encourage obedience
 - (D) eliminate distraction.
- 1043.** Which of the following approaches most strongly rejects the idea that human na-

- ture is determined by past and present events?
- (A) Freud psychoanalysis
(B) Behaviourism
(C) Structuralism (D) Humanism.
- 1044.** The method that will allow cause and effect inferences to be made is the
- (A) correctional study
(B) survey
(C) experiment
(D) case study.
- 1045.** Research conducted by watching other people's behaviour as they go about their normal routine is called
- (A) Naturalistic observation
(B) Participant observation
(C) Survey (D) Case study.
- 1046.** In an unbalanced, or skewed, distribution, which measure of central tendency is least biased ?
- (A) Mean (B) Median
(C) Mode (D) Range.
- 1047.** Suppose you wish to study the effects of room temperature on job productivity on an assembly line. The independent variable in this study would be
- (A) the temperature in the room
(B) the number of people present
(C) the number of goods assembled
(D) how hot or cold people said they felt.
- 1048.** Suppose that, generally speaking, the taller people are, the more they weight. Which of the following is the best estimate of the correlation coefficient between height and weight ?
- (A) +.63 (B) -.11
(C) 0.00 (D) -.77.
- 1049.** The average person goes through about how many sleep cycles in as typical night's sleep ?
- (A) One (B) Five
(C) Eight (D) Hundreds.
- 1050.** The major difference between stimulants and alcohol is that stimulants
- (A) breathing slows
(B) skin resistance to electrical conductivity rises
(C) blood pressure falls.
(D) oxygen consumption increases.
- 1051.** Every time you get drunk you are outgoing and introduce yourself to people. When you are sober, you don't remember any of them. This is an example of
- (A) inhibitory conflict
(B) consolidation memory
(C) alcoholic blackout
(D) korsakoff's syndrome.
- 1052.** A major function of the hypothalamus is to
- (A) control attention
(B) coordinate body movements
(C) maintain a steady internal environment for the body
(D) transfer information from one part of the brain to another.
- 1053.** You volunteer for an experiment in which electrodes are attached to your scalp and you are asked to think about certain memories. The researcher is probably using the technique of
- (A) EEG recording
(B) PET scan
(C) CAT scan (D) MRI.
- 1054.** Blaine has head injury that is causing him difficulty in regulating breathing and blood flow. His face is contorting, he is salivating heavily, and he is making chewing movements. You conclude the accident caused injury to his
- (A) Amygdala (B) Cerebellum
(C) Medulla
(D) Left frontal lobe.
- 1055.** According to the principle of sensory adaptation, our sensory systems are particularly
- (A) duration (B) intensity
(C) quality (D) change.
- 1056.** Generalising from an experiment described in the text, if you hypnotised a person and suggested to the person that he was blind in his left eye, you would cause him to
- (A) be truly blind in that eye
(B) be able to see with that eye only when his right eye was closed

- (C) be able to see with that eye but deny the sight
- (D) admit to being able to see if confronted with evidence that the eye was working.
- 1057.** In most cases of classical conditioning, the
- (A) CS occurs just before the UCS
- (B) CS occurs just after the UCS
- (C) CR occurs before the CS
- (D) CS and UCS occur randomly.
- 1058.** The tendency for an extinguished response to recur after a rest from the experimental procedure is called
- (A) superstitious behaviour
- (B) discrimination
- (C) spontaneous recovery
- (D) behaviour modification.
- 1059.** A rat learns that he will reinforce only when a red high in his cage is on. This is an example of
- (A) generalisation
- (B) stimulus control
- (C) continuous reinforcement
- (D) backward conditioning.
- 1060.** Which of the following statements regarding the rods and cones is false ?
- (A) They both sense light
- (B) There are more types of cones than rods
- (C) Rods are concentrated in the fovea
- (D) Both (B) & (C) are false
- 1061.** The phenomenon of perceiving patterns according to closeness in position is called
- (A) simplicity (B) grouping
- (C) proximity (D) continuity.
- 1062.** Suppose Mary wants to know if she can learn Spanish by playing some language tapes while she sleeps. Her question concerns the general topic of _____ and according to the text, the answer would most likely _____
- (A) subliminal perception; may be
- (B) subliminal perception; No
- (C) signal detection; may be
- (D) signal detection; no.
- 1063.** You suspect you have a hearing loss, so you go to a specialist who fits you with earphones and tells you to rise your right hand whenever you hear a tone. The specialist is testing your
- (A) just noticeable differences
- (B) sensory threshold
- (C) Weber's ratio
- (D) adaptation.
- 1064.** Suppose you believe that hearing is set up the same way as a harmonica: The note you hear depends on where on the instrument you blow. You probably subscribe to one of these theories.
- (A) Frequency theory
- (B) Volley theory
- (C) Opponent process theory
- (D) Place theory.
- 1065.** How would you extinguish an apparently conditioned response ?
- (A) Stop reinforcement
- (B) Prevent the response
- (C) Punish the response
- (D) Remove the stimulus.
- 1066.** You teach a child to be afraid of the dog next door because it bites children, but now the child becomes afraid of all dogs. This is an example of
- (A) discrimination
- (B) generalisation
- (C) stimulus control
- (D) a discriminative stimulus.
- 1067.** You drive around all day in a strange city looking for a particular building. You finally find it. Now your friend wants you to go to another building. Even though you weren't looking for it, you can drive directly to the second building, since you remember it from earlier in the day. This example is similar to
- (A) latent learning
- (B) stimulus control
- (C) modeling
- (D) discrimination training.
- 1068.** The memory that lasts about twenty seconds and is maintained by acoustic rehearsal is
- (A) sensory memory

- (B) long-term memory
(C) short-term memory
(D) procedural memory.
- 1069.** Shallow processing memory is distinguished from deep processing memory by
(A) part of the brain in which storage occurs
(B) kind of material being remembered
(C) amount of material being remembered
(D) kind of rehearsal used to remember.
- 1070.** Anterograde amnesia refers to the loss of
(A) sensory memory
(B) things stored in long-term memory
(C) memory for new information
(D) recognition and recall.
- 1071.** When a person remembers sentences, he or she generally remembers
(A) the entire sentence verbatim
(B) the propositions
(C) the nouns but not the adjectives
(D) the prepositions.
- 1072.** You meet John, who is forty-five years old, intelligent, dedicated, verbal, concerned about other people's welfare, and wealthy. You conclude that he is more likely to be a doctor than a truck driver. You are using the heuristic called
(A) probabilistic inference
(B) utility analysis
(C) availability
(D) representatives.
- 1073.** All normal humans have twenty-three pairs of
(A) genes
(B) chromosomes
(C) DNA molecules
(D) executive control structures.
- 1074.** The process of changing pre-existing schemas to conform to new information is called
(A) assimilation
(B) accommodation
(C) egocentrism
(D) overregulatization.
- 1075.** A memory expert remembers long list of words by imagining that he walking down a familiar street and at each step placing an object on the side walk. To remember the list, he simply walks down the street and picks up the objects. This technique is an example of
(A) déjà vu
(B) method of loci
(C) peg word system
(D) eidetic imagery.
- 1076.** Marelene is raped as a child but has no memory of this traumatic event. What theory best accounts for this loss?
(A) Motivated forgetting
(B) Interference
(C) Consolidation
(D) Decay.
- 1077.** A major difficulty we encounter in problem-solving is the cognitive limitations imposed by
(A) short-term memory
(B) long-term memory
(C) both short-term and long-term memory
(D) our perceptual systems.
- 1078.** Another name for a rule of thumb problem-solving strategy is
(A) algorithm (B) heuristic
(C) mental
(D) subgoal analysis.
- 1079.** The tendency to view an object as being usable only for its customary use is called
(A) mental set
(B) construct
(C) an algorithm
(D) functional fixedness.
- 1080.** Bobby broke a lamp while he was trying to dust the shade. He fear that his mother will be very upset and punish him for breaking the lamp. That he ignores the fact of his trying to be helpful suggests that he is in which of Kohlberg's stages ?
(A) Identify diffusion
(B) Pre-conventional
(C) Conventional
(D) Post-conventional.

- 1081.** Generally speaking, the sympathetic division of the nervous system dominates during _____; the parasympathetic dominates during _____.
(A) Sleep; wakefulness
(B) Emergency; relaxation
(C) Work; play
(D) Joy; anger.
- 1082.** Which of the following is one of the dimensions on the circle of emotions ?
(A) Inherited-to-learned emotions
(B) High-to-low-activation
(C) Internal-to-external sources
(D) Verbal-to-verbal display.
- 1083.** Julie is an experiment where every 30 seconds a loud tone is sounded. For the first several tone bursts, Julie jumped when she heard the tone. Gradually, though, she came to ignore the tone and hardly noticed it when it sounded. This is best thought of as an example of
(A) assimilation
(B) metacognition
(C) accommodation
(D) habituation.
- 1084.** Jenny calls all animals with fur and four legs "doggie". Now she learns that some of these animals are bears and some are bunnies. Her learning here is best thought of as an example of
(A) Habituation
(B) Metacognition
(C) Accommodation
(D) Assimilation.
- 1085.** Which of the following is an example of telegraphic speech ?
(A) I must have it now (accompanied by a gesture)
(B) I'd like that
(C) Doggy bye-bye
(D) We went to the store.
- 1086.** According to Erik Erikson, the stage of trust versus mistrust corresponds to which of the following ages ?
(A) One
(B) Three through five
(C) Adolescence
(D) Middle Adulthood.
- 1087.** According to Ainsworth, a parent who responds slowly or with only partial effectiveness to an infant's distress is likely to produce a child with which of the following kinds of attachment ?
(A) Partial attachment
(B) Negative attachment
(C) Disattachment
(D) Anxious attachment.
- 1088.** Which of the following characteristics does not appear to be associated with children who react badly to their parent's divorce ?
(A) Low self-esteem
(B) Independent personalities
(C) Difficult temperaments
(D) A tendency to blame themselves for their parents' problems
- 1089.** Which of the following is crucial for the explanation given by social learning theory of how gender roles are acquired?
(A) That parents are permissive
(B) That parents are authoritarian
(C) That parents treat all children the same
(D) That parents behave differently towards different sex children.
- 1090.** The comparison theory of happiness says that you are happiest when
(A) you are less fortunate than others around you
(B) you are as fortunate as others around you
(C) you are more fortunate than others around you
(D) you have what you need.
- 1091.** Marlene believes that emotion is produced by the combination of physiological factors and situational cues. Her theory most closely corresponds to
(A) cognitive arousal theory
(B) Cannon-Bard theory
(C) James-Lange theory
(D) facial feedback theory.
- 1092.** Suppose you are coping with a crisis by are not following the stages that health care professionals believe to be beneficial. According to the test, your response should be considered

- (A) normal (B) retarded
(C) deviant (D) neurotic.
- 1093.** The basic assumptions of Humanistic approach
(a) People are naturally striving to be creative and happy.
(b) It is possible to specify an universal set of values that will provide people with moral anchor enabling them to decide what is right or wrong and good or bad.
(c) It believes in uniqueness of individuals and they should be trained to realise their inner potentials.
(d) All the innate potentials and growth mechanisms can be studied scientifically similar to natural success.
- Codes :**
(A) (a) and (b) (B) (b) and (c)
(C) (b) and (d) (D) (c) and (d)
- 1094.** The Hebb experiment on sensory deprivation reported in the text was _____ drive reduction theories and _____ optimal level theories.
(A) supportive of; supportive of
(B) supportive of; contradictory to
(C) contradictory to; supportive of
(D) unrelated to; supportive of.
- 1095.** According to Eysenck, as compared to introverts, extroverts have _____ levels of internal stimulation and seek _____ levels of external stimulation.
(A) higher; the same
(B) lower; the same
(C) higher; lower
(D) lower; higher.
- 1096.** Maria gets married, but is terribly distressed on her wedding night to find that she is unable to engage in sexual intercourse because of intense and painful muscle spasms around her vagina. If this condition is serious and persists, it would best be thought of as an example of
(A) vaginismus
(B) primary orgasmic dysfunction
(C) secondary orgasmic dysfunction
(D) premature orgasmic dysfunction.
- 1097.** A group of scores against which an individual's test score is compared is called a
(A) norm (B) reference group
(C) positive reference group
(D) standard.
- 1098.** About 95 per cent of the American population scores between _____ and _____ on a standardized intelligence test.
(A) 85 and 115 (B) 70 and 130
(C) 55 and 145 (D) 30 and 170.
- 1099.** The more _____ the environments of a group of people being studies, the _____ the genetic contribution will be to variations in intelligence.
(A) dissimilar; more average
(B) similar; lower
(C) dissimilar; higher
(D) similar; higher.
- 1100.** Selecting test items that are answered differently by "normal" and "mentally ill" people and using them to test for possible mental illness is the rationale behind which of the following tests ?
(A) Rorschach Inkblot Test
(B) MMPI
(C) K-ABC (D) TAT.
- 1101.** Suppose you wish to know whether the test you have developed to place children in an advanced math class is consistently measuring the same characteristic. You would want to assess the test's
(A) norms (B) validity
(C) standardisation (D) reliability.
- 1102.** Most studies of stop-smoking programmes find a relapse rate of about
(A) 5 per cent (B) 20 per cent
(C) 70 per cent (D) 95 per cent.
- 1103.** Surgeon General Koop's conclusion found that _____ women experience psychological distress after an abortion; the APA's conclusion was that _____ women experience significant psychological distress.
(A) many; many (B) few; few
(C) many; few (D) few; many.
- 1104.** Generally speaking, how are optimism and pessimism related to stress ?

- (A) Optimists experience more stress than pessimists
 - (B) Pessimists experience more stress than optimists
 - (C) Optimists and pessimists experience the same amount of stress, but optimists report less conflict as a result
 - (D) Optimists and pessimists experience the same amount of stress, but pessimists report less conflict as a result.
- 1105.** Even though Lynn is very intelligent and has excellent grades in college, she is not interested in going to medical school because "it would probably be too difficult." Which of the following constructs would probably describe Lynn ?
- (A) High fear of success
 - (B) High fear of failure
 - (C) Low need for achievement
 - (D) High need for achievement.
- 1106.** According to Freud, which of the following is the major motivating force in personality ?
- (A) Unconscious (B) Conscious
 - (C) Ego (D) Conscience.
- 1107.** The intense desire to take the place of the same-sex parent in the affections of the parent of the opposite sex is called
- (A) the Oedipus conflict
 - (B) reaction formation
 - (C) sublimation
 - (D) the pleasure principle.
- 1108.** Which of the following does not describe Freud's view of women as compared to men ?
- (A) Morally and culturally inferior
 - (B) Envious of the male penis
 - (C) Having noid
 - (D) Less productive than males.
- 1109.** According to McAdams' theory of psychobiography, the major human drive is one for
- (A) self-coherence
 - (B) self-actualisation
 - (C) superiority
 - (D) reproductive success.
- 1110.** James believes that the major motivation in person's life is to strive for perfection. In this respect, he is most like
- (A) Sigmund Freud
 - (B) Alfred Adler
 - (C) Carl Jung (D) Karen Horney.
- 1111.** Marty believes that humans have as in-born potential for growth, creativity, and spontaneity. He would best be classified as advocating
- (A) the psychoanalytic approach
 - (B) trait theory
 - (C) social cognitive
 - (D) the humanistic approach.
- 1112.** In psychoanalysis, transference refers to:
- (A) repression
 - (B) projection of feelings about others into therapist
 - (C) psychoanalysis
 - (D) free association.
- 1113.** Some of Freud's followers who later broke away (such as Jung and Adler) placed much more emphasis than Freud on the concept of
- (A) unconscious (B) superego
 - (C) ego (D) id.
- 1114.** Regular vigorous exercise seems to have what effect on stress ?
- (A) It increases it
 - (B) It decreases it
 - (C) The effect it has varies from person to person
 - (D) It has no effect on stress.
- 1115.** Whenever Roger gets angry, he goes out for a long run to "let off some steam." Roger's behaviour is best thought of as an example of
- (A) catharsis
 - (B) contingency contracting
 - (C) stimulus control
 - (D) shaping.
- 1116.** The therapist most likely to treat the symptoms rather than an underlying problem is the
- (A) psychoanalytic therapist
 - (B) family-systems therapist

- (C) learning theory therapist
(D) biological therapist.
- 1117.** Which of the following statements regarding use of the DSM-II-R for diagnosis of mental illness is false ?
(A) Its usefulness probably outweighs the hazards of its use
(B) It currently views homosexuality as a category of abnormal behaviour
(C) Its categories change with changing times and attitudes
(D) Its new edition is more specific in describing the various disorder.
- 1118.** The disorder characterised by extreme swings in mood from sadness to elation is called
(A) schizophrenia
(B) dissociative disorder
(C) depression
(D) manic-depressive disorder.
- 1119.** Bart is jumpy, irritable, fearful, has trouble sleeping, and cannot concentrate on his work. An amateur diagnosis would be that Bart suffers from
(A) generalised anxiety disorder
(B) obsessions
(C) paranoid schizophrenia
(D) a phobia.
- 1120.** Julia is so afraid of riding in an elevator that she will shop in stores only on the ground floor. She would be classified as
(A) obsessive (B) dysthymic
(C) phobic (D) compulsive.
- 1121.** Bill is arrested for murdering his mother, wife, and two children and for trying to kill himself. He reports that he feels no sadness or guilt over his actions. Your diagnosis of Bill's condition is
(A) manic depressive
(B) schizophrenia
(C) borderline personality disorder
(D) sociopathic personality.
- 1122.** In a study described in the text, white subjects were asked to determine whether target words were or were not associated with the words "black" and "white". Which of the following best summarize the result of that study ?
(A) Subjects made more negative associations to the word "white".
(B) Subjects made more negative associations to the word "black".
(C) Students responded faster when positive traits were associated with the word "black".
(D) Students responded faster when positive traits were associated with the word "white".
- 1123.** Suppose you are in charge of a sixth grade class in which the children are very prejudiced against two Vietnamese student's in the class. According to the text, the best way to reduce the prejudice is to
(A) move the two Vietnamese students out of the classroom
(B) lecture the class about the evils of prejudice, especially racism
(C) create a positive situation in which all students are forced to cooperate
(D) ask the students individually to be nice to the Vietnamese
- 1124.** In Milgram's obedience experiment, subjects believed that Milgram was studying
(A) obedience (B) learning
(C) aggression (D) conformity.
- 1125.** The technique that involves asking for a small favour before asking for a large one is called the
(A) that's not --all technique.
(B) foot-in- the-door phenomenon
(C) door-in-the-face phenomenon
(D) low-ball phenomenon.
- 1126.** Berkowitz argues that which of the following factors is most closely related to the experience of aggression ?
(A) Altruism (B) Hostility
(C) Withdrawal
(D) Negative feelings.
- 1127.** According to the text's discussion of pornography and aggression, the Surgeon General's recommendation called for ____; and the social scientists' recommendation called for _____.
(A) more education; stricter laws limiting access to pornographic materials

- (B) stricter enforcement obscenity laws; more education
(C) doing nothing; stricter enforcement of obscenity laws
(D) redefining of obscenity; stricter enforcement of the obscenity laws we have.
- 1128.** According to the text, psychosurgery is occasionally used as a treatment of last resort for all of the following disorder except
(A) severe depression
(B) obsessive compulsive disorder
(C) manic episodes in manic-depression
(D) schizophrenia.
- 1129.** Which of the following drugs has proved to be effective in the treatment of obsessive compulsive disorder ?
(A) Xanax, an antianxiety drug
(B) Prozac, an antidepressant
(C) Thorazine, an antipsychotic
(D) Lithium.
- 1130.** Mr. Alvarey makes his three-year-old daughter sit on the basement step whenever she is naughty so that she is away from any opportunity for reinforcement. His technique is best thought of as an example of
(A) aversive conditioning
(B) time out
(C) participant modelling
(D) self-instructional training.
- 1131.** Which of the following serves as genetic messengers carrying information from the nucleus of the cell the cell protoplasm where proteins are manufactured?
(A) m-RNA (B) DNA
(C) Amino acids (D) transfer RNA.
- 1132.** The following trait(s) is (are) determined by dominant genes.
(A) Blood type (B) Dwarfism
(C) Albinism
(D) Both (A) and (B)
- 1133.** Which of the following statements is false ?
(A) Mutations are usually for the worse
(B) Most mutant genes are dominant
(C) Mutation consists of changes in the rungs of the DNA ladder-involving deletions, additions, or modifications of the base chemicals.
(D) Congenital malformations, possibly including cancer, are associated with faulty DNA instructions due to gene mutations.
- 1134.** The following is (are) the constitutional reaction tendency (tendencies) that are apparent soon after birth
(A) activity level (vigour)
(B) sensitivity and emotional disposition
(C) adaptability
(D) all of the above
- 1135.** Studies on the crucial importance of culture in shaping development have been done by
(A) Margaret Mead
(B) Sigmund Freud
(C) Max Weber (D) all of the above.
- 1136.** The 'self-image' incorporates the individuals' perception of
(A) self-identity (B) self-evaluation
(C) self-deal (D) all the above.
- 1137.** The 'level of aspiration' is used to connote
(A) a person's aspirations in relation to his abilities
(B) a person's aspirations in relation to his environmental opportunities
(C) a person's aspirations in relation to his abilities and environmental opportunities
(D) a person's aspirations in relation to the, maximum social status attainable in the society.
- 1138.** If the needed stimulation or opportunity for activity is lacking at critical period, then given functions
(A) may not appear
(B) may be slower in making their appearance
(C) may be only partially adequate
(D) either of the above may happen.
- 1139.** The sex of the child is decided by
(A) sperm cell
(B) egg cell

- (C) either sperm cell or egg cell
(D) destiny.
- 1140.** Which of the following is true ?
(A) The odds for a boy baby are slightly greater than those for a girl baby
(B) The odds for a girl baby are slightly greater than those for a boy baby
(C) The odds for a girl baby are the same as those for a boy baby
(D) Investigations comparing the odds for a boy baby and those for a girl baby have not yet been made.
- 1141.** Aberrations involving sex-linked traits are
(A) less likely to be manifested in females
(B) less likely to be manifested in males
(C) males and females are equally susceptible to them
(D) fatal in nature and the foetus usually does not survive due to these genetic aberrations.
- 1142.** The following is an example of subculture found in the Indian culture.
(A) Khasis of Assam
(B) Parsis in Gujarat
(C) Coorgs
(D) The civil servants
(E) All the above
- 1143.** During infancy and early childhood, the social heritage is transmitted to the child
(A) almost exclusively through the family
(B) through the family and the school
(C) through the peer group
(D) through the family and the mass media like radio and television.
- 1144.** Which of the following statements is false ?
(A) Peer groups transmit the wider culture
(B) The peer group serves as a reality check
(C) The peer world helps children and youth to achieve emotional independence
(D) Most peer groups are in conflict with the adult world because of the so called 'generation-gap.'
- 1145.** During the course of development, 'conscience' develops through the process of
(A) identification and development of inner controls
(B) reinforcement
(C) imitation and modelling of adults responses
(D) instructions.
- 1146.** Which of the following statements is correct ?
(A) Ducklings will follow the first moving object they see and thereafter show preference for this object and others like it
(B) Ducklings will follow the mother when they see it since they instinctively know they are safe in doing so
(C) Ducklings do not follow the first moving object when they see it since they tend to be very cautious
(D) Ducklings can be imprinted only to the members of their own species.
- 1147.** Match the following
- | | |
|-----------------------|---|
| (I) Riesen | (1) Critical period studies in intellectual development |
| (II) Harlow & studies | (2) Critical period Harlow in sensory motor development |
| (III) Skeels | (3) Critical period studies in social attachment |
- The following pairs match
(A) (I) - (2); (II) - (3); (III) - (1)
(B) (I) - (1); (II) - (2); (III) - (3)
(C) (I) - (3); (II) - (2); (III) - (1)
(D) (I) - (2); (II) - (1); (III) - (3).
- 1148.** Language used as a means for thinking and communication is
(A) a set of symbols
(B) a set of images
(C) a set of concepts
(D) a set of generalizations.
- 1149.** A stress reaction does not involve
(A) constriction of pupils
(B) increased blood sugar

- (C) increased blood pressure
(D) increased muscle tone.
- 1150.** "Stop smoking or get lung cancer" is an example of
(A) approach-avoidance conflict
(B) approach-approach conflict
(C) avoidance-avoidance conflict
(D) double approach-avoidance conflict.
- 1151.** In the above question, in case the individual makes a decision and moves towards the object
(A) the approach gradient increases and no consequent difficulty arises
(B) the level of the non-chosen gradient decreases and no consequent difficulty arises
(C) the avoidance gradient increases and pushes the individual into conflict again
(D) both (A) and (B) are correct.
- 1152.** The overtly indulged child
(A) enters readily into warm human relationships but exploits them for his own purposes
(B) is usually rebellious in dealing with authority
(C) often finds it difficult to accept present frustrations in the interests of long range goals
(D) all the above are true.
- 1153.** Which of the following is false ? Unrealistic demands of parents may lead to
(A) rigid, restricted personalities
(B) misidentification
(C) the child developing high self-esteem
(D) a series of self-devaluating failures and the child eventually resists or rebels.
- 1154.** A mother complains that her son does not show his love for her but testily rejects his attempts to show affection with the statement that he is too grown up to be acting like a child. The son experiences
(A) a double-bind phenomenon
(B) rejection
(C) an unrealistic demand from his mother
(D) restriction.
- 1155.** If a child of an alcoholic parent learns to abhor alcohol, the alcoholic parent has served as a
(A) positive model
(B) negative model
(C) neutral identity with little influence on the child's behaviour
(D) a child of an alcoholic parent will also become an alcoholic .
- 1156.** Motivation is a concept that represents the energy which moves a person to actively satisfy
(A) physical needs
(B) psychological needs
(C) social needs
(D) all the above.
- 1157.** The primary needs are basically
(A) psychological in origin
(B) psychological needs
(C) social needs
(D) all the above.
- 1158.** Stress is experienced when
(A) internal homeostatic balance is disrupted
(B) there is loss of significant objects *e.g.* a body part
(C) there is injury or threat of injury
(D) there is frustration
(E) all the above.
- 1159.** In case of anxiety
(A) emotion is related to a real thing
(B) the event responsible for the anxiety is said to be repressed
(C) the person is aware of the threatening object
(D) the event responsible for the anxiety is said to be generalized.
- 1160.** The statement 'I am nervous' is
(A) a subjective evidence of anxiety
(B) an objective evidence of anxiety
(C) an objective evidence of fear
(D) an indication of phobia.
- 1161.** Which of the following is not a physiological correlate of anxiety ?
(A) GEM to urinary dysfunction
(B) Functional gastrointestinal disorders
(C) Hyperventilation syndrome

- (D) Gross over activity of sympathetic nervous system.
- 1162.** In terms of etiology, anxiety is usually viewed as repressed or forgotten
(A) fear (B) guilt
(C) sorrow (D) grief.
- 1163.** With regard to children who have debilitating and terminal illness, their greatest fear is
(A) death (B) pain
(C) separation from parents
(D) being anesthetized.
- 1164.** When six to twelve months of infant confront any one who is a non-mother, they experience
(A) castration anxiety
(B) separation anxiety
(C) stranger anxiety
(D) anticipatory anxiety.
- 1165.** The situation in which a person encounters threat from an authority figure such as a supervisor, represents an extension of the concept of
(A) castration anxiety
(B) separation anxiety
(C) stranger anxiety
(D) anticipatory anxiety.
- 1166.** Which is the correct sequence ?
(A) Frustration - Anger - Hostility
(B) Anger - Frustration - Hostility
(C) Goals blocked - Anger - Frustration
(D) Goals blocked - Frustration - Anger.
- 1167.** Which of the following statements is false ?
(A) Anger is not necessarily destructive, but more aggressive in nature
(B) Hostility pervades the person's entire behavioural repertoire
(C) Hostility is relatively short-lived if the frustrating stimulus is removed
(D) Hostility may or may not be related to a specific frustrating stimulus or condition.
- 1168.** The patient states "I feel uptight about my new job". This is an example of
(A) objective evidence of anxiety
(B) subjective evidence of anxiety
(C) inappropriate affect
(D) repression of affect.
- 1169.** In grieving and mourning process we know that at different ages the mourning can take different forms. Which of the following is an incorrect statement?
(A) The infant might protest, deny, and detach itself away
(B) A child in the latency stage would probably be jocular and perhaps hypomanic
(C) A child in the adolescent stage might turn to antisocial acting out
(D) Middle-age persons quite frequently turn to hypochondriacally symptoms
- 1170.** Guilt is
(A) sadness in the face of external environmental disapproval
(B) shame mixed with hostility
(C) mixture of sadness plus anger turned
(D) none of the above.
- 1171.** A stimulus is a
(A) mechanical change
(B) physical change
(C) chemical change
(D) any of the above.
- 1172.** The most important feature in a stimulus is the element of
(A) intensity (B) change
(C) wavelength/pitch
(D) depth.
- 1173.** A sense organ or receptor is
(A) selectively sensitive to some types of change in its environment and not to others
(B) sensitive to all types of changes in its environment
(C) a specialized part of the body
(D) both (a) and (c) are correct
- 1174.** The receiving cells of the retina used for twilight vision or light of low intensity are
(A) rod cells (B) cone cells
(C) fovea (D) blind spot.
- 1175.** The area of sharpest vision in daylight is called
(A) fovea (B) receptor
(C) blind spot (D) optic nerve.

- 1176.** One is temporarily blinded by the glare of oncoming headlights. This is due to
(A) some cells in the retina are killed
(B) defect of eye-adaptation
(C) the eye gets temporarily de-focused
(D) the glare of the headlight makes us blink and shut the eyes for some time.
- 1177.** In any 24-hour period the greatest percentage of car accident occur during
(A) the afternoon
(B) the twilight hour
(C) the night (D) the forenoon
- 1178.** In complete night vision (a moonless night with clouds obscuring the stars) the colours seen are
(A) red, yellow, green and blue although they appear blacker and greyer
(B) green and blue although they appear to have lighten somewhat
(C) various shaded of blue
(D) all the colours are seen as shaded of black, grey and white.
- 1179.** If you have been staring at a bright object intently and it is removed suddenly, you will continue to see images of the object. This is an example of
(A) an optical illusion
(B) an afterimage
(C) a visual purple effect
(D) Purkinje phenomenon.
- 1180.** Match the following: The afterimage of
(1) red (i) yellowish
(2) blue (ii) greenish
(3) black (iii) white or greyish
(iv) bluish
(A) (1) - (i); (2) - (ii), (3) - (iv) are correct matches indicating the colours of images and their afterimages
(B) (1) - (ii); (2) - (i), (3) - (iii) are correct matches indicating the colours of images and their afterimages
(C) (1) - (iii); (2) - (i), (3) - (ii) are correct matches indicating the colours of images and their afterimages
(D) (1) - (i); (2) - (ii), (3) - (iii) are correct matches indicating the colours of images and their afterimages
- 1181.** As age increases, hearing acuity decreases for
(A) very low frequencies
(B) relatively low frequencies
(C) higher frequencies
(D) frequencies between 1,000 and 8,000 vibration per second.
- 1182.** A sound wave is
(A) an electromagnetic wave
(B) an electrostatic wave
(C) a pressure wave
(D) a magneto-dynamic wave.
- 1183.** The fundamental units of the auditory receptors corresponding to the rods and cones in vision are
(A) the hair cells in the organ of corti
(B) the particles in the semicircular canal
(C) the bones of the middle ear
(D) the hair cells in the oval shaped window of the inner ear.
- 1184.** The human ear responds to sound waves having a frequency from
(A) 1 per second to 10,000 vibrations per second
(B) 100 per second to 100,000 vibrations per second
(C) 25 per second to 20,000 vibrations per second
(D) 25 per second to 200,000 vibrations per second.
- 1185.** Intense cold (e.g., playing with snow-balls for an hour) will
(A) activate the cold spots on the surface of the skin
(B) deactivate the cold spots on the surface of the skin
(C) activate the hot as well as the cold spots on the surface of the skin
(D) deactivate the hot spots on the surface of the skin.
- 1186.** The sense of 'balance' is controlled by
(A) the semicircular canals
(B) the vestibule
(C) the organ of corti
(D) all the above.
- 1187.** The 'awareness of body movements' is called
(A) kinesthesia (B) vertigo
(C) dynamics (D) consciousness.

- 1188.** The phenomenon of hypnotism indicates the effect of _____ on perception.
(A) mind (B) desire
(C) background factors
(D) previous occurrence.
- 1189.** Events are remembered in the form of
(A) images (B) words only
(C) deeds (D) any of the above.
- 1190.** Which of the following statements is false ?
(A) Visual imagery, the best-developed type, is nevertheless not especially useful for precise reproduction
(B) Very rapid forgetting occurs in the first few hours after learning, followed by more gradual forgetting
(C) Most everyday forgetting can be explained by the atrophy theory of forgetting (decay of memory traces) rather than by the interference theory of forgetting
(D) The interference or confusion of present learning by subsequent learning is called retroactive inhibition.
- 1191.** The following can produce a sensory phenomena
(A) stimulation of sensory nerve endings of kinesthesia
(B) toxic conditions e.g., various drugs
(C) lesions of visual cortex
(D) migraine headaches
- 1192.** The shadow of a leaf on a wall at night is perceived as a tarantula spider. This is an example of
(A) illusion (B) hallucination
(C) disillusion (D) delusion.
- 1193.** Which of the following is not expected after one hour of sensory deprivation ?
(A) Anxiety (B) Depression
(C) Hostility (D) Fugue.
- 1194.** The simplest form of remembering is
(A) recall (B) recognition
(C) performance
(D) reproduction of material once learned.
- 1195.** Loss of memory for events occurring 24 hours prior to being knocked unconscious from an automobile accident is an example of
(A) intergraded amnesia
(B) retrograde amnesia
(C) paramnesia
(D) patchy amnesia.
- 1196.** The recovery of lost memory abruptly implies that the memory defect was
(A) psychogenic in origin
(B) organic in origin
(C) static in origin
(D) faked response of the patient.
- 1197.** Verbal abilities are under the executive control of
(A) the left hemisphere
(B) the right hemisphere
(C) cerebellum (D) vocal chords.
- 1198.** 'Idiot savants' are individuals who are
(A) with IQs in the range of 'retarded' but show dramatic excellence for some singular talent
(B) with IQ in the range 21 to 35
(C) with IQ in the range of genius but who have singular outstanding deficit
(D) with IQ below 20.
- 1199.** Finding a telephone number in a telephone book and having the number available for a few seconds represents
(A) immediate memory
(B) short-term memory
(C) iconic memory
(D) confabulation.
- 1200.** The following is the correct sequence of the stages involved in creative thinking
(A) preparation - orientation - illumination - verification
(B) preparation - incubation - illumination - orientation - verification
(C) orientation - preparation - incubation - illumination - verification
(D) orientation - preparation - illumination - incubation - verification.
- 1201.** While reviewing an electroencephalographic tracking from an adult patient with a long history of epilepsy, a physician notes considerable delta activity.

- The patient is most likely
(A) has diffuse brain damage
(B) is entering a toxic deuterium
(C) fell asleep during the tracing
(D) may have a space-occupying lesion.
- 1202.** An adopted child's IQ correlates
(A) higher with that of biological parents than with the IQ of adoptive parents
(B) higher with that of adoptive parents than with the IQ of biological parents
(C) higher with that of his peers than with the IQ of his natural siblings
(D) both (B) and (C) are true.
- 1203.** The following represents chromosomal disorders in man
(A) Turner's Syndrome
(B) Klinefelter's Syndrome
(C) Supermates (D) Superfemales.
- 1204.** The holding of irrational beliefs despite evidence that they have no basis in reality is called
(A) a psychogenic fugue state
(B) a hallucination
(C) a delusion
(D) a conversion reaction.
- 1205.** The antipsychotic drugs are most likely to be used for
(A) active schizophrenia
(B) passive schizophrenia
(C) both active and passive schizophrenia
(D) neither active nor passive schizophrenia.
- 1206.** Suppose you are terribly afraid of spiders. Your therapist suggests you try to relax as she brings increasingly more fearful "spider-related things" things before you. This therapy is most similar to
(A) systematic desensitisation
(B) psychoanalysis
(C) pschoanalysis
(D) self-instructional training.
- 1207.** According to the Diagnostic and Statistical Manual of Mental Disorders, Edition II (DSM-II) an IQ of 65 classifies a person as
(A) borderline mental retardation
(B) mild mental retardation
(C) moderated mental retardation
(D) severe mental retardation.
- 1208.** The outcome of the learning effort depends upon
(A) motivation (B) intellectual
(C) strategy or method of attack
(D) all the above.
- 1209.** Which of the following statements is false?
(A) Intelligence level correlates highly with creativity
(B) Evaluative and inductive thinking correlates positively with creativity
(C) The "self-assertiveness" trait correlates with high scores on the various psychological tests of creativity
(D) A number of investigators have questioned the use of psychological tests for measuring creativity.
- 1210.** Deductive thinking refers to thinking in which
(A) the individual goes beyond the present information, adding new elements that are not inherent in the known facts
(B) an inference or conclusion is derived logically from information on hand
(C) the individual judges the suitability or appropriateness of an idea
(D) all the above.
- 1211.** Creativity correlates most highly with
(A) inductive and deductive thinking
(B) deductive thinking
(C) inductive and evaluative thinking
(D) inductive, deductive and evaluative thinking.
- 1212.** Which of the following statements is false?
(A) Creativity correlates highly with academic achievement because a creative student usually finds his way through and solves problems easily
(B) According to psychoanalytic theory, creativity is a form of neuroticism, in which instinctual drives have been diverted into creative activities
(C) Most current investigators believe that personality maladjustment is ordinarily a handicap to creativity

- (D) According to psychological studies, it was found that leading artists and scientists shared one trait in common: the willingness to work long and hard.
- 1213.** Which of the following is the correct sequence of Piaget's four stages of cognitive development ?
- (A) Proportional stage—sensorimotor stage—concrete operational stage—formal operational stage
- (B) Proportional stage—sensorimotor stage—formal operational stage—concrete operational stage
- (C) Sensorimotor stage—proportional stage—formal operational stage—concrete operational stage
- (D) Sensorimotor stage—preoperational stage—concrete operational stage—formal operational stage.
- 1214.** The 'id'
- (A) is completely altruistic and unconcerned with reality
- (B) can generate images and wishes related to need gratification
- (C) is not much concerned with immediate gratification
- (D) all the above are true.
- 1215.** The superego consists of the following element(s)
- (A) conscience (B) ego-ideal
- (C) conscience and ego-idea
- (D) ego in interaction with id
- 1216.** Which of the following statements is correct according to Freud's theory of personality ?
- (A) The 'superego' and 'id' operate through the 'ego' system
- (B) the 'ego' and 'id' operate through the 'superego' system
- (C) The 'id' 'ego' and 'superego' all operate independently. This is also the reason why an individual's behaviour is so ambivalent
- (D) The 'ego' and 'superego' operated through the 'id' system.
- 1217.** Personality adjustment requires
- (A) adequate resolution of conflict between id demands and superego demands
- (B) adoption of defence mechanism
- (C) reaction formation so that the individual asserts himself in a hostile environment
- (D) all the above are true.
- 1218.** In Freud's theory of personality there is a certain degree of correspondence between
- (1) unconscious-id
- (2) preconscious-id
- (3) conscious-ego
- (4) conscious-superego.
- Which of the alternative is correct ?
- (A) Both (1) and (3)
- (B) Both (1) and (4)
- (C) Both (2) and (3)
- (D) Only (1).
- 1219.** 'Cigarette smoking' according to Freud's psychoanalysis would reflect a certain degree of
- (A) oral fixation
- (B) anal fixation
- (C) fixation in the latency stage
- (D) fixation in the genital stage.
- 1220.** Freud's 'primitive drives' are
- (A) constructive drives
- (B) destructive and aggressive urges
- (C) constructive drives as well as destructive and aggressive urges
- (D) ambiguous, undefined drives which propel man to doom.
- 1221.** The concept of 'libido' refers to
- (A) the basic energy of life
- (B) the death instinct in man
- (C) the anxious reactions to the norms of the society
- (D) the uncertain part of man's personality.
- 1222.** The 'id' operates in terms of
- (A) pleasure principle
- (B) secondary process
- (C) reality principle
- (D) both (A) and (B) are correct.
- 1223.** A daughter's unconscious incestuous desires towards her father are referred as
- (A) oedipal complex

- (B) electra complex
(C) eros complex
(D) thanatos complex.
- 1224.** The latency phase psychoanalytic theory occurs at the same time as which of Erikson's stages ?
(A) Trust vs mistrust
(B) Identity vs role confusion
(C) Industry vs inferiority
(D) Autonomy vs shame and doubt.
- 1225.** In Erikson's theory of psychological tasks, which of the following is not a correct task ?
(A) Integrity vs despair
(B) Industry vs inferiority
(C) Intimacy vs Isolation
(D) Identity vs shame and doubt.
- 1226.** All of the statements below regarding adolescence (age 13 to 18) are true except
(A) in Erikson's framework, the basic task to be resolved is to develop affiliation with others
(B) one-third of teenage marriages according to American statistics, end in divorce within 4 years
(C) are in Piaget's stage of formal operations
(D) must maintain a sense of identity in the face of rapid changes.
- 1227.** All of the following are considered to be functions of 'ego' except
(A) personal values
(B) defence mechanisms
(C) object relations
(D) reality testing
(e) thought processes.
- 1228.** The conditioning of fear in a human infant was demonstrated in an experiment by
(A) Mary Cover Jones
(B) Watson and Rayner
(C) Skinner
(D) Watson.
- 1229.** Such conditioned fears once developed
(A) can be eliminated by behaviour therapy
(B) can never be eliminated
(C) become worse and worse spreading to other objects and cannot be eliminated
(D) eliminate automatically with time.
- 1230.** The 'law of effect' was formulated by
(A) Thorndike (B) Skinner
(C) Pavlov (D) Watson.
- 1231.** In classical conditioning, response is
(A) elicited (B) emitted
(C) eradicated (D) abetted.
- 1232.** In learning to press a lever to obtain food, the animal has to undergo
(A) classical conditioning
(B) respondent conditioning
(C) operant conditioning
(D) both (B) and (C) are correct.
- 1233.** Among human beings, learning of language as a means of communication is a case of
(A) printing
(B) classical conditioning
(C) operant conditioning
(D) Pavlovian conditioning.
- 1234.** In learning theory, grades in an academic course would be an example of
(A) acquisition
(B) generalization
(C) primary reinforcers
(D) secondary reinforcers
- 1235.** A student did poorly in an examination which was given in a specific room. The student is afraid he will fail the course. On the next occasion he walks into the room he experiences fear. This is an example of
(A) classical conditioning learning
(B) social learning
(C) cognitive learning
(D) inhibition learning
- 1236.** 'Reinforcement' is essential in case of
(A) respondent conditioning
(B) operant conditioning
(C) both operant and respondent conditioning
(D) innate learning.

- 1237.** Reinforcers may be
(A) pleasant (positive)
(B) aversive (negative)
(C) neutral
(D) both (A) and (B) are true.
- 1238.** 'Negative reinforcement' involves
(A) avoiding the aversive stimulus
(B) receiving the aversive stimulus
(C) punishment
(D) the concept of 'negative reinforcement' does not exist as reinforcers by definition are always positive.
- 1239.** Which of the following is a primary drive ?
(A) Hunger
(B) Self-esteem
(C) Social approval
(D) All the above.
- 1240.** According to behaviouristic model, the following process(es) is(are) based on an elaboration of the basic discrimination operation
(A) perceiving
(B) forming concepts
(C) solving-problems and making decisions
(D) all the above.
- 1241.** Which of the following personality theorist is not associated with the humanistic (self) man ?
(A) Allport (B) Rogers
(C) Murray (D) Jung.
- 1242.** Who amongst the following is not related to the existential approach in viewing man ?
(A) Kerkegaard (B) Satre
(C) Wertheimer (D) Heidegger.
- 1243.** Which of the following statements is false ? The existential model
(A) emphasizes the uniqueness of the individual
(B) emphasizes man's freedom for self-direction and self-fulfilment
(C) stresses on man's quest for values and meaning
(D) places considerably less faith in the inner experiencing of the individual and more faith in modern science for dealing with man's deepest problems.
- 1244.** The following is an existential theme
(A) freedom, choice and courage
(B) meaning, value and obligation
(C) the encounter with nothingness
(D) (A), (B) and (C) are all existential themes.
- 1245.** The humanistic model views man as having
(A) strong propensities for overcoming irrationality
(B) positive and rational propensities
(C) freedom for self-direction
(D) only (A) and (C) are correct
- 1246.** According to the humanistic model of man, man is
(A) self-aware (B) evaluative
(C) future-oriented
(D) capable of resisting environmental influences
- 1247.** The humanistic model assumes that
(A) man can to a large extent be understood in terms of external stimulus conditions alone
(B) internal psychological structures and processes also have a causal influence on a person's thinking, feeling and action
(C) both (A) and (B) are true
(D) both (A) and (B) are false
- 1248.** According to Rogers
(A) perceived threat to the self is followed by defence—including the narrowing and rigidification of perception and coping behaviour
(B) maladjustive behaviour results when the individual feels too threatened to acknowledge his own experiences and to be himself
(C) the individual's tendencies are towards growth and wholeness, and when his experiences and alternatives are clarified, he chooses the path of personal growth
(D) all the above statements are true according to Roger's theory of personality.

- 1249.** The Thematic Apperception Test (TAT) is an example of what type of psychological test ?
 (A) Intelligence (B) Achievement
 (C) Interest (D) Personality.
- 1250.** Development in the infant is characterised by milestones of appearance of selected behaviours. Which of the following is(are) milestone(s) ?
 (A) Social smiling (B) Stranger anxiety
 (C) Verbalizations (D) All the above.
- 1251.** Match the following
- | | |
|---------------------|---|
| (i) Eugene Bleuler | (1) Father of Psychiatry |
| (ii) James Braid | (2) Introduced the term schizophrenia |
| (iii) Sigmund Freud | (3) Coined the term hypnosis |
| (iv) Johann Weyer | (4) Elaborated the mechanism of defence |
| (v) Erik Erikson | (5) Emphasized cultural influences in human behaviour |
- Which of the following indicates the right combinations ?
 (A) (i)-(1); (ii)-(3); (iii)-(4); (iv)-(2); (v)-(5)
 (B) (i)-(3); (ii)-(2); (iii)-(4); (iv)-(1); (v)-(5)
 (C) (i)-(2); (ii)-(3); (iii)-(4); (iv)-(1); (v)-(5)
 (D) (i)-(4); (ii)-(2); (iii)-(3); (iv)-(1); (v)-(5).
- 1252.** Social smiling appears at about
 (A) two months (B) a week
 (C) six months (D) a year.
- 1253.** At the age of two, a child usually responds with
 (A) "No!" to any request
 (B) "Yes!" to any request
 (C) shows indecision when a request is made
 (D) shows vacillation when a request is made.
- 1254.** Gender identity is usually learned and set by the age of
 (A) 1 year (B) 3 years
 (C) 7 years (D) 10 years
- 1255.** Which of the following statements is false ? In case of maternal deprivation and if no mother substitute is provided, the infant
 (A) can enter an extremely withdrawn condition and subsequently die
 (B) can develop the ability to form close interpersonal relations in later life as a reaction formation to his childhood deprivation
 (C) can demonstrate retardation in cognitive, language and motor development
 (D) can become unresponsive and withdrawn from all social interactions.
- 1256.** Any persistent attitude is likely to serve the function(s) of
 (i) object appraisal
 (ii) social adjustment
 (iii) externalization
 Which of the following is true ?
 (A) Only (i) and (ii)
 (B) Only (ii) and (iii)
 (C) Only (i) and (iii)
 (D) (i), (ii) and (iii).
- 1257.** In general, the research on how terminating and unwanted pregnancy in the first trimester affects the psychological health to the mother can best be summarized as
 (A) It causes few short-term problems but frequently problems develop over the long-term
 (B) most women experience major problems in coping at the time, but nearly all work them out satisfactorily with time
 (C) abortion causes significant short-term and long-term psychological problems
 (D) abortion does not pose a psychological hazard for most women.
- 1258.** According to the psychoanalytic perspective, psychological disorders are produced by
 (A) brain injury
 (B) biochemical imbalances in the brain
 (C) the isolation produced by modern society

- (D) unresolved conflicts in childhood.
- 1259.** Hypochondriasis and conversion disorder both fall into the general category called
- personality disorder
 - dissociative disorders
 - anxiety disorders
 - somatoform disorders.
- 1260.** The milder but more chronic form of manic-depressive disorder is called
- borderline personality disorder
 - narcissistic personality disorder
 - cyclothymia
 - dysthymia.
- 1261.** The logic displayed by a preschool child e.g. the sun came up because it's time for me to get up, reflects
- egocentricity
 - narcissism
 - extension of self to outer phenomena
 - causal relationship.
- 1262.** As the child learns to speak, nouns, verbs and adjectives appear in the following sequence
- nouns—adjectives—verbs
 - adjectives—verbs—nouns
 - adjectives—nouns—verbs
 - nouns—verbs—adjectives
 - verbs—adjectives—nouns.
- 1263.** For persons with normal and higher intelligence
- there is positive and significant correlation between the rapidity with which children talk and intelligence
 - there is no correlation between the rapidity with which children talk and intelligence
 - there is high correlation between the rapidity with which children talk and intelligence
 - there is a low and negative correlation between the rapidity with which children talk and intelligence.
- 1264.** The stages in moral development were formulated by
- | | |
|--------------|-------------|
| (A) Erikson | (B) Piaget |
| (C) Kohlberg | (D) Terkel. |
- 1265.** Which of the following statements according to New Comb would be true ? In interpersonal relations
- there is a strong trend towards subjective balance from the beginning, with increasing trends towards objective balance developing over time
 - there is a strong trend towards objective balance in the beginning with increasing trends towards objects balance developing over time
 - the subjective balance and the objective balance has to be there right from the beginning
 - none of the above is true.
- 1266.** An attitude
- must have definite orientations in the world of object (or values)
 - must be an altogether automatic and routine type of conduct
 - is simply a social instinct
 - all the above are true.
- 1267.** Attitudes can be formed through
- integration differentiation
 - trauma
 - all the above
 - none of the above
- 1268.** Attitudes
- are mental depositions with a child is born with
 - can be formed through imitation of parents, teachers or peers
 - neither (A) nor (B) is true
 - both (A) and (B) are true.
- 1269.** Which of the following pertains to theories of attitudinal change ?
- Heider's theory of balance
 - Newcomb's theory
 - Festinger's cognitive dissonance theory
 - All of the above
- 1270.** In case a person is highly ego-involved with an issue
- his 'latitude of acceptance' will be narrower than the accompanying latitude of rejection
 - his 'latitude of rejection' will be narrower than the accompanying 'latitude of acceptance'

- (C) his 'latitude of acceptance' will overlap with his 'latitude of rejection'
- (D) his 'latitude of acceptance' can be either narrower or broader than the accompanying 'latitude of rejection'.
- 1271.** Maximal persuasive effects are to be expected when the position advocated in the communication falls near the boundary of the recipient's latitude of acceptance. This can be viewed in terms of
- (A) assimilation effect
- (B) contrast effect
- (C) marginality effect
- (D) uncertainty effect.
- 1272.** If *p* dissolved the experienced unity of *o* by introducing a cognitive differentiation that segregates the aspect of '*o*' as a liar from the rest of *o*
- (A) there would be 2 triads which are balanced
- (B) there would be 2 triads which are unbalanced
- (C) one of the 2 triads formed would still remain unbalanced
- (D) such cognitive differentiation is not possible.
- 1273.** 'Opinion change' depends on the nature of
- (A) the communicator
- (B) the communication
- (C) the audience
- (D) all the above.
- 1274.** The following is an example of an unorganized group (collectivity)
- (A) family
- (B) an audience
- (C) citizen of a nation
- (D) all the above.
- 1275.** Individuals attain group membership
- (A) by birth (B) by choice
- (C) by invitation
- (D) by mutual consent
- 1276.** Individuals can be most intimate in
- (A) a dyad (B) a triad
- (C) a subgroup (D) a supergroup.
- 1277.** Which of the following statements is false?
- (A) Triads always have the potential to move into the 'drama triangle' with the roles of the persecutor, victim and rescuer
- (B) The persons in the triangle of a triad assume different roles at different times
- (C) In a group, every individual's needs are completely met
- (D) A group of more than three persons tends to subdivide and form relations to at least one other person in the group which are afflictive (attraction) and differentiating (repulsion).
- 1278.** A group which assembles mainly to support a given identity is called a
- (A) sentient group
- (B) reference group
- (C) task-oriented group
- (D) membership group.
- 1279.** A neurosurgical team doing surgery is
- (A) a task group focused on the process of how things are occurring
- (B) a task group focused on content decision outcome issues
- (C) a sentient group focused on content decision outcome issues
- (D) a sentient group which is process oriented.
- 1280.** Not knowing that one is ill, represents
- (A) disorientation to person
- (B) disorientation to time
- (C) disorientation to situation
- (D) disorientation with respect to affect.
- 1281.** Which of the following is most actively involved in the phenomena of "attention"?
- (A) Hypothalamus
- (B) Cerebellum
- (C) Reticular Activating System (RAS)
- (D) Medulla Oblongata.
- 1282.** When the person assumes a totally "new life" without being aware of a different earlier life, the person is said to be in
- (A) dream state
- (B) fugue state
- (C) twilight state
- (D) somnambulism state.

- 1283.** "One's own telephone number" represents a
(A) conscious material
(B) unconscious material
(C) preconscious material
(D) subconscious material.
- 1284.** Which of the following statements is false? In case of a democratic leadership style, the group
(A) displays high morale
(B) performs consistently
(C) has sporadic high performance when the leader is present not when absent production
(D) shows greater participation in group activities.
- 1285.** Prejudices
(A) involve distortion of time small element of truth
(B) are directed towards a subgroup of people
(C) involves over-generalisation and oversimplification
(D) all the above.
- 1286.** Which of the following defence mechanisms is involved in the phenomena of scape goatting ?
(A) Interjection (B) Projection
(C) Intellectualization
(D) Suppression.
- 1287.** The groups with which an individual identifies, or with which he would like to be identified are called his
(A) reference groups
(B) subgroups
(C) in-groups
(D) membership groups.
- 1288.** According to the Diagnostic and Statistical Manual of Mental Disorder Edition II (DSM-II), an IQ of 65 classifies a person as
(A) borderline mental retardation
(B) mild-mental retardation
(C) moderate mental retardation
(D) severe mental retardation.
- 1289.** All of the following are examples of psychophysiology responses to anxiety except
(A) excessive perspiration
(B) tension headaches, constriction in chest, backache
(C) aphasia, apraxia
(D) transient systolic hypertension.
- 1290.** The concept of defence mechanisms was elaborated by
(A) Eugene Blitter
(B) Sigmund Freud
(C) Erik Erikson (D) James Braid.
- 1291.** Which of the following defence mechanisms would you naturally expect to occur as a psychological concomitant of physical illness, where the person is placed in a hospital ?
(A) Isolation (B) Regression
(C) Compensation (D) Interjection.
- 1292.** Which of the following defence mechanisms is always pathological ?
(A) Retinalisation (B) Isolation
(C) Denial (D) Conversion.
- 1293.** The following are the dimensions of orientation.
(A) time and place
(B) time and situation
(C) time, place and situation
(D) person, time, place and situation
- 1294.** Patient with anorexia nervosa
(A) more commonly are adolescent boys
(B) rarely die from this disorder
(C) frequently report feeling fat when objectively report feeling fat when objectively emaciated
(D) complain bitterly of nausea
- 1295.** The following are the types of anxiety
(A) Reality anxiety
(B) Neurotic anxiety
(C) Moral anxiety
(D) Existential anxiety
- 1296.** Neurotic anxiety arises when
(A) the individual's id impulses threaten to breakthrough his ego controls and result in behaviour that will lead to his punishment
(B) the individual does something or contemplates doing something or conflict with his super ego and arouses feelings of guilt

- (C) the individual is confronted by dangers
(D) all the above result in neurotic anxiety.
- 1297.** A 14-year-old male is brought into the psychiatrist's office by his mother. They have had an argument over the length of his hair. The body reacts to the psychiatrist in a hostile argumentative fashion even though the psychiatrist has reasonably long hair, a beard, and has not provoked the reaction from the young man. The phenomena which is occurring is
(A) counter transference
(B) transference
(C) acting-out
(D) regression.
- 1298.** In a crisis management of persons who abuse drugs, there are certain drugs of abuse from which persons often die they are abruptly withdrawn from the substance. Which of the following preparations constitutes a danger of death from abrupt withdrawal ?
(A) Stimulants (B) Psychedelics
(C) Sedative hypnotics
(D) Opiates.
- 1299.** At what age do adults begin to ask these questions What have I done with my life ? What have I accomplished ?
(A) Thirty (B) Forty
(C) Fifty-five (D) Sixty-five.
- 1300.** A psychiatric house officer is seeing a 30-year-old man in the emergency room. He has been hospitalised previously for a schizophrenic disorder, paranoid type. On this occasion he presents all of the following symptoms. Which is most suggestive of a delirium (acute brain syndrome) ?
(A) Visual hallucinations
(B) Speech bordering on incoherence
(C) Persecutory delusions
(D) Concrete thinking
- 1301.** All of the following substances produce a withdrawal syndrome except
(A) barbiturates (B) opioids
(C) cocaine (D) caffeine.
- 1302.** The intoxication most likely confused with an acute schizophrenic disorder is that produced by
(A) Barbiturates (B) Cocaine
(C) Amphetamines (D) alcohol
(E) Lysergic acid diethylamide (LSD)
- 1303.** The possibility of suicide must be considered in all of the following. In which is it least likely ?
(A) Dementia with a depression
(B) Alcoholism
(C) Delirium (acute brain syndrome)
(D) Schizophreniform disorder
(e) Bipolar disorder, mixed type.
- 1304.** Identify the most accurate statements regarding schizophrenic disorders.
(A) Process, schizophrenia has a better diagnosis than reactive schizophrenia
(B) hebephrenia is more common than catatonia
(C) schizophrenia affects about 2 per cent of the population according to studies in America.
(D) Schizophrenia with a chronic course shows deterioration towards dementia.
- 1305.** A genetic component probably plays a role in the genesis of all the following mental disorders. However, the evidence is least compelling in
(A) schizophrenia
(B) manic-depressive illness
(C) anti-social personality disorder
(D) schizoid personality disorder.
- 1306.** In which of the following are psychological factors probably least significant etiologically ?
(A) Dermatitis factitia
(B) Neurodermatitis
(C) Herpes (D) Acne.
- 1307.** All of the following statements about electroshock therapy are correct except
(A) the most commonly-accepted indication is major depression
(B) risk of mortality in patients with depression is considerably more than the risk of mortality in such patients who are untreated

- (C) the most common complication is an organic brain disorder
(D) the muscle relaxants have decreased the incidence of compression fractures of the spine.
- 1308.** Which of the following emphasized family and culture, and the broader challenges faced by individuals in each of the eight major stages of development ?
(A) Sigmund Freud
(B) Erik Erikson
(C) Albert Bandura
(D) Lawrence Kohlberg.
- 1309.** Heredity and environment interact significantly; inherited tendencies can be altered within
(A) Temperamental limits
(B) Hereditary possibilities
(C) Bidirectional influences
(D) The reaction range.
- 1310.** Which of the following appears to be a major difference between infants in low-quality day care and infants raised at home ?
(A) Day-care infants formed attachments to their teachers rather than their mothers
(B) Day-care infants formed stronger attachments
(C) Day-care infants formed attachment later in their development
(D) Day-care infants formed less secure attachment.
- 1311.** Some obese people and people who are starving behave in a somewhat similar manner. Which of the following is not one of their common characteristics ?
(A) They are highly sensitive to food cues
(B) They are easily frustrated and upset
(C) They have very high energy levels
(D) They have high levels of free fatty acids in their blood.
- 1312.** John refuses to jaywalk because he is afraid he will be stopped by a police officer and given a ticket. According to Kohlberg's theory of moral development, John is functioning at which of the following levels ?
(A) Conventional
(B) Law and order
(C) Pre-conventional
(D) Post-conventional.
- 1313.** In an emergency, which of the following would not typically occur ?
(A) Blood moves from the stomach to the skeletal muscles
(B) Sugar is released into the bloodstream
(C) The pupils of the eyes dilate
(D) Breathing becomes shallower and slower as the person concentrates on the emergency ?
- 1314.** The results of a study conducted on the emotionality of men with severed spinal cords indicated that
(A) emotions are unrelated to the somatic nervous system
(B) emotions are unrelated to the autonomic nervous system
(C) a fully functioning nervous system contributes to emotion
(D) spinal cord injuries intensify negative emotions and reduce positive emotions.
- 1315.** On Schacter's and Singer's experiment with the "vitamin" injections, which groups of subjects imitated the Cohort's behaviour ?
(A) Informed and misinformed
(B) Misinformed only
(C) Placebo and informed
(D) Ignorant and misinformed.
- 1316.** According to the text, which of the following beliefs about crisis reactions is probably true ?
(A) There are universal reaction to crisis
(B) People go through stages as they respond to crisis
(C) The response to crisis varies widely from person to person.
(D) Crises are ultimately resolved.
- 1317.** Bob believes different situations produce different physiological response patterns. These, in turn, around their corresponding emotions. Bob's theory most closely corresponds to
(A) cognitive arousal theory

- (B) cognitive appraisal theory
(C) Cannon-Bard theory
(D) James-Lange theory.
- 1318.** An incentive differs from a drive in that it involves
(A) an expectation
(B) humans, not animals
(C) animals, not humans
(D) biology but not learning.
- 1319.** Compare to that of animals, the human sexual response is more controlled by
(A) hormones
(B) preoptic region of the hypothalamus
(C) reticular formation
(D) cerebral cortex.
- 1320.** In playing a ringtoss game, people with a high need for achievement generally
(A) stand close to the target
(B) stand at an intermediate distance from the target
(C) stand far from the target
(D) refuse to play.
- 1321.** You are going to take a very important test and want to achieve the level of arousal that will maximize your performance. You conclude the level-off arousal you should strive for is
(A) zero
(B) a low level (to reduce tension)
(C) a moderate amount
(D) a high level.
- 1322.** Which of the following defence mechanism underlies all others ?
(A) Projection (B) Fixation
(C) Regression (D) Repression.
- 1323.** Most of the psychoanalytic theorists who broke away from Freud came to place more emphasis than Freud on the
(A) Ego (B) Id
(C) Unconscious (D) Superego.
- 1324.** Eysenck concluded that personality structure varies on three dimensions; neuroticism versus emotional stability, introversion versus extroversion, and psychoticism. Underlying these traits are differences in
(A) drive
(B) neurological activity levels
(C) early experiences in the environment
(D) schemas and problem-solving strategies
- 1325.** Jack finds any mention of male femininity offensive. His therapist suggests that, in actuality, this may be Jack's way of reducing anxiety from his true fear of being a homosexual. If so, Jack is using the defence mechanism of
(A) reaction formation
(B) displacement
(C) projection (D) regression.
- 1326.** Which of the following tests measures intelligence individually and reports a single IQ score ?
(A) Stanford-Binet
(B) Scholastic Aptitude Test
(C) Wechsler Intelligence Scale for children
(D) Thematic Apperception Test.
- 1327.** Janet is 10-year-old. Her IQ is 120. According to Binet's original IQ formula, what is Janet's mental age ?
(A) 6 (B) 8
(C) 10 (D) 12.
- 1328.** You are taking a test that asks you whether you like, dislike, or are indifferent to activities such as playing with children, working with your hands, and so on. You conclude you are taking the
(A) Strong-Campbell Interest Inventory
(B) Wechsler Adult Intelligence Scale
(C) Thematic Apperception Test
(D) System of Multicultural Pluralistic Assessment.
- 1329.** The gradual cutting back of rewards for appropriate behaviour is called
(A) self-monitoring
(B) self-reinforcement
(C) stimulus control
(D) shaping.
- 1330.** The SRSS assigns stress to events in one's life that involve
(A) only extreme changes
(B) negative changes
(C) both positive and negative changes
(D) both changes and unchanging conditions.

- 1331.** The text reports a study in which one group of nursing home residents were told that they were responsible for their arrangements and a second group were told that they would be taken care of. The results of this study showed that
- (A) The group who took care of themselves were happier and healthier
 - (B) The group who were taken care of were happier and healthier
 - (C) The first group who were happier but less healthy
 - (D) The second group were happier but less healthy.
- 1332.** Standing your ground and fighting for what you want is a strategy of coping called
- (A) playful problem-solving
 - (B) distancing
 - (C) positive reappraisal
 - (D) confrontation coping.
- 1333.** Sally's roommate is continually doing favour for people—to a point where she gets frustrated because she has no time of her own. According to Horney, she is moving
- (A) against people
 - (B) towards people
 - (C) away from people
 - (D) with people.
- 1334.** A person with an IQ score of 28 and whose functioning is at this same level would be considered
- (A) severely retarded
 - (B) mildly retarded
 - (C) profoundly retarded
 - (D) moderately retarded.
- 1335.** The major problem with Arthur Jensen's conclusion about the heritability of intelligence is
- (A) he failed to consider that black Americans have not been in this country as long as white Americans
 - (B) he over emphasised the crucial role of environment as a contributor to intelligence
 - (C) he suggested that the factors responsible for variation of intelligence within groups could also explain variation between groups
 - (D) he based his findings on old measures and tests of intelligence, which have very little validity in today's world.
- 1336.** Research indicates that the member of a schizophrenic's family who is most likely to be schizophrenic is
- (A) a parent
 - (B) a sibling
 - (C) an identical twin
 - (D) a fraternal twin.
- 1337.** The vocalisation of whatever thought comes to mind in whatever order, without self-censorship, logical structure, or interruption from the therapist is called
- (A) free association
 - (B) transactional analysis
 - (C) client-entered therapy
 - (D) psychoanalysis.
- 1338.** According to client centred therapy, the unreasonable expectations that are sometimes placed on people in order for them to receive affection are called
- (A) free associations
 - (B) transference
 - (C) unconditional positive regard
 - (D) conditional positive regard.
- 1339.** Paradoxical intervention involves
- (A) pairing an aversive stimulus with a maladaptive stimulus
 - (B) getting the client to imagine a positive self-image
 - (C) putting the client in a forced-positive interaction
 - (D) imagining the feared object.
- 1340.** Which of the following statements is/are correct ?
- (A) The right side of the brain is the creative side
 - (B) The right and left brains are specialized to handle different kinds of information
 - (C) Language tasks are always handled by the left brain
 - (D) All the above.
- 1341.** In reproduction of bodily cells (by process of mitosis), the number of chromosomes

- (A) remains the same
(B) is doubled
(C) is halved (D) is quadrupled.
- 1342.** The process in which the chromosome pairs separate and half of to 'daughter' cells is called
(A) cytokinesis (B) mitosis
(C) meiosis
(D) such a thing normally never happen.
- 1343.** The Y-chromosome carries
(A) a few dominant genes for sexual characteristics and few recessive genes
(B) only a few genes recessive for sexual characteristics
(C) genes for colour blindness
(D) a few genes dominant for sexual characteristics but no recessive genes.
- 1344.** Environmental influences
(A) can change the genetic make up
(B) cannot change the IQ level of an individual
(C) can change the utilization of the inherited potential
(D) all the above are true.
- 1345.** Genetic factors play an important role in determining
(A) intelligence
(B) emotional stability
(C) schizophrenia
(D) all the above.
- 1346.** The following characteristics is not carried by recessive genes
(A) Brown-eyes
(B) Baldness
(C) Susceptibility to poison ivy
(D) Haemophilia.
- 1347.** The following characteristics is not sex linked.
(A) Colour blindness
(B) Albinism
(C) Baldness (D) Haemophilia.
- 1348.** Down's Syndrome occurs as a result of
(A) a deficit of one chromosome on pair 23
(B) a deficit of one chromosome on pair 21
(C) an extra chromosome on pair 23
(D) an extra chromosome on pair 21.
- 1349.** The condition whereby a female has only one X-chromosome is known as
(A) Turner's Syndrome
(B) Klinefelter' Syndrome
(C) Down's Syndrome
(D) none of the above.
- 1350.** The methods-used for studying the heritability of traits in human being is by means of
(A) selective breeding
(B) fraternal and identical twin studies
(C) identical twin studies
(D) all the above.
- 1351.** Diabetes is caused by
(A) hereditary factors only
(B) environmental factors
(C) interaction between hereditary and environment
(D) the cause of diabetes is not certain.
- 1352.** A child if deprived of emotional attachments during the first three years of life
(A) develops strong affectional relationship in the later life
(B) can still develop normal social relationship in later life
(C) develops a secure personality in the later stages of life
(D) cannot develop normal affectional relationships in later life.
- 1353.** Riesen's experiment on early deprivation showed that dark reared monkeys required exposure in light before they could develop the proper response, but it also illustrated that they required much less time to learn than newborn monkeys once the goggles were removed. This experiment thus illustrated
(A) the role of experience
(B) the role of maturation
(C) the role of critical period in development
(D) the role of experience and maturation.
- 1354.** Maturation
(A) is an innately governed sequence of physical changes that does not

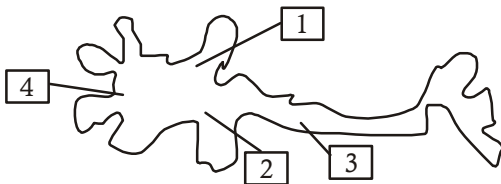
- depend on particular environmental events
- (B) is an innately governed sequence of physical changes that can be altered by environmental events
- (C) cannot be accelerated or impeded by the quality of environment
- (D) both (a) and (b) are true.
- 1355.** To have an accelerated development of the infant, it is best to
- (A) provide as much stimulation as possible
- (B) provide highly-enriched environments
- (C) provide stimulation appropriate to the level of maturation
- (D) all the above.
- 1356.** An individual with an XXY 23rd chromosome is
- (A) physically a female with marked masculine characteristics
- (B) physically a male with marked feminine characteristics
- (C) a homosexual
- (D) an enunch.
- 1357.** Which of the following statements is true regarding super males ?
- (A) Super males have an extra Y- chromosome (type XXY)
- (B) They are taller than average
- (C) They are usually homosexuals
- (D) They reach sexual maturity early and may have a higher than normal sex drive.
- 1358.** Traits like intelligence, height and emotionality that show continuous variations are determined by
- (A) many sets of genes
- (B) single genes
- (C) not determined by any gene but by environmental influences only
- (D) may be determined by a single gene or by many sets of genes.
- 1359.** The notation of a critical period in development indicates
- (A) the role of maturation
- (B) the role of environment
- (C) the role of hereditary
- (D) all the above.
- 1360.** Motor development is
- (A) a process, the sequence of which can be altered by environmental stimulation
- (B) largely influenced by the culture in which the child is reared
- (C) primarily a maturational process
- (D) all the above.
- 1361.** The conduction of a nervous impulse
- (A) does not depend on the presence or absence of the myelin sheath
- (B) is more rapid in non-myelinated fibers than in myelinated fibers
- (C) is the same in both myelinated and non-myelinated fibres
- (D) is more rapid in myelinated fibres than in non-myelinated fibres.
- 1362.** Nerve impulse in the human body travels at a speed
- (A) equal to the velocity of light (3×10^5 km per second)
- (B) 3 to 300 km per hour
- (C) 30 to 30,000 km per hour
- (D) none of the above.
- 1363.** A resting neuron is in a state of
- (A) zero potential (B) action potential
- (C) resting potential
- (D) infinite potential.
- 1364.** Stimulating early environment
- (A) helps in later intellectual development
- (B) helps in development of social relationships
- (C) provides the background necessary to cope with the environment at a later age
- (D) all the above.
- 1365.** Jean Piaget's studies have focused on
- (A) the cognitive or intellectual development of children
- (B) the development of speech in children
- (C) the development of social relationships
- (D) all the above.
- 1366.** According to Piaget, a child largely learns conservation concepts during the following stage of intellectual development.
- (A) Sensorimotor stage

- (B) Concrete operational stage
(C) Preoperational stage
(D) Formal operational stage.
- 1367.** The following conservation concept is most difficult to conceive. The conservation of
(A) mass (B) volume
(C) number (D) weight.
- 1368.** A child can think in abstract terms and reason by hypothesis in
(A) formal operational stage
(B) concrete operational stage
(C) pre-operational stage
(D) both concrete and formal operational stages.
- 1369.** Attachment of an infant to its mother is because it sees her largely as a source of
(A) food (B) security
(C) affection
(D) food and affection.
- 1370.** On reunion, babies who show anxious attachments to their mothers
(A) immediately seek close contact and continue to show a need for closeness
(B) show ambivalent behaviour
(C) show avoidant behaviour
(D) may show avoidant or ambivalent.
- 1371.** The nervous impulse is transmitted across the synapse by
(A) receptor activity
(B) perceptual processes
(C) afferent activity
(D) neural conduction.
- 1372.** Anxious attachment develops as a result of
(A) insensitive mothering
(B) repeated separations
(C) both (A) and (B) are correct
(D) no separations.
- 1373.** During transmission along an axon, the nerve impulse
(A) continuously diminishes in size
(B) travels with the speed of light
(C) continuously increases in size
(D) does not diminish in size.
- 1374.** The transmission of an impulse in a neuron depends on
(A) the all-or-none principle
(B) graded potential
(C) both (A) and (B) are correct
(D) neither (A) nor (B) are correct.
- 1375.** The decision of the neuron to fire or not depends on potentials in the dendrites or cell bodies that are
(A) all-or-none potentials
(B) can be either all-or-none potentials
(C) graded potentials
(D) none of the above is true.
- 1376.** The information concerning stimulus strength is conveyed by
(A) the size of the action potential
(B) the frequency with which an individual neuron fires
(C) the number of neurons that are excited
(D) only (B) and (C) are correct.
- 1377.** The nerves that carry sensory information from the periphery of the body and the internal organs to the spinal cord and brain are called
(A) afferent nerves
(B) peripheral nerves
(C) efferent nerves
(D) connector nerves.
- 1378.** The central nervous system consists of
(A) the brain
(B) the brain and the spinal cord
(C) the brain, the spinal cord and the autonomic system
(D) the brain and the somatic system.
- 1379.** The following statements is true about the autonomic system
(A) it helps in voluntary movements
(B) it plays an important role in carrying messages regarding pain, pressure and temperature variations
(C) it plays an important role in emotions
(D) all the above.
- 1380.** Recent studies have shown that as the animal becomes more proficient in learning to perform a new task
(A) synapses change in both number and size

- (B) synapses change in number only
(C) synapses change in size only
(D) synapses change neither in size nor in number.
- 1381.** The likelihood that a given neuron will fire an all-or-none action potential at any given moment is determined by
(A) the excitatory synapses only
(B) the constant interplay of excitatory and inhibitory synapses
(C) the inhibitory synapses
(D) none of the above.
- 1382.** Some of the mood-altering drugs (_____?_____?_____? tranquillizers and LSD) probably create effect by
(A) changing the activity of the synapses
(B) blocking the release of some synaptic transmitters and thus allowing fewer messages to get through
(C) accumulating in certain brain cells, where they may act like certain neural transmitters and over stimulate the cells
(D) either of the above alternatives are true.
- 1383.** A three-neuron reflex are
(A) consists of the afferent neuron, the connector neuron and the efferent neuron
(B) consists of the afferent neuron, the dorsal root ganglion and the afferent neuron
(C) is found even in the simplest reflex
(D) both (B) and (C) are correct.
- 1384.** Messages regarding pain, pressure and temperature variations are carried by
(A) automatic system
(B) somatic system
(C) spinal cord
(D) central nervous system.
- 1385.** In the case of a paraplegic accident victim when the spinal cord is severed from the stem
(A) knee jerk response may still function
(B) erection of the penis may still function
(C) both knee jerk response and the erection of the penis are possible
(D) neither knee jerk response nor the erection of the penis are possible.
- 1386.** The knee jerk
(A) is a reflex response
(B) it can occur solely in the spinal cord without any assistance from the brain
(C) it is normally modulated by the messages from the higher nervous system in addition to its basic mechanism built into the spinal cord
(D) all the above statements are correct.
- 1387.** The endocrine activity is regulated by
(A) cerebellum (B) hypothalamus
(C) thalamus (D) brain stem.
- 1388.** Stimulation of the Reticular Activating System by a current with a more rapidly changing wave form
(A) produces pain
(B) always induces sleeps
(C) produces pleasure
(D) always awakens the sleeping animal.
- 1389.** If you see a person who is unable to carry out an intended sequence of actions, and a small distraction makes him forget what they have set to do, he is most likely to have
(A) a lesion in the limbic system
(B) a lesion in the Reticular Activating System
(C) A lesion in the hypothalamus
(D) none of the above.
- 1390.** The animals higher up in the phylogenetic scale had a cerebral cortex which is
(A) more smooth
(B) more wrinkled and convoluted
(C) confined to a small region of the brain
(D) lighter and thinner.
- 1391.** The development of the cerebral cortex depends on
(A) the organism's genetic make up only
(B) the organism's early experience only
(C) the organism's place on the phylogenetic scale
(D) all the above.
- 1392.** In case of man, the amount of cortex corresponding to
(A) hands and fingers is much larger than that devoted to the feet and toes

- (B) feet and toes is much larger than that devoted to the hand's finger
- (C) feet and toes and to hands and fingers is approximately the same amount
- (D) the tongue is the maximum.
- 1393.** The temporal lobe of the brain contains
- (A) the auditory area
- (B) the auditory area and the visual association area
- (C) the visual area
- (D) the striate area.
- 1394.** Which of the following statements is false ?
- (A) speech in most cases is controlled by the left hemisphere
- (B) Broca's area is involved in the production of speech sounds
- (C) Broca's area is concerned with understanding speech
- (D) In some left-handed people, speech centres may be located in the right hemisphere or equally divided between both the right and the left hemisphere.
- 1395.** The layer of the brain which came latest in the evolutionary development is
- (A) the central core
- (B) the limbic system
- (C) the cerebrum
- (D) all of the layers developed at the same stage of evolutionary history.
- 1396.** The following play an important role in emotions
- (A) cerebrum (B) hypothalamus
- (C) limbic system
- (D) limbic system and hypothalamus.
- 1397.** Routine actions like walking, dancing, and producing speech sounds
- (A) are programmed into the cerebellum and occur automatically without having to think
- (B) occur at the level of the spinal cord only
- (C) require thinking and the involvement of cerebral cortex
- (D) require the involvement of the hypothalamus and the limbic system.
- 1398.** When a continuous electric current of a certain constant voltage is sent through electrodes implanted in the reticular formation of cat or dog
- (A) the animal goes into coma
- (B) the animal if asleep is awakened
- (C) sleep is produced
- (D) the animal experiences pleasure.
- 1399.** The following is false about the left and the right hemispheres of the brain.
- (A) The left and the right hemispheres are symmetrical in their functions
- (B) The left hemisphere controls written and spoken language and mathematical calculations
- (C) The right hemisphere is involved in spatial construction and pattern sense
- (D) The left and the right hemispheres are asymmetrical in their functions.
- 1400.** In monkeys, lesions in the frontal lobes
- (A) produce deficits in the ability to recognize and discriminate different forms
- (B) destroy the ability to solve a delayed response problem
- (C) produce deficit in production of speech
- (D) produce deficit in understanding of speech.
- 1401.** Severe epileptic seizures have been successfully controlled by
- (A) cutting the corpus callosum
- (B) by removing the brain tissue in the area where epileptic disturbance is focused
- (C) both (A) and (B) are true
- (D) by removing the Broca's area.
- 1402.** The revelations about human brain are largely through studies on
- (A) patients (army veterans) with brain injury
- (B) patients who have had surgical ablations to remove tumours or control epilepsy
- (C) electrical or chemical stimulation
- (D) all the above.

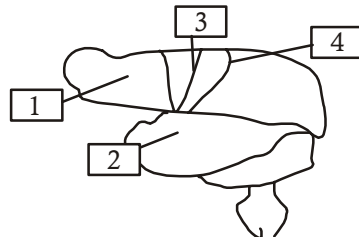
- 1403.** The hypothalamus
 (A) controls both the anterior and the posterior though in different way
 (B) controls the anterior pituitary
 (C) controls the posterior pituitary
 (D) does not control the pituitary gland.
- 1404.** Hair and fat distribution in human body is controlled mainly by the secretions from the
 (A) adrenal glands (B) gonads
 (C) pituitary (D) thyroid.
- 1405.** Steroids
 (A) promote the release of sugar stored in the liver so that the body has energy for quick action
 (B) are sex hormones and determine the sexual behaviour
 (C) are released as a result of stimulation of adrenal cortex when the later is acted on by a hormone secreted by the pituitary which in turn was stimulated by noradrenalin
 (D) both (A) and (C) are correct.
- 1406.** Specialized cells for receiving various types of stimulation are
 (A) neurons (B) receptors
 (C) effectors (D) glial cells.
- 1407.** In the diagram of a neuron below, which of the numbered locations corresponds to the axon ?



- (A) 1 (B) 2
 (C) 3 (D) 4.
- 1408.** What kind of cells carry information from the brain to the muscles and glands ?
 (A) Sensory neurons
 (B) Glial cells
 (C) Motor neurons
 (D) Inter-neurons.
- 1409.** The major function of the glial cells is to
 (A) manufacture RNA
 (B) stimulate hormones
 (C) speed neural conduction
 (D) nourish and support neurons.
- 1410.** The visual area is present in
 (A) the occipital lobe
 (B) the cerebral hemisphere
 (C) the striate area
 (D) all the above are true.
- 1411.** The following is false of the split brain subjects.
 (A) Then corpus callosum is cut
 (B) They have difficulty in performing day-to-day functions
 (C) They ordinarily function without any difficulty in day-to-day life situations
 (D) The subjects is aware only of what is going on in his left hemisphere because it is that hemisphere alone that can communicate through language.
- 1412.** The left hemisphere is
 (A) concerned to a large extent with spatial and pattern sense
 (B) the minor hemisphere
 (C) the major hemisphere
 (D) centre of musical ability.
- 1413.** Lesions in the visual association area cause
 (A) deficits in the ability to recognize and discriminate different forms
 (B) loss of visual acuity
 (C) delayed-response deficit
 (D) only (A) and (B) are true.
- 1414.** Dwarfism occurs as a result of
 (A) under secretion of an anterior pituitary hormone
 (B) over secretion of an anterior pituitary hormone
 (C) environmental influence
 (D) inadequate protein intake especially during early childhood.
- 1415.** The pituitary gland is the master endocrine gland because
 (A) it produces the largest number of different hormones
 (B) it controls the secretion of several other endocrine glands
 (C) both (a) and (b) are correct

- (D) it controls the timing and amount of body growth.
- 1416.** When the interior of a neuron is electrically positive in relation to the exterior, the neuron is in a state known as
(A) polarization
(B) resting potential
(C) action potential
(D) graded potential.
- 1417.** Which of the following is most clearly responsible for speeding the conduction of a neural impulse ?
(A) Myelin (B) Glial cells
(C) Inter-neurons (D) Hormones.
- 1418.** Which of the following would usually be controlled primarily by the right cerebral hemispheres ?
(A) Language
(B) The right half of the body
(C) Analytical thinking
(D) Spatial relationships.
- 1419.** You are walking home at night from the library. Someone jumps out of the bushes in front of you and acts as though he is going to hurt you. Your body mobilizes for action and in so doing calls into play the
(A) sympathetic division
(B) para-sympathetic division
(C) hypothalamus
(D) somatic system.
- 1420.** Children sometimes play a game where they load little beans into a carefully balanced cup until finally one of the beans tips the cup over and all the beans spill out. This example is most similar in concept to which the following.
(A) Graded potentials
(B) Action potentials
(C) All-or-none response
(D) Nodes of Ranvier.
- 1421.** A tumour growing in John's head has stopped his growth at a height of only four feet nine inches. This tumour is probably located near his
(A) thyroid
(B) limbic system
(C) pituitary
(D) reticular formation.
- 1422.** Mary is having problems with the regulation of her metabolism. If the doctor suspects an endocrine disorder, he or she will probably first examine Mary's
(A) adrenal glands (B) ovaries
(C) thyroid gland (D) pons.
- 1423.** Which part of the human brain would be most similar to the brain of a snake?
(A) The limbic system
(B) The central core
(C) The forebrain
(D) The cerebral hemisphere.
- 1424.** The practice of studying people or animals who have suffered from brain damage is called the technique of
(A) lesioning
(B) clinical observation
(C) electrical stimulation
(D) chemical stimulation.
- 1425.** A major function of the hypothalamus is to
(A) control attention
(B) coordinate body movements
(C) maintain a steady internal environment for the body
(D) transfer information from one part to the other.
- 1426.** Neural conduction can be speeded up if the neural impulse can "jump" from one gap on the axon to the next. The gaps are called
(A) myelin sheaths
(B) graded potentials
(C) all-or-none responses
(D) nodes of Ranvier.
- 1427.** A synaptic cleft is a
(A) gap between two neurons
(B) kind of neurotransmitter
(C) neural insulator
(D) change in electrical potential.
- 1428.** Whether or not neural conduction from one neuron to the next occurs depends primarily on
(A) refraction
(B) resting potential
(C) neurotransmitters
(D) stimulation.

- 1429.** Poisons which act by occupying the neuromuscular receptors work by mimicking the effects of
(A) neuropeptides
(B) acetylcholine
(C) endogenous opioids
(D) Parkinson's disease.
- 1430.** The messages conducted in the nervous system are _____; in the endocrine system they are _____.
(A) electrochemical; chemical
(B) chemical; electrochemical
(C) physical; chemical
(D) electrical; physical.
- 1431.** The "master gland" which manufactures hormones which affect most of the other glands and which regulates growth is the
(A) adrenal cortex
(B) adrenal medulla
(C) pituitary gland
(D) thyroid gland.
- 1432.** The gland that controls such diverse activities as the level of sugar in the blood and the secretion of the sex hormone androgen is called
(A) thyroid (B) gonads
(C) adrenal cortex
(D) adrenal medulla.
- 1433.** The pleasure center discovered by Olds and Milner is located in the
(A) spetal area
(B) hippocampus
(C) right cerebral hemisphere
(D) amygdala.
- 1434.** The somatosensory cortex is located in the
(A) limbic system
(B) parietal lobe
(C) right cerebral hemisphere
(D) occipital cortex.
- 1435.** You volunteer for an experiment in which electrodes are attached to your scalp and you are asked to think about certain memories. The researcher is probably using the technique of
(A) EEG recording (B) PET scan
(C) CAT scan (D) MRI.
- 1436.** Blaine has head injury that is causing him difficulty in regulating breathing and blood flow. His face is contorting, he is salivating heavily, and he is making chewing movements. You conclude the accident caused injury to his
(A) amygdala
(B) cerebellum
(C) medulla
(D) hypothalamus.
- 1437.** A boy is brought into a hospital suffering from head injuries. The most noticeable symptom is his inability to sleep regularly for eight hours. You suspect damage to the
(A) cerebellum
(B) reticular formation
(C) thalamus
(D) hypothalamus.
- 1438.** Suppose you lesion a portion of a male rat's brain. The rat now exhibits inappropriate mating behaviours, has difficulty in feeding, and engages in inappropriate fight and flight responses. You conclude you have damaged the rat's
(A) right cerebral hemisphere
(B) limbic system
(C) hypothalamus (D) pons.
- 1439.** A man's brain is damaged in such a way that he has no control over his body movements. The damage is probably located in the area identified by number ____ in the diagram below.



- (A) 1 (B) 2
(C) 3 (D) 4.

- 1440.** Cutting through a person's corpus callosum causes the person to
(A) die
(B) be unable to speak or understand language

- (C) have difficulty coordinating certain kinds of activities
(D) become blind.
- 1441.** Persons having difficulty with language and speech following an accident that resulted in injury to the brain are most likely to have sustained damage in the
(A) right cerebral hemisphere
(B) left cerebral hemisphere
(C) right cerebral hemisphere if they are a male and left cerebral hemisphere if they are a female
(D) I have no idea what you are talking about.
- 1442.** In the Featurd Study linking brain structure to schizophrenia it was found that when compared to the non-szhizophrenic twin, the szhizophrenic twin was more likely to have
(A) A different genotype
(B) Enlarged ventricles in the brain
(C) A less placid autonomic nervous system
(D) A more stressful environment.
- 1443.** In carrying out the “fight or flight” response, the role of supervisor is assigned to the
(A) adrenal gland
(B) pituitary gland
(C) hypothalamus
(D) parasympathetic nervous system.
- 1444.** A person’s current weight and height could be said to exemplify his or her
(A) genotype
(B) phenotype
(C) both the above
(D) None of the above.
- 1445.** Which of the following kinds of studies can truly demonstrate that specific traits are indeed inherited ?
(A) Gamily studies (B) Twin studies
(C) Adoption studies
(D) None of the above.
- 1446.** Neurons that are specialized to communicate directly with the muscles of the body are called
(A) sensory neurons
(B) motor neurons
(C) actuator neurons
(D) muscle neurons.
- 1447.** Which part of the neuron has the responsibility for receiving information from other neurons ?
(A) The cell body (B) The soma
(C) The axon (D) The dendrites.
- 1448.** The myelin sheath serves to
(A) permit faster transmission of the neural impulse
(B) keep neural impulses on the right track
(C) both the above
(D) none of the above.
- 1449.** Suppose I believe that any particular part of the brain has a specific function but is interrelated with the other brain areas. This belief is essentially that expressed by
(A) Olds and Milner in their work with the pleasure centre
(B) Lashley’s position that the entire cortex is involved in activity
(C) the principle of localization and integration
(D) the principle of laterlization.
- 1450.** A women declares that she wants to wear a flowery dress, but she reaches for a solid blue dress. Her behaviour would be similar to that of someone who has
(A) Parkinson’s disease
(B) Alsheimer’s disease
(C) a severed corpus callosum
(D) damaged wernicke’s area.
- 1451.** The woman who says she wants to wear one dress but reaches for another would best be treated with
(A) L-DOPA
(B) hormone therapy
(C) brain tissue implant
(D) relearning.
- 1452.** Most neurons are involved in transmitting information
(A) from one neuron to another
(B) from the outside world to the brain
(C) from the brain to the muscles
(D) none of the above.
- 1453.** Current evidence indicates that schizophrnia results from

- (A) genetic factors
(B) environmental factors
(C) multiple causes that involve both genetic and environmental factors completely unknown factors
(D) completely unknown factors.
- 1454.** Psychology as a science can be said to be
(A) empirical
(B) rational
(C) analytic
(D) both (B) and (C).
- 1455.** The change in the polarity of a neuron that results from the inflow of positively charged ions and the outflow of negatively charged ions is called the
(A) presynaptic potential
(B) postsynaptic potential
(C) synaptic potential
(D) none of the above.
- 1456.** The task of passing a message from one neuron to another is actually carried out by
(A) the myelin sheath
(B) the glia cells
(C) the action potential
(D) neuro-transmitters.
- 1457.** Which of the following neurotransmitters can only have inhibitory effect at receptor sites ?
(A) GABA (B) Dopamine
(C) Limbic system (D) Forebrain.
- 1458.** Which of the following techniques is often used by neurosurgeons to map the brain when performing brain surgery ?
(A) EEG recordings
(B) ESB
(C) Lesioning (D) All the above.
- 1459.** The seat of emotion is to be found in the
(A) reticular formation
(B) hind brain
(C) limbic system
(D) forebrain.
- 1460.** Therapist suggested to Sunil that while trying to give up the habit of washing hands unnecessarily, he should wear a rubber band around his wrist and snap it everytime he feels like washing his hands. What therapy he is using ?
(1) Systematic desensitization
(2) Aversion therapy
(3) Flooding
(4) Modelling
- 1461.** We sometimes attempt to solve problems by applying the techniques that worked in similar situations in the past. What type of method are we using ?
(A) Trial and error (B) Analogy
(C) Algorithms (D) Heuristics
- 1462.** The fact that a nearer object may obscure a more distant object from the view is due to :
(A) Figure and ground
(B) Proximity
(C) Interposition
(D) Closure
- 1463.** Match List-I with List-II and indicate your answer with the help of codes given below :
- | List - I
(Phenomenon) | List - II
(Psychologist) |
|----------------------------------|-------------------------------------|
| p. Learned helplessness | 1. Bandura |
| q. Cognitive map | 2. Seligman |
| r. Observational learning | 3. Wolpe |
| s. Systematic desensitization | 4. Tolman |
- Codes :**
- | | p | q | r | s |
|-----|----------|----------|----------|----------|
| (A) | 2 | 4 | 1 | 3 |
| (B) | 3 | 4 | 2 | 1 |
| (C) | 1 | 3 | 4 | 2 |
| (D) | 4 | 2 | 3 | 1 |
- 1464.** In a 3×4 contingency table, using chi-square test, the degrees of freedom associated with the chisquare will be :
(A) 12 (B) 9
(C) 7 (D) 6
- 1465.** Match List-I with List-II and indicate your answer using the codes given below :
- | List - I
(Function) | List - II
(Area of the Brain) |
|--------------------------------|--|
| p. Visuo-spatial function | 1. Hypothalamus |

- q. Social norms 2. Cerebellum
 r. Releasing factors 3. Orbito-frontal cortex
 s. Balance 4. Parietal cortex

Codes :

	p	q	r	s
(A)	2	1	4	3
(B)	1	2	3	4
(C)	4	3	2	1
(D)	4	3	1	2

- 1466.** Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using the codes below :

Assertion (A) : In Edwards Personal Preference Schedule (EPPS), forced choice format is used.

Reason (R) : Forced choice format reduces acquiescence response set.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

- 1467.** Why do individuals with high need for achievement quit early when faced by very difficult task ?

- (A) Get tired easily
 (B) Have poor perseverance
 (C) Have high anxiety
 (D) Have high need for efficiency

- 1468. Assertion (A) :** According to R. B. Cattell, source traits are important in the understanding of personality.

Reason (R) : Oblique rotation is used for uncovering source traits.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true

- 1469.** Which of the following represents the Tetrad equation as given by Spearman ?

(A) $r_{12} \times r_{34} = r_{14} \times r_{23}$

(B) $r_{13} \times r_{24} = r_{14} \times r_{23}$

(C) $r_{14} \times r_{23} = r_{13} \times r_{24}$

(D) All of these

- 1470.** What among the following define a problem space ?

1. An initial state
 2. A goal state
 3. A set of operations

(A) 1, 3 (B) 1, 2

(C) 2, 3 (D) 1, 2, 3

- 1471.** A generalized picture or model of an object or event, learned in the past and stored in memory, is :

- (A) perceptual constancy
 (B) memory trace
 (C) prototype
 (D) cognitive map

- 1472. Assertion (A) :** We are able to recognize the similarity and differences of two objects.

Reason (R) : Dissociable subsystems operate simultaneously in each cerebral hemisphere.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 (C) (A) is true but (R) is false
 (D) Both (A) and (R) are false

- 1473.** What is the correct sequence of events given below ?

1. Releasing of neurotransmitters
 2. Post synaptic potential
 3. Binding on receptor sites
 4. Synthesis of neurotransmitters

Codes :

(A) 1 2 3 4 (B) 4 3 2 1

(C) 4 1 3 2 (D) 2 4 1 3

- 1474. Assertion (A) :** The amplitude of the action potential in a neuron does not go beyond + 40 mV.

Reason (R) : Neurons function on “all or none” basis.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

1475. A man reflects his anger from his boss to his child, is an example of :

(A) Projection

(B) Reaction Formation

(C) Displacement

(D) Regression

1476. Speed and power tests can be distinguished on the basis of :

1. Time limit

2. Difficulty level

3. Verbal/Non-verbal content

Codes :

(A) 1 and 2

(B) 1 and 3

(C) 2 and 3

(D) 1, 2 and 3

1477. Natural concepts are based on :

(A) prototypes

(B) a set of rules/properties

(C) schemas

(D) propositions

1478. Match List-I with List-II and indicate your answer using the codes given below :

List - I

(Presentation of reinforcement)

List - II

(Type of reinforcement)

p. An unpleasant stimulus is presented to decrease behaviour

1. Positive reinforcement

q. An unpleasant stimulus is removed to increase behaviour

2. Negative reinforcement

r. A pleasant stimulus is presented to increase behaviour

3. Positive punishment

s. A pleasant stimulus is removed to decrease behaviour

4. Negative punishment

Codes :

p q r s

(A) 1 2 3 4

(B) 2 3 4 1

(C) 4 1 3 2

(D) 3 2 1 4

1479. Short term memory is capable of holding approximately how many items of unrelated information ?

(A) one

(B) two to four

(C) five to nine

(D) ten to twelve

1480. Learning of fears is best explained by :

(A) Operant conditioning

(B) Classical conditioning

(C) Observational learning

(D) Latent learning

1481. Choose the correct sequence of emergence of the following theories of emotion :

1. James – Lange

2. Cannon – Bard

3. Schachter and Singer

4. Darwin

(A) 1 3 2 4

(B) 4 3 2 1

(C) 4 1 2 3

(D) 1 2 3 4

1482. What is true of drives ?

1. Internal states

2. Aroused by tension

3. Underlying need

4. Guided by external stimuli

(A) 1, 2 and 4

(B) 1, 2 and 3

(C) 2, 3 and 4

(D) 1, 2, 3 and 4

1483. Intellectual functioning is mainly in the domain of which lobe of the Cortex ?

(A) Occipital lobe

(B) Frontal lobe

(C) Parietal lobe

(D) Temporal lobe

1484. A nerve cell that responds to a vertical line but not to horizontal line is an example of

(A) linear perspective system

(B) feature detector

(C) pattern analyser

(D) selection and organization process

1485. The visual signals from each retina reach striate cortex via which of the following?

- (A) Lateral geniculate nucleus
- (B) Medial geniculate nucleus
- (C) Infero-temporal cortex
- (D) Posterior parietal cortex

1486. Observational learning involves :

1. Attention and retention
2. Attention and Reinforcement
3. Production processes and motivation
4. Reinforcement and motivation

- (A) 1 and 3 only (B) 2 and 4 only
(C) 3 and 4 only (D) 1 and 2 only

1487. Which of the following factors reduce the accuracy of eyewitness testimony ?

1. Suggestibility
2. Source monitoring errors
3. Interference
4. Decay of memory traces
5. Illusion of outgroup homogeneity

Codes :

- (A) 1, 2 and 3 (B) 1, 3 and 4
(C) 1, 2 and 5 (D) 2, 3 and 5

1488. Microsleep occurs mainly due to one of the following :

- (A) Sleep disorder
- (B) Drugged state
- (C) Sleep deprivation
- (D) Dreaming state

1489. Self-Efficacy is

- (A) A person's mental model of his or her ability and attitudes.
- (B) A generalised evaluative attitude toward the self that influences both moods and behaviour
- (C) The process by which a person comes to know or perceive the personal attributes
- (D) A set of beliefs that one can perform adequately in a particular situation.

1490. Which one of the following is a product in Guilford's model ?

- (A) Evaluation
- (B) Cognition
- (C) Transformation
- (D) Symbols

Instructions for Questions 1491 to 1494 :

Read the following information and answer the four questions that follow :

A researcher wanted to test the hypothesis that a given drug would adversely affect the rate of learning. A group of 40 subjects was employed in the research. Each subject was required to learn a task upto two errorless trials. Then each subject was given a specified drug dosage (1mg/body weight in kilogram) and thirty minutes after, the subject was required to learn another equated learning task upto two errorless trials. The number of trials required to learn each task followed a normal distribution. The mean number of trials required to learn the tasks were 24 and 20 only for normal and 'drugged conditions', respectively. The two conditions had comparable standard deviations. The mean difference was tested by a suitable statistical test and the difference was found to be significant at .01 level.

1491. What can you conclude about the researcher's hypothesis ?

- (A) The hypothesis has been verified
- (B) The hypothesis has been rejected
- (C) The hypothesis has been partially verified
- (D) The data are inadequate to evaluate researcher's hypothesis.

1492. What statistical test appears to be suitable for evaluating mean differences ?

- (A) Independent samples t test
- (B) One-way ANOVA for independent groups
- (C) Paired samples t-test
- (D) Chi-square test

1493. The dependent variable in this experiment is :

- (A) Drug dosage
- (B) Trials required to learn the task
- (C) Body weight
- (D) Time interval between the drug administration and learning the second task

1494. The above information suggests that :

1. Parametric model should be followed in data analysis
2. Non-parametric model should be followed in data analysis.
3. The scores under two conditions appear to have homogeneity of variance

4. The scores under two conditions appear to be heterogeneous for variance.

Codes :

- (A) 1 and 3 only (B) 1 and 4 only
(C) 2 and 3 only (D) 2 and 4 only

- 1495.** Railroad tracks that seem to draw closer together as they recede into the distance are an example of

- (A) linear perspective
(B) relative size
(C) binocular vision
(D) feature detection

- 1496.** Memory for particular events is called :

- (A) semantic (B) sensory
(C) episodic (D) procedural

- 1497.** A test of Abstract Thinking had the norms in terms of normalized T scores (Mean = 50, SD = 10). If the subject received a T score of 60, what would be his percentile rank ?

- (A) 16 (B) 60
(C) 66 (D) 84

- 1498.** Match List-I with List-II and indicate your answer using the codes given below :

List - I
(Childhood experience)

List - II
(Personality characteristic)

- | | |
|---|---------------------|
| p. Little gratification in oral stage | 1. Overly dependent |
| q. Excessive gratification in oral stage | 2. Impulsive |
| r. Overly harsh toilet training in Anal stage | 3. Hostile |
| s. Relieved toilet training in Anal stage | 4. Compulsive |

Codes :

- | | p | q | r | s |
|-----|----------|----------|----------|----------|
| (A) | 2 | 4 | 3 | 1 |
| (B) | 1 | 3 | 4 | 2 |
| (C) | 1 | 4 | 2 | 3 |
| (D) | 2 | 3 | 1 | 4 |

- 1499.** Multiple Abstract Variance Analysis is used to study the effect of

- (A) one independent variable on more than one dependent variables
(B) more than one independent variables on one dependent variable
(C) more than one independent variables on more than one dependent variables
(D) None of these

- 1500.** Match List-I with List-II and indicate your answer using the codes given below :

List - I
(Brief description)

List - II
(Type of reinforcement)

- | | |
|--|-------------------|
| p. reinforcement occurs after a set time period | 1. fixed ratio |
| q. reinforcement occurs after a set number of responses interval | 2. variable |
| r. reinforcement occurs after a varying time period | 3. fixed interval |
| s. reinforcement occurs after a varying number of responses | 4. variable ratio |

Codes :

- | | p | q | r | s |
|-----|----------|----------|----------|----------|
| (A) | 3 | 1 | 2 | 4 |
| (B) | 1 | 4 | 3 | 2 |
| (C) | 4 | 3 | 1 | 2 |
| (D) | 2 | 4 | 3 | 1 |

- 1501.** Which of the following effectors is mainly responsible for instrumental (motor) response ?

- (A) Smooth muscles
(B) Cardiac muscles
(C) Skeletal muscle
(D) Adrenal gland

- 1502.** A process following learning during which a memory becomes more firmly established is called :

- (A) over-learning
(B) consolidation

- (C) transfer period
(D) constructive processing

1503. Assertion (A) : A large body of research indicates that violence and aggression is learned through observation.

Reason (R) : Learning through observation takes place due to exposure to similar situation and experienced reinforcement.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) Both (A) and (R) are false.

1504. Some steps in test construction are listed below :

1. Internal consistency assessment
2. Item analysis
3. Test-retest reliability
4. Item writing

Usually, the above steps are carried out in the following sequence.

- (A) 4, 2, 1, 3 (B) 4, 1, 3, 2
(C) 2, 4, 1, 3 (D) 4, 3, 2, 1

1505. What is the correct order of the following conflicts occurring in an individual's life ?

1. Autonomy V/s self doubt
2. Identity V/s Role confusion
3. Competence V/s Inferiority
4. Intimacy V/s Isolation

Codes :

- (A) 1 2 3 4 (B) 2 1 4 3
(C) 1 3 2 4 (D) 3 2 4 1

1506. Which of the following attitude scaling method/s has/have been proposed by L.L. Thurstone ?

1. Paired comparison
2. Summated ratings
3. Scalogram
4. Equal appearing intervals

Codes :

- (A) 2 only (B) 4 only
(C) 1 and 4 only (D) 2 and 3 only

1507. Match List-I with List-II and indicate your answer using the given codes :

**List - I
(Validity type)**

- p. Predictive validity
q. Concurrent validity
r. Construct validity
s. Content validity

**List - II
(Brief description)**

1. Covering all the important areas in a given course of study
2. Yielding theoretically consistent results
3. High relationship between the new test and the established test of same trait
4. Forecasting some significant behaviour on the basis of test scores

Codes :

- | | p | q | r | s |
|-----|---|---|---|---|
| (A) | 3 | 4 | 2 | 1 |
| (B) | 4 | 3 | 1 | 2 |
| (C) | 2 | 1 | 4 | 3 |
| (D) | 4 | 3 | 2 | 1 |

1508. During emotions skin conductance changes occur due to the activity of some of the following :

1. Sweat glands
2. Adrenal glands
3. Parasympathetic nervous system

Codes :

- (A) 1 and 2 (B) 1 and 3
(C) 2 and 3 (D) 1, 2 and 3

1509. What among the following is true for mental set ?

1. It can enhance the quality and speed of perceiving and problem solving under some conditions.
2. It can enhance the quality and speed of perceiving and problem solving in new situations.
3. It may inhibit or distort the quality of mental activities in new situations.
4. It may inhibit or distort the quality of mental activities under same conditions.

Codes :

- (A) 1 and 2 only (B) 2 and 3 only
(C) 1 and 3 only (D) 3 and 4 only

1510. Match the following :

List – I

List – II

- | | |
|--------------------|---|
| 1. Thinking | p. Mental activity through which we transform available information in order to reach conclusions |
| 2. Reasoning | q. An activity that involves the manipulation of mental representation of various features of the external world. |
| 3. Decision making | r. Processing information in various ways in order to move toward desired goals. |
| 4. Problem solving | s. The process of choosing between two or more alternatives on the basis of information about them. |

Codes :

- | | p | q | r | s |
|-----|----------|----------|----------|----------|
| (A) | 4 | 3 | 2 | 1 |
| (B) | 2 | 1 | 4 | 3 |
| (C) | 3 | 4 | 2 | 1 |
| (D) | 2 | 3 | 1 | 4 |

1511. In an experiment the stimuli were presented to the subjects in certain order and after the presentation of stimuli their order was disturbed. In the test phase the subjects were needed to set them in the presented order. Which of the following methods was used in this experiment ?

- (A) Recognition
(B) Relearning

- (C) Reconstruction
(D) Recall

1512. Given below are two statements, one labelled as Assertion (A) and the other labelled as Reason (R). Indicate your answer using the codes given below :

Assertion (A) : If your teacher embarrasses you for asking a question in class, it is less likely that you would ask questions in class in future.

Reason (R) : Negative reinforcement reduces the occurrence of a behaviour.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

1513. What is the correct sequence of sympathetic and parasympathetic division given below?

- (A) Thoracic – Sacral – Cranial – Cervical – Lumbar
(B) Cranial – Lumbar – Thoracic – Cervical – Sacral
(C) Sacral – Cervical – Thoracic – Cranial – Lumbar
(D) Sacral – Cranial – Cervical – Thoracic – Lumbar

1514. The human ear can generally hear sounds ranging from

- (A) 20 – 1000 Hz
(B) 10 – 2000 Hz
(C) 20 – 2000 Hz
(D) 1000 – 5000 Hz

1515. Match List-I with List-II and indicate your answer using codes given below :

List – I

(Personality Theories)

List – II

(Focus)

- | | |
|------------------|-------------------------|
| p. Behaviourists | 1. Self-concepts |
| q. Psychodynamic | 2. Habits |
| r. Biological | 3. Childhood fixations |
| s. Humanistic | 4. Inheritance of genes |

Codes :

	p	q	r	s
(A)	1	3	4	2
(B)	2	4	3	1
(C)	2	3	4	1
(D)	1	4	2	3

- 1516.** The chronological age that most typically corresponds to a given level of performance is called
 (A) Intelligence Quotient
 (B) Maturation
 (C) Mental age
 (D) None of the above
- 1517.** The role of unconscious in creative thinking is known as the process of
 (A) Verification (B) Illumination
 (C) Inoculation (D) Incubation
- 1518.** Which factor is not much related to happiness ?
 (A) High self-esteem
 (B) Educational level
 (C) Close friendship or satisfactory marriage
 (D) Meaningful religious faith
- 1519.** Which is the correct order of 'Products' given by Guilford ?
 (A) Units, Relations, Classes, Systems, Implications, Transformations
 (B) Units, Classes, Systems, Relations, Implications, Transformations
 (C) Units, Classes, Relations, Systems, Transformations, Implications
 (D) Units, Classes, Systems, Relations, Transformations, Implications
- 1520.** Which is not the obstacle to problem solving ?
 (A) Confirmation bias
 (B) Fixation
 (C) Mental Set
 (D) Heuristics
- 1521.** Given below are two statements, one labelled as Assertion (A), and the other labelled as Reason (R). Indicate your answer using the codes given below :

Assertion (A) : Algorithm is a logical rule that guarantees solving a particular problem.

Reason (R) : One can find the solution even to complex problems by following step by step procedure.

Codes :

- (A) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (B) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (C) (A) is true, but (R) is false.
 (D) (A) is false, but (R) is true.

- 1522.** Which of the following statements concerning reinforcement is correct ?
 (A) Learning is most rapid with partial reinforcement, but continuous reinforcement produces the greatest resistance to extinction.
 (B) Learning is most rapid with continuous reinforcement but partial reinforcement produces the greatest resistance to extinction.
 (C) Learning is the fastest and resistance to extinction is the greatest after continuous reinforcement.
 (D) Learning is the fastest and resistance to extinction is the greatest following partial reinforcement.
- 1523.** The sentence "Blue Jeans wear false smiles" has correct ____ but incorrect _____.
 (A) Morphemes; phonemes
 (B) Phonemes; morphemes
 (C) Semantics; syntax
 (D) Syntax; semantics

- 1524.** Match the following lists according to the types and characteristics of thinking :

List - I**(Types of Thinking)**

- p. Autistic thinking
 q. Realistic thinking
 r. Convergent thinking
 s. Creative thinking

List - II**(Characteristics)**

1. Reasoning
 2. Drawing facts
 3. Unusual uses
 4. Fantasy

Codes :

	p	q	r	s
(A)	1	3	4	2
(B)	3	1	2	4
(C)	2	1	3	4
(D)	4	1	2	3

- 1525.** The main function of autoclitic behaviour is to
1. Qualify responses
 2. Express relations
 3. Providing a grammatical framework for verbal behaviour

Codes :

- (A) 1 only (B) 2 and 3 only
(C) 1 and 3 only (D) 1, 2 and 3

- 1526.** The phi-phenomenon, stroboscopic motion, induced motion and autokinetic motion are all

- (A) Pictorial cues
(B) Apparent motion
(C) Double images
(D) Non-verbal cues

- 1527.** Which one of the following is the most important feature of the defense mechanism of Rationalization ?

- (A) Going back to an earlier stage of development.
(B) Justifying one's actions.
(C) Magically atoning for certain acts that give rise to guilt.
(D) Attributing one's emotions to other persons.

- 1528.** "People control basic anxiety by moving toward, away from, and against others." Which group of personality psychologists theorized this statement ?

- (A) Psychoanalytic
(B) Neo-Freudian
(C) Social learning
(D) Cognitive

- 1529.** A psychotic person would probably score the highest on which MMPI-2 scale ?

- (A) Depression (B) Hysteria
(C) Schizophrenia
(D) Mania

- 1530.** Match List – I with List – II and indicate your answer by using the codes given below the lists :

**List – I
(Part of eye)**

- p. Pupil
q. Iris

**List – II
(Function)**

1. Focus images on the retina.
2. Begin the processing of visual information.

r. Lens

s. Retina

3. Allows light to enter.

4. Controls the size of the pupil.

Codes :

	p	q	r	s
(A)	1	2	3	4
(B)	3	4	1	2
(C)	1	3	4	2
(D)	4	2	1	3

- 1531.** Learning by imitating others' behaviour is called _____ learning. The researcher best known for studying this type of learning is

- (A) Observational : Bandura
(B) Secondary : Pavlov
(C) Observational : Watson
(D) Secondary : Skinner

- 1532.** As we enter a movie theatre from bright light the visual sensitivity increases and within 5-10 minutes we are able to see under low levels of illumination. This is due to

- (A) Visual acuity
(B) Dark adaptation
(C) Saturation
(D) Transduction

- 1533.** Which of the following internal conditions determine perceptual process ?

1. Intelligence 2. Emotion
3. Motivation 4. Information
(A) 1 and 2 only (B) 2 and 3 only
(C) 2 and 4 only (D) 1 and 4 only

- 1534.** Wechsler Adult Intelligence Scale has eleven subtests out of which

- (A) five are verbal and six are performance.
(B) six are verbal and five are performance.
(C) seven are verbal and four are performance.
(D) four are verbal and seven are performance.

- 1535.** What is the correct sequence of memory processes ? Indicate your answer using the codes given below :

1. Encoding 2. Storage
3. Attention 4. Retrieval

Codes :

- (A) 3, 1, 2, 4 (B) 2, 3, 1, 4
 (C) 1, 3, 2, 4 (D) 3, 2, 1, 4

- 1536.** Match the monocular cues with their explanations and select the correct answer using the code given below :

List – I**(Cues)**

p. Relative size

q. Linear

perspective

r. Motion Parallax

s. Texture gradient

List – II**(Explanation)**

1. Straight lines seem to join together as they become more distant.

2. An object changes position on the retina as the head moves.

3. If two objects are of same size, the one producing the smaller retinal image is farther away.

4. The texture of a surface appears smoother as distance increases.

Codes :

- | | p | q | r | s |
|-----|----------|----------|----------|----------|
| (A) | 3 | 1 | 2 | 4 |
| (B) | 1 | 3 | 4 | 2 |
| (C) | 4 | 2 | 3 | 1 |
| (D) | 2 | 4 | 1 | 3 |

- 1537.** Read the following two statements, Assertion (A) and Reason (R) and indicate your answer using the codes given below :

Assertion (A) : Biderman's 'recognition by components' theory states that objects can be thought of as being composed of basic building blocks called Geons.

Reason (R) : The relationship between features and geons does not exist. Geons are not composed of features such as edges, corners etc.

Codes :

- (A) (A) and (R) are false.

(B) (A) is false, but (R) is true.

(C) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(D) (A) is true, but (R) is false.

- 1538.** Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using the codes given below :

Assertion (A) : People just coldly store and retrieve bits of data.

Reason (R) : Psychologists recognize that people reconstruct their own version of the past.

Codes :

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(B) Both (A) and (R) are correct, but (R) is not the correct explanation of (A).

(C) (A) is false, but (R) is true.

(D) (A) is true, but (R) is false.

- 1539.** Ventromedial nucleus, a central area on the underside of the hypothalamus, lead to

(A) Excessive eating

(B) Stop-eating

(C) Stop sex

(D) Excessive obesity

- 1540.** Read each of the following two statements : Assertion (A) and Reason (R) and indicate your answer using the codes below :

Assertion (A) : According to interference theory of forgetting, forgetting is caused due to the intervening task which occurs between original task and its recall.

Reason (R) : The intervening task weaken the memory trace.

Codes :

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

- 1541.** Match List – I with List – II and indicate your answer using the codes given below :

**List – I
(Emotion)**

- p. Fear
q. Anger
r. Depression

**List – II
(Nerves Activity)**

1. Parasympathetic arousal
2. Sympathetic arousal
3. Sympathetic and parasympathetic arousal
4. Psychosomatic

Codes :

	p	q	r
(A)	2	3	1
(B)	1	2	3
(C)	2	1	3
(D)	3	2	1

- 1542.** Read the following two statements, Assertion (A) and Reason (R) and indicate your answer using the codes given below :

Assertion (A) : The high achievement motivation displayed by children has emotional roots.

Reason (R) : Highly motivated children often have parents who encourage their independence from an early age and praise and reward them for their successes.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true and (R) is not the correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

- 1543.** A person, otherwise limited in mental ability, has amazing specific skills. This is because of

- (A) Academic under-achievement
(B) Practical Intelligence
(C) Academic over-achievement
(D) Savant Syndrome

- 1544.** A man borrows money and forgets to pay it back because paying back is painful. This is an example of

- (A) Repression
(B) Reaction Formation
(C) Regressia
(D) Rationalization

- 1545.** Whenever daughter performs well in her studies she receives a chocolate from parents but her brother is given a chocolate only when he gets the highest marks in the class. This is a paradigm of

- (A) Positive and negative reinforcement.
(B) Primary and secondary reinforcement.
(C) Continuous and secondary reinforcement.
(D) Continuous and partial reinforcement.

- 1546.** The location of the optic nerve exit point from the retina of the eye is called as

- (A) Bipolar cell (B) Fovea
(C) Blind spot (D) Cone

- 1547.** For an intelligence test, while computing item-remainder correlations for item analysis, we compute

- (A) Phi-coefficient
(B) Spearman rho
(C) Tetrachoric correlation
(D) Point-biserial correlation

- 1548.** Which of the following personality psychologist does not fall in the group of humanistic approaches of personality ?

- (A) Carl Rogers
(B) Rollo May
(C) Martin Saligman
(D) Abraham Maslow

- 1549.** A subject has a T score (Mean = 50; SD = 10) of 40 on an abstract reasoning test. The corresponding percentile rank would be

- (A) 16 (B) 34
(C) 40 (D) 84

- 1550.** Which of the following tests are considered to be culture-fair tests of intelligence ?

1. Cattell's Culture-Fair Test of Intelligence
2. Raven's Progressive Matrices
3. Wechsler's Intelligence Scale for Children
4. Goodenough-Harris Test

Codes :

- (A) 1 only
(B) 1 and 2 only

- (C) 1, 2 and 3 only
(D) 1, 2 and 4 only
- 1551.** The information in terms of sound is maintained in
(A) Semantic code
(B) Acoustic code
(C) Iconic code
(D) Information code
- 1552.** Which of the following personality tests are considered to be the projective in nature ?
1. Thematic Apperception Test
2. Kent-Rosenoff Word Association Test
3. NEO Five-Factor Inventory
4. Rotter's Incomplete Sentence Blank
(A) 1 only
(B) 1 and 2 only
(C) 1, 2 and 4 only
(D) 2, 3 and 4 only
- 1553.** Read each of the following two statements Assertion (A) and Reason (R) and indicate your answer using the codes below :
- Assertion (A) :** Thalamus and Hypothalamus play an important role in emotions.
Reason (R) : Thalamus and Hypothalamus anatomically lie close each other.
Codes :
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(C) (A) is true, (R) is false.
(D) (A) is false, (R) is true.
- 1554.** Taylor conceptualized anxiety as a drive and hypothesized that subjects scoring higher on her anxiety scale would be conditioned faster than the low-scoring subjects. The experimental studies confirmed this hypothesis for eyelid conditioning. The results provide evidence for the
(A) predictive validity of the anxiety scale.
(B) construct validity of the anxiety scale.

(C) concurrent validity of the anxiety scale.
(D) content validity of the anxiety scale.

- 1555.** Given below are two statements, one labelled as Assertion (A) and the other labelled as Reason (R). Indicate your answer using the codes given below :

Assertion (A) : According to Flynn, performance on IQ tests has substantially increased around the world at all age levels.

Reason (R) : Performance on IQ tests is because of the interaction between genetics and environment.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

Question Nos. 1556 to 1560 :

Read the following paragraph and answer the five questions which follow :

The concept of variable is of central importance in research. In psychological research, variables can be classified in several ways. Edwards has classified variables in three types – stimulus variables, behavioural variables and organismic variables. From another perspective, variables have been classified into two types – the continuous variables and discrete variables. The discrete variables are further subclassified as real discrete variables and artificially discrete variables.

The variables in research can be defined in various ways, the two common types of definitions being – constitutive definitions and operational definitions. The operationalization of variables is of crucial importance in carrying out actual research. The tentative statements of expected relationship between two or more variables are called hypotheses. The research hypotheses may be directional or non-directional. Needless to say that the hypotheses constitute to be the powerful tools in psychological research.

1556. Which of the following variable/s can be labelled as organismic variable/s ?

1. Task complexity
2. Intelligence
3. Errors made while learning task
4. Heart rate

Codes :

- (A) 4 only (B) 2 and 4 only
(C) 1, 2 and 4 only
(D) 2, 3 and 4 only

1557. Which one of the following is an example of an artificially discrete variable?

- (A) Number of children in a family
(B) Nationality
(C) Sex
(D) Examination results in terms of pass and fail

1558. Which of the following is/are operational definitions of intelligence ?

1. Intelligence ratings by an observing psychologist.
2. Intelligence as an ability to think critically, solve problems and adapt to the novel situations.
3. Intelligence as a score on Raven's Progressive matrices.

Codes :

- (A) 2 only (B) 3 only
(C) 1 and 3 only (D) 1, 2, 3

1559. 'Males are more extraverted than females' is an example of

- (A) Null hypothesis
(B) Non-directional hypothesis
(C) Directional hypothesis
(D) Unverifiable hypothesis

1560. Which of the following represent/s stylistically correct version/s of a hypothesis?

1. To study the relationship between intelligence and academic achievement.
2. Is there a relationship between intelligence and academic achievement?
3. Intelligence and academic achievement are positively correlated.
4. Intelligence and academic achievement are unrelated.

Codes :

- (A) 1 only (B) 3 only
(C) 1 and 2 only (D) 3 and 4 only

1561. Substantially higher performance scores than verbal scores on the subtests of Wechsler Adult Intelligence Scale (WAIS) indicates

- (A) Gender differences
(B) Learning difficulties
(C) Genetic influences
(D) Cultural biases

1562. Ruchi remembers that when she was eight year old she was whimsical. This memory of Ruchi is called

- (A) Episodic memory
(B) Semantic memory
(C) Sensory memory
(D) Amnesic memory

1563. Which is the most basic and common obstacle to problem solving ?

- (A) Confirmation bias
(B) Fixation
(C) Functional fixedness
(D) Mental set

1564. The child who says "Milk gone" is engaging in _____. This type of utterance demonstrates that children are actively experimenting with rules of _____.

- (A) Babbling; syntax
(B) Telegraphic speech; syntax
(C) Babbling ; semantics
(D) Telegraphic speech; semantics

1565. Problem solving comprises of four stages. Choose the correct sequence of stages

- (A) Incubation, preparation, verification, illumination
(B) Preparation, incubation, illumination, verification
(C) Incubation, preparation, illumination, verification
(D) Preparation, illumination, incubation, verification

1566. Match the following lists :

List - I
(Author)

List - II
(Focus/Emphasis)

- | | |
|--------------------------|------------------------|
| a. Kagan & Haveman, 1979 | 1. Mental manipulation |
|--------------------------|------------------------|

- b. Silverman, 1978 2. Solution of problem
 c. Whittaker, 1970 3. Mediating process
 d. Humphrey, 1963 4. Goal directed process

Codes :

	a	b	c	d		a	b	c	d
(A)	1	2	4	3	(B)	1	2	3	4
(C)	1	3	2	4	(D)	1	4	3	2

- 1567.** Given below are two statements, one labelled as Assertion (A) and the other labelled as Reason (R).

Assertion (A) : Thinking about objects brings change in attitudes.

Reason (R) : Attitudes toward complex issues and objects are typically a mixture of positive and negative feelings.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (C) (A) is true, but (R) is false.
 (D) (A) is false, but (R) is true.

- 1568.** Match List – I with List – II and indicate your answer using the codes given below :

List – I (Memory Phenomena)	List – II (Brief Description)
a. Semantic memory	1. Memory for events that occur in a particular time, place or context.
b. Priming	2. Memory for skills and habits.
c. Episodic memory	3. Memory for general knowledge and facts about the world.
d. Procedural memory	4. Information that people already have in storage is activated to help them remember new information better and faster.

Codes :

	a	b	c	d		a	b	c	d
(A)	3	2	4	1	(B)	4	3	2	1
(C)	3	4	1	2	(D)	2	4	3	1

- 1569.** A subject is presented four non-sense syllabus one by one each for two seconds. After presentation of the four items, the subject is asked to count backward aloud by threes from some number for fifteen seconds and after that he/she is asked to recall the non-sense syllabus. What memory storage is being measured in this experiment ?

- (A) Sensory memory storage
 (B) Working memory storage
 (C) Short term memory storage
 (D) Long term memory storage

- 1570.** When the previously learned task affects the retention of task being currently acquired, the phenomenon is referred to as :

- (A) Retroactive interference
 (B) Proactive interference
 (C) Retroactive effect
 (D) Proactive effect

- 1571. Assertion (A) :** Positive psychology draws its strength from humanistic psychology.

Reason (R) : Both humanistic and positive psychology believe in positive human qualities.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true, but (R) is not the explanation of (A).
 (C) (A) is true, but (R) is false.
 (D) (A) is false, but (R) is true.

- 1572.** The basic difference between classical conditioning and instrumental learning is of

- (A) Presentation of stimulus
 (B) Emitting of response
 (C) Temporal contiguity
 (D) Mechanism

- 1573.** Major criticisms of Thurston's theory of intelligence are based on

- I. Use of subjective measures
 II. Restricted heterogeneity in sample

III. Method of factor analysis

- (A) I and II are correct.
- (B) I and III are correct.
- (C) II and III are correct.
- (D) I, II and III are correct.

- 1574.** Given below are two statements, one labelled as Assertion (A) and the other labelled as Reason (R). Indicate your answer using the codes given below :

Assertion (A) : Heritability of intelligence explains variations due to genetics for individuals within a given population.

Reason (R) : Earlier the children from deprived families were adopted, the higher their intelligence score will be.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (C) (A) is true, but (R) is false.
- (D) (A) is false, but (R) is true.

- 1575.** Read each of the following two statements Assertion (A) and Reason (R) and indicate your answer using the codes given below :

Assertion (A) : Psychoactive drugs affects the nervous systems to cause change in perception or mood.

Reason (R) : Reticular activating system is related to perception and mood.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (C) (A) is true, but (R) is false.
- (D) (A) is false, but (R) is true.

- 1576.** Match List - I with List - II and indicate your answer using the codes given below :

List - I
(Glands)

- a. Pituitary
- b. Adrenal Cortex
- c. Pineal
- d. Adrenal Medulla

List - II
(Hormones)

- 1. Melatonin
- 2. Adrenaline
- 3. Prolactin
- 4. Corticosteroids

Codes :

- | a | b | c | d | a | b | c | d |
|-------|---|---|---|-------|---|---|---|
| (A) 2 | 1 | 3 | 4 | (B) 1 | 3 | 2 | 4 |
| (C) 4 | 3 | 1 | 2 | (D) 3 | 4 | 1 | 2 |

- 1577.** What is the correct sequence of ear parts given below ?

- (A) Eardrum - Hammer - Anvil - Stirrup - Oval window
- (B) Oval window - Hammer - Eardrum - Anvil - Stirrup
- (C) Eardrum - Anvil - Stirrup - Hammer - Oval window
- (D) Oval window - Anvil - Hammer - Stirrup - Eardrum

- 1578.** We can cope with stress by becoming aware of our irrational, upsetting thoughts and replacing them with rational, calming thoughts. This view represents.

- (A) Cognitive behavioural approach
- (B) Psychoanalytical approach
- (C) Socio-cultural approach
- (D) Bio-social approach

- 1579.** An instrumental response is conditioned only when organism interprets the reinforcement as being controlled by its response. This view is supported by

- (A) Tolman
- (B) Skinner
- (C) Seligman
- (D) Beck

- 1580.** When a child gets two chocolates once every week for its performance in the class it is an instance of

- (A) Fixed ratio, variable interval
- (B) Fixed ratio, fixed interval
- (C) Variable ratio, fixed interval
- (D) Variable ratio, variable interval

- 1581.** Match List - I with List - II and indicate your answer using the codes given below :

List - I

(Learning term) (Brief explanation)

- a. Positive reinforcement

- 1. Removal of a pleasant or appetitive stimulus after a behavioural response.

- b. Classical

- 2. Delivery of a pleasant conditioning

- or appetitive stimulus that follows a behavioural response.
- c. Punishment 3. Presenting a neutral stimulus after a stimulus that usually elicits a specific response.
- d. Omission training 4. Presentation of an unpleasant or aversive stimulus after a behavioural response.

Codes :

a	b	c	d	a	b	c	d
(A) 2	3	4	1	(B) 4	1	3	2
(C) 1	4	2	3	(D) 3	4	1	2

- 1582.** Match List – I (Psychologist) with List – II (Test) :

List – I (Name of List – II

Psychologist) (Name of Test)

- | | |
|------------------------|--|
| a. J.B. Rotter | 1. NEO-Personality inventory |
| b. C. Robert Cloninger | 2. Hardy Personality |
| c. Suzanne Kobasa | 3. Locus of control |
| d. P.T. Costa | 4. Temperament and character inventory |

Codes :

a	b	c	d	a	b	c	d
(A) 1	2	3	4	(B) 3	4	2	1
(C) 3	4	1	2	(D) 3	2	4	1

- 1583.** In a study on 'Career Aspirations of Students', the respondents were required to state their career aspirations in about sixty words. This would be an example of
- (A) Fixed-alternative item
(B) Scale item
(C) Open-end item
(D) Checklist item

- 1584.** Read each of the following two statements-Assertion (A) and Reason (R) and indicate your answer using the codes given below :

Assertion (A) : In Ponzo illusion and Muller-Lyer illusion, linear perspective is misapplied to the display.

Reason (R) : According to Gregory perception is only a function of stimulus characteristics.

Codes :

(A) (A) is true and (R) is false.

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is false and (R) is true.

(D) Both (A) and (R) are true and (R) is the correct explanation of (A).

- 1585.** Match each of the following organizational laws with its meaning. Select the correct answer using the code given below :

List – I

(Law)

a. Closure

b. Proximity

c. Similarity

d. Simplicity

List – II

(Meaning)

1. Elements close together are grouped together.

2. Patterns are perceived in the most basic, direct manner possible.

3. Groupings are made in terms of complete figures.

4. Elements similar in appearance are grouped together.

Codes :

a	b	c	d	a	b	c	d
(A) 1	4	2	3	(B) 2	3	1	4
(C) 4	2	3	1	(D) 3	1	4	2

- 1586.** Which of the following is not one of the Big Five personality factors ?

(A) Submissiveness

(B) Agreeableness

(C) Extroversion

(D) Openness to experience

- 1587.** Read the following two statements Assertion (A) and Reason (R) and indicate your answer using the codes given below :

Assertion (A) : The level of arousal for optimal performance varies for different tasks.

Reason (R) : Too little arousal can be non-motivating, too high arousal can be disruptive.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true and (R) is not the correct explanation of (A).
 (C) (A) is true, but (R) is false.
 (D) (A) is false, but (R) is true.

1588. Match List – I with List – II and indicate your answer using the codes given below :

List – I		List – II	
(Behaviour)		(Expression mode)	
a. Tone		1. Facial	
b. Crying		2. Glanduar	
c. Laugh		3. Varbal	
d. Eyebrow		4. Muscular	

Codes :

a	b	c	d	a	b	c	d
(A) 3	2	4	1	(B) 1	4	3	2
(C) 2	4	3	1	(D) 2	1	3	4

1589. A slow graded electrical potential produced by a receptor cell in response to a physical stimulus is

- (A) Receptor potential
 (B) Generator potential
 (C) Transduction (D) Arousal

1590. This area contains ‘Somoto Sensory Cortex’

- (A) Parietal lobe (B) Frontal lobe
 (C) Temporal lobe (D) Occipital lobe

1591. Give the correct sequence of the location of following glands in human body from top to bottom

- (A) Thyroid – Adrenal – Pancreas – Pituitary
 (B) Thyroid – Pancreas – Pituitary – Adrenal
 (C) Pituitary – Adrenal – Thyroid – Pancreas
 (D) Pituitary – Thyroid – Adrenal – Pancreas

1592. A complex behaviour that is rigidly patterned throughout a species and is unlearned is called

- (A) Imprinting (B) Instinct
 (C) Emotion (D) Feeling

1593. Our tendency to perceive objects as unchanging despite changes in sensory input is an illustration of

- (A) Figure-Ground relationship
 (B) Perceptual constancy
 (C) Binocular cues
 (D) Linear perspective

1594. Given below are two statements, one labelled as Assertion (A) and the other labelled as Reason (R) :

Assertion (A) : As compared to other methods of psychological research, experimental method is usually considered to be best suited for studying cause and effect relationship.

Reason (R) : Secondary variance is more efficiently controlled in experimental method.

In the context of the above two statements, which one of the following conclusion is correct ?

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true, but (R) is not the explanation of (A).
 (C) (A) is true, but (R) is false.
 (D) (A) is false, but (R) is true.

1595. Consider the following three types of psychological researches :

1. Field experiment
2. Laboratory experiment
3. Ex post facto study

Arrange these typical investigations in descending order in terms of the researcher’s ability to control secondary variance.

- (A) 3, 1, 2 (B) 2, 1, 3
 (C) 3, 2, 1 (D) 1, 2, 3

1596. What is the correct sequence of development of the following personality tests:

1. Maudsley Personality Inventory
2. Minnesota Multiphasic Personality Inventory-2

3. Millon Clinical Multiaxial Inventory
4. NEO PI-3

Codes :

- (A) 1, 2, 3, 4 (B) 4, 3, 2, 1
(C) 2, 4, 1, 3 (D) 1, 3, 2, 4

- 1597.** Perceiving one thing in relation to another when both are presented simultaneously is called

- (A) Simultaneous perception
(B) Figure-Ground perception
(C) Consecutive perception
(D) Successive perception

- 1598.** Verbal behaviour that is reinforced when someone else's verbal response is repeated verbatim is called

- (A) Autoclitic behaviour
(B) Echoic behaviour
(C) Reinforced behaviour
(D) Operant behaviour

- 1599.** Which of the following are essential to Spearman's methodology of theory development ?

- I. Tetral equation
II. Tetral difference
III. Specific factor
IV. Neurological basis of intelligence

Codes :

- (A) I, II and III (B) I, III and IV
(C) II, III and IV (D) I, II and IV

- 1600.** Dissociable sub-systems operate simultaneously in each hemisphere and help us recognize similarity at one level and differences at another level in the perceived pattern. This system consists of

1. Top-down processing system
2. Abstract category sub-system
3. Bottom-up processing system
4. Specific-exemplar sub-system

Codes :

- (A) 1 and 2 only (B) 3 and 4 only
(C) 2 and 4 only (D) 1 and 4 only

- 1601.** Based on Eysenckian theory of personality, what is the correct sequence levels of personality from bottom to top ?

1. Habit level 2. Trait level
3. Type level 4. S.R. level

Codes :

- (A) 1, 2, 3, 4 (B) 4, 3, 2, 1
(C) 4, 1, 2, 3 (D) 2, 4, 1, 3

- 1602.** Speed and power tests cannot be differentiated on the basis of

- I. Time limit
II. Verbal content
III. Non-verbal content
IV. Difficulty level

Codes :

- (A) I and IV (B) II and IV
(C) II and III (D) I, III and IV

- 1603.** According to two factors of Retroactive Interference, RI is caused by two factors : One is unlearning of the first task during the time one is engaged in the second task. What is the second factor?

- (A) Unlearning of the responses of second task.
(B) Competition of the first task responses with the second task responses that inhibits the recall responses of the first task at the time of recall.
(C) Inability to recall responses from first task.
(D) Decaying of memory-traces of responses of first task.

- 1604.** One of the difficulty with the survey method is

- (A) inability to examine changes over time.
(B) that subjects may give dishonest or inaccurate responses.
(C) the relatively small number of subjects used.
(D) the impossibility of getting a representative sample.

- 1605.** A cell in a contingency table had an obtained frequency of 16 and an expected frequency of 25. What would be the contribution of this cell to the total chi-square value?

- (A) 3.24 (B) 5.06
(C) 9.00 (D) 81.00

Read the following paragraph and answer the **five** questions which follow :

The notion of true score is at the heart of classical reliability theory. This model proposes that the obtained score on a variable is a sum of true score and the random error of measurement, the

two components being independent. Using this model, various ways of computing reliability have been developed. One of the way of examining the reliability of a test is to test the homogeneity of test items or the internal consistency. The most widely used method of estimating the reliability is splithalf technique, also called as odd-even reliability. In this context, Spearman Brown correction is important. Kuder-Richardson reliability and Chronbach Alpha are the two other methods of studying internal consistency. Another important type of reliability is the interscorer reliability. This reliability is not useful for objective tests, but for certain types it is quite important. There are also other types of reliability such as parallel form reliability and test-retest reliability, the latter concerned with temporal stability of the test results.

1606. Reliability can be defined as

- (A) $\frac{\text{True score variance}}{\text{Total variance}}$
 (B) $\frac{\text{Error variance}}{\text{Total variance}}$
 (C) $\frac{\text{True score variance}}{\text{Error variance}}$
 (D) $\frac{\text{Error variance}}{\text{Total score variance}}$

1607. For a newly developed test, the scores on odd and even part correlated by 0.4. After applying Spearman-Brown formula
 (A) The corrected reliability would be 0.8.
 (B) The corrected reliability would be 0.4.
 (C) The corrected reliability would be greater than 0.4 and lesser than 0.8.
 (D) The corrected reliability would be lesser than 0.4.

1608. Which of the following statement is true for tests with dichotomous items ?
 (A) Chronbach Alpha is greater than Kuder-Richardson reliability.
 (B) Chronbach Alpha is smaller than Kuder-Richardson reliability.
 (C) Chronbach Alpha and Kuder-Richardson reliability are equal.
 (D) Chronbach Alpha may be greater or smaller than Kuder-Richardson reliability.

1609. For which of the following types of tests interscorer reliability is relevant ?

1. Objective personality inventories
2. Projective tests
3. Achievement tests with descriptive items
4. Intelligence tests

Codes :

- (A) 1 and 2 only (B) 2 and 4 only
 (C) 1 and 3 only (D) 2 and 3 only

1610. If two groups are administered the same test with a time gap of eight weeks, we would be able to compute

- (A) Test-retest reliability
 (B) Parallel form reliability
 (C) Interscorer reliability
 (D) None of the above reliabilities can be computed.

1611. The visible spectrum wavelength ranges from

- (A) 380 to 760 nanometres
 (B) 200 to 700 nanometres
 (C) 400 to 1000 nanometres
 (D) 100 to 1000 nanometres

1612. Shyam does not like to be in family gatherings but his mother has asked him to attend it. He decides to tell a lie that he has examination on that day however he finds it very distasteful. Shyam is experiencing ____.

- (A) Double avoidance conflict
 (B) Double approach conflict
 (C) Approach-avoidance conflict
 (D) Double approach-double avoidance conflict

1613. Given below are two statements, one labelled as Assertion (A) and the other labelled as Reason (R). Indicate your answer using the code given below :

Assertion (A) : Computational approach involves a rigorous, mathematically oriented analysis of certain aspects of visual perception derived largely from computer simulation and artificial intelligence.

Reason (R) : Neurophysiological approach argues that sensory and perceptual phenomena are best explained by known neural and physiological mechanisms serving sensory structures.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are correct, but (R) is not the correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

1614. Having seen or heard a stimulus once may facilitate our recognizing it on a latter occasion, even if we are not aware that this is happening. This phenomena is referred to as

- (A) Primacy effect
(B) Recency effect
(C) Practice effect
(D) Priming effect

1615. Given below are the two statements, the first labelled as Assertion (A) and the other labelled as Reason (R). Indicate your answer using the codes given below :

Assertion (A) : Naturalistic observation is the highly preferred method in ethological research.

Reason (R) : Naturalistic observation is free from observer-bias.

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true, but (R) is not the correct explanation.
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

1616. There are various kinds of cells in human body. Which one of the following is not a neuron ?

- (A) Sensory (B) Glia
(C) Inter (D) Motor

1617. Given below are two statements, the first labelled as Assertion (A) and the other labelled as Reason (R). Indicate your answer using the codes given below :

Assertion (A) : During action potential, there is change in polarization of ions across neuronal membrane.

Reason (R) : All the negative ions are transported outside the neuronal membrane.

Codes :

- (A) Both (A) & (R) are true and (R) is the correct explanation of (A).
(B) Both (A) & (R) are true, but (R) is not the correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

1618. Individual differences in intelligence are due to

- (A) only genetic factors
(B) only environmental factors
(C) genetic and economic factors
(D) many factors including genetic and environmental

1619. Put the parts of the Brain in sequence, going from higher to lower level of brain

- I. Metencephalon
II. Mesencephalon
III. Diencephalon
IV. Telencephalon
V. Myelencephalon

Codes :

- (A) I, II, III, IV, V (B) II, IV, I, III, V
(C) IV, III, II, I, V (D) III, II, I, V, IV

1620. Arrange the following layers of Retina in order of light travelling from environment to back of the eyeball :

- I. Ganglion cells
II. Cone receptors
III. Horizontal cells
IV. Bipolar cells

Codes :

- (A) I, II, III, IV (B) II, III, IV, I
(C) IV, II, III, I (D) I, IV, III, II

1621. _____ psychologists work to improve efficiency of people in business.

- (A) cognitive
(B) developmental
(C) industrial/organizational
(D) physiological

1622. Psychology is the science of

- (A) behaviour and mental processes
(B) objective introspection
(C) inductive reasoning
(D) emotions

1623. A hypothesis is

- (A) the independent variable

- (B) an explanation of a phenomenon
(C) a testable prediction derived from a theory
(D) the dependant variable.
- 1624.** The research method used by Frued was
(A) correlational method
(B) naturalistic observation
(C) survey research
(D) case study method
- 1625.** The amount of association between two or more variable is
(A) correlation
(B) naturalistic observation
(C) reliability
(D) synchronicity
- 1626.** Gestalt theory emphasized
(A) a flow of consciousness
(B) the atoms of thought
(C) environmental stimuli
(D) our tendency to see patterns
- 1627.** Short fibers that branch out from cell body and pick up the incoming message are called
(A) dendrites (B) axons
(C) nerves (D) terminals
- 1628.** The thyroid gland controls
(A) glucose absorption
(B) emotions
(B) metabolism (D) sexuality
- 1629.** The ability of the eye to distinguish fine details is called
(A) visual dilation
(B) visual acuity
(C) visual sensitivity
(D) adaptation
- 1630.** A disorder called _____ results when fluid pressure builds up inside the eye and causes damage to optic nerve.
(A) prospagnia (B) achromatospia
(C) dyslexia (D) glaucoma
- 1631.** Optical illusion result from distortion in
(A) transduction (B) sensation
(C) perception (D) adaptation
- 1632.** Our general method for dealing with environment is known as
(A) intelligence
(B) perceptual style
(C) personality
(D) cognitive style
- 1633.** External stimuli that lead to goal-directed behavior are called
(A) drives (B) needs
(C) incentive (D) reciprocal
- 1634.** Most psychologist believe that aggressions
(A) an innate biological response to frustration
(B) linked to sexual drive
(C) a learned response
(D) a drive that builds up over time and must be released
- 1635.** The process of detecting ,translating and transmitting from external environment to brain is called
(A) perception
(B) sensation
(C) selective attention
(D) adaptation
- 1636.** According to Alfred Adler ,the prime motivating force in a person's life is
(A) physical gratification
(B) existential anxiety
(C) striving for superiority
(D) the need for power
- 1637.** Which of the following learning mechanism dopes b.f. Skinner see as being the major means by which behaviour is learned?
(A) classical conditioning
(B) operant conditioning
(C) observational conditioning
(D) insight learning
- 1638.** Which of the following approaches to personality is least deterministic?
(A) the humanistic approach
(B) the psychoanalytical approach
(C) skinner's approach
(D) the behavioural approach
- 1639.** Which of the following did Carl Rogers believe fosters a congruent self-concept?
(A) conditional love
(B) appropriate role models

- (C) immediate-need gratification
(D) unconditional love
- 1640.** Stereotypes are
(A) special types of schemas that are part of people's shared cultural background
(B) widely held beliefs that people have certain characteristics because of their membership in a particular group
(C) equivalent to prejudice
(D) both (A) and (B)
- 1641.** In developmental psychology, the term nature refers to _____ factors that influence development.
(A) Biological
(B) Environmental
(C) Physical
(D) All of these
- 1642.** In the study of development, the most important biological factor is _____ the systematic physical growth of the body, including the nervous system.
(A) Maturation
(B) Hormones
(C) Growth factors
(D) Environment
- 1643.** A biologically determined period in the life of some animals during which certain forms of learning can take place most easily is called a
(A) Stage
(B) Milestone
(C) Critical period
(D) Landmark
- 1644.** _____ is distinguished for major stages of cognitive development, the sensorimotor, the preoperational stage, the concrete operational stage and the formal operational stage.
(A) Jean Piaget
(B) John Bowlby
(C) Sigmund Freud
(D) Harry Harlow
- 1645.** Kohlberg's theory of moral development was criticized by Gilligan primarily because
(A) The data did not support his conclusions
(B) It was based on a study of boys only
(C) It was done so long ago
(D) His subjects consisted of urban children only
- 1646.** The rapid increase in weight and height that occurs around the onset of puberty is known as the
(A) Maturation stage
(B) Physical development stage
(C) Adolescent growth spurt
(D) Menarche
- 1647.** The _____ stage is characterized by an ability to use abstract concepts
(A) Concrete operational
(B) Formal operational
(C) Preoperational
(D) Operational
- 1648.** We are more likely to experience which of the following during adolescence than at any other time in our lives?
(A) Moodiness
(B) Parent-child conflicts
(C) Risky behaviour
(D) All of these
- 1649.** A therapist who encourages clients to relate their dreams and searches for the unconscious roots of their problems is drawn from
(A) Psychoanalysis
(B) Humanistic therapies
(C) Person-centered therapy
(D) Cognitive therapy
- 1650.** Compared to psychoanalytic therapists, humanistic therapists are more likely to emphasize
(A) Hidden or repressed feelings
(B) Childhood experiences
(C) Psychological disorders
(D) Self-Fulfillment & growth
- 1651.** Behaviour therapies apply learning principles to
(A) Identify and treat the underlying causes of dysfunction
(B) Improve learning and insight
(C) Eliminate the unwanted behaviour
(D) Improve communication and social sensitivity

- 1652.** An example of a cognitive therapy is rational-emotive therapy, which is developed by
 (A) Carl Rogers
 (B) Joseph Wolpe
 (C) Albert Ellis
 (D) Allen Bergin
- 1653.** The explicit and implicit rules that a society establishes to govern conduct are referred to as
 (A) Norms (B) Culture
 (C) Morality (D) Conventions
- 1654.** The violation of a society's explicit and implicit norms can be viewed as not normal in the sense that the violation is
 (A) Deviant
 (B) Distressful
 (C) Dysfunctional
 (D) A danger to one's self or to others
- 1655.** Which aspect of the definition of abnormality includes unhappiness?
 (A) Distress
 (B) Deviance
 (C) Dysfunction
 (D) Danger to self or others
- 1656.** Which of the following is not true regarding people with a mental illness?
 (A) They frequently are in considerable distress
 (B) Society views such people as in need of help
 (C) They usually pose a danger to themselves or others
 (D) They usually have behaviour that is in some way different
- 1657.** Dementia is a condition that is characterized by
 (A) General paresis
 (B) Physical ailments
 (C) A general intellectual decline
 (D) The loss of contact with reality
- 1658.** The ancient Greek physicians characterized hysteria as
 (A) Euphoric behaviour
 (B) Uncontrollable sobbing
 (C) An inability to control emotional responses
 (D) A physical ailment without any physical cause
- 1659.** What model of mental illness did most people hold during the middle ages?
 (A) The moral model
 (B) The medical model
 (C) The psychogenic model
 (D) None of these
- 1660.** The first medical practitioner to specialize in mental illness was
 (A) Johan Weyer (B) William Tuke
 (C) Benjamin Rush
 (D) Sigmund Freud
- 1661.** Psychology may best be described as the scientific study of ____ and ____.
 (A) Mental states, physical states
 (B) Thoughts, emotions
 (C) Behavior, mental processes
 (D) Mental health, mental illness
- 1662.** Freud believed that adult problems usually
 (A) Result in Freudian slip
 (B) Result in bad dreams
 (C) Can be traced back to critical stage during childhood
 (D) Are the result of poor behavior
- 1663.** Gestalt theory emphasizes
 (A) A flow of consciousness
 (B) The atoms of thought
 (C) Environmental stimuli
 (D) Our tendency to see pattern
- 1664.** Whereas the _____ asked what happens when an organism does something; the _____ asked how and why.
 (A) Functionalist, behaviorist
 (B) structuralist, introspectionist
 (C) structuralist, functionalist
 (D) Functionalist, structuralist
- 1665.** The _____ lobe is to hearing as the occipital lobe is to vision.
 (A) Frontal (B) Temporal
 (C) Parietal (D) cerebeller
- 1666.** Reflexes are usually controlled by the
 (A) Medulla (B) Frontal lobe
 (C) Spinal cord (D) None of these
- 1667.** A part of the brain that sends signals "alert" to higher centers of the brain in response to incoming messages is

- (A) Limbic system
 - (B) Reticular formation
 - (C) amygdala
 - (D) None of these
- 1668.** Perception of the brightness of a color in affected mainly by
- (A) The amplitude of light waves
 - (B) The wavelength of light waves
 - (C) The purity of light waves
 - (D) The saturation of light waves
- 1669.** Which of the following is not a clue for depth perception?
- (A) Interposition
 - (B) Orientation
 - (C) Linear perspective
 - (D) Reduced clarity
- 1670.** Psychophysics is the study of
- (A) Perceptual illness
 - (B) The psychological perception of physical stimuli
 - (C) Depth perception
 - (D) Movement perception
- 1671.** Perceptual constancies are
- (A) Illusion in which we perceive something that does not correspond to the sensory information
 - (B) Confusing to an individual rather than helping him determine what really exists
 - (C) Likely inborn and not subject to leaning
 - (D) An aid in perceiving a stable and consistent world
- 1672.** Which of the following is a subdivision of the autonomic nervous system?
- (A) Both the sympathetic and Para-sympathetic nervous system
 - (B) Only sympathetic nervous system
 - (C) Only the parasympathetic nervous system
 - (D) Brain and spinal cord
- 1673.** Any stimulus that follows a behavior and increases the likelihood that the behavior will be repeated is called a
- (A) Cue
 - (B) Situational stimulus
 - (C) reinforcer
 - (D) Punisher
- 1674.** Conditioned response may be eliminated by withdrawing reinforcement. This is known as
- (A) Stimulus generalization
 - (B) Extinction
 - (C) Discrimination
 - (D) Spontaneous recovery
- 1675.** The ability to learn by observing a model or receiving instructions, without reinforcement, is called
- (A) Cognitive learning
 - (B) Contingency
 - (C) Social learning
 - (D) Instrumental learning
- 1676.** Analysis of avoidance learning suggest that many phobias are acquired through _____ conditioning.
- (A) Classical
 - (B) Operant
 - (C) Reinforcement
 - (D) Intermittent
- 1677.** The James-Lange or body reaction theory of emotion says
- (A) You feel emotion then a bodily reaction
 - (B) You react with your body first then you feel emotion
 - (C) The somatic nervous system is the seat of emotion
 - (D) Emotions and visceral reactions are simultaneous
- 1678.** The frustration-aggression hypothesis
- (A) Assume that aggression is basic instinct
 - (B) Claims that frustration and aggression are both instinctive
 - (C) Assumes that frustration produces aggression
 - (D) Was developed by social learning theorist
- 1679.** In Freud's theory of personality
- (A) The ego obeys the reality principle
 - (B) The id operates by secondary process
 - (C) The superego obeys the pleasure principle
 - (D) The ego operates by primary process thinking
- 1680.** Which of the following approaches to personality is least deterministic?

- (A) The humanistic approach
(B) The psychoanalytic approach
(C) The social learning approach
(D) None of these
- 1681.** The child's increasing skill at using his muscles is due chiefly to
(A) Maturation
(B) Learning
(C) Pushing by parents
(D) An opportunity to exercise
- 1682.** The stage of prenatal development during which the developing organism is most vulnerable to injury is the
(A) Zygotic stage
(B) Germinal stage
(C) Fetal stage
(D) Embryonic stage
- 1683.** Which of the following is part of the psychosocial domain?
(A) Motor skills
(B) Memory
(C) Judgment
(D) Style of behaving
- 1684.** It would be more accurate to say that maturation
(A) Can be delayed by illness or poor nutrition
(B) Is dramatically accelerated by good nutrition
(C) Is greatly accelerated by good health care and delayed by illness
(D) Is not affected by any environmental factors because it is genetic
- 1685.** The first two weeks of life are referred to as
(A) The period of infancy
(B) The preoperational stage
(C) Early childhood
(D) The neonatal period
- 1686.** An individual becomes a member of social group through the process of
(A) Aging (B) Socialization
(C) Learning (D) Maturation
- 1687.** When a researcher tests several groups of people at the same time each group at a different age, he is conducting a(n) _____ study.
(A) Observational
(B) Longitudinal
(C) Cross-sectional
(D) Sequential
- 1688.** Piaget's stage for infancy is
(A) Concrete operations
(B) Formal operation
(C) Preoperational thought
(D) Sensory motor thoughts
- 1689.** Job satisfaction is enhanced by tasks that are
(A) Challenging but not overwhelming
(B) Very easy to accomplish
(C) Overwhelming and rewarding
(D) Unrelated to one's personal needs
- 1690.** Basic characteristics of tasks that result in social loafing is that they are
(A) Additive
(B) Subtractive
(C) Multiplicative
(D) Negative
- 1691.** The process by which a trained professional uses psychological methods to help people with psychological problem is known as
(A) Psychiatry
(B) Psychoanalysis
(C) Psychotherapy
(D) Psychosurgery
- 1692.** Albert Ellis and Aron Beck are names associated with the _____ therapy approach.
(A) Gestalt
(B) Cognitive
(C) Behavioral
(D) Phenomenological
- 1693.** An affective disorder in which a person swings from one mood extreme to another is classified as
(A) Manic (B) Depressive
(C) Unipolar (D) Bipolar
- 1694.** The leading cause of mental retardation is believed to be
(A) Fetal alcohol syndrome
(B) Organic brain syndrome
(C) Environmental factors
(D) Inherited traits

- 1695.** People who consistently come up with _____ explanations of events are more prone to depression.
(A) Overly optimistic
(B) Pessimistic
(C) Delusional
(D) dysthemic
- 1696.** According to Piaget the process of building mental representation of the world through direct interaction with it is
(A) Adaptation
(B) Egocentrism
(C) metacognition
(D) Conservation
- 1697.** Intelligence can be defined as
(A) Knowledge of a great many facts
(B) The ability to get good grades in school
(C) The ability to think abstractly and learn from experience
(D) All the factors that make one person different from another
- 1698.** When people are _____ they have lost touch with reality.
(A) Manic (B) Neurotic
(C) Psychotic (D) Psychopathic
- 1699.** Which of the following clinical procedures are based, in part on classical conditioning?
(A) Transference
(B) Systematic desensitization
(C) Token economy
(D) Two chair technique
- 1700.** Illnesses that seem to result from an interaction of physical and psychological factors are called
(A) Hysterical
(B) Psychosomatic
(C) Somatic
(D) None of the above
- 1701.** In which way behaviour is different from mental processes?
(A) Introspection
(B) Observation right
(C) Research (D) Psychoanalysis
- 1702.** Who is particularly interested in studying patterns of behaviour, beliefs and values which are shared by people?
(A) Social Psychologist
(B) Experimental Psychologist
(C) Clinical Psychologist
(D) Psycho Analyst
- 1703.** Who believed that mind is composed of senses, ideas, images and feelings?
(A) William James (B) Titcher
(C) Carl Rogers (D) Wundt
- 1704.** Which area of hypothalamus is responsible for satiety behaviour?
(A) Lateral hypothalamus
(B) Ventromedial hypothalamus
(C) Hyper Phobia
(D) Antidiuretic hormone
- 1705.** Which perspectives of psychology focuses on determining the extent to which psychological characteristics such as memory, intelligence are influenced by heredity?
(A) Neuroscience (B) Biogenic
(C) cognitive (D) socio-cultural
- 1706.** Which of the following relates to disorders in comprehension or production of speech?
(A) Dysphasia (B) Aphasia
(C) Broca's area (D) All of these
- 1707.** Which lobe of the brain is involved in planning, decision-making and some aspects of language?
(A) Frontal (B) Parietal
(C) Temporal (D) Occipital
- 1708.** Which neurotransmitter plays an important role in the experience of pleasure?
(A) Serotonin
(B) Dopamine
(C) Nor-epinephrine
(D) Acetylcholine
- 1709.** The hormone cortisol produced by adrenal gland is responsible for
(A) Reducing the body activity to take rest
(B) Activating the body to prepare for stressful situation
(C) Balancing the homeostasis
(D) Planning and decision-making
- 1710.** The process through which a learned response stops occurring due to the non-availability of reinforcement is called

- (A) Generalization
(B) Punishment
(C) Spontaneous Recovery
(D) Extinction
- 1711.** When a teacher appreciates his students every time they ask questions with the hope that this will enable them to handle more technical questions in future, he is using conditioning?
(A) Classical conditioning
(B) Shaping
(C) Modeling (D) Reinforcing
- 1712.** IN classical conditioning, the organism creates an association between
(A) Two Situations
(B) Two Consequences
(C) Two Stimuli
(D) Two Responses
- 1713.** Which of the following is not a characteristic of self-actualised people according to Maslow?
(A) They live creatively and fully using their potentials
(B) They have efficient perceptions of reality, self and others
(C) They are sensitive to fake and dishonest
(D) They feel ashamed of their negative emotions
- 1714.** According to Freud, the process through which individuals release their excessive aggressive energy is called
(A) Dream Work
(B) PsychoAnalysis
(C) Catharsis
(D) Slips of tongue
- 1715.** A layer of nerve cells at the back of eye which converts light waves into electrical impulses is
(A) Fovea (B) Retina
(C) Cornea (D) Iris
- 1716.** The process through which brain uses the two retinal images into one dimensional perception is called
(A) Depth perception
(B) Stereopsis
(C) Binocular cues
(D) Retinal Disparity
- 1717.** Which of the following is not an attribute of extroversion in the BIG FIVE FACTORS theory of personality?
(A) Sociable (B) Talkative
(C) Self-conscious (D) Affectionate
- 1718.** Which part of personality opposes the irrational desires of id according to Freud's Psychoanalytic Theory of Personality?
(A) Ego (B) Psychic energy
(C) Super Ego (D) Libido
- 1719.** A psychological test is good only when it measures what it is supposed to measure. This statement refers to which of the following?
(A) Degree to which it is objective, precise and short
(B) Degree to which it is consistent across several administrations
(C) Degree to which it is norm based
(D) Degree to which it is psychometrically sound
- 1720.** Which aspect of social interactions is measured by the theory of planned behaviour?
(A) Attachment
(B) Socialization
(C) Group formation
(D) Attitudes
- 1721.** A strategy used by the ego to defend itself against the anxiety provoked by the conflict of everyday life refers to
(A) Defense mechanism
(B) Displacement
(C) Rationalization
(D) Ego-ideal
- 1722.** Development is about to
(A) Age (B) Gender
(C) Change (D) Genes
- 1723.** The self-image formed during adolescence that integrates and individual's ideas of what he or she is and want to be refers to
(A) Ego-ideal (B) Ego-identity
(C) Crisis (D) Ego
- 1724.** The theory of how people explain other's behaviour by attributing it either to internal disposition or to external situation refers to

- (A) Attribution theory
(B) Dispositional theory
(C) Situational theory
(D) Attributing causality
- 1725.** The behavioural approach is to reward and punishment as the cognitive approach is to
(A) Biological processes
(B) The humanistic struggle for self fulfillment
(C) individual genetic make up
(D) Internal processes of the mind or mental events
- 1726.** Which of the following perspectives of psychology is most concerned with the individuals reaching their maximum potential?
(A) Behavioural
(B) Cognitive
(C) Humanistic
(D) Psychoanalysis
- 1727.** Jobs can be redesigned to make the work more interesting, challenging and to increase responsibility and opportunities for achievement
(A) Job Satisfaction
(B) Job specification
(C) Job enrichment
(D) Job description
- 1728.** The basic difference between obsession and compulsion is
(A) The former involves preoccupation with certain thoughts while the later involves preoccupation of certain actions
(B) The former is a mood disorder whereas the later is an anxiety disorder.
(C) The former is diagnosed along with the Axis III of DSM-IV whereas the later is diagnosed along with Axis II of DSM-IV
(D) Actually, there is no difference between the two disorders.
- 1729.** The process by which egg is ripened and released is
(A) Mitosis (B) Fertilization
(C) Ovulation (D) Implantation
- 1730.** The process by which we learn not to respond to similar stimuli in identical manner is
(A) Generalization (B) Extinction
(C) Discrimination (D) All of these
- 1731.** Which method of gathering information about the brain indicates overall activity of the brain?
(A) Positron Emission Tomography
(B) Electroencephalogram
(C) Electrical stimulation
(D) Studying damage to the brain
- 1732.** The physiological and psychological response to stress referred as
(A) Transition (B) Stressors
(C) Strain (D) Hypertension
- 1733.** The behavioural technique for reducing anxiety in which patients practice relaxation while visualising provoking situations of increasing intensity refers
(A) Symptom substitution
(B) Time-out
(C) Systematic Desensitization
(D) Token economy
- 1734.** The ability to acquire information from the culture such as vocabulary and the kind of info learned in the schools is an example of
(A) Fluid intelligence
(B) Componential intelligence
(C) Crystallised intelligence
(D) Contextual intelligence
- 1735.** Critics of Freudian psychoanalysis maintain that this form of therapy
(A) does not put enough emphasis on sex
(B) can be harmful by tolerating
(C) is good mostly for treating phobias
(D) is too quick to be effective
- 1736.** When Maya first saw violent movie, she was scared and disgusted. But now she has seen several such movies, she is barely affected by them. Maya has experienced
(A) Habituation (B) Displacement
(C) Cultivation (D) Catharsis
- 1737.** A person's IQ initially represented the ratio of a person's _____ age divided by his or her _____ age.

- (A) Chronological, mental
(B) Mental, chronological
(C) Real, Intentional
(D) Chronological , neurological
- 1738.** Which of the following is true regarding the relative influence of genes and the environment of behaviour?
(A) Environment is the primary influence throughout life
(B) Genes are the primary influence before birth, and environment is the primary influence following birth.
(C) Genes are primary influence on the development of physical structures and environment factors are a great influence on the development of behavioural capacities.
(D) In no sense can either genes or environment are dormant unit the approach of puberty.
- 1739.** Why do similarities in the IQ of raised-together siblings tend to disappear as the children grow older?
(A) Sibling learn to over-compensate for deficiencies in each other's abilities
(B) The genes that differentiate intellectual abilities are dormant until the approach of puberty
(C) As children mature and become increasingly autonomous, they each tend to create a unique environment
(D) All of the above
- 1740.** The release of emotions is called
(A) Behaviour change
(B) Insight
(C) Mastery (D) Catharsis
- 1741.** What procedure is used by experimenters to determine whether a difference between conditions of an experiment is large enough for us to have confidence in its validity?
(A) Correlation coefficient
(B) Scientific intuition
(C) Statistical analysis
(D) None of the above
- 1742.** If one finds a positive correlation between degree of coffee drinking and the likelihood of heart attacks. One can conclude that
(A) Coffee drinking causes heart attack
(B) Individuals prone to heart attacks are predisposed to drink a lot of coffee
(C) An active life style of certain people causes heart attack
(D) None of the above
- 1743.** Someone whose corpus callosum has been cut will experience difficulty in
(A) Naming an object held in the left hand
(B) Naming an object held in the right hand
(C) Recognizing an object held in the left hand
(D) Recognizing an object held in the right hand
- 1744.** Taste depends
(A) On one's taste buds and the smell, temperature and texture of food
(B) Only on one's taste buds
(C) Only on smell and texture of food
(D) None of these
- 1745.** Smoking during pregnancy is associated with
(A) High intelligence in the fetus, since the fetus must be clever to keep the cigarette burning
(B) Low birth weight
(C) High mortality
(D) Both (B) and (C)
- 1746.** The application of psychological theories, methods and techniques to solve practical human problems describes an area of psychology known as
(A) Social psychology
(B) Para-psychology
(C) Applied psychology
(D) Humanistic psychology
- 1747.** Which of the following is not one of the factors, which tend to hinder the fairness of intelligence tests for lower class children?
(A) Inappropriate norms
(B) Items requiring certain experience
(C) Test assume test-taking skills
(D) Test givers bias the result

- 1748.** As E scores of I-E scale show a general trend upward
(A) People show more reserve suspicion and withdrawal
(B) Assassinations and events like Watergate and Vietnam increase
(C) The ability for people to openly express themselves decrease
(D) Both A and C
- 1749.** In ancient times, who supposedly inhibited the body and soul of a “crazy” person?
(A) God (B) A saint
(C) The devil (D) None of these
- 1750.** “Double approach-avoidance” conflict is
(A) Approaching a problem from two different angles
(B) A doubly strong tendency to approach an object
(C) Being faced with a choice between two equally attractive goals
(D) Choosing between two equally attractive objects such that the choice of one means giving up the other
- 1751.** “Delusion” refer to
(A) Thoughts or beliefs that have no basis in reality
(B) Feelings that have no basis in reality
(C) Behaviors that have no basis in reality
(D) None of these
- 1752.** The basic assumption of humanistic theories include
(A) Humans are basically good and worthy
(B) Humans are basically bad and unworthy and thus have to be reconditioned using learning principles
(C) There is a natural growth process that can be blocked by bad condition
(D) Both A and C
- 1753.** Rotter’s social learning theory emphasizes the following
(A) Behavioral potential
(B) Expectancy
(C) A reinterpretation of Freud’s theory
(D) Both A and B
- 1754.** “Human beings respond to their subjective cognitions about their world rather than to the objective environment” is an argument for the effectiveness of
(A) Primal therapy
(B) Freudian therapy
(C) Cognitive therapy
(D) Rankian therapy
- 1755.** How can morality covertly enter the therapy session?
(A) By having ministers take over the role of therapist
(B) By the use of techniques to alter a sexual preference
(C) By allowing an insane person into therapy
(D) By allowing philosophers of ethics to become therapists
- 1756.** Research on the role of reinforcement in insight therapy has found that
(A) It does not play a role
(B) It does play a role
(C) It has an effect only on dependent patients
(D) It plays a role in highly structured therapy situation
- 1757.** The notion that aggression can be reduced by allowing angry individuals to engage harmless activities that allow them to “blow off self esteem” is
(A) Frustration-aggression hypothesis
(B) Displacement hypothesis
(C) Sublimation hypothesis
(D) Catharsis hypothesis
- 1758.** A person behavior is usually attributed to external causes when all but one of the following conditions exist
(A) Others act in the same way
(B) The person acts the same way at other times
(C) The person acts differently in other situations
(D) The person seems aware of the environment
- 1759.** A study of gastric ulceration in response to stress produced by shock found that the most ulceration was shown by rates who

- (A) Only heard a tone but received no shock
 (B) Received varying levels of shock
 (C) Could predict the onset of shock
 (D) Could not predict the onset of shock
- 1760.** Studies of crowding have found that crowding
 (A) Depends only on physical density
 (B) Affects mainly females
 (C) Can intensify feelings
 (D) Has negative effects only among the elderly
- 1761.** Developmental psychologists believe that two factors that influence human development are
 (A) Motivation and emotion
 (B) Self and others
 (C) Genetic makes up and experience
 (D) Rewards and punishments
- 1762.** Motor skills are largely a result of
 (A) Learning
 (B) Maturation process
 (C) Practice
 (D) Observing other
- 1763.** In Piaget's theory, the first two years of life are called the _____ stages.
 (A) Paralinguistic
 (B) Exploratory
 (C) Sensorimotor
 (D) Preoperational
- 1764.** Learning theories explain attachment of infants to their parents in terms of
 (A) Conditioning
 (B) Observational learning
 (C) The maturation of perceptual skills
 (D) Cognitive development
- 1765.** Freud was among the first to suggest that abnormal behavior
 (A) Can have a hereditary basis
 (B) Is not the result of demonic possession
 (C) Is psychology caused
 (D) Can result from biological factors
- 1766.** Which of the following is not a common symptom of the depression?
 (A) Insomnia
 (B) Delusions
 (C) Poor appetite
 (D) Lethargy
- 1767.** Gradual exposure to actual feared situation is called
 (A) Cognitive desensitization
 (B) In vivo desensitization
 (C) Flooding
 (D) Breaking of resistance
- 1768.** Rotational-emotive therapy is a type of
 (A) Psychoanalysis
 (B) Client-centered therapy
 (C) Cognitive-behavior therapy
 (D) Behavior therapy
- 1769.** The concept of intelligence is closely related to
 (A) Motivation (B) Learning
 (C) Perception (D) Cognition
- 1770.** Most IQ tests assess
 (A) Academic motivation
 (B) Convergent thinking
 (C) Perceptual motor skills
 (D) Creativity
- 1771.** Addictive disorders include
 (A) Alcoholism and drug addiction
 (B) Overeating
 (C) Sociopathology
 (D) All the above
- 1772.** Seizures, confusions, delusions and hallucinations are symptoms of
 (A) Advanced alcoholism
 (B) Delirium tremens
 (C) Alcoholic withdrawal
 (D) All of the above
- 1773.** The central concept in Gestalt therapy is
 (A) Awareness
 (B) Self-fulfillment
 (C) Self-control
 (D) Desensitization
- 1774.** The techniques used in behavior modification
 (A) Stress interpersonal interactions
 (B) Employ the principal of learning
 (C) Are capable to a very limited range of psychological problems
 (D) All involve some sort of operant conditioning

- 1775.** Research has suggested that compulsive behavior persists because
(A) It reduces anxiety
(B) There is some underlying conflict
(C) Others begin to expect it
(D) It diverts the attention of the individual from the problem
- 1776.** A perceptual experience, which is not grounded in reality, is called a/an
(A) Delusion
(B) Illusionary images
(C) Hallucinations
(D) Spontaneous discharge of sensory neurons
- 1777.** The hallucinations of schizophrenic are most likely to be
(A) Auditory (B) Visual
(C) Tactual (D) Olfactory
- 1778.** Among people with severe mood disorder, _____ is most common.
(A) Mania
(B) Depression
(C) Manic-depression
(D) Euphoria
- 1779.** Rogers believes that all of us are born with
(A) Unconditional positive regard
(B) A drive for self fulfillment
(C) A sense of individuality and uniqueness
(D) A variety of incongruence, which must be resolved in infancy and early childhood
- 1780.** Tests that employ real life problems that the examinee is likely to face on the job are called
(A) Job tasks
(B) Valid tests
(C) Situational tests
(D) Projective techniques
- 1781.** The group in an experiment which receives no treatment is called
(A) Control group
(B) Experimental group
(C) No group (D) None of these
- 1782.** The portion of your nervous system which controls breathing and digestion is
(A) Axon (B) Autonomic
(C) Linear circuit (D) None of these
- 1783.** The color, smell and feeling of the flowers are relayed through what part of brain
(A) Acetylcholine (B) Thalamus
(C) Motor area (D) None of these
- 1784.** A hungry person would find food to be a
(A) Primary reinforcer
(B) Secondary reinforcer
(C) None of these
- 1785.** A psychologist explains you that learning can be best defined in terms of underlying thought process. What theory is being stated
(A) Social learning
(B) Cognitive learning
(C) Trial and error
(D) None of these
- 1786.** According to Maslow, a person with no job, no friends and no house can be self actualized
(A) 100 percent yes
(B) 100 percent no
(C) None of these
(D) All of the above
- 1787.** A need to establish and maintain relationship with other people
(A) Achievement motivation
(B) Need for affiliation
(C) None of these
(D) All of the above
- 1788.** Term n-ach was introduced by
(A) Mc Clelland (B) Jung
(C) Maslow (D) None of these
- 1789.** Repression refers to
(A) Primary defense mechanism
(B) Regression
(C) Frustration
(D) None of the above
- 1790.** Enduring dimensions of personality characteristics differentiating people from another is called
(A) Factor analysis (B) Trait
(C) Determinism (D) None of these
- 1791.** The theory which suggests that people learn attitude by observing their own behavior is

- (A) Operant conditioning
(B) CR
(C) Self-perception
(D) None of the above
- 1792.** Theory of cognitive dissonance is offered by
(A) Skinner (B) Bandura
(C) Festinger (D) None of these
- 1793.** The ability to view the world in 3 dimensions and perceive distance is
(A) Depth perception (B) Illusion
(C) Delusion (D) None of these
- 1794.** According to Bandura, one is socialized by
(A) Punishment (B) Observation
(C) Food (D) None of these
- 1795.** Crowding causes increase in heart rate and the level of adrenaline
(A) No (B) Yes
(C) Both (D) None
- 1796.** Studies demonstrate that greater genetic similarity is associated with greater similarity of attitudes
(A) True (B) False
(C) Both (D) None
- 1797.** A theory that states that emotions are a joint result of non-specific physiological arousal and interpretation of the arousal is called
(A) Two-factor theory of emotion
(B) James-Lange theory of emotion
(C) Bem's theory of emotion
(D) None of these
- 1798.** Maintenance of an internal biological balance is called
(A) Instinct (B) Need
(C) Homeostasis (D) None of these
- 1799.** The part of personality that provides a buffer between the id and the outside world is
(A) Super-ego (B) Ego
(C) Ego-ideal (D) None of these
- 1800.** Specialized cells of nervous system carry
(A) Messages
(B) Reflexes
(C) Both (A) and (B)
(D) None of the above
- 1801.** The combination of responses or ideals in novel way is called
(A) Exploration (B) Creativity
(C) Thinking (D) None of these
- 1802.** Group intelligence testing is better than individual testing
(A) True (B) False
(C) Both (D) None
- 1803.** Birth typically occurs after
(A) Forty weeks conception
(B) Thirty eight weeks conception
(C) Four weeks conception
(D) None of these
- 1804.** Longitudinal researches investigate
(A) Behavior through times as subject age
(B) Behavior of different ages are compared
(C) None of these
(D) All of the above
- 1805.** The cause of abnormal behavior is only physiological, it is explained by
(A) Psycho model
(B) Medical model
(C) Psychosocial model
(D) None of the above
- 1806.** One abnormal behavior was associated with witch craft
(A) True (B) False
(C) Both (D) None
- 1807.** A feeling of apprehension or tension is
(A) Frustration (B) Panic
(C) Anxiety (D) None of these
- 1808.** Which of the following treatments deal with phobias by gradual exposure?
(A) Super ego control
(B) Systematic desensitization
(C) Reinforcement
(D) Dreams
- 1809.** The operation of the brain's neurotransmitters is altered because
(A) Use of LDS
(B) Use of alcohol
(C) None of these
(D) All of the above
- 1810.** Researches show that poor performance is the result of

- (A) Poor conditions
(B) Job satisfaction
(C) Both (A) and (B)
(D) None of these
- 1811.** Low job satisfaction is likely to bring about high absenteeism
(A) False (B) True
(C) Both (D) None
- 1812.** The term propinquity explains
(A) Geographical proximity
(B) Why people affiliate with one another
(C) Both (A) and (B)
(D) None of these
- 1813.** Group think refers to
(A) Deterioration of mental efficiency
(B) Deterioration of physical energy
(C) Think tank
(D) None of these
- 1814.** Main functions of stimulants are
(A) To alleviate tension
(B) To provide energy and alertness
(C) Prescribed for insomnia
(D) None of these
- 1815.** Chromosomal anomalies can be recognized by slanting eyes and flat nose
(A) Down's syndrome
(B) PKU
(C) Langdon Down's syndrome
(D) All of the above
- 1816.** Characters disorders, such as obsessive-compulsive behavior are developed due to disturbed parent-child relation in the
(A) Later Oedipal phase
(B) Paranoid stage
(C) Libido (D) None of these
- 1817.** According to the Piaget, from 2 to 7 years of age, language is developed slowly and gradually this stage is called
(A) Preoperational stage
(B) Egocentric stage
(C) Sensorimotor stage
(D) None of these
- 1818.** Speech like but meaningless sound appear between 3 months to one year is termed as
(A) Syntax (B) Babble
(C) Conversation (D) None of these
- 1819.** The ability to deal with new problems and encounters is technically called as
(A) Crystallized intelligence
(B) G or g-factor of intelligence
(C) Fluid intelligence
(D) None of these
- 1820.** The trichroic theory of intelligence suggests that there are
(A) Four components
(B) Six components
(C) Three components
(D) None of these
- 1821.** The junction where the axon of a sending neuron communicates with a receiving neuron is called the
(A) Reuptake site (B) Receptor site
(C) Synapse (D) Axon terminal
- 1822.** The autonomic nervous system differs from the somatic nervous system in that its operation is largely
(A) Involuntary
(B) Voluntary
(C) Controlled by the brain
(D) Controlled by the spinal chord
- 1823.** The most fundamental principle of perceptual organization is called the
(A) Figure-ground relationship
(B) Volley principle
(C) Dark adaptation phenomenon
(D) Law of closure
- 1824.** Which of the following phenomena was studied by parlor?
(A) Maturation
(B) Animal cognition
(C) Operant conditioning
(D) Classical conditioning
- 1825.** What can occur if a person believes that a connection exists between an act and its consequences when there is no relationship between the two?
(A) Classical conditioning
(B) Superstitious behavior
(C) Shaping
(D) Sequential learning

- 1826.** Advertisers place beautiful people or likeable places and objects with the products they are trying to sell because these items
 (A) Distract from the disadvantages of the products
 (B) Cause pleasant feelings to be evoked
 (C) Are part of the products' basic qualities
 (D) Are just elements of scenery
- 1827.** The body's natural tendency to maintain a state of internal balance or equilibrium is called
 (A) Arousal
 (B) Opponent process
 (C) Homeostasis
 (D) Instinct
- 1828.** Concerns with meeting standards of excellence and accomplishing difficult tasks refer to need for
 (A) Affiliation (B) Achievement
 (C) Power (D) Apperception
- 1829.** Which of the following condition is not associated with prolonged sensory deprivation?
 (A) Inability to concentrate
 (B) A satisfying-relaxed feeling
 (C) Hallucinations
 (D) Confusion
- 1830.** Presence of others may interfere with performance due to
 (A) Social inhabitation
 (B) Social loafing
 (C) Distraction (D) All of these
- 1831.** Psychology has been defined by psychologists as
 (A) The study of behavior
 (B) The study of mental activity
 (C) The science that studies behavior and mental process
 (D) None of these
- 1832.** When we look at a distant object, we usually judge its size by
 (A) Object size
 (B) Perspective size
 (C) A compromise between object size and perspective size
 (D) Retinal size
- 1833.** The view that we are born with the ability to perceive the way we do it held by
 (A) Sensory psychologists
 (B) Nativists
 (C) Empiricists
 (D) Contemporary psychologists
- 1834.** The ability to focus on stimuli in which we are interested while resisting distracting stimuli is called
 (A) Concentrated attending
 (B) Stimulus focusing
 (C) Selective attention
 (D) Structured perceiving
- 1835.** In Freud's theory of personality
 (A) The ego obeys the reality principle
 (B) The id operates by secondary process thinking
 (C) The super-ego obeys the pleasure principle
 (D) The ego operates by primary process thinking
- 1836.** Trait theory has been criticized on the ground that
 (A) Traits may be highly dependent on the situation
 (B) Moderator variable are often very influential
 (C) They often do not specify how traits are organized within the personality
 (D) All of the (A), (B) and (C)
- 1837.** When we receive mixed information about a person, we tend to base our impression on the information that is
 (A) Favorable (B) Unfavorable
 (C) Received first (D) Received last
- 1838.** Most of the social psychological research on attitude change has been generated by theories concerning
 (A) Consistency in attitudes and behavior
 (B) Cognitive dissonance
 (C) Self-perception
 (D) Attribution
- 1839.** The public opinion survey is
 (A) A passive record of opinion
 (B) Limited to what the public believes at one moment in time

- (C) Generally ignored by successful politicians
(D) Increasingly helping to shape opinion as well as measure it
- 1840.** The objectivity of science lies in
(A) The capability of scientists to avoid the prejudices of their society
(B) The choice of question studied
(C) Its methodology
(D) All of these
- 1841.** What are the problems associated with low birth weight?
(A) Susceptibility to infection
(B) Difficulty in maintaining body temperature
(C) Susceptibility to infection
(D) Difficulty with breathing, sucking swallowing and digestion
- 1842.** According to Piaget, the major accomplishment of the sensorimotor stage is
(A) Abstract thinking
(B) Egocentrism
(C) Centration
(D) Object permanence
- 1843.** During the stage of concrete operations children
(A) Understand the concept of reversibility
(B) Do not yet understand the concept of conversation
(C) Are able to solve abstract problems
(D) None of these
- 1844.** Studies show that an "authoritarian family" tends to produce an adolescent who is
(A) Dependent and obedient
(B) Self-reliant
(C) Independent but deserved
(D) Surface-complaint but rebellious underneath
- 1845.** Child-rearing methods in Pakistan
(A) Differ little from those in other countries
(B) Differ from one social class to another
(C) Have changed very little over the past fifty years
(D) Are now pretty much the same from one social class to the next
- 1846.** According to Erickson's theory, the struggle during adolescence is
(A) Intimacy versus isolation
(B) Initiative versus role confusion
(C) Competence versus inferiority
(D) None of the above
- 1847.** A major defining characteristic of personality disorder is that they
(A) Are psychological, more from society's view than from that of the person's who have them
(B) Reflect a lack of contrast with reality
(C) Are comparatively easy to treat
(D) Are frequently relative, i.e. short term responses to stress
- 1848.** The behavioral interpretation of depression relates it to "learned happiness." The most obvious symptom of this condition is
(A) Passivity
(B) Increased appetite
(C) Excessive-aggressiveness when threatened
(D) An increase in random maladaptive behavior
- 1849.** Behavior therapists believe that insight is
(A) A worthwhile goal
(B) Non sufficient for behavior change
(C) Not necessary for behavior change
(D) All of these
- 1850.** The suggested cause of abnormal behavior from the cognitive perspective is
(A) Faulty learning
(B) Early childhood experiences
(C) Unconscious unresolved conflicts
(D) Faulty thinking
- 1851.** The most frequently occurring major psychological disorder is
(A) Phobias (B) Schizophrenia
(C) Depression (D) Bipolar disorder
- 1852.** Person-centered therapy is best described as
(A) Confrontive (B) Structured
(C) Nondirective (D) Objective
- 1853.** Systematic desensitization is used in treatment of

- (A) Schizophrenia
(B) Mood disorders
(C) Phobias
(D) Somatoform disorders
- 1854.** Guilford's structure of intellect model of intelligence is notable because
(A) It separate operations from content and product
(B) It rejects the idea of gender feature
(C) It yields 180 unique intellectual factors
(D) All of the above
- 1855.** The most important criterion of whether an individual should be considered retarded is that individual's
(A) Social competence
(B) Mental age
(C) IQ
(D) Ability to learn to speak
- 1856.** Improved job satisfaction result in _____ in work performance and _____ in turnover.
(A) Decrease : decrease
(B) Increase no : change
(C) Increase : increase
(D) No change: decrease
- 1857.** Some employers allow their employees to create their own sceduals within set parameters. This is called
(A) Job sharing (B) Job enrichment
(C) Time sharing (D) Flextime
- 1858.** All of the following are some of the major sources of work stress EXCEPT
(A) Little control
(B) Shift work
(C) High decision latitude
(D) Discrimination
- 1859.** Bonuses, recognition awards, praise and time off improve motivation through the application of
(A) Performance appraisal
(B) Benefits
(C) Reinforcement
(D) Objective criteria
- 1860.** The Binet-Simon scale was adapted for American use by
(A) Thurston (B) Terman
(C) Wechsler (D) Binet
- 1861.** The term phenomenology refers to
(A) A science of perception
(B) Detailed analysis of our sensations
(C) "Native" description of our immediate experience
(D) Scientific study of "queer" occurrences
- 1862.** The degree of concentration or dilution of color is known as its
(A) Brightness (B) Hue
(C) Saturation (D) Volume
- 1863.** Differences in perception among species depend upon
(A) Habitat
(B) Complexity of nervous system
(C) Receptor mechanism
(D) All of the above
- 1864.** Psychophysics is the study of the relationship between
(A) Sensory attributes and physical energy
(B) Perception and physiological process
(C) Psychology and physics
(D) None of these
- 1865.** The concept of adaptation was introduced by
(A) Brown (B) Helson
(C) Wertheimer (D) Kohler
- 1866.** Convergence of the eyes in controlled primarily by
(A) Extrinsic muscle of the eye
(B) Curvature of lens
(C) Ciliary muscle of eye
(D) Muscles of iris
- 1867.** Factors predisposing an individual toward feeling of jealousy include
(A) Self-hate
(B) Insecurity in self-conception
(C) Lack of intensity in feeling of love
(D) All of the above
- 1868.** The so-called primary emotion includes
(A) Pain, fear, hate, love
(B) Shame, guilt, anger, grief
(C) Joy, fear, anger, grief
(D) All of the above
- 1869.** Coping behavior is
(A) Often unconscious

- (B) A form of habitual behavior
(C) Always goal-directed
(D) Usually carried out directly
- 1870.** Deficiency motivation is to abundance motivation as
(A) Desire is to need
(B) Innate is to learned
(C) Primary is to secondary
(D) Need is to desire
- 1871.** The concept of homeostasis most closely associated with the name
(A) Richter (B) Cannon
(C) Levin (D) Warden
- 1872.** McClelland's method of measuring the achievement motive made use of
(A) Self-ratings
(B) Fantasy
(C) Person's estimates of level of aspiration
(D) Activity level
- 1873.** Sources of frustration include
(A) Environmental lacks
(B) Environmental obstacles
(C) Incompatibles motives
(D) All of the above
- 1874.** The "local" stimulus theory of hunger is associated with the name of
(A) Cannon (B) Beach
(C) Ingram (D) Anderson
- 1875.** The Cannon-Bard theory of emotions emphasis the role of the
(A) Skeletal muscles
(B) Thalamus and hypothalamus
(C) Hormonal system
(D) All of the above
- 1876.** An insufficient supply of thyroid hormone can result in
(A) Increased metabolic rate
(B) Cretinism
(C) Increased emotionality
(D) None of these
- 1877.** The so-called "master gland" which helps control the activities of various ductless glands is the
(A) Adrenal medulla
(B) Thyroid
(C) Anterior pituitary
(D) Gonad
- 1878.** The term "reinforcement" refers to the pairing of
(A) CS and US (B) CR and UR
(C) US and UR (D) CS and CR
- 1879.** One of the following is not a method of measuring retention
(A) Recognition (B) Relearning
(C) Recall (D) Reminiscence
- 1880.** When new acquisitions improve retention of the old, we speak of
(A) Positive transfer
(B) Negative transfer
(C) Reproductive facilitation
(D) Reproductive interference
- 1881.** The social distance scale is mainly a measure of
(A) Ethnic attitude
(B) Cohesiveness of a group
(C) Upward social mobility
(D) Social class differences
- 1882.** A person has attitudes towards
(A) All possible objects
(B) All objects which are perceived by him
(C) All objects which exist in his own life-space
(D) None of these
- 1883.** So-called "functional" and "organic" psychoses are
(A) Really both functional in origin
(B) Really both organic in origin
(C) Clearly differentiated as to origin
(D) Not clearly distinguishable
- 1884.** Which one of the following is not listed as a classification of psychoneurosis?
(A) Phobic reactions
(B) Conversion reactions
(C) Neurasthenic reactions
(D) Delusional reactions
- 1885.** Psychoneurosis tends to be
(A) Severe than the psychoses
(B) Milder than neuroses
(C) Milder than psychoses
(D) About the same severity as the psychoses
- 1886.** In essence the defense mechanisms are
(A) Defense against anxiety

- (B) Ways of convincing others of one's rightness
(C) Escape reactions
(D) Conscious self-deception
- 1887.** The Thematic Apperception Test is a
(A) Situational test
(B) Performance test
(C) Projective test
(D) Personality inventory
- 1888.** Psychosomatic disorders are best illustrated by
(A) Asthma caused by anxiety
(B) Neurotic anxiety
(C) Insomnia
(D) Over-eating
- 1889.** Dependence, passivity and greediness are regarded as traits found in the so-called
(A) Anal character (B) Phallic stage
(C) Genital stage (D) Oral character
- 1890.** Probably the most insidious form of defense mechanism is
(A) Reaction formation
(B) Defensive identification
(C) Projection
(D) Withdrawal
- 1891.** Endomorphy is to eotomorphy as
(A) Love of comfort is to social inhabitation
(B) Social inhabitation is to love of comfort
(C) Love of comfort is to need for action
(D) Need for action is to social inhabitation
- 1892.** Personality inventories are
(A) Subjectively scored
(B) Objectively scored
(C) Always scored on a priori basis
(D) Always scored on an empirical basis
- 1893.** The difficulty with type theories of personality is that they are
(A) Too simple
(B) Too contradictory
(C) Too arbitrary
(D) All of these
- 1894.** The concept of introversion and extroversion was advanced by
(A) Spranger (B) Kretshmer
(C) Jaensch (D) Jung
- 1895.** The term "germ cell" when used in genetics refers to
(A) Somatic cells
(B) Body cells
(C) Ova and sperm cells
(D) Both (A) and (B)
- 1896.** The overall bodily and mental growth curves for man
(A) Diverge at puberty age
(B) Run parallel course
(C) Show drop at puberty age
(D) Both (A) and (B)
- 1897.** As the individual continues his growth after birth, the neural cell in his brain
(A) Change their chemical composition
(B) Increase in number
(C) Both (A) and (B)
(D) None of these
- 1898.** With the onset of puberty, the rate of growth of
(A) All bodily organs slow down
(B) Intellectual development slows down
(C) Both (A) and (B)
(D) Neither (A) and (B)
- 1899.** To mark the onset and termination of adolescence
(A) Physiological measures are usually used for both boys and girls
(B) Physiological measures are usually used for girls but not for boys
(C) Physiological measures are usually used for the onset but not for the end
(D) Physiological measurements are usually used for both boys and girls
- 1900.** The prototype of the modern intelligence test for children was first developed by
(A) Stanford (B) Terman
(C) Binet (D) Wechsler
- 1901.** The peripheral nervous system is made up of
(A) Axons and dendrites
(B) The brain and the spinal cord
(C) The autonomic and somatic nervous systems

- (D) The sympathetic and parasympathetic nervous systems
- 1902.** The adrenal hormones which also appear to function as neurotransmitters in the brain are
- (A) Dopamine and serotonin
(B) Progesterone and testosterone
(C) Epinephrine and nor epinephrine
(D) Insulin and toberlone
- 1903.** The detection of physical energy emitted or reflected by physical objects: it occurs when energy in the external environment or the body stimulates receptors in the sense organs
- (A) Cognition (B) Sensation
(C) Perception (D) Adaptation
- 1904.** Which of the following is a dimension of visual experience?
- (A) Loudness (B) Pitch
(C) Timbre (D) Brightness
- 1905.** All human senses evolved for the same purpose to
- (A) Increase our ability to enjoy the environment
(B) Make up more aware our feelings and internal state
(C) Provide information that could improve our ability to communicate
(D) Help us survive
- 1906.** The fovea and blind spot are both features of the
- (A) Cornea (B) Lens
(C) Retina (D) Pupil
- 1907.** Which of the following is the best example of a primary reinforcing?
- (A) Food (B) Money
(C) Pay check (D) New clothes
- 1908.** Which of the following could be attributed to Tolman's research?
- (A) Discovery of the law of effect
(B) The concept of cognitive map
(C) The negative effect of punishment
(D) The role of classical conditioning in the formation of phobic disorders
- 1909.** Which of the following factors influence the impact of interference on learning?
- (A) The nature of the material being learned
(B) The organization and meaningfulness of the information learned
(C) The type of activity that flows studying
(D) All of the above
- 1910.** Which of the following psychologists developed the need/drive theory of emotion?
- (A) Clark Hull
(B) Abraham Maslow
(C) Robert Plutchik
(D) Water Cannon
- 1911.** Which of the following orders of Maslow's hierarchy of need is correct?
- (A) Safety, physiological, belongingness, self-actualization, esteem
(B) Belongingness, esteem, safety, physiological, self-actualization
(C) Physiological, belongingness, safety, esteem, self-actualization
(D) Physiological, safety, belongingness, esteem, self-actualization
- 1912.** Central traits
- (A) Were suggested by a behaviorist to explain behavior
(B) Organize and control behavior across many situations
(C) Are more situations specific than they are secondary traits
(D) Are seen in growth-oriented individuals
- 1913.** Norms are influenced by
- (A) The surrounding culture
(B) The surrounding sub-culture
(C) The context of the situation
(D) All of the above
- 1914.** Discrepancies between attitudes lead to
- (A) Prejudice
(B) Cognitive dissonance
(C) The fundamental attribution error
(D) Relative deprivation
- 1915.** Many of the neo Freudians have modified Freud's original theory of personality to give more emphasis to
- (A) Childhood sexuality
(B) Social and cultural factors
(C) Differences between the sexes
(D) Unconscious forces

- 1916.** Motivation is defined as inferred process within a person that causes organism to move toward
(A) A goal
(B) All other organisms
(C) A homeostatic state
(D) Higher levels in the hierarchy of needs
- 1917.** In which motivational conflicts are must chosen by “the lesser of two evils”
(A) Approach-Approach
(B) Approach-Avoidance
(C) Avoidance-Avoidance
(D) Multiple approach-avoidance
- 1918.** Which of the following identified universal symbolic images that appear in myths, art, dreams, and other expressions of the collective unconscious?
(A) Freud (B) Adler
(C) Jung (D) Horney
- 1919.** Raymond B Cattell advanced the study of personality traits by using a statistical method called
(A) Locus of control
(B) Identification
(C) Factor analysis
(D) Phrenology
- 1920.** Perfection is a problem that is most associated with the
(A) Id (B) Ego
(C) Superego (D) Ego and superego
- 1921.** Which of the following represent correct sequence?
(A) Zygote, fetus, embryo, neonate, infant
(B) Zygote, embryo, neonate, fetus, infant
(C) Embryo, zygote, fetus, neonate, infant
(D) Zygote, embryo, fetus, neonate, infant
- 1922.** Maturation refers to development that occurs as a result of
(A) Genetically determined signals
(B) Instinctive process
(C) Homeostatic imbalances
(D) An interaction between biology and environment
- 1923.** Which of the following is not one of the aspects of development studied by psychologists?
(A) Universal aspects of development
(B) Individual differences in development
(C) Cultural differences in development
(D) None of these
- 1924.** Which of the following is most likely to lead to high job satisfaction?
(A) Matching personality with occupation
(B) Promotion practices based on seniority
(C) Having an out going personality
(D) Working alone
- 1925.** Which of the following improves worker skills and reduces frustration and stress?
(A) Job training
(B) Job analysis
(C) Access to an employee assistance program
(D) An organizational culture
- 1926.** The assessment tools upon which clinicians depends gather information about their clients belong in which of the following categories.
(A) Interviews (B) Tests
(C) Observations
(D) All of these choices
- 1927.** Which of the following tests is likely to have the highest reliability?
(A) The TAT (B) The Rorschach
(C) The MMPI
(D) The Draw-a-Person test
- 1928.** Clinicians attempting to systematically assess the abnormal behavior of a client will frequently
(A) Rely primarily on personality tests
(B) Employ a battery of tests
(C) Rely primarily on projective tests
(D) Rely primarily on neuropsychological tests
- 1929.** The best definition of psychotherapy includes
(A) A sufferer, a healer and a systematic interaction between the two

- (B) A healer that uncovers unconscious conflicts and heals the personality
(C) The removal of symptoms about which the patient complains
(D) The use of talking and drugs about which the patient complains
- 1930.** Psycho dynamic therapists believe that maladaptive behavior stems from
(A) Disturbed thought patterns
(B) Inappropriate learning patterns
(C) A failure to live up one's full potential
(D) Emotional trauma experienced in childhood
- 1931.** Which of the following therapists make intensive use of role playing?
(A) Client-centered therapists
(B) Psychoanalytic therapists
(C) Existential therapists
(D) Gestalt therapists
- 1932.** Client-centered therapy is most consistent with the
(A) Behavioral approach
(B) Cognitive approach
(C) Psychodynamic approach
(D) Humanistic approach
- 1933.** Illnesses that seem to result from an interaction of physical and psychological factors are called
(A) Hysterical (B) Psychosomatic
(C) Somatic
(D) Conversion disorders
- 1934.** Chromosomal mapping studies have found that
(A) Schizophrenia does not appear to have genetic components
(B) Schizophrenia does appear to have a genetic component
(C) Adoption studies do not reveal anything about schizophrenia
(D) There is a single cause of schizophrenia
- 1935.** The leading cause of mind retardation is believed to be
(A) Fetal alcohol syndrome
(B) Organic brain syndrome
(C) Environmental factors
(D) Inherited traits
- 1936.** Therapists who focus on altering faulty thought process are
(A) Cognitive therapists
(B) Humanistic therapists
(C) Biological therapists
(D) Psychodynamic therapists
- 1937.** In general children who score well on IQ tests have parents who
(A) Pay for tutors and special schools
(B) Have very high IQs
(C) Punish them if they fail to do it well
(D) Spend time with them and actively encourage their development
- 1938.** One hallmark of projective tests is that they
(A) Are multiphase
(B) Are criterion referenced
(C) Require responses to ambiguous stimuli
(D) Measure only one trait
- 1939.** When psychological tests are used in personal selection, an important concern regarding the tests is
(A) Cost
(B) Ease of administration
(C) Validity
(D) Objectivity
- 1940.** A conversion disorder is characterized by
(A) Functional impairment of a limb or sensory ability with no apparent physical cause
(B) A constant fear of becoming seriously ill
(C) Frequently vague complaints of physical symptoms
(D) None of these
- 1941.** By the perceptual organization of stimuli the person's span of attention is
(A) Increased (B) Decreased
(C) Divided (D) Unaffected
- 1942.** Figural after-effects pertain to distortion
(A) In memory of figures
(B) Caused by ambiguity of figures
(C) Caused by prolonged fixation of figures
(D) Caused by oscillation of attention

- 1943.** One of the following is not a monocular depth cue
(A) Interposition
(B) Relative moment
(C) Convergence
(D) Accommodation
- 1944.** The cerebellum and cerebrum are
(A) Functionally independent
(B) Bilaterally symmetrical
(C) Found in the mid-brain
(D) In front of the central fissure
- 1945.** Factors predisposing an individual towards feeling of jealousy include
(A) Lack of intensity of feeling of love
(B) Insecurity of self-conception
(C) Self-hate
(D) All of these
- 1946.** Copying behavior is
(A) Often unconscious
(B) A form of habitual behavior
(C) Always goal-directed
(D) Usually carried out with minimum attention
- 1947.** Self-assertive behavior is an example primarily of
(A) An abundance motive
(B) A stimulation motive
(C) A survival motive
(D) A security motive
- 1948.** The concept of homeostasis most closely associated with the name of
(A) Richter (B) Cannon
(C) Levin (D) Warden
- 1949.** The approach-approach conflict involves
(A) A stable equilibrium
(B) An unstable equilibrium
(C) An oscillation of (A) and (B)
(D) Neither (A) nor (B), (C)
- 1950.** The two parts of autonomic nervous system are
(A) Voluntary and involuntary
(B) Cerebrospinal and cortical
(C) Thalamus and hypothalamus
(D) Sympathetic and parasympathetic
- 1951.** Which one of the following is not the law of perceptual grouping?
(A) Proximity
(B) Similarity constancy
(C) Constancy (D) Continuation
- 1952.** In Pavlovian conditioning the learned response is
(A) CS (B) CR
(C) UCS (D) UCR
- 1953.** Thorndike's "Law of Effect" was an early form of the present day principle of
(A) Extinction
(B) Contiguity
(C) Reinforcement
(D) Trial and error
- 1954.** When new acquisition improves retention of the old, we speak of
(A) Positive transfer
(B) Negative transfer
(C) Reproductive facilitation
(D) Reproductive interference
- 1955.** Of the following one is not a measure of retention
(A) Recognition (B) Recall
(C) Relearning (D) Reminiscence
- 1956.** Typically the curve obtained in discrimination learning in animals is
(A) Positively accelerated
(B) Linearly accelerated
(C) Negatively accelerated
(D) S-shaped
- 1957.** Development of any skill depends intimately on
(A) Muscle facility
(B) Feedback
(C) Abstractions
(D) Kinesthetic stimulation
- 1958.** The theory of personality in which "basic anxiety" is the primary concept was associated with the name
(A) Fromm
(B) Sullivan
(C) Horney
(D) None of the above
- 1959.** In the perception of people, feedback will serve to
(A) Correct an incorrect initial perception
(B) Confirm an accurate initial perception

- (C) Stabilize and inaccurate initial perception
(D) Nay of the above would be true
- 1960.** The essential feature of client-centered therapy is
(A) The giving of reassurance to the client
(B) The facilitating the client to arrive at his own insight about his problem
(C) Its directive character
(D) It's emphasis upon searching for deep unconscious aspects of the problem
- 1961.** Studies on maturation and learning have indicated
(A) That environmental stimulation may effect growth
(B) That some factor may affect normal development
(C) That the two follow independent course
(D) Both (A) and (B) above
- 1962.** The over-all bodily and mental curves for man
(A) Diverge at the age of puberty
(B) Run parallel course
(C) Show a drop at the age of puberty
(D) Both (B) and (C)
- 1963.** Psychosomatic disorders are best illustrated by
(A) Asthma caused by anxiety
(B) Neurotic anxiety
(C) Depression caused by tuberculosis
(D) Over-eating
- 1964.** Which one of the following is not listed as classification of psychoneurosis?
(A) Phobic reactions
(B) Conversion reactions
(C) Delusional reactions
(D) Neurasthenic reactions
- 1965.** So-called "functional" and "organic" psychoses are
(A) Really both functional in origin
(B) Really both organic in origin
(C) Clearly differentiated as to the origin
(D) Not clearly distinguishable
- 1966.** Psychodrama is one of the psychotherapeutic techniques which basically involve situations representing
(A) Ego blocking
(B) Superego blocking
(C) Primary ego
(D) All of the above
- 1967.** Psychodynamic therapist believes that maladaptive behavior stems from
(A) Emotional trauma experienced in childhood
(B) A failure to live up to one's full potential
(C) Disturbed thought patterns
(D) Inappropriate learning patterns
- 1968.** The semantic differential scale is used mainly to measure
(A) Values (B) Opinions
(C) Attitudes (D) Interests
- 1969.** While selecting a psychological test for personnel selection, main concern regarding the test is
(A) Cost
(B) Ease of administration
(C) Validity
(D) Objectively
- 1970.** The concept of introversion and extroversion was advanced by
(A) Spranger (B) Jung
(C) Kretchmer (D) Jeansch
- 1971.** Emergency reactions refer to
(A) Individual's attempt to cope with an emergency
(B) Bodily changes accompanying emotions of anger and fear
(C) Feeling of startle when suddenly stimulated
(D) Ways that the individual seeks to achieve emotional control
- 1972.** Studies show that an "authoritarian" family tends to produce an adolescent who is
(A) Dependent and obedient
(B) Independent but reserved
(C) Surface-compliant but rebellious underneath
(D) Self reliant

- 1973.** The non-directive approach characterizes
(A) Psychoanalysis
(B) Behavior modification
(C) Client-centered therapy
(D) Hypnosis
- 1974.** Child-rearing practices in Pakistan
(A) Differ little from those in other countries
(B) Differ from one social class to another
(C) Have changed very little over past 50 years
(D) Are now pretty much the same from one social class to the next
- 1975.** In Maslow's theory of personality the main concept is that of
(A) Functional autonomy
(B) Acquired drives
(C) Interpersonal relations
(D) Self-actualization
- 1976.** According to Erickson's theory, the struggle during adolescence is
(A) Intimacy vs. isolation
(B) Initiative vs. guilt
(C) Identity vs. confusion
(D) Competence vs. inferiority
- 1977.** According to clinical psychologists unconscious is
(A) A construct (B) A place
(C) A thing (D) An observation
- 1978.** Disagreeing with Freud, Erickson emphasized more on
(A) Conflict (B) Racial memory
(C) Adulthood (D) Early habits
- 1979.** The prototype of the modern intelligence test for children was developed by
(A) Terman (B) Wechsler
(C) Binet (D) Stanford
- 1980.** Thurston's method of attitude scaling, the key factor is the use of
(A) Numbers (B) Judges
(C) Reinforcement
(D) Positive and negative items
- 1981.** The term "Phenomenology" refers to
(A) The science of perception
(B) The scientific study of "queer" occurrences
(C) The "naïve" description of our immediate experiences
(D) None of the above
- 1982.** The degree of concentration or dilution of a colour is known as its
(A) Hue (B) Saturation
(C) Volume (D) Brightness
- 1983.** Difference in perception among species depends upon
(A) Habitat
(B) Complexity of nervous system
(C) Receptor mechanisms
(D) All of the above
- 1984.** As compared with adult's perception, the child's perception tends to be
(A) Less susceptibility to illusion
(B) Longer in time-perspectives
(C) More fully characterized by constancy
(D) More fully affected by physiognomic properties
- 1985.** As compared with the figure, the ground appears to be;
(A) Better structured
(B) Continuous and unbroken
(C) Better localized
(D) More salient in attention
- 1986.** The concept of adaptation level was introduced by
(A) Brown (B) Wertheimer
(C) Kohler (D) Helson
- 1987.** By the perceptual organization of stimuli the person's span of attention is
(A) Increased (B) Decreased
(C) Devided (D) Unaffected
- 1988.** The cue of accommodation is especially effective for
(A) Long distances
(B) Short distances
(C) Moving objects
(D) All of the above
- 1989.** The area of the brain where all the fibers from the skin senses get connected before entering cerebral cortex is the

- (A) Cerebellum
 - (B) Spinal Cord
 - (C) Thalamus
 - (D) Reticular formation
- 1990.** The so called primary emotions include
- (A) joy, fear, anger, grief
 - (B) pain, fear, hate, love
 - (C) shame, guilt, anger, grief
 - (D) all of the above
- 1991.** Factors predisposing an individual towards feeling of jealousy include
- (A) Insecurity in self-conception
 - (B) Lack of intensity of feelings of love
 - (C) Self hate
 - (D) All of the above
- 1992.** Behaviour episodes are
- (A) Organized sequences of behaviour
 - (B) Characterized by a constant direction of action
 - (C) Often overlapping
 - (D) All of the above
- 1993.** Self-assertive behaviour is an example primarily of
- (A) a stimulation motive
 - (B) a security motive
 - (C) an abundance motive
 - (D) a survival motive
- 1994.** The emergency-control branch of human nervous system is
- (A) Sympathetic
 - (B) Parasympathetic
 - (C) Cerebrospinal
 - (D) Ventromedial
- 1995.** The "local" stimulus theory of hunger is associated with the name of
- (A) Anderson (B) Beach
 - (C) Cannon (D) Helson
- 1996.** Examples of ductless glands are
- (A) Gonads, salivary glands, adrenals
 - (B) tear glands, salivary glands, gonads
 - (C) Thyroid, pituitary, tear gland
 - (D) Adrenal, thyroid, gonads
- 1997.** In conditioning studies CR and UR
- (A) are usually, but not always, identical
 - (B) are rarely identical
 - (C) Have not similarity
 - (D) Are always identical
- 1998.** A curve which shows an overall negative slope may show
- (A) negative acceleration
 - (B) positive acceleration
 - (C) linear acceleration
 - (D) any one of the above
- 1999.** The development of any kind of skill depends intimately upon;
- (A) Muscle facility
 - (B) Abstract knowledge
 - (C) Feedback
 - (D) Kinesthetic stimulation
- 2000.** Aphasia seems to be correlated with lesions to the
- (A) Temporal lobes of brain
 - (B) Motor Cortex of brain
 - (C) Cerebellar cortex of brain
 - (D) Frontal lobes of brain
- 2001.** The four main goals of psychological research are
- (A) Description, experimentation, prediction and control
 - (B) Description, explanation, assessment and manipulation
 - (C) Description, prediction, assessment and manipulation
 - (D) Description, explanation, control and validity
- 2002.** A verifiable scientific finding is one that can be
- (A) Observed in real life
 - (B) Proven impossible to test
 - (C) Proven valid, accurate and indisputable
 - (D) Replicated with the same result
- 2003.** An independent variable is
- (A) The only variable of interest
 - (B) A variable that is independently verified
 - (C) A variable whose value depends on that of the dependent variable
 - (D) The variable that is manipulated by experiment
- 2004.** The main regions of brain are the
- (A) Hind brain, forebrain, midbrain

- (B) Cerebral cortex, forebrain, hind brain and midbrain
 - (C) Forebrain, midbrain and anterior region
 - (D) Central hemisphere, and left hemisphere
- 2005.** Of the three different types of the neurons, sensory neurons are unique in that they
- (A) Carry impulses from the brain and spinal cord to parts of the body that then respond to the impulse.
 - (B) Receive information from the receptor cells and send this information to the brain or spinal cord.
 - (C) Reside exclusively in the peripheral nervous system has two functions, to send signals to motor neurons, and to receive signals from receptors.
 - (D) None of these
- 2006.** The four parts of the neurons are
- (A) Axon, cell body, myelin sheath and dendrites
 - (B) Axon, dendrites, nucleus and terminal buttons
 - (C) Axon, cell body, dendrites and terminal buttons
 - (D) Axon, dendrites, myelin sheath and nodes of Ranvier
- 2007.** Effectors and receptor cells are different in
- (A) Their composition and speed of conduction
 - (B) The direction in which they carry nerve impulses
 - (C) The corresponding branch of the nervous system to which they belong
 - (D) The effectors never connect with the brain, while receptors receive information from the brain.
- 2008.** Three dimensional perception of the world is referred to as
- (A) Stereopsis (B) Stroboscopy
 - (C) Multiple perspectives
 - (D) None of the above
- 2009.** Perceptual constancy refers to
- (A) Perception of an object remaining the same even when our immediate sensation of the object changes.
 - (B) Perceptual tendency to group objects together on the basis of their similarity.
 - (C) Perception of an object's changing even though the object stays the same
 - (D) Tendency to close up, incomplete objects into already existing perceptual shapes
- 2010.** Gestaltists refer to the fact that we perceive smooth flowing forms more readily than discrete forms as
- (A) Similarity
 - (B) Symmetry
 - (C) Holistic perception
 - (D) Continuity
- 2011.** Classical conditioning is a process whereby
- (A) Certain response result from certain stimulus presentations
 - (B) The CS is always presented before the UCS
 - (C) Actions that are rewarded tend to be strengthened
 - (D) Learning occurs when a stimulus is paired with a certain response
- 2012.** Edward Tolman's experiments with rats showed that
- (A) Learning could be achieved without reinforcement
 - (B) Rats could learn second order conditioning in the absence of food
 - (C) All rats behaviour could be analyzed by stimulus – response contingencies
 - (D) None of the above
- 2013.** Albert Bandura's social learning theory holds that
- (A) Learning is best in group settings
 - (B) Learning can be achieved by observing and modeling another person's behaviour
 - (C) Mere exposure to an aggressive scene always later elicits aggressive behaviour
 - (D) Social learning takes place at an unconscious level
- 2014.** According to Maslow's hierarchy of needs, individuals

- (A) May bypass a level in their quest for self actualization
(B) Must satisfy self-esteem needs before achieving belongingness and love needs
(C) Always end their lives self actualized
(D) None of the above
- 2015.** Research by Edward Deci has suggested that extrinsic motivators are more effective when they are
(A) Stated up front so that the individual knows what he or she will be receiving
(B) Tangible and easily identifiable
(C) Received immediately before the task is accomplished
(D) Intangible, such as verbal praise or smile
- 2016.** Goals can help us motivated by all the following except
(A) Focusing our attention
(B) Encouraging alternative ways of achieving goals
(C) Allowing us the freedom to be spontaneous in our activities
(D) None of the above
- 2017.** Psychodynamic determinism refers to
(A) Behaviour that is ruled by forces over which we have no control
(B) Behaviour that is preconscious in origin
(C) Id impulses that will forever remain unfulfilled
(D) The delimiting characteristics of the superego
- 2018.** According to Freudian, dream terminology, condensation refers to
(A) Repressed urges that find disguised outlets for expression
(B) The bizarre, irrational quality of dreams
(C) The process whereby unacceptable thoughts or impulses are combined into a single dream image
(D) The process whereby one thing may stand for another in dream interpretation
- 2019.** Group popularization is an affect whereby
(A) Extreme opinions in a group become moderate with increased group interaction
(B) Group members moderate positions become polarized with increased group interaction
(C) Group decisions represent an exaggeration of the group members initial positions
(D) The group members' main concern is to maintain unanimity and group cohesion
- 2020.** Groupthink is unlikely to occur when which of the following factors is present?
(A) There is a high degree of stress in the decision-making process
(B) There are clear objectives in the decision making process
(C) The group is homogeneous in composition
(D) The group is ideologically isolated
- 2021.** The study of the hereditary transmission of attributes is
(A) Heritability, biological traits
(B) Genetics, genes
(C) Adaptability, dominant and recessive traits
(D) Mutation, differentiation
- 2022.** A male born with an extra Y chromosome, XYY, shows some evidence of increased aggressiveness and typically scores lower than normal on intelligence tests
(A) Turner's syndrome
(B) Supermale Syndrome
(C) XXX syndrome
(D) Klinefelter's syndrome
- 2023.** The material of which chromosomes are composed is called
(A) Deoxyribonucleic acid
(B) Genes
(C) Atoms
(D) Myelin
- 2024.** Maturation refers to
(A) The attainment of successive stages of cognitive development.

- (B) Relatively stable changes in an individual's thought and behavior or a result of biological processor of aging.
- (C) Relatively stable changes in an individual's thought and behavior as a result of accumulating experience.
- (D) The development of an individual thought and behavior due to the interactions of biological and environment
- 2025.** According to one definition, behavior is abnormal if it is
- (A) Labeled as abnormal by the society in which the individual lives
- (B) Not under conscious control by the individual
- (C) Statistically typical
- (D) None of the above
- 2026.** Anxiety disorders may be characterized by
- (A) Psychotic symptoms
- (B) Low affect
- (C) Telegraphic thoughts
- (D) Somatic symptoms
- 2027.** Two examples of mood disorders are
- (A) Major depression and generalized anxiety disorder
- (B) Obsessive compulsive disorder and manic-depressive disorder
- (C) Major depression and dipolar disorder
- (D) Major depression and obsessive compulsive disorder
- 2028.** An individual cannot remember events during an immediately following a serious airplane crash
- (A) Dissociative amnesia
- (B) Dissociative fugue
- (C) Retrograde amnesia
- (D) Fractured amnesia
- 2029.** A type of schizophrenia characterized by hallucinations, delusions, and diffuse psychological organization is referred to as
- (A) Undifferentiated
- (B) Disorganized
- (C) Residual
- (D) Fragmented
- 2030.** An individual who is identified as having a borderline personality disorder shows
- (A) Persecutory thoughts
- (B) Exhibitionistic tendencies
- (C) Instability in mood and social relations
- (D) Apathy and indifference to the opinions of others
- 2031.** Which symptoms is not typically characteristic of an autistic child?
- (A) Impairment in communication
- (B) Hyperactivity
- (C) Restricted range of interest
- (D) Social isolation
- 2032.** The Stanford-Binet, WAIS-R and WISC are known to have very high reliability. This means that
- (A) A pretest has been given to a representative
- (B) The test yields consistent results, for example-retesting
- (C) The test measures what it is supposed to measure
- (D) The result of the test will be distributed on a bell shaped curve
- 2033.** The experience that has the clearest, most profound effect on intellectual development
- (A) Being enrolled in Head Start Program
- (B) Growing up in an economically disadvantaged home or neighbourhood
- (C) Being raised in a very neglected home
- (D) Being exposed to very stimulating toys and lessons in infancy
- 2034.** A therapist who encourages clients to relate their dreams and searches its roots is
- (A) Psychoanalysis
- (B) Humanistic therapy
- (C) Person-centered therapy
- (D) Cognitive therapy

- 2035.** The most enthusiastic or optimistic view of Psychotherapy comes from
(A) Outcome research
(B) Psychologist Hans Eyesenck
(C) Reports of clinicians
(D) None of the above
- 2036.** An approach that seeks to alleviate poverty and to other stresses that put people at high risk for developing psychological disorders is
(A) Biomedical therapy
(B) The humanistic approach
(C) Empathy and active listening
(D) None of the above
- 2037.** If a person washes his hands 100 times a day, the behavior will be labeled disordered because it is
(A) Unjustified and maladaptive
(B) An atypical behavior
(C) Not explained by the medical model
(D) Harmful and disturbing to others
- 2038.** The DSM-111-R has improved reliability because it helps mental health workers base their diagnosis on
(A) A few well-defined categories
(B) In-depth history of the patient
(C) The patient's observable behaviour
(D) The theories of Pinel, Freud and others
- 2039.** An episode of intense dread, which is typically accompanied by feelings of terror and dizziness, chest pain etc. is called
(A) Generalized or chronic anxiety
(B) A social phobia
(C) A panic attack
(D) An obsessive fear
- 2040.** Amnesia, fugue and multiple personality involve gaps in awareness, for example, sudden loss of memory or change of identity. These Psychological disorders are called
(A) Anxiety disorders
(B) Dissociative disorders
(C) Mood disorders
(D) Memory disorders
- 2041.** The three basic components of prejudice are Effective
(A) Internal and External
(B) Real and Imagined
(C) Descriptive and Prescriptive
(D) Cognitive and Behavioral
- 2042.** Which of the following is an example of extrinsic motivation?
(A) Doing a good job because of the sense of satisfaction when you are finished
(B) Doing a good job in order to get a pay raise
(C) Working hard because of the pride you take in your work
(D) Doing a good job because you enjoy your work
- 2043.** Sources of frustration include
(A) Environmental lacks
(B) Environmental obstacles
(C) incompatible motives
(D) All of these
- 2044.** Which of the following is NOT a characteristic of perception?
(A) Perception is an active process
(B) Perception is a highly selective process
(C) Perception is unaffected by motivation
(D) Perception is so automatic that we are not aware of it
- 2045.** The first task in psychology is to carefully observe and objectively describe
(A) Animals
(B) Behaviour
(C) Social Institution
(D) Symptoms of abnormality
- 2046.** Homeostasis is maintained by the _____ which also plays an important role in emotions.
(A) Hypothalamus
(B) Thalamus
(C) Limbic System
(D) Reticular Activating System
- 2047.** Some of the fibers from each eye cross over to the opposite brain hemisphere at the

- (A) Optic Chaisma
(B) Optic Nerve
(C) Blind Spot (D) Fovea
- 2048.** Which one of the following is not a dimension of the auditory experience?
(A) Loudness (B) Pitch
(C) Saturation (D) Timbre
- 2049.** The best studied perceptual constancies include
(A) Brightness, focus, angle
(B) Shape, location, size
(C) Colour, texture, content
(D) Size, age, weight
- 2050.** Behaviour according to Operant conditioning theory is controlled by
(A) Its consequences
(B) Free will
(C) Knowledge
(D) Instinct
- 2051.** Experts generally agree that the primary emotions include
(A) Fear, Anger and Joy
(B) Disgust, anger and shame
(C) Love, joy and sadness
(D) Fear, anger and hatred
- 2052.** According to Freud, most of the things, we do, think and feel are caused by
(A) Conditioned response to our surroundings
(B) Unconscious forces of which we are not aware
(C) A strong desire to grow and develop
(D) The beliefs and ideas we have about the world
- 2053.** One of the factors Alfred Adler believed to be important in shaping personality is
(A) Difficulty with toilet training
(B) Birth order in the family
(C) Unconditional positive regard
(D) The collective unconscious
- 2054.** Which of the following orders of Maslow's hierarchy of needs is correct
(A) Safety, physiological, belongingness, self actualization, esteem
(B) Belongingness, esteem, physiological, safety, self-actualization
(C) Physiological, belongingness, safety, esteem, self-actualization
(D) Physiological, safety, belongingness, esteem, self-actualization
- 2055.** The statement that psychology is empirical means
(A) Psychology is different from other sciences
(B) The methods of psychology are based on controlled experiments and on observations made with great precision and objectivity
(C) Scientific psychology originated in Greek and roman ages
(D) None of the above
- 2056.** Many of the Neo-Freudians have modified Freud's original theory of personality to give more emphasis to
(A) Child sexuality
(B) Social and cultural factors
(C) Difference between sexes
(D) Unconscious forces
- 2057.** Norms can be best defined as
(A) Standards and expectations shared by members of a society
(B) rules of law
(C) Statistically normal behaviour
(D) Universal rules of conduct
- 2058.** The kind of conflict experienced by a student who simultaneously wants to watch a late night movie on television and get a good night's sleep is best described as
(A) Approach-avoidance
(B) Double approach-avoidance
(C) Approach-Approach
(D) Avoidance-Avoidance
- 2059.** Axon and dendrites are part of a
(A) Myelin Sheath
(B) Cell body
(C) Neuron (D) Node
- 2060.** Motion pictures are based on
(A) The gamma phenomenon
(B) The principle of illusionary circle
(C) The autokinetic illusion
(D) Stroboscopic vision

- 2061.** According to Alfred Adler, helplessness of the infant accounts for
(A) feelings of inferiority
(B) compensatory strivings
(C) A need to strive for superiority
(D) All of these
- 2062.** Jean Piaget's theory of cognitive development states
(A) Children are egocentric during the stage of pre operational thought
(B) Object permanence is developed during stage of concrete operational thought
(C) When children proceed from the stage of pre operational thought to the stage of formal operations, they achieve sense of conservation and reversibility
(D) Child development is largely culturally determined
- 2063.** Autonomous functions of the ego include all of the following except
(A) Perception (B) Repression
(C) Language (D) Intelligence
- 2064.** Which of the following statement applies to the unconscious?
(A) Its elements are inaccessible to the conscious
(B) Its closely related to the pleasure principle
(C) Its closely related to instincts
(D) All of these
- 2065.** A gambler lost 1000 rupees at a race-track but he was not upset. He said, "I would have spent the money on any other thing any way". Which defense mechanism does his statement involve?
(A) Reaction formation
(B) Undoing
(C) Denial (D) Rationalization
- 2066.** Fixed false belief based on incorrect inferences about external reality is called
(A) Delusion (B) Obsession
(C) Overvalued Idea
(D) Phobia
- 2067.** A simple schizophrenia is characterised by
(A) An insidious loss of drive and ambition
(B) Persistent Delusions
(C) Persistent Hallucinations
(D) All of these
- 2068.** Which of the following instances suggest psychotherapy would be of no use?
(A) A person who completely lacks remorse for past doing
(B) A person with an inability to form emotional attachment
(C) A person whose life situation cannot be modified
(D) A person who is a close friend of the analyst
- 2069.** The most frequently cited characteristic among children with ADHD is
(A) Perceptual-motor impairment
(B) Disorder of memory and thinking
(C) Hyperactivity
(D) Disorder of speech and learning
- 2070.** In operant conditioning
(A) Continuous reinforcement schedule is least susceptible to extinction
(B) Negative reinforcement is a type of punishment
(C) The process is related to trial and error learning
(D) Shaping occurs when responses are coincidentally paired to a reinforcer
- 2071.** Ashi memorized her shopping list. When she got to the store she found out that she had forgotten many of the items in the middle of the list. This is an example of
(A) Inappropriate coding
(B) Retrograde Amnesia
(C) Proactive Interference
(D) The serial position effect
- 2072.** Positive reinforcement _____ the rate of responding; Negative reinforcement _____ the rate of responding.
(A) Increases; decreases
(B) Decreases; increases
(C) Increases; increases
(D) Decreases; decreases
- 2073.** Hormones are to endocrine system as _____ are to nervous system.

- (A) Nerves (B) Synapses
(C) Neurotransmitters
(D) Action Potential
- 2074.** The thalamus can be characterised as
(A) A regulatory system
(B) The conscious switch of the brain
(C) A relay system
(D) A bridge between two cerebral hemispheres
- 2075.** Chomsky proposed that children learn a language
(A) One's thought determines the nature of one's language
(B) Language and thought are separate and independent processes
(C) Language and thought interact, each influencing the other
(D) Because they possess an innate language acquisition device
- 2076.** According to James-Lange Theory of Emotion
(A) The experiences of emotions depends upon autonomic arousal and on one's cognitive interpretation of that arousal
(B) Different patterns of autonomic arousal lead to the experience of different emotions
(C) Emotions occur when the thalamus sends signals simultaneously to the cortex and to the autonomic nervous system
(D) Emotions develop because of their adaptive value
- 2077.** When the development of the same subject is studied over a period of time, the study is called a
(A) cross sectional study
(B) longitudinal study
(C) life history study
(D) sequential study
- 2078.** The process of excluding from the consciousness the ideas and feelings which cause guilt and shame is known as
(A) Projection (B) Repression
(C) Reaction formation
(D) Regression
- 2079.** The type of learning associated with Skinner is
(A) Classical Conditioning
(B) Operant Conditioning
(C) Respondent Conditioning
(D) Observation Learning
- 2080.** The English Scientist who considered measuring intelligence by measuring head size
(A) Galton (B) Binet
(C) Terman (D) Wechsler
- 2081.** In psychology, case studies are used to
(A) draw conclusions, about individual behavior on the basis of group finding
(B) draw general conclusions about behaviour of the client
(C) assess heritability of individual
(D) show importance of case studies.
- 2082.** Psychology is concerned with the study of
(A) how group exchange resources
(B) overt behavior only
(C) mind and behavior
(D) how aggregate behavior is specified
- 2083.** An independent variable is
(A) the only variable of interest
(B) a variable that is independently verified
(C) a variable whose value depends on dependent variable
(D) the variable that is manipulated by experimenter
- 2084.** Introspection is
(A) a scientific method used to study dreams
(B) a psychotherapeutic technique where the patient gives a running account of internal dialogues
(C) a process used to uncover symbolic representation in dream
(D) the analysis of the content of our own thoughts.
- 2085.** The law of effect states that
(A) close temporal contiguity is the key characteristic in forming an association between stimulus and response.

- (B) A response will be strengthened if the organism is rewarded for that response
- (C) effect is strong if the effect is long lasting.
- (D) Response generalize across similar surroundings.
- 2086.** Structuralism and functionalism are similar in that both
- (A) emphasize the analysis of mental structures to explain behavior
- (B) emphasize the fluid, personal nature of consciousness.
- (C) Regard psychology as the science of conscious experience.
- (D) Emphasize the observation of behavior rather than consciousness.
- 2087.** The school of psychology that argues that nearly all behaviour is a result of conditioning and that the environment shapes behavior is labeled
- (A) getalt, wertheimer
- (B) psychoanalytic, freud
- (C) behaviorism, tichner
- (D) behaviorism, Watson
- 2088.** Chemical which are produced in the brain, and facilitate communication between nerve cells are called
- (A) neurotransmitter
- (B) plasmas
- (C) syntactic structures
- (D) adrenocorticoids
- 2089.** Psychologists, who typically apply psychological principles to diagnose and treat emotional problems, including illness and marital and family conflict are in the subfield called
- (A) biological
- (B) clinical and counseling
- (C) social and personality
- (D) school and educational
- 2090.** The first step in any research project is generating a
- (A) theory
- (B) experiment
- (C) hypothesis
- (D) None of the above
- 2091.** The discipline that deals with sampling data from a population and then drawing inferences about the population from the sample is
- (A) hypothesis testing
- (B) statistics
- (C) multivariate experimentation
- (D) independent sampling
- 2092.** When two variables are correlated, variation in one?
- (A) may possibly cause variation in the other.
- (B) Can only be inferred with social validation
- (C) Can be cause of variation in the other only if the correction is positive.
- (D) Always suggests a causal relationship with the other.
- 2093.** In the early stages of research, the most efficient way of making progress toward explaining phenomenon is to
- (A) videotape
- (B) precisely measure
- (C) survey
- (D) observe
- 2094.** A scholarly summary of a body of research on some topic is called
- (A) case history
- (B) literature review
- (C) meta-analysis
- (D) theory
- 2095.** The principle governing the ethical treatment of human participants in research essentially states that the hazards anticipated in the research project should be not grater than those encountered in the daily activities.
- (A) informed consent
- (B) right to privacy
- (C) minimal risk
- (D) informed participants
- 2096.** S-R psychology is an approach associated with the perspective
- (A) behavioral (B) psychoanalytic
- (C) cognitive
- (D) subjectivist

- 2097.** Psychologists who are interested in the perception of motion, part-whole relationship and in how people judge size were identified with what school of psychology?
 (A) psychoanalytic
 (B) gestalt
 (C) behavioral
 (D) structural
- 2098.** The center of Freud's theory is the thoughts, attitudes, impulses, wishes and motivations of which we are unaware
 (A) conditioning concept
 (B) unconsciousness
 (C) biological motivation proposition
 (D) multivariate construct
- 2099.** Relating overt behavior to electrical and chemical events taking place inside the body is characteristic of which contemporary perspective?
 (A) behavioral (B) subjectivist
 (C) biological (D) cognitive
- 2100.** The subjectivist perspective in psychology
 (A) is most like cognitive one
 (B) is generally not influenced by cultural and emotional variables
 (C) has been most pervasive
 (D) all of these
- 2101.** A child is creating new schemata to account for new information, Piaget calls this process
 (A) assimilation
 (B) accommodation
 (C) operations
 (D) functional autonomy
- 2102.** Children begin to understand and use abstractions during which of the Piaget stages of cognitive development
 (A) oral
 (B) preoperational
 (C) state of formal operations
 (D) secondary stage
- 2103.** Which theoretical viewpoint emphasizes the importance of learning for understanding development
 (A) Freudian
 (B) information processing theory
 (C) ecological theory
 (D) behavioral theory
- 2104.** Behavior is governed by its consequences describe
 (A) operant conditioning
 (B) psychological crisis
 (C) supply demand theory
 (D) the Yerkes-Dodson law
- 2105.** Which of the following statements is incorrect
 (A) genetic influences affect children more than adults
 (B) even if genetic factors underlie a particular behavior, it is subject to change
 (C) traits such as extroversion and introversion are influenced by genes
 (D) intelligence has a genetic basis
- 2106.** If a characteristic requires only one gene to show itself the characteristic is considered
 (A) dominant
 (B) co-dominant
 (C) incompletely dominant
 (D) recessive
- 2107.** A genetic disorder in which the production of mucus affects the respiratory system is
 (A) Huntington's disease
 (B) Down syndrome
 (C) cystic fibrosis
 (D) sickle cell anemia
- 2108.** Aptitudes are
 (A) preferences to perform certain activities
 (B) high level of achievement in an area of endeavour
 (C) the abilities of an individual to learn in specific endeavour
 (D) skills one brings to a given task
- 2109.** In psychological research, a _____ is defined as an entity that can occur with different values.
 (A) hypothesis
 (B) measurement system

- (C) variable
(D) experiment group
- 2110.** Ability to exercise precise control over a variable is what distinguishes the _____ Method from other methods of scientific observation.
(A) control group identification
(B) randomized selection
(C) hypothesis testing
(D) experimental
- 2111.** Consideration of participants in psychological research as _____ in the research enterprise is a central principle of ethical research today.
(A) informed participants
(B) full partners
(C) willing participants
(D) co-investigators
- 2112.** Maturation refers to
(A) the attainment of successive stages of cognitive development
(B) relatively stable changes in an individual's thought or behavior as a result of a biological process of aging.
(C) Relatively stable changes in an individual's thought or behaviour as a result of accumulating experience
(D) The development of an individual's thought and behavior due to interaction of biological and environmental factors
- 2113.** Learning refers to
(A) the attainment of successive stages of cognitive development
(B) changes in an individual's thought or behavior as a result of biological process of aging
(C) changes in an individual's thought and behavior as result of accumulating experience
(D) The development of an individual's thought and behavior due to interaction of biological and environmental factors
- 2114.** Psychodynamic determinism refers to
(A) behavior that is ruled by forces over which we have no control
(B) behavior that is preconscious in origin
(C) id impulses that will forever remain unfulfilled
(D) the delimiting characteristics of superego
- 2115.** According to Freudian dream terminology, condensation refers to
(A) repressed urges that find disguised outlets for expression
(B) the bizarre, irrational quality of dream
(C) the process whereby unacceptable thoughts or impulses are combined into a single dream image
(D) process where one thing may stand for another in dream interpretation
- 2116.** According to one definition, behavior is abnormal if
(A) labeled as abnormal by the society in which the individual lives
(B) not under conscious control by individual statistically typical
(C) statistically typical
(D) adaptive to the individual
- 2117.** Axis III in DSM-IV address
(A) the major abnormal disorders
(B) primary personality disorders
(C) physical disorders
(D) the severity of psychological symptoms
- 2118.** An individual who is identified as having a borderline personality disorder shows
(A) perceptive thoughts
(B) exhibitionistic tendencies
(C) instability in mood and social relations
(D) apathy and indifference to opinion of others
- 2119.** Clinical interview typically includes
(A) a follow up evaluation and assessment after therapy is terminated
(B) the initial diagnosis of a client's psychological functioning
(C) the psycho physiological functioning
(D) the client's initial and final evaluation

- 2120.** Two explicitly directive psychotherapeutic approaches are
(A) behavior therapy and psychodynamic therapy
(B) behavior therapy and humanistic therapy
(C) humanistic therapy and existential therapy
(D) cognitive therapy and behavior therapy
- 2121.** Mean, Median and Mode are
(A) Measures of deviation
(B) Ways of sampling
(C) Measures of control tendency
(D) None of the above
- 2122.** Research is
(A) Searching again and again
(B) Finding solution to any problem
(C) Working in a scientific way to search for truth of any problem
(D) None of the above
- 2123.** Which of the following is the first step in starting the research process?
(A) Searching sources of information to locate problem.
(B) Survey of related literature
(C) Identification of problem
(D) Searching for solutions to the problem
- 2124.** A common test in research demands much priority on
(A) Reliability (B) Usability
(C) Objectivity
(D) All of the above
- 2125.** Action research means
(A) A longitudinal research
(B) An applied research
(C) A research initiated to solve an immediate problem
(D) A research with socioeconomic objective
- 2126.** A reasoning where we start with certain particular statements and conclude with a universal statement is called
(A) Deductive Reasoning
(B) Inductive Reasoning
(C) Abnormal Reasoning
(D) Transcendental Reasoning
- 2127.** Which of the following variables cannot be expressed in quantitative terms?
(A) Socio-economic Status
(B) Marital Status
(C) Numerical Aptitude
(D) Professional Attitude
- 2128.** The essential qualities of a researcher are
(A) Spirit of free enquiry
(B) Reliance on observation and evidence
(C) Systematization or theorizing of knowledge
(D) All the above
- 2129.** In the process of conducting research 'Formulation of Hypothesis' is followed by
(A) Statement of Objectives
(B) Analysis of Data
(C) Selection of Research Tools
(D) Collection of Data
- 2130.** A research paper is a brief report of research work based on
(A) Primary Data only
(B) Secondary Data only
(C) Both Primary and Secondary Data
(D) None of the above
- 2131.** One of the following is not an open source software
(A) DSpace (B) Windows
(C) Green-stone (D) Linux
- 2132.** Classification of all types of libraries has been made by-
(A) IFLA (B) UNISIST
(C) UNESCO (D) INSDOC
- 2133.** Reading centre's, story hours, exhibitions and reading to literature are forms of
(A) Extension service
(B) Service of a public library
(C) Both (A) and (B)
(D) None of the above
- 2134.** Informal self education is possible in what kind of library?
(A) National Library
(B) Public Library
(C) Specific Library
(D) College Library

- 2135.** _____ is a process of information.
(A) Books (B) CD-ROM
(C) Computers (D) None of these
- 2136.** Feedback mechanism is a part of which service?
(A) Reprography (B) CAS
(C) Translation service
(D) SDI
- 2137.** What is the collection of terms or records in MARC called?
(A) System (B) Network
(C) Website (D) Database
- 2138.** What is Bibliometry?
(A) Function of Library Network
(B) Information Management Service
(C) Information Management Tool
(D) Library Service
- 2139.** Microchip was invented by
(A) Microsoft (B) IBM
(C) DELL (D) Intel
- 2140.** Information is
(A) Raw Data
(B) Processed Data
(C) Input data
(D) Organized data
- 2141.** Conference proceedings are considered as _____ documents.
(A) Conventional (B) Primary
(C) Secondary (D) Tertiary
- 2142.** RSS feed is a tool of
(A) Graphic design (B) Web 1.0
(C) Web 2.0 (D) Architecture
- 2143.** An appropriate source to find out descriptive information is
(A) Bibliography (B) Directory
(C) Encyclopedia (D) Dictionary
- 2144.** One of the following search engine is exclusively meant for scientific information
(A) Google (B) Yahoo
(C) SCIRUS (D) Altavista
- 2145.** Technological Gatekeeper is
(A) A formal method of giving current awareness service
(B) A method of technology assessment and evaluation
(C) A process of transfer of technology
(D) An informal mechanism of keeping user informed of relevant development
- 2146.** The Farmington plan is associated with
(A) Library Legislation
(B) Library Cataloguing
(C) Library Cooperation
(D) Library Indexing Service
- 2147.** UNESCO assisted Model Public Library in India is located at
(A) Kolkata (B) Delhi
(C) Mumbai
(D) Chennai
- 2148.** Mark the 'odd one out'
(A) Cow-Calf principle
(B) Principle of osmosis
(C) Wall picture principle
(D) Whole organ principle
- 2149.** Shelf list facilitates
(A) Classification
(B) Weeding out
(C) Stock verification
(D) Documentation
- 2150.** Questionnaire is a
(A) Research method
(B) Measurement technique
(C) Tool for data collection
(D) Data analysis technique
- 2151.** A periodical evaluation of an employee is done through
(A) Job rotation
(B) Performance appraisal
(C) Refresher course
(D) Work guide
- 2152.** "Controlled Group" is a term used in
(A) Survey research
(B) Historical research
(C) Experimental research
(D) Descriptive research
- 2153.** 'Noise' in Information Retrieval is due to
(A) Precision
(B) Recall
(C) Relevant information
(D) Redundant information

- 2154.** What is the relationship between ISBD and cataloguing codes ?
 (A) They are not related at all
 (B) Cataloguing codes will include bibliographic description
 (C) ISBD includes cataloguing rules
 (D) ISBD can replace cataloguing rules
- 2155.** Tagging in web 2.0 application is called
 (A) Taxonomy (B) Folksonomy
 (C) Syndication (D) Directory
- 2156.** Inductive logic proceeds from
 (A) General to General
 (B) Particular to General
 (C) General to Particular
 (D) Particular to Particular
- 2157.** Which of the following is not a "Graphic representation" ?
 (A) Pie Chart (B) Bar Chart
 (C) Table (D) Histogram
- 2158.** The oldest and the largest Library Association in the world is
 (A) ALA (B) LA
 (C) IFLA (D) IASLIC
- 2159.** Which of the following is not covered under Intellectual Property Rights ?
 (A) Copyrights (B) Patents
 (C) Trade Marks (D) Thesaurus
- 2160.** Ontology is
 (A) An Indexing Method
 (B) Classification of Internet based documents
 (C) Cataloguing of Internet based documents
 (D) Documentation service
- 2161.** Social psychology is the scientific study of how people ____, ____, and ____ one another.
 (A) understand, feel about, act toward
 (B) perceive, think about, act toward
 (C) think about, influence, relate to
 (D) observe, influence, conflict with
- 2162.** Social psychology began to emerge as the vibrant field it is today during
 (A) the depression of the early 1930s when researchers examined the effects of deprivation on aggression and altruism
 (B) world War I when psychologists conducted studies of social conflict and cooperation
 (C) World War II when researchers performed studies of persuasion and soldier morale
 (D) the Korean War when psychologists examined the effects of brainwashing on prisoners of War
- 2163.** In comparison to the sociologist, the social psychologist
 (A) is more likely to study the social causes of behavior
 (B) is more likely to study individuals than groups
 (C) gives less attention to our internal functioning
 (D) relies more heavily on correlational research
- 2164.** In comparison to personality psychology, social psychology
 (A) has a shorter history
 (B) is more concerned with the biological causes of behavior
 (C) is more likely to use case studies in theory development
 (D) has greater concern for differences between individuals
- 2165.** In comparison to personality psychology, social psychology
 (A) focuses more on our common humanity
 (B) has more famous, well-known theorists
 (C) is an older specialty within the discipline of psychology
 (D) provides more "grand" or comprehensive theories of human functioning
- 2166.** According to the text, which word best describes the relationship between different disciplines that study behavior?
 (A) supplementary (B) contradictory
 (C) complementary (D) competing
- 2167.** The text states that as a scientific discipline, social psychology
 (A) is superior to those disciplines which assume a more subjective approach to the study of human nature

- (B) can assist in explaining the meaning of life
- (C) is one perspective from which we can view and better understand human nature
- (D) offers explanations for human nature that often contradict the claims of other disciplines
- 2168.** Most social-psychological research is conducted either in the field or in the _____ and is either correlational or _____.
- (A) clinic; survey
- (B) laboratory; experimental
- (C) laboratory; survey
- (D) clinic; experimental
- 2169.** Survey researchers obtain a representative group
- (A) through random assignment
- (B) by selecting at least 2000 respondents to be interviewed
- (C) by taking a random sample
- (D) either through telephone books or automobile registrations
- 2170.** The great strength of _____ is that it tends to occur in real-world settings where it can examine questions regarding important factors like race, sex, and social status.
- (A) correlational research
- (B) field experimentation
- (C) laboratory experimentation
- (D) quasi-experimentation
- 2171.** Which of the following distinguishes the correlational method from experimentation?
- (A) the correlational method uses a smaller group of subjects
- (B) the correlational method enables researchers to study social attitudes
- (C) no attempt is made to systematically manipulate one or more factors with the correlational method
- (D) the findings from the correlational method are more likely to be contaminated by the experimenter's values
- 2172.** To determine whether changing one variable (like education) will produce changes in another (like income), we need to conduct _____ research.
- (A) survey
- (B) correlational
- (C) experimental
- (D) statistical
- 2173.** Random assignment means that each person taking part in an experiment must
- (A) have an equal chance of being in a given condition in the experiment
- (B) be assigned to all the conditions of the experimental treatment
- (C) be randomly selected from the larger population
- (D) be given random responses to the experimenter's questions
- 2174.** An experiment has mundane realism if
- (A) the experimental task is similar to tasks in everyday life
- (B) it involves and absorbs people
- (C) it is conducted in the field
- (D) the experimenter deceives subjects
- 2175.** The experimental factor that the experimenter manipulates is called the _____ variable.
- (A) dependent (B) control
- (C) independent (D) experimental
- 2176.** In an experiment, the variable being measured is called the _____ variable.
- (A) control (B) independent
- (C) experimental (D) dependent
- 2177.** The experimental method is used in about _____ of all social-psychological research studies.
- (A) one-fourth (B) one-half
- (C) three-fourths (D) nine-tenths
- 2178.** Random assignment is necessary to
- (A) insure mundane realism
- (B) insure informed consent
- (C) rule out pre-existing differences between subjects in different experimental conditions
- (D) avoid the naturalistic fallacy

- 2179.** Which of the following is false according to the text?
 (A) experimenters standardize their instructions to subjects to minimize demand characteristics
 (B) major psychological organizations have developed a number of ethical principles to guide investigators
 (C) informed consent is an important ethical principle to be followed in conducting research
 (D) deception should never be used in conducting research
- 2180.** Hypotheses are best characterized as
 (A) axioms (B) principles
 (C) predictions (D) conclusions
- 2181.** According to the text, _____ tends to make people overconfident about the validity of their judgments and predictions.
 (A) the fundamental attribution error
 (B) illusory correlation
 (C) the naturalistic fallacy
 (D) the hindsight bias
- 2182.** In an experimental study of the effects of failure on self-esteem, self-esteem would be the
 (A) control condition
 (B) independent variable
 (C) dependent variable
 (D) experimental condition
- 2183.** Widely held ideas and values that we take for granted and that help us make sense of our world are called
 (A) pluralistic assumptions
 (B) cultural traditions
 (C) social heuristics
 (D) social representations
- 2184.** The text suggests that adjectives such as "self-actualized," "mature," and "well-adjusted" demonstrate
 (A) how psychological concepts have hidden values
 (B) how psychological concepts are individualistic
 (C) an inordinate concern with mental health
 (D) how personality psychologists are more influential than social psychologists
- 2185.** According to the text, the fact that human thinking always involves interpretation
 (A) provides a valid reason for dismissing science
 (B) is precisely why we need scientific analysis
 (C) is a reason for preferring experimental over correlational research
 (D) has been more frequently recognized by those in the sciences than by those in the humanities
- 2186.** How prior opinions control interpretations is illustrated by
 (A) Hastorf and Cantril's study of reactions to a university football game
 (B) Teigen's study of the "I-Knew-It-All-Along" phenomenon
 (C) the Literary Digest's erroneous prediction of the winner of the 1936 presidential election
 (D) Bachman and O' Malley's study of the relationship between self-esteem and achievement
- 2187.** The naturalistic fallacy provides an example of
 (A) the hindsight bias
 (B) how values penetrate the work of the scientist
 (C) how commonsense notions are often wrong
 (D) how naturalistic observation is unable to answer questions about cause-effect relationships
- 2188.** In comparison to North American social psychologists, European social psychologists tend to give more attention to the _____ levels of explanation.
 (A) intrapersonal and interpersonal
 (B) intergroup and societal
 (C) interpersonal and intergroup
 (D) intrapersonal and societal
- 2189.** According to social psychologist Fathali Moghaddam, countries that compose social psychology's second world include
 (A) Canada and France

- (B) Great Britain and Germany
(C) Cuba and Canada
(D) Nigeria and France
- 2190.** Who of the following is most likely to study how religious attitudes develop within the typical individual?
(A) a sociologist
(B) a social worker
(C) a social psychologist
(D) a theologian
- 2191.** Who of the following would be most likely to study how the political attitudes of middle-class people differ from those of lower-class people?
(A) a personality psychologist
(B) a social psychologist
(C) a social biologist (D) a sociologist
- 2192.** You would like to know the relationship between the number of psychology courses people take and their interpersonal sensitivity. You survey university students to determine how much psychology they have taken and then have them complete a test of social sensitivity. Finally you plot the relationship. This is an example of
(A) a laboratory experiment
(B) a field experiment
(C) a correlational study
(D) participant observation
- 2193.** The telephone company wants to survey its 100,000 customers. Four proposals for sampling the customers are being considered. Which would you recommend?
(A) interview every 75th person listed in the telephone directory
(B) mail a questionnaire to all 100,000 customers and assume at least 1200 will respond
(C) interview the people in every 50th residence from a postal listing of all addresses
(D) interview those 1000 persons with the highest phone bills
- 2194.** A research psychologist manipulates the level of fear in human subjects in the laboratory and then examines what effect the different levels of fear have on the subjects' reaction times. In this study, reaction time is the _____ variable.
(A) dependent
(B) correlational
(C) independent
(D) experimental
- 2195.** In conducting a study of conformity, the experimenters decide to tape-record the instructions that are to be presented to all subjects. Their decision is most likely an attempt to minimize the effect of
(A) hindsight bias
(B) mundane realism
(C) naturalistic fallacy
(D) demand characteristics
- 2196.** A researcher finds that Canadians bathe on the average of once a day. He concludes that an educational program is necessary to encourage more frequent bathing by those who bathe less than once a day. The researcher is probably guilty of
(A) hindsight bias
(B) the "I-Knew-It-All-Along" phenomenon
(C) illusory correlation
(D) the naturalistic fallacy
- 2197.** Which of the following research methods would be most effective in demonstrating that the presence of others improves our performance of a task?
(A) an experiment
(B) correlational study
(C) a survey
(D) a field study
- 2198.** Which of the following techniques would be the most effective way of investigating the relationship between the political preferences and the age of Canadian citizens?
(A) an experiment
(B) a case study
(C) a correlational study
(D) participant observation

- 2199.** A negative correlation between degree of wealth and the likelihood of being involved in criminal activity would indicate that
- (A) poverty makes people more likely to commit a crime
 - (B) the poor are more likely to be involved in criminal activity than are the wealthy
 - (C) being involved in criminal activity usually prevents people from accumulating wealth
 - (D) all of the above are necessarily true
- 2200.** In a research study investigating the effects of stress on the desire to affiliate, half of the participants complete an easy test of mental ability and half complete a difficult test. What technique should the investigators use to ensure that any posttest differences in the group's desire to affiliate actually result from the differences in test difficulty?
- (A) random sampling
 - (B) random assignment
 - (C) replication
 - (D) none of the above

ANSWER SHEET

1. (C)	2. (A)	3. (C)	4. (A)	5. (C)	6. (A)	7. (B)	8. (D)	9. (C)	10. (D)
11. (A)	12. (C)	13. (D)	14. (B)	15. (B)	16. (A)	17. (C)	18. (C)	19. (D)	20. (A)
21. (B)	22. (C)	23. (C)	24. (D)	25. (C)	26. (A)	27. (C)	28. (B)	29. (B)	30. (A)
31. (A)	32. (C)	33. (B)	34. (A)	35. (C)	36. (A)	37. (A)	38. (A)	39. (C)	40. (B)
41. (C)	42. (B)	43. (A)	44. (B)	45. (C)	46. (B)	47. (B)	48. (C)	49. (D)	50. (C)
51. (C)	52. (C)	53. (C)	54. (B)	55. (C)	56. (D)	57. (A)	58. (D)	59. (C)	60. (A)
61. (B)	62. (D)	63. (C)	64. (A)	65. (C)	66. (A)	67. (D)	68. (C)	69. (D)	70. (B)
71. (A)	72. (A)	73. (B)	74. (B)	75. (C)	76. (C)	77. (D)	78. (C)	79. (A)	80. (C)
81. (C)	82. (D)	83. (B)	84. (B)	85. (B)	86. (B)	87. (A)	88. (D)	89. (B)	90. (C)
91. (C)	92. (B)	93. (D)	94. (B)	95. (C)	96. (A)	97. (D)	98. (B)	99. (C)	100. (C)
101. (B)	102. (C)	103. (C)	104. (B)	105. (A)	106. (D)	107. (C)	108. (D)	109. (A)	110. (A)
111. (B)	112. (C)	113. (A)	114. (D)	115. (A)	116. (C)	117. (A)	118. (D)	119. (B)	120. (C)
121. (D)	122. (D)	123. (D)	124. (C)	125. (C)	126. (C)	127. (A)	128. (B)	129. (D)	130. (B)
131. (B)	132. (C)	133. (B)	134. (B)	135. (C)	136. (A)	137. (C)	138. (C)	139. (A)	140. (D)
141. (C)	142. (C)	143. (D)	144. (C)	145. (D)	146. (C)	147. (D)	148. (D)	149. (C)	150. (D)
151. (A)	152. (A)	153. (B)	154. (A)	155. (B)	156. (A)	157. (C)	158. (D)	159. (D)	160. (D)
161. (B)	162. (B)	163. (D)	164. (C)	165. (D)	166. (A)	167. (C)	168. (B)	169. (B)	170. (B)
171. (A)	172. (D)	173. (A)	174. (C)	175. (C)	176. (D)	177. (A)	178. (B)	179. (A)	180. (A)
181. (B)	182. (A)	183. (A)	184. (C)	185. (A)	186. (B)	187. (C)	188. (D)	189. (A)	190. (B)
191. (C)	192. (A)	193. (A)	194. (A)	195. (A)	196. (A)	197. (A)	198. (A)	199. (A)	200. (A)
201. (B)	202. (C)	203. (A)	204. (D)	205. (D)	206. (A)	207. (C)	208. (B)	209. (C)	210. (A)
211. (B)	212. (C)	213. (D)	214. (B)	215. (B)	216. (C)	217. (A)	218. (D)	219. (B)	220. (D)
221. (A)	222. (B)	223. (C)	224. (C)	225. (D)	226. (A)	227. (C)	228. (D)	229. (B)	230. (D)
231. (A)	232. (C)	233. (A)	234. (D)	235. (A)	236. (B)	237. (D)	238. (A)	239. (B)	240. (C)
241. (A)	242. (A)	243. (B)	244. (C)	245. (D)	246. (B)	247. (D)	248. (D)	249. (A)	250. (A)
251. (D)	252. (B)	253. (A)	254. (C)	255. (D)	256. (B)	257. (C)	258. (A)	259. (B)	260. (B)
261. (C)	262. (C)	263. (A)	264. (D)	265. (B)	266. (A)	267. (D)	268. (D)	269. (B)	270. (A)
271. (B)	272. (A)	273. (B)	274. (D)	275. (C)	276. (D)	277. (D)	278. (B)	279. (A)	280. (D)
281. (B)	282. (A)	283. (B)	284. (D)	285. (A)	286. (D)	287. (A)	288. (C)	289. (A)	290. (D)
291. (B)	292. (D)	293. (A)	294. (D)	295. (B)	296. (C)	297. (D)	298. (D)	299. (C)	300. (C)
301. (C)	302. (C)	303. (C)	304. (D)	305. (C)	306. (B)	307. (B)	308. (C)	309. (C)	310. (B)
311. (D)	312. (A)	313. (B)	314. (A)	315. (C)	316. (A)	317. (B)	318. (B)	319. (C)	320. (D)
321. (B)	322. (D)	323. (A)	324. (B)	325. (A)	326. (C)	327. (A)	328. (C)	329. (B)	330. (A)
331. (B)	332. (D)	333. (D)	334. (B)	335. (A)	336. (A)	337. (A)	338. (A)	339. (A)	340. (A)
341. (B)	342. (C)	343. (D)	344. (C)	345. (D)	346. (A)	347. (B)	348. (B)	349. (A)	350. (A)
351. (A)	352. (C)	353. (D)	354. (A)	355. (B)	356. (C)	357. (B)	358. (C)	359. (A)	360. (B)
361. (C)	362. (D)	363. (A)	364. (C)	365. (D)	366. (A)	367. (A)	368. (C)	369. (D)	370. (B)
371. (C)	372. (B)	373. (B)	374. (B)	375. (D)	376. (A)	377. (B)	378. (C)	379. (D)	380. (A)

381. (C)	382. (B)	383. (C)	384. (A)	385. (B)	386. (B)	387. (D)	388. (A)	389. (B)	390. (C)
391. (C)	392. (A)	393. (C)	394. (A)	395. (B)	396. (D)	397. (B)	398. (B)	399. (D)	400. (A)
401. (B)	402. (D)	403. (A)	404. (C)	405. (B)	406. (D)	407. (A)	408. (B)	409. (C)	410. (C)
411. (C)	412. (A)	413. (C)	414. (B)	415. (B)	416. (D)	417. (A)	418. (D)	419. (D)	420. (A)
421. (D)	422. (B)	423. (A)	424. (C)	425. (D)	426. (D)	427. (C)	428. (D)	429. (D)	430. (A)
431. (B)	432. (A)	433. (B)	434. (A)	435. (C)	436. (A)	437. (B)	438. (A)	439. (A)	440. (D)
441. (B)	442. (C)	443. (A)	444. (A)	445. (A)	446. (A)	447. (A)	448. (A)	449. (A)	450. (A)
451. (A)	452. (C)	453. (B)	454. (C)	455. (C)	456. (D)	457. (D)	458. (A)	459. (C)	460. (D)
461. (B)	462. (C)	463. (D)	464. (A)	465. (B)	466. (D)	467. (A)	468. (C)	469. (A)	470. (B)
471. (C)	472. (A)	473. (B)	474. (C)	475. (D)	476. (D)	477. (B)	478. (B)	479. (B)	480. (A)
481. (D)	482. (B)	483. (C)	484. (A)	485. (B)	486. (C)	487. (C)	488. (A)	489. (A)	490. (B)
491. (D)	492. (A)	493. (B)	494. (D)	495. (B)	496. (C)	497. (D)	498. (D)	499. (C)	500. (D)
501. (D)	502. (B)	503. (B)	504. (B)	505. (D)	506. (D)	507. (C)	508. (B)	509. (D)	510. (D)
511. (A)	512. (B)	513. (C)	514. (D)	515. (D)	516. (C)	517. (A)	518. (A)	519. (C)	520. (B)
521. (D)	522. (B)	523. (D)	524. (C)	525. (D)	526. (D)	527. (C)	528. (B)	529. (B)	530. (C)
531. (B)	532. (D)	533. (D)	534. (A)	535. (D)	536. (D)	537. (B)	538. (C)	539. (B)	540. (C)
541. (A)	542. (D)	543. (C)	544. (D)	545. (A)	546. (A)	547. (A)	548. (A)	549. (B)	550. (A)
551. (C)	552. (D)	553. (B)	554. (C)	555. (C)	556. (A)	557. (A)	558. (D)	559. (B)	560. (D)
561. (A)	562. (A)	563. (B)	564. (A)	565. (B)	566. (B)	567. (D)	568. (A)	569. (B)	570. (D)
571. (D)	572. (C)	573. (B)	574. (A)	575. (C)	576. (B)	577. (C)	578. (C)	579. (B)	580. (C)
581. (C)	582. (D)	583. (A)	584. (B)	585. (B)	586. (C)	587. (D)	588. (A)	589. (A)	590. (B)
591. (C)	592. (D)	593. (D)	594. (A)	595. (B)	596. (D)	597. (A)	598. (C)	599. (D)	600. (C)
601. (A)	602. (B)	603. (C)	604. (D)	605. (B)	606. (C)	607. (B)	608. (A)	609. (D)	610. (D)
611. (A)	612. (C)	613. (D)	614. (A)	615. (D)	616. (A)	617. (B)	618. (C)	619. (D)	620. (A)
621. (B)	622. (C)	623. (A)	624. (B)	625. (D)	626. (C)	627. (C)	628. (D)	629. (A)	630. (B)
631. (C)	632. (D)	633. (C)	634. (C)	635. (C)	636. (D)	637. (B)	638. (A)	639. (A)	640. (A)
641. (D)	642. (A)	643. (B)	644. (C)	645. (C)	646. (C)	647. (A)	648. (B)	649. (D)	650. (A)
651. (D)	652. (A)	653. (B)	654. (C)	655. (C)	656. (C)	657. (A)	658. (C)	659. (C)	660. (A)
661. (C)	662. (B)	663. (B)	664. (D)	665. (D)	666. (A)	667. (B)	668. (D)	669. (C)	670. (A)
671. (B)	672. (C)	673. (B)	674. (D)	675. (D)	676. (A)	677. (D)	678. (A)	679. (A)	680. (C)
681. (B)	682. (C)	683. (A)	684. (C)	685. (B)	686. (B)	687. (B)	688. (D)	689. (D)	690. (C)
691. (B)	692. (A)	693. (C)	694. (D)	695. (B)	696. (D)	697. (A)	698. (A)	699. (C)	700. (C)
701. (C)	702. (B)	703. (A)	704. (B)	705. (A)	706. (A)	707. (C)	708. (C)	709. (B)	710. (D)
711. (A)	712. (D)	713. (C)	714. (D)	715. (C)	716. (B)	717. (A)	718. (C)	719. (D)	720. (C)
721. (A)	722. (B)	723. (C)	724. (D)	725. (A)	726. (C)	727. (D)	728. (A)	729. (B)	730. (C)
731. (C)	732. (D)	733. (B)	734. (D)	735. (A)	736. (B)	737. (D)	738. (B)	739. (C)	740. (D)
741. (A)	742. (A)	743. (B)	744. (C)	745. (C)	746. (D)	747. (A)	748. (A)	749. (D)	750. (A)
751. (B)	752. (A)	753. (B)	754. (C)	755. (A)	756. (B)	757. (C)	758. (A)	759. (C)	760. (D)
761. (A)	762. (C)	763. (B)	764. (B)	765. (C)	766. (D)	767. (C)	768. (C)	769. (D)	770. (B)
771. (D)	772. (D)	773. (B)	774. (C)	775. (B)	776. (A)	777. (D)	778. (A)	779. (B)	780. (B)
781. (B)	782. (B)	783. (B)	784. (C)	785. (B)	786. (C)	787. (D)	788. (B)	789. (A)	790. (A)
791. (D)	792. (C)	793. (B)	794. (C)	795. (B)	796. (B)	797. (D)	798. (D)	799. (C)	800. (A)

801. (D)	802. (D)	803. (D)	804. (A)	805. (B)	806. (C)	807. (B)	808. (A)	809. (B)	810. (D)
811. (C)	812. (A)	813. (D)	814. (D)	815. (B)	816. (C)	817. (C)	818. (A)	819. (B)	820. (C)
821. (A)	822. (B)	823. (B)	824. (C)	825. (D)	826. (C)	827. (D)	828. (C)	829. (D)	830. (D)
831. (A)	832. (C)	833. (B)	834. (C)	835. (D)	836. (B)	837. (C)	838. (B)	839. (C)	840. (A)
841. (B)	842. (D)	843. (A)	844. (C)	845. (D)	846. (A)	847. (B)	848. (C)	849. (A)	850. (C)
851. (D)	852. (A)	853. (B)	854. (A)	855. (B)	856. (C)	857. (D)	858. (B)	859. (C)	860. (D)
861. (D)	862. (B)	863. (D)	864. (A)	865. (B)	866. (D)	867. (A)	868. (B)	869. (D)	870. (A)
871. (B)	872. (C)	873. (D)	874. (B)	875. (A)	876. (C)	877. (C)	878. (C)	879. (C)	880. (A)
881. (A)	882. (B)	883. (B)	884. (D)	885. (D)	886. (D)	887. (A)	888. (C)	889. (C)	890. (B)
891. (B)	892. (D)	893. (C)	894. (A)	895. (A)	896. (A)	897. (C)	898. (A)	899. (C)	900. (B)
901. (B)	902. (D)	903. (A)	904. (A)	905. (A)	906. (A)	907. (A)	908. (C)	909. (C)	910. (B)
911. (B)	912. (C)	913. (C)	914. (C)	915. (A)	916. (A)	917. (B)	918. (C)	919. (D)	920. (D)
921. (C)	922. (B)	923. (B)	924. (A)	925. (D)	926. (A)	927. (C)	928. (C)	929. (A)	930. (D)
931. (A)	932. (A)	933. (A)	934. (D)	935. (B)	936. (A)	937. (B)	938. (C)	939. (C)	940. (A)
941. (A)	942. (C)	943. (D)	944. (C)	945. (D)	946. (B)	947. (C)	948. (A)	949. (C)	950. (B)
951. (D)	952. (A)	953. (C)	954. (B)	955. (C)	956. (D)	957. (C)	958. (A)	959. (B)	960. (A)
961. (D)	962. (D)	963. (B)	964. (B)	965. (A)	966. (C)	967. (B)	968. (C)	969. (D)	970. (C)
971. (D)	972. (A)	973. (B)	974. (A)	975. (A)	976. (D)	977. (C)	978. (B)	979. (C)	980. (B)
981. (B)	982. (A)	983. (C)	984. (D)	985. (A)	986. (B)	987. (D)	988. (A)	989. (B)	990. (D)
991. (B)	992. (D)	993. (A)	994. (A)	995. (D)	996. (C)	997. (D)	998. (C)	999. (B)	1000. (C)
1001. (C)	1002. (D)	1003. (D)	1004. (D)	1005. (C)	1006. (D)	1007. (A)	1008. (B)	1009. (A)	1010. (C)
1011. (A)	1012. (D)	1013. (D)	1014. (B)	1015. (B)	1016. (A)	1017. (B)	1018. (B)	1019. (C)	1020. (A)
1021. (D)	1022. (C)	1023. (D)	1024. (B)	1025. (D)	1026. (C)	1027. (A)	1028. (C)	1029. (B)	1030. (D)
1031. (A)	1032. (D)	1033. (A)	1034. (C)	1035. (B)	1036. (A)	1037. (B)	1038. (C)	1039. (D)	1040. (D)
1041. (D)	1042. (C)	1043. (D)	1044. (C)	1045. (A)	1046. (B)	1047. (A)	1048. (A)	1049. (B)	1050. (B)
1051. (D)	1052. (C)	1053. (A)	1054. (A)	1055. (C)	1056. (C)	1057. (A)	1058. (C)	1059. (B)	1060. (D)
1061. (C)	1062. (B)	1063. (B)	1064. (D)	1065. (D)	1066. (B)	1067. (A)	1068. (C)	1069. (D)	1070. (C)
1071. (A)	1072. (B)	1073. (B)	1074. (B)	1075. (B)	1076. (A)	1077. (A)	1078. (B)	1079. (D)	1080. (D)
1081. (B)	1082. (B)	1083. (D)	1084. (C)	1085. (C)	1086. (A)	1087. (A)	1088. (B)	1089. (D)	1090. (C)
1091. (C)	1092. (A)	1093. (A)	1094. (C)	1095. (D)	1096. (A)	1097. (D)	1098. (A)	1099. (B)	1100. (B)
1101. (B)	1102. (D)	1103. (C)	1104. (B)	1105. (A)	1106. (A)	1107. (A)	1108. (A)	1109. (A)	1110. (B)
1111. (D)	1112. (B)	1113. (C)	1114. (B)	1115. (C)	1116. (D)	1117. (C)	1118. (D)	1119. (A)	1120. (C)
1121. (D)	1122. (B)	1123. (D)	1124. (C)	1125. (B)	1126. (B)	1127. (D)	1128. (B)	1129. (B)	1130. (B)
1131. (A)	1132. (D)	1133. (B)	1134. (D)	1135. (A)	1136. (D)	1137. (C)	1138. (D)	1139. (A)	1140. (C)
1141. (A)	1142. (C)	1143. (A)	1144. (D)	1145. (A)	1146. (A)	1147. (A)	1148. (A)	1149. (A)	1150. (C)
1151. (D)	1152. (D)	1153. (C)	1154. (A)	1155. (B)	1156. (D)	1157. (A)	1158. (C)	1159. (D)	1160. (B)
1161. (A)	1162. (C)	1163. (A)	1164. (C)	1165. (C)	1166. (D)	1167. (D)	1168. (C)	1169. (B)	1170. (C)
1171. (D)	1172. (B)	1173. (D)	1174. (A)	1175. (A)	1176. (A)	1177. (A)	1178. (B)	1179. (B)	1180. (D)
1181. (C)	1182. (C)	1183. (A)	1184. (C)	1185. (C)	1186. (A)	1187. (A)	1188. (A)	1189. (D)	1190. (C)
1191. (D)	1192. (A)	1193. (D)	1194. (C)	1195. (B)	1196. (B)	1197. (A)	1198. (A)	1199. (A)	1200. (C)
1201. (C)	1202. (A)	1203. (A)	1204. (C)	1205. (C)	1206. (A)	1207. (B)	1208. (D)	1209. (A)	1210. (B)
1211. (C)	1212. (A)	1213. (D)	1214. (B)	1215. (C)	1216. (A)	1217. (A)	1218. (A)	1219. (A)	1220. (C)

1221. (A)	1222. (A)	1223. (B)	1224. (C)	1225. (D)	1226. (A)	1227. (A)	1228. (B)	1229. (A)	1230. (A)
1231. (A)	1232. (C)	1233. (C)	1234. (D)	1235. (A)	1236. (C)	1237. (D)	1238. (A)	1239. (A)	1240. (D)
1241. (D)	1242. (C)	1243. (D)	1244. (D)	1245. (C)	1246. (C)	1247. (B)	1248. (D)	1249. (D)	1250. (D)
1251. (C)	1252. (A)	1253. (A)	1254. (B)	1255. (B)	1256. (D)	1257. (D)	1258. (D)	1259. (D)	1260. (C)
1261. (A)	1262. (D)	1263. (B)	1264. (C)	1265. (A)	1266. (A)	1267. (C)	1268. (B)	1269. (D)	1270. (A)
1271. (A)	1272. (B)	1273. (D)	1274. (D)	1275. (D)	1276. (B)	1277. (C)	1278. (A)	1279. (C)	1280. (D)
1281. (C)	1282. (B)	1283. (C)	1284. (D)	1285. (B)	1286. (D)	1287. (D)	1288. (B)	1289. (C)	1290. (B)
1291. (B)	1292. (B)	1293. (D)	1294. (C)	1295. (B)	1296. (C)	1297. (B)	1298. (A)	1299. (D)	1300. (B)
1301. (C)	1302. (C)	1303. (C)	1304. (C)	1305. (C)	1306. (C)	1307. (C)	1308. (B)	1309. (B)	1310. (D)
1311. (A)	1312. (B)	1313. (C)	1314. (D)	1315. (D)	1316. (D)	1317. (D)	1318. (B)	1319. (C)	1320. (B)
1321. (B)	1322. (C)	1323. (A)	1324. (A)	1325. (A)	1326. (A)	1327. (D)	1328. (D)	1329. (A)	1330. (D)
1331. (A)	1332. (A)	1333. (B)	1334. (A)	1335. (C)	1336. (C)	1337. (C)	1338. (D)	1339. (D)	1340. (D)
1341. (A)	1342. (C)	1343. (D)	1344. (C)	1345. (D)	1346. (A)	1347. (B)	1348. (D)	1349. (A)	1350. (B)
1351. (C)	1352. (D)	1353. (D)	1354. (A)	1355. (C)	1356. (B)	1357. (B)	1358. (A)	1359. (D)	1360. (C)
1361. (D)	1362. (B)	1363. (C)	1364. (D)	1365. (A)	1366. (B)	1367. (B)	1368. (A)	1369. (B)	1370. (C)
1371. (A)	1372. (C)	1373. (D)	1374. (B)	1375. (C)	1376. (D)	1377. (A)	1378. (B)	1379. (C)	1380. (A)
1381. (B)	1382. (D)	1383. (A)	1384. (D)	1385. (C)	1386. (D)	1387. (B)	1388. (D)	1389. (A)	1390. (B)
1391. (A)	1392. (B)	1393. (B)	1394. (C)	1395. (C)	1396. (D)	1397. (A)	1398. (C)	1399. (A)	1400. (B)
1401. (C)	1402. (D)	1403. (A)	1404. (C)	1405. (D)	1406. (B)	1407. (C)	1408. (C)	1409. (D)	1410. (D)
1411. (B)	1412. (C)	1413. (C)	1414. (A)	1415. (C)	1416. (C)	1417. (A)	1418. (D)	1419. (A)	1420. (C)
1421. (A)	1422. (C)	1423. (A)	1424. (B)	1425. (C)	1426. (D)	1427. (A)	1428. (C)	1429. (B)	1430. (A)
1431. (C)	1432. (C)	1433. (A)	1434. (B)	1435. (A)	1436. (A)	1437. (B)	1438. (C)	1439. (A)	1440. (C)
1441. (B)	1442. (B)	1443. (C)	1444. (C)	1445. (D)	1446. (B)	1447. (D)	1448. (C)	1449. (B)	1450. (C)
1451. (D)	1452. (A)	1453. (C)	1454. (A)	1455. (D)	1456. (D)	1457. (A)	1458. (B)	1459. (C)	1460. (B)
1461. (B)	1462. (C)	1463. (A)	1464. (D)	1465. (D)	1466. (C)	1467. (D)	1468. (A)	1469. (D)	1470. (D)
1471. (C)	1472. (A)	1473. (C)	1474. (B)	1475. (C)	1476. (A)	1477. (A)	1478. (D)	1479. (C)	1480. (B)
1481. (C)	1482. (B)	1483. (B)	1484. (B)	1485. (A)	1486. (A)	1487. (C)	1488. (C)	1489. (D)	1490. (C)
1491. (B)	1492. (C)	1493. (B)	1494. (A)	1495. (A)	1496. (C)	1497. (D)	1498. (B)	1499. (D)	1500. (A)
1501. (C)	1502. (B)	1503. (C)	1504. (A)	1505. (C)	1506. (C)	1507. (D)	1508. (A)	1509. (C)	1510. (B)
1511. (C)	1512. (C)	1513. (C)	1514. (C)	1515. (C)	1516. (C)	1517. (C)	1518. (B)	1519. (C)	1520. (D)
1521. (B)	1522. (B)	1523. (D)	1524. (D)	1525. (D)	1526. (B)	1527. (B)	1528. (B)	1529. (C)	1530. (B)
1531. (A)	1532. (B)	1533. (B)	1534. (B)	1535. (A)	1536. (A)	1537. (D)	1538. (C)	1539. (B)	1540. (B)
1541. (A)	1542. (B)	1543. (D)	1544. (A)	1545. (D)	1546. (C)	1547. (D)	1548. (C)	1549. (A)	1550. (D)
1551. (C)	1552. (C)	1553. (C)	1554. (B)	1555. (B)	1556. (B)	1557. (D)	1558. (C)	1559. (C)	1560. (D)
1561. (B)	1562. (A)	1563. (B)	1564. (B)	1565. (B)	1566. (A)	1567. (A)	1568. (C)	1569. (C)	1570. (B)
1571. (B)	1572. (D)	1573. (C)	1574. (B)	1575. (C)	1576. (D)	1577. (A)	1578. (A)	1579. (C)	1580. (B)
1581. (A)	1582. (B)	1583. (C)	1584. (A)	1585. (D)	1586. (A)	1587. (B)	1588. (A)	1589. (A)	1590. (A)
1591. (D)	1592. (B)	1593. (B)	1594. (A)	1595. (B)	1596. (D)	1597. (B)	1598. (B)	1599. (A)	1600. (C)
1601. (C)	1602. (C)	1603. (B)	1604. (B)	1605. (A)	1606. (A)	1607. (C)	1608. (C)	1609. (D)	1610. (D)
1611. (A)	1612. (A)	1613. (B)	1614. (D)	1615. (C)	1616. (B)	1617. (C)	1618. (D)	1619. (C)	1620. (D)
1621. (C)	1622. (A)	1623. (C)	1624. (D)	1625. (A)	1626. (D)	1627. (A)	1628. (B)	1629. (B)	1630. (D)
1631. (C)	1632. (D)	1633. (C)	1634. (A)	1635. (B)	1636. (C)	1637. (B)	1638. (A)	1639. (D)	1640. (B)

1641. (A)	1642. (C)	1643. (C)	1644. (A)	1645. (B)	1646. (C)	1647. (B)	1648. (D)	1649. (A)	1650. (D)
1651. (C)	1652. (C)	1653. (A)	1654. (A)	1655. (A)	1656. (B)	1657. (C)	1658. (C)	1659. (D)	1660. (D)
1661. (C)	1662. (C)	1663. (D)	1664. (D)	1665. (B)	1666. (C)	1667. (B)	1668. (D)	1669. (B)	1670. (D)
1671. (D)	1672. (A)	1673. (C)	1674. (B)	1675. (C)	1676. (A)	1677. (B)	1678. (C)	1679. (A)	1680. (A)
1681. (A)	1682. (D)	1683. (D)	1684. (A)	1685. (D)	1686. (B)	1687. (C)	1688. (D)	1689. (A)	1690. (A)
1691. (C)	1692. (B)	1693. (D)	1694. (D)	1695. (B)	1696. (A)	1697. (C)	1698. (C)	1699. (B)	1700. (B)
1701. (B)	1702. (A)	1703. (B)	1704. (B)	1705. (D)	1706. (A)	1707. (A)	1708. (C)	1709. (D)	1710. (B)
1711. (D)	1712. (B)	1713. (D)	1714. (C)	1715. (A)	1716. (B)	1717. (C)	1718. (A)	1719. (A)	1720. (D)
1721. (A)	1722. (A)	1723. (B)	1724. (A)	1725. (D)	1726. (C)	1727. (C)	1728. (A)	1729. (A)	1730. (C)
1731. (A)	1732. (D)	1733. (C)	1734. (B)	1735. (B)	1736. (A)	1737. (B)	1738. (D)	1739. (B)	1740. (D)
1741. (A)	1742. (B)	1743. (B)	1744. (B)	1745. (D)	1746. (C)	1747. (D)	1748. (A)	1749. (C)	1750. (D)
1751. (A)	1752. (A)	1753. (D)	1754. (B)	1755. (D)	1756. (B)	1757. (B)	1758. (C)	1759. (C)	1760. (C)
1761. (C)	1762. (A)	1763. (C)	1764. (A)	1765. (B)	1766. (B)	1767. (B)	1768. (C)	1769. (D)	1770. (D)
1771. (D)	1772. (D)	1773. (A)	1774. (B)	1775. (B)	1776. (C)	1777. (A)	1778. (B)	1779. (B)	1780. (C)
1781. (A)	1782. (B)	1783. (B)	1784. (A)	1785. (B)	1786. (B)	1787. (B)	1788. (D)	1789. (A)	1790. (B)
1791. (C)	1792. (C)	1793. (A)	1794. (B)	1795. (B)	1796. (A)	1797. (A)	1798. (C)	1799. (B)	1800. (C)
1801. (B)	1802. (A)	1803. (B)	1804. (B)	1805. (B)	1806. (A)	1807. (C)	1808. (B)	1809. (A)	1810. (A)
1811. (B)	1812. (C)	1813. (A)	1814. (B)	1815. (A)	1816. (D)	1817. (A)	1818. (B)	1819. (C)	1820. (C)
1821. (C)	1822. (A)	1823. (A)	1824. (A)	1825. (A)	1826. (B)	1827. (C)	1828. (B)	1829. (A)	1830. (D)
1831. (C)	1832. (C)	1833. (B)	1834. (C)	1835. (A)	1836. (D)	1837. (C)	1838. (A)	1839. (A)	1840. (D)
1841. (D)	1842. (D)	1843. (A)	1844. (A)	1845. (B)	1846. (A)	1847. (A)	1848. (D)	1849. (A)	1850. (D)
1851. (D)	1852. (A)	1853. (C)	1854. (D)	1855. (A)	1856. (C)	1857. (D)	1858. (C)	1859. (C)	1860. (B)
1861. (C)	1862. (C)	1863. (D)	1864. (B)	1865. (D)	1866. (A)	1867. (B)	1868. (C)	1869. (A)	1870. (C)
1871. (B)	1872. (C)	1873. (D)	1874. (A)	1875. (B)	1876. (C)	1877. (C)	1878. (D)	1879. (D)	1880. (A)
1881. (D)	1882. (C)	1883. (C)	1884. (A)	1885. (A)	1886. (A)	1887. (D)	1888. (B)	1889. (D)	1890. (D)
1891. (C)	1892. (B)	1893. (A)	1894. (D)	1895. (C)	1896. (A)	1897. (B)	1898. (A)	1899. (A)	1900. (B)
1901. (C)	1902. (A)	1903. (B)	1904. (D)	1905. (D)	1906. (C)	1907. (A)	1908. (A)	1909. (C)	1910. (A)
1911. (D)	1912. (A)	1913. (D)	1914. (B)	1915. (B)	1916. (A)	1917. (A)	1918. (A)	1919. (C)	1920. (C)
1921. (D)	1922. (D)	1923. (D)	1924. (A)	1925. (C)	1926. (D)	1927. (C)	1928. (D)	1929. (A)	1930. (D)
1931. (A)	1932. (D)	1933. (B)	1934. (A)	1935. (D)	1936. (A)	1937. (B)	1938. (A)	1939. (D)	1940. (A)
1941. (A)	1942. (C)	1943. (B)	1944. (A)	1945. (D)	1946. (A)	1947. (C)	1948. (B)	1949. (C)	1950. (D)
1951. (A)	1952. (A)	1953. (C)	1954. (C)	1955. (B)	1956. (D)	1957. (D)	1958. (D)	1959. (B)	1960. (D)
1961. (A)	1962. (A)	1963. (B)	1964. (D)	1965. (A)	1966. (D)	1967. (A)	1968. (B)	1969. (D)	1970. (B)
1971. (C)	1972. (D)	1973. (B)	1974. (B)	1975. (D)	1976. (A)	1977. (D)	1978. (C)	1979. (C)	1980. (D)
1981. (D)	1982. (D)	1983. (D)	1984. (C)	1985. (B)	1986. (C)	1987. (D)	1988. (A)	1989. (B)	1990. (A)
1991. (D)	1992. (D)	1993. (D)	1994. (A)	1995. (A)	1996. (A)	1997. (B)	1998. (D)	1999. (C)	2000. (B)
2001. (A)	2002. (D)	2003. (D)	2004. (A)	2005. (B)	2006. (C)	2007. (B)	2008. (D)	2009. (A)	2010. (D)
2011. (D)	2012. (D)	2013. (B)	2014. (B)	2015. (D)	2016. (D)	2017. (A)	2018. (A)	2019. (D)	2020. (D)
2021. (B)	2022. (B)	2023. (A)	2024. (B)	2025. (D)	2026. (D)	2027. (C)	2028. (A)	2029. (B)	2030. (C)
2031. (B)	2032. (B)	2033. (D)	2034. (A)	2035. (D)	2036. (D)	2037. (A)	2038. (A)	2039. (B)	2040. (D)
2041. (D)	2042. (B)	2043. (D)	2044. (C)	2045. (B)	2046. (B)	2047. (A)	2048. (D)	2049. (B)	2050. (A)
2051. (A)	2052. (B)	2053. (B)	2054. (D)	2055. (D)	2056. (B)	2057. (A)	2058. (C)	2059. (C)	2060. (D)

2061. (A)	2062. (A)	2063. (D)	2064. (A)	2065. (D)	2066. (A)	2067. (D)	2068. (D)	2069. (C)	2070. (B)
2071. (D)	2072. (C)	2073. (C)	2074. (B)	2075. (D)	2076. (A)	2077. (B)	2078. (B)	2079. (B)	2080. (A)
2081. (B)	2082. (C)	2083. (D)	2084. (D)	2085. (B)	2086. (C)	2087. (D)	2088. (A)	2089. (B)	2090. (D)
2091. (B)	2092. (A)	2093. (D)	2094. (B)	2095. (C)	2096. (A)	2097. (B)	2098. (B)	2099. (C)	2100. (B)
2101. (B)	2102. (C)	2103. (D)	2104. (A)	2105. (A)	2106. (A)	2107. (C)	2108. (C)	2109. (C)	2110. (D)
2111. (A)	2112. (B)	2113. (C)	2114. (A)	2115. (C)	2116. (A)	2117. (C)	2118. (C)	2119. (C)	2120. (D)
2121. (D)	2122. (C)	2123. (C)	2124. (D)	2125. (C)	2126. (B)	2127. (C)	2128. (D)	2129. (C)	2130. (C)
2131. (B)	2132. (C)	2133. (C)	2134. (B)	2135. (B)	2136. (D)	2137. (D)	2138. (C)	2139. (D)	2140. (D)
2141. (B)	2142. (C)	2143. (C)	2144. (C)	2145. (D)	2146. (C)	2147. (B)	2148. (B)	2149. (C)	2150. (C)
2151. (B)	2152. (C)	2153. (D)	2154. (B)	2155. (B)	2156. (B)	2157. (C)	2158. (A)	2159. (D)	2160. (B)
2161. (C)	2162. (C)	2163. (B)	2164. (A)	2165. (A)	2166. (C)	2167. (C)	2168. (B)	2169. (C)	2170. (A)
2171. (C)	2172. (C)	2173. (A)	2174. (A)	2175. (C)	2176. (D)	2177. (C)	2178. (C)	2179. (D)	2180. (C)
2181. (D)	2182. (C)	2183. (D)	2184. (A)	2185. (B)	2186. (A)	2187. (B)	2188. (B)	2189. (B)	2190. (C)
2191. (D)	2192. (C)	2193. (A)	2194. (A)	2195. (D)	2196. (D)	2197. (A)	2198. (C)	2199. (B)	2200. (B)

MODEL PAPER (Revised Pattern)

Note : This paper contains **hundred (100)** Objective type questions, each question carrying **two (2)** marks. Attempt all of them.

1. Key issues in the study of environmental effects on intelligence are :

(a) Reaction Range
(b) Proximal and Distal Causation
(c) Colinearity
(d) Environment Sphere

Code :

(A) (a), (b) and (c) only
(B) (a), (b) and (d) only
(C) (b), (c) and (d) only
(D) (a), (c) and (d) only

2. The factors that operate with deep levels of processing are :

(a) Distinctiveness
(b) Sameness
(c) Elaboration
(d) Physical contours

Code :

(A) (a) and (b) only
(B) (b) and (c) only
(C) (b) and (d) only
(D) (a) and (c) only

3. Which one of the following is not the component of Goleman's Concept of Emotional Intelligence ?

(A) Knowing our own emotions
(B) Managing our own emotions
(C) Analytical ability
(D) Interpersonal skills

4. For a scale, norms were developed using T scores (Mean = 50, S.D. = 10). A subject got a T score of 60. What would be his PR ?

(A) 16 (B) 34
(C) 60 (D) 84

5. Match List-I with List-II

List-I

List-II

(a) Aptitude	(1) Two factor theory of intelligence
(b) Hereditary	(2) WAIS
(c) Spearman	(3) Special ability
(d) Measurement of Intelligence	(4) Genes

Code :

	(a)	(b)	(c)	(d)
(A)	3	4	1	2
(B)	4	3	2	1
(C)	3	1	2	4
(D)	1	2	3	4

6. According to Atkinson's theory, the tendency to approach an achievement task is a function of

(A) The motive for success.
(B) The subjective probability of success on the task.
(C) The incentive value of reaching the goal.
(D) All of the above.

7. Match List-I with List-II and select the correct answer using the codes given below the list.

List-I

(EEG Record)

List -II (State of the individual)

(a) Alpha	(1) Deep sleep
(b) Beta	(2) Alert
(c) Delta	(3) Emotional state
(d) Theta	(4) Relaxed

Code :

	(a)	(b)	(c)	(d)
(A)	3	1	2	4
(B)	4	2	1	3
(C)	1	2	3	4
(D)	4	1	2	3

8. Memories of historical facts are to _____ memory, as memories of your breakfast this morning are to _____ memory.
- (A) episodic ; procedural
(B) semantic ; episodic
(C) procedural ; semantic
(D) long-term ; short term
9. According to Tulving the 3 systems of memory are
- (A) Immediate memory, short term memory and long term memory.
(B) Working memory, meta memory and short term memory.
(C) Procedural memory, semantic memory and episodic memory.
(D) Sensory memory, mnemonics and secondary memory.
10. Match List-I (training in) with List-II (Helps in learning) and select the correct answer using the codes given below :
- | List-I | List-II |
|----------------|-----------------|
| (a) Piano | (1) Rote memory |
| (b) Sanskrit | (2) English |
| (c) Latin | (3) Typing |
| (d) Recitation | (4) Hindi |
- Code :**
- | (a) | (b) | (c) | (d) |
|-------|-----|-----|-----|
| (A) 3 | 4 | 2 | 1 |
| (B) 3 | 4 | 1 | 2 |
| (C) 4 | 3 | 1 | 2 |
| (D) 4 | 3 | 2 | 1 |
11. Who provided for the first time the powerful set of pictorial cues ?
- (A) Koffka
(B) Wertheimer
(C) Kohler
(D) Gibson
12. Sohan met with a head injury in a road accident. Since the accident, Sohan is unable to form new memories. He remembers many things which he did before accident but he cannot remember what he said fifteen minutes ago. The part of Sohan's brain, which was most probably injured was
- (A) brain stem
(B) hippocampus
(C) spinal cord
(D) reticular formation
13. The depressed spot in the retina which occupies the center of the visual field in which images are focussed most sharply is called
- (A) fovea
(B) cornea
(C) iris
(D) optic nerve
14. A psychologist developed a new test of numerical ability. He correlated the new test with an established test of numerical ability using a sample of 400 subjects. The obtained correlation of 0.28 between the two tests is statistically significant at 0.01 level. For the new test, this provides an evidence of
- (A) Satisfactory concurrent validity
(B) Unsatisfactory concurrent validity
(C) Satisfactory predictive validity
(D) Unsatisfactory predictive validity
15. Match List-I (psychologists) with list-II (theories) and select the correct answer using the codes given below the list.
- | List-I
(Psychologists) | List-II
(Theories) |
|---------------------------|---|
| (a) Sternberg | (1) Sampling Theory of Intelligence |
| (b) Jensen | (2) Level I and Level II Theory of Intelligence |
| (c) J.P. Das | (3) Pass Model of Intelligence |
| (d) Thomson | (4) Triarchic Theory of Intelligence |
- Code:**
- | (a) | (b) | (c) | (d) |
|-------|-----|-----|-----|
| (A) 3 | 2 | 4 | 1 |
| (B) 1 | 4 | 2 | 3 |
| (C) 4 | 2 | 3 | 1 |
| (D) 2 | 4 | 1 | 3 |
16. Which of the following represents thinking process ?
- (A) Tension
(B) Imitation

- (C) Sub-vocal speech
(D) Debate
17. Which theory of emotion implies that people can change their emotions simply by changing the way they label their arousal ?
(A) The James-Lange Theory
(B) The Cannon-Bard Theory
(C) Schachter's two-Factor Theory
(D) Opponent-Process Theory
18. Training designed to improve problem solving ability emphasizes
(A) Putting two and two-together
(B) Implicit categorizing
(C) Solution of multiple choice problem
(D) Ideational flexibility
19. Problems that require a common object to be used in an unusual way may be difficult to solve because of
(A) Mental set
(B) Irrelevant information
(C) Unnecessary constraints
(D) Functional fixedness
20. When persons first see something in their environment, such as a tree, this image first enters
(A) echoic memory
(B) iconic memory
(C) short term memory
(D) long term memory
21. For conditioning to occur, the proper order of events is
(A) US – UR – CR
(B) CS – CR – UR
(C) CS – US – UR
(D) UR – US – CS
22. When the light is seen to travel from one point to another, even though there was no stimulation in the intervening region, is not called
(A) Stroboscopic movement
(B) Phi phenomenon
(C) Apparent movement
(D) Induced movement
23. Location in Rorschach test is concerned with
(A) Whole/part
(B) Fantasy/emotion
(C) Colour/movement
(D) Figure/landscape
24. Choose the correct order in which human needs are to be satisfied.
1. Esteem needs
2. Need for self actualization
3. Love and belongingness needs
4. Aesthetic needs
Codes :
(A) 4, 3, 2, 1 (B) 3, 1, 4, 2
(C) 1, 3, 4, 2 (D) 3, 2, 4, 1
25. Dinner Plates on a table are seen as round even though from the angle of viewing they are oval. This is an example of
(A) Size constancy
(B) Shape constancy
(C) Brightness constancy
(D) Colour constancy
26. A procedure by which the application or removal of a stimulus decreases the strength of a behaviour are both examples of
(A) Negative reinforcement
(B) Punishment
(C) Positive reinforcement
(D) Secondary reinforcement
27. The theory which attempted to explain both the structure as well as processes of long term memory is known as
(A) TLC theory of Quillian
(B) ACT theory of Anderson
(C) Information processing theory of Atkinson and Shiffrin
(D) Dual coding theory of Pavio.
28. Match the items on List-I with the items on List-II.
- | List-I | List-II |
|--------------------|------------------------|
| (a) Iconic memory | (1) Baddadey |
| (b) Schema theory | (2) Muller & Pilzecker |
| (c) Working memory | (3) Sperling |
| (d) Consolidation | (4) Bart Lett theory |
- Code:**
- | | (a) | (b) | (c) | (d) |
|-----|------------|------------|------------|------------|
| (A) | 1 | 3 | 4 | 2 |
| (B) | 3 | 4 | 1 | 2 |
| (C) | 2 | 3 | 1 | 4 |
| (D) | 4 | 1 | 2 | 1 |

29. Stages in creative thinking are
 (A) Preparation, incubation, illumination, evaluation, revision.
 (B) Incubation, preparation, illumination, evaluation, revision.
 (C) Preparation, illumination, incubation, evaluation, revision.
 (D) Illumination, preparation, incubation, revision, evaluation
30. Which one of the following is the correct sequence of motivational set ?
 (A) Goal-directed behaviour, attainment of goal, drive, satisfaction.
 (B) Drive, satisfaction, attainment of goal, goal directed behaviour.
 (C) Drive, goal-directed behaviour, attainment of goal, satisfaction
 (D) Goal-directed behaviour, drive, satisfaction, attainment of goal
31. Evidence for hereditary influence can be gathered from
 (A) Twin studies
 (B) Adoption studies
 (C) Socio-economic status
 (D) Study of biological parents and children
32. In the neuron, which one of the following carries information away from the cell body?
 (A) Glial cells
 (B) Dendrite
 (C) Astrocyte
 (D) Axon
33. Which one of the following concepts is not part of Behaviouristic theories of personality ?
 (A) Reciprocal Determinism
 (B) Self Efficacy
 (C) Collective Unconscious
 (D) Reinforcement/Punishment
34. Which one of the following is not a trait theory of personality proposed through factor analysis ?
 (A) Eysenck's theory
 (B) Cattell's theory
 (C) McCrae and Costa's theory
 (D) Allport's theory
35. Two schedules of reinforcement that produce the highest rates of response are
 (A) Continuous and fixed interval
 (B) Fixed interval and variable interval
 (C) Variable interval and variable ratio
 (D) Fixed ratio and variable ratio
36. Procedural memory would be affected by damage to the
 (A) Cerebrum
 (B) Cerebellum
 (C) Motor cortex
 (D) Limbic system
37. In the familiar Carrot and Stick notion, which one of the following theories of motivation focuses more directly on carrot ?
 (A) Equity theory
 (B) Arousal theory
 (C) Expectancy theory
 (D) Needs hierarchy theory
38. The procedure which is frequently used in concept learning of children involves
 (A) Card sorting task
 (B) Ravan's Progressive matrices
 (C) Children's apperception test
 (D) Differential aptitude test
39. Which one of the following sub-tests is not part of Wechsler Adult Intelligence Scale?
 (A) Information (B) Vocabulary
 (C) Creativity (D) Digit symbol
40. Match List-I with List-II and select the correct answer using the codes given below the lists.
- | List-I | | List-II | |
|--|------------|-------------------------|------------|
| (Brain functions) | | (Brain parts) | |
| (p) Controls body balance | | (1) Pons | |
| (q) Maintains breathing and heart beat | | (2) Medulla | |
| (r) Co-ordinates the muscle movement | | (3) Reticular formation | |
| (s) Activates the brain to produce arousal | | (4) Cerebellum | |
| (p) | (q) | (r) | (s) |
| (A) 1 | 4 | 2 | 3 |
| (B) 2 | 4 | 1 | 3 |
| (C) 3 | 4 | 2 | 1 |
| (D) 4 | 2 | 1 | 3 |

Instructions for Qn. Nos. 41 to 45 : There are a pair of statements labelled as Assertion (A) and Reason (R). Read the pair of statements and answer the questions using the following codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (C) (A) is true but (R) is false.
- (D) (A) is false (R) is true.

41. Assertion (A) : Fixation is a kind of behavioural disorder.

Reason (R) : Fixation originates from unresolved conflicts during childhood.

42. Assertion (A) : Paired comparison scaling is not suitable for scaling large number of items.

Reason (R) : With large number of items, the number of possible pairs of items would be very large and experimentally inconvenient.

43. Assertion (A) : The phenomena of perceptual organisation, movement, depth, and perceptual constancies support what we see, but not how to explain.

Reason (R) : Perceptual ability is innate, we are designed to perceive the way we do.

44. Assertion (A) : Personality disturbances are due to the exposure of children to unhealthy role models.

Reason (R) : This happens because children's behaviour/personality is shaped through observational learning.

45. Assertion (A) : Edward's Personality Preference Schedule uses the forced-choice format.

Reason (R) : Forced-choice format prevents acquiescence response set.

Read the following paragraph and answer the five questions (Q. 46 to Q. 50) that follow :

A researcher hypothesized that a given drug would adversely affect the reaction time. Forty subjects were tested for discriminatory reaction time. Each subject was then given a prescribed drug dosage (5 mg/kg body weight) and after appropriate time gap, the subject was re-

tested for discriminatory reaction time. The time-interval between a 'ready signal' and actual presentation of the stimulus was randomly varied from half a second to three seconds. For evaluating the mean difference on the reaction time, the researcher considered various statistical options like the t test, analysis of variance, chi-square test etc. and employed the appropriate technique/s. The mean difference was statistically significant (Mean without drug condition = 240 m.s., SD = 20 ; and mean with drug condition = 210 m.s., SD = 18). The statistical examination further revealed that the difference between the two standard deviations was insignificant. During the course of research, the researcher suspected that the effect of drug dosage would be different for discriminatory and simple reaction times. For this purpose researcher considered the different experimental designs options such as matched two-group design, factorial design, analysis of covariance design, quasiexperimental design etc. Finally researcher selected the appropriate design to carry out the new investigation.

46. On the basis of above information, what can you conclude about the researcher's hypothesis ?

- (A) The researcher's hypothesis is rejected.
- (B) The researcher's hypothesis is partially verified.
- (C) The researcher's hypothesis is accepted.
- (D) The data are inadequate to evaluate the researcher's hypothesis.

47. For evaluating the mean differences, which of the following techniques would be appropriate ?

1. t test for independent samples.
 2. Repeated measures ANOVA
 3. Chi-square test
 4. Paired samples t test.
- (A) 4 only
 - (B) 2 and 4 only
 - (C) 1, 2 and 3 only
 - (D) 2, 3 and 4 only

48. Which of the following variables/ is/are not controlled variables ?

1. Drug to body weight ratio.
2. Foreperiod

3. Type of reaction
4. Reaction time
(A) 4 only (B) 1 and 4 only
(C) 2 and 4 only (D) 1, 2 and 4 only

49. On the basis of above passage, what would you conclude about the variance ?
(A) The two conditions have homogeneity of variance.
(B) The two conditions have heterogeneity of variance.
(C) We cannot conclude about the homogeneity of variance from the above data.
(D) It is necessary to apply an additional test to infer about homogeneity of variance.
50. For studying whether the effect of drug would be different for discriminatory and simple reactions, the appropriate design would be
(A) Quasi-experimental design
(B) Matched two-group design
(C) Analysis of covariance design
(D) Factorial design
51. Which of the following cannot be inferred from correlations ?
(A) Cause and effect relationship
(B) Covariance between variables
(C) Direction of relationship
(D) Percentage of explained variance
52. Experimental and Quasi Experimental cannot be distinguished on the basis of
(A) Manipulation of independent variable.
(B) Random assignment of subjects to experimental conditions.
(C) Studying causal relationship.
(D) Applying statistical analysis.
53. Which of the following tests assumes the sample size to be large ?
(A) Chi-square test
(B) Kalmogrov-Smirnov test
(C) Run-test
(D) None of these

54. Match List – I with List – II.

List – I

- (a) Experiment

List – II

1. Structured V/s Unstructured

- (b) Interview
(c) Observation
(d) Cross-cultural studies
2. Participant V/s Non-participant
3. Etic V/s Emic
4. True V/s Quasi

Codes :

	(a)	(b)	(c)	(d)
(A)	1	4	3	2
(B)	1	3	4	2
(C)	4	1	2	3
(D)	4	1	3	2

55. **Assertion (A) :** We cannot remember all the experiences as they are not organised.
Reason (R) : We can remember the effects of experiences as they are organized.
(A) Both (A) and (R) are true and (R) is the reason of (A).
(B) Both (A) and (R) are true, but (R) is *not* the correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.
56. The hemisphere involved in the perception and expression of negative emotions.
(A) Right hemisphere
(B) Left hemisphere
(C) Both Right and Left hemisphere
(D) Left or Right hemisphere depends upon intensity
57. Which is the content theory of motivation ?
(A) Equity theory
(B) Goal setting theory
(C) Needs Hierarchy theory
(D) Valence-Expectancy theory
58. Match List – I with List – II and indicate your answer by using the codes given below the list.

List – I

- (a) Expectancy theory
(b) Equity theory
(c) Goal-setting theory
(d) ERG

List – II

1. Desire to achieve a particular goal.
2. Making choices based on perceived expectations.
3. Motivation based on existence, relatedness and growth needs.
4. Motivation influ-

enced by perception of how fairly one is treated.

Codes :

	(a)	(b)	(c)	(d)
(A)	2	4	1	3
(B)	1	3	2	4
(C)	2	3	4	1
(D)	3	4	2	1

59. Match List – I with List – II and select the correct answer using the codes given below the lists.

List – I

- (a) Psychoanalytic approach
- (b) Instinct theory
- (c) Growth and deficiency motivation
- (d) Humanistic approach

List – II

- 1. Carl Rogers
- 2. Maslow
- 3. Freud
- 4. McDougall

Codes :

	(a)	(b)	(c)	(d)
(A)	4	3	2	1
(B)	3	4	2	1
(C)	1	4	2	3
(D)	2	3	1	4

60. Which of the following is not used for measuring intelligence ?
- (A) Reaction time experiments
 - (B) Projective tests
 - (C) Psychometric tests
 - (D) Performance tests
61. Speed and Power tests cannot be distinguished on the basis of following :
- (A) Time limit
 - (B) Length of test
 - (C) Difficulty level
 - (D) Verbal or non-verbal contents
62. A general relation perceiving capacity which represent one's potential intelligence, which is independent of socialization and education is
- (A) Crystallized intelligence
 - (B) Fluid intelligence
 - (C) Abstract intelligence
 - (D) General intelligence factor
63. Which of the following researchers found psychometric evidence for Freud's concepts ?

- (A) Eysenck
- (B) Cattell
- (C) Guilford
- (D) None of these

64. Inferiority complex is associated with

- (a) feeling of inadequacy
- (b) powerlessness
- (c) experiences of all children
- (d) helplessness

- (A) (a), (b) and (c)
- (B) (b), (c) and (d)
- (C) (a), (b) and (d)
- (D) (c) and (d)

65. The part of the autonomic nervous system, when activated, is responsible for the body's 'fight or flight' sensation.

- (A) Brain stem
- (B) Limbic system
- (C) Sympathetic nervous system
- (D) Para-sympathetic nervous system

66. Match List – I with List – II.

List – I

List – II

- | | |
|--------------|--|
| (a) McKellar | 1. Assimilation and accommodation. |
| (b) Freud | 2. Rational thinking. |
| (c) Piaget | 3. Pre-conscious, conscious and unconscious. |
| (d) Guilford | 4. Divergent thinking. |

Codes :

	(a)	(b)	(c)	(d)
(A)	1	2	3	4
(B)	2	3	1	4
(C)	3	4	2	1
(D)	4	1	3	2

67. Creativity is measured on the dimension of

- (A) originality
- (B) applicability
- (C) flexibility
- (D) all of the above

68. Match List – I with List – II.

List – I

List – II

- | | |
|--|--------------------------|
| (a) Ability to think ahead. | 1. Syllogistic reasoning |
| (b) Inference drawn from one particular to another particular. | 2. Planning |

- (c) Evaluation of whether a conclusion follows on from two premises.
 (d) Drawing conclusions from several observations.
 reasoning

Codes :

	(a)	(b)	(c)	(d)
(A)	1	4	2	3
(B)	2	3	1	4
(C)	4	2	1	3
(D)	3	1	2	4

69. Psychoanalytic theory basically is concerned with
 (a) analysis of unconscious motives.
 (b) psycho-sexual development.
 (c) structure of personality in terms of id, ego and superego.
 (d) ego-identity.
 (A) (b), (c) and (d) (B) (a), (b) and (c)
 (C) (b) and (c) (D) (a), (c) and (d)

70. Given below are two statements, one labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Type A persons show anger, hostility, time urgency and depression.

Reason (R) : Type A persons are more prone to cardio-vascular diseases.
 In the context of the above two statements, which one of the following is correct ?

Codes :

- (A) (A) is correct, but (R) is wrong.
 (B) Both (A) and (R) are correct.
 (C) (A) is wrong, but (R) is correct.
 (D) Both (A) and (R) are wrong.

71. In verbal conditioning experiments the following variables are controlled
 (a) Nature of verbal tasks
 (b) Nature of reinforcement
 (c) Meaningfulness of material
 (d) Familiarity of material
 (A) (a) and (b) are correct.
 (B) (b) and (c) are correct.
 (C) (c) and (d) are correct.
 (D) (a) and (d) are correct.

72. Indicate the correct sequence of stages in creative thinking

- (a) Incubation
 (b) Preparation
 (c) Evaluation
 (d) Illumination
 (A) (a), (c), (d), (b)
 (B) (b), (a), (d), (c)
 (C) (a), (b), (c), (d)
 (D) (d), (a), (b), (c)

73. Match List – I with List – II.

List – I (Obstacles in problem solving)

List – II (Names of Psychologists)

- | | |
|--------------------------|-------------------|
| (a) Functional fixedness | 1. Duncker |
| (b) Mental set | 2. Luchins |
| (c) Analogies | 3. Glik & Holyoak |
| (d) Heuristics | 4. Newell & Simon |

Codes :

	(a)	(b)	(c)	(d)
(A)	1	4	2	3
(B)	2	3	4	1
(C)	4	1	2	3
(D)	1	2	3	4

74. With the help of classical conditioning emotional responses

- (A) are learned best
 (B) are equal to other responses
 (C) are learned very slowly
 (D) are learned only when US (Unconditioned Stimulus) is powerful

75. Given below are two statements, one labelled as Assertion (A) and the other labelled as Reason (R).

Assertion (A) : Humans are motivated by needs beyond basic biological survival.

Reason (R) : Individuals strive to achieve self improvement.

In the context of the above two statements, which one of the following is correct ?

Codes :

- (A) Both (A) and (R) are true and (R) is the correct explanation.
 (B) Both (A) and (R) are true, but (R) is not the correct explanation.

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

76. According to Witkin the 'fieldindependent subjects are

(A) More analytical

(B) More influenced by the frame

(C) More conforming

(D) Sensitive to context

77. Match List – I with List – II with most appropriate combinations.

List – I

List – II

- | | |
|--|------------------------------------|
| (a) Internal-consistency reliability. | 1. Product moment correlation |
| (b) Split-half reliability. | 2. Spearman-Brown prophecy formula |
| (c) Relationship between reliability and length of test. | 3. Rulona-formula |
| (d) Test-retest reliability | 4. Kudor-Richardson formula |

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-----|-----|-----|-----|
| (A) | 4 | 2 | 3 | 1 |
| (B) | 4 | 2 | 1 | 3 |
| (C) | 4 | 3 | 2 | 1 |
| (D) | 1 | 3 | 2 | 4 |

78. Which of the following is not a criterion based validity ?

(A) Concurrent validity

(B) Face validity

(C) Factorial validity

(D) Predictive validity

79. While constructing a test on Anxiety, Item Discrimination means

(A) Persons high and low on Anxiety will score differently on that item.

(B) The scores of two individuals will be different on that item irrespective of their being high or low on Anxiety.

(C) The scores of two persons who are high and low on Anxiety will have same score on that item.

(D) Person high on Anxiety will score higher on that item and nothing can

be said about the person low on Anxiety.

80. **Assertion (A)** : Culture fair intelligence test measures fluid intelligence.

Reason (R) : Fluid intelligence is biologically determined.

Codes :

(A) (A) is true, but (R) is false.

(B) (A) is false, but (R) is true.

(C) Both (A) and (R) are true, but (R) is *not* the correct explanation.

(D) Both (A) and (R) are true and (R) is the correct explanation.

81. **Assertion (A)** : Personality traits measured by 16 PF are source traits.

Reason (R) : Cattell used oblique rotation to identify personality traits.

Codes :

(A) Both (A) and (R) are true and (R) is the correct explanation.

(B) (A) is true, but (R) is false.

(C) (A) is false, but (R) is true.

(D) Both (A) and (R) are true, but (R) is *not* the correct explanation.

82. When number of independent variables in an experiment are increased, what will be the changes in the treatment and error components of total variance ?

(A) Error component will decrease and treatment component will increase.

(B) Error component will decrease and there will be no effect on treatment component.

(C) Error component will increase and treatment component will increase.

(D) Error component will increase and there will be no effect on treatment component.

83. Match the following :

List – I

List – II

- | | |
|---------------|------------------------------------|
| (a) Marbe | 1. Inter-item association |
| (b) Delse | 2. Frequency effect in free recall |
| (c) Zeller | 3. Effect of failure on memory |
| (d) Zeigarnik | 4. Recall of interrupted task |

Codes :

	(a)	(b)	(c)	(d)
(A)	4	3	2	1
(B)	2	1	3	4
(C)	3	2	1	4
(D)	1	4	2	3

84. If meaningfulness is taken on X axis and percentage of correct responses on Y axis, the curve will be
 (A) negatively accelerated
 (B) positively accelerated
 (C) inverted U type
 (D) S type
85. If a subject is supposed to learn a list and relearn it in reverse order,
 (a) Recall method leads to positive transfer.
 (b) Anticipation method leads to no transfer.
 (c) Recall method leads to negative transfer.
 (d) Anticipation method leads to negative transfer.
 (A) (a) & (b) are correct.
 (B) (a) & (d) are correct.
 (C) (c) & (d) are correct.
 (D) (b) & (c) are correct.
86. Capacity of sensory memory is
 (A) always less than short term & long term memory.
 (B) less than long term memory, but more than short term memory.
 (C) equal to short term memory, but less than long term memory.
 (D) more or less than short term memory depending on nature of the stimulus.
87. Perception of figure & ground depends on
 (a) size of two areas
 (b) completeness of two areas
 (c) brightness of two areas
 (d) exposure time
 (A) (a), (b) & (d)
 (B) (b), (c) & (d)
 (C) (a), (b), & (c)
 (D) (b) only
88. Bem, in his self-perception theory, believes we arrive at our attitudes by

- (A) A dissonance reduction
 (B) Trying to explain our behaviour
 (C) Being reinforced for them
 (D) Learning them from our parents.

89. Broadbent's model suggests that early selection of items to be attended is done on the basis of

- (A) Physical characteristics
 (B) Semantic characteristics
 (C) Interest
 (D) Need

90. Match the items of List – I with List – II.

List – I (Researcher)	List – II (Field of Research)
(a) Hebb	1. Transcational functionalism
(b) Adler	2. Adaptation level
(c) Ames	3. Goal perception
(d) Koffka	4. Cell assembly

Codes :

	(a)	(b)	(c)	(d)
(A)	2	3	4	1
(B)	3	2	1	4
(C)	4	3	1	2
(D)	1	4	2	3

91. Ponzo illusion increases with
 (a) age
 (b) learning
 (c) size of the stimulus
 (d) brightness of the stimulus
 (A) (a) & (b)
 (B) (c) & (d)
 (C) (b) & (d)
 (D) (c) & (b)
92. Match List – I and List – II.

List – I (Brain Physiology)	List – II (Role in motivation)
(a) Medial forebrain bundle	1. General arousal
(b) Hypothalamus	2. Cognitive side of motivation
(c) Reticular activating system	3. Aggression and sex
(d) Cerebral cortex	4. Pleasure area

Codes :

	(a)	(b)	(c)	(d)
(A)	4	3	1	2
(B)	2	1	3	4
(C)	3	2	4	1
(D)	1	4	2	3

- 93. Assertion (A) :** In case of blind Braille readers, increase in sensory cortex surface area devoted to right forefinger is seen.

Reason (R) : Brain is very flexible and adjust the location of function if environment demands it.

Codes :

- (A) Assertion and Reason both are true and Reason is the explanation of Assertion.
 (B) Assertion and Reason both are true, but Reason is not the explanation of Assertion.
 (C) Assertion is true and Reason is false.
 (D) Assertion is false, but Reason is true.

- 94.** Hypothalamus plays an important role in

- (a) motivation
 (b) homeostasis
 (c) emotional arousal
 (d) memory

- (A) (a), (b) and (d)
 (B) (b), (c) and (d)
 (C) (a), (b) and (c)
 (D) (b) and (d) only

- 95.** When physiological changes are treated as independent variable in learning experiment ?

- (A) A drug is injected to the subject before learning.
 (B) A drug is injected to the subject after learning.
 (C) Physiological changes resulting from learning are recorded.
 (D) Physiological changes correlated with forgetting are recorded.

- 96. Assertion (A) :** In some researches it was found that immediate serial recall of verbal items varying in meaning was facilitated with visual input for Chinese subjects but auditory for American subjects.

Reason (R) : Possibility of visual versus auditory confusion varies according to nature of a particular language.

Codes :

- (A) Assertion and Reason both are true and Reason is the correct explanation of Assertion.
 (B) Assertion and Reason both are true, but Reason is not the correct explanation of Assertion.
 (C) Assertion is true, but Reason is false.
 (D) Assertion and Reason both are false.

- 97. Assertion (A) :** Most functions of brain are contra laterally controlled, i.e. information of one side of body is received by hemisphere on opposite side of brain.

Reason (R) : Many functions are duplicated in both hemispheres, though there are differences in function.

Codes :

- (A) Assertion is true and Reason is true and Reason is the correct explanation of Assertion.
 (B) Assertion is true and Reason is also true, but Reason is not the correct explanation of Assertion.
 (C) Assertion is true, but Reason is false.
 (D) Assertion is false, but Reason is true.

- 98.** Once Pavlov was called to his student's laboratory to watch classical conditioning with dog. However, the dog did not salivate irrespective of appropriate procedures. The reason was

- (A) induction effect
 (B) external inhibition
 (C) inhibition of inhibition
 (D) internal inhibition

- 99.** Match List – I with List – II.

List – I	List – II
(a) Trace conditioning	1. When CS and US occurs together.
(b) Delayed conditioning	2. CS appears first, it disappears then US appears.
(c) Simultaneous conditioning	3. CS appears first but also present when the US appears.
(d) Backward conditioning	4. When US is presented first and after the disappearance of US, CS is presented.

Codes :

	(a)	(b)	(c)	(d)
(A)	2	1	3	4
(B)	1	2	3	4
(C)	4	3	1	2
(D)	3	1	4	2

100. According to conceptual peg hypothesis, recall of paired associates depend upon the type of concept. There, highest to lowest order of recall is

- (A) concrete-concrete, abstract-concrete, concrete-abstract, abstract-abstract.
 (B) concrete-concrete, concrete-abstract, abstract-concrete, abstract-abstract.
 (C) concrete-abstract, concrete-concrete, abstract-concrete, abstract-abstract.
 (D) concrete-concrete, abstract-abstract, abstract-concrete, concrete-abstract.

ANSWER SHEET

1. (A)	2. (D)	3. (C)	4. (B)	5. (A)	6. (D)	7. (C)	8. (B)	9. (C)	10. (A)
11. (B)	12. (B)	13. (A)	14. (B)	15. (C)	16. (D)	17. (A)	18. (C)	19. (B)	20. (B)
21. (C)	22. (D)	23. (A)	24. (D)	25. (B)	26. (D)	27. (C)	28. (B)	29. (A)	30. (A)
31. (D)	32. (D)	33. (C)	34. (C)	35. (A)	36. (B)	37. (D)	38. (A)	39. (C)	40. (D)
41. (B)	42. (C)	43. (A)	44. (A)	45. (A)	46. (D)	47. (C)	48. (C)	49. (D)	50. (A)
51. (D)	52. (B)	53. (B)	54. (B)	55. (C)	56. (D)	57. (C)	58. (A)	59. (C)	60. (B)
61. (D)	62. (D)	63. (A)	64. (C)	65. (D)	66. (B)	67. (A)	68. (B)	69. (A)	70. (D)
71. (B)	72. (B)	73. (D)	74. (D)	75. (B)	76. (B)	77. (A)	78. (D)	79. (D)	80. (C)
81. (A)	82. (C)	83. (B)	84. (C)	85. (C)	86. (D)	87. (C)	88. (A)	89. (B)	90. (C)
91. (C)	92. (A)	93. (A)	94. (B)	95. (B)	96. (B)	97. (B)	98. (C)	99. (A)	100. (C)

PREVIOUS YEARS' PAPERS

June, 2015 PSYCHOLOGY PAPER-II

Note : This paper contains **fifty** (50) multiple choice questions, each question carrying **two** (2) marks. Attempt all of them.

- Partial reinforcement is considered better than continuous reinforcement because
 - the learning occurs rapidly
 - the learning occurs slowly
 - the extinction occurs rapidly
 - the extinction occurs slowly
- Match List - I with List - II and indicate your answer using the codes given below :

List-I (Test)	List - II (Dimension / Trait assessed)
(a) EPPS	(i) State anxiety
(b) EPQ - R	(ii) Psychoticism
(c) NEO - PI - R	(iii) Achievement motivation
(d) STAI	(iv) Openness to experience

Codes :

	(a)	(b)	(c)	(d)
(1)	(iii)	(ii)	(iv)	(i)
(2)	(i)	(ii)	(iv)	(iii)
(3)	(iii)	(iv)	(ii)	(i)
(4)	(i)	(iv)	(ii)	(iii)
- Which of the following is correctly arranged?
 - Sensory memory / short / term memory / forgetting
 - Long / term memory / sensory memory / short-term memory
 - Sensory memory / short / term memory / long-term memory
 - Episodic memory / short term memory / long-term memory
- Arrange the four basic characteristics of instincts as per Freud's psychoanalytic theory in terms of sequence.
 - an impetus
 - a source
 - an object
 - an aim

Chose the correct code for sequence of events

 - (b) (a) (d) (c)
 - (c) (b) (a) (d)
 - (b) (d) (a) (c)
 - (a) (b) (c) (d)
- History of intelligence Testing can be discribed in following sequence
 - Galton J.M. Caltell Benet - Simon Wechsler
 - J.M. Cattell Stern-Terman Benet - Simon Wechsler
 - Wister Benet - Simon Spearman R.B. Cattell
 - Galton J.M. Cattell Spearman Wechsler
- Bartlett described memory as a :
 - Constructive mental process
 - Reproductive mental process
 - Active mental process
 - A process of retention
- Deterioration Quotient (DQ) was first used in which intelligence tests ?
 - Benet - Simon Test

- (2) Galton - Cattell Test
(3) Raven Progressive Matrices
(4) Wechsler Adult Intelligence Test
8. According to Hull - Spence theory, anxiety is emotionally based drive and therefore, it can be hypothesized that high anxious subjects would be faster on eyelid conditioning than the low anxious subjects. Using Taylor Manifest Anxiety Scale (MAS), a researcher confirmed this hypothesis. This provides evidence for the :
- (a) Convergent and discriminant validity of the MAS
(b) Validity of the Hull - Spence theory
(c) Generalizability of the MAS
(d) Construct validity of the MAS
(1) (b) only (2) (a) and (b) only
(3) (b) and (c) only (4) (b) and (d) only
9. The best home environment for intellectual development is one :
- (1) In which child is encouraged to work out problems with guidance and support from parents
(2) In which parents dictate the child what to do
(3) In which child is left on his own
(4) In which father is absent to impose his authority
10. According to interference theory, which one of the following causes forgetting ?
- (1) Failure of storage
(2) Failure of encoding
(3) Failure of retrieval
(4) Response Competition
11. The evidence that sleep is a biological motive comes from _____ in lengthy sleep deprived subjects.
- (1) rebound effect of slow wave sleep
(2) rebound effect of REM sleep
(3) insomnia
(4) enhanced stage - I and stage - II sleep
12. On a Wechsler type intelligence scale, the subject has a deviation IQ 115. What would be his T score (Mean = 50, SD = 10) ?
- (1) 60 (2) 65
(3) 68 (4) 84
13. Complex behavioural sequences which are species - typical behaviours are triggered by
- (1) releasers in the environment
(2) learned behaviours
(3) needs
(4) incentives
14. The following areas are involved in human vision
- (a) Prestriate cortex
(b) Striate cortex
(c) Inferotemporal cortex
(d) Posterior parietal cortex
- Choose the correct option :
- (1) (a) and (b) (2) (a), (b) and (c)
(3) (b), (c) and (d) (4) (a), (b), (c) and (d)
15. Which one of the following is not correct in regard of semantic memory ?
- (1) It is relatively stable
(2) There is interference in it
(3) It is free from acquisition context
(4) It has chunking
16. Self - modelling means :
- (1) A type of learning in which individuals watch themselves behaving in a situationally appropriate manner and then show the same behaviours later on.
(2) Type of learning in which individuals learn new behaviour
(3) Type of learning in which individuals learn to develop self-beliefs
(4) Type of learning in which individuals learn through guided - participation from significant others.
17. Match List - I and List - II and indicate your correct choice among the codes given below
- | List - I
(Type of
reinforcer) | List - II
(Description) |
|--|--|
| (a) Punisher | (i) acquired satisfier |
| (b) Positive reinforcer | (ii) removing aversive stimulus |

- (c) Negative reinforcer (iii) presenting reward
(d) Secondary reinforcer (iv) aversive consequence

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|------|
| (1) | (iv) | (iii) | (ii) | (i) |
| (2) | (i) | (iii) | (ii) | (iv) |
| (3) | (iii) | (iv) | (ii) | (i) |
| (4) | (i) | (ii) | (iii) | (iv) |

18. While travelling in a train, nearby objects appear to move faster in the opposite direction than do far away objects this phenomena is called :
- (1) Phi - Phenomena
(2) Motion parallax
(3) Autokinetic movement
(4) Perception of movement
19. Which of the following is correctly arranged ?
- (1) Identify - set goals - explore - alternatives - select - evaluate - implement
(2) Identify - explore - set goals - alternatives - select - implement - evaluate
(3) Select - explore - identify - set goals - alternatives - implement - evaluate
(4) Explore - identify - select - set goals - alternatives - evaluate - implement

20. Read each of the following two statements - Assertion (A) and Reason (R) and indicate your answer using the codes given below

Assertion (A) : According to the true score model in the Classical Test Theory (CTT), for a population or a large sample, Mean of the obtained scores = Mean of the true scores.

Reason (R) : In the CTT, the errors of measurement are assumed to be independently distributed with a mean and variance equal to zero.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)

- (2) Both (A) and (R) are true, but (R) is **not** the correct explanation of (A)
(3) (A) is true, but (R) is false
(4) (A) is false, but (R) is true

21. Match List-I and List-II, and select the correct answer by choosing from codes given below :

List - I

List - II

- | | |
|-----------------|---|
| (a) Libido | (i) Human beings strive to return to an inorganic state of balance that preceded life, in which there is no painful struggle to satisfy biological needs. |
| (b) Thantos | (ii) Psychic and pleasurable feelings associated with gratification of life instincts. |
| (c) Suppression | (iii) A person's refusal to perceive an unpleasant environment in external reality. |
| (d) Denial | (iv) Individual's active and conscious attempt to stop anxiety provoking thoughts by simply not thinking about them. |

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|-------|
| (1) | (i) | (ii) | (iii) | (iv) |
| (2) | (iii) | (i) | (ii) | (iv) |
| (3) | (ii) | (i) | (iv) | (iii) |
| (4) | (iv) | (iii) | (ii) | (i) |

22. Read each of the following two statements - Assertion (A) and Reason (R) and indicate your answer using the codes given below :

Assertion (A). Retinal disparity is an important monocular cue in depth perception.

Reason (R). Convergence is a muscular cue in depth perception.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)

- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
 (3) (A) is true, but (R) is false
 (4) (A) is false, but (R) is true
23. As per Schachter and Singer's theory of emotions, one shall experience emotions based on situational clues when
- there is a physiological arousal
 - there are cognitions / explanations available for arousal
 - there is no physiological arousal
 - there are no proper explanations available for one's arousal

Chose the correct code :

- (1) (a) and (b) (2) (a) and (d)
 (3) (b) and (c) (4) (c) and (d)
24. Spearman Inferred General Intelligence on the basis of
- Positive Manifold in intercorrelation matrix
 - Tetrad differences in equation
 - unequal scores of same subjects on intelligence tests
 - Reports from the subjects

Codes :

- (1) (a) and (c) (2) (b) and (d)
 (3) (a) and (b) (4) (c) and (d)
25. The parasympathetic nerves project from :
- the lumbar and thoracic regions of the spinal cord
 - the brain and sacral region of the spinal cord
 - the brain and the lumbar region of the spinal cord
 - the lumbar and the sacral regions of the spinal cord

26. Read each of the following two statements - Assertion (A) and Reason (R) and indicate your answer using the codes given below :
- Assertion (A) :** People who learn more and more lists of words on successive days have more and more difficulty learning each new list next day.

Reason (R) : People have limited capacity to learn verbal material.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
 (3) (A) is true, but (R) is false
 (4) (A) is false, but (R) is true
27. Read each of the following two statements - Assertion (A) and Reason (R) and indicate your answer using the codes given below :
- Assertion (A) :** A synapse does not necessarily result into action potential in post-synaptic neuron.
- Reason (R) :** Synapses produce inhibitory post-synaptic potential also in post-synaptic neuron.
- Codes :
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
 (3) (A) is true, but (R) is false
 (4) (A) is false, but (R) is true
28. Which of the following is not the way we acquire concepts ?
- Naturally acquired
 - By Discrimination Learning
 - Innate
 - By Definition
29. Match List - I with List - II and select the correct answer by choosing from the codes given below :

List - I

- (a) Functional fixedness

List - II

- (i) A cognitive process that employ appropriate novelty to originality that suits some purpose.
 (ii) A problem solving phenomenon in which people seeing alternative uses for common objects.
- (b) Back tracking

- (c) Creativity (iii) A problem solving technique that involves keeping track of solving process where in assumptions are made ; so they may later be changed.
- (d) Working backward (iv) A problem solving technique that identifies the final goals and the steps in reverse order.

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|------|
| (1) | (i) | (ii) | (iii) | (iv) |
| (2) | (ii) | (iii) | (iv) | (i) |
| (3) | (iii) | (i) | (ii) | (iv) |
| (4) | (ii) | (iii) | (i) | (iv) |

30. Visual cliff is a laboratory method for testing depth perception in :

- | | |
|-------------|----------------------|
| (1) Infants | (2) Children |
| (3) Adults | (4) Old - age people |

31. Read each of the following two statements - Assertion (A) and Reason (R) and indicate your answer using the codes given below :
Assertion (A) : Surface traits are simply a collection of trait elements which obviously go together in many different individuals and circumstances. Intercorrelated trait elements or variables cluster together to form a surface trait.

Reason (R) : Source traits are underlying factors that control the variation in surface clusters.

Codes :

- | |
|--|
| (1) Both (A) and (R) are true and (R) is the correct explanation of (A) |
| (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A) |
| (3) (A) is true, but (R) is false |
| (4) (A) is false, but (R) is true |
32. Match List - I with List - II and select the correct answer by choosing correct code :

List-I

- (a) -67 mv
 (b) + 45 mv
 (c) - 70 mv
 (d) - 72 mv

List - II

- (i) Polarized
 (ii) Hyperpolarized
 (iii) Depolarized
 (iv) Completely depolarized

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|------|
| (1) | (iii) | (ii) | (i) | (iv) |
| (2) | (iii) | (iv) | (i) | (ii) |
| (3) | (ii) | (iv) | (iii) | (i) |
| (4) | (iv) | (iii) | (i) | (ii) |

33. Read each of the following two statements - Assertion (A) and Reason (R) and indicate your answer using the codes given below :
Assertion (A) : In problem solving process, people generally evolve / select the optimal solutions of a problem.

Reason (R) : A number of solutions are identified to choose from.

Codes :

- | |
|--|
| (1) Both (A) and (R) are true and (R) is the correct explanation of (A) |
| (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A) |
| (3) (A) is true, but (R) is false |
| (4) (A) is false, but (R) is true |
34. Sometimes old information facilitates our learning of new information. The phenomenon is known as _____
- | |
|----------------------------|
| (1) Consolidation effect |
| (2) Transfer effect |
| (3) Proactive interference |
| (4) Excitatory effect |
35. A man judged to be six feet tall when standing at ten feet away has 'X' size of retinal image where as at twenty feet, it has X/2 retinal size image. How tall shall he be perceived at a distance of five feet ?
- | |
|--------------------------------------|
| (1) 6 feet |
| (2) $6 \times \frac{10 - 5}{2}$ feet |

- (3) $6 \times \frac{10 \pm 5}{2}$ feet
 (4) Size can not be judged
36. The first attempt to both theorise and measure intelligence was made by
 (1) J.M. Cattell (2) Galton
 (3) Binet (4) Spearman
37. Read each of the following two statements - Assertion (A) and Reason (R) and indicate your answer using the codes given below :
Assertion (A) : According to Maslow, after the gratification of basic needs we move in positive direction, but it is not automatic.
Reason (R) : We often fear our best side, our talents, our finest impulses, our creativeness.
Codes :
 (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true, but (R) is **not** the correct explanation of (A)
 (3) (A) is true, but (R) is false
 (4) (A) is false, but (R) is true
38. A fifth test was added to an old established selection test battery of four tests. Two multiple correlations, one between a suitable external criterion and three tests of the old battery, and the other between the same criterion and four tests of the new battery were computed. The significant change in R^2 would provide an evidence for the :
 (1) predictive validity of the old battery
 (2) incremental validity of the fifth test
 (3) predictive validity of the new battery
 (4) concurrent validity of the fifth test
39. Which of the following stages is called the period of problem solving behaviour ?
 (1) Secondary Circular Reaction
 (2) Reflex
 (3) Tertiary Reaction
 (4) Primary Circular Reaction
40. The rate of responding is found to be higher in the condition of :
 (1) Variable-interval schedule of reinforcement
 (2) Fixed-ratio schedule of reinforcement
 (3) Variable-ratio schedule of reinforcement
 (4) Both in (2) and (3)
41. Read the following statements and indicate which are correct.
 (a) Archetypes are themes that have existed in all cultures throughout history.
 (b) Personal unconscious is the region next to the ego which consists of all the forgotten experiences that have lost their intensity
 (c) Masculine Archetype in men is called animus.
 (d) Persona is a compromise between demands of the environment and necessities of an individual.
Codes :
 (1) (a), (b) and (c)
 (2) (b), (c) and (d)
 (3) (a), (b) and (d)
 (4) (a), (c) and (d)
42. Trace the correct sequence of the following events for audition
 (a) synapses in primary auditory cortex
 (b) synapse in superior olives
 (c) synapse in the inferior colliculi
 (d) synapse in the cochlear nuclei
 (e) synapse in medial geniculate nucleus
Codes :
 (1) (d) (b) (c) (e) and (a)
 (2) (d) (c) (b) (e) and (a)
 (3) (b) (c) (e) (d) and (a)
 (4) (c) (b) (e) (d) and (a)
43. Our brains construct our perceptions because :
 (a) Sensations are disassembled into information bits
 (b) Neuro muscular cues from the greater inward turn when the eyes view a near object

- (c) Brain reassembles information into its own functional model
- (d) Perception is merely projecting the world onto our brain

Codes :

- (1) (a) (b) (c)
- (2) (b) (c) (d)
- (3) (a) (c) (d)
- (4) (a) (b) (d)

44. Match List - I and List - II and select the correct answer from the codes given below :

List - I

List-II

- | | |
|-----------------------|---|
| (a) Self-Schema | (i) A generalised evaluation of self |
| (b) Self-esteem | (ii) When a person doubts one's abilities to perform a task |
| (c) Self-presentation | (iii) Organisation of informations about self |
| (d) Self-handicapping | (iv) The way one thinks of presenting oneself before public |

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|------|
| (1) | (iv) | (ii) | (iii) | (i) |
| (2) | (ii) | (iii) | (i) | (iv) |
| (3) | (iii) | (i) | (iv) | (ii) |
| (4) | (i) | (iii) | (ii) | (iv) |

45. Which of the following factors reduce the accuracy of eyewitness testimony ?

- (a) suggestibility
- (b) Source monitoring errors
- (c) Interference
- (d) Illusion of out group homogeneity

Codes :

- (1) (a) and (b)
- (2) (a), (b) and (d)
- (3) (a), (c) and (d)
- (4) (b), (c) and (d)

Instructions for Q. Nos. 46 to 50 :

Read the following paragraph and answer the five questions that follow :

Psychology, as a scientific study of behaviour, employs several diverse research methods to collect and analyse data. In the early phase of modern psychology, experimental method has been highly popular. The early psychologists, like Fechner and Ebbirghaus, were well-known experimentalists. Experimental method allows the active manipulation of the independent variable (IV) Ronald Fisher and others systematized the area of experimental designs. Due to the success of controlled laboratory experiments, field experiments were also carried out where the IV is manipulated in the natural or field setting. Especially due to the need for policy and program evaluation research, quasi-experimental designs have also been developed. Various non-experimental methods, including surveys and correlational research, are also employed in psychology. The survey method employs interviews, both structured and unstructured, and questionnaires. Both open-end and close-end questions and scale items can be used in interview schedules and questionnaires. Different types of observations, including naturalistic observation, participant and non-participant observation, are also used in certain areas like ethology. In the qualitative research tradition, case studies are widely used. Special research approaches, for example, ethnographic research, have been developed for specialized areas like cross-cultural psychology. In addition, psychologists use various measurement techniques including standardized objective tests, projective methods, situational tests and even physiological measures. The use of such diverse methods and measurement techniques has enriched the science of psychology as well as our understanding of human and animal behaviour including the cognitive processes.

46. Which of the following are considered as quasi-experimental designs ?

- (a) Time series designs
- (b) Regression-discontinuity design
- (c) Survey
- (d) Non-equivalent control group design

Codes :

- (1) (d) only
 (2) (a) and (d) only
 (3) (a), (b) and (d) only
 (4) (b), (c) and (d) only
47. Which one of the following is a limitation of case study ?
 (1) It does not provide indepth information of the subject
 (2) It is difficult to generalize from the case study
 (3) It is useful only in clinical and counselling psychology
 (4) It uses diverse methods and sources to collect data about the subject
48. Read each of the following two statements - Assertion (A) and Reason (R) and indicate your answer using the codes given below :
Assertion (A) : Experimental method, as compared to other methods, is more suitable for studying cause - and - effect relationship.
Reason (R) : As compared to other methods, experimental method permits more efficient control of extraneous variance.
- Codes :**
 (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true, but (R) is **not** the correct explanation of (A)
 (3) (A) is true, but (R) is false
 (4) (A) is false, but (R) is true
49. The emic - etic distinction is emphasized in:
 (1) quasi-experimental research
 (2) ethological research
 (3) case studies
 (4) ethnographic research
50. In a survey of students, each student was asked to state his views on the relevance of present educational system in not more than 60 words. This is an example of :
 (1) Scale item
 (2) Closed - end question
 (3) Check - list item
 (4) Open - end question

Answer Sheet

1. (4)	2. (1)	3. (3)	4. (3)	5. (1)	6. (1)	7. (4)	8. (4)	9. (1)	10. (4)
11. (1)	12. (1)	13. (1)	14. (4)	15. (3)	16. (1)	17. (1)	18. (2)	19. (2)	20. (3)
21. (3)	22. (4)	23. (2)	24. (3)	25. (2)	26. (3)	27. (2)	28. (3)	29. (4)	30. (1)
31. (2)	32. (4)	33. (1)	34. (2)	35. (1)	36. (2)	37. (1)	38. (2)	39. (3)	40. (4)
41. (3)	42. (1)	43. (3)	44. (3)	45. (2)	46. (3)	47. (2)	48. (1)	49. (4)	50. (4)

June, 2015
PSYCHOLOGY
PAPER-III

Note : This paper contains **seventy five (75)** objective type questions of **two (2)** marks each. **All** questions are compulsory.

1. Which of the following is an application of operant conditioning ?
 - (1) Aversive conditioning
 - (2) Behaviour modification
 - (3) Systematic desensitization
 - (4) Transfer of learning
2. The basic assumptions of Humanistic approach
 - (a) People are naturally striving to be creative and happy.
 - (b) It is possible to specify an universal set of values that will provide people with moral anchor enabling them to decide what is right or wrong and good or bad.
 - (c) It believes in uniqueness of individuals and they should be trained to realise their inner potentials.
 - (d) All the innate potentials and growth mechanisms can be studied scientifically similar to natural success.

Codes :

(1) (a) and (b)	(2) (b) and (c)
(3) (b) and (d)	(4) (c) and (d)
3. Deductive and inductive reasoning refers to
 - (a) Universal to universal
 - (b) Universal to particular
 - (c) Particular to particular
 - (d) Particular to universal

Codes :

(1) (a), (b)	(2) (c), (a)
(3) (b), (d)	(4) (d), (c)
4. According to Wallas creative performance completes through following sequence
 - (1) Preparation Illumination Incubation Verification
 - (2) Problem funding Problem Identification Incubation Insight
 - (3) Problem identification Preparation Incubation Illumination
 - (4) Preparation Incubation Illumination Verification
5. Read each of the following two statements :
Assertion (A) and Reason (R) and indicate your answer using the codes given below :
Assertion (A) : Gf and Gc correlate positively with each other across the age levels but with moderate magnitude.
Reason (R) : Gf and Gc have different growth patterns. Gc grows rapidly in childhood and adolescence whereas Gf grows rapidly during adulthood.

Codes :

 - (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
6. In absorptive phase of energy metabolism, what is the position of insulin and glucagon hormones ?
 - (a) High insulin
 - (b) High glucagon
 - (c) Low insulin
 - (d) Low glucagon

Codes :

- (1) (a) and (b) (2) (b) and (d)
 (3) (c) and (d) (4) (a) and (d)

7. Pygmalion effect is
 (1) Exceptional fall in the performance of a student as a result of teacher's expectations
 (2) Exceptional progress made by a student as a result of teacher's expectations
 (3) Exceptional fall in the performance of a student because of peer competition
 (4) Exceptional progress made by a student as a result of peer competition
8. Which one of the following semantic dimensions are measured by the "semantic differential scale" ?
 (1) Evaluation and potency
 (2) Potency and action
 (3) Evaluation and action
 (4) Evaluation, potency and action
9. Conversion Disorder (Hysteria) is now placed in the category of
 (1) Anxiety Disorders
 (2) Anxiety related Disorders
 (3) Somatoform Disorders
 (4) Psychological Factors Affecting Physical conditions
10. Suppose Merry wants to know if she can learn Spanish by playing some language tapes while she sleeps. Her question concerns the general topic of _____ and according to the text, the answer would most likely be _____.
 (1) Subliminal perception; may be
 (2) Signal detection; no
 (3) Subliminal perception; no
 (4) Signal detection; may be
11. Read each of the following two statements :
 Assertion (A) and Reason (R) and indicate your answer using the codes given below.
Assertion (A) : In the randomized two-group design, if the mean difference between the two groups is analysed both by t and F tests, F is always larger than t.

Reason (R) : In the randomized two group design, the $t^2 = F$.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
12. Which one of the following laid down the social foundation of organizational behaviour ?
 (1) Viteles (2) Munsturberg
 (3) Taylor (4) Mayo
13. Which theories of personality are considered third-force-movement in contemporary Psychology ?
 (1) Psycho analysis
 (2) Behaviour is tic
 (3) Humanistic/ Existential
 (4) Cognitive
14. Read each of the following two statements :
 Assertion (A) and Reason (R) and indicate your answer using the codes given below :
Assertion (A) : Ego-involved learners are driven by performance goals.
Reason (R) : Ego-involved learners are pre-occupied with them serves and are concerned with their evaluation by others.
Codes :
 (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
15. According to Piaget, which is the primary motivating force behind development :
 (1) Accommodation
 (2) Assimilation
 (3) Equilibration
 (4) Adaptation

16. Read each of the following two statements :
Assertion (A) and Reason (R) and indicate your answer using the codes given below :

Assertion (A) : Method of loci helps keeping memories intact and retrievable.

Reason (R) : Familiar loci are already organized in our long term memories.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
17. Which one of the following biases play important role in stereotyping and prejudice ?
- (1) False consensus
 - (2) Illusory correlation
 - (3) Priming
 - (4) Confirmation
18. Which of the following glands does secrete sex steroids that are released by the gonads?
- (1) Adrenal medulla
 - (2) Anterior pituitary
 - (3) Adrenal cortex
 - (4) Pancreas
19. Which one of the following was not an assumption of the foundation of "Scientific Management" propounded by Taylor ?
- (1) Interest of Employer and employees are the same.
 - (2) Employees mostly want high wages / salary.
 - (3) Employer want a low labour cost.
 - (4) Management of incentives has great value.
20. Which one of the following proposed the "ABX" theory of dyadic attraction ?
- (1) Header
 - (2) Jones
 - (3) Newcomb
 - (4) Lombardo
21. The statement, "Normalcy and Abnormalcy differ in degree not in kind" represents the
- (1) Deviance Criteria
 - (2) Personal Discomfort Criteria
 - (3) Statistical Criteria
 - (4) Social Discomfort Criteria
22. According to signal detection theory detection of stimuli depends on :
- (a) Their physical energy
 - (b) Internal factors like relative costs
 - (c) Absolute threshold
 - (d) Internal factors like benefits with detecting their presence
- Codes :**
- (1) (a), (b), (c)
 - (2) (b), (c), (d)
 - (3) (c), (a), (d)
 - (4) (a), (b), (d)
23. Which of the following are the sources of error in social cognition ?
- (a) Counterfactuals thinking
 - (b) Thought suppression
 - (c) Magical thinking
 - (d) Judging by resemblance
- Codes :**
- (1) (a) and (b)
 - (2) (a), (b) and (c)
 - (3) (a), (c) and (d)
 - (4) (c) and (d)
24. Match the items of List - I with the items of List - II. Select the correct response from the answer codes given below :
- | List - I | List-II |
|------------------------|--|
| (a) Latent learning | (i) Learning a response to terminate an aversive stimulus. |
| (b) Cognitive learning | (ii) Occurrence of learning is not evident in behaviour until later. |
| (c) Escape learning | (iii) A change in the way of processing informations as a result of experiences. |
| (d) Avoidance learning | (iv) Learning a response to avoid an aversive stimulus |

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|------|-------|
| (1) | (iii) | (ii) | (iv) | (i) |
| (2) | (ii) | (iii) | (i) | (iv) |
| (3) | (i) | (iv) | (ii) | (iii) |
| (4) | (iv) | (iii) | (i) | (ii) |

25. The diagram below is an example of :

The boy hit the ball over the fence

- (1) Constituents
 - (2) Linguistic Universals
 - (3) Morphemes
 - (4) Underlying representation
26. Creativity is considered to be grounded in Right Hemisphere because :
- (a) Left Hemisphere specialises in sequential and analytical processes.
 - (b) Right Hemisphere specialises in global, parallel and holistic processes.
 - (c) Left Hemisphere is involved in generation of novel ideas.
 - (d) Creativity involves divergent thinking that is function of right hemisphere.

Codes :

- | | | | |
|-----|------------------|-----|------------------|
| (1) | (a), (b) and (d) | (2) | (b), (c) and (a) |
| (3) | (d), (b) and (c) | (4) | (b), (c) and (d) |

27. Read each of the following two statements : Assertion (A) and Reason (R) and indicate your answer using the codes given below :

Assertion (A) : Being awake disrupts the homeostasis of the body.

Reason (R) : Sleep is terminated by a return to homeostasis.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
28. Which one of the following can be primarily employed to assess creativity ?
- (1) Abstract thinking.
 - (2) Convergent thinking.

(3) Divergent thinking.

(4) Symbolic thinking.

29. Which of the following psychologists first empirically established that social stress distort physical health ?

- (1) Selye
- (2) Lazarus and Folkman
- (3) Holmes and Rahe
- (4) Srivastava and Pestonjee

30. Which of the following two styles of leadership were identified in Ohio State Leadership studies ?

- (1) Employees - centered and production - centered
- (2) Initiating structure and people oriented
- (3) Initiating structure and consideration for relationship
- (4) Authoritarian and democratic

31. Read each of the following two statements : Assertion (A) and Reason (R) and indicate your answer using the codes given below :

Assertion (A) : The recognition - by - components (RBC) model provides a method for recognising three dimensional objects across variations in new points.

Reason (R) : Any three dimensional object can be generally described according to its parts and their spatial relations.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

32. Which one of the following may not be put in the category of "Absolute Standard" methods of performance appraisal ?

- (1) Critical incidents method
- (2) Behaviourally Anchored Rating method
- (3) Ranking method
- (4) Checklist method

33. Match List - I with List - II and select the correct answer using the codes given below :

List - I (Designs)**List - II (Features)**

- | | |
|--|--|
| (a) Randomized block design | (i) Each subject participates under all experimental conditions. |
| (b) Factorial design | (ii) Analyses the data for matched groups |
| (c) Single-factor repeated measures design | (iii) Suitable for studying the interactions among the independent variables |
| (d) Design with a covariate | (iv) Employs statistical control |

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|------|------|
| (1) | (iii) | (ii) | (iv) | (i) |
| (2) | (iii) | (ii) | (i) | (iv) |
| (3) | (ii) | (iii) | (i) | (iv) |
| (4) | (ii) | (iii) | (iv) | (i) |

34. Cultures of honour are characterized by strong norms indicating that aggression is :

- (1) an inappropriate response to insults to one's honour.
- (2) an appropriate response to regards to one's honour.
- (3) an appropriate response to insults to one's honour.
- (4) an inappropriate response to regards to one's honour.

35. Discovery and understanding of our own abilities bring happiness but also bring fear of new responsibilities and duties. It is called :

- (1) Electra complex
- (2) Jonah complex
- (3) Motives conflict
- (4) Motive to avoid success

36. Memory for surprising and consequential events (emotional public events) are known as

- (1) Photographic memory
- (2) Flash bulb memory

- (3) Iconic memory
- (4) Implicit memory

37. Intimacy, Passion and commitment are the components of :

- (1) Romantic Love
- (2) Companionate Love
- (3) Infatuated Love
- (4) Consummate Love

38. Which of the following are innate behaviours :

- (a) Reflexes
- (b) Taxes
- (c) Fixed Action Patterns
- (d) Genotype

Codes :

- (1) (a) and (c) only
- (2) (a), (b) and (c) only
- (3) (a), (c) and (d) only
- (4) (a), (b), (c) and (d)

39. Which one of the following proposed the "response inhibition" theory of extinction ?

- (1) Hull
- (2) Guthrie
- (3) Tolman
- (4) Deese

40. Read each of the following two statements : Assertion (A) and Reason (R) and indicate your answer using the codes given below :

Assertion (A) : An infant often acts as if objects stop existing when they are moved out of his perceptual range

Reason (R) : The infant has not yet developed the object concept.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

41. Rollo May's views on personality development can be described in following sequence

- (1) Physical Dependence - Psychological dependence - Rebellion - ordinary consciousness of self - creative consciousness of self.

- (2) Physical dependence - Rebellion - ordinary consciousness of self - self actualization
- (3) Psychological dependence - physical dependence - ordinary consciousness of self - creative consciousness of self
- (4) Physical dependence - psychological dependence - ordinary consciousness of self - Rebellion - creative consciousness of self.

42. We tend to attribute to another's behaviour to external causes when information is of
- (a) High consensus
- (b) Low distinctiveness
- (c) High distinctiveness
- (d) High consistency

Codes :

- (1) (a), (b) (2) (a), (c) and (d)
- (3) (b), (c) and (d) (4) (a), (b) and (d)

43. According to Kohlberg, _____ and _____ are the major reasons of moral behaviour during the stage of conventional morality
- (1) Gaining acceptance and Getting rewards
- (2) Avoiding censure by others and Achieving justice
- (3) Gaining acceptance and Avoiding censure by others
- (4) Avoiding censure by others and Achieving justice

44. Match the List - I and List - II; and select the **correct** answer from the codes given below :

List - I	List - II
(a) Sufficient cause	(i) A condition, existence of which guarantee for the genesis of a mental disorder
(b) Reinforcing cause	(ii) A condition that tends to maintain maladaptive behaviour that has already occurred

- (c) Contributory cause (iii) A condition that occurs relatively in early life and may not show its effect for many years
- (d) Distal cause (iv) A condition that increases the probability of a disorder

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|-------|
| (1) | (ii) | (iv) | (iii) | (i) |
| (2) | (i) | (ii) | (iv) | (iii) |
| (3) | (iv) | (iii) | (ii) | (i) |
| (4) | (iii) | (i) | (iv) | (ii) |

45. Match List-I with List-II and choose your answer by using the codes given below :

List-I (Name)	List - II (Feature)
(a) Template Matching Model	(i) The extent to which Spatial relations deviate from the prototype.
(b) Feature Matching Model	(ii) Description of three dimensional objects according to their parts and spatial relations.
(c) Recognition by-component Model	(iii) Visual analysis detects colours and edges
(d) Configuration Model	(iv) Recognition of bar codes and finger prints.

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|------|-------|-------|------|
| (1) | (i) | (ii) | (iii) | (iv) |
| (2) | (i) | (iii) | (iv) | (ii) |
| (3) | (iv) | (ii) | (iii) | (i) |
| (4) | (iv) | (iii) | (i) | (ii) |

46. Which of the following statements are true about the principal components analysis ?

- (a) Unities (values of 1) are retained in the diagonal spaces of the correlation matrix.
- (b) The first principal component explains maximum variance.

- (c) Principal components analysis follows common factor model.
 (d) Principal components analysis yields oblique unrotated factors.

Codes :

- (1) (a) and (b) only
 (2) (a) and (d) only
 (3) (b) and (c) only
 (4) (a), (b) and (d) only
47. Find the correct sequence among the following oxytocin related events
 (a) Carrying down by axons
 (b) Contraction of uterine muscles
 (c) Synthesis in paraventricular nucleus of hypothalamus
 (d) Secretion by neurosecretory cells in posterior pituitary

Codes :

- (1) (a), (d), (c), (b) (2) (c), (a), (d), (b)
 (3) (a), (b), (d), (c) (4) (d), (a), (c), (b)
48. Match List - I with List - II and select the correct answer by choosing correct code :

List - I (Concept) List - II (Explanation)

- | | |
|------------------------------|---|
| (a) Situated Learning | (i) Internalize knowledge and skills developed in interaction with others |
| (b) Social Negotiation | (ii) Aspect of learning process that relies on collaboration with others and respect for different perspectives |
| (c) Appropriate | (iii) Skills are tied to the settings in which they are learned and are difficult to apply in new settings |
| (d) Cognitive Apprenticeship | (iv) A relationship in which a less experienced learner acquires knowledge and skills in interaction with expert. |

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|------|-------|------|
| (1) | (i) | (ii) | (iii) | (iv) |
| (2) | (iii) | (iv) | (i) | (ii) |
| (3) | (iii) | (ii) | (i) | (iv) |
| (4) | (i) | (iv) | (iii) | (ii) |

49. Which one of the following concepts is equivalent to the concept of "habit family hierarchy" proposed by Hull ?
 (1) Drive reduction
 (2) Cognitive map
 (3) Habit strength
 (4) Means-end remedies
50. A psychotic disorder in which a person's functioning is not markedly impaired, whether is behaviour obviously odd or bizarre and free from hallucinations is called
 (1) Schizo-affective Disorder
 (2) Schizophreniform Disorder
 (3) Delusional Disorder
 (4) Shared Psychotic Disorder
51. What is not typical of Atkinson and Shiffrin model of memory ?
 (1) STM is engaged in complex cognitive activities.
 (2) STM is gateway to LTM.
 (3) It is a sequential model
 (4) STM controls and enhances information to LTM
52. According to Erikson, adequate training, sufficient education and good models during the age range 6 to 11 years will lead to
 (1) Initiative (2) Identity
 (3) Industriousness (4) Autonomy
53. Read each of the following two statements Assertion (A) and Reason (R) and indicate your answer using the codes given below
- Assertion (A) :** People's fantasies would reflect their achievement concerns.
Reason (R) : All people strive to succeed by excelling at any task where evaluation is possible.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

54. Match List - I and List - II and select the correct answer by choosing from the codes given below :

List - I	List - II
(Method)	(Effect on behaviour)
(a) Electrical stimulation of satiety centre	(i) Cessation of eating
(b) Electrical stimulation of hunger centre	(ii) Initiation of eating
(c) Lesion of satiety centre	(iii) Hyperphagia
(d) Lesion of hunger centre	(iv) Aphagia

Codes :

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (ii) | (i) | (iv) | (iii) |
| (2) | (i) | (ii) | (iv) | (iii) |
| (3) | (ii) | (i) | (iii) | (iv) |
| (4) | (i) | (ii) | (iii) | (iv) |

55. Which of the following statements are true about simple bivariate linear regression?
- (a) The b coefficient is a slope of the regression line.
 - (b) The b coefficient denotes change in Y, when X changes by 1 unit.
 - (c) The errors of prediction have a mean and variance of zero.
 - (d) If the b coefficient is significant, the product moment correlation between X and Y is also significant.

Codes :

- | | | | |
|-----|------------------|-----|------------------|
| (1) | (a) and (b) | (2) | (c) and (d) |
| (3) | (a), (b) and (c) | (4) | (a), (b) and (d) |

56. People vary in their perceptual-cognitive styles, which may be due to
- (a) The degree to which their perceptions are flexible

- (b) Aspects of their personality
- (c) Rigid belief
- (d) Field dependence or independence
- (e) Motives

Codes :

- | | | | |
|-----|--------------------|-----|--------------------|
| (1) | (a), (b), (c), (d) | (2) | (b), (c), (d), (e) |
| (3) | (c), (d), (e), (a) | (4) | (e), (a), (b), (d) |

57. Which of the following fit into declarative memories ?

- (a) Semantic memory
- (b) Short term memory
- (c) Episodic memory
- (d) Procedural memory

Codes :

- (1) (a) and (c)
- (2) (b) and (d)
- (3) (a), (b) and (c)
- (4) (a), (b), (c) and (d)

58. Read each of the following two statements : Assertion (A) and Reason (R) and indicate your answer using the codes given below :

Assertion (A) : In operant conditioning the rat presses the bar spontaneously.

Reason (R) : Shaping is a procedure in which rewards gradually guide an animal's behaviour toward a desired goal.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

59. According to cognitive approach. The role of a teacher is to

- (a) Correct wrong answers
- (b) Correct misconceptions
- (c) Facilitate and Supervise
- (d) Teach effective strategies

Codes :

- | | | | |
|-----|-------------|-----|-------------|
| (1) | (a) and (c) | (2) | (a) and (d) |
| (3) | (b) and (c) | (4) | (b) and (d) |

60. Match List - I and List - II and indicate your correct choice by using codes given below :

List - I (Theories of motivation)

List - II (Characteristic)

- | | |
|-------------------------------|--------------------------|
| (a) Drive theories | (i) Pulling by goal |
| (b) Incentive theories | (ii) Information |
| (c) Opponent-process theories | (iii) Hedonic |
| (d) Cognitive (social) | (iv) Pushing toward goal |

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|------|-------|-------|------|
| (1) | (i) | (ii) | (iii) | (iv) |
| (2) | (iv) | (i) | (iii) | (ii) |
| (3) | (ii) | (i) | (iii) | (iv) |
| (4) | (iv) | (iii) | (i) | (ii) |

61. Match the List - I and List - II and select the correct answer from codes given below :

List - I

List - II

- | | |
|---------------|---|
| (a) Mitwelt | (i) A person exists in a given particular place at a particular time. Being conscious of it one can choose the direction of one's life. |
| (b) Dasein | (ii) The experiences and meaning we share with others through our relationships |
| (c) Umwelt | (iii) Unique presence of human beings of self-awareness and self relatedness. |
| (d) Eigenwelt | (iv) The biological and natural environment for human beings and animals including biological needs, drives, and instincts. |

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|------|-------|-------|
| (1) | (i) | (iv) | (iii) | (ii) |
| (2) | (iii) | (ii) | (i) | (iv) |
| (3) | (ii) | (i) | (iv) | (iii) |
| (4) | (iv) | (ii) | (iii) | (i) |

62. Intelligence is necessary but not sufficient condition for creativity. It represents :

- (1) Threshold theory
- (2) Certification theory
- (3) Interference theory
- (4) Collaborative theory

63. In which of the following disorders a person has two or more different personalities, and either of which is dominant at a time ?

- (1) Dissociative amnesia
- (2) Split personality
- (3) Dissociative identity disorder
- (4) Schizophrenia.

64. What is not true of facial feedback hypothesis of emotions ?

- (1) Acting as another acts does not help us feel what another feels.
- (2) Facial expressions influence our emotional experience.
- (3) To be empathic mimic other's emotional expressions.
- (4) Professional actors exhibit expression specific autonomic nervous system changes.

65. Which is not a characteristic symptom of Narcolepsy ?

- (1) Cataplexy
- (2) Sleep paralysis
- (3) Nocturnal myoclonus
- (4) Hypnagogic hallucinations

66. In a study, twenty-five subjects performed the same task under three conditions-relaxed, moderately stressed, and highly stressed. The suitable analysis of variance was carried out. The degrees of freedom for the conditions X subjects interaction (used as an estimate of error variance) would be :

- | | |
|--------|--------|
| (1) 24 | (2) 48 |
| (3) 72 | (4) 74 |

67. Which one of the following is not included in Hackman and Oldham's Job characteristics model of work motivation ?

- (1) Variety of skill
- (2) Significance of the task

- (3) Quality of job life
(4) Autonomy
68. Which is the correct sequence of stages involved in creative thinking ?
- (1) Preparation - orientation - illumination - verification, incubation.
 - (2) Preparation - incubation - illumination - orientation - verification.
 - (3) Orientation - preparation - incubation - illumination - verification.
 - (4) Orientation - preparation - illumination - incubation - verification.
69. Which of the following is not correct of the Piaget's theory
- (1) Disequilibrium terminates in structural change
 - (2) Individual stages of development cannot be skipped or omitted from the sequence.
 - (3) Stages of equilibrium do not last as long as stages of disequilibrium
 - (4) Later stages are independent of the elements and experiences of early stages
70. Match the List-I and List-II and select the correct answer by choosing from the codes given below :

List - I

List - II

- | | |
|----------------------------|---|
| (a) Knowing one's emotions | (i) Having skills in managing emotions of others |
| (b) Motivating oneself | (ii) Handling feelings and emotions so they are appropriate |
| (c) Managing Emotions | (iii) Marshalling emotions in service of goal |
| (d) Handling Relationships | (iv) Monitoring feelings from moment to moment |

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|------|-------|-------|------|
| (1) | (iv) | (iii) | (ii) | (i) |
| (2) | (i) | (ii) | (iii) | (iv) |
| (3) | (i) | (iii) | (iv) | (ii) |
| (4) | (ii) | (iii) | (i) | (iv) |

Instructions for Question numbers 71 to 75.

Read the following paragraph and answer the five questions that follows :

The most popular method of measuring attitudes is to administer the attitude scale. Although Bogardus Social Distance Scale (1925) is one of the first attitude scales, a major breakthrough in attitude scaling was made by L.L. Thurstone who developed the law of comparative judgement (LCJ) and also proposed the three scaling methods - paired comparison, equal appearing intervals, and successive intervals. In the paired comparison, every statement / stimulus is paired with every other statement / stimulus. The computational methods of paired comparison are most extensively developed and use Thurstone's LCJ directly for this purpose. Since the paired comparison method has limitations in scaling large number of statements / stimuli, the two other methods - equal appearing intervals, and successive intervals - were developed by Thurstone. The method of successive intervals was computationally laborious in the pre-computer era and hence it was not so popular. In the classical scaling tradition Guttman developed the scalogram technique and Edwards developed the scale discrimination technique. Likert's development of summated ratings represent an important landmark in attitude measurement. It follows the traditional psychometric model. Aiken and Groth - Marnatt (2009) pointed out that other scaling techniques, less frequently used for attitude assessment, include Osgood's semantic differential, Q-sort, magnitude estimation, expectancy value scaling, facet analysis, multidimensional scaling etc.

71. If eleven attitude statements are to be scaled by the method of paired comparison, what would be the number of pairs ?
- (1) 11
 - (2) 55
 - (3) 110
 - (4) 121
72. In which of the scaling method, item-residual correlations are frequently calculated ?
- (1) Equal appearing intervals
 - (2) Paired comparison

- (3) Successive intervals
(4) Summated ratings
73. Read each of the following two statements :
Assertion (A) and **Reason (R)** and indicate your answer using the codes given below :
Assertion (A) : In method of equal appearing intervals, median is often used as a measure of scale value.
Reason (R) : In normal distribution mean, median and mode are equal.
Codes :
 (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
74. Which one of the following technique usually yields multidimensional attitude scales?
 (1) Summated ratings
 (2) Paired comparison
 (3) Equal appearing intervals
 (4) Scalogram
75. Which of the following technique/s is/are well-known for being associated with cumulative scales ?
 (a) Bogardus Social Distance Scales
 (b) Likert's summated ratings
 (c) Guttman's scalogram analysis
 (d) Osgood's semantic differential
Codes :
 (1) (a) only (2) (a) and (b) only
 (3) (a) and (c) only (4) (a), (c) and (d) only

Answer Sheet

1. (2)	2. (1)	3. (3)	4. (4)	5. (3)	6. (4)	7. (2)	8. (4)	9. (3)	10. (3)
11. (4)	12. (4)	13. (3)	14. (1)	15. (3)	16. (2)	17. (2)	18. (3)	19. (2)	20. (3)
21. (3)	22. (4)	23. (4)	24. (2)	25. (1)	26. (1)	27. (2)	28. (3)	29. (3)	30. (3)
31. (1)	32. (3)	33. (3)	34. (3)	35. (2)	36. (2)	37. (4)	38. (2)	39. (1)	40. (1)
41. (1)	42. (2)	43. (3)	44. (2)	45. (1)	46. (1)	47. (2)	48. (3)	49. (2)	50. (3)
51. (1)	52. (3)	53. (3)	54. (4)	55. (4)	56. (4)	57. (1)	58. (4)	59. (4)	60. (2)
61. (3)	62. (1)	63. (3)	64. (1)	65. (3)	66. (2)	67. (3)	68. (3)	69. (4)	70. (1)
71. (2)	72. (4)	73. (2)	74. (1)	75. (3)					

December, 2015

PSYCHOLOGY

PAPER-II

Note : This paper contains **fifty (50)** multiple choice questions, each question carrying **two (2)** marks. Attempt all of them.

- Motion sickness can occur when :
 - one loses his vestibular sense.
 - one relies only on visual information.
 - one's visual information and vestibular system's information are congruent.
 - one's visual system and vestibular system are in conflict.

Codes :

 - (a) only
 - (a) and (b)
 - (d) only
 - (a) and (d)
- Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using the codes given below :
Assertion (A): Iconic store is a discrete visual sensory store.
Reason (R): Because information is believed to be stored in the form of visual codes.
Codes :
 - Both (A) and (R) are true and (R) is the correct explanation of (A).
 - Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (A) is true, but (R) is false.
 - (A) is false, but (R) is true.
- Backward Conditioning occurs when :
 - CS and US are presented simultaneously
 - CS is presented first and US is presented before the termination of CS
 - CS is presented first and US is presented after the termination of CS
 - US is presented first and CS is presented after the termination of US
- Which among the following defines problem space ?
 - An initial state
 - A goal state
 - A set of operations

Codes :

 - (a) and (c)
 - (a) and (b)
 - (b) and (c)
 - (a), (b) and (c)
- The James-Lange theory and the cognitive theory of emotion disagree on whether
 - specific brain centers are involved in specific emotions.
 - bodily feedback determines which emotion is felt.
 - individuals can judge their emotions accurately.
 - there is no any biological involvement in human emotions.
- Match **List-I** with **List-II** and select the **correct** answer by choosing from the codes given below :

List-I	List-II
(a) Binet	(i) Intelligent people often have intelligent relatives
(b) Broca	(ii) Performance based intelligence tests eliminate cultural bias
(c) Wechsler	(iii) Correlation between intelligence and brain size
(d) Galton	(iv) Intelligence is a collection of mental abilities

Codes :

 - (a) (b) (c) (d)
 - (a) (b) (c) (d)
 - (a) (b) (c) (d)
 - (a) (b) (c) (d)
- Biologically based emotional and behavioural tendencies that are evident in early childhood represent :

- (1) Trait (2) Type
(3) Temperament (4) Style
8. In Jungian theory, the idea that energy is automatically redistributed in the psyche in order to achieve equilibrium or balance depicts :
- (1) Principle of Equivalence
(2) Principle of Synchronicity
(3) Principle of Entrophy
(4) Principle of Opposites
9. After the recent terrorist attacks in France, a psychologist proposed a study wherein he would interview two hundred French citizens to find whether their attitude towards fate as a consequence of the attack. This study can best be labelled as :
- (1) attitude change experiment
(2) field experiment
(3) ex post facto study
(4) psychometric study
10. Match **List-I** with **List-II** and select the **correct** answer by choosing from the codes given below :

List-I	List-II
(Cutaneous receptors)	(Responds to)
(a) Free nerve endings	(i) Sudden displacement of skin
(b) Ruffini endings	(ii) Gradual skin indentation
(c) Merkel's disks	(iii) Gradual skin stretch
(d) Pacinian corpuscles	(iv) Temperature change and pain

Codes :

- | | | | | | | | | | |
|-----|-------|------|-------|------|-----|------|-------|------|-----|
| (a) | (b) | (c) | (d) | (a) | (b) | (c) | (d) | | |
| (1) | (i) | (ii) | (iii) | (iv) | (2) | (ii) | (iii) | (iv) | (i) |
| (3) | (iii) | (iv) | (i) | (ii) | (4) | (iv) | (iii) | (ii) | (i) |

11. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using the codes given below :
Assertion (A): A tree shall appear in same perceived size whether one views it from near or far point.
Reason (R): The size of retinal image remains invariant.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.
12. Which of the following is **correct** while comparing classical and operant conditioning?
- (1) Operant conditioning takes place before reinforcement while classical conditioning takes place after reinforcement.
(2) Operant conditioning takes place as a result of some voluntary action while classical conditioning takes place without choice.
(3) In operant conditioning, response is elicited while in classical conditioning it is emitted.
(4) In operant conditioning magnitude of the response is the index of conditioning while in classical conditioning it is the rate of response.
13. Words learned under water are recalled :
- (1) Better on land than under water
(2) Better above land (sky) than on land
(3) Better above land (sky) than under water
(4) Better under water than on land
14. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using the codes given below :
Assertion (A): People overlook flaws if the overall issue is perceived as important.
Reason (R): People have some biases.
- Codes :**
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true
15. Simi is totally blind, but she follows a 24-hr day-night cycle like the rest of us, and seems comfortably adapted to it. Studies on animals suggest she relies on :

- (a) a built-in rhythm in her Supra-Chiasmatic Nucleus (SCN).
 (b) non-visual receptors in her eyes.
 (c) social activity (d) clocks
- Codes :**
 (1) (b) only (2) (d) only
 (3) (c) and (d) (4) (a), (c) and (d)
16. Flynn effect is :
 (1) High IQ score of a generation in comparison to succeeding generation.
 (2) High IQ score of a generation in comparison to preceding generation.
 (3) A person's increase in accumulated knowledge with increase in age.
 (4) A person's change in accumulated knowledge with increase in age.
17. Which of the following sequence is **correct** ?
 (1) Oral phase Anal phase Latency
 Phallic Genital
 (2) Anal Oral Phallic Latency
 Genital
 (3) Oral Anal Phallic Latency
 Genital
 (4) Oral Phallic Anal Genital
 Latency
18. Which one of the following is considered as a non-parametric alternative to one-way ANOVA for independent samples ?
 (1) Kruskal - Wallis H test
 (3) Kolmogorov - Smirnov test
 (2) Mann - Whitney U test
 (4) Friedman's ANOVA
19. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using the codes given below :
Assertion (A): Action potentials normally travel along axons in one direction.
Reason (R): The portions of an axon over which an action potential has just travelled are left momentarily refractory.
Codes :
 (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
20. Match **List-I** with **List-II** and select the **correct** answer by choosing from the codes given below :
- | List-I (Cues) | List-II (Perception) |
|-----------------------------|---|
| (a) Linear perspective | (i) Our two eyes observe objects from slightly different positions. |
| (b) Aerial perspective | (ii) Parallel lines appears to converge at distance. |
| (c) Atmospheric perspective | (iii) Objects below the horizon appear near. |
| (d) Retinal disparity | (iv) Farther away objects are less distinctly seen. |
- Codes :**
 (a) (b) (c) (d) (a) (b) (c) (d)
 (1) (i) (ii) (iii) (iv) (2) (ii) (iii) (iv) (i)
 (3) (iii) (iv) (i) (ii) (4) (iv) (iii) (ii) (i)
21. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using the codes given below :
Assertion (A): A desirable behaviour of an individual can be effectively used as a reference for another less desirable activity.
Reason (R): Premack has suggested that all responses should be thought of as potential reinforcers.
Codes :
 (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
22. Which memory is the result of instrumental / motor learning processes ?
 (1) Semantic memory
 (2) Episodic memory
 (3) Sensory memory
 (4) Procedural memory

23. If you begin with a problem and come up with many different solutions; you are using, which is a form of creative thinking; contrarily sometimes you come up with correct solution by using thinking.

(1) convergent, inductive
(2) convergent, deductive
(3) convergent, divergent
(4) divergent, convergent

24. Choose the **correct** formula to measure Body Mass Index (BMI).

(1) Weight in kg squared / height in meters
(2) Weight in kg squared / Height in meters squared
(3) Weight in kg / height in meters
(4) Weight in kg / height in meters squared

25. Which of the following are **not** identified by Thurstone ?

(a) Induction (b) Spatial
(c) Symbolic (d) Semantic

Codes :

(1) (a) and (b) (2) (c) and (d)
(3) (a) and (c) (4) (a), (b) and (c)

26. Match the items of **List-I** with the items of **List-II**. Select the **correct** answer from the answer codes given below :

List-I (Concept) List-II (Theorist)

(a) Rational Functions (i) Freud
(b) Impetus (ii) Jung
(c) Masculine Protest (iii) Adler
(d) Hyper competitiveness (iv) Horney

Codes :

(a) (b) (c) (d) (a) (b) (c) (d)
(1) (i) (ii) (iii) (iv) (2) (ii) (i) (iii) (iv)
(3) (iii) (ii) (iv) (i) (4) (iv) (iii) (i) (ii)

27. Match **List-I** with **List-II** and indicate your answer using the codes given below :

List-I (Types of variable) List-II (Examples)

(a) Continuous variable (i) Examination results in terms of pass and fail
(b) Real discrete variable (ii) Voice pitch

(c) Artificially discrete variable (iii) Intelligence
(d) Qualitative variable (iv) Nationality

Codes :

(a) (b) (c) (d) (a) (b) (c) (d)
(1) (iii) (iv) (i) (ii) (2) (iii) (i) (iv) (ii)
(3) (ii) (iv) (i) (iii) (4) (ii) (i) (iv) (iii)

28. In which lobe of the cerebral cortex, the primary olfactory cortex lies ?

(1) Frontal lobe (2) Temporal lobe
(3) Occipital lobe (4) Parietal lobe

29. "Factor Analysis is used as a means of confirming a structured model than as a means of discovering a structural model". This is true of which theory of intelligence ?

(1) Spearman (3) Jensen
(2) Thurstone (4) Guilford

30. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using the codes given below :

Assertion (A): One focuses on the discovery of truth and interests that are empirical, critical and rational.

Reason (R): A person high on economic values focuses on usefulness and practicality.

Codes :

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.

31. A supervisor, who was asked to rate his subordinates on the basis of his observations, found an employee who did not take a medical leave in that year. On the basis of this observation, the supervisor rated the employee as 'medically fit', 'highly efficient' and 'meticulous'. This is an example of :

(1) lineancy error
(2) constant error
(3) tendency to give extreme rating
(4) halo effect

32. Which of the following parts of the basal ganglia constitute the striatum ?

- (a) Amygdala (b) Caudate
(c) Putamen (d) Globus Pallidus

Codes :

- (1) (a) and (b) (2) (a) and (c)
(3) (b) and (c) (4) (c) and (d)

33. Among the laws of perceptual grouping, the law of simplicity is a tendency to :

- (1) perceive simple patterns in terms of complex shapes
(2) perceive complex patterns in terms of simpler shapes
(3) perceive simpler shapes only
(4) perceive complex shapes only

34. What is the role of positive and negative reinforcement ?

- (1) To increase the likelihood that responses preceding both will be repeated.
(2) To decrease the likelihood that responses preceding negative reinforcement will be repeated.
(3) To increase the likelihood that responses preceding only positive reinforcement will be repeated.
(4) To ensure that there are no negative consequences following the behaviour.

35. Match **List-I** with **List-II** and select the **correct** answer by choosing from the codes given below :

List-I

List-II

- | | |
|---------------------------|---|
| (a) Sensory memory | (i) Information that passes through a gate and transferred to short term memory |
| (b) Attention | (ii) Information subjected to deep processing |
| (c) Short term memory | (iii) Temporary storage of sensory information |
| (d) Elaborative rehearsal | (iv) Brief storage of information |

Codes :

- (a) (b) (c) (d) (a) (b) (c) (d)
(1) (i) (ii) (iii) (iv) (2) (iii) (i) (iv) (ii)
(3) (iii) (iv) (i) (ii) (4) (iv) (iii) (ii) (i)

36. While forming a concept one chooses to take one hypothesis and selects many of its features to quickly form the concept at once :

- (1) Successive scanning
(2) Conservative focusing
(3) Focus gambling
(4) Simultaneous scanning

37. Which among the following are the final outcome of curiosity ?

- (a) Drives to explore
(b) Increases arousal
(c) Gains information and resources
(d) Reduction in tension

Codes :

- (1) (b) and (c) (2) (b), (c) and (d)
(3) (a) and (d) (4) (a), (b) and (c)

38. Read the statement and choose the **correct** answer out of given options :

"If one hand is placed in a container of hot water and other in cold water. Then both hands are placed in a container of lukewarm water." Most people experience :

- (1) The hand initially placed in hot water feels the lukewarm water as cool whereas hand initially placed in cold water feels it as hot.
(2) The hand initially placed in hot water feels the lukewarm water as hot whereas hand initially placed in cold water feels it as cold.
(3) Both the hands will have feeling of equal hotness but more than temperature of lukewarm water.
(4) Both the hands will have feeling of equal coldness lower than temperature of lukewarm water.

39. An accused is harassed and tortured until he confesses to the crime is an example of :

- (1) Negative reinforcement
(2) Punishment

- (3) Primary reinforcement
(4) Secondary reinforcement
40. Which of the following are **not** Mnemonic devices ?

- (a) Categorical clustering
(b) Pegword system
(c) Consolidation (d) Rehearsal

Codes :

- (1) (a) and (b) (2) (b) and (c)
(3) (a) and (c) (4) (c) and (d)

41. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using the codes given below :
Assertion (A): Advertising and media campaigns are based frequently on the availability heuristic.

Reason (R): Consumers recall instances of being satisfied and expect this outcome to recur.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.

42. Match **List-I** with **List-II** and select the **correct** answer by choosing from the codes given below :

List-I (Measures)	List-II (Body part)
(a) EMG	(i) Sweat glands
(b) EEG	(ii) Iris
(c) GSR	(iii) Muscles
(d) Pupilometry	(iv) Cortical neurons

Codes :

- (a) (b) (c) (d) (a) (b) (c) (d)
(1) (i) (ii) (ii) (iv) (2) (ii) (iii) (iv) (i)
(3) (iii) (iv) (i) (ii) (4) (iv) (iii) (ii) (i)

43. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using the codes given below :
Assertion (A): Value of h^2 (heritability) derived for a particular population is not the correct estimation for other populations.

Reason (R): Different populations have similar genotypes but they experience different environment.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.

44. What is the **correct** sequence an auditory stimulus takes to reach the primary auditory cortex ?

- (1) Cochlear nuclei - Superior olive - Inferior colliculus - Medial geniculate nucleus
(2) Cochlear nuclei - Inferior colliculus - Superior olive - Medial geniculate nucleus
(3) Superior olive - Cochlear nuclei - Inferior colliculus - Medial geniculate nucleus
(4) Medial geniculate nucleus - Inferior colliculus - Superior olive - Cochlear nucleus

45. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using the codes given below :
Assertion (A): Non-equivalent control groups are often encountered in quasi-experimental research.

Reason (R): Quasi-experimental designs are highly useful in policy and program evaluation research.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.

Instruction for Q.Nos. 46 to 50 :

Read the following paragraph and answer the five questions which follow.

A psychometrician wanted to develop a new test of Abstract Thinking. He prepared

eighty items, presumably assessing abstract thinking. Each item had five alternative answers, one of them being correct. The psychometrician carried out item analysis by using item-remainder correlations. Item analysis retained sixty items. To examine internal consistency among the items, inter-item correlations were inspected. Then the split-half reliability coefficient, Kuder - Richardson reliability coefficient, and Cronbach's alpha coefficient were obtained for the sixty item test. The analysis yielded the Cronbach's alpha coefficient of 0.90. Since the test was fairly long and the alpha coefficient was also quite high, an attempt was made to develop two short versions of equal length by randomly dividing the test items. The 60 item version was, then, correlated with the already established abstract reasoning test, using $N = 500$. The correlation between the new and an established test was 0.24, significant at .01 level.

46. In the above context, which one of the following correlation should be computed to obtain item-remainder correlations ?
- (1) Point biserial correlation
 - (2) Phi-coefficient
 - (3) Rank difference correlation
 - (4) Tetrachoric correlation
47. Which one of the following statement would be **true** in the above context ?
- (1) Kuder - Richardson reliability coefficient would also be .90, but the split-half reliability coefficient may differ from .90.
 - (2) Split-half reliability coefficient would also be .90, but the Kuder - Richardson reliability coefficient may differ from .90.
 - (3) Both Kuder - Richardson reliability coefficient and split-half reliability coefficient would be .90.
 - (4) The correlation between odd and even parts of the test would be .90.
48. To obtain inter-item correlations, which one of the following correlation coefficient should be used in the above analysis ?
- (1) Biserial correlation
 - (2) Point biserial correlation
 - (3) Phi-coefficient
 - (4) Rank difference correlation
49. What is the expected Cronbach alpha for the short versions, referred to in the paragraph ?
- (1) 0.45
 - (2) 0.67
 - (3) 0.82
 - (4) 0.90
50. Which one of the following conclusion can be drawn from the data provided above? The new Abstract Thinking Test has :
- (1) Satisfactory convergent validity
 - (2) Satisfactory divergent validity
 - (3) Satisfactory concurrent validity
 - (4) Unsatisfactory concurrent validity

Answer Sheet

1. (3)	2. (3)	3. (4)	4. (4)	5. (2)	6. (4)	7. (3)	8. (3)	9. (3)	10. (4)
11. (3)	12. (2)	13. (4)	14. (1)	15. (1)	16. (2)	17. (3)	18. (4)	19. (1)	20. (2)
21. (1)	22. (4)	23. (3)	24. (4)	25. (2)	26. (2)	27. (1)	28. (2)	29. (4)	30. (2)
31. (4)	32. (3)	33. (2)	34. (1)	35. (2)	36. (3)	37. (1)	38. (1)	39. (4)	40. (4)
41. (1)	42. (3)	43. (3)	44. (1)	45. (2)	46. (1)	47. (1)	48. (3)	49. (3)	50. (4)

December, 2015

PSYCHOLOGY

PAPER-III

Note : This paper contains **seventy five (75)** objective type questions of **two (2)** marks each. **All** questions are compulsory.

1. The factors that influence social behaviour and thoughts in certain ways are called :
 - (1) Independent variables
 - (2) Confounding variables
 - (3) Dependent variables
 - (4) Mediating variables

2. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using the codes given below :
Assertion (A): In pre-conventional stage children understand morality on the basis of social customs and values.

Reason (R): In pre-conventional stage, children make decisions based on whether they will be rewarded or punished.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
 - (3) (A) is true, but (R) is false
 - (4) (A) is false, but (R) is true
3. Research shows that the following does **not** contribute to teacher effectiveness :
 - (1) Communication skills
 - (2) Use of ICT
 - (3) Years of experience
 - (4) Use of student feedback
 4. Match **List - I** with **List - II** and select the **correct** answer by choosing from codes given below :

List - I

- (a) Graphic rating scales

List - II

- (i) Evaluation method where employees are ranked from best to worst.

- (b) Behaviourally anchored rating scales (ii) Evaluator rates employees on classification of some qualities.
- (c) Group order ranking (iii) Evaluator rates performance factors on an incremental basis.
- (d) Individual ranking (iv) Evaluator rates actual job related behaviour along a continuum.

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|------|
| (1) | (i) | (ii) | (iii) | (iv) |
| (2) | (iv) | (iii) | (i) | (ii) |
| (3) | (iii) | (iv) | (ii) | (i) |
| (4) | (ii) | (iv) | (iii) | (i) |

5. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using the codes given below :
Assertion (A): Obsessive compulsive disorder is a disorder of thought and is thus classified in the schizophrenia spectrum in DSM-5.

Reason (R): Evidence base shows that a combination of pharmacotherapy and psychotherapy yields good results in OCD therapy.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 - (3) (A) is true but (R) is false
 - (4) (A) is false but (R) is true
6. Phonemic restoration as we perceive speech in a noisy party is an example of :

- (1) Top down processing
 - (2) Bottom up processing
 - (3) Subliminal perception
 - (4) Supraliminal perception
7. Feeling touchy or hypersensitive following an upsetting experience is a form of :
- (1) Imprinting
 - (2) Habituation
 - (3) Sensitization
 - (4) Reflexive Behaviour
8. Entry of information into long term memory depends upon :
- (a) Maintenance rehearsal
 - (b) Elaborative rehearsal
 - (c) Method of loci
 - (d) Primary memory
- Codes :**
- (1) (a) and (b)
 - (2) (a) and (c)
 - (3) (b) and (d)
 - (4) (b) and (c)
9. Because it has the external features associated with the concept of dog, a wolf is perceived as a dog. This is an example of :
- (1) Centration
 - (2) Equilibration
 - (3) Object permanence
 - (4) Prototype
10. Which of the following is the final destination for much of the brain's information about emotion before action is taken ?
- (1) Amygdala
 - (2) Anterior cingulate cortex
 - (3) Pre-frontal cortex
 - (4) Hypothalamus
11. Cattell propounded gf-gc theory of intelligence mainly based on :
- (1) First order factors of abilities
 - (2) Second order factors of abilities
 - (3) Third order factors of abilities
 - (4) Higher order factors of abilities
12. Which of the following are elements in Carl Rogers' person centred theory of personality and psychotherapy ?
- (a) ideal self
 - (b) congruence
 - (c) positive regard
 - (d) dissonance
- Codes :**
- (1) (a), (b) and (c)
 - (2) (b), (c) and (d)
 - (3) (a), (c) and (d)
 - (4) (a), (b) and (d)
13. Which one of the following statements is true ?
- (1) The 'Global Five' factors in personality refer to the second-order factors of 16 PF, latest edition.
 - (2) The 'Global Five' factors refer to the factors assessed by the NEO-Five Factor Inventory, latest edition.
 - (3) The 'Global Five' factors refer to the factors assessed by Goldberg's bipolar adjectives.
 - (4) The 'Global Five' factors are derived from Eysenck Personality Profiler, latest edition.
14. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using the codes given below :
Assertion (A): Castration does not eliminate sexual behaviour completely in mammals, including humans, of both sexes.
Reason (R): Hormones do not play role in the regulation of sexual behaviour.
- Codes :**
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
 - (3) (A) is true, but (R) is false
 - (4) (A) is false, but (R) is true
15. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using codes given below :
Assertion (A): Helpful individuals perceive the world as fair and predictable in which good behaviour is rewarded and bad behaviour is punished.

- Reason (R):** People believe in a just world.
- Codes :**
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
 - (3) (A) is true, but (R) is false
 - (4) (A) is false, but (R) is true
16. Which of the following statements best defines maturation ?
- (1) It is directly based on social-cognitive learning
 - (2) It is an automatic biological development of the body that naturally unfolds over time
 - (3) It does not take place in human beings
 - (4) It is the basis of all physical and psychological developments
17. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using codes given below :
- Assertion (A):** Leaders exert powerful effects on followers through the attributions followers make about them.
- Reason (R):** Leaders have self confidence and environmental sensitivity.
- Codes :**
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
 - (3) (A) is true, but (R) is false
 - (4) (A) is false, but (R) is true
18. The following disorders are coded on Axis I of a five axis diagnosis :
- (a) V codes
 - (b) personality disorders
 - (c) clinical conditions
 - (d) intellectual deficit
- Codes :**
- (1) (a) and (c) only
 - (2) (b) and (d) only
 - (3) (a) and (d) only
 - (4) (b) and (c) only
19. Four important elements in the theory of signal detection are :
- (1) errors, hits, speed and accuracy
 - (2) speed, accuracy, power and false alarm
 - (3) hit, miss, correct rejection and false alarm
 - (4) acuity, efficacy, speed and power
20. In Hullian System the term Hierarchy of Responses is used to depict:
- (a) If the first innate response pattern does not alleviate a need, another pattern will occur.
 - (b) If the second innate response pattern does not reduce the need another pattern will occur.
 - (c) If the third innate response pattern does not work, still another pattern will occur and so on.
 - (d) If none of the innate behaviour pattern is effective in reducing the need, the organism will stop responding.
- Codes :**
- (1) (a), (b) and (c)
 - (2) (a), (c) and (d)
 - (3) (b), (c) and (d)
 - (4) (a), (b) and (d)
21. Jia had written a list to do her week end work, but incidently left it home while going out for her first activity. Trying to remember the list, Jia remembers what was at the beginning of the list and what was at the end but not those things in the middle. This is an example of :
- (1) Encoding specificity effect
 - (2) Flash bulb memory
 - (3) Serial position effect
 - (4) Tip-of-the tongue effect
22. How does one choose the option with the highest expected utility while making rational decision making ? Find the **correct** sequence :
- (a) Assigning utilities
 - (b) Weighting / multiplying the utilities by the likelihoods

- (c) Evaluating the likelihood of alternatives
(d) Assessing the consequences
Codes :
(1) (a), (d), (b), (c)
(2) (a), (c), (b), (d)
(3) (a), (d), (c), (b)
(4) (c), (d), (a), (b)
23. Brief stress enhances the activity of immune system as evidenced by the action of
(1) macrophages (2) T cells
(3) B cells (4) all the above
24. Torrance Tests of Creative Thinking (TTCT) are mainly indexing :
(1) Creative Products
(2) Creative Processes
(3) Creative Personality
(4) Creative Persuasion
25. Personality assessments in forensic settings use the following psychometric and behavioural measures :
(a) Galvanic Skin Response (GSR)
(b) Millon Clinical Multiaxial Inventory (MCMI)
(c) Continuous Performance Test (CPT)
(d) Sixteen Personality Factors Test (16 PF)
Codes :
(1) (a) and (b)
(2) (c) and (d)
(3) (a), (b) and (c)
(4) (b), (c) and (d)
26. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using codes given below :
Assertion (A): Paired comparison method is not considered suitable for scaling large number of attitude scale statements.
Reason (R): For computing scale values, paired comparison method uses Thurstone's Law of comparative judgement.
Codes :
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
(3) (A) is true, but (R) is false
(4) (A) is false, but (R) is true
27. The following are found to be significant factors in educational achievement
(a) teacher - student ratio
(b) medium of instruction
(c) rehearsal and practice
(d) environmental stimulation
Codes :
(1) (a) and (b) only
(2) (a) and (c) only
(3) (a), (c) and (d) only
(4) (b), (c) and (d) only
28. Which one of the scaling method is more likely to yield multidimensional attitude scale ?
(1) Guttman scalogram analysis
(2) Paired comparison method
(3) Equal appearing intervals method
(4) Method of summated rating
29. When is aspiration a method of choice to make a lesion ?
(1) in a deeper brain area
(2) in an area of cerebral cortex
(3) an irreversible lesion
(4) in underlying white matter
30. Trace the sequence of events turning default female sex of the fetus into male :
(a) SRY gene produces Sry protein
(b) Presence of Y chromosome
(c) Testosterone masculinize Wolffian duct into seminal vesicles
(d) Primitive gonad develop into testes
(e) Mullerian inhibiting hormone defeminizes the fetus
Codes :
(1) (b), (a), (d), (e), (c)
(2) (b), (d), (e), (a), (c)
(3) (b), (d), (a), (e), (c)
(4) (d), (b), (a), (c), (e)
31. A psychologist was administering a projective test which involved a word association task.

She tended to nod and smile every time a plural word was given by the respondent. The following testing bias was seen :

- (1) favoritism
- (2) test wiseness
- (3) selective reinforcement
- (4) gender of assessor

32. Match **List - I** and **List - II** and select the **correct** answer by choosing from codes given below

List - I (Measure of Personality Assessment)	List - II (Scoring element)
---	------------------------------------

- | | |
|------------------------|------------------------|
| (a) MMPI | (i) V, X, Y, Z scales |
| (b) MCMI | (ii) Needs and presses |
| (c) TAT | (iii) Populars |
| (d) Rorschach Inkblots | (iv) L, F, K scales |

Codes :

- | | | | |
|-----------|-------|-------|-------|
| (a) | (b) | (c) | (d) |
| (1) (iii) | (ii) | (i) | (iv) |
| (2) (ii) | (iv) | (iii) | (i) |
| (3) (i) | (iii) | (iv) | (ii) |
| (4) (iv) | (i) | (ii) | (iii) |

33. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using codes given below :

Assertion (A): gf and gc correlate positively with each other with moderate magnitude.

Reason (R): Cattell used second - order factor solutions for the identification of gf and gc were oblique.

Codes :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
- (3) (A) is true, but (R) is false
- (4) (A) is false, but (R) is true

34. The underlying motivation for acts of aggression is .

- (1) intent to harm (2) anger
- (3) retaliation
- (4) need to exert control

35. Which of the following is a characteristic of Chomsky's theory ?

- (1) Development of language is between three and five years of age
- (2) Language development is dependent on the reinforcements received by the child
- (3) Children acquire mistakes in language by observing their parents and others around them
- (4) Children have an innate mental grammar

36. Match **List - I** with **List - II** and select the **correct** answer by choosing from the codes given below :

- | | |
|---------------------------------------|--|
| List - I | List - II |
| (a) Physical trauma distressing life. | (i) The cognitive and emotional upheaval that follows a highly |
| (b) Psychological trauma | (ii) Most of us remember nothing before the age of about 3 to 4 years. |
| (c) Infantile amnesia | (iii) It may cause permanent amnesia if particular areas of the brain are damaged. |
| (d) Repression | (iv) Distressing mental content is transferred from the conscious to the unconscious level of awareness. |

Codes :

- | | | | |
|-----------|-------|-------|------|
| (a) | (b) | (c) | (d) |
| (1) (i) | (ii) | (iii) | (iv) |
| (2) (ii) | (iii) | (iv) | (i) |
| (3) (iii) | (i) | (iv) | (ii) |
| (4) (iv) | (iii) | (ii) | (i) |

37. Match the items of **List - I** with items of **List - II** and select the **correct** answer from the answers given in codes :

**List - I
(Concept)**

- (a) Vicarious Trial and Error
(b) Hypothetical Deduction
(c) Cumulative Recording
(d) Reciprocal Determination

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|------|-------|-------|-------|
| (1) | (i) | (ii) | (iii) | (iv) |
| (2) | (i) | (iii) | (iv) | (ii) |
| (3) | (ii) | (iii) | (i) | (iv) |
| (4) | (iv) | (i) | (ii) | (iii) |

38. The centre - surround organization of ganglion cells that enhance contrast at edges in a visual environment is well illustrated by :

- (1) Craik bands
(2) Mach bands
(3) Luminance bands
(4) Optic bands

39. Match List - I with List - II and select the correct answer by choosing from the codes given below :

List - I (Diagnoses)

- (a) General Medical conditions
(b) GAF (Global Assessment Functioning)
(c) Vcodes
(d) Personality disorders

List - II (Axis used in Multi-axial system)

- (i) V
(ii) I
(iii) II
(iv) III

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|-------|
| (1) | (ii) | (iii) | (iv) | (i) |
| (2) | (iii) | (iv) | (i) | (ii) |
| (3) | (iv) | (i) | (ii) | (iii) |
| (4) | (i) | (ii) | (iii) | (iv) |

40. The growth needs of ERG theory are represented by various needs in Maslow's model:

- (1) Physiological and social needs
(2) Safety needs and esteem needs

**List - II
(Theorists)**

- (i) Skinner
(ii) Tolman
(iii) Hull
(iv) Bandura

- (3) Social needs and esteem needs
(4) Esteem needs and self-actualization needs

41. In Bloom's taxonomy of educational objectives, the following appear in ascending order :

- | | | | | |
|-----|------------|------------|------------|------------|
| (1) | Remember | Understand | Apply | Analyze |
| (2) | Apply | Analyze | Understand | Remember |
| (3) | Understand | Apply | Remember | Analyze |
| (4) | Analyze | Remember | Apply | Understand |

42. Failure to develop consistent identity results in ____.

- (1) Inferiority
(2) Role confusion
(3) Stagnation
(4) Social isolation

43. Helpers respond to the needs of a victim because they want to accomplish something rewarding; this may be referred as :

- (1) Negative state - relief model
(2) Empathic - joy hypothesis
(3) Empathy - altruism hypothesis
(4) Pluralistic empathy

44. The philosophy of Advaita Vedanta in respect of shaping beliefs does **not** distinguish between :

- (a) Self and non - self
(b) Human and nature
(c) Human and society

Codes :

- (1) (a) and (b)
(2) (a) and (c)
(3) (b) and (c) (4) (a), (b) and (c)

45. Which is the **correct** order of the emergence of following Psycho-Social strengths ?

- (1) Love, Fidelity, Wisdom, Care
(2) Care, Fidelity, Love, Wisdom
(3) Fidelity, Love, Care, Wisdom
(4) Care, Fidelity, Wisdom, Love

46. From the following list of tools used by the career guidance counsellor, choose the odd one out :

(1) DAT (2) GATB
(3) DBDA (4) SVIB

47. Job design comprises of different aspects :

(a) Job enlargement
(b) Job enrichment
(c) Job rotation

Codes :

(1) (a) and (b) (2) (a) and (c)
(3) (b) and (c) (4) (a), (b) and (c)

48. The following are disorders that are included in the anxiety disorders spectrum :

(a) Panic disorder
(b) Body dysmorphic disorder
(c) PTSD
(d) Fugue

Codes :

(1) (a) and (b) only
(2) (a) and (c) only
(3) (a), (b) and (d)
(4) (b), (c) and (d)

49. Detecting movement in spite of camouflage is easier for people who are :

(1) field dependent
(2) field independent
(3) both field dependent and field independent
(4) neither field dependent nor field independent

50. Read the following statement and decides :
"If a rat that had previously learned to traverse a maze or a runway to obtain food is placed directly into a non - empty goal box, it will stop traversing the maze or runway on subsequent trials". What it exemplifies ?

(1) Latent Learning
(2) Place Learning
(3) Response Extinction
(4) Latent Extinction

51. Match **List - I** with **List - II** and select the **correct** answer by choosing from the codes given below :

List - I

List - II

(a) Encoding Failure	(i) Information that is not accessed evades from the storage over a time
(b) Proactive interference	(ii) Information is not attended to encoding
(c) Decay	(iii) Newer information interferes with the retrieval of older information
(d) Retroactive interference	(iv) Older information already in memory interferes with the learning of new information

Codes :

	(a)	(b)	(c)	(d)
(1)	(i)	(ii)	(iii)	(iv)
(2)	(i)	(iii)	(iv)	(ii)
(3)	(ii)	(iii)	(iv)	(i)
(4)	(ii)	(iv)	(i)	(iii)

52. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using codes given below :

Assertion (A): Language affects our thought process.

Reason (R): Language affects our ability to store information.

Codes :

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
(3) (A) is true, but (R) is false
(4) (A) is false, but (R) is true

53. Trace the right sequence of hypothetical series of events by McClelland in the context of n - Ach :

- (a) n - Ach in children
- (b) Independence training by Parents
- (c) Economic development
- (d) Self reliance values

Codes :

- (1) (a), (b), (d), (c)
- (2) (d), (b), (a), (c)
- (3) (a), (d), (b), (c)
- (4) (d), (b), (c), (a)

54. Match the items of **List - I** with the items of **List - II** and select the **correct** answer from answer codes given below :

List-I (Concept)	List - II (Description)
(a) Little-C	(i) Everyday creativity
(b) Big-C	(ii) Creativity in Eminents
(c) Pro-C	(iii) Creativity in a chosen profession
(d) Mini-C	(iv) Personal creativity

Codes :

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (i) | (ii) | (iii) | (iv) |
| (2) | (ii) | (i) | (iv) | (iii) |
| (3) | (iii) | (ii) | (i) | (iv) |
| (4) | (i) | (ii) | (iv) | (iii) |

55. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using the codes given below :

Assertion (A): Projective techniques are difficult to standardize and norm.

Reason (R): Projective techniques are not timed tests.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
- (3) (A) is true, but (R) is false
- (4) (A) is false, but (R) is true

56. A 19 year old boy was administered the Wechsler Intelligence Scale for children by the clerical staff of a busy psychologist who later wrote the test report. A xerox copy of

the test material was shared with the child's parents to ensure that he performs well. The following ethical principles of testing were violated.

- (a) use of appropriate test
- (b) professional qualification of tester
- (c) copyright
- (d) culture fair testing

Codes :

- (1) (a) and (c) only
- (2) (b) and (c) only
- (3) (a), (b) and (c) only
- (4) All of the above

57. Match List - I with List - II and select the correct answer by choosing from the codes given below :

List - I (Sleep phenomenon)	List - II (EEG display)
(a) Stage -1 sleep	(i) Single large - ve wave followed by large + ve wave
(b) Stage - 3 sleep	(ii) Low voltage high frequency EEG signal
(c) Sleep spindle	(iii) Occasional presence of delta waves
(d) K - complex	(iv) Waxing and waning burst of 12-14 Hz wave

Codes :

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (i) | (iv) | (iii) | (ii) |
| (2) | (ii) | (iii) | (iv) | (i) |
| (3) | (iii) | (iv) | (i) | (ii) |
| (4) | (iv) | (iii) | (ii) | (i) |

58. Emergent stage -1 sleep is characterized by :

- (a) sleep - spindle
- (b) electromyographic changes
- (c) electro-oculographic changes
- (d) K - complex

Codes :

- (1) (a) and (b)
 - (2) (a) and (d)
 - (3) (a), (c) and (d)
 - (4) (b), (c) and (d)
59. Self is conceptualized in Western and Eastern perspectives on personality, respectively, as :

- (1) free and deterministic
- (2) instinctive and learned
- (3) ideal and real
- (4) independent and interdependent

60. Match the items of **List - I** and **List - II** and select the **correct** answer from the answers given below :

List - I (Scholars)	List - II (Concepts)
(a) Cattell	(i) Assortative Mating
(b) Jensen	(ii) Investment Hypothesis
(c) Sternberg	(iii) Wisdom
(d) Spearman	(iv) Positive Manifold

Codes :

(a)	(b)	(c)	(d)	(a)	(b)	(c)	(d)
(1) (i)	(ii)	(iii)	(iv)	(2) (ii)	(i)	(iv)	(iii)
(3) (iii)	(iv)	(ii)	(i)	(4) (ii)	(i)	(iii)	(iv)

61. Match **List - I** with **List - II** and select the **correct** answer by choosing from the codes given below :

List - I (Tools/Theory)	List - II (Features)
(a) Kelley's Personal Construct Theory	(i) 11 categories of scoring TAT
(b) Murray's TAT	(ii) 6 subcategories of needs and goals
(c) McClelland and Atkinson's measure of n-Ach	(iii) 20 needs
(d) Rotter's scale	(iv) One postulate and 11 corollaries

Codes :

(a)	(b)	(c)	(d)	(a)	(b)	(c)	(d)
(1) (iv)	(iii)	(ii)	(i)	(2) (ii)	(iii)	(iv)	(i)
(3) (iii)	(iv)	(i)	(ii)	(4) (iv)	(iii)	(i)	(ii)

62. Match **List - I** with **List - II** and select the **correct** answer by choosing from the codes given below :

List - I (Concept)	List - II (Description)
(a) Phonemes	(i) Smallest meaningful combination of sounds

- (b) Syntax (ii) Effective and appropriate use of language in social context
- (c) Morphemes (iii) Basic sound of consonants and vowels
- (d) Pragmatics (iv) Combination of words to make meaningful phrases and sentences

Codes :

(a)	(b)	(c)	(d)	(a)	(b)	(c)	(d)
(1) (i)	(ii)	(iii)	(iv)	(2) (ii)	(iii)	(iv)	(i)
(3) (iii)	(iv)	(i)	(ii)	(4) (iv)	(iii)	(ii)	(i)

63. Read each of the following two statements - **Assertion (A)** and **Reason (R)** and indicate your answer using codes given below :

Assertion (A): The forgetting of a memory is caused by the disappearance of its engram over a period of time.

Reason (R): Once the engram has disappeared, the memory no longer exists anywhere in the memory system.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
- (3) (A) is true, but (R) is false
- (4) (A) is false, but (R) is true

64. Which is the **correct** sequence of Reinforcement schedules in terms of their effectiveness in response rate and resistance to extinction during learning experiments ?

- (1) Variable Ratio Fixed Ratio Variable Interval Fixed Interval Continuous Reinforcement schedule
- (2) Fixed Ratio Variable Ratio Fixed Interval Variable Interval Continuous Reinforcement schedule
- (3) Continuous Reinforcement schedule Variable Ratio Fixed Ratio Variable Interval Fixed Interval

- (4) Variable Ratio Variable Interval
Fixed Ratio Fixed Interval Continuous Reinforcement schedule
65. When a Background Interference Procedure was used experimentally, the following findings became apparent ?
- (1) Field independent persons did better than field dependent persons
 - (2) Field dependent persons did better than field independent persons
 - (3) Both field dependent and field independent persons did equally well
 - (4) BIP enhanced the performance of all subjects
66. People with moderate mental retardation would have an IQ in the range of :
- (1) 10 to 19 (2) 35 to 54
 - (3) 55 to 70 (4) 20 to 34
67. The following is **not** typically found in a school guidance counsellor's office :
- (1) Sand pit
 - (2) Lie detector
 - (3) Psychometric tests
 - (4) Puppets and play dough
68. The following social factors play a role in educational settings :
- (a) co-operation (b) competition
 - (c) peer influence (d) role modelling
- Codes :**
- (1) (a) and (b) only (2) (b) and (c) only
 - (3) (c) and (d) only (4) All of the above
69. According to Piaget the children are able to grasp the concept of object permanence and conservation in :
- (1) Sensory motor stage
 - (2) Pre-operational stage
 - (3) Concrete operational stage
 - (4) Formal operations stage
70. Match **List - I** with **List - II**, and select the **correct** answer by choosing from the codes given below :

List - I

- (a) Symbolic social influence

List - II

- (i) A form of social influence in

which one person orders to perform some action

- (b) Normative social influence (ii) Social influence based on the desire to be correct
- (c) Informational social influence (iii) Social influence based on the desire to be liked
- (d) Obedience (iv) Social influence resulting from the mental representation of others or our relationship

Codes :

- | | (a) | (b) | (c) | (d) | | (a) | (b) | (c) | (d) |
|-----|-------|------|-------|------|-----|------|-------|------|-----|
| (1) | (i) | (ii) | (iii) | (iv) | (2) | (ii) | (iii) | (iv) | (i) |
| (3) | (iii) | (iv) | (i) | (ii) | (4) | (iv) | (iii) | (ii) | (i) |

Instructions for Q.Nos. 71 to 75.

Read the following paragraph and answer the five questions which follow :

Paragraph : Experimental personality research was extremely popular after 1950 for about three to four decades. In experimental personality research, one of the major focus is to study how the effect of experimentally manipulated independent variable is moderated by personality variable/s. In one of such study, the researcher wanted to study whether the effect of Interpolated learning (A) is moderated by Anxiety (B). The researcher hypothesized that the interpolated learning would adversely affect the recall of original learning more in high anxious subjects than low anxious subjects. Using a standardized Anxiety scale, from an initial pool of two hundred subjects, fifty high and fifty low anxious subjects were obtained. Both, high as well as low anxious groups were further divided into two groups-Experimental group and Control group. The experimental groups first learned List A of CVC trigrams, then learned another List - List B of CVC trigrams with similar association values, and then tested

for Recall (A). The subjects in control groups learned List A. Then they received the cancellation task for the specific period (equal to the average time, taken by the corresponding experimental group to learn List B). The subjects were then tested for the Recall of (A). Groupwise means and standard deviations for recall scores were obtained.

The standard deviations were found comparable across the groups. Treating recall scores as the dependent variable, the appropriate ANOVA was carried out. The main effect of A was found significant at 0.05 level. Similarly, the F-ratio, defined as ' $MS_{A \times B} / MS_E$ ' was found statistically significant at 0.01 level.

71. Which one of the following design is used in the above research ?
- (1) Multigroup design with single I.V.
 - (2) 2×2 factorial design
 - (3) 2×2 mixed factorial design
 - (4) Repeated measures design
72. Which one of the following statement is **true** in the context of above study ?
- (1) The assumption of homogeneity of variance was satisfied.
 - (2) The assumption of homogeneity of variance was not satisfied.
 - (3) The correction for heterogeneity of variance was required.
 - (4) The data are insufficient to evaluate the assumption of homogeneity of variance.
73. On the basis of the information provided about the main effect of A, it can be concluded that :
- (1) Interpolated learning had no effect on the recall of A.
 - (2) Interpolated learning caused retroactive interference.
 - (3) Interpolated learning caused retroactive facilitation.
 - (4) Either retroactive interference or retroactive facilitation has occurred.
74. What would be the degrees of freedom for the numerator and denominator of the F-ratio defined as ' $MS_{A \times B} / MS_E$ ' ?
- (1) 1 and 96 respectively
 - (2) 1 and 98 respectively
 - (3) 2 and 97 respectively
 - (4) 3 and 96 respectively
75. Which of the following conclusion/s can be drawn about the researcher's hypothesis?
- (a) Researcher's hypothesis has been verified.
 - (b) Researcher's hypothesis has not been verified.
 - (c) Full data have not been presented which is necessary to evaluate the researcher's hypothesis.
 - (d) Anxiety moderated the effect of interpolated learning.
- Codes :**
- (1) (a) only
 - (2) (a) and (d) only
 - (3) (c) and (d) only
 - (4) (b) and (d) only

Answer Sheet

1. (4)	2. (4)	3. (3)	4. (3)	5. (4)	6. (2)	7. (3)	8. (4)	9. (4)	10. (3)
11. (2)	12. (1)	13. (1)	14. (3)	15. (1)	16. (2)	17. (1)	18. (1)	19. (3)	20. (1)
21. (3)	22. (4)	23. (4)	24. (2)	25. (1)	26. (2)	27. (3)	28. (4)	29. (2)	30. (1)
31. (3)	32. (4)	33. (2)	34. (4)	35. (4)	36. (3)	37. (3)	38. (4)	39. (3)	40. (4)
41. (1)	42. (2)	43. (2)	44. (4)	45. (3)	46. (4)	47. (4)	48. (2)	49. (2)	50. (4)
51. (4)	52. (2)	53. (2)	54. (1)	55. (2)	56. (3)	57. (2)	58. (2)	59. (4)	60. (4)
61. (4)	62. (3)	63. (1)	64. (1)	65. (1)	66. (2)	67. (4)	68. (4)	69. (2)	70. (4)
71. (2)	72. (1)	73. (4)	74. (1)	75. (3)					

July, 2016
PSYCHOLOGY
 PAPER-II

Note : This paper contains **fifty (50)** multiple choice questions, each question carrying **two (2)** marks. Attempt all of them.

1. A procedure that establishes a sequence of responses which lead to a reward following the final response and then working backwards is called
 - (1) Shaping
 - (2) Stimulus control
 - (3) Chaining
 - (4) Stimulus generalization
2. In a study on scaling of attitude items, eleven statements were included. What would be the number of pairs of attitude statements ?
 - (1) 21
 - (2) 55
 - (3) 110
 - (4) 121
3. Who first attempted to describe personality traits in terms of Lexican descriptors ?
 - (1) Ross Stagner
 - (2) Allport and Odbert
 - (3) R.B. Cattell
 - (4) Coasta and McCrae
4. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :
Assertion (A) : Classical conditioning is quick when UCS is presented immediately after CS rather than before.
Reason (R) : According to cognitive perspective, classical conditioning occurs because CS provides expectancy about the coming of UCS.
Codes :
 - (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
5. Which of the following have been explained by Freud in his psycho-analytic theory of motivation ?
 - a. Homeostasis
 - b. Drives
 - c. Hedonism
 - d. Instincts**Codes :**
 - (1) a, b and d only.
 - (2) a, c and d only.
 - (3) b, c and d only.
 - (4) a, b and c only.
6. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I (Concept)	List – II (Explanation)
a. Trait	i. Specific emotional and behavioural qualities of personality that are thought to be biologically based
b. Type	ii. A collection of highly interconnected traits or dispositions which determine personality integration
c. System	iii. Clustering of many stable and enduring dispositions or qualities.
d. Temperament	iv. Consistent pattern of emotional and behavioural reactions a person displays across variety of situations.

Codes :

	a	b	c	d
(1)	i	ii	iii	iv
(2)	ii	i	iii	iv
(3)	iii	ii	iv	i
(4)	iv	iii	ii	i

7. Which of the following is correctly arranged?
- Encoding – retrieval – storage
 - Encoding – storage – retrieval
 - Storage – encoding – retrieval
 - Retrieval – encoding – storage
8. Which one of the following does not contribute to memory?
- Interesting / disinteresting learning material
 - Goal behind learning
 - Gender
 - Some rest after learning
9. In terms of efficiency in scanning information, which one of the following strategies is considered to be the best?
- Conservative focusing
 - Focus gambling
 - Simultaneous scanning
 - Successive scanning
10. In retrograde amnesia, which of the following kind of problem is seen?
- Problems with recalling previously learned material
 - Problems with face recognition.
 - Difficulty finding routes.
 - Poor encoding of verbal material.
11. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : Extraverted individuals are sociable, impulsive and like excitement.

Reason (R) : Brains of extraverts react more slowly and weakly to stimuli which cause them to seek excitement by approaching the environment.

Codes :

- Both (A) and (R) are true and (R) is the correct explanation of (A).
- Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (A) is true, but (R) is false.
- (A) is false, but (R) is true.

12. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I
(Concepts)

a. Personality
Structure

b. Personality
Process

c. Personality
Development

d. Psychopathology
and Change

List – II
(Explanation)

i. Psychological reactions and motives that change dynamically over relatively brief periods.

ii. Stable and enduring qualities that define the individual and distinguish individuals from one another.

iii. People sometimes change and sometimes resist change or are unable to change accordingly

iv. Everybody comes up to be an unique individual by adulthood

Codes :

	a	b	c	d
(1)	i	ii	iii	iv
(2)	ii	iii	iv	i
(3)	ii	i	iv	iii
(4)	iii	i	ii	iv

13. Match List – I with List – II and indicate your answer using the codes given below :

List – I (Concepts)

a. Artificial
concepts

List – II (Explanation)

i. Representation of
objects

- b. Natural concepts ii. Clearest objects in the physical world
- c. Prototypes iii. Defined by set of rules
- d. Visual images iv. Not based on precise properties

Codes :

	a	b	c	d
(1)	ii	iii	iv	i
(2)	iii	iv	ii	i
(3)	iii	ii	iv	i
(4)	iv	iii	i	ii

14. According to Master and Johnson following is the correct sequence of human sexual response :

- | | | |
|----------------|------------|---------|
| (1) Excitement | Orgasm | Plateau |
| Resolution | | |
| (2) Excitement | Plateau | Orgasm |
| Resolution | | |
| (3) Excitement | Arousal | Orgasm |
| Resolution | | |
| (4) Arousal | Excitement | Orgasm |
| Resolution | | |

15. In graphical representations showing stress levels on X axis and performance on Y axis, best performance can be expected at

- (1) Lower end of X axis
- (2) Middle portion of X axis
- (3) Higher end of X axis
- (4) None of the above

16. Lesions of the ventromedial nuclei in the hypothalamus produce

- (1) Anorexia (2) Obesity
- (3) Low level of Leptin
- (4) Hypoinsulinemia

17. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I
(Attitude Scaling Technique)

- a. Paired comparison

List – II
(Computational Feature)

- i. Computes median category as scale values

- b. Scalogram ii. Computes Z-scores as estimates of scale values
- c. Equal appearing intervals iii. Computes coefficient of reproducibility
- d. Summated ratings iv. Computes routine measures of item analysis like item-remainder correlation to select items

Codes :

	a	b	c	d
(1)	iii	ii	i	iv
(2)	i	iv	ii	iii
(3)	ii	iii	i	iv
(4)	ii	iii	iv	i

18. When the action potential reaches the end of the axon terminals, it causes the release of

- (1) An electric spark that sets off the next neuron
- (2) Positively charged ions that excite the next cell
- (3) Negatively charged ions that inhibit the next cell.
- (4) Neurotransmitters that excite or inhibit the next cell.

19. According to Freud following are the characteristics of instinct :

- a. A source in some bodily deficit.
- b. An aim that focuses on gratification of need.
- c. An impetus that propels the person to act.
- d. An object through which instinct achieves its aim.

Codes :

- (1) a and b only
- (2) a, b and d only
- (3) b, c and d only
- (4) a, b, c and d

20. Guilford initially started his work extending Thurston's theory of primary abilities but later on diverged on the grounds :

- Obtained factors were not primary (independent) rather related to each other
- Thurston's factor analysis was mainly oriented to get simple structure
- It was difficult to name the abilities (factors)
- Guilford also found second-order and higher-order factors.

Codes :

- a and b only.
- b and c only.
- a, b and c only.
- a, b, c and d

21. AIDS attack a number of very different population; young gay men, intravenous drug users, haemophiliacs, Haitians, infants and recipients of blood transfusion, but commonly among them is the decrease in T lymphocytes, so it is an example of

- Inductive reasoning
- Deductive reasoning
- Analogical reasoning
- Syllogism

22. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I

- Reintegrative memory
- Recall
- Recognition
- Relearning

List – II

- Familiarity that we have met it before
- There is some residue from the past to learn easily
- Activity of some performance learned in the past
- To check details of the recovery of events in the personal past

Codes :

- | | a | b | c | d |
|-----|----|-----|-----|-----|
| (1) | iv | iii | i | ii |
| (2) | iv | i | iii | ii |
| (3) | i | ii | iv | iii |
| (4) | i | iv | ii | iii |

23. A 40 years old person is having the mental age of 20 years. What would be his IQ ?

- 100
- 50
- 80
- 125

24. Muller-Lyer illusion is likely to occur more in cultures characterised by

- People living in rectangular and square buildings
- People living in buildings which are round and have few corners
- People living in open spaces
- People living in caves

25. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : According to Classical Test Theory (CTT), reliability can be conceptualized as the ratio of true score variance to the observed score variance.

Reason (R) : According to the CTT, the random error of measurement is assumed to be uncorrelated with the true score.

Codes :

- Both (A) and (R) are true and (R) is the correct explanation of (A).
- Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (A) is true, but (R) is false.
- (A) is false, but (R) is true.

26. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I

(System)

- Visual system
- Auditory system

List – II

(Components)

- Hammer, pinna, oval window
- Cornea, vitreous humour, aqueous humour

- c. Neural system iii. GABA, Serotonin, Dopamine
 d. Messenger System iv. Glial cells, myelin, Neurillemma

Codes :

- | | a | b | c | d |
|-----|-----|-----|-----|-----|
| (1) | i | ii | iii | iv |
| (2) | ii | iii | iv | i |
| (3) | iii | ii | i | iv |
| (4) | ii | i | iv | iii |

27. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : Organisms learn to perform responses that produce positive reinforcers.

Reason (R) : Positive reinforcement strengthens responses that permit escape from or avoidance of stimulus.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.

28. One prominent class of theorists of deductive reasoning believe that deduction depends on _____ of inference akin to those of _____.

- (1) informal rules; logical calculus
 (2) formal rules; analytical calculus
 (3) procedural rules; logical calculus
 (4) formal rules; logical calculus

29. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : Ideas lose their original identity in fusion into more complex ideas.

Reason (R) : Mind sometimes is analogous to chemical laws.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

30. What type of cell is responsible for the reproduction of other cells of the body ?

- a. Blood cells b. Stem cells
 c. Neurons d. Basal cells

Codes :

- (1) a only (2) b only
 (3) c & d only (4) d & a only

31. If the bones of the middle ear begins to deteriorate, the person will suffer difficulty in

- a. Loudness b. Timbre
 c. Conduction d. Stimulation

Codes :

- (1) a only (2) b and c
 (3) c only (4) c and d

32. What is the correct sequence of products in Guilford's Struct of Intellect Model (SOI) ?

- (1) Unit Class System Relations
 Implication Transformation
 (2) Class Unit Systems Relation
 Transformation Implication
 (3) Unit Class Relation System
 Transformation Implication
 (4) Unit Relation Class System
 Transformation Implication

33. The light passes through the eye in which of the following sequence ?

- (1) Cornea, Pupil, Lens, Retina
 (2) Cornea, Lens, Pupil, Retina
 (3) Lens, Cornea, Pupil, Retina
 (4) Pupil, Cornea, Lens, Retina

34. Which one of the following statement is true in the context of item analysis ?

- (1) Item-total correlation = Item-remainder correlation
 (2) Item-total correlation is greater than or equal to item-remainder correlation.
 (3) Item-total correlation is smaller than or equal to item-remainder correlation.
 (4) Item-total correlation cannot be negative.

35. Match List – I with List – II and select the correct answer by choosing from the codes given below :

**List – I
(Concept)**

- a. Mental Age
b. Intelligence Quotient
c. Mental Tests
d. Tetrad Equation

**List – II
(Psychologists)**

- i. Stern
ii. Binet
iii. Spearman
iv. J.M. Cattell

Codes :

- | | a | b | c | d |
|-----|-----|-----|-----|-----|
| (1) | i | ii | iii | iv |
| (2) | ii | i | iv | iii |
| (3) | iii | ii | i | iv |
| (4) | iv | iii | ii | i |

36. The difference in images in the two eyes is greater for objects that are close and smaller for distant objects is an example of

- (1) Binocular Disparity
(2) Convergence
(3) Accommodation
(4) Relative Size

37. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : Information is transferred from STM to LTM.

Reason (R) : Information maintained in the rehearsal buffer is prevented from decaying in STM.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.

38. Arrange the parts of C.N.S. in ascending order :

- I. Thalamus
II. Medulla oblongata
III. Spinal cord
IV. Frontal lobe

Codes :

- (1) I, II, III, IV (2) II, I, III, IV
(3) IV, II, I, III (4) III, II, I, IV

39. Shaping in instrumental conditioning means

- (1) Teaching a complex sequence of behaviours by first shaping the final response in the sequence and then working backwards
(2) A continuous reinforcement schedule is used for establishing new behaviours.
(3) It involves reinforcing successive approximations of the final desired behaviour.
(4) Consistent occurrence of a behaviour in the presence of discriminative stimulus.

40. According to Trichromatic theory of colour vision, which of the following are the three types of cones ?

- (1) Red, Yellow and Blue
(2) Red, Blue and Green
(3) Yellow, Blue and Green
(4) Yellow, Red and Blue

41. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : gf and gc correlate positively across the age levels with moderate magnitude.

Reason (R) : gf being neural energy is invested in the cultivation of gc and both are having different growth patterns.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.

42. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : The creators of three dimensional movies simulate convergence by pho-

tographing a scene with two cameras placed a few inches apart.

Reason (R) : Convergence is a binocular cue for perceiving depth by the extent to which the eyes converge inwards when looking at an object.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
43. In heightened emotional states, the following functions are NOT triggered by sympathetic Nervous System activity :
- perspiration
 - heightened pulse rate
 - digestion
 - pupillary dilation

Codes :

- (1) c alone (2) both c and d
 (3) a, b and c (4) a, c and d
44. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I (Type of Conditioning)	List – II (Explanation)
a. Delayed conditioning	i. The onset of UCS precedes the onset of the CS and the UCS goes off before CS comes on
b. Trace conditioning	ii. The CS and UCS come on and go off at the same time
c. Simultaneous conditioning	iii. The onset of CS precedes the onset of the UCS and the CS goes off before the onset of UCS
d. Backward conditioning	iv. The onset of the CS precedes the onset of the UCS

and the CS remains on for the initial part of UCS

Codes :

- | | a | b | c | d |
|-----|----------|----------|----------|----------|
| (1) | i | ii | iv | iii |
| (2) | iv | ii | iii | i |
| (3) | iv | iii | i | ii |
| (4) | iv | iii | ii | i |

45. Which of the following statement / s is / are true, when the Eysenck Personality Questionnaire (EPQ or EPQ-R) and Costa and McCrae's NEO-PI-3 are compared ?
- EPQ-R Psychoticism (P) scale is uncorrelated with Five-Factor Agreeableness (A) and Conscientiousness (C) Scale.
 - P scale is positively correlated with A and C scales.
 - P scale is negatively correlated with A and C scales.
 - EPQ-R has Lie scale, NEO-PI-3 has no lie scale.
 - Both EPQ-R and NEO-PI-3 have lie scale.

Codes :

- (1) a and d only (2) b and e only
 (3) c and d only (4) a and e only

Instructions (Questions 46 to 50) :

Read the following paragraph and answer the five questions which follow :

An environmental psychologist hypothesized that the noise would adversely affect the performance on mental tasks. One hundred and fifty subjects, 75 boys and 75 girls, studying in the twelfth standard, in the age range of 17 to 19 years and in the IQ range of 85 to 115, constituted the initial pool of subjects. Three groups of equal size were formed by random assignment. Group I performed under high noise condition (75 decibels), Group II performed under moderate noise condition (60 decibels), and Group III performed under silent condition. Each subject was instructed to solve 45 simple nu-

merical problems in the allotted time of 90 minutes. The mean number of correctly solved problems were twenty-four, twenty and fifteen respectively. The intergroup mean differences were statistically significant.

46. The variable 'intelligence', in the above study, can be labelled as
- Behavioural variable
 - Organismic variable
 - Continuous variable
 - Controlled variable

Codes :

- (1) a and d only (2) b and c only
(3) a, c and d only (4) b, c and d only

47. Which one of the following conclusion can be drawn on the basis of the above study ?

- The researcher's hypothesis has been accepted.
- The researcher's hypothesis has been rejected.
- The researcher's hypothesis has been partially accepted.
- Inadequate data to evaluate researcher's hypothesis.

48. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : In the above study, the subjects have been randomly assigned to the three groups.

Reason (R) : Random assignment of the subjects enables the experimenter to manipulate the independent variable effectively.

Codes :

- Both (A) and (R) are true and (R) is the correct explanation of (A).
- Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (A) is true, but (R) is false.
- (A) is false, but (R) is true.

49. Match List – I with List – II in the context of the above study, select the correct answer by choosing from the codes given below :

List – I

List – II

(Types of Variable) (Variables)

- | | |
|-------------------------|--|
| a. Independent variable | i. Subject's education |
| b. Dependent variable | ii. Subject's gender |
| c. Controlled variable | iii. Noise level |
| d. Discrete variable | iv. Number of numerical problems correctly solved by subject |

Codes :

- | | a | b | c | d |
|-----|-----|-----|----|----|
| (1) | iii | iv | i | ii |
| (2) | iv | iii | i | ii |
| (3) | iv | iii | ii | i |
| (4) | iii | iv | ii | i |

50. In the above study, the researcher's hypothesis would be labelled as

- directional
- non-directional
- Causal
- correlational

Codes :

- (1) a only (2) a and c only
(3) b and c only (4) b and d only

Answer Sheet

1. (3)	2. (2)	3. (2)	4. (1)	5. (2)	6. (4)	7. (2)	8. (3)	9. (3)	10. (1)
11. (1)	12. (3)	13. (2)	14. (2)	15. (2)	16. (2)	17. (3)	18. (4)	19. (4)	20. (4)
21. (1)	22. (1)	23. (4)	24. (1)	25. (1)	26. (4)	27. (3)	28. (4)	29. (1)	30. (2)
31. (3)	32. (3)	33. (1)	34. (2)	35. (2)	36. (1)	37. (2)	38. (4)	39. (3)	40. (2)
41. (1)	42. (4)	43. (1)	44. (4)	45. (3)	46. (4)	47. (2)	48. (3)	49. (1)	50. (2)

July, 2016
PSYCHOLOGY
 PAPER-III

Note : This paper contains **seventy five (75)** objective type questions of **two (2)** marks each. **All** questions are compulsory.

1. Amit met with an accident and later turned into impulsive, irresponsible and less concerned with the consequences of his actions. He must have suffered damage in
 - (1) Parietal lobe (2) Frontal lobe
 - (3) Occipital lobe (4) Temporal lobe
 2. Effective teaching may include conceptual scaffolding, which is best described by the following statement :
 - (1) introducing practice sessions after every module.
 - (2) gradually fading support as student proficiency increases.
 - (3) punishing student for incorrect answers.
 - (4) encouraging brighter students to help weaker students.
 3. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : Individuals high in achievement motivation are more attracted to tasks of intermediate difficulty.

Reason (R) : Individuals high in achievement motivation are highly motivated to avoid failure.

Codes :

 - (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
 4. A social worker, appointed to assist the psychiatrist, used to administer and interpret Rorschach test. This is an
 - (1) ethically incorrect practice
 - (2) ethically correct practice
 - (3) ethically correct practice provided the results are used only for diagnostic purpose.
 - (4) ethically correct practice provided the results are kept confidential and are used only for diagnostic purpose.
 5. Facial feedback hypothesis means
 - (a) Your facial expression affects other's emotions.
 - (b) Other's facial expression affects your emotions.
 - (c) Facial expression provides feedback to the brain about emotions.
 - (d) All the three
- Codes :**
- (1) (a) only (2) (a) and (b)
 - (3) (c) only (4) (d) and (c)
6. The following instrument can not be utilized in the measurement of emotions :
 - (1) Sphygmomano meter
 - (2) Galvanic skin response
 - (3) Digital thermometer
 - (4) Pupillary measurement camera
7. Match the items of List – I with List – II :

List – I (Disorder)	List – II (Typical Symptoms)
a. PTSD	i. Pill rolling movement
b. Major depressive disorder	ii. Flashbacks
c. Parkinson's Disease	iii. Echolalia
d. Catatonic Schizophrenia	iv. Appetite loss

Codes :

	a	b	c	d
(1)	i	ii	iii	iv
(2)	iv	iii	ii	i
(3)	iii	i	iv	ii
(4)	ii	iv	i	iii

8. Arrange in sequence the stage of personal-ity development according to Rollo May
- (1) Innocence ordinary consciousness of self rebellion '!' creative consciousness of self
 - (2) Innocence rebellion ordinary consciousness of self creative consciousness of self
 - (3) Ordinary consciousness of self innocence rebellion '!' creative consciousness of self
 - (4) Rebellion innocence ordinary consciousness of self creative consciousness of self
9. The rapid change in performance as the size of reinforcement in varied is called
- (1) Crespi effect
 - (2) Disequilibrium hypothesis
 - (3) Premack principal
 - (4) Programmed learning
10. Creativity has been conceptualized in terms of following :
- (a) Creativity as a product
 - (b) Creativity as a process
 - (c) Creativity as a potential
 - (d) Creativity as perseverance

Codes :

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (a), (b) and (c) only
- (4) (a), (b) and (d) only

11. In Solomon Asch's study on conformity, subjects conformed to the Group approximately what percent of time
- (1) 25% (2) 35%
 - (3) 45% (4) over 50%
12. Perceived fairness of the amount and rewards among individuals refers to

_____ while perceived fairness of the ways used for rewards and pays refers to _____.

- (1) Organizational justice, Distributive justice
- (2) Equity, Organizational justice
- (3) Distributive justice; Procedural justice
- (4) Procedural justice; Distributive justice

13. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I**(Verbal Responses)**

- a. Mand
- b. Tact
- c. Echoic behaviour
- d. Autoclitic behaviour

List – II**(Explanations)**

- i. Provides grammatical framework for verbal behaviour
- ii. Repeating someone else's utterances
- iii. Verbal behaviour of naming things
- iv. Verbal command that is reinforced when the listener carries out the command

Codes :

	a	b	c	d
(1)	iv	iii	ii	i
(2)	iii	iv	i	ii
(3)	iii	iv	ii	i
(4)	iv	iii	i	ii

14. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I**(Scale)**

- a. Structural scales
- b. Lambda
- c. Abasement

List – II**(Test)**

- i. Rorschach Inkblot Test
- ii. Zukerman-Kuhlman Personality Questionnaire
- iii. California Psychological Inventory

- d. Impulsive Sensation Seeking iv. Thematic Apperception Test

Codes :

	a	b	c	d
(1)	i	ii	iii	iv
(2)	iii	i	iv	ii
(3)	ii	iii	i	iv
(4)	iv	i	ii	iii

15. The concept of _____ is vital in the understanding of fractional antedating goal response

- (1) Drive stimulus reduction
(2) Reactive Inhibition
(3) Secondary Reinforcement
(4) Conditioned Inhibition

16. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I
(Term in factor analysis)

List – II
(Example/Brief explanation)

- | | |
|------------------------------------|--|
| a. Factor extraction method | i. Correlations among varimax rotated factors |
| b. Cattell's scree | ii. Promax |
| c. Factor rotation method | iii. Principal axes |
| d. Inter-factor correlation matrix | iv. Correlation among oblique-rotated factors |
| | v. Method to decide number of factors for rotation |

Codes :

	a	b	c	d
(1)	iii	v	ii	iv
(2)	iii	v	ii	i
(3)	v	iii	ii	iv
(4)	ii	iii	v	iv

17. Which of the following identity status describes the individual who has been exploring but not yet committed to self chosen values and goals ?

- (1) Identity achievement
(2) Identity foreclosure
(3) Identity diffusion
(4) Identity moratorium

18. Scientific study of Creativity is considered to be started by

- (1) Galton (2) Guilford
(3) Torrance (4) Mednick

19. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I
(Concepts)

List – II
(Explanation)

- | | |
|---------------------------|--|
| a. Metacomponents | i. Determining which elements in a set of information are relevant for one's purpose and which are not. |
| b. Selective Encoding | ii. Marshalling emotions in the service of a goal |
| c. Motivating Oneself | iii. Higher-order mental processes being used to guide problem solving efforts. |
| d. Practical Intelligence | iv. Using mental components to help one to adapt, shape, or select an environment that is appropriate for oneself. |

Codes :

	a	b	c	d
(1)	i	ii	iii	iv
(2)	iii	i	ii	iv
(3)	ii	iii	iv	i
(4)	iv	iii	i	ii

20. Culture exerts strong effect on memory through the operation of cultural schema. One's memory is influenced by learning in
(1) Own culture effect

- (2) Culture effect due to enculturation
(3) Acculturation effect
(4) Socialization effect
21. Evolutionary Social Psychology Research seeks to investigate :
- (1) Changes in human behaviour
(2) Seeks to investigate the potential role of genetic factors on social behaviour
(3) Interrelationships between people of two groups of different social environment
(4) Focuses on individual's behaviour.
22. Which one of the following is the right explanation of flash bulb memory ?
- (1) New, important, helpful social events
(2) Unspecific, important, common events
(3) Specific, surprising, unimportant events
(4) Unspecific, important, common events
23. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :
- Assertion (A) :** Heteronomous children focus on the objective consequences rather than intentions.
Reason (R) : Moral development in a heteronomous child is limited.
- Codes :**
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.
24. Match List – I with List – II and select the correct answer by choosing from the codes given below :
- | List – I
(Psychologist) | List – II
(Phenomenon) |
|----------------------------|------------------------------|
| a. Ebbinghaus | i. Some tasks to complete |
| b. Bartlett | ii. Short Stories Use |
| c. Murray | iii. Use of Meaningful Words |
- d. Lewin
- iv. Use of Non-sense Syllables
- Codes :**
- | | a | b | c | d |
|-----|-----|-----|-----|----|
| (1) | i | ii | iii | iv |
| (2) | ii | iii | iv | i |
| (3) | iii | iv | i | ii |
| (4) | iv | iii | ii | i |
25. Which of the following sequences is correct with respect to effective teaching ?
- (1) Match student's level specify learning goals emphasize self comparison suggest improvements and model problem solving.
(2) Specify learning goals emphasize self-comparison suggest improvements and model problem solving match student's level.
(3) Suggest improvements and model problem solving match student's level specify learning goals emphasize self-comparison
(4) Emphasize self-comparison match student's level specify learning goals suggest improvements and model problem solving.
26. Which of the following techniques can be used to study activities in a living human brain ?
- (a) Computerized Tomography
(b) Ablation and Lesioning
(c) Stereotoxic Procedures
(d) Functional Magnetic Resonance Imaging
- (1) only (a) and (d)
(2) only (c) and (d)
(3) only (b), (c) and (d)
(4) only (a), (c) and (d)
27. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :
- Assertion (A) :** Preoperational child has irreversible thought.
Reason (R) : The preoperational child fails to perform conservation tasks.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.

28. Sleep disorders can be categorized as

- (1) Paraphilias and Dysfunctions
 (2) Insomnia and Hypersomnia
 (3) Parasomnias and Dyssomnias
 (4) Serotonergic and Dopaminergic

29. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I (Goals in educational achievement)**List – II (Description)**

- | | |
|----------------------------|---|
| a. Mastery | i. Solving problem at hand gains importance |
| b. Task involvement | ii. Students intend to maximize ability, regardless of short term failure |
| c. Performance improvement | iii. Students focus on how they are judged by others. |
| d. Ego involvement | iv. Students intend to increase competence |

Codes :

- | | a | b | c | d |
|-----|-----|-----|-----|-----|
| (1) | ii | i | iv | iii |
| (2) | i | ii | iii | iv |
| (3) | iii | iv | ii | i |
| (4) | iv | iii | ii | i |

30. Which of the following phenomena result from failures in attentional capture ?

- (a) Inattentional blindness
 (b) Change detection
 (c) Flicker paradigm

Code :

- (1) only (A) and (c)

(2) only (A) and (b)

(3) only (b) and (c)

(4) (a), (b) and (c)

31. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : Individuals high in self-esteem have a relatively weak desire for social rewards.

Reason (R) : Social anxiety and loneliness are both associated with unrewarding social interactions.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.

32. Which of the following statement / s is / are true about culture-fair tests ?

- (a) They attempt to minimize the use of language.
 (b) Suitable different versions of the test are devised for different cultural groups.
 (c) They are composed of materials to which all sociocultural groups have been presumably exposed.
 (d) They attempt to assess the effects of culture on people's intellectual and creative skills.

(1) (a) only

(2) (a) and (c) only

(3) (a), (b) and (c) only

(4) (b), (c), (d) only

33. Which is the correct sequence of the stages involved in creative thinking ?

- (1) Preparation; orientation; illumination; incubation; verification
 (2) Preparation; incubation; illumination; orientation; verification
 (3) Orientation; preparation; incubation; illumination; verification
 (4) Orientation; preparation; illumination; incubation; verification

34. Selective attention in vision and audition respectively have been dubbed as

- (1) 'optic' and 'haptic'
- (2) 'spotlight' and 'gateway'
- (3) 'spatial' and 'acoustic'
- (4) 'central' and 'peripheral'

35. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I
(Intervention
model)

List – II
(Used primarily for)

- | | |
|--|--------------------------------|
| a. Eye Movement
Desensitization
and Retraining | i. Ventilation |
| b. Client Centred
Therapy | ii. Cognitive
Restructuring |
| c. Rational Emotive
Behaviour Therapy | iii. Post-traumatic
Stress |
| d. Behaviour
Therapy | iv. Desensitization |

Codes :

- | a | b | c | d |
|---------|-----|-----|----|
| (1) ii | iii | iv | i |
| (2) i | ii | iii | iv |
| (3) iii | i | ii | iv |
| (4) iv | iii | ii | i |

36. Which is not the part of "on the job training" ?

- (1) Orientation training
- (2) Job instruction training
- (3) Role playing
- (4) Apprentice training

37. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I

List – II

- | | |
|----------------|--|
| a. Symbolizing | i. Employee's belief
that how success-
fully they can ac-
complish the task
in future. |
| b. Forethought | ii. Employees pro-
cess visual experi- |

ences that guide
for their future ac-
tions

- | | |
|--------------------|--|
| c. Self-regulatory | iii. Employees plan
their actions an-
ticipating their de-
sired performance
level |
| d. Self-reflective | iv. Employees ac-
tions by setting in-
ternal standards
for their perfor-
mance. |

Codes :

- | | a | b | c | d |
|-----|-----|-----|-----|-----|
| (1) | i | ii | iii | iv |
| (2) | ii | iii | iv | i |
| (3) | iii | iv | i | ii |
| (4) | iv | i | ii | iii |

38. "Pygmalion effect" in educational settings suggests :

- (1) exceptional progress by student as a result of high teacher expectations.
- (2) biases creeping into summative assessment
- (3) authenticity factors in classroom assessment.
- (4) influences on curricular structure.

39. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I
(Psychologist)

List – II
(Concept/approach)

- | | |
|-------------------|--------------------------------------|
| a. Vygotsky | i. Ethological ap-
proach |
| b. Piaget | ii. Ecological ap-
proach |
| c. Bronfenbrenner | iii. Zone of proximal
development |
| d. Bowlby | iv. Epistemological
approach |

Codes :

- | | a | b | c | d | | a | b | c | d |
|-----|-----|----|-----|----|-----|-----|---|----|-----|
| (1) | iii | iv | ii | i | (2) | iii | i | ii | iv |
| (3) | i | iv | iii | ii | (4) | iv | i | ii | iii |

40. Which of the following is not a diagnosable sleep disorder :

- (1) Somnambulism (2) Somniloquy
(3) Sleep terror (4) Insomnia

41. A field experiment at the Bronx Zoo illustrated the potential importance of _____ indicating that an individual has legitimate authority. Fill in the blank from given below.

- (1) Gender
(2) Uniforms
(3) Age
(4) An authoritarian personality

42. Self-actualizers are characterised as

- (a) They have more efficient perception of reality than non-actualisers.
(b) They show greater acceptance of themselves, others, and nature than do non-actualizers.
(c) They tend to be more concerned with undertaking tasks that will benefit themselves.
(d) They also have some weaknesses.

Codes :

- (1) (a), (b) and (c) only
(2) (a), (b) and (d) only
(3) (b), (c) and (d) only
(4) (a), (c) and (d) only

43. Thinking is a complex set of collaborations between _____ and _____ representations and processes.

- (1) Linguistic; semantic
(2) Linguistic; verbal
(3) Semantic; non-linguistic
(4) Linguistic; non-linguistic

44. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I

- a. Autistic thinking
b. Creative thinking
c. Deterministic thinking
d. Cognitive thinking

List – II

- i. Planning
ii. Fantasy
iii. Illusion
iv. Dreams

Codes :

- | | a | b | c | d |
|-----|----|-----|-----|-----|
| (1) | i | ii | iii | iv |
| (2) | iv | iii | ii | i |
| (3) | iv | i | ii | iii |
| (4) | i | iv | iii | ii |

45. Signal detection depends upon

- (1) motivation and expectations
(2) signal intensity and noise
(3) sensitivity and response criterion
(4) absolute and relative thresholds

46. Which of the following does not present with symptoms of mental retardation ?

- (1) Cri du chat syndrome
(2) Down's syndrome
(3) Fugue
(4) Trisomy-21

47. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using the codes given below :

Assertion (A) : Neurons cannot fire during absolute refractory period.

Reason (R) : A neuron can release only one neurotransmitter.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.

48. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using the codes given below :

Assertion (A) : When method of equal appearing intervals is employed to scale attitude items, median is usually used as an estimate of scale value.

Reason (R) : Under normal distribution, mean, median and mode are identical.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).

- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.

49. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I

(Book)

- a. Toward a Psychology of Being
 b. A Way of Being
 c. Man's Search for Meaning
 d. Love and Will

List – II

(Author)

- i. May
 ii. Maslow
 iii. Rogers
 iv. Frankl

Codes :

- | | a | b | c | d |
|-----|-----|-----|-----|-----|
| (1) | i | ii | iii | iv |
| (2) | ii | iii | iv | i |
| (3) | iii | ii | i | iv |
| (4) | iv | i | ii | iii |

50. Therapist suggested to Sunil that while trying to give up the habit of washing hands unnecessarily, he should wear a rubber band around his wrist and snap it everytime he feels like washing his hands. What therapy he is using ?

- (1) Systematic desensitization
 (2) Aversion therapy
 (3) Flooding
 (4) Modelling

51. The need for _____ produces active, controlling social behaviour while need for _____ produces more passive, less controlling social behaviour.

- (1) affiliation; self-esteem
 (2) confidence; intimacy
 (3) affiliation; intimacy
 (4) intimacy; affiliation

52. According to Triarchic Theory of Intelligence, arrange the correct functional sequence of metacomponents

- (1) Identification of problem defining the givens, goals & obstacles choosing

appropriate strategy selection of lower order processes selection of mental representation allocation of mental resources monitoring evaluation.

(2) Identification of problem defining the givens, goals & obstacles selection of lower order processes choosing appropriate strategy selecting a mental representation allocating mental resources monitoring evaluation.

(3) Identification of problem choosing appropriate strategy defining givens, goals and obstacles allocating mental resources monitoring selecting mental representation evaluation

(4) Defining the givens, goals and obstacles identification of problem selection of lower order processes choosing appropriate strategy selecting a mental representation allocating mental resources monitoring evaluation

53. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I

(Method of student discipline)

- a. Positive reinforcement
 b. Negative reinforcement
 c. Presentation punishment
 d. Removal punishment

List – II

(Description)

- i. decreasing behaviour by presenting aversive stimulus after it
 ii. strengthening behaviour by removing aversive stimulus when behaviour occurs
 iii. decreasing behaviour by removing pleasant stimulus after behaviour occurs.
 iv. strengthening behaviour by presenting desired stimulus after behaviour occurs.

Codes :

	a	b	c	d
(1)	i	ii	iii	iv
(2)	ii	iv	i	iii
(3)	iv	ii	i	iii
(4)	iii	i	ii	iv

54. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : The child always carries water bottle with him while he goes to school.

Reason (R) : The child wants to satisfy his primary need.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

55. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : According to Rogers, ideal condition for the development of a healthy self concept and movement toward becoming fully functioning is unconditional positive regard.

Reason (R) : With unconditional positive regard there is incongruence between true self and experience.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

56. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : Managers differ across cultures in their interpersonal styles and skills.

Reason (R) : Personal values held by a manager shape his perception of a situation.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

57. Which among the following neurotransmitters are found throughout central nervous system ?

- (a) Acetylcholine (b) Norepinephrine
- (c) Dopamine (d) Serotonin
- (e) GABA

Codes :

- (1) (a) and (c) and (d)
- (2) (c) and (d) and (e)
- (3) (b) and (c) and (d)
- (4) (a) and (d) and (e)

58. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : As per Guilford's SOI model, intelligence can be taken as superset of creativity.

Reason (R) : Creativity involves divergent production that is just one of five operations of intellect.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

59. In which one of the following scaling methods, Law of Comparative Judgement is used clearly while computing scale values ?

- (1) Summated ratings
- (2) Equal appearing intervals
- (3) JND scales
- (4) Paired Comparison

60. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : Japanese subjects detect background changes while U.S. subjects detect focal object changes.

Reason (R) : Eastern cultures encourage a holistic view, Western cultures de-emphasize context.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
61. The model of memory most frequently cited consists of Short Term Memory (STM) which serve(s) as gateway to Long Term Memory (LTM). This model was propounded by
- (1) Ebbinghaus
 - (2) Craik and Lockhart
 - (3) Tulving
 - (4) Atkinson and Shiffring
62. Which of the following disorders generally show a childhood onset ?
- (a) Delusional disorder
 - (b) Asperger's syndrome
 - (c) Attention-deficit hyperactivity disorder
 - (d) Mild Cognitive Impairment
- Codes :**
- (1) (c) and (d) only
 - (2) (b) and (c) only
 - (3) (a), (b) and (c) only
 - (4) (b), (c) and (d) only
63. The phenomenon of clinging to one's initial conceptions after the basis on which they were formed and had been discredited refers to :
- (1) Belief bias
 - (2) Belief perseverance
 - (3) Over confidence
 - (4) Framing decisions
64. Long Term Potentiality involves
- (1) release of nitric oxide
 - (2) increase in cell bodies
 - (3) increased number of NMDA (N-methyl-d-aspartic acid) receptors
 - (4) increased sensitivity of NMDA (N-methyl-d-aspartic acid) receptors
65. Mechanistic and Cognitive approaches to motivation differ in the extent to which _____ are invoked to account for the initiation, direction, intensity and persistence of goal directed behaviour.
- (1) Dynamic traits
 - (2) Higher mental processes
 - (3) Drives
 - (4) Instincts
66. Which of the following are the correct explanations of discrimination and stimulus generalizations ?
- (a) Learning to respond to one stimulus but not to the other similar stimuli.
 - (b) Learning to respond to one stimulus but not to the other different stimuli.
 - (c) Stimuli similar to the one used during conditioning also elicit the conditioned response.
 - (d) Stimuli different to the one used during conditioning also elicit the conditioned response.
- Codes :**
- (1) (a) and (c)
 - (2) (a) and (d)
 - (3) (b) and (c)
 - (4) (b) and (d)
67. Training and development offer competitive advantage to a firm by
- (a) removing performance deficiency
 - (b) minimizing accidents and damage
 - (c) making employees stay long
 - (d) meeting employees personal needs
- Codes :**
- (1) (a), (c), (d)
 - (2) (b), (c), (d)
 - (3) (a), (b), (d)
 - (4) (a), (b), (c)
68. Which of the following is the term for describing a child's one word utterances ?
- (1) Over extensions

- (2) Under extensions
- (3) Holophrases
- (4) Telegraphic speech

69. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : People have tendency to recall experiences that are consistent with their current moods.

Reason (R) : Moods and memories are congruent.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
70. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I
(Approach to perception)

List – II
(Description)

- | | |
|---------------------|--|
| a. Computational | i. fragments of sensory information form reality |
| b. Constructivist | ii. direct reception of information from environment |
| c. Ecological | iii. separate recognition of visual features |
| d. Feature analysis | iv. manipulate incoming signals to understand world |

Codes :

- | | a | b | c | d |
|-----|-----|-----|-----|-----|
| (1) | i | ii | iii | iv |
| (2) | iv | i | ii | iii |
| (3) | iii | iv | i | ii |
| (4) | ii | iii | iv | i |

Instructions for questions 71 to 75 :

Read the following paragraph and answer the five questions which follow :

An industrial psychologist wanted to study the job efficiency of the company executives in relation to M.B.A. marks, selection test scores, evaluations scores at the end of inhouse training. Job efficiency correlated with M.B.A. marks ($r = 0.40$), selection test scores ($r = 0.50$) and the evaluation scores at the end of inhouse training. ($r = 0.60$). All the three correlations were statistically significant. The multiple regression was computed using job efficiency as the criterion and the other three variables as the predictors. The three predictors explained 56.25 percent of variance in the job efficiency. The industrial psychologist further thought that the communication skills assessment would enhance the predictability of job efficiency. As such, the communication skills of the respondents were also assessed, which correlated with job efficiency by 0.35, again a statistically significant value. However when communication skills were added as the fourth predictor in the multiple regression, the change in R^2 (squared multiple correlation) was found to be insignificant.

71. What percent of variance in job efficiency is explained by evaluation scores at the end of training ?
- (1) 36 percent (2) 40 percent
(3) 60 percent (4) 64 percent
72. What would be the multiple correlation between the job efficiency and the three predictors (M.B.A. marks, selection test scores, and the evaluations scores at the end of inhouse training) ?
- (1) 0.50 (2) 0.65
(3) 0.75 (4) Inadequate data
73. Which one of the following conclusion can be drawn on the basis of above study ?
- (a) All the four predictors (M.B.A marks, selection test scores, evaluation scores at the end of inhouse training, and the

- communication skills) are individually valid.
- (b) Out of the above four predictors, all except communication skills, are individually valid.
- (c) Communication skills assessment has satisfactory incremental validity.
- (d) Communication skills assessment has unsatisfactory incremental validity.
- (1) (b) only
- (2) (a) and (c) only
- (3) (a) and (d) only
- (4) (b) and (d) only
74. The b coefficient obtained in multiple regression is
- (1) Partial correlation coefficient
- (2) Correlation coefficient
- (3) Regression coefficient
- (4) Partial regression coefficient
75. Which one of the following conclusion can be drawn on the basis of above paragraph ?
- (1) All the four predictors (M.B.A. marks, selection test scores, evaluation scores at the end of inhouse training and communication skills) are uncorrelated with each other.
- (2) Out of the four predictors, at least some are correlated with each other.
- (3) Communication skills scores are uncorrelated with other predictors.
- (4) The four predictors theoretically appear to be correlated, but we cannot infer about their intercorrelations from the data provided.

Answer Sheet

1. (2)	2. (2)	3. (1)	4. (1)	5. (3)	6. (3)	7. (4)	8. (2)	9. (1)	10. (3)
11. (2)	12. (3)	13. (1)	14. (2)	15. (3)	16. (1)	17. (4)	18. (2)	19. (2)	20. (1)
21. (2)	22. (1)	23. (1)	24. (4)	25. (1)	26. (4)	27. (1)	28. (3)	29. (1)	30. (4)
31. (2)	32. (2)	33. (3)	34. (2)	35. (3)	36. (3)	37. (2)	38. (1)	39. (1)	40. (2)
41. (4)	42. (2)	43. (4)	44. (3)	45. (3)	46. (3)	47. (3)	48. (2)	49. (2)	50. (2)
51. (3)	52. (2)	53. (3)	54. (1)	55. (3)	56. (1)	57. (4)	58. (1)	59. (4)	60. (1)
61. (4)	62. (2)	63. (2)	64. (1)	65. (2)	66. (1)	67. (4)	68. (3)	69. (1)	70. (2)
71. (1)	72. (3)	73. (3)	74. (4)	75. (2)					

January, 2017
PSYCHOLOGY
PAPER-II

Note : This paper contains **fifty (50)** multiple choice questions, each question carrying **two (2)** marks. Attempt all of them.

1. Which among the following are the products in Guilford's structure of intellect model?

(a) Systems (b) Transformation
 (c) Reasoning (d) Interpretation

Codes :

- (1) (a) and (b) only
 (2) (a), (b) and (c) only
 (3) (a), (b) and (d) only
 (4) (a), (b), (c) and (d)

2. According to Jung, which of the following is the deepest and darkest part of collective unconscious and contains all of the animal instincts

(1) Anima (2) Animus
 (3) Shadow (4) Persona

3. In classical conditioning progress of conditioning can be examined by indexing Conditioned Response (CR) in terms of

(a) Quantity (b) Latency
 (c) Rate (d) Density

Codes :

- (1) (a), (b) and (d) only
 (2) (b), (c), and (d) only
 (3) (a), (b), and (c) only
 (4) (a), (c), and (d) only

4. Which of the following intelligence tests are applicable to an illiterate adult ?

(a) Wechsler Adult Intelligence Scale
 (b) Alexander Pass Along Test
 (c) Coloured Progressive Matrices
 (d) Koh's Block Design Test

Codes :

- (1) (a) only (2) (b) and (d) only
 (3) (a) and (c) only (4) (b), (c) and (d) only

5. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I

(Variable type)

- a. Artificially discrete variable
 b. Continuous variable
 c. Qualitative variable
 d. Real discrete variable

List – II

(Example)

1. Neuroticism
 2. Voice pitch
 3. Gender
 4. Admission test results in terms of admitted / not-admitted

Codes :

	a	b	c	d
(1)	3	1	2	4
(2)	4	1	2	3
(3)	4	2	1	3
(4)	3	4	2	1

6. Length and Intensity of Lightwave play an important role in

- (1) Perception of colour and brightness
 (2) Perception of colour and illumination
 (3) Perception of constancy and brightness
 (4) Perception of brightness and illumination

7. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I

(Concept)

- a. Rods

List – II

(Explanation)

- i. Two optic nerves from two retina join each other for a short distance.

- b. Cones ii. Do not distinguish colours.
 c. Blind spot iii. Good detail vision.
 d. Optic Chiasm iv. Optic Nerves exit the eyes.

Codes :

- | | a | b | c | d |
|-----|-----|-----|-----|----|
| (1) | ii | iii | i | iv |
| (2) | ii | iii | iv | i |
| (3) | iii | ii | i | iv |
| (4) | i | ii | iii | iv |

8. Geeta studied for sociology exam and then studied for psychology exam. Certain concepts and theories are similar in psychology and sociology. While taking sociology exam, she finds that she keeps getting confused with psychology theorists. Geeta's problem is most likely due to
- (1) Decay of memory trace
 - (2) Proactive interference
 - (3) Retroactive interference
 - (4) Encoding failure
9. Which of the following sequence is correct in case of the curve of forgetting ?
- (1) Percent recall of words is greatest immediately after learning Percent recall decreases rapidly within the first hour Percent recall further decreases rapidly after 30 days.
 - (2) Percent recall of words is greatest immediately after learning Percent recall decreases rapidly within the first hour Forgetting levels off after the first hour.
 - (3) Percent recall of words is greatest after two days Forgetting levels off within 10 – 15 days Percent recall decreases further after 30 days.
 - (4) Percent recall of words is greatest after 2 days Percent recall decreases rapidly 20 days Forgetting levels off after 30 days.
10. Consider the following three types of investigations :

- (a) Expost facto field studies,
- (b) Laboratory experiments, and
- (c) Field experiments,

Arrange the above three types of investigations in ascending order in terms of researcher's ability to control secondary variance.

- (1) (c), (b), (a) (2) (b), (c), (a)
- (3) (b), (a), (c) (4) (a), (c), (b)

11. Read each of the following two statements – Assertion (A) and Reason (R) and indicate your answer using codes given below :

Assertion (A) : People often attain their visual experiences after many years of blindness.

Reason (R) : Most of the knowledge of the world had come to them through the sense of touch.

Codes :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

12. In Pavlovian conditioning paradigm the strongest and most rapid conditioning occurs in

- (1) Simultaneous conditioning
- (2) Trace conditioning
- (3) Short-delay conditioning
- (4) Long-delay conditioning

13. Read each of the following two statements – Assertion (A) and Reason (R) ; and indicate your answer using codes given below :

Assertion (A) : The information processing approach is rooted in structuralism.

Reason (R) : It attempts to identify the basic capacities and processes we use in cognition.

Codes :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

- (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
14. Utilitarianism provided the description of _____ of thought and behaviour.
 (1) what (2) when
 (3) why (4) how
15. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :
Assertion (A) : Naturalistic observation is a very popular research method in ethological research.
Reason (R) : Naturalistic observation is free from observer bias.
Codes :
 (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
16. Meaningfulness of verbal material can be indexed in terms of
 (1) Associative Reaction Time
 (2) Associative Value
 (3) Single Response Free Association
 (4) Single Response Controlled Association
17. Facial Feedback Hypothesis supports
 (1) Canon-Bard Theory of Emotion
 (2) James-Lange Theory of Emotion
 (3) Two-Factor Theory of Emotion
 (4) Opponent Process Theory of Emotion
18. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :
Assertion (A) : In some cases people attempt to reach the goal slowly and hesitantly.
Reason (R) : In approach – avoidance conflict, a single goal has positive valence as well as negative valence.
Codes :
 (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
19. Which of the following is/are the limitation/s of the survey method ?
 (a) Difficulties in obtaining representative samples.
 (b) Unsuitable for subjects with low educational levels.
 (c) Socially desirable responding.
 (d) Can not be used in longitudinal research
Codes :
 (1) (d) only
 (2) (b) and (c) only
 (3) (a) and (c) only
 (4) (a), (b) and (c) only
20. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :
Assertion (A) : In cross-sectional research to study development process of intelligence; age and cohorts are confounded.
Reason (R) : According to Flynn effect, each successive generation of people is more intelligent than the preceding one.
Codes :
 (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
21. Chronometric measurement of intelligence involves the use of which of the following :
 (1) Reaction time apparatus
 (2) Flicker-Fusion apparatus
 (3) ECG Bio-feedback apparatus
 (4) EMG Bio-feedback apparatus
22. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : The individual is left with little or no energy for cathexis (APM) after satisfying a particular need.

Reason (R) : Cathexis is the investment of psychic energy in the thoughts of objects or processes that will satisfy a need.

Codes :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
23. Network of auditory pathways can be arranged in which of following sequence :
- (1) Auditory Nerve Cochlear Nuclei
Superior olives Lateral Leminiscus
Inferior Colliculi Medial Geniculate Nuclei Primary Auditory Cortex.
 - (2) Cochlear Nuclei Auditory Nerve
Lateral Leminiscus Superior Olives
Inferior Colliculi Medial Geniculate Nuclei Primary Auditory Cortex.
 - (3) Auditory Nerve Cochlear Nuclei
Lateral Leminiscus Inferior Colliculi
Superior Colliculi Medial Geniculate Nuclei Auditory Cortex
 - (4) Cochlear Nuclei Auditory Nerve
Inferior Colliculi Superior Colliculi
Lateral Leminiscus Medial Geniculate Nuclei Auditory Cortex.
24. The tendency for parallel lines to appear to converge on each other is termed as
- (1) Motion parallax
 - (2) Ponzo illusion
 - (3) Linear perspective
 - (4) Aerial perspective
25. In a contingency table, one of the cell has an obtained frequency of 20 and an expected frequency of 30. What would be the contribution of this cell towards the total Chi-square value ?
- (1) 3.33 (2) 5.00
 - (3) 10.00 (4) 20.00

26. Match List – I with List – II and select the correct answer by choosing from the codes given below :

**List – I
(Concept)**

**List – II
(Explanation)**

- | | |
|-------------------------|---|
| a. Analogical reasoning | i. Processes that allow to make predictions about events / objects. |
| b. Deductive reasoning | ii. Using inferences to add new knowledge. |
| c. Inductive reasoning | iii. Using means-end analysis. |
| d. Physical reasoning | iv. Drawing valid conclusions from premises. |

Codes :

- | | a | b | c | d |
|-----|-----|-----|-----|----|
| (1) | i | ii | iii | iv |
| (2) | ii | iii | iv | i |
| (3) | iii | iv | ii | i |
| (4) | iv | iii | ii | i |

27. Match List – I with List – II and select the correct answer by choosing from the codes given below :

**List – I
(Concept)**

**List – II
(Explanation)**

- | | |
|---------------------------|---|
| a. Cortical Mosaic | i. Cortex is constantly being excited or inhibited depending on what the organism is experiencing. |
| b. Dynamic Stereotype | ii. Extinction that occurs after a previously excitatory CS is repeatedly presented without reinforcer. |
| c. Conditioned Inhibition | iii. Tendency of organisms to attend to and explore novel stimuli that occurs in their environment. |

- d. Orienting Reflex
- iv. Events consistently occurring in the environment come to have neurological representation making the responses rapid and automatic to the familiar environment.

Codes :

a	b	c	d	a	b	c	d
(1) ii	iii	i	iv	(2) i	iv	ii	iii
(3) iii	iv	i	ii	(4) iv	ii	iii	i

28. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I (Concept) List – II (Explanation)

- | | |
|----------------------|--|
| a. Drive | i. A state of deficiency or excess of something in the organism, the fulfilment or elimination of which is essential for survival and balance. |
| b. Yerkes-Dodson Law | ii. A state of psychological readiness that motivates the organism to act for the fulfilment of need and reduction of tension. |
| c. Power Motive | iii. The level of arousal beyond which performance begins to decline. |
| d. Need | iv. Tendency or capacity to produce intended effects on the behaviour or emotions of other persons. |

Codes :

a	b	c	d	a	b	c	d
(1) i	ii	iii	iv	(2) ii	iii	iv	i
(3) iii	iv	i	ii	(4) iv	iii	ii	i

29. Match List – I with List – II and select the correct answer by choosing from the codes given below :

**List – I
(Concept)**

**List – II
(Explanation)**

- | | |
|-------------------|---|
| a. Syntelity | i. Dynamic source trait with an environmental origin. |
| b. Self-sentiment | ii. Description of traits that characterize a group. |
| c. Metaerg | iii. Causal relationship between Ergs, sentiments and Attitudes. |
| d. Subsidiation | iv. The most powerful sentiment which organizes the entire personality. |

Codes :

a	b	c	d	a	b	c	d
(1) iv	ii	i	iii	(2) ii	iv	i	iii
(3) iv	iii	ii	i	(4) ii	iv	iii	i

30. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : Minor frustrations will add up to produce considerable frustration but not to aggression.

Reason (R) : Disruption of goal-directed behaviour usually leads to frustration.

Codes :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.

31. Which of the following are not nuclei of Thalamus ?

- (a) Internal Lamina
 (b) Anterior Nucleus
 (c) Ventromedial Nucleus
 (d) Paraventricular Nucleus

Codes :

- (1) (a) and (b) only (2) (b) and (c) only
 (3) (c) and (d) only (4) (a) and (c) only

32. Short-term memory holds information for
 (a) 50 seconds
 (b) 30 seconds or less
 (c) more than 60 seconds
 (d) between 50 and 60 seconds

Codes :

- (1) (c) only (2) (a) only
 (3) (b) only (4) (a) and (d)

33. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I**(Concept)**

a. Horizontal-
Vertical illusion

b. Atmospheric
perspective

c. Convergence

d. Figure-ground

List – II**(Explanation)**

i. Our eyes turn inward toward one-another in order to see close objects.

ii. What we perceive often is based more on what goes on in our mind than what is in front of our eyes.

iii. Tendency to perceive objects high in our visual field as more distant.

iv. The farther away objects are, the less distinctly they are seen.

Codes :

- | | a | b | c | d |
|-----|-----|-----|----|-----|
| (1) | i | ii | ii | iv |
| (2) | ii | iii | iv | i |
| (3) | iii | iv | i | ii |
| (4) | iv | i | ii | iii |

34. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : Participants of a study were shown novel geometric shapes that served

as targets. Then they were shown fragments of these shapes. Participants were quicker to recognize the fragments as part of the original target.

Reason (R) : We do not use Gestalt principles in our everyday perception.

Codes :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.

35. In problem solving if you follow rules that reduce the number of operations or allow you to take short cuts, you are using :

- (1) Algorithm (2) Heuristic
 (3) Strategy (4) Skill

36. Anticipation Method can be used in

- (a) Free Recall
 (b) Serial Learning
 (c) Verbal Discrimination
 (d) Paired Associates Learning

Codes :

- (1) (a), (b) and (c) only
 (2) (b), (c) and (d) only
 (3) (a) and (b) only
 (4) (b) and (d) only

37. Gestalt Psychologist Max Wertheimer attempted to train children to think in a logical and systematic way where he used several figures such as

- (a) Squares and rectangles
 (b) Parallelogram
 (c) Triangles
 (d) Circles

Codes :

- (1) (a) & (b) only (2) (b) and (c) only
 (3) (c) and (d) only (4) (a), (b) and (c) only

38. Which of the following sequence describes the energy mobilization in body ?

- (1) Fasting phase Absorptive phase
 Cephalic phase

- (2) Cephalic phase Absorptive phase
Fasting phase
- (3) Absorptive phase Fasting phase
Cephalic phase
- (4) Fasting phase Cephalic phase
Absorptive phase
39. According to Bandura, which of the following cognitive mechanisms are adopted by a person to act contrary to his or her moral principles without experiencing self-contempt :
- (a) Moral justification
(b) Dehumanization
(c) Displacement of Responsibility
(d) Diffusion of Responsibility

Codes :

- (1) (a), (b) and (c) only
(2) (a), (b) and (d) only
(3) (b), (c) and (d) only
(4) (a), (b), (c) and (d)
40. In Guilford's struct of Intellect Model, obtained abilities have been named in which of the following sequence :
- (1) Operations, contents, products
(2) Operations; products; contents
(3) Contents; operations; products
(4) Products; contents; operations
41. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :
- Assertion (A) :** Differences between intelligence levels of monozygotic twins are attributed to environmental influences.
- Reason (R) :** Monozygotic twins have identical genetically determined reaction range.
- Codes :**
- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.
42. Match List – I with List – II and select the correct answer by choosing from the codes given below :

List – I
(Explanation)

List – II
(Concept)

- | | |
|---|---------------------|
| a. Remembering meaning of words, properties of objects. | i. Sensory memory |
| b. Helping information to be brought back to mind when it cannot be recalled spontaneously. | ii. Semantic memory |
| c. Remembering specific experiences of the past. | iii. Retrieval cues |
| d. Retaining representation of sensory input for brief periods of time. | iv. Episodic memory |

Codes :

- | | a | b | c | d |
|-----|-----|-----|-----|----|
| (1) | i | ii | iii | iv |
| (2) | ii | iii | iv | i |
| (3) | iii | ii | i | iv |
| (4) | iv | ii | iii | i |

43. Information in short-term memory enters long-term storage through
- (1) selective attention
(2) elaborate rehearsal
(3) learning
(4) information-processing
44. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :
- Assertion (A) :** A neural membrane is double-layered.
- Reason (R) :** Lipid molecules turn their tails toward each other.
- Codes :**
- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.
45. The inputs from our sensory receptors are structured by
- (1) Selective principles
(2) Gestalt principles

- (3) Analytical principles
- (4) Learning principles

Instructions for Questions 46 to 50 :

Read the following paragraph and answer the five questions which follow :

A researcher working in the area of human intelligence wanted to develop a new verbal test of intelligence. For this purpose, he wrote 200 objective multiple-choice items. Each item had four alternatives, one of which was right and the three remaining ones were wrong. He administered this initial version to a sample of 400 subjects. Using P_{27} and P_{73} as the cut-off points, he obtained two groups – Low Scoring and High Scoring respectively. The item difficulty index and the item discrimination index were computed for each of the 200 items. To assess whether each item significantly discriminates between Low scoring and High scoring groups, appropriate statistical significance tests were also employed. This item analysis process retained 150 items. The final version of 150 items was administered to a fresh sample of 750 subjects. The appropriate measures of internal consistency reliability were obtained. The test norms were developed using normalized T scores (mean = 100, SD = 20) and the Wechsler type deviation IQ scores. To assess the dimensionality of verbal intelligence, the researcher also carried out the factor analysis of item scores.

46. Which one of the following test of statistical significance can be employed in the above study to evaluate whether the item significantly discriminates between the Low scoring and High scoring groups ?
- (1) Chi-square
 - (2) Point-biserial correlation
 - (3) t test
 - (4) Biserial correlation
47. Which of the following reliability coefficients can be computed in the above study to assess the internal consistency ?
- (a) Odd-even reliability coefficient
 - (b) Kuder-Richardson reliability coefficient
 - (c) Cronbach alpha coefficient
 - (d) Interscorer reliability coefficient
- Codes :**
- (1) (a) only
 - (2) (a) and (c) only
 - (3) (a), (b) and (c) only
 - (4) (b), (c) and (d) only
48. For the above test, what is the expected percentile rank for the T score of 120 ?
- (1) 60
 - (2) 68
 - (3) 75
 - (4) 84
49. For the above test, a T score of 80 would correspond to the deviation IQ of
- (1) 68
 - (2) 80
 - (3) 85
 - (4) 115
50. In the above study, which type of correlations can be employed to carry out the factor analysis of the test items ?
- (1) Biserial correlations
 - (2) Phi coefficients
 - (3) Point biserial correlations
 - (4) Rank order correlations

Answer Sheet

1. (1)	2. (3)	3. (3)	4. (2)	5. (2)	6. (1)	7. (2)	8. (3)	9. (2)	10. (4)
11. (1)	12. (3)	13. (1)	14. (3)	15. (3)	16. (2)	17. (2)	18. (1)	19. (3)	20. (1)
21. (1)	22. (4)	23. (1)	24. (3)	25. (1)	26. (3)	27. (2)	28. (2)	29. (2)	30. (4)
31. (3)	32. (3)	33. (3)	34. (3)	35. (2)	36. (4)	37. (4)	38. (2)	39. (4)	40. (1)
41. (1)	42. (2)	43. (2)	44. (1)	45. (2)	46. (1)	47. (3)	48. (4)	49. (3)	50. (2)

January, 2017
PSYCHOLOGY
 PAPER-III

Note : This paper contains **seventy five (75)** objective type questions of **two (2)** marks each. **All** questions are compulsory.

1. The full form of ROC is
 - (1) Receiver-Operating Characteristic
 - (2) Response-Operating Characteristic
 - (3) Receiver Operating Curve
 - (4) Response Operating Curve
2. Match List-I with List-II and select the correct answer by choosing from the codes given below :

List – I
(Concepts)

a. Engram

b. Explicit memory

c. Long-term potentiation

d. Implicit memory

List - II
(Explanations)

i. Engaging in conscious recollection

ii. Changes in dendrites and in the sensitivity of synapse through repeated stimulation

iii. Recollecting something without being consciously aware

iv. The physical trace of memory in the brain

Codes :

- | | a | b | c | d |
|-----|----------|----------|----------|----------|
| (1) | i | ii | iii | iv |
| (2) | iv | i | ii | iii |
| (3) | ii | iii | iv | i |
| (4) | iii | i | ii | iv |

3. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : An infant often acts as if objects stop existing when they are moved out of his perceptual range.

Reason (R) : The infant has not yet developed the object concept.

Codes :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

4. Mechanistic view of motivation can be characterised as :

- (a) S-O-R explanation.
- (b) Direction of behaviour is determined by bonds between stimulus and response or habits.
- (c) Probability of elicitation of various habits depend upon prior reinforcement contingencies.
- (d) Behaviour once elicited persists until the removal of sustaining stimuli.

Codes :

- (1) (a), (b) and (c) only
- (2) (a), (c) and (d) only
- (3) (b), (c) and (d) only
- (4) (a), (b) and (d) only

5. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : In terms of SOI model, intelligence can be taken as a superset of creativity.

Reason (R) : Divergent production is just one of the operations of intellect.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).

- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
6. Which of the following is true of the threshold hypothesis regarding intelligence creativity relationship ?
 (1) There is a 'U' type relationship between intelligence and creativity.
 (2) There is 'inverted U' type relationship between intelligence and creativity.
 (3) The relationship between the two is 'ill-defined' above a specific level of intelligence.
 (4) The relationship between the two is ill defined below a specific level of intelligence.
7. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :
Assertion (A) : According to Rollo May, to grow psychologically, our values must be constantly challenged.
Reason (R) : Rollo May proposed that Neurotic anxiety occurs when our values are challenged.
- Codes :**
 (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
8. Match List-I with List-II and select the correct answer by choosing from the codes given below :
- | List – I
(Concept) | List – II
(Explanation) |
|------------------------------|--|
| a. Monism | i. Brain functions as an undifferentiated whole |
| b. Dualism | ii. Mind and body consists of the same substance |
- c. Equipotentiality iii. Body is material and mind is non-material
 d. Reductionism iv. Understanding of behavioural phenomenon in terms of biological principles
- Codes :**
- | | a | b | c | d |
|-----|----------|----------|----------|----------|
| (1) | i | ii | iii | iv |
| (2) | ii | iii | i | iv |
| (3) | iv | iii | ii | i |
| (4) | iii | i | iv | ii |
9. The correlation matrix was factor-analyzed by the three methods listed below and the same number of factors were extracted in each case :
 a. Principal components method
 b. Maximum likelihood method
 c. Principal axes method
 Arrange the above methods in descending order in terms of variance condensed :
 (1) b, c and a (2) b, a and c
 (3) c, a and b (4) a, c and b
10. Which of the following are lesion techniques?
 a. Aspiration Lesions
 b. Radioactive frequency lesions
 c. Cryogenic Blockade
 d. Bregma
- Codes :**
 (1) a, b and d only (2) b, c and d only
 (3) a, b and c only (4) a, b, c and d
11. People tend to be in better physical health and are better able to cope with stress when they are provided with
 a. Physical comfort
 b. Moral comfort
 c. Psychological comfort
 d. Ideological comfort
- Codes :**
 (1) a & b only (2) b & c only
 (3) c & d only (4) a & c only

12. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : Leadership is a process where people merely make an attribution about other individuals.

Reason (R) : People regard leader as a change agent.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
13. Besides psychology which Behavioural Science contributed in the development of Organizational Psychology as a field of study:
- | | |
|----------------------|-----------------|
| a. Sociology | b. Anthropology |
| c. Political Science | d. Social work |

Codes :

- (1) a & b only (2) a, b & d only
 (3) b, c & d only (4) a, b & c only
14. Which of the following is not a positive symptom of schizophrenia ?
- Disorganized speech
 - Bizarre behaviour
 - Flat affect
 - Hallucinations

Codes :

- (1) d only (2) a and b
 (3) c only (4) a and d
15. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : People belonging to a particular group have some common perceptual patterns.

Reason (R) : Personal experiences influence one's perception.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

16. A study shows that children from poor families drew large circles to represent a coin than the children from rich families. This is an example of

- Perceptual constancy
- Perceptual distortion
- Perceptual set
- Perceptual adaptation

17. Match List-I with List-II and select the correct answer by choosing from the codes given below :

List – I

List – II

- | | |
|----------------------|---|
| a. Hit | i. Signal is present but response is No. |
| b. Miss | ii. Signal is present and response is Yes. |
| c. False Alarm | iii. Signal is absent and the response is No. |
| d. Correct Rejection | iv. Signal is absent but the response is Yes. |

Codes :

- | | a | b | c | d |
|-----|----|-----|-----|-----|
| (1) | iv | i | iii | ii |
| (2) | ii | i | iv | iii |
| (3) | ii | i | iii | iv |
| (4) | iv | iii | ii | i |

18. People often cannot remember where they heard what they heard. Sometimes people think they saw things they did not see or heard things they did not hear. This is the case of

- Bias
- Misattribution
- Suggestibility
- Blocking

19. Memory retrieval is

- Reconstructive
- Constructive
- Unconscious
- Symbolic

Codes :

- (1) (a) only (2) (b) only
 (3) (a) and (b) (4) (c) and (d)

20. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : Words that are logically connected have been found to be recalled more easily than the words that were concretely connected.

Reason (R) : The deeper the level of processing, the higher the level of recall achieved.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
21. Analogical reasoning consists of the following processes :
- a. Retrieval b. Mapping
 - c. Evaluation d. Abstraction
 - e. Predictions
- Codes :**
- (1) a, b and d only (2) a, c and d only
 - (3) a, b, d and e only
 - (4) All the five
22. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :
- Assertion (A) :** Patients with Amygdala damage are unable to recognise fear in others' facial expression.
- Reason (R) :** Looking at fearful faces activates Amygdala.
- Codes :**
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
23. In Sternberg's triarchic theory, the type of intelligence that involves the ability to think critically and analytically is called
- (1) componential intelligence
 - (2) contextual intelligence
 - (3) crystallized intelligence
 - (4) experiential intelligence
24. Who among the following has given Regression Argument to support racial differences in intelligence ?
- (1) H.J. Eysenck (2) M.W. Eysenck
 - (3) Jensen (4) Burt and Vernon
25. Creative thinking is a combination of flexibility in thinking and reorganization in understanding to produce ____ and ____.
- (1) new ideas; solutions
 - (2) new ideas; conclusions
 - (3) innovative ideas; conclusions
 - (4) innovative ideas; solutions
26. Which of the following components of creativity are scored in the "Picture construction" sub-test of non-verbal Torrance Test of Creative Thinking (Non-verbal) ?
- a. Originality b. Elaboration
 - c. Fluency d. Flexibility
- Codes :**
- (1) a and b only (2) a, b and d only
 - (3) b and d only (4) a, b and c only
27. The fear of one's own potential greatness and the ambivalent feelings toward greatness in others is called
- (1) Proprium striving
 - (2) Self-exonerating mechanism
 - (3) Fluctuating asymmetry
 - (4) Jonah-complex
28. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :
- Assertion (A) :** Pituitary gland is frequently referred to as the master gland.
- Reason (R) :** Most of pituitary hormones are tropic hormones.
- Codes :**
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.

29. Match List-I with List-II and select the correct answer by choosing from the codes given below :

**List – I
(Variables)**

a. Confirmity

b. Compliance

c. Intense
indoctrination

d. Social norms

**List – II
(Explanations)**

i. Social influence involving direct request from others

ii. Process in which members of extreme groups accept the beliefs and rules in unquestioning way

iii. Rules indicating how individuals are expected to behave in a specific situation

iv. Social influence in which individuals change their attitude/behaviour to go along with existing norms

Codes :

	a	b	c	d
(1)	iv	iii	ii	i
(2)	i	iii	ii	iv
(3)	iv	i	ii	iii
(4)	iii	i	ii	iv

30. Match List-I with List-II and select the correct answer by choosing from the codes given below :

**List – I
(Concept)**

a. Cohesiveness and conformity

b. Norms and social influence

**List – II
(Process)**

i. Social influence based on individual's desire to be liked by others

ii. Social influence based on individual's desire to be correct-to possess the accurate perceptions of the social world

c. Individuation iii. The need to be distinguishable from others in some respect

d. Information and social influence iv. Accepting influence of those we like

Codes :

	a	b	c	d
(1)	iv	iii	i	ii
(2)	iii	iv	ii	i
(3)	ii	iii	i	iv
(4)	iv	i	iii	ii

31. The degree to which variations in environment can affect the development of individuals with the particular genotype is

(1) Canalization (2) Co-dominance
(3) Reaction range (4) Ritualization

32. Match List-I with List-II and select the correct answer by choosing from the codes given below :

**List – I
(Theorist)**

a. Piaget

b. Kohlberg

c. Bronfenbrenner

d. Vygotsky

**List – II
(Explanation)**

i. Children construct knowledge through social interaction with others.

ii. Children are embedded in a number of environmental systems and influences

iii. Children construct knowledge by transforming, organizing and reorganizing previous knowledge.

iv. Children's moral thoughts are advanced through discussions with others.

Codes :

	a	b	c	d
(1)	i	ii	iii	iv
(2)	ii	iv	iii	i
(3)	iii	ii	i	iv
(4)	iii	iv	ii	i

33. Three groups of subjects were formed with random assignment. Group-I was tested under 'No drug' condition, Group-II was tested under 'Low drug dosage' condition, and Group-III was tested under 'High drug dosage' condition. Each group was required to learn two lists, a list of CVC trigrams and a list of three-lettered meaningful words. The dependent variable was the number of items recalled after ten minutes after learning each list. This experiment employs

- (1) Randomized three group design
- (2) Randomized six group design
- (3) 3×2 mixed factorial design
- (4) Randomized 3×2 factorial design

34. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : Teachers need to teach children the use of multiple strategies : finding out what works well, when and where.

Reason (R) : Most children are not consciously aware of a variety of strategies they can employ while learning.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

35. Which of the following sequence is correct in case of steps in a goal-setting program for students ?

- (1) Specify the objectives to be accomplished Define goal difficulty
Specify the time span for assessment of performance Discuss goal attainment strategies
- (2) Discuss goal attainment strategies
Define goal difficulty Specify the objectives to be accomplished Specify time span for assessment of performance.
- (3) Specify the time span for assessment of performance Discuss goal attainment

strategies Specify the objectives to be accomplished Define goal difficulty

- (4) Define goal difficulty Discuss goal attainment strategies Specify time span for assessment of performance
Specify the objectives to be accomplished.

36. The field of Industrial Psychology was created / developed by

- (1) F.B. Taylor (2) H. Munsterberg
- (3) Elton Mayo (4) H. Mintzberg

37. Match List-I with List-II and select the correct answer by choosing from the codes given below :

List – I

(Therapy)

- a. Systematic desensitization
- b. Social-Learning Therapy
- c. Cognitive Therapy
- d. Cognitive Behaviour Modification

List – II

(Therapist)

- i. Albert Bandura
- ii. Meichenbaum
- iii. Joseph Wolpe
- iv. Aaron Beck

Codes :

- | | a | b | c | d |
|-----|-----|-----|-----|----|
| (1) | i | ii | iii | iv |
| (2) | iii | i | iv | ii |
| (3) | ii | iii | iv | i |
| (4) | iv | iii | ii | i |

38. A procedure that provides information to the learner in small steps, guarantees immediate feedback and allows the learner to determine the pace with which to complete the task. It is called

- (1) Auto shaping (2) Chaining
- (3) Programmed learning
- (4) Continuous contracting

39. Behaviour modification by the application of operant conditioning consists of the following :

- a. Token economy b. Punishment
- c. Time-out
- d. Withdrawal of reinforcement

Codes :

- (1) a & b only (2) b & c only
(3) a & d only (4) a & c only

40. Match List-I with List-II and select the correct answer by choosing from the codes given below :

List – I

List – II

- | | |
|---------------------------|--|
| a. Observational learning | i. Learning where an object is located |
| b. Place learning | ii. A response that can be observed by others |
| c. Overt responding | iii. The process whereby information is acquired by attending to events in the environment |
| d. Latent learning | iv. Learning that takes place independent of reinforcement |

Codes :

- | | | | |
|----------|----------|----------|----------|
| a | b | c | d |
| (1) ii | iv | iii | i |
| (2) iii | i | ii | iv |
| (3) ii | i | iii | iv |
| (4) iii | iv | ii | i |

41. According to self-determination theory _____ and _____ form the basis for intrinsic motivation.

- (1) competence ; relatedness
(2) competence ; autonomy
(3) control ; autonomy
(4) control ; competence

42. Match List-I with List-II and select the correct answer by choosing from the codes given below :

List – I

List – II

(Concept)

(Explanation)

- | | |
|-------------------------|--|
| a. Fixed Action Pattern | i. When two incompatible response tendencies are simultaneously aroused, the |
|-------------------------|--|

organisms may display behaviours that appear to be irrelevant to the situation.

- | | |
|--------------------------|--|
| b. Displacement Activity | ii. Highly stereotyped responses that are assumed to be genetically programmed reactions that occur in the presence of a particular 'sign' stimulus. |
| c. Hedonic Bias | iii. A state in which a person performs an action because the act itself is satisfying or rewarding internally. |
| d. Intrinsic motivation | iv. Tendency to take more credit for success than to take responsibility for failure. |

Codes :

- | | | | | | | | |
|----------|----------|----------|----------|----------|----------|----------|----------|
| a | b | c | d | a | b | c | d |
| (1) i | ii | iii | iv | (2) ii | i | iv | iii |
| (3) iii | iv | ii | i | (4) iv | iii | i | ii |

43. Match List-I with List-II and select the correct answer by choosing from the codes given below :

List – I

List – II

(Concept)

(Meaning)

- | | |
|---------------|---------------------------------------|
| a. Dasein | i. Physical, objective world |
| b. Umwelt | ii. Human interactions |
| c. Eigen welt | iii. Individual as being in the world |
| d. Mitwelt | iv. Self-awareness |

Codes :

- | | | | |
|----------|----------|----------|----------|
| a | b | c | d |
| (1) i | iii | iv | ii |
| (2) i | iii | ii | iv |
| (3) iii | i | iv | ii |
| (4) iii | i | ii | iv |

44. A group of 25 subjects were tested on reaction time under three conditions – low stressful, moderately stressful, and highly stressful. The data were analysed by suitable ANOVA. What would be the degrees of freedom for 'stressfulness $\times$ subjects' interaction ?

(1) 2 (2) 3
(3) 24 (4) 48

45. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : Most Experimental studies of lesion effects are studies of bilateral than unilateral lesions.

Reason (R) : Behavioural effects of unilateral lesions in all brain structures can be difficult to detect.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.
46. An act, that helps or is designed to help others regardless of the helper's motive is called:
- (1) Prosocial Behaviour
(2) Altruism
(3) Nurturance
(4) Social obligation
47. Difference between what one can accomplish with or without help is called
- (1) Zone of proximal development
(2) Resilience
(3) Identity fusion
(4) Transition
48. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : According to Adler, a child's birth order is important in shaping his / her personality.

Reason (R) : There are unique experiences associated with birth order of every child.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.

49. Punishment given frequently to young students leads to

- (1) improved interpersonal relationships
(2) desirable behaviours
(3) better academic performance
(4) fear, anger and worry

50. Persons with genetic "markers" for schizophrenia have a physical vulnerability to the disorder but will not develop schizophrenia unless they are exposed to critical times in development. This is according to

- (1) Biological model
(2) Social learning model
(3) Neurobiological model
(4) Stress-vulnerability model

51. The influence of surrounding environment on perception is called

- (1) Hoffding function
(2) Priming effect
(3) Law of Pragnaz
(4) Context effect

52. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : Hull proposed that Reaction Potential (SER) is the product of Habit Strength (SHR) and Drive (D) from this some testable prediction can be made about the rate of eye lid conditioning in relation to drive.

Reason (R) : Hull's theory is based on Hypothetico-Deductive model which involved deducting testable consequences based on postulates.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).

- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
53. Long Term Potentiation means facilitation of synaptic transmission following :
 (1) High-frequency electrical stimulation applied to pre-synaptic neurons.
 (2) Low-frequency electrical stimulation applied to pre-synaptic neurons.
 (3) High-frequency electrical stimulations applied to post-synaptic neurons.
 (4) Low-frequency electrical stimulation applied to post-synaptic neurons.
54. Which one of the following is NOT a mnemonic device ?
 (1) Method of Loci
 (2) Method of reinforcement
 (3) Categorical clustering
 (4) Pegword system
55. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :
Assertion (A) : Second order factors can be obtained from promax rotated first-order factors.
Reason (R) : Promax rotation is a method of oblique rotation.
Codes :
 (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
56. In an experiment one group of subjects was asked to estimate the product of $8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$ in 5 seconds. The other group was asked to estimate the product of $1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8$ in the same time. The former group attained a larger estimate than the later group. The biasing effect involved in this case is because of
 (1) Availability of information
 (2) Adjustment

- (3) Representativeness
 (4) Gambler's fallacy

57. Match List-I with List-II and select the correct answer by choosing from the codes given below :

List – I (Phenomenon)	List – II (Explanation)
a. Thinking	i. Process of choosing between two or more alternatives on the basis of information about them
b. Reasoning	ii. Processing information in various ways to move towards a desired goal
c. Decision making	iii. Mental activity through which we transform available information in order to reach conclusion
d. Problem solving	iv. An activity that involves the manipulation of mental representation of various features of the external world

Codes :

- | | | | | | | | |
|---------|----|-----|----|--------|-----|----|-----|
| a | b | c | d | a | b | c | d |
| (1) i | ii | iii | iv | (2) ii | iii | iv | i |
| (3) iii | iv | ii | i | (4) iv | i | ii | iii |

58. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : In general, members of joint family tend to be low on stress than those of nucleus family.

Reason (R) : Social support increases the stress responses.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

- (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
59. Rogers use Q-sort technique to identify –
 (1) Internal and external frame of reference
 (2) Real self and Ideal self
 (3) Conditional and unconditional positive regard
 (4) Conditions of worth and worthlessness
60. Which of the following are the characteristics of Nomothetic approach to personality?
 a. Emphasis on variations in a person
 b. Interpretation of test scores
 c. Group norms
 d. General laws of personality
- Codes :**
 (1) a, c and d only (2) b, c and d only
 (3) a, b and d only (4) a, b, c and d
61. Sensory-specific satiety takes place in
 (1) Papillae
 (2) Thalamus
 (3) Nucleus of Solitary Tract (NST)
 (4) Insula
62. The multiple correlation of academic achievement with Numerical ability, Abstract reasoning, and Verbal ability was found to be 0.6. Which of the following statement / s would be true in this context ?
 a. None of the three predictors would correlate more than 0.6 with academic achievement.
 b. The three predictors together would explain 36 percent variance in academic achievement.
 c. At least, one predictor would correlate 0.6 with academic achievement.
 d. The three predictors would explain 60 percent of variance in academic achievement.
- (1) b only (2) a and b only
 (3) b and d only (4) c and d only
63. The different schemas had an impact on social cognitive, which refer to
 a. Persons b. Groups
 c. Roles d. Events

Codes :

- (1) a, b, c only (2) b, c and d only
 (3) a, b and d only (4) a, c and d only

64. Match List-I with List-II and select the correct answer by choosing from the codes given below :

List – I**(Test/Experiment)**

- a. Strange Situation Test
 b. Visual-Cliff Test
 c. Bodo-Doll Experiment
 d. Heinz Dilemma

List – II**(Concept)**

- i. Observation Learning
 ii. Morality
 iii. Attachment
 iv. Depth perception

Codes :

- | | a | b | c | d | | a | b | c | d |
|-----|-----|----|----|-----|-----|-----|----|----|-----|
| (1) | ii | i | iv | iii | (2) | iii | i | iv | ii |
| (3) | iii | iv | i | ii | (4) | ii | iv | i | iii |

65. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : According to Vygotsky, human thinking and reasoning are products of social activity.

Reason (R) : Every cognitive process appears first on social plane and then on psychological plane after internalization.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
66. The most effective way for dealing with bullying in schools is
 (1) to give harsh punishment to those students who indulge in such behaviour.
 (2) to provide counselling to students who are potential bullies and clearly communicate to them that such behaviour will not be tolerated.

- (3) to isolate such students and ban them from participation in any extra-curricular activity
 (4) to ignore the problem and continue to focus only on good students

67. Match List-I with List-II and select the correct answer by choosing from the codes given below :

List – I

a. Variable-pay program

b. Piece-rate pay plan

c. Profit-sharing plan

d. Gain sharing

List – II

i. An incentive plan for the improvement of group productivity linked with total amount of money allocation.

ii. Compensation distribution based on certain formula around company's profitability

iii. Workers are paid a fixed sum for each unit of production

iv. Portion of employees' pay based on individual / organizational measure of performance

Codes :

- | | a | b | c | d |
|-----|-----|-----|-----|----|
| (1) | i | ii | iii | iv |
| (2) | ii | iii | iv | i |
| (3) | iii | iv | i | ii |
| (4) | iv | iii | ii | i |

68. McClelland's theory basically focuses on ____ needs whilst Maslow's theory focuses on ____ needs.

- (1) two ; four (2) three ; four
 (3) three ; five (4) two ; five

69. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : A large proportion of Down Syndrome cases can be identified in prena-

tal diagnosis through the use of amniocentesis.

Reason (R) : The amniocentesis procedure reveals that the fetus does not have the defect, and the parents are spared months of needless anxiety.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.

70. Vijay was working as executive manager in a multinational company. His team members always found it difficult to get along with him. Vijay always craved for excessive admiration and attention, took undue advantage of others, was very arrogant and preoccupied with fantasies of unlimited success and power. He seems to be suffering from

- (1) Borderline personality disorder
 (2) Narcissistic personality disorder
 (3) Histrionic Personality disorder
 (4) Schizoid personality disorder

Instructions for Questions 71 to 75 :

Read the following paragraph and answer the five questions which follow :

Scaling in psychology can be classified into two categories – psychophysical scaling and psychological scaling. Although psychological scaling had its origin in Fechner's method of first choices and the work on colour preference, the major work in psychological scaling began in 1920's in the context of attitude measurement. Bogardus Social Distance Scale (1925) is one of such first attempts. A major breakthrough in attitude scaling occurred when Thurstone developed his Law of Comparative Judgement (LCJ) and also proposed the three scaling methods-paired comparison, equal appearing intervals, and successive intervals. In the paired comparison, every statement / stimulus is paired with every other statement / stimulus. The computational methods of paired comparison are most extensively de-

veloped, some of them using Thurstone's LCJ directly for this purpose. Such pairing of statements/stimuli is not involved in methods of Equal Appearing Intervals (EAI) and Successive Intervals (SI). Moreover, the distributions of the categories, assigned by the judges to each attitude statement in EAI and SI need not be normal; in fact the distributions for several statements are skewed. This situation requires some computational precautions. Moreover, the method of successive intervals was computationally quite laborious in pre-computer era and hence it was not so popular. In the classical scaling tradition Guttman developed the scalogram technique and Edwards developed the scale discrimination technique. Likert's development of summated ratings represent an important landmark in attitude measurement. It follows the traditional psychometric model. Aiken and Groth-Marnatt (2009) concluded that other scaling techniques, less frequently used for attitude measurement, include Osgood's semantic differential, Q-sort, expectancy value rating, magnitude estimation, multidimensional scaling etc.

71. The scale values of the attitude statements, obtained by analyzing paired comparison data following Thurstone's LCJ, would yield
 (1) Nominal Scale (2) Ordinal scale
 (3) Interval scale (4) Ratio scale

72. Read each of the following two statements – Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : The method of paired comparison is not suitable for scaling the large number of attitude statements.

Reason (R) : In method of paired comparison, every attitude statement is paired with every other attitude statement.

Codes :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
73. Which one of the following can be used to find the scale values of attitude statements, when method of equal appearing intervals is used ?
 (1) Mean
 (2) Standard deviation
 (3) Median
 (4) Quartile deviation
74. Fill in the blank with the most appropriate alternative. It would be desirable to carry out the factor analysis of attitude scales obtained by method of _____.
 (1) paired comparison
 (2) equal appearing intervals
 (3) successive intervals
 (4) summated ratings
75. In studying attitudes towards the outgroup, a series of bipolar adjective scales were used. Most probably, the researcher used
 (1) Bogardus Social Distance Scale
 (2) Osgood's semantic differential
 (3) Guttman's scalogram
 (4) Attitude scale obtained by method of equal appearing intervals

Answer Sheet

1. (1)	2. (2)	3. (1)	4. (3)	5. (1)	6. (3)	7. (3)	8. (2)	9. (4)	10. (3)
11. (4)	12. (1)	13. (4)	14. (3)	15. (1)	16. (3)	17. (2)	18. (2)	19. (3)	20. (1)
21. (4)	22. (1)	23. (1)	24. (3)	25. (4)	26. (1)	27. (4)	28. (1)	29. (4)	30. (4)
31. (3)	32. (4)	33. (3)	34. (1)	35. (1)	36. (2)	37. (2)	38. (3)	39. (4)	40. (2)
41. (2)	42. (2)	43. (3)	44. (4)	45. (3)	46. (1)	47. (1)	48. (1)	49. (4)	50. (4)
51. (4)	52. (2)	53. (1)	54. (2)	55. (1)	56. (2)	57. (2)	58. (3)	59. (2)	60. (2)
61. (3)	62. (2)	63. (4)	64. (3)	65. (1)	66. (2)	67. (4)	68. (3)	69. (1)	70. (2)
71. (3)	72. (1)	73. (3)	74. (4)	75. (2)					

November, 2017

PSYCHOLOGY

PAPER-II

Note : This paper contains fifty (50) multiple choice questions, each question carrying two (2) marks. Attempt all of them.

1. Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I (Concept)	List - II (Description)
(a) Type	(i) Consistent style of behaviour or emotion, a person displays across the situations
(b) Trait	(ii) A cluster of many dispositions
(c) System	(iii) Enduring qualities that define the individual and distinguish individuals from one another
(d) Structure	(iv) A collection of highly interconnected parts of personality whose overall behaviour reflects them being organised

Codes :

- | | | | | |
|-----|------------|------------|------------|------------|
| | (a) | (b) | (c) | (d) |
| (1) | (i) | (ii) | (iii) | (iv) |
| (2) | (ii) | (i) | (iv) | (iii) |
| (3) | (iii) | (ii) | (i) | (iv) |
| (4) | (iv) | (iii) | (ii) | (i) |
2. Which of the following is true for Tip_of_the Tongue (TOT) state ?
- Feeling of Knowing
 - Feeling that memory is available
 - Retrievable memory
 - Maintenance recoding

Code :

- (1) (a) and (b) only (2) (a) and (c) only
(3) (b) and (d) only (4) (c) and (d) only
3. Read each of the following two statements - Assertion (A) and Reason (R); and indicate your answer using code given below :
- Assertion (A) :** Conditioning done using partial reinforcement schedule decreases resistance to change.
- Reason (R) :** Partial reinforcement schedules influence rate of responding.
- Code :**
- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
4. Arrange the clinical scales of MMPI - II in correct sequence.
- (1) Depression Hypochondriasis Hysteria Psychopathic Deviate Masculinity-Femininity Paranoia Psychasthenia Schizophrenia Hypomania Social Introversion
 - (2) Hypochondriasis Depression Hysteria Psychopathic Deviate Masculinity-Femininity Paranoia Psychasthenia Schizophrenia Hypomania Social Introversion
 - (3) Social Introversion Hypomania Hypochondriasis Depression Hysteria Psychopathic Deviate Masculinity-Femininity Paranoia Psychasthenia Schizophrenia

- (4) Social introversion Hypomania
Schizophrenia Psychasthenia
Paranoia Masculinity-Femininity
Psychopathic deviate Hysteria Depression Hypochondriasis
5. Allport defined personality as :
- (1) Response style
 - (2) Social stimulus value
 - (3) Intervening variable
 - (4) Thinking style
6. Read each of the following two statements - Assertion (A) and Reason (R); and indicate your answer using code given below :
- Assertion (A) :** Increase in synaptic strength is due to the simultaneous stimulation of pre - synaptic and post - synaptic neurons.
- Reason (R) :** Long term Potentiation can be induced in the laboratory by stimulating the pre - synaptic neurons with pulses of high frequency of electricity for few seconds.
- Code :**
- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
7. Concepts allow us to identify things without _____.
- (1) Learning (2) Relearning
 - (3) Recognition (4) Remembering
8. Below given are some concepts. Indicate which is the correct sequence in terms of levels of increasing abstraction ?
- Concepts are :
- (a) Negation concept
 - (b) Alternative denial concept
 - (c) Disjunction concept
- Code :**
- (1) (a), (b), (c) (2) (a), (c), (b)
 - (3) (b), (a), (c) (4) (c), (b), (a)
9. Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I
(Description)

List - II
(Concept)

- | | |
|---|------------------------|
| (a) Transformation of the order of items-transfer | (i) Non - specific |
| (b) Transformation in contents of items | (ii) Learning to learn |
| (c) Use of coding devices and mediators | (iii) Coding |
| (d) Learning how to perform the task | (iv) Clustering |

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|-------|
| (1) | (iii) | (iv) | (i) | (ii) |
| (2) | (iv) | (iii) | (ii) | (i) |
| (3) | (ii) | (i) | (iv) | (iii) |
| (4) | (i) | (ii) | (iii) | (iv) |

10. Binocular cues depend on :
- (a) Combination of the images in the left and right retinas
 - (b) The way two eyes work together
 - (c) Image in either right or left eye
 - (d) Difference between the images in the two eyes
- Code :**
- (1) (a), (b) and (d) only
 - (2) (a), (c) and (d) only
 - (3) (c), (a) and (b) only
 - (4) (b), (c) and (d) only
11. The correct order of the stages of Master's and Johnson's sexual response cycle is :
- (1) plateau, excitement, orgasm, resolution
 - (2) excitement, plateau, orgasm, resolution
 - (3) excitement, orgasm, plateau, resolution
 - (4) excitement, plateau, orgasm, refractory
12. Where is Broca's area located ?
- (1) Frontal Lobe
 - (2) Occipital Lobe
 - (3) Parietal Lobe
 - (4) Temporal Lobe

13. Read each of the following two statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Generally, performance is optimal when arousal is low.

Reason (R) : On easy tasks performance becomes optimal.

Code :

- (1) Both (A) and (R) are true, (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
14. Motivation is best understood as a state that :
 (1) Reduces a drive
 (2) Aims at satisfying a biological need
 (3) Energises an organism to act
 (4) Energises and directs behaviour
15. Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I (Concept)	List - II (Phenomenon)
(a) Functional fixedness hypothesis	(i) A kind of set that would make the images of all material almost similar
(b) Functional equivalence hypothesis	(ii) Mental manipulation performed on images of objects made to be similar to those that would be performed on actual objects
(c) Functional manipulative hypothesis	(iii) A kind of set that would point thoughts in the wrong direction
(d) Functional object imagery	(iv) Mental manipulation performed on images of objects which are very similar to those that would be performed on actual objects

Codes :

	(a)	(b)	(c)	(d)
(1)	(i)	(ii)	(iii)	(iv)
(2)	(ii)	(iii)	(iv)	(i)
(3)	(iii)	(iv)	(ii)	(i)
(4)	(iv)	(iii)	(i)	(ii)

16. Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I (Concept)	List - II (Explanation)
(a) Procedural memory	(i) Memory for facts, words, dates, names of objects
(b) Declarative memory	(ii) Combining several letters, words, phrases into one "stack" of information
(c) Episodic memory	(iii) Memory for typing, driving, riding a bicycle.
(d) Recoding	(iv) Memory about "what", "where" and "when" of our lives

Codes :

	(a)	(b)	(c)	(d)
(1)	(i)	(ii)	(iii)	(iv)
(2)	(iii)	(i)	(iv)	(ii)
(3)	(iv)	(iii)	(ii)	(i)
(4)	(ii)	(iv)	(iii)	(i)

17. Purposive manipulation of reinforcement contingencies, which encourage desirable behaviour, is called :
 (1) Contingency management
 (2) Contingency contracting
 (3) Contingent reinforcement
 (4) Non-contingent reinforcement
18. Which of the following depth cue is based on muscular feedback ?
 (1) Texture gradients
 (2) Motion parallax
 (3) Accommodation
 (4) Light and Shadow

19. Nomothetic approach in personality research was mainly emphasised by :
(a) Allport (b) Skinner
(c) Cattell (d) Eysenk
Code :
(1) (a) and (c) only
(2) (a) and (b) only
(3) (b) and (c) only
(4) (c) and (d) only
20. Read each of the following two statements - Assertion (A) and Reason (R); and indicate your answer using code given below :
Assertion (A) : Spearman proposed the existence of specific factor of intelligence.
Reason (R) : It was observed that tetrad differences are always equal to zero.
Code :
(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.
21. Which theory of emotion implies that every emotion is associated with a unique physiological reaction ?
(1) James - Lange theory
(2) Cannon - Bard theory
(3) Two factor theory
(4) Opponent - process theory
22. Read each of the following two statements - Assertion (A) and Reason (R); and indicate your answer using code given below :
Assertion (A) : Some people perceive wolf as dog.
Reason (R) : External features in a category of concepts are associated with each other.
Code :
(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.
23. The frequency with which an operant response occurs before it is systematically reinforced, is called :
(1) Rate of responding
(2) Operant level
(3) Orienting reflex
(4) Autoclitic behaviour
24. An important factor which enables one to adapt to inverted vision is :
(1) Orientation
(2) Active movement
(3) New learning
(4) Frame of reference
25. _____ defined intelligence as the ability to 'see relations and to deduce correlates'.
(1) Binet (2) Spearman
(3) Terman (4) Boring
26. Which of the following statements are true about Eysenck Personality Questionnaire-Revised (EPQ - R) and the NEO - Personality Inventory - 3 (NEO - PI - 3) ?
(a) Both EPQ - R and NEO - PI - 3 have Extraversion and Neuroticism scales.
(b) Eysenckian Psychoticism scale correlates positively with Agreeableness, and negatively with conscientiousness scales of NEO - PI - 3.
(c) Eysenckian Psychoticism scale correlates negatively with Agreeableness and conscientiousness scales of NEO - PI - 3.
(d) Only EPQ - R has lie scale; NEO - PI - 3 does not have lie scale.
Code :
(1) (a) and (b) only
(2) (a) and (c) only
(3) (a), (b) and (c) only
(4) (a), (c) and (d) only
27. Afferent and Efferent nerves are found in which of the following ?
(1) Sympathetic Nervous System
(2) Para - Sympathetic Nervous System
(3) Somatic Nervous System
(4) Central Nervous System

28. Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I (Type of Test)	List - II (Example)
(a) Forced choice personality inventory	(i) Minnesota Multiphasic Personality Inventory (MMPI - 2)
(b) Ipsative Scale	(ii) Rosenzweig Picture Frustration Study
(c) Personality inventory with a lie scale	(iii) Edwards Personal Preference Schedule (EPPS)
(d) Projective test	(iv) Allport - Vernon - Lindzey's Study of Values

Codes :

(a)	(b)	(c)	(d)
(1) (iv)	(iii)	(ii)	(i)
(2) (iii)	(iv)	(i)	(ii)
(3) (iii)	(i)	(iv)	(ii)
(4) (iv)	(ii)	(iii)	(i)

29. Which is the correct sequence of parts of ear through which sound travels in the ear ?

- (1) Tympanic membrane Stapes Incus Malleus Cochlea
- (2) Tympanic membrane Incus Stapes Malleus Cochlea
- (3) Malleus Incus Stapes Cochlea Tympanic membrane
- (4) Tympanic membrane Malleus Incus Stapes Cochlea

30. The 1986 Symposium on intelligence identified the main loci of intelligence :

- (a) Intelligence within the individual
- (b) Intelligence within the environment
- (c) Intelligence within the interaction between the individual and environment
- (d) Intelligence to be measured by psychometric measures

Code :

- (1) (a) and (b) only (2) (c) and (d) only
- (3) (a), (b) and (c) only
- (4) (b), (c) and (d) only

31. Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I (Terms)	List - II (Explanation)
(a) Anorexia nervosa	(i) Sudden death occurs as terrified person's parasympathetic nervous system over reacts due to which heart slows down to stop
(b) Bulimia nervosa	(ii) A resting period after orgasm, during which a person cannot achieve another orgasm
(c) Refractory period	(iii) An eating disorder in which a person significantly becomes underweight.
(d) Voodoo Curse	(iv) An eating disorder marked by over eating.

Codes :

(a)	(b)	(c)	(d)
(1) (i)	(ii)	(iii)	(iv)
(2) (ii)	(iii)	(iv)	(i)
(3) (iii)	(iv)	(i)	(ii)
(4) (iii)	(iv)	(ii)	(i)

32. The most commonly overlooked and obvious reason for forgetting is :

- (1) Encoding specificity
- (2) Encoding failure
- (3) Lack of sleep
- (4) Faulty eating habits

33. Which of the following sequence is correctly arranged ?

- (1) Incoming information Rehearsal buffer Selective attention
- (2) Rehearsal Buffer Selective attention Incoming information
- (3) Selective attention Rehearsal Buffer Incoming information
- (4) Incoming information Selective attention Rehearsal buffer

34. Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I (Concept)	List - II (Explanation)
(a) Inattentional blindness	(i) Medium point of one's personal frame of reference
(b) Habituation	(ii) Ingrained patterns of organisation and attention
(c) Adaption level	(iii) Failure to perceive a stimulus that is in plain view, but not the focus of attention
(d) Perceptual habits	(iv) Tendency to respond less to predictable and unchanging stimuli

Codes :

(a)	(b)	(c)	(d)
(1) (iii)	(iv)	(i)	(ii)
(2) (ii)	(i)	(iii)	(iv)
(3) (iii)	(ii)	(iv)	(i)
(4) (iv)	(iii)	(ii)	(i)

35. Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I (Tests)	List - II (Authors)
(a) Cognitive Assessment Battery	(i) Cattell, RB
(b) Deviation IQ	(ii) Terman
(c) Cognitive Assessment System	(iii) Wechsler
(d) Stanford - Binet Scales	(iv) Naglieri and Das

Codes :

(a)	(b)	(c)	(d)
(1) (i)	(ii)	(iii)	(iv)
(2) (ii)	(iii)	(iv)	(i)
(3) (iv)	(iii)	(i)	(ii)
(4) (i)	(iii)	(iv)	(ii)

36. Read each of the following two statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Some persons engage in activities such as collecting things, being good listeners, or being gullible persons.

Reason (R) : In Freudian psychoanalysis, it is assumed that fixation at oral stage results in the development of oral incorporative character.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

37. A researcher wants to scale nine attitude items by the method of paired comparison. What would be the number of pairs of the attitude items ?

- (1) 81
- (2) 72
- (3) 36
- (4) 9

38. Which of the following are regulated by Sympathetic Nervous System ?

- (a) Dilation of pupils
- (b) Decreased activity of Adrenal Gland
- (c) Digestion is stimulated
- (d) Increased Heart Rate
- (e) Dryness in mouth

Code :

- (1) (a), (b) and (c) only
- (2) (b) and (c) only
- (3) (a), (d) and (e) only
- (4) (a), (c), (d) and (e) only

39. Which of the following can cause errors in reasoning ?

- (a) Premise phrasing manner
- (b) Foreclosure effects
- (c) Confirmation bias

Code :

- (1) (a) and (b)
- (2) (a) and (c)
- (3) (b) and (c)
- (4) (a), (b) and (c)

40. Read each of the following two statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : The names of objects, days of the week, the seasons and other general facts are all quite lasting and we do not forget them.

Reason (R) : Most of our basic factual knowledge about the world is almost totally immune to forgetting.

Code :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
41. According to Skinner, which of the following are the characteristics of 'Respondent Behaviour' ?
- (a) It is elicited. (b) It is emitted.
 - (c) It is conditioned
 - (d) It is unconditioned.

Code :

- (1) (a) and (c) only
 - (2) (b) and (c) only
 - (3) (a), (c) and (d) only
 - (4) (b), (c) and (d) only
42. Read each of the following two statements - Assertion (A) and Reason (R); and indicate your answer using code given below :
- Assertion (A) :** To use a computer, we must learn to pay attention to specific stimuli, such as icons, commands and signals.
- Reason (R) :** Learning creates perceptual habits that affect our daily experience.
- Code :**
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
43. Arrange Thurston's primary abilities in correct sequence :

- (1) Verbal Comprehension; Verbal Fluency; Numerical Ability; Memory; Perceptual Speed; Inductive Reasoning, Spatial Visualization
- (2) Verbal Fluency; Verbal Comprehension; Numerical Ability; Memory; Perceptual Speed; Spatial Visualization; Inductive Reasoning
- (3) Numerical Ability; Verbal Comprehension; Verbal Fluency; Memory; Inductive Reasoning; Spatial Visualization; Perceptual Speed
- (4) Spatial Visualization; Inductive Reasoning; Memory; Perceptual Speed; Numerical Ability; Verbal Fluency; Verbal Comprehension

44. Read each of the following two statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : According to the Classical Test Theory (CTT), the true score component and the error component of the observed score are assumed to be uncorrelated.

Reason (R) : According to the CTT, the error component is assumed to be distributed with a mean of zero.

Code :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
45. Arish, on a Wechsler type of scale, has a DIQ (deviation IQ) of 85. His percentile rank would be :
- | | |
|--------|--------|
| (1) 16 | (2) 34 |
| (3) 68 | (4) 85 |

Instructions for questions 46 to 50.

Read the following paragraph and answer the five questions which follow :

A psychologist, working in the area of Learning and Memory hypothesized that a particular drug would adversely affect the learning rate. One hundred and fifty subjects, 75 boys and 75 girls, studying in the first year of a degree college, in

the age range of 18 to 20 years, were available for the study. It was ensured that all the subjects were in the DIQ (deviation IQ) range of 85 to 115. Three groups - A, B and C - of equal size were formed by random assignment. Group A received high dosage of drug, Group B received moderate dosage of drug, and Group C received only placebo. A list of 15 CVC trigrams, each having an association value in the range of 50 to 70 was already prepared. Each subject was required to learn the list upto two errorless trials using method of anticipation on a memory drum. The mean trials required to learn the list were 12, 15 and 20 respectively. The intergroup mean differences were statistically significant at .01 level.

46. Which one of the following conclusions can be drawn on the basis of the above study ?

- (1) The researcher's hypothesis has been verified.
- (2) The researcher's hypothesis has been rejected.
- (3) The researcher's hypothesis has been partially accepted.
- (4) Inadequate data to evaluate the hypothesis.

47. In the above study, 'association value' can be labelled as :

- (a) Continuous variable
- (b) Controlled variable
- (c) Organismic variable
- (d) Stimulus variable

Code :

- (1) (a) and (b) only
- (2) (a) and (c) only
- (3) (a), (b) and (c) only
- (4) (a), (b) and (d) only

48. Read each of the following two statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : In the above study, the subjects have been randomly assigned to the three groups.

Reason (R) : The independent variable is manipulated more effectively with random assignment of subjects.

Code :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

49. Match List - I with List - II in the context of above study and select the correct answer by choosing from the code given below :

List - I

List - II

- | | |
|--------------------------|--|
| (a) Independent variable | (i) Intelligence |
| (b) Dependent variable | (ii) Drug dosage |
| (c) Controlled variable | (iii) Gender |
| (d) Discrete variable | (iv) Trials required to learn the list |

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|------|------|-------|-------|
| (1) | (iv) | (ii) | (iii) | (i) |
| (2) | (iv) | (ii) | (i) | (iii) |
| (3) | (ii) | (iv) | (iii) | (i) |
| (4) | (ii) | (iv) | (i) | (iii) |

50. In the above study, the variance caused by Drug dosage can be labelled as :

- (1) Primary variance
- (2) Secondary variance
- (3) Controlled variance
- (4) Error variance

Answer Sheet

1. (2)	2. (1)	3. (4)	4. (2)	5. (3)	6. (1)	7. (2)	8. (1)	9. (2)	10. (1)
11. (2)	12. (1)	13. (1)	14. (4)	15. (4)	16. (2)	17. (1)	18. (3)	19. (4)	20. (3)
21. (1)	22. (1)	23. (2)	24. (2)	25. (2)	26. (4)	27. (3)	28. (2)	29. (4)	30. (3)
31. (4)	32. (2)	33. (4)	34. (1)	35. (4)	36. (1)	37. (3)	38. (3)	39. (4)	40. (1)
41. (3)	42. (2)	43. (1)	44. (2)	45. (1)	46. (2)	47. (4)	48. (3)	49. (4)	50. (1)

November, 2017

PSYCHOLOGY

PAPER-III

Note : This paper contains **seventy five (75)** objective type questions of **two (2)** marks each. **All** questions are compulsory.

- The act in which people help others without any personal interest, motive, reason is called :
 - Altruism
 - Nurturance
 - Prosocial Behaviour
 - Social Obligation
- Read each of the following two statements - Assertion (A) and Reason (R) ; and indicate your answer using code given below :
Assertion (A) : Mean level of fluid intelligence (gf) is higher in young adults than older adults.
Reason (R) : Crystallized intelligence increases throughout life span.
Code :
 - Both (A) and (R) are true, and (R) is the correct explanation of (A).
 - Both (A) and (R) are true; but (R) is not the correct explanation of (A).
 - (A) is true, but (R) is false.
 - (A) is false, but (R) is true.
- The multiple correlation ranges between :
 - 0.50 to +0.50
 - 1.00 to +1.00
 - Zero to 1.00
 - There is a fixed range for bivariate correlation, but not for multiple correlation
- Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I (Concept) (a) Viewer - centered representation	List - II (Explanation) (i) The individual stores a representation of the object, independent of its appearance
---	---
- | | |
|---|--|
| (b) Structuralist view of perception

(c) Object - centered representation

(d) Constructivist view of perception | (ii) The individual builds a cognitive understanding of the object

(iii) The individual stores the way, the object looks to him or her

(iv) Simple sensations constitute the building blocks of perceived form |
|---|--|
- Codes :**

(a)	(b)	(c)	(d)
(1) (i)	(ii)	(iii)	(iv)
(2) (iii)	(iv)	(i)	(ii)
(3) (ii)	(i)	(iv)	(iii)
(4) (iv)	(iii)	(ii)	(i)
- According to Gibson's theory of perception :
 - We directly perceive the environment. We do not need the aid of complex thought processes.
 - We do not directly perceive the environment. We need the aid of complex thought processes.
 - We perceive the environment using our intelligence.
 - We do not connect what we perceive, to what we have stored in our mind.
- According to Bandura, which of the following is the correct sequence of processes involved in observation learning ?

- (1) Attention Retention Cognitive Rehearsal Corrective Action
- (2) Attention Cognitive Rehearsal Corrective Action Retention
- (3) Attention Corrective Action Cognitive Rehearsal Retention
- (4) Corrective Action Attention Cognitive Rehearsal Retention
7. Read each of the following two statements - Assertion (A) and Reason (R) ; and indicate your answer using code given below :
- Assertion (A) :** Lewin asserted that avoidance-avoidance conflicts are comparatively more stable.
- Reason (R) :** Opposing forces tend to remain in state of equilibrium.
- Code :**
- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.
8. Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I
(Concept)

- (a) Intrinsic motivation
- (b) Achievement motivation
- (c) Self - handi-capping

List - II
(Description)

- (i) "I am really not feeling too good today. I may not be able to perform well".
- (ii) "I love to drive very fast, no matter what happens".
- (iii) "For me, accomplishing difficult tasks and meeting high standards of excellence is the top most priority".

- (d) Sensation-seeking

- (iv) "I go for swimming everyday. I just love it. I take out time from my tight study schedule".

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|-------|
| (1) | (i) | (ii) | (iii) | (iv) |
| (2) | (ii) | (i) | (iv) | (iii) |
| (3) | (iv) | (iii) | (i) | (ii) |
| (4) | (iii) | (iv) | (ii) | (i) |

9. Identity status of individuals who are exploring alternatives in an effort to find values and goals to guide their life is called :
- (1) Identity Diffusion
- (2) Identity Foreclosure
- (3) Moratorium
- (4) Identity Confusion
10. Sex chromosome anomaly (XXY) results into the development of :
- (1) Klinefelter's Syndrome
- (2) Down Syndrome
- (3) Turner's Syndrome
- (4) Phenylketonuria
11. Which of the following is associated with memory for emotionally charged events ?
- (a) Thalamus (b) Hypothalamus
- (c) Amygdala (d) Temporal lobe
- Code :**
- (1) (a) and (b) only (2) (c) only
- (3) (b) and (d) only (4) (d) only
12. "Development is the result of Individual's active participation in the development process in interaction with important environmental influences". This is the assumption of which of the following approaches ?
- (1) Maturation / Biological
- (2) Psychodynamic
- (3) Behavioural
- (4) Cognitive
13. A person, who reports that he oftenly feels above his physical body. It is an example of :

- (1) Derealization
(2) Depersonalization
(3) Delusion
(4) Identity Disorientation
14. Humanistic theories of personality are characterised as :
- (a) Having optimistic ideas about human nature
(b) Emphasize personal responsibility
(c) Focus on present than on past
(d) Believe that fate is an important determinant of human behaviour
- Code :**
- (1) (a) and (b) only
(2) (c) and (d) only
(3) (a), (b) and (c) only
(4) (b), (c) and (d) only
15. The following statements represent the work of Noam Chomsky, choose the correct one from the code given below :
- (a) He called himself a Cartesian linguistic
(b) He analyzed syntax
(c) He wrote about transformational grammar
(d) He developed an artificial language called 'Yerkish'
- Code :**
- (1) (a), (b) and (c) (2) (a), (b) and (d)
(3) (b), (c) and (d) (4) (a), (c) and (d)
16. Which of the following is the evidence that REM sleep specifically enhances consolidation ?
- (a) REM increases after learning
(b) REM deprivation interferes with learning
(c) Performance improves following REM sleep
- Code :**
- (1) (a) and (b) only (2) (a) and (c) only
(3) (b) and (c) only (4) (a), (b) and (c)
17. Read each of the following two statements - Assertion (A) and Reason (R) ; and indicate your answer using code given below.
- Assertion (A) :** Managers make judgements about employee's dispositions in various situations.
- Reason (R) :** Inferences are drawn on the basis of observations.
- Code :**
- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.
18. Arrange the following correlation coefficients in descending order in terms of their strength of relationship, as defined by common variance :
- (1) -0.8 0.7 -0.5 0.4
(2) 0.7 0.4 -0.5 -0.8
(3) 0.4 -0.5 0.7 -0.8
(4) -0.8 -0.5 0.4 0.7
19. A mother is telling her child to eat vegetables before getting dessert. It is an example of :
- (1) Reward - Contrast Principle
(2) Delay Conditioning
(3) Premack Principle
(4) Crespi Effect
20. Read each of the following two statements - Assertion (A) and Reason (R) ; and indicate your answer using code given below :
- Assertion (A) :** Teacher's motivating style determines student's motivation for reading
- Reason (R) :** To enhance student's interest in reading, teacher may offer a reward for reading along with helpful feedback.
- Code :**
- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.
21. The levels of processing proposed by Craik and Tulving follow the specific sequence in

progressive order of depth ; choose the correct.

- (1) Acoustic Semantic Physical
- (2) Physical Acoustic Semantic
- (3) Semantic Physical Acoustic
- (4) Acoustic Physical Semantic

22. DSM - III is considered qualitatively different and superior to its predecessors, because :

- (a) It is multiaxial
- (b) It deleted misnomer words
- (c) It introduced coding system
- (d) It used experimental data

Code :

- (1) (a), (b) and (c) only
- (2) (a), (c) and (d) only
- (3) (b), (c) and (d) only
- (4) (a), (b), (c) and (d)

23. According to Jensen what percent (%) of variance in intelligence is explained by genetic factors :

- (1) 20 to 30% (2) 55 to 65%
- (3) 80 to 90% (4) 40 to 50%

24. Read each of the following two statements - Assertion (A) and Reason (R) ; and indicate your answer using code given below :

Assertion (A) : Westerners tend to focus mostly on the figure when viewing a scene and much less on the ground whereas East Asians pay greater attention to the overall context.

Reason (R) : Culture affects the way we direct our attention to notice or ignore things around us.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

25. Read each of the following two statements - Assertion (A) and Reason (R) ; and indicate your answer using code given below :

Assertion (A) : Autoshaaping involves S - S learning.

Reason (R) : Some behaviours are innate responses to a stimulus and because of temporal contiguity between them, they acquire stimulus related properties.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

26. Emotions can be characterised as :

- (a) Giving rise to affective experiences
- (b) Stimulating generation of cognitive explanations
- (c) Triggering a variety of internal neurobiological changes
- (d) Always eliciting rewarding behaviour

Code :

- (1) (a), (b) and (c) (2) (a), (c) and (d)
- (3) (b), (c) and (d) (4) (a), (b), (c) and (d)

27. Read each of the following two statements - Assertion (A) and Reason (R) ; and indicate your answer using code given below :

Assertion (A) : Every cigarette packet has a verbal and pictorial warning about negative consequences of smoking.

Reason (R) : Inconsistent cognitions have the motivational properties.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

28. When students study and practice hard with interest and enjoyment, it is called :

- (1) Emotional engagement
- (2) Cognitive engagement
- (3) Behavioral engagement
- (4) Healthy adjustment

29. If the teacher presents the extrinsic motivator in a controlled way, then
- (1) It will enhance competence of students
 - (2) It will interfere with autonomy of students
 - (3) It will enhance their overall learning
 - (4) It will enhance the autonomy of students
30. According to the Connectionist view, memories are :
- (1) Abstract concepts
 - (2) Knowledge structures
 - (3) Interconnections among neurons
 - (4) Schemas
31. Read each of the following two statements - Assertion (A) and Reason (R) ; and indicate your answer using code given below :
- Assertion (A) :** The dynamic process of parent - child interaction during Phallic stage leads to the solidification of Ego.
- Reason (R) :** Child's successful resolution of Oedipus Complex leads to appropriate gender role identification.
- Code :**
- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
32. A patient in Mental Hospital somehow procured a needle and thread, and while sleeping, she tried to stitch her lips together, so that someone does not pour poison in her mouth when asleep. Most probably she suffers from :
- (1) Bipolar Disorder
 - (2) Hebephrenic Schizophrenia
 - (3) Delusional Disorder
 - (4) Borderline Personality Disorder
33. John, the chess master selects his next move by considering the moves that it would threaten the opponent's queen whilst his opponent, a chess playing computer, selects its next move by considering all possible moves John is using _____ whilst the computer is using _____.
- (1) Algorithm ; heuristic
 - (2) Prototype ; mental set
 - (3) Mental set ; prototype
 - (4) Heuristic ; algorithm
34. Today social psychology studies :
- (a) Social cognition
 - (b) Ethnic Diversity
 - (c) Multiculturalism
- Code :**
- (1) (a) and (b)
 - (2) (a) and (c)
 - (3) (b) and (c)
 - (4) (a), (b) and (c)
35. Read each of the following two statements - Assertion (A) and Reason (R) ; and indicate your answer using code given below :
- Assertion (A) :** People perform in accordance to the social positions they hold.
- Reason (R) :** People select relevant information from tremendous amount of social information.
- Code :**
- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
36. The sequence of different dimensions of Job Characteristics Model are :
- (1) Skill variety, task significance, autonomy, task identity, feedback
 - (2) Skill variety, task identity, task significance, autonomy, feedback
 - (3) Task identity, task significance, autonomy, skill variety, feedback
 - (4) Autonomy, task identity, skill variety, task significance, feedback
37. Fill in the blank by selecting appropriate alternative. Harlow's classic studies with baby monkeys and wire or cloth artificial mothers have demonstrated the importance of _____ in the early development.

- (1) Regular feeding
(2) Classical conditioning
(3) Freedom to explore
(4) Physical contact
38. A person with Schizophrenia was found to repeat whatever words the psychiatric social worker was speaking. This behaviour can be labelled as :
- (1) Echopraxia (2) Neologism
(3) Word Salad (4) Echolalia
39. Which of the following are not Factor Analytically derived personality inventories ?
- (a) 16 PF
(b) Comrey Personality Scales
(c) Jenkins Activity Survey
(d) Edwards Personal Preference Schedule
- Code :**
- (1) (a) and (b) only
(2) (c) and (d) only
(3) (a), (b) and (c) only
(4) (b), (c) and (d) only
40. In which parts of brain the feeling of hunger and full are felt ?
- (1) Hypothalamus ; Cerebral Cortex
(2) Lateral hypothalamus ; Ventromedial hypothalamus
(3) Lateral hypothalamus ; Cerebral Cortex
(4) Cerebral Cortex ; Ventromedial hypothalamus
41. As per Goleman "Motivating Oneself" component of emotional intelligence includes :
- (a) Marshalling emotions in the service of goal
(b) Delaying gratification and stifling impulsiveness
(c) Being able to get into 'flow state'
(d) Monitoring feelings from moment to moment
- Code :**
- (1) (a) and (b) only
(2) (c) and (d) only
(3) (a), (b) and (c) only
(4) (b), (c) and (d) only
42. Read each of the following two statements - Assertion (A) and Reason (R) ; and indicate your answer using code given below :
- Assertion (A) :** In Time Series Design, autocorrelation is a routine feature.
Reason (R) : Autocorrelation is efficiently handled by statistical method ARIMA
- Code :**
- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.
43. An example of Feature Analysis is :
- (1) Phonics (2) Words
(3) Context of words
(4) Meaning of words
44. Read each of the following two statements - Assertion (A) and Reason (R) ; and indicate your answer using code given below :
- Assertion (A) :** According to Dollard and Miller, observational learning will occur only if organism's overt behaviour is followed by reinforcement.
Reason (R) : Copying behaviour involves guiding of one person's behaviour by another person.
- Code :**
- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.
45. Which of the following is the final destination for much of the brain's information about emotion, before action is taken ?
- (1) Amygdale
(2) Anterior Cingulate Cortex
(3) Pre - frontal cortex
(4) Hypothalamus

46. Read each of the following two statements - Assertion (A) and Reason (R) ; and indicate your answer using code given below :

Assertion (A) : You can remember vividly what you wore on your first date.

Reason (R) : Memories of emotionally significant events cannot be accurate at all.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

47. Read each of the following two statements - Assertion (A) and Reason (R) ; and indicate your answer using code given below :

Assertion (A) : Cattell incorporated factors, Q_1 , Q_2 , Q_3 , and Q_4 in 16 PF

Reason (R) : Some factors appeared only in L - data, not in Q - data

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

48. Read each of the following two statements - Assertion (A) and Reason (R) ; and indicate your answer using code given below :

Assertion (A) : Naturalistic decision making involves decisions as they occur in every day life

Reason (R) : The decisions required in many situations often have serious ramifications for those who make decisions.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

49. Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I

(Phenomenon)

List - II

(Explanation)

- | | |
|-------------------------|--|
| (a) Over - extension | (i) The use of single word to represent whole phrase or sentence |
| (b) Under-extension | (ii) The tendency to apply a word to more objects or actions than it actually represents |
| (c) Holophrastic speech | (iii) Speech marked by reliance on nouns and verbs, while omitting other parts of speech including articles and propositions |
| (d) Telegraphic speech | (iv) The tendency to apply a word to fewer objects or actions than it actually represents |

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|------|-------|------|-------|
| (1) | (i) | (iii) | (ii) | (iv) |
| (2) | (ii) | (iv) | (i) | (iii) |
| (3) | (i) | (iv) | (ii) | (iii) |
| (4) | (ii) | (iii) | (i) | (iv) |

50. Which part of Central Nervous System controls much of the Endocrine System by regulating the Pituitary Gland ?

- (1) Amygdala
- (2) Hypothalamus
- (3) Hippocampus
- (4) Medulla

51. Cognitive structures that represent an organised collection of knowledge about people, events and concepts are called :

- (1) Images
- (2) Schemas
- (3) Social influence
- (4) Disposition

52. Which is not the part of Equity Theory ?

- (1) Over payment

- (2) Under payment
- (3) Linear payment
- (4) Equitable payment

53. Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I (Dimensions)	List - II (Behaviour)
(a) Delegating	(i) High task and high relationship
(b) Participating	(ii) High task and low relationship
(c) Selling	(iii) Low relationship and low task
(d) Telling	(iv) High relationship and low task

Codes :

- | (a) | (b) | (c) | (d) |
|-----------|-------|-------|------|
| (1) (i) | (ii) | (iii) | (iv) |
| (2) (ii) | (iii) | (iv) | (i) |
| (3) (iii) | (iv) | (ii) | (i) |
| (4) (iv) | (iii) | (ii) | (i) |

54. Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I (Concept)	List - II (Description)
(a) Successful Intelligence	(i) Ability to cope with the problems of day-to-day life
(b) Practical Intelligence	(ii) Ability to achieve one's goals of life across given cultural contexts, capitalizing on strengths and correcting weaknesses to adapt with the environment
(c) Mating Intelligence	(iii) Ability to cope with novel problems and situations
(d) Creative Intelligence	(iv) Ability to understand the outcomes of mind's reproductive system

Codes :

- | (a) | (b) | (c) | (d) |
|-----------|-------|-------|-------|
| (1) (i) | (ii) | (iii) | (iv) |
| (2) (ii) | (i) | (iv) | (iii) |
| (3) (iii) | (iv) | (ii) | (i) |
| (4) (iv) | (iii) | (i) | (ii) |

55. How many 'three way interactions' can be studied in a $4 \times 3 \times 3$ factorial design ?

- (1) 1 only
- (2) 4 only
- (3) 12 only
- (4) 3 only

56. In a verbal memory experiment, the two independent variables, 'meaningfulness of material' and 'retention interval' were employed, each varied at three levels. The number of words correctly recalled was the dependent variable. The sum of squares due to meaningfulness of material is 270. The Mean square for the meaningfulness of material would be :

- (1) 30.0
- (2) 67.5
- (3) 90.0
- (4) 135.0

57. In 'Signal Detection Theory', which of the following determinants are assumed ?

- (a) Observer's sensitivity
- (b) Sensory threshold
- (c) Response bias
- (d) Decision bias

Code :

- (1) (a) only
- (2) (a) and (c) only
- (3) (a), (c) and (d) only
- (4) (b) and (c) only

58. According to Tolman, which among the following are the "Individual Differences" variables which interact with independent variables to produce behaviour :

- (a) Age
- (b) Heredity
- (c) Previous Training
- (d) Special Endocrine, Drug or Vitamin conditions

Code :

- (1) (a) and (c) only
- (2) (a), (c) and (d) only
- (3) (b), (c) and (d) only
- (4) (a), (b), (c) and (d)

59. Which of the following theories represent the mechanistic metaphor of motivation ?

- (a) Psychoanalytic
- (b) Ethological
- (c) Sociobiological
- (d) Expectancy - value

Code :

- (1) (a), (b) and (c) only
- (2) (b), (c) and (d) only
- (3) (a), (c) and (d) only
- (4) (a), (b), (c) and (d)

60. According to Meichenbaum (1977), the steps involved in cognitive behaviour modification are :

- (1) Covert self instruction overt external guidance overt self guidance cognitive modeling faded overt self-guidance
- (2) Cognitive modeling faded overt self-guidance overt external guidance Covert self instruction overt self-guidance
- (3) Covert self instruction faded overt self - guidance overt external guidance Cognitive modeling overt self - guidance
- (4) Cognitive modeling overt external guidance overt self - guidance faded overt self - guidance Covert self-instruction

61. Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I

(Concept)

- (a) Interactive Images

List - II

(Description)

- (i) Visualizing walking around an area with distinctive landmarks that you know well and link them
- (ii) "one is a bun" "Two is a shoe"

- (c) Method of Loci

- (d) Pegword System

(iii) If you have to remember to buy socks, apples and a pair of scissors, you might imagine cutting a socks that has an apple stuffed in it with a pair of scissors

(iv) Devising a word or expression in which each of its letters stands for a certain other word or concept

Codes :

- | | (a) | (b) | (c) | (d) |
|-----|-------|-------|-------|------|
| (1) | (i) | (ii) | (iii) | (iv) |
| (2) | (iii) | (iv) | (i) | (ii) |
| (3) | (ii) | (iii) | (iv) | (i) |
| (4) | (iv) | (iii) | (ii) | (i) |

62. Which of the following concept, according to which each mental event has a unique time of ascendancy, the plan for which is contained in the organism's gene ?

- (1) Epigenesis (2) Maturation
- (3) Genotype (4) Phenotype

63. Rollo May's description of personality development can be explained in which of following sequences :

- (1) Innocence Dependence Rebellion
Ordinary Consciousness of self
Creative Consciousness of self
- (2) Dependence Innocence Rebellion
Ordinary Consciousness of self
Creative Consciousness of self
- (3) Innocence Rebellion Dependence
Ordinary Consciousness of self
Creative Consciousness of self
- (4) Dependence Innocence Rebellion
Creative Consciousness of Self
Ordinary Consciousness of self

64. Match List - I with List - II and select the correct answer by choosing from the code given below :

**List - I
(Test)**

- (a) Personal Orientation Inventory
(b) Q - sort
(c) House - Tree - Person Test
(d) Objective Analytic Tests Battery

Codes :

- | (a) | (b) | (c) | (d) | (a) | (b) | (c) | (d) |
|-----|-------|------|-------|------|-----|------|-------|
| (1) | (i) | (ii) | (iii) | (iv) | (2) | (ii) | (i) |
| (3) | (iii) | (iv) | (ii) | (i) | (4) | (iv) | (iii) |

65. Because of the lightning speed, computers can retrieve and manipulate stored data faster than people, but the human brain beats the computer hands down when it comes to :
- (1) Using heuristics
(2) Following algorithms
(3) Serial processing
(4) Simultaneous processing
66. Which of the following is the sleep disorder in which the person is physically active during REM sleep ?
- (1) Narcolepsy (2) Cataplexy
(3) Somnambulism
(4) REM Sleep Behaviour Disorder
67. Match List - I with List - II and select the correct answer by choosing from the code given below :

**List - I
(Technique)**

- (a) CT
(b) MRI
(c) PET
(d) fMRI

**List - II
(Description)**

- (i) Detects increase in oxygen levels during neural activity
(ii) Measures variations in hydrogen concentration in brain tissue
(iii) Forms 3 - D image of brain by combining X - rays of cross sections of brain
(iv) Image produced by emission from injected

substances that have been made radioactive

Codes :

- | (a) | (b) | (c) | (d) | (a) | (b) | (c) | (d) |
|-----|------|------|-------|------|-----|-------|------|
| (1) | (iv) | (i) | (iii) | (ii) | (2) | (iii) | (ii) |
| (3) | (iv) | (ii) | (iii) | (i) | (4) | (iii) | (i) |

68. Match List - I with List - II and select the correct answer by choosing from the code given below :

List - I**(Phenomenon)**

- (a) Just world phenomenon
(b) Foot-in-door phenomenon
(c) Social trap
(d) Mere exposure phenomenon

List - II**(Explanation)**

- (i) Tendency to have first agreed to a small request and to comply with a larger request
(ii) A situation where conflicting parties are being caught for protecting self interest
(iii) The phenomenon where repeatedly facing novel stimuli increases their liking
(iv) The belief that world is just and the people get what they deserve to get and vice - versa

Codes :

- | (a) | (b) | (c) | (d) | (a) | (b) | (c) | (d) |
|-----|-------|------|-------|------|-----|------|-------|
| (1) | (i) | (ii) | (iii) | (iv) | (2) | (ii) | (iii) |
| (3) | (iii) | (ii) | (i) | (iv) | (4) | (iv) | (i) |

69. Organizational behaviour focusses on :

- (a) How to improve productivity
(b) Reduce absenteeism and turnover
(c) Increase employees job satisfaction and commitment
(d) Increase decision making ability

Code :

- | | | | |
|-----|---------------|-----|---------------|
| (1) | (a), (b), (d) | (2) | (b), (c), (d) |
| (3) | (a), (b), (c) | (4) | (a), (c), (d) |

70. Fluid general intelligence (gf) includes the abilities :

- (a) to perceive relationship among stimulus patterns
- (b) to comprehend implications
- (c) to draw inferences from relationships
- (d) to acquire quantitative knowledge

Code :

- (1) (a) and (b) only (2) (c) and (d) only
- (3) (a), (b) and (c) only
- (4) (b), (c) and (d) only

Instructions for Q. 71 to 75

Read the following paragraph and answer the five questions that follow :

A personality psychologist wanted to factor analyze the important personality traits. He selected personality scales of 78 important personality traits, presumed to cover most of the important characteristics. These scales were administered to a sample of eight hundred and fifty respondents. The intercorrelations among the 78 traits were computed and the 78×78 correlation matrix was obtained. Retaining unities in the diagonal spaces of the correlation matrix, unrotated factors were extracted. Cattell's scree plot favored fifteen factor solution.

The fifteen factors together condensed 72 percent of variance. The first factor condensed 18 percent variance. The 15 factors were rotated and the inter-factor correlations were examined. Since some of the inter - factor correlations were substantial, the psychologist decided to carry out the second order factor analysis .

71. Which one of the following method of factor extraction has been, most likely, employed in the above study ?

- (1) Maximum likelihood
- (2) Unweighted least square
- (3) Principal components method
- (4) Principal axis method

72. What percent of variance would be condensed by the second unrotated factor in the above study ?

- (1) 18 percent
- (2) Less than 18 percent
- (3) More than 18 percent
- (4) Inadequate data

73. Cattell's scree plot has :

- (1) Order of eigen value on X - axis and actual eigen value on Y - axis
- (2) Order of eigen value on X - axis and the covariance on Y - axis
- (3) Actual eigen value on X - axis and order of eigen value on Y - axis
- (4) Actual eigen value on X - axis and covariance on Y - axis

74. Out of the following four methods of factor rotation, which method /s has / have, most likely, been used in the above study ?

- (a) Varimax (b) Quartimax
- (c) Promax (d) Oblimin

Code :

- (1) (a) and (b) only (2) (a) and (c) only
- (3) (b) and (c) only (4) (c) and (d) only

75. What percentage of total variance will be explained by the 15 factors after rotation ?

- (1) Less than 72 percent
- (2) 72 percent
- (3) More than 72 percent
- (4) Inadequate data

Answer Sheet

1. (3)	2. (2)	3. (3)	4. (2)	5. (1)	6. (1)	7. (3)	8. (3)	9. (3)	10. (1)
11. (2)	12. (4)	13. (2)	14. (3)	15. (1)	16. (4)	17. (1)	18. (1)	19. (3)	20. (1)
21. (2)	22. (1)	23. (3)	24. (*)	25. (1)	26. (1)	27. (1)	28. (1)	29. (2)	30. (3)
31. (4)	32. (3)	33. (4)	34. (4)	35. (2)	36. (2)	37. (4)	38. (4)	39. (2)	40. (2)
41. (3)	42. (2)	43. (1)	44. (2)	45. (3)	46. (3)	47. (2)	48. (1)	49. (2)	50. (2)
51. (2)	52. (3)	53. (4)	54. (2)	55. (1)	56. (4)	57. (2)	58. (4)	59. (1)	60. (4)
61. (2)	62. (1)	63. (2)	64. (2)	65. (4)	66. (4)	67. (2)	68. (4)	69. (3)	70. (3)
71. (3)	72. (2)	73. (1)	74. (4)	75. (2)					

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PAPER-II

Note : This paper contains **hundred (100)** objective type questions of **two (2)** marks each. **All** questions are **compulsory**.

1. In Erikson's theory, which psycho-social crises precede and succeed Autonomy v/s Shame :

- (1) Basic trust v/s mistrust and Industry v/s Inferiority.
- (2) Initiative v/s Guilt and Industry v/s Inferiority.
- (3) Industry v/s Inferiority and Identity v/s Role confusion.
- (4) Basic trust v/s mistrust and Initiative v/s Guilt.

2. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Horner found high level of motive to avoid success or fear of success in females.

Reason (R) : As a result of success, the threat of social rejection and fear concerning perceived lack of femininity are aroused in women.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

3. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I (Concept)	List - II (Explanation)
(a) Life Space	(i) Unlearned, fixed stereotyped pattern of activity.
(b) Instinctual Behaviour	(ii) Psychological reality depicting the totality of facts that de-

termine behaviour at a moment in time. It counts of environment as perceived by the person.

- | | |
|----------------------|--|
| (c) Vacuum Behaviour | (iii) Tendency to resume the previously unfinished task during intervening period. |
| (d) Task Resumption | (iv) Behaviour patterns that appear when the sign stimulus is not identifiable. |

Codes :

- | | |
|-------------------------|-------------------------|
| (a) (b) (c) (d) | (a) (b) (c) (d) |
| (1) (i) (ii) (iii) (iv) | (2) (ii) (i) (iv) (iii) |
| (3) (iii) (iv) (i) (ii) | (4) (iv) (iii) (ii) (i) |

4. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Perceived fairness of the interpersonal treatment is used to determine organizational outcomes.

Reason (R) : People expect informational transparency and respect in organizations.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

5. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I (Concept)	List - II (Explanation)
(a) Episodic retrieval	(i) Physical environment at retrieval

- matching that at encoding.
- (b) Context-dependent retrieval (ii) Mood at retrieval matching that at encoding.
- (c) State-dependent retrieval (iii) Retrieval stimulated by hints from the external and internal environment.
- (d) Cue-dependent retrieval (iv) Subsequent reactivation of stored memory traces.

Codes :

- | | | | | | | | |
|-----|-------|------|-------|------|-----|------|-------|
| (a) | (b) | (c) | (d) | (a) | (b) | (c) | (d) |
| (1) | (i) | (ii) | (iii) | (iv) | (2) | (ii) | (iv) |
| (3) | (iii) | (iv) | (ii) | (i) | (4) | (iv) | (i) |
| | | | | | | (ii) | (iii) |

6. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Rational persuasion refers to the use of logical arguments and facts to persuade others for achieving a desired result.

Reason (R) : Managers use several techniques to influence others in organizations.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

7. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I
(Concept)

- (a) Counter-conditioning

List - II
(Description)

- (i) After conditioning, a new CS(B) when paired with already conditioned stimulus CS(A) and presented as a compound stimulus, little or no conditioning occurs to CS(B).

- (b) Blocking (ii) Elimination of undesirable behaviour by pairing CS with US other than the one that originally reinforced that behaviour.
- (c) Over-shadowing (iii) Inhibition of conditioned response caused by conditioned emotional response.
- (d) Conditioned Suppression (iv) While using compound CS where one of its components is more salient than the other, only the salient component is conditioned.

Codes :

- | | | | | | | | |
|-----|------|-------|-------|------|-----|------|-------|
| (a) | (b) | (c) | (d) | (a) | (b) | (c) | (d) |
| (1) | (ii) | (i) | (iii) | (iv) | (2) | (ii) | (i) |
| (3) | (iv) | (iii) | (i) | (ii) | (4) | (iv) | (iii) |
| | | | | | | (ii) | (i) |

8. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I (Concept)

- (a) Convergence

List - II (Explanation)

- (i) If you see a distant mountain range on a crystal - clear day, it looks like it was only a few miles away.

- (b) Frame of reference

- (ii) When the two images are fused into one overall image.

- (c) Stereoscopic vision

- (iii) When we look at a distant object, the lines of vision from our eyes are parallel. Looking at a nearby object, the eyes turn in.

- (d) Aerial perspective

- (iv) An internal perspective relative to which events are perceived.

Codes :

- (a) (b) (c) (d) (a) (b) (c) (d)
 (1) (i) (ii) (iii) (iv) (2) (iii) (iv) (ii) (i)
 (3) (ii) (iii) (iv) (i) (4) (iv) (i) (iii) (ii)

9. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Perception remains constant although the proximal sensation changes.

Reason (R) : Our perceptual system has mechanisms that adjust our perception of the proximal stimulus.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
10. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I
(Concept)

(a) Physiological Psychology

(b) Neuropsychology

(c) Psychophysiology

(d) Cognitive Neuroscience

List - II
(Description)

(i) Studies the psychological effects of brain damage in human patients.

(ii) Studies the neural mechanisms of behaviour by manipulating the nervous systems of non-human animals in controlled experiments.

(iii) Studies the relation between physiological activity and psychological processes in human subjects by non-invasive physiological recording.

(iv) Studies the neural mechanisms of human cognition, ma-

inly by using functional brain imaging techniques.

Codes :

- (a) (b) (c) (d) (a) (b) (c) (d)
 (1) (i) (ii) (iii) (iv) (2) (iii) (iv) (i) (ii)
 (3) (ii) (i) (iv) (iii) (4) (iv) (iii) (ii) (i)

11. Which of the following explains the Threshold Hypothesis regarding Intelligence - Creativity relationship ?

- (1) There is a positive correlation between Intelligence and Creativity.
 (2) There is a positive correlation between Intelligence and Creativity upto a particular level of Intelligence.
 (3) There is a positive correlation between Intelligence and Creativity beyond a particular level of Intelligence.
 (4) There is a negative correlation between Intelligence and Creativity beyond a particular level of Intelligence.

12. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I
(Concept)

(a) Concordance

(b) Penetrance

(c) Prevalence

List - II
(Explanation)

(i) Percentage of cases in which, if specific gene is present, a particular trait, characteristic or disease will actually manifest itself in the fully developed organism.

(ii) Relationship between twins or other family members with respect to a given trait.

(iii) Number of new cases of specific condition or disease that arise during a particular period of time.

- (d) Incidence (iv) Frequency of occurrence of a given condition among a certain population at a particular point of time.

Codes :

- (a) (b) (c) (d) (a) (b) (c) (d)
 (1) (i) (ii) (iii) (iv) (2) (iii) (iv) (i) (ii)
 (3) (ii) (i) (iv) (iii) (4) (iv) (iii) (ii) (i)

13. Which of the following significant changes have been incorporated in DSM-5 ?

- (a) The title of the manual is not DSM-V, but DSM-5, a deliberate shift from traditional Roman numerals used in previous editions toward Arabic numerals.
 (b) Multiaxial Assessment System - a central feature of DSM since its introduction in DSM - III in 1980, has been dropped altogether from DSM-5.
 (c) DSM-5 authors have overhauled the manual to emphasise neuropsychological or biological roots of mental disorders.
 (d) DSM-5 authors have removed 5 of 10 personality disorders previously included in that section. Dropped disorders are paranoid, schizoid, histrionic, dependent and narcissistic personality disorders.

Code :

- (1) (a), (b), (c) are correct; (d) is incorrect.
 (2) (b), (c) and (d) are correct; (a) is incorrect.
 (3) (a) and (b) are correct; (c) and (d) are incorrect.
 (4) (c) and (d) are correct; (a) and (b) are incorrect.

14. Which one of the trait theories / models is / are NOT based on factor analytic methodology ?

- (a) Allport's Theory
 (b) Costa and McCrae's Model
 (c) Goldberg's Model
 (d) Eysenck's Theory

Code :

- (1) (a) only (2) (a) and (c) only
 (3) (b) and (c) only (4) (b) and (d) only

15. The factors that operate with deep levels of processing are :

- (a) Distinctiveness (b) Sameness
 (c) Elaboration (d) Physical contours

Code :

- (1) (a) and (b) only (2) (b) and (c) only
 (3) (b) and (d) only (4) (a) and (c) only

16. Observation learning involves following processes. Arrange them in the correct sequence :

- (a) Attentional processes
 (b) Retention processes
 (c) Production processes
 (d) Motivational processes

Code :

- (1) (a) (b) (c) (d)
 (2) (a) (d) (b) (c)
 (3) (d) (a) (b) (c)
 (4) (a) (b) (d) (c)

17. Key issues in the study of environmental effects on intelligence are :

- (a) Reaction Range
 (b) Proximal and Distal Causation
 (c) Colinearity
 (d) Environment Sphere

Code :

- (1) (a), (b) and (c) only
 (2) (a), (b) and (d) only
 (3) (b), (c) and (d) only
 (4) (a), (c) and (d) only

18. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : According to two factor theory, cognitive appraisal of cues in the environment helps the individual label the emotions.

Reason (R) : An individual is aroused when he confronts any emotion producing event.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
19. Whether we can be affected by stimuli that remain outside our conscious awareness has been examined by researches using a technique called :
- (1) Cuing (2) Visualization
 (3) Imagery (4) Priming
20. Which of the following sequences depicts Erikson's stages of development correctly ?
- (1) Oral Sensory Muscular anal Locomotor genital Latency Adolescence Adulthood.
 (2) Muscular anal Oral Sensory Locomotor genital Latency Adolescence Adulthood.
 (3) Oral Sensory Muscular anal Latency Locomotor genital Adolescence Adulthood.
 (4) Muscular anal Locomotor genital Oral Sensory Latency Adolescence Adulthood.
21. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I (Concept)	List - II (Explanation)
(a) Magnocellular System	(i) Discrimination of fine detail and color
(b) Parvocellular System	(ii) Specialised for brightness contrast and for movement.
(c) Fourier Analysis	(iii) Neurons that fire most when they receive input from both ears at the same time.
(d) Coincidence Detectors	(iv) Analysis of a complex wave into sine wave components.

Codes :

- (a) (b) (c) (d) (a) (b) (c) (d)
 (1) (i) (ii) (iii) (iv) (2) (iii) (iv) (i) (ii)
 (3) (ii) (i) (iv) (iii) (4) (iv) (iii) (ii) (i)

22. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I (Concept)	List - II (Explanation)
(a) Inductive thought	(i) Thought that is intuitive and haphazard.
(b) Logical thought	(ii) Thought that applies a general set of rules to specific situations.
(c) Deductive thought	(iii) Drawing conclusions on the basis of principles of learning.
(d) Illogical thought	(iv) Thinking in which a principle is inferred from a series of specific examples.

Codes :

- (a) (b) (c) (d) (a) (b) (c) (d)
 (1) (i) (ii) (iii) (iv) (2) (ii) (iii) (iv) (i)
 (3) (iii) (iv) (i) (ii) (4) (iv) (iii) (ii) (i)

23. Which of the following personality questionnaires have a lie scale ?

- (a) Edward Personality Preference Schedule
 (b) Eysenck Personality Questionnaire
 (c) Minnesota Multiphasic Personality Questionnaire - 2
 (d) Rotter's Locus of Control Scale.

Code :

- (1) (a) and (b) only (2) (a) and (c) only
 (3) (b) and (c) only (4) (a), (b) and (c) only

24. Read the following statements and describe which are correct :

- (a) Archetypes are themes that have existed in all cultures through history.
 (b) Personal unconscious is the region next to ego which consists of all the forgotten experiences that have lost their intensity.
 (c) Masculine archetype in men is called animus.
 (d) Persona is a compromise between demands of the environment and necessities of individuals.

Code :

- (1) (a), (b) and (c) are correct ; (d) is not correct.
- (2) (b), (c) and (d) are correct ; (a) is not correct.
- (3) (a), (b) and (d) are correct ; (c) is not correct.
- (4) (a), (c) and (d) are correct ; and (b) is not correct.

25. George Sperling (1960) studied iconic memory using :

- (a) Partial-report procedure
- (b) Whole-report procedure
- (c) Self-report procedure
- (d) Backward procedure

Code :

- (1) (a) only (2) (d) only
- (3) (a) and (c) only (4) (b) and (d) only

26. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I	List - II
(Sub Test)	(Test)
(a) Visual Puzzles	(i) WAIS - IV
(b) Famous people	(ii) Kaulfman Adult Intelligence Test (KAIT).
(c) Matrix Task	(iii) Cognitive Assessment System (CAS)
(d) Matrix Patterns	(iv) Otis-Lenon Test

Codes :

- | | | | | | | | | | |
|-----|-------|------|-------|------|-----|------|-------|------|-------|
| (a) | (b) | (c) | (d) | (a) | (b) | (c) | (d) | | |
| (1) | (i) | (ii) | (iii) | (iv) | (2) | (ii) | (i) | (iv) | (iii) |
| (3) | (iii) | (iv) | (i) | (ii) | (4) | (iv) | (iii) | (ii) | (i) |

27. Which of the following terms explains the Reinforcement which is given only if a specific response is made ?

- (1) Concurrent Reinforcement
- (2) Incremental Reinforcement
- (3) Contingency Contract
- (4) Contingent Reinforcement

28. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : A person scored high on Hope of Success (as measured by TAT) and high on Fear of Failure (as measured by Mandler and Sarason's TAQ). As per Atkinson's theory, his resultant achievement motivation will be intermediate.

Reason (R) : As per Atkinson, the resultant achievement motivation is a product of Hope of Success and Fear of Failure.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

29. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Personality factors such as D, J, K and P are not incorporated in Cattell's 16 PF test.

Reason (R) : Some personality factors appeared only in Life - data (L-data) measures.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

30. During which stage most of the creative thinking occurs ?

- (1) Preparation
- (2) Incubation
- (3) Illumination
- (4) Evaluation

31. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I	List - II
(Concept)	(Explanation)
(a) Disjunctive Concept	(i) A class of objects that have two or more features in common.

- (b) Prototype (ii) Showing relationship between object and surroundings.
- (c) Conjunctive Concept (iii) Showing the presence of at least one of several possible features.
- (d) Relational Concept (iv) An ideal model referring to a particular concept.

Codes :

- | | | | | | | | |
|----------|-------|-------|------|-----------|-------|-----|------|
| (a) | (b) | (c) | (d) | (a) | (b) | (c) | (d) |
| (1) (i) | (ii) | (iii) | (iv) | (2) (iii) | (iv) | (i) | (ii) |
| (3) (ii) | (iii) | (iv) | (i) | (4) (iv) | (iii) | (i) | (ii) |

32. Chimpanzee Sarah's one of the most outstanding achievements was the construction of sentences involving :

- (1) Negation
 (2) Conditional relationship
 (3) Adult grammar
 (4) Unprompted questions

33. In a typical 'Learning' experiment which of the following are used as dependent variables ?

- (a) Probability of Response
 (b) Latency
 (c) Rate of Presentation
 (d) Trials to Extinction

Code :

- (1) (a) and (b) only
 (2) (b) and (d) only
 (3) (a), (b) and (c) only
 (4) (a), (b) and (d) only

34. Explanation of the moon illusion given by _____ is, "The horizon seems more distant than the night sky."

- (1) Perceptual learning hypothesis
 (2) Expectancy hypothesis
 (3) Apparent - distance hypothesis
 (4) Frame of reference hypothesis

35. Absorptive phase of energy mobilization involves the following events. Arrange them in the correct sequence.

- (a) Glucose increases.
 (b) Parasympathetic activation.

- (c) Pancreas secretes insulin.
 (d) Glucose enters the body cells.
 (e) Glucose is stored in liver and muscles as glycogen.
 (f) Fat stored in adipose cells as triglycerides.

Code :

- (1) (a) (b) (c) (e) (d) (f)
 (2) (a) (b) (c) (d) (e) (f)
 (3) (a) (c) (b) (d) (f) (e)
 (4) (a) (d) (b) (f) (e) (c)

36. In a multiple regression analysis, the three predictors explained 49 percent variance in the criterion variable. What would be the value of multiple correlation ?

- (1) 0.30 (2) 0.49
 (3) 0.51 (4) 0.70

37. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : The analysis of time series designs often involve the problem of auto correlation.

Reason (R) : Quasi - experiments, as compared to laboratory experiments, do not control secondary variance efficiently.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.

38. Future oriented performance appraisal techniques consist of :

- (a) MBO (b) 360° Appraisal
 (c) Psychological Appraisal
 (d) BARS

Code :

- (1) (a), (b) and (d) (2) (a), (b) and (c)
 (3) (b), (c) and (d) (4) (a), (c) and (d)

39. In group counselling, what is the correct sequence of stages of group development ?

- (a) Forming (b) Norming
 (c) Mourning (d) Storming
 (e) Performing

Code :

- (1) (b) (c) (a) (e) (d)
 (2) (a) (d) (b) (e) (c)
 (3) (a) (c) (e) (b) (d)
 (4) (d) (a) (b) (c) (e)

40. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Pre-operational child is incapable of performing operations.

Reason (R) : Pre-operational child cannot mentally re-arrange a sequence of events into reverse order.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.

41. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : For extraversion, monozygotic twins correlated more than the dizygotic twins.

Reason (R) : Extraversion has genetic basis.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true, but (R) is not the correct Explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.

42. During autobiographical recall, we often reveal a _____, as a consequence, our memory of the past may be _____.

- (1) inconsistency bias; distorted
 (2) consistency bias; distorted
 (3) inconsistency bias; enhanced
 (4) consistency bias; enhanced

43. Persons with a lesion at the ventromedial nucleus are likely to :

- (1) Overeat
 (2) Undereat

- (3) Eat normally
 (4) Eat normally or undereat

44. The percepts are based on :

- (a) What we sense (b) What we know
 (c) What we infer (d) What we conform

Code :

- (1) (a) and (b) only
 (2) (b) and (c) only
 (3) (a), (b) and (c) only
 (4) (a), (b) and (d) only

45. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I

(Concept)

(a) Phrenology

(b) Selective Breeding

(c) Plasticity

(d) Twin studies

List - II

(Explanation)

(i) Field that locates areas of the brain responsible for specific aspects of emotional and behaviour functioning.

(ii) An approach to establish genetic - behaviour relationship through mating of successive generations with a particular trait.

(iii) Ability of parts of neurobiological system to change temporarily and for extended period time.

(iv) An approach to establish of genetic - behaviour relationship through the comparison degree of similarity among identical twins, fraternal twins, and non-twins siblings.

Codes :

- (a) (b) (c) (d) (a) (b) (c) (d)
 (1) (i) (ii) (iii) (iv) (2) (ii) (i) (iv) (iii)
 (3) (iii) (iv) (i) (ii) (4) (iv) (iii) (ii) (i)

Instructions for Questions 46 to 50.

Read the following paragraph and answer the five questions which follow :

A researcher in experimental psychology hypothesized that a particular 'Reward System' for fast reactions would improve the reaction time. To verify this hypothesis, he selected forty subjects in the age range of 25 to 35 years from an organization. Following the standard reaction time experiment, he measured the simple reaction time without the reward system. While taking each trial in the experiment, the time-interval between the ready signal and the onset of stimulus was randomly varied between two to five seconds. In the second part of the experiment, the reward system was introduced and the simple reaction time was measured again for the same subjects. The mean reaction time before the introduction of reward system was 210 milliseconds and the mean reaction time after the introduction of reward system was 250 milliseconds. The difference between the two means was statistically significant at .01 level.

46. In the above study, the researcher's hypothesis is :

- (1) Correlational hypothesis
- (2) Directional hypothesis
- (3) Non-directional hypothesis
- (4) Null hypothesis

47. Which of the following statistical techniques can be used for evaluating the mean differences ?

- (a) Independent samples *t*-test
- (b) Paired samples *t*-test
- (c) Mann-Whitney U-test
- (d) Repeated measures ANOVA

Code :

- (1) (b) only
- (2) (a) and (b) only
- (3) (b) and (d) only
- (4) (a), (b) and (d) only

48. Which one of the following conclusions can be drawn on the basis of the above study ?

- (1) The researcher's hypothesis has been accepted.
- (2) The researcher's hypothesis has been rejected.
- (3) The researcher's hypothesis has been partially accepted.

(4) Inadequate hypothesis to verify the hypothesis.

49. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : In the above study, the foreperiods were randomly varied.

Reason (R) : Random variation of the foreperiods reduces the false reactions.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

50. In this experiment, reaction time can be labelled as :

- (a) Continuous variable
- (b) Dependent variable
- (c) Stimulus variable
- (d) Behavioural variable

Code :

- (1) (a) only
- (2) (a) and (b) only
- (3) (a), (b) and (c) only
- (4) (a), (b) and (d) only

51. In Vertebrates, myelin is produced in the brain and spinal cord by _____, and in rest of nervous system by _____.

- (1) Oligodendrocytes ; Schwann cells
- (2) Schwann cells ; Oligodendrocytes
- (3) Astrocytes ; Microglia
- (4) Microglia ; Astrocytes

52. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Algorithm is a logical rule that guarantees solving a particular problem.

Reason (R) : One can find the solution even to complex problems by following step by step procedure.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (3) (A) is true, but (R) is false.
 (4) (A) is false, but (R) is true.
53. Match List - I with List - II and select the correct answer by choosing from code given below :
- | List - I
(Personality tests) | List - II
(Dimensions covered) |
|---|---|
| (a) EPQ-R | (i) Need for power |
| (b) NEO-PI-3 | (ii) Psychoticism |
| (c) TAT | (iii) Anxiety as a state |
| (d) STAI | (iv) Openness to experience |
- Codes :**
- | (a) | (b) | (c) | (d) | (a) | (b) | (c) | (d) |
|-----|------|------|-------|-------|-----|------|------|
| (1) | (ii) | (iv) | (i) | (iii) | (2) | (iv) | (ii) |
| (3) | (ii) | (iv) | (iii) | (i) | (4) | (i) | (iv) |
54. The basic speech sounds are called _____; and the meaningful smallest units of speech are called _____.
 (1) Phonemes; Fixation
 (2) Phonemes; Lexigram
 (3) Morphemes; Phonemes
 (4) Phonemes; Morphemes
55. Which of the following concept / s is / are not found in Freudian Psychoanalysis ?
 (a) Unconscious motivation
 (b) Collective unconscious
 (c) Striving for superiority
 (d) Instincts
- Code :**
 (1) (a) and (b) only (2) (b) and (c) only
 (3) (c) and (d) only (4) (a), (b) and (d) only
56. Tests of declarative memory are termed as _____ tests, whereas tests of non-declarative memory are termed as _____ tests.
 (1) Long-term memory; explicit memory
 (2) Short-term memory; implicit memory
 (3) Implicit memory; explicit memory
 (4) Explicit memory; implicit memory
57. Which method is used for memorising a poem ?
 (1) Massed practice
 (2) Distributed practice
 (3) Serial memorising
 (4) Syntactic memorising
58. _____ makes neurons stand out visibly depicting which neurons are active; whereas _____ uses antibodies attached to a dye to identify cellular components such as receptors, neurotransmitters, or enzymes.
 (1) Immunocytochemistry; Autoradiography
 (2) Autoradiography; Immunocytochemistry
 (3) Myelin staining; Nissl Staining
 (4) Nissl Staining; Myelin Staining
59. Which of the following are predictors of good prognosis in Schizophrenia.
 (a) Positive symptoms
 (b) Late onset
 (c) Acute onset
 (d) No precipitating factors
- Code :**
 (1) (a), (b) and (c) are correct; (d) is incorrect
 (2) (b), (c) and (d) are correct; (a) is incorrect
 (3) (c), (d) and (b) are correct; (a) is incorrect
 (4) (a), (c) and (d) are correct; (b) is incorrect
60. The body's response to danger is triggered by the release of _____ by the _____ glands.
 (1) Acetylcholine; adrenal
 (2) Epinephrine and norepinephrine; adrenal
 (3) Acetylcholine; pituitary
 (4) Epinephrine and norepinephrine; pituitary
61. Which of the following is the characteristic of Mastery-Oriented Students ?
 (1) They see ability as improvable.
 (2) They feel competent when they succeed.
 (3) They feel worthless when they fail.
 (4) They set unrealistically high goals.
62. A child's cognitive transition from external influences to internal thoughts occurs in four stages. What is the correct order of this transition ?

- (a) Naive Stage
- (b) Natural / Primitive Stage
- (c) Ego-centric speech Stage
- (d) Ingrowth Stage

Code :

- (1) (a) (b) (c) (d)
- (2) (b) (a) (c) (d)
- (3) (a) (b) (d) (c)
- (4) (b) (a) (d) (c)

63. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I (Psychologist)	List - II (Concept)
(a) Bruner	(i) Conservation
(b) Vygotsky	(ii) Iconic mode of representation
(c) Sternberg	(iii) Scaffolding
(d) Piaget	(iv) Componential Analysis

Codes :

- | | |
|-------------------------|-------------------------|
| (a) (b) (c) (d) | (a) (b) (c) (d) |
| (1) (ii) (iii) (iv) (i) | (2) (iv) (ii) (iii) (i) |
| (3) (ii) (iii) (i) (iv) | (4) (iii) (ii) (iv) (i) |

64. Which of the following issues are addressed in the field of Personality Psychology :

- (a) Human Universals
- (b) Individual Differences
- (c) Uniqueness
- (d) Modification of Behaviour

Code :

- (1) (a) and (b) Only
- (2) (b) and (c) Only
- (3) (a), (b) and (c) Only
- (4) (b), (c) and (d) Only

65. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I (Psychologist)	List - II (Concept)
(a) Thorndike	(i) Equipotentiality premise
(b) Pavlov	(ii) Cortical Mosaic
(c) Wertheimer	(iii) Neural Bond
(d) Seligman	(iv) Isomorphism

Codes :

- | | |
|-------------------------|-------------------------|
| (a) (b) (c) (d) | (a) (b) (c) (d) |
| (1) (i) (iv) (iii) (ii) | (2) (ii) (iii) (iv) (i) |
| (3) (iii) (ii) (iv) (i) | (4) (iv) (i) (ii) (iii) |

66. In case of visual stimulus "Where" pathways in the brain are mainly responsible for processing :

- (a) Location
- (b) Colour
- (c) Motion
- (d) Shape

Code :

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (c) and (d) only
- (4) (a) and (c) only

67. Which of the following sequences describes the Cranial Nerves in ascending order ?

- (1) Optic Trochlear Abducens
Auditory - Vestibular Vagus
- (2) Trochlear Optic Auditory -
Vestibular Abducens Vagus
- (3) Optic Abducens Trochlear
Vagus Auditory - Vestibular
- (4) Abducens Trochlear Optic
Auditory - Vestibular Vagus

68. Many psychologists played pivotal role in the emergence of the field of Social Psychology. Arrange them in correct chronological sequence.

- (1) F. Allport; William McDougall; Kurt Lewin; Muzafer Sherif.
- (2) Kurt Lewin; William McDougall; Muzafer Sherif; F. Allport.
- (3) William McDougall; F. Allport; Kurt Lewin; Muzafer Sherif.
- (4) F. Allport; William McDougall; Muzafer Sherif; Kurt Lewin.

69. Kinship Selection Theory suggests that we help others who are related to us because this increases the likelihood that our _____ shall be _____ to future _____.

- (1) Character; transferred; population
- (2) Genes; transmitted; population.
- (3) Genes; transmitted; generation.
- (4) Genes; transferred; population.

70. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Results of experimental studies of bilateral lesion effects are considered more promising than those of unilateral lesions.

Reason (R) : Behaviour effects of bilateral lesions are milder than those of unilateral lesions.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

71. Which of the following are the basic forms of intelligence in Triarchic Theory ?

- (a) Dimensional
- (b) Componential
- (c) Contextual
- (d) Experiential

Code :

- (1) (a), (b) and (c) only
- (2) (a), (b) and (d) only
- (3) (b), (c) and (d) only
- (4) (a), (c) and (d) only

72. In a single factor repeated measures design, the F-ratio, evaluating the effect of independent variable (treatment) is evaluated as :

- (1) $MS_{\text{treatment}} / MS_{\text{subjects}}$
- (2) $MS_{\text{treatment}} / MS_{\text{treatment} \times \text{subjects}}$
- (3) $MS_{\text{treatment}} / (MS_{\text{subjects}} + (MS_{\text{treatment} \times \text{subjects}}))$
- (4) $MS_{\text{treatment}} / MS_{\text{total}}$

73. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I (Leadership style)	List - II (Description)
(a) Task Management	(i) Less focus on both production and workers
(b) Team Management	(ii) More Focus on both production and workers

- (c) Impoverished Management (iii) Less focus on production but more on workers
- (d) Country Club Management (iv) More focus on production but less on workers

Codes :

- | | | | | | | | |
|-----|-------|------|-------|-----|------|-------|------|
| (a) | (b) | (c) | (d) | (a) | (b) | (c) | (d) |
| (1) | (i) | (ii) | (iii) | (2) | (iv) | (iii) | (ii) |
| (3) | (iii) | (iv) | (i) | (4) | (iv) | (ii) | (i) |

74. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Behaviour therapy helps people to respond to life situations in the way they would like to respond.

Reason (R) : Positive therapeutic relationship is a necessary but not sufficient condition for effective behaviour therapy.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

75. Decentration and reversible thought processes are the characteristics of which of the following stage ?

- (1) Sensory - Motor
- (2) Pre - Operational
- (3) Concrete - Operational
- (4) Formal Operations

76. Match List - I with List - II and select the correct answer by choosing from codes given below :

List - I (Concept)	List - II (Explanation)
(a) Central executive	(i) Helps us in playing videogames, jigsaw puzzles, etc.
(b) Visuospatial sketchpad	(ii) Helps us in solving new problems and plan future activities.

- (c) Episodic buffer (iii) Helps us decide what to do next and what not to do.
- (d) Phonological loop (iv) Helps us in reading and mathematical calculations.

Codes :

- (a) (b) (c) (d) (a) (b) (c) (d)
- (1) (i) (ii) (iii) (iv) (2) (iii) (i) (ii) (iv)
- (3) (iv) (iii) (i) (ii) (4) (ii) (iii) (iv) (i)

77. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using codes given below :

Assertion (A) : Working memory is like a workbench where material is constantly being handled, combined and transformed.

Reason (R) : Both new material and old material retrieved from long-term memory are held in working memory.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

78. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Galton developed anthropometric tests of intelligence.

Reason (R) : Psychometric tests were low in reliability and validity.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

79. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Proprioceptive stimuli result from the firing of the kinesthetic receptors in the muscles, joints and tendons of the body.

Reason (R) : A response is conditioned to the stimuli present prior to the ingestion of a primary reinforcer.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

80. Activation of anterior portions of both hemispheres is associated with the _____ of emotions, while activation of the posterior portions of the hemispheres is associated with _____.

- (1) expressions; moods
- (2) valence; arousal
- (3) intensity; expressions
- (4) arousal; valence

81. Signal detection theory identifies two distinct processes in sensory detection :

- (a) sensory process
- (b) decision process
- (c) motivational process
- (d) response bias

Code :

- (1) (a) and (b) only (2) (c) and (d) only
- (3) (a) and (c) only (4) (a) and (d) only

82. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Visual Simple Reaction Time is more than Auditory Simple Reaction Time.

Reason (R) : Visual system involves more complex photochemical processes.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (3) (A) is true, but (R) is false.
- (4) (A) is false, but (R) is true.

83. Creative thinking is different from routine problem solving in respect of :

- (a) Fluency (b) Flexibility
(c) Originality (d) Day dreaming

Code :

- (1) (a) and (b) (2) (b) and (c)
(3) (c) and (d) (4) (a), (b) and (c)

84. The Wechsler type deviation IQ of 115 corresponds to which value of percentile rank?

- (1) 68 (2) 84
(3) 90 (4) 9

85. Match List - I with List - II and select the correct answer by choosing from code given below :

List - I
(Term)

List - II
(Description)

- | | |
|------------------------------|--|
| (a) Conformity | (i) Rules indicating how individuals are expected to behave in a specific situation. |
| (b) Compliance | (ii) Social influence involving direct request from other |
| (c) Social norms | (iii) Social influence where in individuals change their attitude to go along with existing norms. |
| (d) Intense indoctrinization | (iv) Process in which members of extreme group accept the beliefs and rules in unquestioning way. |

Codes :

- | | |
|-------------------------|-------------------------|
| (a) (b) (c) (d) | (a) (b) (c) (d) |
| (1) (i) (ii) (iii) (iv) | (2) (ii) (iii) (iv) (i) |
| (3) (iii) (iv) (i) (ii) | (4) (iii) (iv) (ii) (i) |

86. Spatial processing of location relies on :

- (a) Dorsal pathway
(b) Inferior temporal cortex
(c) Subcortical pathway
(d) Ventral pathway

Code :

- (1) (a) only
(2) (d) only
(3) (b) and (c) only
(4) (a) and (d) only

87. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Jensen has put forward Regression argument in favour of his theory.

Reason (R) : Children of parents having very high intelligence levels will have below average intelligence scores.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.

88. Handling feelings appropriately, ability to soothe oneself, and ability to shake off rampant anxiety are the characteristics of which of the components of Emotional Intelligence as proposed by Goleman ?

- (1) Knowing one's emotions
(2) Managing emotions
(3) Motivating oneself
(4) Handling relationships

89. Which of the following new disorders have been introduced in DSM-5 ?

- (a) Premenstrual Dysphoric Disorder
(b) Mild Neurocognitive Disorder
(c) Disruptive Mood Dysregulation Disorder
(d) Road Rage Behaviour Disorder

Code :

- (1) (a), (b) and (c) are correct; (d) is incorrect.
(2) (a), (b) and (d) are correct; (c) is incorrect.
(3) (a), (c) and (d) are correct; (b) is incorrect.
(4) All (a), (b), (c) and (d) are correct.

90. In selection process, Guilford - Zimmerman Test assesses :

- (1) Mathematical and Verbal skills.
(2) Mental and Personal profile.
(3) Extent of temperamental adaptability.
(4) Analytical and Verbal Skills.

91. 'Mechanistic Behaviour Analyses' are characterised as :

- (a) Behaviour is instigated by the onset of external or internal stimuli.
- (b) Direction of behaviour is determined by S-R bonds or habits.
- (c) Behaviour directed toward the goal persists till the achievement of goal.
- (d) Events related to the goal achievement are encoded, categorised, and transformed into a belief.

Code :

- (1) (a) and (b) correct ; (c) and (d) are incorrect.
 - (2) (b) and (c) correct ; (a) and (d) are incorrect.
 - (3) (a), (b) and (c) are correct ; (d) is incorrect.
 - (4) (b), (c) and (d) are correct ; (a) is incorrect.
92. Guilford (1981) in his modified SOI model mentioned hierarchical structure of intellectual abilities reporting that there are _____ first order factors ; _____ second order factors; and _____ third order factors.
- (1) 150 ; 85 ; 20 (2) 120 ; 80 ; 20
(3) 150 ; 85 ; 16 (4) 180 ; 75 ; 18

93. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : Children have remarkable ability to create mental representation of the world in the form of language.

Reason (R) : Reasoning and problem solving are the ways of manipulating the ideas.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
94. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :
- Assertion (A) :** Forgetting of everyday events in young people is not an indication of a poor memory.

Reason (R) : Ineffective encoding due to failure to attend to an event while it is happening results in failure to remember.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.
95. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :
- Assertion (A) :** Premack principle, based on Probability-Differential Hypothesis, states that the opportunity to engage in a frequently occurring activity can be used to reinforce less frequently activity.
- Reason (R) :** According to Probability-Differential Hypothesis, restricted access to a response makes that response reinforcing and excessive access to that response makes that response punishing.
- Code :**
- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - (3) (A) is true, but (R) is false.
 - (4) (A) is false, but (R) is true.

Instructions for questions 96 to 100 :

Read the following paragraph and answer the five questions which follow :

An educational psychologist wanted to develop an achievement test in physics for graduate students. After going through the usual test construction procedure, he retained 100 items; each item having five alternatives, one of them being correct. The psychologist wanted to examine the factor structure of the resulting achievement test. Using a sample of 1,000 subjects, the psychologist obtained the inter-item correlations. He obtained the initial estimates of communality and then decided to factor analyze the inter-item correlations matrix by following the common-factor model.

He extracted the unrotated factors and retained the first nine factors for rotation. The nine unrotated factors explained 67.5 percent of variance. After factor rotation, all the inter-factor correlations were found to be zero. The factor structure of the achievement test of physics was found to be interpretable.

96. The inter-item correlations would be :

- (1) Biserial correlations
- (2) Kendall's tau
- (3) Phi-coefficients
- (4) Point-biserial correlations

97. Read each of the following statements - Assertion (A) and Reason (R); and indicate your answer using code given below :

Assertion (A) : For the factor analysis of item scores, Common-Factor model is quite often more suitable.

Reason (R) : Item scores are quite often less reliable than the psychometrically measured trait variables.

Code :

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

98. Which one of the following method of factor extraction, the psychologist would NOT use in the present study ?

- (1) Principal axes method
- (2) Principal components method
- (3) Psychometric (alpha) factor analysis
- (4) Maximum likelihood method

99. Which method /s of rotation has / have most probably been used in the present study ?

- (a) Oblimin
- (b) Promax
- (c) Quartimax
- (d) Varimax

Code :

- (1) (a) only
- (2) (c) only
- (3) (c) and (d) only
- (4) (b), (c) and (d) only

100. The second-order factor analysis of the nine rotated factors is not recommended in the present study because :

- (1) There are only nine first-order factors.
- (2) The interfactor correlations are zero.
- (3) The factors obtained by common-factor model are not subjected to second-order factor analysis.
- (4) Unrotated factors are more suitable for second-order factor analysis.

Answer Sheet

1. (4)	2. (1)	3. (2)	4. (1)	5. (4)	6. (1)	7. (2)	8. (2)	9. (1)	10. (3)
11. (2)	12. (3)	13. (3)	14. (1)	15. (4)	16. (1)	17. (1)	18. (2)	19. (4)	20. (1)
21. (3)	22. (4)	23. (3)	24. (3)	25. (1)	26. (1)	27. (4)	28. (3)	29. (1)	30. (2)
31. (2)	32. (2)	33. (4)	34. (3)	35. (2)	36. (4)	37. (2)	38. (2)	39. (2)	40. (1)
41. (1)	42. (2)	43. (1)	44. (3)	45. (1)	46. (2)	47. (3)	48. (2)	49. (1)	50. (4)
51. (1)	52. (1)	53. (1)	54. (4)	55. (2)	56. (4)	57. (2)	58. (2)	59. (4)	60. (2)
61. (1)	62. (2)	63. (1)	64. (3)	65. (3)	66. (4)	67. (1)	68. (3)	69. (3)	70. (2)
71. (3)	72. (2)	73. (4)	74. (2)	75. (3)	76. (2)	77. (2)	78. (3)	79. (1)	80. (2)
81. (1)	82. (1)	83. (4)	84. (2)	85. (3)	86. (1)	87. (3)	88. (2)	89. (1)	90. (3)
91. (3)	92. (3)	93. (2)	94. (1)	95. (3)	96. (3)	97. (1)	98. (2)	99. (3)	100. (2)